

Geometry at the ICM

278 Problems, Their Histories, and Open Frontiers

**A research survey of the Geometry collection at icmconjectures.com
Frozen dataset edition: 14 August 2026**

Research compilation prepared with Codex

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Preface

This volume organizes all 278 records classified as *Geometry* in the public ICM Conjectures database as it stood on 14 August 2026. Its aim is simultaneously historical and forward-looking: each record contains the problem statement, its originating ICM source, the literature and progress narrative supplied by the database, and every separately listed remaining open problem. The organization follows the database’s 11 geometry subfields, while the audit appendices make the coverage independently checkable.

The International Mathematical Union’s proceedings archive spans ICM volumes from 1893 through 2022 and provides the primary historical backbone for the collection: <https://www.mathunion.org/icm/proceedings/1920>. The present survey is not an official publication of the IMU, and the ICM Conjectures database is a separate project. The distinction matters: this is a research compilation of a live, editable dataset, not a declaration of mathematical consensus.

How to read this volume

Each numbered section corresponds to exactly one website record. The blue box reproduces the mathematical statement retained by the dataset. It is followed by the database’s historical and bibliographic account, including the stated verdict and discussion of what has and has not been proved. Red-edged boxes contain descendant questions explicitly entered in the website’s “Some remaining open problems” field. The final primary-source pointer preserves the originating ICM URL.

The label *partially resolved* should be read literally. It may mean that one implication is known, that special dimensions or regularity classes are settled, that a formerly broad program now has a precise surviving core, or that strong estimates fall short of a sharp conjectural bound. It is not a uniform measure of distance from a solution.

Editorial and verification status

The source payload marks 5 of the 278 Geometry records as reviewed, 272 as unreviewed, and 1 as under review. This volume exposes that field for every record rather than hiding it. URLs, DOIs, and arXiv pointers are retained verbatim in the scholarly narratives and also extracted into a machine-readable link inventory. Bibliographic presence is therefore auditable, but inclusion in this edition must not be confused with independent peer review of every status claim.

Scope at a glance

Measure	Count
Geometry records	278
Subfields	11
Source-year span	1893–2022
Partially resolved records	149
Open or open-ended records	129
Resolved records	0
Unclassified records	0
Dataset-reviewed records	5
Dataset-unreviewed records	272
Separately listed remaining open problems	393
Records with at least one separate open problem	226
Unique original ICM source records	167
Unique URLs retained in the audit inventory	2205

The 11 subfields and their record counts are:

Subfield	Records
Global/Riemannian geometry	78
Discrete and polyhedral geometry	35
Classical and local differential geometry	33
Symplectic and contact geometry	31
Metric and topological geometry	28
Manifolds and global analysis	21
Geometric flows and geometric PDE	19
Convex geometry	12
Applications of geometry / global analysis	8
Incidence, axiomatic, and classical geometry	8
Singularity theory	5
Total	278

A historical map of the collection

From classical configurations to global structures

The earliest records belong to a period in which geometry was being recast through topology, Lie theory, and the calculus of variations. Problems about curves, surfaces, rigidity, and incidence retain classical formulations, yet their later histories increasingly depend on global invariants. This is one reason the collection resists a simple chronology: a question that looks local may ultimately be governed by index theory, a moduli space, or the topology of a transformation group.

Curvature as topology, analysis, and obstruction

Across the Riemannian and global-analysis chapters, curvature conditions serve three roles. They are geometric hypotheses controlling comparison and compactness; analytic coefficients in nonlinear PDE; and topological obstructions detected through characteristic classes or operator indices. The Hopf-type sign questions, scalar-curvature rigidity problems, and positive-curvature classification programs illustrate how these roles interlock. Progress often consists of proving the predicted conclusion under spin, symmetry, dimension, or regularity hypotheses while the unrestricted statement persists.

The symplectic and categorical turn

Late twentieth-century ICM lectures introduced a cluster of questions in which the relevant geometric object is no longer only a manifold, but a category, field theory, or enumerative package attached to it. Floer theory, Gromov–Witten invariants, homological mirror symmetry, and gauge-theoretic constructions appear throughout the collection. In this setting “solution” can require equivalence between independently developed frameworks, together with foundations strong enough to compare all cases in the original scope.

Metric geometry and quantitative modernity

The metric chapter shows a complementary movement: remove linear or smooth structure, then ask how much of it can be recovered quantitatively. Embedding distortion, dimension reduction, coarse assembly, and the structure of metric limits replace exact isomorphism by controlled approximation. Here the unresolved part is often a sharp exponent, a dimension-free constant, an endpoint, or the passage from average to worst-case control.

Why old problems remain mathematically active

The records collectively demonstrate that longevity does not imply stagnation. A conjecture may generate techniques that settle entire neighboring theories without deciding the original endpoint. Conversely, a new conceptual framework may reveal that an old statement contains several logically distinct claims. The problem-by-problem format preserves these differences and makes the open frontier explicit rather than compressing it into a single binary status.

Method and conventions

Frozen source

The complete JSON response from <https://icmconjectures.com/api/records> was downloaded on 14 August 2026. The source reported dataset version 1786281211823164465-21207818 and an update timestamp of 2026-08-09T13:13:31+00:00. The full payload and the extracted Geometry-only snapshot are included in the project.

Completeness rule

A record is in scope if and only if its API field `category` equals `Geometry`. The resulting count is 278. Records are ordered first by the website’s subfield order and then by source year and record identifier. Each API identifier occurs once in the body and once in the master coverage table.

Status normalization

For tabulation only, an early status phrase in each progress narrative is normalized to *Partially resolved*, *Open/open-ended*, *Resolved*, or *Unclassified*. No mathematical inference is made from this normalization. The full wording of the underlying assessment is retained in the corresponding record.

Citation convention

The original ICM source URL is displayed in the metadata box and the primary-source pointer. Later literature is presented locally inside each problem’s research record because the evidence is problem-specific and often explains exactly which theorem proves which subcase. DOI and arXiv URLs are left clickable. The source catalogue appendix consolidates original ICM sources, while the machine-readable link inventory covers every extracted URL.

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Part I.

The 278 Geometry Problems

1. Global/Riemannian geometry

This chapter forms the largest group in the collection. Its records run from global questions about curvature, topology, and rigidity to Kähler geometry, comparison geometry, geometric group actions, and scalar curvature. A recurring historical pattern is the replacement of a qualitative geometric question by an analytic or variational criterion: curvature is linked to index theory, canonical metrics to stability, and geometric limits to quantitative regularity. Several problems have celebrated solutions in important special cases while their dimension-free or fully general forms remain open.

Coverage: 78 website records; **source years:** 1958–2022; **separately listed descendant open problems:** 103.

1.1. 1. DIFFERENTIAL GEOMETRY AND INTEGRAL GEOMETRY [1958-geom-7]

Record: 1958-geom-7

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1958: DIFFERENTIAL GEOMETRY AND INTEGRAL GEOMETRY

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1958/ICM1958.ocr.pdf#page=505>

1.1.1. Problem statement

Let M^n be a compact differentiable n -manifold. For an immersion $x : M^n \rightarrow E^{n+N}$ into a Euclidean space, let B_ν be the bundle of unit normal vectors to the immersed submanifold $x(M^n)$. Fix an origin in E^{n+N} , let S_0 be the unit sphere centered at that origin, and let $T : B_\nu \rightarrow S_0$ send each unit normal vector to the endpoint of the parallel unit vector based at the origin. Define the total curvature $T_x(M^n)$ of the immersion x to be the volume of the image $T(B_\nu)$, with each point counted with multiplicity equal to the number of its preimages under T , divided by the volume of S_0 . Define

$$T(M^n) = \min_x T_x(M^n),$$

where the minimum is taken over all immersions x of M^n into Euclidean spaces. Prove or disprove that, for every compact differentiable manifold M^n , the minimum total curvature $T(M^n)$ is equal to the minimum total number of cells in any subdivision of M^n

as a cell complex.

1.1.2. Historical background, literature, and current progress

1.1.3. Verdict: partially resolved

1.1.3.1. Precise problem

The statement is the Chern-Lashof total absolute curvature problem. For a compact smooth differentiable manifold M , one averages, over unit normal directions, the number of preimages of the spherical normal map of a Euclidean immersion. Equivalently, for generic height functions, this total absolute curvature is the mean number of critical points of the height functions. The question is whether the best possible value over Euclidean immersions of M is exactly the smallest possible number of cells in a cell decomposition of M , usually expressed in the later literature as the Morse number $\mu(M)$, the minimum number of critical points of a Morse function on M .

1.1.3.2. Verdict

Partially resolved. The literature proves major versions of the Chern-Lashof problem, especially the infimum/Morse-number formulation and high-dimensional fixed-ambient variants, but I did not find a published result that covers the exact statement as written for every compact differentiable manifold with an actual minimum over all immersions. In particular, the modern references generally formulate the invariant as an infimum, and attainment is a separate tightness/minimal-immersion question.

1.1.3.3. Literature checked

1. On the total curvature of immersed manifolds

Authors: Shiing-Shen Chern and Richard K. Lashof.

Venue: *American Journal of Mathematics* 79(2), 306-318.

Year: 1957.

DOI/URL: <https://doi.org/10.2307/2372684>.

Role: Original source for the Euclidean total curvature theory. It proves foundational Chern-Lashof lower bounds, including lower bounds by topology and the convex hypersurface characterization in the minimal sphere case, but it does not settle the full all-manifolds equality asserted in the problem statement.

2. On the total curvature of immersed manifolds, II

Authors: Shiing-Shen Chern and Richard K. Lashof.

Venue: *Michigan Mathematical Journal* 5(1), 5-12.

Year: 1958.

DOI/URL: <https://doi.org/10.1307/mmj/1028998005>.

Role: Continuation of the original work and the source context for the Chern-Lashof conjectural comparison between total curvature and the minimal cell/Morse count. It provides foundational inequalities and special cases, not a full general proof.

3. Minimal total absolute curvature for immersions

Author: Nicolaas H. Kuiper.

Venue: *Inventiones Mathematicae* 10, 209-238.

Year: 1970.

DOI/URL: <https://doi.org/10.1007/BF01403250>.

Role: Develops the tight/minimal total absolute curvature viewpoint. The abstract explicitly says that many manifolds have no tight smooth immersion in any Euclidean space, so attainment of the optimal lower bound is not automatic in the exact “minimum” sense.

4. Geometry in total absolute curvature theory

Author: Nicolaas H. Kuiper.

Venue: *Perspectives in Mathematics*, Oberwolfach 1984, Birkhäuser, Basel, pp. 377-392.

Year: 1984.

URL: https://repo-archives.ihes.fr/FONDS_IHES/I_Prepublications/KUIPER/1972-1991/M_84_08/M_84_08_web.pdf.

Role: States the Morse-number formulation: if $\gamma(M)$ is the minimum number of critical points of a smooth nondegenerate real function, then the infimum of total absolute curvature over Euclidean embeddings is known to be $\gamma(M)$. This supports the infimum version, not universal attainment of a minimum.

5. Total absolute curvature and embedded Morse numbers

Author: R. W. Sharpe.

Venue: *Journal of Differential Geometry* 28(1), 59-92.

Year: 1988.

DOI/URL: <https://doi.org/10.4310/jdg/1214442160>.

Role: Computes infima of total absolute curvature for smooth embeddings in fixed ambient situations under mild hypotheses, using embedded Morse theory. This is substantial progress but has hypotheses and does not by itself cover the unrestricted all-dimensional minimum statement.

6. A proof of the Chern-Lashof conjecture in dimensions greater than five

Author: R. W. Sharpe.

Venue: *Commentarii Mathematici Helvetici* 64, 221-235.

Year: 1989.

DOI/URL: <https://doi.org/10.1007/BF02564672>.

Role: Proves that, for smooth manifolds of dimension $m > 5$, the best possible lower bound for total absolute curvature is the Morse number $\mu(M)$. The dimension restriction means it is not a full all-dimensional resolution of the statement in the problem statement.

7. Taut immersions into complete Riemannian manifolds

Authors: Chuu-Lian Terng and Gudlaugur Thorbergsson.

Venue: *Tight and Taut Submanifolds*, Mathematical Sciences Research Institute Publications 32, Cambridge University Press, pp. 181-228.

Year: 1997.

URL: <https://library.slmath.org/books/Book32/files/terng.pdf>.

Role: Survey/context reference. It states that Chern-Lashof's proof gives $\tau(M) \geq \mu(M)$ and that Sharpe proved $\tau(M) = \mu(M)$ under mild dimension assumptions, confirming that the broad literature treats the available theorem as conditional rather than as a fully general minimum-attainment theorem.

8. The circular Morse-Smale characteristic of closed surfaces

Authors: Dorin Andrica, Dana Mangra, and Cornel Pintea.

Venue: *Bulletin Mathématique de la Société des Sciences Mathématiques de Roumanie* 57(105), no. 3, 235-242.

Year: 2014.

URL: https://ssmr.ro/bulletin/pdf/57-3/articol_1.pdf.

Role: Modern reference stating the Chern-Lashof conjecture as the assertion that the Morse-Smale characteristic is the infimum of total curvatures over Euclidean embeddings and noting that it holds for several manifolds. This supports the conclusion that related special cases are known, but not an exact full-resolution claim for the problem as written.

1.1.3.4. What remains unresolved for this problem

The sources above justify strong partial progress: the optimal lower bound/infimum is identified with the Morse number in important formulations, and Sharpe proves a high-dimensional version. However, the problem asks for a minimum over all immersions of every compact differentiable manifold and equality with the minimum cell count. The literature I found does not provide a theorem proving that exact minimum-attainment statement for all compact differentiable manifolds. Because the modern formulation is usually an infimum and tight/minimal immersions may fail to exist for many manifolds, the exact statement is not fully resolved by the cited literature.

1.1.4. Remaining open problems recorded by the database

1.1.4.1. Open problem 1

Prove the Chern–Lashof conjecture for closed smooth 5-manifolds: show that the infimum of the total absolute curvatures of their Euclidean immersions equals their Morse number (the minimum number of critical points of a Morse function). This is the first dimension not covered by Sharpe’s theorem for dimensions greater than five.

Reference supplied by the database. R. W. Sharpe, “A proof of the Chern–Lashof conjecture in dimensions greater than five,” *Commentarii Mathematici Helvetici* **64** (1989), no. 2, 221–235, especially Theorem A (which proves the conjecture only for $m > 5$), DOI: <https://doi.org/10.1007/BF02564672>. The conjecture continued to be stated without a general low-dimensional solution in Dorin Andrica, Dana Mangra, and Cornel Pinteau, “The circular Morse–Smale characteristic of closed surfaces,” *Bulletin Mathématique de la Société des Sciences Mathématiques de Roumanie* **57(105)** (2014), no. 3, 235–242, introduction, [article PDF](#).

1.1.5. Primary-source pointer

The originating ICM record is *DIFFERENTIAL GEOMETRY AND INTEGRAL GEOMETRY* (1958). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1958/ICM1958.ocr.pdf#page=505>.

1.2. 2. The Hopf Product Conjecture [1970-geom-12]

Record: 1970-geom-12

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: reviewed (reviewed flag: yes)

Original source: ICM 1970: Differential geometry; its past and its future

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1970.1/ICM1970.1.ocr.pdf#page=79>

1.2.1. Problem statement

Does $S^2 \times S^2$, the product of two 2-dimensional spheres, admit a Riemannian metric of strictly positive sectional curvature?

1.2.2. Historical background, literature, and current progress

1.2.3. Verdict: open

1.2.3.1. Precise problem

The problem asks the classical Hopf product problem: whether the smooth manifold $S^2 \times S^2$ admits any Riemannian metric of strictly positive sectional curvature.

The expected conjectural answer in the literature is negative, but I found no published or otherwise citable theorem that proves nonexistence for all Riemannian metrics on $S^2 \times S^2$, and no construction of such a positive-curvature metric.

1.2.3.2. Current status

The exact unrestricted problem remains open. Recent survey literature still describes it as open in general, while noting that it is settled in special metric classes. The correct status is therefore **Open**.

1.2.3.3. Original source and status references

- Title: *Differential Geometry; Its Past and Its Future*
- Authors: Shiing-Shen Chern; records a question asked by H. Hopf
- Venue/year: Actes, Congrès international des mathématiciens, Nice 1970, Tome 1, pp. 41-53
- DOI/arXiv/URL: <https://www.hiroleetanaka.com/pdfs/2014-fall-1-230a-icm1970-chern-differential-geometry.pdf>
- Relevance: Problem I asks exactly whether the product of two 2-dimensional spheres admits a Riemannian metric of strictly positive curvature.

- Title: *Bounds on Sectional Curvature and Interactions with Topology*
- Author: Manuel Amann
- Venue/year: Jahresbericht der Deutschen Mathematiker-Vereinigung 123, 27-55, 2021; published online 2020
- DOI/arXiv/URL: <https://doi.org/10.1365/s13291-020-00225-x>
- Relevance: States the Hopf conjecture for $S^2 \times S^2$, says it is settled only in special metric classes, and says it is widely open in general.
- Title: *Harmonic 2-forms and positively curved 4-manifolds*
- Authors: Kefeng Liu and Jianming Wan
- Venue/year: Science China Mathematics 64, 1613-1620, 2021; published online 2020
- DOI/arXiv/URL: <https://doi.org/10.1007/s11425-019-1641-9>; arXiv:1909.00110
- Relevance: The introduction says the Hopf conjecture $S^2 \times S^2$ has no positive sectional curvature metric is still widely open except for special cases.

1.2.3.4. Special cases progress

- Title: *On the topology of positively curved 4-manifolds with symmetry*
- Authors: Wu-Yi Hsiang and Bruce Kleiner
- Venue/year: Journal of Differential Geometry 29(3), 615-621, 1989
- DOI/arXiv/URL: <https://doi.org/10.4310/jdg/1214443064>
- Result and limitation: A compact simply connected positively curved 4-manifold with an effective isometric circle action is homeomorphic to S^4 or $\mathbb{CP}^2$. Since $S^2 \times S^2$ is neither, this rules out positive-curvature metrics on $S^2 \times S^2$ with nontrivial circle symmetry. It does not rule out metrics with finite or trivial isometry group.
- Title: *A Remark On Positively Curved 4-Manifolds*
- Author: Christian Bär
- Venue/year: Differential Geometry and its Applications 4(1), 71-75, 1994
- DOI/arXiv/URL: [https://doi.org/10.1016/0926-2245\(94\)90009-4](https://doi.org/10.1016/0926-2245(94)90009-4); PDF: https://www.math.uni-potsdam.de/fileadmin/user_upload/Prof-Geometrie/Dokumente/Publikationen/hopf.pdf
- Result and limitation: If a compact oriented connected 4-manifold has indefinite intersection form, then it admits no metric with $K \geq 1$ and $\|\nabla R\|_{L^\infty} \leq 2/\pi$. This applies to $S^2 \times S^2$, but only under the extra derivative bound.
- Title: *Geometrically formal 4-manifolds with nonnegative sectional curvature*
- Author: Christian Bär
- Venue/year: Communications in Analysis and Geometry 23, 479-497, 2015
- DOI/arXiv/URL: <https://doi.org/10.4310/CAG.2015.v23.n3.a2>; arXiv:1212.1325
- Result and limitation: Shows that the Hopf product conjecture - namely, the nonexistence

of a positive-sectional-curvature metric on $S^2 \times S^2$ - holds for formal metrics and related harmonic-form length conditions. This is a restriction on the metric class and does not cover arbitrary metrics on $S^2 \times S^2$.

- Title: *Positive biorthogonal curvature on $S^2 \times S^2$*
- Author: Renato G. Bettiol
- Venue/year: Proceedings of the American Mathematical Society 142(12), 4341-4353, 2014
- DOI/arXiv/URL: <https://doi.org/10.1090/S0002-9939-2014-12173-3>; arXiv:1210.0043
- Result and limitation: Constructs metrics on $S^2 \times S^2$ with positive biorthogonal curvature (and in fact positive Ricci curvature and positive distance curvature $\sec_\theta$ for all $\theta > 0$), a package Bettiol describes as intermediate between admitting $\text{Ric} > 0$ and admitting $\sec > 0$.
- Title: *Harmonic 2-forms and positively curved 4-manifolds*
- Authors: Kefeng Liu and Jianming Wan
- Venue/year: Science China Mathematics 64, 1613-1620, 2021; published online 2020
- DOI/arXiv/URL: <https://doi.org/10.1007/s11425-019-1641-9>; arXiv:1909.00110
- Result and limitation: If a compact oriented 4-manifold has positive sectional curvature and all harmonic 2-forms satisfy $|\nabla \phi|^2 \geq 2|d|\phi||^2$, a Kato-type inequality, then its intersection form is definite. Since $S^2 \times S^2$ is indefinite, this would rule it out under the assumption of the Kato-type inequality. The inequality is not known for arbitrary positive-curvature metrics.
- Title: *Conformal Metrics to a Product or Doubly Warped Product on $S^2 \times S^2$ and the Hopf Conjecture*
- Authors: Thierno Seck and Athoumane Niang
- Venue/year: Journal of Mathematics 2022, Article ID 6268017, 5 pp., 2022
- DOI/arXiv/URL: <https://doi.org/10.1155/2022/6268017>
- Result and limitation: Proves no positive sectional curvature within two metric classes: conformal metrics to the product metric and doubly warped product metrics. These classes do not include all Riemannian metrics on $S^2 \times S^2$.
- Title: *Metrics Induced by $\mathbb{R}^{6+k}$ on the Graph of k -Smooth Functions and the Hopf Conjecture*
- Authors: Thierno Seck, Athoumane Niang, and Adama Thiandoum
- Venue/year: In *Nonlinear Analysis, Geometry and Applications*, Trends in Mathematics, Birkhäuser, pp. 285-298, 2024
- DOI/arXiv/URL: https://doi.org/10.1007/978-3-031-52681-7_13
- Result and limitation: By Perdomo's Theorem A (Isr. J. Math. 156 (2006)), every Riemannian metric on $S^2 \times S^2$ is, up to homothety, isometric to a graph metric over the standard embedding $S^2 \times S^2 \subset \mathbb{R}^6$ in sufficiently high codimension; since homothety preserves the sign of sectional curvature, the preprint's claimed theorem — that no graph of an $\mathbb{R}^k$ -valued function over $S^2 \times S^2$ has positive sectional curvature — would, if correct, resolve the conjecture in full. The claim is equivalent in strength to the conjecture itself; it appears only in an unrefereed preprint, has not been validated, and the refereed literature

through 2026 continues to treat the conjecture as open. We therefore do not count it. (Perdomo's published Theorem B establishes the graph statement only for scalar-valued F over the product metric.)

1.2.3.5. What remains unresolved

It remains unknown whether an arbitrary Riemannian metric on $S^2 \times S^2$ can have strictly positive sectional curvature everywhere, or whether such metrics are impossible without extra hypotheses such as symmetry, pinching/derivative bounds, formality-type conditions, Kato inequalities, conformal/warped-product form, or graph-induced form.

1.2.4. Remaining open problems recorded by the database

1.2.4.1. Open problem 1

Let ϕ be a harmonic 2-form on a compact Riemannian 4-manifold of positive sectional curvature. Must the strengthened Kato inequality $|\nabla \phi|^2 \geq 2|d\phi|^2$ hold?

Reference supplied by the database. Kefeng Liu and Jianming Wan, "Harmonic 2-forms and positively curved 4-manifolds," *Science China Mathematics* **64** (2021), no. 7, 1613–1620, explicit question at the end of Section 4; Theorem 1.1 proves that an affirmative answer would imply definiteness and hence the Hopf product conjecture, DOI: <https://doi.org/10.1007/s11425-019-1641-9>, [arXiv:1909.00110](https://arxiv.org/abs/1909.00110).

1.2.5. Primary-source pointer

The originating ICM record is *Differential geometry; its past and its future* (1970). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1970.1/ICM1970.1.ocr.pdf#page=79>.

1.3. 3. Differential geometry; its past and its future [1970-geom-13]

Record: 1970-geom-13

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1970: Differential geometry; its past and its future

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1970.1/ICM1970.1.ocr.pdf#page=79>

1.3.1. Problem statement

Prove that if M is a compact orientable Riemannian manifold of even dimension n , and $\chi(M)$ denotes its Euler-Poincare characteristic, then:

- if all sectional curvatures of M are nonnegative, then $\chi(M) \geq 0$;
- if all sectional curvatures of M are nonpositive, then $\chi(M) \geq 0$ when $n \equiv 0 \pmod{4}$, and $\chi(M) \leq 0$ when $n \equiv 2 \pmod{4}$.

1.3.2. Historical background, literature, and current progress

1.3.3. Verdict: partially resolved

1.3.3.1. Precise problem

This is the Hopf/Chern-Hopf Euler characteristic sign problem for compact even-dimensional Riemannian manifolds whose sectional curvatures have a fixed sign:

- if $sec \geq 0$, prove $chi(M) \geq 0$;
- if $sec \leq 0$, prove $(-1)^{(n/2)}chi(M) \geq 0$, equivalently $chi(M) \geq 0$ for $n \equiv 0 \pmod{4}$ and $chi(M) \leq 0$ for $n \equiv 2 \pmod{4}$.

The problem asks for both halves, in all compact orientable even dimensions.

1.3.3.2. Status assessment

The problem is not fully resolved. The nonnegative sectional-curvature half remains open in general: the positive-curvature Hopf conjecture, a special case of $sec \geq 0$, is still described in modern literature as open in dimensions 6 and higher. There is substantial partial progress in dimension 4 and under additional hypotheses such as symmetry, Kahler structure, higher-order curvature positivity, or special topological/geometric assumptions.

For the nonpositive half, there is a recent arXiv preprint by Fang and Wang that explicitly claims the full nonpositive case. Its Theorem 1.9, as stated, has hypotheses and conclusion matching the nonpositive half of this problem. However, even accepting that citable preprint

as progress, it does not address the $\sec \geq 0$ half, so it cannot fully resolve the stated combined statement.

1.3.3.3. Key literature and progress

Original/source formulation

- Title: Differential geometry: its past and its future
- Authors: Shiing-Shen Chern
- Venue: Actes du Congrès International des Mathématiciens 1970, Vol. 1, Gauthier-Villars, Paris, pp. 41-53
- Year: 1971
- DOI/arXiv/URL: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1970.1/ICM1970.1.ocr.pdf> ; alternate PDF <https://www.hiroleetanaka.com/pdfs/2014-fall-230a-icm1970-cherndifferential-geometry.pdf>
- Result/explanation: Problem III and Conjecture is exactly this problem's question: for compact orientable even-dimensional M , Chern asks for the Euler characteristic sign under $\sec \geq 0$ and $\sec \leq 0$, and notes that it was known for $n = 4$ and constant sectional curvature.

Low-dimensional result

- Title: On curvature and characteristic classes of a Riemannian manifold
- Authors: Shiing-Shen Chern
- Venue: Abhandlungen aus dem Mathematischen Seminar der Universität Hamburg, 20, 117-126
- Year: 1955
- DOI/arXiv/URL: <https://doi.org/10.1007/BF02960745>
- Result/explanation: Gives the Chern-Gauss-Bonnet curvature approach used in the 4-dimensional case; Chern's 1970 ICM article cites this as part of the known $n = 4$ resolution.
- Title: Some implications of the generalized Gauss-Bonnet theorem
- Authors: Richard L. Bishop and Samuel I. Goldberg
- Venue: Transactions of the American Mathematical Society, 112(3), 508-535
- Year: 1964
- DOI/arXiv/URL: <https://doi.org/10.1090/S0002-9947-1964-0163271-8>
- Result/explanation: Theorem 1.1 proves that a compact oriented 4-manifold with non-negative or nonpositive sectional curvature has nonnegative Euler characteristic, with strict positivity in the strictly positive or strictly negative cases. This resolves the problem statement in dimension 4, but not in higher even dimensions.

Obstruction to the naive Gauss-Bonnet-integrand approach

- Title: Positive sectional curvatures does not imply positive Gauss-Bonnet integrand

- Authors: Robert Geroch
- Venue: Proceedings of the American Mathematical Society, 54, 267-270
- Year: 1976
- DOI/arXiv/URL: <https://doi.org/10.1090/S0002-9939-1976-0390961-8>
- Result/explanation: Constructs a 6-dimensional algebraic curvature tensor with positive sectional curvatures but negative Gauss-Bonnet integrand. This is not a counterexample to Hopf's conjecture, but it shows why pointwise positivity of the Gauss-Bonnet integrand cannot prove the general result.

Higher-order/special curvature conditions

- Title: Euler-Poincare characteristic and higher order sectional curvature. I
- Authors: Chuan-Chih Hsiung and Kenneth Michael Shiskowski
- Venue: Transactions of the American Mathematical Society, 305(1), 113-128
- Year: 1988
- DOI/arXiv/URL: <https://doi.org/10.1090/S0002-9947-1988-0920149-3>
- Result/explanation: Treats Hopf's conjecture under higher-order sectional-curvature assumptions. The paper itself describes the stated Hopf conjecture as a long-standing open conjecture, so it is partial progress only.

Kahler and related nonpositive-curvature progress

- Title: Kahler hyperbolicity and L2-Hodge theory
- Authors: Mikhail Gromov
- Venue: Journal of Differential Geometry, 33(1), 263-292
- Year: 1991
- DOI/arXiv/URL: <https://doi.org/10.4310/jdg/1214446039>
- Result/explanation: Proves Euler characteristic sign results for Kahler hyperbolic manifolds using L2-Hodge theory. This covers important Kahler/negatively curved situations but assumes additional Kahler-type structure and does not prove the full Riemannian statement.
- Title: Kahler parabolicity and the Euler number of compact manifolds of non-positive sectional curvature
- Authors: Jianguo Cao and Frederico Xavier
- Venue: Mathematische Annalen, 319, 483-491
- Year: 2001
- DOI/arXiv/URL: <https://doi.org/10.1007/PL00004444>
- Result/explanation: Proves that if a compact $2n$ -dimensional Riemannian manifold has nonpositive sectional curvature and is homeomorphic to a Kahler manifold, then $(-1)^n \chi(M) \geq 0$. The extra hypothesis "homeomorphic to a Kahler manifold" means this is not the full nonpositive Riemannian case.

Symmetry results for positive/nonnegative side

- Title: On the Hopf conjecture with symmetry
- Authors: Lee Kennard
- Venue: Geometry & Topology, 17(1), 563-593
- Year: 2013
- DOI/arXiv/URL: <https://doi.org/10.2140/gt.2013.17.563> ; arXiv:1203.3808
- Result/explanation: Proves the positive-curvature Hopf conjecture under additional torus-symmetry assumptions. The introduction states that the conjecture holds in dimensions 2 and 4 but remains open in higher dimensions. Thus it is partial progress, not a full resolution of the problem.

Modern survey evidence that the nonnegative case remains open

- Title: Riemannian manifolds with positive sectional curvature
- Authors: Wolfgang Ziller
- Venue: Geometry of Manifolds with Non-negative Sectional Curvature, Lecture Notes in Mathematics 2110, Springer, pp. 1-19
- Year: 2014
- DOI/arXiv/URL: <https://www2.math.upenn.edu/~wziller/papers/SurveyMexico.pdf>
- Result/explanation: Lists Hopf's conjectures, including "A compact manifold with $\sec \geq 0$ has non-negative Euler characteristic" and the positive-curvature even-dimensional version, as conjectures.
- Title: Bounds on Sectional Curvature and Interactions with Topology
- Authors: Manuel Amann
- Venue: Jahresbericht der Deutschen Mathematiker-Vereinigung, 123, 27-55
- Year: 2021
- DOI/arXiv/URL: <https://doi.org/10.1365/s13291-020-00225-x>
- Result/explanation: A modern survey explaining that the Hopf conjecture for nonnegative curvature would follow from the Bott-Grove-Halperin conjecture. This places the nonnegative case among still-conjectural consequences rather than resolved theorems.

Claimed full nonpositive-curvature preprint

- Title: On non-elliptic symplectic manifolds
- Authors: Shouwen Fang and Hongyu Wang
- Venue: arXiv preprint, arXiv:1807.00326v4
- Year: submitted 2018, last revised 2024
- DOI/arXiv/URL: <https://doi.org/10.48550/arXiv.1807.00326> ; <https://arxiv.org/abs/1807.00326>
- Result/explanation: Theorem 1.9 states: if (M^{2n}, g) is a closed Riemannian manifold with strictly negative sectional curvature, then $(-1)^n \chi(M^{2n}) > 0$; if it has nonpositive sectional curvature, then $(-1)^n \chi(M^{2n}) \geq 0$. The hypotheses and conclusion, as stated, cover the nonpositive half of this problem. The preprint does not address the nonnegative half, so it does not fully resolve the combined problem.

1.3.3.4. What remains unresolved

The all-dimensional $\sec \geq 0 \Rightarrow \chi(M) \geq 0$ assertion remains unresolved. Since strictly positive sectional curvature is a special case of nonnegative sectional curvature, a complete proof of the stated nonnegative half would also settle the classical positive-curvature Hopf conjecture in dimensions 6 and higher; modern papers and surveys still treat that as open except under additional assumptions. Therefore the problem is partially resolved, not fully resolved.

1.3.4. Remaining open problems recorded by the database

1.3.4.1. Open problem 1

Prove the first unresolved dimensional case of the positive-curvature Hopf conjecture: every closed orientable Riemannian 6-manifold with strictly positive sectional curvature has strictly positive Euler characteristic.

Reference supplied by the database. Lee Kennard, “On the Hopf conjecture with symmetry,” *Geometry & Topology* **17** (2013), no. 1, 563–593, introduction (stating the positive-curvature Euler-characteristic conjecture and that it is known in dimensions two and four), DOI: <https://doi.org/10.2140/gt.2013.17.563>, [arXiv:1203.3808](https://arxiv.org/abs/1203.3808). Current-status confirmation: Manuel Amann, “Bounds on sectional curvature and interactions with topology,” *Jahresbericht der Deutschen Mathematiker-Vereinigung* **123** (2021), 27–55, DOI: <https://doi.org/10.1365/s13291-020-00225-x>.

1.3.4.2. Open problem 2

Prove the six-dimensional nonnegative-curvature case: every closed orientable Riemannian 6-manifold with nonnegative sectional curvature has nonnegative Euler characteristic.

Reference supplied by the database. Shiing-Shen Chern, “Differential geometry: its past and its future,” in *Actes du Congrès International des Mathématiciens (Nice, 1970)*, Tome 1, Gauthier-Villars, Paris, 1971, pp. 41–53, Problem III (the nonnegative-curvature Euler-sign conjecture, noting the four-dimensional case), [ICM proceedings PDF](#). Current-status confirmation: Wolfgang Ziller, “Riemannian manifolds with positive sectional curvature,” in *Geometry of Manifolds with Non-negative Sectional Curvature*, Lecture Notes in Mathematics 2110, Springer, Cham, 2014, pp. 1–19 (list of Hopf conjectures), DOI: https://doi.org/10.1007/978-3-319-06373-7_1.

1.3.4.3. Open problem 3

Prove the Bott–Grove–Halperin conjecture: every closed simply connected Riemannian manifold with nonnegative sectional curvature is rationally elliptic, i.e. $\dim_{\mathbb{Q}}(\pi_*(M) \otimes \mathbb{Q}) < \infty$. This would in particular imply the nonnegative-curvature Euler-characteristic inequality.

Reference supplied by the database. Karsten Grove, Burkhard Wilking, and Joseph Yeager, with an appendix by Steve Halperin, “Almost non-negative curvature and rational ellipticity in cohomogeneity two,” *Annales de l’Institut Fourier* **69** (2019), no. 7, 2921–2939, conjecture in the introduction, DOI: <https://doi.org/10.5802/aif.3340>. Current-status confirmation: Manuel Amann, “Bounds on sectional curvature and interactions with topology,” *Jahresbericht der Deutschen Mathematiker-Vereinigung* **123** (2021), 27–55, Conjecture (Bott–Grove–Halperin), DOI: <https://doi.org/10.1365/s13291-020-00225-x>.

1.3.4.4. Open problem 4

Prove the Singer L^2 -cohomology conjecture for closed nonpositively curved manifolds: the reduced L^2 -homology of the universal cover vanishes in every degree except possibly the middle degree. This would imply the conjectured sign of the Euler characteristic on the nonpositive-curvature side.

Reference supplied by the database. Michael W. Davis and Boris Okun, “Vanishing theorems and conjectures for the ℓ^2 -homology of right-angled Coxeter groups,” *Geometry & Topology* **5** (2001), 7–74, Singer Conjecture and its implication for the Hopf–Chern sign conjecture, DOI: <https://doi.org/10.2140/gt.2001.5.7>, [arXiv:math/0102104](https://arxiv.org/abs/math/0102104). Current-status confirmation: Laurențiu Maxim, “On singular generalizations of the Singer–Hopf conjecture,” *Mathematische Nachrichten* **296** (2023), no. 11, 5232–5241, Conjecture 1.1 and introduction, DOI: <https://doi.org/10.1002/mana.202200322>.

1.3.5. Primary-source pointer

The originating ICM record is *Differential geometry; its past and its future* (1970). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1970.1/ICM1970.1.ocr.pdf#page=79>.

1.4. 4. Geometric Aspects of the Generalized Plateau Problem [1974-geom-20]

Record: 1974-geom-20

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1974: Geometric Aspects of the Generalized Plateau Problem

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1974.2/ICM1974.2.ocr.pdf#page=15>

1.4.1. Problem statement

Prove or disprove that if M is a compact simply connected Riemannian n -manifold whose sectional curvatures are strictly $1/4$ -pinched, in the sense that after a constant rescaling of the metric

$$\frac{1}{4} < K(\sigma) \leq 1$$

for every tangent 2-plane $\sigma \subset T_x M$, then for every p with $1 \leq p \leq n - 1$, there is no nonzero rectifiable p -current V with compact support in M that is closed and stable; equivalently, there is no such V satisfying $dV = 0$, having first variation of mass equal to zero, and having second variation of mass nonnegative under all smooth deformations preserving its boundary.

1.4.2. Historical background, literature, and current progress

1.4.3. Verdict: partially resolved

1.4.3.1. Precise problem

This is the Lawson-Simons stable-current conjecture: if M^n is compact, simply connected, and strictly $1/4$ -pinched after rescaling,

$$1/4 < K(\sigma) \leq 1,$$

then for every $1 \leq p \leq n - 1$ there should be no nonzero closed stable rectifiable p -current in M . Equivalently, no nonzero compactly supported rectifiable p -current with $dV = 0$, zero first variation of mass, and nonnegative second variation under all smooth boundary-preserving deformations should exist.

1.4.3.2. Current status

I found no published or otherwise citable theorem proving this full statement for all compact simply connected strictly $1/4$ -pinched Riemannian manifolds and all p . Recent literature still describes the Lawson-Simons conjecture in full generality as open. The problem is nevertheless

not “fully open” in the sense of having no progress: there are substantial special cases and results with extra hypotheses.

The differentiable sphere theorem of Brendle-Schoen shows that such an M is diffeomorphic to a space form, and hence to S^n under simple connectivity, but this does not prove the stable-current statement. Absence of stable null-homologous currents is a variational/geometric assertion, not a consequence of the topology alone.

1.4.3.3. Progress and relevant literature

- H. Blaine Lawson, Jr.; James Simons, “On stable currents and their application to global problems in real and complex geometry,” *Annals of Mathematics* (2) 98, no. 3, 427-450, 1973. DOI: 10.2307/1970913. URL: <https://annals.math.princeton.edu/1973/98-3/p03>.

This is the foundational paper. It proves nonexistence of stable integral currents, currents mod p , and stable varifolds on the round Euclidean sphere, and proves related extrinsic nonexistence theorems for submanifolds of spheres. It is also cited by later papers as the source of the conjecture. These results cover important model cases but not arbitrary strictly $1/4$ -pinched metrics.

- H. Blaine Lawson, Jr., “Geometric aspects of the generalized Plateau problem,” *Proceedings of the International Congress of Mathematicians, Vancouver 1974*, vol. 2, Canadian Mathematical Congress, 1975, pp. 7-13. URL: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1974.2/ICM1974.2.ocr.pdf>.

This ICM article is the original source listed in the dataset. It formulates the conjectural nonexistence of stable closed currents on compact simply connected manifolds with strictly quarter-pinched sectional curvature after surveying stable currents on spheres and projective spaces.

- Ralph Howard, “The nonexistence of stable submanifolds, varifolds, and harmonic maps in sufficiently pinched simply connected Riemannian manifolds,” *Michigan Mathematical Journal* 32, no. 3, 321-334, 1985. DOI/URL: <https://doi.org/10.1307/mmj/1029003241>.

Howard proves nonexistence under sufficiently strong pinching constants depending on the dimension and current/submanifold dimension. Later summaries note these constants are strictly larger than $1/4$, so this is partial progress and does not reach the exact $1/4$ -pinched range.

- Mario J. Micallef; John Douglas Moore, “Minimal two-spheres and the topology of manifolds with positive curvature on totally isotropic two-planes,” *Annals of Mathematics* (2) 127, no. 1, 199-227, 1988. DOI: 10.2307/1971420. URL: <https://annals.math.princeton.edu/1988/127-1/p06>.

Their minimal-two-sphere instability and index estimates apply to the two-sphere case under the relevant positive isotropic curvature consequences of quarter pinching. This is a low-dimensional/special-domain result, not the full stable-current statement.

- Yi-Bing Shen; Hui-Qun Xu, “On the nonexistence of stable minimal submanifolds in positively pinched Riemannian manifolds,” in *Geometry and Topology of Submanifolds, X* (Beijing/Berlin, 1999), World Scientific, 2000, pp. 274-283. MR 1801920.

This is repeatedly cited as a partial result toward Lawson-Simons, especially for extra hypotheses such as ambient hypersurface settings. Modern discussions do not treat it as resolving the full conjecture.

- Yi-Bing Shen; Qun He, “On stable currents and positively curved hypersurfaces,”

Proceedings of the American Mathematical Society 129, no. 1, 237-246, 2001. MR 1784025.

This proves nonexistence of stable currents or stable varifolds for complete pinched hypersurfaces in real space forms with nonnegative constant curvature. The hypersurface/space-form assumptions are extra hypotheses not present in the problem statement.

- **Ze-Jun Hu; Guo-Xin Wei, “On the nonexistence of stable minimal submanifolds and the Lawson-Simons conjecture,”** *Colloquium Mathematicum* 96, no. 2, 213-223, 2003. DOI: 10.4064/cm96-2-6. URL: <https://eudml.org/doc/284869>.

This proves no stable compact minimal submanifolds when the ambient compact Riemannian manifold satisfies certain curvature bounds and isometrically immerses as a hypersurface in Euclidean space or the unit sphere. This is important evidence but has extrinsic immersion assumptions and concerns minimal submanifolds, so it does not cover arbitrary rectifiable currents in arbitrary strictly $1/4$ -pinched manifolds.

- **Xueshan Zhang, “On the nonexistence of stable currents in submanifolds of a Euclidean space,”** *Tohoku Mathematical Journal* 56, no. 4, 491-499, 2004. DOI: 10.2748/tmj/1113246746. URL: https://www.jstage.jst.go.jp/article/tmj/56/4/56_4_491/_article/-char/ja.

Zhang states that the original Lawson-Simons conjecture is still open, and proves stable-current nonexistence under additional extrinsic principal-curvature/sectional-curvature inequalities for submanifolds of Euclidean or nonnegative space forms. These hypotheses are not implied by the stated intrinsic pinching assumption.

- **Ralph Howard; S. Walter Wei, “On the existence and nonexistence of stable submanifolds and currents in positively curved manifolds and the topology of submanifolds in Euclidean spaces,”** in *Geometry and Topology of Submanifolds and Currents, Contemporary Mathematics* 646, American Mathematical Society, 2015, pp. 127-167. MR 3384978.

This later survey/article is cited by recent papers as collecting existence and nonexistence results for stable submanifolds and currents in positively curved settings. It provides context and partial results, not a full proof of the strict $1/4$ -pinched conjecture.

- **Giada Franz; Federico Trinca, “On the Stability of Minimal Submanifolds in Conformal Spheres,”** *Journal of Geometric Analysis* 33, article 335, 2023. DOI: 10.1007/s12220-023-01398-4. arXiv:2211.01106. URL: <https://link.springer.com/article/10.1007/s12220-023-01398-4>.

This recent paper explicitly states that the Lawson-Simons conjecture in full generality is still open. It proves that an n -dimensional Riemannian sphere conformal to the round sphere and δ -pinched contains no closed stable minimal k -submanifold for $2 \leq k \leq n - \delta^{-1}$. For $\delta > 1/4$, this verifies part of the conjecture only for conformal spheres and only in the indicated dimension range.

- **Hang Chen, “A conformal invariant and its application to the nonexistence of minimal submanifolds,”** arXiv:2310.09724, version 2, 2024. URL: <https://arxiv.org/abs/2310.09724>.

This preprint also states that the Lawson-Simons conjecture is still open and proves further nonexistence theorems using a conformal invariant under nonnegative sectional curvature assumptions. The added conformal-invariant smallness and dimension-range restrictions make it partial progress, not a full resolution.

1.4.3.4. What remains unresolved

What remains open is the exact intrinsic statement for every compact simply connected strictly $1/4$ -pinched Riemannian n -manifold and every p , $1 \leq p \leq n - 1$, with no conformal-to-round, stronger-pinching, hypersurface, extrinsic principal-curvature, low-dimensional, or smooth-submanifold restriction. The literature found gives partial positive evidence but does not prove the full absence of nonzero closed stable rectifiable p -currents in all cases asked by the problem statement.

1.4.4. Remaining open problems recorded by the database

1.4.4.1. Open problem 1

Let (S^n, g) have sectional curvatures $\frac{1}{4} < K \leq 1$. Can it contain a closed stable minimal surface of arbitrary topology (equivalently, can one prove instability when the two-dimensional minimal surface is not assumed to be a two-sphere)?

Reference supplied by the database. Daren Cheng, Chao Li, and Christos Mantoulidis, “Topics in Differential Geometry,” lecture notes for Richard Schoen’s Stanford Math 286 course, Spring 2015, Conjecture 5.4 and the paragraph following it (“otherwise open, even when Σ is a general two dimensional surface”), https://web.stanford.edu/~rchlch/notes/Schoen_286_spring_2015.pdf; current partial-result/status reference: Giada Franz and Federico Trinca, “On the Stability of Minimal Submanifolds in Conformal Spheres,” The Journal of Geometric Analysis 33 (2023), no. 10, Article 335, <https://doi.org/10.1007/s12220-023-01398-4>.

1.4.5. Primary-source pointer

The originating ICM record is *Geometric Aspects of the Generalized Plateau Problem* (1974). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1974.2/ICM1974.2.ocr.pdf#page=15>.

1.5. 5. Some Recent Advances in the Multidimensional Parametric Calculus of Variations [1974-geom-23]

Record: 1974-geom-23

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1974: Some Recent Advances in the Multidimensional Parametric Calculus of Variations

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1974.2/ICM1974.2.ocr.pdf#page=239>

1.5.1. Problem statement

For every compact Riemannian manifold N and every integer k with $0 < k < \dim N$, does N contain a closed k -dimensional submanifold $M \subset N$ whose mean curvature vector is identically zero at every point of M ?

1.5.2. Historical background, literature, and current progress

1.5.3. Verdict: partially resolved

1.5.3.1. Precise problem checked

The problem asks whether every compact Riemannian manifold N^n , for every integer k with $0 < k < n$, contains a closed smooth k -dimensional submanifold $M \subset N$ whose mean curvature vector vanishes identically. Equivalently, it asks for a closed smooth minimal submanifold of every positive dimension below the ambient dimension, in every compact ambient Riemannian manifold.

1.5.3.2. Literature status

The problem is not fully resolved in this smooth-submanifold form. The literature gives major partial results: closed geodesics for $k = 1$, generalized stationary integral varifolds in all dimensions, and smooth embedded minimal hypersurfaces in low ambient dimensions. However, the all- N , all- k , smooth closed submanifold statement is not covered by the known general theorems. The main unresolved gap is higher codimension and/or high-dimensional smooth regularity: stationary varifolds or singular hypersurfaces are not the same as closed smooth submanifolds.

1.5.3.3. Key sources and progress

1. Title: "Some Recent Advances in the Multidimensional Parametric Calculus of Variations"
Authors: William K. Allard

Venue: Proceedings of the International Congress of Mathematicians, Vancouver 1974, Canadian Mathematical Congress, 1975, pp. 231-236

DOI/arXiv/URL: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1974.2/ICM1974.2.occr.pdf>

Relevance: This appears to be the original source for the question. Allard states the problem of whether every compact Riemannian manifold contains a closed submanifold of prescribed dimension with zero mean curvature, and says the answer is not known in general. The same source records Almgren's generalized existence theorem and Pitts' then-new 3-dimensional result, so it already distinguishes smooth submanifolds from rectifiable/generalized solutions.

2. Title: "Variational problems on closed manifolds"

Authors: L. A. Lyusternik and A. I. Fet

Venue: Doklady Akademii Nauk SSSR (N.S.) 81, 1951, pp. 17-18

DOI/arXiv/URL: no DOI/arXiv found in live search

Relevance: This is the classical closed-geodesic theorem. It gives the $k = 1$ analogue: every compact Riemannian manifold has a nonconstant closed geodesic, hence a one-dimensional minimal object. It does not address $k \geq 2$.

3. Title: "The theory of varifolds: a variational calculus in the large for the k -dimensional area integrand"

Authors: Frederick J. Almgren, Jr.

Venue: Institute for Advanced Study / Princeton mimeographed notes, 1964/1965

DOI/arXiv/URL: <https://albert.ias.edu/20.500.12111/8025>

Relevance: Almgren's min-max theory gives nonzero stationary integral k -varifolds in compact Riemannian manifolds. This is broad progress for all k , but a stationary integral varifold may be singular and therefore does not prove the existence of a closed smooth k -submanifold.

4. Title: "Existence and Regularity of Minimal Surfaces on Riemannian Manifolds"

Authors: Jon T. Pitts

Venue: Mathematical Notes, vol. 27, Princeton University Press / University of Tokyo Press, 1981

DOI/arXiv/URL: <https://www.jstor.org/stable/j.ctt7zv66w>

Relevance: Pitts proves a general existence theorem for nonzero stationary integral varifolds with an almost-minimizing property, and a regular embedded minimal hypersurface theorem in a restricted dimension range. This is not the full statement because the all-dimensional result is in the varifold category, while the smooth regular theorem is codimension one and dimension-restricted.

5. Title: "Regularity of stable minimal hypersurfaces"

Authors: Richard Schoen and Leon Simon

Venue: Communications on Pure and Applied Mathematics 34(6), 1981, pp. 741-797

DOI/arXiv/URL: <https://doi.org/10.1002/cpa.3160340603>

Relevance: Together with Almgren-Pitts min-max theory, this gives closed minimal hypersurfaces with optimal regularity: smooth outside a singular set of codimension at least 7. In ambient dimension at most 7 this yields a smooth embedded closed minimal hypersurface. In higher ambient dimensions it allows singularities, so it does not resolve the smooth-submanifold question as stated.

6. Title: "Existence of infinitely many minimal hypersurfaces in closed manifolds"

Authors: Antoine Song

Venue: Annals of Mathematics 197(3), 2023, pp. 859-895

DOI/arXiv/URL: <https://doi.org/10.4007/annals.2023.197.3.1> ;

arXiv:1806.08816

Relevance: Song proves that every closed Riemannian manifold of dimension 3 through 7 contains infinitely many smoothly embedded closed minimal hypersurfaces. This is a strong resolution of the hypersurface case in low ambient dimensions, but it covers only $k = n - 1$ and $3 \leq n \leq 7$.

7. Title: “A viscosity method in the min-max theory of minimal surfaces”

Authors: Tristan Riviere

Venue: Publications Mathematiques de l’IHES 126, 2017, pp. 177-246

DOI/arXiv/URL: <https://doi.org/10.1007/s10240-017-0094-z>

Relevance: This is important progress for two-dimensional minimal surfaces/min-max in arbitrary codimension. It does not prove the existence of a closed smooth k -dimensional minimal submanifold for every k in every compact Riemannian manifold.

8. Title: “A proof of the multiplicity one conjecture for min-max minimal surfaces in arbitrary codimension”

Authors: Alessandro Pigati and Tristan Riviere

Venue: Duke Mathematical Journal 169(11), 2020, pp. 2005-2044

DOI/arXiv/URL: <https://doi.org/10.1215/00127094-2020-0002>

Relevance: This advances the two-dimensional arbitrary-codimension min-max theory, but it is not an all-dimensions existence theorem for smooth closed k -submanifolds.

9. Title: “Minimal submanifolds from the abelian Higgs model”

Authors: Alessandro Pigati and Daniel Stern

Venue: Inventiones Mathematicae 223, 2021, pp. 1027-1095

DOI/arXiv/URL: <https://doi.org/10.1007/s00222-020-01000-6>

Relevance: This produces codimension-two stationary integral varifold limits from a variational gauge-theoretic model. It is significant higher-codimension progress, but the output is again a generalized minimal object and does not settle the smooth closed submanifold existence question for all k .

1.5.3.4. What remains unresolved

No citable result found in the live search proves that for every compact Riemannian N^n and every $1 < k < n - 1$ there is a closed smooth k -dimensional minimal submanifold $M \subset N$. Nor do the high-dimensional hypersurface results prove the stated smooth-submanifold conclusion in ambient dimension at least 8, because the general min-max object may have a singular set. Therefore the exact problem is only partially resolved.

1.5.4. Remaining open problems recorded by the database

1.5.4.1. Open problem 1

Prove that Almgren’s higher-codimension min-max varifolds have the same regularity as area-minimizing integral currents; in particular, prove the corresponding regularity statement when the min-max submanifolds are two-dimensional.

Reference supplied by the database. Da Rong Cheng and Xin Zhou, “Existence of classical minimal surfaces in 4 and 5-manifolds,” arXiv:2606.24754 [math.DG], 2026, Introduc-

tion, subsection “Context” (the explicitly stated higher-codimension regularity conjecture).
<https://arxiv.org/abs/2606.24754>

1.5.5. Primary-source pointer

The originating ICM record is *Some Recent Advances in the Multidimensional Parametric Calculus of Variations* (1974). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1974.2/ICM1974.2.ocr.pdf#page=239>.

1.6. 6. Eigenvalues of the Laplacian and Invariants of Manifolds [1974-geom-300]

Record: 1974-geom-300

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1974: Eigenvalues of the Laplacian and Invariants of Manifolds

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1974.1/ICM1974.1.ocr.pdf#page=243>

1.6.1. Problem statement

For each integer $n \geq 2$ and each $r > 0$, let $(S^n(r), g_{\text{round}})$ be the round n -sphere of radius r . Determine whether every smooth closed oriented n -dimensional Riemannian manifold (M, g) whose Laplace–Beltrami operator on smooth functions has the same eigenvalues, counted with multiplicity, as that of $(S^n(r), g_{\text{round}})$ must be isometric to $(S^n(r), g_{\text{round}})$.

1.6.2. Historical background, literature, and current progress

The problem is **unresolved**. Tanno’s theorem gives an affirmative answer for every radius in dimensions $2 \leq n \leq 6$, but no proof or counterexample is known in any dimension $n \geq 7$. Consequently, the quantification over all $n \geq 2$ in the stated problem is not completely settled.

- Shukichi Tanno, “Eigenvalues of the Laplacian of Riemannian Manifolds,” *Tohoku Mathematical Journal* 25 (1973), 391–403, <https://doi.org/10.2748/tmj/1178241341>. Theorem B proves that a compact orientable Riemannian manifold isospectral on functions to a round sphere is isometric to that sphere when $2 \leq n \leq 6$. Rescaling the metric covers every radius $r > 0$ in the present statement.
- Shukichi Tanno, “A Characterization of the Canonical Spheres by the Spectrum,” *Mathematische Zeitschrift* 175 (1980), 267–274, <https://doi.org/10.1007/BF01163028>. This paper explicitly states that the function-spectrum problem remains open for $n \geq 7$ and proves an all-dimensional local result: an isospectral metric sufficiently close to a constant-curvature metric is round. It also shows that every isospectral deformation of the round metric on S^n is trivial. These are local rigidity results, not a global answer in high dimensions.
- Vijay K. Patodi, “Curvature and the Eigenforms of the Laplace Operator,” *Journal of Differential Geometry* 5 (1971), 233–249, <https://doi.org/10.4310/jdg/1214429791>. Proposition 4.1 characterizes round spheres in every dimension when one knows both the function spectrum and the spectrum of the Hodge Laplacian on 1-forms. The extra 1-form spectrum is not among the hypotheses of the stated problem.
- Renato G. Bettiol, Emilio A. Lauret, and Paolo Piccione, “The First Eigenvalue of a Homogeneous CROSS,” *The Journal of Geometric Analysis* 32 (2022), Article 76, <https://doi.org/10.1007/s12220-021-00826-7>, [arXiv:2001.08471](https://arxiv.org/abs/2001.08471). Theorem C proves that two compact rank-one symmetric spaces equipped with homogeneous met-

rics are isospectral if and only if they are isometric. Thus the round sphere is spectrally unique within this homogeneous CROSS class in every dimension, but the arbitrary manifold and arbitrary metric allowed in the stated problem need not belong to that class.

- Emilio A. Lauret and Juan Sebastián Rodríguez, “Spectrally Distinguishing Symmetric Spaces II,” *Extracta Mathematicae* 40 (2025), 91–120, <https://doi.org/10.17398/2605-5686.40.1.91>, [arXiv:2411.06886](https://arxiv.org/abs/2411.06886). The introduction records that spectral uniqueness of a round sphere among orientable Riemannian manifolds remains unknown in dimensions at least 7; the paper therefore works in the narrower homogeneous setting.
- Leonardo F. Cavenaghi, João Marcos do Ó, and Lloann D. Sperança, “Hearing Exotic Smooth Structures,” *Advanced Nonlinear Studies* 25 (2025), 1112–1126, <https://doi.org/10.1515/ans-2023-0151>, [arXiv:2402.10106](https://arxiv.org/abs/2402.10106). The authors explicitly note that spectral uniqueness of the round 7-sphere is still unknown. Their equalities concern basic spectra associated with group actions, rather than the full function spectrum required here, so they do not provide a counterexample.

Thus the established global result exhausts $n \leq 6$, while local rigidity, additional Hodge spectral data, or homogeneity settle only stronger or restricted variants in higher dimensions. The unrestricted cases $n \geq 7$ remain the gap.

1.6.3. Remaining open problems recorded by the database

1.6.3.1. Open problem 1

For each $r > 0$, let $(S^7(r), g_{\text{round}})$ be the round 7-sphere of radius r . Determine whether every smooth closed oriented 7-dimensional Riemannian manifold (M, g) whose Laplace–Beltrami operator on smooth functions has the same eigenvalues, counted with multiplicity, as that of $(S^7(r), g_{\text{round}})$ must be isometric to $(S^7(r), g_{\text{round}})$.

Reference supplied by the database. Leonardo F. Cavenaghi, João Marcos do Ó, and Lloann D. Sperança, “Hearing Exotic Smooth Structures,” *Advanced Nonlinear Studies* 25, no. 4 (2025), 1112–1126, open question identified in the discussion preceding Section 2, <https://doi.org/10.1515/ans-2023-0151>, [arXiv:2402.10106](https://arxiv.org/abs/2402.10106).

1.6.4. Primary-source pointer

The originating ICM record is *Eigenvalues of the Laplacian and Invariants of Manifolds* (1974). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1974.1/ICM1974.1.ocr.pdf#page=243>.

1.7. 7. The Generalized Cartan-Hadamard Conjecture [1978-geom-25]

Record: 1978-geom-25

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: reviewed (reviewed flag: yes)

Original source: ICM 1978: Isoperimetric inequalities and eigenvalues of the Laplacian

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1978.1/ICM1978.1.ocr.pdf#page=435>

1.7.1. Problem statement

In dimension 2, if D is a simply connected domain on a Riemannian surface, $A = \text{area}(D)$, $L = \text{length}(\partial D)$, K is the Gauss curvature, and $M = \sup_D K$, then

$$L^2 \geq 4\pi A - MA^2,$$

with equality holds only when $K \equiv M$ on D and D is a geodesic disk.

Determine whether the higher-dimensional isoperimetric analogue holds for domains in Riemannian manifolds of dimension $n \geq 3$.

Determine whether the corresponding sharp isoperimetric comparison remains valid in dimension $n \geq 3$, using n -dimensional volume for D , $(n-1)$ -dimensional measure for ∂D , and the appropriate upper sectional-curvature bound in place of M , when the ambient Riemannian curvature is not constant.

1.7.2. Historical background, literature, and current progress

1.7.3. Verdict: partially resolved

1.7.3.1. Precise problem checked

The problem asks for the higher-dimensional analogue of the Bol-Fiala-Weil surface inequality

$$L^2 \geq 4\pi A - (\sup_D K)A^2$$

for a simply connected surface domain. In modern terminology the natural nonpositive-curvature formulation is the generalized Cartan-Hadamard isoperimetric comparison: if M^n is complete and simply connected with sectional curvature $K \leq \kappa \leq 0$, whether every domain $\Omega \subset M$ satisfy

$$|\partial\Omega| \geq I_{n,\kappa}(|\Omega|),$$

where $I_{n,\kappa}$ is the isoperimetric profile of the simply connected space form of constant curvature κ .

1.7.3.2. Current literature status

The problem is not fully resolved in the generality asked. The generalized Cartan-Hadamard comparison is known in important cases, but no published or otherwise citable theorem I found covers all $n \geq 3$, all volumes, and the full variable sectional-curvature comparison. Moreover, the naive higher-dimensional analogue for arbitrary Riemannian balls/topological balls is false.

Key references and what they prove

1. Robert Osserman, "The isoperimetric inequality", *Bulletin of the American Mathematical Society* 84, no. 6 (1978), 1182-1238, DOI: 10.1090/S0002-9904-1978-14553-4.
 - Records the two-dimensional variable-curvature inequalities, including the form saying that a simply connected surface domain has boundary length at least that of the model constant-curvature disk with curvature $\sup_D K$. This is the surface result whose higher-dimensional analogue is at issue.
2. Robert Osserman, "Isoperimetric inequalities and eigenvalues of the Laplacian", *Proceedings of the International Congress of Mathematicians, Helsinki 1978*, Vol. I, pp. 435-442, Acad. Sci. Fennica, Helsinki, 1980.
 - Original source connected to the dataset entry; discusses these isoperimetric inequalities in relation to eigenvalue estimates.
3. Andre Weil, "Sur les surfaces a courbure negative", *Comptes Rendus de l'Academie des Sciences de Paris* 182 (1926), 1069-1071.
 - Proves the $n = 2$, $\kappa = 0$ Cartan-Hadamard/Weil surface case.
4. Gerrit Bol, "Isoperimetrische Ungleichungen fur Bereiche auf Flächen", *Jahresbericht der Deutschen Mathematiker-Vereinigung* 51 (1941), 219-257.
 - Gives the sharp two-dimensional comparison for $\kappa \neq 0$, i.e. the surface model-space comparison underlying the stated formula.
5. Christopher B. Croke, "A sharp four-dimensional isoperimetric inequality", *Commentarii Mathematici Helvetici* 59, no. 2 (1984), 187-192.
 - Proves the sharp Euclidean Cartan-Hadamard inequality in dimension $n = 4$ for $\kappa = 0$. This is a major special case, not the full variable-curvature comparison for all dimensions or all κ .
6. Bruce Kleiner, "An isoperimetric comparison theorem", *Inventiones Mathematicae* 108, no. 1 (1992), 37-47.
 - Proves the generalized Cartan-Hadamard comparison in dimension $n = 3$ for all $\kappa \leq 0$. This fully resolves the three-dimensional Cartan-Hadamard version but not $n \geq 4$ for $\kappa < 0$, nor all higher dimensions.
7. Frank Morgan and David L. Johnson, "Some sharp isoperimetric theorems for Riemannian manifolds", *Indiana University Mathematics Journal* 49, no. 3 (2000), 1017-1041.
 - Establishes sharp small-volume isoperimetric results under curvature hypotheses. These are local/small-domain results and do not resolve the global problem.

8. Benoit R. Kloeckner and Greg Kuperberg, “The Cartan-Hadamard conjecture and the Little Prince”, *Revista Matematica Iberoamericana* 35, no. 4 (2019), 1195-1258, DOI: 10.4171/RMI/1082, arXiv:1303.3115.
 - States the generalized Cartan-Hadamard conjecture explicitly as the comparison for complete simply connected n -manifolds with $K \leq \kappa \leq 0$. Their results give unified proofs in dimensions 2 and 4 for $\kappa = 0$, partial $n = 4$ results for $\kappa < 0$ with an additional chord/volume smallness condition, and related positive-curvature versions under unique-geodesic and volume restrictions.
 - They also prove counterexamples to a naive extension: for every $\varepsilon > 0$, there are Riemannian 3-balls with nearly pinched negative curvature, bounded boundary volume, and arbitrarily large volume. Thus topology plus an upper curvature bound alone cannot be the right higher-dimensional analogue.
9. Mohammad Ghomi and Joel Spruck, “Total Curvature and the Isoperimetric Inequality in Cartan-Hadamard Manifolds”, *Journal of Geometric Analysis* 32, article 50 (2022), DOI: 10.1007/s12220-021-00801-2, arXiv:1908.09814.
 - Develops total-curvature comparison machinery and applications to the isoperimetric problem in nonpositive curvature. The accepted abstract does not claim a full proof of the Cartan-Hadamard conjecture; later work still treats the all-dimensional general case as open.
10. Mohammad Ghomi and John Ioannis Stavroulakis, “A Local Isoperimetric Inequality for Balls with Nonpositive Curvature”, *International Mathematics Research Notices* 2026, no. 6, rnag043, DOI: 10.1093/imrn/rnag043, arXiv:2505.11971.
 - Proves a sharp local version: small perturbations of the Euclidean ball metric to nonpositive curvature do not reduce the isoperimetric ratio. This is local and does not settle the global comparison.
11. Mohammad Ghomi, “Isoperimetric and total curvature inequalities in Cartan-Hadamard manifolds with nullity”, arXiv:2605.24638 (2026 preprint).
 - Proves the Euclidean Cartan-Hadamard isoperimetric inequality for Cartan-Hadamard manifolds whose nullity index is at least $n - 3$. This gives new high-dimensional classes but still imposes extra structure and does not cover arbitrary variable sectional curvature.

1.7.3.3. What remains unresolved

No citable result found proves the full sharp higher-dimensional model-space comparison for all domains in all dimensions $n \geq 3$ under only the appropriate upper sectional-curvature bound. The exact Cartan-Hadamard/generalized Cartan-Hadamard problem remains open beyond the known low-dimensional and special-geometry cases, especially for the full $\kappa < 0$ comparison in higher dimensions and for unrestricted $n \geq 5$. The broader naive version without ambient Cartan-Hadamard or unique-geodesic hypotheses is known to be false.

1.7.4. Remaining open problems recorded by the database

1.7.4.1. Open problem 1

Let (M^5, g) be a complete simply connected Riemannian 5-manifold with nonpositive sectional curvature. Prove that every bounded sufficiently regular domain $\Omega \subset M$ satisfies the sharp Euclidean isoperimetric inequality

$$|\partial\Omega|^5 \geq 5^5 \omega_5 |\Omega|^4,$$

with equality only for a Euclidean ball in a flat region.

Reference supplied by the database. Felix Schulze, “Optimal Isoperimetric Inequalities for Surfaces in Any Codimension in Cartan–Hadamard Manifolds,” *Geometric and Functional Analysis* 30 (2020), 255–288, Introduction, Cartan–Hadamard Conjecture and the statement that the cases beyond dimensions (2,3) and the flat (4)-dimensional case remain open. DOI: 10.1007/s00039-020-00522-8. <https://doi.org/10.1007/s00039-020-00522-8>

1.7.4.2. Open problem 2

Let (M^4, g) be complete and simply connected with sectional curvature bounded above by $-\kappa < 0$. Prove that every bounded sufficiently regular domain $\Omega \subset M$ has boundary 3-volume at least that of the geodesic ball in the 4-dimensional space form of constant sectional curvature $-\kappa$ having the same 4-volume as Ω , with the corresponding rigidity in the equality case.

Reference supplied by the database. Felix Schulze, “Optimal Isoperimetric Inequalities for Surfaces in Any Codimension in Cartan–Hadamard Manifolds,” *Geometric and Functional Analysis* 30 (2020), 255–288, Introduction, strengthened Cartan–Hadamard conjecture and the statement that only a partial result is known when $(n=4)$ and $\kappa \neq 0$. DOI: 10.1007/s00039-020-00522-8. <https://doi.org/10.1007/s00039-020-00522-8>

1.7.5. Primary-source pointer

The originating ICM record is *Isoperimetric inequalities and eigenvalues of the Laplacian* (1978). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1978.1/ICM1978.1.ocr.pdf#page=435>.

1.8. 8. The role of partial differential equations in differential geometry [1978-geom-28]

Record: 1978-geom-28

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1978: The role of partial differential equations in differential geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1978.1/ICM1978.1.ocr.pdf#page=237>

1.8.1. Problem statement

Let (M^n, g) be a compact Riemannian manifold whose scalar curvature is nonnegative at every point. Suppose that M is topologically covered by Euclidean space, equivalently that its universal cover is homeomorphic to $\mathbb{R}^n$. Must (M, g) be flat, i.e. have identically zero Riemannian curvature tensor?

1.8.2. Historical background, literature, and current progress

1.8.3. Status check

Verdict: partially resolved.

1.8.3.1. Problem identified

The question is Yau's scalar-curvature rigidity problem: if a compact Riemannian manifold (M^n, g) has scalar curvature $Sc_g \geq 0$ and its universal cover is homeomorphic to $\mathbb{R}^n$, must the full Riemann curvature tensor vanish identically?

Equivalently, this asks for scalar-curvature rigidity of closed manifolds topologically covered by Euclidean space. Such manifolds are aspherical, so the question is closely tied to the Gromov-Lawson/Schoen-Yau conjecture that closed aspherical manifolds admit no positive scalar curvature metric.

1.8.3.2. Why the exact problem is not fully resolved

No published or otherwise citable theorem found covers all dimensions and all compact manifolds whose universal cover is homeomorphic to $\mathbb{R}^n$. Recent literature still states that the aspherical positive-scalar-curvature conjecture is widely open in dimensions greater than five. Since an all-dimensional solution of that conjecture, together with standard scalar-flat rigidity, would imply an affirmative answer to this problem, the problem remains open in general, especially in dimensions $n \geq 6$.

The problem is not fully open, however: substantial classes are known, including all dimensions for enlargeable manifolds and all closed aspherical manifolds in dimensions 3, 4, 5.

1.8.3.3. Relevant literature and progress

Original source Title: The Role of Partial Differential Equations in Differential Geometry

Authors: Shing-Tung Yau

Venue/year: Proceedings of the International Congress of Mathematicians, Helsinki 1978, Acad. Sci. Fennica, Helsinki, 1980, pp. 237-250

URL: https://archive.ymsc.tsinghua.edu.cn/pacm_download/59/11114-Shing-Tung_Yau_217.pdf

Relevance: On p. 242 of the proceedings version, Yau asks: "If a compact manifold with nonnegative scalar curvature is covered by the euclidean space topologically, is it a flat manifold?" This is the exact problem statement.

Scalar-flat rigidity reduction Title: Scalar curvature and conformal deformation of Riemannian structure

Authors: Jerry L. Kazdan and F. W. Warner

Venue/year: Journal of Differential Geometry 10 (1975), 113-134

DOI/URL: <https://doi.org/10.4310/jdg/1214432678>

Relevance: Provides the scalar-curvature deformation background used in the standard reduction: if a closed manifold has a metric with $Sc \geq 0$ and $Sc > 0$ somewhere, one obtains a positive scalar curvature metric.

Title: Une stratification de l'espace des structures riemanniennes

Author: Jean-Pierre Bourguignon

Venue/year: Compositio Mathematica 30 (1975), no. 1, 1-41

URL: <http://eudml.org/doc/89245>

Relevance: Contains the scalar-flat deformation rigidity often used in this context: on a closed manifold admitting no positive scalar curvature metric, a nonnegative-scalar-curvature metric must be Ricci-flat.

Title: The structure of compact Ricci-flat Riemannian manifolds

Authors: Arthur E. Fischer and Joseph A. Wolf

Venue/year: Journal of Differential Geometry 10 (1975), 277-288

DOI/URL: <https://doi.org/10.4310/jdg/1214432794>

Relevance: Gives the compact Ricci-flat structure theorem. Combined with the Cheeger-Gromoll splitting theorem, a compact Ricci-flat manifold whose universal cover is $\mathbb{R}^n$ is flat. Thus positive-scalar-curvature obstructions for the underlying topology yield the desired nonnegative-scalar-curvature rigidity.

Enlargeable cases Title: Spin and scalar curvature in the presence of a fundamental group.

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Authors: Mikhail Gromov and H. Blaine Lawson, Jr.

Venue/year: Annals of Mathematics (2) 111 (1980), no. 2, 209-230

DOI/URL: <https://doi.org/10.2307/1971198>

Relevance: Theorem A proves that compact enlargeable manifolds admit no positive scalar curvature metric and that any metric with nonnegative scalar curvature on such a manifold is flat. This covers tori and many standard aspherical examples, but enlargeability is an extra hypothesis and does not cover every compact manifold topologically covered by $\mathbb{R}^n$.

Dimension 3 Title: Existence of incompressible minimal surfaces and the topology of three-dimensional manifolds with nonnegative scalar curvature

Authors: Richard Schoen and Shing-Tung Yau

Venue/year: Annals of Mathematics (2) 110 (1979), no. 1, 127-142

DOI/URL: <https://doi.org/10.2307/1971247>

Relevance: Foundational minimal-surface obstruction results for 3-manifolds with nonnegative/positive scalar curvature.

Title: Positive scalar curvature and the Dirac operator on complete Riemannian manifolds

Authors: Mikhail Gromov and H. Blaine Lawson, Jr.

Venue/year: Publications Mathématiques de l’IHES 58 (1983), 83-196

DOI/URL: <https://doi.org/10.1007/BF02953774>

Relevance: Later accounts cite these results, together with Schoen-Yau’s 3-manifold work, as verifying the closed aspherical case in dimension 3. Hence the problem statement is settled in dimension 3.

Dimensions 4 and 5 Title: Generalized soap bubbles and the topology of manifolds with positive scalar curvature

Authors: Otis Chodosh and Chao Li

Venue/year: Annals of Mathematics (2) 199 (2024), no. 2, 707-740

DOI/arXiv/URL: <https://doi.org/10.4007/annals.2024.199.2.3> ; arXiv:2008.11888, <https://arxiv.org/abs/2008.11888>

Relevance: Proves that for $n \in \{4, 5\}$, a closed aspherical n -manifold admits no positive scalar curvature metric. The final arXiv version states the rigidity form as Theorem 2: any metric of nonnegative scalar curvature on such a manifold is flat. Since universal cover $\mathbb{R}^n$ implies aspherical, this resolves the question in dimensions 4 and 5.

Title: No metrics with Positive Scalar Curvatures on Aspherical 5-Manifolds

Author: Misha Gromov

Venue/year: arXiv preprint, 2020

arXiv/URL: arXiv:2009.05332, <https://arxiv.org/abs/2009.05332>

Relevance: Independently proves the no-positive-scalar-curvature statement for closed aspherical 5-manifolds via μ -bubble methods. This is consistent with, but not broader than, the later published Chodosh-Li result for the problem.

Recent status and remaining gap Title: Positive scalar curvature metrics and aspherical summands

Authors: Shuli Chen, Jianchun Chu and Jintian Zhu

Venue/year: Accepted by Duke Mathematical Journal; arXiv preprint, submitted 2023, final version 2025

arXiv/URL: arXiv:2312.04698, <https://arxiv.org/abs/2312.04698>

Relevance: The introduction summarizes the current status: dimension 3 was verified by Gromov-Lawson, dimensions 4 and 5 by Chodosh-Li and Gromov, but “the aspherical conjecture is still widely open in dimensions greater than five.” The paper proves further noncompact connected-sum results for dimensions 3, 4, 5, not the all-dimensional compact rigidity asked in this problem.

1.8.3.4. What remains unresolved

The remaining unresolved part is the full all-dimensional statement for arbitrary compact n -manifolds, $n \geq 6$, whose universal cover is homeomorphic to $\mathbb{R}^n$. Known positive results require dimension bounds or extra hypotheses such as enlargeability, spin/index-theoretic assumptions, nonpositive sectional curvature, or special group-theoretic conditions. No source found proves that every compact manifold topologically covered by $\mathbb{R}^n$ and carrying a metric with $Sc \geq 0$ must be flat.

1.8.4. Remaining open problems recorded by the database

1.8.4.1. Open problem 1

Prove that no closed aspherical (6)-manifold admits a Riemannian metric of positive scalar curvature.

Reference supplied by the database. Liam Mazurowski, Tongrui Wang, and Xuan Yao, “On the Topology of Manifolds with Positive Intermediate Curvature,” *Advances in Mathematics* 485 (2026), Article 110731, Introduction, $K(\pi, 1)$ conjecture in dimension (6) (equivalently the unresolved $(n=6, m=5)$ scalar-curvature case of the paper’s conjecture). DOI: 10.1016/j.aim.2025.110731; arXiv:2503.13815. <https://doi.org/10.1016/j.aim.2025.110731>

1.8.5. Primary-source pointer

The originating ICM record is *The role of partial differential equations in differential geometry* (1978). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1978.1/ICM1978.1.ocr.pdf#page=237>.

1.9. 9. The role of partial differential equations in differential geometry [1978-geom-29]

Record: 1978-geom-29

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1978: The role of partial differential equations in differential geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1978.1/ICM1978.1.ocr.pdf#page=237>

1.9.1. Problem statement

Does every four dimensional compact simply-connected Riemannian manifold with zero Ricci curvature admit a Kähler structure?

1.9.2. Historical background, literature, and current progress

1.9.3. Status check

Verdict: partially resolved.

1.9.3.1. Precise problem

The problem asks whether every compact simply-connected 4-dimensional Riemannian manifold (M, g) with $\text{Ric}(g) = 0$ must admit a Kähler structure compatible with the Ricci-flat metric. In dimension 4 this is essentially the question whether such a metric must be locally hyperkähler, equivalently have reduced/special holonomy $SU(2) = \text{Sp}(1)$, rather than generic holonomy $SO(4)$.

1.9.3.2. Original source

- Title: “The Role of Partial Differential Equations in Differential Geometry”
- Author: Shing-Tung Yau
- Venue/year: Proceedings of the International Congress of Mathematicians, Helsinki 1978; published by Acad. Sci. Fennica, Helsinki, 1980, pp. 237-250
- DOI/arXiv/URL: https://archive.ymsc.tsinghua.edu.cn/pacm_download/59/11114-Shing-Tung_Yau_217.pdf
- Relevance: Yau states exactly the question: “Does every four dimensional compact simply-connected Riemannian manifold with zero Ricci curvature admit a Kahler structure?” He immediately records only partial positive evidence: Hitchin’s observation covers K3 surfaces and, more generally, compact spin manifolds with nonzero index. This is not a full solution of all simply-connected compact Ricci-flat 4-manifolds.

1.9.3.3. Partial progress and related literature

- Title: “Harmonic spinors”
- Author: Nigel J. Hitchin
- Venue/year: Advances in Mathematics 14 (1974), 1-55
- DOI/arXiv/URL: DOI 10.1016/0001-8708(74)90021-8; author listing: <https://people.maths.ox.ac.uk/hitchin/files/pubs.html>
- Relevance: The Lichnerowicz formula implies that on a scalar-flat/Ricci-flat spin manifold, harmonic spinors forced by nonzero Dirac index are parallel. In the present 4-dimensional simply-connected setting, this forces special holonomy and hence a hyperkähler/Kähler metric. This proves the nonzero-index spin cases mentioned by Yau, but leaves spin zero-index and non-spin possibilities outside its scope.
- Title: “On compact four-dimensional Einstein manifolds”
- Author: Nigel J. Hitchin
- Venue/year: Journal of Differential Geometry 9 (1974), 435-442
- DOI/arXiv/URL: DOI 10.4310/jdg/1214432419
- Relevance: Hitchin’s 4-dimensional Einstein results, together with the spinorial argument above, give strong restrictions and the K3/equality cases, but they do not exclude all compact simply-connected Ricci-flat 4-manifolds with generic holonomy.
- Title: “On the Ricci curvature of a compact Kähler manifold and the complex Monge-Ampère equation, I”
- Author: Shing-Tung Yau
- Venue/year: Communications on Pure and Applied Mathematics 31 (1978), 339-411
- DOI/arXiv/URL: DOI 10.1002/cpa.3160310304
- Relevance: Yau’s solution of the Calabi conjecture constructs Ricci-flat Kähler metrics on compact Kähler manifolds with vanishing first Chern class, in particular on K3 surfaces. This supplies the main known simply-connected compact 4-dimensional examples, but it is an existence theorem under a pre-existing Kähler/complex hypothesis, not a converse for arbitrary Ricci-flat Riemannian 4-manifolds.
- Title: “Conformal geometry of Ricci flat 4-manifolds”
- Author: Mitsuhiro Itoh
- Venue/year: Kodai Mathematical Journal 17 (1994), 179-200
- DOI/arXiv/URL: DOI 10.2996/kmj/1138039959
- Relevance: For the underlying K3 smooth 4-manifold, Ricci-flat metrics are shown to be Ricci-flat Kähler/hyperkähler and the moduli is described by period data. This resolves the question on the K3 diffeomorphism type, but assumes the manifold is K3 and therefore does not settle the general question.
- Title: “On noncollapsed almost Ricci-flat 4-manifolds”
- Authors: Vitali Kapovitch and John Lott
- Venue/year: American Journal of Mathematics 141 (2019), 737-755
- DOI/arXiv/URL: arXiv:1606.09041, <https://arxiv.org/abs/1606.09041>

- Relevance: Under noncollapsing, diameter/volume normalization, spin, nonzero A -hat genus, and sufficiently small Ricci curvature, the manifold is diffeomorphic to $K3$; the paper also notes that known closed Ricci-flat 4-manifolds are flat or finitely covered by $K3$. These are strong stability/near-Ricci-flat results, but they impose extra hypotheses and do not classify all exact Ricci-flat simply-connected 4-manifolds.
- Title: “Gluing Eguchi-Hanson Metrics and a Question of Page”
- Authors: Simon Brendle and Nikolaos Kapouleas
- Venue/year: Communications on Pure and Applied Mathematics 70 (2017), 1366-1401
- DOI/arXiv/URL: DOI 10.1002/cpa.21678; arXiv:1405.0056, <https://arxiv.org/abs/1405.0056>
- Relevance: Page asked whether the Kummer construction could be modified by reversing orientations of some Eguchi-Hanson pieces; success would have produced compact Einstein/Ricci-flat metrics that are neither Kähler nor self-dual. Brendle and Kapouleas find a nonvanishing obstruction in a highly symmetric mixed-orientation configuration. This is negative evidence against one proposed route to a counterexample, not a proof of the general statement.
- Title: “Higher order obstructions to the desingularization of Einstein metrics”
- Author: Tristan Ozuch
- Venue/year: Cambridge Journal of Mathematics 9 (4) (2022), 901-976
- DOI/arXiv/URL: arXiv:2012.13316, <https://arxiv.org/abs/2012.13316>
- Relevance: Ozuch studies higher-order obstructions to desingularizing Einstein orbifolds and applies them to Page’s proposed Ricci-flat but non-Kähler Kummer-type construction, finding many obstructions. Again, this obstructs a major construction strategy but does not rule out every possible generic-holonomy Ricci-flat 4-manifold.
- Title: “Supersymmetry, Ricci Flat Manifolds and the String Landscape”
- Author: Bobby Samir Acharya
- Venue/year: Journal of High Energy Physics 2020, article 128; arXiv version 2020
- DOI/arXiv/URL: arXiv:1906.06886, <https://arxiv.org/abs/1906.06886>
- Relevance: This survey formulates the broader conjecture that compact simply-connected Ricci-flat manifolds have special holonomy, explicitly noting the open possibility of compact simply-connected Ricci-flat manifolds with generic holonomy. In dimension 4 it identifies the expected special-holonomy case as $K3$ and records remaining unsettled cases such as possible Ricci-flat metrics on $S^2 \times S^2$ or other zero-index/non-spin topologies.
- Title: “Geometry of Gravitational Instantons”
- Authors: Mingyang Li and Song Sun
- Venue/year: arXiv preprint, 2026
- DOI/arXiv/URL: arXiv:2606.10443, <https://arxiv.org/abs/2606.10443>
- Relevance: This current survey states a closely equivalent open problem: “Is there a compact simply-connected Ricci-flat 4-manifold which is not diffeomorphic to the $K3$ manifold?” It notes that the Page/Gibbons-Pope mixed-orientation Kummer approach has only produced negative obstruction results. This confirms that the

literature still treats the general compact simply-connected Ricci-flat 4-dimensional classification/Kähler question as unresolved.

1.9.3.4. Applicability check

The available positive theorems do not cover the full statement in the problem statement. The K3 and spin nonzero-index cases imply a Kähler/hyperkähler structure, and Yau's theorem constructs Ricci-flat Kähler examples once a suitable complex/Kähler structure is already present. However, none of the cited results proves that every compact simply-connected Ricci-flat 4-manifold has K3 topology or reduced holonomy. The remaining obstruction is exactly the possible existence of a compact simply-connected Ricci-flat 4-manifold with generic holonomy $SO(4)$ (or, equivalently in this setting, a Ricci-flat metric not forced to be Kähler/hyperkähler). Current literature treats this as an open possibility and supplies only partial restrictions and failed construction attempts.

Therefore the problem is partially resolved, not fully resolved.

1.9.4. Remaining open problems recorded by the database

1.9.4.1. Open problem 1

Does the smooth (4)-sphere S^4 admit a Ricci-flat Riemannian metric?

Reference supplied by the database. Matthias Wink (joint work with Jan Nienhaus), "Einstein Metrics on Spheres," in Aaron Naber, André Neves, Eleonora Di Nezza, and Burkhard Wilking (organizers), "Geometrie," *Oberwolfach Reports* 21 (2024), No. 28, pp. 1604–1606, explicit open question on p. 1604. DOI: 10.4171/OWR/2024/28. <https://doi.org/10.4171/OWR/2024/28>

1.9.5. Primary-source pointer

The originating ICM record is *The role of partial differential equations in differential geometry* (1978). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1978.1/ICM1978.1.ocr.pdf#page=237>.

1.10. 10. Extension Problems in Several Complex Variables [1978-geom-303]

Record: 1978-geom-303

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1978: Extension Problems in Several Complex Variables

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1978.2/ICM1978.2.ocr.pdf#page=165>

1.10.1. Problem statement

Prove that if X and Y are homotopy-equivalent compact Kähler manifolds of the same complex dimension $n \geq 2$, each equipped with a Kähler metric of negative Riemannian sectional curvature, then X and Y are either biholomorphic or conjugate biholomorphic.

1.10.2. Historical background, literature, and current progress

The problem is **unresolved**. The assertion is known in complex dimension 2 and under curvature hypotheses stronger than negative Riemannian sectional curvature, but no verified result covers the stated problem in all dimensions $n \geq 2$.

- Yum-Tong Siu, “The Complex-Analyticity of Harmonic Maps and the Strong Rigidity of Compact Kähler Manifolds,” *Annals of Mathematics* 112 (1980), 73–111, DOI [10.2307/1971321](https://doi.org/10.2307/1971321). Theorem 2 proves the desired biholomorphic-or-conjugate-biholomorphic conclusion when the curvature tensor of one manifold is **strongly negative in Siu’s sense**; Siu explicitly notes that this condition is stronger than negative Riemannian sectional curvature. Thus this is a major special case, not a proof under the exact hypotheses.
- Fangyang Zheng, “First Pontrjagin Form, Rigidity and Strong Rigidity of Nonpositively Curved Kähler Surface of General Type,” *Mathematische Zeitschrift* 220 (1995), 159–169, DOI [10.1007/BF02572607](https://doi.org/10.1007/BF02572607). Zheng proves the strong-rigidity result needed for compact Kähler surfaces; consequently the exact problem is settled for $n = 2$. It does not cover complex dimensions $n \geq 3$.
- Bing-Long Chen and Xiaokui Yang, “Compact Kähler Manifolds Homotopic to Negatively Curved Riemannian Manifolds,” *Mathematische Annalen* 370 (2018), 1477–1489, DOI [10.1007/s00208-017-1521-7](https://doi.org/10.1007/s00208-017-1521-7), arXiv:1609.06918. The authors state Yau’s conjecture with negative Riemannian sectional curvature, identify Zheng’s result as its complete solution in dimension 2, and prove that any compact Kähler manifold homotopy equivalent to a compact negatively curved Riemannian manifold has ample canonical bundle and a negative Kähler–Einstein metric; they also prove an almost-complex hyperbolicity result. These conclusions impose strong restrictions on both X and Y but do not identify their complex structures.
- Georgios Daskalopoulos and Chikako Mese, “Rigidity of Teichmüller Space,” *Inventiones*

Mathematicae 224 (2021), 791–916, DOI 10.1007/s00222-020-01020-2. Its historical account again describes the negative-sectional-curvature assertion as Yau’s conjecture and cites Siu’s theorem only for the stronger “strong negative curvature” case.

Accordingly, the remaining gap is the biholomorphic or conjugate-biholomorphic conclusion in complex dimensions $n \geq 3$ when the available metrics are assumed only to have negative Riemannian sectional curvature. No proof or counterexample covering that gap was found.

1.10.3. Remaining open problems recorded by the database

No separate descendant problem is listed in the website’s “Some remaining open problems” field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

1.10.4. Primary-source pointer

The originating ICM record is *Extension Problems in Several Complex Variables* (1978). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1978.2/ICM1978.2.ocr.pdf#page=165>.

1.11. 11. Some recent developments in complex differential geometry [1983-geom-32]

Record: 1983-geom-32

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1983: Some recent developments in complex differential geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1983.1/ICM1983.1.ocr.pdf#page=349>

1.11.1. Problem statement

Every complete non-compact Kähler manifold of complex dimension n with positive sectional curvature is biholomorphic to $\mathbb{C}^n$.

1.11.2. Historical background, literature, and current progress

1.11.3. Status: partially resolved

1.11.3.1. Precise problem

The problem asks the Greene-Wu/Yau weak uniformization problem: whether every complete noncompact Kähler manifold of arbitrary complex dimension n , with positive real sectional curvature, must be biholomorphic to $\mathbb{C}^n$.

This is not merely the better-known positive holomorphic bisectional curvature version. Results for bisectional curvature with extra hypotheses such as maximal volume growth do not settle the problem statement, because the statement assumes no volume-growth, curvature-decay, bounded-curvature, or finite-Chern-number condition.

1.11.3.2. Verdict

Partially resolved. The statement is now proved in complex dimension 2, and dimension 1 is classical, but I found no citable theorem proving the full arbitrary-dimensional statement under only positive sectional curvature.

1.11.3.3. Progress and key literature

1. Yum-Tong Siu, "Some Recent Developments in Complex Differential Geometry." Proceedings of the International Congress of Mathematicians, Warsaw 1983, Vol. 1, PWN/North-Holland, 1984, pp. 287-297. URL: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1983.1/ICM1983.1.ocr.pdf>

This is the dataset's original source context. Later papers identify the positive-sectional-curvature statement as the Greene-Wu/Yau weak form of Yau's uniformization conjecture.

2. **Ved Datar, Vamsi Pritham Pingali, Harish Seshadri, "Uniformisation of complete Kähler surfaces with positive sectional curvature."** arXiv preprint, 2026. arXiv:2604.11220. DOI/URL: <https://doi.org/10.48550/arXiv.2604.11220>

Theorem 1 proves: if (X, ω) is a complete noncompact Kähler surface with positive sectional curvature, then X is biholomorphic to $\mathbb{C}^2$. This exactly resolves the problem statement in complex dimension 2, with no extra assumptions at infinity. It does not cover $n \geq 3$.

3. **Ved V. Datar, Vamsi Pritham Pingali, Harish Seshadri, "The top Yau-Yang conjecture for Kähler manifolds with positive sectional curvature."** arXiv preprint, 2026. arXiv:2606.19806. DOI/URL: <https://doi.org/10.48550/arXiv.2606.19806>

Theorem 1.1 proves finite top Ricci mass $\int_X \text{Ric}_\omega^n < \infty$ for complete noncompact Kähler n -manifolds with positive sectional curvature. Theorem 1.2 gives quasiprojectivity under bounded sectional curvature. This is strong all-dimensional progress, but quasiprojectivity is weaker than being biholomorphic to $\mathbb{C}^n$. The paper explicitly notes that a higher-dimensional Ramanujam-type rigidity result is a challenging open problem.

4. **Ved Datar, Vamsi Pritham Pingali, Harish Seshadri, "The complex Monge-Ampere equation and an application to uniformisation of surfaces."** arXiv preprint, 2025. arXiv:2511.06849. DOI/URL: <https://doi.org/10.48550/arXiv.2511.06849>

Proves the surface case under positive and bounded sectional curvature by showing finiteness of the square of the Ricci form and applying Chen-Zhu/Ramanujam. This was superseded for the unbounded surface case by arXiv:2604.11220.

5. **Bing-Long Chen, Xi-Ping Zhu, "Positively curved complete noncompact Kähler manifolds."** Acta Mathematica Scientia 29B(4), 829-845, 2009. DOI: 10.1016/S0252-9602(09)60073-4.

Gives earlier partial uniformization results using finite analytic Chern number/top Ricci integral hypotheses; in complex dimension 2, combined with Ramanujam-type criteria, these imply $\mathbb{C}^2$ under additional assumptions. These hypotheses are not part of the problem.

6. **Bing-Long Chen, Xi-Ping Zhu, "Yau's uniformization conjecture for manifolds with non-maximal volume growth."** Acta Mathematica Scientia 38B(5), 1468-1484, 2018. DOI: 10.1016/S0252-9602(18)30828-2.

Surveys progress and proves partial results under non-maximal/minimal volume-growth and finite-Chern-type conditions. These are extra hypotheses and do not settle the arbitrary-dimensional positive-sectional statement.

7. **Gang Liu, "On Yau's uniformization conjecture."** Cambridge Journal of Mathematics 7(1-2), 33-70, 2019. DOI: 10.4310/CJM.2019.v7.n1.a2. arXiv:1606.08958.

Theorem 1.1 proves that a complete noncompact Kähler manifold with nonnegative bi-sectional curvature and maximal volume growth is biholomorphic to $\mathbb{C}^n$. This is important progress for Yau's bi-sectional-curvature conjecture, but maximal volume growth is an extra assumption not present in the problem statement.

8. **Man-Chun Lee, Luen-Fai Tam, “Chern-Ricci flows on noncompact complex manifolds.”** Journal of Differential Geometry 115(3), 529-564, 2020. DOI: 10.4310/jdg/1594260018. arXiv:1708.00141.

Gives an independent Chern-Ricci-flow proof of the maximal-volume-growth case with nonnegative bisectional curvature. Again, maximal volume growth is an extra hypothesis.

1.11.3.4. What remains unresolved

For complex dimensions $n \geq 3$, I found no published or otherwise citable theorem proving that every complete noncompact Kähler n -manifold with positive sectional curvature is biholomorphic to $\mathbb{C}^n$ without additional hypotheses. Current all-dimensional results reach finite top Ricci mass and quasiprojectivity under boundedness assumptions, but they do not imply $\mathbb{C}^n$ in higher dimensions.

1.11.4. Remaining open problems recorded by the database

1.11.4.1. Open problem 1

Let (X^n, ω) be a complete noncompact Kähler manifold with positive sectional curvature, with no boundedness assumption on the sectional curvature. Prove that X is biholomorphic to a quasiprojective variety.

Reference supplied by the database. Ved V. Datar, Vamsi P. Pingali, and Harish Seshadri, “The top Yau–Yang conjecture for Kähler manifolds with positive sectional curvature,” arXiv preprint (2026), arXiv:2606.19806, discussion immediately following Theorem 1.2 (the authors state that removing the bounded-sectional-curvature assumption is work in progress). <https://arxiv.org/abs/2606.19806>

1.11.4.2. Open problem 2

Let $n > 2$, and let X^n be a quasiprojective complex manifold admitting a complete Kähler metric of positive sectional curvature. Prove that X is biholomorphic to $\mathbb{C}^n$.

Reference supplied by the database. Ved V. Datar, Vamsi P. Pingali, and Harish Seshadri, “The top Yau–Yang conjecture for Kähler manifolds with positive sectional curvature,” arXiv preprint (2026), arXiv:2606.19806, discussion immediately following Theorem 1.2 (identified as the open higher-dimensional Ramanujam-type problem under curvature hypotheses). <https://arxiv.org/abs/2606.19806>

1.11.5. Primary-source pointer

The originating ICM record is *Some recent developments in complex differential geometry* (1983). The source link retained by the dataset is <https://www.mathunion.org/fileadmin>

[n/ICM/Proceedings/ICM1983.1/ICM1983.1.ocr.pdf#page=349.](#)

1.12. 12. Some recent developments in complex differential geometry [1983-geom-33]

Record: 1983-geom-33

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1983: Some recent developments in complex differential geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1983.1/ICM1983.1.ocr.pdf#page=349>

1.12.1. Problem statement

Prove that every complete noncompact Kähler manifold whose holomorphic bisectional curvature is positive is a Stein manifold.

1.12.2. Historical background, literature, and current progress

1.12.3. Verdict: partially resolved

1.12.3.1. Precise problem

The problem asks whether every complete noncompact Kahler manifold with strictly positive holomorphic bisectional curvature is necessarily a Stein manifold. This is the Greene-Wu/Siu Stein-type problem for positive bisectional curvature. It is weaker than Yau's uniformization conjecture, which asks whether such a manifold is biholomorphic to $\mathbb{C}^n$.

1.12.3.2. Current status

I did not find a published or otherwise citable result proving the full statement without extra hypotheses. Recent literature continues to treat the corresponding Stein/strong-Stein step under only $BK > 0$ as unknown. The problem is therefore not fully resolved. It is not completely untouched, however: several papers prove Steinness or the stronger conclusion $M \cong \mathbb{C}^n$ after adding volume growth, curvature decay, a pole/Busemann-exhaustion condition, or stronger positive sectional curvature.

1.12.3.3. Original/source literature

- Title: Some Recent Developments in Complex Differential Geometry
- Author: Yum-Tong Siu
- Venue: Proceedings of the International Congress of Mathematicians, Warszawa, 1983, pp. 287-297
- Year: 1983
- DOI/arXiv/URL: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1983.1/ICM1983.1.ocr.pdf>

- Relevance: This is the source named by the dataset. Later papers identify the Stein question for complete noncompact Kahler manifolds with positive bisectional curvature as a conjecture of Greene-Wu and Siu.

1.12.3.4. Partial progress

- Title: C infinity convex functions and manifolds of positive curvature
- Authors: R. E. Greene and H. Wu
- Venue: Acta Mathematica 137, 209-245
- Year: 1976
- DOI/arXiv/URL: <https://doi.org/10.1007/BF02392418>
- Result: Greene and Wu proved Steinness for complete noncompact Kahler manifolds under the stronger hypothesis of positive Riemannian sectional curvature.
- Why partial: Positive sectional curvature is a stronger geometric assumption than positive holomorphic bisectional curvature. This does not cover all manifolds in the problem statement.
- Title: On Kahler Manifolds of positive bisectional curvature and a theorem of Hartogs
- Authors: R. E. Greene and H. Wu
- Venue: Abhandlungen aus dem Mathematischen Seminar der Universitat Hamburg 47, 171-185
- Year: 1978
- DOI/arXiv/URL: <https://doi.org/10.1007/BF02941360>
- Result: Foundational work on function theory under positive bisectional curvature; frequently cited in later discussions of the Stein conjecture.
- Why partial: It does not prove that every complete noncompact Kahler manifold with positive bisectional curvature is Stein.
- Title: An embedding theorem of complete Kahler manifolds with positive bisectional curvature onto affine algebraic varieties
- Author: Ngaiming Mok
- Venue: Bulletin de la Societe Mathematique de France 112, 197-258
- Year: 1984
- DOI/arXiv/URL: <https://doi.org/10.24033/bsmf.2005>; <https://www.numdam.org/articles/10.24033/bsmf.2005/>
- Result: Mok explicitly states the conjecture that a noncompact complete Kahler manifold of positive bisectional curvature is Stein. He proves Steinness and affine-algebraic embedding under added hypotheses, including Euclidean volume lower growth and quadratic scalar-curvature decay.
- Why partial: The statement has no volume-growth or curvature-decay assumptions.
- Title: On complete noncompact Kahler manifolds with positive bisectional curvature
- Authors: Bing-Long Chen and Xi-Ping Zhu

- Venue: Mathematische Annalen 327, 1-23
- Year: 2003
- DOI/arXiv/URL: <https://doi.org/10.1007/s00208-003-0385-1> ;
arXiv:math/0211370, <https://arxiv.org/abs/math/0211370>
- Result: The paper proves that complete noncompact Kahler manifolds with positive bisectional curvature satisfying suitable growth conditions are biholomorphic to pseudoconvex domains of $\mathbb{C}^n$, and obtains stronger surface uniformization results under additional geometric hypotheses.
- Why partial: The theorem requires extra growth/curvature assumptions not present in the problem statement.
- Title: Plurisubharmonic functions and the structure of complete Kahler manifolds with nonnegative curvature
- Authors: Lei Ni and Luen-Fai Tam
- Venue: Journal of Differential Geometry 64(3), 457-524
- Year: 2003
- DOI/arXiv/URL: <https://doi.org/10.4310/jdg/1090427001> ;
arXiv:math/0304096, <https://arxiv.org/abs/math/0304096>
- Result: Corollary 4.1 states that a complete Kahler manifold with nonnegative holomorphic bisectional curvature is Stein if it has maximum volume growth or has a pole.
- Why partial: Positive holomorphic bisectional curvature implies nonnegative bisectional curvature, but the statement does not assume maximum volume growth or existence of a pole.
- Title: On Yau's uniformization conjecture
- Author: Gang Liu
- Venue: Cambridge Journal of Mathematics 7, no. 1-2, 33-70
- Year: 2019
- DOI/arXiv/URL: https://archive.intlpress.com/site/pub/files/_fulltext/journals/cjm/2019/0007/0001/CJM-2019-0007-0001-a002.pdf
- Result: Proves the stronger conclusion that a complete noncompact Kahler manifold with nonnegative bisectional curvature and maximal volume growth is biholomorphic to $\mathbb{C}^n$.
- Why partial: Maximal volume growth is an additional hypothesis.
- Title: Chern-Ricci flows on noncompact complex manifolds
- Authors: Man-Chun Lee and Luen-Fai Tam
- Venue: Journal of Differential Geometry 115(3), 529-564
- Year: 2020
- DOI/arXiv/URL: <https://doi.org/10.4310/jdg/1594260018>
- Result: Gives another proof, by Chern-Ricci flow methods, that the maximal-volume-growth case is biholomorphic to $\mathbb{C}^n$.
- Why partial: Again, the maximal volume growth assumption is not part of the problem.

- Title: The complex Monge-Ampere equation and an application to uniformisation of surfaces
- Authors: V. Datar, V. P. Pingali, and H. Seshadri
- Venue: arXiv preprint
- Year: 2025
- DOI/arXiv/URL: arXiv:2511.06849, <https://arxiv.org/abs/2511.06849>
- Result: The introduction notes that positive sectional curvature gives Steinness by Greene-Wu, but that neither Steinness nor the related exhaustion-function conclusion is known under the weaker assumption $BK > 0$. It proves related Monge-Ampere/psh-weight results under added hypotheses.
- Why partial/status evidence: This is recent evidence that the full bisectional-curvature-only Stein problem remains open.
- Title: Uniformisation of complete Kahler surfaces with positive sectional curvature
- Authors: V. Datar, V. P. Pingali, and H. Seshadri
- Venue: arXiv preprint
- Year: 2026
- DOI/arXiv/URL: arXiv:2604.11220, <https://arxiv.org/abs/2604.11220>
- Result: The paper discusses Yau's noncompact positive bisectional-curvature conjecture and states that proving a strong Stein-type exhaustion under $BK > 0$ is presently open; it proves results for surfaces under positive sectional curvature or additional assumptions.
- Why partial/status evidence: Its hypotheses are stronger or include extra conditions, and it does not resolve the exact statement.

1.12.3.5. What remains unresolved

No cited result proves that positive holomorphic bisectional curvature alone forces the existence of a strictly plurisubharmonic exhaustion, or otherwise proves Steinness, for every complete noncompact Kahler manifold. The unresolved gap is precisely the removal of the extra hypotheses used in known results: maximal volume growth, curvature decay, a pole/Busemann exhaustion, positive sectional curvature, bounded curvature, or surface-specific assumptions.

1.12.4. Remaining open problems recorded by the database

1.12.4.1. Open problem 1

Let (X, ω) be a complete noncompact Kähler manifold with positive holomorphic bisectional curvature. Must the cohomology at infinity $H_{\infty}^1(X; \mathbb{R})$ vanish?

Reference supplied by the database. Ved Datar, Vamsi Pritham Pingali, and Harish Seshadri, "Uniformisation of complete Kähler surfaces with positive sectional curvature," arXiv preprint (2026), arXiv:2604.11220, Section 8.3, "Topological aspects," concluding question. <https://arxiv.org/abs/2604.11220>

1.12.5. Primary-source pointer

The originating ICM record is *Some recent developments in complex differential geometry* (1983). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1983.1/ICM1983.1.ocr.pdf#page=349>.

1.13. 13. The geometry of Teichmüller space [1983-geom-35]

Record: 1983-geom-35

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1983: The geometry of Teichmüller space

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1983.1/ICM1983.1.ocr.pdf#page=727>

1.13.1. Problem statement

For a fixed closed surface S of genus g , let T_g be its Teichmüller space, the space of conformal structures on S modulo conformal equivalence by maps isotopic to the identity. Let μ, ν be measured geodesic laminations on S that together fill S , meaning that no essential simple closed curve is disjoint from both. Define the line of minima

$$L_{\mu, \nu} = \{X_t : 0 < t < 1\},$$

where, for each $t \in (0, 1)$, X_t is the unique point of T_g minimizing

$$X \mapsto t l_\mu(X) + (1 - t) l_\nu(X),$$

and $l_\lambda(X)$ denotes the geodesic length of the measured lamination λ in the structure X . Prove or disprove that any two distinct points of T_g lie on exactly one such line of minima.

1.13.2. Historical background, literature, and current progress

1.13.3. Status check

Verdict: partially resolved.

1.13.3.1. Precise problem

For a fixed closed surface S , Kerckhoff's line of minima associated to a filling pair of measured laminations μ, ν is the set of minimizers of positive linear combinations of the length functions l_μ and l_ν . The problem asks whether every two distinct points of Teichmüller space T_g lie on exactly one such line of minima. Since rescaling μ and ν only changes the parametrization, the relevant uniqueness is uniqueness of the line as a subset of T_g , equivalently of the projective filling pair up to the standard interchange/reparametrization ambiguity.

1.13.3.2. Current status

The problem is not fully resolved. Kerckhoff proved the existence half: any two points of Teichmüller space lie on at least one line of minima. I found no published or citable result

proving the missing uniqueness assertion for all closed surfaces and all pairs of points, nor a counterexample disproving it. Later literature develops the geometry, boundary behavior, quasi-geodesic properties, and geodesic-current projection framework for lines of minima, but these results do not imply uniqueness of a line through arbitrary two points of T_g .

1.13.3.3. Key literature and progress

1. Steven P. Kerckhoff, "The geometry of Teichmuller space", Proceedings of the International Congress of Mathematicians, Warsaw 1983, Vol. 1, PWN, Warsaw, 1984, pp. 665-678. URL: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1983.1/ICM1983.1.ocr.pdf>

This is the original ICM source context for the question about geometric analogues in Teichmuller space.

2. Steven P. Kerckhoff, "Lines of minima in Teichmuller space", Duke Mathematical Journal 65, no. 2 (1992), 187-213. DOI: 10.1215/S0012-7094-92-06507-0. URL: <https://doi.org/10.1215/S0012-7094-92-06507-0>

Kerckhoff proved that for filling measured laminations the positive linear combination of length functions has a unique minimizer for each parameter, giving a well-defined line of minima. His Theorem 3.6/Corollary 3.7 proves existence of a line of minima between every pair of points in $T(S)$. This is substantial partial progress, but it is not the full statement because it does not prove uniqueness of that line.

3. Caroline Series, "Kerckhoff's lines of minima in Teichmuller space", in Handbook of Teichmuller Theory, Volume III, IRMA Lectures in Mathematics and Theoretical Physics 17, EMS Press, 2012, pp. 123-153. DOI: 10.4171/103-1/4. URL: <https://ems.press/books/irma/113/contents>

Series surveys Kerckhoff's theory and explicitly records the status: lines of minima for a fixed lamination foliate $T(S)$, any two points lie on at least one line, and whether that line is unique is unknown. This confirms that the existence theorem did not settle the exact uniqueness problem.

4. Raquel Diaz and Caroline Series, "Limit points of lines of minima in Thurston's boundary of Teichmuller space", Algebraic & Geometric Topology 3 (2003), 207-234. DOI: 10.2140/agt.2003.3.207. arXiv: math/0303108. URL: <https://arxiv.org/abs/math/0303108>

This paper studies limiting behavior at Thurston's boundary. It proves convergence for maximal uniquely ergodic laminations and different behavior for rational laminations. These results clarify boundary behavior of a given line, but they do not address uniqueness of a line through two interior points of T_g .

5. Young-Eun Choi, Kasra Rafi, and Caroline Series, "Lines of Minima and Teichmuller Geodesics", Geometric and Functional Analysis 18 (2008), 698-754. DOI: 10.1007/s00039-008-0675-6. arXiv: math/0605135. URL: <https://arxiv.org/abs/math/0605135>

This paper compares a fixed line of minima with the Teichmuller geodesic determined by the same filling pair, characterizing short curves and estimating distances. It proves strong structure for individual lines but not uniqueness through arbitrary point pairs.

6. Young-Eun Choi, Kasra Rafi, and Caroline Series, "Lines of minima are uniformly quasi-geodesic", Pacific Journal of Mathematics 237, no. 1 (2008), 21-44. DOI:

10.2140/pjm.2008.237.21. arXiv: 0706.2044. URL: <https://arxiv.org/abs/0706.2044>

This proves that every line of minima is a Teichmüller quasi-geodesic with uniform constants depending only on the surface topology. This is further geometric progress, but it is compatible with either uniqueness or non-uniqueness through two points.

7. Yuki Iguchi, “Lines of minima with no end in Thurston’s boundary of Teichmüller space”, *Conformal Geometry and Dynamics* 16 (2012), 22-43. DOI: 10.1090/S1088-4173-2012-00240-8. URL: <https://www.ams.org/ecgd/2012-16-02/S1088-4173-2012-00240-8/>

This gives examples of lines of minima with no limit in the Thurston compactification and locates their limit sets in a simplex. It concerns asymptotic behavior, not the uniqueness of a line through two prescribed interior points.

8. Sebastian Hensel and Jenya Sapir, “A projection from filling currents to Teichmüller space”, *Proceedings of the American Mathematical Society* 151 (2023), 3621-3633. DOI: 10.1090/proc/16311. arXiv: 2109.14768. URL: <https://arxiv.org/abs/2109.14768>

Hensel and Sapir generalize Kerckhoff’s minimizer construction to a mapping-class-group-equivariant, continuous, proper projection from projective filling geodesic currents to $T(S)$. Their result recovers/generalizes the existence of unique minimizers for filling currents, but it does not prove the injectivity or two-point uniqueness needed to conclude that two arbitrary points determine exactly one Kerckhoff line of minima.

1.13.3.4. What remains unresolved

The unresolved part is the uniqueness clause: given two distinct points $X, Y \in T_g$, determine whether there is at most one line of minima $L_{\mu, \nu}$ containing both. The literature proves existence and many structural properties of lines of minima, but I found no theorem covering all genera and all point pairs that proves this uniqueness, and no published counterexample that would settle the question negatively.

1.13.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website’s “Some remaining open problems” field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

1.13.5. Primary-source pointer

The originating ICM record is *The geometry of Teichmüller space* (1983). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1983.1/ICM1983.1.ocr.pdf#page=727>.

1.14. 14. E. A., Almost Lie groups [1986-geom-290]

Record: 1986-geom-290

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1986: E. A., Almost Lie groups

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1986.1/ICM1986.1.ocr.pdf#page=663>

1.14.1. Problem statement

Determine the exact value of

$$d_{\min} = \inf\{\text{diam}(S^3, g) : g \text{ is a smooth Riemannian metric on } S^3, |K_g(\sigma)| \leq 1 \text{ for every tangent 2-plane } \sigma\}.$$

where $K_g(\sigma)$ denotes the sectional curvature of g on σ .

1.14.2. Historical background, literature, and current progress

Verdict: unresolved.

The exact value of $d_{\min}$ is not known. The verified literature gives the explicit bracket

$$\exp\left(-\frac{1}{2} \exp(\exp 9)\right) \leq d_{\min} \leq \pi,$$

but does not determine either equality. The upper bound is attained by the unit round metric on S^3 . For the lower bound, Theorem 1.5 of Buser and Karcher states that a compact n -manifold satisfying $|K| < \varepsilon_n \text{diam}^{-2}$, with $\varepsilon_n = \exp(-\exp(\exp(n^2)))$, has a fundamental group containing a torsion-free nilpotent normal subgroup of rank n . If an admissible metric on S^3 had diameter below $\sqrt{\varepsilon_3}$, this theorem would contradict $\pi_1(S^3) = 0$. This proves only the displayed very small positive lower bound.

Ernst A. Ruh, “Almost Lie Groups,” in *Proceedings of the International Congress of Mathematicians, Berkeley, California, USA, 1986*, Vol. 1, American Mathematical Society, 1987, pp. 663-666, [official proceedings PDF](#). Ruh explicitly formulates this S^3 extremal problem, records $d_{\min} \leq \pi$ and an almost-flat lower estimate, and says that the optimal answer was unknown.

Peter Buser and Hermann Karcher, “Gromov’s Almost Flat Manifolds,” *Asterisque* 81, Societe mathematique de France, 1981, 154 pp., [Numdam](#). Their Theorem 1.5 supplies the quantitative constant used above. The underlying structural results are Mikhail Gromov, “Almost Flat Manifolds,” *Journal of Differential Geometry* 13 (1978), 231-241, <https://doi.org/10.4310/jdg/1214434488>, and Ernst A. Ruh, “Almost Flat Manifolds,” *Journal of Differential Geometry* 17 (1982), 1-14, <https://doi.org/10.4310/jdg/1214436698>. These results force sufficiently small-diameter, two-sided curvature-bounded manifolds to be infranil; they do not compute the threshold for S^3 .

Jyh-Yang Wu, “The Topological Spectrum of a Smooth Closed Manifold,” *Indiana University Mathematics Journal* 44 (1995), 511-534, <https://doi.org/10.1512/iumj.1995>.

44.1999. Wu defines the same minimal-diameter invariant for every closed manifold and proves a dimension-dependent gap theorem characterizing the zero case by infranilmanifolds. Since S^3 is not infranil, this confirms positivity but gives no exact value.

A. V. Podobryaev, “Diameter of SU_2 for a Left-Invariant Axisymmetric Riemannian Metric,” *Mathematical Notes* 103 (2018), 846–851, <https://doi.org/10.1134/S0001434618050188>, [arXiv:1710.02945](https://arxiv.org/abs/1710.02945). This paper gives an exact diameter formula for Berger spheres, the axisymmetric left-invariant subfamily on S^3 , but it does not cover arbitrary Riemannian metrics and therefore does not resolve the stated infimum.

Thus established progress consists of positivity, the round upper bound, general almost-flat gap theorems, and exact calculations in a restricted homogeneous family. The gap between the explicit lower bound and π , for arbitrary smooth metrics satisfying $|K| \leq 1$, remains unresolved.

1.14.3. Remaining open problems recorded by the database

1.14.3.1. Open problem 1

Let

$$SU(2) = \{U \in M_2(\mathbb{C}) : U^*U = I_2, \det U = 1\}$$

and let

$$X_1 = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad X_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad X_3 = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$$

be a basis of its Lie algebra $\mathfrak{su}(2)$. For positive real numbers a, b, c , let $g_{(a,b,c)}$ be the left-invariant Riemannian metric on $SU(2)$ for which $\{aX_1, bX_2, cX_3\}$ is an orthonormal basis at the identity. Determine $\text{diam}(SU(2), g_{(a,b,c)})$ explicitly as a function of a, b, c .

Reference supplied by the database. Emilio A. Lauret, “Diameter of $SU(2)$ endowed with a left-invariant metric,” MathOverflow, Question 281481, 2017, <https://mathoverflow.net/questions/281481/diameter-of-mathrmsu2-endowed-with-a-left-invariant-metric>.

1.14.4. Primary-source pointer

The originating ICM record is E. A., *Almost Lie groups* (1986). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1986.1/ICM1986.1.ocr.pdf#page=663>.

1.15. 15. W., Manifolds of nonpositive sectional curvature and manifolds without conjugate points [1986-geom-304]

Record: 1986-geom-304

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1986: W., Manifolds of nonpositive sectional curvature and manifolds without conjugate points

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1986.1/ICM1986.1.ocr.pdf#page=586>

1.15.1. Problem statement

Determine whether every compact Riemannian manifold M without conjugate points is flat (that is, has identically zero sectional curvature) if the measure-theoretic entropy $h_\mu(g^t)$ of its geodesic flow $g^t : SM \rightarrow SM$, $g^t(v) = \dot{\gamma}_v(t)$, on the unit tangent bundle SM , with respect to the normalized Liouville measure μ , is zero.

1.15.2. Historical background, literature, and current progress

Verdict: unresolved.

No proof or counterexample is known for the exact implication with only the assumptions of compactness, absence of conjugate points, and zero entropy of normalized Liouville measure. Ballmann explicitly posed this implication and recorded it as unknown. Freire and Mañé's entropy formula identifies h_μ with the Liouville average of the trace of the unstable Green–Riccati operator, but without a no-focal-points or curvature-sign assumption that operator need not have the pointwise positivity needed to turn a zero average into flatness.

Established progress is as follows.

- Alexandre Freire and Ricardo Mañé, *On the Entropy of the Geodesic Flow in Manifolds without Conjugate Points*, **Inventiones Mathematicae** 69 (1982), 375–392, <https://doi.org/10.1007/BF01389360>. They prove the entropy formula for compact manifolds without conjugate points (and identify topological entropy with volume entropy). This is the foundational reduction, not a flatness theorem in the full generality of the question.
- Werner Ballmann, *Manifolds of Nonpositive Sectional Curvature and Manifolds without Conjugate Points*, in **Proceedings of the International Congress of Mathematicians, Berkeley 1986**, Vols. 1–2, American Mathematical Society, 1987, pp. 484–490, [official proceedings PDF](#). Ballmann states the exact zero-Liouville-entropy question and says it is unknown.
- Werner Ballmann and Maciej P. Wojtkowski, *An Estimate for the Measure Theoretic Entropy of Geodesic Flows*, **Ergodic Theory and Dynamical Systems** 9 (1989), 271–279, <https://doi.org/10.1017/S014338570000496X>. For compact manifolds of nonpositive sectional curvature they prove a curvature lower bound for h_μ ; equality $h_\mu = 0$ forces the curvature operator, hence all sectional curvature, to vanish. Thus the problem is resolved affirmatively under the additional hypothesis $K \leq 0$.

- Gerhard Knieper, *Mannigfaltigkeiten ohne konjugierte Punkte*, **Bonner Mathematische Schriften** 168, Universität Bonn, 1986, MR 0851010, [University of Bonn series catalogue](#); see also Rafael O. Ruggiero and Vladimir A. Rosas Meneses, *On the Pesin Set of Expansive Geodesic Flows in Manifolds with No Conjugate Points*, **Bulletin of the Brazilian Mathematical Society** 34 (2003), 263–274, <https://doi.org/10.1007/s00574-003-0012-5>. Knieper proves that a compact manifold without conjugate points and with bounded asymptote is a torus when its metric entropy is zero. Dmitri Burago and Sergei Ivanov, *Riemannian Tori without Conjugate Points Are Flat*, **Geometric and Functional Analysis** 4 (1994), 259–269, <https://doi.org/10.1007/BF01896241>, then supplies flatness. This resolves the bounded-asymptote case, which includes manifolds without focal points, but not all manifolds without conjugate points.
- Sergio Romaña, *Curvature Rigidity Through Level Sets of Lyapunov Exponents in Geodesic Flows*, arXiv:2507.02482 (2025), [arXiv:2507.02482](https://arxiv.org/abs/2507.02482). Theorem 1.1 proves flatness for manifolds without focal points when the zero-Lyapunov-exponent level set has full measure for a fully supported invariant measure. Applied to Liouville measure via Pesin’s formula, this again settles a stronger-hypothesis case, not the general no-conjugate-points case.
- Javier Echevarría Cuesta and James Marshall Reber, *Thermostats without Conjugate Points*, **Ergodic Theory and Dynamical Systems** 46 (2026), 1494–1520, <https://doi.org/10.1017/etds.2026.10285>. They still describe the stronger Freire–Mañé zero-topological-entropy rigidity statement as a conjecture in general (while explaining the surface case). Since zero Liouville entropy is weaker than zero topological entropy, this does not close the present gap.

What remains is precisely to control the cancellation and low-regularity phenomena of the Green bundles for arbitrary compact manifolds without conjugate points. The cited results do not remove the extra curvature, no-focal-points, bounded-asymptote, or regularity hypotheses, so they do not collectively resolve the stated problem.

1.15.3. Remaining open problems recorded by the database

1.15.3.1. Open problem 1

Determine whether every compact Riemannian manifold M of dimension at least 3 without conjugate points is flat if the topological entropy of its geodesic flow $g^t : SM \rightarrow SM$, $g^t(v) = \dot{\gamma}_v(t)$, on the unit tangent bundle SM is zero.

Reference supplied by the database. Javier Echevarría Cuesta and James Marshall Reber, *Thermostats without Conjugate Points*, **Ergodic Theory and Dynamical Systems** 46 (2026), no. 6, 1494–1520, Section 1.2 (Freire–Mañé conjecture, discussion preceding Theorem 1.9), <https://doi.org/10.1017/etds.2026.10285>.

1.15.4. Primary-source pointer

The originating ICM record is W., *Manifolds of nonpositive sectional curvature and manifolds without conjugate points* (1986). The source link retained by the dataset is <https://www.>

mathunion.org/fileadmin/ICM/Proceedings/ICM1986.1/ICM1986.1.ocr.pdf#page=586.

1.16. 16. Collapsing Riemannian Manifolds and Its Applications [1990-geom-305]

Record: 1990-geom-305

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1990: Collapsing Riemannian Manifolds and Its Applications

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1990.1/ICM1990.1.ocr.pdf#page=579>

1.16.1. Problem statement

Prove that for every integer $n > 1$ there exists a constant $\varepsilon_n > 0$, depending only on n , such that every closed n -dimensional Riemannian manifold M satisfying

1. $\pi_k(M) = 0$ for every integer $k > 1$,
2. $|K_M| \leq 1$, where K_M denotes the sectional curvature of M , and
3. $\pi_1(M)$ contains no subgroup isomorphic to $\mathbb{Z}^2$,

has $\text{Vol}(M) > \varepsilon_n$.

1.16.2. Historical background, literature, and current progress

The problem is **unresolved**. It asks for a dimension-only positive gap in Gromov's minimal volume for all closed aspherical manifolds whose fundamental groups contain no $\mathbb{Z}^2$. No proof or counterexample covering every dimension and every metric with $|K_M| \leq 1$ was found.

The following results give the principal verified progress.

- Kenji Fukaya, *Collapsing Riemannian Manifolds and Its Applications*, in *Proceedings of the International Congress of Mathematicians, Kyoto 1990*, Mathematical Society of Japan, 1991, pp. 491-500, Conjecture 3.6 ([official proceedings PDF](#)). Theorem 3.5 proves the required volume lower bound when an additional bound $\text{Diam}(M) \leq D$ is imposed (with the constant then depending on n and D); Conjecture 3.6 is precisely the request to remove that diameter hypothesis. The same bounded-diameter result originates in Fukaya's *A Compactness Theorem of a Set of Aspherical Riemannian Orbifolds*, in *A Fete of Topology*, Academic Press, 1988, pp. 391-413, DOI <https://doi.org/10.1016/B978-0-12-480440-1.50022-8>.
- S. V. Buyalo, *Volume and Fundamental Group of a Manifold of Nonpositive Curvature*, *Mathematics of the USSR-Sbornik* **50** (1985), no. 1, 137-150, DOI <https://doi.org/10.1070/SM1985v050n01ABEH002737>. Theorem 1.4 gives a dimension-only volume gap in the nonpositive-curvature setting: sufficiently small volume forces rank at least two, hence the relevant fundamental group contains a rank-two free abelian subgroup. Thus the original assertion is known for the additional hypothesis $K_M \leq 0$.
- Jianguo Cao and Xiaoyang Chen, *Minimal Volume and Simplicial Norm of Visibility*

n-Manifolds and Compact 3-Manifolds, 2008, arXiv:0812.3353. Their Theorem B records the geometrization consequence that a closed aspherical 3-manifold with no $\mathbb{Z}^2$ subgroup is hyperbolic, and their volume estimate gives $\text{MinVol}(M) \geq \|M\|/24$. Together with the dimension-three lower gap for volumes of closed hyperbolic manifolds, this settles the original problem in dimension 3 (dimension 2 follows directly from surface classification and Gauss-Bonnet).

- Roman Sauer, *Amenable Covers, Volume and L^2 -Betti Numbers of Aspherical Manifolds*, *Journal für die reine und angewandte Mathematik* **2009** (2009), no. 636, 47-92, DOI <https://doi.org/10.1515/CRELLE.2009.082>. Sauer proves dimension-dependent volume bounds in terms of L^2 -Betti numbers and shows that all such Betti numbers vanish below a universal minimal-volume threshold. This is substantive partial progress, but the hypothesis that $\pi_1(M)$ contains no $\mathbb{Z}^2$ does not provide the required nonvanishing invariant, so it does not settle the stated problem.
- Sergio Zamora, *Fundamental Groups of Aspherical Manifolds That Collapse*, 2021, arXiv:2101.06267, Theorems 1-2 and Question 8. With a uniform upper diameter bound and a Ricci lower bound, Zamora proves noncollapse for aspherical manifolds with non-elementary hyperbolic fundamental groups. Question 8 explicitly asks whether the diameter hypothesis can be removed. This later, narrower group-theoretic case remains posed as open and does not cover general groups having no $\mathbb{Z}^2$.

Consequently, the established bounded-diameter, nonpositive-curvature, low-dimensional, and invariant-dependent results do not exhaust the hypotheses of the problem. The remaining gap is a dimension-only volume bound in arbitrary dimension for mixed-sign sectional curvature, without any diameter bound or additional algebraic/topological invariant.

1.16.3. Remaining open problems recorded by the database

1.16.3.1. Open problem 1

Determine whether there exist an integer $d > 1$ and a sequence (M_i, g_i) of closed d -dimensional Riemannian manifolds such that $\pi_k(M_i) = 0$ for every $k > 1$, each fundamental group $\pi_1(M_i)$ is a non-elementary word-hyperbolic group (that is, it is word-hyperbolic and not virtually cyclic), $\text{Ric}_{g_i} \geq -1$, and $\text{Vol}(M_i, g_i) \rightarrow 0$ as $i \rightarrow \infty$.

Reference supplied by the database. Sergio Zamora, “Fundamental Groups of Aspherical Manifolds That Collapse,” preprint, 2021, Question 8, arXiv:2101.06267, <https://arxiv.org/abs/2101.06267>.

1.16.4. Primary-source pointer

The originating ICM record is *Collapsing Riemannian Manifolds and Its Applications* (1990). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1990.1/ICM1990.1.ocr.pdf#page=579>.

1.17. 17. Metric and Topological Measurements of Manifolds [1990-geom-307]

Record: 1990-geom-307

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1990: Metric and Topological Measurements of Manifolds

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1990.1/ICM1990.1.ocr.pdf#page=599>

1.17.1. Problem statement

Fix an integer $n \geq 2$, a real number ℓ , and a constant $D > 0$. Construct and characterize a structural decomposition for every sequence (M_i) of closed, connected Riemannian n -manifolds such that

$$\sec_{M_i} \geq \ell, \quad \text{diam}(M_i) \leq D, \quad \text{vol}(M_i) \rightarrow 0.$$

After passing to a subsequence, (M_i) converges in the Gromov–Hausdorff topology to a compact metric space X of dimension strictly less than n ; this limit is an inner metric space (the distance between two points is the infimum of the lengths of curves joining them) and is an Alexandrov space with curvature bounded below by ℓ in the sense of Toponogov triangle comparison. The decomposition must describe the manifolds M_i , for all sufficiently large i , in terms of the geometry and singular structure of an arbitrary such limit X , and thereby give a corresponding structural description of every closed, connected Riemannian n -manifold M satisfying

$$\sec_M \geq \ell, \quad \text{diam}(M) \leq D, \quad \text{vol}(M) \leq \varepsilon$$

when $\varepsilon > 0$ is sufficiently small in terms of n , ℓ , and D .

The decomposition must extend the following smooth-limit case. If X is a Riemannian manifold, then, for all sufficiently large i , there is an almost-Riemannian fibration $M_i \rightarrow X$, meaning a smooth surjective submersion whose differential on vectors orthogonal to the fibers has length distortion tending uniformly to 1 as $i \rightarrow \infty$. Moreover, for every fiber F , some finite covering space of F admits an almost-Riemannian fibration onto a flat torus of dimension

$$b_1(F) = \dim H^1(F; \mathbb{R}).$$

1.17.2. Historical background, literature, and current progress

Status: unresolved.

The exact problem asks for a structural decomposition in every dimension and for every compact singular Alexandrov limit. No such general theory is currently known. The literature supplies a smooth-base fibration theorem, complete low-dimensional classifications, local restrictions for arbitrary singular limits, and global fibrations under additional regularity as-

sumptions on the limit, but these results do not cover arbitrary singular Alexandrov spaces in arbitrary dimension.

Key established results are as follows.

- Takao Yamaguchi, “Collapsing and pinching under a lower curvature bound,” *Annals of Mathematics* 133 (1991), 317–357, DOI [10.2307/2944340](https://doi.org/10.2307/2944340). This proves the benchmark case in the statement: when the lower-dimensional limit is a smooth Riemannian manifold, the collapsing manifolds admit almost-Riemannian fibrations over it; the fibers have the stated virtual torus-fibration property. It does not treat general singular limits.
- Grigori Perelman, “Collapsing with No Proper Extremal Subsets,” in *Comparison Geometry*, MSRI Publications 30, Cambridge University Press, 1997, 149–156, DOI [10.1017/9781009701891.007](https://doi.org/10.1017/9781009701891.007). If the limit has no proper extremal subsets, Perelman obtains the Serre homotopy exact sequence associated with a regular fiber. The paper explicitly presents an actual fiber-bundle conclusion only as an expectation and proves this weaker homotopy statement, so it is not the required decomposition even under its extra hypothesis.
- Vitali Kapovitch, “Restrictions on collapsing with a lower sectional curvature bound,” *Mathematische Zeitschrift* 249 (2005), 519–539, [arXiv:math/0110087](https://arxiv.org/abs/math/0110087). For an arbitrary singular Alexandrov limit X , Theorem 1.1 shows that a fixed-radius ball above any $x \in X$ is simply homotopy equivalent to a finite CW complex of dimension at most $n - \dim X$. This is substantive local topological control, but it gives neither singular fibers nor a compatible global decomposition.
- Takashi Shioya and Takao Yamaguchi, “Collapsing Three-Manifolds Under a Lower Curvature Bound,” *Journal of Differential Geometry* 56 (2000), 1–66, DOI [10.4310/jdg/1090347524](https://doi.org/10.4310/jdg/1090347524), and Takao Yamaguchi, “Collapsing 4-manifolds under a lower curvature bound,” revised preprint (2024), [arXiv:1205.0323](https://arxiv.org/abs/1205.0323). These classify the topology and generalized fiber structures of collapsing closed orientable manifolds in dimensions three and four, respectively. They are low-dimensional special cases and do not extend to arbitrary n .
- Tadashi Fujioka, “A fibration theorem for collapsing sequences of Alexandrov spaces,” *Journal of Topology and Analysis* 15 (2023), 265–298, DOI [10.1142/S179352532150028X](https://doi.org/10.1142/S179352532150028X), and “Collapsing to Alexandrov spaces with isolated mild singularities,” *Differential Geometry and its Applications* 85 (2022), article 101951, DOI [10.1016/j.difgeo.2022.101951](https://doi.org/10.1016/j.difgeo.2022.101951), [arXiv:2108.11030](https://arxiv.org/abs/2108.11030). These extend locally trivial fibrations beyond smooth limits under quantitative regularity of spaces of directions, and then to limits having only finitely many weakly strained exceptional points. The latter paper still explicitly asks for extensions to all weakly strained limits and to two-dimensional limits without proper extremal subsets; its hypotheses exclude general singular X .
- Hongzhi Huang, Xian-Tao Huang, Jikang Wang, and Xingyu Zhu, “Fibrations, the First Betti Number, and Almost Nonnegative Ricci Curvature,” preprint (2026), [arXiv:2605.24380](https://arxiv.org/abs/2605.24380). Theorem 1.4 refines Yamaguchi’s smooth-limit result by showing that the fiber itself, rather than merely a finite cover, fibers over its b_1 -dimensional torus. Its base is still a smooth manifold satisfying a Reifenberg-type regularity condition, so it does not address arbitrary singular Alexandrov limits.

Thus the remaining gap is precisely the requested compatible global description of singular fibers and their assembly over all strata of an arbitrary compact Alexandrov limit, uniformly in general dimension. None of the cited partial results covers that case, separately or in combination.

1.17.3. Remaining open problems recorded by the database

1.17.3.1. Open problem 1

Fix integers $n > k > 0$ and a real number κ . Let (M_j) be a sequence of closed n -dimensional Riemannian manifolds with sectional curvature at least κ that converges in the Gromov–Hausdorff topology to a compact k -dimensional Alexandrov space X with curvature at least κ . Assume that every point $p \in X$ is weakly k -strained, meaning that its space of directions $\Sigma_p X$ contains $k + 1$ directions whose pairwise angles are all greater than $\pi/2$. Prove that, for all sufficiently large j , the manifold M_j admits a locally trivial fibration over X .

Reference supplied by the database. Tadashi Fujioka, “Collapsing to Alexandrov spaces with isolated mild singularities,” *Differential Geometry and its Applications* 85 (2022), article 101951, Problem 1.3. DOI: <https://doi.org/10.1016/j.difgeo.2022.101951>; arXiv:2108.11030, <https://arxiv.org/abs/2108.11030>.

1.17.3.2. Open problem 2

Fix an integer $n > 2$ and a real number κ . Let (M_j) be a sequence of closed n -dimensional Riemannian manifolds with sectional curvature at least κ that converges in the Gromov–Hausdorff topology to a compact two-dimensional Alexandrov space X with curvature at least κ . Assume that, for every $p \in X$, the space of directions $\Sigma_p X$ is a circle of length greater than π . Prove that, for all sufficiently large j , the manifold M_j admits a locally trivial fibration over X .

Reference supplied by the database. Tadashi Fujioka, “Collapsing to Alexandrov spaces with isolated mild singularities,” *Differential Geometry and its Applications* 85 (2022), article 101951, Problem 1.4. DOI: <https://doi.org/10.1016/j.difgeo.2022.101951>; arXiv:2108.11030, <https://arxiv.org/abs/2108.11030>.

1.17.4. Primary-source pointer

The originating ICM record is *Metric and Topological Measurements of Manifolds* (1990). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1990.1/ICM1990.1.ocr.pdf#page=599>.

1.18. 18. An algorithm to recognize the 3 -sphere [1994-geom-51]

Record: 1994-geom-51

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1994: An algorithm to recognize the 3 -sphere

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.1/ICM1994.1.ocr.pdf#page=673>

1.18.1. Problem statement

Prove that, for each fixed genus, analytic minimal surfaces in a bumpy Riemannian metric have uniformly bounded index of instability, where a bumpy metric is one for which minimal surfaces have no Jacobi fields, equivalently are isolated critical points of the area functional.

1.18.2. Historical background, literature, and current progress

1.18.3. Verdict: partially resolved

1.18.3.1. Precise problem checked

The problem asks whether, for a fixed genus, closed analytic minimal surfaces in a bumpy Riemannian metric have a uniform upper bound on Morse index (index of instability). I interpret the intended setting, following the Pitts-Rubinstein/Hass-Norbury-Rubinstein literature, as closed embedded minimal surfaces in a closed Riemannian 3-manifold. A bumpy metric means that closed minimal surfaces have no nontrivial Jacobi fields.

I did not find a published or citable theorem that proves this exact statement for all bumpy analytic metrics, nor a citable counterexample satisfying both the bumpy and analytic hypotheses. The literature gives strong related results, but the exact bumpy/analytic fixed-genus question remains unresolved.

1.18.3.2. Progress and near-results

1. Original/source context

- **Title:** *An algorithm to recognize the 3-sphere*
- **Author:** J. Hyam Rubinstein
- **Venue/year:** Proceedings of the International Congress of Mathematicians, Zurich 1994, Birkhauser, 1995
- **DOI/URL:** DOI https://doi.org/10.1007/978-3-0348-9078-6_54; Springer URL: https://link.springer.com/chapter/10.1007/978-3-0348-9078-6_54
- **Relevance:** This is the source for this problem context for the Pitts-Rubinstein program relating bounded Morse index for fixed-genus minimal surfaces to 3-manifold

recognition/spherical space form questions. It does not itself provide a full resolution of the stated analytic bumpy-metric statement.

2. Fixed-genus counterexamples for smooth metrics, but not bumpy/analytic as stated

- **Title:** *Minimal spheres of arbitrarily high Morse index*
- **Authors:** Joel Hass, Paul Norbury, J. Hyam Rubinstein
- **Venue/year:** Communications in Analysis and Geometry 11 (2003), no. 3, 425-439
- **DOI/arXiv/URL:** DOI <https://doi.org/10.4310/CAG.2003.v11.n3.a2>; arXiv math/0206286; URL <https://arxiv.org/abs/math/0206286>
- **Result:** Constructs a smooth Riemannian metric on any 3-manifold with embedded minimal spheres (genus zero) of arbitrarily high Morse index.
- **Applicability check:** This disproves any unrestricted smooth-metric version of a fixed-genus index bound. It does **not** resolve the problem statement as stated, because the theorem does not assert that the constructed metric is bumpy or analytic. The paper also notes that in positive Ricci curvature fixed-genus compactness/index bounds reappear, so the high-index phenomenon is not automatically preserved by perturbation.

3. Genus-one high-index examples

- **Title:** *Metrics without Morse index bounds*
- **Authors:** Tobias H. Colding, Nancy Hingston
- **Venue/year:** Duke Mathematical Journal 119 (2003), no. 2, 345-365
- **DOI/arXiv/URL:** DOI <https://doi.org/10.1215/S0012-7094-03-11925-0>; arXiv math/0210290; URL <https://arxiv.org/abs/math/0210290>
- **Result:** Gives metrics with embedded minimal tori of arbitrarily high Morse index.
- **Applicability check:** This covers fixed genus one for specially constructed smooth metrics. The authors discuss bumpy metrics and genericity, but the theorem does not establish a bumpy analytic counterexample to this problem.

4. All fixed genera for smooth metrics, including positive scalar curvature cases

- **Title:** *Singular limit laminations, Morse index, and positive scalar curvature*
- **Authors:** Tobias H. Colding, Camillo De Lellis
- **Venue/year:** Topology 44 (2005), no. 1, 25-45
- **DOI/arXiv/URL:** DOI <https://doi.org/10.1016/j.top.2004.01.007>; arXiv math/0208100; URL <https://arxiv.org/abs/math/0208100>
- **Result:** For any 3-manifold and any nonnegative genus, constructs metrics with embedded minimal surfaces of that genus whose Morse indices go to infinity; on manifolds admitting positive scalar curvature, the metric can be chosen with positive scalar curvature.
- **Applicability check:** This is the closest negative result to this problem and shows that fixed genus alone cannot bound index. However, the theorem is for constructed smooth metrics and does not impose the stated bumpy or analytic hypotheses.

5. Area plus genus gives index bounds

- **Title:** *Comparison between second variation of area and second variation of energy of a minimal surface*
- **Authors:** Norio Ejiri, Mario Micallef
- **Venue/year:** Advances in Calculus of Variations 1 (2008), no. 3, 223-239

- **DOI/arXiv/URL:** DOI <https://doi.org/10.1515/ACV.2008.009>; arXiv 0708.2188; URL <https://arxiv.org/abs/0708.2188>
- **Result:** Bounds the Morse index of a minimal surface in terms of its area, genus, and ambient geometry.
- **Applicability check:** This proves the desired index bound under an additional area bound. The statement has no area bound, and known examples show area cannot generally be controlled by genus in arbitrary metrics.

6. Bumpy metrics with fixed genus and area bound

- **Title:** *Embedded minimal surfaces without area bounds in 3-manifolds*
- **Authors:** Tobias H. Colding, William P. Minicozzi II
- **Venue/year:** in *Geometry and Topology: Aarhus*, Contemporary Mathematics 258, American Mathematical Society, 2000
- **DOI/arXiv/URL:** AMS URL <https://www.ams.org/books/conm/258>
- **Result used in later literature:** For a closed orientable 3-manifold with a bumpy metric, fixed genus and fixed area bound give only finitely many closed embedded minimal surfaces.
- **Applicability check:** Together with Ejiri-Micallef this yields an index bound when area is bounded. It does not settle the problem statement because the problem asks for a genus-only bound.

7. Positive Ricci curvature bumpy metrics: fixed genus is controlled, but this is an extra hypothesis

- **Title:** *Existence of minimal surfaces of arbitrarily large Morse index*
- **Authors:** Haozhao Li, Xin Zhou
- **Venue/year:** *Calculus of Variations and Partial Differential Equations* 55 (2016), article 64
- **DOI/arXiv/URL:** DOI <https://doi.org/10.1007/s00526-016-1007-6>; arXiv 1504.00970; URL <https://arxiv.org/abs/1504.00970>
- **Result:** For a closed 3-manifold with a bumpy metric of positive Ricci curvature, there are embedded minimal surfaces of arbitrarily large Morse index, and for each index bound there are only finitely many such surfaces. The paper also recalls that positive Ricci curvature gives area bounds for fixed genus via Choi-Wang/Yang-Yau, so fixed-genus families are finite in the bumpy positive-Ricci setting.
- **Applicability check:** This is important partial progress. It proves unbounded index occurs in bumpy metrics only with genus going to infinity in this setting; it does not answer the general bumpy analytic fixed-genus problem.

8. Positive scalar curvature and bounded-index compactness/finiteness

- **Title:** *Minimal hypersurfaces with bounded index*
- **Authors:** Otis Chodosh, Daniel Ketover, Davi Maximo
- **Venue/year:** *Inventiones Mathematicae* 209 (2017), no. 3, 617-664
- **DOI/arXiv/URL:** DOI <https://doi.org/10.1007/s00222-017-0717-5>; arXiv 1509.06724; URL <https://arxiv.org/abs/1509.06724>
- **Result:** In closed 3-manifolds with positive scalar curvature, a uniform index bound implies uniform area and genus bounds; for bumpy positive-scalar-curvature metrics, bounded-index minimal surfaces are finite.
- **Applicability check:** This is the converse direction from the problem statement: it starts with an index bound and derives topology/area control. It does not prove that fixed genus forces an index bound.

9. Generic finiteness with bounded index under positive scalar curvature

- **Title:** *Generic finiteness of minimal surfaces with bounded Morse index*
- **Author:** Alessandro Carlotto
- **Venue/year:** *Annali della Scuola Normale Superiore di Pisa, Classe di Scienze* 17 (2017), no. 3, 1153–1171
- **DOI/arXiv/URL:** DOI https://doi.org/10.2422/2036-2145.201510_003; arXiv 1509.07101; URL <https://arxiv.org/abs/1509.07101>
- **Result:** For bumpy positive-scalar-curvature metrics, minimal surfaces with a uniform Morse-index bound are finite, up to the stated projective-plane obstruction; for strongly bumpy metrics the conclusion holds for any closed 3-manifold with positive scalar curvature.
- **Applicability check:** This is another bounded-index-to-finiteness result under extra curvature/genericity assumptions. It does not establish the stated genus-to-index implication.

10. Recent related bumpy-metric preprint

- **Title:** *Closed minimal surfaces of index one in Riemannian manifolds*
- **Authors:** Fernando C. Marques, Andre Neves
- **Venue/year:** arXiv preprint, 2026
- **DOI/arXiv/URL:** arXiv 2606.03583; URL <https://arxiv.org/abs/2606.03583>
- **Result:** Gives new existence results for closed minimal hypersurfaces of index one, including bumpy-metric contexts with unbounded volume under topological assumptions.
- **Applicability check:** This is current related evidence that bumpy metrics do not by themselves impose area compactness. It does not settle the problem statement because the problem is about bounding index from fixed genus.

1.18.3.3. What remains unresolved

The unresolved point is exactly the bumpy/analytic fixed-genus case. Existing smooth counterexamples show fixed genus alone is not enough, but the key counterexamples do not verify the bumpy and analytic hypotheses. Existing positive results prove the desired boundedness only after adding hypotheses such as an area bound, positive Ricci curvature, or other positive-scalar-curvature/bounded-index assumptions. I found no citable theorem proving that every bumpy analytic metric has a uniform Morse-index bound for each fixed genus, and no citable counterexample with a bumpy analytic metric and a fixed-genus sequence of closed minimal surfaces with Morse index tending to infinity.

1.18.4. Remaining open problems recorded by the database

1.18.4.1. Open problem 1

For a generic Riemannian metric on S^3 , prove that the Morse indices of all closed embedded minimal tori are uniformly bounded.

Reference supplied by the database. Tobias H. Colding and Nancy Hingston, “Metrics without Morse index bounds,” *Duke Mathematical Journal* **119** (2003), no. 2, 345–365, especially

the introduction's discussion of the Pitts–Rubinstein fixed-topology/generic-metric question for embedded minimal tori in S^3 , DOI: <https://doi.org/10.1215/S0012-7094-03-11925-0>, [arXiv:math/0210290](https://arxiv.org/abs/math/0210290).

1.18.5. Primary-source pointer

The originating ICM record is *An algorithm to recognize the 3 -sphere* (1994). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.1/ICM1994.1.ocr.pdf#page=673>.

1.19. 19. Positive scalar curvature metrics - Existence and classification questions [1994-geom-59]

Record: 1994-geom-59

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1994: Positive scalar curvature metrics - Existence and classification questions

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.1/ICM1994.1.ocr.pdf#page=697>

1.19.1. Problem statement

Prove that the following stable positive scalar curvature statement holds. Let M be an n -dimensional spin manifold with fundamental group π . Let $u : M \rightarrow B\pi$ be the classifying map of the universal covering of M , and let $\alpha(M) = \alpha(M, u) \in KO_n(C_r^*\pi)$ denote the index of the Dirac operator. Fix a Bott manifold B , meaning a simply connected 8-dimensional spin manifold with $\alpha(B) = b$, where b is the generator of $KO_8(\mathbb{R}) \cong \mathbb{Z}$. Then the product $M \times B^k$, for a sufficiently large integer k , admits a positive scalar curvature metric if and only if $\alpha(M) = 0$.

1.19.2. Historical background, literature, and current progress

1.19.3. Verdict: partially resolved

1.19.3.1. Precise problem

The statement is the stable Gromov-Lawson-Rosenberg conjecture for closed connected spin manifolds, usually stated in dimensions at least 5. For a closed spin manifold M with fundamental group π , classifying map $u : M \rightarrow B\pi$, and Rosenberg/Dirac index

$$\alpha(M, u) \in KO_n(C_r^*(\pi)),$$

the question asks whether

$$M \times B^k$$

admits a positive-scalar-curvature metric for some sufficiently large k , where B is an 8-dimensional Bott manifold, if and only if $\alpha(M, u) = 0$.

The necessity direction is standard: positive scalar curvature forces the Rosenberg index to vanish, and multiplication by the Bott manifold identifies the index under Bott periodicity. The difficult part is the converse for arbitrary π .

1.19.3.2. Progress in the literature

1. Stephan Stolz, "Positive Scalar Curvature Metrics – Existence and Classification Questions", Proceedings of the International Congress of Mathematicians, Zurich 1994,

- Birkhauser, 1995, pp. 625-636. DOI: https://doi.org/10.1007/978-3-0348-9078-6_56. This ICM article is the source context for the stable formulation of the positive scalar curvature existence problem using the Dirac/Rosenberg index and Bott periodicity. It does not prove the full arbitrary- π statement.
2. Jonathan Rosenberg and Stephan Stolz, “A ‘stable’ version of the Gromov-Lawson conjecture”, in *The Cech Centennial, Contemporary Mathematics* 181, American Mathematical Society, 1995, pp. 405-418. DOI: <https://doi.org/10.1090/conm/181/02046>; arXiv:dg-ga/9407002; PDF: <https://math.umd.edu/~jmr/StableGLConj4.pdf>. This paper formulates the Stable Conjecture explicitly: M stably admits a positive-scalar-curvature metric, meaning $M \times B^k$ has one for some k , if and only if $\alpha(M, u) = 0$. It proves important cases:
 - Corollary 4.4: the stable conjecture holds if the assembly map $KO_n(B\pi) \rightarrow KO_n(C^*(\pi))$ is injective.
 - Theorem 4.6: the stable conjecture holds for finite groups π . These are genuine partial resolutions, but the hypotheses do not cover arbitrary fundamental groups.
 3. Jonathan Rosenberg and Stephan Stolz, “Metrics of positive scalar curvature and connections with surgery”, in *Surveys on Surgery Theory, Vol. 2, Annals of Mathematics Studies* 149, Princeton University Press, 2001, pp. 353-386. DOI: <https://doi.org/10.1515/9781400865215-010>; PDF: <https://www.math.umd.edu/~jmr/psc.pdf>. Conjecture 4.17 is the same stable Gromov-Lawson-Rosenberg conjecture with a Bott manifold. Theorem 4.18 proves it under injectivity of the ordinary assembly map, Theorem 4.19 records the finite fundamental group case, and Theorem 4.20 states the broader conditional result: the stable conjecture holds for π if the Baum-Connes assembly map $KO_*^\pi(E\pi) \rightarrow KO_*(C^*(\pi))$ is injective. This still leaves groups for which this injectivity is not known.
 4. Matthias Kreck and Wolfgang Lueck, *The Novikov Conjecture: Geometry and Algebra*, Oberwolfach Seminar notes, 2004, URL: <https://webhomes.maths.ed.ac.uk/~v1ranick/papers/owsem.pdf>. Section 22.5 states the stable Gromov-Lawson-Rosenberg conjecture using the real reduced group C^* -algebra $C_r^*(G; \mathbb{R})$. Theorem 22.15 says that injectivity of the real Baum-Connes assembly map $KO_n(E_{\text{FIN}}G) \rightarrow KO_n(C_r^*(G; \mathbb{R}))$ implies the stable conjecture for all closed spin manifolds of dimension at least 5 with fundamental group G . This verifies that the conditional theorem matches the reduced-real C^* -algebra formulation in the problem statement, but again it is conditional.
 5. James F. Davis and Wolfgang Lueck, “The topological K-theory of certain crystallographic groups”, *Journal of Noncommutative Geometry* 7 (2013), 373-431. DOI: <https://doi.org/10.4171/JNCG/121>; arXiv:1004.2660. For certain crystallographic groups $\Gamma = \mathbb{Z}^n \rtimes \mathbb{Z}/p$, the paper computes real and complex K-theory and proves the unstable Gromov-Lawson-Rosenberg conjecture. This is stronger than the stable statement for those special groups, but it is far from the arbitrary- π statement.
 6. Rudolf Zeidler, “Width, Largeness and Index Theory”, *SIGMA* 16 (2020), 127, 15 pages. DOI: <https://doi.org/10.3842/SIGMA.2020.127>; arXiv:2008.13754. Section 2 restates the stable GLR conjecture in essentially the stated Bott-manifold form and summarizes the current general theorem: Stolz’s result proves the conjecture whenever the real Baum-Connes assembly map is injective. Zeidler notes that this gives many known cases and that any counterexample would also be a counterexample to Baum-Connes. This confirms that the unconditional arbitrary-group statement is still not a theorem in the literature.

7. Noe Barcenás and Mario Velasquez, “Positive scalar curvature and crystallographic fundamental groups”, *New York Journal of Mathematics* 32 (2026), 676-690; arXiv:2503.11931; URL: <https://nyjm.albany.edu/j/2026/32-27.html>. This recent paper studies positive and negative results for the unstable GLR conjecture in crystallographic group families and cites Davis-Pearson, Davis-Lueck, and Hughes as sources for positive stable GLR results under Baum-Connes and extra group-theoretic hypotheses. It supports the current picture: progress is substantial for many classes, but remains group-dependent.

1.19.3.3. Hypothesis check

The literature contains results that match the conclusion of the problem only after adding hypotheses such as:

- π finite;
- injectivity of the relevant real Baum-Connes assembly map;
- membership in particular group classes where that assembly result or stronger GLR statements are known.

The problem asks for the statement for arbitrary fundamental group π . None of the citable results found removes the assembly-injectivity or special-group assumptions. Therefore the exact all-cases statement is not fully resolved.

1.19.3.4. What remains unresolved

The remaining open part is the converse direction for arbitrary discrete fundamental groups:

$$\alpha(M, u) = 0 \implies M \times B^k \text{ has a positive-scalar-curvature metric for some } k.$$

It is known for many groups and would follow from the relevant real Baum-Connes injectivity in general, but no published or otherwise citable theorem proves it for all π .

1.19.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website’s “Some remaining open problems” field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

1.19.5. Primary-source pointer

The originating ICM record is *Positive scalar curvature metrics - Existence and classification questions* (1994). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.1/ICM1994.1.ocr.pdf#page=697>.

1.20. 20. Applications of dynamics to compact manifolds of negative curvature [1994-geom-308]

Record: 1994-geom-308

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1994: Applications of dynamics to compact manifolds of negative curvature

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.2/ICM1994.2.ocr.pdf#page=491>

1.20.1. Problem statement

Let (M, g) be a closed, connected Riemannian manifold with negative sectional curvature, and let $(\tilde{M}, \tilde{g})$ be its universal Riemannian cover. Let $\partial\tilde{M}$ be the geometric boundary of $\tilde{M}$, whose points are asymptotic-equivalence classes of geodesic rays. For each $x \in \tilde{M}$, let $S_x\tilde{M}$ be the unit tangent sphere and let $\tau_x : S_x\tilde{M} \rightarrow \partial\tilde{M}$ map a unit tangent vector to the endpoint at infinity of its geodesic ray. Define the visibility measure λ_x to be the pushforward by τ_x of the spherical Lebesgue measure on $S_x\tilde{M}$.

For each $x \in \tilde{M}$, let ω_x be the harmonic measure on $\partial\tilde{M}$: for every continuous function $f : \partial\tilde{M} \rightarrow \mathbb{R}$, the unique harmonic extension u_f of f to $\tilde{M}$ satisfies

$$u_f(x) = \int_{\partial\tilde{M}} f(\xi) d\omega_x(\xi).$$

Let

$$H = \lim_{R \rightarrow \infty} \frac{1}{R} \log \text{Vol}_{\tilde{g}}(B_{\tilde{M}}(x, R))$$

be the volume entropy of $\tilde{M}$, which is independent of x . For $x, y \in \tilde{M}$ and $\xi \in \partial\tilde{M}$, define the Busemann function by

$$b_{x,\xi}(y) = \lim_{t \rightarrow \infty} (d_{\tilde{g}}(y, \gamma_{x,\xi}(t)) - t),$$

where $\gamma_{x,\xi}$ is the unit-speed geodesic ray from x to ξ . Let $(\nu_x)_{x \in \tilde{M}}$ be a Margulis–Patterson density on $\partial\tilde{M}$, so that

$$\frac{d\nu_y}{d\nu_x}(\xi) = \exp(-Hb_{x,\xi}(y))$$

for all $x, y \in \tilde{M}$ and $\xi \in \partial\tilde{M}$.

Within each family (λ_x) , (ω_x) , and (ν_x) , the measures are mutually absolutely continuous as the base point x varies. Denote the resulting measure classes on $\partial\tilde{M}$ by λ , ω , and ν , respectively. Prove or disprove that if $\omega = \lambda$ or $\omega = \nu$, then (M, g) is locally symmetric.

1.20.2. Historical background, literature, and current progress

Verdict: unresolved.

For closed negatively curved surfaces, both alternatives in the problem are known to force constant curvature. Katok and Ledrappier proved the harmonic–visibility case, while Ledrappier proved the harmonic–Margulis–Patterson (equivalently, harmonic–Bowen–Margulis) case. No counterexample is known in higher dimension, but the available rigidity theorems do not prove either implication there.

The two higher-dimensional gaps are recorded explicitly in the literature. Hasselblatt states as Conjecture 20.27 that coincidence of the harmonic and visibility measure classes should imply local symmetry in higher dimension. Ledrappier observes that coincidence of the harmonic and Patterson–Sullivan classes is equivalent to equality $h = \ell H$ in the Guivarc’h inequality for Brownian motion, and states that in dimension greater than two it is not known whether this equality forces local symmetry. Thus the exact problem is not completely resolved.

Established stronger or special-case results do not fill the gap. If the relevant boundary measures are proportional at every base point (rather than merely mutually absolutely continuous), their Radon–Nikodym formulas force constant mean curvature of horospheres, and known asymptotic-harmonicity rigidity gives local symmetry. The surface results also give a complete answer in dimension two. Neither result upgrades equality of measure classes to the required pointwise proportionality in arbitrary dimension.

Key references:

- Anatole Katok, “Four Applications of Conformal Equivalence to Geometry and Dynamics,” *Ergodic Theory and Dynamical Systems* 8* (1988), 139–152, DOI <https://doi.org/10.1017/S0143385700009391>. This supplies the two-dimensional conformal/entropy rigidity used for the harmonic–visibility conclusion.
- François Ledrappier, “Harmonic Measures and Bowen–Margulis Measures,” *Israel Journal of Mathematics* 71 (1990), 275–287, DOI <https://doi.org/10.1007/BF02773746>. It proves, in particular, that on a closed negatively curved surface equivalence of the harmonic and Bowen–Margulis boundary classes forces constant curvature, and it proves stronger pointwise-equality/asymptotic-harmonicity criteria in all dimensions.
- Chengbo Yue, “Rigidity and Dynamics Around Manifolds of Negative Curvature,” *Mathematical Research Letters* 1 (1994), 123–147, DOI <https://doi.org/10.4310/MRL.1994.v1.n2.a1>. It formulates coincidence of any two of the harmonic, Liouville/visibility, and Bowen–Margulis classes as a rigidity conjecture and records the stronger pointwise-measure results.
- François Ledrappier, “Applications of Dynamics to Compact Manifolds of Negative Curvature,” in S. D. Chatterji (ed.), *Proceedings of the International Congress of Mathematicians, Zürich 1994*, Birkhäuser, 1995, pp. 1195–1202, DOI https://doi.org/10.1007/978-3-0348-9078-6_113. This is the source survey; it states the surface theorems, the pointwise-proportionality theorem, and the remaining measure-class question.
- Boris Hasselblatt, “Problems in Dynamical Systems and Related Topics,” in *Dynamics, Ergodic Theory, and Geometry*, MSRI Publications 54, Cambridge University Press, 2007, pp. 273–324, Conjecture 20.27, [direct article URL](#). It explicitly lists the higher-dimensional harmonic–visibility implication as the Sullivan conjecture and distinguishes it from the known pointwise-proportionality theorem.
- François Ledrappier, “Volume Entropy Rigidity of Non-Positively Curved Symmetric Spaces,” preprint, 2011, arXiv:[1110.0414](https://arxiv.org/abs/1110.0414). Its introduction identifies $h = \ell H$ with the Sullivan measure-rigidity problem, says that the converse is unknown above dimension two, and contrasts it with other entropy equalities that do characterize symmetric spaces.

- Barbara Schapira, “Gibbs Measures in Hyperbolic Geometry and Dynamics,” IHÉS lecture notes, 2025, [direct URL](#). These later notes still present coincidence of the canonical Gibbs, harmonic, and Liouville measures as the Sullivan conjecture and record only the two-dimensional solution.

1.20.3. Remaining open problems recorded by the database

1.20.3.1. Open problem 1

Let (M, g) be a closed, connected Riemannian manifold of dimension greater than 2 with negative sectional curvature, and let $(\tilde{M}, \tilde{g})$ be its universal Riemannian cover. Let $\partial\tilde{M}$ be the geometric boundary of $\tilde{M}$. For each $x \in \tilde{M}$, let $\tau_x : S_x\tilde{M} \rightarrow \partial\tilde{M}$ map a unit tangent vector to the endpoint at infinity of its geodesic ray, and let λ_x be the pushforward by τ_x of spherical Lebesgue measure on $S_x\tilde{M}$.

For each $x \in \tilde{M}$, let ω_x be harmonic measure on $\partial\tilde{M}$, characterized by

$$u_f(x) = \int_{\partial\tilde{M}} f(\xi) d\omega_x(\xi)$$

for every continuous function $f : \partial\tilde{M} \rightarrow \mathbb{R}$, where u_f is the unique harmonic extension of f to $\tilde{M}$. Within each family (λ_x) and (ω_x) , the measures are mutually absolutely continuous as x varies; denote their measure classes by λ and ω , respectively. Prove or disprove that if $\omega = \lambda$, then (M, g) is locally symmetric.

Reference supplied by the database. Boris Hasselblatt, “Problems in Dynamical Systems and Related Topics,” in Boris Hasselblatt (ed.), *Dynamics, Ergodic Theory, and Geometry: Dedicated to Anatole Katok*, Mathematical Sciences Research Institute Publications 54, Cambridge University Press, 2007, pp. 273–324, Conjecture 20.27, <https://library.slmath.org/books/Book54/files/10prob.pdf>.

1.20.3.2. Open problem 2

Let (M, g) be a closed, connected Riemannian manifold of dimension greater than 2 with negative sectional curvature, and let $(\tilde{M}, \tilde{g})$ be its universal Riemannian cover with geometric boundary $\partial\tilde{M}$. For each $x \in \tilde{M}$, let ω_x be harmonic measure on $\partial\tilde{M}$, characterized by

$$u_f(x) = \int_{\partial\tilde{M}} f(\xi) d\omega_x(\xi)$$

for every continuous function $f : \partial\tilde{M} \rightarrow \mathbb{R}$, where u_f is the unique harmonic extension of f to $\tilde{M}$.

Let

$$H = \lim_{R \rightarrow \infty} \frac{1}{R} \log \text{Vol}_{\tilde{g}}(B_{\tilde{M}}(x, R))$$

be the volume entropy. For $x, y \in \tilde{M}$ and $\xi \in \partial\tilde{M}$, let

$$b_{x,\xi}(y) = \lim_{t \rightarrow \infty} (d_{\tilde{g}}(y, \gamma_{x,\xi}(t)) - t),$$

where $\gamma_{x,\xi}$ is the unit-speed geodesic ray from x to ξ . Let $(\nu_x)_{x \in \widetilde{M}}$ be a Margulis–Patterson density satisfying

$$\frac{d\nu_y}{d\nu_x}(\xi) = \exp(-Hb_{x,\xi}(y)).$$

Within each family (ω_x) and (ν_x) , the measures are mutually absolutely continuous as x varies; denote their measure classes by ω and ν , respectively. Prove or disprove that if $\omega = \nu$, then (M, g) is locally symmetric.

Reference supplied by the database. François Ledrappier and Lin Shu, “Entropy Rigidity of Symmetric Spaces without Focal Points,” *Transactions of the American Mathematical Society* 366, no. 7 (2014), 3805–3820, discussion following Theorem 1.1, DOI 10.1090/S0002-9947-2014-06016-9, arXiv:1205.1070.

1.20.4. Primary-source pointer

The originating ICM record is *Applications of dynamics to compact manifolds of negative curvature* (1994). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.2/ICM1994.2.ocr.pdf#page=491>.

1.21. 21. Minimal foliations and laminations [1994-geom-338]

Record: 1994-geom-338

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1994: Minimal foliations and laminations

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.1/ICM1994.1.ocr.pdf#page=525>

1.21.1. Problem statement

Let $n \geq 2$, let $T^n = \mathbb{R}^n / \mathbb{Z}^n$, and let g be a Riemannian metric on T^n . Suppose that, for every primitive class $v \in H_{n-1}(T^n; \mathbb{Z})$, meaning every nonzero class that is not divisible by any integer greater than 1, there exists a foliation of T^n whose leaves are embedded $(n-1)$ -dimensional tori, each representing v after a consistent choice of orientation, and each having zero mean curvature with respect to g . Determine whether g must be flat.

1.21.2. Historical background, literature, and current progress

The problem is **unresolved**. No proof or counterexample covering all dimensions $n \geq 2$ is known. In particular, a June 2026 revision of Nguyen's preprint explicitly restates the simultaneous-foliation hypothesis as Bangert's flatness conjecture and leaves it open.

The case $n = 2$ is resolved affirmatively. A minimal foliation by closed curves is calibrated, so its leaves are shortest closed geodesics in their primitive homology class. Bangert proved that, on a Riemannian two-torus, differentiability of the stable norm in a rational direction is equivalent to the existence of such a foliation; irrational directions are automatically differentiability directions. Thus the hypothesis gives global differentiability, the associated minimizing geodesics have no conjugate points, and Hopf's theorem implies that the metric is flat. This does not settle the higher-dimensional cases.

The strongest higher-dimensional progress is variational. Auer and Bangert proved differentiability of the codimension-one stable norm along the maximal integer-generated subspace containing a given class, in particular full differentiability at totally irrational classes. Nguyen introduced the barrier $\Delta W_r = \omega(r) - S(r)$, where S is the stable norm and ω is a one-parameter min-max width, and proved (for $3 \leq n \leq 7$) that a primitive integral class r admits a foliation by closed homologically area-minimizing hypersurfaces exactly when $\Delta W_r = 0$. Consequently, in those dimensions the hypothesis of the problem is equivalent to simultaneous vanishing of all primitive integral barriers. Nguyen does not prove that this forces flatness and explicitly identifies that implication as the outstanding conjecture.

Key references:

- Victor Bangert, "Minimal Foliations and Laminations," in *Proceedings of the International Congress of Mathematicians, Zürich 1994*, Birkhäuser, 1995, pp. 453–464. DOI: https://doi.org/10.1007/978-3-0348-9078-6_38. This is the source that poses the exact rigidity problem.

- Victor Bangert, “Geodesic Rays, Busemann Functions and Monotone Twist Maps,” *Calculus of Variations and Partial Differential Equations* 2 (1994), 49–63. DOI: <https://doi.org/10.1007/BF01234315>. Its rational-direction differentiability criterion supplies the key two-dimensional reduction from geodesic foliations to global differentiability of the stable norm.
- Eberhard Hopf, “Closed Surfaces Without Conjugate Points,” *Proceedings of the National Academy of Sciences of the United States of America* 34 (1948), 47–51. DOI: <https://doi.org/10.1073/pnas.34.2.47>. Hopf’s theorem gives flatness of a Riemannian two-torus without conjugate points.
- Franz Auer and Victor Bangert, “Differentiability of the Stable Norm in Codimension One,” *American Journal of Mathematics* 128 (2006), 215–238. DOI: <https://doi.org/10.1353/ajm.2006.0002>; arXiv: [math/0403235](https://arxiv.org/abs/math/0403235). Theorem 1.1 establishes the codimension-one differentiability result described above, but no flatness conclusion.
- Hoan Nguyen, “A Min-Max Gap Characterization of Minimal Foliations on the Torus,” preprint, 2026. arXiv: [2605.12428v2](https://arxiv.org/abs/2605.12428v2). Proposition B.1 gives the integral-class foliation/barrier equivalence, and Section 10.1 explicitly records Bangert’s flatness implication as still conjectural.

1.21.3. Remaining open problems recorded by the database

1.21.3.1. Open problem 1

Let $3 \leq n \leq 7$, let $T^n = \mathbb{R}^n / \mathbb{Z}^n$ carry a Riemannian metric g , and let S denote the stable norm on $H_{n-1}(T^n; \mathbb{R})$. For a primitive class $r \in H_{n-1}(T^n; \mathbb{Z})$, let $\omega(r)$ be its one-parameter Almgren-Pitts width: the infimum, over all sweepouts by integral $(n-1)$ -cycles representing r whose associated loop represents the fundamental class $[T^n]$ under the Almgren isomorphism, of the maximum mass attained in the sweepout. Set $\Delta W_r = \omega(r) - S(r)$. A nonzero class $\alpha \in H_{n-1}(T^n; \mathbb{R})$ is totally irrational if it is contained in no proper real subspace spanned by integral homology classes, and a sequence of primitive integral classes (r_k) converges in direction to α if $r_k / S(r_k) \rightarrow \alpha / S(\alpha)$. Prove or disprove that, for every totally irrational α , the limit $\lim_{k \rightarrow \infty} \Delta W_{r_k}$ exists and has the same value for every sequence (r_k) converging in direction to α .

Reference supplied by the database. Hoan Nguyen, “A Min-Max Gap Characterization of Minimal Foliations on the Torus,” preprint, 2026, Section 10.1, arXiv:2605.12428v2, <https://arxiv.org/abs/2605.12428v2>.

1.21.4. Primary-source pointer

The originating ICM record is *Minimal foliations and laminations* (1994). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.1/ICM1994.1.ocr.pdf#page=525>.

1.22. 22. Curvature Contents of Geometric Spaces [1998-geom-63]

Record: 1998-geom-63

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: reviewed (reviewed flag: yes)

Original source: ICM 1998: Curvature Contents of Geometric Spaces

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.2/ICM1998.2.ocr.pdf#page=383>

1.22.1. Problem statement

For a given space X , such as a Riemannian manifold, an Alexandrov space, or a specified family of such spaces, fix a curvature notion $\mathcal{K}$, for example Gaussian curvature on surfaces, scalar curvature, Ricci curvature, sectional curvature, or Alexandrov curvature. Also fix an admissible geometric class $\mathcal{A}$, such as metrics with normalized volume, normalized diameter, fixed conformal class, prescribed topology, or other constraints removing trivial scaling.

The curvature content or curvature capacity of X relative to $\mathcal{K}$ and $\mathcal{A}$ may be defined by

$$\sup\{k \in \mathbb{R} : \exists g \in \mathcal{A} \text{ such that } \mathcal{K}_g \geq k\}.$$

Equivalently, it is the supremum, over all admissible metrics or structures, of the lower bound for the chosen curvature. The problem is to determine whether this supremum is finite, whether it is attained, and what rigidity occurs at equality.

1.22.2. Historical background, literature, and current progress

1.22.3. Status: open-ended program

1.22.3.1. Precise problem

The problem is not a single formal conjecture with a fixed invariant. In Lohkamp's original formulation it is a programmatic question: for a fixed geometric setting and a chosen curvature notion (Gaussian curvature on surfaces; scalar, Ricci, sectional, or Alexandrov curvature in higher-dimensional or metric-space settings), should one be able to define a natural "curvature content" or "capacity" of a fixed manifold, or specified family of spaces, measuring the maximal amount of positive/lower-bounded curvature the space can carry, while curvature-decreasing deformations should have no analogous lower obstruction.

Thus the exact full question would require a citable definition and theorem covering all the listed curvature notions and their relevant constraints, including Riemannian manifolds and Alexandrov spaces.

1.22.3.2. Literature checked and current progress

- **Title: Curvature Contents of Geometric Spaces**
 Authors: Joachim Lohkamp
 Venue: Proceedings of the International Congress of Mathematicians, Vol. II (Berlin, 1998), Documenta Mathematica Extra Vol. II, 381-388
 Year: 1998
 DOI/arXiv/URL: <https://emis.de/ft/10232>
 Role: Original source. It explicitly presents “curvature content” as an intuitive notion rather than a sharp definition, and it surveys special evidence: Gauss-Bonnet/uniformization for surfaces, scalar curvature deformation and positive mass rigidity, Ricci comparison and rigidity results, hyperbolization, Alexandrov curvature, and Gromov’s Betti-number bound.
- **Title: Existence and conformal deformation of metrics with prescribed Gaussian and scalar curvatures**
 Authors: Jerry L. Kazdan, Frank W. Warner
 Venue: Annals of Mathematics 101(2), 317-331
 Year: 1975
 DOI/arXiv/URL: <https://doi.org/10.2307/1970993>
 Role: Gives fundamental prescribed-curvature/flexibility results for Gaussian and scalar curvature. This supports parts of the “no lower bound/decreasing curvature is flexible” theme, but it does not define or resolve a general capacity theory.
- **Title: A direct approach to the determination of Gaussian and scalar curvature functions**
 Authors: Jerry L. Kazdan, Frank W. Warner
 Venue: Inventiones Mathematicae 28, 227-230
 Year: 1975
 DOI/arXiv/URL: <https://doi.org/10.1007/BF01425558>
 Role: Further prescribed scalar/Gaussian curvature results. Partial progress only.
- **Title: Metrics of negative Ricci curvature**
 Authors: Joachim Lohkamp
 Venue: Annals of Mathematics 140(3), 655-683
 Year: 1994
 DOI/arXiv/URL: <https://doi.org/10.2307/2118620>
 Role: Establishes strong flexibility for negative Ricci curvature, a key part of the curvature-decreasing side of the program. It does not settle the positive-capacity side in general.
- **Title: Curvature h-principles**
 Authors: Joachim Lohkamp
 Venue: Annals of Mathematics 142(3), 457-498
 Year: 1995
 DOI/arXiv/URL: <https://doi.org/10.2307/2118552>
 Role: Gives broad flexibility/deformation results for curvature conditions. This is substantial partial progress toward the “no lower bound” direction, not a full capacity theorem.
- **Title: Scalar curvature and hammocks**
 Authors: Joachim Lohkamp
 Venue: Mathematische Annalen 313, 385-407
 Year: 1999

DOI/arXiv/URL: <https://doi.org/10.1007/s002080050266>

Role: Develops scalar curvature deformation ideas used in Lohkamp's discussion. Partial progress for scalar curvature only.

- **Title: On the proof of the positive mass conjecture in general relativity**
 Authors: Richard Schoen, Shing-Tung Yau
 Venue: Communications in Mathematical Physics 65, 45-76
 Year: 1979
 DOI/arXiv/URL: <https://doi.org/10.1007/BF01940959>
 Role: Positive mass rigidity is one of the scalar-curvature obstructions behind Lohkamp's "maximal content" discussion. It applies in the asymptotically flat scalar-curvature setting, not to all curvature contents.
- **Title: Proof of the positive mass theorem. II**
 Authors: Richard Schoen, Shing-Tung Yau
 Venue: Communications in Mathematical Physics 79, 231-260
 Year: 1981
 DOI/arXiv/URL: <https://doi.org/10.1007/BF01942062>
 Role: Further positive mass theorem work; partial scalar-curvature rigidity input.
- **Title: Sharp estimates and the Dirac operator**
 Authors: Marcelo Llarull
 Venue: Mathematische Annalen 310, 55-71
 Year: 1998
 DOI/arXiv/URL: <https://doi.org/10.1007/s002080050136>
 Role: Proves scalar-curvature rigidity of the round sphere under metric domination hypotheses. This is a sharp capacity-type result for a special scalar-curvature comparison problem, not the full problem.
- **Title: Metric Inequalities with Scalar Curvature**
 Authors: Misha Gromov
 Venue: Geometric and Functional Analysis 28, 645-726
 Year: 2018
 DOI/arXiv/URL: <https://doi.org/10.1007/s00039-018-0453-z>
 Role: Provides modern scalar-curvature inequalities and constraints under lower scalar curvature bounds. It advances scalar-curvature capacity-type questions but does not close the general program.
- **Title: Four Lectures on Scalar Curvature**
 Authors: Misha Gromov
 Venue: arXiv preprint; later in *Perspectives in Scalar Curvature*
 Year: 2019 / 2023
 DOI/arXiv/URL: arXiv:1908.10612, <https://arxiv.org/abs/1908.10612>
 Role: Survey of known scalar-curvature constraints and open directions. It confirms active partial progress rather than a completed general theory.
- **Title: Lower bounds on Ricci curvature and the almost rigidity of warped products**
 Authors: Jeff Cheeger, Tobias H. Colding
 Venue: Annals of Mathematics 144(1), 189-237
 Year: 1996
 DOI/arXiv/URL: <https://doi.org/10.2307/2118589>
 Role: Ricci lower-bound rigidity/comparison result, relevant to the Ricci-capacity examples cited by Lohkamp. It has specific hypotheses and does not resolve the full capacity principle.

- **Title: Minimal entropy and Mostow's rigidity theorems**
 Authors: Gerard Besson, Gilles Courtois, Sylvestre Gallot
 Venue: Ergodic Theory and Dynamical Systems 16(4), 623-649
 Year: 1996
 DOI/arXiv/URL: <https://doi.org/10.1017/S0143385700009019>
 Role: Gives sharp entropy/volume rigidity results for locally symmetric negatively curved spaces; relevant to rigidity and curvature bounds in special settings, not a full curvature-content theory.
- **Title: Hyperbolization of polyhedra**
 Authors: Michael W. Davis, Tadeusz Januszkiewicz
 Venue: Journal of Differential Geometry 34(2), 347-388
 Year: 1991
 DOI/arXiv/URL: <https://doi.org/10.4310/jdg/1214447212>
 Role: Provides hyperbolization constructions, a main source of curvature-decreasing/topology-changing examples in the sectional/Alexandrov discussion. Partial progress only.
- **Title: Strict hyperbolization**
 Authors: Ruth M. Charney, Michael W. Davis
 Venue: Topology 34(2), 329-350
 Year: 1995
 DOI/arXiv/URL: [https://doi.org/10.1016/0040-9383\(94\)00027-I](https://doi.org/10.1016/0040-9383(94)00027-I)
 Role: Produces negatively curved spaces in the geodesic/Alexandrov sense and supports the decreasing-curvature side of the sectional/Alexandrov part. It does not give a general positive-capacity theorem.
- **Title: Curvature, diameter and Betti numbers**
 Authors: Mikhail Gromov
 Venue: Commentarii Mathematici Helvetici 56, 179-195
 Year: 1981
 DOI/arXiv/URL: <https://doi.org/10.1007/BF02566208>
 Role: Gives an upper bound for Betti numbers under lower sectional curvature and diameter bounds. Lohkamp identifies this as one concrete sectional-curvature capacity notion, but it is one normalized invariant, not the full program.
- **Title: A. D. Alexandrov spaces with curvature bounded below**
 Authors: Yuri Burago, Mikhail Gromov, Grigori Perelman
 Venue: Russian Mathematical Surveys 47(2), 1-58
 Year: 1992
 DOI/arXiv/URL: <https://doi.org/10.1070/RM1992v047n02ABEH000877>
 Role: Foundational Alexandrov-space theory. It supplies the metric-space framework in which such capacity questions can be posed, but it does not settle the broad principle.
- **Title: Scalar curvature, isoperimetric collapse and General Relativity in the Constant Mean Curvature gauge**
 Authors: Martin Reiris
 Venue: arXiv preprint
 Year: 2010
 DOI/arXiv/URL: arXiv:1002.3302, <https://arxiv.org/abs/1002.3302>
 Role: Later work explicitly states that a suitable scalar curvature capacity is still “to be found” in that context. This is evidence that even the scalar-curvature part is not closed in full generality.

1.22.3.3. Why the full statement is not resolved

The searched literature proves many sharp and influential results for particular curvature notions, normalizations, dimensions, comparison maps, topology classes, or metric-space categories. However, no source found gives a single precise definition of “curvature content/capacity” and a theorem proving the asserted upper-bound/no-lower-bound principle simultaneously for Gaussian, scalar, Ricci, sectional, and Alexandrov curvature under the broad “chosen constraints.” The original article itself avoids a sharp definition, and later scalar-curvature surveys and preprints still treat related capacity-type questions as active.

Therefore the exact problem is not fully resolved. It is also not fully open, because substantial special cases and components of the principle are established.

1.22.3.4. Verdict

open-ended program

1.22.4. Remaining open problems recorded by the database

1.22.4.1. Open problem 1

Let (X, g) be a compact Riemannian manifold with boundary Y and scalar curvature $R_g \geq \sigma$. Is $\int_Y H_g d\mu_{g|_Y}$ bounded above by a constant depending only on the boundary metric $(Y, g|_Y)$ and σ ?

Reference supplied by the database. Yuguang Shi, Wenlong Wang, and Guodong Wei, “Total Mean Curvature of the Boundary and Nonnegative Scalar Curvature Fill-ins,” *Journal für die reine und angewandte Mathematik (Crelle’s Journal)* 2022 (2022), no. 784, 215–250, Conjecture 1.1 (attributed there to M. Gromov, “Four Lectures on Scalar Curvature,” p. 23 of the 2019 preprint; published version in *Perspectives in Scalar Curvature*, Vol. 1, World Scientific, 2023, p. 232). DOI: 10.1515/crelle-2021-0072, arXiv:2007.06756.

Current partial results: Christian Bär, “Upper Bound for the Total Mean Curvature of Spin Fill-ins,” preprint, arXiv:2601.06713 (2026), proves the conjectured bound for spin fill-ins with mean curvature bounded below, $H \geq -\eta$, with the constant depending additionally on η (in particular, the conjecture holds as stated for spin fill-ins with $H \geq 0$). Independently and concurrently, Georg Frenck, Bernhard Hanke, and Sven Hirsch, “Surgery and Total Mean Curvature,” preprint, arXiv:2601.10617 (2026), prove the same η -dependent bound by surgery methods, without the spin assumption when $\dim Y \leq 6$ and $\eta = 0$. Whether the dependence on a lower mean curvature bound can be removed — i.e., the conjecture in full generality — remains open (see the concluding remarks in Bär’s preprint).

1.22.5. Primary-source pointer

The originating ICM record is *Curvature Contents of Geometric Spaces* (1998). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.2/ICM1998.2.ocr.pdf#page=383>.

1.23. 23. Dimension Theory and Large Riemannian Manifolds [1998-geom-64]

Record: 1998-geom-64

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1998: Dimension Theory and Large Riemannian Manifolds

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.2/ICM1998.2.ocr.pdf#page=425>

1.23.1. Problem statement

Prove that a closed aspherical manifold, that is, a closed manifold whose universal cover is contractible, does not admit a Riemannian metric whose scalar curvature is everywhere positive.

1.23.2. Historical background, literature, and current progress

1.23.3. Verdict: partially resolved

1.23.3.1. Precise problem

The problem asks whether every closed aspherical smooth manifold, equivalently every closed manifold with contractible universal cover, admits no Riemannian metric whose scalar curvature is everywhere positive. This is the usual Gromov-Lawson / Schoen-Yau aspherical positive scalar curvature conjecture. The statement is all-dimensional and has no spin, orientability, curvature-model, finite asymptotic dimension, Novikov, or group-theoretic hypothesis.

1.23.3.2. Status check

The problem is not fully resolved. The literature proves the statement in low dimensions, most notably through dimension 5, and proves many higher-dimensional cases under additional hypotheses. However, recent papers still explicitly treat the conjecture as widely open in dimensions greater than 5. No citable theorem found in the search proves the full all-dimensional statement for arbitrary closed aspherical manifolds.

1.23.3.3. Key literature and progress

Source and formulation

- Title: *Positive scalar curvature and the Dirac operator on complete Riemannian manifolds*
- Authors: Mikhail Gromov and H. Blaine Lawson, Jr.
- Venue/year: Publications Mathématiques de l'IHES 58, 83-196, 1983

- DOI/URL: <https://doi.org/10.1007/BF02953774>; https://www.numdam.org/item/PMIHES_1983__58__83_0/
- Relevance: Foundational source for the Gromov-Lawson positive scalar curvature program. Later literature cites it for the three-dimensional closed aspherical case and for broad obstruction methods. It does not prove the all-dimensional conjecture.
- Title: *Manifolds of positive scalar curvature*
- Authors: Jonathan Rosenberg and Stephan Stolz
- Venue/year: In *Algebraic Topology and Its Applications*, Springer, 1994, pp. 241-267
- URL: <https://www.math.umd.edu/~jmr/RosenbergStolz1994.pdf>
- Relevance: Records the explicit conjecture as Conjecture 6.1: no closed aspherical manifold admits a metric of positive scalar curvature. It also explains that many cases follow from strong Novikov-type hypotheses or good geometric properties, so these are conditional/special-case results rather than a complete resolution.
- Title: *Dimension theory and large Riemannian manifolds*
- Author: Alexander N. Dranishnikov
- Venue/year: Proceedings of the International Congress of Mathematicians, Berlin 1998, Doc. Math. Extra Vol. II, 423-432
- URL: <https://www2.math.ethz.ch/EMIS/journals/DMJDMV/xvol-icm/06/Dranishnikov.MAN.ps.gz>
- Relevance: This matches the dataset source context and discusses large-scale dimension ideas around scalar curvature and aspherical manifolds. It is background/source material, not a complete proof of the conjecture.

Low-dimensional resolved cases

- Title: *Generalized soap bubbles and the topology of manifolds with positive scalar curvature*
- Authors: Otis Chodosh and Chao Li
- Venue/year: *Annals of Mathematics* 199(2), 707-740, 2024
- DOI/arXiv/URL: <https://doi.org/10.4007/annals.2024.199.2.3>; arXiv:2008.11888
- Result: Theorem 2 proves that for n in $\{4,5\}$, a smooth closed aspherical n -manifold does not admit a smooth Riemannian metric with positive scalar curvature; any nonnegative scalar curvature metric is flat.
- Scope check: This fully covers dimensions 4 and 5, but only those dimensions.
- Title: *No metrics with Positive Scalar Curvatures on Aspherical 5-Manifolds*
- Author: Misha Gromov
- Venue/year: arXiv preprint, 2020
- DOI/arXiv/URL: <https://doi.org/10.48550/arXiv.2009.05332>; arXiv:2009.05332
- Result: Gives an independent mu-bubble argument implying no positive scalar curvature metrics on compact aspherical manifolds in dimensions up to 5.

- Scope check: Citable preprint evidence for the low-dimensional cases, but still not an all-dimensional theorem.
- Title: *Classifying sufficiently connected PSC manifolds in 4 and 5 dimensions*
- Authors: Otis Chodosh, Chao Li, and Yevgeny Liokumovich
- Venue/year: Geometry & Topology 27, 1635-1655, 2023
- DOI/arXiv: <https://doi.org/10.2140/gt.2023.27.1635> ; arXiv:2105.07306
- Result: Proves a mapping version: if N is a closed 4- or 5-manifold admitting a nonzero degree map to a closed aspherical manifold, then N admits no Riemannian metric with positive scalar curvature.
- Scope check: Stronger than the problem statement in dimensions 4 and 5, but still dimension-limited.

Higher-dimensional partial results

- Title: *On hypersphericity of manifolds with finite asymptotic dimension*
- Author: Alexander N. Dranishnikov
- Venue/year: Transactions of the American Mathematical Society 355(1), 155-167, 2003
- DOI/URL: <https://doi.org/10.1090/S0002-9947-02-03115-X> ; <https://www.jstor.org/stable/3072948>
- Result: Gives the finite-asymptotic-dimension case: a closed aspherical manifold whose fundamental group has finite asymptotic dimension admits no positive scalar curvature metric.
- Scope check: This is all-dimensional only after adding the extra hypothesis on $\pi_1(M)$, so it does not settle arbitrary closed aspherical manifolds.
- Title: *Two-dimension vanishing, splitting and positive scalar curvature*
- Author: Xingyu Zhu
- Venue/year: Mathematische Annalen 392, 2635-2656, 2025
- DOI/URL: <https://doi.org/10.1007/s00208-025-03139-0>
- Result: Reiterates the finite-asymptotic-dimension consequence as Theorem 2.7 and places it among current partial progress.
- Scope check: Again, the theorem has an extra finite-asymptotic-dimension assumption and is therefore partial for this problem.
- Title: *Positive scalar curvature metrics and aspherical summands*
- Authors: Shuli Chen, Jianchun Chu, and Jintian Zhu
- Venue/year: arXiv:2312.04698, submitted 2023, revised 2025; accepted by Duke Mathematical Journal
- DOI/arXiv: <https://doi.org/10.48550/arXiv.2312.04698> ; arXiv:2312.04698
- Result: Proves noncompact/aspherical-summand extensions in dimensions 3, 4, and 5, and surveys the state of the aspherical conjecture.

- Scope check: The paper explicitly states that despite progress through dimension 5, the aspherical conjecture remains widely open in dimensions greater than 5.

1.23.3.4. What remains unresolved

The unresolved part is the full statement for arbitrary closed aspherical manifolds in dimensions $n \geq 6$, without added assumptions such as spin, finite asymptotic dimension, strong Novikov/Baum-Connes hypotheses, nonpositive sectional curvature, enlargeability, special domination maps, or other geometric/group-theoretic restrictions.

1.23.4. Remaining open problems recorded by the database

1.23.4.1. Open problem 1

Prove that no closed aspherical 6-manifold admits a Riemannian metric of positive scalar curvature.

Reference supplied by the database. Liam Mazurowski, Tongrui Wang, and Xuan Yao, “On the Topology of Manifolds with Positive Intermediate Curvature,” *Advances in Mathematics* **485** (2026), Article 110731, Introduction, the $K(\pi, 1)$ conjecture in dimension 6 (equivalently the unresolved $(n, m) = (6, 5)$ scalar-curvature case of the paper’s conjecture), DOI: 10.1016/j.aim.2025.110731, arXiv:2503.13815, <https://doi.org/10.1016/j.aim.2025.110731>.

1.23.5. Primary-source pointer

The originating ICM record is *Dimension Theory and Large Riemannian Manifolds* (1998). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.2/ICM1998.2.ocr.pdf#page=425>.

1.24. 24. Curvature-Decreasing Maps are Volume-Decreasing [1998-geom-74]

Record: 1998-geom-74

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1998: Curvature-Decreasing Maps are Volume-Decreasing

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.2/ICM1998.2.ocr.pdf#page=341>

1.24.1. Problem statement

Let X be a compact n -dimensional manifold, with $n \geq 4$, and suppose that X admits a negatively curved locally symmetric Riemannian metric g_0 . Is every Einstein metric on X homothetic to g_0 , i.e. equal to g_0 up to pullback by a diffeomorphism of X and multiplication by a positive constant?

1.24.2. Historical background, literature, and current progress

1.24.3. Status check

Verdict: partially resolved.

1.24.3.1. Precise problem

The problem asks for global Einstein rigidity of compact negatively curved locally symmetric manifolds in all dimensions $n \geq 4$: if X admits a negatively curved locally symmetric metric g_0 , must every smooth Riemannian Einstein metric on the same smooth manifold be $\lambda\phi^*g_0$ for some $\lambda > 0$ and diffeomorphism $\phi: X \rightarrow X$?

This includes, in particular, the subcase of closed real hyperbolic n -manifolds for every $n \geq 4$.

1.24.3.2. Progress and literature

1. Sylvestre Gallot, "Curvature-Decreasing Maps are Volume-Decreasing", Proceedings of the International Congress of Mathematicians, Berlin 1998, Vol. II, Documenta Mathematica Extra Volume II (1998), 339-348. URL: <https://ems.press/content/book-chapter-files/27163>.

Gallot states the general question as an open conjectural problem after explaining that in dimension four the real and complex hyperbolic cases can be handled by global topological/curvature identities. The source records Theorem 3.6 for real hyperbolic 4-manifolds and Theorem 3.8 for complex hyperbolic 4-manifolds, but it does not prove the all- n statement.

2. Gerard Besson, Gilles Courtois, and Sylvestre Gallot, “Entropies et rigidites des espaces localement symetriques de courbure strictement negative”, *Geometric and Functional Analysis* 5 (1995), 731-799. DOI: <https://doi.org/10.1007/BF01897050>.

Their entropy-rigidity theorem is the key input behind the real hyperbolic four-dimensional uniqueness result. In Gallot’s ICM account, it yields: if a compact 4-manifold admits a real hyperbolic metric, then that metric is the only Einstein metric up to homothety. This covers only the real hyperbolic $n = 4$ part of the problem.

3. Claude LeBrun, “Einstein Metrics and Mostow Rigidity”, *Mathematical Research Letters* 2 (1995), 1-8. URL: <https://www.math.stonybrook.edu/~claudelmo.pdf>.

Theorem 2 proves that if M^4 is a compact quotient of complex hyperbolic 2-space $\mathbb{C}H^2$, then every Einstein metric is $\lambda\phi^*g_0$. This covers the remaining negatively curved locally symmetric case in real dimension four, but not higher dimensions.

4. Michael T. Anderson, “A survey of Einstein metrics on 4-manifolds”, in *Advances in Lectures in Mathematics* 14, International Press, 2010, 1-39. arXiv:0810.4830, URL: <https://arxiv.org/abs/0810.4830>.

Corollary 4.4 restates LeBrun’s complex-hyperbolic 4-manifold uniqueness, and the survey also presents the Besson-Courtois-Gallot entropy rigidity theorem. This is a useful secondary source confirming the four-dimensional scope of the known global uniqueness results.

5. Norihito Koiso, “Rigidity and stability of Einstein metrics: the case of compact symmetric spaces”, *Osaka Journal of Mathematics* 17 (1980), no. 1, 51-73. URL: <https://projecteuclid.org/journals/osaka-journal-of-mathematics/volume-17/issue-1/Rigidity-and-stability-of-Einstein-metrics---the-case/ojm/1200772808.pdf>.

Koiso proves infinitesimal/local rigidity results for Einstein metrics on locally symmetric spaces of non-compact type without two-dimensional factors. This is important local progress, but it does not imply the global uniqueness asked in the problem statement.

6. Richard H. Bamler, “Stability of symmetric spaces of noncompact type under Ricci flow”, *Geometric and Functional Analysis* 25 (2015), 342-416. arXiv:1011.4267, URL: <https://arxiv.org/abs/1011.4267>.

Corollary 1.3 gives a nearby Einstein-metric uniqueness result under small C^0 perturbations for many locally symmetric spaces of noncompact type, with exclusions including hyperbolic and complex-hyperbolic factors. This is local/stability progress only, not a solution of the global question.

7. Ursula Hamenstadt and Frieder Jackel, “Negatively curved Einstein metrics on Gromov-Thurston manifolds”, arXiv:2411.12956, submitted 2024, revised 2025. DOI: <https://doi.org/10.48550/arXiv.2411.12956>.

The paper constructs, for every $n \geq 4$, closed manifolds with negatively curved Einstein metrics but no locally symmetric metric. These examples are not counterexamples to this problem because their manifolds do not admit a locally symmetric metric. The introduction explicitly asks whether an Einstein metric on any closed hyperbolic manifold has constant curvature, showing that the hyperbolic higher-dimensional subcase remains an active unresolved question.

8. Ursula Hamenstadt, “The geometry of branched coverings of hyperbolic manifolds”, arXiv:2605.01027, 2026. DOI: <https://doi.org/10.48550/arXiv.2605.01027>.

01027.

This 2026 survey-style preprint reviews branched covers of hyperbolic manifolds and again uses the known fact that closed hyperbolic 4-manifolds have a unique Einstein metric up to scaling and isometry. It does not claim a higher-dimensional uniqueness theorem for closed hyperbolic manifolds.

1.24.3.3. What remains unresolved

No published or citable theorem found in the live literature search covers the full statement for every compact negatively curved locally symmetric X of every dimension $n \geq 4$. The four-dimensional cases are resolved: real hyperbolic 4-manifolds by Besson-Courtois-Gallot methods, and complex hyperbolic surfaces by LeBrun. There are also local/infinitesimal rigidity and Ricci-flow stability results in several settings.

However, the problem statement includes closed real hyperbolic n -manifolds for $n \geq 5$, and recent literature still phrases the question whether every Einstein metric on a closed hyperbolic manifold has constant curvature as open. Therefore the exact problem is not fully resolved; it is partially resolved.

1.24.4. Remaining open problems recorded by the database

1.24.4.1. Open problem 1

Must every Einstein metric on a closed real hyperbolic manifold have constant sectional curvature?

Reference supplied by the database. Ursula Hamenstädt and Frieder Jäkel, “Negatively curved Einstein metrics on Gromov–Thurston manifolds,” arXiv preprint (2024; revised 2025), unnumbered Question in the introduction (first question); arXiv: [2411.12956](https://arxiv.org/abs/2411.12956), DOI: <https://doi.org/10.48550/arXiv.2411.12956>.

1.24.5. Primary-source pointer

The originating ICM record is *Curvature-Decreasing Maps are Volume-Decreasing* (1998). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.2/ICM1998.2.ocr.pdf#page=341>.

1.25. 25. Weyl Manifolds and Gaussian Thermostats [2002-geom-76]

Record: 2002-geom-76

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Weyl Manifolds and Gaussian Thermostats

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=498>

1.25.1. Problem statement

Prove that, for every integer $n \geq 2$, no n -dimensional torus T^n admits a Weyl structure whose Weyl sectional curvature is negative on every tangent 2-plane. Equivalently, prove that there is no Weyl structure for which $\widehat{K}(\Pi) < 0$ for every tangent 2-plane Π , where, for any orthonormal basis $\{X, Y\}$ of Π ,

$$\widehat{K}(\Pi) = \langle \widehat{R}_a(X, Y)Y, X \rangle = K(\Pi) - |E_\perp|^2 - \operatorname{div}_\Pi E,$$

$K(\Pi)$ is the Riemannian sectional curvature, $E_\perp$ is the component of the vector field E orthogonal to Π , and $\operatorname{div}_\Pi E = \langle \nabla_X E, X \rangle + \langle \nabla_Y E, Y \rangle$.

1.25.2. Historical background, literature, and current progress

Verdict: partially resolved.

1.25.2.1. Precise problem

The problem asks whether, for every integer $n \geq 2$, the smooth torus T^n admits no Weyl structure with strictly negative Weyl sectional curvature on every tangent 2-plane. In the notation used by Wojtkowski, a Weyl structure represented by a Riemannian metric g and vector field E has

$$\widehat{K}(\Pi) = K(\Pi) - |E_\perp|^2 - \operatorname{div}_\Pi E,$$

and the question is whether $\widehat{K}(\Pi) < 0$ for all tangent 2-planes Π is impossible on all tori.

1.25.2.2. Original source and status of the problem

Title: Weyl Manifolds and Gaussian Thermostats

Author: Maciej P. Wojtkowski

Venue/year: Proceedings of the International Congress of Mathematicians, Beijing 2002, Vol. III, Higher Education Press, Beijing, 2002, pp. 511-523

arXiv/URL: arXiv:math/0304461, <https://arxiv.org/abs/math/0304461>

Relevant result: Section 5 states Conjecture 5.3: "There are no Weyl structures of negative sectional curvature on n -dimensional tori." The paper also records Proposition 5.2, a local

obstruction near the homogeneous Weyl structure on a torus, and notes that the $n = 2$ case follows because in dimension 2 the Weyl curvature in the Gauduchon gauge is the Gaussian curvature. On T^2 , Gauss-Bonnet rules out everywhere negative Gaussian curvature in that gauge. This resolves the two-dimensional case but explicitly leaves the higher-dimensional conjecture open.

Title: W-flows on Weyl manifolds and Gaussian thermostats

Author: Maciej P. Wojtkowski

Venue/year: Journal de Mathematiques Pures et Appliquees (9) 79(10), 953-974, 2000

DOI/URL: [https://doi.org/10.1016/S0021-7824\(00\)00176-8](https://doi.org/10.1016/S0021-7824(00)00176-8)

Relevant result: This is the earlier foundational paper relating Weyl connections and Gaussian thermostats and giving dynamical consequences of negative Weyl curvature. It supplies background and motivation, but it does not prove the torus nonexistence conjecture in all dimensions.

1.25.2.3. Partial progress

Title: Rigidity of some Weyl manifolds with nonpositive sectional curvature

Author: Maciej P. Wojtkowski

Venue/year: Proceedings of the American Mathematical Society 133(11), 3395-3402, 2005

DOI/URL: <https://doi.org/10.1090/S0002-9939-05-07809-3> ; author page/PDF: <https://archive.math.arizona.edu/maciejw/>

Relevant result: The paper classifies locally metric Weyl connections with nonpositive sectional curvatures on n -tori and on $S^1 \times S^{n-1}$ with the standard conformal structures. For $S^1 \times S^{n-1}$ it excludes other nonpositive Weyl connections; for T^n the analogous conclusion is proved in the narrower integrable class. Later summaries of this paper state that, on tori with the flat Gauduchon metric, the only nonpositive examples are the constant-vector-field ones. These examples are nonpositive but not strictly negative because planes containing the constant field have zero Weyl curvature. This is important partial progress, but it does not cover arbitrary conformal/Gauduchon metrics or arbitrary Weyl connections on T^n .

Title: Hopf type rigidity for thermostats

Authors: Yernat M. Assylbekov and Nurlan S. Dairbekov

Venue/year: Ergodic Theory and Dynamical Systems 34(6), 1761-1769, 2014

DOI/arXiv/URL: <https://doi.org/10.1017/etds.2013.35> ; arXiv:1204.4380, <https://arxiv.org/abs/1204.4380>

Relevant result: The paper proves a Hopf-type rigidity theorem for magnetic/Gaussian thermostats on T^2 without conjugate points. This is consistent with, and gives another route to, rigidity phenomena for the two-dimensional Gaussian thermostat/Weyl setting. It does not address T^n for $n \geq 3$.

Title: Homogeneous Weyl connections of non-positive curvature

Authors: Gabriela Tereszkievicz and Maciej P. Wojtkowski

Venue/year: Annals of Global Analysis and Geometry 51, 209-229, 2017

DOI/arXiv/URL: <https://doi.org/10.1007/s10455-016-9526-0> ; arXiv:1506.08176, <https://arxiv.org/abs/1506.08176>

Relevant result: This paper studies homogeneous Weyl connections with nonpositive sectional curvature. Its introduction explicitly says that in dimensions 3 and higher the authors do not know examples of Weyl connections with negative sectional curvatures beyond small perturbations of negatively curved Levi-Civita connections. It also summarizes the torus situation: constant vector fields on flat tori give nonpositive examples; previous work gives no other nonpositive Weyl connections with flat Gauduchon metric; and only partial results are

known toward arbitrary Gauduchon metrics. The main theorem proves that homogeneous stretched-nonpositive Weyl connections modeled on unimodular Lie groups are locally product/parallel. These hypotheses are substantially narrower than the problem.

Title: Stretched non-positive Weyl connections on solvable Lie groups

Authors: Maciej Bochenski, Piotr Jastrzebski, and Aleksy Tralle

Venue/year: Annali di Matematica Pura ed Applicata 203, 1463-1481, 2024

DOI/arXiv/URL: <https://doi.org/10.1007/s10231-023-01409-9> ; arXiv:2207.00318, <https://arxiv.org/abs/2207.00318>

Relevant result: This recent paper classifies invariant stretched-nonpositive Weyl connections in several solvable Lie group settings, including a full classification in dimension 4 for unimodular solvable Lie groups. It explicitly does not address the harder problem of finding/classifying general Weyl connections of nonpositive curvature with more interesting dynamics. It therefore supplies current partial progress in homogeneous/SNP cases but not a solution to the torus conjecture.

Title: Thermostats without conjugate points

Authors: Javier Echevarria Cuesta and James Marshall Reber

Venue/year: Ergodic Theory and Dynamical Systems 46(6), 1494-1520, 2026

DOI/arXiv/URL: <https://doi.org/10.1017/etds.2026.10285> ; arXiv:2501.01923, <https://arxiv.org/abs/2501.01923>

Relevant result: This is current related literature on thermostat flows and no-conjugate-points phenomena. It treats general thermostats on surfaces and shows that Hopf-type rigidity does not extend to all thermostat functions on T^2 . Because the problem concerns the specific Weyl/Gaussian thermostat curvature condition and all dimensions, this does not resolve the stated torus problem.

1.25.2.4. What remains unresolved

I did not find any published or otherwise citable theorem proving that every T^n , $n \geq 3$, admits no Weyl structure with $\hat{K}(\Pi) < 0$ for every tangent 2-plane, without additional assumptions such as standard/flat conformal structure, locally metric or integrable Weyl connection, homogeneous/invariant connection, flat Gauduchon metric, or stretched-nonpositive behavior.

Thus the full all-dimensional statement is not fully resolved. The current literature resolves $n = 2$ and several restricted higher-dimensional settings, but the arbitrary higher-dimensional torus case remains open.

1.25.3. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

1.25.4. Primary-source pointer

The originating ICM record is *Weyl Manifolds and Gaussian Thermostats* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=498>.

1.26. 26. Geometry and Nonlinear Analysis [2002-geom-81]

Record: 2002-geom-81

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Geometry and Nonlinear Analysis

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.1/ICM2002.1.ocr.pdf#page=474>

1.26.1. Problem statement

Determine whether every compact differentiable manifold M of dimension at least 5 admits a Riemannian metric g such that $\text{Ric}(g) = \lambda g$ for some constant λ .

1.26.2. Historical background, literature, and current progress

1.26.3. Status check

Verdict: partially resolved.

1.26.3.1. Precise question

Does every compact smooth differentiable manifold M of dimension at least 5 admit a Riemannian Einstein metric, i.e. a metric g satisfying $\text{Ric}(g) = \lambda g$ for some constant λ ?

1.26.3.2. Literature checked

1. Shing-Tung Yau, "Review of geometry and analysis," Asian Journal of Mathematics 4 (2000), no. 1, 235-278; also in Mathematics: Frontiers and Perspectives, Amer. Math. Soc., 2000, 353-401. URL: https://www.intlpress.com/site/pub/files/_fulltext/journals/ajm/2000/0004/0001/AJM-2000-0004-0001-a016.pdf. Yau formulates the broad problem as: classify compact smooth manifolds admitting Einstein metrics, and asks whether M always admits one when $\dim M > 4$. This is the source formulation, not a resolution.
2. Arthur L. Besse, Einstein Manifolds, Ergebnisse der Mathematik und ihrer Grenzgebiete (3), vol. 10, Springer, 1987; reprint DOI: <https://doi.org/10.1007/978-3-540-74311-8>. This standard reference records the lack of known topological restrictions in dimensions greater than 4. It gives background and examples, but no universal existence theorem or high-dimensional counterexample.
3. Andrea Sambusetti, "Einstein manifolds and obstructions to the existence of Einstein metrics," Rendiconti di Matematica e delle sue Applicazioni, Serie VII, 18 (1998), 131-149. URL: <https://www1.mat.uniroma1.it/ricerca/rendiconti/ARCHIVIO/1998%281%29/131-149.pdf>. Sambusetti explicitly states the question

“Does every compact manifold of dimension greater than 4 have an Einstein metric?” and notes that no example was known of a compact manifold of dimension greater than 4 admitting no Einstein metric. This supports that the full problem was unresolved at that time.

4. Jeffrey S. Case, “Global obstructions to conformally Einstein metrics in dimension six,” *New York Journal of Mathematics* 28 (2022), 1085-1098; arXiv:2207.00645. URL: <https://nyjm.albany.edu/j/2022/28-45p.pdf>. Case constructs global conformal invariants that obstruct conformally Einstein metrics in dimension six and defines a possible diffeomorphism invariant obstruction to Einstein metrics. However, the paper states that no closed six-manifold is currently known for which the resulting diffeomorphism invariant is finite, let alone nonpositive, and that it is unclear whether the results can produce a closed six-manifold without an Einstein metric. Thus this is partial obstruction technology, not a counterexample to the universal question.
5. Charles P. Boyer, Krzysztof Galicki, and Janos Kollar, “Einstein metrics on spheres,” *Annals of Mathematics* (2) 162 (2005), no. 1, 557-580; DOI: <https://doi.org/10.4007/annals.2005.162.557>; arXiv:math/0309408. The paper constructs many Sasaki-Einstein metrics on odd-dimensional standard and exotic spheres. This proves many positive special cases in dimensions at least 5, but only for restricted diffeomorphism types.
6. Janos Kollar, “Einstein metrics on connected sums of $S^2 \times S^3$,” *Journal of Differential Geometry* 75 (2007), no. 2, 259-272; DOI: <https://doi.org/10.4310/jdg/1175266266>; arXiv:math/0402141. Kollar constructs infinitely many families of Einstein metrics on connected sums of arbitrarily many copies of $S^2 \times S^3$. This gives a large class of compact 5-manifolds with Einstein metrics, but it does not cover all compact 5-manifolds.
7. Charles P. Boyer and Krzysztof Galicki, “Sasakian geometry, hypersurface singularities, and Einstein metrics,” *Supplemento ai Rendiconti del Circolo Matematico di Palermo* (2), Suppl. 75 (2005), 57-87; arXiv:math/0405256. URL: <https://math.unm.edu/~cboyer/papers/srni.pdf>. Corollary 59 states that every compact smooth simply connected 5-manifold which is spin and has torsion-free $H_2(M, \mathbb{Z})$ admits a Sasakian-Einstein metric. This is a strong 5-dimensional positive result, but it assumes simply connected, spin, and torsion-free second homology.
8. Yuchen Liu, Taro Sano, and Luca Tasin, “Infinitely many families of Sasaki-Einstein metrics on spheres,” *Journal of Differential Geometry* 130 (2025), no. 1, 1-26; DOI: <https://doi.org/10.4310/jdg/1747062009>; arXiv:2203.08468. The paper proves infinitely many families of Sasaki-Einstein metrics on every odd-dimensional standard sphere of dimension at least 5 and on all odd-dimensional exotic spheres that bound parallelizable manifolds. This is further positive progress for special manifolds, not a universal theorem.

1.26.3.3. Assessment

the exact question remains unresolved. I found no published or otherwise citable theorem proving that all compact smooth manifolds of dimension at least 5 admit Einstein metrics, and no published or otherwise citable compact manifold of dimension at least 5 proved to admit no Einstein metric. The known results give many positive special cases and some sign-restricted or conformal-class obstruction technology, but they do not settle the full all-manifolds question.

What remains open is the simultaneous treatment of all possible Einstein constants $\lambda > 0$, $\lambda = 0$, and especially $\lambda < 0$ for arbitrary compact smooth manifolds in dimension at least 5.

1.26.4. Remaining open problems recorded by the database

1.26.4.1. Open problem 1

Does every closed smooth six-dimensional manifold admit an Einstein metric?

Reference supplied by the database. Jeffrey S. Case, “Global obstructions to conformally Einstein metrics in dimension six,” *New York Journal of Mathematics* **28** (2022), 1085–1098, discussion following Theorem 1.6 (the author states that it is unclear whether the developed obstructions can produce a closed six-manifold admitting no Einstein metric); [journal article](#); arXiv: [2207.00645](#).

1.26.4.2. Open problem 2

For a closed six-manifold M , define

$$\mathcal{L}_1(M) = \sup_{g \in \text{Met}(M)} \int_M \left(|\nabla W_g|^2 + 12E_s^t W_{tijk} W^{sijk} - 64|C_g|^2 \right) d\text{vol}_g,$$

where $W, E = \text{Ric} - \frac{R}{6}g$, and C are the Weyl, trace-free Ricci, and Cotton tensors. Is $\mathcal{L}_1(M)$ finite for any closed six-manifold M ; in particular, can it be nonpositive?

Reference supplied by the database. Jeffrey S. Case, “Global obstructions to conformally Einstein metrics in dimension six,” *New York Journal of Mathematics* **28** (2022), 1085–1098, discussion following Theorem 1.3; [journal article](#); arXiv: [2207.00645](#).

1.26.5. Primary-source pointer

The originating ICM record is *Geometry and Nonlinear Analysis* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.1/ICM2002.1.ocr.pdf#page=474>.

1.27. 27. Geometry and Nonlinear Analysis [2002-ag-91]

Record: 2002-ag-91

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Geometry and Nonlinear Analysis

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.1/ICM2002.1.ocr.pdf#page=474>

1.27.1. Problem statement

Let M be a compact Kähler manifold with positive first Chern class $c_1(M) > 0$, and let $\eta(M)$ denote the Lie algebra of holomorphic vector fields on M . Suppose that $\eta(M) = \{0\}$. Prove that M admits a Kähler-Einstein metric if and only if M is asymptotically Chow stable with respect to its anticanonical polarization.

1.27.2. Historical background, literature, and current progress

1.27.3. Verdict: partially resolved

1.27.3.1. Precise problem

The statement is Yau's proposed Chow-stability version of the Kähler-Einstein criterion for smooth Fano manifolds: if M is a compact Kähler/Fano manifold with $c_1(M) > 0$ and no nonzero holomorphic vector fields $\eta(M) = 0$, is

$$M \text{ Kähler-Einstein} \iff (M, -K_M) \text{ asymptotically Chow stable?}$$

Here $\eta(M) = 0$ makes the automorphism group discrete/finite in the Fano setting, so the relevant Chow notion is ordinary asymptotic Chow stability rather than relative stability.

1.27.3.2. Current literature status

1. Original formulation / source.

Title: *Geometry and nonlinear analysis*

Author: Gang Tian

Venue: Proceedings of the ICM, Beijing 2002, vol. 1, pp. 475-493

Year: 2002

arXiv/DOI/URL: arXiv:math/0212404, <https://doi.org/10.48550/arXiv.math/0212404>

Role: Tian explicitly records Yau's conjecture in the form: if $\eta(M) = 0$, then M has a Kähler-Einstein metric iff M is asymptotically Chow stable. Tian notes that Donaldson's theorem gives only a partial answer and that comparing Chow stability with CM-stability is a further problem.

2. Proved direction: Kähler-Einstein implies asymptotic Chow stability.

Title: *Scalar Curvature and Projective Embeddings, I*

Author: Simon K. Donaldson

Venue: Journal of Differential Geometry 59(3), pp. 479-522

Year: 2001

DOI/URL: <https://doi.org/10.4310/jdg/1090349449>

Result used: Donaldson proves that a polarized manifold with a constant scalar curvature Kähler metric and discrete automorphism group is asymptotically Chow stable. Taking $L = -K_M$, a Kähler-Einstein metric is cscK in $c_1(-K_M)$, and $\eta(M) = 0$ gives the discrete automorphism hypothesis. This proves the “only if” direction of the problem statement.

3. Why the converse is not obtained from known Chow-to-K implications.

Title: *A study of the Hilbert-Mumford criterion for the stability of projective varieties*

Authors: Julius Ross and Richard P. Thomas

Venue: Journal of Algebraic Geometry 16(2), pp. 201-255

Year: 2007

DOI/arXiv/URL: 10.1090/S1056-3911-06-00461-9; arXiv:math/0412519

Result used: Their Theorem 3.9 and ensuing discussion give the standard implication chain

asymptotically Chow stable $\Rightarrow$ asymptotically Chow semistable $\Rightarrow$ K -semistable,

but not K -stable or K -polystable. Thus ordinary asymptotic Chow stability, as currently related to K -stability in general, does not trigger the Kähler-Einstein existence theorem.

4. Modern replacement theorem: K -stability, not Chow stability.

Title: *Kähler-Einstein metrics on Fano manifolds. I-III*

Authors: Xiuxiong Chen, Simon Donaldson, Song Sun

Venue: Journal of the American Mathematical Society 28(1), pp. 183-197, 199-234, 235-278

Year: 2015

DOI/arXiv/URL: 10.1090/S0894-0347-2014-00799-2; 10.1090/S0894-0347-2014-00800-6; 10.1090/S0894-0347-2014-00801-8; arXiv:1211.4566, arXiv:1211.4571, arXiv:1302.0282

Related later proof: Robert Berman, Sébastien Boucksom, Mattias Jonsson, *A variational approach to the Yau-Tian-Donaldson conjecture*, Journal of the American Mathematical Society 34(3), pp. 605-652, 2021, DOI 10.1090/jams/964, arXiv:1509.04561.

Role: These works resolve the Fano Yau-Tian-Donaldson criterion using K -polystability (and, with finite automorphism group, K -stability/uniform K -stability). They do not prove that asymptotic Chow stability implies K -polystability, so they do not settle the stated Chow-stability statement.

5. Near miss: a different “asymptotic stability” notion.

Title: *Mahler measures, Stable Pairs, and the Global coercive estimate for the Mabuchi Functional*

Author: Sean Timothy Paul

Venue: arXiv preprint

Year: 2021

arXiv/DOI/URL: arXiv:2105.01240, <https://doi.org/10.48550/arXiv.2105.01240>

Role: Corollary 1.1 states that an anticanonically polarized manifold with finite automorphism group is Kähler-Einstein iff it is “asymptotically stable.” However, the paper defines this through stable pairs involving the resultant and hyperdiscriminant with a Mahler-metric distance, not through the ordinary Chow point. This does not resolve the exact statement.

6. Special/related toric progress, not covering this problem.

Title: *Asymptotic Chow Semistability Implies Ding Polystability for Gorenstein Toric Fano Varieties*

Author: Naoto Yotsutani

Venue: *Mathematics* 11(19), Article 4114

Year: 2023

DOI/arXiv/URL: 10.3390/math11194114; arXiv:1711.10113

Role: Gives a toric/Gorenstein Fano result from asymptotic Chow semistability to Ding polystability in the toric setting. Smooth positive-dimensional toric Fano manifolds have nontrivial holomorphic torus vector fields, and the theorem is not a general result for all $\eta(M) = 0$ Fano manifolds.

Title: *Asymptotic Chow stability of uniformly K-stable toric varieties*

Authors: King Leung Lee and Naoto Yotsutani

Venue: arXiv preprint

Year: 2024; revised 2026

arXiv/DOI/URL: arXiv:2405.06883, <https://doi.org/10.48550/arXiv.2405.06883>

Role: Proves toric sufficient criteria for uniformly K-stable polarized toric varieties to be asymptotically Chow polystable under vanishing Futaki-Ono obstruction. This is a toric/extra-hypothesis result and does not prove the stated all-Fano converse.

7. Cautionary counterexamples outside the stated hypotheses.

Title: *An example of an asymptotically Chow unstable manifold with constant scalar curvature*

Authors: Hajime Ono, Yuji Sano, Naoto Yotsutani

Venue: *Annales de l'Institut Fourier* 62(4), pp. 1265-1287

Year: 2012

DOI/arXiv/URL: 10.5802/aif.2722; arXiv:0906.3836

Role: Gives Kähler-Einstein/cscK toric examples that are not asymptotically Chow semistable when the automorphism group is not discrete. These examples do not refute the problem statement because $\eta(M) \neq 0$, but they show why the no-vector-field hypothesis is essential.

1.27.3.3. Verdict

The problem is **partially resolved**. The implication

$$\text{Kähler-Einstein} \Rightarrow \text{asymptotically Chow stable}$$

is proved by Donaldson under exactly the stated no-holomorphic-vector-field hypothesis. The converse

$$\text{asymptotically Chow stable} \Rightarrow \text{Kähler-Einstein}$$

is not fully resolved in the published/citable literature found: known general implications from Chow stability reach only K -semistability, while the Kähler-Einstein existence theorem requires K -polystability/ K -stability. Related CM/stable-pair or toric results use different stability notions or extra hypotheses and therefore do not cover all cases in the problem statement.

1.27.4. Remaining open problems recorded by the database

1.27.4.1. Open problem 1

For a smooth anticanonically polarized Fano manifold $(M, -K_M)$ with $\eta(M) = 0$, determine whether asymptotic Chow stability forces CM-stability (equivalently, in the modern formulation, K-stability rather than only K-semistability), by comparing the Chow and CM polarizations on the relevant Hilbert scheme.

Reference supplied by the database. Gang Tian, “Geometry and Nonlinear Analysis,” in *Proceedings of the International Congress of Mathematicians, Beijing 2002*, Vol. I, Higher Education Press, Beijing, 2002, pp. 475–493, discussion following Yau’s asymptotic-Chow-stability conjecture on p. 480 (“a very interesting problem ... to compare the Chow stability with the CM-stability”). arXiv: [math/0212404](https://arxiv.org/abs/math/0212404); identifier: <https://doi.org/10.48550/arXiv.math/0212404>. See also Sean Timothy Paul and Gang Tian, “Analysis of Geometric Stability,” *International Mathematics Research Notices* **2004** (2004), no. 48, 2555–2591, which identifies the difference between the Chow and CM polarizations without proving the required stability implication. DOI: <https://doi.org/10.1155/S1073792804131759>; arXiv: [math/0404223](https://arxiv.org/abs/math/0404223).

1.27.5. Primary-source pointer

The originating ICM record is *Geometry and Nonlinear Analysis* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.1/ICM2002.1.ocr.pdf#page=474>.

1.28. 28. Geometry and Nonlinear Analysis [2002-ag-92]

Record: 2002-ag-92

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Geometry and Nonlinear Analysis

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.1/ICM2002.1.ocr.pdf#page=474>

1.28.1. Problem statement

Let (M, g) be a complete noncompact Calabi-Yau manifold of complex dimension n with quadratic curvature decay, meaning that for some base point $p \in M$ there are constants $C, R_0 > 0$ such that $|\mathrm{Rm}_g|(x) \leq C d_g(p, x)^{-2}$ whenever $d_g(p, x) \geq R_0$, and with Euclidean volume growth, meaning that there are constants $c, C' > 0$ such that $cR^{2n} \leq \mathrm{Vol}_g(B_g(p, R)) \leq C'R^{2n}$ for all sufficiently large R . Prove that M admits a compactification by a divisor at infinity: there exist a compact complex manifold $\bar{M}$ and an effective divisor $D \subset \bar{M}$ such that $M = \bar{M} \setminus D$, the divisor D is ample near itself (equivalently, the line bundle $\mathcal{O}_{\bar{M}}(D)$ is positive on a neighborhood of D), and the anticanonical bundle of $\bar{M}$ is proportional to the divisor class of D with coefficient greater than one, namely $K_{\bar{M}}^{-1} = \alpha[D]$ in $\mathrm{Pic}(\bar{M}) \otimes_{\mathbb{Z}} \mathbb{R}$ for some $\alpha > 1$, where $[D]$ denotes the class of $\mathcal{O}_{\bar{M}}(D)$.

1.28.2. Historical background, literature, and current progress

1.28.3. Verdict: partially resolved

1.28.3.1. Problem identified

The problem is the refined Yau/Tian compactification problem for complete noncompact Calabi-Yau manifolds: if a complete Calabi-Yau manifold has Euclidean volume growth and quadratic curvature decay, must it be the complement of a positive divisor at infinity in a compact complex manifold, with the anticanonical class proportional to that divisor by a coefficient greater than one?

The original source I found is:

- **Title:** Geometry and nonlinear analysis
- **Author:** Gang Tian
- **Venue:** Proceedings of the International Congress of Mathematicians, Vol. I, Beijing 2002, Higher Education Press
- **Year:** 2002
- **Pages:** 475-493
- **arXiv/URL:** arXiv:math/0212404, <https://arxiv.org/abs/math/0212404>
- **Problem location:** Around pp. 480-481 Tian states the refined conjecture: a complete Calabi-Yau manifold with quadratic curvature decay and Euclidean volume growth

should be of the form $\bar{M} \setminus D$, with D ample near itself and $K_M^{-1} = \alpha[D]$ for some $\alpha > 1$.

1.28.3.2. Main current progress

- **Title:** No semistability at infinity for Calabi-Yau metrics asymptotic to cones
- **Authors:** Song Sun and Junsheng Zhang
- **Venue:** *Inventiones Mathematicae* 233, no. 1
- **Year:** 2023
- **Pages:** 461-494
- **DOI/arXiv/URL:** DOI 10.1007/s00222-023-01187-4; arXiv:2208.05098, <https://arxiv.org/abs/2208.05098>
- **Relevant result:** Under exactly the metric hypotheses used here, namely Euclidean volume growth and quadratic curvature decay for a complete Calabi-Yau manifold, Theorem 1.2 proves polynomial asymptotic convergence to the unique Calabi-Yau cone at infinity. The paper also proves the “no semistability at infinity” theorem identifying the intermediate Donaldson-Sun/Liu cone with the metric cone.
- **Applicability:** This removes the main analytic gap between the stated assumptions and the polynomially asymptotically conical setting classified by Conlon-Hein.
- **Title:** Classification of asymptotically conical Calabi-Yau manifolds
- **Authors:** Ronan J. Conlon and Hans-Joachim Hein
- **Venue:** *Duke Mathematical Journal* 173, no. 5
- **Year:** 2024
- **Pages:** 947-1015
- **DOI/arXiv/URL:** DOI 10.1215/00127094-2023-0030; arXiv:2201.00870, <https://arxiv.org/abs/2201.00870>
- **Relevant result:** The main Existence and Classification theorems classify smooth complete Calabi-Yau manifolds asymptotic to a Calabi-Yau cone at a polynomial rate. In the proof of classification, for an AC Calabi-Yau manifold M , they construct compactifications $X_i = M \cup D_i$ by admissible divisors, prove $K_{X_i} + \alpha_i[D_i]$ is trivial with $\alpha_i > 1$, prove projectivity, and show the corresponding Remmert reduction has an ample compactifying divisor.
- **Applicability:** Combined with Sun-Zhang, this gives a full algebro-geometric classification under the stated metric assumptions and proves the desired compactification statement in the projective orbifold/admissible-divisor sense.
- **Title:** On sharp rates and analytic compactifications of asymptotically conical Kähler metrics
- **Author:** Chi Li
- **Venue:** *Duke Mathematical Journal* 169, no. 8
- **Year:** 2020
- **Pages:** 1397-1483

- **DOI/arXiv/URL:** DOI 10.1215/00127094-2019-0073; arXiv:1405.2433, <https://arxiv.org/abs/1405.2433>
- **Relevant result:** Provides analytic compactification tools for asymptotically conical complex/Kähler manifolds. Conlon-Hein use and extend this compactification technology, including to orbifold divisors.
- **Title:** Compactification of certain Kähler manifolds with nonnegative Ricci curvature
- **Author:** Gang Liu
- **Venue:** Advances in Mathematics 382
- **Year:** 2021
- **Article:** 107652
- **DOI/URL:** DOI 10.1016/j.aim.2021.107652
- **Relevant result:** Removes a technical hypothesis from Donaldson-Sun's two-step degeneration theory in the complete Calabi-Yau setting with Euclidean volume growth and quadratic curvature decay; used by Sun-Zhang to place the stated manifolds in the algebraic degeneration framework.

1.28.3.3. Why this is not marked fully resolved

The literature now essentially resolves the problem in the modern algebraic/orbifold formulation: under the stated metric assumptions the manifold is polynomially AC, classified by Conlon-Hein, and compactifies by an admissible positive divisor with $K + \alpha D = 0$, $\alpha > 1$.

However, the problem statement as written asks for a **compact complex manifold** $\bar{M}$. The Conlon-Hein compactification produced in the general case is explicitly a **compact complex orbifold** $X_i = M \cup D_i$ or a normal projective variety with an admissible/orbifold divisor. I did not find a published theorem proving that, in all dimensions and for all possible asymptotic cones allowed by the problem statement, this can always be upgraded to a smooth compact complex manifold while preserving the exact anticanonical proportionality and positivity statement.

Therefore the exact statement is not fully resolved as stated, but it has very strong partial resolution: complete classification and the desired compactification in the projective orbifold/admissible-divisor sense.

1.28.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

1.28.5. Primary-source pointer

The originating ICM record is *Geometry and Nonlinear Analysis* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.1/ICM2002.1.ocr.pdf#page=474>.

1.29. 29. Geometry and Nonlinear Analysis [2002-ag-93]

Record: 2002-ag-93

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Geometry and Nonlinear Analysis

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.1/ICM2002.1.ocr.pdf#page=474>

1.29.1. Problem statement

Let (M_i, g_i) be a sequence of compact real n -dimensional Einstein manifolds with the same special holonomy and uniformly bounded diameters, converging in the Gromov-Hausdorff topology to a compact metric space M_∞ . Let $S \subset M_\infty$ be the singular set in the Cheeger-Tian compactness theorem: S is a rectifiable closed subset, $M_\infty \setminus S$ is a smooth manifold carrying an Einstein metric g_∞ with the same holonomy as the g_i , and M_∞ is the metric completion of $M_\infty \setminus S$ for the distance induced by g_∞ .

Prove that S admits a stratification $S = \coprod_{0 \leq i \leq n-4} S_i$ such that each stratum S_i is a smooth manifold of real dimension at most i . If the manifolds (M_i, g_i) are Kähler-Einstein, prove that the odd strata do not occur separately: $S_{2j+1} = S_{2j}$ for every j for which these strata are defined.

1.29.2. Historical background, literature, and current progress

1.29.3. Verdict: partially resolved

1.29.3.1. Precise problem

The question is the regularity/stratification problem for the singular set in noncollapsed Gromov-Hausdorff limits of compact Einstein manifolds with fixed special holonomy. Starting from the Cheeger-Tian compactness picture, where the limit is smooth Einstein with the same holonomy away from a closed rectifiable singular set S of codimension at least 4, the proposed strengthening asks for a decomposition

$$S = \coprod_{0 \leq i \leq n-4} S_i$$

with each S_i a smooth manifold of real dimension at most i . In the Kähler-Einstein case it also asks for the parity improvement $S_{2j+1} = S_{2j}$.

The distinctive statement appears in Gang Tian's ICM article, where after the Cheeger-Tian compactness theorem he writes that the fundamental remaining problem is the regularity of S , conjecturing the above smooth stratification and the Kähler-Einstein parity statement.

1.29.3.2. Original/source literature

- **Geometry and nonlinear analysis**; Gang Tian; *Proceedings of the International Congress of Mathematicians, Vol. I (Beijing, 2002)*, Higher Education Press, 2002, pp. 475-493; arXiv:math/0212404, <https://arxiv.org/abs/math/0212404>. This is the closest match to the wording and explicitly formulates the conjectural smooth stratification of S and the Kähler-Einstein parity $S_{2j+1} = S_{2j}$.
- **Degeneration of Einstein metrics and metrics with special holonomy**; Jeff Cheeger; *Surveys in Differential Geometry*, Vol. VIII, International Press, 2003, pp. 29-73; DOI: 10.4310/SDG.2003.v8.n1.a2. This is the related special-holonomy degeneration source cited in later work.
- **Anti-Self-Duality of Curvature and Degeneration of Metrics with Special Holonomy**; Jeff Cheeger and Gang Tian; *Communications in Mathematical Physics* 255 (2005), 391-417; DOI: 10.1007/s00220-004-1279-0, <https://link.springer.com/article/10.1007/s00220-004-1279-0>. Proves smoothness off a closed codimension ≥ 4 subset for noncollapsed special-holonomy limits and gives further structure supporting, but not proving, the full smooth-stratification conjecture.

1.29.3.3. Current progress

- **On the singularities of spaces with bounded Ricci curvature**; Jeff Cheeger, Tobias H. Colding, and Gang Tian; *Geometric and Functional Analysis* 12(5) (2002), 873-914; DOI: 10.1007/PL00012649. Establishes major structural restrictions on Ricci-limit singularities. In the Kähler-Ricci setting, tangent cones split Euclidean factors in even real dimensions, which gives the expected no-odd-strata behavior for the standard Cheeger-Colding stratification.
- **Regularity of Einstein manifolds and the codimension 4 conjecture**; Jeff Cheeger and Aaron Naber; *Annals of Mathematics* 182(3) (2015), 1093-1165; DOI: 10.4007/annals.2015.182.3.5, <https://annals.math.princeton.edu/wp-content/uploads/annals-v182-n3-p05-p.pdf>. Solves the codimension-4 conjecture for noncollapsed limits with two-sided Ricci bounds: the singular set has Hausdorff/Minkowski dimension at most $n - 4$, and the limit is smooth away from a closed codimension-4 set. This proves the optimal dimension bound but not smooth manifold strata.
- **L^2 Curvature Bounds on Manifolds with Bounded Ricci Curvature**; Wenshuai Jiang and Aaron Naber; *Annals of Mathematics* 193(1) (2021), 107-222; DOI: 10.4007/annals.2021.193.1.2; arXiv:1605.05583, <https://arxiv.org/abs/1605.05583>. Proves L^2 curvature bounds and that, for noncollapsed two-sided Ricci limits, the singular set is $(n - 4)$ -rectifiable with uniformly bounded $\mathcal{H}^{n-4}$ -measure; at $\mathcal{H}^{n-4}$ -a.e. singular point the tangent cone is unique and of orbifold-cone type $\mathbb{R}^{n-4} \times C(S^3/\Gamma)$.
- **Rectifiability of singular sets of noncollapsed limit spaces with Ricci curvature bounded below**; Jeff Cheeger, Wenshuai Jiang, and Aaron Naber; *Annals of Mathematics* 193(2) (2021), 407-538; DOI: 10.4007/annals.2021.193.2.2, <https://annals.math.princeton.edu/2021/193-2/p02>. Proves the Cheeger-Colding strata S^k are k -rectifiable for all k ; under two-sided Ricci bounds it recovers $(n - 4)$ -rectifiability and tangent-cone uniqueness almost everywhere. Rectifiability is substantially weaker than a stratification by smooth manifolds at every point.
- **Gromov-Hausdorff limits of Kähler manifolds and algebraic geometry**; Simon Donaldson and Song Sun; *Acta Mathematica* 213(1) (2014), 63-106; DOI: 10.1007/s11511-014-0116-3, <https://arxiv.org/abs/1206.2609>. For polarized noncollapsed Kähler limits with two-sided Ricci bounds, proves the metric limit is homeomorphic to a

normal projective variety; in the Kähler-Einstein case the metric singular set matches the algebraic singular set. This gives complex-analytic stratifications in this Kähler-Einstein setting, hence no odd real-dimensional strata there, but it is not a theorem for all special holonomy types.

- **Gromov-Hausdorff limits of Kähler manifolds and algebraic geometry, II**; Simon Donaldson and Song Sun; *Journal of Differential Geometry* 107(2) (2017), 327–371; DOI: 10.4310/jdg/1506650422; arXiv:1507.05082, <https://arxiv.org/abs/1507.05082>. Gives further algebro-geometric structure of tangent cones in Kähler-Einstein limits.
- **Gromov-Hausdorff limits of Kähler manifolds with Ricci curvature bounded below**; Gang Liu and Gábor Székelyhidi; *Geometric and Functional Analysis* 32(2) (2022), 236–279; DOI: 10.1007/s00039-022-00594-8, <https://arxiv.org/abs/1804.08567>. Extends the Donaldson-Sun algebraic picture to polarized Kähler manifolds with Ricci bounded below, showing normal projective limits and metric singularities given by analytic subvarieties. Again, this is Kähler-specific and does not cover arbitrary special holonomy.

1.29.3.4. What remains unresolved

The literature fully resolves several important consequences of the conjecture: optimal codimension 4, rectifiability and finite measure of the top singular set, almost-everywhere uniqueness and orbifold-cone structure of tangent cones, and strong complex-analytic structure in polarized Kähler/Kähler-Einstein cases. However, I did not find a published theorem proving that every singular set S arising from all compact Einstein limits with the same special holonomy admits a global decomposition into smooth manifold strata S_i of the requested dimensions.

The gap is not just terminological: k -rectifiability and almost-everywhere tangent cone structure do not imply that each stratum is a smooth manifold, and the Kähler algebraic/analytic results do not cover non-Kähler special holonomy such as G_2 , $\text{Spin}(7)$, or quaternionic-Kähler settings. Therefore the exact statement is not fully resolved.

1.29.4. Remaining open problems recorded by the database

1.29.4.1. Open problem 1

Let (M_i^n, g_i) be a noncollapsed sequence of compact manifolds with special holonomy as in the Cheeger–Tian compactness theorem, and suppose in addition that the characteristic numbers $C(M_i^n)$ defined by Cheeger and Tian are uniformly bounded. Prove that the singularities of the Gromov–Hausdorff limit are of orbifold type outside a closed subset of real codimension at least 6.

Reference supplied by the database. Jeff Cheeger and Gang Tian, “Anti-Self-Duality of Curvature and Degeneration of Metrics with Special Holonomy,” *Communications in Mathematical Physics* 255 (2005), no. 2, 391–417. Main conjecture stated in the abstract and developed in the paper. DOI: <https://doi.org/10.1007/s00220-004-1279-0>.

1.29.4.2. Open problem 2

Let (X, d) be a noncollapsed Gromov–Hausdorff limit of Einstein manifolds whose Einstein constants are uniformly bounded. Prove that the metric tangent cone at every point of X is unique, without assuming that one tangent cone has a smooth cross-section.

Reference supplied by the database. Tobias Holck Colding and William P. Minicozzi II, “On Uniqueness of Tangent Cones for Einstein Manifolds,” *Inventiones Mathematicae* **196** (2014), no. 3, 515–588. The general uniqueness question in the Introduction and its local Einstein-limit form surrounding Theorem 0.3; the paper proves uniqueness under the additional hypothesis that one tangent cone has a smooth cross-section. DOI: <https://doi.org/10.1007/s00222-013-0474-z>; arXiv: 1206.4929.

1.29.4.3. Open problem 3

Let (X, d) be a noncollapsed Gromov–Hausdorff limit of Einstein n -manifolds. Prove that there is a closed subset $\mathcal{S}_{\text{orb}} \subset X$ of Hausdorff codimension at least 5 such that $X \setminus \mathcal{S}_{\text{orb}}$ is a smooth Einstein orbifold, whose orbifold singular set is a countable union of totally geodesic codimension-4 submanifolds and is locally modeled at every such point on $\mathbb{R}^{n-4} \times \mathbb{R}^4/\Gamma$ for a finite group $\Gamma \subset O(4)$.

Reference supplied by the database. Yiqi Huang and Tristan Ozuch, “Regularity of Einstein 5-Manifolds via 4-Dimensional Gap Theorems,” preprint, 2025 (revised 2026), Question 1.29 (“Codimension-5 conjecture”; numbered Question 1.29 in the arXiv version’s §1.3.2). arXiv: 2512.21317.

1.29.5. Primary-source pointer

The originating ICM record is *Geometry and Nonlinear Analysis* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.1/ICM2002.1.ocr.pdf#page=474>.

1.30. 30. Some New Developments of Realization of Surfaces into $\mathbb{R}^3$ [2002-geom-253]

Record: 2002-geom-253

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Some New Developments of Realization of Surfaces into $\mathbb{R}^3$

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=154>

1.30.1. Problem statement

Determine whether every C^∞ Riemannian metric g of nonnegative Gaussian curvature on $M = S^2$, and every complete C^∞ Riemannian metric g of nonnegative Gaussian curvature on $M = \mathbb{R}^2$, admits a global isometric embedding $F : (M, g) \rightarrow \mathbb{R}^3$ of class $C^{2,\alpha}$ for some $0 < \alpha < 1$. In particular, determine whether this holds when g is real-analytic.

1.30.2. Historical background, literature, and current progress

The problem is **unresolved**. No proof or counterexample was found for the exact assertion, including its real-analytic subcase. The unresolved issue is regularity across the set where the Gaussian curvature vanishes; the general existence results give only $C^{1,1}$ embeddings.

- Pengfei Guan and Yan Yan Li, “The Weyl problem with nonnegative Gauss curvature,” *Journal of Differential Geometry* 39 (1994), 331–342, DOI <https://doi.org/10.4310/jdg/1214454874>. Their main theorem proves that every C^4 metric on S^2 with $K \geq 0$ has a global $C^{1,1}$ isometric embedding in $\mathbb{R}^3$. This is the strongest general sphere existence theorem, but it falls short of $C^{2,\alpha}$.
- Jia-Xing Hong and Claude Zuily, “Isometric embedding of the 2-sphere with non negative curvature in $\mathbb{R}^3$,” *Mathematische Zeitschrift* 219 (1995), 323–334, DOI <https://doi.org/10.1007/BF02572368>. This independently proves the same general $C^{1,1}$ sphere result by a different method; it does not improve regularity at $K = 0$ to $C^{2,\alpha}$.
- Jia-Xing Hong, “Isometric embedding in $\mathbb{R}^3$ of complete noncompact nonnegatively curved surfaces,” *Manuscripta Mathematica* 94 (1997), 271–286, [EuDML](#). Theorem A proves that every complete C^4 nonnegatively curved metric on $\mathbb{R}^2$ has a global $C^{1,1}$ isometric embedding, smooth on the positive-curvature locus. Thus the plane case has the same unresolved regularity gap on its zero-curvature set.
- Panagiota Daskalopoulos, “The regularity of solutions in degenerate geometric problems,” *Surveys in Differential Geometry* 19 (2014), 83–110, [direct article PDF](#). Section 4 explicitly retains as a question whether a smooth nonnegatively curved metric on S^2 has a global $C^{2,\alpha}$ (or $C^{2,1}$) isometric embedding and explains the reduction near curvature zeros to degenerate Monge–Ampère regularity. It also reports $C^{2,\delta}$ regularity for the model equation $\det D^2u = |x|^\beta$ with $\beta > 0$, a substantive special model but not a solution for arbitrary curvature-zero sets.
- Xumin Jiang, “Isometric embedding with nonnegative Gauss curvature under the graph setting,” *Discrete and Continuous Dynamical Systems* 39 (2019), 3463–3477,

DOI <https://doi.org/10.3934/dcds.2019143>, arXiv:1806.06189. This proves local $C^{2,1}$ regularity for a $C^{1,1}$ embedding of a C^4 nonnegatively curved metric under an additional graph-coordinate hypothesis. That extra hypothesis is not available for every global embedding in the problem, so the result is partial rather than a resolution.

Consequently, the known results cover global existence in both the sphere and complete-plane cases only at $C^{1,1}$ regularity, while higher regularity is known under additional structural assumptions. The general $C^{2,\alpha}$ conclusion, even for analytic metrics, remains open.

1.30.3. Remaining open problems recorded by the database

1.30.3.1. Open problem 1

Determine whether every real-analytic Riemannian metric g of nonnegative Gaussian curvature on S^2 admits a global isometric embedding $F : (S^2, g) \rightarrow \mathbb{R}^3$ of class $C^{2,\alpha}$ for some $0 < \alpha < 1$.

Reference supplied by the database. Jiaying Hong, “Some New Developments of Realization of Surfaces into $\mathbb{R}^3$,” Proceedings of the International Congress of Mathematicians, Beijing 2002, Vol. III, Higher Education Press, 2002, pp. 155–166, arXiv:math/0304391, <https://arxiv.org/abs/math/0304391>.

1.30.3.2. Open problem 2

Determine whether every complete real-analytic Riemannian metric g of nonnegative Gaussian curvature on $\mathbb{R}^2$ admits a global isometric embedding $F : (\mathbb{R}^2, g) \rightarrow \mathbb{R}^3$ of class $C^{2,\alpha}$ for some $0 < \alpha < 1$.

Reference supplied by the database. Jiaying Hong, “Some New Developments of Realization of Surfaces into $\mathbb{R}^3$,” Proceedings of the International Congress of Mathematicians, Beijing 2002, Vol. III, Higher Education Press, 2002, pp. 155–166, arXiv:math/0304391, <https://arxiv.org/abs/math/0304391>.

1.30.4. Primary-source pointer

The originating ICM record is *Some New Developments of Realization of Surfaces into $\mathbb{R}^3$* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=154>.

1.31. 31. Some New Developments of Realization of Surfaces into $\mathbb{R}^3$ [2002-geom-254]

Record: 2002-geom-254

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Some New Developments of Realization of Surfaces into $\mathbb{R}^3$

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=154>

1.31.1. Problem statement

Identify a non-vacuous geometric condition, not itself phrased in terms of the existence of an embedding, on a smooth geodesically complete Riemannian surface (M, g) with strictly negative Gaussian curvature that guarantees the existence of a smooth isometric embedding $F : (M, g) \rightarrow (\mathbb{R}^3, g_{\text{Eucl}})$; that is, F is a smooth embedding and $F^*g_{\text{Eucl}} = g$.

1.31.2. Historical background, literature, and current progress

The problem is **unresolved** in the stated smooth-embedding sense. The literature contains important intrinsic sufficient conditions for global smooth **isometric immersions**, and it contains explicit examples of complete embedded negatively curved surfaces, but no source located proves that the intrinsic conditions make the resulting immersion injective. Thus these results do not supply the smooth embedding required in the statement.

The principal partial results are as follows.

- **Jiaying Hong**, “Realization in $\mathbb{R}^3$ of Complete Riemannian Manifolds with Negative Curvature,” *Communications in Analysis and Geometry* 1 (1993), no. 4, 487–514. [Direct article PDF](#). Theorem A assumes that (M, g) is smooth, complete, simply connected, and negatively curved, together with monotone decay of $|K|$ faster than inverse-square in geodesic polar coordinates and bounds on specified angular derivatives of $\log |K|$. Its conclusion is a global smooth isometric **immersion** in $\mathbb{R}^3$, not an embedding.
- **Wentao Cao, Feimin Huang, and Dehua Wang**, “Isometric Immersion of Complete Surfaces with Slowly Decaying Negative Gauss Curvature,” *arXiv preprint* (2016), [arXiv:1605.09491](#). [arXiv:1605.09491](#). Theorem 1.1 relaxes Hong’s rapid-decay regime: under an integrability condition on $\sqrt{|K|}$, controlled derivatives, and related hypotheses in geodesic polar coordinates, a complete simply connected negatively curved surface has a global smooth isometric immersion. Again, injectivity is not proved.
- **Wentao Cao, Qing Han, Feimin Huang, and Dehua Wang**, “The Isometric Immersion of Negatively Curved Surfaces with Finite Total Curvature,” *arXiv preprint* (2023; revised 2024), [arXiv:2308.02832](#). [arXiv:2308.02832](#). Theorem 1.1 treats complete simply connected surfaces of finite total Gaussian curvature whose curvature has a controlled decomposition into a radial monotone factor and an oscillatory factor. It proves a global smooth solution of the Gauss–Codazzi system and hence a global smooth isometric im-

mersion. The paper itself describes the general theory as incomplete, and its theorem does not promote the immersion to an injective embedding.

- **Chris Connell and Mohammad Ghomi, “Topology of Negatively Curved Real Affine Algebraic Surfaces,”** *Journal für die reine und angewandte Mathematik* 2008 (2008), no. 624, 1–26, DOI: <https://doi.org/10.1515/CRELLE.2008.078>. This constructs an explicit smooth complete embedded negatively curved quartic surface, homeomorphic to a twice-punctured torus, and recalls classical embedded examples such as the hyperbolic paraboloid and one-sheeted hyperboloid. These are substantive existence examples, but they do not give an intrinsic sufficient criterion for embedding a given abstract complete negatively curved metric.

There are also sharp obstructions. **David Hilbert, “Ueber Flächen von constanter Gausscher Krümmung,”** *Transactions of the American Mathematical Society* 2 (1901), no. 1, 87–99, DOI: <https://doi.org/10.1090/S0002-9947-1901-1500557-5>, excludes a complete regular immersion of constant negative curvature. **N. V. Efimov, “Generation of Singularities on Surfaces of Negative Curvature,”** *Matematicheskii Sbornik (N.S.)* 64(106) (1964), no. 2, 286–320, [MathNet](#), extends nonexistence to curvature bounded above by a negative constant. Since an embedding is an immersion, these also exclude embeddings in those regimes.

What remains is a genuinely intrinsic, non-circular geometric hypothesis under which a global smooth isometric realization is known to be injective. The existing curvature-decay and finite-total-curvature results stop at immersion, so the exact problem in the statement is not completely answered.

1.31.3. Remaining open problems recorded by the database

1.31.3.1. Open problem 1

Determine whether there exists a smooth embedded surface $\Sigma \subset B^3$, where $B^3 = \{x \in \mathbb{R}^3 : |x| < 1\}$ is the open unit ball, such that the Riemannian metric on Σ induced by the Euclidean metric is complete and the Gaussian curvature of Σ is strictly negative at every point.

Reference supplied by the database. Mohammad Ghomi, “Open Problems in Geometry of Curves and Surfaces,” author-hosted survey, last revised 2019, Problem 7.2 (attributed there to Jacques Hadamard, 1898), <https://ghomi.math.gatech.edu/Papers/op.pdf>.

1.31.3.2. Open problem 2

Determine whether there exists a smooth embedded surface $\Sigma \subset \mathbb{R}^3$ such that its induced Riemannian metric is complete, its Gaussian curvature is strictly negative everywhere, its Euler characteristic is negative, and Σ is contained between two parallel planes.

Reference supplied by the database. Mohammad Ghomi, “Open Problems in Geometry of Curves and Surfaces,” author-hosted survey, last revised 2019, Problem 7.3, <https://ghomi.math.gatech.edu/Papers/op.pdf>.

1.31.4. Primary-source pointer

The originating ICM record is *Some New Developments of Realization of Surfaces into R^3* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=154>.

1.32. 32. Some New Developments of Realization of Surfaces into $\mathbb{R}^3$ [2002-geom-255]

Record: 2002-geom-255

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Some New Developments of Realization of Surfaces into $\mathbb{R}^3$

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=154>

1.32.1. Problem statement

Let $\bar{D}$ be the closed unit disk and let g be a smooth Riemannian metric on $\bar{D}$ whose Gaussian curvature is strictly positive. Suppose that the geodesic curvature k_g of ∂D , computed with respect to the inward-pointing unit conormal, is strictly negative everywhere and satisfies

$$\int_{\partial D} |k_g| ds_g < 2\pi.$$

Determine whether every such $(\bar{D}, g)$ admits a smooth isometric embedding $\mathbf{r} = (x, y, z) : \bar{D} \rightarrow \mathbb{R}^3$ such that

$$\mathbf{r}(\partial D) \subset \{(x, y, z) \in \mathbb{R}^3 : z = 0\},$$

equivalently, $z = 0$ on ∂D .

1.32.2. Historical background, literature, and current progress

The problem is **unresolved**. No verified source gives either a smooth planar-boundary isometric embedding for every metric satisfying the stated hypotheses or a counterexample within the strict-sign class $K > 0, k_g < 0$, and $\int_{\partial D} |k_g| ds_g < 2\pi$.

The key literature is as follows.

- Jiaying Hong, “Some New Developments of Realization of Surfaces into $\mathbb{R}^3$,” *Proceedings of the International Congress of Mathematicians, Beijing 2002*, Vol. III, Higher Education Press, 2002, pp. 155–166; arXiv:math/0304391, <https://arxiv.org/abs/math/0304391>. Hong derives $\int_{\partial D} |k_g| ds < 2\pi$ as a necessary condition for a C^2 planar-boundary embedding and explicitly identifies the everywhere-negative case as open. The paper also gives embedded spherical crowns with $k_g < 0$, so existence is known for particular metrics, and records a countable family of such crowns that are not infinitesimally rigid; neither result settles universal existence.
- Jiaying Hong, “Darboux Equations and Isometric Embedding of Riemannian Manifolds with Nonnegative Curvature in $\mathbb{R}^3$,” *Chinese Annals of Mathematics, Series B* 20(2) (1999), 123–136, DOI: <https://doi.org/10.1142/S0252959999000163>. This proves global smooth solvability of the homogeneous Darboux Dirichlet problem and

yields smooth planar-boundary embeddings in the positive-boundary-curvature regime. It does not cover $k_g < 0$.

- Qing Han and Jia-Xing Hong, “Boundary Value Problems,” Chapter 11 in *Isometric Embedding of Riemannian Manifolds in Euclidean Spaces*, Mathematical Surveys and Monographs 130, American Mathematical Society, 2006, pp. 225–247, DOI: <https://doi.org/10.1090/surv/130/11>. The chapter proves the planar-boundary theorem for positive geodesic curvature and, in its notes, still states as open the case where the boundary geodesic curvature is negative and its total absolute curvature is less than 2π . Thus it matches the exact sign and integral conditions here but supplies no resolution.
- Chunhe Li and Jiaying Hong, “Infinitesimal Nonrigidity of Convex Surfaces with Planar Boundary,” *Science in China Series A: Mathematics* 48(4) (2005), 513–521, DOI: <https://doi.org/10.1360/04ys0003>. The authors construct a countable family of positively curved surfaces of revolution with planar boundary, extending beyond a hemisphere, that possess nontrivial infinitesimal deformations preserving the boundary plane. This rigorously explains why the standard continuity method can lose openness in the negative- k_g regime, but infinitesimal nonrigidity is neither nonexistence nor a solution of the nonlinear embedding problem.
- Qing Han, Jiaying Hong, and Gengeng Huang, “Compactness of Alexandrov–Nirenberg Surfaces,” *Communications on Pure and Applied Mathematics* 70(9) (2017), 1706–1753, DOI: <https://doi.org/10.1002/cpa.21686>; arXiv:1402.2388, <https://arxiv.org/abs/1402.2388>. This proves compactness, under quantitative bounds, for already embedded surfaces satisfying $K > 0$ in the interior, $K = 0$ transversely on the boundary, and total curvature 4π . The authors describe existence as a further goal. These hypotheses lie on a degenerate equality boundary of the present problem, rather than covering its strict conditions $K > 0$ on $\bar{D}$ and $\int |k_g| < 2\pi$.

Consequently, the established results provide examples, the complete opposite-sign theorem, a precise linearized obstruction, and compactness in a neighboring degenerate class, while the universal smooth existence question in the statement remains open.

1.32.3. Remaining open problems recorded by the database

No separate descendant problem is listed in the website’s “Some remaining open problems” field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

1.32.4. Primary-source pointer

The originating ICM record is *Some New Developments of Realization of Surfaces into R^3* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=154>.

1.33. 33. Weyl Manifolds and Gaussian Thermostats [2002-geom-276]

Record: 2002-geom-276

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Weyl Manifolds and Gaussian Thermostats

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=498>

1.33.1. Problem statement

Prove that for every finite-dimensional unimodular Lie group G with $\dim G \geq 2$, there is no pair consisting of a left-invariant Riemannian metric g on G and a left-invariant vector field E on G for which the Weyl connection

$$\hat{\nabla}_X Y = \nabla_X Y + g(E, Y)X + g(E, X)Y - g(X, Y)E,$$

where ∇ is the Levi-Civita connection of g , has strictly negative Weyl sectional curvature on every tangent two-plane. More explicitly, for $p \in G$, a two-dimensional subspace $\Pi \subset T_p G$, and a g -orthonormal basis X, Y of Π , let $E_\perp$ be the g -orthogonal projection of E_p onto $\Pi^\perp$, and define

$$\hat{K}(\Pi) = K_g(\Pi) - \|E_\perp\|_g^2 - g(\nabla_X E, X) - g(\nabla_Y E, Y),$$

where $K_g(\Pi)$ is the Riemannian sectional curvature of g on Π . Prove that it is impossible to have $\hat{K}(\Pi) < 0$ for every $p \in G$ and every two-plane $\Pi \subset T_p G$.

1.33.2. Historical background, literature, and current progress

The problem is **unresolved**. No published proof or counterexample covering all finite-dimensional unimodular Lie groups was found. The available rigidity and classification results impose additional hypotheses or cover restricted classes, and therefore do not prove the full assertion.

The problem is Conjecture 5.5 in Maciej P. Wojtkowski, “Weyl Manifolds and Gaussian Thermostats,” *Proceedings of the International Congress of Mathematicians, Beijing 2002*, Vol. III, Higher Education Press, 2002, pp. 511–523; [arXiv:math/0304461](https://arxiv.org/abs/math/0304461). Wojtkowski records the flat-torus and SOL examples with non-positive curvature and zero-curvature planes, then conjectures that strict negativity is impossible on every unimodular Lie group.

Gabriela Tereszkievicz and Maciej P. Wojtkowski, “Homogeneous Weyl Connections of Non-positive Curvature,” *Annals of Global Analysis and Geometry* 51 (2017), 209–229, <https://doi.org/10.1007/s10455-016-9526-0>, proves substantial special cases. Theorem 7.2 (also stated as Theorem 4.3) shows that on a unimodular Lie group a left-invariant vector field satisfying conditions (W1) and (W2), hence in particular an invariant stretched

non-positive (SNP) field, must be parallel; such connections have zero-curvature planes and cannot be strictly negative. Theorems 5.1 and 5.3 and Corollary 5.4 analyze non-positive connections on a class of almost-abelian solvable extensions. However, the paper explicitly says that even the full three-dimensional unimodular non-positive classification was not obtained; SNP is strictly stronger than requiring one fixed Weyl connection to be non-positive or negative.

Maciej Bocheński, Piotr Jastrzębski, and Aleksy Tralle, “Stretched Non-positive Weyl Connections on Solvable Lie Groups,” *Annali di Matematica Pura ed Applicata* 203 (2024), 1463–1481, <https://doi.org/10.1007/s10231-023-01409-9>, classifies invariant SNP connections for four-dimensional unimodular solvable groups and gives broader structural results for solvable groups. The authors expressly distinguish this from the harder problem of ordinary non-positive invariant Weyl curvature and do not resolve the present strict-negativity conjecture.

The latest direct follow-up found is Tomasz Dudek and Aleksy Tralle, “On Non-positive Weyl Connections on Lie Groups,” preprint (2026), [arXiv:2607.17210](https://arxiv.org/abs/2607.17210). It claims the conjectured SOL-only result for three-dimensional unimodular completely solvable groups, but its formal Theorem 1 is stated for the paper’s SNP condition rather than for the exact fixed-connection strict-negativity assertion here. In any event, it does not cover the remaining three-dimensional unimodular groups or any general higher-dimensional class and therefore cannot resolve the present problem.

The dimension-two case is elementary: a two-dimensional unimodular Lie algebra is abelian, so every left-invariant metric is flat and every left-invariant field has zero divergence; consequently the sole Weyl sectional curvature is zero. What remains open is the asserted impossibility for arbitrary unimodular Lie groups in every dimension at least three, without the SNP, almost-abelian, complete-solvability, or other additional restrictions above.

1.33.3. Remaining open problems recorded by the database

1.33.3.1. Open problem 1

Let G be a three-dimensional unimodular Lie group, let g be a nonflat left-invariant Riemannian metric on G , and let E be a left-invariant vector field. Define the Weyl connection

$$\widehat{\nabla}_X Y = \nabla_X Y + g(E, Y)X + g(E, X)Y - g(X, Y)E,$$

where ∇ is the Levi-Civita connection of g . For a tangent two-plane Π with a g -orthonormal basis X, Y , let $E_\perp$ be the component of E orthogonal to Π , and define

$$\widehat{K}(\Pi) = K_g(\Pi) - \|E_\perp\|_g^2 - g(\nabla_X E, X) - g(\nabla_Y E, Y).$$

Prove that if $\widehat{K}(\Pi) \leq 0$ for every tangent two-plane Π , then the Lie algebra of G is isomorphic to $\mathfrak{so}\mathfrak{l}$, the Lie algebra with a basis U, V, W satisfying $[U, V] = 0$, $[W, U] = U$, and $[W, V] = -V$.

Reference supplied by the database. Gabriela Tereszkieicz and Maciej P. Wojtkowski, “Homogeneous Weyl Connections of Non-positive Curvature,” *Annals of Global Analysis and Geometry* 51 (2017), 209–229, three-dimensional conjecture stated in Section 4, <https://doi.org/10.1007/s10455-016-9526-0>.

1.33.4. Primary-source pointer

The originating ICM record is *Weyl Manifolds and Gaussian Thermostats* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=498>.

1.34. 34. Complex Hyperbolic Triangle Groups [2002-geom-277]

Record: 2002-geom-277

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Complex Hyperbolic Triangle Groups

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.2/ICM2002.2.occr.pdf#page=327>

1.34.1. Problem statement

Let $p \leq q \leq r$ belong to $\{2, 3, \dots\} \cup \{\infty\}$ and satisfy

$$\frac{1}{p} + \frac{1}{q} + \frac{1}{r} < 1,$$

where $1/\infty = 0$. Let

$$\Gamma(p, q, r) = \langle I_p, I_q, I_r \mid I_p^2 = I_q^2 = I_r^2 = (I_q I_r)^p = (I_r I_p)^q = (I_p I_q)^r = 1 \rangle,$$

with any relation whose exponent is ∞ omitted. Consider the canonical real one-parameter family of representations

$$\rho_t : \Gamma(p, q, r) \longrightarrow \mathrm{PU}(2, 1)$$

in which I_p, I_q, I_r are mapped to order-two complex reflections in the three complex-geodesic sides of a complex-hyperbolic (p, q, r) triangle, parametrized and oriented so that $\rho_0(\Gamma(p, q, r))$ preserves a totally real, totally geodesic copy of the real hyperbolic plane in the complex hyperbolic plane $\mathbf{H}_{\mathbb{C}}^2$.

Define

$$W_A = I_p I_r I_q I_r, \quad W_B = I_p I_q I_r,$$

and let I be the maximal closed interval of parameters containing 0 for which neither $\rho_t(W_A)$ nor $\rho_t(W_B)$ is elliptic. Let t_+ be the endpoint of I reached by increasing the parameter from 0. For $W \in \Gamma(p, q, r)$, write $W_t = \rho_t(W)$ and define its translation length by

$$\lambda(W_t) = \inf_{x \in \mathbf{H}_{\mathbb{C}}^2} d(x, W_t x),$$

where d is the complex-hyperbolic distance.

Prove that, for every element $W \in \Gamma(p, q, r)$ of infinite order and all $0 \leq s \leq t \leq t_+$,

$$\lambda(W_t) \leq \lambda(W_s).$$

1.34.2. Historical background, literature, and current progress

The problem is **unresolved** as of 3 August 2026. No citable proof or counterexample was found for the stated monotonicity of the translation length of every infinite-order element, uniformly

for every admissible (p, q, r) .

Richard Evan Schwartz first stated the assertion as Conjecture 5.4 in *Complex Hyperbolic Triangle Groups*, Proceedings of the International Congress of Mathematicians, Beijing 2002, Vol. II, Higher Education Press, 2002, pp. 339-350 ([arXiv:math/0304268](https://arxiv.org/abs/math/0304268)). Pierre Will's later survey *Two-generator groups acting on the complex hyperbolic plane*, in *Handbook of Teichmüller Theory*, Vol. VI, IRMA Lectures in Mathematics and Theoretical Physics 27, European Mathematical Society, 2016, pp. 275-334 (<https://doi.org/10.4171/161-1/10>), reproduces the exact translation-length claim as Conjecture 4 rather than reporting a resolution.

The strongest direct progress is Schwartz's *Monotonicity in Complex Hyperbolic Triangle Groups*, unpublished preprint, 2025 ([direct PDF](#)). It formulates a related "First Monotonicity Conjecture" asserting monotonic decrease of the Goldman discriminant of every loxodromic word. Schwartz says that this probably implies translation-distance monotonicity but explicitly notes that this implication was not checked. The paper reports non-rigorous numerical support (modulo roundoff error) for 219 finite triples, 49,140 words of length at most 14, and a 32-subinterval parameter mesh. For a finite triple with at least one even entry, it also describes infinitely many regular trees in totally real slices and establishes decreasing distances between corresponding vertices in each such tree; it sketches the analogous discriminant claim only for words preserving one of these trees. These results neither prove the stated translation-length inequality nor cover arbitrary words, triples with no even entry, parameters continuously rather than on a mesh, cases with an entry 2, or cases with an entry ∞ .

Anna Pratoussevitich, *Traces in Complex Hyperbolic Triangle Groups*, *Geometriae Dedicata* 111 (2005), 159-185 (<https://doi.org/10.1007/s10711-004-1493-0>; [arXiv:math/0402153](https://arxiv.org/abs/math/0402153)), gives a combinatorial Laurent-polynomial formula for the trace of an arbitrary word as a function of the angular invariant and derives some discreteness and non-discreteness consequences. This supplies relevant algebraic machinery but no sign theorem strong enough to establish the required monotonicity. Thus the universal quantifier over all infinite-order W and all admissible signatures remains open.

1.34.3. Remaining open problems recorded by the database

1.34.3.1. Open problem 1

Let $3 \leq n_1 \leq n_2 \leq n_3$ be integers satisfying

$$\frac{1}{n_1} + \frac{1}{n_2} + \frac{1}{n_3} < 1,$$

and let

$$\Gamma(n_1, n_2, n_3) = \langle I_1, I_2, I_3 \mid I_1^2 = I_2^2 = I_3^2 = (I_2 I_3)^{n_1} = (I_3 I_1)^{n_2} = (I_1 I_2)^{n_3} = 1 \rangle.$$

Consider the positive half of the canonical one-parameter deformation

$$\rho_t : \Gamma(n_1, n_2, n_3) \longrightarrow \mathrm{PU}(2, 1), \quad 0 \leq t \leq 1,$$

by order-two complex reflections in the sides of a complex-hyperbolic (n_1, n_2, n_3) triangle. Parametrize this family monotonically so that ρ_0 preserves a totally real copy of the real hyperbolic plane and $[0, 1]$ is the maximal interval for which every element of the form $\rho_t(I_a I_b I_c)$ or $\rho_t(I_a I_b I_a I_c)$, with a, b, c pairwise distinct, is loxodromic for $0 \leq t < 1$.

For $g \in \mathrm{PU}(2, 1)$, choose any lift $\tilde{g} \in \mathrm{SU}(2, 1)$, set $z = \mathrm{tr}(\tilde{g})$, and define the Goldman discriminant

$$\Delta(g) = |z|^4 - 8 \operatorname{Re}(z^3) + 18|z|^2 - 27.$$

This definition is independent of the chosen lift. For $W \in \Gamma(n_1, n_2, n_3)$, write $W_t = \rho_t(W)$.

Prove that, for every W for which W_0 is loxodromic, the function

$$t \mapsto \Delta(W_t)$$

is nonincreasing on $[0, 1]$.

Reference supplied by the database. Richard Evan Schwartz, “Monotonicity in Complex Hyperbolic Triangle Groups,” unpublished preprint, 16 October 2025, First Monotonicity Conjecture, https://www.math.brown.edu/reschwar/Papers/triangle_mono.pdf.

1.34.4. Primary-source pointer

The originating ICM record is *Complex Hyperbolic Triangle Groups* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.2/ICM2002.2.ocr.pdf#page=327>.

1.35. 35. Geometry and Nonlinear Analysis [2002-geom-280]

Record: 2002-geom-280

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Geometry and Nonlinear Analysis

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.1/ICM2002.1.ocf.pdf#page=474>

1.35.1. Problem statement

Prove that, up to an overall constant rescaling of the Riemannian metric, every connected complete quaternion-Kähler manifold of positive scalar curvature is isometric to a Wolf space, that is, to a compact irreducible Riemannian symmetric space equipped with its standard quaternion-Kähler metric.

1.35.2. Historical background, literature, and current progress

The problem is **unresolved**. It is the LeBrun-Salamon conjecture. No proof or counterexample is known in arbitrary dimension; in particular, real dimensions 20 and higher are not covered without additional hypotheses.

Established progress:

- Claude LeBrun and Simon Salamon, “Strong rigidity of positive quaternion-Kähler manifolds,” *Inventiones Mathematicae* **118** (1994), 109–132, <https://doi.org/10.1007/BF01231528>. Their strong-rigidity results reduce any possible nonsymmetric example to the exceptional topological case in which $\pi_2(M)$ is finite and contains nontrivial 2-torsion. They do not eliminate that case. Closely related finiteness is proved in Claude LeBrun, “A Finiteness Theorem for Quaternionic-Kähler Manifolds with Positive Scalar Curvature,” *Contemporary Mathematics* **154** (1993), 89–101, [arXiv:alg-geom/9302005](https://arxiv.org/abs/alg-geom/9302005): in each fixed dimension there are only finitely many such manifolds up to homothety.
- Nigel J. Hitchin, “Kählerian Twistor Spaces,” *Proceedings of the London Mathematical Society* (3) **43** (1981), no. 1, 133–150, <https://doi.org/10.1112/plms/s3-43.1.133>, proves the conjecture in real dimension 4. Yat-Sun Poon and Simon M. Salamon, “Eight-dimensional quaternionic Kähler manifolds with positive scalar curvature,” *Journal of Differential Geometry* **33** (1991), 363–378, <https://doi.org/10.4310/jdg/1214446322>, prove it in real dimension 8.
- Jarosław Buczyński and Jarosław A. Wiśniewski, with an appendix by Andrzej Weber, “Algebraic torus actions on contact manifolds,” *Journal of Differential Geometry* **121** (2022), no. 2, 227–289, <https://doi.org/10.4310/jdg/1659987892>, [arXiv:1802.05002](https://arxiv.org/abs/1802.05002). Theorem 1.1 proves that every positive quaternion-Kähler manifold of real dimension 12 or 16 is a Wolf space. Together with the preceding results, this settles every real dimension at most 16, but no higher dimension unconditionally.
- Manuel Amann, “Partial Classification Results for Positive Quaternion Kähler Manifolds,” *International Journal of Mathematics* **23** (2012), no. 2, 1250038, <https://doi.org/10.1155/2012/1250038>.

[//doi.org/10.1142/S0129167X12500383](https://doi.org/10.1142/S0129167X12500383), [arXiv:0911.4587](https://arxiv.org/abs/0911.4587). In real dimension 20, Theorem A proves the conjecture only under the extra assumptions $\widehat{A}(M) = 0$ and $\dim \text{Isom}(M) \notin \{15, 22, 29\}$; these restrictions leave the full 20-dimensional case open.

- Gianluca Occhetta, Eleonora A. Romano, Luis E. Solá Conde, and Jarosław A. Wiśniewski, “High rank torus actions on contact manifolds,” *Selecta Mathematica (N.S.)* **27** (2021), article 10, <https://doi.org/10.1007/s00029-021-00621-w>, [arXiv:2004.05971](https://arxiv.org/abs/2004.05971). Via twistor spaces, their theorem proves the conjecture when the reductive contact-automorphism group has rank at least $\max\{2, (n-3)/2\}$ for a twistor space of complex dimension $2n+1$; it does not cover low-symmetry cases.
- Peter Petersen and Matthias Wink, “Tachibana-type theorems and special holonomy,” *Annals of Global Analysis and Geometry* **61** (2022), 847–868, <https://doi.org/10.1007/s10455-022-09837-1>. Theorem C and its corollary prove symmetry under an additional nonnegativity condition on the quaternion-Kähler curvature operator.
- Simon Brendle and Uwe Semmelmann, “Quaternionic-Kähler manifolds with nonnegative sectional curvature,” preprint (2025), [arXiv:2506.22181](https://arxiv.org/abs/2506.22181). They prove that a compact positive quaternion-Kähler manifold with nonnegative sectional curvature is symmetric. This resolves that curvature-restricted special case, but positive scalar curvature alone does not imply their sectional-curvature hypothesis.

Thus the known results neither produce a nonsymmetric manifold nor prove that every connected complete positive quaternion-Kähler manifold in the remaining higher-dimensional, low-symmetry, unrestricted-curvature cases is a Wolf space.

1.35.3. Remaining open problems recorded by the database

1.35.3.1. Open problem 1

Prove that, up to an overall constant rescaling of the Riemannian metric, every connected complete 20-dimensional quaternion-Kähler manifold of positive scalar curvature is isometric to a Wolf space, that is, to a compact irreducible Riemannian symmetric space equipped with its standard quaternion-Kähler metric.

Reference supplied by the database. Manuel Amann, “Partial Classification Results for Positive Quaternion Kähler Manifolds,” *International Journal of Mathematics* **23** (2012), no. 2, article 1250038, [doi:10.1142/S0129167X12500383](https://doi.org/10.1142/S0129167X12500383), [arXiv:0911.4587](https://arxiv.org/abs/0911.4587).

1.35.3.2. Open problem 2

Prove that the $\widehat{A}$ -genus of every connected complete quaternion-Kähler manifold M of positive scalar curvature whose second homotopy group $\pi_2(M)$ is finite is zero.

Reference supplied by the database. Manuel Amann, “Partial Classification Results for Positive Quaternion Kähler Manifolds,” *International Journal of Mathematics* **23** (2012), no. 2, article 1250038, [doi:10.1142/S0129167X12500383](https://doi.org/10.1142/S0129167X12500383), [arXiv:0911.4587](https://arxiv.org/abs/0911.4587).

1.35.4. Primary-source pointer

The originating ICM record is *Geometry and Nonlinear Analysis* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.1/ICM2002.1.ocr.pdf#page=474>.

1.36. 36. Weyl Manifolds and Gaussian Thermostats [2002-geom-311]

Record: 2002-geom-311

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Weyl Manifolds and Gaussian Thermostats

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=498>

1.36.1. Problem statement

Let M be a simply connected manifold equipped with a homogeneous Weyl structure $([g], \hat{\nabla})$. Thus $[g]$ is a conformal class of Riemannian metrics, $\hat{\nabla}$ is a torsion-free affine connection such that, for every $g \in [g]$, there exists a one-form φ satisfying

$$\hat{\nabla}g = -2\varphi \otimes g,$$

and a Lie group acts transitively on M while preserving both $[g]$ and $\hat{\nabla}$. Assume that a discrete group of automorphisms of this Weyl structure acts freely and properly discontinuously on M , with compact quotient.

Fix $g \in [g]$, let E be the vector field g -dual to φ , and, for every two-plane $\Pi \subset T_x M$ with a g -orthonormal basis X, Y , define

$$\tilde{K}_g(\Pi) = K_g(\Pi) - \|E_{\Pi^\perp}\|_g^2 - (g(\nabla_X E, X) + g(\nabla_Y E, Y)),$$

where $K_g(\Pi)$ is the sectional curvature of g , ∇ is the Levi-Civita connection of g , and $E_{\Pi^\perp}$ is the component of E orthogonal to Π . Suppose that $\tilde{K}_g(\Pi) < 0$ for every $x \in M$ and every two-plane $\Pi \subset T_x M$; the sign of this quantity is independent of the choice of $g \in [g]$.

Prove that $\hat{\nabla}$ is the Levi-Civita connection of a homogeneous Riemannian metric in $[g]$, and that M , equipped with this metric, is a Riemannian symmetric space.

1.36.2. Historical background, literature, and current progress

The exact assertion remains **unresolved**. It is Conjecture 5.4 in the original source, and no later proof or counterexample covering all homogeneous Weyl structures with a compact quotient and strictly negative Weyl sectional curvature was found. The Riemannian case is classical, but the available Weyl-geometric rigidity results impose additional hypotheses (a fixed conformal model, stretched non-positivity, a Lie-group model, solvability, or low dimension) and do not imply the statement as written.

Maciej P. Wojtkowski, "Weyl Manifolds and Gaussian Thermostats," *Proceedings of the International Congress of Mathematicians, Beijing 2002*, Vol. III, Higher Education Press, 2002, pp. 511–523, Conjecture 5.4, [arXiv:math/0304461](https://arxiv.org/abs/math/0304461). This is the proposing source. It records that the analogous Riemannian compact-factor result is known and asks whether every negatively curved

homogeneous Weyl structure with a compact factor must likewise be Riemannian symmetric.

Maciej P. Wojtkowski, “Rigidity of Some Weyl Manifolds with Nonpositive Sectional Curvature,” *Proceedings of the American Mathematical Society* 133 (2005), 3395–3402, <https://doi.org/10.1090/S0002-9939-05-07809-3>. This paper classifies nonpositively curved Weyl connections for the standard conformal structures on tori and on $S^1 \times S^{n-1}$, completely in the latter case and under integrability restrictions in the torus case. These are fixed special models and do not exhaust the homogeneous compact-factor setting of the conjecture.

Gabriela Tereszkievicz and Maciej P. Wojtkowski, “Homogeneous Weyl Connections of Nonpositive Curvature,” *Annals of Global Analysis and Geometry* 51 (2017), 209–229, <https://doi.org/10.1007/s10455-016-9526-0>, [arXiv:1506.08176](https://arxiv.org/abs/1506.08176). Its main rigidity theorem proves that a homogeneous **stretched non-positive** Weyl field on a homogeneous space modeled on a unimodular Lie group is parallel. Stretched non-positivity requires non-positivity for every sufficiently large rescaling of the Weyl field and is not a consequence of strict negativity for the single connection in the present problem, so this theorem does not resolve the conjecture.

Maciej Bocheński, Piotr Jastrzębski, and Aleksy Tralle, “Stretched Non-positive Weyl Connections on Solvable Lie Groups,” *Annali di Matematica Pura ed Applicata* 203 (2024), 1463–1481, <https://doi.org/10.1007/s10231-023-01409-9>, [arXiv:2207.00318](https://arxiv.org/abs/2207.00318). The authors classify invariant stretched-non-positive connections on four-dimensional unimodular solvable groups and give a structural classification in the solvable setting. They explicitly distinguish this from the harder problem of invariant non-positive Weyl connections and state that no examples of negatively curved Weyl connections on compact manifolds of dimension at least three are known. Thus their results are substantive partial progress but leave the exact strict-negative homogeneous compact-factor problem open.

Tomasz Dudek and Aleksy Tralle, “On Non-positive Weyl Connections on Lie Groups,” preprint, 2026, [arXiv:2607.17210](https://arxiv.org/abs/2607.17210). This very recent work treats left-invariant connections on three-dimensional completely solvable unimodular Lie groups and confirms the relevant three-dimensional classification only in that restricted class. It continues to present the general homogeneous problem as open and does not address arbitrary homogeneous spaces with compact quotient.

What remains is precisely the missing general rigidity step: from homogeneity, existence of a compact quotient, and strict negativity for one Weyl connection, prove exactness of the Weyl connection and then symmetry of the resulting homogeneous Riemannian metric. None of the cited additional-hypothesis results supplies that step in full generality.

1.36.3. Remaining open problems recorded by the database

1.36.3.1. Open problem 1

Let G be a unimodular Lie group of dimension at least two, let g be a left-invariant Riemannian metric on G , and let E be a left-invariant vector field. Write ∇ for the Levi-Civita connection of g , and define the left-invariant Weyl connection

$$\widehat{\nabla}_X Y = \nabla_X Y + g(E, Y)X + g(E, X)Y - g(X, Y)E.$$

For a two-plane $\Pi \subset T_x G$ with a g -orthonormal basis X, Y , define

$$\widehat{K}_g(\Pi) = K_g(\Pi) - \|E_{\Pi^\perp}\|_g^2 - (g(\nabla_X E, X) + g(\nabla_Y E, Y)).$$

Prove that no such left-invariant Weyl structure has $\widehat{K}_g(\Pi) < 0$ for every $x \in G$ and every two-plane $\Pi \subset T_x G$.

Reference supplied by the database. Maciej P. Wojtkowski, “Weyl Manifolds and Gaussian Thermostats,” *Proceedings of the International Congress of Mathematicians, Beijing 2002*, Vol. III, Higher Education Press, 2002, pp. 511–523, Conjecture 5.5. arXiv:math/0304461, <https://arxiv.org/abs/math/0304461>.

1.36.3.2. Open problem 2

Let G be a connected three-dimensional unimodular Lie group, let g be a left-invariant Riemannian metric on G , and let E be a left-invariant vector field. Define

$$\widehat{\nabla}_X Y = \nabla_X Y + g(E, Y)X + g(E, X)Y - g(X, Y)E,$$

where ∇ is the Levi-Civita connection of g , and define the Weyl sectional curvature of a two-plane Π , with g -orthonormal basis X, Y , by

$$\widehat{K}_g(\Pi) = K_g(\Pi) - \|E_{\Pi^\perp}\|_g^2 - (g(\nabla_X E, X) + g(\nabla_Y E, Y)).$$

Prove that if $\widehat{K}_g(\Pi) \leq 0$ for every two-plane Π , then either g is flat and E is parallel, or G is isomorphic to the three-dimensional solvable Lie group

$$\text{Sol} = \mathbb{R} \ltimes \mathbb{R}^2, \quad t \cdot (x, y) = (e^t x, e^{-t} y).$$

Reference supplied by the database. Gabriela Tereszkievicz and Maciej P. Wojtkowski, “Homogeneous Weyl Connections of Non-positive Curvature,” *Annals of Global Analysis and Geometry* 51 (2017), 209–229, unnumbered conjecture in Section 4. DOI: 10.1007/s10455-016-9526-0; arXiv:1506.08176, <https://arxiv.org/abs/1506.08176>.

1.36.4. Primary-source pointer

The originating ICM record is *Weyl Manifolds and Gaussian Thermostats* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=498>.

1.37. 37. Index Iteration Theory for Symplectic Paths with Applications to Nonlinear Hamiltonian... [2002-geom-312]

Record: 2002-geom-312

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Index Iteration Theory for Symplectic Paths with Applications to Nonlinear Hamiltonian Systems

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.2/ICM2002.2.ocr.pdf#page=293>

1.37.1. Problem statement

Prove or disprove that every compact Riemannian manifold of dimension at least two admits infinitely many nonconstant closed geodesics that are geometrically distinct, meaning that no two have the same image in the manifold.

1.37.2. Historical background, literature, and current progress

Verdict: unresolved.

The exact assertion is the closed-geodesics conjecture for arbitrary Riemannian metrics. It is not known even for arbitrary Riemannian metrics on S^n when ($n \geq 3$). More generally, Contreras and Mazzucchelli explicitly state that the conjecture remains open in dimension at least three for closed manifolds having the rational cohomology of a compact rank-one symmetric space S^n , $\mathbb{C}P^n$, $\mathbb{H}P^n$, or $\text{Ca}P^2$. Thus the known theorems do not cover every manifold and every metric in the statement, and no counterexample is known.

Established partial progress includes the following.

- Detlef Gromoll and Wolfgang Meyer, "Periodic Geodesics on Compact Riemannian Manifolds," *Journal of Differential Geometry* **3** (1969), 493–510, <https://doi.org/10.4310/jdg/1214429070>, prove that unbounded Betti numbers of the free loop space force infinitely many geometrically distinct closed geodesics. Micheline Vigué-Poirrier and Dennis Sullivan, "The Homology Theory of the Closed Geodesic Problem," *Journal of Differential Geometry* **11** (1976), 633–644, <https://doi.org/10.4310/jdg/1214433729>, show in the simply connected finite-complex setting that these rational Betti numbers are unbounded precisely when the rational cohomology algebra is not generated by one element. Together these results prove the conjecture for every simply connected closed manifold outside the one-generator rational-cohomology class.
- Victor Bangert and Nancy Hingston, "Closed Geodesics on Manifolds with Infinite Abelian Fundamental Group," *Journal of Differential Geometry* **19** (1984), 277–282, <https://doi.org/10.4310/jdg/1214438679>, prove the full all-metrics assertion when the fundamental group is infinite abelian, including the difficult infinite-cyclic case.
- Victor Bangert, "On the Existence of Closed Geodesics on Two-Spheres," *International*

Journal of Mathematics **4** (1993), 1–10, <https://doi.org/10.1142/S0129167X93000029>, combined with John Franks, “Geodesics on S^2 and Periodic Points of Annulus Homeomorphisms,” *Inventiones Mathematicae* **108** (1992), 403–418, <https://doi.org/10.1007/BF02100612>, proves the assertion for every Riemannian (2)-sphere. Christian Lange, “On the Existence of Closed Geodesics on (2)-Orbifolds,” *Pacific Journal of Mathematics* **294** (2018), 453–472, <https://doi.org/10.2140/pjm.2018.294.453>, proves the stronger statement for every compact Riemannian (2)-orbifold and therefore confirms the conjecture for all closed surfaces.

- Hans-Bert Rademacher and Iskander A. Taimanov, “Closed Geodesics on Connected Sums and (3)-Manifolds,” *Journal of Differential Geometry* **120** (2022), 557–573, <https://doi.org/10.4310/jdg/1649953350>, prove that every Riemannian metric on a compact (3)-manifold with infinite fundamental group has infinitely many geometrically distinct closed geodesics. This still leaves finite-fundamental-group three-manifolds such as S^3 outside the unconditional result.
- Gonzalo Contreras and Marco Mazzucchelli, “Closed Geodesics and the First Betti Number,” *Proceedings of the London Mathematical Society* (3) **131** (2025), no. 3, Paper No. e70085, <https://doi.org/10.1112/plms.70085>, prove that on every closed manifold of dimension at least two with nonzero first Betti number, an open dense set of C^k Riemannian metrics ($2 \leq k \leq \infty$) has infinitely many closed geodesics of arbitrarily large length. Their introduction also records the higher-dimensional compact-rank-one-symmetric-space cohomology cases as still open. This generic-metric theorem does not settle the assertion for every metric.

The remaining gap is therefore genuine: arbitrary metrics on the higher-dimensional one-generator rational-cohomology cases are not covered. In particular, proving the statement for all Riemannian S^n , ($n \geq 3$), would already be new, while a single metric in this range with only finitely many geometric closed-geodesic images would disprove the original assertion.

1.37.3. Remaining open problems recorded by the database

1.37.3.1. Open problem 1

Prove or disprove that every compact Riemannian manifold M without boundary, of dimension at least three, whose rational cohomology ring $H^*(M; \mathbb{Q})$ is isomorphic as a graded algebra to the rational cohomology ring of one of the compact rank-one symmetric spaces S^k for some integer $k \geq 3$, $\mathbb{C}P^k$ for some integer $k \geq 2$, $\mathbb{H}P^k$ for some integer $k \geq 1$, or $\text{Ca}P^2$, admits infinitely many nonconstant closed geodesics with pairwise distinct images in M .

Reference supplied by the database. Gonzalo Contreras and Marco Mazzucchelli, “Closed Geodesics and the First Betti Number,” *Proceedings of the London Mathematical Society* (3) **131** (2025), no. 3, Paper No. e70085, Section 1.1 (“Background”), <https://doi.org/10.1112/plms.70085>.

1.37.4. Primary-source pointer

The originating ICM record is *Index Iteration Theory for Symplectic Paths with Applications to Nonlinear Hamiltonian Systems* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.2/ICM2002.2.ocr.pdf#page=293>.

1.38. 38. Collapsed Riemannian Manifolds with Bounded Sectional Curvature [2002-geom-313]

Record: 2002-geom-313

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Collapsed Riemannian Manifolds with Bounded Sectional Curvature

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.2/ICM2002.2.ocr.pdf#page=311>

1.38.1. Problem statement

Let $n \geq 2$, let $0 < \delta \leq 1$, and let M^n be a closed smooth manifold that admits a Riemannian metric whose sectional curvatures lie in $[\delta, 1]$. Prove that there exists a constant $\varepsilon > 0$, depending only on δ and the homotopy type of M , such that every Riemannian metric g on M satisfying

$$\delta \leq \sec_g \leq 1$$

has injectivity radius $\text{inj}(M, g) \geq \varepsilon$. Equivalently, prove that the infimum of $\text{inj}(M, g)$ over all such δ -pinched metrics g on M is positive.

1.38.2. Historical background, literature, and current progress

The statement is **unresolved**. It is the Klingenberg-Sakai-Yau injectivity-radius conjecture. The general even-dimensional case follows from Klingenberg's injectivity-radius estimate, and the three-dimensional case is known, but no result found settles all odd dimensions greater than three. In particular, the available noncollapsing theorems in higher odd dimensions impose additional stability or topological hypotheses, so they do not prove the statement for every closed manifold and every δ -pinched metric on it.

Key literature and established progress:

- Wilhelm Klingenberg, "Contributions to Riemannian Geometry in the Large," *Annals of Mathematics* 69 (1959), 654-666, <https://doi.org/10.2307/1970029>. Klingenberg's estimate gives $\text{inj}(M, g) \geq \pi$ for a compact simply connected even-dimensional manifold with $0 < \sec_g \leq 1$; together with the standard covering argument, this accounts for the even-dimensional part of the conjecture.
- Yu. D. Burago and V. A. Toponogov, "On Three-Dimensional Riemannian Spaces with Curvature Bounded Above," *Mathematical Notes* 13 (1973), 526-530, <https://doi.org/10.1007/BF01163962>. They obtain a lower injectivity-radius bound for closed simply connected three-manifolds from the relevant lower sectional-curvature and upper Ricci-curvature bounds, covering the simply connected three-dimensional case; passage to a fixed finite quotient gives the bound for a fixed homotopy type.
- Anton Petrunin, Xiaochun Rong, and Wilderich Tuschmann, "Collapsing vs. Positive Pinching," *Geometric and Functional Analysis* 9 (1999), 699-735, <https://doi.org/>

[10.1007/s000390050100](https://doi.org/10.1007/s000390050100). Theorem 0.3 proves the conjecture for stable collapsing sequences (equivalently, for metrics whose distance functions have a fixed upper bound), and Theorem 0.4 rules out uniformly positively pinched collapse along a continuous one-parameter family. Neither hypothesis is automatic for an arbitrary sequence in the statement.

- Anton Petrunin and Wilderich Tuschmann, “Diffeomorphism Finiteness, Positive Pinching, and Second Homotopy,” *Geometric and Functional Analysis* 9 (1999), 736-774, <https://doi.org/10.1007/s000390050101>. Theorem 0.3 proves a uniform lower injectivity-radius bound in every dimension for simply connected manifolds with finite π_2 , even under the weaker bounds $\text{Ric} \geq \delta > 0$ and $\text{sec} \leq 1$. It leaves the cases with infinite π_2 open.
- Fuquan Fang and Xiaochun Rong, “The Second Twisted Betti Number and the Convergence of Collapsing Riemannian Manifolds,” *Inventiones Mathematicae* 150 (2002), 61-109, <https://doi.org/10.1007/s00222-002-0230-2>. Corollary 0.2 proves the higher odd-dimensional assertion when the fundamental group has uniformly bounded exponent and the relevant second twisted Betti number vanishes. These are extra hypotheses absent from the statement.
- Wilderich Tuschmann, “Moduli Spaces of Metrics with Lower or Pinched Curvature Bounds and Their Compactifications,” *Journal of Mathematical Study* 58 (2025), 175-187, <https://doi.org/10.4208/jms.v58n2.25.04>. This recent survey continues to present only the bounded, stable, and continuous-collapse versions and discusses the original conjecture through the still-hypothetical existence of an arbitrary collapsing sequence of δ -pinched metrics.

Thus substantial special cases are known, but the unrestricted odd-dimensional case, and hence the exact all-dimensional statement, remains open.

1.38.3. Remaining open problems recorded by the database

1.38.3.1. Open problem 1

Let B and F be compact simply connected smooth manifolds, and let $T^k = (S^1)^k$. Suppose that there are infinitely many pairwise nonisomorphic principal T^k -bundles over B whose total spaces are all diffeomorphic to F . Let $T^r \rightarrow E \rightarrow F$ be the universal torus bundle of F , meaning the unique principal torus bundle whose total space E is simply connected and has finite second homotopy group $\pi_2(E)$. Prove that the image of the natural action of the diffeomorphism group $\text{Diff}(E)$ on the integral cohomology ring $H^*(E; \mathbb{Z})$ is infinite.

Reference supplied by the database. Anton Petrunin and Wilderich Tuschmann, “Diffeomorphism Finiteness, Positive Pinching, and Second Homotopy,” *Geometric and Functional Analysis* 9(4) (1999), 736-774, Conjecture 5.4, <https://doi.org/10.1007/s000390050101>.

1.38.4. Primary-source pointer

The originating ICM record is *Collapsed Riemannian Manifolds with Bounded Sectional Curvature* (2002). The source link retained by the dataset is <https://www.mathunion.org/fi>

[leadadmin/ICM/Proceedings/ICM2002.2/ICM2002.2.ocr.pdf#page=311.](#)

1.39. 39. Non Linear Elliptic Theory and the Monge-Ampere Equation [2002-geom-314]

Record: 2002-geom-314

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Non Linear Elliptic Theory and the Monge-Ampere Equation

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.1/ICM2002.1.ocr.pdf#page=182>

1.39.1. Problem statement

Let M^n be a smooth manifold of dimension $n > 4$, let g be a Riemannian metric on M , and let

$$A_g = \text{Ric}_g - \frac{R_g}{2(n-1)}g,$$

where Ric_g and R_g are the Ricci tensor and scalar curvature of g . For $1 \leq k \leq n$, let $\sigma_k(A_g)$ denote the k -th elementary symmetric polynomial of the eigenvalues of the endomorphism $g^{-1}A_g$, and let Γ_k^+ be the connected component containing the positive orthant of

$$\{\lambda \in \mathbb{R}^n : \sigma_k(\lambda) > 0\}.$$

For smooth conformal metrics $\tilde{g} = e^{2w}g$, determine the curvature, topological, and conformal-geometric consequences of the following conditions, separately and in combination:

$$\lambda(\tilde{g}^{-1}A_{\tilde{g}}(x)) \in \Gamma_k^+ \quad \text{for every } x \in M,$$

and

$$\sigma_k(A_{\tilde{g}}) = c$$

for a constant c , where $\lambda(\tilde{g}^{-1}A_{\tilde{g}})$ is the eigenvalue vector of $\tilde{g}^{-1}A_{\tilde{g}}$. In particular, characterize the restrictions these conditions impose on M and its conformal class and determine when such conformal metrics exist.

1.39.2. Historical background, literature, and current progress

The problem is **unresolved**. The literature has developed a substantial theory of positive Γ_k -curvature and the σ_k -Yamabe equation, but it does not give the requested curvature, topology, and conformal classification for arbitrary $n > 4$, k , and smooth manifolds. In particular, even after restricting to closed manifolds whose conformal class already contains a Γ_k^+ -admissible metric, existence of a positive constant- σ_k metric remains open in the non-locally-conformally-flat range $2 < k < n/2$. The unrestricted noncompact problem in the statement is still less complete. Most later papers normalize the Schouten tensor as $A/(n-2)$; this leaves Γ_k^+ unchanged and only rescales the constant in the equation.

Key partial results are as follows.

- **“Some properties of the Schouten tensor and applications to conformal geometry,”** Pengfei Guan, Jeff Viaclovsky, and Guofang Wang, *Transactions of the American Mathematical Society* 355 (2003), 925–933, DOI <https://doi.org/10.1090/S0002-9947-02-03132-X>. Theorem 1 proves pointwise that Γ_k^+ -admissibility for $k \geq n/2$ forces positive Ricci curvature, with an explicit Ricci lower bound in terms of σ_k . Corollary 1 classifies the compact locally conformally flat case with nonnegative Γ_k -curvature for $k \geq n/2$, and in the strictly positive case gives a spherical space form. These are strong curvature and topology consequences, but require the indicated range and, for the classification, local conformal flatness.
- **“Schouten tensor and some topological properties,”** Pengfei Guan, Chang-Shou Lin, and Guofang Wang, *Communications in Analysis and Geometry* 13 (2005), 887–902, DOI <https://doi.org/10.4310/CAG.2005.v13.n5.a2>. Theorem 1 gives ranges of vanishing Betti numbers for compact locally conformally flat Γ_k^+ -manifolds with $k < n/2$, while Theorem 2 proves preservation under connected sum. Thus it supplies important lower- k topological information, but not a classification and not analogous vanishing results without local conformal flatness.
- **“The Yamabe problem for higher order curvatures,”** Weimin Sheng, Neil S. Trudinger, and Xu-Jia Wang, *Journal of Differential Geometry* 77 (2007), 515–553, DOI <https://doi.org/10.4310/jdg/1193074903>. The main theorem solves the constant- σ_k problem on a compact manifold, assuming the conformal class is admissible and the equation is variational. This covers $k = 2$ in every dimension and all k in the locally conformally flat case. For $k \geq 3$, the variational assumption is precisely the major restriction in the general non-locally-conformally-flat setting.
- **“Prescribing symmetric functions of the eigenvalues of the Ricci tensor,”** Matthew J. Gursky and Jeff A. Viaclovsky, *Annals of Mathematics* 166 (2007), 475–531, DOI <https://doi.org/10.4007/annals.2007.166.475>. This proves existence and compactness for a broad class of prescribed fully nonlinear conformal curvature equations on closed admissible manifolds in the positive-Ricci regime, and in particular solves the admissible σ_k -Yamabe problem for $k > n/2$.
- **“A compactness theorem for a fully nonlinear Yamabe problem under a lower Ricci curvature bound,”** YanYan Li and Luc Nguyen, *Journal of Functional Analysis* 266 (2014), 3741–3771, DOI <https://doi.org/10.1016/j.jfa.2013.08.004>. Theorem 1.5 supplies existence (and, away from the round sphere, compactness) whenever the associated cone has $\mu_1^+ \leq 1$, including Γ_k^+ for $k \geq n/2$. The paper explicitly states that its general existence Problem 1.1 remains largely open for $2 < k < n/2$.
- **“Compact manifolds with positive Γ_2 -curvature,”** Boris Botvinnik and Mohammed Labbi, *Differential Geometry and its Applications* 37 (2014), 1–16, DOI <https://doi.org/10.1016/j.difgeo.2014.09.002>. A surgery theorem and Theorem C show that, for every $n \geq 6$, every finitely presented group occurs as the fundamental group of a closed n -manifold with positive Γ_2 -curvature. This demonstrates that low- k topology is much less rigid than the $k \geq n/2$ positive-Ricci regime and leaves further fundamental-group questions open.
- **“On the σ_2 -Nirenberg problem on S^2 ,”** YanYan Li, Han Lu, and Siyuan Lu, *Journal of Functional Analysis* 283 (2022), article 109606, DOI <https://doi.org/10.1016/j.jfa.2022.109606>. Its literature review records the present dividing line for closed manifolds: existence is proved for $k \geq n/2$, for $k = 2$, or under local conformal flatness, while existence remains open for $2 < k < n/2$ on general compact manifolds. This unresolved subcase is contained in the problem stated here, so the partial results

cannot collectively resolve it.

1.39.3. Remaining open problems recorded by the database

1.39.3.1. Open problem 1

Let n and k be integers satisfying $2 < k < n/2$, and let (M^n, g) be a smooth closed Riemannian manifold that is not locally conformally flat. Define

$$A_g = \text{Ric}_g - \frac{R_g}{2(n-1)}g,$$

and let Γ_k^+ be the connected component containing the positive orthant of $\{\lambda \in \mathbb{R}^n : \sigma_k(\lambda) > 0\}$, where σ_k is the k -th elementary symmetric polynomial. Assume that the eigenvalues of $g^{-1}A_g$ lie in Γ_k^+ at every point of M . Prove or disprove that there exist $w \in C^\infty(M)$ and a constant $c > 0$ such that $\tilde{g} = e^{2w}g$ satisfies

$$\lambda(\tilde{g}^{-1}A_{\tilde{g}}) \in \Gamma_k^+ \quad \text{and} \quad \sigma_k(A_{\tilde{g}}) = c$$

everywhere on M .

Reference supplied by the database. YanYan Li and Luc Nguyen, “A compactness theorem for a fully nonlinear Yamabe problem under a lower Ricci curvature bound,” *Journal of Functional Analysis* 266 (2014), 3741–3771, Problem 1.1 and the discussion following it. DOI: <https://doi.org/10.1016/j.jfa.2013.08.004>; arXiv:1212.0460.

1.39.3.2. Open problem 2

Determine whether every finitely presented group π can be realized as the fundamental group of a smooth closed five-dimensional manifold M admitting a Riemannian metric g with positive Γ_2 -curvature; that is, with

$$\sigma_1(\lambda(g^{-1}A_g)) > 0 \quad \text{and} \quad \sigma_2(\lambda(g^{-1}A_g)) > 0$$

at every point of M , where

$$A_g = \text{Ric}_g - \frac{R_g}{8}g$$

and σ_1, σ_2 are the first two elementary symmetric polynomials. If not every finitely presented group can be so realized, characterize the restrictions on $\pi_1(M)$.

Reference supplied by the database. Boris Botvinnik and Mohammed Labbi, “Compact manifolds with positive Γ_2 -curvature,” *Differential Geometry and its Applications* 37 (2014), 1–16, remark following Theorem C in Section 4.2. DOI: <https://doi.org/10.1016/j.difgeo.2014.09.002>; arXiv:1305.5313.

1.39.3.3. Open problem 3

Let (M^n, g_0) be a smooth closed Riemannian manifold of dimension $n > 4$, let $2 \leq k \leq n$, and define the Schouten tensor

$$S_{g_0} = \frac{1}{n-2} \left(\text{Ric}_{g_0} - \frac{R_{g_0}}{2(n-1)} g_0 \right).$$

Let Γ_k^+ be the connected component containing the positive orthant of $\{\lambda \in \mathbb{R}^n : \sigma_k(\lambda) > 0\}$. Suppose that $u \in C^0(M)$ is a viscosity solution of

$$\lambda \left(g_0^{-1} \left(S_{g_0} + \nabla_{g_0}^2 u + du \otimes du - \frac{1}{2} |du|_{g_0}^2 g_0 \right) \right) \in \partial \Gamma_k^+$$

on M . Determine whether every such solution belongs to $C^{1,1}(M)$, and determine whether any two such solutions on (M, g_0) differ by an additive constant.

Reference supplied by the database. Jonah A. J. Duncan and Luc Nguyen, “Differential inclusions for the Schouten tensor and nonlinear eigenvalue problems in conformal geometry,” *Advances in Mathematics* 432 (2023), article 109263, Remark 1.8. DOI: <https://doi.org/10.1016/j.aim.2023.109263>; arXiv:2208.00523.

1.39.4. Primary-source pointer

The originating ICM record is *Non Linear Elliptic Theory and the Monge-Ampere Equation* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.1/ICM2002.1.ocr.pdf#page=182>.

1.40. 40. Collapsed Riemannian Manifolds with Bounded Sectional Curvature [2002-geom-315]

Record: 2002-geom-315

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Collapsed Riemannian Manifolds with Bounded Sectional Curvature

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.2/ICM2002.2.ocr.pdf#page=311>

1.40.1. Problem statement

Let M^n be a closed smooth manifold admitting an F -structure $\mathcal{F} = \{(V_i, U_i, T^{k_i})\}$. Thus $\{U_i\}$ is a locally finite open cover of M ; for each i , $\pi_i : V_i \rightarrow U_i$ is a finite normal covering and the torus T^{k_i} acts effectively on V_i ; the action is compatible with the deck group in the sense that there is an action of the deck group on T^{k_i} by automorphisms for which $\gamma(t \cdot v) = \gamma(t) \cdot \gamma(v)$; and, whenever $U_i \cap U_j \neq \emptyset$, the restrictions of V_i and V_j over $U_i \cap U_j$ have a common finite covering on which the lifted T^{k_i} - and T^{k_j} -actions commute. An $\mathcal{F}$ -orbit is a minimal subset of M saturated under the projections of all these local torus actions, and the rank of $\mathcal{F}$ is the minimum of the dimensions of its $\mathcal{F}$ -orbits. Assume that $\mathcal{F}$ has positive rank.

Prove that every Riemannian metric g on M with nonpositive sectional curvature admits an abelian local splitting structure, in the following sense. Let $(\tilde{M}, \tilde{g})$ be the universal Riemannian covering of (M, g) , let $p : \tilde{M} \rightarrow M$ be the covering map, and let Γ be its deck-transformation group. There must exist a Γ -invariant open dense subset $\Omega \subseteq \tilde{M}$ and, for every $x \in \Omega$, an open neighborhood $W_x \subseteq \tilde{M}$, a Riemannian manifold N_x , an integer $r_x \geq 1$, and a specified isometry

$$W_x \cong N_x \times \mathbb{R}^{r_x},$$

such that the assignment is Γ -equivariant: the splitting assigned at γx is the translate by γ of the splitting assigned at x for every $\gamma \in \Gamma$. The connected Euclidean leaves determined by the $\mathbb{R}^{r_x}$ -factors must have the additional properties that the image under p of every such leaf is a closed embedded flat submanifold of M , and, whenever the images of two such leaves intersect, one of those images is contained in the other.

1.40.2. Historical background, literature, and current progress

The problem is **unresolved**. The literature proves that every nonpositively curved metric on such a manifold has a local splitting structure, but it does not prove in full generality that the structure is abelian, namely that the relevant flats satisfy the required commutation/nesting property. No counterexample was found.

Jianguo Cao, Jeff Cheeger, and Xiaochun Rong, "Splittings and Cr -structures for manifolds with nonpositive sectional curvature," *Inventiones Mathematicae* **144** (2001), 139–167, DOI: [ht](https://doi.org/10.1007/s002220100051)

[tps://doi.org/10.1007/PL00005800](https://doi.org/10.1007/PL00005800). This proves an abelian local splitting structure when the particular normalized nonpositively curved metric is uniformly sufficiently thin (its injectivity radius is below a dimension-dependent constant everywhere), and constructs a compatible Cr -structure. The hypothesis that the smooth manifold admits some positive-rank F -structure does not force an arbitrary nonpositively curved metric on it to satisfy this quantitative thinness condition.

Jianguo Cao, Jeff Cheeger, and Xiaochun Rong, “Local splitting structures on nonpositively curved manifolds and semirigidity in dimension 3,” *Communications in Analysis and Geometry* **12** (2004), 389–415, DOI: <https://doi.org/10.4310/CAG.2004.v12.n1.a17>. Theorem A gives a local splitting structure for every nonpositively curved metric whenever an F -structured closed manifold maps to the target with nonzero degree; taking the identity map gives the nonabelian part of the present assertion. The paper proves abelianness when the F -structure is pure (Corollary 4.5), under the injectivity and orbit-dimension assumptions of Theorem B, and without those extra assumptions in dimension three (Theorem C). Its Conjecture 4.2 leaves the unrestricted higher-dimensional case open.

Xiaochun Rong, “Collapsed Riemannian Manifolds with Bounded Sectional Curvature,” *Proceedings of the International Congress of Mathematicians, Beijing 2002*, Vol. II, 323–338 (2002; arXiv version 2003), arXiv:math/0304267, <https://arxiv.org/abs/math/0304267>. Section 2 states the exact positive-rank F -structure assertion as a conjecture and records verification in dimension three and additional special cases.

Xiaochun Rong, “Collapsed Manifolds with Bounded Sectional Curvature and Applications,” *Surveys in Differential Geometry* **11** (2006), 1–24, DOI: <https://doi.org/10.4310/SDG.2006.v11.n1.a2>. Theorem 6.6 again records the general local-splitting result and explicitly identifies the remaining problem as proving the abelian intersection/commutation condition (6.2).

Thus the strongest established result covers the existence of nontrivial Euclidean local factors for every metric, and abelianness for pure structures, the special injective high-orbit-rank class, dimension three, or a quantitatively collapsed nonpositively curved metric. What remains unresolved is abelianness for an arbitrary positive-rank, possibly mixed F -structure in unrestricted dimension and for every nonpositively curved metric on the underlying manifold.

1.40.3. Remaining open problems recorded by the database

1.40.3.1. Open problem 1

Let M be a closed smooth manifold. Suppose that M admits a Riemannian metric g_0 of nonpositive sectional curvature with an abelian local splitting structure: on the universal Riemannian cover $(\tilde{M}, \tilde{g}_0)$ there is a deck-equivariant assignment, on an open dense subset, of neighborhoods with specified isometric product splittings $W_x \cong N_x \times \mathbb{R}^{r_x}$, where $r_x \geq 1$, such that every connected Euclidean leaf projects to a closed embedded flat submanifold of M and, whenever two projected leaves intersect, one is contained in the other. Determine whether every Riemannian metric g of nonpositive sectional curvature on M must also admit an abelian local splitting structure with these properties.

Reference supplied by the database. Xiaochun Rong, “Collapsed Riemannian Manifolds with Bounded Sectional Curvature,” *Proceedings of the International Congress of Mathemati-*

cians, Beijing 2002, Vol. II, 323–338, Higher Education Press, Beijing, 2002, Question (2.2.1); arXiv:math/0304267, <https://arxiv.org/abs/math/0304267>.

1.40.3.2. Open problem 2

Let M be a closed smooth manifold carrying a positive-rank Cr -structure. Thus M has an injective F -structure with an atlas $\{(V_i, U_i, T^{k_i})\}$ such that each finite normal covering $V_i \rightarrow U_i$ is diffeomorphic to $D_i \times T^{k_i}$, the torus T^{k_i} acts by multiplication on the second factor, and, whenever $U_i \cap U_j \neq \emptyset$ with $i \neq j$, one has $k_i < k_j$ or $k_j < k_i$; injectivity means that the inclusion of every F -orbit into M induces an injective homomorphism on fundamental groups. Determine whether every Riemannian metric g of nonpositive sectional curvature on M admits a local splitting structure compatible with this Cr -structure. Here a local splitting structure consists of a deck-equivariant assignment, on an open dense subset of the universal Riemannian cover $(\tilde{M}, \tilde{g})$, of neighborhoods with specified isometric splittings $W_x \cong N_x \times \mathbb{R}^{r_x}$ with $r_x \geq 1$; compatibility means that there is a diffeomorphism of M carrying each projected Euclidean leaf of the local splittings onto an orbit of the Cr -structure while preserving their natural flat affine structures.

Reference supplied by the database. Xiaochun Rong, “Collapsed Riemannian Manifolds with Bounded Sectional Curvature,” *Proceedings of the International Congress of Mathematicians, Beijing 2002*, Vol. II, 323–338, Higher Education Press, Beijing, 2002, Question (2.2.2); arXiv:math/0304267, <https://arxiv.org/abs/math/0304267>.

1.40.4. Primary-source pointer

The originating ICM record is *Collapsed Riemannian Manifolds with Bounded Sectional Curvature* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.2/ICM2002.2.ocr.pdf#page=311>.

1.41. 41. Applications of loop group factorization to geometric soliton equations [2006-geom-91]

Record: 2006-geom-91

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2006: Applications of loop group factorization to geometric soliton equations

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.2/ICM2006.2.ocr.pdf#page=937>

1.41.1. Problem statement

Let $x_1, \dots, x_n$ be the standard coordinates on $\mathbb{R}^n$, and let $F = (f_{ij})$ be a global smooth symmetric solution of the $U(n)/O(n)$ -system on $\mathbb{R}^n$, with $f_{ii} = 0$ for all i . Determine whether there exist global smooth functions $b_1, \dots, b_n : \mathbb{R}^n \rightarrow (0, \infty)$ satisfying the first-order system

$$\partial_{x_j} b_i = f_{ij} b_j, \quad i \neq j,$$

such that the associated diagonal Riemannian metric

$$ds^2 = \sum_{i=1}^n b_i^2 dx_i^2$$

is complete.

1.41.2. Historical background, literature, and current progress

this problem is fully open

1.41.3. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

1.41.4. Primary-source pointer

The originating ICM record is *Applications of loop group factorization to geometric soliton equations* (2006). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.2/ICM2006.2.ocr.pdf#page=937>.

1.42. 42. Finiteness of arithmetic Kleinian reflection groups [2006-geom-340]

Record: 2006-geom-340

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2006: Finiteness of arithmetic Kleinian reflection groups

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.2/ICM2006.2.ocr.pdf#page=961>

1.42.1. Problem statement

For an n -dimensional Riemannian orbifold O , let $|O|$ denote its underlying topological space. For $m \geq n$, a continuous map $\varphi : |O| \rightarrow \mathbb{S}^m$ is called piecewise conformal if it is piecewise a conformal immersion. Let g_{round} be the standard round metric on $\mathbb{S}^m$, and let $\text{Mob}(\mathbb{S}^m)$ be the group of conformal transformations of $\mathbb{S}^m$. Define

$$V_c(m, \varphi) = \sup_{\mu \in \text{Mob}(\mathbb{S}^m)} \text{Vol}(O, (\mu \circ \varphi)^* g_{\text{round}}),$$

$$V_c(m, O) = \inf_{\substack{\varphi: |O| \rightarrow \mathbb{S}^m \\ \varphi \text{ piecewise conformal}}} V_c(m, \varphi), \quad V_c(O) = \lim_{m \rightarrow \infty} V_c(m, O).$$

An elliptic n -orbifold is a quotient $\mathbb{S}^n/\Gamma$, equipped with its quotient spherical metric, where Γ is a finite subgroup of $O(n+1)$.

Prove that, for every positive integer n , there exists a finite constant $K(n)$, depending only on n , such that

$$V_c(O) \leq K(n)$$

for every elliptic n -orbifold O .

1.42.2. Historical background, literature, and current progress

The assertion is **unresolved**. The literature gives a bound depending polynomially on the order of the finite group, but no bound depending only on the dimension is known in general.

- Ian Agol, “Finiteness of arithmetic Kleinian reflection groups,” *Proceedings of the International Congress of Mathematicians, Madrid 2006*, Vol. II, EMS Press, 2007, pp. 951–960, Conjecture 7.1, DOI: <https://doi.org/10.4171/022-2/45>, arXiv: [math/0512560](https://arxiv.org/abs/math/0512560). This is the proposing source. Its Lemma 3.1 shows that every finite subgroup of $O(3)$ has index at most 4 in a finite reflection group; together with the conformal-volume covering inequality, this proves the required uniform bound for elliptic 2-orbifolds (and the 1-dimensional case follows similarly from the classification of finite subgroups of $O(2)$). It does not cover all dimensions.
- Ian Agol, Mikhail Belolipetsky, Peter Storm, and Kevin Whyte, “Finiteness of arithmetic hyperbolic reflection groups,” *Groups, Geometry, and Dynamics* **2** (2008), no. 4, 481–498, DOI: <https://doi.org/10.4171/GGD/47>, arXiv: [math/0612132](https://arxiv.org/abs/math/0612132). Theorem 3.1

proves for every finite $\Gamma < O(n + 1)$ that

$$V_c(\mathbb{S}^n/\Gamma) < C(n)|\Gamma|^{2n}.$$

This establishes finiteness for each individual quotient and is the strongest general estimate located, but its dependence on $|\Gamma|$ leaves the requested uniform constant $K(n)$ unproved. The paper explicitly repeats the order-independent statement as a conjecture.

- Mikhail Belolipetsky, “Arithmetic hyperbolic reflection groups,” *Bulletin of the American Mathematical Society* **53** (2016), no. 3, 437–475, DOI: <https://doi.org/10.1090/bull/1530>, arXiv: [1506.03111](https://arxiv.org/abs/1506.03111). The survey reviews the conformal-volume method and states that outside the low-dimensional cases the conformal volume of the relevant spherical quotients is not known to be bounded. No later proof or counterexample resolving the all-dimensional assertion was found.

Thus the remaining gap is precisely the removal of the factor depending on $|\Gamma|$ for arbitrary finite subgroups $\Gamma < O(n + 1)$, beginning in dimensions not covered by the finite-subgroup/reflection-group argument.

1.42.3. Remaining open problems recorded by the database

1.42.3.1. Open problem 1

For an n -dimensional Riemannian orbifold O , let $|O|$ denote its underlying topological space. For $m \geq n$, call a continuous map $\varphi : |O| \rightarrow \mathbb{S}^m$ piecewise conformal if it is piecewise a conformal immersion. Let g_{round} be the standard round metric on $\mathbb{S}^m$, and let $\text{Möb}(\mathbb{S}^m)$ be the group of conformal transformations of $\mathbb{S}^m$. Define

$$V_c(m, \varphi) = \sup_{\mu \in \text{Möb}(\mathbb{S}^m)} \text{Vol}(O, (\mu \circ \varphi)^* g_{\text{round}}),$$

$$V_c(m, O) = \inf_{\substack{\varphi: |O| \rightarrow \mathbb{S}^m \\ \varphi \text{ piecewise conformal}}} V_c(m, \varphi), \quad V_c(O) = \lim_{m \rightarrow \infty} V_c(m, O).$$

An elliptic n -orbifold is a quotient $\mathbb{S}^n/\Gamma$, equipped with its quotient spherical metric, where Γ is a finite subgroup of $O(n + 1)$.

Prove that, for every integer n with $1 \leq n \leq 16$, there exists a finite constant $K(n)$, depending only on n , such that $V_c(O) \leq K(n)$ for every elliptic n -orbifold O .

Reference supplied by the database. Ian Agol, “Finiteness of arithmetic Kleinian reflection groups,” *Proceedings of the International Congress of Mathematicians, Madrid 2006*, Vol. II, EMS Press, 2007, pp. 951–960, Conjecture 7.1 and the immediately preceding discussion; DOI: <https://doi.org/10.4171/022-2/45>; arXiv:math/0512560.

1.42.4. Primary-source pointer

The originating ICM record is *Finiteness of arithmetic Kleinian reflection groups* (2006). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.2/ICM2006.2.ocr.pdf#page=961>.

1.43. 43. Metaphors in Systolic Geometry [2010-geom-114]

Record: 2010-geom-114

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2010: Metaphors in Systolic Geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=751>

1.43.1. Problem statement

Fix $r > 0$. For a Riemannian metric g on the n -dimensional torus T^n and a point $p \in T^n$, let $\text{Sc}_r(p)$ denote the macroscopic scalar curvature at scale r : if $\tilde{V}(p, r)$ is the volume, in the universal cover $(\tilde{T}^n, \tilde{g})$, of the ball of radius r centered at a lift of p , then $\text{Sc}_r(p)$ is the number S such that $\tilde{V}(p, r)$ equals the volume of a ball of radius r in the simply connected n -dimensional space of constant scalar curvature S .

Prove or disprove that there is no Riemannian metric g on T^n for which $\text{Sc}_r(p) > 0$ at every point $p \in T^n$. Equivalently, prove or disprove that for every Riemannian metric g on T^n , the universal cover $(\tilde{T}^n, \tilde{g})$ contains a ball of radius r and volume at least $\omega_n r^n$, where ω_n is the Euclidean volume of the unit n -ball.

1.43.2. Historical background, literature, and current progress

1.43.3. Status check: Generalized Geroch conjecture

Verdict: partially resolved.

1.43.3.1. Precise problem checked

The problem asks for the sharp macroscopic-scalar-curvature analogue of the Geroch conjecture for tori. For every dimension n , every Riemannian metric g on T^n , and every fixed scale $r > 0$, must the Riemannian universal cover $(\tilde{T}^n, \tilde{g})$ contain some radius- r ball of volume at least $\omega_n r^n$? Equivalently, can T^n carry a metric whose macroscopic scalar curvature at scale r is positive at every point?

This is the sharp Euclidean-ball constant statement. A nonsharp lower bound $c(n)r^n$, even for all r , is not enough to resolve it.

1.43.3.2. Source and formulation

- Title: Large Riemannian manifolds Authors: Mikhail Gromov Venue: Curvature and Topology of Riemannian Manifolds, Katata 1985, Lecture Notes in Mathematics 1201, Springer Year: 1986 DOI/URL: <https://doi.org/10.1007/BFb0075649> Role: Original source for Gromov's large-manifold conjectures behind this problem.

- Title: Metaphors in systolic geometry Authors: Larry Guth Venue: Proceedings of the International Congress of Mathematicians 2010, World Scientific, pp. 745-768 Year: 2010 URL: <https://math.mit.edu/~lguth/Exposition/icmguth.pdf> Role: States the exact problem in the form used in this problem: “Fix $r > 0$. The n -dimensional torus does not admit a metric with $Sc_r > 0$.” Guth also states the equivalent formulation that the universal cover contains a radius- r ball of volume at least $\omega_n r^n$, and says the conjecture was wide open at that time.

1.43.3.3. Main partial progress

- Title: Systolic inequalities and minimal hypersurfaces Authors: Larry Guth Venue: Geometric and Functional Analysis 19, 1688-1692 Year: 2010 arXiv/URL: arXiv:0903.5299, <https://arxiv.org/abs/0903.5299> Result used for status: Guth proves that if $R < \frac{1}{2} \text{Sys}(T^n, g)$, then some radius- R ball in T^n has volume at least $c_n R^n$. Why it is only partial: the constant is c_n , not the sharp Euclidean ω_n , and the statement is tied to the systole scale rather than the exact universal-cover statement for every prescribed r .
- Title: Volumes of balls in large Riemannian manifolds Authors: Larry Guth Venue: Annals of Mathematics 173(1), 51-76 Year: 2011 DOI/arXiv/URL: 10.4007/annals.2011.173.1.2; arXiv:math/0610212; <https://arxiv.org/abs/math/0610212> Result used for status: Corollary 3 proves that if M is a closed aspherical Riemannian n -manifold, then the supremal volume $V(R)$ of radius- R balls in the universal cover satisfies $V(R) \geq \delta(n)R^n$ for all R . This applies to T^n . Why it is only partial: $\delta(n)$ is an unspecified positive dimensional constant and is not shown to equal ω_n . Thus the theorem gives a qualitative/nonsharp macroscopic lower bound but not the sharp Euclidean threshold.
- Title: Volume and macroscopic scalar curvature Authors: Sabine Braun and Roman Sauer Venue: Geometric and Functional Analysis 31, 1321-1376 Year: 2021 DOI/arXiv/URL: 10.1007/s00039-021-00588-y; arXiv:2012.08999; <https://arxiv.org/abs/2012.08999> Result used for status: Theorem 1.3 proves that for every rationally essential Riemannian d -manifold and every $R > 0$, the universal cover contains a radius- R ball of volume $> \epsilon(d)R^d$. Since T^n is rationally essential, this applies to the torus. Hypothesis check: the torus is closed, oriented, and rationally essential, so the theorem covers the topological class in the problem statement. Why it is only partial: Braun-Sauer explicitly frame the theorem as a nonsharp macroscopic cousin; the exact conjecture would follow only if $\epsilon(d)$ could be chosen to be the Euclidean d -ball volume ω_d , which their theorem does not prove.
- Title: Uryson width and volume Authors: Panos Papasoglu Venue: Geometric and Functional Analysis 30, 574-587 Year: 2020 DOI/URL: 10.1007/s00039-020-00533-5; <https://link.springer.com/article/10.1007/s00039-020-00533-5> Result used for status: Gives a short proof and related width-volume approach to Guth-type nonsharp lower bounds. Why it is only partial: it does not replace the dimensional constants by the sharp Euclidean volume ω_n .
- Title: Linear bounds for constants in Gromov’s systolic inequality and related results Authors: Alexander Nabutovsky Venue: Geometry & Topology 26(7), 3123-3142 Year: 2022 DOI/arXiv/URL: 10.2140/gt.2022.26.3123; arXiv:1909.12225; <https://arxiv.org/abs/1909.12225> Result used for status: Improves constants in Guth/Gromov-type systolic and ball-volume estimates. Why it is only partial: improving nonsharp constants still does not prove the Euclidean sharp constant in the generalized Geroch

conjecture.

- Title: Macroscopic scalar curvature and local collapsing Authors: Stephane Sabourau Venue: Annales Scientifiques de l'Ecole Normale Supérieure (4) 55(4), 919-936 Year: 2022 arXiv/URL: arXiv:2006.00663; <https://arxiv.org/abs/2006.00663> Result used for status: Proves strong lower bounds for balls in universal covers of negatively curved manifolds under small-volume hypotheses and interprets them via macroscopic scalar curvature. Why it is only partial: the hypotheses and conclusion concern negatively curved/aspherical settings with different comparison constants and do not prove the sharp torus statement.
- Title: Macroscopic scalar curvature and areas of cycles Authors: Hannah Alpert and Kei Funano Venue: Geometric and Functional Analysis 27(4), 727-743 Year: 2017 DOI/arXiv/URL: 10.1007/s00039-017-0414-3; arXiv:1705.02923; <https://arxiv.org/abs/1705.02923> Result used for status: Develops Guth's macroscopic-scalar-curvature methods for area lower bounds of homology classes in $\Sigma \times S^1$. Why it is only partial: it is about hyperbolic products and cycle areas, not the sharp ball-volume lower bound for all metrics on T^n .
- Title: Macroscopic scalar curvature and codimension 2 width Authors: Hannah Alpert, Alexey Balitskiy, and Larry Guth Venue: Journal of Topology and Analysis 16(6), 979-987 Year: 2024 DOI/arXiv/URL: 10.1142/S1793525323500024; arXiv:2112.04594; <https://arxiv.org/abs/2112.04594> Result used for status: Gives width conclusions in dimension 3 under strong positive macroscopic-curvature and local acyclicity assumptions. Why it is only partial: the added hypotheses and width conclusion do not imply the exact $\omega_n r^n$ torus ball-volume statement.
- Title: Urysohn width and macroscopic scalar curvature Authors: Aditya Kumar and Balarka Sen Venue: arXiv preprint Year: 2026 arXiv/URL: arXiv:2601.14669; <https://arxiv.org/abs/2601.14669> Result used for status: Constructs examples with positive macroscopic scalar curvature in the context of circle bundles and shows that a macroscopic Urysohn-width analogue is false in dimensions at least four. Why it is only partial: these examples are not metrics on T^n satisfying the stated condition, and the result does not settle the generalized Geroch conjecture for tori.

1.43.3.4. Applicability check and remaining gap

The closest theorems that apply directly to T^n are Guth's aspherical-manifold ball-volume theorem and Braun-Sauer's rationally essential manifold theorem. Their hypotheses do cover the torus, and their conclusions hold at every scale after scaling. However, both conclusions have an unspecified dimensional positive constant rather than the sharp Euclidean constant ω_n . Since the problem asks exactly whether some ball has volume at least $\omega_n r^n$, these theorems do not fully resolve the problem.

I found no published or otherwise citable result proving the full sharp statement for all n , all $r > 0$, and all Riemannian metrics on T^n . The current literature therefore gives substantial nonsharp and related progress, but the exact generalized Geroch conjecture in this problem remains unresolved.

1.43.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

1.43.5. Primary-source pointer

The originating ICM record is *Metaphors in Systolic Geometry* (2010). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=751>.

1.44. 44. Metaphors in Systolic Geometry [2010-geom-115]

Record: 2010-geom-115

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2010: Metaphors in Systolic Geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=751>

1.44.1. Problem statement

Prove that if M^n is a closed n -dimensional hyperbolic manifold with hyperbolic Riemannian metric hyp , then for any Riemannian metric g on the underlying manifold M such that

$$\text{Sc}(g)(x) \geq \text{Sc}(\text{hyp})(x)$$

for every point $x \in M$, the Riemannian volumes satisfy

$$\text{Vol}(M, g) \geq \text{Vol}(M, \text{hyp}).$$

1.44.2. Historical background, literature, and current progress

1.44.3. Verdict: partially resolved

1.44.3.1. Precise problem

The problem asks for Schoen's hyperbolic volume comparison conjecture: for a closed hyperbolic Riemannian manifold (M^n, g_{hyp}) , does every smooth Riemannian metric g on the same underlying manifold satisfying

$$\text{Scal}_g \geq \text{Scal}_{g_{\text{hyp}}} = -n(n-1)$$

pointwise necessarily satisfy

$$\text{Vol}(M, g) \geq \text{Vol}(M, g_{\text{hyp}})?$$

For $n = 2$ this follows from Gauss-Bonnet. The substantive conjecture is for $n \geq 3$, and the full all-dimensional statement remains unresolved.

1.44.3.2. Status check

I found no citable theorem that proves the full statement for all closed hyperbolic n -manifolds and all smooth metrics satisfying only the scalar-curvature lower bound. Recent papers still describe the higher-dimensional case as open or only partially confirmed. The literature gives important special cases: dimension 3, metrics close to the hyperbolic metric, stronger Ricci-curvature hypotheses, conformal/self-dual cases in dimension 4, and macroscopic analogues. These do not imply the full statement because they add hypotheses not present in the conjecture or replace scalar curvature by a different large-scale condition.

1.44.3.3. Key literature and progress

1. Richard M. Schoen, “Variational theory for the total scalar curvature functional for Riemannian metrics and related topics”, in *Topics in Calculus of Variations*, Lecture Notes in Mathematics 1365, Springer, 1989, pp. 120-154. DOI: 10.1007/BFb0089180. URL: <https://link.springer.com/chapter/10.1007/BFb0089180>. This is the original source for Schoen’s conjecture on the Yamabe invariant of closed hyperbolic manifolds. Later literature identifies the volume-comparison formulation in this problem as equivalent to the hyperbolic Yamabe-invariant conjecture.
2. G. Besson, G. Courtois, and S. Gallot, “Volume et entropie minimale des espaces localement symétriques”, *Inventiones Mathematicae* 103 (1991), 417-445. DOI: 10.1007/BF01239520. URL: <https://link.springer.com/article/10.1007/BF01239520>. This gives local/entropy rigidity results used as partial evidence for Schoen’s conjecture. As summarized in later sources, it verifies the scalar-curvature volume comparison for metrics C^2 -close to the hyperbolic metric, but this closeness assumption is not in the problem statement.
3. G. Besson, G. Courtois, and S. Gallot, “Entropies et rigidités des espaces localement symétriques de courbure strictement négative”, *Geometric and Functional Analysis* 5 (1995), 731-799. DOI: 10.1007/BF01897050. URL: <https://link.springer.com/article/10.1007/BF01897050>. This proves a global minimal-entropy/rigidity theorem. Together with Bishop-Gromov it yields the desired volume lower bound under the stronger assumption $\text{Ric}_g \geq -(n-1)g$, but Ricci lower bounds are much stronger than the scalar-curvature lower bound in the problem statement.
4. Michael T. Anderson, “Scalar curvature and geometrization conjectures for 3-manifolds”, in *Comparison Geometry*, MSRI Publications 30, Cambridge University Press, 1997, pp. 49-82. URL: <https://library.slmath.org/books/Book30/files/anderson.pdf>. Anderson formulates the three-dimensional scalar-curvature/Yamabe-invariant picture. In dimension 3, Perelman’s Ricci-flow and geometrization work implies the corresponding closed hyperbolic 3-manifold volume comparison: if $R_g \geq -6$, then $\text{Vol}(M, g) \geq \text{Vol}(M, g_{\text{hyp}})$.
5. Michael T. Anderson, “Canonical metrics on 3-manifolds and 4-manifolds”, *Asian Journal of Mathematics* 10(1) (2006), 127-163. DOI: 10.4310/AJM.2006.v10.n1.a8. arXiv: math/0511034. URL: <https://arxiv.org/abs/math/0511034>. This records the post-Perelman 3-dimensional sigma/Yamabe picture. It supports the conclusion that the conjecture is resolved in dimension 3, but it does not settle dimensions $n \geq 4$.
6. Ian Agol, Peter A. Storm, and William P. Thurston, with appendix by Nathan Dunfield, “Lower bounds on volumes of hyperbolic Haken 3-manifolds”, *Journal of the American Mathematical Society* 20(4) (2007), 1053-1077. DOI: 10.1090/S0894-0347-07-00564-4. arXiv: math/0506338. URL: <https://arxiv.org/abs/math/0506338>. This uses Perelman’s work to prove volume inequalities for 3-manifolds under scalar-curvature/minimal-boundary hypotheses. It is important three-dimensional progress but not a proof of the all-dimensional closed-manifold conjecture.
7. Larry Guth, “Metaphors in systolic geometry”, in *Proceedings of the International Congress of Mathematicians 2010*, Vol. II, Hindustan Book Agency, 2010, pp. 745-768. DOI: 10.1142/9789814324359_0072. URL: <https://math.mit.edu/~lguth/Exposition/icmguth.pdf>. Guth presents the Schoen conjecture as open and notes that in dimensions $n \geq 5$ no scalar-curvature-only lower volume bound was known at

that time. He proves a non-sharp macroscopic analogue involving volumes of balls in the universal cover, not the pointwise scalar-curvature hypothesis itself.

8. **Wei Yuan**, “Volume comparison with respect to scalar curvature”, *Analysis & PDE* 16(1) (2023), 1-34. DOI: 10.2140/apde.2023.16.1. arXiv: 1609.08849. URL: <https://arxiv.org/abs/1609.08849>. Yuan states Schoen’s hyperbolic volume comparison conjecture explicitly and proves that it is equivalent to the hyperbolic Yamabe-invariant conjecture. The paper proves local volume comparison near strictly stable Einstein metrics and recovers the Besson-Courtois-Gallot local result for closed hyperbolic manifolds. It also states that the conjecture is known in dimension 3, while in higher dimensions the known results require C^2 -closeness or stronger Ricci hypotheses.
9. **Florent Balacheff and Stephane Karam**, “Macroscopic Schoen conjecture for manifolds with nonzero simplicial volume”, *Transactions of the American Mathematical Society* 372(10) (2019), 7071-7086. DOI: 10.1090/tran/7765. arXiv: 1709.10049. URL: <https://www.ams.org/tran/2019-372-10/S0002-9947-2019-07765-6/>. This proves macroscopic/non-sharp versions related to Guth’s framework. It concerns large-scale ball-volume hypotheses and simplicial volume, so it is partial evidence rather than a proof of the scalar-curvature conjecture in the problem statement.
10. **Sabine Braun and Roman Sauer**, “Volume and macroscopic scalar curvature”, *Geometric and Functional Analysis* 31 (2021), 1321-1376. DOI: 10.1007/s00039-021-00588-y. URL: <https://link.springer.com/article/10.1007/s00039-021-00588-y>. This proves macroscopic cousins of scalar-curvature conjectures, replacing scalar-curvature lower bounds by upper bounds on volumes of 1-balls in the universal cover. The authors explicitly describe Guth’s volume theorem as a non-sharp macroscopic cousin of Schoen’s conjecture, not a resolution of the original conjecture.
11. **Demetre Kazaras, Antoine Song, and Kai Xu**, “Scalar curvature and volume entropy of hyperbolic 3-manifolds”, *Journal of the European Mathematical Society* (2025), published online first. DOI: 10.4171/JEMS/1710. arXiv: 2312.00138. URL: <https://ems.press/journals/jems/articles/14299127>. This constructs metrics on closed hyperbolic 3-manifolds with scalar curvature at least -6 but volume entropy greater than 2, disproving a proposed entropy route to the 3-dimensional volume comparison. It does not refute Schoen’s conjecture; rather, it shows that the Besson-Courtois-Gallot entropy method cannot be extended from Ricci lower bounds to scalar lower bounds in this way.
12. **Jialong Deng**, “Scalar Curvature in Dimension 4”, arXiv preprint, 2025. arXiv: 2512.13528. DOI: 10.48550/arXiv.2512.13528. URL: <https://arxiv.org/abs/2512.13528>. This recent preprint gives partial dimension-4 confirmations, for example for self-dual, anti-self-dual, or locally conformally flat metrics on closed oriented hyperbolic 4-manifolds with scalar curvature bounded below by -12 . These are extra geometric restrictions on the metric and therefore do not prove the arbitrary-metric statement in dimension 4.

1.44.3.4. What remains unresolved

The unresolved part is the exact scalar-curvature-only comparison for arbitrary smooth metrics on closed hyperbolic n -manifolds in dimensions $n \geq 4$:

$$\text{Scal}_g \geq -n(n-1) \implies \text{Vol}(M, g) \geq \text{Vol}(M, g_{\text{hyp}}).$$

No result found removes the extra assumptions appearing in the known higher-dimensional results. Therefore the problem should be classified as **partially resolved**, not fully resolved.

1.44.4. Remaining open problems recorded by the database

1.44.4.1. Open problem 1

Let (M^4, g_{hyp}) be a closed oriented hyperbolic four-manifold. Prove that every Riemannian metric g on M satisfying $\text{Scal}_g \geq -12$ has $\text{Vol}(M, g) \geq \text{Vol}(M, g_{\text{hyp}})$, without imposing local conformal flatness, self-duality, anti-self-duality, or closeness to g_{hyp} .

Reference supplied by the database. Jialong Deng, “Scalar Curvature in Dimension 4,” arXiv preprint, 2025, §1 and Theorem A (explicitly treats the unrestricted dimension-four Schoen conjecture as open and proves only additional-hypothesis cases; no separately numbered conjecture). arXiv:2512.13528. <https://arxiv.org/abs/2512.13528>

1.44.5. Primary-source pointer

The originating ICM record is *Metaphors in Systolic Geometry* (2010). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=751>.

1.45. 45. Metaphors in Systolic Geometry [2010-geom-116]

Record: 2010-geom-116

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2010: Metaphors in Systolic Geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=751>

1.45.1. Problem statement

Let $r > 0$. Let (M^n, hyp) be a closed hyperbolic n -manifold, where hyp denotes its hyperbolic Riemannian metric. For a Riemannian metric h on M and a point $p \in M$, define the macroscopic scalar curvature $\text{Sc}_r^h(p)$ at scale r as follows: let $\tilde{V}_h(p, r)$ be the volume of the ball of radius r centered at any lift of p in the universal cover of (M, h) , and let $\tilde{V}_S(r)$ be the volume of the ball of radius r in the simply connected n -dimensional space of constant curvature and scalar curvature S ; then $\text{Sc}_r^h(p)$ is the number S such that $\tilde{V}_h(p, r) = \tilde{V}_S(r)$.

Prove that if g is a Riemannian metric on M such that $\text{Sc}_r^g(p) \geq \text{Sc}_r^{\text{hyp}}(p)$ for every $p \in M$, equivalently, every radius- r ball in the universal cover of (M, g) has volume at most the volume of a radius- r ball in the universal cover of (M, hyp) , then

$$\text{Vol}(M, g) \geq \text{Vol}(M, \text{hyp}).$$

1.45.2. Historical background, literature, and current progress

1.45.3. Verdict: partially resolved

1.45.3.1. Precise problem checked

The problem asks for the sharp macroscopic Schoen/Guth conjecture for closed hyperbolic manifolds:

For every closed hyperbolic n -manifold (M, hyp) , every Riemannian metric g on M , and every fixed radius $r > 0$, if every radius- r ball in the universal cover of (M, g) has volume at most the radius- r ball in H^n , equivalently $\text{Sc}_r^g(p) \geq \text{Sc}_r^{\text{hyp}}(p)$ for all p , then

$$\text{Vol}(M, g) \geq \text{Vol}(M, \text{hyp}).$$

Equivalently, by contrapositive, if $\text{Vol}(M, g) < \text{Vol}(M, \text{hyp})$, then for every $r > 0$ there must be some radius- r ball in the universal cover of (M, g) whose volume is larger than the hyperbolic radius- r ball. This is Guth's sharp Conjecture 2 / generalized Schoen conjecture.

1.45.3.2. Status

No citable paper found proves this exact sharp statement. The literature gives substantial partial progress, but all available theorems found either use a non-sharp small constant in place of 1, restrict the radius range, add extra hypotheses such as negative sectional curvature, or prove only an asymptotic large-radius conclusion.

1.45.3.3. Key literature and what it proves

1. Larry Guth, “Metaphors in systolic geometry”, Proceedings of the International Congress of Mathematicians, Vol. II, Hindustan Book Agency, 2010, pp. 745-768. DOI: https://doi.org/10.1142/9789814324359_0072. URL: <https://math.mit.edu/~lguth/Exposition/icmguth.pdf>.

This is the source containing the stated “Generalized Schoen conjecture” formulation: for every $r > 0$, $Sc_r(g) \geq Sc_r(hyp)$ should imply $Vol(M, g) \geq Vol(M, hyp)$. Guth explicitly describes this conjecture as far out of reach. The same exposition records only a non-sharp theorem: for $r = 1$, the unit-ball upper bound in the universal cover implies $Vol(M, g) \geq c(n) Vol(M, hyp)$, with a dimensional constant $c(n)$, not the sharp constant 1.

2. Larry Guth, “Volumes of balls in large Riemannian manifolds”, Annals of Mathematics 173(1), 51-76, 2011. DOI: <https://doi.org/10.4007/annals.2011.173.1.2>. arXiv: math.DG/0610212. URL: <https://annals.math.princeton.edu/2011/173-1/p02>.

Guth states Conjecture 2: if (M, hyp) is closed hyperbolic and $Vol(M, g) < Vol(M, hyp)$, then $V_{\tilde{M}, g}(R) > V_{H^n}(R)$ for all R . This is the contrapositive form of the sharp statement. Theorem 2 proves only the non-sharp case: for each dimension n there is $\delta(n) > 0$ such that $Vol(M, g) < \delta(n) Vol(M, hyp)$ forces a unit ball in the universal cover to be larger than the unit ball in H^n . Guth also notes that variants for other fixed radii have constants depending on the radius and that the constants deteriorate; this does not resolve the sharp conjecture.

3. Gerard Besson, Gilles Courtois, and Sylvestre Gallot, “Minimal entropy and Mostow’s rigidity theorems”, Ergodic Theory and Dynamical Systems 16(4), 623-649, 1996. DOI: <https://doi.org/10.1017/S0143385700009019>. URL: <https://www.cambridge.org/core/journals/ergodic-theory-and-dynamical-systems/article/minimal-entropy-and-mostows-rigidity-theorems/7B7F706676FAC80C728846B8A29AFF00>.

The minimal entropy theorem implies the required ball-volume conclusion only for all sufficiently large radii, with the threshold depending on the metric g , when $Vol(M, g) < Vol(M, hyp)$. This is a sharp asymptotic/large-scale result, but it does not imply the stated fixed arbitrary radius statement, especially not small radii.

4. Steve Karam, “Growth of balls in the universal cover of surfaces and graphs”, Transactions of the American Mathematical Society 367(8), 5355-5373, 2015. DOI: <https://doi.org/10.1090/S0002-9947-2015-06189-3>. arXiv: 1304.3567. URL: <https://arxiv.org/abs/1304.3567>.

Karam proves a uniform non-sharp version in dimension 2: for closed hyperbolic surfaces, $Area(M, h) \leq \delta Area(M, hyp)$ implies $V_{\tilde{M}, h}(R) \geq V_{H^2}(R)$ for every $R \geq 1$. This is important partial progress, but it is only dimension 2, uses a small constant δ

instead of 1, and covers $R \geq 1$ rather than all $R > 0$.

5. Florent Balacheff and Steve Karam, “Macroscopic Schoen conjecture for manifolds with nonzero simplicial volume”, Transactions of the American Mathematical Society 372(10), 7071-7086, 2019. DOI: <https://doi.org/10.1090/tran/7765>. arXiv: 1709.10049. URL: <https://arxiv.org/abs/1709.10049>.

Theorem 1.2 gives a non-sharp simplicial-volume inequality for ratios of universal-cover ball volumes for $R \geq 1$. Theorem 1.3 gives, for a closed hyperbolic manifold with an auxiliary metric of negative sectional curvature and sufficiently small volume $\text{Vol}(M, g) < \beta_n \text{Vol}(M, \text{hyp})$, the conclusion that for every $R \geq 1$ some universal-cover R -ball is larger than the hyperbolic one. This has both a non-sharp constant and an extra negative-curvature hypothesis, and it does not cover small radii.

6. Sabine Braun and Roman Sauer, “Volume and macroscopic scalar curvature”, Geometric and Functional Analysis 31, 1321-1376, 2021. DOI: <https://doi.org/10.1007/s00039-021-00588-y>. arXiv: 2012.08999. URL: <https://arxiv.org/abs/2012.08999>.

Theorem 1.1 proves that an upper bound on volumes of unit balls in the universal cover gives $\|M\| \leq \text{const}(d, V_1) \text{Vol}(M)$. For a closed hyperbolic manifold this implies a Guth-type volume lower bound only up to a dimensional constant. The paper explicitly describes Guth’s theorem as the non-sharp macroscopic cousin of Schoen’s conjecture because of this dimensional constant.

7. Stephane Sabourau, “Macroscopic scalar curvature and local collapsing”, Annales Scientifiques de l’Ecole Normale Supérieure 55(4), 919-936, 2022. DOI: <https://doi.org/10.24033/asens.2509>. arXiv: 2006.00663. URL: <https://arxiv.org/abs/2006.00663>.

Theorem 1.4 proves that, for a closed hyperbolic n -manifold and any metric g , if $\text{Vol}(M, g)$ is smaller than a dimensional absolute constant δ'_n , then $V_{\widetilde{M}}(r) \geq V_{H^n}(r)$ for every $r \geq 1$. This gives uniform radius control and removes the negative-curvature hypothesis, but the volume bound is not the sharp relative bound $\text{Vol}(M, \text{hyp})$ and the result still covers only $r \geq 1$.

8. Hannah Alpert, “Growth in the universal cover under large simplicial volume”, Journal of Topology and Analysis 18(4), 1033-1042, 2026. DOI: <https://doi.org/10.1142/S1793525325500025>. arXiv: 2402.04932. URL: <https://arxiv.org/abs/2402.04932>.

The main theorem proves that if $(\text{Vol}(M, g))^2 / \|M\|_{\Delta} < \delta'_n$, then $V_r(\widetilde{M}) \geq V_r(H^n)$ for all $r \geq 1$. The introduction explicitly calls Guth’s sharp statement, $\text{Vol}(M, g) < \text{Vol}(M, \text{hyp})$ implying the ball-volume conclusion for all r , the “most ambitious conjecture” and describes the result as partial progress. Thus this paper also does not resolve the problem statement.

1.45.3.4. What remains unresolved

The unresolved part is exactly the sharp threshold and full scale range:

- replace all small constants $c(n)$, $\delta(n)$, β_n , or absolute δ'_n by the sharp constant 1 relative to $\text{Vol}(M, \text{hyp})$;
- allow every Riemannian metric g , without extra curvature assumptions;
- prove the conclusion for every fixed $r > 0$, including small radii, not only $r = 1$,

$r \geq 1$, or sufficiently large r .

Therefore the correct status for the problem is **partially resolved**, not fully resolved.

1.45.4. Remaining open problems recorded by the database

1.45.4.1. Open problem 1

For every integer $n \geq 3$, prove that there exists $\delta_n > 0$ with the following property. If M is a closed, oriented, connected n -manifold admitting a hyperbolic metric and g is another Riemannian metric on M such that

$$\frac{\text{Vol}(M, g)}{\|M\|_{\Delta}} < \delta_n,$$

where $\|M\|_{\Delta}$ is the simplicial volume, then for every $r \geq 1$ the universal cover of (M, g) contains an r -ball whose volume is at least the volume of an r -ball in $\mathbb{H}^n$.

Reference supplied by the database. Hannah Alpert, “Growth in the Universal Cover under Large Simplicial Volume,” *Journal of Topology and Analysis* 18, no. 4 (2026), 1033–1042, Conjecture 4 (attributed to a suggestion of Larry Guth; the paper notes that only $n = 2$ is known). DOI: 10.1142/S1793525325500025. arXiv:2402.04932. <https://doi.org/10.1142/S1793525325500025>

1.45.5. Primary-source pointer

The originating ICM record is *Metaphors in Systolic Geometry* (2010). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=751>.

1.46. 46. Riemannian Manifolds of Positive Curvature [2010-geom-118]

Record: 2010-geom-118

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: reviewed (reviewed flag: yes)

Original source: ICM 2010: Riemannian Manifolds of Positive Curvature

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.1/ICM2010.1.pdf#page=442>

1.46.1. Problem statement

Prove that for every even integer n , if (M^n, g) is a compact n -dimensional Riemannian manifold whose sectional curvature is positive at every point and in every two-dimensional tangent plane, then the Euler characteristic $\chi(M)$ is positive.

1.46.2. Historical background, literature, and current progress

1.46.3. Verdict: partially resolved

1.46.3.1. Precise problem

This problem asks for the positive-sectional-curvature Euler-characteristic Hopf conjecture:

For every even integer n , every compact n -dimensional Riemannian manifold with positive sectional curvature in every tangent 2-plane has positive Euler characteristic.

Equivalently, for every closed positively curved Riemannian manifold M of even dimension, prove $\chi(M) > 0$.

1.46.3.2. Status

The literature does not fully resolve this exact all-even-dimensional statement. The unrestricted conjecture is known in low dimensions and under additional hypotheses such as large symmetry, but no published or otherwise citable result found proves the statement for all compact positively curved even-dimensional manifolds.

The problem is therefore partially resolved, not fully resolved. It is also not best recorded as “fully open”, because dimensions 2 and 4 and many special classes are known.

What remains unresolved is the general case in even dimensions at least 6 without extra hypotheses such as homogeneity, cohomogeneity bounds, or torus-symmetry assumptions.

1.46.3.3. Original source and current-status references

1. Title: Differentialgeometrie und topologische Gestalt Authors: Heinz Hopf Venue: Jahresbericht der Deutschen Mathematiker-Vereinigung 41, 209-229 Year: 1932 DOI/arXiv/URL: <https://eudml.org/doc/145926> Role: Original source for Hopf's questions in global differential geometry; later literature cites this as the source of the positive-curvature Euler-characteristic conjecture.
2. Title: Riemannian Manifolds with Positive Sectional Curvature Authors: Wolfgang Ziller Venue: Survey article; available from the author's webpage Year: 2010/2014 survey version DOI/arXiv/URL: <https://www2.math.upenn.edu/~wziller/papers/SurveyMexico.pdf> Role: Lists Hopf's conjecture that an even-dimensional manifold with positive curvature has positive Euler characteristic, and surveys progress via positive curvature and symmetry. It does not report a general solution.
3. Title: On the Hopf conjecture with symmetry Authors: Lee Kennard Venue: Geometry & Topology 17, 563-593 Year: 2013 DOI/arXiv/URL: DOI 10.2140/gt.2013.17.563; arXiv:1203.3808; <https://arxiv.org/abs/1203.3808> Role: States that the Hopf conjecture holds in dimensions 2 and 4 but remains open in higher dimensions. Proves a partial result: if n is divisible by 4 and M has positive sectional curvature with an effective isometric torus action of rank at least $2 \log_2(n)$, then $\chi(M) > 0$. This has an extra symmetry hypothesis and does not resolve the full statement.

1.46.3.4. Partial progress

1. Title: A simple intrinsic proof of the Gauss-Bonnet formula for closed Riemannian manifolds Authors: Shiing-Shen Chern Venue: Annals of Mathematics 45, 747-752 Year: 1944 DOI/arXiv/URL: <https://doi.org/10.2307/1969302> Role: Provides the intrinsic Gauss-Bonnet formula. In dimension 2, positive sectional curvature is positive Gaussian curvature, so Gauss-Bonnet gives $\chi(M) > 0$.
2. Title: On the curvature and characteristic classes of a Riemannian manifold Authors: Shiing-Shen Chern Venue: Abhandlungen aus dem Mathematischen Seminar der Universität Hamburg 20, 117-126 Year: 1955 DOI/arXiv/URL: DOI 10.1007/BF02960745 Role: Standard reference in the Hopf-conjecture literature for curvature and characteristic classes. Later papers report that the algebraic Hopf conjecture was settled in dimension 4 by Milnor, unpublished, as cited through Chern; this supports the known four-dimensional case but does not extend to all dimensions.
3. Title: On the connectivity of spaces of positive curvature Authors: John L. Synge Venue: The Quarterly Journal of Mathematics (Oxford) os-7, 316-320 Year: 1936 <https://doi.org/10.1093/qmath/os-7.1.316> Role: Proves that a compact even-dimensional manifold with positive sectional curvature has fundamental group 0 or $\mathbb{Z}_2$ (simply connected when orientable). With Poincaré duality this gives $b_1 = b_3 = 0$, hence $\chi = 2 + b_2 > 0$ in dimension 4. This is a topological route to the four-dimensional case, independent of positivity of the Gauss-Bonnet-Chern integrand, and therefore not affected by Geroch's dimension-6 counterexample to the algebraic approach.
4. Title: Positive sectional curvature does not imply positive Gauss-Bonnet integrand Authors: Robert Geroch Venue: Proceedings of the American Mathematical Society 54, 267-270 Year: 1976 DOI/arXiv/URL: DOI 10.1090/S0002-9939-1976-0390961-8 Role: Shows that the stronger algebraic strategy fails in dimension 6: positive sectional curvature of an algebraic curvature tensor need not make the Gauss-Bonnet integrand positive. This

is not a counterexample to Hopf's global conjecture, but explains why Gauss-Bonnet integrand positivity does not solve the higher-dimensional cases.

5. Title: The Hopf conjecture for manifolds with low cohomogeneity or high symmetry rank
 Authors: Thomas Puttmann and Catherine Searle Venue: Proceedings of the American Mathematical Society 130, no. 1, 163-166 Year: 2002 DOI/arXiv/URL: DOI 10.1090/S0002-9939-01-06039-7; <https://arxiv.org/abs/1207.4086> Role: Proves positive Euler characteristic for compact even-dimensional positively curved manifolds under low cohomogeneity or high symmetry rank assumptions. In particular, their Theorem 1 assumes an isometric compact Lie group action with $k - (\text{rank } G - \text{rank } H) \leq 5$; their Theorem 2 assumes a torus action of specified rank. These extra hypotheses do not cover the full statement.
6. Title: Splitting of torus representations and applications in the Grove symmetry program
 Authors: Lee Kennard, Michael Wiemeler, and Burkhard Wilking Venue: arXiv preprint Year: 2021 DOI/arXiv/URL: DOI 10.48550/arXiv.2106.14723; arXiv:2106.14723; <https://arxiv.org/abs/2106.14723> Role: Proves the Hopf conclusion under the extra assumption that the isometry group has rank at least 5. This is strong progress in the Grove symmetry program, but it is conditional on symmetry and does not imply the unrestricted conjecture.
7. Title: An improved four-periodicity theorem and a conjecture of Hopf with symmetry
 Authors: Jan Nienhaus Venue: arXiv preprint / PhD thesis Year: 2022 DOI/arXiv/URL: DOI 10.48550/arXiv.2211.13151; arXiv:2211.13151; <https://arxiv.org/abs/2211.13151> Role: Improves the symmetry result to effective isometric T^4 -actions: every even-dimensional closed positively curved manifold with such an action has $\chi(M) > 0$. The T^4 -action hypothesis is an additional assumption, so this does not resolve the full statement.
8. Title: Positive curvature, torus symmetry, and matroids Authors: Lee Kennard, Michael Wiemeler, and Burkhard Wilking Venue: Journal of the European Mathematical Society (JEMS) Year: 2025 DOI/arXiv/URL: DOI 10.4171/JEMS/1656; arXiv:2212.08152; <https://arxiv.org/abs/2212.08152> Role: Recent progress on positive sectional curvature with torus symmetry, using matroid methods and torus-representation structure. Its theorems retain torus-symmetry and isotropy hypotheses, so they do not prove the full Hopf Euler-characteristic conjecture.

1.46.3.5. Applicability check

None of the cited higher-dimensional results covers every compact even-dimensional positively sectional-curved Riemannian manifold. The main modern positive results require extra assumptions such as a large torus action, low cohomogeneity, rank bounds on the isometry group, or special isotropy structure. These hypotheses are not present in the problem statement.

Thus the exact statement remains unresolved in general, with known partial cases and conditional progress only.

1.46.4. Remaining open problems recorded by the database

1.46.4.1. Open problem 1

Let p be a prime, let M be a topological space, and suppose that a nonzero class $x \in H^k(M; \mathbb{Z}_p)$ induces cohomological periodicity up to degree pk and has minimal degree among all classes with this property. Prove that:

1. if $p = 2$, then $k \in \{1, 2, 4, 8\}$, and if x induces periodicity up to degree $3k$, then $k \neq 8$;
2. if $p > 2$, then $k = 2\lambda$ for some $\lambda \mid (p - 1)$.

Here, a class $x \in H^k(M; \mathbb{Z}_p)$ induces periodicity up to degree c if multiplication by x is surjective $H^i(M; \mathbb{Z}_p) \rightarrow H^{i+k}(M; \mathbb{Z}_p)$ for $0 \leq i < c - k$ and injective for $0 < i \leq c - k$.

Reference supplied by the database. Lee Kennard, “On the Hopf Conjecture with Symmetry,” *Geometry & Topology* **17** (2013), 563–593, Conjecture 6.3 and the discussion immediately following it, DOI: <https://doi.org/10.2140/gt.2013.17.563>, arXiv: 1203.3808.

1.46.5. Primary-source pointer

The originating ICM record is *Riemannian Manifolds of Positive Curvature* (2010). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.1/ICM2010.1.pdf#page=442>.

1.47. 47. Riemannian Manifolds of Positive Curvature [2010-geom-119]

Record: 2010-geom-119

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2010: Riemannian Manifolds of Positive Curvature

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.1/ICM2010.1.pdf#page=442>

1.47.1. Problem statement

Call a Riemannian manifold (M, g) strictly δ -pinched in the pointwise sense if, for every point $p \in M$ and every pair of two-dimensional planes $\pi_1, \pi_2 \subset T_p M$, the sectional curvatures satisfy $0 < \delta K(\pi_1) < K(\pi_2)$, where $K(\pi)$ denotes the sectional curvature of π .

Prove that the following two assertions hold:

1. For every even integer n , there exists a real number $\delta_1(n) \in (0, 1/4)$ such that if (M, g_0) is a compact, simply connected Riemannian manifold of dimension n that is strictly $\delta_1(n)$ -pinched in the pointwise sense, then M is homeomorphic to S^n or diffeomorphic to a compact symmetric space of rank one.
2. For each integer $n \geq 4$, there exists a real number $\delta_2(n) \in (0, 1/4)$ such that if (M, g_0) is a compact, simply connected Riemannian manifold of dimension n that is strictly $\delta_2(n)$ -pinched in the pointwise sense, then M is diffeomorphic to S^n or to a compact symmetric space of rank one.

1.47.2. Historical background, literature, and current progress

1.47.3. Verdict: partially resolved

1.47.3.1. Precise problem checked

The problem asks for sub-quarter *pointwise* pinching constants. In the notation of the problem, the hypothesis is that for each point p , all sectional curvatures of two-planes in $T_p M$ have ratio $K_{\min}(p)/K_{\max}(p) > \delta_i(n)$, with $\delta_i(n) < 1/4$. This is weaker than a global pinching bound of the form $\sec_g \in [1/4 - \varepsilon, 1]$ after one normalization.

The requested conclusions are:

- for every even n , such pointwise $\delta_1(n)$ -pinching implies M is homeomorphic to S^n or diffeomorphic to a compact rank-one symmetric space;
- for every $n \geq 4$, such pointwise $\delta_2(n)$ -pinching implies M is diffeomorphic to S^n or to a compact rank-one symmetric space.

1.47.3.2. Literature found

1. Classification of Almost Quarter-Pinched Manifolds

Peter Petersen and Terence Tao, *Proceedings of the American Mathematical Society* 137 (2009), no. 7, 2437-2440. DOI: <https://doi.org/10.1090/S0002-9939-09-09802-5>. arXiv: 0807.1900. URL: <https://arxiv.org/abs/0807.1900>

Theorem 1.1 proves that for each n there is $\varepsilon(n) > 0$ such that any simply connected n -manifold with sectional curvatures globally in $[1/4 - \varepsilon, 1]$ is diffeomorphic to a sphere or compact rank-one symmetric space. This resolves the analogous *global* almost quarter-pinching problem and strengthens the older even-dimensional Berger conclusion. It does not cover the problem statement exactly, because the statement assumes only pointwise pinching.

2. Manifolds with 1/4-pinched Curvature are Space Forms

Simon Brendle and Richard Schoen, *Journal of the American Mathematical Society* 22 (2009), no. 1, 287-307. DOI: <https://doi.org/10.1090/S0894-0347-08-00613-9>. arXiv: 0705.0766. URL: <https://arxiv.org/abs/0705.0766>

Theorem 1 proves the differentiable sphere theorem under strictly pointwise 1/4-pinched sectional curvatures. This is a pointwise theorem, but it is an endpoint theorem with pinching constant 1/4, not a sub-quarter gap theorem for some $\delta(n) < 1/4$.

3. Classification of manifolds with weakly 1/4-pinched curvatures

Simon Brendle and Richard Schoen, *Acta Mathematica* 200 (2008), 1-13. DOI: <https://doi.org/10.1007/s11511-008-0022-7>. arXiv: 0705.3963. URL: <https://arxiv.org/abs/0705.3963>

The main theorem classifies compact locally irreducible manifolds M for which $M \times \mathbb{R}^2$ has nonnegative isotropic curvature, and gives the weak pointwise 1/4-pinched classification. Again, this is endpoint pointwise progress, not the requested $\delta(n) < 1/4$ pointwise result.

4. Sur les variétés riemanniennes pincées juste au-dessous de 1/4

Marcel Berger, *Annales de l'Institut Fourier* 33 (1983), no. 2, 135-150. URL: https://www.numdam.org/item/AIF_1983__33_2_135_0/

Berger proved an even-dimensional almost quarter-pinching result with the weaker conclusion that the manifold is homeomorphic to a sphere or diffeomorphic to a compact rank-one symmetric space. Petersen-Tao explicitly cite this as the earlier even-dimensional result. This is part of the global almost quarter-pinching line, not a full pointwise sub-quarter solution.

5. A sphere theorem with a pinching constant below 1/4

Uwe Abresch and Wolfgang T. Meyer, *Journal of Differential Geometry* 44 (1996), no. 2, 214-261. URL: <https://projecteuclid.org/journals/journal-of-differential-geometry/volume-44/issue-2/A-sphere-theorem-with-a-pinching-constant-below-1-4/10.4310/jdg/1214458996.full>

This gives related below-1/4 sphere-theorem progress in the global pinching tradition, but it does not supply the full pointwise classification asked in the problem statement.

1.47.3.3. Status assessment

The closest complete theorem is Petersen-Tao, but its hypothesis is a global sectional-curvature interval $[1/4 - \varepsilon, 1]$. A compact pointwise δ -pinched metric need not satisfy any such global

δ -pinching after a single rescaling, because the curvature scale can vary from point to point. Therefore Petersen-Tao does not formally imply the pointwise statement.

The Brendle-Schoen theorems are pointwise, but they handle the exact $1/4$ endpoint, not the existence of a dimension-dependent constant below $1/4$ whose weaker pointwise hypothesis still forces the same classification.

I did not find a published or otherwise citable theorem that proves the full statement exactly as written for pointwise $\delta_i(n) < 1/4$. The problem is therefore not fully resolved as stated. The current literature gives strong partial progress: full resolution for global almost quarter-pinching and endpoint pointwise $1/4$ -pinching, but not the sub-quarter pointwise gap demanded here.

1.47.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

1.47.5. Primary-source pointer

The originating ICM record is *Riemannian Manifolds of Positive Curvature* (2010). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.1/ICM2010.1.pdf#page=442>.

1.48. 48. Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries [2010-geom-286]

Record: 2010-geom-286

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2010: Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=836>

1.48.1. Problem statement

Let $\mathbb{E}^3(\kappa, \tau)$ be a simply connected homogeneous Riemannian three-manifold with four-dimensional isometry group, base curvature $\kappa \leq 0$, and bundle curvature τ , where $\kappa - 4\tau^2 \neq 0$. Let

$$\pi : \mathbb{E}^3(\kappa, \tau) \rightarrow \mathcal{M}^2(\kappa)$$

be its canonical Riemannian fibration over the complete simply connected surface $\mathcal{M}^2(\kappa)$ of constant Gaussian curvature κ .

Classify all oriented constant-mean-curvature immersions $X : \Sigma \rightarrow \mathbb{E}^3(\kappa, \tau)$ whose induced metric on Σ is complete, whose constant mean curvature H is critical in the sense that

$$H^2 = -\frac{\kappa}{4},$$

and whose Jacobi operator

$$L = \Delta + \|B\|^2 + \text{Ric}(N, N)$$

is stable, meaning that

$$-\int_{\Sigma} f Lf \, dA \geq 0$$

for every $f \in C_c^\infty(\Sigma)$. Here Δ and dA are the Laplace–Beltrami operator and area element of the induced metric, B is the second fundamental form of X , N is its globally defined unit normal compatible with the chosen orientation, and Ric is the ambient Ricci tensor. Prove that every such immersion has one of the following two images:

1. an entire vertical graph, for which $\pi \circ X : \Sigma \rightarrow \mathcal{M}^2(\kappa)$ is a diffeomorphism; or
2. a vertical cylinder $\pi^{-1}(\gamma)$, where $\gamma \subset \mathcal{M}^2(\kappa)$ is a complete curve of constant geodesic curvature $2H$, with the sign determined by the chosen orientation.

1.48.2. Historical background, literature, and current progress

Verdict: unresolved.

The exact classification is still open. In the included case $\kappa = 0$ (hence $\tau \neq 0$ and $H = 0$), it becomes the conjecture that every complete stable minimal surface in the Riemannian Heisen-

berg group Nil_3 is either an entire vertical graph or a vertical plane. Meeks, Pérez, and Tinaglia state precisely this assertion as Conjecture 15.28, the “Strong-Halfspace Conjecture in Nil_3 .” No later proof or counterexample covering this case, and therefore no resolution of the full $\mathbb{E}^3(\kappa, \tau)$ statement, was found.

Established progress is substantial but conditional:

- Isabel Fernández and Pablo Mira, “Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries,” *Proceedings of the International Congress of Mathematicians 2010*, Vol. II, World Scientific, 2011, pp. 830–861, DOI https://doi.org/10.1142/9789814324359_0076, arXiv:1004.4752. Section 6.4 formulates the full assertion as a conjecture and notes the parabolic case.
- José M. Manzano, Joaquín Pérez, and M. Magdalena Rodríguez, “Parabolic stable surfaces with constant mean curvature,” *Calculus of Variations and Partial Differential Equations* **42** (2011), 137–152, DOI <https://doi.org/10.1007/s00526-010-0383-6>, arXiv:0910.5373. Corollary 3.3 proves the requested classification when the complete stable surface is parabolic: at critical mean curvature it is an entire vertical graph or the stated vertical cylinder.
- José M. Manzano and M. Magdalena Rodríguez, “On Complete Constant Mean Curvature Vertical Multigraphs in $\mathbb{E}(\kappa, \tau)$,” *Journal of Geometric Analysis* **25** (2015), 336–346, DOI <https://doi.org/10.1007/s12220-013-9431-8>, arXiv:1206.1578. They prove that a complete CMC immersion transverse to the vertical Killing field is a vertical graph; together with the critical-CMC multigraph theorem cited there, it is entire. This does not prove that an arbitrary stable surface is transverse or vertical.
- José M. Manzano and Barbara Nelli, “Height and Area Estimates for Constant Mean Curvature Graphs in $\mathbb{E}(\kappa, \tau)$ -Spaces,” *Journal of Geometric Analysis* **27** (2017), 3441–3473, DOI <https://doi.org/10.1007/s12220-017-9810-7>, arXiv:1504.05239. Corollary 4 obtains the cylinder conclusion for $\tau \neq 0$ under the additional hypothesis that the squared vertical angle function is integrable; the extra hypothesis is not part of the present problem.
- William H. Meeks III, Joaquín Pérez, and Giuseppe Tinaglia, “Constant Mean Curvature Surfaces,” *Surveys in Differential Geometry* **21** (2016), 179–287, DOI <https://doi.org/10.4310/SDG.2016.v21.n1.a6>. Conjecture 15.28 restates the unresolved Nil_3 case exactly.

Thus the parabolic, transverse, and certain integrable-angle cases are known. The unresolved part consists of general complete stable critical-CMC immersions of hyperbolic conformal type for which stability alone has not been shown to force the vertical Jacobi function to vanish identically or have a fixed sign.

1.48.3. Remaining open problems recorded by the database

1.48.3.1. Open problem 1

Let Nil_3 be the three-dimensional Heisenberg group equipped with its standard left-invariant Riemannian metric, and let $\pi : \text{Nil}_3 \rightarrow \mathbb{R}^2$ be its canonical Riemannian submersion. Classify all oriented minimal immersions $X : \Sigma \rightarrow \text{Nil}_3$ whose induced metric is complete and whose Jacobi operator $L = \Delta + \|B\|^2 + \text{Ric}(N, N)$ is stable, in the sense that $-\int_{\Sigma} f L f \, dA \geq 0$ for every $f \in C_c^\infty(\Sigma)$. Prove that every such immersion is either an entire vertical graph, meaning that $\pi \circ X : \Sigma \rightarrow \mathbb{R}^2$ is a diffeomorphism, or a vertical

plane $\pi^{-1}(\ell)$ over a straight line $\ell \subset \mathbb{R}^2$.

Reference supplied by the database. William H. Meeks III, Joaquín Pérez, and Giuseppe Tinaglia, “Constant Mean Curvature Surfaces,” *Surveys in Differential Geometry* **21** (2016), 179–287, Conjecture 15.28 (“Strong-Halfspace Conjecture in Nil_3 ”), DOI: <https://doi.org/10.4310/SDG.2016.v21.n1.a6>.

1.48.3.2. Open problem 2

Let Nil_3 be the three-dimensional Heisenberg group with its standard left-invariant Riemannian metric, and let $\pi : \text{Nil}_3 \rightarrow \mathbb{R}^2$ be its canonical Riemannian submersion. Prove that no entire minimal vertical graph $X : \Sigma \rightarrow \text{Nil}_3$, for which $\pi \circ X : \Sigma \rightarrow \mathbb{R}^2$ is a diffeomorphism, has at most quadratic intrinsic area growth; explicitly, prove that there do not exist $p \in \Sigma$ and $C > 0$ such that $\text{Area}(B_\Sigma(p, R)) \leq CR^2$ for every $R \geq 1$, where $B_\Sigma(p, R)$ is the radius- R geodesic ball in the induced metric.

Reference supplied by the database. José M. Manzano, Joaquín Pérez, and M. Magdalena Rodríguez, “Parabolic Stable Surfaces with Constant Mean Curvature,” *Calculus of Variations and Partial Differential Equations* **42** (2011), 137–152, Remark 3.4, DOI: <https://doi.org/10.1007/s00526-010-0383-6>; arXiv:0910.5373.

1.48.4. Primary-source pointer

The originating ICM record is *Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries* (2010). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=836>.

1.49. 49. Locally Homogeneous Geometric Manifolds [2010-geom-317]

Record: 2010-geom-317

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2010: Locally Homogeneous Geometric Manifolds

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=723>

1.49.1. Problem statement

Let $n \geq 1$, and let M be a closed, connected, oriented n -dimensional affine manifold; that is, M has an atlas of coordinate charts into $\mathbb{R}^n$ whose transition maps are locally restrictions of elements of the affine group $\text{Aff}(\mathbb{R}^n) = \text{GL}(n, \mathbb{R}) \ltimes \mathbb{R}^n$. Let $\text{dev} : \tilde{M} \rightarrow \mathbb{R}^n$ be a developing map on the universal cover, and let $h : \pi_1(M) \rightarrow \text{Aff}(\mathbb{R}^n)$ be its holonomy representation, so that $\text{dev}(\gamma x) = h(\gamma) \text{dev}(x)$ for all $\gamma \in \pi_1(M)$ and $x \in \tilde{M}$. If $L : \text{Aff}(\mathbb{R}^n) \rightarrow \text{GL}(n, \mathbb{R})$ denotes the homomorphism taking an affine transformation to its linear part, prove or disprove that dev is a diffeomorphism from $\tilde{M}$ onto $\mathbb{R}^n$ (equivalently, the induced flat torsion-free affine connection on M is geodesically complete) if and only if $(L \circ h)(\pi_1(M)) \subseteq \text{SL}(n, \mathbb{R})$ (equivalently, M admits a nowhere-vanishing parallel volume form).

1.49.2. Historical background, literature, and current progress

The exact problem is **unresolved**. No accepted theorem or counterexample covers all closed oriented affine manifolds in all dimensions. In particular, the implication from parallel volume to completeness remains open even in real dimension (3); in higher dimensions the converse implication for arbitrary complete compact affine manifolds is also not known in full generality (it would follow from the general Auslander conjecture, which is known only under additional hypotheses and in bounded dimension).

Established progress includes the following.

- David Fried, William M. Goldman, and Morris W. Hirsch, "Affine manifolds with nilpotent holonomy," *Commentarii Mathematici Helvetici* **56** (1981), 487–523, DOI <https://doi.org/10.1007/BF02566225>. Theorem 6.11 (and Theorem 8.4) proves the full equivalence for compact affine manifolds with nilpotent affine holonomy.
- William M. Goldman and Morris W. Hirsch, "Affine manifolds and orbits of algebraic groups," *Transactions of the American Mathematical Society* **295** (1986), 175–198, DOI <https://doi.org/10.1090/S0002-9947-1986-0831195-0>. They prove completeness implies parallel volume for virtually solvable holonomy, and prove the converse when the solvable holonomy has sufficiently small solvable rank, as well as for homogeneous affine manifolds and manifolds with a rational Riemannian metric.
- Yves Carrière, "Autour de la conjecture de L. Markus sur les variétés affines," *Inventiones Mathematicae* **95** (1989), 615–628, DOI <https://doi.org/10.1007/BF013938>

94. The discompactness theorem proves the parallel-volume-to-completeness implication when the linear holonomy has discompactness at most (1); this includes compact flat Lorentzian manifolds and settles the conjecture in dimension (2).
- Kyeonghee Jo and In Kang Kim, “On the Markus conjecture in convex case,” *Annals of Global Analysis and Geometry* **60** (2021), 911–940, DOI <https://doi.org/10.1007/s10455-021-09796-z>, arXiv:1809.07918. Theorem 1.1 proves completeness for compact convex affine manifolds with parallel volume when the identity component of the affine automorphism group of the developing domain has nonempty boundary limit set; Theorem 1.2 covers every convex case of dimension at most (5).
 - Raphaël V. Alexandre, “Affine three-manifolds with centralizing holonomy,” preprint (2023), arXiv:2303.16665, [direct URL](https://arxiv.org/abs/2303.16665). Theorem 3.1 proves completeness in dimension (3) when the unimodular affine holonomy centralizes a nonidentity volume-preserving affine transformation; its corollaries cover holonomy with nontrivial center, nondiscrete automorphism group, and oriented Seifert fibrations. The introduction explicitly records that the unrestricted dimension-(3) case remains open.
 - Malek Hanounah and Lilia Mehidi, “On completeness of foliated structures, and null Killing fields,” *Mathematische Annalen* **392** (2025), 619–658, DOI <https://doi.org/10.1007/s00208-025-03095-9>. This proves further unimodular lightlike holonomy cases (Theorems 1.10–1.11) but explicitly leaves the compact pseudo-Riemannian lightlike groups $\mathit{mathsf{L}}_{\mathit{mathsf{fu}}}(n, 1)$ open.

For the other implication, Herbert Abels, Gregory A. Margulis, and Gregory A. Soifer, “The Auslander conjecture for dimension less than 7,” preprint (revised 2020), arXiv:2011.12788, [direct URL](https://arxiv.org/abs/2011.12788), proves that affine crystallographic groups in dimensions (<7) are virtually solvable. Combined with Goldman–Hirsch, this gives “complete implies parallel volume” in those dimensions, but it does not cover arbitrary dimension.

A 2025 first-version preprint by Mihail Cocos, “A Riemannian Characterization of Compact Affine Manifolds with Parallel Volume,” arXiv:2506.14778, claims the open implication. It is not an accepted resolution: the crucial assertion in Section 3.3 that a tensor conformal to an arbitrary auxiliary metric is parallel is merely asserted and does not follow from the preceding local construction. Later refereed literature above continues to identify special cases as open. Thus the remaining gap is both the unrestricted parallel-volume-to-completeness implication (already in dimension (3)) and, for the stated all-dimensional equivalence, the unrestricted high-dimensional converse.

1.49.3. Remaining open problems recorded by the database

1.49.3.1. Open problem 1

Let M be a closed, connected, oriented 3-dimensional affine manifold with developing map $\text{dev} : \widetilde{M} \rightarrow \mathbb{R}^3$ and holonomy representation $h : \pi_1(M) \rightarrow \text{Aff}(\mathbb{R}^3)$. If every transformation in $h(\pi_1(M))$ has linear part in $\text{SL}(3, \mathbb{R})$, prove or disprove that dev is a diffeomorphism from $\widetilde{M}$ onto $\mathbb{R}^3$.

Reference supplied by the database. Raphaël V. Alexandre, “Affine three-manifolds with centralizing holonomy,” preprint, 2023, Introduction (the dimension-three Markus conjecture is explicitly identified as still open), arXiv:2303.16665, <https://arxiv.org/abs/2303.16665>.

1.49.3.2. Open problem 2

Let $n \geq 6$, and let M be a closed, connected, oriented n -dimensional affine manifold whose developing map is a diffeomorphism $\text{dev} : \tilde{M} \rightarrow \Omega$ onto a convex domain $\Omega \subseteq \mathbb{R}^n$. If the linear holonomy of M lies in $\text{SL}(n, \mathbb{R})$, prove or disprove that $\Omega = \mathbb{R}^n$.

Reference supplied by the database. Kyeonghee Jo and Inkang Kim, “On the Markus conjecture in convex case,” *Annals of Global Analysis and Geometry* 60 (2021), 911–940, Introduction and Theorems 1.1–1.2 (the unrestricted convex case is identified as unresolved, while dimensions at most 5 are proved), DOI: 10.1007/s10455-021-09796-z; arXiv:1809.07918, <https://arxiv.org/abs/1809.07918>.

1.49.3.3. Open problem 3

For $n \geq 1$, let

$$G_n = \left\{ x \mapsto \begin{pmatrix} 1 & \alpha^\top \\ 0 & A \end{pmatrix} x + b : \alpha \in \mathbb{R}^{n+1}, A \in \text{SO}(n, 1), b \in \mathbb{R}^{n+2} \right\} \subseteq \text{Aff}(\mathbb{R}^{n+2}),$$

where $\text{SO}(n, 1)$ preserves a nondegenerate quadratic form of signature $(n, 1)$. Prove or disprove that every closed, connected, oriented manifold with a $(G_n, \mathbb{R}^{n+2})$ -structure is complete; equivalently, prove or disprove that its developing map is a diffeomorphism from its universal cover onto $\mathbb{R}^{n+2}$.

Reference supplied by the database. Malek Hanounah and Lilia Mehidi, “On completeness of foliated structures, and null Killing fields,” *Mathematische Annalen* 392 (2025), 619–658, Section 4.1 (completeness of compact $(L_u(n, 1), \mathbb{R}^{n+2})$ -manifolds is explicitly identified as open), DOI: 10.1007/s00208-025-03095-9, <https://doi.org/10.1007/s00208-025-03095-9>.

1.49.3.4. Open problem 4

Let B be a compact, connected n -dimensional manifold with an affine atlas whose transition maps are locally of the form $x \mapsto Ax + b$, where $A \in \text{GL}(n, \mathbb{Z})$ and $b \in \mathbb{Z}^n$. Determine whether every such integral-integral affine manifold is affinely isomorphic to a quotient $\mathbb{R}^n/\Gamma$, where Γ is a discrete group of integral-integral affine transformations acting freely and properly discontinuously on $\mathbb{R}^n$.

Reference supplied by the database. Mark D. Hamilton, Yael Karshon, and Takahiko Yoshida, “Integral-integral affine geometry, geometric quantization, and Riemann–Roch,” *Journal of Geometry and Physics* 222 (2026), article 105745, Section 3 (unnumbered question following the examples), DOI: 10.1016/j.geomphys.2025.105745; arXiv:2411.10348, <https://arxiv.org/abs/2411.10348>.

1.49.4. Primary-source pointer

The originating ICM record is *Locally Homogeneous Geometric Manifolds* (2010). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=723>.

1.50. 50. Quasiregular Mappings, Curvature & Dynamics [2010-geom-318]

Record: 2010-geom-318

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2010: Quasiregular Mappings, Curvature & Dynamics

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.3/ICM2010.3.pdf#page=260>

1.50.1. Problem statement

Let $n \geq 2$, let $\mathbb{H}^n$ be hyperbolic n -space, and let

$$M = \mathbb{H}^n / \Gamma_M \quad \text{and} \quad N = \mathbb{H}^n / \Gamma_N$$

be hyperbolic n -manifolds, where $\Gamma_M, \Gamma_N \leq \text{Isom}^+(\mathbb{H}^n)$ are torsion-free, discrete, non-elementary groups. A map $f : M \rightarrow N$ is quasiregular if it is nonconstant, belongs locally to the Sobolev class $W^{1,n}$, and there is a constant $K < \infty$ such that

$$\|Df(x)\|^n \leq K J_f(x)$$

for almost every $x \in M$, where $\|Df(x)\|$ is the operator norm of the differential and $J_f(x)$ is its Jacobian determinant with respect to the hyperbolic metrics. Prove or disprove that there is no quasiregular map $f : M \rightarrow N$ that is simultaneously proper (the preimage of every compact subset of N is compact), branched (not locally injective), and π_1 -injective (the induced homomorphism $f_* : \pi_1(M) \rightarrow \pi_1(N)$ is injective).

1.50.2. Historical background, literature, and current progress

The exact problem remains **unresolved** as of 4 August 2026. The known theorems rule out such maps in substantial cases, but no citable result was found that covers arbitrary hyperbolic quotients in every dimension ($n \geq 2$).

The main partial result is Martin R. Bridson, Aimo Hinkkanen, and Gaven J. Martin, “Quasiregular self-mappings of manifolds and word hyperbolic groups,” *Compositio Mathematica* **143** (2007), 1613–1622, <https://doi.org/10.1112/S0010437X07003028>. Theorem 3.1 proves that a proper quasiregular map with finite-kernel induced homomorphism is a finite-sheeted covering whenever at least one limit set has topological dimension at least $(n-2)$, or at least one limit set has Hausdorff dimension $(n-1)$. Consequently, Corollary 3.2 settles all closed hyperbolic (n) -manifolds, Corollary 3.3 settles the stated convex-cocompact case with more than one non-cuspidal boundary component, and Theorem 3.4 settles all hyperbolic (3) -manifolds with non-elementary fundamental group. Theorem 3.1 also covers $(n=2)$, since a non-elementary limit set is nonempty and hence has topological dimension at least $(0=n-2)$.

Gaven J. Martin, “Quasiregular Mappings, Curvature & Dynamics,” in *Proceedings of the International Congress of Mathematicians 2010 (Hyderabad)*, World Scientific, 2011, 260–276, <https://doi.org/10.1112/S0010437X07003028>.

[//doi.org/10.1142/9789814324359_0105](https://doi.org/10.1142/9789814324359_0105), restates these results and explicitly presents the unrestricted assertion as only “possibly true.” It supplies no theorem beyond the cases above.

A later related result is Aapo Kauranen, Rami Luisto, and Ville Tengvall, “On proper branched coverings and a question of Vuorinen,” *Bulletin of the London Mathematical Society* **54** (2022), 145–160, <https://doi.org/10.1112/blms.12565>, [arXiv:1904.12645](https://arxiv.org/abs/1904.12645). It proves injectivity results for proper branched coverings of Euclidean balls in dimension (3) and in the unbranched case, and notes the hyperbolic-manifold work as related. Its hypotheses do not imply the assertion here and it does not treat the remaining hyperbolic cases.

Thus the unresolved gap is the existence or nonexistence of a proper, branched, π_1 -injective quasiregular map between arbitrary hyperbolic (n)-manifolds for ($n \geq 4$) when neither limit set satisfies the topological- or Hausdorff-dimension hypotheses of Bridson–Hinkkanen–Martin. No substantive later progress on that exact gap was found.

1.50.3. Remaining open problems recorded by the database

1.50.3.1. Open problem 1

Let $n \geq 4$, let $\mathbb{H}^n$ be hyperbolic n -space, and let $M = \mathbb{H}^n/\Gamma_M$ and $N = \mathbb{H}^n/\Gamma_N$ be hyperbolic n -manifolds, where $\Gamma_M, \Gamma_N \leq \text{Isom}^+(\mathbb{H}^n)$ are torsion-free, discrete, non-elementary groups. A map $f : M \rightarrow N$ is quasiregular if it is nonconstant, belongs locally to $W^{1,n}$, and satisfies $\|Df(x)\|^n \leq KJ_f(x)$ almost everywhere for some $K < \infty$. Prove or disprove that there is no quasiregular map $f : M \rightarrow N$ that is simultaneously proper, branched, and π_1 -injective.

Reference supplied by the database. Martin R. Bridson, Aimo Hinkkanen, and Gaven J. Martin, “Quasiregular self-mappings of manifolds and word hyperbolic groups,” *Compositio Mathematica* **143** (2007), no. 6, 1613–1622, unnumbered question following Corollary 3.3 and preceding Theorem 3.4, <https://doi.org/10.1112/S0010437X07003028>.

1.50.4. Primary-source pointer

The originating ICM record is *Quasiregular Mappings, Curvature & Dynamics* (2010). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.3/ICM2010.3.pdf#page=260>.

1.51. 51. Reconstruction of Collapsed Manifolds [2010-geom-319]

Record: 2010-geom-319

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2010: Reconstruction of Collapsed Manifolds

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=905>

1.51.1. Problem statement

Let n be a positive integer and let $D > 0$. Let (M_i) be a sequence of closed n -dimensional Riemannian manifolds satisfying

$$\sec(M_i) \geq -1 \quad \text{and} \quad \text{diam}(M_i) \leq D,$$

and suppose that $d_{GH}(M_i, X) \rightarrow 0$ for a compact Alexandrov space X of curvature at least -1 with $\dim X < n$.

A stratified singular fibration over X means a continuous surjection $f : M_i \rightarrow X$ for which there is a finite filtration by closed subsets

$$X = S_0 \supset S_1 \supset \cdots \supset S_\ell = \emptyset$$

such that every stratum $S_j \setminus S_{j+1}$ is a topological manifold and the restriction

$$f^{-1}(S_j \setminus S_{j+1}) \rightarrow S_j \setminus S_{j+1}$$

is a locally trivial fiber bundle; the fiber type is locally constant along each stratum but may change from one stratum to another.

Prove that, for every sufficiently large i , there exists a stratified singular fibration

$$f_i : M_i \rightarrow X.$$

1.51.2. Historical background, literature, and current progress

The assertion is **unresolved**. No citable result was found that constructs the stated finite-stratum, locally trivial singular fibration for every collapsed sequence under only a lower sectional-curvature bound and an upper diameter bound. Recent literature continues to describe such a structure over an arbitrary singular Alexandrov limit as expected rather than proved.

Key evidence and partial progress:

- Takao Yamaguchi, "Reconstruction of Collapsed Manifolds," *Proceedings of the International Congress of Mathematicians 2010, Hyderabad*, Vol. III, 2010, pp. 905–917 ([direct proceedings PDF](#)). Conjecture 4.7 proposes a singular fibration in a generalized sense in precisely the general-dimensional, lower-sectional-curvature collapse setting. The paper

records singular-fiber descriptions in dimensions three and four, but does not prove the conjecture in arbitrary dimension.

- Takao Yamaguchi, “Collapsing and Pinching under a Lower Curvature Bound,” *Annals of Mathematics* 133 (1991), 317–357, DOI [10.2307/2946563](https://doi.org/10.2307/2946563). The fibration theorem gives a locally trivial bundle when the limit has the required regularity (in particular, no relevant singular points). It does not handle arbitrary singular Alexandrov limits.
- Takashi Shioya and Takao Yamaguchi, “Collapsing Three-Manifolds under a Lower Curvature Bound,” *Journal of Differential Geometry* 56 (2000), 1–66, DOI [10.4310/jdg/1090347524](https://doi.org/10.4310/jdg/1090347524); and Takao Yamaguchi, “Collapsing 4-Manifolds under a Lower Curvature Bound,” preprint, 2012, [arXiv:1205.0323](https://arxiv.org/abs/1205.0323). These describe the topology and generalized singular-fiber structures in the low-dimensional orientable cases, but they do not cover unrestricted dimension and hypotheses in the present statement.
- Tadashi Fujioka, “A Fibration Theorem for Collapsing Sequences of Alexandrov Spaces,” *Journal of Topology and Analysis* 15 (2023), 265–298, DOI [10.1142/S179352532150028X](https://doi.org/10.1142/S179352532150028X). This proves local triviality under quantitative weak-singularity assumptions together with a lower bound on volumes of spaces of directions. Those additional assumptions are absent from the problem.
- Tadashi Fujioka, “Collapsing to Alexandrov Spaces with Isolated Mild Singularities,” *Differential Geometry and its Applications* 85 (2022), Article 101951, DOI [10.1016/j.difgeo.2022.101951](https://doi.org/10.1016/j.difgeo.2022.101951). Theorem 1.1 gives a locally trivial fibration when all but finitely many points of the limit are strained and each exceptional point is weakly strained. This is a stronger conclusion for a restricted class of limits, not the arbitrary singular limit required here.
- Tadashi Fujioka, “Application of Good Coverings to Collapsing Alexandrov Spaces,” *Pacific Journal of Mathematics* 316 (2022), 335–365, DOI [10.2140/pjm.2022.316.335](https://doi.org/10.2140/pjm.2022.316.335). For limits with no proper extremal subsets, it constructs long exact homotopy and cohomological spectral sequences and extends corresponding information to primitive extremal subsets. These homotopy-level conclusions do not construct the global stratified locally trivial map required in the statement.
- Tadashi Fujioka, “Euler Characteristics of Collapsing Alexandrov Spaces,” *Arnold Mathematical Journal* 10 (2024), 463–472, DOI [10.1007/s40598-024-00246-9](https://doi.org/10.1007/s40598-024-00246-9). This proves the expected stratum-by-stratum Euler-characteristic formula using regular fibers, while explicitly noting that there is still no general-dimensional theory and that a global singular fibration over arbitrary singular strata remains expected.

Thus the literature supplies ordinary fibrations for regular or mildly singular limits, homotopy and invariant-level substitutes over broader extremal strata, and low-dimensional classifications. The missing case is exactly the unrestricted singular Alexandrov limit in arbitrary dimension, where compatibility and local triviality of fibers across a finite stratification have not been established.

1.51.3. Remaining open problems recorded by the database

1.51.3.1. Open problem 1

Let n and k be positive integers with $k < n$, and let $D > 0$. Let (M_i) be a sequence of closed n -dimensional Riemannian manifolds satisfying

$$\sec(M_i) \geq -1 \quad \text{and} \quad \text{diam}(M_i) \leq D,$$

and suppose that $d_{GH}(M_i, X) \rightarrow 0$ for a compact k -dimensional Alexandrov space X of curvature at least -1 . Assume that every point $p \in X$ is weakly k -strained, meaning that its space of directions $\Sigma_p X$ contains $k+1$ directions whose pairwise angles are all greater than $\pi/2$.

Prove that, for every sufficiently large i , there exists a locally trivial fiber bundle

$$f_i : M_i \rightarrow X.$$

Reference supplied by the database. Tadashi Fujioka, “Collapsing to Alexandrov Spaces with Isolated Mild Singularities,” *Differential Geometry and its Applications* 85 (2022), Article 101951, Problem 1.3, DOI: <https://doi.org/10.1016/j.difgeo.2022.101951>; arXiv:2108.11030.

1.51.3.2. Open problem 2

Let $n > 2$ be an integer and let $D > 0$. Let (M_i) be a sequence of closed n -dimensional Riemannian manifolds satisfying

$$\sec(M_i) \geq -1 \quad \text{and} \quad \text{diam}(M_i) \leq D,$$

and suppose that $d_{GH}(M_i, X) \rightarrow 0$ for a compact two-dimensional Alexandrov space X of curvature at least -1 . Assume that X has no proper extremal subsets, equivalently, that for every $p \in X$ the space of directions $\Sigma_p X$ is a metric circle of length greater than π .

Prove that, for every sufficiently large i , there exists a locally trivial fiber bundle

$$f_i : M_i \rightarrow X.$$

Reference supplied by the database. Tadashi Fujioka, “Collapsing to Alexandrov Spaces with Isolated Mild Singularities,” *Differential Geometry and its Applications* 85 (2022), Article 101951, Problem 1.4, DOI: <https://doi.org/10.1016/j.difgeo.2022.101951>; arXiv:2108.11030.

1.51.4. Primary-source pointer

The originating ICM record is *Reconstruction of Collapsed Manifolds* (2010). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=905>.

1.52. 52. Reconstruction of Collapsed Manifolds [2010-geom-320]

Record: 2010-geom-320

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2010: Reconstruction of Collapsed Manifolds

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=905>

1.52.1. Problem statement

Prove or disprove the following assertion: for every positive integer n and every closed n -dimensional Riemannian manifold M with nonnegative sectional curvature,

$$\tau_n(M) \leq 2^n.$$

Here a *contractible essential covering* of M is specified by a finite rooted tree whose root represents M and whose other vertices are labeled by metric balls in M , subject to the following conditions. The balls labeling the children of the root cover M . If a nonterminal vertex is labeled by a ball B , then there is a smaller concentric metric ball $\tilde{B} \subset B$ that is isotopic to B as a subset of M , and the balls labeling the children of that vertex are contained in B and cover $\tilde{B}$. Every ball labeling a terminal vertex is contractible. The depth of such a covering is the maximum number of edges in a path from the root to a terminal vertex, and, for a positive integer m , $\tau_m(M)$ is the minimum number of terminal vertices among all contractible essential coverings of M having depth at most m .

1.52.2. Historical background, literature, and current progress

The problem is **unresolved**. The literature proves that $\tau_n(M)$ is bounded by some constant depending only on n , but it does not establish the proposed sharp constant 2^n for every closed nonnegatively curved Riemannian n -manifold and does not provide a counterexample.

- Takao Yamaguchi, "Reconstruction of Collapsed Manifolds," *Proceedings of the International Congress of Mathematicians 2010*, Vol. II, Hindustan Book Agency, 2010, pp. 899–913, DOI: [10.1142/9789814324359_0079](https://doi.org/10.1142/9789814324359_0079). Corollary 5.4 proves the weaker uniform estimate $\tau_n(M) \leq C_n$ for some dimension-dependent constant. Conjecture 5.5 is the exact assertion in the statement. The paper also verifies $\tau_n \leq 2^n$ for the displayed families of increasingly collapsed flat tori and almost-flat nilmanifolds, while showing that depth less than n cannot be uniformly sufficient for those families.
- Takao Yamaguchi, "Collapsing and Essential Coverings," preprint, 2012, [arXiv:1205.0441](https://arxiv.org/abs/1205.0441). Theorem 1.1 and Theorem 4.4 prove the more general non-sharp bound $\tau_n(M) \leq N(n, D)$ for compact n -dimensional Alexandrov spaces with curvature at least -1 and diameter at most D . Conjecture 4.8 explicitly retains $\tau_n(M) \leq 2^n$ as a conjecture even for the broader class of compact nonnegatively curved Alexandrov spaces; Remark 4.6 confirms that the terminal balls may be taken to be n -disks in the Riemannian case.

- Tadashi Fujioka, “Uniform Boundedness on Extremal Subsets in Alexandrov Spaces,” *Journal of Geometric Analysis* 35 (2025), Article 46, DOI: [10.1007/s12220-024-01864-7](https://doi.org/10.1007/s12220-024-01864-7), [arXiv:1809.00603](https://arxiv.org/abs/1809.00603). Theorem 3.7 reviews and supplies a self-contained account of Yamaguchi’s uniform essential-covering bound. Remark 4.5 still identifies the sharp 2^n estimate as Yamaguchi’s conjecture; the paper’s new uniform bounds for extremal subsets do not prove that conjecture.

Thus the strongest general result remains existence of an unspecified dimension-dependent bound. The unresolved gap is to prove the sharp bound 2^n for every manifold in the statement or to construct a nonnegatively curved closed Riemannian counterexample.

1.52.3. Remaining open problems recorded by the database

No separate descendant problem is listed in the website’s “Some remaining open problems” field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

1.52.4. Primary-source pointer

The originating ICM record is *Reconstruction of Collapsed Manifolds* (2010). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=905>.

1.53. 53. Reconstruction of Collapsed Manifolds [2010-geom-321]

Record: 2010-geom-321

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2010: Reconstruction of Collapsed Manifolds

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=905>

1.53.1. Problem statement

Prove or disprove that, for every positive integer n , every connected, compact n -dimensional Riemannian manifold M without boundary and with nonnegative sectional curvature, and every field F ,

$$\sum_{i=0}^n \dim_F H_i(M; F) \leq 2^n,$$

where $H_i(M; F)$ denotes the i th singular homology group of M with coefficients in F .

1.53.2. Historical background, literature, and current progress

The exact statement is **unresolved**. No proof or counterexample is known for arbitrary closed nonnegatively curved manifolds in all dimensions and with coefficients in every field. In particular, the literature still treats the sharp value 2^n as conjectural; results over $\mathbb{Q}$ or under symmetry assumptions do not cover possible torsion detected by positive-characteristic fields.

- Mikhael Gromov, “Curvature, diameter and Betti numbers,” *Commentarii Mathematici Helvetici* **56** (1981), 179–195, [direct PDF](#). Gromov proves that the total Betti number over an arbitrary field is bounded by some constant depending only on n . He explicitly identifies 2^n , the total Betti number of the flat n -torus, as the expected optimal bound, but his estimate is much larger.
- Uwe Abresch, “Lower curvature bounds, Toponogov’s theorem, and bounded topology. II,” *Annales scientifiques de l’École Normale Supérieure* (4) **20** (1987), 475–502, <https://doi.org/10.24033/asens.1541>. Abresch substantially improves Gromov’s explicit arbitrary-field estimate (modern surveys quote the resulting coarse bound as 10^{10n^4}), but it remains far above 2^n and therefore does not resolve the problem.
- Takao Yamaguchi, “Reconstruction of Collapsed Manifolds,” *Proceedings of the International Congress of Mathematicians, Hyderabad 2010*, Vol. II, Hindustan Book Agency, 2010, 899–913, https://doi.org/10.1142/9789814324359_0079. After recalling Gromov’s uniform bound for every coefficient field, this survey states the precise 2^n total-Betti-number assertion as an open conjecture.
- Yves Félix, Stephen Halperin, and Jean-Claude Thomas, *Rational Homotopy Theory*, Graduate Texts in Mathematics 205, Springer, 2001, <https://doi.org/10.1007/97>

8-1-4613-0105-9. Theorem 33.2 gives $b_i(X; \mathbb{Q}) \leq \binom{n}{i}$ for a rationally elliptic space of formal dimension n , hence total rational Betti number at most 2^n . This proves the desired inequality over $\mathbb{Q}$ for rationally elliptic manifolds, but the Bott–Grove–Halperin conjecture that every simply connected closed nonnegatively curved manifold is rationally elliptic remains open, and rational ellipticity does not settle arbitrary field coefficients.

- Karsten Grove, Burkhard Wilking, and Joseph Yeager, “Almost non-negative curvature and rational ellipticity in cohomogeneity two,” *Annales de l’Institut Fourier* **69** (2019), 2921–2939, <https://doi.org/10.5802/aif.3340>. They prove rational ellipticity for simply connected almost nonnegatively curved manifolds carrying an isometric action of cohomogeneity at most two. Combined with the preceding rational-homotopy bound, this establishes the 2^n inequality over $\mathbb{Q}$ in that restricted class only.
- Oliver Goertsches and Michael Wiemeler, “Non-negatively curved GKM orbifolds,” *Mathematische Zeitschrift* **300** (2022), 2007–2036, <https://doi.org/10.1007/s00209-021-02853-0>. Corollary 1.2 proves a sharper rational total-Betti-number bound for orientable nonnegatively curved GKM_4 orbifolds and explicitly describes it as a verification of Gromov’s conjecture in that special case.
- Manuel Amann, “Bounds on Sectional Curvature and Interactions with Topology,” *Jahresbericht der Deutschen Mathematiker-Vereinigung* **123** (2021), 27–55, <https://doi.org/10.1365/s13291-020-00225-x>. This later survey continues to list the stronger degreewise bounds $b_k(M) \leq \binom{n}{k}$, which imply the stated total bound, as Gromov’s conjecture and explains that rational ellipticity would imply them.

Thus the established general result is only $\sum_i \dim_F H_i(M; F) \leq C(n)$ for a large explicit $C(n)$. Sharp 2^n bounds are known for important rationally elliptic or symmetric subclasses, chiefly with rational coefficients, but the unrestricted manifold and arbitrary-field cases remain open.

1.53.3. Remaining open problems recorded by the database

1.53.3.1. Open problem 1

Prove that, for every positive integer n and every connected, compact n -dimensional Riemannian manifold M without boundary and with positive sectional curvature,

$$\sum_{i=0}^n \dim_{\mathbb{Q}} H_i(M; \mathbb{Q}) \leq 2^n,$$

where $H_i(M; \mathbb{Q})$ denotes the i th singular homology group of M with rational coefficients.

Reference supplied by the database. Shing-Tung Yau, “Topics on Geometric Analysis,” *Surveys in Differential Geometry* **17** (2012), 459–474, Section 1, item B, [doi:10.4310/SDG.2012.v17.n1.a11](https://doi.org/10.4310/SDG.2012.v17.n1.a11), <https://doi.org/10.4310/SDG.2012.v17.n1.a11>.

1.53.4. Primary-source pointer

The originating ICM record is *Reconstruction of Collapsed Manifolds* (2010). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=905>.

1.54. 54. Riemannian Manifolds of Positive Curvature [2010-geom-322]

Record: 2010-geom-322

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2010: Riemannian Manifolds of Positive Curvature

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.1/ICM2010.1.pdf#page=442>

1.54.1. Problem statement

Prove that if (M, g) is a compact Riemannian manifold of dimension $n \geq 4$ with positive isotropic curvature, meaning that

$$R(e_1, e_3, e_1, e_3) + R(e_1, e_4, e_1, e_4) + R(e_2, e_3, e_2, e_3) + R(e_2, e_4, e_2, e_4) - 2R(e_1, e_2, e_3, e_4) > 0$$

at every point $p \in M$ for every orthonormal four-frame $\{e_1, e_2, e_3, e_4\} \subset T_p M$, where R is the Riemann curvature tensor of g , then the fundamental group $\pi_1(M)$ contains a free subgroup F of finite index.

1.54.2. Historical background, literature, and current progress

The exact statement is **unresolved**. It is the Gromov–Fraser conjecture that the fundamental group of every compact manifold with positive isotropic curvature is virtually free. Established results cover dimension 4 and, through recent Ricci-flow work and preprints, the range $n \geq 9$, but not all dimensions $n \geq 4$; the remaining gap is $5 \leq n \leq 8$.

The main established progress is as follows.

- Ailana M. Fraser, “Fundamental groups of manifolds with positive isotropic curvature,” *Annals of Mathematics* 158 (2003), 345–354, DOI <https://doi.org/10.4007/annals.2003.158.345>. For $n \geq 5$, Fraser proves that $\pi_1(M)$ contains no subgroup isomorphic to $\mathbb{Z} \oplus \mathbb{Z}$. This is a necessary consequence of virtual freeness, but it does not prove virtual freeness.
- Ailana Fraser and Jon Wolfson, “The fundamental group of manifolds of positive isotropic curvature and surface groups,” *Duke Mathematical Journal* 133 (2006), 325–334, DOI <https://doi.org/10.1215/S0012-7094-06-13325-2>. For $n \geq 5$, they strengthen Fraser’s obstruction by excluding every closed orientable surface group as a subgroup of $\pi_1(M)$. This still falls short of classifying $\pi_1(M)$ as virtually free.
- Bing-Long Chen, Siu-Hung Tang, and Xi-Ping Zhu, “Complete classification of compact four-manifolds with positive isotropic curvature,” *Journal of Differential Geometry* 91 (2012), 41–80, DOI <https://doi.org/10.4310/jdg/1343133700>. Their complete four-dimensional classification expresses every such manifold as a connected sum of spherical pieces and quotients of $S^3 \times \mathbb{R}$; consequently its fundamental group is virtually free. Thus the exact assertion is proved for $n = 4$.

- **Simon Brendle**, “Ricci flow with surgery on manifolds with positive isotropic curvature,” *Annals of Mathematics* 190 (2019), 465–559, DOI <https://doi.org/10.4007/annals.2019.190.2.2>. Brendle proves a classification for $n \geq 12$ under the additional hypothesis that there is no nontrivial incompressible $(n - 1)$ -dimensional space form.
- **Hong Huang**, “Classification of compact manifolds with positive isotropic curvature,” preprint (2023; version 8, 2025), arXiv:2305.18154. Huang removes Brendle’s incompressibility hypothesis for $n \geq 12$, obtaining a connected-sum classification that implies a finite-index free subgroup. Huang also explains that the same argument applies in dimensions $9 \leq n \leq 11$ once combined with Chen’s newer pinching estimate.
- **Zhengnan Chen**, “Manifolds with positive isotropic curvature of dimension at least nine,” preprint (2024; version 3, 2026), arXiv:2410.21078. Chen extends the needed Ricci-flow pinching estimate from $n \geq 12$ to $n \geq 9$, proving the classification in that range under Brendle’s incompressibility hypothesis and recording Huang’s removal of that hypothesis. The corresponding estimate and classification remain unavailable in dimensions $5 \leq n \leq 8$.

Gabriele La Nave’s unpublished manuscript “Macroscopic dimension and fundamental group of manifolds with positive isotropic curvature” (2013), arXiv:1303.5096 claims the full virtually-free conclusion. It has not been accepted as a resolution: subsequent work above continues to identify and prove only dimensional cases of the Gromov–Fraser conjecture, and current literature explicitly leaves the lower-dimensional range open. Accordingly, that isolated claim is not sufficient for a resolved verdict.

1.54.3. Remaining open problems recorded by the database

1.54.3.1. Open problem 1

Prove that if (M, g) is a compact Riemannian manifold of dimension $n \geq 4$ such that

$$R(e_1, e_3, e_1, e_3) + R(e_1, e_4, e_1, e_4) + R(e_2, e_3, e_2, e_3) + R(e_2, e_4, e_2, e_4) - 2R(e_1, e_2, e_3, e_4) > 0$$

at every point $p \in M$ for every orthonormal four-frame $\{e_1, e_2, e_3, e_4\} \subset T_p M$, where R is the Riemann curvature tensor of g , then there exists a finite-index subgroup $\Gamma \leq \pi_1(M)$ such that no finitely generated subgroup $H \leq \Gamma$ has exactly one end, where the number of ends of H is the number of ends of any Cayley graph of H with respect to a finite generating set.

Reference supplied by the database. Ailana Fraser and Jon Wolfson, “The Fundamental Group of Manifolds of Positive Isotropic Curvature and Surface Groups,” *Duke Mathematical Journal* 133 (2006), no. 2, 325–334, weaker conjecture in the Introduction, DOI: <https://doi.org/10.1215/S0012-7094-06-13325-2>.

1.54.3.2. Open problem 2

Prove that for every $\alpha > 0$ there exists a constant $C(\alpha) > 0$ with the following property. Let (M, g) be a complete Riemannian manifold of dimension $n \geq 4$ whose sectional curvature is bounded above and whose injectivity radius is bounded below by a positive constant. Suppose that

$$R(e_1, e_3, e_1, e_3) + R(e_1, e_4, e_1, e_4) + R(e_2, e_3, e_2, e_3) + R(e_2, e_4, e_2, e_4) - 2R(e_1, e_2, e_3, e_4) \geq \alpha$$

at every point of M for every orthonormal four-frame $\{e_1, e_2, e_3, e_4\}$. If γ is any smooth simple closed curve in M that bounds a disk in M , then $\text{fillrad}(\gamma) \leq C(\alpha)$, where

$$\text{fillrad}(\gamma) = \sup\{r > 0 : \gamma \text{ does not bound a disk in } N_r(\gamma)\}$$

and $N_r(\gamma) = \{x \in M : d(x, \gamma) \leq r\}$.

Reference supplied by the database. Mohan Ramachandran and Jon Wolfson, “Fill Radius and the Fundamental Group,” *Journal of Topology and Analysis* 2 (2010), no. 1, 99-107, Conjecture 1.2, DOI: <https://doi.org/10.1142/S1793525310000288>.

1.54.4. Primary-source pointer

The originating ICM record is *Riemannian Manifolds of Positive Curvature* (2010). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.1/ICM2010.1.pdf#page=442>.

1.55. 55. Volume Comparison via Boundary Distances [2010-geom-323]

Record: 2010-geom-323

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2010: Volume Comparison via Boundary Distances

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=775>

1.55.1. Problem statement

Prove that the following holds. Let (M, g) be a compact orientable n -dimensional Riemannian manifold with boundary such that ∂M is strictly convex, every geodesic segment in M realizes the Riemannian distance between its endpoints, and no geodesic in M has conjugate points. Let (M', g') be any compact orientable n -dimensional Riemannian manifold with boundary, and let $\phi : \partial M \rightarrow \partial M'$ be a fixed identification. If

$$d_{M'}(\phi(x), \phi(y)) \geq d_M(x, y) \quad \text{for all } x, y \in \partial M,$$

where d_M and $d_{M'}$ denote the Riemannian distance functions induced on M and M' , respectively, prove that

$$\text{Vol}(M', g') \geq \text{Vol}(M, g).$$

Moreover, prove that equality holds only if there exists a Riemannian isometry $F : (M, g) \rightarrow (M', g')$ satisfying $F|_{\partial M} = \phi$.

1.55.2. Historical background, literature, and current progress

The exact problem is **unresolved**. It is Sergei Ivanov's strict (or unique) minimal-filling conjecture for simple Riemannian manifolds. None of the known results covers an arbitrary simple metric together with every compact orientable competitor of arbitrary topology, and the equality-rigidity conclusion is likewise unknown in that generality.

Key evidence and partial progress:

- Sergei Ivanov, "Volume Comparison via Boundary Distances," *Proceedings of the International Congress of Mathematicians 2010*, vol. II, 2010, pp. 769–784, Conjecture 1.6⁺, DOI https://doi.org/10.1142/9789814324359_0073, arXiv:1004.2505. This source states the exact assertion as the conjecture that every simple manifold is a unique minimal filling, up to an isometry fixing the boundary.
- Dmitri Burago and Sergei Ivanov, "Boundary Rigidity and Filling Volume Minimality of Metrics Close to a Flat One," *Annals of Mathematics* 171 (2010), 1183–1211, Theorem 2, DOI <https://doi.org/10.4007/annals.2010.171.1183>. This proves the full volume inequality and its equality-rigidity conclusion when the metric is sufficiently C^2 -close to a Euclidean metric, with no topological restriction on the orientable competitor.

- Dmitri Burago and Sergei Ivanov, “Area Minimizers and Boundary Rigidity of Almost Hyperbolic Metrics,” *Duke Mathematical Journal* 162 (2013), 1205–1248, Theorem 1.6, DOI <https://doi.org/10.1215/00127094-2142529>, arXiv:1011.1570. This proves strict minimal filling, hence the exact inequality and equality case, for metrics sufficiently close to a real hyperbolic metric.
- Sergei Ivanov, “Local Monotonicity of Riemannian and Finsler Volume with Respect to Boundary Distances,” *Geometriae Dedicata* 164 (2013), 83–96, Theorem 1, DOI <https://doi.org/10.1007/s10711-012-9760-y>, arXiv:1109.4091. For every simple Riemannian metric, this proves the volume inequality when both metrics lie in a sufficiently small C^∞ neighborhood on the same disc. It does not treat arbitrary competitors and gives no general equality-rigidity conclusion.
- Sergei Ivanov, “Filling Minimality of Finslerian 2-Discs,” *Proceedings of the Steklov Institute of Mathematics* 273 (2011), 176–190, DOI <https://doi.org/10.1134/S0081543811040079>, arXiv:0910.2257. In dimension two this proves global minimality for competitors homeomorphic to the disc. It explicitly leaves competitors of arbitrary genus outside its result.
- Yuping Ruan, “Filling Volume Minimality and Boundary Rigidity of Metrics Close to a Negatively Curved Symmetric Metric,” *Mathematische Annalen* 391 (2025), 1065–1122, DOI <https://doi.org/10.1007/s00208-024-02941-6>, arXiv:2201.09175. This proves strict minimal filling near real, complex, quaternionic, and Cayley hyperbolic metrics, enlarging the special classes for which the exact conclusion holds.
- Joseph Briggs and Chris Wells, “A Discrete View of Gromov’s Filling Area Conjecture,” preprint, 2026, arXiv:2602.17859, Theorem 1.2. This recent work still records the arbitrary-topology circle-filling problem as unresolved and proves only the universal lower bound 1.36π , below the conjectured 2π . That open limit case underscores the unresolved global-topology obstruction already present in dimension two.

Consequently, the strongest results establish the conjecture for substantial geometric neighborhoods and establish weaker local or topology-restricted forms for general simple metrics, but the unrestricted assertion in the statement remains open.

1.55.3. Remaining open problems recorded by the database

1.55.3.1. Open problem 1

Prove that the following holds. Let (M, g) be a compact orientable Riemannian surface with boundary such that ∂M is strictly convex, every geodesic segment in M realizes the Riemannian distance between its endpoints, and no geodesic in M has conjugate points. Let (M', g') be any compact orientable Riemannian surface, of arbitrary genus, and let $\phi : \partial M \rightarrow \partial M'$ be a fixed identification. If

$$d_{M'}(\phi(x), \phi(y)) \geq d_M(x, y) \quad \text{for all } x, y \in \partial M,$$

then

$$\text{Area}(M', g') \geq \text{Area}(M, g).$$

Reference supplied by the database. Sergei Ivanov, “Local Monotonicity of Riemannian and Finsler Volume with Respect to Boundary Distances,” *Geometriae Dedicata* 164 (2013), no. 1, 83–96, Introduction (identifying the arbitrary-genus two-dimensional case as open), DOI:

<https://doi.org/10.1007/s10711-012-9760-y>; arXiv:1109.4091, <https://arxiv.org/abs/1109.4091>.

1.55.3.2. Open problem 2

Prove that every simple compact Riemannian manifold of dimension $n \geq 3$ is boundary rigid. Explicitly, let (M, g) be a compact n -dimensional Riemannian manifold with strictly convex boundary such that every geodesic segment in M realizes the Riemannian distance between its endpoints and no geodesic in M has conjugate points. Let (M', g') be another compact n -dimensional Riemannian manifold, and let $\phi : \partial M \rightarrow \partial M'$ be a fixed identification satisfying

$$d_{M'}(\phi(x), \phi(y)) = d_M(x, y) \quad \text{for all } x, y \in \partial M.$$

Prove that there exists a Riemannian isometry $F : (M, g) \rightarrow (M', g')$ such that $F|_{\partial M} = \phi$.

Reference supplied by the database. René Michel, “Sur la rigidité imposée par la longueur des géodésiques,” *Inventiones Mathematicae* 65 (1981/82), 71–83 (Michel’s boundary-rigidity conjecture), DOI: <https://doi.org/10.1007/BF01389295>.

1.55.3.3. Open problem 3

Prove that the following holds. Let C be the Riemannian circle of length 2π , equipped with its intrinsic distance d_C . If (Σ, g) is a compact orientable Riemannian surface and $\phi : C \rightarrow \partial \Sigma$ is an identification such that

$$d_{\Sigma}(\phi(x), \phi(y)) = d_C(x, y) \quad \text{for all } x, y \in C,$$

then

$$\text{Area}(\Sigma, g) \geq 2\pi.$$

Reference supplied by the database. Mikhael Gromov, “Filling Riemannian Manifolds,” *Journal of Differential Geometry* 18 (1983), no. 1, 1–147, §2.2 (Gromov’s filling-area conjecture), DOI: <https://doi.org/10.4310/jdg/1214509283>.

1.55.4. Primary-source pointer

The originating ICM record is *Volume Comparison via Boundary Distances* (2010). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=775>.

1.56. 56. Extremal Kähler metrics [2014-geom-122]

Record: 2014-geom-122

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: Extremal Kähler metrics

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=1035>

1.56.1. Problem statement

Let M be a compact complex projective manifold and let $L \rightarrow M$ be an ample line bundle, so that (M, L) is a polarized manifold. Suppose that M has no non-zero holomorphic vector fields. Prove that M admits a constant scalar curvature Kähler metric whose Kähler class is the first Chern class $c_1(L)$ if and only if the polarized pair (M, L) is K -stable in Donaldson's sense: for every test-configuration χ for (M, L) , meaning a $\mathbf{C}^*$ -equivariant flat polarized family $(M, \mathcal{L})$ over $\mathbf{C}$ with generic fiber isomorphic to (M, L^r) for some exponent r , the Futaki invariant satisfies $\text{Fut}(\chi) > 0$ whenever the norm $\|\chi\|$ is positive.

1.56.2. Historical background, literature, and current progress

1.56.3. Verdict: partially resolved

1.56.3.1. Precise problem checked

The problem asks for the general constant-scalar-curvature Kähler form of the Yau-Tian-Donaldson conjecture:

For every smooth compact complex projective polarized manifold (M, L) with no non-zero holomorphic vector fields, does M admit a cscK metric in $c_1(L)$ if and only if (M, L) is K -stable in Donaldson's original test-configuration sense, i.e. $\text{Fut}(\chi) > 0$ for every test configuration of positive norm?

This is stronger, as stated, than the currently proved general equivalences with uniform K -stability/non-Archimedean coercivity.

1.56.3.2. Status

I did not find a published or otherwise citable result proving the exact statement in the problem statement for all smooth polarized manifolds with discrete automorphism group. The literature proves major pieces:

- cscK existence implies K -stability/ K -polystability under the stated discrete-automorphism hypothesis;
- several important special cases are resolved;

- the full general existence theorem is known under uniform K-stability or equivalent non-Archimedean coercivity-type hypotheses.

What remains unresolved for the problem as written is the converse from plain Donaldson K-stability, with only strict positivity on individual positive-norm test configurations, to existence of a cscK metric in $c_1(L)$. Current general theorems require a uniform lower bound/non-Archimedean coercivity condition, and the sources explicitly distinguish this from ordinary K-stability.

1.56.3.3. Key literature checked

1. **Title:** Scalar curvature and stability of toric varieties
Authors: S. K. Donaldson
Venue/year: Journal of Differential Geometry 62 (2002), 289-349
DOI/arXiv/URL: DOI 10.4310/jdg/1090950195; projecteuclid.org/euclid.jdg/1090950195
Relevance: Introduces Donaldson's test configurations and the K-stability framework for polarized varieties, and formulates the cscK/stability expectation in the toric setting. This is part of the source of the stated "Donaldson's sense" condition.
2. **Title:** Lower bounds on the Calabi functional
Authors: S. K. Donaldson
Venue/year: Journal of Differential Geometry 70 (2005), 453-472
DOI/arXiv/URL: DOI 10.4310/jdg/1143643312
Relevance: Gives the lower-bound mechanism relating the Calabi functional to Donaldson-Futaki invariants. It supports the necessary direction but does not prove the full converse.
3. **Title:** K-stability of constant scalar curvature Kähler manifolds
Authors: J. Stoppa
Venue/year: Advances in Mathematics 221 (2009), 1397-1408
DOI/arXiv/URL: DOI 10.1016/j.aim.2009.02.009; arXiv:0803.4095
Relevance: Proves that a polarized manifold admitting a cscK metric and having finite automorphism group is K-stable. This covers the "only if" direction of the problem under the no-holomorphic-vector-fields hypothesis, but not the converse.
4. **Title:** Kähler-Einstein metrics and stability
Authors: X. X. Chen, S. Donaldson, and S. Sun
Venue/year: International Mathematics Research Notices 2014, no. 8, 2119-2125, plus the detailed series in Journal of the American Mathematical Society 28 (2015)
DOI/arXiv/URL: DOI 10.1093/imrn/rns275; arXiv:1210.7494 and related preprints
Relevance: Resolves the Fano Kähler-Einstein version of Yau-Tian-Donaldson. This is a landmark special case, not the full cscK polarized-manifold statement in the problem statement.
5. **Title:** An Introduction to Extremal Kähler Metrics
Author: G. Székelyhidi
Venue/year: Graduate Studies in Mathematics 152, American Mathematical Society, 2014
DOI/arXiv/URL: AMS book page, ISBN 978-1-4704-1047-4
Relevance: Surveys the cscK/extremal Yau-Tian-Donaldson problem and records the general cscK converse as a central conjectural problem rather than a theorem.
6. **Title:** On the constant scalar curvature Kähler metrics, I: A priori estimates; II: Existence results

Authors: X. X. Chen and J. Cheng

Venue/year: Journal of the American Mathematical Society 34 (2021), 909-936 and 937-1009

DOI/arXiv/URL: DOI 10.1090/jams/967 and DOI 10.1090/jams/966

Relevance: Proves deep analytic existence results for cscK metrics in terms of properness/coercivity of the Mabuchi K-energy. This is a major analytic component of YTD, but it is not the plain Donaldson K-stability criterion.

7. **Title:** Uniform stability of twisted constant scalar curvature Kähler metrics
Author: R. Dervan
Venue/year: International Mathematics Research Notices 2016, no. 15, 4728-4783
DOI/arXiv/URL: DOI 10.1093/imrn/rnv291; arXiv:1412.0648
Relevance: Introduces the minimum norm and formulates uniform K-stability as the stability notion analogous to coercivity of the Mabuchi functional. This paper explicitly shows why the pointwise positivity is not the same as the quantitative condition used in later existence theorems.
8. **Title:** Uniform K-stability, Duistermaat-Heckman measures and singularities of pairs
Authors: S. Boucksom, T. Hisamoto, and M. Jonsson
Venue/year: Annales de l'Institut Fourier 67 (2017), no. 2, 743-841
DOI/arXiv/URL: DOI 10.5802/aif.3096; arXiv:1504.00682
Relevance: Develops non-Archimedean and Duistermaat-Heckman tools for uniform K-stability. Its role in the literature is precisely to bridge the gap between pointwise positivity of Donaldson-Futaki invariants and analytic coercivity.
9. **Title:** Uniform K-stability and asymptotics of energy functionals in Kähler geometry
Authors: S. Boucksom, T. Hisamoto, and M. Jonsson
Venue/year: Journal of the European Mathematical Society 21 (2019), 2905-2944
DOI/arXiv/URL: DOI 10.4171/JEMS/894; arXiv:1603.01026
Relevance: Relates uniform K-stability/non-Archimedean functionals to slopes of analytic energy functionals. This is a key bridge toward uniform YTD results, but again uses uniform stability rather than the pointwise Donaldson K-stability.
10. **Title:** Geodesic rays and stability in the cscK problem
Author: C. Li
Venue/year: Annales scientifiques de l'École normale supérieure, accepted version on arXiv, 2021
DOI/arXiv/URL: arXiv:2001.01366
Relevance: Reduces the uniform Yau-Tian-Donaldson conjecture for cscK metrics to a Boucksom-Jonsson regularization conjecture and proves sufficient uniform stability conditions for cscK existence. This is important progress but is explicitly uniform, not the stated ordinary Donaldson K-stability.
11. **Title:** On the Yau-Tian-Donaldson conjecture for weighted cscK metrics
Authors: S. Boucksom and M. Jonsson
Venue/year: arXiv preprint, 2025
DOI/arXiv/URL: arXiv:2509.15016; <https://arxiv.org/abs/2509.15016>
Relevance: Establishes a version of the Yau-Tian-Donaldson conjecture for cscK and weighted extremal metrics using non-Archimedean pluripotential theory. The formulation is not the plain Donaldson positivity criterion.
12. **Title:** From complex to non-Archimedean geometry: an approach to the YTD conjecture
Authors: S. Boucksom and M. Jonsson
Venue/year: arXiv preprint/SRI notes, 2026

DOI/arXiv/URL: arXiv:2602.10800; <https://arxiv.org/abs/2602.10800>

Relevance: Surveys and sketches the non-Archimedean approach to a version of YTD for cscK metrics. This confirms the current modern formulation through non-Archimedean/uniform conditions.

13. **Title:** A solution to the Yau-Tian-Donaldson Conjecture

Author: A. Trusiani

Venue/year: arXiv preprint, 2026

DOI/arXiv/URL: arXiv:2605.30063; <https://arxiv.org/abs/2605.30063>

Relevance: Gives a recent citable claimed solution of the general YTD correspondence in uniform/non-Archimedean terms. The abstract and introduction distinguish K-polystability from the stronger uniform conditions, and state that existence implies K-polystability while the converse is not expected in that non-uniform form. Therefore this does not resolve the exact statement with only Donaldson's pointwise K-stability.

1.56.3.4. Why this is not fully resolved

The stated stability hypothesis is:

$$\text{Fut}(\chi) > 0 \quad \text{for every positive-norm test configuration } \chi.$$

This is pointwise positivity. The general existence theorems I found require a quantitative estimate of the form

$$\text{Fut}(\chi) \geq \delta \|\chi\| \quad \text{or an equivalent non-Archimedean/coercivity condition}$$

for some uniform $\delta > 0$, possibly modulo automorphisms. Since the statement assumes no non-zero holomorphic vector fields, the automorphism issue is simplified, but the distinction between pointwise K-stability and uniform K-stability remains the decisive gap. I found no citable theorem proving that the plain Donaldson K-stability is equivalent to uniform K-stability for all smooth polarized manifolds with discrete automorphism group, nor a theorem deriving cscK existence directly from the pointwise condition in full generality.

Therefore the current status of the exact problem is **partially resolved**, not fully resolved.

1.56.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

1.56.5. Primary-source pointer

The originating ICM record is *Extremal Kähler metrics* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=1035>.

1.57. 57. New applications of Min-max Theory [2014-geom-128]

Record: 2014-geom-128

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: New applications of Min-max Theory

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=957>

1.57.1. Problem statement

Prove or disprove that, for every integer $n \geq 3$, among all non-totally geodesic minimal hypersurfaces in the unit round sphere

$$S^n = \{x \in \mathbb{R}^{n+1} : |x| = 1\},$$

the least possible $(n-1)$ -dimensional volume is attained by the following standard product hypersurface:

if $n = 2m$, then the minimizer is

$$S^{m-1}\left(\sqrt{\frac{m-1}{2m-1}}\right) \times S^m\left(\sqrt{\frac{m}{2m-1}}\right) \subset S^{2m} = S^n;$$

if $n = 2m - 1$, then the minimizer is

$$S^{m-1}\left(\frac{1}{\sqrt{2}}\right) \times S^{m-1}\left(\frac{1}{\sqrt{2}}\right) \subset S^{2m-1} = S^n.$$

Here $S^k(r)$ denotes the k -sphere of radius r , and each product is included in the unit sphere by the standard product embedding.

1.57.2. Historical background, literature, and current progress

1.57.3. Verdict: partially resolved

1.57.3.1. Precise problem checked

The problem asks for the sharp second-volume gap for closed non-totally geodesic minimal hypersurfaces in the unit round sphere. In the notation more common in the literature, for a hypersurface $M^d \subset S^{d+1}$, the conjecture says that the least possible d -volume among non-totally geodesic closed minimal hypersurfaces is the volume of the balanced Clifford hypersurface

$$S^k(\sqrt{k/d}) \times S^{d-k}(\sqrt{(d-k)/d}), \quad k = \lfloor d/2 \rfloor.$$

With $d = n - 1$, this is exactly the pair of products stated in the problem statement for ambient S^n . This is usually called Yau's conjecture or the Solomon-Yau/Solomon conjecture on the second lowest volume of minimal hypersurfaces in spheres.

1.57.3.2. Current status

No published or otherwise citable result found proves the full statement for all $n \geq 3$. The ambient S^3 case is resolved, and several higher-dimensional restricted cases are resolved, but the unrestricted embedded higher-dimensional case remains open. Therefore the problem is partially resolved, not fully resolved.

Recent literature still treats the higher-dimensional problem as partial. For example, Ding-Ge-Li (2025) explicitly says that for hypersurface dimension $d \geq 3$ “there are many partial results,” then lists rotational, asymptotic/cone, antipodal-symmetric, and nonembedded cases rather than a complete theorem.

1.57.3.3. Progress and relevant literature

1. Problem/source statement

- Title: *New applications of min-max theory*
- Author: Andre Neves
- Venue/year: Proceedings of the International Congress of Mathematicians, Seoul 2014, Vol. II, 2014
- arXiv/URL: arXiv:1409.7537, <https://arxiv.org/abs/1409.7537>
- Relevance: Section “Old questions” states exactly Solomon’s conjecture in the stated ambient-sphere notation, including the even and odd Clifford products. It records the S^3 case as confirmed by Marques-Neves and mentions only partial higher-dimensional progress.

2. Full solution only in ambient S^3

- Title: *The Min-Max theory and the Willmore conjecture*
- Authors: Fernando C. Marques and Andre Neves
- Venue/year: Annals of Mathematics 179 (2014), no. 2, 683-782
- DOI/arXiv/URL: DOI 10.4007/annals.2014.179.2.6; arXiv:1202.6036; <https://arxiv.org/abs/1202.6036>
- Result: Theorem B proves that every embedded closed minimal surface in S^3 of genus at least one has area at least $2\pi^2$, with equality only for the Clifford torus. Together with Almgren’s theorem that a minimal two-sphere in S^3 is totally geodesic and the Li-Yau lower bound for nonembedded surfaces, this proves the stated $n = 3$ case.

3. Nonembedded case in all dimensions

- Title: *Volume gap for minimal submanifolds in spheres*
- Authors: Jianquan Ge and Fagui Li
- Venue/year: arXiv preprint, updated 2024
- arXiv/URL: arXiv:2210.04654, <https://arxiv.org/abs/2210.04654>
- Result: Theorem 1.2 and Corollary 1.3 give a multiplicity/volume lower bound implying that if the volume is at most the Clifford value then the hypersurface must be embedded. The authors state this as verifying the nonembedded case of Yau’s conjecture. This does not settle embedded hypersurfaces in dimensions $d \geq 3$.

4. Independent nonembedded-case method

- Title: *Weighted monotonicity theorems and applications to minimal surfaces of $\mathbb{H}^n$ and S^n*
- Author: Manh Tien Nguyen

- Venue/year: Transactions of the American Mathematical Society 376 (2023), no. 8, 5899-5921
- DOI/arXiv/URL: DOI 10.1090/tran/8949; arXiv:2105.12625; <https://arxiv.org/abs/2105.12625>
- Result: Corollary 4.8 and Remark 4.9 show that a nonembedded minimal hypersurface in S^n has volume at least twice the equator volume, which is enough to confirm Yau's/Solomon's conjectured lower bound for nonembedded hypersurfaces. Again this leaves the embedded higher-dimensional case.

5. Antipodal-symmetric embedded case

- Title: *Isoperimetry and volume preserving stability in real projective spaces*
- Author: Celso Viana
- Venue/year: Journal of Differential Geometry 125 (2023), no. 1, 187-205
- DOI/arXiv/URL: DOI 10.4310/jdg/1695236595; arXiv:1907.09445; <https://arxiv.org/abs/1907.09445>
- Result: Theorem 1.4 proves a Willmore-type inequality for compact embedded hypersurfaces separating S^{d+1} into two antipodal-invariant regions, with equality only for the balanced Clifford hypersurface. Remark 1.5 says this confirms Solomon's conjecture in the class of minimal hypersurfaces with antipodal symmetry. The antipodal-invariance hypothesis is an extra restriction, so this is not the full statement.

6. Rotational hypersurfaces

- Title: *n-dimensional area of minimal rotational hypersurfaces in spheres*
- Authors: Oscar Perdomo and Guoxin Wei
- Venue/year: Nonlinear Analysis 125 (2015), 241-250
- DOI/URL: DOI 10.1016/j.na.2015.05.017
- Result: Verifies the conjectured lower bound for many compact minimal rotational hypersurfaces, reported in later papers as covering dimensions $2 \leq d \leq 100$ in that restricted rotational class.
- Title: *Area of minimal hypersurfaces in the unit sphere*
- Authors: Qing-Ming Cheng, Guoxin Wei, and Yuting Zeng
- Venue/year: Asian Journal of Mathematics 25 (2021), no. 2, 183-194
- DOI/arXiv/URL: DOI 10.4310/AJM.2021.v25.n2.a2; arXiv:1907.07314; <https://arxiv.org/abs/1907.07314>
- Result: Theorem 1.1 proves the Yau/Solomon conjecture for compact minimal rotational hypersurfaces.
- Title: *Area of Minimal Hypersurfaces in the Unit Sphere (II)*
- Authors: Qing-Ming Cheng, Junqi Lai, and Guoxin Wei
- Venue/year: Journal of Geometric Analysis 34 (2024), article 369
- DOI/URL: DOI 10.1007/s12220-024-01808-1; <https://doi.org/10.1007/s12220-024-01808-1>
- Result: Gives sharper area estimates for compact minimal rotational hypersurfaces and confirms conjectures from the preceding rotational papers. These theorems have a rotational-symmetry hypothesis, so they do not imply the unrestricted result.

7. Asymptotic/cone-type partial progress

- Title: *Sharp lower bounds on density for area-minimizing cones*
- Authors: Tom Ilmanen and Brian White
- Venue/year: Cambridge Journal of Mathematics 3 (2015), no. 1-2, 1-18
- DOI/URL: DOI 10.4310/CJM.2015.v3.n1.a1

- Result: Later papers summarize this as proving an asymptotic form of the conjecture for topologically nontrivial embedded hypersurfaces whose cones are area-minimizing. This has strong extra hypotheses and gives an asymptotic/cone statement, not the exact all-hypersurface Clifford minimizer statement in the problem statement.

8. Recent volume-gap improvements, still not the conjecture

- Title: *Volume gap between the minimal submanifold and the unit sphere*
- Authors: Weiran Ding, Jianquan Ge, and Fagui Li
- Venue/year: International Journal of Mathematics, 2025; arXiv preprint 2024
- DOI/arXiv/URL: DOI 10.1142/S0129167X25500296; arXiv:2412.01288; <https://arxiv.org/abs/2412.01288>
- Result: Improves Cheng-Li-Yau type volume gaps above the equator. Its introduction still describes the Solomon-Yau/Yau conjecture for $d \geq 3$ as having “many partial results” and lists the restricted results above. The gap it proves is not the sharp Clifford-hypersurface gap required by this problem.

1.57.3.4. What remains unresolved

For ambient spheres S^N with $N \geq 4$ (equivalently hypersurface dimension $d = N - 1 \geq 3$), no cited theorem found proves that every closed embedded non-totally geodesic minimal hypersurface has volume at least that of the balanced Clifford hypersurface, with equality only for that Clifford hypersurface, without extra assumptions such as rotational symmetry, antipodal symmetry, nonembeddedness, curvature conditions, or area-minimizing-cone hypotheses.

1.57.4. Remaining open problems recorded by the database

1.57.4.1. Open problem 1

Prove that every closed non-totally-geodesic minimal hypersurface $M^3 \subset S^4$ satisfies

$$\text{Vol}_3(M) \geq \text{Vol}_3\left(S^1\left(\sqrt{\frac{1}{3}}\right) \times S^2\left(\sqrt{\frac{2}{3}}\right)\right),$$

with equality only for that Clifford hypersurface up to an ambient isometry.

Reference supplied by the database. Qing-Ming Cheng, Guoxin Wei, and Yuting Zeng, “Area of minimal hypersurfaces in the unit sphere,” *Asian Journal of Mathematics* **25**(2) (2021), 183–194, introduction (the Solomon–Yau least-area conjecture, explicitly treated there only for rotational hypersurfaces). DOI: <https://doi.org/10.4310/AJM.2021.v25.n2.a2>. The fact that the conjecture remains open in every hypersurface dimension $n \geq 3$, hence in the displayed $M^3 \subset S^4$ case, is explicitly recorded in Jingwen Chen, “Symmetric mean curvature flow on the n -sphere,” *Nonlinear Analysis* **239** (2024), Article 113437, DOI: <https://doi.org/10.1016/j.na.2023.113437>.

1.57.5. Primary-source pointer

The originating ICM record is *New applications of Min-max Theory* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=957>.

1.58. 58. New applications of Min-max Theory [2014-geom-129]

Record: 2014-geom-129

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: New applications of Min-max Theory

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=957>

1.58.1. Problem statement

Prove that for every Riemannian metric on the 3-sphere S^3 , there exist at least five pairwise distinct embedded minimal surfaces in (S^3, g) that are tori.

1.58.2. Historical background, literature, and current progress

1.58.3. Verdict: partially resolved

1.58.3.1. Precise problem

The problem asks whether every smooth Riemannian metric g on the 3-sphere S^3 admits at least five pairwise distinct closed embedded minimal surfaces of genus one, i.e. embedded minimal tori, in (S^3, g) . Distinct means distinct embedded surfaces, not just different parametrizations or multiplicities.

1.58.3.2. Literature checked

1. New applications of Min-max Theory

Author: Andre Neves

Venue/year: Proceedings of the International Congress of Mathematicians, Seoul 2014, Vol. II, pp. 939-957; arXiv preprint, 2014

arXiv/URL: arXiv:1409.7537, <https://arxiv.org/abs/1409.7537>

Relevant statement: In the open-problems section Neves records White's conjecture: any metric on a 3-sphere has five distinct minimal embedded tori, and notes that White proved it for small perturbations of the round metric. This identifies the problem statement as White's enumerative minimal-torus conjecture, not merely a positive-Ricci or near-round statement.

2. Every three-sphere of positive Ricci curvature contains a minimal embedded torus

Author: Brian White

Venue/year: Bulletin of the American Mathematical Society (N.S.) 21(1), 71-75, 1989

DOI/URL: 10.1090/S0273-0979-1989-15765-0; <https://projecteuclid.org/journals/bulletin-of-the-american-mathematical-society-new-series/volume-21/issue-1/Every-three-sphere-of-positive-Ricci-curvature-contains-a-minimal/bams/1183555126.f>

ull

Relevant result: White proves existence of at least one embedded minimal torus for metrics on S^3 with positive Ricci curvature. This is important progress but it gives one torus and assumes positive Ricci curvature, so it does not imply the stated five-torus statement for arbitrary metrics.

3. The Space of Minimal Submanifolds for Varying Riemannian Metrics

Author: Brian White

Venue/year: Indiana University Mathematics Journal 40(1), 161-200, 1991

DOI/URL: 10.1512/iumj.1991.40.40008; <https://iumj.org/article/3473/>

Relevant result: Later papers cite White's perturbation theorem from this work as proving that every sufficiently round metric on S^3 has at least five embedded minimal tori, and a generic sufficiently round metric has at least nine. This covers a neighborhood of the round metric, not all Riemannian metrics on S^3 .

4. Existence of 5 minimal tori in 3-spheres of positive Ricci curvature

Authors: Adrian Chun-Pong Chu and Yangyang Li

Venue/year: arXiv preprint, 2024; current arXiv version dated July 31, 2025

arXiv/DOI/URL: arXiv:2409.09315; 10.48550/arXiv.2409.09315; <https://arxiv.org/abs/2409.09315>

Relevant theorem: Theorem A states that every smooth Riemannian 3-sphere of positive Ricci curvature contains at least five embedded minimal tori. This proves White's conjectured lower bound in the positive-Ricci class. It still has the extra hypothesis $\text{Ric} > 0$, so it does not cover arbitrary metrics on S^3 .

5. Existence of embedded minimal tori in three-spheres with positive Ricci curvature

Authors: Xingzhe Li and Zhichao Wang

Venue/year: arXiv preprint, 2024

arXiv/DOI/URL: arXiv:2409.10391; 10.48550/arXiv.2409.10391; <https://arxiv.org/abs/2409.10391>

Relevant theorem: Theorem 1.1 proves that a positive-Ricci metric on S^3 has at least four embedded minimal tori, and a bumpy positive-Ricci metric has at least nine. This independently confirms substantial positive-Ricci progress but also does not imply the arbitrary-metric five-torus statement.

6. An enumerative min-max theorem for minimal surfaces

Authors: Adrian Chun-Pong Chu, Yangyang Li, and Zhihan Wang

Venue/year: arXiv preprint, 2026

arXiv/URL: arXiv:2601.01736, <https://arxiv.org/abs/2601.01736>

Relevance: The introduction summarizes the current landscape: Poincaré's 2-sphere geodesic conjecture is fully solved; Yau's four-sphere conjecture is solved for bumpy metrics and positive Ricci curvature; and White's five-torus conjecture is proved for positive-Ricci 3-spheres. The paper's own enumerative theorem is also stated under positive Ricci curvature, so it does not remove the extra curvature hypothesis.

1.58.3.3. Conclusion

the full statement is not fully resolved. The strongest directly relevant result found is Chu–Li's theorem proving the exact lower bound of five embedded minimal tori for every S^3 metric with positive Ricci curvature. White's earlier work gives one torus under positive Ricci curvature and five tori only for sufficiently round metrics. Li–Wang give related positive-Ricci and bumpy-positive-Ricci lower bounds.

What remains unresolved is the full arbitrary-metric case: no published or citable preprint found proves that every Riemannian metric on S^3 , without positive Ricci curvature, bumpiness, or closeness to the round metric, contains at least five pairwise distinct embedded minimal tori.

1.58.4. Remaining open problems recorded by the database

1.58.4.1. Open problem 1

Prove that every bumpy Riemannian metric on S^3 , without any curvature assumption, admits at least five pairwise distinct embedded minimal tori.

Reference supplied by the database. Brian White, “Every three-sphere of positive Ricci curvature contains a minimal embedded torus,” *Bulletin of the American Mathematical Society* **21**(1) (1989), 71–75, where White proposed the five-minimal-tori conjecture. DOI: <https://doi.org/10.1090/S0273-0979-1989-15765-0>. Current status: Adrian Chun-Pong Chu and Yangyang Li, “Existence of 5 minimal tori in 3-spheres of positive Ricci curvature,” preprint, revised 2025, proves the conjecture under positive Ricci curvature but leaves the general case—including bumpy metrics with no curvature assumption—open, [arXiv:2409.09315](https://arxiv.org/abs/2409.09315).

1.58.5. Primary-source pointer

The originating ICM record is *New applications of Min-max Theory* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=957>.

1.59. 59. Non-negatively curved manifolds and Tits geometry [2014-geom-130]

Record: 2014-geom-130

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: reviewed (reviewed flag: yes)

Original source: ICM 2014: Non-negatively curved manifolds and Tits geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=885>

1.59.1. Problem statement

Prove that there is no Riemannian metric with positive sectional curvature on the product $\mathbb{S}^2 \times \mathbb{S}^2$, where $\mathbb{S}^2$ denotes the 2-sphere. More generally, prove that if M and N are simply-connected connected closed (compact without boundary) positive-dimensional manifolds, then the product manifold $M \times N$ admits no Riemannian metric with positive sectional curvature; likewise, prove that the underlying smooth manifold of a compact symmetric space of rank at least two admits no Riemannian metric with positive sectional curvature.

1.59.2. Historical background, literature, and current progress

1.59.3. Verdict: partially resolved

1.59.3.1. Precise problem

The problem asks whether any Riemannian metric with strictly positive sectional curvature can exist on the smooth manifold $S^2 \times S^2$. It also asks the broader Hopf product/symmetric-space question: no nontrivial product $M \times N$ of closed positive-dimensional manifolds should admit positive sectional curvature, and no compact symmetric space of rank at least two should admit such a metric on its underlying smooth manifold.

1.59.3.2. Status

The full statement is not resolved. In particular, the basic case $S^2 \times S^2$ remains open in the published literature, so the product and rank-at-least-two symmetric-space assertions cannot be counted as fully proved. The literature contains substantial partial results, and some recent papers describe the generalized product conjecture as open.

1.59.3.3. Key literature and progress

1. Heinz Hopf, "Differentialgeometrie und topologische Gestalt", Jahresbericht der Deutschen Mathematiker-Vereinigung 41, 209-229, 1932. <https://eudml.org/doc/145926>. This is the historical source for Hopf-type curvature/topology

- questions; later sources identify the $S^2 \times S^2$ positive-curvature problem as Hopf's product conjecture.
2. Wolfgang Ziller, "Riemannian manifolds with positive sectional curvature", in *Geometry of Manifolds with Non-negative Sectional Curvature*, Lecture Notes in Mathematics 2110, Springer, 2014, pp. 1-19. DOI: 10.1007/978-3-319-06373-7_1. Ziller states exactly the relevant conjectural package: no positive sectional curvature on $S^2 \times S^2$, more generally no positive curvature on products of compact manifolds or on symmetric spaces of rank at least two. This is a survey/status source, not a resolution.
 3. Jean-Pierre Bourguignon, "Some constructions related to H. Hopf's conjecture on product manifolds", in *Differential Geometry, Proceedings of Symposia in Pure Mathematics 27*, American Mathematical Society, 1975, pp. 33-37. MR 0380906. This gives a local obstruction near the product metric on $S^2 \times S^2$. It does not exclude arbitrary metrics on $S^2 \times S^2$.
 4. Alan Weinstein, "Positively curved n -manifolds in R^{n+2} ", *Journal of Differential Geometry* 4(1), 1-4, 1970. DOI: 10.4310/jdg/1214429270. This rules out a positive-curvature metric on $S^2 \times S^2$ induced by an immersion in R^6 . It is an extrinsic special case, not a theorem about all intrinsic metrics.
 5. Wu-Yi Hsiang and Bruce Kleiner, "On the topology of positively curved 4-manifolds with symmetry", *Journal of Differential Geometry* 29(3), 615-621, 1989. DOI: 10.4310/jdg/1214443064. Their theorem implies that a simply connected positively curved 4-manifold with an isometric circle action is homeomorphic to S^4 or $\mathbb{CP}^2$. Thus $S^2 \times S^2$ cannot carry a positive-curvature metric with an isometric circle action. This leaves metrics with no such symmetry open.
 6. Lee Kennard, "Positively curved Riemannian metrics with logarithmic symmetry rank bounds", *Commentarii Mathematici Helvetici* 89(4), 937-962, 2014. DOI: 10.4171/CMH/340. Kennard proves rational-cohomology obstructions for positively curved manifolds with large symmetry rank and gives evidence for the symmetric-space version of Hopf's conjecture. The theorem assumes large isometric torus symmetry and does not cover arbitrary metrics.
 7. Manuel Amann and Lee Kennard, "On a generalized conjecture of Hopf with symmetry", *Compositio Mathematica* 153(2), 313-322, 2017. DOI: 10.1112/S0010437X16008150; arXiv:1402.7255. The abstract states the generalized conjecture for nontrivial products and proves evidence in the presence of symmetry, specifically excluding positively curved metrics on $N \times N$ with a certain degree of torus symmetry. This is a partial result because of the symmetry hypothesis and because it does not cover arbitrary $M \times N$.
 8. Oscar Perdomo, "Metrics on products of surfaces with non-positive sectional curvature", *Israel Journal of Mathematics* 156, 65-71, 2006. DOI: 10.1007/BF02773824. Perdomo proves that certain graph metrics over products of compact surfaces do not have positive sectional curvature. This is an extrinsic/graph-metric result, not a proof for all Riemannian metrics on $S^2 \times S^2$.
 9. Thierno Seck and Athoumane Niang, "Conformal Metrics to a Product or Doubly Warped Product on $S^2 \times S^2$ and the Hopf Conjecture", *Journal of Mathematics* 2022, Article ID 6268017, 2022. DOI: 10.1155/2022/6268017. The paper proves the conjecture for conformal metrics to the product metric and doubly warped product metrics on $S^2 \times S^2$, and explicitly says the generalized nontrivial-product conjecture is open. These are metric-class restrictions, so the full problem remains open.
 10. Athoumane Niang, Thierno Seck, and Adama Thiandoum, "Proof of the Hopf Conjec-

ture”, ResearchGate preprint, posted 2024, URL: https://www.researchgate.net/publication/377694075_PROOF_OF_THE_HOPF_CONJECTURE. This item claims to prove Hopf’s conjecture, but the theorem stated there concerns metrics induced on graphs $\{(x, F(x)) : x \in S^2 \times S^2\} \subset R^{6+k}$. That conclusion does not by itself cover all Riemannian metrics on $S^2 \times S^2$; an arbitrary metric need not be represented as such a graph over the standard $S^2 \times S^2 \subset R^6$. I therefore do not count it as a citable resolution of the problem.

1.59.3.4. What remains unresolved

No published or otherwise reliable citable theorem found in the search proves that every Riemannian metric on $S^2 \times S^2$ has some nonpositive sectional 2-plane. Since this basic special case is still unresolved, the full assertion for all compact products $M \times N$ and for all compact symmetric spaces of rank at least two is also unresolved. The known results above are genuine partial progress but all impose extra hypotheses such as proximity to the product metric, extrinsic graph/immersion form, conformal or warped product form, or nontrivial isometric symmetry.

1.59.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website’s “Some remaining open problems” field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

1.59.5. Primary-source pointer

The originating ICM record is *Non-negatively curved manifolds and Tits geometry* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=885>.

1.60. 60. The Geometry of Ricci Curvature [2014-geom-135]

Record: 2014-geom-135

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: The Geometry of Ricci Curvature

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=929>

1.60.1. Problem statement

Let $n \in \mathbb{N}$ and $v > 0$. Suppose that (M_j^n, g_j, p_j) is a sequence of complete pointed n -dimensional Riemannian manifolds converging in the pointed Gromov-Hausdorff sense to a pointed metric space (X, d, p) , with $\text{Ric}_{g_j} \geq -(n-1)$ and $\text{Vol}_{g_j}(B_1(p_j)) > v$ for every j . For $x \in X$, a tangent cone at x means any pointed Gromov-Hausdorff limit of rescalings $(X, r_i^{-1}d, x)$ along a sequence $r_i \rightarrow 0$. Prove that the set of points $x \in X$ at which there is more than one tangent cone, up to pointed isometry, has Hausdorff dimension at most $n - 3$.

1.60.2. Historical background, literature, and current progress

1.60.3. Verdict: partially resolved

1.60.3.1. Precise problem

The problem asks whether, for every noncollapsed pointed Gromov-Hausdorff limit of complete Riemannian n -manifolds with $\text{Ric} \geq -(n-1)$, the locus of points with more than one tangent cone has Hausdorff dimension at most $n - 3$. Equivalently: are tangent cones unique outside a codimension-three subset?

1.60.3.2. Verdict

This is not fully resolved in the generality stated. It is the codimension-three tangent-cone uniqueness conjecture for noncollapsed lower-Ricci limit spaces. Published general results give substantial partial progress, especially uniqueness outside an $\mathcal{H}^{n-2}$ -measure-zero set, but that does not imply Hausdorff dimension at most $n - 3$.

1.60.3.3. Literature checked

1. The geometry of Ricci curvature

Author: Aaron Naber

Venue/year: Proceedings of the International Congress of Mathematicians, Seoul 2014, Vol. II, Kyung Moon Sa, Seoul, 2014, pp. 911-937

URL: https://kyungmoon.com/shop/item.php?it_id=1759215764

Relevance: This is the source family for the problem. Later papers cite this ICM article as one of the places where the codimension-three tangent-cone uniqueness problem is formulated.

2. Conjectures and Open Questions on the Structure and Regularity of Spaces with Lower Ricci Curvature Bounds

Author: Aaron Naber

Venue/year: SIGMA Symmetry Integrability Geom. Methods Appl. 16 (2020), Paper No. 104

DOI/arXiv/URL: DOI 10.3842/SIGMA.2020.104; arXiv:2010.10031; <https://arxiv.org/abs/2010.10031>

Relevance: Conjecture 2.7 states exactly that if $(M_i^n, g_i, p_i) \rightarrow (X, d, p)$, $\text{Ric} \geq -(n-1)$, and $\text{Vol}(B_1(p_i)) > v > 0$, then tangent cones of X are unique away from a set of codimension three. The paper explicitly says the known result at that time was only uniqueness away from a set of $\mathcal{H}^{n-2}$ -measure zero.

3. Rectifiability of singular sets of noncollapsed limit spaces with Ricci curvature bounded below

Authors: Jeff Cheeger, Wenshuai Jiang, Aaron Naber

Venue/year: Annals of Mathematics (2) 193 (2021), no. 2, 407-538

DOI/arXiv/URL: DOI 10.4007/annals.2021.193.2.2; arXiv:1805.07988; <https://annals.math.princeton.edu/2021/193-2/p02>

Progress: For the same noncollapsed lower-Ricci hypotheses as in the problem statement, they prove that each stratum S^k is k -rectifiable and that for $\mathcal{H}^k$ -a.e. $x \in S^k$, every tangent cone is k -symmetric. As an application, tangent cones are unique $\mathcal{H}^{n-2}$ -a.e.

Why this is only partial: An exceptional set of zero $\mathcal{H}^{n-2}$ -measure can still have Hausdorff dimension $n-2$. The problem asks for Hausdorff dimension at most $n-3$, so this theorem does not imply the full statement.

4. Characterization of Tangent Cones of Noncollapsed Limits with Lower Ricci Bounds and Applications

Authors: Tobias H. Colding, Aaron Naber

Venue/year: Geometric and Functional Analysis 23 (2013), 134-148

DOI/arXiv/URL: DOI 10.1007/s00039-012-0202-7; arXiv:1108.3244; <https://arxiv.org/abs/1108.3244>

Progress/evidence: Shows that tangent cones in the exact noncollapsed lower-Ricci setting need not be unique at a point, including examples with nonunique and even non-homeomorphic tangent cones. This explains why the conjecture concerns the size of the nonuniqueness locus rather than pointwise uniqueness.

5. Examples of tangent cones of non-collapsed Ricci limit spaces

Author: Philipp Reiser

Venue/year: arXiv preprint, 2024

DOI/arXiv/URL: arXiv:2409.11954; <https://arxiv.org/abs/2409.11954>

Progress/evidence: Extends Colding-Naber type examples by constructing additional possible cross sections of tangent cones at the same point in noncollapsed Ricci limit spaces. This is further evidence about possible nonuniqueness phenomena, but it is not a dimension bound for the full nonuniqueness set.

6. Gromov-Hausdorff limits of Kähler manifolds with Ricci curvature bounded below

Authors: Gang Liu, Gábor Székelyhidi

Venue/year: Geometric and Functional Analysis 32 (2022), no. 2, 236-279

arXiv/URL: arXiv:1804.08567; <https://arxiv.org/abs/1804.08567>

Progress: Gives stronger structure in the Kähler setting; Naber's 2020 survey notes that

the Kähler case resolves the analogous refined stratification conjectures.

Why this is only partial for this problem: The statement has no Kähler, complex-analytic, or polarization hypothesis, so these results do not cover the full statement.

7. Regularity of Einstein 5-manifolds via 4-dimensional gap theorems

Authors: Yiqi Huang, Tristan Ozuch

Venue/year: arXiv preprint, 2025

arXiv/URL: arXiv:2512.21317; <https://arxiv.org/abs/2512.21317>

Progress: Proves uniqueness of tangent cones on the top stratum for noncollapsed limits of Einstein 5-manifolds and related bounded-Ricci settings.

Why this is only partial for this problem: It assumes dimension five and Einstein/bounded Ricci hypotheses, while the problem asks for arbitrary dimension and only a lower Ricci bound.

1.60.3.4. What remains unresolved

No citable result found proves that the full nonuniqueness locus has Hausdorff dimension at most $n - 3$ for arbitrary noncollapsed lower-Ricci limit spaces. The remaining gap is precisely to upgrade the known $\mathcal{H}^{n-2}$ -a.e. uniqueness/rectifiability theory to a genuine codimension-three Hausdorff-dimension bound for all points with nonunique tangent cones.

1.60.4. Remaining open problems recorded by the database

1.60.4.1. Open problem 1

In a noncollapsed three-dimensional Ricci limit space, prove that the set of points admitting two nonisometric tangent cones has Hausdorff dimension zero.

Reference supplied by the database. Aaron Naber, “Conjectures and Open Questions on the Structure and Regularity of Spaces with Lower Ricci Curvature Bounds,” *Symmetry, Integrability and Geometry: Methods and Applications (SIGMA)* **16** (2020), Article 104, Conjecture 2.7, specialized to $n = 3$. DOI: <https://doi.org/10.3842/SIGMA.2020.104>.

1.60.5. Primary-source pointer

The originating ICM record is *The Geometry of Ricci Curvature* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=929>.

1.61. 61. The Geometry of Ricci Curvature [2014-geom-136]

Record: 2014-geom-136

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: The Geometry of Ricci Curvature

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=929>

1.61.1. Problem statement

Let n be fixed, let $v > 0$, and let (X, d, p) be the pointed Gromov-Hausdorff limit of pointed n -dimensional Riemannian manifolds (M_j^n, g_j, p_j) such that $\text{Ric}_{g_j} \geq -(n-1)$ and $\text{Vol}_{g_j}(B_1(p_j)) > v$ for all j . Prove that there exists a subset $E \subset X$ of Hausdorff dimension at most $n-3$ such that for every $x \in X \setminus E$, the tangent cone of X at x is unique and isometric to $\mathbb{R}^{n-2} \times C(S^1(r_x))$ for some radius $r_x \leq 1$, where $S^1(r_x)$ is the circle of radius r_x and $C(S^1(r_x))$ denotes its metric cone.

1.61.2. Historical background, literature, and current progress

1.61.3. Verdict: partially resolved

1.61.3.1. Precise problem

The statement asks for the codimension-three uniqueness conjecture for noncollapsed Ricci-limit spaces: if X is a pointed Gromov-Hausdorff limit of n -manifolds with $\text{Ric} \geq -(n-1)$ and a uniform lower volume bound, then outside a subset of Hausdorff dimension at most $n-3$, every point should have a unique tangent cone, and that cone should be the two-dimensional flat cone model

$$\mathbb{R}^{n-2} \times C(S^1(r_x)), \quad r_x \leq 1.$$

Regular points are included in this form by taking $r_x = 1$.

1.61.3.2. Verdict

This problem is partially resolved, but I did not find a citable published or preprint result proving the exact statement in full generality.

The main gap is the size of the exceptional set. The best general theorem found gives the same tangent-cone model and uniqueness outside a set of zero $\mathcal{H}^{n-2}$ -measure. That does not imply Hausdorff dimension at most $n-3$: an $\mathcal{H}^{n-2}$ -null set can still have Hausdorff dimension strictly larger than $n-3$, or even equal to $n-2$.

1.61.3.3. Key literature and progress

1. The Geometry of Ricci Curvature

Author: Aaron Naber

Venue/year: Proceedings of the International Congress of Mathematicians, Seoul 2014, Vol. II, Kyung Moon Sa, 2014, pp. 911-937

URL: <https://dokumen.pub/proceedings-of-the-international-congress-of-mathematicians-seoul-august-13-21-2014-korea-vol-2-9788961058032-8961058037-9788961058049-8961058056-8961058061-9788961058070-896105807x.html>

Relevance: Conjecture 6.2 in this ICM article is the source matching the problem statement: after removing a set of Hausdorff dimension at most $n - 3$, tangent cones should be unique and equal to $\mathbb{R}^{n-2} \times C(S_r^1)$.

2. Rectifiability of singular sets of noncollapsed limit spaces with Ricci curvature bounded below

Authors: Jeff Cheeger, Wenshuai Jiang, Aaron Naber

Venue/year: Annals of Mathematics 193 (2021), no. 2, 407-538

DOI/arXiv: DOI 10.4007/annals.2021.193.2.2; arXiv:1805.07988

Relevance: Theorem 1.15 gives the strongest general partial result found. It proves that for a noncollapsed limit space with Ricci curvature bounded below there is a subset $\tilde{S}$ with $\mathcal{H}^{n-2}(\tilde{S}) = 0$ such that at every point outside $\tilde{S}$, tangent cones are unique and are of the form $\mathbb{R}^{n-2} \times C(S_r^1)$. This covers the desired tangent-cone classification and uniqueness, but with the weaker exceptional-set bound $\mathcal{H}^{n-2} = 0$, not $\dim_{\mathcal{H}} \leq n - 3$.

3. Conjectures and Open Questions on the Structure and Regularity of Spaces with Lower Ricci Curvature Bounds

Author: Aaron Naber

Venue/year: SIGMA 16 (2020), Paper No. 104

DOI/arXiv: DOI 10.3842/SIGMA.2020.104; arXiv:2010.10031

Relevance: This survey/open-problems article records the same codimension-three tangent-cone uniqueness assertion as a conjectural statement and explicitly distinguishes it from the known $\mathcal{H}^{n-2}$ -measure-zero result of Cheeger-Jiang-Naber. It is a citable status reference that the full Hausdorff-dimension- $\leq n - 3$ statement remained open after the Annals result.

4. Characterization of tangent cones of noncollapsed limits with lower Ricci bounds and applications

Authors: Tobias H. Colding, Aaron Naber

Venue/year: Geometric and Functional Analysis 23 (2013), 134-148

DOI/arXiv: DOI 10.1007/s00039-012-0201-2; arXiv:1108.3244

Relevance: Gives foundational structure and examples around tangent cones of noncollapsed lower-Ricci limits, including phenomena showing why uniqueness is delicate. It does not prove the codimension-three uniqueness statement.

5. Topological regularity and stability of noncollapsed spaces with Ricci curvature bounded below

Authors: Elia Brue, Alessandro Pigati, Daniele Semola

Venue/year: arXiv preprint, 2024

DOI/arXiv: arXiv:2405.03839

Relevance: Provides later progress on topological regularity/stability for noncollapsed Ricci-limit spaces, but it does not prove the metric tangent-cone uniqueness assertion outside a Hausdorff-dimension- $\leq n - 3$ set in the general Riemannian lower-Ricci

setting.

1.61.3.4. What remains unresolved

To fully resolve the problem statement, one still needs to improve the general exceptional set from $\mathcal{H}^{n-2}$ -measure zero to Hausdorff dimension at most $n - 3$, or otherwise prove that every possible bad point in the top-dimensional singular stratum is contained in such a codimension-three set. The available general literature does not establish that improvement.

1.61.4. Remaining open problems recorded by the database

1.61.4.1. Open problem 1

Let X^n be a noncollapsed Ricci limit space. Prove that there is a set $E \subset X$ with $\dim_{\mathcal{H}} E \leq n - 3$ such that, at every $x \in X \setminus E$, every tangent cone is $(n - 2)$ -symmetric; equivalently, every tangent cone splits isometrically as

$$\mathbb{R}^{n-2} \times C(Z)$$

for some compact metric space Z . No uniqueness of the tangent cone is required.

Reference supplied by the database. Aaron Naber, “Conjectures and Open Questions on the Structure and Regularity of Spaces with Lower Ricci Curvature Bounds,” *Symmetry, Integrability and Geometry: Methods and Applications (SIGMA)* **16** (2020), Article 104, Conjecture 2.16 with $k = n - 3$. DOI: <https://doi.org/10.3842/SIGMA.2020.104>. Its open status is reaffirmed in Elia Bruè, Alessandro Pigati, and Daniele Semola, “Topological regularity and stability of noncollapsed spaces with Ricci curvature bounded below,” preprint (2024), Remark 1.5, [arXiv:2405.03839](https://arxiv.org/abs/2405.03839).

1.61.5. Primary-source pointer

The originating ICM record is *The Geometry of Ricci Curvature* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=929>.

1.62. 62. The Geometry of Ricci Curvature [2014-geom-137]

Record: 2014-geom-137

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: The Geometry of Ricci Curvature

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=929>

1.62.1. Problem statement

Prove that if $n \in \mathbb{N}$ and $v > 0$, and if (M_j^n, g_j, p_j) is a sequence of pointed n -dimensional Einstein Riemannian manifolds such that $|\text{Ric}_{g_j}| \leq n-1$ and $\text{Vol}_{g_j}(B_1^{g_j}(p_j)) > v$ for every j , which converges in the pointed Gromov-Hausdorff sense to the pointed metric space (X, d, p) , then X is homeomorphic to a real analytic variety.

1.62.2. Historical background, literature, and current progress

1.62.3. Verdict: partially resolved

1.62.3.1. Precise problem

The problem is Naber's structural conjecture for noncollapsed pointed Gromov-Hausdorff limits of Einstein manifolds: for every dimension n and noncollapsing constant $v > 0$, if (M_j^n, g_j, p_j) are Einstein with $|\text{Ric}_{g_j}| \leq n-1$ and $\text{Vol}(B_1(p_j)) > v$, and if they converge to (X, d, p) , must the metric space X be homeomorphic to a real analytic variety?

1.62.3.2. Verdict

Partially resolved. I did not find a theorem that covers the full statement in all dimensions for arbitrary Riemannian Einstein manifolds. The literature proves strong regularity and several special cases, but the general real-analytic-variety conclusion remains open.

1.62.3.3. Original source

- Title: "The Geometry of Ricci Curvature"
- Author: Aaron Naber
- Venue/year: Proceedings of the International Congress of Mathematicians - Seoul 2014, Vol. II, Kyung Moon Sa, Seoul, 2014, pp. 911-937
- DOI/arXiv/URL: no DOI found; proceedings text visible at <https://dokumen.pub/proceedings-of-the-international-congress-of-mathematicians-seoul-august-13-21-2014-korea-vol-2-9788961058032-8961058037-9788961058049-8961058045-9788961058056-89610>

[58053-9788961058063-8961058061-9788961058070-896105807x.html](https://doi.org/10.4007/annals.2015.182.3.5)

- Relevance: Conjecture 6.4 states exactly the problem: a noncollapsed Gromov-Hausdorff limit of Einstein manifolds with $|Ric| \leq n - 1$ should be homeomorphic to a real analytic variety. Naber notes progress for compact Kahler manifolds with positive first Chern class, but does not prove the general conjecture.

1.62.3.4. Main partial progress

- Title: “Regularity of Einstein manifolds and the codimension 4 conjecture”
- Authors: Jeff Cheeger and Aaron Naber
- Venue/year: Annals of Mathematics 182 (2015), no. 3, 1093-1165
- DOI/arXiv/URL: <https://doi.org/10.4007/annals.2015.182.3.5> ; arXiv:1406.6534 ; <https://annals.math.princeton.edu/2015/182-3/p05>
- What it proves: For noncollapsed manifolds with bounded Ricci curvature, including Einstein manifolds, a Gromov-Hausdorff limit is smooth away from a closed subset of codimension 4; it also gives curvature estimates.
- Why it is not a full resolution: Smoothness off a codimension-4 set and curvature estimates do not identify the entire metric space topologically with a real analytic variety.
- Title: “ L^2 curvature bounds on manifolds with bounded Ricci curvature”
- Authors: Wenshuai Jiang and Aaron Naber
- Venue/year: Annals of Mathematics 193 (2021), no. 1, 107-222
- DOI/arXiv/URL: <https://doi.org/10.4007/annals.2021.193.1.2> ; arXiv:1605.05583 ; <https://annals.math.princeton.edu/2021/193-1/p02>
- What it proves: Under the same bounded-Ricci and noncollapsing hypotheses, the singular set of a limit is $(n-4)$ -rectifiable with uniform Hausdorff measure estimates, and for H^{n-4} -a.e. singular point the tangent cone is uniquely $R^{n-4} \times C(S^3/\Gamma_x)$.
- Why it is not a full resolution: Rectifiability and almost-everywhere tangent-cone structure are weaker than a global homeomorphism to a real analytic variety.
- Title: “Rectifiability of singular sets of noncollapsed limit spaces with Ricci curvature bounded below”
- Authors: Jeff Cheeger, Wenshuai Jiang, and Aaron Naber
- Venue/year: Annals of Mathematics 193 (2021), no. 2, 407-538
- DOI/arXiv/URL: <https://doi.org/10.4007/annals.2021.193.2.2> ; arXiv:1805.07988 ; <https://annals.math.princeton.edu/2021/193-2/p02>
- What it proves: For noncollapsed lower-Ricci limit spaces, the strata of the singular set are rectifiable and have strong measure-theoretic structure.
- Why it is not a full resolution: The theorem applies even more generally than Einstein limits, but its conclusion is geometric-measure-theoretic, not a real analytic variety structure.

- Title: “Regularity of Einstein 5-manifolds via 4-dimensional gap theorems”
- Authors: Yiqi Huang and Tristan Ozuch
- Venue/year: arXiv preprint, arXiv:2512.21317, v1 2025 and v2 2026
- DOI/arXiv/URL: <https://arxiv.org/abs/2512.21317>; <https://arxiv.org/html/2512.21317>
- What it proves: In dimension 5, for noncollapsed Einstein limits, the orbifold-regular set is a real-analytic Einstein 5-orbifold and is homeomorphic to a real-analytic variety. The paper explicitly describes this as confirming Naber’s Conjecture 6.4 away from extreme points in dimension 5.
- Why it is not a full resolution: The result is dimension 5 and excludes the orbifold-singular/extreme-point set. The paper formulates further open questions/conjectures about the remaining singular set in dimension 5 and about higher-dimensional orbifold regularity.

1.62.3.5. Special cases with stronger conclusions

- Title: “Ricci curvature bounds and Einstein metrics on compact manifolds”
- Author: Michael T. Anderson
- Venue/year: Journal of the American Mathematical Society 2 (1989), no. 3, 455-490
- DOI/arXiv/URL: <https://doi.org/10.1090/S0894-0347-1989-0999661-1>
- Relevance: Part of the four-dimensional orbifold compactness theory for Einstein metrics.
- Limitation: It is a four-dimensional compactness/orbifold result, not the all-dimensional Riemannian Einstein-limit statement.
- Title: “On a construction of coordinates at infinity on manifolds with fast curvature decay and maximal volume growth”
- Authors: Shigetoshi Bando, Atsushi Kasue, and Hiraku Nakajima
- Venue/year: Inventiones Mathematicae 97 (1989), 313-349
- DOI/arXiv/URL: <https://doi.org/10.1007/BF01389045>
- Relevance: Provides analytic tools used in four-dimensional Einstein/ALE/orbifold compactness and removable singularity theory.
- Limitation: It does not prove the all-dimensional real-analytic-variety conclusion.
- Title: “Gromov-Hausdorff limits of Kahler manifolds and algebraic geometry”
- Authors: Simon Donaldson and Song Sun
- Venue/year: Acta Mathematica 213 (2014), no. 1, 63-106
- DOI/arXiv/URL: <https://doi.org/10.1007/s11511-014-0116-3>; arXiv:1206.2609
- What it proves: In the polarized Kahler setting with suitable noncollapsing and two-sided Ricci bounds, limit spaces acquire normal projective algebraic variety structures.
- Why it is not a full resolution: The statement has no Kahler, complex, polarization, com-

pactness, or diameter hypothesis; it asks about arbitrary Riemannian Einstein limits.

- Title: “Gromov-Hausdorff limits of Kahler manifolds and algebraic geometry, II”
- Authors: Simon Donaldson and Song Sun
- Venue/year: Journal of Differential Geometry 107 (2017), no. 2, 327-371
- DOI/arXiv/URL: <https://doi.org/10.4310/jdg/1506650422> ; arXiv:1507.05082
- What it proves: Further algebraic structure results for Kahler-Einstein limits and their tangent cones.
- Why it is not a full resolution: It remains within the Kahler/algebraic framework and does not cover general Riemannian Einstein manifolds.
- Title: “Gromov-Hausdorff limits of Kahler manifolds with Ricci curvature bounded below”
- Authors: Gang Liu and Gabor Székelyhidi
- Venue/year: Geometric and Functional Analysis 32 (2022), no. 2, 236-279
- DOI/arXiv/URL: <https://doi.org/10.1007/s00039-022-00594-8> ; arXiv:1804.08567
- What it proves: Noncollapsed Gromov-Hausdorff limits of polarized Kahler manifolds with Ricci curvature bounded below are homeomorphic to normal projective varieties, extending Donaldson-Sun beyond two-sided Ricci bounds in the polarized Kahler setting.
- Why it is not a full resolution: Again, the polarized Kahler hypotheses are extra assumptions absent from the problem statement.

1.62.3.6. Current obstruction

The remaining unresolved part is the full global topology/analytic structure of arbitrary noncollapsed Einstein limit spaces in all dimensions. Known results control the regular set, codimension, rectifiability, curvature integrability, and some tangent cones; special cases are orbifolds or complex algebraic varieties. None of these results, including the 2026 Huang-Ozuch preprint, proves that every all-dimensional Riemannian Einstein limit satisfying only the stated hypotheses is homeomorphic to a real analytic variety.

1.62.4. Remaining open problems recorded by the database

1.62.4.1. Open problem 1

Let (X, d) be a Gromov-Hausdorff limit of noncollapsed Einstein 5-manifolds. Prove that the non-orbifold set $\mathcal{S}_{\text{orb}} = X \setminus \mathcal{R}_{\text{orb}}$ is countable. Equivalently, away from a countable set, (X, d) should be a smooth real-analytic Einstein 5-orbifold, with singularities locally of the form $\mathbb{R} \times \mathbb{R}^4/\Gamma$ along countably many geodesics.

Reference supplied by the database. Yiqi Huang and Tristan Ozuch, “Regularity of Einstein 5-manifolds via 4-dimensional gap theorems,” arXiv preprint (2025; revised 2026), Conjecture 1.24 (“Codimension-5 conjecture in dimension 5”), arXiv:2512.21317v2 [math.DG], DOI: 10.48550/arXiv.2512.21317, <https://arxiv.org/abs/2512.21317>.

1.62.5. Primary-source pointer

The originating ICM record is *The Geometry of Ricci Curvature* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=929>.

1.63. 63. Non-negatively curved manifolds and Tits geometry [2014-geom-294]

Record: 2014-geom-294

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: Non-negatively curved manifolds and Tits geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=885>

1.63.1. Problem statement

Prove that for every integer $n > 1$ there exists a constant $\delta(n) > 0$, depending only on n , such that every compact smooth n -dimensional manifold that admits a Riemannian metric of positive sectional curvature also admits a Riemannian metric g with the following property: after multiplying g by a positive constant so that the global maximum of its sectional curvature is 1, one has

$$\delta(n) < \sec_g(p, \sigma) \leq 1$$

for every point p of the manifold and every two-dimensional linear subspace $\sigma \subset T_p M$.

1.63.2. Historical background, literature, and current progress

The exact assertion is **unresolved**. It is the uniform pinching conjecture attributed to Marcel Berger. A current survey states that it is true in dimensions 2 and 3 and is otherwise widely open. Thus no dimension-only constant is known for arbitrary compact positively curved manifolds in every dimension $n \geq 4$, and no counterexample is known.

Key references and established progress:

- Wilderich Tuschmann, *Moduli Spaces of Metrics with Lower or Pinched Curvature Bounds and Their Compactifications*, **Journal of Mathematical Study** 58(2) (2025), 175–187, DOI <https://doi.org/10.4208/jms.v58n2.25.04>. Conjecture 5 is the exact uniform pinching assertion (up to the immaterial choice of strict versus weak lower inequality, since a weak positive bound may be decreased). Tuschmann records that uniformization and Hamilton's theorem settle dimensions 2 and 3, while the conjecture is widely open in all higher dimensions.
- Richard S. Hamilton, *Three-manifolds with positive Ricci curvature*, **Journal of Differential Geometry** 17(2) (1982), 255–306, DOI <https://doi.org/10.4310/jdg/1214436922>. The main theorem proves that a compact 3-manifold admitting a metric of positive Ricci curvature also admits a metric of constant positive sectional curvature. Positive sectional curvature implies positive Ricci curvature, so the problem holds in dimension 3 (and classical surface uniformization gives dimension 2).
- Anton Petrunin, Xiaochun Rong, and Wilderich Tuschmann, *Collapsing vs. posi-*

tive pinching, **Geometric and Functional Analysis** 9(4) (1999), 699–735, DOI <https://doi.org/10.1007/s000390050100>. The paper constructs uniformly positively pinched collapsing sequences among (7)-dimensional Eschenburg spaces and (13)-dimensional Bazaikin spaces and proves bounded and continuous forms of the Klingenberg–Sakai noncollapsing conjecture. This verifies that major infinite families of known positively curved manifolds can retain a positive uniform pinching bound, but it does not treat all positively curved manifolds in either dimension.

- Thomas Püttmann, *Optimal pinching constants of odd-dimensional homogeneous spaces*, **Inventiones Mathematicae** 138(3) (1999), 631–684, DOI <https://doi.org/10.1007/s002220050355>. Püttmann computes optimal pinching within broad invariant metric classes on the Berger and Aloff–Wallach spaces, obtaining $1/37$ for the 13-dimensional Berger space and for the relevant invariant metrics on $W_{1,1}^7$, as well as explicit values for other Aloff–Wallach spaces. These are substantive special-family calculations, not a bound for arbitrary positively curved manifolds.

What remains is precisely the unrestricted higher-dimensional case: for each $n \geq 4$, one must control the best achievable pinching ratio uniformly over every diffeomorphism type of compact n -manifold that admits some positively curved metric. Existing low-dimensional classifications, symmetry assumptions, homogeneous or biquotient calculations, and results for selected infinite families do not exhaust that class.

1.63.3. Remaining open problems recorded by the database

1.63.3.1. Open problem 1

Prove that there exists a constant $\delta(4) > 0$ such that every compact smooth four-dimensional manifold that admits a Riemannian metric of positive sectional curvature also admits a Riemannian metric g for which, after rescaling so that the global maximum of the sectional curvature is 1, one has $\delta(4) < \sec_g(p, \sigma) \leq 1$ for every point p and every two-dimensional linear subspace $\sigma \subset T_p M$.

Reference supplied by the database. Wilderich Tuschmann, “Moduli Spaces of Metrics with Lower or Pinched Curvature Bounds and Their Compactifications,” *Journal of Mathematical Study* 58(2) (2025), 175–187, Conjecture 5 and the subsequent open-status statement, DOI: <https://doi.org/10.4208/jms.v58n2.25.04>.

1.63.3.2. Open problem 2

For each integer $k \geq 3$, let $G = S^3 \times S^3$, where S^3 is the group of unit quaternions, and define

$$H = \langle (i, i), (j, -j) \rangle \subset G,$$

$$K^- = \{(e^{i\theta}, e^{i\theta}) : \theta \in \mathbb{R}\} H, \quad K_k^+ = \{(e^{(2k-1)j\theta}, e^{(2k+1)j\theta}) : \theta \in \mathbb{R}\} H.$$

Since $K^-/H \cong K_k^+/H \cong S^1$, let P_k be the compact cohomogeneity-one seven-manifold obtained from the group diagram $H \subset \{K^-, K_k^+\} \subset G$, equivalently

$$P_k = (G \times_{K^-} D^2) \cup_{G/H} (G \times_{K_k^+} D^2),$$

where each quotient circle acts on D^2 by the standard rotation action and the two disk bundles are glued along their common boundary G/H . Determine whether every P_k , $k \geq 3$, admits a G -invariant Riemannian metric of positive sectional curvature.

Reference supplied by the database. Karsten Grove, “Developments around Positive Sectional Curvature,” *Surveys in Differential Geometry* 13 (2009), 117–133, Problem 7, DOI: <https://doi.org/10.4310/SDG.2008.v13.n1.a4>; arXiv:0902.4419.

1.63.4. Primary-source pointer

The originating ICM record is *Non-negatively curved manifolds and Tits geometry* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=885>.

1.64. 64. Einstein 4-manifolds and singularities [2014-geom-324]

Record: 2014-geom-324

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: Einstein 4-manifolds and singularities

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=871>

1.64.1. Problem statement

Prove or disprove that every complete, simply connected, Ricci-flat Riemannian 4-manifold (M, g) with exactly one asymptotically locally Euclidean (ALE) end is hyper-Kähler. Here the ALE hypothesis means that there exist a compact set $K \subset M$, a radius $R > 0$, and a finite subgroup $\Gamma \subset SO(4)$ acting freely on S^3 such that $M \setminus K$ is diffeomorphic to $(\mathbb{R}^4 \setminus B_R)/\Gamma$, and under this diffeomorphism g is asymptotic at infinity to the metric induced by the Euclidean metric on $\mathbb{R}^4$. The hyper-Kähler conclusion means that the holonomy of g is contained in $Sp(1) \cong SU(2)$, equivalently that M admits three parallel complex structures satisfying the quaternionic relations and compatible with g . Consequently, such a manifold would, up to isometry, belong to Kronheimer's classification of hyper-Kähler ALE 4-manifolds.

1.64.2. Historical background, literature, and current progress

Verdict: unresolved.

No proof or counterexample is known for the full statement. The issue is usually called the Bando–Kasue–Nakajima conjecture. The most recent sources below still describe it as open and impose additional topological, geometric, symmetry, or analytic hypotheses. In particular, simple connectivity alone is not known to imply the spin and end-group hypotheses used by Nakajima, nor is an arbitrary manifold in the statement known to have one of the topologies, complex structures, symmetries, or deformation bounds covered by later results.

Established partial progress:

- Shigetoshi Bando, Atsushi Kasue, and Hiraku Nakajima, *On a construction of coordinates at infinity on manifolds with fast curvature decay and maximal volume growth*, **Inventiones Mathematicae** 97 (1989), 313–349, <https://doi.org/10.1007/BF01389045>. This is the foundational ALE analysis associated with the conjecture, but it does not classify all simply connected Ricci-flat ALE four-manifolds.
- Hiraku Nakajima, *Self-duality of ALE Ricci-flat 4-manifolds and positive mass theorem*, in **Recent Topics in Differential and Analytic Geometry**, Advanced Studies in Pure Mathematics 18-I (1990), 385–396, <https://doi.org/10.2969/aspm/01810385>. Corollary 3.3 proves the desired hyper-Kähler conclusion when the manifold is spin and the group at infinity satisfies $\Gamma \subset SU(2)$. These are extra hypotheses not present in the problem.

- Peter B. Kronheimer, *The construction of ALE spaces as hyper-Kähler quotients*, **Journal of Differential Geometry** 29 (1989), 665–683, <https://doi.org/10.4310/jdg/1214443066>, and *A Torelli-type theorem for gravitational instantons*, **Journal of Differential Geometry** 29 (1989), 685–697, <https://doi.org/10.4310/jdg/1214443067>. These construct and classify the hyper-Kähler ALE class; they do not prove that a general Ricci-flat ALE metric is hyper-Kähler.
- Michael T. Lock and Jeff A. Viaclovsky, *Quotient singularities, eta invariants, and self-dual metrics*, **Geometry & Topology** 20 (2016), 1773–1806, <https://doi.org/10.2140/gt.2016.20.1773>, [arXiv:1501.03234](https://arxiv.org/abs/1501.03234). Theorem 1.5 proves the conjecture when the underlying manifold is diffeomorphic to the minimal resolution of $\mathbb{C}^2/\Gamma$, or to an iterated blow-up of such a resolution, for $\Gamma \subset U(2)$; it then forces $\Gamma \subset SU(2)$ and the metric to be hyper-Kähler.
- Mingyang Li, *On 4-dimensional Ricci-flat ALE manifolds*, preprint (2023, revised 2024), [arXiv:2304.01609](https://arxiv.org/abs/2304.01609). It gives a conformal compactification correspondence for Hermitian non-Kähler Ricci-flat ALE metrics and proves that, when $\Gamma \subset SU(2)$, the only such metric is Eguchi–Hanson with the opposite complex orientation. This does not cover metrics with generic Weyl tensor or general end group.
- Bernardo Araneda and James Lucietti, *All toric Hermitian ALE gravitational instantons*, preprint (2025), [arXiv:2510.09291](https://arxiv.org/abs/2510.09291). It proves that the only smooth toric Hermitian non-Kähler ALE instanton is Eguchi–Hanson, thereby proving the conjecture only within the toric Hermitian class.
- Partha Ghosh, *Mass and rigidity in almost Kähler geometry*, preprint (2026), [arXiv:2603.08627](https://arxiv.org/abs/2603.08627). It proves that a four-dimensional Ricci-flat almost-Kähler manifold with maximal volume growth and L^2 curvature is Kähler. ALE metrics meet the analytic conditions, but the existence of a compatible almost-Kähler structure is an additional assumption.
- Michael B. Law, *An analytical characterization of Eguchi–Hanson space and its higher-dimensional analogs*, preprint (2026), [arXiv:2604.23410](https://arxiv.org/abs/2604.23410). Theorem 1.2 classifies Ricci-flat ALE four-spaces with group at infinity $\mathbb{Z}_2$ only under the additional bound $\dim \ker_{L^2} \Delta_L \leq 3$, obtaining Eguchi–Hanson or the flat orbifold.
- Hari K. Kunduri and James Lucietti, *On the existence of toric ALE and ALF gravitational instantons*, preprint (2026), [arXiv:2604.15159](https://arxiv.org/abs/2604.15159). Theorem 1.1 constructs a unique toric ALE solution for each admissible rod structure only up to possible conical singularities. The paper explicitly leaves open whether non-Hermitian examples can be made everywhere smooth, so these singular solutions neither prove nor disprove the stated smooth-manifold problem.

Thus the uncovered case consists of general smooth, complete, simply connected Ricci-flat ALE four-manifolds without the additional spin/end-group, topology, Hermitian or almost-Kähler, toric, or small- L^2 -kernel assumptions. No citable smooth counterexample has been found.

1.64.3. Remaining open problems recorded by the database

1.64.3.1. Open problem 1

Prove or disprove that, up to isometry and constant rescaling, the Eguchi–Hanson metric equipped with the orientation opposite to its hyper-Kähler orientation is the only complete Ricci-flat asymptotically locally Euclidean (ALE) oriented Riemannian 4-manifold of Type II. Here ALE means that outside a compact set the manifold is diffeomorphic to $(\mathbb{R}^4 \setminus B_R)/\Gamma$ for a finite subgroup $\Gamma \subset SO(4)$ acting freely on S^3 and that the metric is

asymptotic to the induced Euclidean quotient metric; Type II means that the self-dual Weyl-curvature endomorphism $W^+ : \Lambda_+^2 \rightarrow \Lambda_+^2$ is nowhere zero and has exactly two distinct eigenvalues at every point. No restriction is imposed on the group Γ at infinity.

Reference supplied by the database. Mingyang Li, *On 4-dimensional Ricci-flat ALE manifolds*, arXiv preprint, 2023 (revised 2024), unnumbered Conjecture in the Introduction, arXiv:2304.01609, <https://arxiv.org/abs/2304.01609>.

1.64.3.2. Open problem 2

Determine whether there exists a complete, simply connected, Ricci-flat asymptotically locally Euclidean (ALE) Riemannian 4-manifold (M, g) with an effective isometric action of the torus T^2 and with the following property for both choices of orientation on M : the corresponding self-dual Weyl-curvature endomorphism $W^+ : \Lambda_+^2 \rightarrow \Lambda_+^2$ has three distinct eigenvalues on a dense open subset of M . Here ALE means that outside a compact set M is diffeomorphic to $(\mathbb{R}^4 \setminus B_R)/\Gamma$ for a finite subgroup $\Gamma \subset SO(4)$ acting freely on S^3 and that g is asymptotic to the induced Euclidean quotient metric.

Reference supplied by the database. Mingyang Li and Song Sun, *Geometry of gravitational instantons*, Surveys in Differential Geometry (to appear), 2026, Question 2, arXiv:2606.10443, <https://arxiv.org/abs/2606.10443>.

1.64.4. Primary-source pointer

The originating ICM record is *Einstein 4-manifolds and singularities* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=871>.

1.65. 65. Hyperbolic orbifolds of small volume [2014-geom-325]

Record: 2014-geom-325

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: Hyperbolic orbifolds of small volume

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=855>

1.65.1. Problem statement

Prove that for every integer $n \geq 2$ with $n \neq 3$ and every compact, orientable hyperbolic n -manifold M , there exists a complete, orientable, noncompact hyperbolic n -manifold M_1 of finite volume such that

$$\text{Vol}(M_1) < \text{Vol}(M).$$

1.65.2. Historical background, literature, and current progress

The problem is **unresolved**. The full assertion is known in dimensions $n = 2, 4, 6$, but no result found covers arbitrary compact hyperbolic n -manifolds in every remaining dimension. In particular, the strongest general high-dimensional comparison is restricted to *arithmetic* compact manifolds, so it does not establish the stated assertion for nonarithmetic M .

- Ruth Kellerhals and Thomas Zehrt, “The Gauss–Bonnet Formula for Hyperbolic Manifolds of Finite Volume,” *Geometriae Dedicata* 84 (2001), 49–62, DOI <https://doi.org/10.1023/A:1010352829732>. Their Gauss–Bonnet formula and Theorem 1.2 show that an orientable compact even-dimensional hyperbolic manifold has even Euler characteristic. Together with a finite-area cusped surface of Euler characteristic -1 , this proves the required comparison for $n = 2$.
- John G. Ratcliffe and Steven T. Tschantz, “The Volume Spectrum of Hyperbolic 4-Manifolds,” *Experimental Mathematics* 9 (2000), 101–125, DOI <https://doi.org/10.1080/10586458.2000.10504640>. They construct complete noncompact hyperbolic 4-manifolds of Euler characteristic 1, hence of the least possible finite volume. Since compact orientable examples have even Euler characteristic, this proves the assertion for $n = 4$.
- Brent Everitt, John G. Ratcliffe, and Steven T. Tschantz, “Right-Angled Coxeter Polytopes, Hyperbolic Six-Manifolds, and a Problem of Siegel,” *Mathematische Annalen* 354 (2012), 871–905, DOI <https://doi.org/10.1007/s00208-011-0744-2>. They construct orientable noncompact hyperbolic 6-manifolds with Euler characteristic -1 , the least possible volume, proving the assertion for $n = 6$.
- Mikhail Belolipetsky and Vincent Emery, “Hyperbolic Manifolds of Small Volume,” *Documenta Mathematica* 19 (2014), 801–814, DOI <https://doi.org/10.4171/DM/464>, arXiv:1310.2270. Conjecture 1 is the exact problem, and Theorem 2 proves it only when M is arithmetic and $n \geq 30$. The paper explicitly notes that extending this to all

manifolds would require the still-open minimal-volume conjecture or a different argument.

- Giovanni Italiano, Bruno Martelli, and Matteo Migliorini, “Hyperbolic 5-Manifolds That Fiber over S^1 ,” *Inventiones Mathematicae* 231 (2023), 1–38, DOI <https://doi.org/10.1007/s00222-022-01141-w>, arXiv:2105.14795. This later work identifies the two-cusped Ratcliffe–Tschantz manifold of volume $7\zeta(3)/4$ as the smallest hyperbolic 5-manifold currently known, not as a proved volume minimizer. Thus even dimension 5 remains outside the established cases of the exact problem.

Consequently, the unresolved gap consists of arbitrary (especially nonarithmetic) compact manifolds in the admissible dimensions other than 2, 4, 6; the arithmetic theorem for $n \geq 30$ is substantive partial progress but does not exhaust the hypotheses of the problem.

1.65.3. Remaining open problems recorded by the database

1.65.3.1. Open problem 1

Determine all integers $n > 4$ for which there exists a complete hyperbolic n -manifold of finite volume with exactly one cusp.

Reference supplied by the database. Edoardo Rizzi, “Some Cusp-Transitive Hyperbolic 4-Manifolds,” *Geometriae Dedicata* 219 (2025), Article 61, Question 1.5. DOI: <https://doi.org/10.1007/s10711-025-01010-9>.

1.65.4. Primary-source pointer

The originating ICM record is *Hyperbolic orbifolds of small volume* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=855>.

1.66. 66. Non-negatively curved manifolds and Tits geometry [2014-geom-326]

Record: 2014-geom-326

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: Non-negatively curved manifolds and Tits geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=885>

1.66.1. Problem statement

Prove that every compact Riemannian manifold M without boundary whose sectional curvature satisfies $\sec_M \geq 0$ everywhere has nonnegative Euler characteristic, $\chi(M) \geq 0$. Prove moreover that if M is even-dimensional and $\sec_M > 0$ everywhere, then $\chi(M) > 0$.

1.66.2. Historical background, literature, and current progress

The problem is **unresolved**. It is the Euler-characteristic form of Hopf's conjecture. Odd-dimensional closed manifolds have Euler characteristic zero, and the curvature assertions are known in dimensions (2) and (4), but neither assertion is proved for arbitrary even-dimensional manifolds in dimensions at least (6). In particular, all currently verified higher-dimensional results impose additional curvature, symmetry, or topology hypotheses, so they do not cover the statement as written.

Key established progress includes:

- Chuan-Chih Hsiung and Kenneth Michael Shiskowski, "Euler-Poincaré characteristic and higher order sectional curvature. I," *Transactions of the American Mathematical Society* **305** (1988), 113–128. DOI: <https://doi.org/10.1090/S0002-9947-1988-0920149-3>. The paper records the theorem of Milnor and Chern proving the nonnegative-curvature assertion in dimension (4), and proves the higher-dimensional sign conclusion only under an extra higher-order sectional-curvature condition.
- Thomas Püttmann and Catherine Searle, "The Hopf conjecture for manifolds with low cohomogeneity or high symmetry rank," *Proceedings of the American Mathematical Society* **130** (2002), 163–166. DOI: <https://doi.org/10.1090/S0002-9939-01-06039-7>. They prove both the positive- and nonnegative-curvature conclusions when an isometric compact-group action with principal isotropy (H), acting group (G), and cohomogeneity (k) satisfies $k - (\text{rank } G - \text{rank } H) \leq 5$.
- Lee Kennard, "On the Hopf Conjecture with Symmetry," *Geometry & Topology* **17** (2013), 563–593. DOI: <https://doi.org/10.2140/gt.2013.17.563>. This proves the positive-curvature conclusion under a logarithmic lower bound on the rank of an isometric torus action and explicitly notes that the unrestricted conjecture is open above dimension (4).
- Jan Nienhaus, "An improved four-periodicity theorem and a conjecture of Hopf with symmetry," arXiv preprint (2022), [arXiv:2211.13151](https://arxiv.org/abs/2211.13151). Corollary 3.3 proves $\chi(M) > 0$ for

every closed even-dimensional positively curved manifold admitting an effective isometric T^4 -action. The torus-action assumption is not present in the problem and remains a genuine gap.

- Burkhard Wilking, “Nonnegatively and positively curved manifolds,” in *Metric and Comparison Geometry, Surveys in Differential Geometry* **11**, International Press, 2007, 25–62. Direct URL: [survey](#). Conjecture 5.2 states the two sign assertions together and explains that the Bott–Grove–Halperin rational-ellipticity conjecture would imply the nonnegative-curvature part; that stronger conjectural route is itself unresolved.

Thus the literature supplies strong special cases but no proof or counterexample covering all compact boundaryless (M) under only the sectional-curvature hypotheses in the statement.

1.66.3. Remaining open problems recorded by the database

1.66.3.1. Open problem 1

Prove that every compact, orientable, six-dimensional Riemannian manifold M without boundary whose sectional curvature satisfies $\sec_M \geq 0$ everywhere has nonnegative Euler characteristic, $\chi(M) \geq 0$.

Reference supplied by the database. Chuan-Chih Hsiung and Kenneth Michael Shiskowski, “Euler-Poincaré characteristic and higher order sectional curvature. I,” *Transactions of the American Mathematical Society* **305** (1988), 113–128; the Hopf conjecture stated in the abstract and introduction, with only dimension 4 identified as settled without the paper’s extra higher-order-curvature hypothesis. DOI: <https://doi.org/10.1090/S0002-9947-1988-0920149-3>

1.66.3.2. Open problem 2

Prove that every compact six-dimensional Riemannian manifold M without boundary whose sectional curvature satisfies $\sec_M > 0$ everywhere has positive Euler characteristic, $\chi(M) > 0$.

Reference supplied by the database. Xiaochun Su and Yusheng Wang, “The Hopf conjecture for positively curved manifolds with discrete abelian group actions,” *Differential Geometry and its Applications* **26** (2008), no. 3, 313–322; Introduction and the explicit six-dimensional fixed-point-component hypothesis used in Section 3. DOI: <https://doi.org/10.1016/j.difgeo.2007.11.023>

1.66.3.3. Open problem 3

Prove that every compact, connected, even-dimensional Riemannian manifold M without boundary that has positive sectional curvature everywhere and admits an effective isometric action of the circle group S^1 has positive Euler characteristic, $\chi(M) > 0$.

Reference supplied by the database. Oliver Knill, “On a theorem of Grove and Searle,” arXiv preprint (2020), Section 1.1 (the open statement $\mathcal{K}_{2d} \subset \mathcal{D}$ for the class with circle symmetry). arXiv:2006.11973, <https://arxiv.org/abs/2006.11973>

1.66.4. Primary-source pointer

The originating ICM record is *Non-negatively curved manifolds and Tits geometry* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=885>.

1.67. 67. Non-negatively curved manifolds and Tits geometry [2014-geom-327]

Record: 2014-geom-327

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: Non-negatively curved manifolds and Tits geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=885>

1.67.1. Problem statement

Let a compact Lie group G act isometrically on a compact, simply connected Riemannian manifold M with nonnegative sectional curvature. Assume that the action is hyperpolar of cohomogeneity at least 2: there is a complete, flat, isometrically immersed submanifold $\Sigma \subset M$, called a section, such that Σ meets every G -orbit and, at every point of intersection, is perpendicular to that orbit; moreover, $\dim(M/G) = \dim \Sigma \geq 2$.

For each singular point $p \in \Sigma$, the slice representation of the isotropy group G_p on the normal space to the orbit $G \cdot p$ is polar with section $T_p \Sigma$. Let W_p denote its finite Weyl group, acting on $T_p \Sigma$ as a linear reflection group, and let W be the group of isometries of Σ generated by the corresponding reflections as p ranges over the singular points of Σ . Let $\tilde{\Sigma}$ be the universal cover of Σ , and let $\tilde{W}$ be the affine Coxeter group generated by the lifts to $\tilde{\Sigma}$ of the reflections in W . Assume that $\tilde{W}$ is irreducible, meaning that it cannot be written as a direct product of two nontrivial Coxeter groups.

Prove that the G -action on M is equivariantly diffeomorphic to a quotient of a polar action on a Riemannian symmetric space. More precisely, prove that there exist a Riemannian symmetric space S , a compact Lie group L acting freely on S , and a polar action of $G \times L$ on S such that the induced G -action on S/L is equivariantly diffeomorphic to the given G -action on M .

1.67.2. Historical background, literature, and current progress

The exact assertion remains unresolved. No proof or counterexample covering every irreducible affine Coxeter type and every cohomogeneity at least 2 was found.

The strongest direct result is Theorem 6.3 of Fuquan Fang, “Non-negatively curved manifolds and Tits geometry,” in *Proceedings of the International Congress of Mathematicians, Seoul 2014*, Vol. II, Kyung Moon Sa, 2014, pp. 867–880 ([official ICM program page](#)). It proves the required principal-bundle/quotient conclusion for connected chamber systems of affine types $\tilde{A}_n$, $\tilde{D}_n$, $\tilde{E}_6$, $\tilde{E}_7$, and $\tilde{E}_8$ (rank at least 3). The same article explicitly leaves the double-bond types $\tilde{B}_n$, $\tilde{C}_n$, and $\tilde{F}_4$ untreated, so it does not prove the full statement.

Fuquan Fang and Karsten Grove, “Reflection groups in non-negative curvature,” *Journal of Differential Geometry* **102** (2016), no. 2, 179–205, <https://doi.org/10.4310/jdg/1453910453>, classify complete nonnegatively curved manifolds with cocompact reflection-group actions and prove the Coxeter splitting theorem used to define irreducibility. They restate the

broader polar-action assertion as a conjecture and do not classify the ambient polar G -actions or establish the required symmetric-space quotient.

Fuquan Fang, Karsten Grove, and Gudlaugur Thorbergsson, “Tits Geometry and Positive Curvature,” *Acta Mathematica* **218** (2017), no. 1, 1–53, <https://doi.org/10.4310/ACTA.2017.v218.n1.a1>, prove the analogous classification under positive sectional curvature. They again state the nonnegative-curvature assertion as a conjecture and explain that the nonnegative case requires substantially different affine-building arguments; their theorem therefore does not cover the zero-curvature cases in the present statement.

Yi Shi, “The dual foliation of polar actions on nonnegatively curved manifolds,” *Geometriae Dedicata* **220** (2026), article 7, <https://doi.org/10.1007/s10711-025-01062-x>, [arXiv:2509.24155](https://arxiv.org/abs/2509.24155), proves that such a polar manifold reduces to a horizontally connected, totally geodesic polar dual leaf with the same orbit space, and otherwise admits a G -equivariant Riemannian submersion onto a homogeneous space with totally geodesic fibers. This is structural progress for nonnegative curvature, but it neither constructs the symmetric space S and free L -action required here nor settles the remaining affine Coxeter types.

Thus the established literature covers important simply-laced cases and gives reflection-group and dual-foliation structure, but the full quotient classification, in particular the untreated irreducible double-bond affine cases, remains open.

1.67.3. Remaining open problems recorded by the database

1.67.3.1. Open problem 1

Let a compact Lie group G act isometrically and hyperpolarly on a compact, simply connected Riemannian manifold M with nonnegative sectional curvature and with cohomogeneity at least 2. Thus there is a complete, flat, isometrically immersed section $\Sigma \subset M$ that meets every G -orbit orthogonally. Let $\widetilde{W}$ be the affine Coxeter group on the universal cover of Σ generated by the lifts of the reflections arising from the polar slice representations at singular points of Σ , and assume that $\widetilde{W}$ has irreducible affine Coxeter type $\widetilde{B}_n$.

Prove that there exist a Riemannian symmetric space S , a compact Lie group L acting freely on S , and a polar action of $G \times L$ on S such that the induced G -action on S/L is equivariantly diffeomorphic to the given G -action on M .

Reference supplied by the database. Fuquan Fang, “Non-negatively curved manifolds and Tits geometry,” *Proceedings of the International Congress of Mathematicians, Seoul 2014*, Vol. II, Kyung Moon Sa, Seoul, 2014, pp. 867–880, Remark 6.5. <https://www.mathunion.org/fileadmin/IMU/ICM2014/offline/en/program/scientific/topics.html>

1.67.3.2. Open problem 2

Let a compact Lie group G act isometrically and hyperpolarly on a compact, simply connected Riemannian manifold M with nonnegative sectional curvature and with cohomogeneity at least 2. Thus there is a complete, flat, isometrically immersed section $\Sigma \subset M$ that meets every G -orbit orthogonally. Let $\widetilde{W}$ be the affine Coxeter group on the universal cover of Σ generated by the lifts of the reflections arising from the polar slice representations at singular points of Σ , and assume that $\widetilde{W}$ has irreducible affine Coxeter type $\widetilde{B}_n$.

geneity at least 2. Thus there is a complete, flat, isometrically immersed section $\Sigma \subset M$ that meets every G -orbit orthogonally. Let $\tilde{W}$ be the affine Coxeter group on the universal cover of Σ generated by the lifts of the reflections arising from the polar slice representations at singular points of Σ , and assume that $\tilde{W}$ has irreducible affine Coxeter type $\tilde{C}_n$.

Prove that there exist a Riemannian symmetric space S , a compact Lie group L acting freely on S , and a polar action of $G \times L$ on S such that the induced G -action on S/L is equivariantly diffeomorphic to the given G -action on M .

Reference supplied by the database. Fuquan Fang, “Non-negatively curved manifolds and Tits geometry,” *Proceedings of the International Congress of Mathematicians, Seoul 2014*, Vol. II, Kyung Moon Sa, Seoul, 2014, pp. 867–880, Remark 6.5. <https://www.mathunion.org/fileadmin/IMU/ICM2014/offline/en/program/scientific/topics.html>

1.67.3.3. Open problem 3

Let a compact Lie group G act isometrically and hyperpolarly on a compact, simply connected Riemannian manifold M with nonnegative sectional curvature and with cohomogeneity at least 2. Thus there is a complete, flat, isometrically immersed section $\Sigma \subset M$ that meets every G -orbit orthogonally. Let $\tilde{W}$ be the affine Coxeter group on the universal cover of Σ generated by the lifts of the reflections arising from the polar slice representations at singular points of Σ , and assume that $\tilde{W}$ has irreducible affine Coxeter type $\tilde{F}_4$.

Prove that there exist a Riemannian symmetric space S , a compact Lie group L acting freely on S , and a polar action of $G \times L$ on S such that the induced G -action on S/L is equivariantly diffeomorphic to the given G -action on M .

Reference supplied by the database. Fuquan Fang, “Non-negatively curved manifolds and Tits geometry,” *Proceedings of the International Congress of Mathematicians, Seoul 2014*, Vol. II, Kyung Moon Sa, Seoul, 2014, pp. 867–880, Remark 6.5. <https://www.mathunion.org/fileadmin/IMU/ICM2014/offline/en/program/scientific/topics.html>

1.67.4. Primary-source pointer

The originating ICM record is *Non-negatively curved manifolds and Tits geometry* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=885>.

1.68. 68. Liouville equations from a variational point of view [2014-geom-336]

Record: 2014-geom-336

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: Liouville equations from a variational point of view

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.3/ICM2014.3.pdf#page=357>

1.68.1. Problem statement

Let (Σ, g) be a smooth compact Riemannian surface without boundary, let $p_1, \dots, p_m \in \Sigma$ be distinct points, and let $\alpha_1, \dots, \alpha_m > -1$. Determine general conditions on (Σ, g) , the marked points p_j , and the weights α_j that obstruct the existence of a conformal metric $\tilde{g} = e^{2w}g$ on $\Sigma \setminus \{p_1, \dots, p_m\}$ having constant Gaussian curvature $K \in \mathbb{R}$ and a conical singularity of total angle $2\pi(1 + \alpha_j)$ at each p_j . Equivalently, derive general criteria under which there is no constant $K \in \mathbb{R}$ and no distributional solution w , smooth away from the marked points and satisfying

$$w(x) = \alpha_j \log d_g(x, p_j) + O(1) \quad \text{as } x \rightarrow p_j,$$

of

$$-\Delta_g w + K_g = K e^{2w} - 2\pi \sum_{j=1}^m \alpha_j \delta_{p_j} \quad \text{on } \Sigma,$$

where Δ_g is the Laplace–Beltrami operator, K_g is the Gaussian curvature of g , d_g is the Riemannian distance, and δ_{p_j} is the Dirac measure at p_j .

1.68.2. Historical background, literature, and current progress

Verdict: unresolved.

Write $\beta_j = 1 + \alpha_j > 0$ and $\chi_{\text{con}} = \chi(\Sigma) + \sum_j (\beta_j - 1) = \chi(\Sigma) + \sum_j \alpha_j$. Conical Gauss–Bonnet forces the sign of K to be the sign of χ_{con} . The cases $\chi_{\text{con}} < 0$ and $\chi_{\text{con}} = 0$ are completely understood: the hyperbolic or flat conical metric exists for every prescribed conformal structure, marked divisor, and admissible angle vector (with the flat Gauss–Bonnet equality), and is unique after the usual normalization. Thus the unresolved part of the stated problem is the spherical regime $\chi_{\text{con}} > 0$, especially when one or more cone angles exceed 2π .

Substantial obstruction and existence results are known, but they do not characterize all fixed data $(\Sigma, g, p_j, \alpha_j)$. In particular, angle-only existence on some conformal surface is weaker than existence for the prescribed conformal class and prescribed marked points in the statement. Current work continues to obtain classifications only in special configurations, local deformation results, or existence for selected divisors. No verified collection of results exhausts all spherical fixed-data cases.

Key references and precise progress:

- Marc Troyanov, *Prescribing Curvature on Compact Surfaces with Conical Singularities*, **Transactions of the American Mathematical Society** 324(2) (1991), 793–821, <https://doi.org/10.1090/S0002-9947-1991-1005085-9>. Troyanov proves existence and uniqueness in the negative case, treats the flat case, and proves fixed-conformal-class existence in the spherical subcritical region $0 < \chi_{\text{con}} < 2 \min\{1, \beta_1, \dots, \beta_m\}$. These results leave the critical and supercritical spherical regimes.
- Feng Luo and Gang Tian, *Liouville Equation and Spherical Convex Polytopes*, **Proceedings of the American Mathematical Society** 116(4) (1992), 1119–1129, <https://doi.org/10.1090/S0002-9939-1992-1137227-5>. For the sphere with all cone angles below 2π , they give the necessary-and-sufficient inequality and uniqueness. This is a complete fixed-point result only for that small-angle regime.
- Alexandre Eremenko, *Metrics of Positive Curvature with Conic Singularities on the Sphere*, **Proceedings of the American Mathematical Society** 132(11) (2004), 3349–3355, <https://doi.org/10.1090/S0002-9939-04-07439-8>. Eremenko gives a necessary-and-sufficient condition for the sphere with exactly three prescribed cone angles. Three marked points carry no conformal moduli, so this does not settle four or more prescribed points or positive genus.
- Gabriele Mondello and Dmitri Panov, *Spherical Metrics with Conical Singularities on a 2-Sphere: Angle Constraints*, **International Mathematics Research Notices** 2016(16) (2016), 4937–4995, <https://doi.org/10.1093/imrn/rnv300>. They derive sharp holonomy obstructions on angle vectors and construct non-coaxial spherical metrics when the constraints are strict. The singular points and marked conformal class are not prescribed, and the equality/coaxial locus requires separate results.
- Gabriele Mondello and Dmitri Panov, *Spherical Surfaces with Conical Points: Systole Inequality and Moduli Spaces with Many Connected Components*, **Geometric and Functional Analysis** 29(4) (2019), 1110–1193, <https://doi.org/10.1007/s00039-019-00506-3>. Their Theorem A proves that the positive-genus Gauss–Bonnet condition suffices for existence in *some* conformal class, whereas Theorem D produces open sets of angle data for which a chosen conformal class has no spherical representative. Problem 1.14 explicitly retains the fixed-conformal-class problem; these results demonstrate that the missing dependence on (g, p_j) is genuine.
- Rafe Mazzeo and Xuwen Zhu, *Conical Metrics on Riemann Surfaces, II: Spherical Metrics*, **International Mathematics Research Notices** 2022(12) (2022), 9044–9113, <https://doi.org/10.1093/imrn/rnab011>. They develop local deformation theory for large cone angles, identify the spectral obstruction $2 \in \text{spec}(\Delta)$, and construct local solution spaces by allowing cone points to split. Their results are local and explicitly leave gaps in global spherical existence for prescribed conic data.
- Yu Feng, Jijian Song, and Bin Xu, *Existence and Non-uniqueness of Cone Spherical Metrics with Prescribed Singularities on a Compact Riemann Surface with Positive Genus*, **Journal of Geometric Analysis** 36 (2026), article 58, <https://doi.org/10.1007/s12220-025-02300-0>. For cone angles in $2\pi\mathbb{Z}_{>1}$, they prove existence and nonuniqueness for certain divisors chosen inside complete linear systems and describe positive-dimensional subsets of representable divisors. This is recent progress for positive genus, but it does not decide an arbitrary prescribed divisor.

What remains unresolved is a general necessary-and-sufficient obstruction theory in the spherical regime for an arbitrary prescribed conformal surface, arbitrary marked points, and arbi-

trary weights $\alpha_j > -1$, particularly with large cone angles. Hence the exact broad problem in the statement is not completely resolved.

1.68.3. Remaining open problems recorded by the database

1.68.3.1. Open problem 1

Let X be a compact Riemann surface of genus $g \geq 2$, and let

$$D = \sum_{j=1}^m n_j p_j$$

be an effective divisor of odd degree $d = \sum_j n_j > 2g - 2$, where the points p_j are distinct and the multiplicities n_j are positive integers. Prove that there are only finitely many conformal metrics of constant Gaussian curvature 1 on $X \setminus \{p_1, \dots, p_m\}$ having a conical singularity of total angle $2\pi(n_j + 1)$ at each p_j .

Reference supplied by the database. Lingguang Li, Jijian Song, and Bin Xu, “Irreducible Cone Spherical Metrics and Stable Extensions of Two Line Bundles,” *Advances in Mathematics* 388 (2021), article 107854, Further Discussion following Corollary 2.3. <https://doi.org/10.1016/j.aim.2021.107854>

1.68.3.2. Open problem 2

Let X be a compact Riemann surface of genus $g \geq 2$, and let L be a generic holomorphic line bundle in $\text{Pic}^\ell(X)$, where $\ell \leq -g/2$. Prove that there exist a degree-zero holomorphic line bundle $J \in \text{Pic}^0(X)$ and a short exact sequence of holomorphic vector bundles

$$0 \rightarrow L \rightarrow J \oplus J^{-1} \rightarrow L^{-1} \rightarrow 0.$$

Reference supplied by the database. Yu Feng, Jijian Song, and Bin Xu, “Existence and Non-uniqueness of Cone Spherical Metrics with Prescribed Singularities on a Compact Riemann Surface with Positive Genus,” *Journal of Geometric Analysis* 36 (2026), article 58, Conjecture 6.1. <https://doi.org/10.1007/s12220-025-02300-0>

1.68.4. Primary-source pointer

The originating ICM record is *Liouville equations from a variational point of view* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.3/ICM2014.3.pdf#page=357>.

1.69. 69. Analytic tools for the study of flows and inverse problems [2018-geom-138]

Record: 2018-geom-138

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Analytic tools for the study of flows and inverse problems

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol3-ver1-eb.pdf#page=972>

1.69.1. Problem statement

Let (M_1, g_1) and (M_2, g_2) be Riemannian surfaces with common boundary $\partial M_1 = \partial M_2$. Assume that each boundary is strictly convex, that neither surface has conjugate points, and that the trapped set of each geodesic flow is hyperbolic, where the trapped set is the set of unit tangent vectors whose corresponding geodesics remain in the manifold for all forward and backward time.

For $i = 1, 2$, let SM_i be the unit tangent bundle. For $y = (x, v) \in \partial SM_i$, let $\gamma_{x,v}(t) = \exp_x^{g_i}(tv)$, and let $\tau_{g_i}^+(x, v) \in [0, \infty]$ be the maximal non-negative time for which this geodesic remains defined in the manifold. Set

$$\Gamma_-^{(i)} = \{y \in \partial SM_i : \tau_{g_i}^+(y) = +\infty\}.$$

For $y = (x, v) \in \partial SM_i \setminus \Gamma_-^{(i)}$, define the scattering map

$$\sigma_{g_i}(x, v) = (\gamma_{x,v}(\tau_{g_i}^+(x, v)), \dot{\gamma}_{x,v}(\tau_{g_i}^+(x, v))) \in \partial SM_i.$$

The lens data of (M_i, g_i) is the pair $(\tau_{g_i}^+, \sigma_{g_i})$.

Prove or disprove that if the full lens data agree,

$$\tau_{g_1}^+ = \tau_{g_2}^+ \quad \text{and} \quad \sigma_{g_1} = \sigma_{g_2},$$

then there exists a diffeomorphism $\psi : M_1 \rightarrow M_2$ fixing the boundary such that

$$\psi^* g_2 = g_1.$$

1.69.2. Historical background, literature, and current progress

1.69.3. Verdict: partially resolved

1.69.3.1. Precise problem

The problem asks for smooth lens rigidity for compact Riemannian surfaces with strictly convex boundary, no conjugate points, and hyperbolic trapped set (often called surfaces of Anosov type): if two such metrics have the same ordinary/unmarked lens data, namely the same travel-time function and scattering map on the boundary unit bundle, must they be isometric by a

diffeomorphism fixing the boundary? Equivalently, after Guillarmou's scattering-rigidity theorem gives a boundary-fixing conformal diffeomorphism, does the travel-time part of the lens data force the remaining conformal factor to be trivial?

1.69.3.2. Status

I did not find a published or otherwise citable theorem proving this full smooth unmarked statement. The closest current formulation is Conjecture 1.7 in Cekic–Guillarmou–Lefeuvre, which is exactly the unmarked lens rigidity problem for manifolds of Anosov type. That paper explicitly lists only partial answers and says, in the smooth two-dimensional setting of Guillarmou's theorem, that recovering the conformal factor remains open. Therefore the full statement is not fully resolved.

1.69.3.3. Main partial progress

1. Lens rigidity for manifolds with hyperbolic trapped set

Colin Guillarmou. *Journal of the American Mathematical Society* 30 (2017), no. 2, 561–599. DOI: <https://doi.org/10.1090/jams/865>. arXiv: <https://arxiv.org/abs/1412.1760>.

This is the source theorem behind the problem. It proves deformation lens rigidity under the no-conjugate-points/hyperbolic-trapped-set assumptions and, for surfaces, proves that scattering data determine the topology and conformal class. It does not determine the conformal factor from the full lens data in the general smooth case.

2. Local lens rigidity for manifolds of Anosov type

Mihajlo Cekic, Colin Guillarmou, Thibault Lefeuvre. *Analysis & PDE* 17 (2024), no. 8, 2737–2795. DOI: <https://doi.org/10.2140/apde.2024.17.2737>. arXiv: <https://arxiv.org/abs/2204.02476>.

The paper formulates the exact unmarked problem as Conjecture 1.7 and proves local rigidity: for a fixed Anosov-type metric in dimension 2, any sufficiently close smooth metric with the same lens data is boundary-fixing isometric to it. This is only a local result near a reference metric, not the full global smooth statement.

3. Marked boundary rigidity for surfaces of Anosov type

Alena Erchenko, Thibault Lefeuvre. *Mathematische Zeitschrift* 306, article 36 (2024). DOI: <https://doi.org/10.1007/s00209-024-03433-8>. arXiv: <https://arxiv.org/abs/2305.06893>.

This proves that two smooth surfaces of Anosov type with the same marked boundary distance are isometric. Marked boundary/lens data records the homotopy class or equivalently data on the universal cover. That is stronger/different data than the ordinary lens data in this problem, so it does not fully resolve the question.

4. Riemannian Anosov extension and applications

Dong Chen, Alena Erchenko, Andrey Gogolev. *Journal de l'Ecole polytechnique - Mathématiques* 10 (2023), 945–987. DOI: <https://doi.org/10.5802/jep.237>. arXiv: <https://arxiv.org/abs/2009.13665>.

Corollary B proves marked lens rigidity in a conformal class under the same Anosov-type hypotheses. This directly addresses conformal factors, but assumes marked lens equivalence, not just the unmarked lens data in the problem statement.

5. Scattering rigidity for analytic metrics

Yannick Guedes Bonthonneau, Colin Guillarmou, Malo Jezequel. *Cambridge Journal of*

Mathematics 12 (2024), no. 1, 165–222. DOI: <https://doi.org/10.4310/CJM.2024.v12.n1.a2>. arXiv: <https://arxiv.org/abs/2201.02100>.

This proves that for real analytic manifolds of Anosov type with analytic strictly convex boundary, the scattering map alone determines the metric up to boundary-fixing isometry. The analyticity assumption is an extra hypothesis absent from the problem statement, so it is only partial progress for the smooth problem.

6. Lens rigidity with trapped geodesics in two dimensions

Christopher B. Croke, Pilar Herreros. *Asian Journal of Mathematics* 20 (2016), no. 1, 47–58. arXiv: <https://arxiv.org/abs/1108.4938>.

This proves lens rigidity for specific two-dimensional trapped examples, including negatively curved cylinders with convex boundary, but not for all smooth surfaces with no conjugate points and hyperbolic trapped set.

7. Boundary and lens rigidity for non-convex manifolds

Colin Guillarmou, Marco Mazzucchelli, Leo Tzou. *American Journal of Mathematics* 143 (2021), no. 2, 533–575. DOI: <https://doi.org/10.1353/ajm.2021.0012>. arXiv: <https://arxiv.org/abs/1711.10059>.

This proves lens rigidity for non-trapping compact Riemannian surfaces with no conjugate points. The problem statement specifically allows trapping with hyperbolic trapped set, so this does not cover the full statement.

1.69.3.4. What remains unresolved

The missing step is global smooth rigidity from ordinary/unmarked full lens data for all surfaces with strictly convex boundary, no conjugate points, and hyperbolic trapped set. Existing theorems resolve marked data, analytic metrics, local neighborhoods of a fixed metric, non-trapping cases, or special topologies/curvature examples, but no located result removes all of these extra hypotheses.

1.69.4. Remaining open problems recorded by the database

1.69.4.1. Open problem 1

Let M be a compact surface with strictly convex boundary, no conjugate points, and hyperbolic trapped set. Suppose g_1 and g_2 are metrics of this type in the same conformal class and have identical lens data. Prove that their conformal factors agree up to a diffeomorphism fixing ∂M (equivalently, prove global lens rigidity within a fixed conformal class).

Reference supplied by the database. Mihajlo Cekić, Colin Guillarmou, and Thibault Lefeuvre, “Local lens rigidity for manifolds of Anosov type,” *Analysis & PDE* 17 (2024), no. 8, 2737–2795, discussion after Conjecture 1.7 (item (2): recovering the conformal factor is still open), DOI: 10.2140/apde.2024.17.2737, <https://doi.org/10.2140/apde.2024.17.2737>; arXiv:2204.02476, <https://arxiv.org/abs/2204.02476>.

1.69.5. Primary-source pointer

The originating ICM record is *Analytic tools for the study of flows and inverse problems* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol3-ver1-eb.pdf#page=972>.

1.70. 70. Global surfaces of section for Reeb flows in dimension three and beyond [2018-geom-298]

Record: 2018-geom-298

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Global surfaces of section for Reeb flows in dimension three and beyond

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/uploads/ICM-2018-vol2-ver1-eb.pdf#page=1048>

1.70.1. Problem statement

Let g range over all smooth Riemannian metrics on the two-sphere S^2 . For each such metric, let $\ell_{\min}(S^2, g)$ denote the length of the shortest nonconstant closed geodesic on (S^2, g) , and define the systolic ratio by

$$\rho_{\text{sys}}(S^2, g) = \frac{\ell_{\min}(S^2, g)^2}{\text{Area}(S^2, g)}.$$

Prove that

$$\sup_g \rho_{\text{sys}}(S^2, g) = 2\sqrt{3}.$$

1.70.2. Historical background, literature, and current progress

The problem is **unresolved**. The best verified global bounds remain

$$2\sqrt{3} \leq \sup_g \rho_{\text{sys}}(S^2, g) \leq 32.$$

The lower bound comes from smooth Riemannian metrics approximating the singular Calabi–Croke doubled equilateral triangle, whose systolic ratio is $2\sqrt{3}$. The upper bound is Rotman’s theorem $\ell_{\min}(S^2, g) \leq 4\sqrt{2} \sqrt{\text{Area}(S^2, g)}$. These estimates leave a large gap, so the claimed equality is neither proved nor disproved. A 2025 paper still explicitly identifies 32 as the best-known general bound and $2\sqrt{3}$ as the conjectured optimum.

Key references and partial progress:

- Christopher B. Croke, “Area and the length of the shortest closed geodesic,” *Journal of Differential Geometry* 27 (1988), 1–21. DOI: <https://doi.org/10.4310/jdg/1214441646>. Croke proved a universal area bound for the shortest closed geodesic on every Riemannian two-sphere and described the Calabi–Croke construction underlying the lower bound, but his upper constant was not sharp.
- Regina Rotman, “The length of a shortest closed geodesic and the area of a 2-dimensional sphere,” *Proceedings of the American Mathematical Society* 134 (2006), 3041–3047. DOI: <https://doi.org/10.2307/4090111>

[tps://doi.org/10.1090/S0002-9939-06-08297-9](https://doi.org/10.1090/S0002-9939-06-08297-9). Theorem 0.1 proves the current best general estimate $\ell_{\min} \leq 4\sqrt{2}\sqrt{\text{Area}}$, equivalently $\rho_{\text{sys}} \leq 32$.

- Stéphane Sabourau, “Local extremality of the Calabi–Croke sphere for the length of the shortest closed geodesic,” *Journal of the London Mathematical Society* 82 (2010), 549–562. DOI: <https://doi.org/10.1112/jlms/jdq045>; arXiv: 0907.2223. This proves local extremality of the singular Calabi–Croke sphere among fixed-area Riemannian spheres with conical singularities, including in the Lipschitz topology. It does not give the required global upper bound for all smooth metrics.
- Alberto Abbondandolo, Barney Bramham, Umberto L. Hryniewicz, and Pedro A. S. Salomão, “A systolic inequality for geodesic flows on the two-sphere,” *Mathematische Annalen* 367 (2017), 701–753. DOI: <https://doi.org/10.1007/s00208-016-1385-2>; arXiv: 1410.7790. For positively curved metrics with pinching ratio $\delta > (4 + \sqrt{7})/8$, the paper proves the stronger inequality $\rho_{\text{sys}} \leq \pi$, with equality exactly for Zoll metrics. This only covers a restricted class.
- Alberto Abbondandolo, Barney Bramham, Umberto L. Hryniewicz, and Pedro A. S. Salomão, “Sharp systolic inequalities for Riemannian and Finsler spheres of revolution,” *Transactions of the American Mathematical Society* 374 (2021), 1815–1845. DOI: <https://doi.org/10.1090/tran/8233>; arXiv: 1808.06995. For every Riemannian sphere of revolution it proves $\rho_{\text{sys}} \leq \pi$, again with equality exactly in the Zoll case; rotational symmetry is an additional hypothesis.
- Yuhang Liu, “Lengths of closed geodesics for certain warped product metrics and nearly round metrics on spheres,” *Journal of Inequalities and Applications* 2025, article 85 (2025). DOI: <https://doi.org/10.1186/s13660-025-03323-5>. The introduction records 32 as the best-known general two-sphere bound and treats only special warped-product and nearly round metrics, confirming that the unrestricted equality remains open.

The Finsler counterexamples in the literature do not resolve this problem because the quantifier here is restricted to smooth Riemannian metrics.

1.70.3. Remaining open problems recorded by the database

1.70.3.1. Open problem 1

Let g range over all smooth Riemannian metrics on S^2 with positive Gaussian curvature. Write $K_{\min}(g)$ and $K_{\max}(g)$ for the minimum and maximum of the Gaussian curvature, respectively, and call g 1/4-pinched if

$$\frac{K_{\min}(g)}{K_{\max}(g)} \geq \frac{1}{4}.$$

For each such metric, let $\ell_{\min}(S^2, g)$ be the length of its shortest nonconstant closed geodesic and define

$$\rho_{\text{sys}}(S^2, g) = \frac{\ell_{\min}(S^2, g)^2}{\text{Area}(S^2, g)}.$$

Prove or disprove that

$$\sup_{g \text{ is } 1/4\text{-pinched}} \rho_{\text{sys}}(S^2, g) = \pi.$$

Reference supplied by the database. Pedro A. S. Salomão and Umberto L. Hryniewicz, “Global surfaces of section for Reeb flows in dimension three and beyond,” in *Proceedings of the International Congress of Mathematicians—Rio de Janeiro 2018, Vol. II: Invited Lectures*, World Scientific, 2018, pp. 941–967, Question 2. DOI: https://doi.org/10.1142/9789813272880_0083; arXiv:1712.01925.

1.70.4. Primary-source pointer

The originating ICM record is *Global surfaces of section for Reeb flows in dimension three and beyond* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/uploads/ICM-2018-vol2-ver1-eb.pdf#page=1048>.

1.71. 71. Analytic tools for the study of flows and inverse problems [2018-geom-328]

Record: 2018-geom-328

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Analytic tools for the study of flows and inverse problems

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol3-ver1-eb.pdf#page=972>

1.71.1. Problem statement

Let M be a compact smooth manifold diffeomorphic to a closed ball, and let g_1 and g_2 be smooth Riemannian metrics on M . Assume that, for each $i \in \{1, 2\}$, the boundary ∂M is strictly convex with respect to g_i and (M, g_i) has no conjugate points. Let $d_{g_i} : M \times M \rightarrow [0, \infty)$ be the Riemannian distance induced by g_i . Prove that if

$$d_{g_1}(x, y) = d_{g_2}(x, y) \quad \text{for all } x, y \in \partial M,$$

then there exists a diffeomorphism $\psi : M \rightarrow M$ such that $\psi|_{\partial M} = \text{Id}_{\partial M}$ and $\psi^*g_2 = g_1$.

1.71.2. Historical background, literature, and current progress

Verdict: unresolved.

The statement is Michel's boundary rigidity conjecture for all simple Riemannian manifolds. It is completely proved in dimension two, but the unrestricted smooth problem in dimensions $n \geq 3$ remains open. The higher-dimensional results below impose additional genericity, proximity, curvature, or convex-foliation hypotheses and therefore do not cover every pair of metrics allowed by the statement.

- René Michel, *Sur la rigidité imposée par la longueur des géodésiques*, *Inventiones Mathematicae* 65 (1981/82), 71-83, DOI <https://doi.org/10.1007/BF01389295>. This paper introduced the simple-manifold boundary rigidity conjecture in the form used here.
- Leonid Pestov and Gunther Uhlmann, *Two dimensional compact simple Riemannian manifolds are boundary distance rigid*, *Annals of Mathematics* 161 (2005), 1093-1110, DOI <https://doi.org/10.4007/annals.2005.161.1093>. This proves the full assertion for every simple surface.
- Plamen Stefanov and Gunther Uhlmann, *Boundary rigidity and stability for generic simple metrics*, *Journal of the American Mathematical Society* 18 (2005), 975-1003, DOI <https://doi.org/10.1090/S0894-0347-05-00494-7>, arXiv: [math/0408075](https://arxiv.org/abs/math/0408075). This proves global uniqueness for a generic class of simple metrics, including the real-analytic case, and local stability near such metrics; genericity or closeness is extra to the present statement.
- Dmitri Burago and Sergei Ivanov, *Boundary rigidity and filling volume minimality of metrics*

close to a flat one, *Annals of Mathematics* 171 (2010), 1183-1211, DOI <https://doi.org/10.4007/annals.2010.171.1183>. This proves boundary rigidity for an open class of metrics sufficiently C^2 -close to a Euclidean metric, not for arbitrary simple metrics.

- Plamen Stefanov, Gunther Uhlmann, and Andras Vasy, *Local and global boundary rigidity and the geodesic X-ray transform in the normal gauge*, *Annals of Mathematics* 194 (2021), 1-95, DOI <https://doi.org/10.4007/annals.2021.194.1.1>, arXiv: [1702.03638](https://arxiv.org/abs/1702.03638). This gives local recovery near strictly convex boundary points and global rigidity under a suitable strictly convex foliation, with curvature and no-focal-points corollaries. A general simple manifold need not satisfy that extra global foliation condition.
- Gabriel P. Paternain, Mikko Salo, and Gunther Uhlmann, *Geometric Inverse Problems: With Emphasis on Two Dimensions*, Cambridge Studies in Advanced Mathematics 204, Cambridge University Press, 2023, DOI <https://doi.org/10.1017/9781009039901>. This recent monograph presents the complete simple-surface theorem and the higher-dimensional partial theory; it does not report a resolution for arbitrary smooth simple metrics in dimensions $n \geq 3$.

Thus the established cases do not exhaust the hypotheses of the statement, and the remaining gap is boundary rigidity for arbitrary smooth simple Riemannian metrics in dimensions at least three.

1.71.3. Remaining open problems recorded by the database

1.71.3.1. Open problem 1

Let (M, g) be a compact smooth Riemannian manifold of dimension $n \geq 3$, diffeomorphic to a closed ball, with strictly convex boundary and no conjugate points. For a smooth symmetric covariant 2-tensor field f on M , define its geodesic X-ray transform by

$$I_2 f(\gamma) = \int_0^\tau f_{\gamma(t)}(\dot{\gamma}(t), \dot{\gamma}(t)) dt$$

for every maximal unit-speed geodesic segment $\gamma : [0, \tau] \rightarrow M$ with endpoints on ∂M . Determine whether $I_2 f(\gamma) = 0$ for every such γ implies that there exists a smooth 1-form v on M , with $v|_{\partial M} = 0$, such that $f = d^s v$, where

$$(d^s v)(X, Y) = \frac{1}{2}((\nabla_X v)(Y) + (\nabla_Y v)(X))$$

and ∇ is the Levi-Civita connection of g .

Reference supplied by the database. Gabriel P. Paternain, Mikko Salo, and Gunther Uhlmann, *Tensor tomography: Progress and challenges*, *Chinese Annals of Mathematics, Series B* 35(3) (2014), 399-428, Section 11, open problem 2. DOI: <https://doi.org/10.1007/s11401-014-0834-z>; arXiv: [1303.6114](https://arxiv.org/abs/1303.6114).

1.71.3.2. Open problem 2

Let (M, g) be a compact smooth Riemannian manifold of dimension $n \geq 3$, diffeomorphic to a closed ball, with strictly convex boundary and no conjugate points. Determine whether there must exist a smooth function $\varphi \in C^\infty(M)$ whose Riemannian Hessian is positive definite everywhere, that is,

$$\text{Hess}_g \varphi(v, v) > 0$$

for every point $x \in M$ and every nonzero vector $v \in T_x M$.

Reference supplied by the database. Hanming Zhou, *The weighted X-ray transform and applications*, in Eric Todd Quinto, Plamen D. Stefanov, and Gunther Uhlmann (eds.), *Microlocal Analysis and Inverse Problems in Tomography and Geometry*, De Gruyter, 2024, pp. 167-186, open question following Theorem 2.5. DOI: <https://doi.org/10.1515/9783111337487-005>.

1.71.4. Primary-source pointer

The originating ICM record is *Analytic tools for the study of flows and inverse problems* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol3-ver1-eb.pdf#page=972>.

1.72. 72. Conformal geometry on four manifolds - Noether lecture [2018-geom-329]

Record: 2018-geom-329

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Conformal geometry on four manifolds - Noether lecture

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=130>

1.72.1. Problem statement

Let $k \geq 2$, and let $\{g_i^+\}$ be a sequence of oriented conformally compact Einstein metrics on the interior of the four-dimensional ball B^4 , satisfying

$$\text{Ric}_{g_i^+} = -3g_i^+.$$

Suppose that the conformal infinity of g_i^+ is $(S^3, [h_i])$. For each i , choose a Yamabe representative $h_i \in [h_i]$ of nonnegative type, and assume that $\{h_i\}$ is contained in a compact subset of $C^{k+3}(S^3)$ and that there is a constant $c_1 > 0$ such that

$$Y(S^3, [h_i]) \geq c_1$$

for every i , where $Y(S^3, [h_i])$ denotes the Yamabe constant of the conformal class $[h_i]$. Let r_i be the geodesic defining function associated with h_i , and let

$$g_i^* = e^{2w_i} g_i^+$$

be the Fefferman–Graham compactification determined by

$$-\Delta_{g_i^+} w_i = 3, \quad w_i - \log r_i \rightarrow 0 \quad \text{at } S^3,$$

so that $g_i^*|_{S^3} = h_i$. For any four-dimensional Riemannian metric g , define

$$A_g = \text{Ric}_g - \frac{R_g}{6} g,$$

and let $\sigma_2(A_g)$ be the second elementary symmetric function of the eigenvalues of the endomorphism $g^{-1}A_g$.

Prove or disprove that the condition

$$\int_{B^4} \sigma_2(A_{g_i^*}) dv_{g_i^*} > 0$$

for every i implies that, for every $\alpha \in (0, 1)$, every subsequence of $\{g_i^*\}$ has a further subsequence that converges in $C^{k+2, \alpha}(\overline{B^4})$ after pullback by diffeomorphisms of $\overline{B^4}$ whose restrictions to S^3 are the identity.

1.72.2. Historical background, literature, and current progress

The exact implication is **unresolved**. No proof or counterexample covering mere positivity of $\int_{B^4} \sigma_2(A_{g_i^*}) dv_{g_i^*}$ was found.

Sun-Yung A. Chang, *Conformal Geometry on Four Manifolds*, in *Proceedings of the International Congress of Mathematicians 2018*, Vol. I, World Scientific, 2018, pp. 119–146, [arXiv:1809.06339](https://arxiv.org/abs/1809.06339), explicitly asks whether the near-maximal hypothesis in the perturbative compactness theorem can be weakened to positivity and identifies the missing step as a connection between positive renormalized volume and the local nonconcentration condition on boundary T -curvature.

Sun-Yung A. Chang and Yuxin Ge, *Compactness of Conformally Compact Einstein Manifolds in Dimension 4*, *Advances in Mathematics* **340** (2018), 588–652, <https://doi.org/10.1016/j.aim.2018.10.010>, prove compactness under additional nonconcentration hypotheses on the boundary S -tensor or T -curvature (Theorems 1.1 and 1.7). These hypotheses are not shown to follow from positive total σ_2 .

Sun-Yung A. Chang, Yuxin Ge, and Jie Qing, *Compactness of Conformally Compact Einstein 4-Manifolds II*, *Advances in Mathematics* **373** (2020), Article 107325, <https://doi.org/10.1016/j.aim.2020.107325>, remove auxiliary interior Yamabe-constant assumptions from the preceding nonconcentration theorems. Their Theorem 1.3 also gives compactness when the total Weyl energy is sufficiently small, equivalently on B^4 when the total σ_2 is sufficiently close to its hyperbolic maximum. Neither result covers arbitrary positive total σ_2 .

Michael T. Anderson, *Einstein Metrics with Prescribed Conformal Infinity on 4-Manifolds*, *Geometric and Functional Analysis* **18** (2008), 305–366, <https://doi.org/10.1007/s00039-008-00668-5>, states a stronger properness theorem which, if its boundary estimates apply as claimed, would imply compactness on B^4 without the σ_2 assumption. However, Chang and Ge, *On Conformally Compact Einstein Manifolds*, *Revista de la Union Matematica Argentina* **64** (2022), no. 1, <https://doi.org/10.33044/revuma.3156>, explicitly report that key estimates in Anderson’s argument remain unclear and continue to state compactness only with the extra S - or T -nonconcentration condition. Thus the asserted properness result does not supply a presently verified resolution of the exact problem.

The later paper Sun-Yung A. Chang, Yuxin Ge, Xiaoshang Jin, and Jie Qing, *Perturbation Compactness and Uniqueness for a Class of Conformally Compact Einstein Manifolds*, *Advanced Nonlinear Studies* **24** (2024), 247–278, <https://doi.org/10.1515/ans-2023-0124>, extends perturbative compactness methods in general dimensions but still requires small total Weyl curvature or conformal infinity close to the round sphere. The gap between these stronger assumptions and mere positivity remains.

1.72.3. Remaining open problems recorded by the database

1.72.3.1. Open problem 1

Let $k \geq 5$, and let $\{g_i^+\}$ be a sequence of oriented conformally compact Einstein metrics on the interior of B^4 , with $\text{Ric}_{g_i^+} = -3g_i^+$ and conformal infinities $(S^3, [h_i])$. For each i , let $h_i \in [h_i]$ be a Yamabe representative of nonnegative type. Assume that $\{h_i\}$ is contained in a compact subset of $C^{k+3}(S^3)$ and that $Y(S^3, [h_i]) \geq c_1 > 0$ for every i . Let $g_i^* = e^{2w_i} g_i^+$ be the Fefferman–Graham compactification satisfying

$$-\Delta_{g_i^*} w_i = 3, \quad g_i^*|_{S^3} = h_i,$$

and define its boundary T -curvature by

$$T_i = \frac{1}{12} \left. \frac{\partial R_{g_i^*}}{\partial n_i} \right|_{S^3},$$

where n_i is the outward unit normal to S^3 with respect to g_i^* . For a four-dimensional metric g , set $A_g = \text{Ric}_g - \frac{R_g}{6}g$, and let $\sigma_2(A_g)$ be the second elementary symmetric function of the eigenvalues of $g^{-1}A_g$.

Determine whether

$$\int_{B^4} \sigma_2(A_{g_i^*}) dv_{g_i^*} > 0 \quad \text{for every } i$$

implies the uniform local nonconcentration condition

$$\liminf_{r \rightarrow 0} \inf_i \inf_{x \in S^3} \int_{B_{h_i}(x, r)} T_i dv_{h_i} \geq 0,$$

where $B_{h_i}(x, r)$ is the geodesic ball of radius r in (S^3, h_i) .

Reference supplied by the database. Sun-Yung Alice Chang, “Conformal Geometry on Four Manifolds,” in *Proceedings of the International Congress of Mathematicians, Rio de Janeiro 2018*, Vol. I, B. Sirakov, P. N. de Souza, and M. Viana (eds.), World Scientific, 2018, pp. 119–146, Section 5 following Theorem 5.3, arXiv:1809.06339, <https://arxiv.org/abs/1809.06339>.

1.72.4. Primary-source pointer

The originating ICM record is *Conformal geometry on four manifolds - Noether lecture* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=130>.

1.73. 73. Emergent complex geometry [2022-geom-161]

Record: 2022-geom-161

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2022: Emergent complex geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-562-9-volume-4.pdf#page=99>

1.73.1. Problem statement

Let X be a Fano manifold, i.e. a compact complex manifold whose anticanonical line bundle $-K_X$ is positive (ample). For each positive integer k , set

$$N_k := \dim_{\mathbb{C}} H^0(X, -kK_X).$$

Choose a basis $s_1^{(k)}, \dots, s_{N_k}^{(k)}$ of $H^0(X, -kK_X)$ and define the Slater determinant

$$(\det S^{(k)})(x_1, \dots, x_{N_k}) := \det(s_i^{(k)}(x_j))_{i,j=1}^{N_k},$$

a holomorphic section of $-kK_{X^{N_k}}$. On the complement of its zero locus, $|\det S^{(k)}|^{-2/k}$ defines a measure on X^{N_k} , and put

$$Z_{N_k} := \int_{X^{N_k}} |\det S^{(k)}|^{-2/k}.$$

Say that X is Gibbs stable at level k if $Z_{N_k} < \infty$, and Gibbs stable if this holds for all sufficiently large k . When $Z_{N_k} < \infty$, the corresponding canonical N_k -point process on X is the symmetric probability measure

$$\mu^{(N_k)} := Z_{N_k}^{-1} |\det S^{(k)}|^{-2/k}$$

on X^{N_k} ; after normalization it is independent of the chosen basis. For a configuration $(x_1, \dots, x_{N_k})$ sampled from this process, its empirical measure is

$$\delta_{N_k} := \frac{1}{N_k} \sum_{j=1}^{N_k} \delta_{x_j}.$$

Prove that X admits a unique Kähler-Einstein metric ω_{KE} with positive Ricci curvature, $\text{Ric } \omega_{\text{KE}} = \omega_{\text{KE}}$, if and only if X is Gibbs stable. Furthermore, prove that if X is Gibbs stable, then, as $k \rightarrow \infty$ (equivalently $N_k \rightarrow \infty$), the empirical measures δ_{N_k} of the above canonical point processes converge in probability to the normalized volume form $dV_{\text{KE}} := \omega_{\text{KE}}^{\dim_{\mathbb{C}} X} / \int_X \omega_{\text{KE}}^{\dim_{\mathbb{C}} X}$ of ω_{KE} .

1.73.2. Historical background, literature, and current progress

1.73.3. Verdict: partially resolved

1.73.3.1. Precise problem

The problem asks for Berman's Gibbs-stability conjecture for smooth Fano manifolds: for the canonical Slater-determinant point processes associated to sections of $-kK_X$, prove that X admits a unique positive Ricci Kähler-Einstein metric if and only if the partition functions Z_{N_k} are finite for all sufficiently large k , and, under this Gibbs stability hypothesis, prove convergence in probability of the empirical measures to the normalized Kähler-Einstein volume form.

1.73.3.2. Literature checked

1. **Kähler-Einstein metrics, canonical random point processes and birational geometry**
Author: Robert J. Berman
Venue: *Algebraic Geometry: Salt Lake City 2015*, Proceedings of Symposia in Pure Mathematics 97.1, AMS, 2018, pp. 29-73
DOI/arXiv/URL: DOI <https://doi.org/10.1090/pspum/097.1/01669>; arXiv 1307.3634, <https://arxiv.org/abs/1307.3634>
Status contribution: This is the original source of the exact problem. Conjecture 6.1 states the equivalence between asymptotic Gibbs stability and a unique Kähler-Einstein metric, plus convergence of the empirical measures. Theorem 6.3 proves only the stronger hypothesis "strong Gibbs stability" implies existence of a unique Kähler-Einstein metric. Thus it is partial progress, not a full resolution.
2. **Emergent complex geometry**
Author: Robert J. Berman
Venue: *Proceedings of the International Congress of Mathematicians 2022*, Vol. 4, EMS Press, 2023, pp. 2456-2483
DOI/arXiv/URL: DOI <https://doi.org/10.4171/ICM2022/59>; <https://ems.press/content/book-chapter-files/33209>
Status contribution: Restates the problem as Conjecture 2.3. It also records that Gibbs stability implies finite automorphism group, introduces the uniform Gibbs version, and states Theorem 2.5: uniform Gibbs stability implies uniform K-stability. The article explicitly treats the equivalence and empirical-measure convergence as conjectural.
3. **On Berman-Gibbs stability and K-stability of $\mathbb{Q}$ -Fano varieties**
Author: Kenton Fujita
Venue: *Compositio Mathematica* 152(2), 2016, pp. 288-298
DOI/arXiv/URL: DOI <https://doi.org/10.1112/S0010437X1500768X>; Cambridge Core URL <https://www.cambridge.org/core/journals/compositio-mathematica/article/on-bermangibbs-stability-and-kstability-of-mathbbqfano-varieties/15DF205CC7BF193578B9B4AC09686933>
Status contribution: Proves that Berman-Gibbs stability or semistability implies the corresponding K-stability or K-semistability. Together with Yau-Tian-Donaldson results this gives one direction toward existence of Kähler-Einstein metrics, but it does not prove the converse Kähler-Einstein/unique Kähler-Einstein implies Gibbs stability, and it does not prove empirical-measure convergence.
4. **On the K-stability of Fano varieties and anticanonical divisors**

Authors: Kento Fujita and Yuji Odaka

Venue: *Tohoku Mathematical Journal*, Second Series 70(4), 2018, pp. 511-521

DOI/arXiv/URL: DOI <https://doi.org/10.2748/tmj/1546570823>; arXiv 1602.01305, <https://arxiv.org/abs/1602.01305>

Status contribution: Develops anticanonical-divisor criteria and relates them to Berman-Gibbs stability; in the survey/source literature this is cited for the implication from uniform Gibbs stability to uniform K-stability. This is a stronger-hypothesis/one-direction result and does not settle ordinary Gibbs stability equivalence or the point-process convergence statement.

5. Measure Preserving Holomorphic Vector Fields, Invariant Anti-Canonical Divisors and Gibbs Stability

Author: Robert J. Berman

Venue: *Analysis Mathematica* 48, 2022, pp. 347-375

DOI/arXiv/URL: DOI <https://doi.org/10.1007/s10476-022-0154-6>; arXiv 2201.03325, <https://arxiv.org/abs/2201.03325>

Status contribution: Proves that Gibbs stability rules out non-trivial holomorphic vector fields. This explains why the conjecture is phrased with uniqueness/finite automorphism behavior, but it is not a proof of the full equivalence or convergence.

6. The probabilistic vs the quantization approach to Kähler-Einstein geometry

Author: Robert J. Berman

Venue: *Mathematische Annalen* 388, 2024, pp. 4383-4404

DOI/arXiv/URL: DOI <https://doi.org/10.1007/s00208-023-02627-5>; arXiv 2109.06575, <https://arxiv.org/abs/2109.06575>

Status contribution: Theorem 1.2 and Corollary 1.3 give a direct analytic proof that uniform Gibbs stability implies existence of a unique Kähler-Einstein metric. The paper explicitly identifies this as one direction of the conjectural picture and notes that ordinary Gibbs stability is weaker than uniform Gibbs stability. It does not prove the converse or the empirical-measure convergence under ordinary Gibbs stability.

7. Gibbs polystability of Fano manifolds, stability thresholds and symmetry breaking

Authors: Rolf Andreasson, Robert J. Berman, Ludvig Svensson

Venue: arXiv preprint, 2025

DOI/arXiv/URL: arXiv 2511.16173, <https://arxiv.org/abs/2511.16173>

Status contribution: Extends the framework to log Fano manifolds with non-discrete automorphism groups by introducing Gibbs polystability. The paper states the polystable analogue as conjectural, proves several results for log Fano curves and some examples, and proves a strong-uniform Gibbs-polystability implication. It is significant progress but not a full resolution of the all-dimensional smooth Fano statement.

8. Open problems in K-stability of Fano varieties

Authors: Chenyang Xu and Ziquan Zhuang

Venue: arXiv preprint, 2026

DOI/arXiv/URL: arXiv 2601.15576v2, <https://arxiv.org/abs/2601.15576>

Status contribution: Section 2.3, “Gibbs stability”, describes the Fano Gibbs-stability picture as largely conjectural. It states the expected equivalence with K-stability as Conjecture 15 and says only one direction is known. It also says the empirical-measure convergence to the Kähler-Einstein volume form is conjectural.

1.73.3.3. What is resolved

- Gibbs stability or uniform Gibbs stability has been shown to imply algebraic K-stability or uniform K-stability in the cited algebraic work.
- Strong or uniform Gibbs stability implies existence of a unique Kähler-Einstein metric by the cited Berman results, and also via K-stability plus the Yau-Tian-Donaldson theorem.
- Gibbs stability is known to force discreteness/absence of holomorphic vector fields.
- Various special or modified settings are understood, including general type cases, log Fano curves, and polystable/symmetry-broken variants.

1.73.3.4. What remains unresolved

the exact statement is not fully proved in the literature. In particular, the converse direction that every Fano manifold with a unique positive Ricci Kähler-Einstein metric is Gibbs stable in the ordinary sense remains open in general. The convergence in probability of the canonical empirical measures to the Kähler-Einstein volume form under ordinary Gibbs stability also remains conjectural. Current results either prove only one direction, assume stronger stability such as strong or uniform Gibbs stability, treat special cases such as curves, or formulate broader polystable versions that remain conjectural.

1.73.4. Remaining open problems recorded by the database

1.73.4.1. Open problem 1

For a smooth Fano variety X , let

$$\gamma(X) = \liminf_{m \rightarrow \infty} \text{lct}\left(X^{N_m}, \frac{1}{m}D_m\right), \quad N_m = h^0(X, -mK_X),$$

where D_m is the determinantal divisor on X^{N_m} defined by a basis of $H^0(X, -mK_X)$. Is $\gamma(X) > 1$ equivalent to K-stability, and is $\gamma(X) \geq 1$ equivalent to K-semistability?

Reference supplied by the database. Chenyang Xu and Ziquan Zhuang, “Open Problems in K-Stability of Fano Varieties,” arXiv preprint, 2026, §2.3, Conjecture 15 (attributed to Berman), arXiv: [2601.15576](https://arxiv.org/abs/2601.15576). The underlying Gibbs-stability conjecture originates in Robert J. Berman, “Kähler–Einstein Metrics, Canonical Random Point Processes and Birational Geometry,” in *Algebraic Geometry: Salt Lake City 2015*, Proceedings of Symposia in Pure Mathematics **97.1**, American Mathematical Society, 2018, pp. 29–73, Conjecture 6.1, DOI: <https://doi.org/10.1090/pspum/097.1/01669>, arXiv: [1307.3634](https://arxiv.org/abs/1307.3634).

1.73.4.2. Open problem 2

For a smooth Fano variety X whose stability threshold satisfies $\delta(X) \leq 1$, is the Gibbs threshold equal to the stability threshold, $\gamma(X) = \delta(X)$?

Reference supplied by the database. Chenyang Xu and Ziquan Zhuang, “Open Problems in K-Stability of Fano Varieties,” arXiv preprint, 2026, §2.3, the explicit expectation immediately following Conjecture 15 that $\delta(X) = \gamma(X)$ when $\delta(X) \leq 1$, together with the statement that only the inequality $\delta(X) \geq \gamma(X)$ is known, arXiv: [2601.15576](#). The known one-sided inequality is due to Kento Fujita, “On Berman–Gibbs Stability and K-Stability of $\mathbb{Q}$ -Fano Varieties,” *Compositio Mathematica* **152** (2016), no. 2, 288–298, DOI: <https://doi.org/10.1112/S0010437X1500768X>, arXiv: [1501.00248](#).

1.73.5. Primary-source pointer

The originating ICM record is *Emergent complex geometry* (2022). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-562-9-volume-4.pdf#page=99>.

1.74. 74. Emergent complex geometry [2022-geom-162]

Record: 2022-geom-162

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2022: Emergent complex geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-562-9-volume-4.pdf#page=99>

1.74.1. Problem statement

Prove that, for a polarized projective algebraic manifold (X, L) , where X is a projective algebraic manifold and L is a holomorphic line bundle over X whose first Chern class $c_1(L)$ contains a Kähler form, there exists a Kähler metric whose Kähler form represents $c_1(L)$ and whose scalar curvature is constant if and only if (X, L) is K -polystable: for every test configuration $\rho = (\mathcal{X}, \mathcal{L})$ of (X, L) , that is, a $\mathbb{C}^*$ -equivariant degeneration over $\mathbb{C}$ restricting over $\mathbb{C}^*$ to the trivial polarized family $(X \times \mathbb{C}^*, L)$, with normal total space $\mathcal{X}$ and relatively ample $\mathbb{Q}$ -line bundle $\mathcal{L}$, the Donaldson-Futaki invariant satisfies $DF(\rho) \geq 0$, and equality occurs only when $\mathcal{X}$ is biholomorphic to $X \times \mathbb{C}$.

1.74.2. Historical background, literature, and current progress

1.74.3. Verdict: partially resolved

1.74.3.1. Precise problem

The problem asks for the general cscK form of the Yau-Tian-Donaldson conjecture for every smooth polarized projective manifold (X, L) : existence of a constant scalar curvature Kähler metric in $c_1(L)$ should be equivalent to the classical Donaldson-Futaki K -polystability condition tested on normal relatively ample test configurations, with equality only for product test configurations.

1.74.3.2. Status check

This exact statement is not fully resolved. The forward implication, cscK implies classical K -polystability, is known. Several important special cases are known, including the Fano Kähler-Einstein case and toric surfaces. Very recent work gives a full correspondence for stronger uniform/equivariant/non-Archimedean stability notions, but those hypotheses are not the same as the classical K -polystability condition. In fact, current papers explicitly distinguish these stronger notions from classical K -polystability and say the converse from classical K -polystability to cscK is not expected in this generality.

1.74.3.3. Key literature and applicability

1. Simon K. Donaldson, “Scalar curvature and stability of toric varieties”, *Journal of Differential Geometry* 62 (2002), no. 2, 289-349. DOI: 10.4310/jdg/1090950195. URL: <https://projecteuclid.org/journals/journal-of-differential-geometry/volume-62/issue-2/Scalar-Curvature-and-Stability-of-Toric-Varieties/10.4310/jdg/1090950195.full>
 - Introduced the algebro-geometric K -stability framework for polarized varieties in the cscK setting and formulated the expected link with constant scalar curvature Kahler metrics. This is a source for the problem, not a full proof in all cases.
2. Simon K. Donaldson, “Constant Scalar Curvature Metrics on Toric Surfaces”, *Geometric and Functional Analysis* 19 (2009), 83-136. DOI: 10.1007/s00039-009-0714-y. arXiv:0805.0128. URL: <https://arxiv.org/abs/0805.0128>
 - Proves the existence direction in the toric surface case, assuming the relevant toric K -stability. This is a major special case but does not cover arbitrary polarized manifolds.
3. Jacopo Stoppa, “ K -stability of constant scalar curvature Kahler manifolds”, *Advances in Mathematics* 221 (2009), no. 4, 1397-1408. DOI: 10.1016/j.aim.2009.02.013. arXiv:0803.4095. URL: <https://arxiv.org/abs/0803.4095>
 - Proves that a polarized manifold admitting a cscK metric and having discrete automorphism group is K -stable. This is the forward implication under a discreteness hypothesis, not the converse and not all automorphism cases.
4. Robert J. Berman, Tamas Darvas, Chinh H. Lu, “Regularity of weak minimizers of the K -energy and applications to properness and K -stability”, *Annales Scientifiques de l’Ecole Normale Supérieure* 53 (2020), no. 2, 267-289. DOI: 10.24033/asens.2422. arXiv:1602.03114. URL: <https://arxiv.org/abs/1602.03114>
 - Proves that an ample line bundle (X, L) admitting a cscK metric in $c_1(L)$ is K -polystable. This covers the necessary direction of the problem statement for polarized manifolds, but it does not prove that classical K -polystability implies existence of a cscK metric.
5. Xiuxiong Chen, Simon Donaldson, Song Sun, “Kahler-Einstein metrics on Fano manifolds I-III”, *Journal of the American Mathematical Society* 28 (2015), no. 1, 183-197, 199-234, 235-278. DOI for I: 10.1090/S0894-0347-2014-00799-2; arXiv:1211.4566, arXiv:1212.4714, arXiv:1302.0282. URL: <https://arxiv.org/abs/1302.0282>
 - Resolves the Yau-Tian-Donaldson correspondence for smooth Fano manifolds with the anticanonical polarization. This is a central special case of the problem, but it does not imply the cscK statement for arbitrary polarizations.
6. Xiuxiong Chen and Jingrui Cheng, “On the constant scalar curvature Kahler metrics (I)–A priori estimates”, *Journal of the American Mathematical Society* 34 (2021), no. 4, 909-936, and “On the constant scalar curvature Kahler metrics (II)–Existence results”, *Journal of the American Mathematical Society* 34 (2021), no. 4, 937-1009. DOI for II: 10.1090/jams/966. arXiv:1712.06697 and arXiv:1801.00656. URL: <https://arxiv.org/abs/1801.00656>
 - Gives fundamental analytic existence/properness results for cscK metrics, including that properness/coercivity of the Mabuchi K -energy implies existence. This is a major ingredient in modern YTD work, but properness/coercivity is not the same as the classical Donaldson-Futaki K -polystability condition.

7. **Sebastien Boucksom, Tomoyuki Hisamoto, Mattias Jonsson**, “Uniform K-stability and asymptotics of energy functionals in Kahler geometry”, *Journal of the European Mathematical Society* 21 (2019), no. 9, 2905-2944; erratum *JEMS* 24 (2022), no. 2, 735-736. DOI: 10.4171/JEMS/894; erratum DOI: 10.4171/JEMS/1215. arXiv:1603.01026. URL: <https://arxiv.org/abs/1603.01026>
 - Relates slopes of energy functionals to non-Archimedean invariants and shows, among other results, that coercivity implies uniform K-stability. This supports stronger uniform versions of YTD but does not identify ordinary K-polystability with cscK existence in all cases.
8. **Chi Li**, “Geodesic rays and stability in the cscK problem”, *Annales Scientifiques de l’Ecole Normale Supérieure* 55 (2022), no. 6, 1529-1574. DOI: 10.24033/asens.2523. arXiv:2001.01366. URL: <https://arxiv.org/abs/2001.01366>
 - Proves existence results from strengthened stability notions over models/non-Archimedean objects. These are partial progress toward general cscK YTD, but the assumptions are stronger than classical K-polystability over normal ample test configurations.
9. **Sebastien Boucksom and Mattias Jonsson**, “On the Yau-Tian-Donaldson conjecture for weighted cscK metrics”, arXiv:2509.15016 (2025). DOI: 10.48550/arXiv.2509.15016. URL: <https://arxiv.org/abs/2509.15016>
 - Establishes a YTD-type correspondence for cscK and weighted extremal metrics using non-Archimedean pluripotential theory and stronger $\hat{K}$ -polystability-type notions on the completed non-Archimedean space, not merely the classical Donaldson-Futaki condition on normal test configurations. The introduction explicitly says that $\hat{K}$ -polystability is a stronger version of K-polystability and that the converse from classical K-polystability is not expected in general.
10. **Antonio Trusiani**, “A solution to the Yau-Tian-Donaldson Conjecture through Special Fujita Approximations”, arXiv:2605.30063 (2026). DOI: 10.48550/arXiv.2605.30063. URL: <https://arxiv.org/abs/2605.30063>
 - Proves a uniform/equivariant version: a smooth polarized projective variety admits a cscK metric iff it is $\text{Aut}^\circ(X, L)$ -uniformly K-stable, and relates several strengthened stability notions. The paper itself notes that existence of cscK metrics implies K-polystability, while the converse from classical K-polystability is not expected; therefore it does not prove the exact classical K-polystability equivalence.
11. **Vestislav Apostolov, David M. J. Calderbank, Paul Gauduchon, Christina W. Tonnesen-Friedman**, “Hamiltonian 2-forms in Kahler geometry. III. Extremal metrics and stability”, *Inventiones Mathematicae* 173 (2008), no. 3, 547-601. DOI: 10.1007/s00222-008-0126-x. URL: <https://doi.org/10.1007/s00222-008-0126-x>
 - Gives examples and computations in the extremal/cscK stability landscape that later authors cite as evidence that ordinary K-polystability is too weak for the general cscK existence problem. These examples are diagnostic evidence for the gap between classical and uniform stability, not a full resolution of the stated smooth polarized-manifold statement.
12. **Masafumi Hattori**, “A decomposition formula for J-stability and its applications”, arXiv:2103.04603 (2021). DOI: 10.48550/arXiv.2103.04603. URL: <https://arxiv.org/abs/2103.04603>

- Constructs K -stable but not uniformly K -stable examples in singular/log settings and is cited by recent YTD papers as further evidence that classical K -polystability should not be the final general cscK criterion. The constructions do not by themselves settle the stated smooth polarized projective manifold case.

1.74.3.4. What remains unresolved for this problem

The unresolved point is the converse in exactly the formulation: whether ordinary Donaldson-Futaki K -polystability for all normal relatively ample test configurations, with equality only for products, is sufficient for a cscK metric on every polarized smooth projective manifold. Current complete correspondences use stronger uniform, equivariant, model, or completed non-Archimedean stability notions. Since those hypotheses are not identical to the classical K -polystability, the full statement is not fully resolved.

1.74.4. Remaining open problems recorded by the database

1.74.4.1. Open problem 1

Let P be a Delzant polytope of dimension at least 3, with vanishing extremal affine function (equivalently, vanishing Futaki character). If P is torically K -polystable in Donaldson's classical sense— $\mathcal{L}_P(f) \geq 0$ for every rational convex piecewise-linear f , with equality only for affine f —must the corresponding polarized toric manifold admit a constant-scalar-curvature Kähler metric, equivalently a smooth symplectic potential satisfying the constant-scalar-curvature Abreu equation with Guillemin boundary behaviour?

Reference supplied by the database. Simon K. Donaldson, “Scalar Curvature and Stability of Toric Varieties,” *Journal of Differential Geometry* **62** (2002), no. 2, 289–349; the toric stability/cscK conjecture formulated in the Introduction (and reduced to the Abreu equation in the body of the paper). DOI: <https://doi.org/10.4310/jdg/1090950195>. The two-dimensional case was proved in Simon K. Donaldson, “Constant Scalar Curvature Metrics on Toric Surfaces,” *Geometric and Functional Analysis* **19** (2009), 83–136, DOI: <https://doi.org/10.1007/s00039-009-0714-y>, arXiv: 0805.0128; later all-dimensional existence results assume the stronger uniform stability condition, as in Bohui Chen, An-Min Li, and Li Sheng, “Interior Estimates for the n -Dimensional Abreu’s Equation,” *Advances in Mathematics* **251** (2014), 35–46, DOI: <https://doi.org/10.1016/j.aim.2013.10.022>, and Chi Li, “Geodesic Rays and Stability in the cscK Problem,” *Annales Scientifiques de l’École Normale Supérieure* (4) **55** (2022), no. 6, 1529–1574, DOI: <https://doi.org/10.24033/asens.2523>, arXiv: 2001.01366.

1.74.5. Primary-source pointer

The originating ICM record is *Emergent complex geometry* (2022). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-562-9-volume-4.pdf#page=99>.

1.75. 75. Kähler manifolds with curvature bounded below [2022-geom-163]

Record: 2022-geom-163

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2022: Kähler manifolds with curvature bounded below

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-562-9-volume-4.pdf#page=219>

1.75.1. Problem statement

Prove that if M^n is a complete noncompact Kähler manifold of complex dimension n with positive holomorphic bisectional curvature, meaning that at every point of M , for all nonzero vectors $X, Y \in T^{1,0}M$ at that point,

$$\frac{R(X, \bar{X}, Y, \bar{Y})}{\|X\|^2\|Y\|^2 + |\langle X, \bar{Y} \rangle|^2} > 0,$$

then M is biholomorphic to the complex Euclidean space $\mathbb{C}^n$.

1.75.2. Historical background, literature, and current progress

1.75.3. Verdict: partially resolved

1.75.3.1. Precise problem

This is Yau's uniformization conjecture for complete noncompact Kahler manifolds: if a complete noncompact complex n -dimensional Kahler manifold has positive holomorphic bisectional curvature, then its underlying complex manifold is biholomorphic to $\mathbb{C}^n$. The statement imposes no maximal volume growth, bounded curvature, curvature decay, finite Chern number, topological, or dimension restriction.

1.75.3.2. Status check

The literature does not fully resolve the statement as written. The strongest all-dimensional theorems prove the conclusion after adding maximal volume growth, and several other papers prove special cases under bounded curvature, curvature decay, finite Chern-number, surface, or sectional-curvature hypotheses. These hypotheses are not consequences of the stated assumptions in the published results checked here. Recent literature in 2026 still explicitly describes the full conjecture as open.

1.75.3.3. Key literature and hypothesis comparison

1. Shing-Tung Yau, "A review of complex differential geometry," in *Several Complex Variables and Complex Geometry, Part 2* (Santa Cruz, CA, 1989), Proceedings of Symposia in Pure Mathematics 52, Part 2, American Mathematical Society, 1991, pp. 619-625. DOI: 10.1090/pspum/052.2/1128577.
This is one standard cited source for the conjecture. Later surveys quote it as asking whether every complete noncompact Kahler manifold with positive bisectional curvature is biholomorphic to $\mathbb{C}^n$.
2. Ngaiming Mok, Yum-Tong Siu, and Shing-Tung Yau, "The Poincare-Lelong equation on complete Kahler manifolds," *Compositio Mathematica* 44 (1981), no. 1-3, 183-218. URL: https://www.numdam.org/item/CM_1981__44_1-3_183_0/.
Foundational analytic progress used in later uniformization work, but not a proof of the full conjecture. It supplies techniques under growth hypotheses rather than proving the unrestricted biholomorphism conclusion.
3. Ngaiming Mok, "An embedding theorem of complete Kahler manifolds of positive bisectional curvature onto affine algebraic varieties," *Bulletin de la Societe Mathematique de France* 112 (1984), 197-258. DOI: 10.24033/bsmf.2005. URL: <https://www.numdam.org/articles/10.24033/bsmf.2005/>.
Mok proved affine-algebraic embedding under extra assumptions such as quadratic scalar-curvature decay and Euclidean/maximal volume growth. This is substantial progress but does not imply that every manifold in the problem statement is biholomorphic to $\mathbb{C}^n$.
4. Bing-Long Chen, Siu-Hung Tang, and Xi-Ping Zhu, "A Uniformization Theorem Of Complete Noncompact Kahler Surfaces With Positive Bisectional Curvature," *Journal of Differential Geometry* 67 (2004), no. 3, 519-570. DOI: 10.4310/jdg/1102091357. arXiv: math/0211372, <https://arxiv.org/abs/math/0211372>.
This proves a two-dimensional result under positive and bounded holomorphic bisectional curvature, Euclidean volume growth, and average scalar-curvature decay. It is only a special case and has extra hypotheses.
5. Albert Chau and Luen-Fai Tam, "On the complex structure of Kahler manifolds with nonnegative curvature," *Journal of Differential Geometry* 73 (2006), no. 3, 491-530. DOI: 10.4310/jdg/1146169936. arXiv: math/0504422, <https://arxiv.org/abs/math/0504422>.
The paper proves that a complete noncompact Kahler manifold with nonnegative bounded holomorphic bisectional curvature and maximal volume growth is biholomorphic to $\mathbb{C}^n$. The bounded-curvature and maximal-volume-growth assumptions are extra.
6. Bing-Long Chen and Xi-Ping Zhu, "Yau's uniformization conjecture for manifolds with non-maximal volume growth," *Acta Mathematica Scientia* 38B (2018), no. 5, 1468-1484. DOI: 10.1016/S0252-9602(18)30828-2.
This studies the nonmaximal-volume-growth regime and gives partial results, including conditions involving bounded curvature and finite Chern-number-type hypotheses. It does not prove the unrestricted conjecture.
7. Gang Liu, "On Yau's uniformization conjecture," *Cambridge Journal of Mathematics* 7 (2019), no. 1-2, 33-70. DOI: 10.4310/CJM.2019.v7.n1.a2. arXiv:1606.08958, <https://arxiv.org/abs/1606.08958>.
The main theorem states that if M^n is a complete noncompact Kahler manifold with

nonnegative bisectional curvature and maximal volume growth, then M is biholomorphic to $\mathbb{C}^n$. This confirms Yau's conjecture only in the maximal-volume-growth case. the problem statement has no such assumption.

8. Man-Chun Lee and Luen-Fai Tam, "Chern-Ricci flows on noncompact complex manifolds," *Journal of Differential Geometry* 115 (2020), no. 3, 529-564. DOI: 10.4310/jdg/1594260018. arXiv:1708.00141, <https://arxiv.org/abs/1708.00141>.

This gives another proof that a complete noncompact Kahler manifold with nonnegative bisectional curvature and maximal volume growth is biholomorphic to $\mathbb{C}^n$, explicitly calling it a partial answer to Yau's conjecture. Again, maximal volume growth is extra.

9. Gang Liu, "Kahler manifolds with curvature bounded below," in *Proceedings of the International Congress of Mathematicians 2022*, Vol. 4, EMS Press, 2023, pp. 2576-2592. DOI: 10.4171/ICM2022/6. URL: <https://ems.press/content/book-chapter-files/33214>.

This survey states Conjecture 4 exactly as the problem and records the known progress. It presents the unrestricted statement as a conjecture, while proving and surveying related results such as the maximal-volume-growth case and finite generation questions.

10. Yang Shi, "Minimal Degrees, Volume Growth, and Curvature Decay on Complete Kahler Manifolds," *International Mathematics Research Notices* 2026, Issue 9, rnag080, published 23 April 2026. DOI: 10.1093/imrn/rnag080. arXiv:2511.01263, <https://arxiv.org/abs/2511.01263>.

This 2026 paper states Yau's uniformization conjecture and explicitly says that the full conjecture is still widely open. It describes Liu's result and the Lee-Tam proof as resolving the maximal-volume-growth case, not the full statement.

11. Ved Datar, Vamsi Pritham Pingali, and Harish Seshadri, "Uniformisation of complete Kahler surfaces with positive sectional curvature," arXiv:2604.11220, 2026. URL: <https://arxiv.org/abs/2604.11220>.

This preprint proves that a complete noncompact Kahler surface with positive sectional curvature is biholomorphic to $\mathbb{C}^2$, without assumptions at infinity. It is recent progress on a weaker two-dimensional variant, but it assumes positive sectional curvature rather than the all-dimensional positive bisectional-curvature hypothesis.

1.75.3.4. What remains unresolved

The missing case is the full unrestricted one: complete noncompact Kahler manifolds of arbitrary complex dimension with positive holomorphic bisectional curvature but without assuming maximal volume growth, bounded curvature, curvature decay, finite Chern-number conditions, positive sectional curvature, or auxiliary topology. Existing citable theorems do not cover all such manifolds, so the correct status is partially resolved rather than fully resolved.

1.75.4. Remaining open problems recorded by the database

1.75.4.1. Open problem 1

Prove the complex-surface case of Yau's uniformization conjecture: every complete non-compact Kähler surface with positive holomorphic bisectional curvature is biholomorphic to $\mathbb{C}^2$, without imposing bounded curvature, volume-growth, curvature-decay, topological, or positive-sectional-curvature hypotheses.

Reference supplied by the database. Ved Datar, Vamsi Pritham Pingali, and Harish Seshadri, "Uniformisation of Complete Kähler Surfaces with Positive Sectional Curvature," 2026, Introduction and discussion following Theorem 1 (explicitly contrasting the solved positive-sectional-curvature surface case with the still-unknown positive-holomorphic-bisectional-curvature case), arXiv: [2604.11220](https://arxiv.org/abs/2604.11220). The original conjecture is attributed there to Shing-Tung Yau, "Open Problems in Geometry," in *Differential Geometry: Partial Differential Equations on Manifolds* (Los Angeles, CA, 1990), Proceedings of Symposia in Pure Mathematics 54, Part 1, American Mathematical Society, 1993, pp. 1–28.

1.75.4.2. Open problem 2

Does every complete noncompact Kähler manifold with positive holomorphic bisectional curvature admit a smooth proper plurisubharmonic exhaustion function ρ with bounded gradient (that is, is it strongly Stein in the sense used by Greene and Wu)?

Reference supplied by the database. Ved Datar, Vamsi Pritham Pingali, and Harish Seshadri, "Uniformisation of Complete Kähler Surfaces with Positive Sectional Curvature," 2026, Introduction, discussion immediately after Theorem 1 (identifying strong Steinness as known under positive sectional curvature but unknown under the weaker hypothesis of positive holomorphic bisectional curvature), arXiv: [2604.11220](https://arxiv.org/abs/2604.11220). The positive-sectional-curvature result being contrasted is Robert E. Greene and H. Wu, "On Kähler Manifolds of Positive Bisectional Curvature and a Theorem of Hartogs," *Abhandlungen aus dem Mathematischen Seminar der Universität Hamburg* 47 (1978), 171–185, DOI: <https://doi.org/10.1007/BF02941481>.

1.75.4.3. Open problem 3

Does the complex surface $\mathbb{C} \times \mathbb{C}^*$ admit a complete Kähler metric with positive holomorphic bisectional curvature?

Reference supplied by the database. Ved Datar, Vamsi Pritham Pingali, and Harish Seshadri, "Uniformisation of Complete Kähler Surfaces with Positive Sectional Curvature," 2026, Introduction, discussion immediately after Theorem 1 (where this specific existence question is explicitly identified as unknown), arXiv: [2604.11220](https://arxiv.org/abs/2604.11220).

1.75.5. Primary-source pointer

The originating ICM record is *Kähler manifolds with curvature bounded below* (2022). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-562-9-volume-4.pdf#page=219>.

1.76. 76. Kähler manifolds with curvature bounded below [2022-geom-164]

Record: 2022-geom-164

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2022: Kähler manifolds with curvature bounded below

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-562-9-volume-4.pdf#page=219>

1.76.1. Problem statement

Prove that every complete noncompact Kähler manifold with positive Ricci curvature is biholomorphic to a Zariski open subset of some compact Kähler manifold.

1.76.2. Historical background, literature, and current progress

1.76.3. Verdict: partially resolved

1.76.3.1. Precise problem

The problem asks Yau's compactification question: whether every complete noncompact Kahler manifold whose Ricci curvature is everywhere positive is biholomorphic to a Zariski open subset of some compact Kahler manifold. No assumptions on dimension, bisectional/sectional curvature, volume growth, curvature decay, bounded curvature, finite topology, finite Chern/Ricci integral, or tangent cones are included in the problem statement.

1.76.3.2. Status check

I did not find a published or citable theorem proving this full statement. The closest results prove quasi-projectivity or compactification only after adding substantial hypotheses. Since those hypotheses are not consequences of positive Ricci curvature alone in the cited literature, they do not resolve the exact question.

The problem is not fully open, because there is significant partial progress. The current verdict is therefore **partially resolved**.

1.76.3.3. Key literature and progress

1. S. T. Yau, "S. S. Chern: A great geometer of the Twentieth Century." International Press, 1992. URL not found.

This is cited by Gang Liu's ICM survey as the original source of Question 71, p. 304: complete noncompact Kahler manifolds with positive Ricci curvature should be biholomorphic to a Zariski open set of a compact Kahler manifold.

2. Ngaiming Mok, “An embedding theorem of complete Kähler manifolds of positive bisectional curvature onto affine algebraic varieties.” *Bulletin de la Société Mathématique de France* 112 (1984), 197-258. DOI: 10.24033/bsmf.2005. URL: <https://www.numdam.org/articles/10.24033/bsmf.2005/>
Mok proved affine algebraicity under stronger curvature and asymptotic hypotheses: positive holomorphic bisectional curvature, maximal/Euclidean volume growth, and scalar-curvature decay. This is a compactification-type result, but it does not cover arbitrary positive Ricci curvature.
3. Ngaiming Mok, “An embedding theorem of complete Kähler manifolds of positive Ricci curvature onto quasi-projective varieties.” *Mathematische Annalen* 286 (1990), 373-408. DOI: 10.1007/BF01453581. URL: <https://link.springer.com/article/10.1007/BF01453581>
As summarized explicitly in Liu’s 2021 paper, Mok proved quasi-projectivity for complete noncompact Kahler manifolds with positive Ricci curvature assuming maximal volume growth, quadratic curvature decay $|Rm| \leq C/r^2$, and finite integral $\int_M Ric^n < \infty$. These extra assumptions mean Mok’s theorem is partial progress, not a full solution.
4. Wing-Keung To, “Quasi-projective embeddings of noncompact complete Kähler manifolds of positive Ricci curvature and satisfying certain topological conditions.” *Duke Mathematical Journal* 63(3) (1991), 745-789. DOI: 10.1215/S0012-7094-91-06332-5. URL: <https://projecteuclid.org/journals/duke-mathematical-journal/volume-63/issue-3/Quasi-projective-embeddings-of-noncompact-complete-K%C3%A4hler-manifolds-of-positive/10.1215/S0012-7094-91-06332-5.full>
This gives quasi-projective embeddings for positive-Ricci complete noncompact Kahler manifolds subject to additional topological conditions. The title itself records that the theorem is conditional; it does not establish Yau’s question for all positive-Ricci cases.
5. Bing-Long Chen and Xi-Ping Zhu, “Positively curved complete noncompact Kähler manifolds.” *Acta Mathematica Scientia* 29B(4) (2009), 829-845. DOI: 10.1016/S0252-9602(09)60073-4. arXiv: math/0211373. URL: <https://arxiv.org/abs/math/0211373>
The main theorem proves quasi-projectivity for complete noncompact Kahler manifolds with positive and bounded sectional curvature plus $\int_M Ric^n < \infty$, and gives C^2 in complex dimension 2. This concerns stronger positive sectional curvature and an integral finiteness condition, so it is only special-case progress for the problem.
6. Gang Liu, “Compactification of certain noncompact Kähler manifolds with non-negative Ricci curvature.” *Advances in Mathematics* 382 (2021), article 107652. DOI: 10.1016/j.aim.2021.107652. arXiv:1706.06067. URL: <https://arxiv.org/abs/1706.06067>
Theorem 1.1/3.1 states: if (M^n, p) is complete noncompact Kahler with nonnegative Ricci curvature and maximal volume growth, then in particular M is quasiprojective if Ricci curvature is positive and $|Rm| \leq C/r^2$. This strengthens Mok by removing the finite $\int_M Ric^n$ assumption, but still assumes maximal volume growth and quadratic curvature decay. Liu explicitly says this “supports” Yau’s Question 71, not that it settles it.
7. Gang Liu, “Kähler manifolds with curvature bounded below.” *Proceedings of the International Congress of Mathematicians 2022*, Vol. 4, EMS Press, pp. 2576-2592, 2022/2023. DOI: 10.4171/ICM2022/6. URL: <https://ems.press/content/book-chapter-files/33214>

This survey restates Yau's Question 71 and describes Liu's compactification theorem as supporting it under extra hypotheses. The survey also distinguishes positive Ricci curvature from stronger bisectional/sectional curvature assumptions.

8. **Gang Liu**, "Complete Kähler manifolds with nonnegative Ricci curvature." arXiv preprint, 2024. arXiv:2404.08537. URL: <https://arxiv.org/abs/2404.08537>

This later preprint studies scalar-curvature integrals, volume growth, and rigidity of Ricci-flat Kähler manifolds with nonnegative Ricci curvature. Its abstracted main results do not claim the compactification statement for all positive-Ricci complete noncompact Kähler manifolds.

9. **Ved V. Datar, Vamsi P. Pingali, and Harish Seshadri**, "Uniformisation of complete Kähler surfaces with positive sectional curvature." arXiv preprint, 2026. arXiv:2604.11220. URL: <https://arxiv.org/abs/2604.11220>

This resolves the weak Yau uniformization problem for complex surfaces with positive sectional curvature, proving biholomorphism to C^2 . Positive sectional curvature in complex dimension 2 is much stronger and narrower than positive Ricci curvature in arbitrary dimension, so it does not resolve this problem.

10. **Ved V. Datar, Vamsi P. Pingali, and Harish Seshadri**, "The top Yau–Yang conjecture for Kähler manifolds with positive sectional curvature." arXiv preprint, 2026. arXiv:2606.19806. URL: <https://arxiv.org/abs/2606.19806>

The abstract proves finiteness of the top wedge power of the Ricci form for complete noncompact Kähler manifolds with positive sectional curvature and obtains quasiprojectivity under bounded sectional curvature via Chen-Zhu. Again, this is positive sectional curvature plus boundedness for quasiprojectivity, not the full positive-Ricci problem.

1.76.3.4. What remains unresolved

The missing case is exactly the general one in the problem statement: a complete noncompact Kähler manifold with positive Ricci curvature and no further global control. The literature above does not show that positive Ricci curvature alone implies maximal volume growth, quadratic curvature decay, finite $\int_M Ric^n$, bounded sectional curvature, positive bisectional/sectional curvature, finite topology, or the topological assumptions used in the known embedding theorems. Therefore the full compactification question remains unresolved.

1.76.4. Remaining open problems recorded by the database

1.76.4.1. Open problem 1

Let M^n , $n \geq 3$, be a complete noncompact Kähler manifold with nonnegative Ricci curvature and maximal (Euclidean) volume growth. If its holomorphic bisectional curvature satisfies $BK(x) \geq -C/r(x)^2$, must M be biholomorphic to a Zariski open subset of a Moishezon manifold, without assuming that M has a unique tangent cone at infinity?

Reference supplied by the database. Gang Liu, "Compactification of Certain Noncompact Kähler Manifolds with Nonnegative Ricci Curvature," *Advances in Mathematics* **382** (2021), Article 107652, discussion immediately following Theorem 1.1(II) (explicitly asking to remove

uniqueness of the tangent cone at infinity). DOI: <https://doi.org/10.1016/j.aim.2021.107652>; arXiv: [1706.06067](https://arxiv.org/abs/1706.06067). The complex-surface case is subsequent partial progress: Gang Liu, “Complete Kähler Manifolds with Nonnegative Ricci Curvature II,” 2025, main theorem, arXiv: [2510.10600](https://arxiv.org/abs/2510.10600); no corresponding all-dimensional result was found.

1.76.5. Primary-source pointer

The originating ICM record is *Kähler manifolds with curvature bounded below* (2022). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-562-9-volume-4.pdf#page=219>.

1.77. 77. Mean curvature and variational theory [2022-geom-166]

Record: 2022-geom-166

Subfield: Global/Riemannian geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2022: Mean curvature and variational theory

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-562-9-volume-4.pdf#page=339>

1.77.1. Problem statement

Prove or disprove that every closed Riemannian manifold (M^{n+1}, g) with $3 \leq n+1 \leq 7$ contains, for every constant $c > 0$, at least two distinct closed hypersurfaces whose mean curvature is identically equal to c .

1.77.2. Historical background, literature, and current progress

1.77.3. Verdict: partially resolved

1.77.3.1. Precise problem

The problem asks for the Twin Bubble Conjecture: for every fixed closed Riemannian manifold (M^{n+1}, g) with $3 \leq n+1 \leq 7$ and every prescribed constant $c > 0$, there should be at least two distinct closed hypersurfaces whose mean curvature is identically c . In the recent min-max literature this is usually stated for smooth closed almost-embedded hypersurfaces.

The literature does not currently prove this full fixed-metric, all- c statement. There is substantial progress in special regimes and under extra hypotheses, so the problem is not fully open, but it remains unresolved for arbitrary metrics and arbitrary intermediate values of c .

1.77.3.2. Original source

- Title: Mean curvature and variational theory
- Author: Xin Zhou
- Venue: ICM–International Congress of Mathematicians, Vol. IV, Sections 5-8, EMS Press, Berlin, pages 2696-2717
- Year: 2023
- DOI/URL: DOI 10.4171/ICM2022/27; <https://ems.press/books/standalone/276/5487>
- Relevance: This ICM survey formulates the conjectural statement that every closed manifold (M^{n+1}, g) , $3 \leq n+1 \leq 7$, contains at least two distinct closed CMC hypersurfaces with mean curvature c for any $c > 0$.

1.77.3.3. Progress and non-resolving results

- Title: Min-max theory for constant mean curvature hypersurfaces
- Authors: Xin Zhou and Jonathan J. Zhu
- Venue: *Inventiones Mathematicae* 218, 441-490
- Year: 2019
- DOI/arXiv/URL: DOI 10.1007/s00222-019-00886-1; arXiv:1707.08012; <https://arxiv.org/abs/1707.08012>
- Result: Proves existence of a nontrivial smooth closed almost-embedded CMC hypersurface of any prescribed mean curvature c in an arbitrary closed manifold; for nonzero c the min-max solution has multiplicity one.
- Why not full resolution: This gives one hypersurface for each c , not two.
- Title: Existence of multiple closed CMC hypersurfaces with small mean curvature
- Author: Akashdeep Dey
- Venue: *Journal of Differential Geometry* 125(2), 379-403
- Year: 2023
- DOI/arXiv/URL: DOI 10.4310/jdg/1696432925; arXiv:1910.00989; <https://arxiv.org/abs/1910.00989>
- Result: For every m , there is $c^*(m) > 0$ depending on g such that if $0 < c < c^*(m)$, then (M, g) contains at least m closed c -CMC hypersurfaces with optimal regularity. In dimensions $3 \leq n+1 \leq 7$, this gives smooth hypersurfaces.
- Why not full resolution: Taking $m = 2$ proves the conclusion only for sufficiently small c , not for every $c > 0$.
- Title: Foliation by constant mean curvature spheres
- Author: Rugang Ye
- Venue: *Pacific Journal of Mathematics* 147(2), 381-396
- Year: 1991
- DOI/arXiv/URL: <https://projecteuclid.org/journals/pacific-journal-of-mathematics/volume-147/issue-2/Foliation-by-constant-mean-curvature-spheres/pjm/1102644918.pdf>
- Result: Constructs local foliations by small CMC spheres near nondegenerate critical points of the scalar curvature, giving large-mean-curvature examples under a nondegeneracy hypothesis.
- Why not full resolution: This applies only in the large- c regime and uses extra scalar-curvature hypotheses.
- Title: Constant mean curvature spheres in Riemannian manifolds
- Authors: Frank Pacard and Xingwang Xu
- Venue: *Manuscripta Mathematica* 128, 275-295
- Year: 2009
- DOI/arXiv/URL: DOI 10.1007/s00229-008-0230-7; <https://link.springer.co>

[m/article/10.1007/s00229-008-0230-7](https://doi.org/10.1007/s00229-008-0230-7)

- Result: Proves existence of embedded spheres with large constant mean curvature in any compact Riemannian manifold, partially generalizing Ye's result and giving large- c multiplicity information via Lusternik-Schnirelmann category.
- Why not full resolution: This covers sufficiently large c , not all positive c .
- Title: The half-volume spectrum of a manifold
- Authors: Liam Mazurowski and Xin Zhou
- Venue: Calculus of Variations and Partial Differential Equations 64, article 155
- Year: 2025
- DOI/arXiv/URL: DOI 10.1007/s00526-025-02949-z; <https://link.springer.com/article/10.1007/s00526-025-02949-z>
- Result: Develops the half-volume spectrum and proves, in dimensions $3 \leq n + 1 \leq 7$, existence of half-volume CMC hypersurfaces realizing phase-transition half-volume widths, possibly with additional minimal hypersurface components.
- Why not full resolution: The result controls enclosed volume rather than prescribing the exact mean curvature c .
- Title: Infinitely Many Half-Volume Constant Mean Curvature Hypersurfaces via Min-Max Theory
- Authors: Liam Mazurowski and Xin Zhou
- Venue: arXiv preprint
- Year: 2024
- DOI/arXiv/URL: DOI 10.48550/arXiv.2405.00595; arXiv:2405.00595; <https://arxiv.org/abs/2405.00595>
- Result: For closed manifolds of dimension $3 \leq n + 1 \leq 5$, if the metric is generic or has positive Ricci curvature, there are infinitely many geometrically distinct half-volume CMC hypersurfaces.
- Why not full resolution: This imposes generic or positive Ricci hypotheses, is limited to dimensions up to five, and does not prescribe the mean curvature value c .
- Title: An Alternative for Constant Mean Curvature Hypersurfaces
- Authors: Liam Mazurowski and Xin Zhou
- Venue: arXiv preprint
- Year: 2024
- DOI/arXiv/URL: DOI 10.48550/arXiv.2408.13864; arXiv:2408.13864; <https://arxiv.org/abs/2408.13864>
- Result: For a generic metric on a closed manifold of dimension $3 \leq n + 1 \leq 7$ and every $c > 0$, either there are infinitely many distinct closed almost-embedded hypersurfaces with CMC exactly c , or there are infinitely many closed almost-embedded CMC hypersurfaces with mean curvature less than c enclosing half the volume.
- Why not full resolution: It assumes a generic metric and the second alternative does not produce even two hypersurfaces with mean curvature exactly c .

- Title: Pairs of Embedded Spheres with Pinched Prescribed Mean Curvature
- Authors: Liam Mazurowski and Xin Zhou
- Venue: arXiv preprint
- Year: 2025
- DOI/arXiv/URL: DOI 10.48550/arXiv.2511.08228; arXiv:2511.08228; <https://arxiv.org/abs/2511.08228>
- Result: Proves two embedded two-spheres in the unit three-sphere under a pinching condition on a prescribed mean-curvature function. The introduction explicitly states the Twin Bubble Conjecture and says the relevant two-solution conjectures are still open.
- Why not full resolution: This is a special result for S^3 with a pinching condition and does not cover arbitrary closed Riemannian manifolds or all constants c .
- Title: Existence of multiple constant mean curvature hypersurfaces for varying Riemannian metrics
- Authors: Xiaoxiang Jiao and Wenduo Zou
- Venue: arXiv preprint, version 3
- Year: 2026
- DOI/arXiv/URL: DOI 10.48550/arXiv.2511.19102; arXiv:2511.19102; <https://arxiv.org/abs/2511.19102>
- Result: For a closed Riemannian manifold of dimension $3 \leq n + 1 \leq 7$, if the number of closed c -CMC hypersurfaces in (M, g) is finite, then one can vary the metric to a metric h with strictly more c -CMC hypersurfaces while preserving the old ones, with a quantitative norm bound.
- Why not full resolution: The metric is changed from g to h ; the problem asks for the original fixed metric.

1.77.3.4. What remains unresolved

No citable result found proves that every fixed closed Riemannian manifold (M^{n+1}, g) , $3 \leq n + 1 \leq 7$, has at least two distinct closed CMC hypersurfaces of every prescribed constant $c > 0$. The small- c and large- c regimes are covered by existing work, and generic/perturbative alternatives are known, but the arbitrary fixed metric and arbitrary intermediate prescribed mean curvature case remains open.

1.77.4. Remaining open problems recorded by the database

1.77.4.1. Open problem 1

Let g be an arbitrary Riemannian metric on S^3 . For every $c > 0$, are there at least two distinct immersions $S^2 \rightarrow (S^3, g)$ whose mean curvature is identically c ?

Reference supplied by the database. Liam Mazurowski and Xin Zhou, “Pairs of Embedded Spheres with Pinched Prescribed Mean Curvature,” 2025, Conjecture 1.6 (the modified

two-solution CMC-sphere conjecture, stated after the discussion of the counterexample to the embedded version), arXiv: [2511.08228](https://arxiv.org/abs/2511.08228).

1.77.4.2. Open problem 2

Let (M^{n+1}, g) , $3 \leq n + 1 \leq 5$, have a generic metric, and let $\{\Sigma_p\}$ be the sequence of geometrically distinct half-volume constant-mean-curvature hypersurfaces supplied by the half-volume min-max theory, with $\text{Area}(\Sigma_p) \rightarrow \infty$. Must their constant mean curvatures satisfy $H(\Sigma_p) \rightarrow 0$ as $p \rightarrow \infty$?

Reference supplied by the database. Liam Mazurowski and Xin Zhou, “An Alternative for Constant Mean Curvature Hypersurfaces,” *Annali della Scuola Normale Superiore di Pisa, Classe di Scienze* (to appear; accepted version available in 2026), §1.1, final paragraph (explicitly asking whether the mean curvatures of the half-volume sequence tend to zero), arXiv: [2408.13864](https://arxiv.org/abs/2408.13864). The sequence in the question is constructed in Liam Mazurowski and Xin Zhou, “Infinitely Many Half-Volume Constant Mean Curvature Hypersurfaces via Min-Max Theory,” 2024, arXiv: [2405.00595](https://arxiv.org/abs/2405.00595).

1.77.5. Primary-source pointer

The originating ICM record is *Mean curvature and variational theory* (2022). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-562-9-volume-4.pdf#page=339>.

1.78. 78. Kähler manifolds with curvature bounded below [2022-geom-333]

Record: 2022-geom-333

Subfield: Global/Riemannian geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2022: Kähler manifolds with curvature bounded below

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-562-9-volume-4.pdf#page=219>

1.78.1. Problem statement

Prove that for every complete Ricci-flat Kähler manifold M of finite topological type, there exist a compact complex analytic space $\overline{M}$ and a closed complex analytic subset $D \subset \overline{M}$ such that M is biholomorphic to $\overline{M} \setminus D$.

1.78.2. Historical background, literature, and current progress

The problem is **unresolved**. No result located through 4 August 2026 proves complex-analytic compactifiability from finite topological type alone. Known theorems require substantial extra control at infinity, and no collection of them covers every complete Ricci-flat Kähler manifold of finite topological type.

Key references and partial progress:

- Shing-Tung Yau, “Nonlinear Analysis in Geometry,” *L’Enseignement Mathématique* 33 (1987), 109–158, DOI <https://doi.org/10.5169/seals-87888>. This is the cited source of the finite-topological-type analytic compactification conjecture.
- Gang Liu, “Compactification of Certain Kähler Manifolds with Nonnegative Ricci Curvature,” *Advances in Mathematics* 382 (2021), 107652, DOI <https://doi.org/10.1016/j.aim.2021.107652>. Corollary 1.1 proves the desired conclusion in the special case of maximal (Euclidean) volume growth and quadratic curvature decay: such a Ricci-flat Kähler manifold is a crepant resolution of a normal affine algebraic variety. Theorem 1.1 also gives a Moishezon compactification under related bisectional-curvature decay and tangent-cone hypotheses. Neither finite topological type nor Ricci-flatness alone supplies these hypotheses.
- Chi Li, “On Sharp Rates and Analytic Compactifications of Asymptotically Conical Kähler Metrics,” *Duke Mathematical Journal* 169 (2020), no. 8, DOI <https://doi.org/10.1215/00127094-2019-0073>. The paper proves analytic compactification for complex structures asymptotic at a positive rate to an appropriate complex cone. This establishes another important asymptotically conical special case, not the finite-topological-type statement in general.
- Song Sun and Ruobing Zhang, “Collapsing Geometry of Hyperkähler 4-Manifolds and Applications,” *Acta Mathematica* 232 (2024), no. 2, 325–424, DOI <https://doi.org/10.4310/ACTA.2024.v232.n2.a2>. They classify the asymptotic models of hyperkähler gravitational instantons, meaning complete hyperkähler four-manifolds

with finite curvature energy (equivalently, quadratic curvature decay). This is major progress in complex dimension two but assumes the analytic decay not known to follow from finite topological type.

- Mingyang Li and Song Sun, “Geometry of Gravitational Instantons,” *Surveys in Differential Geometry*, to appear (2026), arXiv:2606.10443. Theorem 4.9 assembles a compactification result for hyperkähler gravitational instantons: for some compatible complex structure they are complements of divisors in algebraic surfaces. The survey still lists as Conjecture 30 that a complete oriented Ricci-flat four-manifold is a gravitational instanton if and only if it has finite topological type. Moreover, Theorem 4.9 does not assert compactification for every preassigned Kähler complex structure.
- Michael T. Anderson, Peter B. Kronheimer, and Claude LeBrun, “Complete Ricci-Flat Kähler Manifolds of Infinite Topological Type,” *Communications in Mathematical Physics* 125 (1989), no. 4, 637–642, DOI <https://doi.org/10.1007/BF01228345>. Their examples obstruct the corresponding statement without the finite-topological-type hypothesis, but do not disprove the problem stated here.

Thus the remaining gap is precisely to obtain a complex-analytic compactification for arbitrary finite-topological-type ends without assuming Euclidean volume growth, curvature decay, finite curvature energy, conical asymptotics, tangent-cone regularity, or a change of complex structure.

1.78.3. Remaining open problems recorded by the database

1.78.3.1. Open problem 1

Prove that every complete, noncompact, oriented Ricci-flat Riemannian 4-manifold (M, g) of finite topological type has quadratic curvature decay: there exist a point $p \in M$ and constants $C, R > 0$ such that $|\text{Riem}_g|(x) \leq C d_g(p, x)^{-2}$ whenever $d_g(p, x) \geq R$.

Reference supplied by the database. Mingyang Li and Song Sun, “Geometry of Gravitational Instantons,” *Surveys in Differential Geometry*, to appear (2026), Conjecture 30, arXiv:2606.10443, <https://arxiv.org/abs/2606.10443>.

1.78.4. Primary-source pointer

The originating ICM record is *Kähler manifolds with curvature bounded below* (2022). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-562-9-volume-4.pdf#page=219>.

2. Discrete and polyhedral geometry

This chapter ranges from face-number and realization problems to packing, covering, convex-position, and incidence questions. Exact extremal constants and dimension dependence are especially prominent. Computation and probabilistic methods frequently establish asymptotic bounds, whereas the original sharp statement remains open even when a great deal is known around it.

Coverage: 35 website records; **source years:** 1904–2022; **separately listed descendant open problems:** 59.

2.1. 79. Über die diskontinuierlichen und nicht-konvexen gleicheckig-gleichflächigen Polyeder [1904-geom-218]

Record: 1904-geom-218

Subfield: Discrete and polyhedral geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1904: Über die diskontinuierlichen und nicht-konvexen gleicheckig-gleichflächigen Polyeder

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1904/ICM1904.ocr.pdf#page=717>

2.1.1. Problem statement

Classify all finite nonconvex polyhedra in Euclidean three-space that are simultaneously isohedral and isogonal in the following sense: any two faces are directly or oppositely congruent, with equal dihedral angles at corresponding edges, and the spherical vertex figures at any two vertices are directly or oppositely congruent, with equal lengths on corresponding incident edges. Restrict the classification to polyhedra of the first class, namely those whose oriented volume is nonzero and hence can be made positive by choosing the orientation.

2.1.2. Historical background, literature, and current progress

The problem is **unresolved**. Brückner's condition is local: faces and spherical vertex figures must be congruent in the stated metric sense. The strongest modern completeness theorem instead assumes that the full Euclidean symmetry group is transitive on the faces and vertices

and also imposes a faithful, planar, nondegenerate realization. Congruence of all faces or vertex figures is weaker than transitivity of the whole polyhedron, and no cited result proves that every positive-volume polyhedron in Brückner's class satisfies the additional modern hypotheses. Thus the modern enumeration does not exhaust the exact class in the statement.

- **Max Brückner**, “Über die gleicheckig-gleichflächigen, diskontinuierlichen und nichtkonvexen Polyeder,” *Nova Acta Leopoldina*, vol. 86, no. 1 (1906), pp. 1–348 and 29 plates. [Direct scan](#). Brückner systematically derives many first-class nonconvex examples, but on p. 162 expressly says that a strict exhaustion proof is lacking because the relevant parameters and core polyhedra vary continuously. This is the foundational partial enumeration, not a completeness proof.
- **D. M. Y. Sommerville**, “Isohedral and Isogonal Generalizations of the Regular Polyhedra,” *Proceedings of the Royal Society of Edinburgh*, vol. 52 (1933), pp. 251–263. DOI: <https://doi.org/10.1017/S0370164600019520>. Sommerville records Brückner's exact congruence-based meanings of equal faces and equal vertices and develops the convex/generalized theory; it does not classify the nonconvex positive-volume class.
- **Branko Grünbaum**, “Are Your Polyhedra the Same as My Polyhedra?,” in *Discrete and Computational Geometry: The Goodman-Pollack Festschrift*, Algorithms and Combinatorics 25, Springer (2003), pp. 461–488. DOI: https://doi.org/10.1007/978-3-642-55566-4_21. Grünbaum formulates a precise abstract-realization framework, documents the inconsistent admissibility conventions in Hess and Brückner, and identifies completeness for full-dimensional symmetry-transitive noble polyhedra as open. This clarifies rather than closes the gap for the historical congruence-based class.
- **Ulrich Mikloweit**, “Exploring Noble Polyhedra With the Program Stella4D,” *Bridges 2020 Conference Proceedings* (2020), pp. 257–264. [Direct proceedings PDF](#). Mikloweit constructs 61 symmetry-transitive noble polyhedra and two infinite families, including many examples absent from Brückner, but explicitly states that the list is not complete.
- **Connor Hill**, “The Complete Set of Noble Polyhedra,” *arXiv* (2026). arXiv: [2607.28711](https://arxiv.org/abs/2607.28711). Hill gives a computer-assisted proof that, under his Section 2 definitions, the only prismatic examples are disphenoids and stephanoids and that there are exactly 146 further nonprismatic noble polyhedra up to similarity. This completely resolves the finite, faithful, planar, nondegenerate, vertex-transitive and face-transitive version, but the paper explicitly excludes nonfaithful realizations, faces revisiting vertices, and adjacent coplanar faces and does not prove equivalence with Brückner's weaker local congruence hypotheses. The remaining gap is therefore the classification of positive-oriented-volume polyhedra satisfying the statement's local face and vertex-figure congruence conditions without those extra symmetry and nondegeneracy assumptions.

2.1.3. Remaining open problems recorded by the database

2.1.3.1. Open problem 1

Classify all finite nonconvex polyhedra in Euclidean three-space whose oriented volume is nonzero, whose faces are pairwise directly or oppositely congruent with equal dihedral angles at corresponding edges, whose spherical vertex figures are pairwise directly or oppositely congruent with equal lengths on corresponding incident edges, and whose

inner core (the innermost bounded cell in the arrangement of the planes containing the faces) is a convex face-transitive polyhedron with full octahedral symmetry.

Reference supplied by the database. Max Brückner, “Über die gleicheckig-gleichflächigen, diskontinuierlichen und nichtkonvexen Polyeder,” *Nova Acta: Abhandlungen der Kaiserlich Leopoldinisch-Carolinischen Deutschen Akademie der Naturforscher*, vol. 86, no. 1, Halle: E. Karras, 1906, pp. 1–348, especially the unproved exhaustion statement for the Hexakisoctaedertypus on pp. 159–162, <https://books.google.com/books?id=oK01AQAAMAAJ&pg=PA162>.

2.1.3.2. Open problem 2

Classify all connected finite nonconvex polyhedra in Euclidean three-space whose oriented volume is nonzero, whose faces are pairwise directly or oppositely congruent with equal dihedral angles at corresponding edges, whose spherical vertex figures are pairwise directly or oppositely congruent with equal lengths on corresponding incident edges, and whose inner core (the innermost bounded cell in the arrangement of the planes containing the faces) is a convex face-transitive polyhedron with full icosahedral symmetry.

Reference supplied by the database. Max Brückner, “Über die gleicheckig-gleichflächigen, diskontinuierlichen und nichtkonvexen Polyeder,” *Nova Acta: Abhandlungen der Kaiserlich Leopoldinisch-Carolinischen Deutschen Akademie der Naturforscher*, vol. 86, no. 1, Halle: E. Karras, 1906, pp. 1–348, especially the explicitly incomplete discussion of continuous first-class polyhedra of the Dyakishekontaedertypus on pp. 310–316, <https://books.google.com/books?id=oK01AQAAMAAJ&pg=PA316>.

2.1.4. Primary-source pointer

The originating ICM record is *Über die diskontinuierlichen und nicht-konvexen gleicheckig-gleichflächigen Polyeder* (1904). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1904/ICM1904.ocr.pdf#page=717>.

2.2. 80. Recent progress in the geometry of numbers [1950-geom-6]

Record: 1950-geom-6

Subfield: Discrete and polyhedral geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1950: Recent progress in the geometry of numbers

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1950.1/ICM1950.1.ocr.pdf#page=174>

2.2.1. Problem statement

For every positive integer n , every full-rank lattice $\Lambda \subset \mathbb{R}^n$ with determinant (covolume) 1, and every point $c = (c_1, \dots, c_n) \in \mathbb{R}^n$, there exists a lattice point $x = (x_1, \dots, x_n) \in \Lambda$ such that

$$\left| \prod_{i=1}^n (x_i - c_i) \right| \leq 2^{-n}.$$

2.2.2. Historical background, literature, and current progress

2.2.3. Verdict: partially resolved

2.2.3.1. Precise problem

The problem asks for the full inhomogeneous Minkowski conjecture on the product of linear forms: for every dimension n , every determinant-one full-rank lattice Λ in $\mathbb{R}^n$, and every translate c in $\mathbb{R}^n$, there should be x in Λ with

$$\left| \prod_i (x_i - c_i) \right| \leq 2^{-n}.$$

This is equivalent to the usual formulation for n real non-homogeneous linear forms. If $\Lambda = A \mathbb{Z}^n$ with $|\det A| = 1$, then the coordinate functions of Az are n real linear forms $L_i(z)$, and the stated inequality becomes

$$\left| \prod_i (L_i(z) - c_i) \right| \leq |\det A| / 2^n.$$

Thus literature proving Minkowski's conjecture for products of n non-homogeneous linear forms in all dimensions would resolve the problem statement. Literature proving only fixed dimensions, estimates with constants worse than 2^{-n} , almost-everywhere results, function-field analogues, or results under extra lattice hypotheses is only partial progress.

2.2.3.2. Current status

No citable source found proves the full all-dimensional real Euclidean statement. The strongest exact published progress I found proves the conjecture through dimension $n \leq 10$.

Therefore the problem is not fully resolved; the unresolved part is the exact statement for real determinant-one lattices in dimensions $n \geq 11$.

2.2.3.3. Key literature and applicability

1. Title: Ueber die Annaeherung an eine reelle Groesse durch rationale Zahlen
 Authors: Hermann Minkowski
 Venue/year: Mathematische Annalen 54 (1901), 91-124
 DOI/URL: <https://doi.org/10.1007/BF01454200> ; <https://eudml.org/doc/157996>
 Role: Original source associated with Minkowski's conjecture. It is historical/original-source evidence, not a modern full resolution.
2. Title: On the inhomogeneous minimum of the product of n linear forms
 Authors: B. J. Birch and H. P. F. Swinnerton-Dyer
 Venue/year: Mathematika 3 (1956), 25-39
 DOI/URL: <https://doi.org/10.1112/S0025579300000863> ; <https://www.cambridge.org/core/journals/mathematika/article/on-the-inhomogeneous-minimum-of-the-product-of-n-linear-forms/1458B4BE234DD5013BC3BC14CAF79711>
 Role: Gives the standard modern formulation and important reduction results. It does not prove the conjecture in all dimensions.
3. Title: Minkowski's conjecture, well-rounded lattices and topological dimension
 Authors: Curtis T. McMullen
 Venue/year: Journal of the American Mathematical Society 18 (2005), 711-734
 DOI/URL: <https://doi.org/10.1090/S0894-0347-05-00483-2> ; <https://people.math.harvard.edu/~ctm/papers/home/text/papers/mink/mink.pdf>
 Role: Proves a major reduction via well-rounded lattices and proves Minkowski's conjecture for $n = 6$, assuming known covering-radius cases through dimension 6. Its Theorem 8.1 says covering-radius bounds for all well-rounded unimodular lattices in dimensions $k \leq n$ imply Minkowski's conjecture in those dimensions. This is a conditional reduction in general, not a full all-dimensional proof.
4. Title: On conjectures of Minkowski and Woods for $n = 7$
 Authors: R. J. Hans-Gill, Madhu Raka, Ranjeet Sehmi
 Venue/year: Journal of Number Theory 129(5), 1011-1033 (2009)
 DOI/URL: <https://doi.org/10.1016/j.jnt.2008.10.020>
 Role: Proves the relevant Woods/Minkowski route in dimension 7. This is a fixed-dimensional result only.
5. Title: On conjectures of Minkowski and Woods for $n = 8$
 Authors: R. J. Hans-Gill, Madhu Raka, Ranjeet Sehmi
 Venue/year: Acta Arithmetica 147(4), 337-385 (2011)
 DOI/URL: <https://doi.org/10.4064/aa147-4-3> ; <https://eudml.org/doc/279840>
 Role: Proves the relevant Woods/Minkowski route in dimension 8. This is a fixed-dimensional result only.
6. Title: On conjectures of Minkowski and Woods for $n = 9$
 Authors: Leetika Kathuria, Madhu Raka
 Venue/year: Proceedings - Mathematical Sciences 126(4), 501-548 (2016)

DOI/arXiv/URL: <https://doi.org/10.1007/s12044-016-0295-2> ;
arXiv:1410.5743 ; <https://arxiv.org/abs/1410.5743>

Role: Proves Woods' conjecture for $n = 9$, from which Minkowski's conjecture follows in dimension 9 via McMullen's result. This does not cover $n \geq 10$.

7. Title: On conjectures of Minkowski and Woods for $n = 10$

Authors: Leetika Kathuria, Madhu Raka

Venue/year: Proceedings - Mathematical Sciences 132(2), article 45 (2022)

DOI/arXiv/URL: <https://doi.org/10.1007/s12044-022-00679-2> ;
arXiv:2009.09992 ; <https://arxiv.org/abs/2009.09992>

Role: Proves Woods' conjecture and hence Minkowski's conjecture for $n = 10$. This is the largest exact-dimensional resolution I found. It proves the problem statement only when the positive integer n is 10, not for every n .

8. Title: Stable and well-rounded lattices in diagonal orbits

Authors: Omri Nisan Solan

Venue/year: Israel Journal of Mathematics 234, 501-519 (2019)

DOI/URL: <https://doi.org/10.1007/s11856-019-1912-4>

Role: Gives a newer diagonal-orbit/stable-lattice reduction and new proofs in dimensions up to 7 under finite covering-radius checks. It is a reduction/partial-progress paper, not an all-dimensional proof.

9. Title: Counterexamples to a conjecture of Woods

Authors: Oded Regev, Uri Shapira, Barak Weiss

Venue/year: Duke Mathematical Journal 166(13), 2443-2446 (2017)

DOI/arXiv/URL: <https://doi.org/10.1215/00127094-2017-0018> ;
arXiv:1604.07644

Role: Disproves Woods' conjecture in high dimensions. This does not disprove or resolve Minkowski's conjecture; it only shows that one sufficient route through Woods cannot work in those dimensions.

10. Title: Counterexamples to the Woods Conjecture in dimensions $d \geq 24$

Authors: Hao Chen, Liqing Xu

Venue/year: Journal de Theorie des Nombres de Bordeaux 31(3), 723-726 (2019)

DOI/URL: <https://doi.org/10.5802/jtnb.1105> ; <https://jtnb.centre-mersenne.org/item/10.5802/jtnb.1105.pdf>

Role: Improves high-dimensional counterexamples to Woods' conjecture to $d \geq 24$. Again, this is not a counterexample to Minkowski's conjecture and does not settle the problem statement.

11. Title: Lecture Notes on Geometry of Numbers

Author: R. J. Hans-Gill

Venue/year: Springer Nature Singapore, 2024

DOI/URL: <https://doi.org/10.1007/978-981-99-9602-5>

Role: Recent survey/book source stating that Minkowski's conjecture is proved for $n \leq 10$ and discussing weaker estimates and open Woods dimensions beyond that. This supports the current-status assessment but does not contain a full all-dimensional proof.

12. Title: On covering radii in function fields

Author: Noy Soffer Aranov

Venue/year: Monatshefte fuer Mathematik 206, 705-734 (2025)

DOI/URL: <https://doi.org/10.1007/s00605-024-02049-8>

Role: Treats function-field analogues and cites the $n \leq 10$ real-dimensional literature. It is not applicable to this problem because the problem statement is about real lattices

in $\mathbb{R}^n$.

2.2.3.4. What remains unresolved

The full statement asks for every positive integer n . Published exact results cover only fixed dimensions up to $n = 10$. For $n \geq 11$, I found estimates, reductions, almost-everywhere statements, and analogues in other settings, but no published or otherwise citable theorem whose hypotheses and conclusion imply

for every unimodular Λ in $\mathbb{R}^n$ and every c in $\mathbb{R}^n$,
there exists x in Λ with $|\prod_{i=1}^n (x_i - c_i)| \leq 2^{-n}$.

Therefore the literature only partially resolves the problem statement.

2.2.4. Remaining open problems recorded by the database

2.2.4.1. Open problem 1

Prove Minkowski's inhomogeneous product-of-linear-forms conjecture in the first unresolved dimension $n = 11$: for every unimodular lattice $\Lambda \subset \mathbb{R}^{11}$ and every $c \in \mathbb{R}^{11}$, there is an $x \in \Lambda$ such that $|\prod_{i=1}^{11} (x_i - c_i)| \leq 2^{-11}$.

Reference supplied by the database. R. J. Hans-Gill, Madhu Raka, and Ranjeet Sehmi, *Lecture Notes on Geometry of Numbers*, University Texts in the Mathematical Sciences, Springer Singapore, 2024, Chapter "Inhomogeneous Problems," pp. 161–193 (Minkowski's conjecture; recorded as proved only for $n \leq 10$), DOI: <https://doi.org/10.1007/978-981-99-9602-5>. The endpoint $n = 10$ is proved in Leetika Kathuria and Madhu Raka, "On conjectures of Minkowski and Woods for $n = 10$," *Proceedings of the Indian Academy of Sciences (Mathematical Sciences)* **132** (2022), Article 45, DOI: <https://doi.org/10.1007/s12044-022-00679-2>, [arXiv:2009.09992](https://arxiv.org/abs/2009.09992).

2.2.4.2. Open problem 2

Prove Woods's covering-radius conjecture in dimension 11: every well-rounded unimodular lattice in $\mathbb{R}^{11}$ has covering radius at most $\sqrt{11}/2$. This is an explicitly proposed intermediate problem that would imply the eleven-dimensional case of Minkowski's conjecture.

Reference supplied by the database. Hao Chen and Liqing Xu, "Counterexamples to the Woods Conjecture in dimensions $d \geq 24$," *Journal de Théorie des Nombres de Bordeaux* **31** (2019), no. 3, 723–726, introduction (explicitly identifying dimensions $10 \leq d \leq 23$ as open at that time), DOI: <https://doi.org/10.5802/jtnb.1105>. The dimension-10 case was subsequently settled by Leetika Kathuria and Madhu Raka, "On conjectures of Minkowski and Woods for $n = 10$," *Proceedings of the Indian Academy of Sciences (Mathematical Sciences)*

132 (2022), Article 45, DOI: <https://doi.org/10.1007/s12044-022-00679-2>, leaving $n = 11$ as the first open dimension; the implication to Minkowski's conjecture is due to Curtis T. McMullen, "Minkowski's conjecture, well-rounded lattices and topological dimension," *Journal of the American Mathematical Society* **18** (2005), no. 3, 711–734, Theorem 8.1, DOI: <https://doi.org/10.1090/S0894-0347-05-00483-2>.

2.2.5. Primary-source pointer

The originating ICM record is *Recent progress in the geometry of numbers* (1950). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1950.1/ICM1950.1.ocr.pdf#page=174>.

2.3. 81. On the application of combinatorial analysis to number theory, geometry and analysis [1970-geom-16]

Record: 1970-geom-16

Subfield: Discrete and polyhedral geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1970: On the application of combinatorial analysis to number theory, geometry and analysis

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1970.3/ICM1970.3.ocr.pdf#page=209>

2.3.1. Problem statement

For every integer $n \geq 3$, let $f(n)$ be the smallest integer with the following property: every set of $f(n)$ points in the plane, with no three of the points collinear, contains n points that are the vertices of a convex n -gon.

Prove or disprove that

$$f(n) = 2^{n-2} + 1$$

for every integer $n \geq 3$.

2.3.2. Historical background, literature, and current progress

2.3.3. Verdict: partially resolved

2.3.3.1. Precise problem

Let $f(n)$ be the least integer such that every set of $f(n)$ points in the plane in general position, meaning no three collinear, contains n points in convex position. The problem asks for the exact value of $f(n)$ for every $n \geq 3$, namely the Erdős-Szekeres happy-ending conjecture

$$f(n) = 2^{n-2} + 1.$$

The lower bound $f(n) \geq 2^{n-2} + 1$ is known, so a full resolution would require proving the matching upper bound $f(n) \leq 2^{n-2} + 1$ for every $n \geq 3$.

2.3.3.2. Status check

The problem is not fully resolved. Exact values are known through $n = 6$, but the first open exact case is $n = 7$, where the conjecture predicts $f(7) = 33$. Recent literature still describes the full conjecture as open. General upper bounds now have the correct exponential order up to a subexponential factor, but they do not imply the exact formula. Several newer papers prove relaxed or restricted versions, but their hypotheses or conclusions do not cover all point sets in the problem statement.

2.3.3.3. Key literature and progress

1. Title: “A combinatorial problem in geometry” Authors: Paul Erdős and George Szekeres Venue/year: *Compositio Mathematica* 2, 463-470, 1935 DOI/arXiv/URL: http://www.numdam.org/item?id=CM_1935__2__463_0 Result/relevance: Original Erdős-Szekeres paper proving finiteness of $f(n)$ and an early upper bound. This is the source of the convex-position problem, but it does not prove the exact formula.
2. Title: “On some extremum problems in elementary geometry” Authors: Paul Erdős and George Szekeres Venue/year: *Annales Universitatis Scientiarum Budapestinensis de Rolando Eötvös Nominatae, Sectio Mathematica* 3-4, 53-62, 1960-1961 DOI/arXiv/URL: no DOI located; standard citation appears in later publisher bibliographies Result/relevance: Gives the classical construction showing $f(n) \geq 2^{n-2} + 1$, the lower bound predicted by the conjecture. This is necessary but not sufficient for the stated equality.
3. Title: “On the application of combinatorial analysis to number theory, geometry and analysis” Authors: Paul Erdős Venue/year: *Actes du Congrès International des Mathématiciens*, Nice 1970, Tome 3, Gauthier-Villars, Paris, 201-210, 1971 DOI/arXiv/URL: https://www.renyi.hu/~p_erdos/1971-13.pdf Result/relevance: Records the problem in the same form: whether there is a smallest $f(n)$, Szekeres’s conjecture $f(n) = 2^{n-2} + 1$, and the then-known exact cases. It is source evidence for the question, not a resolution.
4. Title: “Konvexe Fünfecke in ebenen Punktmengen” Authors: Heiko Harborth Venue/year: *Elemente der Mathematik* 33, 116-118, 1978 DOI/arXiv/URL: DOI 10.5169/seals-32945; <http://eudml.org/doc/141217> Result/relevance: Published proof of the $n = 5$ case, $f(5) = 9 = 2^3 + 1$. This is a special case only.
5. Title: “Computer solution to the 17-point Erdős-Szekeres problem” Authors: George Szekeres and Lindsay Peters Venue/year: *The ANZIAM Journal* 48(2), 151-164, 2006 DOI/arXiv/URL: DOI 10.1017/S144618110000300X; <https://doi.org/10.1017/S144618110000300X> Result/relevance: Computer-assisted proof that every 17-point general-position planar set contains a convex hexagon, so $f(6) = 17 = 2^4 + 1$. This verifies the conjecture through $n = 6$, but not beyond.
6. Title: “Fast Formal Proof of the Erdős-Szekeres Conjecture for Convex Polygons with at Most 6 Points” Authors: Filip Marić Venue/year: *Journal of Automated Reasoning* 62, 301-329, 2019 DOI/arXiv/URL: DOI 10.1007/s10817-017-9423-7; <https://doi.org/10.1007/s10817-017-9423-7> Result/relevance: Formal/SAT-based verification of the conjecture for polygons with at most 6 vertices. This strengthens confidence in the low-dimensional exact cases but still covers only $n \leq 6$.
7. Title: “The Erdős-Szekeres problem on points in convex position - a survey” Authors: Walter Morris and Valeriu Soltan Venue/year: *Bulletin of the American Mathematical Society* 37(4), 437-458, 2000 DOI/arXiv/URL: DOI 10.1090/S0273-0979-00-00877-6; <https://doi.org/10.1090/S0273-0979-00-00877-6> Result/relevance: Survey of known results and variants. Useful background confirming that the main exact problem remained far from solved as of 2000.
8. Title: “Forced Convex n-Gons in the Plane” Authors: Fan R. K. Chung and Ronald L. Graham Venue/year: *Discrete & Computational Geometry* 19, 367-371, 1998 DOI/arXiv/URL: DOI 10.1007/PL00009353; <https://doi.org/10.1007/PL00009353> Result/relevance: First small improvement over the Erdős-Szekeres upper bound, removing the final $+1$ for $n \geq 4$. It gives an upper bound much larger than

$$2^{n-2} + 1.$$

9. Title: “Finding Convex Sets Among Points in the Plane” Authors: Daniel J. Kleitman and Lior Pachter Venue/year: Discrete & Computational Geometry 19, 405-410, 1998 DOI/arXiv/URL: DOI 10.1007/PL00009358; <https://doi.org/10.1007/PL00009358> Result/relevance: Further improves the general upper bound, but still not to the conjectured exact value.
10. Title: “Note on the Erdős-Szekeres Theorem” Authors: Géza Tóth and Pavel Valtr Venue/year: Discrete & Computational Geometry 19, 457-459, 1998 DOI/arXiv/URL: DOI 10.1007/PL00009363; <https://doi.org/10.1007/PL00009363> Result/relevance: Improves the upper bound to $g(n) \leq \binom{2n-5}{n-2} + 2$. This applies to the same point-set problem but remains far above the conjectured exact threshold.
11. Title: “An Improved Upper Bound for the Erdős-Szekeres Conjecture” Authors: Hossein Nassajian Mojarrad and Georgios Vlachos Venue/year: Discrete & Computational Geometry 56, 165-180, 2016 DOI/arXiv/URL: DOI 10.1007/s00454-016-9791-5; arXiv:1510.06255 Result/relevance: Gives a stronger binomial-type upper bound. Its conclusion is still an upper bound larger than $2^{n-2} + 1$, so it does not resolve the exact formula.
12. Title: “On the Erdős-Szekeres convex polygon problem” Authors: Andrew Suk Venue/year: Journal of the American Mathematical Society 30(4), 1047-1053, 2017 DOI/arXiv/URL: DOI 10.1090/jams/869; arXiv:1604.08657 Result/relevance: Proves the asymptotic upper bound $f(n) = 2^{n+o(n)}$, more precisely an upper bound of the form $2^{n+O(n^{2/3} \log n)}$. This is a major near-asymptotic result for the same hypotheses, but it does not imply $f(n) = 2^{n-2} + 1$.
13. Title: “Two extensions of the Erdős-Szekeres problem” Authors: Andreas F. Holmsen, Hossein Nassajian Mojarrad, János Pach, and Gábor Tardos Venue/year: Journal of the European Mathematical Society 22(12), 3981-3995, 2020 DOI/arXiv/URL: DOI 10.4171/JEMS/1000; arXiv:1710.11415 Result/relevance: Improves the general upper-bound error term to $2^{n+O(\sqrt{n \log n})}$ and extends methods to related convexity structures. This still does not give the exact conjectured threshold.
14. Title: “A SAT attack on the Erdős-Szekeres conjecture” Authors: Martin Balko and Pavel Valtr Venue/year: European Journal of Combinatorics 66, 13-23, 2017 DOI/arXiv/URL: DOI 10.1016/j.ejc.2017.06.010 Result/relevance: Refutes a stronger Peters-Szekeres hypergraph-coloring conjecture and develops SAT-based approaches. Since it refutes a stronger auxiliary statement rather than proving the original point-set conjecture, it is partial/computational progress only.
15. Title: “The Erdős-Szekeres Conjecture Revisited” Authors: Jineon Baek and Martin Balko Venue/year: 41st International Symposium on Computational Geometry (SoCG 2025), LIPIcs 332, 13:1-13:15, 2025 DOI/arXiv/URL: DOI 10.4230/LIPIcs.SoCG.2025.13; <https://doi.org/10.4230/LIPIcs.SoCG.2025.13> Result/relevance: The abstract explicitly describes the Erdős-Szekeres conjecture as “still open.” The paper proves a tight relaxed version for split k -gons and proves the conjecture for decomposable sets, but split k -gons are weaker than k points in convex position and decomposability is an extra hypothesis. Therefore these results do not cover all point sets in the problem statement.
16. Title: “Notes on the 33-point Erdős-Szekeres problem” Authors: Dumitru Bogan Venue/year: preprint, December 2025 DOI/arXiv/URL: arXiv:2512.24061; <https://arxiv.org/abs/2512.24061> Result/relevance: Treats $ES(7)$, where the conjecture predicts $ES(7) = 33$, as the first open case. It gives a SAT encoding

and UNSAT certificates for some anchored subfamilies, but not a complete proof that every 33-point general-position set contains a convex heptagon.

2.3.3.4. What remains unresolved

The exact equality $f(n) = 2^{n-2} + 1$ remains unproved for all $n \geq 7$. In particular, the concrete first open case $f(7) = 33$ remains unresolved in citable literature found in this search. The available results either prove only low- n cases, provide weaker general upper bounds, or assume extra structure/replace convex k -gons by relaxed objects.

2.3.4. Remaining open problems recorded by the database

2.3.4.1. Open problem 1

Does every set of 33 points in the plane in general position contain seven points in convex position; equivalently, is $f(7) = 33$?

Reference supplied by the database. Bogdan Dumitru, “Notes on the 33-Point Erdős–Szekeres Problem,” arXiv:2512.24061 [math.CO], 2025, first open case stated in the abstract and §1, <https://doi.org/10.48550/arXiv.2512.24061>.

2.3.5. Primary-source pointer

The originating ICM record is *On the application of combinatorial analysis to number theory, geometry and analysis* (1970). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1970.3/ICM1970.3.ocr.pdf#page=209>.

2.4. 82. Recent results in convexity [1978-geom-238]

Record: 1978-geom-238

Subfield: Discrete and polyhedral geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1978: Recent results in convexity

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1978.1/ICM1978.1.ocr.pdf#page=429>

2.4.1. Problem statement

Let n be a positive integer, and let $\mathcal{A}$ be a family of subsets of Euclidean space $\mathbb{R}^n$ such that $\bigcup_{A \in \mathcal{A}} A = \mathbb{R}^n$ and

$$|A| < \frac{1}{3} \left(\frac{4}{3} \right)^{4n/3}.$$

Prove that there exists a set $A \in \mathcal{A}$ such that, for every real number $d > 0$, there are points $x, y \in A$ with $\|x - y\|_2 = d$.

2.4.2. Historical background, literature, and current progress

The statement is **unresolved**.

Let $\chi_p(\mathbb{R}^n)$ be the least number of colors in a coloring of $\mathbb{R}^n$ such that every color class omits at least one positive distance. Refining any cover to a partition shows that the statement is equivalent to

$$\chi_p(\mathbb{R}^n) \geq \left\lceil \frac{1}{3} \left(\frac{4}{3} \right)^{4n/3} \right\rceil \quad \text{for every } n.$$

No result located proves this bound or supplies a counterexample. Its exponential base is $(4/3)^{4/3} = 1.467523\dots$, whereas the strongest currently recorded asymptotic lower bound is only

$$\chi_p(\mathbb{R}^n) \geq (1.239\dots + o(1))^n.$$

The known upper bound $\chi_p(\mathbb{R}^n) \leq (3 + o(1))^n$ also does not disprove the requested estimate. Thus $1.467523\dots$ lies inside the present exponential gap.

There is an important source discrepancy. Larman's cited 1978 paper actually prints the weaker threshold $\frac{1}{3}(\frac{4}{3})^{3n/4}$, not the exponent $4n/3$ appearing in the supplied ICM statement. The verdict here concerns the requested $4n/3$ statement exactly; it has not been silently replaced by Larman's printed conjecture.

Key references and progress:

- D. G. Larman and C. A. Rogers, "The realization of distances within sets in Euclidean space," *Mathematika* **19** (1972), 1–24, DOI <https://doi.org/10.1112/S0025579300004903>. They prove the critical-configuration principle: a finite M -point distance graph with independence number D forces one member of every cover by fewer

than M/D sets to realize all distances. In particular, they obtain the polynomial threshold $n(n-1)/6$ and an asymptotic upper bound $(3+o(1))^n$ for $\chi_p(\mathbb{R}^n)$.

- D. G. Larman, “A note on the realization of distances within sets in Euclidean space,” *Commentarii Mathematici Helvetici* **53** (1978), 529–535, DOI <https://doi.org/10.1007/BF02566096>. Theorem 3 gives the polynomial threshold $(n-1)(n-2)(n-3)/178200$; the paper then proposes the distinct conjectural threshold $\frac{1}{3}(\frac{4}{3})^{3n/4}$.
- Peter Frankl and Richard M. Wilson, “Intersection theorems with geometric consequences,” *Combinatorica* **1** (1981), 357–368, DOI <https://doi.org/10.1007/BF02579457>. Their intersection theorem yields the first exponential lower bound, $\chi_p(\mathbb{R}^n) \geq (1+o(1))1.2^n$, through the Larman–Rogers critical-configuration principle.
- Andrei M. Raigorodskii, “On the chromatic number of a space,” *Russian Mathematical Surveys* **55** (2000), 351–352, DOI <https://doi.org/10.1070/RM2000v055n02ABEH000281>. Its ternary distance-graph construction improves the relevant finite-configuration ratio to the base 1.239 ...
- Alexander Soifer, *The New Mathematical Coloring Book: Mathematics of Coloring and the Colorful Life of Its Creators*, 2nd ed., Springer, 2024, DOI <https://doi.org/10.1007/978-1-0716-3597-1>. Section 10 records explicitly that $(1.239... + o(1))^n \leq \chi_p(\mathbb{R}^n) \leq (3+o(1))^n$ and identifies the gap as open. Neither endpoint reaches or excludes the exact bound in the statement.

Consequently, the established progress is exponential but remains quantitatively short of the requested lower bound, and no complete resolution of all dimensions is known.

2.4.3. Remaining open problems recorded by the database

2.4.3.1. Open problem 1

Let n be a positive integer, and let $\mathcal{A}$ be a family of subsets of Euclidean space $\mathbb{R}^n$ such that $\bigcup_{A \in \mathcal{A}} A = \mathbb{R}^n$ and

$$|A| < \frac{1}{3} \left(\frac{4}{3} \right)^{3n/4}.$$

Prove that there exists a set $A \in \mathcal{A}$ such that, for every real number $d > 0$, there are points $x, y \in A$ with $\|x - y\|_2 = d$.

Reference supplied by the database. D. G. Larman, “A note on the realization of distances within sets in Euclidean space,” *Commentarii Mathematici Helvetici* **53** (1978), 529–535, Conjecture on p. 529, <https://doi.org/10.1007/BF02566096>.

2.4.3.2. Open problem 2

Prove or disprove that, whenever four subsets A_1, A_2, A_3, A_4 cover the Euclidean plane $\mathbb{R}^2$, at least one of them realizes every positive distance; that is, there exists $i \in \{1, 2, 3, 4\}$ such that, for every real number $d > 0$, there are points $x, y \in A_i$ with $\|x - y\|_2 = d$.

Reference supplied by the database. D. G. Larman, “Recent results in convexity,” *Proceedings of the International Congress of Mathematicians, Helsinki 1978*, Vol. 1, Academia Scientiarum Fennica, Helsinki, 1980, pp. 429–434, Section 6, <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1978.1/ICM1978.1.ocr.pdf>.

2.4.4. Primary-source pointer

The originating ICM record is *Recent results in convexity* (1978). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1978.1/ICM1978.1.ocr.pdf#page=429>.

2.5. 83. Conjectures and open questions in rigidity [1978-geom-302]

Record: 1978-geom-302

Subfield: Discrete and polyhedral geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1978: Conjectures and open questions in rigidity

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1978.1/ICM1978.1.ocr.pdf#page=407>

2.5.1. Problem statement

Determine whether, for every integer $n \geq 3$, every finite triangulation of a closed n -dimensional manifold that is piecewise-linearly embedded in Euclidean space $\mathbb{R}^{n+1}$ is rigid: must every continuous motion of its vertices that preserves the length of every edge preserve all pairwise distances between the vertices (equivalently, be induced by a continuous family of Euclidean rigid motions of $\mathbb{R}^{n+1}$)?

2.5.2. Historical background, literature, and current progress

The problem is **unresolved**. In the standard terminology of higher-dimensional flexible polyhedra, the statement asks whether an embedded flexible polyhedron can exist in Euclidean dimension $d = n + 1 \geq 4$. No such example is currently known, but there is also no theorem proving rigidity for every embedded triangulated closed hypersurface.

The strongest constructive progress drops embeddedness. Hellmuth Stachel constructed flexible realizations of the boundary complex of the four-dimensional cross-polytope in $\mathbb{R}^4$, but these realizations are self-intersecting. Alexander A. Gaifullin subsequently constructed and classified flexible cross-polytopes in Euclidean spaces of every dimension; his Euclidean examples are likewise self-intersecting. Thus these results satisfy the edge-length-preserving and closed-manifold parts of the question but not the piecewise-linear embedding hypothesis.

There is also strong generic evidence for rigidity. Fogelsanger's rigidity theorem, restated with a self-contained proof in the 2024 paper by Cruickshank, Jackson, and Tanigawa, implies that the graph of every connected triangulated $(d - 1)$ -manifold is generically rigid in $\mathbb{R}^d$. The same paper characterizes when these graphs are generically globally rigid. These are generic-realization results and therefore do not exclude exceptional embedded flexible realizations, which are exactly what the problem asks about.

Gaifullin explicitly posed the existence of embedded flexible polyhedra in $\mathbb{R}^d$, $d \geq 4$, as Problem 3.5 in his 2018 survey. Robert Miranda's 2025 dissertation still records the existence of embeddable flexible polyhedra in dimensions at least four as unknown. Consequently, neither a counterexample nor a proof covers even the first required ambient dimension $\mathbb{R}^4$, and the full quantified statement remains open.

Key references:

- Robert Connelly, "Conjectures and Open Questions in Rigidity," *Proceedings of the Inter-*

national Congress of Mathematicians, Helsinki 1978, Vol. 1, Academia Scientiarum Fennica, 1980, pp. 407-414. This is the source of the higher-dimensional rigidity question (Question III.4.C). [Direct PDF](#)

- Hellmuth Stachel, “Flexible Cross-Polytopes in the Euclidean 4-Space,” *Journal for Geometry and Graphics* 4(2) (2000), 159-167. It constructs continuous flexes in $\mathbb{R}^4$, but the cross-polytopes are not embedded. [Direct PDF](#)
- Alexander A. Gaifullin, “Flexible Cross-Polytopes in Spaces of Constant Curvature,” *Proceedings of the Steklov Institute of Mathematics* 286 (2014), 77-113. DOI: <https://doi.org/10.1134/S0081543814060066>; arXiv: [1312.7608](https://arxiv.org/abs/1312.7608). It constructs self-intersecting flexible cross-polytopes in Euclidean spaces of all dimensions and classifies all flexible cross-polytopes.
- Alexander A. Gaifullin, “Flexible Polyhedra and Their Volumes,” in *European Congress of Mathematics, Berlin 2016*, EMS, 2018, pp. 63-83. DOI: <https://doi.org/10.4171/176-1/2>; arXiv: [1605.09316](https://arxiv.org/abs/1605.09316). Conjecture 3.3 addresses self-intersection of Euclidean flexible cross-polytopes, and Problem 3.5 explicitly asks whether embedded flexible polyhedra exist in $\mathbb{R}^d$ for $d \geq 4$.
- James Cruickshank, Bill Jackson, and Shin-ichi Tanigawa, “Global Rigidity of Triangulated Manifolds,” *Advances in Mathematics* 458 (2024), 109953. DOI: <https://doi.org/10.1016/j.aim.2024.109953>; arXiv: [2204.02503](https://arxiv.org/abs/2204.02503). It proves generic global rigidity criteria for graphs of triangulated manifolds and gives a proof of the generic rigidity theorem used above.
- Robert Miranda, *Rigidity and Flexibility of Polyhedral Structures*, Ph.D. dissertation, University of California, Los Angeles, 2025. It states that embeddable flexible polyhedra are still not known in dimensions at least four. [Direct PDF](#)

2.5.3. Remaining open problems recorded by the database

2.5.3.1. Open problem 1

Prove or disprove that, for every integer $d \geq 4$, every flexible realization in $\mathbb{R}^d$ of the boundary complex C_d of the d -dimensional cross-polytope is self-intersecting. Here C_d is the simplicial $(d-1)$ -sphere with vertices $v_1^+, v_1^-, \dots, v_d^+, v_d^-$, whose facets are the d -element sets containing exactly one of v_i^+ and v_i^- for each i ; a realization is flexible if its vertices admit a nontrivial continuous motion preserving every edge length; and it is self-intersecting if its simplexwise-affine realization map $C_d \rightarrow \mathbb{R}^d$ is not injective.

Reference supplied by the database. Alexander A. Gaifullin, “Flexible Polyhedra and Their Volumes,” in *European Congress of Mathematics, Berlin, July 18-22, 2016*, European Mathematical Society, 2018, pp. 63-83, Conjecture 3.3, DOI: <https://doi.org/10.4171/176-1/2>, arXiv:1605.09316 (<https://arxiv.org/abs/1605.09316>).

2.5.4. Primary-source pointer

The originating ICM record is *Conjectures and open questions in rigidity* (1978). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1978.1/ICM1978.1.ocr.pdf#page=407>.

2.6. 84. V. V. , Discrete reflection groups in Lobachevsky spaces and algebraic surfaces [1986-geom-274]

Record: 1986-geom-274

Subfield: Discrete and polyhedral geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1986: V. V. , Discrete reflection groups in Lobachevsky spaces and algebraic surfaces

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1986.1/ICM1986.1.ocr.pdf#page=756>

2.6.1. Problem statement

Let $\mathbb{H}^n$ be real hyperbolic n -space of curvature -1 , and let $\overline{\mathbb{H}^n}$ denote its projective compactification. A generalized crystallographic hyperbolic reflection group (GenCHRG) is a discrete subgroup $G \leq \text{Isom}(\mathbb{H}^n)$ of finite covolume for which there is a normal subgroup $W \triangleleft G$, generated by reflections in hyperplanes, with the following property: if $\mathcal{M}$ is a fundamental chamber for W , then $A = G/W$ acts on $\mathcal{M}$ by symmetries and has a finite-index subgroup preserving a proper totally geodesic subspace of $\mathbb{H}^n$, or a proper projective subspace of $\overline{\mathbb{H}^n}$. An ordinary crystallographic hyperbolic reflection group (CHRG) is the special case in which A is finite; equivalently, the reflection chamber $\mathcal{M}$ has finite hyperbolic volume.

Formulate precise GenCHRG analogues of all the results below and prove them. Each analogue must apply to the reflection chamber $\mathcal{M}$ and the symmetry action of A , and must specialize to the stated CHRG result when A is finite.

1. **Narrow-facet estimates.** For a closed convex polyhedron $Q \subset \mathbb{H}^n$, write $H(F)$ for the supporting hyperplane of a facet F , and write $d(H_1, H_2)$ for the hyperbolic distance between two hyperplanes, with distance 0 when they meet. Prove an analogue of the assertion that Q has a facet N such that

$$d(H(F), H(F')) < \text{arcosh}(7)$$

for every pair of facets F, F' whose supporting hyperplanes both meet $H(N)$. Also prove an analogue of the assertion that Q has $n + 1$ facets $F_1, \dots, F_{n+1}$ such that their supporting hyperplanes are pairwise at distance less than $\text{arcosh}(31)$, have no common point and admit no hyperplane orthogonal to all of them, and cannot be partitioned into two nonempty families such that every hyperplane in one family is orthogonal to every hyperplane in the other.

2. **Combinatorial estimate.** Call an n -dimensional convex polyhedron Q almost simple if a neighborhood of every vertex is a cone over a simple polyhedron. For $0 \leq l \leq k \leq n$, let $\alpha_k^l(Q)$ be the average, over all k -dimensional faces of Q , of the number of their l -dimensional faces. Prove an analogue of the estimate

$$\alpha_k^l(Q) < \binom{n-l}{n-k} \frac{\binom{\lfloor n/2 \rfloor}{l} + \binom{n-\lfloor n/2 \rfloor}{l}}{\binom{\lfloor n/2 \rfloor}{k} + \binom{n-\lfloor n/2 \rfloor}{k}} \quad \left(l \leq k \leq \frac{n+1}{2} \right).$$

3. **Arithmetic finiteness and dimension bounds.** Let K be a totally real number field of degree $N = [K : \mathbb{Q}]$, let O_K be its ring of integers, and let S be a projective O_K -module of rank $n + 1$ equipped with a nondegenerate symmetric O_K -valued bilinear form. Assume that the form has signature $(1, n)$ at one real embedding of K and is definite at every other real embedding. Let $O^+(S)$ be the subgroup of its isometry group preserving one component of the positive cone at the distinguished embedding, and let $W(S) \triangleleft O^+(S)$ be the subgroup generated by all hyperplane reflections belonging to $O^+(S)$. Call S generalized reflective if $O^+(S)$, with reflection subgroup $W(S)$, is a GenCHRG; two such lattices are similar if, after multiplying one form by an element of $K^\times$, they become isometric as O_K -lattices. Establish GenCHRG analogues of all the following conclusions:
- for each fixed $n \geq 2$ and N , only finitely many ground fields K and similarity classes of generalized reflective lattices occur;
 - for all sufficiently large N , one has $n \leq 9$;
 - a CHRG with bounded fundamental chamber has $n < 30$, a CHRG with an unbounded finite-volume fundamental chamber has $n \leq 995$, and an arithmetic CHRG has $n < 30$.
4. **Local estimates at a narrow facet.** Let $\mathcal{M}$ be a reflection chamber, let N be a narrow facet as in part 1, and let V be a vertex of N , possibly in $\partial\mathbb{H}^n$. For every edge $r_i \subset N$ issuing from V , let $V_i \neq V$ be its other endpoint and let N_i be the facet of $\mathcal{M}$ satisfying $r_i \cap N_i = \{V_i\}$. The Coxeter diagram at V is the diagram whose vertices are the facets of $\mathcal{M}$ containing V , with edges determined by their dihedral angles. Prove GenCHRG counterparts of the following assertions, with explicit functions f, g, h tending to infinity with their integer arguments:
- if V is ideal, then two of the hyperplanes $H(N_i)$ are at distance greater than $f(n)$;
 - if V is finite, then two of them are at distance greater than $g(\ell)$, where ℓ is the maximum number of vertices in a connected component of the Coxeter diagram at V ;
 - by iterating the edge-and-opposite-facet construction from the successive endpoints, one obtains two facets whose supporting hyperplanes are at distance greater than $h(t)$, where t is the number of connected components of the Coxeter diagram at V .

The local estimates must give dimension bounds and enough local information near N for classification. In the finite-quotient case they must in particular recover $n < 140$ when V is an ideal vertex with affine Coxeter diagram $\tilde{A}_{n-1}$, and $n < 300$ when the narrow facet N has no ideal vertices.

2.6.2. Historical background, literature, and current progress

unresolved

No citable source located gives GenCHRG counterparts of all four requested packages. The closest direct work proves substantial portions of the generalized program, but it neither states nor proves the exact narrow-facet and local estimates requested here, and its strongest dimension bound for the hyperbolic-type case still depends on the dimension of the invariant subspace.

V. V. Nikulin, “Basis of the Diagram Method for Generalized Reflection Groups in Lobachevsky Spaces and Algebraic Surfaces with Nef Anticanonical Class,” *International Journal of Mathematics* 7(1) (1996), 71–108, DOI <https://doi.org/10.1142/S0129167X96000062>, arXiv [alg-geom/9405011](https://arxiv.org/abs/alg-geom/9405011). This is the principal direct sequel. It develops a diagram method for locally finite, possibly infinite-faced acute hyperbolic polyhedra and divides GenCHRGs into elliptic, parabolic, and hyperbolic types. Theorem 2.3 and Remark 2.5 give the bounds $\dim L < 996$ in the elliptic case, $\dim L < 997$ in the parabolic case, and $\dim L < 996 + \dim T$ in the hyperbolic case relative to an invariant subspace T . Remark 2.4 explicitly says that an absolute bound independent of $\dim T$ is unknown. These results provide generalized combinatorial-diagram machinery and dimension bounds, but not the two stated metric narrow-facet estimates with constants 7 and 31, not the displayed average-face inequality in the required GenCHRG setting, and not explicit functions f, g, h yielding the requested local classification information and the bounds 140 and 300.

V. V. Nikulin, “Reflection Groups in Lobachevskii Spaces and the Denominator Identity for Lorentzian Kac–Moody Algebras,” *Izvestiya: Mathematics* 60(2) (1996), 305–334, DOI <https://doi.org/10.1070/IM1996v060n02ABEH000072>, arXiv [alg-geom/9503003](https://arxiv.org/abs/alg-geom/9503003). Theorem 1.1.3 proves finiteness, for fixed rank, of parabolic reflective hyperbolic lattices up to similarity and isomorphism. Theorem 1.1.3' supplies a bounded connected Gram configuration for restricted parabolic polyhedra, with normalized Gram entries at most 62, and the paper notes the elementary bound $\text{rk } S \leq 43$ while conjecturing the sharp bound 26. This is a strong analogue for the parabolic subclass, not a treatment of every GenCHRG type or all four requested conclusions.

V. V. Nikulin, “K3 Surfaces with Interesting Groups of Automorphisms,” *Journal of Mathematical Sciences* 95(1) (1999), 2028–2048, DOI <https://doi.org/10.1007/BF02169159>, arXiv [alg-geom/9701011](https://arxiv.org/abs/alg-geom/9701011). Theorem 2 proves that, for fixed field degree, only finitely many totally real ground fields support reflective hyperbolic lattices of rank at least three, and that for a fixed rank and fixed field only finitely many such lattices occur up to scaling. Combining its two clauses proves the fixed- n , fixed- N finiteness conclusion in part 3 for the elliptic, parabolic, and hyperbolic types. The paper does not prove the remaining high-degree $n \leq 9$ GenCHRG conclusion or the full geometric and local estimates in the statement.

Mikhail Belolipetsky and John McLeod, “Reflective and Quasi-Reflective Bianchi Groups,” *Transformation Groups* 18(4) (2013), 971–994, DOI <https://doi.org/10.1007/s00031-013-9245-6>, arXiv [1210.2759](https://arxiv.org/abs/1210.2759). The paper completely classifies the reflective and quasi-reflective Bianchi groups and their extensions. In the terminology relevant here, quasi-reflective groups form a low-dimensional parabolic special case, so this is classification progress in dimension three only.

Mikhail Belolipetsky, “Arithmetic Hyperbolic Reflection Groups,” *Bulletin of the American Mathematical Society* 53(3) (2016), 437–475, DOI <https://doi.org/10.1090/bull/1530>, arXiv [1506.03111](https://arxiv.org/abs/1506.03111). Section 9 surveys reflective quotients and records that quasi-reflective groups are known only in bounded dimension, that the Conway example occurs in dimension 25, and that the conjectured sharp upper dimension 25 was still unproved. It also identifies the dimension-three classification above as the only available quantitative spectral result for quasi-reflective groups.

For comparison, Ian Agol, Mikhail Belolipetsky, Peter Storm, and Kevin Whyte, “Finiteness of Arithmetic Hyperbolic Reflection Groups,” *Groups, Geometry, and Dynamics* 2(4) (2008), 481–498, DOI <https://doi.org/10.4171/GGD/47>, arXiv [math/0612132](https://arxiv.org/abs/math/0612132), proves finiteness of conjugacy classes of maximal ordinary arithmetic hyperbolic reflection groups. Its finite-reflective-quotient hypothesis does not cover the generalized groups in this problem.

Thus the established literature covers the generalized diagram framework, dimension estimates depending on GenCHRG type, fixed-rank/field-degree arithmetic finiteness, and selected parabolic classifications. What remains unresolved is the exact combined problem: GenCHRG versions of both narrow-facet assertions, the stated combinatorial average-face bound, all the requested uniform arithmetic and dimension conclusions, and the explicit iterated local estimates with their specified finite-quotient consequences.

2.6.3. Remaining open problems recorded by the database

2.6.3.1. Open problem 1

Determine whether there exists an absolute constant C with the following property. Let $G \leq \text{Isom}(\mathbb{H}^n)$ be a discrete group of finite covolume, let $W \triangleleft G$ be the normal subgroup generated by the hyperplane reflections in G , and let $\mathcal{M}$ be a fundamental chamber for W . Suppose that a finite-index subgroup of G/W , acting by symmetries of $\mathcal{M}$, preserves a proper totally geodesic subspace $T \subset \mathbb{H}^n$ with $0 < \dim T < n - 1$. Must $n < C$, independently of G , n , and $\dim T$?

Reference supplied by the database. V. V. Nikulin, “Basis of the Diagram Method for Generalized Reflection Groups in Lobachevsky Spaces and Algebraic Surfaces with Nef Anticanonical Class,” *International Journal of Mathematics* 7(1) (1996), 71–108, Remark 2.4. DOI: <https://doi.org/10.1142/S0129167X96000062>; arXiv: alg-geom/9405011.

2.6.3.2. Open problem 2

Prove that if $\Gamma \leq \text{Isom}(\mathbb{H}^n)$ is an arithmetic lattice and $R \triangleleft \Gamma$ is the subgroup generated by all hyperplane reflections in Γ , then Γ/R cannot be an infinite affine crystallographic group when $n > 25$. Equivalently, prove that no arithmetic quasi-reflective hyperbolic group exists in dimension greater than 25.

Reference supplied by the database. Mikhail Belolipetsky, “Arithmetic Hyperbolic Reflection Groups,” *Bulletin of the American Mathematical Society* 53(3) (2016), 437–475, Section 9. DOI: <https://doi.org/10.1090/bull/1530>; arXiv: 1506.03111.

2.6.4. Primary-source pointer

The originating ICM record is V. V. , *Discrete reflection groups in Lobachevsky spaces and algebraic surfaces* (1986). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1986.1/ICM1986.1.ocr.pdf#page=756>.

2.7. 85. Combinatorics and convexity [1994-geom-52]

Record: 1994-geom-52

Subfield: Discrete and polyhedral geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1994: Combinatorics and convexity

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.2/ICM1994.2.ocr.pdf#page=659>

2.7.1. Problem statement

Prove or disprove that for every centrally symmetric convex d -dimensional polytope $P \subset \mathbb{R}^d$ with center at the origin (that is, $x \in P$ implies $-x \in P$), every combinatorial automorphism ϕ of P (equivalently, every automorphism of the face lattice of P , acting in particular on the vertices of P) has the following property: for every vertex v of P , $\phi(-v) = -\phi(v)$.

2.7.2. Historical background, literature, and current progress

2.7.3. Verdict: partially resolved / unresolved

2.7.3.1. Precise problem

The problem asks whether the antipodal vertex-pairing of an origin-centrally symmetric convex polytope is determined by its face lattice in the following strong sense: if $P = -P$ and ϕ is any combinatorial automorphism of P , must $\phi(-v) = -\phi(v)$ for every vertex v ? Equivalently, does the central symmetry $v \mapsto -v$ commute with the full combinatorial automorphism group of P ?

2.7.3.2. Status

I did not find a published or otherwise citable result that proves or disproves this statement for all centrally symmetric convex polytopes. The exact face-lattice automorphism question remains unresolved. The literature contains relevant partial and adjacent progress, but the hypotheses/conclusions do not cover the full statement.

2.7.3.3. Relevant literature and progress

1. Combinatorics and Convexity

Author: Gil Kalai

Venue/year: Proceedings of the International Congress of Mathematicians, Vol. 1, 2 (Zurich, 1994), Birkhauser, Basel, 1995, pp. 1363-1374

DOI/URL: https://doi.org/10.1007/978-3-0348-9078-6_131 ;
author PDF: <https://www.ma.huji.ac.il/~kalai/co.pdf>

Relevance: Kalai records Perles' question in essentially the stated form: whether every combinatorial automorphism ϕ of a centrally symmetric d -polytope satisfies $\phi(-v) = -\phi(v)$. This is the original/open-problem source, not a resolution.

2. **Is combinatorial automorphism of symmetric convex polytope always antipodal?**

Author: Fedor Petrov, with comments by Frieder Ladisch and Richard Stanley

Venue/year: MathOverflow, asked 2015, modified 2016

DOI/arXiv/URL: <https://mathoverflow.net/questions/223642/is-combinatorial-automorphism-of-symmetric-convex-polytope-always-antipodal>

Relevance: This is a direct restatement of the same problem. The comments identify the equivalent formulation that the central symmetry should centralize all combinatorial automorphisms and point back to Kalai's ICM article. No answer resolving the problem is given.

3. **On combinatorial and affine automorphisms of polytopes**

Authors: Jurgen Bokowski, Gunter Ewald, Peter Kleinschmidt

Venue/year: Israel Journal of Mathematics 47 (1984), 123-130

DOI/URL: <https://doi.org/10.1007/BF02760511>

Relevance: The paper disproves the older conjecture that every combinatorial automorphism of a convex polytope is induced by an affine transformation. If a combinatorial automorphism of a centrally symmetric polytope is affine/geometric, then it preserves the center and hence the antipodal pairing; this paper shows that this affine-realizability route cannot settle Perles' question in general. It does not itself provide a centrally symmetric counterexample to the problem statement.

4. **Prescribing symmetries and automorphisms for polytopes**

Authors: Egon Schulte, Pablo Soberon, Gordon Ian Williams

Venue/year: In *Polytopes and Discrete Geometry*, Contemporary Mathematics 764, American Mathematical Society, Providence, RI, 2021, pp. 221-233

DOI/arXiv/URL: <https://doi.org/10.1090/conm/764/15359> ; arXiv:1902.05439

Relevance: Theorem 4.1 constructs, for every finite abelian group Γ of even order and involution σ , a centrally symmetric convex polytope with $G(P) = \Gamma(P) = \Gamma$ and σ realized as central symmetry. These are broad positive examples where the combinatorial automorphism group equals the geometric symmetry group, so central symmetry is central. This is only an existence/extra-hypothesis result and does not prove the property for arbitrary centrally symmetric polytopes.

5. **Rigidity, Tensegrity, and Reconstruction of Polytopes Under Metric Constraints**

Author: Martin Winter

Venue/year: International Mathematics Research Notices 2024, no. 9, 7721-7747; published online 2023

DOI/arXiv/URL: <https://doi.org/10.1093/imrn/rnad298> ; arXiv:2302.14194

Relevance: Winter proves several reconstruction/rigidity results and notes a subtle obstruction for centrally symmetric polytopes: David E. Speyer constructed a 12-vertex 4-polytope whose edge graph has an automorphism that does not preserve antipodality. This is strong negative evidence for graph-level variants, but it is explicitly an edge-graph automorphism statement, not a face-lattice automorphism statement. Therefore it does not disprove the problem.

6. **Does the edge-graph of a centrally symmetric polytope determine which vertices are antipodal?**

Author: David E. Speyer, answer to a MathOverflow question by Martin Winter

Venue/year: MathOverflow, 2023

DOI/arXiv/URL: <https://mathoverflow.net/questions/440534/does-the-edge-graph-of-a-centrally-symmetric-polytope-determine-which-vertices-a>

Relevance: Speyer gives the underlying 12-vertex 4-dimensional centrally symmetric polytope with an edge-graph symmetry interchanging non-antipodal vertices. This resolves a stronger-looking graph-only variant negatively, but it does not verify that the symmetry preserves all faces of the polytope. Thus it is not a counterexample to Perles' face-lattice automorphism question.

2.7.3.4. What remains unresolved

No located source proves that every face-lattice automorphism of every centrally symmetric convex polytope preserves antipodal vertex pairs, and no located source gives a centrally symmetric convex polytope with a genuine face-lattice automorphism violating antipodality. The full all-dimensions, all-polytopes statement in the problem remains open.

2.7.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

2.7.5. Primary-source pointer

The originating ICM record is *Combinatorics and convexity* (1994). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.2/ICM1994.2.ocr.pdf#page=659>.

2.8. 86. Combinatorics and convexity [1994-geom-53]

Record: 1994-geom-53

Subfield: Discrete and polyhedral geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1994: Combinatorics and convexity

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.2/ICM1994.2.ocr.pdf#page=659>

2.8.1. Problem statement

For integers $d \geq 1$ and $n \geq d + 1$, let $C_d(n)$ denote the cyclic d -polytope with n vertices, i.e. the convex hull in $\mathbb{R}^d$ of n points on the moment curve $t \mapsto (t, t^2, \dots, t^d)$. Prove or disprove that, if K is any $(d-1)$ -dimensional simplicial complex with n vertices such that for every face $F \in K$ with r vertices the link $\text{lk}_K(F)$ has the same Euler characteristic as the sphere S^{d-r-1} , then for every k with $0 \leq k \leq d-1$, the number $f_k(K)$ of k -dimensional faces of K is at most $f_k(C_d(n))$, the number of k -dimensional faces of $C_d(n)$.

2.8.2. Historical background, literature, and current progress

2.8.3. Verdict: partially resolved

2.8.3.1. Precise problem

The problem asks for Klee's Eulerian-complex extension of Motzkin's Upper Bound Conjecture: if K is an Eulerian $(d-1)$ -dimensional simplicial complex on n vertices, meaning every face link has the Euler characteristic of the corresponding sphere, must

$$f_k(K) \leq f_k(C_d(n)) \quad 0 \leq k \leq d-1$$

hold for all d, n, k ? The $k=0$ case is automatic because both sides have n vertices, so the real content is the componentwise upper bound for all higher face numbers.

2.8.3.2. Verdict

Partially resolved. The exact all-cases statement is not fully proved in the literature I found. Current survey literature states the conjecture in essentially this form and says that, in this generality, it remains wide open. However, many important subclasses are known.

2.8.3.3. Key literature and what it proves

1. Theodore S. Motzkin, "Comonotone curves and polyhedra, Abstract III", Bulletin of the American Mathematical Society 63 (1957), 35. DOI not found; cited in later literature, e.g. Cambridge Core references.

This is the original Motzkin upper-bound conjecture source for convex polytopes. It concerns polytopes, not arbitrary Eulerian complexes.

2. Victor Klee, “A combinatorial analogue of Poincaré’s duality theorem”, *Canadian Journal of Mathematics* 16 (1964), 517-531. DOI: 10.4153/CJM-1964-053-0. URL: <https://www.cambridge.org/core/journals/canadian-journal-of-mathematics/article/combinatorial-analogue-of-poincares-duality-theorem/0F30C9676FAA0E82E9266BC2D97E8805>
Klee introduced the Eulerian-complex framework and Dehn-Sommerville relations for this setting. This supplies structural background but not the full upper-bound theorem.
3. Victor Klee, “On the Number of Vertices of a Convex Polytope”, *Canadian Journal of Mathematics* 16 (1964), 701-720. DOI: 10.4153/CJM-1964-067-6. URL: <https://www.cambridge.org/core/journals/canadian-journal-of-mathematics/article/on-the-number-of-vertices-of-a-convex-polytope/62AF79696185093199301D13A02C5D4F>
Later surveys report that Klee proved the upper-bound assertion for Eulerian complexes with sufficiently many vertices and conjectured it for all Eulerian complexes. This is partial because the problem asks for every $n \geq d + 1$.
4. Peter McMullen, “The maximum numbers of faces of a convex polytope”, *Mathematika* 17(2) (1970), 179-184. DOI: 10.1112/S0025579300002850. URL: <https://www.cambridge.org/core/journals/mathematika/article/maximum-numbers-of-faces-of-a-convex-polytope/13483049ECEDF0F2D760C50BAE8D6F21>
Proves Motzkin’s upper-bound conjecture for convex polytopes. This covers boundary complexes of polytopes but not arbitrary Eulerian complexes.
5. Richard P. Stanley, “The Upper Bound Conjecture and Cohen-Macaulay Rings”, *Studies in Applied Mathematics* 54(2) (1975), 135-142. DOI: 10.1002/sapm1975542135. URL: <https://onlinelibrary.wiley.com/doi/10.1002/sapm1975542135>
Proves the upper-bound theorem via Cohen-Macaulay methods, in particular for simplicial spheres/homology spheres. This does not cover Eulerian complexes that are not Cohen-Macaulay.
6. Isabella Novik, “Upper bound theorems for homology manifolds”, *Israel Journal of Mathematics* 108 (1998), 45-82. DOI: 10.1007/BF02783042. URL: <https://link.springer.com/article/10.1007/BF02783042>
Proves the UBC for all odd-dimensional homology manifolds and for even-dimensional homology manifolds satisfying a Betti-number condition; the known consequences include Eulerian homology manifolds. This is still stronger local topology than an arbitrary Eulerian complex.
7. Patricia Hersh and Isabella Novik, “A short simplicial h -vector and the upper bound theorem”, *Discrete & Computational Geometry* 28(3) (2002), 283-289. DOI: 10.1007/s00454-002-0746-7. arXiv: math/0111302. URL: <https://arxiv.org/abs/math/0111302>
Theorem 1 proves the UBC for a class of odd-dimensional simplicial complexes, including odd-dimensional Eulerian complexes with isolated singularities. This leaves even-dimensional cases and more general singularities open.
8. Isabella Novik and Ed Swartz, “Face numbers of pseudomanifolds with isolated singularities”, *Mathematica Scandinavica* 110(2) (2012), 198-222. DOI: 10.7146/math.scand.a-15204. arXiv: 1004.5100. URL: <https://arxiv.org/abs/1004.5100>
Theorem 4.10 proves the h -number form of the UBC for Eulerian pseudomanifolds with

homologically isolated singularities, assuming $n \geq 3d - 4$. This is partial because it adds both a singularity condition and a vertex lower bound.

9. Steven Klee and Isabella Novik, “Face enumeration on simplicial complexes”, in *Recent Trends in Combinatorics*, IMA Volumes in Mathematics and its Applications 159, Springer, 2016, pp. 653-686. arXiv: 1505.06380. URL: <https://arxiv.org/abs/1505.06380>

Section 4.2 summarizes the state of the UBC for manifolds and Eulerian complexes: Klee proved sufficiently-many-vertices cases; McMullen, Stanley, Novik, Hersch-Novik, and Novik-Swartz prove the subclasses above; but Klee’s conjecture for all Eulerian complexes remains wide open.

10. Isabella Novik, “Face numbers: the upper bound side of the story”, *Proceedings of the International Congress of Mathematicians 2022*, Vol. 6, EMS Press, 2023, pp. 4622-4644. DOI: 10.4171/ICM2022/43. URL: <https://ems.press/content/book-chapter-files/33308>

Conjecture 4.1 states the same Eulerian-complex upper-bound assertion: for an Eulerian simplicial complex Δ of dimension $d - 1$ with n vertices, $f_i(\Delta) \leq f_i(C_d(n))$. The survey explicitly says that in this generality the conjecture remains wide open, while it is known for simplicial spheres, Eulerian manifolds, and some pseudomanifolds with mild singularities.

2.8.3.4. What remains unresolved

No source found proves the full statement for every Eulerian $(d - 1)$ -dimensional simplicial complex with arbitrary singularities and every $n \geq d + 1$. The published/citable results all impose extra hypotheses, such as being a polytope, a simplicial sphere or homology sphere, an Eulerian homology manifold, an odd-dimensional pseudomanifold with isolated singularities, a homologically isolated singularity condition, or sufficiently many vertices. Therefore the exact problem is not fully resolved, but it has substantial partial progress.

2.8.4. Remaining open problems recorded by the database

2.8.4.1. Open problem 1

Prove the Upper Bound Conjecture for even-dimensional Eulerian simplicial pseudomanifolds with isolated singularities: if Δ is such a complex of dimension $d - 1$ on n vertices, prove $f_i(\Delta) \leq f_i(C_d(n))$ for every i .

Reference supplied by the database. Isabella Novik and Ed Swartz, “Face numbers of pseudomanifolds with isolated singularities,” *Mathematica Scandinavica* **110** (2012), no. 2, 198–222, discussion immediately preceding Theorem 4.10 (explicitly identifying the even-dimensional isolated-singularity cases as open), DOI: <https://doi.org/10.7146/math.scand.a-15204>, arXiv:1004.5100.

2.8.5. Primary-source pointer

The originating ICM record is *Combinatorics and convexity* (1994). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.2/ICM1994.2.ocr.pdf#page=659>.

2.9. 87. Combinatorics and convexity [1994-geom-54]

Record: 1994-geom-54

Subfield: Discrete and polyhedral geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1994: Combinatorics and convexity

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.2/ICM1994.2.ocr.pdf#page=659>

2.9.1. Problem statement

Prove or disprove that every centrally symmetric convex d -dimensional polytope $P \subset \mathbb{R}^d$ has at least 3^d nonempty faces in its face lattice, counting P itself as a face but not counting the empty face.

2.9.2. Historical background, literature, and current progress

2.9.3. Verdict: partially resolved

2.9.3.1. Precise problem

This problem is Kalai's 3^d conjecture for centrally symmetric polytopes:

Every centrally symmetric convex d -dimensional polytope P in $\mathbb{R}^d$ has at least 3^d nonempty faces, counting P itself and not counting the empty face.

Some papers use the equivalent convention "count the empty face but not P "; since exactly one face is exchanged in either convention, the numerical bound is the same.

2.9.3.2. Why it is not fully resolved

I found no published or citable result proving the statement for all centrally symmetric d -polytopes in all dimensions. The literature consistently treats the general case as open beyond the known special cases. In particular, Novik's survey says that the 3^d conjecture is known for centrally symmetric simplicial polytopes, by duality for simple polytopes, and for all centrally symmetric polytopes of dimension at most 4, but is "wide open" in all other cases.

Thus the exact statement remains unresolved for arbitrary centrally symmetric d -polytopes in dimensions $d \geq 5$ that are not covered by the special classes below.

2.9.3.3. Progress in the literature

1. Title: The number of faces of centrally-symmetric polytopes Authors: Gil Kalai Venue: Graphs and Combinatorics 5(1), 389-391 Year: 1989 DOI/URL: <https://doi.org/10.1007/BF01788696> Role: Original source of the conjecture. It states the general all-dimensions problem for centrally symmetric convex polytopes.

2. Title: On the number of faces of centrally-symmetric simplicial polytopes Authors: Richard P. Stanley Venue: Graphs and Combinatorics 3, 55-66 Year: 1987 DOI/URL: <https://doi.org/10.1007/BF01788529> Result: Gives lower bounds for face numbers of centrally symmetric simplicial polytopes. These imply Kalai's 3^d bound for centrally symmetric simplicial polytopes; by polarity, the bound also holds for centrally symmetric simple polytopes. Limitation: This does not cover non-simple, non-simplicial centrally symmetric polytopes.
3. Title: Borsuk's theorem and the number of facets of centrally symmetric polytopes Authors: Imre Barany, Laszlo Lovasz Venue: Acta Mathematica Academiae Scientiarum Hungaricae 40, 323-329 Year: 1982 DOI/URL: <https://doi.org/10.1007/BF01903592> Result: Earlier facet-number progress for centrally symmetric polytopes and part of the background leading to the simplicial case. Limitation: It is not a proof of the total-face 3^d conjecture in full generality.
4. Title: On Kalai's Conjectures Concerning Centrally Symmetric Polytopes Authors: Raman Sanyal, Axel Werner, Guenter M. Ziegler Venue: Discrete & Computational Geometry 41, 183-198 Year: 2009 DOI/arXiv: <https://doi.org/10.1007/s00454-008-9104-8>; arXiv:0708.3661 Result: Proves Kalai's Conjecture A, the 3^d conjecture, for all centrally symmetric polytopes of dimension at most 4. It also disproves stronger conjectures B and C in higher dimensions. Limitation: The 3^d result is dimensional: it does not prove the conjecture for arbitrary dimensions $d \geq 5$.
5. Title: Face Numbers of Centrally Symmetric Polytopes Produced from Split Graphs Authors: Ragnar Freij, Matthias Henze, Moritz W. Schmitt, Guenter M. Ziegler Venue: Electronic Journal of Combinatorics 20(2), #P32 Year: 2013 DOI/arXiv: <https://doi.org/10.37236/3315>; arXiv:1201.5790 Result: Proves Kalai's 3^d conjecture for Hansen polytopes of split graphs, with formula $s(H(G)) = 3^d + p_G(C, S)$. Limitation: Hansen polytopes of split graphs are a special family, not all centrally symmetric polytopes.
6. Title: A note on Kalai's 3^d Conjecture Authors: Gregory R. Chambers, Elia Portnoy Venue: arXiv preprint Year: 2022, revised 2023 DOI/arXiv: <https://doi.org/10.48550/arXiv.2211.09215>; arXiv:2211.09215 Result: Proves the conjecture when the polytope is symmetric about d hyperplanes with orthogonal normal vectors, equivalently after an orthogonal change of coordinates, the coordinate-symmetric/unconditional case. Limitation: This is a symmetry-hypothesis special case.
7. Title: Kalai's 3^d conjecture for unconditional and locally anti-blocking polytopes Authors: Raman Sanyal, Martin Winter Venue: Proceedings of the American Mathematical Society 153(01), 279-290 Year: 2025 DOI/arXiv: <https://doi.org/10.1090/proc/16909>; arXiv:2308.02909 Result: Theorem 3 proves $s(P) \geq 3^d$ for unconditional polytopes, with equality exactly for Hanner polytopes. Theorem 4 proves the more general result for proper locally anti-blocking polytopes, with equality exactly for generalized Hanner polytopes. Limitation: Not every centrally symmetric polytope is unconditional or locally anti-blocking.
8. Title: A tale of centrally symmetric polytopes and spheres Author: Isabella Novik Venue: Recent Trends in Algebraic Combinatorics, Association for Women in Mathematics Series 16, Springer, 305-331 Year: 2019 arXiv/URL: arXiv:1711.09310; <https://arxiv.org/abs/1711.09310> Role: Survey source. It records the known cases and explicitly states that the 3^d conjecture is wide open outside centrally symmetric simplicial/simple polytopes and dimension at most 4.

2.9.3.4. Remaining unresolved part

The full statement for arbitrary centrally symmetric convex d -dimensional polytopes remains open in dimensions $d \geq 5$, except for the special classes listed above. Since the literature proves only special cases and no theorem found covers all centrally symmetric polytopes, the correct status is partially resolved.

2.9.4. Remaining open problems recorded by the database

2.9.4.1. Open problem 1

Prove that every centrally symmetric five-dimensional convex polytope has at least $3^5 = 243$ nonempty faces, counting the polytope itself and excluding the empty face.

Reference supplied by the database. Raman Sanyal, Axel Werner, and Günter M. Ziegler, “On Kalai’s conjectures concerning centrally symmetric polytopes,” *Discrete & Computational Geometry* **41** (2009), no. 2, 183–198, Conjecture A (the 3^d -conjecture) and the proof through dimension 4, DOI: <https://doi.org/10.1007/s00454-008-9104-x>, [arXiv:0708.3661](https://arxiv.org/abs/0708.3661). Current open status is also recorded in Raman Sanyal and Martin Winter, “Kalai’s 3^d conjecture for unconditional and locally anti-blocking polytopes,” *Mathematische Zeitschrift* **307** (2024), article 46, DOI: <https://doi.org/10.1007/s00209-024-03505-1>, [arXiv:2308.02909](https://arxiv.org/abs/2308.02909).

2.9.5. Primary-source pointer

The originating ICM record is *Combinatorics and convexity* (1994). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.2/ICM1994.2.ocr.pdf#page=659>.

2.10. 88. Combinatorics and convexity [1994-geom-55]

Record: 1994-geom-55

Subfield: Discrete and polyhedral geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1994: Combinatorics and convexity

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.2/ICM1994.2.ocr.pdf#page=659>

2.10.1. Problem statement

Prove that for every simplicial $(d - 1)$ -dimensional sphere K , the facet-adjacency data determine the entire face structure of K up to combinatorial isomorphism. Equivalently, if one records, for each facet F of K , the d neighboring facets of F that share a codimension-one face with F , then this facet-neighbor puzzle has a unique solution.

2.10.2. Historical background, literature, and current progress

2.10.3. Status check

Verdict: partially resolved.

2.10.3.1. Precise problem

The problem asks whether every simplicial $(d-1)$ -sphere K is uniquely determined, up to combinatorial isomorphism, by its facet-ridge graph: the graph whose vertices are facets of K , with two facets adjacent exactly when they share a codimension-one face. Equivalently, if two simplicial spheres have isomorphic facet-ridge graphs, must the graph isomorphism extend to an isomorphism of simplicial complexes?

This is the standard reconstruction problem for simplicial spheres, formulated in recent papers as the Blind-Mani/Kalai conjecture. It is stronger than the classical polytopal case because it quantifies over all simplicial spheres, not only boundaries of simplicial polytopes and not only shellable spheres.

2.10.3.2. Literature and current progress

1. Title: Puzzles and polytope isomorphisms Authors: Roswitha Blind and Peter Mani-Levitska Venue: Aequationes Mathematicae 34(2-3), 287-297 Year: 1987 DOI/URL: <https://doi.org/10.1007/BF01830678> Relevance: Proves Perles' conjecture that the graph of a simple polytope determines its combinatorial structure. In dual form, this says the facet-ridge graph determines a simplicial polytope. This fully resolves the polytopal special case, but the problem asks about arbitrary simplicial spheres.

2. Title: A simple way to tell a simple polytope from its graph Author: Gil Kalai Venue: Journal of Combinatorial Theory, Series A 49(2), 381-383 Year: 1988 DOI/URL: [https://doi.org/10.1016/0097-3165\(88\)90064-7](https://doi.org/10.1016/0097-3165(88)90064-7) Relevance: Gives a short constructive proof of the Blind-Mani theorem. Again, this proves the dual statement for simplicial polytopes, not for all simplicial spheres.
3. Title: Graphs, skeleta and reconstruction of polytopes Author: Margaret M. Bayer Venue: Acta Mathematica Hungarica 155, 61-73 Year: 2018 DOI/arXiv/URL: <https://doi.org/10.1007/s10474-018-0804-0>, arXiv:1710.00118 Relevance: Survey of reconstruction results. It states the dual consequence of Blind-Mani/Kalai: the face lattice of a simplicial polytope can be reconstructed from its facet-ridge graph. This is background for why the problem is a generalization beyond polytopes.
4. Title: Subword Complexes and Kalai's Conjecture on Reconstruction of Spheres Authors: Cesar Ceballos and Joseph Doolittle Venue: Discrete & Computational Geometry 74, 23-48 Year: 2025 DOI/arXiv/URL: <https://doi.org/10.1007/s00454-025-00733-6>, arXiv:2206.15461 Relevance: States the conjecture in the exact form: for d -dimensional simplicial spheres P, Q , every facet-ridge graph isomorphism $FR(P) \rightarrow FR(Q)$ should uniquely extend to a simplicial isomorphism. Their Theorem 1.3 proves this only for finite-type spherical subword complexes. This is a special family of simplicial spheres, so it is partial progress only.
5. Title: Reconstructing a shellable sphere from its facet-ridge graph Author: Yirong Yang Venue: arXiv preprint Year: 2024, revised 2025 DOI/arXiv/URL: arXiv:2401.04220, <https://arxiv.org/abs/2401.04220> Relevance: Theorem 1.2 proves that the combinatorial structure of a shellable simplicial sphere is determined by its facet-ridge graph. This is the strongest positive result found. It covers all shellable simplicial spheres, hence all simplicial polytopes, but it does not cover arbitrary simplicial spheres. The paper explicitly remarks that it remains open whether the theorem holds for more general classes such as constructible spheres, partitionable spheres, or all simplicial spheres, and explains that the proof uses shellability to get acyclic good orientations.
6. Title: Unshellable triangulations of spheres Author: W. B. R. Lickorish Venue: European Journal of Combinatorics 12(6), 527-530 Year: 1991 DOI/URL: [https://doi.org/10.1016/S0195-6698\(13\)80103-5](https://doi.org/10.1016/S0195-6698(13)80103-5) Relevance: Shows that non-shellable triangulated spheres exist. This confirms that Yang's shellability hypothesis is a real restriction and not automatically satisfied by all simplicial spheres.
7. Title: A minimal counterexample to a strengthening of Perles' conjecture Author: Joseph Doolittle Venue: arXiv preprint Year: 2018 DOI/arXiv/URL: arXiv:1809.00662, <https://arxiv.org/abs/1809.00662> Relevance: Gives counterexamples to a stronger Perles-piece criterion and discusses barriers to extending polytopal reconstruction methods to arbitrary simplicial spheres. It does not disprove the exact reconstruction statement, but it shows that some natural stronger local-recognition approaches fail.

2.10.3.3. What remains unresolved

No published or otherwise citable result found proves that every simplicial sphere is determined by its facet-ridge graph. The literature resolves important subclasses: simplicial polytopes, finite-type spherical subword complexes, and shellable simplicial spheres. The exact all-cases statement in the problem remains open beyond these hypotheses.

Therefore the correct status is partially resolved, not fully resolved.

2.10.4. Remaining open problems recorded by the database

2.10.4.1. Open problem 1

Prove that every constructible simplicial sphere is uniquely determined, up to simplicial isomorphism, by its facet–ridge graph.

Reference supplied by the database. Yirong Yang, “Reconstructing a shellable sphere from its facet-ridge graph,” arXiv preprint (2024; revised 2025), open problem in the concluding paragraph asking whether the main reconstruction theorem extends to constructible spheres, [arXiv:2401.04220v2](https://arxiv.org/abs/2401.04220v2).

2.10.4.2. Open problem 2

Prove that every partitionable simplicial sphere is uniquely determined, up to simplicial isomorphism, by its facet–ridge graph.

Reference supplied by the database. Yirong Yang, “Reconstructing a shellable sphere from its facet-ridge graph,” arXiv preprint (2024; revised 2025), open problem in the concluding paragraph asking whether the main reconstruction theorem extends to partitionable spheres, [arXiv:2401.04220v2](https://arxiv.org/abs/2401.04220v2).

2.10.5. Primary-source pointer

The originating ICM record is *Combinatorics and convexity* (1994). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.2/ICM1994.2.ocr.pdf#page=659>.

2.11. 89. Combinatorics and convexity [1994-geom-241]

Record: 1994-geom-241

Subfield: Discrete and polyhedral geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1994: Combinatorics and convexity

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.2/ICM1994.2.ocr.pdf#page=659>

2.11.1. Problem statement

For each positive integer n , let $B_2^n = \{x \in \mathbb{R}^n : \|x\|_2 \leq 1\}$, and for $A \subseteq \mathbb{R}^n$ let $\text{diam}(A) = \sup\{\|x - y\|_2 : x, y \in A\}$. Define

$$d_n = \inf \left\{ \max_{1 \leq i \leq n+1} \text{diam}(A_i) : B_2^n \subseteq \bigcup_{i=1}^{n+1} A_i \right\}.$$

Prove that $2 - d_n = O(1/n)$ as $n \rightarrow \infty$; equivalently, prove that there are constants $C > 0$ and n_0 such that $d_n \geq 2 - C/n$ for every $n \geq n_0$.

2.11.2. Historical background, literature, and current progress

The problem is unresolved. In the common normalization in which the Euclidean ball has diameter (1), write $u(n) = d_n/2$. The best verified general bounds are

$$1 - \frac{3 \log n}{2n} + O\left(\frac{1}{n}\right) \leq u(n) \leq \begin{cases} \sqrt{\frac{n+1}{n+2}}, & n \text{ even,} \\ \sqrt{\frac{1}{2} + \frac{1}{2} \sqrt{\frac{n-1}{n+3}}}, & n \text{ odd.} \end{cases}$$

Thus the literature proves $d_n \geq 2 - O((\log n)/n)$, while the standard simplicial decomposition gives $d_n \leq 2 - 1/n + O(1/n^2)$. The requested estimate $d_n \geq 2 - C/n$, equivalently $2 - d_n = O(1/n)$, is precisely the remaining logarithmic gap.

- “Von der Zerlegung der Kugel in kleinere Teile,” Hugo Hadwiger, *Gazeta de Matemática* 15, no. 57 (1954), 1–3. Direct URL: <https://gazeta.spm.pt/getArtigo?gid=904>. Hadwiger determines the optimum in dimensions $n \leq 3$, gives a general lower bound for the diameter of a part, and poses the extremal ball-decomposition question.
- “The Decomposition of the n-Sphere and the Boundaries of Plane Convex Domains,” David G. Larman and Nicholas K. Tamvakis, *Annals of Discrete Mathematics* 20, in *Convexity and Graph Theory* (1984), 209–214. DOI: [https://doi.org/10.1016/S0304-0208\(08\)72828-7](https://doi.org/10.1016/S0304-0208(08)72828-7). Their volume argument proves $u(n) \geq 1 - \frac{3 \log n}{2n} + O(1/n)$, the strongest verified asymptotic lower bound found.
- “Some old and new problems in combinatorial geometry I: around Borsuk’s problem,” Gil Kalai, in *Surveys in Combinatorics 2015*, London Mathematical Society Lecture

Note Series 424 (Cambridge University Press, 2015), 147–174. DOI: <https://doi.org/10.1017/CBO9781316106853.005>; arXiv:1505.04952. Section 4.4 defines the same quantity ($u(n)$), records the simplicial $1 - \Omega(1/n)$ upper bound and the Larman–Tamvakis lower bound, and explicitly identifies closing the logarithmic gap as open.

- “Four Classic Problems,” Gábor Fejes Tóth and Włodzimierz Kuperberg, in *Lagerungen, Grundlehren der mathematischen Wissenschaften 360* (Springer, 2023), 303–312. DOI: https://doi.org/10.1007/978-3-031-21800-2_15; arXiv:2202.09863. This later survey gives the parity-dependent simplicial upper bounds above, states their optimality as a conjecture, and still lists only the Larman–Tamvakis lower bound.

2.11.3. Remaining open problems recorded by the database

2.11.3.1. Open problem 1

For $A \subseteq \mathbb{R}^4$, let $\text{diam}(A) = \sup\{\|x - y\|_2 : x, y \in A\}$, and define

$$d_4 = \inf \left\{ \max_{1 \leq i \leq 5} \text{diam}(A_i) : \{x \in \mathbb{R}^4 : \|x\|_2 \leq 1\} \subseteq \bigcup_{i=1}^5 A_i \right\}.$$

Prove or disprove that $d_4 = 2\sqrt{5/6}$, the value attained by the standard regular-simplex decomposition.

Reference supplied by the database. Gil Kalai, “Some old and new problems in combinatorial geometry I: around Borsuk’s problem,” *Surveys in Combinatorics 2015*, London Mathematical Society Lecture Note Series 424, Cambridge University Press, 2015, pp. 147–174, Section 4.4, Problem 4.5, DOI: <https://doi.org/10.1017/CBO9781316106853.005>, arXiv:1505.04952.

2.11.4. Primary-source pointer

The originating ICM record is *Combinatorics and convexity* (1994). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.2/ICM1994.2.ocr.pdf#page=659>.

2.12. 90. Combinatorics and convexity [1994-geom-242]

Record: 1994-geom-242

Subfield: Discrete and polyhedral geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1994: Combinatorics and convexity

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.2/ICM1994.2.ocr.pdf#page=659>

2.12.1. Problem statement

Let d be a nonnegative integer and let A be a nonempty finite subset of $\mathbb{R}^d$. For each integer r with $1 \leq r \leq |A|$, define

$$f(A, r) = \max_{(\Omega_1, \dots, \Omega_r)} \dim \left(\bigcap_{i=1}^r \operatorname{conv}(\Omega_i) \right),$$

where the maximum is over all partitions of A into r nonempty, pairwise disjoint subsets $\Omega_1, \dots, \Omega_r$, conv denotes convex hull, $\dim$ denotes affine dimension, and $\dim \emptyset = -1$. Prove that

$$\sum_{r=1}^{|A|} f(A, r) \geq 0.$$

2.12.2. Historical background, literature, and current progress

Verdict: unresolved.

For each r , let $T_r(A)$ be the set of points that lie in the convex hull of each part of some r -partition of A . This is a finite union of convex polytopes, so $\dim T_r(A) = f(A, r)$. Thus the problem is exactly Kalai's cascade conjecture, not merely a related Tverberg problem. The most recent direct paper found calls the conjecture a long-standing open problem and expressly says that its results do not prove the full conjecture for linear maps. No proof or counterexample covering arbitrary finite configurations in every dimension is known.

Established progress:

- Akiva Kadari, *Generalizations of Tverberg's Theorem in the Plane*, M.Sc. thesis, Hebrew University of Jerusalem, 1990 (in Hebrew; unpublished). This proves the conjecture without a general-position assumption for $d \leq 2$. The result and its scope are recorded in the two surveys below and in Soberon's 2026 paper.
- John R. Reay, *Several Generalizations of Tverberg's Theorem*, **Israel Journal of Mathematics** 34(3), 1979, 238-244 (published 1980), DOI 10.1007/BF02760885. Reay determines the relevant Tverberg-set dimensions for configurations in sufficiently strong general position; consequently the cascade inequality holds under that additional hypothesis.
- Jean-Pierre Roudneff, *New Cases of Reay's Conjecture on Partitions of Points into Simplices with k -Dimensional Intersection*, **European Journal of Combinatorics** 30(8), 2009, 1919-1943, DOI 10.1016/j.ejc.2008.12.015. For ordinary general-position configurations, Roud-

neff proves the critical dimension bound when the partition order is at most 8, when $d \leq 8$, and when the target intersection dimension is 1 and $d \leq 24$. Through the general-position equivalence with Reay's conjecture, these give corresponding cases of the cascade conjecture, but do not cover arbitrary degeneracies or all parameters.

- Pablo Soberón, *Tverberg Cores and Kalai's Cascade Conjecture*, arXiv:2605.21305v1, 2026, [arXiv:2605.21305](https://arxiv.org/abs/2605.21305). Theorem 4 proves the exact cascade inequality for arbitrary finite sets satisfying $\dim T_2(A) \leq 0$. Theorem 3 proves a prime-power topological robust-core consequence of a bound on $\dim T_r$, but the paper states that this does not yield the full linear conjecture.

Current-status sources:

- Imre Barany and Pablo Soberón, *Tverberg's Theorem Is 50 Years Old: A Survey*, **Bulletin of the American Mathematical Society** 55(4), 2018, 459-492, DOI [10.1090/bull/1634](https://doi.org/10.1090/bull/1634). Conjecture 4.14 states the cascade conjecture and relates its general-position form to Reay's dimension conjecture.
- Imre Barany and Gil Kalai, *Helly-Type Problems*, **Bulletin of the American Mathematical Society** 59(4), 2022, 471-502, DOI [10.1090/bull/1753](https://doi.org/10.1090/bull/1753). Conjecture 5.1 states the all-configurations form and describes it as wide open.

What remains is precisely the inequality for unrestricted finite point configurations in dimensions $d \geq 3$ outside the cases above. In particular, even the first implication beyond Radon's theorem, isolated in the 2022 survey and again as a small open case in the 2026 paper, is not known in full generality.

2.12.3. Remaining open problems recorded by the database

2.12.3.1. Open problem 1

Let d be a positive integer and let $A \subset \mathbb{R}^d$ consist of $2d + 2$ points. Prove that at least one of the following holds:

1. There is a partition $A = A_1 \sqcup A_2$ such that $\text{conv}(A_1)$ and $\text{conv}(A_2)$ are d -dimensional simplices whose interiors intersect.
2. There is a partition $A = A_1 \sqcup A_2 \sqcup A_3$ into three nonempty sets such that $\text{conv}(A_1) \cap \text{conv}(A_2) \cap \text{conv}(A_3) \neq \emptyset$.

Reference supplied by the database. Imre Barany and Gil Kalai, *Helly-Type Problems*, Bulletin of the American Mathematical Society 59(4) (2022), 471-502, Section 5.3 (special case of the cascade conjecture), <https://doi.org/10.1090/bull/1753>.

2.12.3.2. Open problem 2

Let d be a positive integer and let $X \subset \mathbb{R}^d$ be a nonempty finite set in general position, meaning that every subset of at most $d + 1$ points is affinely independent. For $1 \leq r \leq |X|$,

let

$$T_r(X) = \bigcup_{(X_1, \dots, X_r)} \bigcap_{i=1}^r \text{conv}(X_i),$$

where the union is over all partitions of X into r nonempty sets. Prove that

$$\sum_{r=1}^{|X|} \dim(\text{conv}(T_r(X))) \geq 0,$$

where $\dim$ denotes affine dimension and $\dim \emptyset = -1$.

Reference supplied by the database. Imre Barany and Pablo Soberon, *Tverberg's Theorem Is 50 Years Old: A Survey*, Bulletin of the American Mathematical Society 55(4) (2018), 459-492, Conjecture 4.15 (Weak Cascade Conjecture), <https://doi.org/10.1090/bull/1634>.

2.12.3.3. Open problem 3

Let d and r be positive integers, let k be an integer with $0 \leq k \leq d$, and let $A \subset \mathbb{R}^d$ be a set of $(d+1)(r-1) + k + 1$ points in general position, meaning that every subset of at most $d+1$ points is affinely independent. Define $T_r(A)$ to be the set of all points $x \in \mathbb{R}^d$ for which there is a partition $A = A_1 \sqcup \dots \sqcup A_r$ into nonempty sets satisfying

$$x \in \bigcap_{i=1}^r \text{conv}(A_i).$$

Prove that $\dim T_r(A) \geq k$, where $\dim$ is Hausdorff dimension.

Reference supplied by the database. John R. Reay, *Several Generalizations of Tverberg's Theorem*, Israel Journal of Mathematics 34(3) (1979), 238-244 (published 1980), Reay's dimension conjecture, <https://doi.org/10.1007/BF02760885>.

2.12.4. Primary-source pointer

The originating ICM record is *Combinatorics and convexity* (1994). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.2/ICM1994.2.ocr.pdf#page=659>.

2.13. 91. Mathematical Snapshots From the Computational Geometry Landscape [1998-geom-244]

Record: 1998-geom-244

Subfield: Discrete and polyhedral geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1998: Mathematical Snapshots From the Computational Geometry Landscape

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.3/ICM1998.3.ocr.pdf#page=367>

2.13.1. Problem statement

For each positive integer n , determine the maximum possible number of unordered pairs $\{p, q\}$ with $p \neq q$ and Euclidean distance $\|p - q\| = 1$, among all sets of n distinct points in the Euclidean plane.

2.13.2. Historical background, literature, and current progress

The problem is **unresolved**. Let $u(n)$ denote the requested maximum. Exact values are now known for every $n \leq 21$, but not for all n : already the proved bounds $60 \leq u(22) \leq 61$ do not coincide. More generally, the best upper-bound exponent remains $4/3$, while recent lower-bound constructions are only of order $n^{1+\delta}$ for a fixed small $\delta > 0$. Thus neither the exact function nor its asymptotic order is known.

Key progress:

- “Unit distances in the Euclidean plane,” Joel H. Spencer, Endre Szemerédi, and William T. Trotter, in *Graph Theory and Combinatorics* (Cambridge, 1983), Academic Press, 1984, pp. 293–303 ([direct bibliographic record](#)). This proves $u(n) = O(n^{4/3})$, which remains the best known general upper-bound exponent.
- “An improved constant factor for the unit distance problem,” Péter Ágoston and Dömötör Pálvolgyi, *Studia Scientiarum Mathematicarum Hungarica* 59(1) (2022), 40–57 ([DOI 10.1556/012.2022.01517](#); [arXiv:2006.06285](#)). Theorem 1 sharpens the explicit form of the preceding upper bound to $u(n) \leq \sqrt[3]{29/4} n^{4/3} < 1.94n^{4/3}$, without improving its exponent; the paper also proves $u(15) = 37$.
- “The Erdős unit distance problem for small point sets,” Boris Alexeev, Dustin G. Mixon, and Hans Parshall, *arXiv preprint* (2024, revised 2025) ([arXiv:2412.11914](#)). Theorem 1 determines $u(n)$ and enumerates all densest unit-distance graphs for every $n \leq 21$. For $22 \leq n \leq 30$ it gives nonmatching upper and lower bounds; in particular, $60 \leq u(22) \leq 61$, so the exact problem remains open even at $n = 22$.
- “An explicit lower bound for the unit distance problem,” Will Sawin, *arXiv preprint* (2026) ([arXiv:2605.20579](#)). Theorem 1 constructs arbitrarily large planar point sets giving $u(n) \geq n^{1.014114}/C$ for an absolute constant C (after an inessential constant adjustment between ordered and unordered pairs). This is a rigorous polynomially superlinear lower bound and supersedes Erdős’s earlier $n^{1+\Omega(1/\log \log n)}$ construction, but it is far

below the upper bound.

- **“Optimizing Explicit Unit-Distance Lower-Bound Certificates,”** Michael T. M. Emmerich, *arXiv preprint* (2026) ([arXiv:2606.03419](https://arxiv.org/abs/2606.03419)). This computational certificate study verifies parameters within Sawin’s criterion supporting the stronger reported consequence $u(n) > n^{1.0152}$ for arbitrarily large n . It improves only the lower-bound exponent and does not close the gap.
- **“Erdős’s Unit Distance Problem and Rigidity,”** János Pach, Orit E. Raz, and József Solymosi, *42nd International Symposium on Computational Geometry (SoCG 2026)*, LIPIcs 367, Article 83 (2026) (DOI [10.4230/LIPIcs.SoCG.2026.83](https://doi.org/10.4230/LIPIcs.SoCG.2026.83); [arXiv:2507.15679](https://arxiv.org/abs/2507.15679)). The authors prove a structural theorem for configurations with nearly $n^{4/3}$ unit distances and show that their explicitly stated Rigidity Conjecture would imply $u(n) = O(n^{4/3}/\log^{1/12} n)$. Because the rigidity input is still conjectural, this is a conditional route rather than an improved unconditional upper bound.

What remains unresolved is to determine $u(n)$ for every n (beginning with $n = 22$) and, even at the coarser asymptotic level, to close the gap between the polynomially superlinear lower bounds and the $O(n^{4/3})$ upper bound.

2.13.3. Remaining open problems recorded by the database

2.13.3.1. Open problem 1

Determine the maximum possible number of unordered pairs of points at Euclidean distance 1 among all sets of 22 distinct points in the Euclidean plane.

Reference supplied by the database. Boris Alexeev, Dustin G. Mixon, and Hans Parshall, “The Erdős unit distance problem for small point sets,” *arXiv preprint*, 2024 (revised 2025), Theorem 1(b), [arXiv:2412.11914](https://arxiv.org/abs/2412.11914), <https://arxiv.org/abs/2412.11914>.

2.13.3.2. Open problem 2

Prove that there is an absolute constant $C > 0$ such that, for every integer $n \geq 3$, every set of n points in the Euclidean plane in convex position, meaning that every point is a vertex of the convex hull of the set, determines at most Cn unordered pairs of points at Euclidean distance 1.

Reference supplied by the database. Amol Aggarwal, “On unit distances in a convex polygon,” *Discrete Mathematics* 338(3) (2015), 88–92, Erdős–Moser conjecture stated in Section 1.1, DOI: [10.1016/j.disc.2014.10.009](https://doi.org/10.1016/j.disc.2014.10.009), [arXiv:1009.2216](https://arxiv.org/abs/1009.2216).

2.13.3.3. Open problem 3

Let $[n] = \{1, \dots, n\}$. Prove that, for every fixed $c > 0$, there is an integer $n_0(c)$ such that the following holds for every $n \geq n_0(c)$. Let $G = ([n], E)$ be a finite simple graph with $|E| \geq cn^{7/6}$, and let $\mathbf{p} = (p_1, \dots, p_n) \in (\mathbb{R}^2)^n$ be a realization of its vertices in the Euclidean plane such that, for every vertex $v \in [n]$, the set $\{p_w : \{v, w\} \in E\}$ is not contained in a line. Then G has a subgraph $G' = (V', E')$ with $|V'| \geq 4$ for which the framework $(G', \mathbf{p}|_{V'})$ is rigid: there is a neighborhood U of $(p_v)_{v \in V'}$ in $(\mathbb{R}^2)^{V'}$ such that, whenever $(q_v)_{v \in V'} \in U$ satisfies $\|q_v - q_w\| = \|p_v - p_w\|$ for every $\{v, w\} \in E'$, there is a Euclidean isometry T with $q_v = T(p_v)$ for every $v \in V'$.

Reference supplied by the database. János Pach, Orit E. Raz, and József Solymosi, “Erdős’s Unit Distance Problem and Rigidity,” 42nd International Symposium on Computational Geometry (SoCG 2026), Leibniz International Proceedings in Informatics 367 (2026), Article 83, Conjecture 7, DOI: 10.4230/LIPIcs.SocG.2026.83, arXiv:2507.15679.

2.13.4. Primary-source pointer

The originating ICM record is *Mathematical Snapshots From the Computational Geometry Landscape* (1998). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.3/ICM1998.3.ocr.pdf#page=367>.

2.14. 92. Mathematical Snapshots From the Computational Geometry Landscape [1998-geom-245]

Record: 1998-geom-245

Subfield: Discrete and polyhedral geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1998: Mathematical Snapshots From the Computational Geometry Landscape

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.3/ICM1998.3.ocf.pdf#page=367>

2.14.1. Problem statement

For each integer $n \geq 3$, determine

$$\Delta(n) = \max_{\substack{P \subset [0,1]^2 \\ |P|=n}} \min_{\substack{Q \subset P \\ |Q|=3}} \text{area}(\text{conv } Q),$$

that is, the largest possible value, over all n -point sets P in the unit square, of the minimum area of a triangle whose three vertices belong to P .

2.14.2. Historical background, literature, and current progress

Unresolved. The exact values of $\Delta(n)$ are not known for all n , and even the asymptotic order of $\Delta(n)$ remains undetermined. The strongest established general bounds currently leave the wide gap

$$c \frac{\log n}{n^2} \leq \Delta(n) \leq C_\varepsilon n^{-7/6+\varepsilon} \quad (\varepsilon > 0).$$

János Komlós, János Pintz, and Endre Szemerédi, “A Lower Bound for Heilbronn’s Problem,” *Journal of the London Mathematical Society* (2) 25 (1982), 13–24, <https://doi.org/10.1112/jlms/s2-25.1.13>. Their probabilistic construction gives the lower bound $\Delta(n) \gtrsim (\log n)/n^2$ (changing between a fixed unit-area disk and the unit square affects only an absolute constant), and in particular disproves Heilbronn’s proposed $O(n^{-2})$ upper bound.

Alex Cohen, Cosmin Pohoata, and Dmitrii Zakharov, “Lower Bounds for Incidences,” *Inventiones Mathematicae* 240 (2025), 1045–1118, <https://doi.org/10.1007/s00222-025-01331-2>; arXiv:2409.07658. Theorem 1.8 proves that, for every $\varepsilon > 0$, every n -point subset of the unit square contains a triangle of area $O_\varepsilon(n^{-7/6+\varepsilon})$, yielding the current strongest general upper bound for $\Delta(n)$.

Dmitrii Zakharov, “Small Triangles,” *Journal of the London Mathematical Society* 113 (2026), e70447, <https://doi.org/10.1112/jlms.70447>. This survey states the current bounds above and emphasizes that they remain far from the widely expected upper bound $\Delta(n) \leq C_\varepsilon n^{-2+\varepsilon}$.

Nathan Sudermann-Merx, “From Computational Certification to Exact Coordinates: Heilbronn’s Triangle Problem on the Unit Square Using Mixed-Integer Optimization,” preprint

(2026), arXiv:2603.11107. This work certifies the optimum for $n = 9$ and gives exact optimal configurations for $5 \leq n \leq 9$, but explicitly leaves certification for $n = 10$ open. Thus the exact small- n results do not resolve the requested function for every n .

2.14.3. Remaining open problems recorded by the database

2.14.3.1. Open problem 1

Determine

$$\Delta(10) = \max_{\substack{P \subset [0,1]^2 \\ |P|=10}} \min_{\substack{Q \subset P \\ |Q|=3}} \text{area}(\text{conv } Q).$$

Reference supplied by the database. Nathan Sudermann-Merx, “From Computational Certification to Exact Coordinates: Heilbronn’s Triangle Problem on the Unit Square Using Mixed-Integer Optimization,” preprint, 2026, Section 7 (“Outlook and final remarks”), arXiv:2603.11107, <https://arxiv.org/abs/2603.11107>.

2.14.3.2. Open problem 2

For each integer $n \geq 3$, let

$$\Delta(n) = \max_{\substack{P \subset [0,1]^2 \\ |P|=n}} \min_{\substack{Q \subset P \\ |Q|=3}} \text{area}(\text{conv } Q).$$

Call an n -point set P *optimal* if its minimum triangle area is $\Delta(n)$, and call a three-point subset $Q \subset P$ *critical* if $\text{area}(\text{conv } Q) = \Delta(n)$. Determine whether there is a function $f(n) \rightarrow \infty$ such that every optimal n -point set has at least $f(n)$ critical three-point subsets.

Reference supplied by the database. Nathan Sudermann-Merx, “From Computational Certification to Exact Coordinates: Heilbronn’s Triangle Problem on the Unit Square Using Mixed-Integer Optimization,” preprint, 2026, Section 6.1 (“Growth of the number of critical triangles”), arXiv:2603.11107, <https://arxiv.org/abs/2603.11107>.

2.14.3.3. Open problem 3

For each integer $n \geq 3$, let

$$\Delta(n) = \max_{\substack{P \subset [0,1]^2 \\ |P|=n}} \min_{\substack{Q \subset P \\ |Q|=3}} \text{area}(\text{conv } Q).$$

Determine whether there is an absolute constant $C > 0$ such that, for every $n \geq 3$ and every n -point set $P \subset [0,1]^2$ whose minimum triangle area equals $\Delta(n)$, the triangle areas

$$\{\text{area}(\text{conv } Q) : Q \subset P, |Q| = 3\}$$

take at most Cn distinct values.

Reference supplied by the database. Nathan Sudermann-Merx, “From Computational Certification to Exact Coordinates: Heilbronn’s Triangle Problem on the Unit Square Using Mixed-Integer Optimization,” preprint, 2026, Section 6.2 (“Clustering of noncritical triangle areas”), arXiv:2603.11107, <https://arxiv.org/abs/2603.11107>.

2.14.4. Primary-source pointer

The originating ICM record is *Mathematical Snapshots From the Computational Geometry Landscape* (1998). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.3/ICM1998.3.ocr.pdf#page=367>.

2.15. 93. Halving Point Sets [1998-geom-246]

Record: 1998-geom-246

Subfield: Discrete and polyhedral geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1998: Halving Point Sets

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.3/ICM1998.3.ocr.pdf#page=473>

2.15.1. Problem statement

Let $d \geq 4$, $n > d$, and $0 \leq j < (n - d)/2$ be integers. Let $P \subset \mathbb{R}^d$ be a set of n points in general position, meaning that no $d+1$ points of P lie in a common affine hyperplane. For $0 \leq i \leq j$, an oriented $(d-1)$ -simplex spanned by d points of P is called an i -facet if exactly i points of P lie in the open positive half-space bounded by its affine hull. Determine whether, for every such d, n, j , the maximum possible total number of i -facets with $0 \leq i \leq j$, over all such sets P , is attained by a set in convex position, that is, by a set P for which every point is a vertex of $\text{conv}(P)$.

2.15.2. Historical background, literature, and current progress

The problem is **unresolved**. In the nontrivial part of the stated range, it is implied by the Generalized Upper Bound Conjecture for sublevels of Eckhoff, Linhart, and Welzl, which predicts the stronger conclusion that a neighborly point configuration (for example, points on the moment curve) maximizes the number of $(\leq j)$ -facets. The most recent result located still describes that conjecture as long-standing and proves only a special corank. No proof that some convex-position set always attains the unrestricted maximum, and no counterexample to that assertion, was found. When $n - d$ is odd and $j = (n - d - 1)/2$, the count is configuration-independent by reversal of orientation; this trivial boundary case does not settle the other parameters.

Key established progress:

- Peter McMullen, "The Maximum Numbers of Faces of a Convex Polytope," *Mathematika* 17(2), 1970, 179-184, <https://doi.org/10.1112/S0025579300002850>. The Upper Bound Theorem proves the desired affirmative conclusion for $j = 0$ in every dimension: the number of 0-facets is maximized by a cyclic polytope, whose vertices are in convex position. It does not control the additional i -facets for $1 \leq i \leq j$.
- Kenneth L. Clarkson and Peter W. Shor, "Applications of Random Sampling in Computational Geometry, II," *Discrete & Computational Geometry* 4, 1989, 387-421, <https://doi.org/10.1007/BF02187740>. Their sampling bounds give $G_j = O(n^{\lfloor d/2 \rfloor} (j+1)^{\lceil d/2 \rceil})$ for fixed d , and point sets on the moment curve attain this order of growth. Thus a convex configuration is asymptotically optimal up to a dimension-dependent constant, but the result neither gives the exact maximum nor proves exact attainment in convex position.
- Uli Wagner, "On a Geometric Generalization of the Upper Bound Theorem," *47th Annual*

IEEE Symposium on Foundations of Computer Science (FOCS 2006), 2006, 635-645, <https://doi.org/10.1109/FOCS.2006.53>. Wagner proves the corresponding upper bound for low-level vertices in arrangements up to a factor of 2 in the affine formulation. This is quantitative progress toward the exact generalized upper-bound conjecture, but the constant-factor gap cannot decide whether an exact maximizer is convex.

- Brett Leroux and Luis Rademacher, “Algebraic k -Sets and Generally Neighborly Embeddings,” *Discrete & Computational Geometry* 67(2), 2022, 605-629, <https://doi.org/10.1007/s00454-021-00340-1>. Theorem 3.12 proves that, for a convex-position set $S \subset \mathbb{R}^d$, its number of k -facets is at most n/d times the extremal number of k -facets for $n - 1$ points in $\mathbb{R}^{d-1}$. This improves upper bounds within the convex class but does not compare that class exactly with all point configurations.
- Elizaveta Streltsova and Uli Wagner, “Levels in Arrangements: Linear Relations, the g -Matrix, and Applications to Crossing Numbers,” *41st International Symposium on Computational Geometry (SoCG 2025)*, LIPIcs 332, Article 75, 2025, <https://doi.org/10.4230/LIPIcs.SocG.2025.75>, related full version [arXiv:2504.07770](https://arxiv.org/abs/2504.07770). Conjecture 5 states the stronger neighborly-configuration upper bound. Theorem 6 proves it for vector configurations of corank 3; after homogenizing point sets, this yields the desired affirmative conclusion when $n = d + 4$ (including the only nontrivial new level $j = 1$). The paper does not cover arbitrary $n - d$, which is the remaining gap in the stated problem.

2.15.3. Remaining open problems recorded by the database

No separate descendant problem is listed in the website’s “Some remaining open problems” field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

2.15.4. Primary-source pointer

The originating ICM record is *Halving Point Sets* (1998). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.3/ICM1998.3.ocr.pdf#page=473>.

2.16. 94. The Sphere Packing Problem [1998-geom-247]

Record: 1998-geom-247

Subfield: Discrete and polyhedral geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1998: The Sphere Packing Problem

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.3/ICM1998.3.ocr.pdf#page=389>

2.16.1. Problem statement

For each integer $n \geq 1$, let Δ_n be the supremum of the upper densities of all packings of congruent closed Euclidean balls in $\mathbb{R}^n$ (packings whose ball interiors are pairwise disjoint), and let $\Delta_n^{(L)}$ be the corresponding supremum restricted to lattice packings, in which the ball centers form a full-rank lattice in $\mathbb{R}^n$. Prove that

$$\Delta_n^{(L)} = \Delta_n$$

for every integer n with $1 \leq n \leq 9$.

2.16.2. Historical background, literature, and current progress

The problem is **unresolved**. The unrestricted sphere-packing problem is rigorously solved in dimensions 1, 2, 3, 8, and 24, so within the range in the statement the equality $\Delta_n^{(L)} = \Delta_n$ is established for $n = 1, 2, 3, 8$. It remains unproved for $n = 4, 5, 6, 7, 9$. A current peer-reviewed account explicitly records that the sphere-packing problem is solved only in dimensions 1, 2, 3, 8, 24, while describing Conway and Sloane's proposed low-dimensional packings as conjectural.

The lattice side is much better understood. The optimal lattice packings were already known through dimension 8; Mathieu Dutour Sikirić and Wessel van Woerden have now completed Voronoi's algorithm in dimension 9, enumerating all 2,237,251,040 perfect lattices and proving in Theorem 1 and Corollary 1 that the laminated lattice Λ_9 is the unique densest lattice packing and that $\gamma_9 = 2$. Thus the remaining issue is unrestricted, possibly nonlattice packing optimality in dimensions 4, 5, 6, 7, 9, not identification of the lattice optima.

The strongest general upper bounds currently recorded for those five dimensions are still strictly above the lattice densities. The rigorous three-point bounds of Cohn, de Laat, and Salmon give, respectively, $0.6361073322 > 0.6168502751$ for $n = 4$, $0.5126451307 > 0.4652576133$ for $n = 5$, $0.4103032819 > 0.3729475456$ for $n = 6$, $0.3211471057 > 0.2952978731$ for $n = 7$, and $0.1911204153 > 0.1457748758$ for $n = 9$, where the number on the right is the density attained by the corresponding optimal lattice. These are substantive quantitative advances but leave a strict gap in every unresolved dimension.

Key references:

- John H. Conway and Neil J. A. Sloane, "What Are All the Best Sphere Packings in Low

Dimensions?”, *Discrete & Computational Geometry* **13** (1995), 383–403. DOI: <https://doi.org/10.1007/BF02574051>. This is the source of the conjectural low-dimensional optimal packings and of the assertion that lattice and unrestricted optima agree through dimension 9.

- Thomas C. Hales, “A Proof of the Kepler Conjecture”, *Annals of Mathematics* **162** (2005), 1065–1185. DOI: <https://doi.org/10.4007/annals.2005.162.1065>; and Thomas Hales et al., “A Formal Proof of the Kepler Conjecture”, *Forum of Mathematics, Pi* **5** (2017), e2. DOI: <https://doi.org/10.1017/fmp.2017.1>, arXiv:1501.02155. These prove unrestricted optimality in dimension 3, where an optimal lattice packing exists.
- Maryna S. Viazovska, “The Sphere Packing Problem in Dimension 8”, *Annals of Mathematics* **185** (2017), 991–1015. DOI: <https://doi.org/10.4007/annals.2017.185.3.7>, arXiv:1603.04246. This proves that the E_8 lattice is optimal among all packings, establishing the $n = 8$ case.
- Henry Cohn, David de Laat, and Andrew Salmon, “Three-Point Bounds for Sphere Packing”, preprint (2022), arXiv:2206.15373. Direct URL: <https://arxiv.org/abs/2206.15373>. The paper proves the three-point bound and rigorously verifies the numerical upper bounds quoted above for dimensions 4–7 and 9, none of which reaches the lattice lower bound.
- Mathieu Dutour Sikirić and Wessel van Woerden, “The Lattice Packing Problem in Dimension 9 by Voronoi’s Algorithm”, preprint, revised 2026, arXiv:2508.20719. Direct URL: <https://arxiv.org/abs/2508.20719>. Theorem 1 and Corollary 1 settle the lattice problem in dimension 9, but the paper explicitly notes that the unrestricted sphere-packing problem is known only in dimensions 1, 2, 3, 8, 24.
- Henry Cohn and Isaac Rajagopal, “Variations on Five-Dimensional Sphere Packings”, *Discrete & Computational Geometry* (2026). DOI: <https://doi.org/10.1007/s00454-026-00841-x>. This current account again states that unrestricted sphere packing is solved only in dimensions 1, 2, 3, 8, 24, gives new conjecturally optimal five-dimensional configurations, and records that full optimality of the D_5 packing remains open.

2.16.3. Remaining open problems recorded by the database

2.16.3.1. Open problem 1

Let $C \subset \mathbb{R}^2$ be an infinite discrete set equipped with a coloring $\chi : C \rightarrow \mathbb{Z}/4\mathbb{Z}$. Assume that every pair of distinct points $x, y \in C$ satisfies

- $\|x - y\| \geq \sqrt{2}$ if $\chi(x) = \chi(y)$;
- $\|x - y\| \geq \sqrt{5}/2$ if $\chi(x) - \chi(y) \equiv \pm 1 \pmod{4}$;
- $\|x - y\| \geq 1$ if $\chi(x) - \chi(y) \equiv 2 \pmod{4}$.

Prove that

$$\limsup_{r \rightarrow \infty} \frac{\#\{x \in C : \|x\| \leq r\}}{\pi r^2} \leq 1.$$

Reference supplied by the database. Henry Cohn and Isaac Rajagopal, “Variations on Five-Dimensional Sphere Packings,” *Discrete & Computational Geometry* (2026), Conjecture 4.1.

DOI: <https://doi.org/10.1007/s00454-026-00841-x>

2.16.4. Primary-source pointer

The originating ICM record is *The Sphere Packing Problem* (1998). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.3/ICM1998.3.ocr.pdf#page=389>.

2.17. 95. The Sphere Packing Problem [1998-geom-248]

Record: 1998-geom-248

Subfield: Discrete and polyhedral geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1998: The Sphere Packing Problem

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.3/ICM1998.3.ocf.pdf#page=389>

2.17.1. Problem statement

Let

$$D_9 = \left\{ x = (x_1, \dots, x_9) \in \mathbb{Z}^9 : \sum_{i=1}^9 x_i \equiv 0 \pmod{2} \right\}$$

and, for each $\theta \in \mathbb{R}$, let

$$D_9^{\theta+} = D_9 \cup \left(D_9 + \left(\frac{1}{2}, \dots, \frac{1}{2}, \frac{\theta}{2} \right) \right),$$

where the first eight coordinates of the translation vector are $1/2$. Place congruent closed balls of radius $1/\sqrt{2}$ at the points of $D_9^{\theta+}$; their interiors are pairwise disjoint. Let Δ_9 be the supremum of the upper densities of all packings of congruent balls in $\mathbb{R}^9$, and let $\Delta_9^{(L)}$ be the corresponding supremum restricted to lattice packings. Prove that, for every $\theta \in \mathbb{R}$, the density of this packing is $\Delta_9 = \Delta_9^{(L)}$.

2.17.2. Historical background, literature, and current progress

The exact problem is **unresolved** as of 3 August 2026. Every member of the displayed family is periodic under D_9 , has two centers per fundamental cell of covolume 2, and hence has density

$$\text{vol}(B_9)2^{-9/2} = \frac{\sqrt{2}\pi^4}{945} \approx 0.1457748758.$$

The new exact lattice result proves that this value is $\Delta_9^{(L)}$, but no theorem proves that it is the unrestricted optimum Δ_9 . In particular, the best verified general upper bound recorded below is approximately 0.191121, strictly above the construction's density, so the assertion $\Delta_9 = \Delta_9^{(L)}$ remains open.

Key references and progress:

- J. H. Conway and N. J. A. Sloane, "What Are All the Best Sphere Packings in Low Dimensions?", *Discrete & Computational Geometry* 13 (1995), 383–403. DOI: <https://doi.org/10.1007/BF02574051>. This paper introduces and analyzes the fluid family $D_9^{\theta+}$, shows that its members have the same density, and presents their global optimality as conjectural rather than proved.
- Alexei Andreanov and Yoav Kallus, "Locally Optimal 2-Periodic Sphere Packings", *Discrete & Computational Geometry* 63 (2020), 182–208. DOI: <https://doi.org/10.1007/s00454-020-00182-2>

[g/10.1007/s00454-019-00150-6](https://doi.org/10.1007/s00454-019-00150-6); arXiv: 1704.08156. The paper proves a general local theory for 2-periodic packings and identifies the nine-dimensional fluid-diamond family based on D_9 as a family of locally optimal 2-periodic point sets attaining the highest known nine-dimensional density. Local optimality does not establish the required global optimum.

- **Henry Cohn, David de Laat, and Andrew Salmon, “Three-Point Bounds for Sphere Packing”, preprint (2022).** arXiv: 2206.15373. Theorem 1.1 and its rigorously certified computations give an unrestricted nine-dimensional sphere-packing upper bound of approximately 0.191121 (Table 1.1), improving earlier bounds but not reaching $\sqrt{2}\pi^4/945$.
- **Mathieu Dutour Sikirić and Wessel van Woerden, “The Lattice Packing Problem in Dimension 9 by Voronoi’s Algorithm”, preprint (2025).** arXiv: 2508.20719. Theorem 1 and Corollary 1 complete the enumeration of all 2,237,251,040 similarity classes of perfect nine-dimensional lattices and prove that Λ_9 is the unique densest lattice, equivalently $\gamma_9 = 2$. Since $D_9^{0+} = \Lambda_9$ and every $D_9^{\theta+}$ has the same density, this resolves the equality with $\Delta_9^{(L)}$, but it does not cover nonlattice packings and therefore does not resolve the full statement.

Thus the strongest exact progress settles the complete lattice component and establishes local optimality within the relevant periodic setting; what remains is to rule out every denser nonlattice packing in $\mathbb{R}^9$.

2.17.3. Remaining open problems recorded by the database

2.17.3.1. Open problem 1

Let $L \subset \mathbb{R}^9$ be a full-rank lattice, let $t \in \mathbb{R}^9$, and suppose that the point set $P = L \cup (L + t)$ has positive minimum distance $d(P)$. Place closed balls of radius $d(P)/2$ at the points of P . Determine whether every such 2-periodic sphere packing has density at most $\sqrt{2}\pi^4/945$, the density attained by the fluid-diamond packings $D_9 \cup (D_9 + ((1/2)^8, \theta/2))$.

Reference supplied by the database. Alexei Andreanov and Yoav Kallus, “Locally Optimal 2-Periodic Sphere Packings,” *Discrete & Computational Geometry* 63(1) (2020), 182–208; question of determining $\phi_{d,2}$ in the Introduction and open questions in Section 5. DOI: <https://doi.org/10.1007/s00454-019-00150-6>; arXiv:1704.08156.

2.17.3.2. Open problem 2

Let $D_8 = \{z \in \mathbb{Z}^8 : \sum_{i=1}^8 z_i \equiv 0 \pmod{2}\}$, let $T = \mathbb{R}^8/D_8$ carry its quotient Euclidean metric and normalized Haar measure, and put $X = \mathbb{R} \times T$ with product metric and product Haar measure. Define

$$C = \{p \in X : d_X(p, 0) \geq \sqrt{2}\}.$$

Let $K(X)$ be the cone of real-valued continuous integrable positive-definite functions on the locally compact abelian group X . Prove or disprove that

$$\inf \left\{ f(0) : f \in K(X), f(p) \leq 0 \text{ for every } p \in C, \int_X f = 1 \right\} = 2.$$

Reference supplied by the database. Andrew Salmon, “Linear Programming Bounds for Fibered Sphere Packings,” preprint (2026), Section 3.2, open question following Theorem 13. arXiv:2607.25254, <https://arxiv.org/abs/2607.25254>.

2.17.3.3. Open problem 3

Let $D_8 = \{z \in \mathbb{Z}^8 : \sum_{j=1}^8 z_j \equiv 0 \pmod{2}\}$. Consider every locally finite point set in $\mathbb{R} \times \mathbb{R}^8 \cong \mathbb{R}^9$ of the form

$$P = \bigcup_{i \in I} (\{b_i\} \times (U_i D_8 + t_i)),$$

where $b_i \in \mathbb{R}$, $U_i \in O(8)$, and $t_i \in \mathbb{R}^8$, and suppose that the distance between any two distinct points of P is at least $\sqrt{2}$. Prove or disprove that the packing of balls of radius $1/\sqrt{2}$ centered at P has upper density at most $\sqrt{2} \pi^4/945$.

Reference supplied by the database. Andrew Salmon, “Linear Programming Bounds for Fibered Sphere Packings,” preprint (2026), Section 3.2, open question following Theorem 13. arXiv:2607.25254, <https://arxiv.org/abs/2607.25254>.

2.17.4. Primary-source pointer

The originating ICM record is *The Sphere Packing Problem* (1998). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.3/ICM1998.3.ocr.pdf#page=389>.

2.18. 96. Halving Point Sets [1998-geom-291]

Record: 1998-geom-291

Subfield: Discrete and polyhedral geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1998: Halving Point Sets

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.3/ICM1998.3.ocr.pdf#page=473>

2.18.1. Problem statement

For each even integer n , the construction of Maria Klawe, Mike Paterson, and Nicholas Pippenger gives a simple allowable sequence on n symbols having $n2^{\Omega(\sqrt{\log n})}$ median transpositions. A simple allowable sequence on n symbols is a sequence of permutations that begins with $(1, 2, \dots, n)$, ends with $(n, n-1, \dots, 1)$, changes at each step by transposing two adjacent symbols, and transposes each unordered pair of symbols exactly once. A transposition is called median if it exchanges the symbols in positions $n/2$ and $n/2 + 1$. Determine whether every allowable sequence arising from this construction is stretchable: that is, whether it can be obtained from a configuration of n points in general position in $\mathbb{R}^2$ by recording, as an oriented line rotates through 180° , the order of the orthogonal projections of the points onto the line. Equivalently, determine whether the dual pseudoline arrangements arising from the construction are combinatorially equivalent to arrangements of straight lines.

2.18.2. Historical background, literature, and current progress

The exact problem is unresolved. No citable proof was found that the Klawe–Paterson–Pippenger (KPP) allowable sequences are stretchable, and no citable nonstretchability obstruction or counterexample was found for this family. The original ICM source explicitly asks whether the KPP construction is realizable. The most detailed later treatment reconstructs the KPP construction and proof, but still states that its realizability is open.

The main literature evidence is as follows.

- Maria M. Klawe, Michael S. Paterson, and Nicholas J. Pippenger, *Inversions with $n^{1+\Omega(1/\sqrt{\log n})}$ transpositions at the median*, unpublished manuscript, 1982. This is the construction whose stretchability is at issue. It is reported and discussed in Jacob E. Goodman and Richard Pollack, *Allowable Sequences and Order Types in Discrete and Computational Geometry*, in *New Trends in Discrete and Computational Geometry*, Algorithms and Combinatorics 10, Springer, 1993, pp. 103–134, DOI [10.1007/978-3-642-58043-7_6](https://doi.org/10.1007/978-3-642-58043-7_6). The construction proves the stated pseudoline/allowable-sequence lower bound but does not prove realizability by points or straight lines.
- Artur Andrzejak and Emo Welzl, *Halving Point Sets*, *Documenta Mathematica*, Extra Volume ICM 1998 III, 1998, pp. 471–478, DOI [10.4171/DMS/1-3/46](https://doi.org/10.4171/DMS/1-3/46). In its section on lower bounds and oriented matroids, this source explicitly identifies stretchability of the KPP construction as open.

- Artur Andrzejak, *On k -Sets and Their Generalizations*, doctoral dissertation, ETH Zurich, Diss. ETH No. 13441, 2000, DOI [10.3929/ethz-a-003879053](https://doi.org/10.3929/ethz-a-003879053). Section 5.3 supplies a careful proof of the KPP pseudoline construction from the difficult-to-read handwritten manuscript. It expressly observes that the later realizable construction of Toth is different and concludes that the KPP realizability question remains open. This is the strongest direct follow-up found, but it gives neither a realization nor a nonrealizability certificate.
- Geza Toth, *Point Sets with Many k -Sets*, *Discrete & Computational Geometry* 26(2), 2001, pp. 187–194, DOI [10.1007/s004540010022](https://doi.org/10.1007/s004540010022). Toth constructs actual planar point sets, hence stretchable line arrangements, with $ne^{\Omega(\sqrt{\log k})}$ k -sets and in particular the corresponding halving-line lower bound. Toth states that this construction appears essentially different from KPP's, so it establishes the desired asymptotic consequence for another family but does not answer the exact stretchability question here.
- Gabriel Nivasch, *An Improved, Simple Construction of Many Halving Edges*, in *Surveys on Discrete and Computational Geometry: Twenty Years Later*, Contemporary Mathematics 453, American Mathematical Society, 2008, pp. 299–305, DOI [10.1090/conm/453/08804](https://doi.org/10.1090/conm/453/08804). This gives a simpler realizable construction and improves the constant in the exponent of Toth's lower bound. It is again an alternative line-arrangement construction, not a realization or obstruction for the KPP sequences.

Thus the asymptotic lower bound sought as a consequence of stretchability is known by independent constructions, but the exact KPP allowable sequences have not been classified as stretchable or nonstretchable. No substantive partial result specifically deciding a subfamily of the KPP construction was found.

2.18.3. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

2.18.4. Primary-source pointer

The originating ICM record is *Halving Point Sets* (1998). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.3/ICM1998.3.ocr.pdf#page=473>.

2.19. 97. Face Numbers of 4-Polytopes and 3-Spheres [2002-geom-77]

Record: 2002-geom-77

Subfield: Discrete and polyhedral geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Face Numbers of 4-Polytopes and 3-Spheres

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=605>

2.19.1. Problem statement

Prove that for every strongly regular cellulation P of the 3-sphere S^3 , meaning a regular cell decomposition of S^3 in which the intersection of any two cells is a face of both cells or is empty, if $f_i(P)$ denotes the number of i -dimensional faces for $i = 0, 1, 2, 3$, and $f_{03}(P)$ denotes the number of vertex-facet incidences, equivalently the number of chains consisting of a 0-dimensional face contained in a 3-dimensional face, then

$$f_{03}(P) - 3(f_0(P) + f_3(P)) \geq -10.$$

Equivalently, whenever the quotient is defined, the complexity

$$C(P) := \frac{f_{03}(P) - 20}{f_0(P) + f_3(P) - 10}$$

satisfies $C(P) \geq 3$.

2.19.2. Historical background, literature, and current progress

2.19.3. Verdict: partially resolved

2.19.3.1. Precise problem

The problem asks whether every strongly regular cellulation P of S^3 satisfies

$$f_{03}(P) - 3(f_0(P) + f_3(P)) \geq -10.$$

Equivalently, for non-simplex cases where the quotient is defined, it asks whether the normalized complexity

$$C(P) = \frac{f_{03}(P) - 20}{f_0(P) + f_3(P) - 10}$$

always satisfies $C(P) \geq 3$. This is the rank-5 / dimension-4 toric g_2 -nonnegativity inequality for the face lattice of a strongly regular CW 3-sphere.

2.19.3.2. Current status

I found no published or otherwise citable result proving this inequality for all strongly regular cellulations of S^3 . The problem is therefore not fully resolved. There is genuine partial progress: the inequality is known for all convex 4-polytope boundaries, hence for the polytopal subclass of strongly regular 3-spheres, and for several related/special classes. Those results do not cover arbitrary strongly regular CW 3-spheres.

2.19.3.3. Key literature and applicability

1. Face numbers of 4-Polytopes and 3-Spheres

Günter M. Ziegler, *Proceedings of the International Congress of Mathematicians, Vol. III (Beijing, 2002)*, Higher Education Press, 2002, pp. 625-634. arXiv:math/0208073, DOI: <http://doi.org/10.48550/arXiv.math/0208073>.

This is the original source for the fatness/complexity formulation. It states the exact inequality $f_{03} - 3(f_0 + f_3) \geq -10$, equivalently $C(P) \geq 3$, identifies it with $g_2^{\text{tor}}(P) \geq 0$, and says it was known for rational polytopes and all convex polytopes but not for strongly regular 3-spheres or Eulerian lattices. Thus it frames the exact unresolved extension.

2. Generalized h -vectors, intersection cohomology of toric varieties, and related results

Richard P. Stanley, in *Commutative Algebra and Combinatorics*, Advanced Studies in Pure Mathematics 11, Kinokuniya/North-Holland, 1987, pp. 187-213. URL: <https://math.mit.edu/~rstan/pubs/pubfiles/69.pdf>.

Stanley proves toric g -vector nonnegativity in the rational polytope setting. This implies the stated inequality for rational convex 4-polytopes, but not for arbitrary strongly regular 3-spheres.

3. Rigidity and the lower bound theorem. I

Gil Kalai, *Inventiones Mathematicae* 88 (1987), 125-151. DOI: <https://doi.org/10.1007/BF01405094>.

Kalai gives a rigidity-theoretic proof of the relevant toric g_2 inequality for general convex polytopes. This covers all 4-polytope boundary complexes, a proper subclass of the strongly regular 3-spheres in the problem statement.

4. Hard Lefschetz theorem for nonrational polytopes

Kalle Karu, *Inventiones Mathematicae* 157 (2004), 419-447. arXiv:math/0112087, DOI: <https://doi.org/10.1007/s00222-004-0358-3>.

Karu proves the Hard Lefschetz theorem for combinatorial intersection cohomology of general polytopes, giving another route to toric g -vector nonnegativity for all convex polytopes. This still does not apply to non-polytopal strongly regular CW 3-spheres.

5. Nonpolytopal Nonsimplicial Lattice Spheres with Nonnegative Toric g -Vector

Louis J. Billera and Eran Nevo, *Discrete & Computational Geometry* 48(4), 1048-1057, 2012. arXiv:1110.0739, DOI: <https://doi.org/10.1007/s00454-012-9449-x>.

This constructs many nonpolytopal nonsimplicial Gorenstein* meet semilattices with nonnegative toric g -vector, supporting Stanley's conjecture. It is partial progress, not a theorem for every strongly regular CW 3-sphere.

6. The flag f -vectors of Gorenstein* order complexes of dimension 3

Satoshi Murai and Eran Nevo, *Proceedings of the American Mathematical Society* 142(5), 1527-1538, 2014. arXiv:1108.0572, DOI: <https://doi.org/10.1090/S0002-9939-2014-11897-1>.

This characterizes rank-5 Gorenstein* poset cd-indices/flag vectors. It does not settle

the problem statement because strongly regular CW spheres impose the additional intersection-property/lattice condition, and the theorem is not a proof of $g_2^{\text{tor}} \geq 0$ for all such lattices.

7. A Lower Bound Theorem for Strongly Regular CW Spheres with up to $2d + 1$ Vertices

Lei Xue, *Discrete & Computational Geometry* 72, 1348-1368, 2024. arXiv:2207.13839, DOI: <https://doi.org/10.1007/s00454-023-00553-6>.

This gives sharp lower bounds for ordinary face numbers of strongly regular CW spheres/pseudomanifolds with few vertices. It is related lower-bound progress but does not prove the vertex-facet incidence inequality for all strongly regular cellulations of S^3 .

2.19.3.4. What remains unresolved

The missing case is the full non-polytopal strongly regular CW 3-sphere case: no searched paper proves that every such cellulation has $f_{03} - 3(f_0 + f_3) \geq -10$, and no searched paper gives a counterexample. The exact problem should therefore be classified as partially resolved, not fully resolved.

2.19.4. Remaining open problems recorded by the database

2.19.4.1. Open problem 1

Let S be a strongly regular 3-sphere that is 2-simple and 2-simplicial (every edge lies in exactly three facets and every 2-face is a triangle), and suppose that S is not the boundary of the 4-simplex. Is its fatness

$$F(S) = \frac{f_1(S) + f_2(S) - 20}{f_0(S) + f_3(S) - 10}$$

always at least 4?

Reference supplied by the database. Philip Brinkmann, *f-Vector Spaces of Polytopes, Spheres, and Eulerian Lattices*, doctoral dissertation, Department of Mathematics and Computer Science, Freie Universität Berlin, 2016, Chapter 1, Conjecture 1 (the text notes that this conjecture would imply $C \geq 3$ for 2-simple 2-simplicial 3-spheres); DOI: <https://doi.org/10.17169/refubium-13151>; repository record: [Refubium](#).

2.19.5. Primary-source pointer

The originating ICM record is *Face Numbers of 4-Polytopes and 3-Spheres* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=605>.

2.20. 98. Face Numbers of 4-Polytopes and 3-Spheres [2002-geom-78]

Record: 2002-geom-78

Subfield: Discrete and polyhedral geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Face Numbers of 4-Polytopes and 3-Spheres

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=605>

2.20.1. Problem statement

Prove that for every strongly regular CW 3-sphere P , meaning a regular cell decomposition of S^3 in which the intersection of any two cells is either empty or a common face of both cells, if $f_i(P)$ denotes the number of i -dimensional faces of P , then the fatness

$$F(P) := \frac{f_1(P) + f_2(P) - 20}{f_0(P) + f_3(P) - 10}$$

satisfies $F(P) \geq \frac{5}{2}$. Equivalently,

$$2(f_1(P) + f_2(P) - 20) \geq 5(f_0(P) + f_3(P) - 10).$$

2.20.2. Historical background, literature, and current progress

2.20.3. Verdict: partially resolved

2.20.3.1. Precise problem

The problem asks whether every strongly regular CW decomposition P of S^3 , i.e. every regular CW 3-sphere with the intersection property/face-poset lattice property, satisfies

$$F(P) = \frac{f_1(P) + f_2(P) - 20}{f_0(P) + f_3(P) - 10} \geq \frac{5}{2},$$

equivalently

$$5f_0(P) - 2f_1(P) - 2f_2(P) + 5f_3(P) \leq 10.$$

2.20.3.2. Status

I did not find any published or citable result proving this inequality for all strongly regular CW 3-spheres, nor any published counterexample. The exact all-cases problem remains unresolved. It is not fully open in the sense that several important subclasses and adjacent forms are known, so the correct verdict is partially resolved.

2.20.3.3. Literature and progress

1. **Face numbers of 4-Polytopes and 3-Spheres.** Günter M. Ziegler. Proceedings of the International Congress of Mathematicians, Beijing 2002, Vol. III, Higher Education Press, pp. 625-636, 2002/2003. arXiv:math/0208073, DOI/URL: <https://doi.org/10.48550/arXiv.math/0208073>.
This is the original source for the normalized fatness parameter used in the problem statement. Ziegler identifies the inequality $\varphi_0 + \varphi_3 \leq 2/5$, equivalently $F \geq 5/2$, as valid for 4-polytopes but explicitly says that its validity for strongly regular 3-spheres is open.
2. **The extended f-vectors of 4-polytopes.** Margaret M. Bayer. Journal of Combinatorial Theory, Series A 44(1), 141-151, 1987. DOI: [https://doi.org/10.1016/0097-3165\(87\)90066-5](https://doi.org/10.1016/0097-3165(87)90066-5).
Bayer's 4-polytope inequalities include the linear inequality $5f_0 - 2f_1 - 2f_2 + 5f_3 \leq 10$, i.e. $F \geq 5/2$, for convex 4-polytopes. This resolves the boundary-sphere-of-a-4-polytope subclass, but not arbitrary strongly regular CW 3-spheres.
3. **Lower-bound theorems for pseudomanifolds.** Tiong-Seng Tay. Discrete & Computational Geometry 13(2), 203-216, 1995. DOI: <https://doi.org/10.1007/BF02574038>.
The simplicial lower-bound theorem implies the fatness inequality for simplicial strongly regular 3-spheres, and by duality for simple cases. This is a proper special case, not the nonsimplicial strongly regular CW case in full.
4. **Fat 4-polytopes and fatter 3-spheres.** David Eppstein, Greg Kuperberg, and Günter M. Ziegler. In *Discrete Geometry: In Honor of W. Kuperberg's 60th Birthday*, Pure and Applied Mathematics 253, Marcel Dekker, pp. 239-265, 2003. arXiv:math/0204007, URL: <https://arxiv.org/abs/math/0204007>.
This constructs strongly regular CW 3-spheres of unbounded fatness. It resolves the upper/unbounded-fatness direction for strongly regular CW 3-spheres, but it gives no lower-bound theorem and no counterexample to $F \geq 5/2$.
5. **A flag vector of a 3-sphere that is not the flag vector of a 4-polytope.** Philip Brinkmann and Günter M. Ziegler. Mathematika 63(1), 260-271, 2017. arXiv:1506.08148, DOI: <https://doi.org/10.1112/S0025579316000267>.
The paper separates flag vectors of strongly regular 3-spheres from those of 4-polytopes, and states that although the lower bound for fatness is known for 4-polytopes, it is not known whether it holds for 3-spheres.
6. **Small f-vectors of 3-spheres and of 4-polytopes.** Philip Brinkmann and Günter M. Ziegler. Mathematics of Computation 87(314), 2955-2975, 2018. arXiv:1610.01028, DOI: <https://doi.org/10.1090/mcom/3300>.
This gives finite enumerations and examples showing that f-vectors of strongly regular 3-spheres differ from f-vectors of 4-polytopes. It explicitly lists "Is the fatness lower bound $F \geq 2.5$ valid for all 3-spheres?" as a key open problem; its enumeration does not cover all cases.
7. **The flag f-vectors of Gorenstein* order complexes of dimension 3.** Satoshi Murai and Eran Nevo. Proceedings of the American Mathematical Society 142(5), 1527-1538, 2014. arXiv:1108.0572, DOI: <https://doi.org/10.1090/S0002-9939-2014-11897-1>.
This characterizes cd-indices of rank-5 Gorenstein* posets/order complexes and a larger regular-CW setting. It does not resolve the problem because the problem concerns the original f-vector of strongly regular CW 3-spheres with the intersection/face-lattice prop-

erty.

8. **A Lower Bound Theorem for Strongly Regular CW Spheres with up to $2d + 1$ Vertices.** Lei Xue. Discrete & Computational Geometry 72(3), 1348-1368, 2024. arXiv:2207.13839, DOI: <https://doi.org/10.1007/s00454-023-00553-6>.

This proves sharp lower bounds on individual face numbers for strongly regular CW spheres in a bounded-vertex regime. It is substantial partial progress on lower-bound phenomena for strongly regular CW spheres, but it does not prove the stated fatness inequality for all strongly regular CW 3-spheres.

9. **Face numbers of shellable CW balls and spheres.** Joshua Hinman. Israel Journal of Mathematics, 2025. arXiv:2409.08427, DOI: <https://doi.org/10.1007/s11856-025-2887-y>.

This proves lower bounds for shellable strongly regular CW balls and spheres, and for CL-shellable plus dual CL-shellable face posets. The conclusions are not the exact global inequality $F \geq 5/2$ for all strongly regular CW 3-spheres.

10. **Fat Shellable Spheres.** Joshua Hinman. Preprint, 2025. arXiv:2509.20771, URL: <https://arxiv.org/abs/2509.20771>.

This constructs arbitrarily fat strongly regular CW 3-spheres that are shellable and dual shellable. It strengthens the upper/unbounded-fatness side, but it does not settle the lower-bound question.

2.20.3.4. What remains unresolved

The missing case is the full nonsimplicial, nonpolytopal class of strongly regular CW 3-spheres. Existing results prove the inequality for convex 4-polytope boundaries and simplicial/simple subclasses, and provide several related lower-bound theorems or enumerations, but no citable theorem found covers every strongly regular CW 3-sphere and implies

$$2(f_1 + f_2 - 20) \geq 5(f_0 + f_3 - 10).$$

2.20.4. Remaining open problems recorded by the database

2.20.4.1. Open problem 1

Let S be a strongly regular 3-sphere that is 2-simple and 2-simplicial (every edge lies in exactly three facets and every 2-face is a triangle), and suppose that S is not the boundary of the 4-simplex. Is its fatness

$$F(S) = \frac{f_1(S) + f_2(S) - 20}{f_0(S) + f_3(S) - 10}$$

always at least 4?

Reference supplied by the database. Philip Brinkmann, *f-Vector Spaces of Polytopes, Spheres, and Eulerian Lattices*, doctoral dissertation, Department of Mathematics and Computer Science, Freie Universität Berlin, 2016, Chapter 1, Conjecture 1; DOI: <https://doi.org/10.17169/refubium-13151>; repository record: [Refubium](#).

2.20.5. Primary-source pointer

The originating ICM record is *Face Numbers of 4-Polytopes and 3-Spheres* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=605>.

2.21. 99. Random Points, Convex Bodies, Lattices [2002-geom-87]

Record: 2002-geom-87

Subfield: Discrete and polyhedral geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Random Points, Convex Bodies, Lattices

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=512>

2.21.1. Problem statement

Let $K \subset \mathbb{R}^d$ be a rational polytope given by m linear inequalities with integral coefficients. Define the size of each inequality to be the number of bits needed to encode it as a binary string, and let ϕ be the sum of these sizes over the m defining inequalities. Let

$$I(K) = \text{conv}(\mathbb{Z}^d \cap K)$$

be the integer convex hull of K , and for a polytope P , let $f_i(P)$ denote the number of i -dimensional faces of P . Prove or disprove that, for every $i = 0, 1, \dots, d-1$,

$$f_i(I(K)) \ll \phi^{d-1},$$

where the implied constant in $\ll$ may depend on d and on the number m of defining inequalities.

2.21.2. Historical background, literature, and current progress

2.21.3. Verdict: partially resolved

2.21.3.1. Precise problem

The problem asks for the following all-face strengthening of the classical integer-hull vertex bound. If a rational polytope $K \subset \mathbb{R}^d$ is given by m integral linear inequalities of total binary encoding size ϕ , and

$$I(K) = \text{conv}(K \cap \mathbb{Z}^d),$$

then for every $i = 0, \dots, d-1$ one wants

$$f_i(I(K)) \leq C_{d,m} \phi^{d-1}.$$

The important point is that this is not just the vertex case $i = 0$; it asks for facets and all intermediate face numbers with the same exponent.

2.21.3.2. Original source

Title: “Random Points, Convex Bodies, Lattices”

Author: Imre Barany

Venue/year: Proceedings of the International Congress of Mathematicians, Beijing 2002, Vol. III, pp. 527-536; arXiv preprint, 2003

arXiv/URL: arXiv:math/0304462, <https://arxiv.org/abs/math/0304462>

Relevant passage: Section 4 defines $I(K)$, recalls the Cook-Hartmann-Kannan-McDiarmid bound for $f_0(I(K))$, and then states that “most likely” the same inequality holds for all $i = 0, \dots, d - 1$. This is exactly the question.

2.21.3.3. Progress found

Title: “On integer points in polyhedra”

Authors: William Cook, Martin Hartmann, Ravindran Kannan, Colin McDiarmid

Venue/year: Combinatorica 12, 27-37 (1992)

DOI/URL: <https://doi.org/10.1007/BF01191202>

Result: Gives an upper bound on the number of vertices of the integer hull of a rational polyhedron in terms of dimension, number of inequalities, and encoding size. In the usual row-size notation this is cited as

$$|\text{vert}(P^I)| \leq 2m^n (6n^2\phi)^{n-1}.$$

Since the maximum row size is at most the total size used in the problem statement, this proves the requested bound for $i = 0$. It does not prove the same exponent for facets or intermediate faces.

Title: “On integer points in polyhedra: A lower bound”

Authors: Imre Barany, Roger Howe, Laszlo Lovasz

Venue/year: Combinatorica 12, 135-142 (1992)

DOI/URL: <https://doi.org/10.1007/BF01204716>

Result: Constructs, for fixed d , rational simplices of size at most ϕ whose integer hulls have $\gg_d \phi^{d-1}$ vertices. This shows the vertex exponent $d - 1$ is sharp, and Barany notes that the same exponent would be best possible for all faces if the all-face statement were true.

Title: “On the number of vertices in integer linear programming problems”

Author: Nikolai Yu. Zolotykh

Venue/year: arXiv preprint, 2006

arXiv/DOI/URL: arXiv:math/0611356, <https://doi.org/10.48550/arXiv.math/0611356>

Result: Surveys and improves bounds for the number of vertices of integer hulls. Its facet discussion only derives weaker polynomial bounds from vertex bounds via McMullen’s general face-number theorem; it does not give the conjectured $C_{d,m}\phi^{d-1}$ bound for all faces.

Title: “New Bounds for the Vertices of the Integer Hull”

Authors: Sebastian Berndt, Klaus Jansen, Kim-Manuel Klein

Venue/year: Symposium on Simplicity in Algorithms (SOSA), SIAM, 2021, pp. 25-36; arXiv:2006.10847

DOI/URL: <https://doi.org/10.1137/1.9781611976496.3>, <https://arxiv.org/abs/2006.10847>

Result: Gives later improved bounds and enumeration results for vertices of integer hulls in canonical integer-programming settings. This confirms continuing progress on vertex complexity, but it does not resolve the all-face bound in the problem statement.

2.21.3.4. What is resolved

The vertex case $f_0(I(K)) \ll_{d,m} \phi^{d-1}$ is fully resolved and sharp. The full problem also follows in dimensions $d \leq 3$: for polygons, edges and vertices are comparable, and for 3-polytopes Euler/incidence relations bound edges and facets linearly in the number of vertices.

2.21.3.5. What remains unresolved

I found no published or otherwise citable theorem proving

$$f_i(I(K)) \ll_{d,m} \phi^{d-1}$$

for every i in arbitrary fixed dimension. In dimensions $d \geq 4$, the known vertex bound plus general polytope face-number estimates gives only a weaker polynomial upper bound with a larger exponent. Thus the exact all-face statement in the problem statement remains unresolved beyond the vertex and low-dimensional consequences.

2.21.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

2.21.5. Primary-source pointer

The originating ICM record is *Random Points, Convex Bodies, Lattices* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=512>.

2.22. 100. Random Points, Convex Bodies, Lattices [2002-geom-88]

Record: 2002-geom-88

Subfield: Discrete and polyhedral geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Random Points, Convex Bodies, Lattices

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=512>

2.22.1. Problem statement

Fix an integer $d \geq 2$ and a constant $D < \infty$. Let C^d denote the class of convex bodies in $\mathbb{R}^d$, and let $\mathbb{Z}^d$ be the standard integer lattice. For a convex body $L \subset \mathbb{R}^d$, let

$$I(L) = \text{conv}(\mathbb{Z}^d \cap L)$$

be its integer convex hull. For a polytope P , let $f_i(P)$ denote the number of i -dimensional faces of P .

Let G be the compact group of Euclidean isometries of $\mathbb{R}^d$ modulo translations by elements of $\mathbb{Z}^d$, equipped with its normalized Haar probability measure, and choose $g \in G$ according to this measure. For $K \in C^d$, define

$$u_K(x) = \text{Vol}(K \cap (2x - K)), \quad x \in K,$$

where $2x - K = \{2x - y : y \in K\}$ is the reflection of K about x , and write

$$K(u_K < t) = \{x \in K : u_K(x) < t\}.$$

Prove or disprove that, for every face dimension $i = 0, 1, \dots, d-1$, uniformly over all $K \in C^d$ for which the ratio of the radius of the smallest Euclidean ball containing K to the radius of the largest Euclidean ball contained in K is at most D , one has

$$\text{Vol } K(u_K < 1) \ll_{d,D} \mathbb{E}_g f_i(I(gK)) \ll_{d,D} \text{Vol } K(u_K < 1)$$

as $\text{Vol } K \rightarrow \infty$, where the Vinogradov constants implicit in $\ll_{d,D}$ depend only on d and D .

2.22.2. Historical background, literature, and current progress

2.22.3. Verdict: partially resolved

2.22.3.1. Problem checked

The problem asks for the conjectural extension of Barany's Theorem 4.3 from vertices to every face dimension: for fixed $d \geq 2$ and bounded roundness D , uniformly over all D -rounded

convex bodies $K \subset \mathbb{R}^d$ with $\text{Vol } K \rightarrow \infty$, prove

$$\text{Vol } K(u_K < 1) \ll_{d,D} \mathbb{E}_g f_i(I(gK)) \ll_{d,D} \text{Vol } K(u_K < 1)$$

for every $i = 0, \dots, d-1$, where $I(gK) = \text{conv}(\mathbb{Z}^d \cap gK)$ and g is a random Euclidean isometry modulo integer translations.

2.22.3.2. Literature and progress

1. Random points, convex bodies, lattices

Author: Imre Barany.

Venue/year: Proceedings of the International Congress of Mathematicians, Beijing 2002, Vol. III, 527-536; arXiv preprint 2003.

DOI/arXiv/URL: arXiv:math/0304462, <https://doi.org/10.48550/arXiv.math/0304462>.

Status contribution: This is the source of Theorem 4.3. It states the two-sided estimate only for $f_0(I(gK))$, the expected number of vertices. Immediately after the theorem it says that the same is probably true for i -dimensional faces, but that there is “no proof in sight.” Thus the source explicitly frames the stated all-face statement as a conjectural extension, not as a proved theorem.

2. The Randomized Integer Convex Hull

Authors: Imre Barany and Jiri Matousek.

Venue/year: Discrete & Computational Geometry 33, 3-25, 2005.

DOI/arXiv/URL: <https://doi.org/10.1007/s00454-003-0836-1>.

Status contribution: This is the main citable source for the randomized integer convex hull theorem. Its abstract and later survey formulations say it estimates the expected number of vertices of $I_L(K)$ for sufficiently round convex bodies of sufficiently large volume. It resolves the stated $i = 0$ case, but it does not prove the estimates for all higher-dimensional faces in general dimension.

3. Random points and lattice points in convex bodies

Author: Imre Barany.

Venue/year: Bulletin of the American Mathematical Society 45(3), 339-365, 2008.

DOI/arXiv/URL: <https://doi.org/10.1090/S0273-0979-08-01210-X>.

Status contribution: The survey restates the randomized integer hull theorem as Theorem 13.3:

$$\text{Vol } K(v < 1) \ll \mathbb{E} f_0(I_L(K)) \ll \text{Vol } K(v < 1)$$

for D -rounded K . It then says that most likely $\mathbb{E} f_j(I_L(K))$ behaves like $\mathbb{E} f_j(K_n)$ for all j , but that a proof looks hopeless at that time. This confirms that the all- j extension was not known in the standard survey literature.

4. The convex hull of the integer points in a large ball

Authors: Imre Barany and David G. Larman.

Venue/year: Mathematische Annalen 312, 167-181, 1998.

DOI/arXiv/URL: <https://doi.org/10.1007/s002080050217>.

Status contribution: This proves all face-number orders for the deterministic integer hull of a large Euclidean ball, $P_r = \text{conv}(\mathbb{Z}^d \cap rB^d)$. This is important related progress, but it is a special deterministic/smooth-body case and does not imply the uniform randomized statement for all D -rounded convex bodies.

5. Expected mean width of the randomized integer convex hull

Authors: Binh Hong Ngoc and Matthias Reitzner.

Venue/year: Mathematika 67(2), 422-433, 2021.

DOI/arXiv/URL: <https://doi.org/10.1112/mtk.12080>; arXiv:2003.06864.

Status contribution: This later paper studies the expected mean width difference and gives related intrinsic-volume upper bounds. It explicitly treats metric/intrinsic-volume quantities rather than f_i -face counts, and it does not resolve the all-face conjecture.

2.22.3.3. What remains unresolved

the full statement is not covered by the literature found. The known theorem proves the vertex case $i = 0$. In dimensions 2 and 3, elementary fixed-dimensional face-vector inequalities make the higher face counts comparable to the number of vertices once the integer hull is full-dimensional. However, for arbitrary fixed d , especially $d \geq 4$, no published or otherwise citable result found proves the required uniform two-sided estimate for every $i = 0, \dots, d - 1$ and every D -rounded convex body.

Therefore the problem is partially resolved, not fully resolved.

2.22.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

2.22.5. Primary-source pointer

The originating ICM record is *Random Points, Convex Bodies, Lattices* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=512>.

2.23. 101. Face Numbers of 4-Polytopes and 3-Spheres [2002-geom-250]

Record: 2002-geom-250

Subfield: Discrete and polyhedral geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Face Numbers of 4-Polytopes and 3-Spheres

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=605>

2.23.1. Problem statement

Let $\mathcal{T}$ be a face-to-face tiling of $\mathbb{R}^3$ by convex 3-dimensional polytopes. Call $\mathcal{T}$ **normal** if there are constants $r, R > 0$ such that every tile of $\mathcal{T}$ has inradius at least r and circumradius at most R . For $i \in \{0, 1, 2, 3\}$ and $\rho > 0$, let $f_i^{\mathcal{T}}(\rho)$ be the number of i -dimensional faces of $\mathcal{T}$ that intersect the interior of the ball of radius ρ centered at the origin. Whenever the limit exists, define the **fatness** of $\mathcal{T}$ by

$$F(\mathcal{T}) = \lim_{\rho \rightarrow \infty} \frac{f_1^{\mathcal{T}}(\rho) + f_2^{\mathcal{T}}(\rho)}{f_0^{\mathcal{T}}(\rho) + f_3^{\mathcal{T}}(\rho)}.$$

Determine whether there is a universal constant $C < \infty$ such that $F(\mathcal{T}) \leq C$ for every normal tiling $\mathcal{T}$ of $\mathbb{R}^3$ for which this limit exists.

2.23.2. Historical background, literature, and current progress

The exact problem remains unresolved. No citable proof of a universal upper bound, and no family of normal tilings of $\mathbb{R}^3$ with unbounded fatness, was found. The gap is the simultaneous production or control of both large average tile complexity and large average incidence at vertices; constructions making only one of these quantities large do not settle the displayed ratio.

The strongest direct lower-bound construction located is a family of normal lattice-transitive tilings whose fatness tends to 6. Thus any universal bound would have to be at least 6, but this gives no unbounded family. Moreover, Ziegler explains that a universal bound for these normal 3-dimensional tilings would imply a bound for the fatness of convex 4-polytopes. That latter problem is itself still reported as open, with the best known convex-polytope constructions having fatness tending to 9; consequently, the implication has not yielded a bound for the tilings.

Key references:

- “Face Numbers of 4-Polytopes and 3-Spheres,” Günter M. Ziegler, *Proceedings of the International Congress of Mathematicians (Beijing 2002)*, Vol. III, Higher Education Press, 2002, pp. 625–634, arXiv:math/0208073. <https://arxiv.org/abs/math/0208073>. Section 7 defines normal tilings and their asymptotic fatness, explicitly asks

the exact question, proves the connection to bounded fatness of 4-polytopes, and reports normal lattice-transitive tilings with fatness arbitrarily close to 6.

- **“Convex Polytopes: Extremal Constructions and f -Vector Shapes,”** Günter M. Ziegler, in *Geometric Combinatorics*, IAS/Park City Mathematics Series 13, American Mathematical Society, 2007, pp. 617–691, DOI 10.1090/pcms/013/10; arXiv:math/0411400. <https://arxiv.org/abs/math/0411400>. Lecture 4 again identifies the 3-dimensional tiling problem and states that either high average numbers of vertices per tile or high average numbers of tiles per vertex can be achieved separately, while simultaneous achievement remains the issue. The same work records convex 4-polytopes with fatness tending to 9, which is progress on the related polytope problem rather than a resolution for tilings.
- **“Projected Products of Polygons,”** Günter M. Ziegler, *Electronic Research Announcements of the American Mathematical Society* 10 (2004), 122–134, DOI 10.1090/S1079-6762-04-00137-4. This constructs 4-polytopes whose fatness tends to 9. It improves the related finite-polytope lower bound but neither proves an upper bound nor constructs normal tilings of unbounded fatness.
- **“How Many Facets on Average can a Tile Have in a Tiling?”**, Nikolai Dolbilin and Masaharu Tanemura, *Forma* 21(3) (2006), 177–196. <https://forma.katachi-jp.com/pdf/2103/21030177.pdf>. The authors construct periodic face-to-face tilings of $\mathbb{R}^3$ whose tiles have arbitrarily many facets, including periodic monotypic and Voronoi examples. This is substantive one-sided progress, but it does not force the fatness ratio to be unbounded because it does not simultaneously establish the required large vertex-to-tile incidence.
- **“Ten Problems in Geometry,”** Moritz W. Schmitt and Günter M. Ziegler, in Marjorie Senechal (ed.), *Shaping Space: Exploring Polyhedra in Nature, Art, and the Geometrical Imagination*, 2nd ed., Springer, 2013, pp. 279–289 and 315–319, DOI 10.1007/978-0-387-92714-5_22. This later problem collection again asks whether a normal face-to-face tiling can simultaneously have many vertices in its tiles and many tiles at its vertices, confirming that the tiling obstruction remained open after the $9 - \varepsilon$ polytope construction.

What remains is to prove one constant C valid for every normal tiling for which the defining limit exists, or to construct a sequence of such tilings with $F(\mathcal{T}) \rightarrow \infty$. No substantive progress beyond finite lower bounds and the separate realization of the two high-incidence phenomena was found.

2.23.3. Remaining open problems recorded by the database

2.23.3.1. Open problem 1

Let $\mathcal{T}$ be a face-to-face tiling of $\mathbb{R}^3$ by convex 3-dimensional polytopes that is invariant under all translations in some full-rank lattice $\Lambda \subset \mathbb{R}^3$. For $i \in \{0, 1, 2, 3\}$, let $f_i(\mathcal{T}/\Lambda)$ be the number of Λ -orbits of i -dimensional faces of $\mathcal{T}$, and define

$$F(\mathcal{T}) = \frac{f_1(\mathcal{T}/\Lambda) + f_2(\mathcal{T}/\Lambda)}{f_0(\mathcal{T}/\Lambda) + f_3(\mathcal{T}/\Lambda)}.$$

Determine whether there is a universal constant $C < \infty$ such that $F(\mathcal{T}) \leq C$ for every such triply periodic tiling $\mathcal{T}$.

Reference supplied by the database. Günter M. Ziegler, “Convex Polytopes: Extremal Constructions and f -Vector Shapes,” in *Geometric Combinatorics*, IAS/Park City Mathematics Series 13, American Mathematical Society, 2007, pp. 617–691, Lecture 4, Section 4.2 (the triply periodic tiling case of the tiling problem), DOI: 10.1090/pcms/013/10; arXiv:math/0411400, <https://arxiv.org/abs/math/0411400>.

2.23.3.2. Open problem 2

Let $\mathcal{T}$ be a face-to-face tiling of $\mathbb{R}^3$ by convex 3-dimensional polytopes, and suppose that its tiles belong to only finitely many congruence classes. For $i \in \{0, 1, 2, 3\}$ and $\rho > 0$, let $f_i^{\mathcal{T}}(\rho)$ be the number of i -dimensional faces of $\mathcal{T}$ that intersect the interior of the ball of radius ρ centered at the origin. Whenever the limit exists, define

$$F(\mathcal{T}) = \lim_{\rho \rightarrow \infty} \frac{f_1^{\mathcal{T}}(\rho) + f_2^{\mathcal{T}}(\rho)}{f_0^{\mathcal{T}}(\rho) + f_3^{\mathcal{T}}(\rho)}.$$

Determine whether there is a universal constant $C < \infty$ such that $F(\mathcal{T}) \leq C$ for every such tiling $\mathcal{T}$ for which this limit exists.

Reference supplied by the database. Günter M. Ziegler, “Convex Polytopes: Extremal Constructions and f -Vector Shapes,” in *Geometric Combinatorics*, IAS/Park City Mathematics Series 13, American Mathematical Society, 2007, pp. 617–691, Lecture 4, Section 4.2 (the finite-congruence-class tiling case of the tiling problem), DOI: 10.1090/pcms/013/10; arXiv:math/0411400, <https://arxiv.org/abs/math/0411400>.

2.23.4. Primary-source pointer

The originating ICM record is *Face Numbers of 4-Polytopes and 3-Spheres* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=605>.

2.24. 102. Discrete Mathematics Methods and Challenges [2002-geom-292]

Record: 2002-geom-292

Subfield: Discrete and polyhedral geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Discrete Mathematics Methods and Challenges

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.1/ICM2002.1.ocr.pdf#page=124>

2.24.1. Problem statement

For each integer $d \geq 1$ and each bounded set $A \subset \mathbb{R}^d$, define its Euclidean diameter by

$$\text{diam}(A) = \sup\{\|x - y\|_2 : x, y \in A\}.$$

Let $m(d)$ be the smallest positive integer such that every bounded set $A \subset \mathbb{R}^d$ containing at least two points can be partitioned into at most $m(d)$ subsets, each of Euclidean diameter strictly smaller than $\text{diam}(A)$. Prove that there exists an absolute constant $c > 1$ such that $m(d) > c^d$ for every integer $d \geq 1$.

2.24.2. Historical background, literature, and current progress

Verdict: unresolved. The quantity $m(d)$ in the statement is the Euclidean Borsuk number, usually denoted $f(d)$ or $b(d)$. No citable result found proves a lower bound $m(d) > c^d$ for any absolute $c > 1$. The strongest known asymptotic lower bound is

$$m(d) \geq (1.225589 \dots + o(1))^{\sqrt{d}},$$

which is subexponential in d . The strongest general asymptotic upper bound remains

$$m(d) \leq (\sqrt{3/2} + o(1))^d.$$

Thus the known results leave a gap between an exponent of order $\sqrt{d}$ and one of order d . In particular, they do not establish the assertion in the statement. A 2025 preprint, published in 2026, explicitly records these as the best bounds for $m(d) = \chi(d, 1)$ and states that even the order of growth of $\log m(d)$ is unknown.

Key references:

- A. M. Raigorodskii, “On a Bound in Borsuk’s Problem,” *Russian Mathematical Surveys* 54(2), 1999, 453–454. DOI: <https://doi.org/10.1070/RM1999v054n02ABEH000146>. The theorem proves the current record lower bound $m(d) \geq R(d)(1.225589 \dots)^{\sqrt{d}}$ with a factor $R(d)$ subexponential on the $\sqrt{d}$ scale, equivalently $m(d) \geq (1.225589 \dots + o(1))^{\sqrt{d}}$.

- **Oded Schramm**, “Illuminating Sets of Constant Width,” *Mathematika* 35(2), 1988, 180–189. DOI: <https://doi.org/10.1112/S0025579300015175>. As a corollary of an illumination bound, this paper proves the general upper estimate $m(d) \leq (\sqrt{3/2} + o(1))^d$.
- **Jean Bourgain and Joram Lindenstrauss**, “On Covering a Set in $\mathbb{R}^N$ by Balls of the Same Diameter,” in *Geometric Aspects of Functional Analysis (1989–90)*, *Lecture Notes in Mathematics* 1469, Springer, 1991, 138–144. DOI: <https://doi.org/10.1007/BFb0089220>. This gives another proof of the same asymptotic upper bound.
- **Arthur I. Bikeev and Andrei M. Raigorodskii**, “On Upper Bounds for the Number of Parts in the Problem of Partitioning Sets into Parts of Smaller Diameter,” *Matematicheskie Zametki* 119(3), 2026, 340–351. DOI: <https://doi.org/10.4213/mzm14820>; arXiv:2508.14578. The introduction identifies the preceding lower and upper estimates as the current best bounds for $m(d)$ and explicitly notes that the order of $\log m(d)$ remains unknown; its new results concern upper bounds for the stricter parameter $\chi(d, b)$ and do not prove the required exponential lower bound at $b = 1$.

2.24.3. Remaining open problems recorded by the database

2.24.3.1. Open problem 1

For each integer $d \geq 1$, let $f(d)$ be the smallest positive integer such that every bounded set $A \subset \mathbb{R}^d$ of positive Euclidean diameter can be partitioned into at most $f(d)$ subsets, each having Euclidean diameter strictly smaller than that of A . Let $g(d)$ be defined in the same way, but with A restricted to finite subsets of $\mathbb{R}^d$. Determine whether $f(d) = g(d)$ for every integer $d \geq 1$.

Reference supplied by the database. Gil Kalai, “Some Old and New Problems in Combinatorial Geometry I: Around Borsuk’s Problem,” Problem 2.2, in *Surveys in Combinatorics 2015*, London Mathematical Society Lecture Note Series 424, Cambridge University Press, 2015, pp. 147–174. DOI: <https://doi.org/10.1017/CBO9781316106853.005>; arXiv:1505.04952.

2.24.3.2. Open problem 2

Let n and k be positive integers, and let $\mathcal{F}$ be a family of k -element subsets of $[n] = \{1, 2, \dots, n\}$ such that $A \cap B \neq \emptyset$ for all distinct $A, B \in \mathcal{F}$. Determine whether $\mathcal{F}$ can always be partitioned into at most n subfamilies such that, within each subfamily, $|A \cap B| \geq 2$ for all distinct members A and B .

Reference supplied by the database. Gil Kalai, “Some Old and New Problems in Combinatorial Geometry I: Around Borsuk’s Problem,” Problem 2.4 (Larman’s conjecture for $t = 1$), in *Surveys in Combinatorics 2015*, London Mathematical Society Lecture Note Series 424, Cambridge University Press, 2015, pp. 147–174. DOI: <https://doi.org/10.1017/CBO9781316106853.005>; arXiv:1505.04952.

2.24.3.3. Open problem 3

Determine the smallest integer $d \geq 1$ for which there exists a bounded set $A \subset \mathbb{R}^d$ of positive Euclidean diameter that cannot be partitioned into at most $d + 1$ subsets, each having Euclidean diameter strictly smaller than that of A .

Reference supplied by the database. Gil Kalai, “Some Old and New Problems in Combinatorial Geometry I: Around Borsuk’s Problem,” Problem 2.13, in *Surveys in Combinatorics 2015*, London Mathematical Society Lecture Note Series 424, Cambridge University Press, 2015, pp. 147-174. DOI: <https://doi.org/10.1017/CBO9781316106853.005>; arXiv:1505.04952.

2.24.4. Primary-source pointer

The originating ICM record is *Discrete Mathematics Methods and Challenges* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.1/ICM2002.1.ocr.pdf#page=124>.

2.25. 103. Discrete Mathematics Methods and Challenges [2002-geom-293]

Record: 2002-geom-293

Subfield: Discrete and polyhedral geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Discrete Mathematics Methods and Challenges

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.1/ICM2002.1.ocr.pdf#page=124>

2.25.1. Problem statement

For each positive integer n , let ℓ_1^n denote $\mathbb{R}^n$ equipped with the norm

$$\|x\|_1 = \sum_{i=1}^n |x_i|.$$

Let $e(\ell_1^n)$ be the maximum cardinality of a set $A \subseteq \mathbb{R}^n$ for which there exists a constant $d > 0$ such that $\|x - y\|_1 = d$ for every pair of distinct points $x, y \in A$. Prove that $e(\ell_1^n) = 2n$ for every positive integer n .

2.25.2. Historical background, literature, and current progress

The statement is unresolved. The vectors $\{\pm e_i : 1 \leq i \leq n\}$ give the lower bound $e(\ell_1^n) \geq 2n$ in every dimension, but the matching upper bound is known only for $n \leq 4$. The best general result remains an upper bound of order $n \log n$; in particular, the gap $2n \leq e(\ell_1^n) \leq Cn \log n$ is not closed for any general $n \geq 5$. A June 2026 paper devoted to Kusner's conjecture explicitly confirms that the ℓ_1 equality is still known only in dimensions at most four.

Key references:

- Richard K. Guy, "An Olla-Podrida of Open Problems, Often Oddly Posed," *The American Mathematical Monthly* 90(3) (1983), 196-200, DOI: <https://doi.org/10.1080/00029890.1983.11971188>. This article records Kusner's conjecture, including the assertion $e(\ell_1^n) = 2n$.
- Hans-Jurgen Bandelt, Victor Chepoi, and Monique Laurent, "Embedding into Rectilinear Spaces," *Discrete & Computational Geometry* 19 (1998), 595-604, DOI: <https://doi.org/10.1007/PL00009370>. The paper proves that the maximum is $2n$ for $n \leq 3$.
- Jack Koolen, Monique Laurent, and Alexander Schrijver, "Equilateral Dimension of the Rectilinear Space," *Designs, Codes and Cryptography* 21 (2000), 149-164, DOI: <https://doi.org/10.1023/A:1008391712305>. Theorem 9 supplies the next exact case, $e(\ell_1^4) = 8$, and the paper leaves the general conjecture open.
- Noga Alon and Pavel Pudlak, "Equilateral Sets in ℓ_p^n ," *Geometric and Functional Analysis* 13(3) (2003), 467-482, DOI: <https://doi.org/10.1007/s00039-003-0418-7>. For $p = 1$, they prove $e(\ell_1^n) \leq Cn \log n$ for an absolute constant C ; this is partial progress and does not yield the conjectured coefficient-free linear bound.

- Hong-Jun Ge, Zixiang Xu, and Yang Zhou, “Kusner’s Conjecture: Exact Values and Linear Bounds,” preprint (2026), arXiv:2606.03987, <https://arxiv.org/abs/2606.03987>. Section 1.1 states that $e(\ell_1^n) = 2n$ is known only for $n \leq 4$; its almost-linear result for $p = 1$ is again $O(n \log n)$, leaving the exact problem unresolved.

2.25.3. Remaining open problems recorded by the database

2.25.3.1. Open problem 1

Let ℓ_1^5 denote $\mathbb{R}^5$ equipped with the norm $\|x\|_1 = \sum_{i=1}^5 |x_i|$, and let $e(\ell_1^5)$ be the maximum cardinality of a set $A \subseteq \mathbb{R}^5$ for which there exists a constant $d > 0$ such that $\|x - y\|_1 = d$ for every pair of distinct points $x, y \in A$. Prove or disprove that $e(\ell_1^5) = 10$.

Reference supplied by the database. Konrad J. Swanepoel, “Equilateral Sets in Finite-Dimensional Normed Spaces,” in *Seminar of Mathematical Analysis*, ed. Daniel Girela Alvarez, Genaro Lopez Acedo, and Rafael Villa Caro, Secretariado de Publicaciones, Universidad de Sevilla, Seville, 2004, pp. 195-237, Problem 1 (Kusner), arXiv:math/0406264, <https://arxiv.org/abs/math/0406264>.

2.25.4. Primary-source pointer

The originating ICM record is *Discrete Mathematics Methods and Challenges* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.1/ICM2002.1.ocr.pdf#page=124>.

2.26. 104. Order and Disorder in Energy Minimization [2010-geom-257]

Record: 2010-geom-257

Subfield: Discrete and polyhedral geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2010: Order and Disorder in Energy Minimization

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.4/ICM2010.4.pdf#page=251>

2.26.1. Problem statement

Let

$$S^2 = \{x \in \mathbb{R}^3 : \|x\| = 1\}.$$

For a five-point subset $C \subset S^2$ and a function $f : (0, 4] \rightarrow \mathbb{R}$, define

$$E_f(C) = \sum_{\{x, y\} \subset C} f(\|x - y\|^2).$$

A function f is *completely monotonic* if it is infinitely differentiable and satisfies

$$(-1)^k f^{(k)}(r) \geq 0 \quad \text{for every integer } k \geq 0 \text{ and every } r \in (0, 4].$$

Define the triangular bipyramid

$$T = \left\{ (0, 0, 1), (0, 0, -1), (1, 0, 0), \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}, 0\right), \left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}, 0\right) \right\},$$

and, for $-1 < h < 1$, define the square pyramid

$$Q_h = \left\{ (0, 0, 1), (\sqrt{1-h^2}, 0, h), (-\sqrt{1-h^2}, 0, h), (0, \sqrt{1-h^2}, h), (0, -\sqrt{1-h^2}, h) \right\}.$$

Prove that for every completely monotonic function $f : (0, 4] \rightarrow \mathbb{R}$, either T or Q_h for some $h \in (-1, 1)$ minimizes E_f among all five-point subsets of S^2 .

2.26.2. Historical background, literature, and current progress

The conjecture is **unresolved**. No citable result found proves that the triangular bipyramid or a square pyramid is a global minimizer for every completely monotonic function of squared chordal distance. In particular, the known results do not cover arbitrary completely monotonic potentials or arbitrary positive mixtures of the special potentials treated in the literature.

Henry Cohn, "Order and Disorder in Energy Minimization," *Proceedings of the International Congress of Mathematicians 2010, Hyderabad*, vol. IV, Hindustan Book Agency, 2010, pp. 2416–2443, arXiv:1003.3053, explicitly states the exact assertion as Conjecture 3.1. The closely related experimental precursor is Brandon Ballinger, Grigoriy Blekherman, Henry Cohn, Noah Giansiracusa, Elizabeth Kelly, and Achill Schürmann, "Experimental Study of

Energy-Minimizing Point Configurations on Spheres,” *Experimental Mathematics* 18(3) (2009), 257–283, DOI: 10.1080/10586458.2009.10129052; it reports numerical evidence for the same two candidate families, not a proof.

Established partial progress includes the following:

- Alexander Tumanov, “Minimal Biquadratic Energy of Five Particles on a 2-Sphere,” *Indiana University Mathematics Journal* 62(6) (2013), 1717–1731, DOI: 10.1512/iumj.2013.62.5148, proves that the triangular bipyramid is the unique minimizer for the biquadratic class he considers (including positive combinations of the auxiliary potentials $4 - r^2$ and $(4 - r^2)^2$). This is a finite-dimensional subclass of completely monotonic potentials, not the full class.
- Richard Evan Schwartz, “Five Point Energy Minimization: A Synopsis,” *Constructive Approximation* 51(3) (2020), 537–564, DOI: 10.1007/s00365-020-09500-7, proves that for the Riesz potential d^{-s} there is a transition value $S \approx 15.048$ such that the triangular bipyramid is the unique minimizer for $0 < s < S$ (and treats the transition endpoint). Thus it verifies a large one-parameter subclass $f(t) = t^{-s/2}$, but not general completely monotonic f .
- Richard Evan Schwartz, “Divide and Conquer: A Distributed Approach to Five Point Energy Minimization,” revised preprint (2025), arXiv:2301.05090v3, gives a computer-assisted proof that a square pyramid is the unique Riesz-energy minimizer for $S < s \leq 15 + 25/512$, in addition to the triangular-bipyramid range below S . It does not establish the claimed dichotomy for all larger Riesz exponents or for arbitrary completely monotonic potentials.
- Andriy V. Bondarenko, D. P. Hardin, and E. B. Saff, “Mesh Ratios for Best-Packing and Limits of Minimal Energy Configurations,” *Acta Mathematica Hungarica* 142(1) (2014), 118–131, DOI: 10.1007/s10474-013-0339-3, proves that as $s \rightarrow \infty$, every limit of five-point Riesz-energy minimizers is the square pyramid with its base on the equator. This asymptotic result does not identify the minimizer at every finite large s .

Consequently, the strongest results cover particular polynomial potentials, a substantial but bounded Riesz-exponent range, and the large-exponent limit. The gap is the universal quantifier over all completely monotonic potentials (and, within the Riesz subfamily, a complete finite-exponent classification beyond the currently proved range).

2.26.3. Remaining open problems recorded by the database

2.26.3.1. Open problem 1

Let

$$S^2 = \{x \in \mathbb{R}^3 : \|x\| = 1\}, \quad E_s(C) = \sum_{\{x,y\} \subset C} \|x - y\|^{-s}$$

for $s > 0$ and every five-point subset $C \subset S^2$. Define

$$T = \left\{ (0, 0, 1), (0, 0, -1), (1, 0, 0), \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}, 0\right), \left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}, 0\right) \right\}$$

and, for $-1 < h < 1$,

$$Q_h = \left\{ (0, 0, 1), (\sqrt{1-h^2}, 0, h), (-\sqrt{1-h^2}, 0, h), (0, \sqrt{1-h^2}, h), (0, -\sqrt{1-h^2}, h) \right\}.$$

Prove that there exists a transition exponent $s_0 \in (15.048, 15.049)$ such that T minimizes E_s among all five-point subsets of S^2 for every $0 < s \leq s_0$, while for every $s > s_0$, some square pyramid Q_h minimizes E_s among all five-point subsets of S^2 .

Reference supplied by the database. Theodor W. Melnyk, Osvald Knop, and William Robert Smith, “Extremal Arrangements of Points and Unit Charges on a Sphere: Equilibrium Configurations Revisited,” *Canadian Journal of Chemistry* 55(10) (1977), 1745–1761. DOI: 10.1139/v77-246.

2.26.3.2. Open problem 2

For each real number $s > 0$, classify, up to orthogonal transformations of $\mathbb{R}^3$ and permutations of the points, all critical points of

$$(x_1, \dots, x_5) \mapsto \sum_{1 \leq i < j \leq 5} \|x_i - x_j\|^{-s}$$

on the configuration manifold

$$\{(x_1, \dots, x_5) \in (S^2)^5 : x_i \neq x_j \text{ whenever } i \neq j\}, \quad S^2 = \{x \in \mathbb{R}^3 : \|x\| = 1\}.$$

Reference supplied by the database. Peter D. Dragnev and Oleg R. Musin, “Log-Optimal $(d + 2)$ -Configurations in d -Dimensions,” *Transactions of the American Mathematical Society, Series B* 10 (2023), 155–170, Section 6.2(a). DOI: 10.1090/btran/118; arXiv:1909.09909.

2.26.4. Primary-source pointer

The originating ICM record is *Order and Disorder in Energy Minimization* (2010). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.4/ICM2010.4.pdf#page=251>.

2.27. 105. Recent results around the diameter of polyhedra [2014-geom-260]

Record: 2014-geom-260

Subfield: Discrete and polyhedral geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: Recent results around the diameter of polyhedra

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.4/ICM2014.4.pdf#page=849>

2.27.1. Problem statement

Determine whether there exists a polynomial $p(d, n)$ such that, for every full-dimensional polyhedron $P \subseteq \mathbb{R}^d$ with at least one vertex and exactly n facets, the diameter of the vertex-edge graph of P is at most $p(d, n)$. Here the vertex-edge graph has the vertices of P as its graph vertices, with two vertices adjacent when they are the endpoints of a one-dimensional face of P , and its diameter is the maximum, over all pairs of vertices, of the length of a shortest path between them.

2.27.2. Historical background, literature, and current progress

The problem is **unresolved**. It is the polynomial Hirsch conjecture. The strongest general upper bound remains quasipolynomial rather than polynomial: Sukegawa proved an upper bound of the form

$$(n - d)^{\log_2 O(d/\log d)}$$

for the maximum vertex-edge diameter of a d -dimensional polyhedron with n facets. No known result replaces the dimension-dependent exponent by an absolute constant. Recent literature still explicitly describes a polynomial bound for the ordinary combinatorial diameter as an open central question; in particular, the problem remains open even for bounded polyhedra (polytopes), so the full class in the statement is not covered.

Key references and progress:

- Noriyoshi Sukegawa, "An Asymptotically Improved Upper Bound on the Diameter of Polyhedra," *Discrete & Computational Geometry* 62 (2019), 690–699. DOI: <https://doi.org/10.1007/s00454-018-0016-y>. This gives the current asymptotically strongest general quasipolynomial bound above, but not a polynomial bound of fixed degree in d and n .
- Francisco Santos, "A Counterexample to the Hirsch Conjecture," *Annals of Mathematics* 176 (2012), 383–412. DOI: <https://doi.org/10.4007/annals.2012.176.1.7>. Santos constructs a 43-dimensional polytope with 86 facets and diameter at least 44, disproving the proposed linear bound $n - d$. This counterexample does not give a superpolynomial diameter and therefore does not disprove the polynomial Hirsch conjecture.
- Nicolas Bonifas, Marco Di Summa, Friedrich Eisenbrand, Nicolai Hähnle, and Martin Niemeier, "On Sub-determinants and the Diameter of Polyhedra," *Discrete & Computa-*

tional Geometry 52 (2014), 102–115. DOI: <https://doi.org/10.1007/s00454-014-9601-x>. For $P = \{x : Ax \leq b\}$ with integral A , this proves $O(\Delta^2 d^4 \log(d\Delta))$, where Δ bounds the absolute values of subdeterminants of A (and a slightly better exponent for bounded P). This is substantive polynomial progress under an additional arithmetic hypothesis, not a bound for arbitrary real polyhedra.

- Daniel Dadush, Zhuan Khye Koh, Bento Natura, and László A. Végh, “On Circuit Diameter Bounds via Circuit Imbalances,” *Mathematical Programming* 206 (2024), 631–662. DOI: <https://doi.org/10.1007/s10107-024-02107-x>. The paper explicitly records the ordinary combinatorial polynomial Hirsch conjecture as open and proves bounds for the relaxed circuit diameter depending on circuit imbalance; circuit steps need not be edges.
- Bento Natura, “Circuit Diameter of Polyhedra is Strongly Polynomial,” preprint (2026), arXiv:2602.06958. This proves an $O(m^2 \log m)$ bound for circuit diameter, resolving the circuit analogue. Because circuit walks may move in circuit directions that are not edges of the vertex-edge graph, it does not resolve the problem stated here.

Thus the remaining gap is between a linear lower bound (including known violations of the specific $n - d$ estimate) and a quasipolynomial general upper bound; no polynomial upper bound and no superpolynomial counterexample is known for the exact class in the statement.

2.27.3. Remaining open problems recorded by the database

2.27.3.1. Open problem 1

Let $d, n \geq 1$, and let $G = (V, E)$ be a graph whose vertices are degree- d monomials in $\mathbb{Z}[x_1, \dots, x_n]$. Suppose that, for every $u, v \in V$, there is a path from u to v in G such that every vertex w on the path is divisible by $\gcd(u, v)$. Prove that the diameter of G is at most $(n - 1)d$.

Reference supplied by the database. Friedrich Eisenbrand, “Recent Results Around the Diameter of Polyhedra,” in *Proceedings of the International Congress of Mathematicians, Seoul 2014*, Vol. IV, Kyung Moon Sa, 2014, pp. 849–861, Conjecture 2.8 (Hähnle conjecture), https://www.mathunion.org/fileadmin/IMU/ICM2014/offline/download/Proceedings_Volume_IV.pdf.

2.27.3.2. Open problem 2

Construct a deterministic simplex algorithm with the following property. Let $A \in \mathbb{R}^{m \times d}$ have full column rank, let $\delta \in (0, 1]$, and suppose that every linearly independent set of d rows of A has the δ -distance property: the Euclidean distance from each row in the set to the span of the other $d - 1$ rows is at least δ times the Euclidean norm of that row. For every $b \in \mathbb{R}^m$ such that $P = \{x \in \mathbb{R}^d : Ax \leq b\}$ has a vertex, every objective vector $c \in \mathbb{R}^d$ whose maximum over P is finite, and every given initial vertex of P , the algorithm must reach a c -maximizing vertex by following edges of P in a number of pivots bounded by a polynomial in d and $1/\delta$, independently of m .

Reference supplied by the database. Friedrich Eisenbrand, “Recent Results Around the Diameter of Polyhedra,” in *Proceedings of the International Congress of Mathematicians, Seoul 2014*, Vol. IV, Kyung Moon Sa, 2014, pp. 849–861, open problem following Theorem 3.3, https://www.mathunion.org/fileadmin/IMU/ICM2014/offline/download/Proceedings_Volume_IV.pdf.

2.27.3.3. Open problem 3

Determine whether there exists a polynomial $q(d, k)$ such that the following holds for every pair of integers $d, k \geq 1$. For every lattice polytope $P \subseteq [0, k]^N$ of affine dimension d , every linear functional $c \in \mathbb{R}^N$ taking distinct values on the vertices of P , and every vertex v of P , there is an edge path $v = v_0, v_1, \dots, v_r$ from v to the unique c -maximizing vertex such that $c(v_{i+1}) > c(v_i)$ for all i and $r \leq q(d, k)$, independently of the ambient dimension N .

Reference supplied by the database. Alexander E. Black, “Small Shadows of Lattice Polytopes,” in *Proceedings of the 2023 Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, SIAM, 2023, pp. 1669–1679, Conjecture 1.8, DOI: 10.1137/1.9781611977554.ch63; arXiv:2204.09129.

2.27.4. Primary-source pointer

The originating ICM record is *Recent results around the diameter of polyhedra* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.4/ICM2014.4.pdf#page=849>.

2.28. 106. Metric dimension reduction A snapshot of the Ribe program [2018-geom-149]

Record: 2018-geom-149

Subfield: Discrete and polyhedral geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Metric dimension reduction A snapshot of the Ribe program

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>

2.28.1. Problem statement

Determine the asymptotic order, up to universal constant factors, of the function $S(\delta)$ as $\delta \in (0, 1/2)$ varies, where $S(\delta)$ is the infimum of all constants $S > 0$ with the following property: for every $n \in \mathbb{N}$, there exist an integer $k \in \mathbb{N}$ with $k \leq S \log n$ and points $y_1, \dots, y_n \in \mathbb{R}^k$ such that, for every pair of distinct indices $i, j \in \{1, \dots, n\}$,

$$1 \leq \|y_i - y_j\|_2 \leq 1 + \delta,$$

where $\|\cdot\|_2$ denotes the Euclidean norm.

2.28.2. Historical background, literature, and current progress

2.28.3. Verdict: partially resolved

2.28.3.1. Precise problem

The problem asks for the asymptotic order, up to universal constant factors as $\delta \rightarrow 0$, of the least constant $S(\delta)$ such that every n admits a δ -almost equilateral n -point set in Euclidean dimension at most $S(\delta) \log n$; equivalently, for each δ , how large the exponential growth rate of the maximum size of a set in $\mathbb{R}^k$ with all distances in $[1, 1+\delta]$ can be.

2.28.3.2. Current progress

1. Metric dimension reduction: A snapshot of the Ribe program

- Authors: Assaf Naor
- Venue/year: Proceedings of the International Congress of Mathematicians–Rio de Janeiro 2018, Vol. I, World Scientific, pp. 759-837, 2018
- DOI/arXiv/URL: arXiv:1809.02376, <https://arxiv.org/abs/1809.02376>
- Relevance: This is the original source of the exact problem considered here, stated as Question 33, “dimension reduction for the vertices of the simplex.” Naor records the known bounds $S(\delta) \leq C/\delta^2$ and $S(\delta) \geq c/(\delta^2 \log(1/\delta))$, and presents the logarithmic gap as unresolved.

2. Extensions of Lipschitz mappings into a Hilbert space

- Authors: William B. Johnson and Joram Lindenstrauss
- Venue/year: Conference in Modern Analysis and Probability, Contemporary Mathematics 26, American Mathematical Society, pp. 189-206, 1984
- DOI/arXiv/URL: available reprint <https://web.stanford.edu/class/cs114/readings/JL-Johnson.pdf>
- Relevance: The Johnson-Lindenstrauss lemma applied to the regular simplex gives the upper bound $S(\delta) \leq C/\delta^2$ after rescaling, since n equidistant points in Hilbert space can be embedded into $O(\delta^{-2} \log n)$ Euclidean dimensions with pairwise distances within factor $1+O(\delta)$.

3. Problems and results in extremal combinatorics-I

- Authors: Noga Alon
- Venue/year: Discrete Mathematics 273(1-3), pp. 31-53, 2003
- DOI/arXiv/URL: [https://doi.org/10.1016/S0012-365X\(03\)00227-9](https://doi.org/10.1016/S0012-365X(03)00227-9)
- Relevance: Gives the near-tight lower-bound method for dimension reduction of the simplex, cited by Naor for this problem. It yields the lower bound with the extra $\log(1/\delta)$ loss.

4. Perturbed Identity Matrices Have High Rank: Proof and Applications

- Authors: Noga Alon
- Venue/year: Combinatorics, Probability and Computing 18(1-2), pp. 3-15, 2009
- DOI/arXiv/URL: <https://doi.org/10.1017/S0963548307008917>; author PDF <https://web.math.princeton.edu/~nalon/PDFS/identity1.pdf>
- Relevance: Proposition 3.1 states that if $P_0, \dots, P_n$ are points in R^k whose mutual distances all lie in $[1, 1+\epsilon]$, with $1/\sqrt{n} \leq \epsilon \leq 1/10$, then $k \geq c\epsilon^{-2} \log n / \log(1/\epsilon)$. This directly implies $S(\delta) \geq c/(\delta^2 \log(1/\delta))$ for the problem. The paper explicitly identifies removal of the $\log(1/\epsilon)$ term as the main open problem for this rank method.

5. Optimal compression of approximate inner products and dimension reduction

- Authors: Noga Alon and Bo'az Klartag
- Venue/year: 58th IEEE Symposium on Foundations of Computer Science (FOCS 2017), pp. 639-650, 2017
- DOI/arXiv/URL: <https://doi.org/10.1109/FOCS.2017.65>; arXiv:1610.00239, <https://arxiv.org/abs/1610.00239>
- Relevance: This later work proves sharp results for approximate distance/inner-product sketches and lower bounds for general Euclidean dimension reduction in several regimes. It does not resolve the present problem because its lower-bound configurations and sketching/general-point-set conclusions do not prove an upper or lower bound matching the exact near-equilateral/simplex existence question.

6. The Johnson-Lindenstrauss Lemma Is Optimal for Linear Dimensionality Reduction

- Authors: Kasper Green Larsen and Jelani Nelson
- Venue/year: ICALP 2016, LIPIcs 55, Article 82, pp. 82:1-82:11, 2016
- DOI/arXiv/URL: <https://doi.org/10.4230/LIPIcs.ICALP.2016.82>

- **Relevance:** Proves optimal $\Omega(\epsilon^{-2} \log n)$ lower bounds for linear maps on suitable point sets, and discusses why the simplex is not known to force this bound in the nonlinear/arbitrary-placement setting. Thus it is related evidence for JL optimality, but it does not imply the full answer to this problem.

2.28.3.3. What remains unresolved

The direct literature gives

$$c/(\delta^2 \log(1/\delta)) \leq S(\delta) \leq C/\delta^2.$$

I did not find a published or otherwise citable theorem proving the missing lower bound $S(\delta) \geq c/\delta^2$, nor a construction improving the upper bound below C/δ^2 for all n . Later sharp JL and sketching results concern general point sets, linear embeddings, additive sketches, or different spherical-code regimes, and their hypotheses/conclusions do not cover exactly the full near-equilateral/simplex problem stated here. The exact asymptotic order of $S(\delta)$ therefore remains unresolved.

2.28.4. Remaining open problems recorded by the database

2.28.4.1. Open problem 1

Is there a universal constant $c > 0$ such that, whenever $0 < \epsilon < 1/2$ and n is sufficiently large as a function of ϵ , every real $n \times n$ matrix $B = (b_{ij})$ with $b_{ii} = 1$ and $|b_{ij}| \leq \epsilon$ for $i \neq j$ satisfies $\text{rank}(B) \geq c\epsilon^{-2} \log n$?

Reference supplied by the database. Noga Alon, “Perturbed Identity Matrices Have High Rank: Proof and Applications,” *Combinatorics, Probability and Computing* 18 (2009), nos. 1–2, 3–15, Theorem 2.1 and §8 (“Concluding remarks”), main open problem (removal of the $\log(1/\epsilon)$ factor); Proposition 3.1 gives the application to almost-equilateral simplex vertices. DOI: 10.1017/S0963548307008917. <https://doi.org/10.1017/S0963548307008917>.

2.28.5. Primary-source pointer

The originating ICM record is *Metric dimension reduction A snapshot of the Ribe program* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>.

2.29. 107. Systems of points with Coulomb interactions [2018-geom-158]

Record: 2018-geom-158

Subfield: Discrete and polyhedral geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Systems of points with Coulomb interactions

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=930>

2.29.1. Problem statement

For the Coulomb jellium crystallization problem in $\mathbb{R}^d$, let $C \subset \mathbb{R}^d$ be an infinite point configuration of density 1 in a uniform neutralizing background, and let $\mathbb{W}(C)$ denote the renormalized Coulomb energy per unit volume obtained from electric potentials H satisfying $-\Delta H = C_d(C - 1)$, after subtracting the divergent self-energy of the point charges.

Prove or disprove that there exists a dimension d_0 such that, for every $d \geq d_0$, no covolume-one lattice $\Lambda \subset \mathbb{R}^d$ minimizes $\mathbb{W}$ among all unit-density configurations.

2.29.2. Historical background, literature, and current progress

2.29.3. Verdict: partially resolved

2.29.3.1. Precise problem checked

The problem asks whether the Coulomb jellium renormalized energy problem has been resolved in high dimension: for unit-density infinite point configurations in a uniform neutralizing background in $\mathbb{R}^d$, with Coulomb renormalized energy $\mathbb{W}$, is it proved that there is a dimension d_0 such that for every $d \geq d_0$, no covolume-one lattice $\Lambda \subset \mathbb{R}^d$ is a global minimizer of $\mathbb{W}$ among all unit-density configurations?

2.29.3.2. Status

I did not find a theorem proving the high-dimensional assertion. In particular, I did not find a published or citable result giving a threshold d_0 , nor even a general theorem showing that all sufficiently high-dimensional Bravais lattices are beaten by non-lattice configurations for the Coulomb renormalized energy $\mathbb{W}$. The exact problem therefore remains unresolved.

The literature does contain substantial partial progress and related crystallization results, including proved lattice minimization in special dimensions. These do not imply the problem statement, because they concern fixed low/special dimensions or restricted classes rather than all sufficiently large d .

2.29.3.3. Relevant literature and progress

1. Title: “Systems of points with Coulomb interactions” Authors: Sylvia Serfaty Venue: Proceedings of the International Congress of Mathematicians 2018; arXiv preprint Year: 2018 arXiv/URL: arXiv:1712.04095, <https://arxiv.org/abs/1712.04095> Result/status contribution: This is the apparent source of the formulation. Section 4.3 states that determining $\min \mathbb{W}$ is open except for one-dimensional analogues; in dimension 2 the triangular lattice is known to minimize only among lattices, while its global optimality is conjectural. It also records that in $d \geq 3$ the minimization of $\mathbb{W}$ restricted to lattices was open except for known strict local minimizers in dimensions 4, 8, and 24. The text explicitly describes the high-dimensional non-lattice expectation as “folklore knowledge”, motivated by sphere packing. This source does not prove the high-dimensional assertion.
2. Title: “Crystallization for Coulomb and Riesz interactions as a consequence of the Cohn-Kumar conjecture” Authors: Mircea Petrache and Sylvia Serfaty Venue: Proceedings of the American Mathematical Society 148, 3047-3057 Year: 2020 DOI/arXiv/URL: DOI 10.1090/proc/15003; arXiv:1908.09714, <https://arxiv.org/abs/1908.09714> Result/status contribution: Theorem 2 proves that the Cohn-Kumar universal optimality conjecture for the relevant lattice implies minimization of the periodic renormalized energies and of $\mathbb{W}$ for Coulomb cases $d \geq 2$ and Riesz cases in the stated range. Corollary 3 applies this in dimensions 8 and 24, where the Cohn-Kumar conjecture is known, and shows that the E_8 lattice and the Leech lattice achieve the corresponding minima. This is genuine progress on Coulomb/Riesz jellium crystallization, but it proves lattice minimizers in dimensions 8 and 24 and does not prove that lattices fail to minimize in all sufficiently high dimensions.
3. Title: “Universal optimality of the E_8 and Leech lattices and interpolation formulas” Authors: Henry Cohn, Abhinav Kumar, Stephen D. Miller, Danylo Radchenko, and Maryna Viazovska Venue: Annals of Mathematics 196 Year: 2022 DOI/arXiv/URL: DOI 10.4007/annals.2022.196.3.1; arXiv:1902.05438, <https://arxiv.org/abs/1902.05438> Result/status contribution: This proves universal optimality of E_8 in dimension 8 and the Leech lattice in dimension 24 for completely monotone interaction potentials. Petrache-Serfaty use this result to obtain Coulomb/Riesz renormalized-energy crystallization in dimensions 8 and 24. It does not address arbitrary high dimensions or prove non-lattice optimality.
4. Title: “Coulomb and Riesz gases: The known and the unknown” Authors: Mathieu Lewin Venue: Journal of Mathematical Physics 63, 061101 Year: 2022 DOI/arXiv/URL: DOI 10.1063/5.0086835; arXiv:2202.09240, <https://arxiv.org/abs/2202.09240> Result/status contribution: This survey summarizes the current crystallization status. Theorem 36 states crystallization results for dimension 1 in specified ranges and for dimensions 8 and 24 for $s \in [d-2, \infty) \setminus \{d\}$, covering the Coulomb exponent $s = d-2$. The surrounding discussion emphasizes that many long-range crystallization questions remain open and that even the two-dimensional Cohn-Kumar conjecture is open. This supports that there is no known theorem resolving the stated high-dimensional non-lattice assertion.
5. Title: “Floating Wigner Crystal and Periodic Jellium Configurations” Author: Asbjorn Christian Drachmann Lauritsen Venue: Journal of Mathematical Physics 62, 063305 Year: 2021 DOI/arXiv/URL: DOI 10.1063/5.0049411; arXiv:2103.07975, <https://arxiv.org/abs/2103.07975> Result/status contribution: This paper develops periodic jellium configurations and related lower bounds, especially in dimension 2. It is relevant

to the renormalized-energy/jellium framework, but it does not prove high-dimensional non-lattice minimization.

2.29.3.4. What remains unresolved

The missing theorem is exactly the quantified high-dimensional statement: existence of d_0 such that, for every $d \geq d_0$, every covolume-one lattice in $\mathbb{R}^d$ fails to be a global minimizer of the Coulomb renormalized energy $\mathbb{W}$ among all unit-density configurations. Existing results prove special-dimensional lattice minimization, restricted lattice-class minimization, local optimality, or structural results for related periodic/jellium energies. None of these results gives the required high-dimensional non-lattice conclusion.

2.29.4. Remaining open problems recorded by the database

2.29.4.1. Open problem 1

Among all covolume-one lattices $L \subset \mathbb{R}^3$, does the body-centered-cubic lattice uniquely minimize the three-dimensional Coulomb jellium energy $\mathbb{W}(L)$ (equivalently, the analytically continued Epstein zeta energy $\zeta_L(1)$)?

Reference supplied by the database. Xavier Blanc and Mathieu Lewin, “The Crystallization Conjecture: A Review,” *EMS Surveys in Mathematical Sciences* **2** (2015), no. 2, 255–306, §2.5–§2.6 and Figure 7 (the explicit still-open prediction that BCC minimizes the three-dimensional Epstein zeta energy for $0 < s < 3/2$, which includes the Coulomb exponent $s = 1$), DOI: <https://doi.org/10.4171/EMSS/13>, arXiv: [1504.01153](https://arxiv.org/abs/1504.01153). The equivalence with lattice jellium energy and the continuing absence of a three-dimensional global lattice-optimality theorem are recorded in Laurent Bétermin, “Three-Dimensional Lattice Ground States for Riesz and Lennard-Jones–Type Energies,” *Studies in Applied Mathematics* **150** (2023), no. 2, 354–415, §1.2 and §2, DOI: <https://doi.org/10.1111/sapm.12533>.

2.29.5. Primary-source pointer

The originating ICM record is *Systems of points with Coulomb interactions* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=930>.

2.30. 108. Systems of points with Coulomb interactions [2018-geom-264]

Record: 2018-geom-264

Subfield: Discrete and polyhedral geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Systems of points with Coulomb interactions

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=930>

2.30.1. Problem statement

For each $n \in \{2, 8, 24\}$, let $\Lambda_n \subset \mathbb{R}^n$ be, respectively, the triangular lattice A_2 , the E_8 root lattice, or the Leech lattice. A periodic point configuration in $\mathbb{R}^n$ is a disjoint union

$$C = \bigcup_{j=1}^N (L + v_j),$$

where $L \subset \mathbb{R}^n$ is a full-rank lattice and $v_j - v_k \notin L$ for $j \neq k$; its point density is $N / \text{vol}(\mathbb{R}^n / L)$. For a function $f : (0, \infty) \rightarrow \mathbb{R}$, define its energy per point by

$$E_f(C) = \frac{1}{N} \sum_{j,k=1}^N \sum_{\substack{x \in L \\ x+v_j-v_k \neq 0}} f(|x+v_j-v_k|^2).$$

Prove that Λ_n is universally optimal for each $n \in \{2, 8, 24\}$: namely, for every completely monotone function $f : (0, \infty) \rightarrow \mathbb{R}$ satisfying

$$(-1)^k f^{(k)}(r) \geq 0 \quad (r > 0, k = 0, 1, 2, \dots)$$

and $f(r) = O(r^{-n/2-\varepsilon})$ as $r \rightarrow \infty$ for some $\varepsilon > 0$, and for every periodic point configuration $C \subset \mathbb{R}^n$ having the same point density as Λ_n , one has

$$E_f(C) \geq E_f(\Lambda_n).$$

2.30.2. Historical background, literature, and current progress

The problem is **unresolved**. The E_8 and Leech-lattice cases are proved, but the triangular-lattice case remains open for general periodic configurations. Since the statement requires all three cases, the two high-dimensional results do not resolve it.

- Henry Cohn and Abhinav Kumar, “Universally optimal distribution of points on spheres,” *Journal of the American Mathematical Society* **20** (2007), 99–148, Conjecture 9.4. DOI: <https://doi.org/10.1090/S0894-0347-06-00546-7>; arXiv: [math/0607446](https://arxiv.org/abs/math/0607446). This is the source of the Euclidean conjecture in precisely the periodic, fixed-density, completely-monotone setting used in the statement.

- Henry Cohn, Abhinav Kumar, Stephen D. Miller, Danylo Radchenko, and Maryna Viazovska, “Universal optimality of the E_8 and Leech lattices and interpolation formulas,” *Annals of Mathematics* **196** (2022), 983–1082, Theorem 1.4. DOI: <https://doi.org/10.4007/annals.2022.196.3.3>; arXiv: [1902.05438](https://arxiv.org/abs/1902.05438). Theorem 1.4 proves universal optimality of E_8 and the Leech lattice against all point configurations of the same density, hence in particular against every periodic competitor in the present problem. The paper explicitly states that the two-dimensional case remains open.
- Hugh L. Montgomery, “Minimal theta functions,” *Glasgow Mathematical Journal* **30** (1988), 75–85, Theorem 1. DOI: <https://doi.org/10.1017/S0017089500007047>. Montgomery proves that the triangular lattice minimizes every Gaussian theta energy among two-dimensional lattices of fixed covolume. Bernstein’s representation of completely monotone functions then gives universal optimality among lattices, but not among general periodic configurations (finite unions of lattice translates), so this does not close the planar case.
- Renaud Coulangen and Achill Schürmann, “Energy minimization, periodic sets and spherical designs,” *International Mathematics Research Notices* **2012** (2012), no. 4, 829–848. DOI: <https://doi.org/10.1093/imrn/rnr048>; arXiv: [1005.4373](https://arxiv.org/abs/1005.4373). They prove a local version of the Cohn–Kumar conjecture among periodic sets, including local universal optimality of A_2 ; global optimality is not obtained.
- Douglas P. Hardin and Nathaniel J. Tenpas, “Universally Optimal Periodic Configurations in the Plane,” *Discrete Analysis* **2025** (2025), Paper No. 26, 63 pp., Theorem 1. DOI: <https://doi.org/10.19086/da.144978>; arXiv: [2307.15822](https://arxiv.org/abs/2307.15822). They prove universal optimality for two restricted planar families: a scaled A_2 configuration among four-point A_2 -periodic configurations and a scaled, rotated A_2 configuration among six-point configurations with a specified rectangular period lattice. Their introduction still identifies unrestricted universal optimality of A_2 as open.

Thus the strongest established planar results cover all lattice competitors, a neighborhood among periodic configurations, and certain fixed-period finite-generator classes. What remains is the global energy inequality for the triangular lattice against every periodic configuration of the same density, for every admissible completely monotone potential.

2.30.3. Remaining open problems recorded by the database

2.30.3.1. Open problem 1

Let

$$A_2 = \mathbb{Z}(1, 0) + \mathbb{Z}\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right) \subset \mathbb{R}^2.$$

A periodic point configuration in $\mathbb{R}^2$ is a disjoint union $C = \bigcup_{j=1}^N (L + v_j)$, where L is a full-rank lattice and $v_j - v_k \notin L$ for $j \neq k$. Its density is $N / \text{vol}(\mathbb{R}^2/L)$, and for $f : (0, \infty) \rightarrow \mathbb{R}$ its energy per point is

$$E_f(C) = \frac{1}{N} \sum_{j,k=1}^N \sum_{\substack{x \in L \\ x+v_j-v_k \neq 0}} f(|x+v_j-v_k|^2).$$

Prove that for every completely monotone function $f : (0, \infty) \rightarrow \mathbb{R}$ such that $f(r) =$

$O(r^{-1-\varepsilon})$ as $r \rightarrow \infty$ for some $\varepsilon > 0$, and every periodic point configuration $C \subset \mathbb{R}^2$ with the same density as A_2 , one has

$$E_f(C) \geq E_f(A_2).$$

Here complete monotonicity means that $f \in C^\infty(0, \infty)$ and $(-1)^k f^{(k)}(r) \geq 0$ for every $r > 0$ and every integer $k \geq 0$.

Reference supplied by the database. Henry Cohn and Abhinav Kumar, “Universally optimal distribution of points on spheres,” *Journal of the American Mathematical Society* **20** (2007), 99–148, Conjecture 9.4 (the $n = 2$ case), DOI: <https://doi.org/10.1090/S0894-0347-06-00546-7>, arXiv: [math/0607446](https://arxiv.org/abs/math/0607446).

2.30.3.2. Open problem 2

Let $C \subset \mathbb{R}^2$ be a locally finite point configuration of asymptotic density 1, identified with the measure $\sum_{p \in C} \delta_p$. For a distributional solution H of

$$-\Delta H = 2\pi \left(\sum_{p \in C} \delta_p - 1 \right),$$

set $g(x) = -\log|x|$ and

$$f_\eta(x) = \max\{g(x) - g(\eta), 0\}, \quad H_\eta = H - \sum_{p \in C} f_\eta(\cdot - p).$$

Define

$$\mathcal{U}(H) = \lim_{\eta \rightarrow 0^+} \left(\limsup_{R \rightarrow \infty} \frac{1}{R^2} \int_{[-R/2, R/2]^2} |\nabla H_\eta|^2 dx + 2\pi \log \eta \right)$$

whenever the limit exists, and define the renormalized Coulomb energy of C by

$$\mathcal{W}(C) = \inf \left\{ \mathcal{U}(H) : -\Delta H = 2\pi \left(\sum_{p \in C} \delta_p - 1 \right) \right\},$$

with the infimum of the empty set understood as $+\infty$.

Prove that the covolume-one triangular lattice is a global minimizer of $\mathcal{W}$ among all locally finite point configurations of asymptotic density 1 in $\mathbb{R}^2$.

Reference supplied by the database. Sylvia Serfaty, “Systems of Points with Coulomb Interactions,” *Proceedings of the International Congress of Mathematicians—Rio de Janeiro 2018*, Vol. I, Plenary Lectures, World Scientific, 2018, pp. 935–977, Conjecture 4.2, DOI: https://doi.org/10.1142/9789813272880_0033, arXiv: [1712.04095](https://arxiv.org/abs/1712.04095).

2.30.3.3. Open problem 3

Let

$$A_2 = \mathbb{Z}(1, 0) + \mathbb{Z}\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right), \quad L = \mathbb{Z}(1, 0) + \mathbb{Z}(0, \sqrt{3}),$$

and let $\Omega = [0, 1) \times [0, \sqrt{3})$ be a fundamental domain for L . Define the eight-point configuration

$$\omega_8^* = \left(\frac{1}{2}A_2\right) \cap \Omega \quad \text{in } \mathbb{R}^2/L.$$

For $a > 0$, define the L -periodic Gaussian potential

$$F_{a,L}(x) = \sum_{v \in L} e^{-a|x+v|^2}$$

and, for an eight-point multiset $\omega = \{x_1, \dots, x_8\} \subset \mathbb{R}^2/L$, define

$$E_{a,L}(\omega) = \sum_{\substack{1 \leq i, j \leq 8 \\ i \neq j}} F_{a,L}(x_i - x_j).$$

Prove that ω_8^* is L -universally optimal: for every $a > 0$ and every eight-point multiset $\omega \subset \mathbb{R}^2/L$,

$$E_{a,L}(\omega) \geq E_{a,L}(\omega_8^*).$$

Reference supplied by the database. Douglas P. Hardin and Nathaniel J. Tenpas, “Universally Optimal Periodic Configurations in the Plane,” *Discrete Analysis* **2025** (2025), Paper No. 26, 63 pp., Section 3.4 (the numerically supported first nontrivial case ω_8^*), DOI: <https://doi.org/10.19086/da.144978>, arXiv: [2307.15822](https://arxiv.org/abs/2307.15822).

2.30.4. Primary-source pointer

The originating ICM record is *Systems of points with Coulomb interactions* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=930>.

2.31. 109. Sharp sphere packings [2018-geom-295]

Record: 2018-geom-295

Subfield: Discrete and polyhedral geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Sharp sphere packings

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/uploads/ICM-2018-vol2-ver1-eb.pdf#page=516>

2.31.1. Problem statement

For each real number $p \geq 1$, define the centrally symmetric convex body

$$K_p = \{(x_1, x_2) \in \mathbb{R}^2 : |x_1|^p + |x_2|^p \leq 1\}.$$

A translative packing of K_p is a collection $\{K_p + t : t \in T\}$, where $T \subset \mathbb{R}^2$ and the interiors of the translates are pairwise disjoint. For $z \in \mathbb{R}^2$ and $R > 0$, let $B(z, R) = \{x \in \mathbb{R}^2 : \|x - z\| \leq R\}$ be the Euclidean ball of radius R centered at z . Define the upper density of the packing by

$$\delta(\{K_p + t : t \in T\}) = \limsup_{R \rightarrow \infty} \sup_{z \in \mathbb{R}^2} \frac{\text{area}\left(\left(\bigcup_{t \in T} (K_p + t)\right) \cap B(z, R)\right)}{\text{area}(B(z, R))},$$

and let $\delta_T(K_p)$ be the supremum of these densities over all translative packings of K_p . For an integrable function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, define its Fourier transform by

$$\hat{f}(y) = \int_{\mathbb{R}^2} f(x) e^{-2\pi i x \cdot y} dx.$$

Call f admissible if there exist constants $C > 0$ and $\varepsilon > 0$ such that

$$|f(x)| \leq C(1 + \|x\|)^{-2-\varepsilon} \quad \text{and} \quad |\hat{f}(x)| \leq C(1 + \|x\|)^{-2-\varepsilon}$$

for every $x \in \mathbb{R}^2$. Let $2K_p = \{2x : x \in K_p\}$, and define

$$\text{LP}(K_p) = \inf_f \left\{ \text{area}(K_p) \frac{f(0)}{\hat{f}(0)} : f \not\equiv 0, \quad f(x) \leq 0 \text{ for every } x \notin 2K_p, \quad \hat{f}(y) \geq 0 \text{ for every } y \in \mathbb{R}^2, \quad \hat{f}(0) > 0 \right\},$$

where the infimum is taken over all admissible real-valued functions f .

Prove that

$$\delta_T(K_p) = \text{LP}(K_p)$$

for every real number $p \geq 1$.

2.31.2. Historical background, literature, and current progress

unresolved

The assertion is not known for all $p \geq 1$. In particular, its special case $p = 2$ is the open question whether the Cohn–Elkies linear-programming bound is sharp for circle packing in $\mathbb{R}^2$. Thus the known optimal hexagonal packing density does not settle the stated equality, because the exact value of the corresponding infinite-dimensional linear program remains unproved.

Established progress:

- Henry Cohn and Noam Elkies, “New upper bounds on sphere packings. I,” *Annals of Mathematics* 157 (2003), 689–714. DOI: <https://doi.org/10.4007/annals.2003.157.689>. Appendix B, Theorem B.1 proves, after taking the infimum over auxiliary functions, the inequality $\delta_T(K_p) \leq \text{LP}(K_p)$ for every $p \geq 1$. Conjecture 7.3 explicitly proposes sharpness in dimension two, but the paper does not prove the reverse inequality.
- László Fejes Tóth, “On the densest packing of convex discs,” *Mathematika* 30 (1983), 1–3. DOI: <https://doi.org/10.1112/S0025579300010354>. This proves that the upper density of a translative packing of a planar convex disc cannot exceed its densest lattice-packing density. It determines the geometric form of the packing optimization for the centrally symmetric bodies K_p , but does not evaluate the Fourier-analytic LP bound.
- Maryna Viazovska, “Sharp sphere packings,” in *Proceedings of the International Congress of Mathematicians—Rio de Janeiro 2018, Vol. II*, World Scientific, 2018, 455–466. DOI: https://doi.org/10.1142/9789813272880_0063. Viazovska reports Cohn and Minton’s numerical study of planar L_p -balls and their conjecture that the resulting LP bounds are sharp; she also states separately that the exact LP solution for equal disks is unknown.
- Henry Cohn, “From sphere packing to Fourier interpolation,” *Bulletin of the American Mathematical Society* 61 (2024), 3–22. DOI: <https://doi.org/10.1090/bull/1813>; arXiv:2407.14999. This later survey confirms that the two-dimensional LP bound still appears sharp but that no proof is known.

Consequently, the literature supplies the universal LP upper bound, the independent reduction of planar translative packing to lattice packing, and numerical evidence for sharpness. What remains is to prove or disprove equality with the LP value; even $p = 2$, and therefore the full continuum of parameters required by the statement, remains unresolved.

2.31.3. Remaining open problems recorded by the database

2.31.3.1. Open problem 1

Let

$$K = \{x \in \mathbb{R}^2 : \|x\| \leq 1\}$$

be the Euclidean unit disk. A translative packing of K is a collection $\{K + t : t \in T\}$, where $T \subset \mathbb{R}^2$ and the interiors of the translates are pairwise disjoint. For $z \in \mathbb{R}^2$ and $R > 0$, let $B(z, R) = \{x \in \mathbb{R}^2 : \|x - z\| \leq R\}$, and let $\delta_T(K)$ be the supremum, over all translative packings of K , of

$$\limsup_{R \rightarrow \infty} \sup_{z \in \mathbb{R}^2} \frac{\text{area}\left(\left(\bigcup_{t \in T} (K + t)\right) \cap B(z, R)\right)}{\text{area}(B(z, R))}.$$

For an integrable function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, define

$$\hat{f}(y) = \int_{\mathbb{R}^2} f(x) e^{-2\pi i x \cdot y} dx.$$

Call f admissible if there exist constants $C > 0$ and $\varepsilon > 0$ such that both $|f(x)|$ and $|\hat{f}(x)|$ are at most $C(1 + \|x\|)^{-2-\varepsilon}$ for every $x \in \mathbb{R}^2$. Define

$$\text{LP}(K) = \inf_f \left\{ \pi \frac{f(0)}{\hat{f}(0)} : f \not\equiv 0, \quad f(x) \leq 0 \text{ for } x \notin 2K, \quad \hat{f}(y) \geq 0 \text{ for all } y \in \mathbb{R}^2, \quad \hat{f}(0) > 0 \right\},$$

where $2K = \{2x : x \in K\}$ and the infimum is over admissible real-valued functions.

Prove that

$$\delta_T(K) = \text{LP}(K).$$

Reference supplied by the database. Maryna Viazovska, “Sharp sphere packings,” in *Proceedings of the International Congress of Mathematicians—Rio de Janeiro 2018, Vol. II*, World Scientific, 2018, 455–466, DOI: https://doi.org/10.1142/9789813272880_0063.

2.31.4. Primary-source pointer

The originating ICM record is *Sharp sphere packings* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/uploads/ICM-2018-vol2-ver1-eb.pdf#page=516>.

2.32. 110. the upper bound side of the story [2022-geom-175]

Record: 2022-geom-175

Subfield: Discrete and polyhedral geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2022: the upper bound side of the story

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-564-3-volume-6.pdf#page=471>

2.32.1. Problem statement

Prove that for every integer $d \geq 4$, if $s(d, n)$ denotes the number of simplicial $(d - 1)$ -spheres with vertex set $[n] = \{1, \dots, n\}$, counted as labeled simplicial complexes up to equality on this fixed vertex set, and if $sn(d, n)$ denotes the number of those simplicial $(d - 1)$ -spheres that are $\lfloor d/2 \rfloor$ -neighborly, meaning that every subset of at most $\lfloor d/2 \rfloor$ vertices is a face, then

$$\lim_{n \rightarrow \infty} \frac{\log sn(d, n)}{\log s(d, n)} = 1.$$

2.32.2. Historical background, literature, and current progress

2.32.3. Verdict: partially resolved

2.32.3.1. Precise problem

For each fixed integer $d \geq 4$, let $s(d, n)$ be the number of labeled simplicial $(d - 1)$ -spheres on the fixed vertex set $[n]$, and let $sn(d, n)$ be the number of those spheres that are $\lfloor d/2 \rfloor$ -neighborly. The question is whether

$$\lim_{n \rightarrow \infty} \frac{\log sn(d, n)}{\log s(d, n)} = 1$$

for every $d \geq 4$. Equivalently, it asks whether neighborly simplicial spheres account for the same logarithmic asymptotic growth as all simplicial spheres, in every dimension covered by the statement.

2.32.3.2. Current status

No citable result found proves this full limit for all $d \geq 4$. The literature gives substantial lower bounds for $sn(d, n)$, but the best known bounds do not imply the required log-ratio statement. Recent surveys explicitly still list the conjecture as out of reach and pose weaker lower-bound problems.

2.32.3.3. Key literature and progress

1. Many triangulated spheres

Authors: Gil Kalai

Venue: *Discrete & Computational Geometry* 3, 1-14

Year: 1988

DOI/URL: <https://doi.org/10.1007/BF02187893>

Role: Original source for the asymptotic enumeration framework for simplicial spheres and for the speculation that there are very many neighborly spheres. Kalai's Section 6.3 phrases the log-ratio question for fixed even d ; later sources state the now-standard all- $d \geq 4$ formulation.

2. Many Neighborly Polytopes and Oriented Matroids

Author: Arnau Padrol

Venue: *Discrete & Computational Geometry* 50, 865-902

Year: 2013

DOI/arXiv/URL: <https://doi.org/10.1007/s00454-013-9544-7>,
arXiv:1202.2810, <https://arxiv.org/abs/1202.2810>

Role: Gives large families of neighborly polytopes and non-realizable neighborly oriented matroids. As summarized in Novik-Zheng and Novik's ICM survey, this implies only $sn(d, n) \geq 2^{\Omega(n \log n)}$ for the neighborly sphere count, for $d \geq 4$. This is partial progress, not the stated limit.

3. Many triangulated odd-dimensional spheres

Authors: Eran Nevo, Francisco Santos, Stedman Wilson

Venue: *Mathematische Annalen* 364, 737-762

Year: 2016

DOI/URL: <https://doi.org/10.1007/s00208-015-1232-x>

Role: Improves lower bounds for the total number $s(d, n)$ in odd sphere dimensions. Together with Kalai, Pfeifle-Ziegler, and Stanley-type upper bounds, current surveys summarize the best general bounds as $2^{\Omega(n^{\lceil \log(d/2) \rceil})} \leq s(d, n) \leq 2^{O(n^{\lceil \log(d/2) \rceil} \log n)}$ for fixed $d \geq 4$. These denominator bounds remain too coarse to settle the log-ratio conjecture.

4. Many neighborly spheres

Authors: Isabella Novik and Hailun Zheng

Venue: *Mathematische Annalen* 388, 969-984

Year: 2024 (published online 2022)

DOI/arXiv/URL: <https://doi.org/10.1007/s00208-022-02538-x>,
arXiv:2104.04476, <https://arxiv.org/abs/2104.04476>

Result: The abstract and Theorem 1.2 state that for every $d \geq 5$, there are at least $2^{\Omega(n^{\lceil \log((d-1)/2) \rceil})}$ $\lfloor (d-1)/2 \rfloor$ -neighborly simplicial $(d-1)$ -spheres with n vertices. The paper explicitly says this improves Padrol's bound but is still far from resolving Kalai's conjecture. It also does not cover the full $d=4$ case in the problem statement.

5. Face numbers: the upper bound side of the story

Author: Isabella Novik

Venue: *Proceedings of the International Congress of Mathematicians 2022*, EMS Press, pp. 4622-4644

Year: 2023

DOI/URL: <https://doi.org/10.4171/ICM2022/43>, <https://ems.press/books/standalone/278/5571>

Role: Survey source for the exact all- $d \geq 4$ formulation as Conjecture 7.1. It states the current bounds on $s(d, n)$, summarizes Padrol and Novik-Zheng, and says that

Conjecture 7.1 is likely out of reach; even proving $sn(d, n) \geq 2^{\Omega(n^{\lfloor d/2 \rfloor})}$ is presented as a more feasible open goal.

6. Neighborly spheres and transversal numbers

Authors: Isabella Novik and Hailun Zheng

Venue: *Open Problems in Algebraic Combinatorics*, Proceedings of Symposia in Pure Mathematics 110, American Mathematical Society, Providence, RI, pp. 113–128

Year: 2024

arXiv/URL: arXiv:2208.02071, <https://arxiv.org/abs/2208.02071>

Role: Open-problems survey. It restates Kalai's conjecture as Conjecture 4.1, identifies Novik-Zheng's $d \geq 5$ lower bound as the current best lower bound on $sn(d, n)$, and then poses the weaker Problem 4.2 asking whether $2^{\Omega(n^{\lfloor d/2 \rfloor})}$ neighborly spheres exist.

2.32.3.4. Why this is not fully resolved

The desired conclusion is a statement about the exact relative logarithmic growth of $sn(d, n)$ and $s(d, n)$ for every $d \geq 4$. The best lower bounds for $sn(d, n)$ are:

- $2^{\Omega(n \log n)}$ for all $d \geq 4$ from Padrol-type constructions;
- $2^{\Omega(n^{\lfloor (d-1)/2 \rfloor})}$ for all $d \geq 5$ from Novik-Zheng.

These do not imply

$$\log sn(d, n) / \log s(d, n) \rightarrow 1.$$

For even d , Novik-Zheng's exponent is one power of n below $\lfloor d/2 \rfloor$. For $d=4$, their theorem does not apply. For odd $d \geq 5$, the exponent matches $\lfloor d/2 \rfloor$, but the known upper bound for $s(d, n)$ still has an additional $\log n$ factor and no theorem identifies the logarithmic growth of $sn(d, n)$ with that of all spheres. Thus the literature gives meaningful partial progress, but not a full resolution of the problem statement.

Related recent work on centrally symmetric neighborly spheres, transversal numbers, missing faces, and small or special families of neighborly spheres concerns different problems or imposes extra structure; it does not prove the full all- $d \geq 4$ log-ratio limit for labeled simplicial spheres.

2.32.4. Remaining open problems recorded by the database

2.32.4.1. Open problem 1

For every fixed even integer $d \geq 4$, are there $2^{\Omega(n^{d/2})}$ distinct simplicial $(d-1)$ -spheres on n vertices that are $d/2$ -neighborly?

Reference supplied by the database. Isabella Novik and Hailun Zheng, “Neighborly spheres and transversal numbers,” in *Open Problems in Algebraic Combinatorics*, Proceedings of Symposia in Pure Mathematics, vol. 110, American Mathematical Society, Providence, RI, 2024, pp. 113–128, Problem 4.2. [arXiv:2208.02071](https://arxiv.org/abs/2208.02071).

2.32.5. Primary-source pointer

The originating ICM record is *the upper bound side of the story* (2022). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-564-3-volume-6.pdf#page=471>.

2.33. 111. The magic of 8 and 24 [2022-geom-269]

Record: 2022-geom-269

Subfield: Discrete and polyhedral geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2022: The magic of 8 and 24

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-559-9-volume-1.pdf#page=511>

2.33.1. Problem statement

For each $d \in \{4, 5\}$, let

$$D_d = \left\{ (x_1, \dots, x_d) \in \mathbb{Z}^d : \sum_{i=1}^d x_i \text{ is even} \right\}.$$

The minimum distance between distinct points of D_d is $\sqrt{2}$, so the closed balls of radius $1/\sqrt{2}$ centered at the points of D_d form a sphere packing in $\mathbb{R}^d$. For a packing $\mathcal{D}$ of congruent closed balls in $\mathbb{R}^d$, define its upper density by

$$\bar{\delta}(\mathcal{D}) = \limsup_{R \rightarrow \infty} \frac{\text{vol}((\bigcup_{B \in \mathcal{D}} B) \cap [-R, R]^d)}{(2R)^d}.$$

Prove or disprove that, for each $d \in \{4, 5\}$, every packing of congruent balls in $\mathbb{R}^d$ has upper density at most that of the packing obtained from D_d .

2.33.2. Historical background, literature, and current progress

The exact problem is **unresolved**. Its assertion is conjunctive, and no established result proves the required upper bound for all sphere packings in $\mathbb{R}^5$. Indeed, a peer-reviewed paper published in July 2026 still states that sphere packing is solved only in dimensions 1, 2, 3, 8, 24. The conjectured densities are

$$\delta(D_4) = \frac{\pi^2}{16} = 0.616850275 \dots, \quad \delta(D_5) = \frac{\pi^2}{15\sqrt{2}} = 0.465257613 \dots$$

Established partial progress is as follows.

- A. Korkine and G. Zolotareff, *Sur les formes quadratiques positives quaternaires*, **Mathematische Annalen** 5 (1872), 581-583, DOI <https://doi.org/10.1007/BF01442912>, and A. Korkine and G. Zolotareff, *Sur les formes quadratiques positives*, **Mathematische Annalen** 11 (1877), 242-292, DOI <https://doi.org/10.1007/BF01442667>. These works prove the optimality (and uniqueness up to similarity) of D_4 and D_5 among lattice packings. They do not cover arbitrary, nonlattice packings.
- Alexei Andreanov and Yoav Kallus, *Locally optimal 2-periodic sphere packings*, **Discrete & Computational Geometry** 63 (2020), 182-208, DOI https://doi.org/10.1007/978-1-4939-9832-7_12.

[7/s00454-019-00150-6](#), arXiv:[1704.08156](#). Their enumeration proves that no 2-periodic packing in dimensions 3, 4, 5 is denser than the optimal lattice. Thus the conjecture holds for the 2-periodic subclass, but this does not bound general periodic packings with more translates or nonperiodic packings.

- Henry Cohn, David de Laat, and Andrew Salmon, *Three-point bounds for sphere packing*, preprint (2022), arXiv:[2206.15373](#). Their Theorem 1.1 and rigorously verified semidefinite-programming certificates give all-packing upper bounds 0.636107333 in dimension 4 and 0.512645131 in dimension 5. Both remain strictly above the D_4 and D_5 densities. A later computational preprint by Rasul Tutunov, Alexandre Maraval, Antoine Grosnit, Xihan Li, Jun Wang, and Haitham Bou-Ammar, *Model-Based and Sample-Efficient AI-Assisted Math Discovery in Sphere Packing* (2025), arXiv:[2512.04829](#), reports the slightly smaller numerical values 0.63610277 and 0.5122645, which also leave substantial gaps.
- Rupert Li, *Dual linear programming bounds for sphere packing via discrete reductions*, **Advances in Mathematics** 460 (2025), Article 110043, DOI <https://doi.org/10.1016/j.aim.2024.110043>, arXiv:[2206.09876](#). The paper proves that the Cohn-Elkies two-point linear-programming bound is not sharp in dimensions 4 and 5, so that method alone cannot prove the conjectured densities.
- David de Laat, Nando M. Leijenhorst, and Willem H. H. de Muinck Keizer, *Optimality and uniqueness of the D_4 root system*, preprint (2024), arXiv:[2404.18794](#). They prove that the 24 roots of D_4 form the unique optimal four-dimensional kissing configuration. This settles an important local problem but not the global density problem in $\mathbb{R}^4$.
- Henry Cohn and Isaac Rajagopal, *Variations on Five-Dimensional Sphere Packings*, **Discrete & Computational Geometry** (2026), DOI <https://doi.org/10.1007/s00454-026-00841-x>, arXiv:[2412.00937](#). They construct and classify additional uniform five-dimensional packings attaining, but not exceeding, the D_5 density. They explicitly note that no optimality theorem is known beyond 2-periodic packings.

A non-peer-reviewed preprint by Deep Bhattacharjee, Ushashi Bhattacharya, and Shounak Bhattacharya, *Sphere Packing in Dimension 4*, Preprints.org (2026), version 8, DOI <https://doi.org/10.20944/preprints202606.0856.v8>, claims a proof for D_4 . It is not presently a verified resolution: a stated global step uses the four-dimensional kissing-number bound of 24 to bound all shell-active neighbors with radii in $[2, 2\sqrt{2})$, although that kissing-number theorem bounds contact neighbors at radius exactly 2. No independent source validating the claimed extension or the completeness of the resulting finite chamber list was found. In any case, the unrestricted D_5 case remains open, so the exact two-dimensional assertion in the problem is not completely resolved.

2.33.3. Remaining open problems recorded by the database

2.33.3.1. Open problem 1

Let $C \subset \mathbb{R}^4$ be the set of centers of a packing of closed unit balls, so that $\|p - q\| \geq 2$ for all distinct $p, q \in C$. For each $p \in C$, define its Voronoi cell by

$$V_p = \{x \in \mathbb{R}^4 : \|x - p\| \leq \|x - q\| \text{ for every } q \in C\}.$$

Prove that every such Voronoi cell satisfies $\text{vol}_4(V_p) \geq 8$.

Reference supplied by the database. Oleg R. Musin, *Towards a Proof of the 24-Cell Conjecture*, Acta Mathematica Hungarica 155(1) (2018), 184-199, 24-cell conjecture, DOI: <https://doi.org/10.1007/s10474-018-0828-5>, arXiv: 1712.04099.

2.33.3.2. Open problem 2

Prove or disprove that every finite set $X \subset S^4 = \{x \in \mathbb{R}^5 : \|x\| = 1\}$ satisfying

$$\langle x, y \rangle \leq \frac{1}{2} \quad \text{for all distinct } x, y \in X$$

has cardinality at most 40.

Reference supplied by the database. Henry Cohn and Isaac Rajagopal, *Variations on Five-Dimensional Sphere Packings*, Discrete & Computational Geometry (2026), Section 2, DOI: <https://doi.org/10.1007/s00454-026-00841-x>, arXiv: 2412.00937.

2.33.3.3. Open problem 3

Let $C \subset \mathbb{R}^2$ be a set equipped with a coloring $c : C \rightarrow \mathbb{Z}/4\mathbb{Z}$ such that, for every pair of distinct points $x, y \in C$,

$$c(x) = c(y) \implies \|x - y\| \geq \sqrt{2},$$

$$c(x) - c(y) \equiv \pm 1 \pmod{4} \implies \|x - y\| \geq \frac{\sqrt{5}}{2},$$

$$c(x) - c(y) \equiv 2 \pmod{4} \implies \|x - y\| \geq 1.$$

Prove that

$$\limsup_{r \rightarrow \infty} \frac{\#\{x \in C : \|x\| \leq r\}}{\pi r^2} \leq 1.$$

Reference supplied by the database. Henry Cohn and Isaac Rajagopal, *Variations on Five-Dimensional Sphere Packings*, Discrete & Computational Geometry (2026), Conjecture 4.1, DOI: <https://doi.org/10.1007/s00454-026-00841-x>, arXiv: 2412.00937.

2.33.4. Primary-source pointer

The originating ICM record is *The magic of 8 and 24* (2022). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-559-9-volume-1.pdf#page=511>.

2.34. 112. The magic of 8 and 24 [2022-geom-270]

Record: 2022-geom-270

Subfield: Discrete and polyhedral geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2022: The magic of 8 and 24

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-559-9-volume-1.pdf#page=511>

2.34.1. Problem statement

For a set of centers $X \subset \mathbb{R}^d$ with $\|x - y\| \geq 2r$ for all distinct $x, y \in X$, define the upper density of the corresponding packing of radius- r balls by

$$\bar{\delta}(X, r) = \limsup_{R \rightarrow \infty} \frac{\text{vol}((\bigcup_{x \in X} B(x, r)) \cap [-R, R]^d)}{(2R)^d}.$$

For $1 \leq d \leq 8$, let

$$L_1 = A_1, \quad L_2 = A_2, \quad L_3 = A_3, \quad L_4 = D_4, \quad L_5 = D_5, \quad L_6 = E_6, \quad L_7 = E_7, \quad L_8 = E_8,$$

where

$$A_d = \left\{ (x_1, \dots, x_{d+1}) \in \mathbb{Z}^{d+1} : \sum_{i=1}^{d+1} x_i = 0 \right\}$$

is regarded as a lattice in its d -dimensional real span,

$$D_d = \left\{ (x_1, \dots, x_d) \in \mathbb{Z}^d : \sum_{i=1}^d x_i \equiv 0 \pmod{2} \right\},$$

and the remaining lattices are defined as follows. Let $e_1, \dots, e_8$ be the standard basis of $\mathbb{R}^8$, let $\gamma = (1/2, \dots, 1/2)$, and set

$$\alpha_i = e_i - e_{i+1} \quad (1 \leq i \leq 6), \quad \alpha_7 = e_6 + e_7, \quad \alpha_8 = -\gamma.$$

Then

$$E_6 = \sum_{i=3}^8 \mathbb{Z} \alpha_i, \quad E_7 = \sum_{i=2}^8 \mathbb{Z} \alpha_i, \quad E_8 = D_8 \cup (D_8 + \gamma),$$

with E_6 and E_7 regarded as lattices in their real spans. For each d , let r_d be half the length of a shortest nonzero vector of L_d .

Prove or disprove that for every $d \in \{1, 2, 3, 4, 5, 6, 7, 8\}$, every packing of congruent balls in $\mathbb{R}^d$ has upper density at most $\bar{\delta}(L_d, r_d)$; equivalently, that the lattice packing L_d is a densest sphere packing in $\mathbb{R}^d$.

2.34.2. Historical background, literature, and current progress

The problem is **unresolved**. The assertion is proved in dimensions 1, 2, 3, 8, but no proof or counterexample is known in dimensions 4, 5, 6, 7. A peer-reviewed status statement published in July 2026 still lists the sphere-packing problem as solved only in dimensions 1, 2, 3, 8, 24.

The resolved components are as follows.

- Dimension 1 is elementary: intervals tile the line, so A_1 has density 1.
- László Fejes Tóth, “Über einen geometrischen Satz,” *Mathematische Zeitschrift* **46** (1940), 83–85, DOI <https://doi.org/10.1007/BF01181430>, proves the planar density bound attained by the hexagonal lattice A_2 .
- Thomas C. Hales, “A proof of the Kepler conjecture,” *Annals of Mathematics* (2) **162** (2005), 1065–1185, DOI <https://doi.org/10.4007/annals.2005.162.1065>, proves that no three-dimensional packing is denser than the face-centered-cubic packing $A_3 = D_3$. Thomas Hales, Mark Adams, Gertrud Bauer, Tat Dat Dang, John Harrison, Le Truong Hoang, Cezary Kaliszyk, Victor Magron, Sean McLaughlin, Tat Thang Nguyen, Quang Truong Nguyen, Tobias Nipkow, Steven Obua, Joseph Pleso, Jason Rute, Alexey Solov'yev, Thi Hoai An Ta, Nam Trung Tran, Thi Diep Trieu, Josef Urban, Ky Vu, and Roland Zumkeller, “A formal proof of the Kepler conjecture,” *Forum of Mathematics, Pi* **5** (2017), e2, DOI <https://doi.org/10.1017/fmp.2017.1>, formally verify the complete proof.
- Maryna S. Viazovska, “The sphere packing problem in dimension 8,” *Annals of Mathematics* (2) **185** (2017), 991–1015, DOI <https://doi.org/10.4007/annals.2017.185.3.7>, proves that no packing in $\mathbb{R}^8$ has density greater than the E_8 lattice packing.

For the unresolved dimensions, the candidate densities and the best currently tabulated rigorous upper bounds are

d	candidate	candidate density	rigorous upper bound
4	D_4	$\pi^2/16 = 0.616850275 \dots$	0.636107332 ...
5	D_5	$\pi^2/(15\sqrt{2}) = 0.465257613 \dots$	0.512645130 ...
6	E_6	$\pi^3/(48\sqrt{3}) = 0.372947545 \dots$	0.410303281 ...
7	E_7	$\pi^3/105 = 0.295297873 \dots$	0.321147105 ...

Henry Cohn, David de Laat, and Andrew Salmon, “Three-point bounds for sphere packing,” arXiv:2206.15373 (2022), <https://arxiv.org/abs/2206.15373>, develop rigorously certified semidefinite three-point bounds and obtain the displayed upper bounds in dimensions 4 through 7. Each bound is strictly larger than the corresponding candidate density, so none settles the required all-packings optimum.

The candidates are nevertheless known to be optimal in important restricted classes. Aleksandr Korkine and Georgy Zolotareff, “Sur les formes quadratiques positives quaternaires,” *Mathematische Annalen* **5** (1872), 581–583, DOI <https://doi.org/10.1007/BF01442912>, and “Sur les formes quadratiques positives,” *Mathematische Annalen* **11** (1877), 242–292, DOI <https://doi.org/10.1007/BF01442667>, establish the lattice optima D_4 and

D_5 . Hans F. Blichfeldt, “The minimum values of positive quadratic forms in six, seven and eight variables,” *Mathematische Zeitschrift* **39** (1935), 1–15, DOI <https://doi.org/10.1007/BF01201341>, establishes the lattice optima E_6, E_7, E_8 . These lattice results do not bound arbitrary non-lattice packings.

Further partial progress in dimension 5 is given by Alexei Andreanov and Yoav Kallus, “Locally optimal 2-periodic sphere packings,” *Discrete & Computational Geometry* **63** (2020), 182–208, DOI <https://doi.org/10.1007/s00454-019-00150-6>, which rules out a denser 2-periodic packing and classifies the densest packings in that restricted class. Henry Cohn and Isaac Rajagopal, “Variations on Five-Dimensional Sphere Packings,” *Discrete & Computational Geometry* (2026), DOI <https://doi.org/10.1007/s00454-026-00841-x>, construct additional non-lattice packings with the same known density and explicitly state that no corresponding global optimality result is known beyond the restricted setting. They also report no resolution in dimensions 6 or 7.

Thus the exact conjunction in the statement fails to have a complete literature answer precisely because the unrestricted cases $d = 4, 5, 6, 7$ remain open.

2.34.3. Remaining open problems recorded by the database

2.34.3.1. Open problem 1

Let

$$D_4 = \{(x_1, x_2, x_3, x_4) \in \mathbb{Z}^4 : x_1 + x_2 + x_3 + x_4 \equiv 0 \pmod{2}\}.$$

For any $X \subset \mathbb{R}^4$ whose distinct points are at distance at least $\sqrt{2}$, define the upper density of the packing of balls of radius $1/\sqrt{2}$ centered at X by

$$\bar{\delta}(X) = \limsup_{R \rightarrow \infty} \frac{\text{vol}\left(\left(\bigcup_{x \in X} B(x, 1/\sqrt{2})\right) \cap [-R, R]^4\right)}{(2R)^4}.$$

Prove or disprove that $\bar{\delta}(X) \leq \pi^2/16$ for every such X , with equality attained when $X = D_4$; equivalently, prove or disprove that D_4 is a densest sphere packing in $\mathbb{R}^4$.

Reference supplied by the database. Andrei Okounkov, “The magic of 8 and 24,” *Proceedings of the International Congress of Mathematicians 2022*, vol. 1, EMS Press, 2022 (published 2023), pp. 492–545, Section 2.1, DOI: <https://doi.org/10.4171/ICM2022/203>.

2.34.3.2. Open problem 2

Let

$$D_5 = \left\{ (x_1, \dots, x_5) \in \mathbb{Z}^5 : \sum_{i=1}^5 x_i \equiv 0 \pmod{2} \right\}.$$

For any $X \subset \mathbb{R}^5$ whose distinct points are at distance at least $\sqrt{2}$, define the upper density of the packing of balls of radius $1/\sqrt{2}$ centered at X by

$$\bar{\delta}(X) = \limsup_{R \rightarrow \infty} \frac{\text{vol}\left(\left(\bigcup_{x \in X} B(x, 1/\sqrt{2})\right) \cap [-R, R]^5\right)}{(2R)^5}.$$

Prove or disprove that $\bar{\delta}(X) \leq \pi^2/(15\sqrt{2})$ for every such X , with equality attained when $X = D_5$; equivalently, prove or disprove that D_5 is a densest sphere packing in $\mathbb{R}^5$.

Reference supplied by the database. Andrei Okounkov, “The magic of 8 and 24,” *Proceedings of the International Congress of Mathematicians 2022*, vol. 1, EMS Press, 2022 (published 2023), pp. 492–545, Section 2.1, DOI: <https://doi.org/10.4171/ICM2022/203>.

2.34.3.3. Open problem 3

Let $\alpha_1, \dots, \alpha_6$ be vectors in $\mathbb{R}^6$ satisfying $\langle \alpha_i, \alpha_i \rangle = 2$ for all i , $\langle \alpha_i, \alpha_j \rangle = -1$ when

$$\{i, j\} \in \{\{1, 2\}, \{2, 3\}, \{3, 4\}, \{3, 5\}, \{5, 6\}\},$$

and $\langle \alpha_i, \alpha_j \rangle = 0$ for all other distinct i, j . Define the E_6 root lattice by $E_6 = \sum_{i=1}^6 \mathbb{Z}\alpha_i$. For any $X \subset \mathbb{R}^6$ whose distinct points are at distance at least $\sqrt{2}$, define

$$\bar{\delta}(X) = \limsup_{R \rightarrow \infty} \frac{\text{vol}\left(\left(\bigcup_{x \in X} B(x, 1/\sqrt{2})\right) \cap [-R, R]^6\right)}{(2R)^6}.$$

Prove or disprove that $\bar{\delta}(X) \leq \pi^3/(48\sqrt{3})$ for every such X , with equality attained when $X = E_6$; equivalently, prove or disprove that E_6 is a densest sphere packing in $\mathbb{R}^6$.

Reference supplied by the database. Andrei Okounkov, “The magic of 8 and 24,” *Proceedings of the International Congress of Mathematicians 2022*, vol. 1, EMS Press, 2022 (published 2023), pp. 492–545, Section 2.1, DOI: <https://doi.org/10.4171/ICM2022/203>.

2.34.3.4. Open problem 4

Let $\alpha_1, \dots, \alpha_7$ be vectors in $\mathbb{R}^7$ satisfying $\langle \alpha_i, \alpha_i \rangle = 2$ for all i , $\langle \alpha_i, \alpha_j \rangle = -1$ when

$$\{i, j\} \in \{\{1, 2\}, \{2, 3\}, \{3, 4\}, \{4, 5\}, \{4, 6\}, \{6, 7\}\},$$

and $\langle \alpha_i, \alpha_j \rangle = 0$ for all other distinct i, j . Define the E_7 root lattice by $E_7 = \sum_{i=1}^7 \mathbb{Z}\alpha_i$. For any $X \subset \mathbb{R}^7$ whose distinct points are at distance at least $\sqrt{2}$, define

$$\bar{\delta}(X) = \limsup_{R \rightarrow \infty} \frac{\text{vol}\left(\left(\bigcup_{x \in X} B(x, 1/\sqrt{2})\right) \cap [-R, R]^7\right)}{(2R)^7}.$$

Prove or disprove that $\bar{\delta}(X) \leq \pi^3/105$ for every such X , with equality attained when $X = E_7$; equivalently, prove or disprove that E_7 is a densest sphere packing in $\mathbb{R}^7$.

Reference supplied by the database. Andrei Okounkov, “The magic of 8 and 24,” *Proceedings of the International Congress of Mathematicians 2022*, vol. 1, EMS Press, 2022 (published 2023), pp. 492–545, Section 2.1, DOI: <https://doi.org/10.4171/ICM2022/203>.

2.34.3.5. Open problem 5

Let $X \subset \mathbb{R}^4$ be the set of centers of a packing of closed unit balls, so that $\|x - z\| \geq 2$ for all distinct $x, z \in X$. For $x \in X$, let

$$V_x = \{y \in \mathbb{R}^4 : \|y - x\| \leq \|y - z\| \text{ for every } z \in X\}$$

be its Voronoi cell. Prove or disprove that every V_x has four-dimensional volume at least the volume of the regular 24-cell circumscribed about a unit ball.

Reference supplied by the database. Oleg R. Musin, “Towards a proof of the 24-cell conjecture,” *Acta Mathematica Hungarica* **155** (2018), no. 1, 184–199, the “24-cell conjecture” in Section 1, DOI: <https://doi.org/10.1007/s10474-018-0828-5>; arXiv:1712.04099.

2.34.4. Primary-source pointer

The originating ICM record is *The magic of 8 and 24* (2022). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-559-9-volume-1.pdf#page=511>.

2.35. 113. The work of Maryna Viazovska [2022-geom-296]

Record: 2022-geom-296

Subfield: Discrete and polyhedral geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2022: The work of Maryna Viazovska

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-559-9-volume-1.pdf#page=101>

2.35.1. Problem statement

Let

$$\Lambda = \left(\frac{2}{\sqrt{3}} \right)^{1/2} \left(\mathbb{Z}(1,0) + \mathbb{Z} \left(\frac{1}{2}, \frac{\sqrt{3}}{2} \right) \right) \subset \mathbb{R}^2$$

be the covolume-one triangular lattice, whose shortest nonzero vectors have length

$$r = \left(\frac{4}{3} \right)^{1/4},$$

and let

$$\Lambda^* = \{y \in \mathbb{R}^2 : \langle x, y \rangle \in \mathbb{Z} \text{ for every } x \in \Lambda\}$$

be its dual lattice. Normalize the Fourier transform by

$$\hat{f}(y) = \int_{\mathbb{R}^2} f(x) e^{-2\pi i \langle x, y \rangle} dx.$$

Construct a radial Schwartz function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ such that

$$f(0) = \hat{f}(0) = 1, \quad f(x) \leq 0 \quad \text{whenever } |x| \geq r, \quad \hat{f}(y) \geq 0 \quad \text{for every } y \in \mathbb{R}^2.$$

Such a function would make the Cohn–Elkies linear-programming bound equal the optimal circle-packing density $\pi/\sqrt{12}$.

Furthermore, for every $\alpha > 0$, construct a radial Schwartz function $f_\alpha : \mathbb{R}^2 \rightarrow \mathbb{R}$ such that

$$f_\alpha(x) \leq e^{-\pi\alpha|x|^2} \quad \text{for every } x \neq 0, \quad \hat{f}_\alpha(y) \geq 0 \quad \text{for every } y \in \mathbb{R}^2,$$

with

$$f_\alpha(x) = e^{-\pi\alpha|x|^2} \quad \text{for every } x \in \Lambda \setminus \{0\}, \quad \hat{f}_\alpha(y) = 0 \quad \text{for every } y \in \Lambda^* \setminus \{0\}.$$

These functions would make the linear-programming energy bound sharp for every Gaussian potential and thereby prove that Λ minimizes energy, among all point configurations of density one, for every potential of the form $p(|x|^2)$, where $p : (0, \infty) \rightarrow \mathbb{R}$ is completely monotone, meaning that

$$(-1)^k p^{(k)}(t) \geq 0$$

for every integer $k \geq 0$ and every $t > 0$.

2.35.2. Historical background, literature, and current progress

Verdict: unresolved.

No citable construction is known for either the sharp two-dimensional Cohn–Elkies function in the statement or the stronger family of Gaussian-energy auxiliary functions. Because the problem asks for both constructions, partial results about circle packing itself, minimization among lattices, local minimality, or restricted periodic competitors do not resolve it.

- Henry Cohn and Noam Elkies, *New upper bounds on sphere packings I*, *Annals of Mathematics* 157 (2003), 689–714. DOI: <https://doi.org/10.4007/annals.2003.157.689>; arXiv: [math/0110009](https://arxiv.org/abs/math/0110009). Conjecture 7.3 asserts that their linear-programming method has an optimal auxiliary function in dimensions (2,8,24). Section 5 derives the complementary-slackness zeros that any sharp function must have. The (d=2) case is

precisely the first construction in the statement after the covolume-one normalization, but the paper does not construct it.

- Hugh L. Montgomery, *Minimal theta functions*, Glasgow Mathematical Journal 30 (1988), 75–85. DOI: <https://doi.org/10.1017/S0017089500007047>. Montgomery proves that the triangular lattice minimizes every Gaussian theta sum among two-dimensional lattices of fixed covolume. By Bernstein’s representation this yields universal energy optimality among lattices for the relevant completely monotone potentials, but it neither treats all density-one point configurations nor constructs the functions required in the statement.
- Henry Cohn, Abhinav Kumar, Stephen D. Miller, Danylo Radchenko, and Maryna Viazovska, *Universal optimality of the E_8 and Leech lattices and interpolation formulas*, Annals of Mathematics 196 (2022), 983–1082. DOI: <https://doi.org/10.4007/annals.2022.196.3.3>; arXiv: [1902.05438](https://arxiv.org/abs/1902.05438). The paper constructs the analogous Gaussian auxiliary functions in dimensions (8) and (24). Section 7 states that sharp planar linear-programming bounds and universal optimality of the hexagonal lattice remain conjectural despite strong numerical evidence, and explains why the direct radial-interpolation strategy used in dimensions (8) and (24) is insufficient in dimension (2).
- Naser Talebizadeh Sardari, *Higher Fourier interpolation on the plane*, preprint (2021). arXiv: [2102.08753](https://arxiv.org/abs/2102.08753). Theorem 1.7 proves that there are infinitely many linearly independent radial Schwartz functions for which both the function and its Fourier transform vanish to order two at all nonzero triangular-lattice radii. This resolves Cohn–Kumar–Miller–Radchenko–Viazovska Conjecture 7.5 and establishes that the shell data do not uniquely determine a planar Schwartz function. It is genuine interpolation progress, but supplies neither required sign inequalities nor the Gaussian majorization/equality conditions.
- Markus Faulhuber, Irina Shafkulovska, and Ilia Zlotnikov, *A note on energy minimization in dimension 2*, Proceedings of the American Mathematical Society, Series B 11 (2024), 664–679. DOI: <https://doi.org/10.1090/bproc/247>; arXiv: [2306.16266](https://arxiv.org/abs/2306.16266). The triangular lattice is shown to outperform several natural periodic competitors, including the honeycomb, for every Gaussian scale. The competitor classes are restricted, so this is not global universal optimality and does not construct the requested auxiliary functions.
- Douglas P. Hardin and Nathaniel J. Tenpas, *Universally Optimal Periodic Configurations in the Plane*, Discrete Analysis 2025:26 (2025), 63 pp. DOI: <https://doi.org/10.19086/da.144978>; arXiv: [2307.15822](https://arxiv.org/abs/2307.15822). The paper proves universal optimality for a scaled A_2 configuration among four-point A_2 -periodic configurations and for a rotated/scaled A_2 configuration among six-point configurations with a specified rectangular period lattice. These restricted periodic theorems do not cover all density-one configurations.
- Thomas Leble, *The hexagonal lattice is universally locally optimal*, preprint (2025). arXiv: [2511.03353](https://arxiv.org/abs/2511.03353). The paper proves local minimality of the hexagonal lattice among all point configurations for every potential that is completely monotone in squared distance, including every Gaussian. Local minimality is substantially weaker than the global minimization and exact auxiliary-function construction requested here.
- Andrew Salmon, *Linear Programming Bounds for Fibered Sphere Packings*, preprint (2026). arXiv: [2607.25254](https://arxiv.org/abs/2607.25254). Section 4 states that no sharp planar magic function is known and formulates its existence as Conjecture 17. The paper proves equivalent or implied sharpness statements for certain fibered packing programs, but leaves the planar scalar conjecture unresolved.

Thus the remaining gap is exactly the global one: construct a radial Schwartz certificate attaining the planar packing bound, and, for every ($\alpha > 0$), construct the stronger Gaussian certificate with the stated interpolation and sign properties. No verified paper found through August 4, 2026 closes either gap.

2.35.3. Remaining open problems recorded by the database

2.35.3.1. Open problem 1

Let

$$\Lambda = \left(\frac{2}{\sqrt{3}} \right)^{1/2} \left(\mathbb{Z}\langle 1, 0 \rangle + \mathbb{Z} \left(\frac{1}{2}, \frac{\sqrt{3}}{2} \right) \right) \subset \mathbb{R}^2$$

be the covolume-one triangular lattice. A periodic point configuration of density one is a set

$$\Gamma = \bigcup_{j=1}^N (t_j + L),$$

where $L \subset \mathbb{R}^2$ is a full-rank lattice, the cosets $t_j + L$ are distinct, and $N / \text{covol}(L) = 1$. For $\alpha > 0$, define its Gaussian energy per point by

$$E_\alpha(\Gamma) = \frac{1}{N} \sum_{j,k=1}^N \sum_{\substack{\ell \in L \\ \ell + t_j - t_k \neq 0}} e^{-\pi\alpha|\ell + t_j - t_k|^2}.$$

Prove that, for every $\alpha > 0$ and every periodic point configuration Γ of density one,

$$E_\alpha(\Lambda) \leq E_\alpha(\Gamma),$$

with equality only when Γ is a Euclidean isometric copy of Λ .

Reference supplied by the database. Markus Faulhuber, Irina Shafkulovska, and Ilia Zlotnikov, “A note on energy minimization in dimension 2,” *Proceedings of the American Mathematical Society, Series B* 11 (2024), 664-679, discussion following Corollary 1.3, DOI: 10.1090/bproc/247, arXiv:2306.16266.

2.35.3.2. Open problem 2

Let

$$A_2 = \mathbb{Z}\langle 1, 0 \rangle + \mathbb{Z} \left(\frac{1}{2}, \frac{\sqrt{3}}{2} \right) \quad \text{and} \quad L = \mathbb{Z}\langle 1, 0 \rangle + \mathbb{Z}\langle 0, \sqrt{3} \rangle.$$

For $a > 0$, define the L -periodic Gaussian potential on $\mathbb{R}^2/L$ by

$$F_{a,L}(x) = \sum_{\ell \in L} e^{-a|x+\ell|^2}.$$

For an eight-point multiset $\omega = \{x_1, \dots, x_8\} \subset \mathbb{R}^2/L$, define

$$E_{a,L}(\omega) = \sum_{\substack{1 \leq i, j \leq 8 \\ i \neq j}} F_{a,L}(x_i - x_j).$$

Prove that the eight-point quotient configuration

$$\omega_8 = \left(\frac{1}{2} A_2 \right) / L$$

minimizes $E_{a,L}$ among all eight-point multisets in $\mathbb{R}^2/L$, for every $a > 0$.

Reference supplied by the database. Douglas P. Hardin and Nathaniel J. Tenpas, “Universally Optimal Periodic Configurations in the Plane,” *Discrete Analysis* 2025:26 (2025), 63 pp., Section 3.4, DOI: 10.19086/da.144978, arXiv:2307.15822.

2.35.3.3. Open problem 3

Let

$$A_2 = \mathbb{Z}(1, 0) + \mathbb{Z}\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right), \quad L = \sqrt{2}A_2, \quad T = \mathbb{R}^2/L,$$

and equip $X = \mathbb{R}^2 \times T$ with the metric

$$d_X((x, \bar{y}), (x', \bar{y}'))^2 = |x - x'|^2 + \min_{\ell \in L} |y - y' - \ell|^2.$$

Give T normalized Haar measure and X the product of Lebesgue measure and Haar measure. Let $\mathcal{A}$ be the set of real-valued, continuous, integrable, positive-definite functions $h : X \rightarrow \mathbb{R}$ such that

$$h(x, \bar{y}) \leq 0 \quad \text{whenever} \quad d_X((x, \bar{y}), (0, \bar{0})) \geq \sqrt{2}, \quad \int_X h = 1.$$

Prove that

$$\inf_{h \in \mathcal{A}} h(0, \bar{0}) = \frac{\sqrt{3}}{2}$$

and that this infimum is attained.

Reference supplied by the database. Andrew Salmon, “Linear Programming Bounds for Fibered Sphere Packings,” preprint (2026), Conjecture 22, arXiv:2607.25254.

2.35.4. Primary-source pointer

The originating ICM record is *The work of Maryna Viazovska* (2022). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-559-9-volume-1.pdf#page=101>.

3. Classical and local differential geometry

This chapter collects problems about submanifolds, minimal and constant-mean-curvature surfaces, local invariants, special coordinate systems, and classical rigidity. The historical record repeatedly shows local differential equations interacting with global completeness, embeddedness, or classification. Modern techniques—geometric measure theory, integrable systems, min–max constructions, and elliptic PDE—have resolved major subcases without eliminating the original classification problems.

Coverage: 33 website records; **source years:** 1908–2022; **separately listed descendant open problems:** 33.

3.1. 114. Les origines, les méthodes et les problèmes de la Géométrie infinitésimale [1908-geom-219]

Record: 1908-geom-219

Subfield: Classical and local differential geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1908: Les origines, les méthodes et les problèmes de la Géométrie infinitésimale

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1908.1/ICM1908.1.ocr.pdf#page=109>

3.1.1. Problem statement

Given an arbitrary smooth regular surface of revolution $S \subset \mathbb{R}^3$, classify, up to Euclidean rigid motions, all smooth immersed surfaces $\tilde{S} \subset \mathbb{R}^3$ that are locally isometric to S , meaning that locally there is a diffeomorphism $F : S \rightarrow \tilde{S}$ that preserves the first fundamental form. In particular, carry out this classification when S is the sphere.

3.1.2. Historical background, literature, and current progress

Verdict: unresolved.

The exact problem asks for all local isometric immersions of every rotational metric, not merely rotational or helicoidal deformations and not merely an existence theorem. No complete classification for an arbitrary profile function is known.

Julius Weingarten's *Sur la deformation des surfaces* (Acta Mathematica **20**, 1897, 159-200; [direct record and text](#)) gives an exhaustive reduction of the prescribed-metric deformation problem to a nonlinear second-order PDE. For a rotational metric, the specialization is the classical Weingarten reduction. Luther P. Eisenhart, *A Treatise on the Differential Geometry of Curves and Surfaces* (Ginn, 1909), Sections 145-151, especially Section 149 ([direct scan](#)), states both directions of this correspondence and then notes that the resulting equation for an arbitrary profile function is not integrated; only special choices are treated. Thus this is a necessary-and-sufficient PDE characterization, not the requested classification of its solutions.

Edouard Goursat, *Sur la methode de Weingarten pour le probleme de la deformation des surfaces*, Bulletin de la Societe Mathematique de France **55** (1927), 5-38, [DOI 10.24033/bsmf.1107](#), further develops that reduction and its integrable special cases. Hynek Baran and Michal Marvan, *Classification of integrable Weingarten surfaces possessing an $sl(2)$ -valued zero curvature representation*, Nonlinearity **23** (2010), 2577-2597, [arXiv:1002.0992](#), classify only the soliton-integrable Weingarten relations satisfying a specified zero-curvature and jet-order criterion. Alvaro Pampano, *A variational characterization of profile curves of invariant linear Weingarten surfaces*, Differential Geometry and its Applications **68** (2020), 101564, [DOI 10.1016/j.difgeo.2019.101564](#), gives a construction for the invariant linear subclass and explicitly describes the general classification of Weingarten surfaces as almost completely open. These results do not exhaust arbitrary rotational metrics or arbitrary isometric immersions.

There is strong complete progress for constant-curvature special cases. David Brander, *Spherical Surfaces*, Experimental Mathematics **25** (2016), 257-272, [DOI 10.1080/10586458.2015.1077359](#), proves that every local surface of constant positive Gaussian curvature is recovered from a harmonic Gauss map into the 2-sphere, with an explicit integration formula, thereby giving a complete local representation for the spherical subcase. David Brander and Martin Svensson, *The geometric Cauchy problem for surfaces with Lorentzian harmonic Gauss maps*, Journal of Differential Geometry **93** (2013), 37-66, [direct journal URL](#), analogously construct all solutions from Cauchy data in the constant-negative-curvature subcase. I. Kh. Sabitov, *Infinitesimal and global rigidity and inflexibility of surfaces of revolution with flattening at the poles*, Sbornik: Mathematics **204** (2013), 1516-1547, [DOI 10.1070/SM2013v204n10ABEH004347](#), proves rigidity and infinitesimal-bending criteria under specific smoothness and pole hypotheses, but does not classify all finite local isometric immersions.

What remains is the arbitrary-profile case, including variable and changing Gaussian curvature: one must solve and classify all solutions of the corresponding Gauss-Codazzi/Weingarten equation and reconstruct every immersion up to rigid motion. The cited constant-curvature, integrable, invariant, and infinitesimal results do not collectively cover that case.

3.1.3. Remaining open problems recorded by the database

3.1.3.1. Open problem 1

Let $\Omega \subset \mathbb{C}$ be a simply connected domain containing 0, let $N : \Omega \rightarrow \mathbb{S}^2$ be a harmonic map, and let $f : \Omega \rightarrow \mathbb{R}^3$, unique up to translation, be defined by $f_z = iN \times N_z$. Assume that f is immersed at 0 and has a fixed-point rotational symmetry of finite order $n \geq 2$: there is a Euclidean rotation R of order n such that $f(e^{2\pi i/n}z) = R(f(z))$ whenever both sides are defined. Determine the possible topological types of the connected component containing 0 of the regular set $\{z \in \Omega : \text{rank}(df_z) = 2\}$, and characterize the singularities

of f on the boundary of that component.

Reference supplied by the database. David Brander, “Spherical Surfaces,” *Experimental Mathematics* 25(3) (2016), 257-272, Section 6.1, “Embedded symmetric topological discs and their natural boundaries,” <https://doi.org/10.1080/10586458.2015.1077359>.

3.1.4. Primary-source pointer

The originating ICM record is *Les origines, les méthodes et les problèmes de la Géométrie infinitésimale* (1908). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1908.1/ICM1908.1.ocr.pdf#page=109>.

3.2. 115. Sulle trasformazioni di Darboux delle superficie d'area minima [1908-geom-220]

Record: 1908-geom-220

Subfield: Classical and local differential geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1908: Sulle trasformazioni di Darboux delle superficie d'area minima

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1908.2/ICM1908.2.ocr.pdf#page=264>

3.2.1. Problem statement

Let $S \subset \mathbb{R}^3$ be a smooth flexible, inextensible surface. A contact element of S is the pair $f = (P, T_P S)$ consisting of a point $P \in S$ and its tangent plane. Suppose that there is a one-parameter family of surfaces S'_λ , each in point-to-point correspondence with S , so that every contact element f of S has a one-parameter family of corresponding contact elements f'_λ of the surfaces S'_λ . Assume that this entire family of contact elements is rigidly attached to f : whenever S undergoes an arbitrary isometric bending in $\mathbb{R}^3$, the family f'_λ is transported with f as an invariant Euclidean configuration. Classify all such data for which, after every isometric bending of S , the resulting three-parameter family of transported contact elements f'_λ remains integrable as a one-parameter family of surfaces S'_λ .

3.2.2. Historical background, literature, and current progress

The problem is **unresolved**. In current terminology it asks for a classification of all three-dimensional integrable rolling distributions of contact elements. The literature gives examples, classifications under additional hypotheses, and generic structural theorems, but no result covers arbitrary rigidly attached one-parameter contact-element families, arbitrary flexible seed surfaces, and every possible degeneracy.

- “Sulle trasformazioni di Darboux delle superficie d'area minima,” Luigi Bianchi, in *Atti del IV Congresso Internazionale dei Matematici (Roma, 6–11 Aprile 1908)*, Vol. II, 1909, pp. 264–273, [direct proceedings PDF](#). Bianchi exhibits the confocal-quadric construction and constructions arising from minimal and constant-curvature surfaces, then explicitly asks for all cases. These are families of solutions, not an exhaustive classification.
- “Sopra una proprietà caratteristica delle congruenze rettilinee di rotolamento,” Luigi Bianchi, *Rendiconti della Reale Accademia dei Lincei*, series 5, vol. 24, 1915, pp. 3–16, [direct article](#); see also §128 of *Transformations of Surfaces*, Luther P. Eisenhart, Princeton University Press, 1923, [full text](#). This classifies the stronger fixed-leaf problem in which one prescribed two-dimensional family of contact elements remains the contact elements of a surface under every bending. It does not cover Bianchi’s 1908 allowance

that the whole three-dimensional family may be refoliated into a new one-parameter family of surfaces.

- **“On Bianchi’s Bäcklund transformation of quadrics,”** Ion I. Dincă, *Journal of Geometry and Physics* 73 (2013), 104–124, DOI <https://doi.org/10.1016/j.geomphys.2013.05.005>, arXiv:1110.5474. The paper proves that the only Bäcklund transformation having a defining auxiliary surface is Bianchi’s transformation for quadrics. This is a rigidity classification of one special subclass, not of all integrable rolling distributions.
- **“On the isometric deformation of surfaces via the Bäcklund transformation,”** Ion I. Dincă, arXiv:1212.2326, 2012 (v6, 2016). Under the additional requirement that the leaves correspond isometrically in a way independent of the seed’s realization, the paper derives a necessary-and-sufficient differential system. It expressly describes this as only a partial fulfillment of the classification program and does not solve or classify all solutions of that system.
- **“On isometric correspondence of leaves,”** Ion I. Dincă, arXiv:1907.13483, 2019 (revised 2024). Theorem 1.4 proves that a generic three-dimensional integrable rolling distribution with seed-independent isometric correspondence of leaves must be a Bäcklund transformation. The theorem adds isometric-leaf correspondence and excludes developable seeds and isotropic developable leaves, so it does not exhaust the problem as stated.
- **“On a new definition of the Bäcklund transformation in the isometric deformation of surfaces,”** Ion I. Dincă, arXiv:2410.01135, 2024 (v8, 2026). The main theorem splits a generic four-dimensional rolling distribution with tangency symmetry into a one-parameter family of generic three-dimensional distributions, again excluding developable seeds and isotropic developable leaves. It is a structural result about a restricted four-dimensional setting and supplies no classification of the unrestricted three-dimensional distributions required here.

Thus the established progress does not classify nongeneric distributions, developable and other degenerate cases, distributions without tangency symmetry, or general distributions whose leaves need not correspond isometrically. Those gaps prevent a complete answer to the exact problem.

3.2.3. Remaining open problems recorded by the database

3.2.3.1. Open problem 1

Determine necessary and sufficient conditions on a two-dimensional Riemannian metric g for the following property. Every local isometric immersion $x_0 : (M, g) \rightarrow \mathbb{C}^3$, where $\mathbb{C}^3$ carries the complex Euclidean bilinear form, admits a nontrivial Bäcklund transformation without a prescribed defining surface: there is a three-dimensional family of contact elements rigidly transported under every rolling of x_0 onto another local isometric immersion of (M, g) , the transported family is always integrable into a one-parameter family of surface leaves, each seed-leaf pair consists of the two focal surfaces of a line congruence for which corresponding asymptotic directions agree, and the induced metrics of the leaves are independent of the chosen isometric immersion of the seed.

Reference supplied by the database. Ion I. Dincă, “On the isometric deformation of surfaces via the Bäcklund transformation,” arXiv:1212.2326 [math.DG], 2012 (version 6, 21 July 2016), §1.2, first program item (“Existence of Bäcklund transformation”), <https://arxiv.org/abs/1212.2326>.

3.2.4. Primary-source pointer

The originating ICM record is *Sulle trasformazioni di Darboux delle superficie d’area minima* (1908). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1908.2/ICM1908.2.ocr.pdf#page=264>.

3.3. 116. Conformal geometry [1912-geom-4]

Record: 1912-geom-4

Subfield: Classical and local differential geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1912: Conformal geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1912.2/ICM1912.2.ocr.pdf#page=81>

3.3.1. Problem statement

Let a and b be regular real analytic arcs through a common vertex V forming a horn angle, i.e. a curvilinear angle of magnitude 0. Suppose that the contact of a and b at V is exactly of order $h - 1$, equivalently that the horn angle has type H_h and the two arcs have h consecutive points in common. Prove or disprove that, whenever h is odd, there exists an analytic curve c through V which is the external bisector of the horn angle, in the sense that a and b are symmetric with respect to c , analogous to the external angle bisector for two straight lines.

3.3.2. Historical background, literature, and current progress

3.3.3. Verdict: partially resolved

3.3.3.1. Precise problem

Kasner's problem is the conformal/Schwarz-reflection bisection problem for a horn angle. Given two regular real analytic arcs through a common vertex V , tangent with exact contact order $h - 1$ (type H_h), decide whether there is a real analytic arc c through V such that Schwarz reflection in c interchanges the two given arcs. After conformal normalization, this is equivalent to asking whether a germ

$$g(u) = u + b_h u^h + \cdots, \quad b_h \neq 0,$$

has an analytic iterative square root

$$f(u) = -u + \cdots, \quad f(f(u)) = g(u),$$

the sign $f'(0) = -1$ corresponding to the external bisector. the assertion is the proposed parity criterion: an external analytic bisector should exist whenever h is odd.

3.3.3.2. Literature and progress

Original source

- **Title:** Conformal geometry
Author: Edward Kasner

Venue: Proceedings of the Fifth International Congress of Mathematicians, Cambridge 1912, vol. 2, Cambridge University Press, pp. 81-90

Year: 1912/1913

URL: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1912.2/ICM1912.2.ocr.pdf>

Relevant result/status: Kasner defines horn angles of type H_h by exact contact order $h - 1$, proves the finite-order invariant statement for horn angles, and states the bisection issue explicitly: type H_2 has an internal but no external bisector, type H_3 has both, and in general if the type is even no external bisector exists while if the type is odd one “probably” exists. Thus the odd case was posed conjecturally, not proved.

Formal solution and uniqueness results

- **Title:** On the Conformal Geometry of Analytic Arcs

Author: G. A. Pfeiffer

Venue: American Journal of Mathematics, vol. 37, no. 4, pp. 395-430

Year: 1915

DOI/URL: <https://doi.org/10.2307/2370214>; <https://archive.org/download/jstor-2370214/2370214.pdf>

Relevant result/status: Pfeiffer reduces the bisection problem to $f(f(u)) = g(u)$. In the tangent horn case $g(u) = u + b, u^r + \dots$, his coefficient calculation gives one formal external solution with $f'(0) = -1$ when the first nonzero term is of odd degree $r = 2p + 1$, and no such formal solution when it is of even degree $r = 2p$. He also proves uniqueness-type statements. This is strong partial progress, but it is formal; the convergence needed for an analytic curve is not proved in the horn/parabolic multiplier-one case.

- **Title:** The Functional Equation $f[f(x)] = g(x)$

Author: G. A. Pfeiffer

Venue: Annals of Mathematics, Second Series, vol. 20, no. 1, pp. 13-22

Year: 1918

DOI/URL: <https://doi.org/10.2307/1967374>; <https://archive.org/download/jstor-1967374/1967374.pdf>

Relevant result/status: This follow-up studies convergence and divergence for $f(f(x)) = g(x)$, but under the irrational neutral condition $|a_1| = 1, a_1'' \neq 1$. That excludes the horn case $g'(0) = 1$. It therefore does not resolve Kasner’s odd-type external-bisector problem.

Modern classification and obstruction results

- **Title:** The classification of curvilinear angles in the complex plane and the groups of $\pm$ holomorphic diffeomorphisms

Author: Isao Nakai

Venue: Annales de la Faculté des Sciences de Toulouse, vol. 7, no. 2, pp. 313-334

Year: 1998

URL: https://www.numdam.org/item/AFST_1998_6_7_2_313_0.pdf

Relevant result/status: Nakai reformulates tangent curvilinear angles through the group generated by Schwarz reflections and uses Écalle-Voronin theory to describe moduli. The paper makes clear that formal equivalence of tangent curvilinear angles need not converge in general. This supports the conclusion that Pfeiffer’s formal odd-degree solution is not by itself an analytic existence theorem.

- **Title:** A Complete Classification for Pairs of Real Analytic Curves in the Complex Plane with Tangential Intersection

Authors: Patrick Ahern and Xianghong Gong

Venue: Journal of Dynamical and Control Systems, vol. 11, pp. 1-71

Year: 2005; erratum 2007

DOI/URL: <https://doi.org/10.1007/s10883-005-0001-7> ; erratum <https://doi.org/10.1007/s10883-007-9015-7>

Relevant result/status: Ahern and Gong give a complete Écalte-Voronin classification for pairs of real analytic curves with tangential intersection and classify related involution/centralizer structures. Their abstract explicitly notes examples that are formally reversible by an antiholomorphic mapping but not by a convergent one. This is modern evidence of analytic obstructions beyond formal parity, but it is not a theorem in the simple form “all odd H_h horn angles have analytic external bisectors.”

- **Title:** The root extraction problem

Author: Christiane Rousseau

Venue: Journal of Differential Equations, vol. 234, no. 1, pp. 110-141

Year: 2007

DOI/URL: <https://doi.org/10.1016/j.jde.2006.12.003>

Relevant result/status: Rousseau studies the N -th iterative root problem for analytic germs and states that, depending on the germ and multiplier, both formal and analytic obstructions can occur. The abstract explicitly says the theory is applied to sectioning a curvilinear angle in N equal parts in conformal geometry. This is directly relevant to bisection as $N = 2$, and again indicates that formal solvability is not the same as analytic bisection.

Special cases and survey-level status

- **Title:** Bisecting Horn Angles

Author: Sergiy Koshkin

Venue: The College Mathematics Journal, vol. 51, no. 2, pp. 124-131

Year: 2020

DOI/arXiv/URL: <https://doi.org/10.1080/07468342.2020.1700745> ; arXiv:2206.09499, <https://arxiv.org/abs/2206.09499>

Relevant result/status: Koshkin gives explicit metric and conformal bisectors for line-circle and circle-circle horn angles. For the general conformal bisection problem he cites Pfeiffer and states that a conformal bisector may not exist at all, while when it does it is unique. These special cases do not cover arbitrary analytic arcs of odd type H_h .

3.3.3.3. What remains unresolved

The literature fully settles only parts of the question: even type has no external formal/analytic candidate in Kasner-Pfeiffer’s sense; odd type has a unique formal external bisector; and special line/circle horn angles have explicit analytic conformal bisectors. Modern Écalte-Voronin/root-extraction work gives analytic moduli and obstruction theory, showing that convergence is a genuine issue.

I did not find a published theorem proving the full statement that every odd-type real analytic horn angle H_h has an analytic external bisector. Nor did I find a citable theorem stated exactly as a complete odd-type horn-bisector classification. Therefore the current status of the exact problem is **partially resolved**, not fully resolved and not fully open.

3.3.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

3.3.5. Primary-source pointer

The originating ICM record is *Conformal geometry* (1912). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1912.2/ICM1912.2.ocr.pdf#page=81>.

3.4. 117. Sur le problème général de la déformation [1920-geom-221]

Record: 1920-geom-221

Subfield: Classical and local differential geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1920: Sur le problème général de la déformation

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1920/ICM1920.ocr.pdf#page=445>

3.4.1. Problem statement

Let V be a four-dimensional complex vector space, and identify the projective null quadric $Q^4 \subset \mathbb{P}(\wedge^2 V)$ with the Grassmannian of lines in $\mathbb{P}(V)$ via the Klein-Pluecker embedding. A surface $f : U \rightarrow Q^4$ is called isothermic if it admits local coordinates that are simultaneously conformal for the conformal structure induced from Q^4 and curvature-line coordinates.

A nondegenerate line congruence $f : U \rightarrow Q^4$ has focal surfaces $x, y : U \rightarrow \mathbb{P}(V)$ when $f(u, v)$ is the line joining $x(u, v)$ to $y(u, v)$ and, for local homogeneous lifts X, Y ,

$$X_v \in \text{span}\{X, Y\}, \quad Y_u \in \text{span}\{X, Y\}.$$

A cyclic linear sequence of length four is a quadruple f_0, f_1, f_2, f_3 for which there are immersed surfaces $x_0, x_1, x_2, x_3 : U \rightarrow \mathbb{P}(V)$, with indices taken modulo 4, such that x_i and x_{i+1} are the focal surfaces of f_i . A three-dimensional conformal subspace of Q^4 is a nonsingular hyperplane section

$$Q_\omega^3 = Q^4 \cap \mathbb{P}(\ker \omega),$$

where $\omega \in \wedge^2 V^*$ is a nondegenerate alternating form.

Classify locally, up to projective transformations of $\mathbb{P}(V)$ and reparametrization of U , all cyclic linear sequences f_0, f_1, f_2, f_3 of length four such that every f_i is isothermic and, for each i , there is a nondegenerate alternating form ω_i with $f_i(U) \subset Q_{\omega_i}^3$.

3.4.2. Historical background, literature, and current progress

Unresolved. No verified source gives a complete local classification under all three simultaneous requirements in the statement: exact four-step closure of the derived line-congruence sequence, the isothermic (equivalently, W -congruence under the Klein correspondence) condition at every member, and containment of every member in a nonsingular linear complex.

Edward S. Hammond, "Periodic Conjugate Nets," *Annals of Mathematics*, Second Series 22 (1921), 238–261, DOI <https://doi.org/10.2307/1967906>, proves necessary and sufficient invariant conditions for an ordinary Laplace sequence of conjugate nets to have period p , including $p = 4$, and gives the corresponding projective-dimension bound. This resolves the unrestricted projective-net closure problem, but it does not impose Cartan's Klein-

quadric, isothermic/ W , and nonspecial-linear-complex conditions.

Buchin Su, “On Certain Periodic Sequences of Laplace of Period Four in Ordinary Space,” *Science Reports of the Tohoku Imperial University, First Series* 25 (1936), 227–256; contemporary abstract at [Japanese Journal of Mathematics](#) 13 (1936), Abstract 36, studies period-four conjugate nets whose diagonal congruences are W -congruences, and in particular treats pairs of such sequences having common diagonal congruences and corresponding conjugate curves. These are substantive period-four and W -congruence results, but the published abstract does not state an exhaustive classification of Cartan’s cyclic quadruples in which all four Klein images belong to nonsingular linear complexes.

Shang-Hsiu Hu, “Laplace Sequences of Surfaces in Projective Space and Two-Dimensional Toda Equations,” *Letters in Mathematical Physics* 57 (2001), 19–32, DOI <https://doi.org/10.1023/A:1017923708829>, classifies the two types of period- n projective Laplace sequences (the second occurring only for even n) through signed two-dimensional Toda equations and gives exact solutions by Darboux transformation. Its period-four case is again a classification of projective Laplace sequences, not of the additional conformal linear-complex subclass in the statement.

Takeshi Sasaki, “Line Congruence and Transformation of Projective Surfaces,” *Kyushu Journal of Mathematics* 60 (2006), 101–243, DOI <https://doi.org/10.2206/kyushujm.60.101>, supplies the strongest explicit partial results. Example 4.15 gives the complete equal-invariant period-four projective-net system

$$(\log h)_{uv} = h - h^{-1}, \quad F_{uv} = hF,$$

together with its compatible second linear equation and a converse reconstruction. Separately, Theorem 8.3 gives an if-and-only-if normal form for two consecutive derived line congruences that both belong to linear complexes: a nonvanishing potential ϕ satisfies

$$(\log \phi)_{uv} = \varepsilon(\phi^2 - k\phi^{-2}), \quad \varepsilon, k \in \{1, -1\},$$

and an explicit compatible linear system reconstructs the congruences; Corollary 8.4 establishes projective two-periodicity for that adjacent pair. The paper does not identify these separate classifications as a complete solution of the exact four-member conformal problem above, nor does it verify exact cyclic closure and the nonsingularity condition for all four members simultaneously.

Thus the literature gives complete classifications after dropping some of the statement’s constraints, and sharp normal forms for adjacent members satisfying the linear-complex constraint. What remains unverified is an exhaustive compatibility and reconstruction theorem imposing all of Cartan’s conditions on the entire exact four-cycle.

3.4.3. Remaining open problems recorded by the database

No separate descendant problem is listed in the website’s “Some remaining open problems” field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

3.4.4. Primary-source pointer

The originating ICM record is *Sur le problème général de la déformation* (1920). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1920/ICM1920.ocr.pdf#page=445>.

3.5. 118. G., Sur les mouvements à deux paramètres doublement décomposables [1924-geom-222]

Record: 1924-geom-222

Subfield: Classical and local differential geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1924: G., Sur les mouvements à deux paramètres doublement décomposables

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1924.1/ICM1924.1.ocr.pdf#page=913>

3.5.1. Problem statement

Classify the surfaces in Euclidean three-space that are swept out by two or more distinct one-parameter families of curves, with all curves within each family congruent to one another under rigid Euclidean motions.

3.5.2. Historical background, literature, and current progress

The problem is **unresolved**. The requested classification allows arbitrary one-parameter rigid Euclidean motions of the generating curves. The literature gives general compatibility equations and a number of substantial classes, but no result exhausts all such surfaces.

- Gabriel Koenigs, “Sur les mouvements doublement décomposables et sur les surfaces qui sont le lieu de deux familles de courbes égales,” *Comptes rendus hebdomadaires des séances de l’Académie des sciences* **157** (1913), 988–991 ([direct volume index](#)); and “Sur les mouvements à deux paramètres doublement décomposables,” *Proceedings of the International Mathematical Congress, Toronto 1924*, vol. I (1928), 913–934 ([proceedings PDF](#)). Koenigs relates each decomposition of a two-parameter rigid motion to a family of congruent curves on every point-trajectory surface, derives the governing differential equations, and constructs nontrivial surfaces with three congruent-curve families and arbitrary functions. He does not prove that these examples exhaust the surfaces in the question.
- Raoul Bricard, “Sur un mouvement plan à deux paramètres, doublement décomposable,” *Nouvelles annales de mathématiques*, 6e série, **2** (1927), 105–109 ([Numdam](#)). Bricard treats a restricted planar-motion case, explicitly says that his earlier examples did not exhaust the question, and supplies another special construction. This is partial progress on Koenigs’s kinematic formulation, not a complete classification of the trajectory surfaces.
- Hellmuth Stachel, “A Family of Multiply Decomposable Euclidean Motions,” in I. Bárány and K. Böröczky (eds.), *Intuitive Geometry (Budapest, 1995)*, Bolyai Society Mathematical Studies **6** (János Bolyai Mathematical Society, 1997), 413–427, ISBN 963-8022-89-2 ([author bibliography](#)). This work constructs a further family of multiply decomposable Euclidean motions. Hellmuth Stachel, Giorgio Figliolini, and Jorge Angeles, “The Logarithmic Spiral and its Spherical Counterpart,” *Journal of Industrial Design and Engineering Graphics* **14**(1) (2019), 41–48 ([direct PDF](#)), proves Bricard’s logarithmic-spiral example and reports that fewer than twenty twofold-decomposable examples in Euclidean two-

or three-space were then known. These results demonstrate continuing case-by-case progress rather than exhaustiveness.

- Daniele Agostini, Türkü Özlüm Çelik, Julia Struwe, and Bernd Sturmfels, “Theta Surfaces,” *Vietnam Journal of Mathematics* **49** (2021), 319–347, <https://doi.org/10.1007/s10013-020-00443-x>. Their Theorem 3 restates Lie’s complete classification of nondegenerate **double-translation** surfaces: they are theta surfaces associated with reduced plane quartics. This settles the important subclass in which both congruent families are generated by translations, but it does not cover rotational, screw, rolling, or general rigid-motion families allowed by the stated problem.

Thus the strongest complete theorem found concerns only the translation subclass. The remaining gap is a classification covering every surface swept out in at least two distinct ways by congruent curves under arbitrary rigid Euclidean motions, including degenerate cases; no such classification appears in the cited classical or current literature.

3.5.3. Remaining open problems recorded by the database

3.5.3.1. Open problem 1

Develop a complete and efficient algorithm that, given an arbitrary rational parametrization of a surface in Euclidean three-space, decides whether the surface is a translation surface and, if so, computes rationally parametrized space curves $c_1(u)$ and $c_2(v)$ together with a rational reparametrization of the surface in the standard form $P(u, v) = \frac{1}{2}(c_1(u) + c_2(v))$.

Reference supplied by the database. Juan Gerardo Alcázar and Georg Muntingh, “Affine Equivalences of Surfaces of Translation and Minimal Surfaces, and Applications to Symmetry Detection and Design,” *Journal of Computational and Applied Mathematics* **411** (2022), Article 114206, Section 5 (“Conclusion and Open Problems”), <https://doi.org/10.1016/j.cam.2022.114206>.

3.5.4. Primary-source pointer

The originating ICM record is G., *Sur les mouvements à deux paramètres doublement décomposables* (1924). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1924.1/ICM1924.1.ocr.pdf#page=913>.

3.6. 119. Riassunto di alcune ricerche di geometria proiettivo-differenziale [1924-geom-223]

Record: 1924-geom-223

Subfield: Classical and local differential geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1924: Riassunto di alcune ricerche di geometria proiettivo-differenziale

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1924.1/ICM1924.1.ocr.pdf#page=855>

3.6.1. Problem statement

Let S be a prescribed surface in projective three-space. Classify all two-parameter congruences of projective lines that admit a nontrivial projective deformation and for which S is the first of the two focal surfaces (the two envelope sheets formed by the focal points of the lines).

3.6.2. Historical background, literature, and current progress

The problem is unresolved. The cited literature develops the local deformation theory, gives necessary conditions and classifications of important special classes, but does not classify, for an arbitrary prescribed surface S , every projectively deformable line congruence in projective three-space having S as its first focal surface.

- Guido Fubini, “Riassunto di alcune ricerche di geometria proiettivo-differenziale,” *Proceedings of the International Mathematical Congress, Toronto, 1924*, vol. 1 (1928), pp. 831–834, <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1924.1/ICM1924.1.ocr.pdf>. Fubini formulates this exact inverse problem as not completely solved. He derives necessary conditions for a projective deformation in terms of the four invariants of the Laplace equations associated with the focal surfaces and their asymptotic directions, but gives no exhaustive construction or classification.
- Élie Cartan, “Sur le problème général de la déformation,” *Proceedings of the International Congress of Mathematicians, Strasbourg, 1920* (1921), pp. 397–406, <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1920/ICM1920.ocr.pdf>. Cartan’s exterior-differential-system analysis proves generic projective rigidity and describes the functional generality of exceptional deformable line congruences. It addresses the direct deformation problem, not the inverse classification with one focal surface prescribed.
- Eduard Čech, “Sur les correspondances asymptotiques entre deux surfaces,” *Atti della Reale Accademia Nazionale dei Lincei, Rendiconti*, 6th ser., vol. 8, part 2 (1928), pp. 484–486 and 552–554, <http://dml.cz/dmlcz/500903>. Čech treats the narrower class of W -congruences, whose focal correspondence preserves asymptotic curves. He proves existence and multiplicity results under relations between two projective invariants, including families for isothermo-asymptotic and R_0 focal surfaces, while explicitly leaving

open the determination of all surfaces that can occur as a focal sheet of several such congruences.

- Eduard Čech, “Déformation projective des congruences W ,” *Czechoslovak Mathematical Journal* 6(81), no. 3 (1956), pp. 401–414, doi:10.21136/CMJ.1956.100205, <https://dml.cz/dmlcz/100205>. Čech generalizes Cartan-type existence results and bounds or constructs projective deformations for special W -congruences. These results neither allow an arbitrary prescribed focal surface nor exhaust congruences outside the W class.
- Vladimír Horák, “Contribution à la déformation projective des congruences de droites,” *Czechoslovak Mathematical Journal* 15(90), no. 2 (1965), pp. 179–204, doi:10.21136/CMJ.1965.100661, <https://dml.cz/dmlcz/100661>. Horák characterizes several focal, point, planar, and projective deformations of nonparabolic congruences with nondegenerate focal surfaces and proves involutivity for a restricted non- W class. In the corresponding W case he obtains a non-involutive system whose prolongation remains technically unresolved; consequently this does not close the prescribed-focal-surface classification.
- Alois Švec, *Projective Differential Geometry of Line Congruences*, Publishing House of the Czechoslovak Academy of Sciences, Prague (1965), <https://books.google.com/books?id=PLC0AAAAIAAJ>. Švec’s synthesis gives a complete treatment of a higher-dimensional deformation problem and collects the classical three-dimensional theory by special types (singular, semisingular, W , and parabolic congruences). Its completeness statement for multidimensional projective spaces does not supply the missing all-surfaces, all-congruences classification in projective three-space.

Thus the established progress consists of rigidity and existence theorems, invariant criteria, and classifications or constructions under additional geometric hypotheses. The remaining gap is the unrestricted inverse problem stated here: no cited result determines all admissible congruences for every prescribed S , nor do the known special cases collectively exhaust all possible S and all congruence types.

3.6.3. Remaining open problems recorded by the database

3.6.3.1. Open problem 1

For each prescribed relation $f(r, s) = 0$, classify all nonruled surfaces S in real projective three-space that are a focal surface of more than one W -congruence satisfying that relation. Here a W -congruence is a two-parameter family of lines with two nonruled focal surfaces whose focal-point correspondence maps asymptotic curves to asymptotic curves. Its invariants r and s are defined at corresponding points as follows: projectively move the second focal surface into first-order contact with the first, pair the two corresponding asymptotic curves through the contact point, and for each pair take the ratio $J(C, \Gamma; A)$ of their curvatures at A , where C and Γ have the same tangent and osculating plane; the resulting two ratios are r and s .

Reference supplied by the database. Eduard Čech, “Sur les correspondances asymptotiques entre deux surfaces,” *Atti della Reale Accademia Nazionale dei Lincei, Rendiconti*, 6th series, vol. 8, part 2 (1928), pp. 484–486 and 552–554, §12, <https://dml.cz/dmlcz/500903>.

3.6.3.2. Open problem 2

For fixed signs $\varepsilon_1, \varepsilon_2 \in \{-1, 1\}$, with $\varepsilon = \varepsilon_1 \varepsilon_2$, classify the local smooth positive solutions $\phi(x, y)$ for which $\phi^2 - \varepsilon \phi^{-2} \neq 0$ and

$$(\log \phi)_{xy} = \phi^2 - \varepsilon \phi^{-2},$$

$$2\phi\phi_x = \left(\frac{\phi^2(\log \phi)_{xx} - \varepsilon_1(\log \phi)_{yy}}{\phi^2 - \varepsilon \phi^{-2}} \right)_y, \quad 2\phi\phi_y = \left(\frac{\phi^2(\log \phi)_{yy} - \varepsilon_2(\log \phi)_{xx}}{\phi^2 - \varepsilon \phi^{-2}} \right)_x.$$

Reference supplied by the database. Takeshi Sasaki, "Line Congruence and Transformation of Projective Surfaces," *Kyushu Journal of Mathematics* 60, no. 1 (2006), pp. 101–243, Theorem 9.1 and Remark 9.2, doi:10.2206/kyushujm.60.101, <https://doi.org/10.2206/kyushujm.60.101>.

3.6.4. Primary-source pointer

The originating ICM record is *Riassunto di alcune ricerche di geometria proiettivo-differenziale* (1924). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1924.1/ICM1924.1.ocr.pdf#page=855>.

3.7. 120. Beispiele von Elementarkurven und Elementarflächen [1928-geom-224]

Record: 1928-geom-224

Subfield: Classical and local differential geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1928: Beispiele von Elementarkurven und Elementarflächen

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1928.4/ICM1928.4.ocr.pdf#page=197>

3.7.1. Problem statement

Let $\Phi \subset \mathbb{R}^3$ be a compact, everywhere convex surface composed of finitely many patches of geometric order two or three, meaning that every line not contained in such a patch meets it in at most two or three points, respectively, counted with multiplicity. Assume that the points and tangent planes of Φ vary continuously; that the radii of curvature of all its normal sections are finite and nonzero and vary continuously with the point; and that the corresponding osculating circles also vary continuously. Suppose that Φ contains no open subset of a sphere and is *cyclic* in the sense that every circle not contained in Φ intersects Φ in at most four points, counted with multiplicity.

Determine whether the hypothesis that Φ has no circle that is a line of curvature is necessary for the conclusion that Φ is an algebraic cyclide or an ellipsoid; equivalently, determine whether that conclusion still holds for every such cyclic ovaloid when circular lines of curvature are allowed.

3.7.2. Historical background, literature, and current progress

The problem remains unresolved. Juel proved the algebraic-cyclide-or-ellipsoid conclusion after excluding circular lines of curvature, explicitly left the necessity of that exclusion open, and described the excluded case as the one in which the two circles through a surface point can coalesce. No later citable source located in searches through August 3, 2026 proves the conclusion or constructs a counterexample under all of Juel's finite-geometric-order, convexity, continuity, and cyclic-order-four hypotheses.

The closest later results do not close this gap:

- Christian Juel, "Beispiele von Elementarkurven und Elementarflächen," *Atti del Congresso Internazionale dei Matematici, Bologna, 3–10 settembre 1928*, vol. IV, 1928, pp. 195–216, [direct proceedings PDF](#). Section 4 proves the conclusion when there is no circular line of curvature and explicitly leaves open whether that final hypothesis is necessary.
- Thomas Ivey, "Surfaces with Orthogonal Families of Circles," *Proceedings of the American Mathematical Society* 123(3) (1995), 865–872, DOI <https://doi.org/10.2307/2160812>. Ivey characterizes smooth surfaces carrying two orthogonal families of circular arcs as Dupin cyclides. Juel assumes only one potentially coalesced circle on the exceptional curvature line, so the existence of two orthogonal families required by Ivey is not available there.

- Mikhail Skopenkov and Rimvydas Krasauskas, “Surfaces Containing Two Circles through Each Point,” *Mathematische Annalen* 373 (2019), 1299–1327, DOI <https://doi.org/10.1007/s00208-018-1739-z>, arXiv [1512.09062](https://arxiv.org/abs/1512.09062). Their Main Theorem 1.1 classifies analytic surfaces with two transversal, analytically varying circle families. It does not cover the nontransversal coalescence on a circular line of curvature, and its hypotheses do not encode Juel’s global cyclic-order-four condition.
- Niels Lubbes, “Surfaces That Are Covered by Two Pencils of Circles,” *Mathematische Zeitschrift* 299 (2021), 1445–1472, DOI <https://doi.org/10.1007/s00209-021-02713-x>, arXiv [1302.6710](https://arxiv.org/abs/1302.6710). This classifies algebraic surfaces covered by at least two circle pencils up to Möbius equivalence. It neither derives two distinct pencils from Juel’s hypotheses in the coalescent case nor treats his possibly nonanalytic elementary surfaces.

Thus the established progress covers the nondegenerate two-family cases under several stronger or different regularity assumptions, while the exact circular-curvature-line case in the statement remains open.

3.7.3. Remaining open problems recorded by the database

No separate descendant problem is listed in the website’s “Some remaining open problems” field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

3.7.4. Primary-source pointer

The originating ICM record is *Beispiele von Elementarkurven und Elementarflächen* (1928). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1928.4/ICM1928.4.ocr.pdf#page=197>.

3.8. 121. Beispiele von Elementarkurven und Elementarflächen [1928-geom-225]

Record: 1928-geom-225

Subfield: Classical and local differential geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1928: Beispiele von Elementarkurven und Elementarflächen

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1928.4/ICM1928.4.ocr.pdf#page=197>

3.8.1. Problem statement

Let $S \subset \mathbb{R}^3$ be a closed regular surface whose tangent planes and normal curvatures vary continuously, whose principal radii of curvature are finite and nonzero, and which contains no open subset of a sphere. Assume that every Euclidean circle not contained in S meets S in at most four points, counted with intersection multiplicity. Suppose that S has a circular line of curvature, that is, a line of curvature that is itself a circle. Determine whether S must be a Dupin cyclide, meaning a surface whose two families of lines of curvature consist of circles; if not, classify all additional possibilities.

3.8.2. Historical background, literature, and current progress

The problem is **unresolved**. The first alternative is false: a non-spherical ellipsoid of revolution satisfies the circle-intersection bound and has circular parallels as curvature lines, but is not a Dupin cyclide. No citable source located gives the requested classification of all closed regular surfaces under the stated finite-circle-order and curvature-continuity hypotheses.

Christian Juel, “Beispiele von Elementarkurven und Elementarflächen,” *Atti del Congresso Internazionale dei Matematici, Bologna 1928*, vol. IV, 195–216 (published 1931), [direct PDF](#). In §4 Juel proves that a cyclic ovaloid with continuous points, tangent planes, and osculating circles is an algebraic cyclide or an ellipsoid when it has no circular line of curvature. He explicitly excludes the circular-curvature-line case and leaves it open; this is precisely the gap addressed by the present problem.

Nobuko Takeuchi, “Cyclides,” *Hokkaido Mathematical Journal* 29 (2000), 119–148, [direct PDF](#). Theorems 6.6 and 7.1 conformally classify algebraic Darboux cyclides (including quadratic degenerations) and determine that they carry between one and six circles through a non-umbilic point. This describes the algebraic candidates but does not prove that every surface in the problem is algebraic.

Fedor Nilov and Mikhail Skopenkov, “A Surface Containing a Line and a Circle Through Each Point Is a Quadric,” *Geometriae Dedicata* 163 (2013), 301–310, DOI <https://doi.org/10.1007/s10711-012-9750-0>, arXiv:1110.2338. Theorem 3.4 proves that a smooth closed sphere- or torus-type surface carrying four continuously varying distinct circles through every point is a cyclide. The problem here assumes only a global four-point intersection bound and one circular curvature line, so that theorem does not apply.

Mikhail Skopenkov and Rimvydas Krasauskas, “Surfaces Containing Two Circles Through Each Point,” *Mathematische Annalen* 373 (2019), 1299–1327, DOI <https://doi.org/10.1007/s00208-018-1739-z>, arXiv:1512.09062. Theorem 1.1 classifies analytic surfaces carrying two transversal, analytically varying circular arcs through every point as Möbius images of Euclidean translational surfaces, Clifford translational surfaces, or Darboux cyclides. Its analyticity and two-transversal-circle-family assumptions are not consequences of the exceptional circular-curvature-line case, and it does not impose or exploit the exact circle-intersection bound in the statement.

Thus there is strong progress after adding algebraicity, analyticity, or several circle families, but no verified result closes the low-regularity exceptional case or classifies all of its additional possibilities.

3.8.3. Remaining open problems recorded by the database

3.8.3.1. Open problem 1

Let $S \subset \mathbb{R}^3$ be a closed strictly convex regular surface whose tangent planes and osculating circles of normal sections vary continuously, whose principal radii of curvature are finite and nonzero, and which contains no open subset of a sphere. Assume that every Euclidean circle not contained in S meets S in at most four points, counted with intersection multiplicity, and that S has a circular line of curvature. Determine whether S must nevertheless be an ellipsoid or a real algebraic Darboux cyclide.

Reference supplied by the database. Christian Juel, “Beispiele von Elementarkurven und Elementarflächen,” *Atti del Congresso Internazionale dei Matematici, Bologna 1928*, vol. IV, 1931, pp. 195–216, §4 (unnumbered open question following the cyclic-ovaloid theorem), <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1928.4/ICM1928.4.ocr.pdf>

3.8.4. Primary-source pointer

The originating ICM record is *Beispiele von Elementarkurven und Elementarflächen* (1928). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1928.4/ICM1928.4.ocr.pdf#page=197>.

3.9. 122. Einige Bemerkungen über Winkelmetrik in Finslerschen Räumen [1932-geom-271]

Record: 1932-geom-271

Subfield: Classical and local differential geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1932: Einige Bemerkungen über Winkelmetrik in Finslerschen Räumen

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1932.2/ICM1932.2.ocr.pdf#page=199>

3.9.1. Problem statement

Let (M, F) be a smooth, strongly convex Finsler manifold of dimension $n \geq 2$. For each $x \in M$, write $F_x = F|_{T_x M}$, and for each $y \in T_x M \setminus \{0\}$ define the fundamental tensor by

$$g_{x,y}(u, v) = \frac{1}{2} \frac{\partial^2}{\partial s \partial t} F_x^2(y + su + tv) \Big|_{s=t=0}, \quad u, v \in T_x M.$$

For every $x \in M$, every two-dimensional linear subspace $P \subset T_x M$, and every oriented smooth regular arc

$$\gamma : [a, b] \rightarrow P \cap \{y \in T_x M : F_x(y) = 1\}$$

of the unit indicatrix in P , define its Bliss angle and Landsberg angle by

$$\Theta_B(\gamma) = \int_a^b F_x(\dot{\gamma}(t)) dt \quad \text{and} \quad \Theta_L(\gamma) = \int_a^b \sqrt{g_{x,\gamma(t)}(\dot{\gamma}(t), \dot{\gamma}(t))} dt,$$

respectively.

Classify all such Finsler manifolds (M, F) for which

$$\Theta_B(\gamma) = \Theta_L(\gamma)$$

for every choice of x , P , and γ . In particular, determine whether this equality property characterizes Riemannian Finsler metrics, namely whether every such F must be of the form

$$F_x(y) = \sqrt{h_x(y, y)}$$

for some Riemannian metric h on M .

3.9.2. Historical background, literature, and current progress

The problem is **unresolved**. Equality for every indicatrix arc is a pointwise condition on every two-dimensional tangent Minkowski norm, so a complete answer requires a classification of all smooth strongly convex indicatrices satisfying the Bliss–Landsberg equality, not merely a result about the Landsberg angle or about a different Finsler angle construction.

The original source explicitly identifies the Riemannian case: centered ellipses form the known

three-parameter family of planar indicatrix solutions. Gołąb reduces the question whether these are all solutions to the classification of periodic solutions of a functional–differential equation of the form

$$f[x + \omega(f, f')] = \Omega(f, f', f''),$$

and states that exhaustiveness was open. Thus the established positive result is that every Riemannian Finsler metric has the required equality; the missing step is either a proof that every admissible indicatrix is elliptical or a smooth strongly convex nonelliptical counterexample.

Later sources verify that the relevant notions remained distinct without closing this gap. Rund’s monograph gives the Bliss and Landsberg constructions in its section on definitions of angle and discusses their different indicatrix arc elements, but gives no classification under their equality. Wu gives a modern polar-coordinate formula for the Landsberg arc element and proves results about total Landsberg angle in Minkowski planes, but does not compare it arc by arc with intrinsic Minkowski (Bliss) length. The nearby theorem of Bielecki and Gołąb characterizing ellipses concerns additivity and symmetry of **Finsler’s** angle, a different angle definition, and therefore does not resolve this problem.

No citable later proof of exhaustiveness, counterexample, or complete classification under the exact hypotheses was found. In particular, the literature located after 1932 provides no substantive partial classification of the exact equality condition beyond the Riemannian family already recorded by Gołąb, so the non-Riemannian case remains open.

Key references:

- Stanisław Gołąb, “Einige Bemerkungen über Winkelmetrik in Finslerschen Räumen,” *Verhandlungen des Internationalen Mathematiker-Kongresses Zürich 1932*, Vol. II, 1932, pp. 178–179. Direct URL: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1932.2/ICM1932.2.ocr.pdf>. This is the proposing source; it records the Riemannian family and the unresolved periodic functional–differential equation.
- Hanno Rund, *The Differential Geometry of Finsler Spaces*, Grundlehren der mathematischen Wissenschaften 101, Springer, 1959, especially Chapter I, Section 7. DOI: <https://doi.org/10.1007/978-3-642-51610-8>. This gives a standard account of the Bliss and Landsberg angle constructions but no solution of their equality classification.
- Adam Bielecki and Stanisław Gołąb, “Sur un problème de la métrique angulaire dans les espaces de Finsler,” *Annales de la Société Polonaise de Mathématique* 18 (1945), 134–144. MR0018965; direct scan: https://jbc.bj.uj.edu.pl/Content/939179/NDIGCZAS094234_1945_001.pdf. Its ellipse characterizations concern additivity and symmetry of Finsler’s angle, not equality of the Bliss and Landsberg arc functionals.
- Bing Ye Wu, “Some Results on the Geometry of Minkowski Plane,” *Archivum Mathematicum* 46 (2010), no. 3, 177–184. MR2735904; direct URL: <https://dml.cz/dmlcz/140786>. This develops the Landsberg angle and total indicatrix length in modern notation but does not address the Bliss–Landsberg equality classification.

3.9.3. Remaining open problems recorded by the database

No separate descendant problem is listed in the website’s “Some remaining open problems” field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

3.9.4. Primary-source pointer

The originating ICM record is *Einige Bemerkungen über Winkelmetrik in Finslerschen Räumen* (1932). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1932.2/ICM1932.2.ocr.pdf#page=199>.

3.10. 123. Über das Anholonomitätsobjekt von Schouten und van Dantzig [1936-geom-227]

Record: 1936-geom-227

Subfield: Classical and local differential geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1936: Über das Anholonomitätsobjekt von Schouten und van Dantzig

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1936.2/ICM1936.2.ocr.pdf#page=170>

3.10.1. Problem statement

Let $U \subset \mathbb{R}^3$ be open, with coordinates x^1, x^2, x^3 , and let $\Omega_{ij}^k : U \rightarrow \mathbb{R}$ be prescribed real-analytic functions satisfying $\Omega_{ji}^k = -\Omega_{ij}^k$ for all $i, j, k \in \{1, 2, 3\}$. For a real-analytic matrix field $A = (A_\lambda^i)_{\lambda,i=1}^3$ with $\det A \neq 0$, write $(A_i^\lambda) = A^{-1}$. Its Schouten–van Dantzig anholonomy coefficients are

$$\frac{1}{2} A_i^\lambda A_j^\mu \left(\frac{\partial A_\mu^k}{\partial x^\lambda} - \frac{\partial A_\lambda^k}{\partial x^\mu} \right),$$

where repeated indices are summed from 1 to 3.

For indices enclosed in square brackets, let $[\lambda\mu\nu]$ denote total antisymmetrization over λ, μ, ν . Successive differentiation of the equations prescribing the anholonomy coefficients yields the necessary algebraic compatibility relations

$$A_{[\lambda}^i A_{\mu}^j \partial_{\nu]} \Omega_{ij}^k - 2\Omega_{ij}^k \Omega_{rs}^j A_{[\lambda}^i A_{\mu}^r A_{\nu]}^s = 0, \quad k = 1, 2, 3.$$

Determine whether local solvability of these compatibility relations by a real-analytic matrix field A with $\det A \neq 0$ is sufficient for local existence of a real-analytic matrix field $\tilde{A} = (\tilde{A}_\lambda^i)$ with $\det \tilde{A} \neq 0$ whose anholonomy coefficients equal the prescribed functions, that is,

$$\Omega_{ij}^k = \frac{1}{2} \tilde{A}_i^\lambda \tilde{A}_j^\mu \left(\frac{\partial \tilde{A}_\mu^k}{\partial x^\lambda} - \frac{\partial \tilde{A}_\lambda^k}{\partial x^\mu} \right) \quad (i, j, k = 1, 2, 3),$$

where $(\tilde{A}_i^\lambda) = \tilde{A}^{-1}$.

3.10.2. Historical background, literature, and current progress

Verdict: unresolved. No citable source located gives either a proof or a counterexample for the exact three-dimensional implication in the statement: the existence of a nondegenerate analytic matrix field satisfying Gołąb's three displayed algebraic compatibility relations is not known to imply, or not imply, a local analytic coframe having the prescribed structure functions.

Gołąb formulated precisely this sufficiency question and left it undecided. For constant prescribed functions, the question has a positive answer: the compatibility relations reduce to the Jacobi identities, and Lie’s local realization theorem supplies a local Maurer–Cartan coframe. This settles only the constant-coefficient special case.

Robert L. Bryant recasts the first $d^2 = 0$ condition for a varying structure function as the “Jacobi manifold” condition. He explicitly notes that this condition is not sufficient in general, while Theorem 4 proves analytic local existence when the image manifold is additionally an *involutive* Jacobi manifold. That theorem neither proves that Gołąb’s three-dimensional relations imply involutivity nor gives a three-dimensional counterexample satisfying the exact fixed-coordinate hypotheses in the statement.

Rui Loja Fernandes and Ivan Struchiner prove local realization when the structure functions and their invariant derivatives together define a Lie algebroid: their equations (3.3)–(3.4) require both the anchor-morphism equation and the Jacobi-type equation. Gołąb assumes only the latter algebraic compatibility after an invertible frame has been chosen; the missing anchor equation is the differential realization equation itself, so this theorem does not close the gap.

Rui Loja Fernandes and Wilmer Smilde identify torsion, curvature, and higher Spencer-cohomology obstructions for general Cartan realization problems. Their Theorems 3.10 and 3.20 give analytic existence after formal integrability (or sufficient Spencer-vanishing hypotheses) has been established. They do not show that Gołąb’s first compatibility relations force all higher obstructions to vanish in dimension three. Thus the literature provides rigorous conditional and special-case progress, but no complete answer to the exact problem.

Key references

- Stanisław Gołąb, “Über das Anholonomitätsobjekt von Schouten und van Dantzig,” in *Comptes rendus du Congrès international des mathématiciens, Oslo, 1936*, vol. II, A. W. Brøgers Boktrykkeri, Oslo, 1937, pp. 153–155. [Official proceedings](#). The paper derives the displayed necessary relations for $n = 3$ and explicitly asks whether they are sufficient.
- Robert L. Bryant, “Notes on Exterior Differential Systems,” Fields Institute workshop notes, 2014, arXiv:1405.3116. [arXiv](#). Section 3 defines Jacobi manifolds, states that the bare Jacobi condition is not generally sufficient, and proves Theorem 4 under the additional involutivity hypothesis; it also records the constant-coefficient Lie-theorem case.
- Rui Loja Fernandes and Ivan Struchiner, “The Classifying Lie Algebroid of a Geometric Structure I: Classes of Coframes,” *Transactions of the American Mathematical Society* **366** (2014), no. 5, 2419–2462, DOI: <https://doi.org/10.1090/S0002-9947-2014-05973-4>, arXiv:1103.5850. Proposition 3.2 identifies the full Lie-algebroid compatibility equations, and the local realization results apply once both of them hold.
- Rui Loja Fernandes and Wilmer Smilde, “Relative Algebroids and Cartan Realization Problems,” arXiv preprint, 2025, arXiv:2503.19233. [arXiv](#). Theorems 3.10 and 3.20 relate formal integrability and analytic realization, while Example 4.19 places Bryant’s Jacobi-manifold condition at the first-obstruction level.

3.10.3. Remaining open problems recorded by the database

No separate descendant problem is listed in the website’s “Some remaining open problems” field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

3.10.4. Primary-source pointer

The originating ICM record is *Über das Anholonomitätsobjekt von Schouten und van Dantzig* (1936). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1936.2/ICM1936.2.ocr.pdf#page=170>.

3.11. 124. MODERN DEVELOPMENT OF SURFACE THEORY [1958-geom-9]

Record: 1958-geom-9

Subfield: Classical and local differential geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1958: MODERN DEVELOPMENT OF SURFACE THEORY

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1958/ICM1958.ocr.pdf#page=67>

3.11.1. Problem statement

Let ds^2 be a prescribed two-dimensional line element defining an intrinsic metric on a surface M , and let S be a convex surface in a three-dimensional space of constant curvature that realizes this metric isometrically (in Euclidean space, this means that some parametrization $\mathbf{x}(u, v)$ of S satisfies $d\mathbf{x}^2 = ds^2$). Determine the minimal conditions on the prescribed line element ds^2 that ensure that the realizing convex surface S is twice differentiable, respectively three times differentiable.

3.11.2. Historical background, literature, and current progress

3.11.3. Verdict: partially resolved

3.11.3.1. Precise problem

The problem asks for a regularity criterion for convex isometric realizations of a prescribed two-dimensional line element. In modern terms: given a two-dimensional intrinsic metric ds^2 on a surface and a convex isometric realization in a three-dimensional space form, determine the weakest conditions on ds^2 which force the realizing surface/embedding to be C^2 , respectively C^3 .

The important point is that this is a minimal-conditions problem, not just an existence theorem for smooth metrics. In a space form of sectional curvature K_0 , the strictly elliptic case corresponds to intrinsic Gauss curvature $K > K_0$; the degenerate convex case allows $K \geq K_0$ and includes flat/developable behavior.

3.11.3.2. Original source

- Title: Modern development of surface theory
- Author: A. D. Alexandrov
- Venue: Proceedings of the International Congress of Mathematicians, Edinburgh 1958, Cambridge University Press, 1960, pp. 3-18
- DOI/arXiv/URL: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1958/ICM1958.ocr.pdf>

- Relevance: Section 6.4 states that known results gave differentiability of the convex surface from stronger differentiability assumptions on the line element, but that finding the minimal conditions ensuring two or three differentiability remained unsolved.

3.11.3.3. Main progress

- Title: On the regularity of a convex surface with a regular metric in Euclidean space
- Author: A. V. Pogorelov
- Venue: Dokl. Akad. Nauk SSSR 139:5, 1056-1058; English translation in Soviet Math. Dokl. 2, 1961
- DOI/arXiv/URL: MathNet listing: <https://www.mathnet.ru/eng/person18697>
- Result/progress: This filled a major gap left in the 1958 discussion. Later summaries state Pogorelov's theorem as: a convex surface with a C^n regular metric, $n \geq 2$, of positive Gauss curvature is $C^{n-1, \alpha}$ for every $0 < \alpha < 1$, and an analytic metric gives an analytic surface. This gives strong sufficient regularity in the strictly positive curvature case, but it has a derivative loss in plain C^n classes and is not a necessary-and-sufficient minimal criterion for all convex realizations.
- Title: Extrinsic Geometry of Convex Surfaces
- Author: A. V. Pogorelov
- Venue: Translations of Mathematical Monographs, vol. 35, American Mathematical Society, 1973; translated from the Russian edition, Nauka, 1969
- DOI/arXiv/URL: AMS book record; also cited in later literature as the standard source
- Result/progress: Gives the general theory of convex surfaces and regularity/rigidity results used in subsequent Weyl problem literature. It supports the positive-curvature regularity theory but does not settle the full minimal-conditions question in the problem statement.
- Title: The Weyl and Minkowski problems in differential geometry in the large
- Author: Louis Nirenberg
- Venue: Communications on Pure and Applied Mathematics 6, 337-394, 1953
- DOI/arXiv/URL: <https://doi.org/10.1002/cpa.3160060303>
- Result/progress: Solved the classical smooth Weyl embedding problem for positively curved metrics on S^2 under high regularity hypotheses, classically C^4 . This is foundational but gives sufficient hypotheses, not the minimal differentiability threshold asked in the problem statement.
- Title: On Weyl's embedding problem
- Author: Erhard Heinz
- Venue: Journal of Mathematics and Mechanics 11, 421-454, 1962
- DOI/arXiv/URL: JSTOR stable URL: <https://www.jstor.org/stable/24900718>
- Result/progress: Settled the C^3 positive-curvature Weyl case for obtaining a C^2 convex embedding in Euclidean space. Later papers describe this as a reduction from Niren-

berg's C^4 requirement to C^3 . This is major progress for the twice differentiable part in the strictly elliptic closed-sphere case, but it is not the full all-cases minimal answer.

- Title: The regularity of convex regions with a metric that is regular in the Holder classes
- Author: I. Kh. Sabitov
- Venue: Sibirsk. Mat. Zh. 17, 907-915, 1976; English translation, Siberian Mathematical Journal 17, 681-687, 1976
- DOI/arXiv/URL: <https://doi.org/10.1007/BF00971678>
- Result/progress: In Holder classes there is no derivative loss: a convex surface with $C^{n,\alpha}$ metric of positive curvature is $C^{n,\alpha}$. Thus $C^{2,\alpha}$ metrics give $C^{2,\alpha}$ convex surfaces and $C^{3,\alpha}$ metrics give $C^{3,\alpha}$ convex surfaces in the positive-curvature setting. This is close to the desired regularity statement but assumes Holder control and positive curvature.
- Title: The connections between the order of smoothness of a surface and its metric
- Authors: I. Kh. Sabitov and S. Z. Shefel'
- Venue: Sibirsk. Mat. Zh. 17, 916-925, 1976; English translation, Siberian Mathematical Journal 17, 687-694, 1976
- DOI/arXiv/URL: <https://doi.org/10.1007/BF00971679>
- Result/progress: Gives converse-type information relating the smoothness of a submanifold and of its induced metric. This helps clarify sharpness within Holder categories, but it is not a complete minimal criterion for arbitrary convex isometric realizations in space forms.
- Title: Regularity of locally convex surfaces
- Author: Friedmar Schulz
- Venue: Bulletin of the Australian Mathematical Society 42:3, 487-497, 1990
- DOI/arXiv/URL: <https://doi.org/10.1017/S0004972700028653>
- Result/progress: Derived interior estimates for locally convex surfaces and relaxed positive-curvature Weyl regularity toward $C^{2,\alpha}$ metric assumptions for C^2 regularity. This is partial progress on the twice differentiable threshold, not a full minimal classification.
- Title: On Weyl's Embedding Problem in Riemannian Manifolds
- Author: Siyuan Lu
- Venue: International Mathematics Research Notices 2020:11, 3229-3259; originally arXiv:1608.07539
- DOI/arXiv/URL: <https://doi.org/10.1093/imrn/rny109>; <https://arxiv.org/abs/1608.07539>
- Result/progress: Theorem 1.3 proves that if g is a C^2 metric on S^2 , D^2g is Dini continuous, and the scalar curvature satisfies $R > 2K$, then (S^2, g) embeds in the three-dimensional space form of curvature K as a C^2 surface. This is one of the strongest modern sufficient criteria for the C^2 part in the strictly elliptic space-form case. It still assumes a Dini modulus and strict curvature and does not prove a necessary-and-sufficient minimal condition.
- Title: The Weyl problem of isometric immersions revisited

- Author: Siran Li
- Venue: Bulletin of the London Mathematical Society 53:1, 220-230, 2021; arXiv:2004.05532
- DOI/arXiv/URL: <https://doi.org/10.1112/blms.12413>; <https://arxiv.org/abs/2004.05532>
- Result/progress: Revisits the Weyl problem and its generalizations, gives sufficient conditions in degenerate settings, and records the strict-curvature consequence that C^3 metrics in suitable ambient settings yield C^3 surfaces. The introduction also notes obstructions in degenerate/nonnegative curvature cases, including metrics real-analytic and positive except at one point that do not admit global C^3 embeddings into R^3 . This reinforces that the full minimal-conditions question is not settled merely by the strict positive-curvature theory.

3.11.3.4. Why this is not fully resolved

The literature fully resolves many important special regimes: smooth or analytic metrics, strictly positive curvature on S^2 , Holder classes, and modern Dini-type sufficient conditions for C^2 embeddings in space forms. These results are too strong or too specialized to answer the problem statement exactly.

What remains unresolved is the complete minimal criterion on the prescribed line element that is necessary and sufficient for C^2 and for C^3 regularity of the realizing convex surface in the full space-form convex-realization problem, including degenerate convex metrics and the transition from strict curvature to nonnegative extrinsic curvature. Current results should therefore be classified as partial progress, not a full resolution.

3.11.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

3.11.5. Primary-source pointer

The originating ICM record is *MODERN DEVELOPMENT OF SURFACE THEORY* (1958). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1958/ICM1958.ocr.pdf#page=67>.

3.12. 125. Differential geometry; its past and its future [1970-geom-14]

Record: 1970-geom-14

Subfield: Classical and local differential geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1970: Differential geometry; its past and its future

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1970.1/ICM1970.1.ocr.pdf#page=79>

3.12.1. Problem statement

Let M be a compact differentiable surface, let $I = (-1, 1)$, and let E^3 denote three-dimensional Euclidean space. Let

$$f : M \times I \rightarrow E^3$$

be a differentiable mapping such that, for each $t \in I$, the map

$$f_t : M \rightarrow E^3, \quad f_t(x) = f(x, t),$$

is an immersion. Suppose that the Riemannian metric ds_t^2 on M induced by the immersion f_t is independent of t . Does there exist, for each $t \in I$, a rigid motion $g(t)$ of E^3 such that

$$f_t(x) = g(t)(f_0(x))$$

for every $x \in M$?

3.12.2. Historical background, literature, and current progress

3.12.3. Verdict: partially resolved

3.12.3.1. Precise problem

This is Chern's Problem VIII from S.-S. Chern, "Differential Geometry; Its Past and Its Future" (ICM Nice 1970): whether every differentiable one-parameter isometric deformation of a compact surface immersed in Euclidean 3-space is trivial, i.e. obtained from the initial immersion by ambient rigid motions.

Equivalently, in the usual smooth/ C^2 interpretation of the classical surface-theory problem, it asks whether closed compact smooth immersed surfaces in $\mathbb{R}^3$ are isometrically flexible.

3.12.3.2. Why the full problem is not resolved

I found no published or otherwise citable theorem that covers arbitrary compact smooth immersed surfaces in $\mathbb{R}^3$ and proves that every one-parameter isometric deformation is induced

by rigid motions, nor a smooth compact counterexample giving such a one-parameter deformation. Modern open-problem accounts still list the existence of a closed C^2 flexible surface in $\mathbb{R}^3$ as open.

the exact question is therefore not fully resolved. The literature contains substantial special-case results and adjacent non-uniqueness results, but they do not imply the full statement/all cases asked by Chern's problem.

3.12.3.3. Key literature and status

Original source

- Title: "Differential Geometry; Its Past and Its Future"
- Author: Shiing-Shen Chern
- Venue: Actes du Congrès International des Mathématiciens, Nice 1970, Tome 1, pp. 41-53
- Year: 1970
- DOI/arXiv/URL: <https://www.hiroleetanaka.com/pdfs/2014-fall-1-230a-icml1970-chern-differential-geometry.pdf>
- Relevant result/status: Chern states Problem VIII exactly in the form of the problem. He notes classical rigidity for compact surfaces under curvature hypotheses, but explicitly says the problem is not known once the curvature condition is dropped.

Convex and nonnegative-curvature special cases

- Title: "Zwei Satze uber die Starrheit der Eiflachen"
- Author: Stephan Cohn-Vossen
- Venue: Nachrichten von der Gesellschaft der Wissenschaften zu Gottingen, Mathematisch-Physikalische Klasse
- Year: 1927
- DOI/arXiv/URL: no DOI found; cited in Chern 1970 and later surveys
- Relevant result/status: Rigidity of closed convex/positively curved surfaces. This supports an affirmative answer only for convex or positive-curvature cases, not arbitrary compact immersed surfaces.
- Title: "Differentialgeometrie geschlossener Flachen im euklidischen Raum"
- Author: K. Voss
- Venue: Jahresbericht der Deutschen Mathematiker-Vereinigung, Vol. 63, pp. 117-136
- Year: 1960-1961
- DOI/arXiv/URL: cited as reference [35] in Chern 1970, URL above
- Relevant result/status: Covers an additional curvature-restricted class only. It does not settle surfaces of mixed curvature or arbitrary immersed compact surfaces.

Open-problem surveys

- Title: "Open Problems in Geometry of Curves and Surfaces"
- Author: Mohammad Ghomi
- Venue: unpublished/open-problems manuscript

- Year: 2017-2019 versions circulated online
- DOI/arXiv/URL: <https://ghomi.math.gatech.edu/Papers/op.pdf>
- Relevant result/status: Section 1.1 states that no closed C^2 , or even $C^{1,1}$, flexible surface in $\mathbb{R}^3$ is known, and asks whether a closed C^2 surface in $\mathbb{R}^3$ can be flexible. This is essentially the smooth regularity form of Chern's one-parameter deformation problem.
- Title: "Review of Geometry and Analysis"
- Author: Shing-Tung Yau
- Venue: Mathematics: Frontiers and Perspectives, American Mathematical Society
- Year: 2000
- DOI/arXiv/URL: https://www.intlpress.com/site/pub/files/_fulltext/journals/ajm/2000/0004/0001/AJM-2000-0004-0001-a016.pdf
- Relevant result/status: Discusses the global rigidity problem for closed surfaces in E^3 as an open classical question and asks for projective/affine formulations. This does not resolve the full problem.

Low-regularity flexibility, not the classical smooth problem

- Title: " C^1 Isometric Imbeddings"
- Author: John Nash
- Venue: Annals of Mathematics, Second Series, Vol. 60, No. 3, pp. 383-396
- Year: 1954
- DOI/arXiv/URL: <https://doi.org/10.2307/1969840>
- Relevant result/status: Nash's C^1 theorem begins the low-regularity flexibility theory. In the C^1 category, isometric embeddings are flexible in ways impossible for C^2 /smooth surfaces, but this does not resolve the classical C^2/C^∞ compact-surface problem.
- Title: "On C^1 -isometric imbeddings. I, II"
- Author: Nicolaas H. Kuiper
- Venue: Indagationes Mathematicae, Vol. 17, pp. 545-556 and 683-689
- Year: 1955
- DOI/arXiv/URL: MR0075640; MathSciNet/Zbl references
- Relevant result/status: Extends Nash's C^1 flexibility. Again, this is a low-regularity counterpoint, not a solution to the smooth/ C^2 version of Chern's problem.

Adjacent non-uniqueness results, not one-parameter flexing

- Title: "Compact Bonnet pairs: isometric tori with the same curvatures"
- Authors: Alexander I. Bobenko, Tim Hoffmann, Andrew O. Sageman-Furnas
- Venue: Publications Mathématiques de l'IHES, Vol. 142, pp. 241-293
- Year: 2025
- DOI/arXiv/URL: <https://doi.org/10.1007/s10240-025-00159-z> ;

arXiv:2110.06335

- Relevant result/status: Constructs two non-congruent smooth/real-analytic immersed tori in $\mathbb{R}^3$ related by a mean-curvature-preserving isometry. This resolves a Bonnet-pair/non-uniqueness problem, but a pair of non-congruent isometric compact immersions is weaker than Chern's requested differentiable one-parameter isometric deformation. The paper itself distinguishes Bonnet pairs from Bonnet families and notes that compact Bonnet families cannot occur in the nonconstant-mean-curvature classification.

Claimed embedded local rigidity result not counted as a resolution

- Title: "Local rigidity of surfaces in space forms"
- Author: Michael T. Anderson
- Venue: arXiv preprint
- Year: 2007, withdrawn 2009
- DOI/arXiv/URL: <https://arxiv.org/abs/0709.3202>
- Relevant result/status: Earlier versions claimed local rigidity of closed embedded surfaces in $\mathbb{R}^3$, which would be only an embedded special case anyway. The arXiv record is withdrawn, so I did not count it as citable resolving literature.

3.12.3.4. What remains unresolved

What remains open is the full smooth/ C^2 compact immersed case with no convexity, nonnegative-curvature, tightness, embeddedness, constant-mean-curvature, or other extra hypotheses: given a differentiable/smooth one-parameter family of compact surface immersions in $\mathbb{R}^3$ with the induced metric independent of the parameter, must the family be induced by ambient rigid motions?

Current verdict for the problem: partially resolved.

3.12.4. Remaining open problems recorded by the database

3.12.4.1. Open problem 1

Does there exist a closed C^2 surface in $\mathbb{R}^3$ which is flexible, i.e. which admits a nontrivial continuous isometric deformation?

Reference supplied by the database. Mohammad Ghomi, "Open Problems in Geometry of Curves and Surfaces," unpublished open-problem manuscript, revised September 2, 2019 (online version current in 2026), Problem 1.1 (attributed there to Leonhard Euler, 1776, and James Clerk Maxwell, 1819), <https://ghomi.math.gatech.edu/Papers/op.pdf>.

3.12.4.2. Open problem 2

Are all smooth tight surfaces in $\mathbb{R}^3$ rigid?

Reference supplied by the database. Mohammad Ghomi, “Open Problems in Geometry of Curves and Surfaces,” unpublished open-problem manuscript, revised September 2, 2019 (online version current in 2026), Problem 1.2 (attributed there to A. D. Alexandrov, 1938, and Louis Nirenberg, 1962), <https://ghomi.math.gatech.edu/Papers/op.pdf>.

3.12.4.3. Open problem 3

Are negatively curved annuli in $\mathbb{R}^3$ bounded by a pair of fixed convex planar curves rigid?

Reference supplied by the database. Mohammad Ghomi, “Open Problems in Geometry of Curves and Surfaces,” unpublished open-problem manuscript, revised September 2, 2019 (online version current in 2026), Problem 1.3, <https://ghomi.math.gatech.edu/Papers/op.pdf>.

3.12.4.4. Open problem 4

Can a negatively curved annulus in $\mathbb{R}^3$, bounded by a pair of convex planar curves, have a closed asymptotic curve?

Reference supplied by the database. Mohammad Ghomi, “Open Problems in Geometry of Curves and Surfaces,” unpublished open-problem manuscript, revised September 2, 2019 (online version current in 2026), unnumbered question immediately following Problem 1.3, <https://ghomi.math.gatech.edu/Papers/op.pdf>; current partial progress: Mohammad Ghomi and Matteo Raffaelli, “Topology of Closed Asymptotic Curves on Negatively Curved Surfaces,” arXiv:2412.19266 [math.DG], 2024 (revised 2025), <https://doi.org/10.48550/arXiv.2412.19266>.

3.12.4.5. Open problem 5

Let Γ be a smooth closed curve immersed in $\mathbb{R}^3$ with a continuous one-to-one binormal vector field B . Must the ribbon (Γ, B) be twisted—that is, must the linking number of Γ and $\Gamma + \varepsilon B$ be nonzero?

Reference supplied by the database. Mohammad Ghomi, “Open Problems in Geometry of Curves and Surfaces,” unpublished open-problem manuscript, revised September 2, 2019 (online version current in 2026), Problem 1.4, <https://ghomi.math.gatech.edu/Papers/op.pdf>; current partial progress: Mohammad Ghomi and Matteo Raffaelli, “Topology of Closed Asymptotic Curves on Negatively Curved Surfaces,” arXiv:2412.19266 [math.DG], 2024 (revised 2025), <https://doi.org/10.48550/arXiv.2412.19266>.

3.12.5. Primary-source pointer

The originating ICM record is *Differential geometry; its past and its future* (1970). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1970.1/ICM1970.1.ocr.pdf#page=79>.

3.13. 126. Isoperimetric inequalities and eigenvalues of the Laplacian [1978-geom-24]

Record: 1978-geom-24

Subfield: Classical and local differential geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1978: Isoperimetric inequalities and eigenvalues of the Laplacian

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1978.1/ICM1978.1.ocr.pdf#page=435>

3.13.1. Problem statement

Prove or disprove that if a minimal surface in Euclidean space $\mathbb{R}^n$ has boundary consisting of several curves, and if A denotes the area of the surface while L denotes the total length of its boundary curves, then

$$L^2 \geq 4\pi A.$$

3.13.2. Historical background, literature, and current progress

3.13.3. Status check

Verdict: partially resolved.

3.13.3.1. Precise problem

The problem asks for the sharp Euclidean isoperimetric inequality for a compact two-dimensional minimal surface in Euclidean space, with boundary a finite union of closed curves. If A is the area of the surface and L is the total length of all boundary components, the question is whether

$$L^2 \geq 4\pi A$$

holds without restrictions on the number of boundary components, the topology of the surface, or the ambient dimension R^n .

3.13.3.2. Source and current progress

1. Title: "Isoperimetric inequalities and eigenvalues of the Laplacian" Authors: Robert Osserman Venue: Proceedings of the International Congress of Mathematicians, Helsinki 1978, pp. 435-442, Acad. Sci. Fennica, Helsinki Year: 1980 DOI/arXiv/URL: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1978.1/ICM1978.1.ocr.pdf> Relevance: This is the apparent source of the problem. Osserman states that the classical inequality holds for minimal surfaces in R^n bounded by a

single curve, and then formulates the several-boundary-curves case as a conjecture. At that time it was known only for doubly connected surfaces, by Osserman-Schiffer in R^3 and Feinberg in arbitrary R^n .

2. Title: "Doubly-connected minimal surfaces" Authors: Robert Osserman and Menahem Schiffer Venue: Archive for Rational Mechanics and Analysis 58, pp. 285-307 Year: 1975 DOI/arXiv/URL: not located in the live search Relevance: Proves the sharp isoperimetric inequality for doubly connected minimal surfaces in R^3 . This is a two-boundary-component/topological special case, not the full several-curve problem.
3. Title: "The isoperimetric inequality for doubly-connected minimal surfaces in R^n " Authors: Jerry M. Feinberg Venue: Journal d'Analyse Mathématique 32, pp. 249-278 Year: 1977 DOI/arXiv/URL: not located in the live search Relevance: Extends the doubly connected case to arbitrary ambient R^n . This still assumes the surface is doubly connected and therefore does not cover arbitrary several boundary curves.
4. Title: "On the isoperimetric inequality for minimal surfaces" Authors: Peter Li, Richard Schoen, Shing-Tung Yau Venue: Annali della Scuola Normale Superiore di Pisa, Classe di Scienze (4) 11(2), pp. 237-244 Year: 1984 DOI/arXiv/URL: https://www.numdam.org/item/ASNSP_1984_4_11_2_237_0/ Relevance: Theorem 1 proves the sharp inequality for compact minimal surfaces in R^n whose boundary is "weakly connected". Theorem 2 proves the sharp inequality in R^3 when the boundary has at most two connected components, without assuming the surface is doubly connected. This is major progress, but the weak-connectedness or at-most-two-components hypotheses do not cover every finite union of boundary curves.
5. Title: "Optimal isoperimetric inequalities" Authors: Frederick J. Almgren, Jr. Venue: Indiana University Mathematics Journal 35, pp. 451-547 Year: 1986 DOI/arXiv/URL: not located in the live search Relevance: Proves sharp isoperimetric inequalities for absolutely area-minimizing integral currents. This covers area-minimizing minimal surfaces, but the problem asks about minimal surfaces in general, not only absolutely area-minimizing ones.
6. Title: "The isoperimetric inequality for a minimal surface with radially connected boundary" Authors: Jaigyoung Choe Venue: Annali della Scuola Normale Superiore di Pisa, Classe di Scienze (4) 17(4), pp. 583-593 Year: 1990 DOI/arXiv/URL: https://www.numdam.org/item/ASNSP_1990_4_17_4_583_0/ Relevance: Theorem 2 proves the sharp inequality for minimal surfaces in R^n whose boundary is radially connected from a point of the surface. Corollary 2 proves that every minimal surface in R^n whose boundary has two connected components satisfies the sharp inequality. Choe also records remaining issues for three or four boundary components and examples not covered by weak or radial connectedness.
7. Title: "On the isoperimetric inequality on a minimal surface" Authors: Andrew Stone Venue: Calculus of Variations and Partial Differential Equations 17, pp. 369-391 Year: 2003 DOI/arXiv/URL: DOI 10.1007/s00526-002-0174-9 Relevance: Gives a formula for the isoperimetric defect of arbitrary minimal surfaces in R^n , and improves the universal non-sharp estimate to $L^2 \geq 2\sqrt{2}\pi A$. Since the constant is below 4π , this is not a resolution of the sharp conjecture.
8. Title: "The isoperimetric inequality for a minimal submanifold in Euclidean space" Authors: Simon Brendle Venue: Journal of the American Mathematical Society 34(2),

pp. 595-603 Year: 2021 DOI/arXiv/URL: DOI 10.1090/jams/969; arXiv:1907.09446
 Relevance: Proves a sharp isoperimetric inequality for compact minimal submanifolds of codimension at most 2. For two-dimensional minimal surfaces this covers arbitrary boundary topology in ambient $\mathbb{R}^3$ and $\mathbb{R}^4$, and surfaces contained in an affine subspace of dimension at most 4. It does not cover the full arbitrary $\mathbb{R}^n$ statement when the ambient codimension is at least 3.

9. Title: “Minimal hypersurfaces and geometric inequalities” Authors: Simon Brendle
 Venue: Annales de la Faculte des sciences de Toulouse: Mathematiques, Serie 6, 32(1), pp. 179-201 Year: 2023 DOI/arXiv/URL: DOI 10.5802/afst.1734; arXiv:2010.03425
 Relevance: A recent survey/exposition. It states the sharp codimension at most 2 result and notes that in codimension 3 or higher only a non-sharp version of the inequality is available.

3.13.3.3. Why this is not fully resolved

Several strong theorems prove the desired $L^2 \geq 4\pi A$ in important subclasses: one boundary curve, two boundary components in arbitrary $\mathbb{R}^n$, weakly connected or radially connected boundary, absolutely area-minimizing surfaces, and arbitrary boundary components in codimension at most 2.

However, I found no citable theorem proving the exact statement for every compact minimal surface in arbitrary Euclidean $\mathbb{R}^n$ with an arbitrary finite number of boundary curves. In particular, for two-dimensional minimal surfaces in ambient $\mathbb{R}^N$ with $N \geq 5$, the codimension is at least 3; Brendle’s published sharp theorem does not cover this range, and recent exposition describes only non-sharp constants there.

What remains unresolved is the full sharp inequality for arbitrary compact minimal surfaces of high codimension with several boundary components, without weak/radial connectedness, area-minimizing, or codimension-at-most-2 hypotheses.

3.13.4. Remaining open problems recorded by the database

3.13.4.1. Open problem 1

Let $S \subset \mathbb{R}^n$ be a compact minimal surface with boundary

$$\partial S = \bigcup_{i=1}^k C_i,$$

where no single component C_i can be separated from all the other boundary components by a hyperplane. If $A = \text{Area}(S)$ and $L_i = \text{Length}(C_i)$, prove that

$$4\pi A \leq \sum_{i=1}^k L_i^2.$$

Reference supplied by the database. Jaigyoung Choe, “Isoperimetric Inequalities of Minimal Submanifolds,” in David Hoffman (ed.), *Global Theory of Minimal Surfaces*, Clay Mathematics Proceedings, Vol. 2, American Mathematical Society, Providence, RI, 2005, pp. 325–369,

Open Problem 2. MR 2167266. <https://www.ams.org/books/cmip/002/>

3.13.5. Primary-source pointer

The originating ICM record is *Isoperimetric inequalities and eigenvalues of the Laplacian* (1978). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1978.1/ICM1978.1.ocr.pdf#page=435>.

3.14. 127. Isoperimetric inequalities and eigenvalues of the Laplacian [1978-geom-26]

Record: 1978-geom-26

Subfield: Classical and local differential geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1978: Isoperimetric inequalities and eigenvalues of the Laplacian

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1978.1/ICM1978.1.ocr.pdf#page=435>

3.14.1. Problem statement

For any integers n and N with $N > n$, does the Euclidean n -dimensional isoperimetric inequality

$$S^n \geq n^n \omega_n V^{n-1}$$

hold for every domain D on an arbitrary n -dimensional minimal variety in $\mathbb{R}^N$, where V is the n -dimensional volume of D , S is the $(n-1)$ -dimensional measure of the boundary ∂D , and ω_n is the volume of the unit ball in $\mathbb{R}^n$?

3.14.2. Historical background, literature, and current progress

3.14.3. Verdict: partially resolved

3.14.3.1. Precise problem

The problem asks whether the sharp Euclidean n -dimensional isoperimetric inequality

$$S^n \geq n^n \omega_n V^{n-1}$$

holds for every domain D in every n -dimensional minimal variety in $\mathbb{R}^N$, for all integers $N > n$. Equivalently, in the smooth compact-with-boundary formulation, it asks whether every compact n -dimensional minimal submanifold/domain $\Sigma^n \subset \mathbb{R}^{n+m}$, in every codimension $m = N - n \geq 1$, satisfies

$$|\partial \Sigma| \geq n \omega_n^{1/n} |\Sigma|^{(n-1)/n}.$$

3.14.3.2. Current literature status

The literature does not appear to fully resolve this exact all-codimension, arbitrary-minimal-variety statement. There is major partial progress:

1. Simon Brendle, "The isoperimetric inequality for a minimal submanifold in Euclidean space", *Journal of the American Mathematical Society* 34(2), 595-603, 2021. DOI: <https://doi.org/10.1090/jams/969>. arXiv:1907.09446, <https://arxiv.org/abs/1907.09446>.

Brendle proves a sharp Sobolev/isoperimetric inequality for compact smooth minimal submanifolds in Euclidean space of codimension at most 2. In particular, his Theorem 3 states that if Σ is a compact n -dimensional minimal submanifold of $\mathbb{R}^{n+2}$, then

$$|\partial\Sigma| \geq n|B^n|^{1/n}|\Sigma|^{(n-1)/n},$$

with equality only for a flat round ball; codimension 1 follows by embedding $\mathbb{R}^{n+1}$ into $\mathbb{R}^{n+2}$. Brendle's general inequality in higher codimension has a codimension-dependent constant and is stated to be sharp only when the codimension is at most 2, so it does not prove the full assertion for all $N > n$.

2. Frederick J. Almgren Jr., "Optimal isoperimetric inequalities", *Indiana University Mathematics Journal* 35(3), 451-547, 1986. DOI: <https://doi.org/10.1512/iumj.1986.35.35028>.

Almgren's optimal filling/isoperimetric inequalities imply the sharp Euclidean inequality for area-minimizing submanifolds/currents in all dimensions and codimensions. This is an important full-codimension special case, but "area-minimizing" is a stronger assumption than "minimal" or "stationary"; it therefore does not cover arbitrary minimal varieties/domains in the problem statement.

3. Jaigyoung Choe, "The isoperimetric inequality for a minimal surface with radially connected boundary", *Annali della Scuola Normale Superiore di Pisa, Classe di Scienze* (4) 17(4), 583-593, 1990. URL: https://www.numdam.org/article/ASNSP_1990_4_17_4_583_0.pdf.

Choe surveys earlier two-dimensional progress and proves the sharp inequality for minimal surfaces satisfying a radial-connectedness condition on the boundary. The paper also records earlier surface cases due to Carleman, Osserman-Schiffer, Feinberg, and Li-Schoen-Yau. These results are restricted to dimension 2 and/or extra topological/boundary hypotheses, so they are not a full solution to this problem.

4. Wenkui Du, Yuchao Yi, and Ziyi Zhao, "Wulff inequality for minimal submanifolds in Euclidean space", arXiv:2412.19063, 2024. URL: <https://arxiv.org/html/2412.19063v1>.

This recent preprint still states the sharp isoperimetric inequality for smooth minimal submanifolds as a conjecture, and summarizes the known status as: Carleman for disk-type minimal surfaces, Almgren for arbitrary-codimension area-minimizing submanifolds, and Brendle for codimension one and two. This supports that the general arbitrary-codimension case is not considered settled in the current literature.

3.14.3.3. What remains unresolved

The full statement asks for every n , every $N > n$, and every domain on an arbitrary n -dimensional minimal variety in $\mathbb{R}^N$. The available resolving theorem of Brendle covers smooth minimal submanifolds only in codimension 1 and 2; Almgren covers all codimensions only with the additional area-minimizing hypothesis; and the older surface literature covers dimension 2 under special assumptions. I found no published or otherwise citable theorem proving the sharp Euclidean inequality for arbitrary minimal varieties/submanifolds in codimension $m \geq 3$, let alone in the full singular "minimal variety" generality.

Therefore the exact problem is **partially resolved**, not fully resolved.

3.14.4. Remaining open problems recorded by the database

3.14.4.1. Open problem 1

Let $S \subset \mathbb{R}^N$ be a compact two-dimensional minimal submanifold with boundary

$$\partial S = \bigcup_{i=1}^k C_i,$$

where no single component C_i can be separated from all the other boundary components by a hyperplane. If $A = \text{Area}(S)$ and $L_i = \text{Length}(C_i)$, prove the strengthened isoperimetric inequality

$$4\pi A \leq \sum_{i=1}^k L_i^2.$$

Reference supplied by the database. Jaigyoung Choe, “Isoperimetric Inequalities of Minimal Submanifolds,” in David Hoffman (ed.), *Global Theory of Minimal Surfaces*, Clay Mathematics Proceedings, Vol. 2, American Mathematical Society, Providence, RI, 2005, pp. 325–369, Open Problem 2. MR 2167266. <https://www.ams.org/books/cmip/002/>

3.14.5. Primary-source pointer

The originating ICM record is *Isoperimetric inequalities and eigenvalues of the Laplacian* (1978). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1978.1/ICM1978.1.ocr.pdf#page=435>.

3.15. 128. Nonlinear control theory and differential geometry [1983-geom-287]

Record: 1983-geom-287

Subfield: Classical and local differential geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1983: Nonlinear control theory and differential geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1983.2/ICM1983.2.ocr.pdf#page=519>

3.15.1. Problem statement

For integers $n > m > 0$, determine the largest integer $\varphi(n, m)$ such that, for every nonempty open set $U \subseteq \mathbb{R}^n$ and every smooth rank- m distribution $D \subset TU$, there exist a nonempty open subset $V \subseteq U$ and a smooth rank- $\varphi(n, m)$ subdistribution $E \subset D|_V$ that is integrable; equivalently, the Lie bracket of any two smooth local sections of E is again a local section of E .

3.15.2. Historical background, literature, and current progress

The problem is **unresolved**. No citable result located determines $\varphi(n, m)$ for every pair $n > m > 0$. In particular, the higher-corank case $n - m \geq 2$ is not covered by the classical one-form normal form, by criteria for a corank-one subdistribution inside a fixed distribution, or by pointwise results about alternating forms.

Established progress includes the following.

- In *Nonlinear Control Theory and Differential Geometry*, Roger W. Brockett, *Proceedings of the International Congress of Mathematicians, Warsaw 1983*, Vol. 2 (published 1984), pp. 1357–1368, MR0804784 ([direct proceedings PDF](#)), Brockett explicitly poses this dimensionality problem without an answer.
- *Size of Tangencies to Non-involutive Distributions*, Zoltan M. Balogh, Cornel Pintea, and Heiner Rohner, *Indiana University Mathematics Journal* 60 (2011), no. 6, 2061–2092, <https://doi.org/10.1512/iumj.2011.60.4489>, proves in Theorem 5.4 that every rank- m hyperplane distribution has, on an open subset, an involutive subdistribution of rank $\lfloor (m + 1)/2 \rfloor$. The Pfaff/contact normal forms show this is sharp, so

$$\varphi(m + 1, m) = \left\lfloor \frac{m + 1}{2} \right\rfloor.$$

Remark 5.6 explicitly says that the authors do not know whether an analogous statement holds in arbitrary higher corank.

- *Exterior Differential Systems*, Robert L. Bryant, S. S. Chern, Robert B. Gardner, Hubert L. Goldschmidt, and Phillip A. Griffiths, MSRI Publications 18, Springer, 1991, <https://doi.org/10.1007/978-1-4613-9714-4>, gives the Pfaff normal form in Theorem II.3.1. It supplies the one-form/hyperplane result above but does not classify general systems of two or more independent one-forms in the smooth category.

- *Contact Systems and Corank One Involutive Subdistributions*, William Pasillas-Lepine and Witold Respondek, *Acta Applicandae Mathematica* 69 (2001), no. 2, 105-128, <https://doi.org/10.1023/A:1013015602007>, Corollary 2.3, gives necessary and sufficient local conditions, in terms of the characteristic distribution, Engel rank, and one additional involutivity test, for a given distribution to contain an involutive subdistribution of corank one. This is a criterion for one candidate rank, not a universal determination of the largest guaranteed rank.
- For any distribution D , its Levi bracket $\Lambda^2 D_x \rightarrow T_x U/D_x$ must vanish on $\Lambda^2 E_x$ for an involutive E . The associated pointwise common-isotropic-subspace problem was studied in *Isotropic Subspaces for Skewforms and Maximal Abelian Subgroups of p -Groups*, Joe Buhler, Ranee Gupta, and Joe Harris, *Journal of Algebra* 108 (1987), no. 1, 269-279, [https://doi.org/10.1016/0021-8693\(87\)90138-4](https://doi.org/10.1016/0021-8693(87)90138-4), and, over non-algebraically-closed fields including $\mathbb{R}$, in *On the Vanishing of Subspaces of Alternating Bilinear Forms*, Rod Gow and Rachel Quinlan, *Linear and Multilinear Algebra* 54 (2006), no. 6, 415-428, <https://doi.org/10.1080/03081080500361264>. These results constrain possible tangent planes but do not integrate a smoothly varying family of such planes into an involutive subdistribution.
- *Turan Problems for Multilinear Maps*, Qiyuan Chen, Zixiang Xu, and Ke Ye, arXiv:2603.00715v2 (2026), [arXiv:2603.00715](https://arxiv.org/abs/2603.00715), determines the corresponding isotropy index for alternating maps over algebraically closed fields. The original problem is over $\mathbb{R}$, and pointwise isotropy remains only a necessary first-order condition; this result therefore does not resolve the differential problem.

Thus the exact values in low or corank-one cases and the available algebraic and geometric criteria are substantive partial results, but the universal function $\varphi(n, m)$ remains undetermined in higher corank.

3.15.3. Remaining open problems recorded by the database

3.15.3.1. Open problem 1

For integers $p \geq 1$ and $q \geq 2$, prove or disprove that, for every nonempty open set $U \subseteq \mathbb{R}^{p+q}$ and every C^1 rank- p distribution $D \subset TU$, there exist a nonempty open subset $V \subseteq U$ and a C^1 rank- $\lfloor (p+q)/(q+1) \rfloor$ subdistribution $E \subset D|_V$ such that $[X, Y](x) \in E_x$ for all $x \in V$ and all C^1 local sections X, Y of E defined near x .

Reference supplied by the database. Zoltan M. Balogh, Cornel Pintea, and Heiner Rohner, “Size of Tangencies to Non-involutive Distributions,” *Indiana University Mathematics Journal* 60 (2011), no. 6, 2061-2092, Remark 5.6, <https://doi.org/10.1512/iumj.2011.60.4489>.

3.15.4. Primary-source pointer

The originating ICM record is *Nonlinear control theory and differential geometry* (1983). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1983.2/ICM1983.2.ocr.pdf#page=519>.

3.16. 129. W. H., Recent progress on the geometry of surfaces in $\mathbb{R}$ and on the use of computer... [1986-geom-41]

Record: 1986-geom-41

Subfield: Classical and local differential geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1986: W. H., Recent progress on the geometry of surfaces in $\mathbb{R}$ and on the use of computer graphics as a research tool

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1986.1/ICM1986.1.ocr.pdf#page=653>

3.16.1. Problem statement

Prove or disprove that for every properly embedded minimal surface $M \subset \mathbb{R}^3$, every intrinsic isometry $f : M \rightarrow M$ of M with its induced Riemannian metric extends to a Euclidean isometry $F : \mathbb{R}^3 \rightarrow \mathbb{R}^3$; that is, $F(M) = M$ and $F|_M = f$.

3.16.2. Historical background, literature, and current progress

3.16.3. Verdict: partially resolved

3.16.3.1. Precise problem

The statement is the Choi–Meeks–White/Meeks Isometry Conjecture for properly embedded minimal surfaces in Euclidean 3-space: if $M \subset \mathbb{R}^3$ is a properly embedded minimal surface and $f : M \rightarrow M$ is an intrinsic isometry for the induced metric, must f be the restriction of an ambient Euclidean isometry F with $F(M) = M$?

This is weaker than the often-cited minimal rigidity formulation, which asks uniqueness up to congruence of isometric minimal immersions, but that stronger formulation implies the extension statement.

3.16.3.2. Status

The exact all-cases statement is not fully resolved in the literature I found. It is proved in several large classes, especially surfaces with more than one end and surfaces of finite genus, but current survey literature still lists the unrestricted statement as a conjecture. The remaining case is essentially one-ended, infinite-genus properly embedded minimal surfaces; more refined reductions point to zero-flux/bounded-curvature/quasi-dilation-periodic scenarios.

3.16.3.3. Key literature and progress

1. Recent progress on the geometry of surfaces in $\mathbb{R}^3$ and on the use of computer graphics as a research tool

Author: William H. Meeks III

Venue/year: Proceedings of the International Congress of Mathematicians, Berkeley 1986, 1987, pp. 551–556.

URL: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1986.1/ICM1986.1.ocr.pdf>

Role: This is the dataset's indicated source area for Meeks' problem. Later surveys identify the question as Meeks' isometry/rigidity conjecture for properly embedded minimal surfaces.

2. Some recent developments in the theory of properly embedded minimal surfaces in $\mathbb{R}^3$

Author: Harold Rosenberg

Venue/year: Séminaire Bourbaki, Vol. 34 (1991–1992), pp. 463–535.

URL: <https://eudml.org/doc/110163>

Role: Records the conjectural status and early partial results: Rosenberg states that Meeks conjectures every symmetry of a properly embedded minimal surface extends to an isometry of $\mathbb{R}^3$, and notes Choi–Meeks–White's proof for surfaces with more than one end and Meeks–Rosenberg's result for doubly periodic surfaces.

3. A rigidity theorem for properly embedded minimal surfaces in $\mathbb{R}^3$

Authors: Hyeon In Choi, William H. Meeks III, Brian White

Venue/year: Journal of Differential Geometry 32 (1990), no. 1, 65–76.

DOI/URL: 10.4310/jdg/1214445037, <https://projecteuclid.org/journals/journal-of-differential-geometry/volume-32/issue-1/A-rigidity-theorem-for-properly-embedded-minimal-surfaces-in-bf/10.4310/jdg/1214445037.pdf>

Result: Proves the extension/rigidity result when the properly embedded minimal surface has more than one end. This is a major partial resolution but does not cover one-ended surfaces.

4. A rigidity theorem for periodic minimal surfaces

Author: Joaquín Pérez

Venue/year: Communications in Analysis and Geometry 7 (1999), no. 1, 95–104.

URL: https://www.intlpress.com/site/pub/files/_fulltext/journals/cag/1999/0007/0001/CAG-1999-0007-0001-a003.pdf

Result: Proves rigidity for periodic properly embedded minimal surfaces with finite-topology quotient, apart from the helicoid as the non-rigid associate-family exception. This is a special periodic case, not the unrestricted problem.

5. The rigidity of embedded constant mean curvature surfaces

Authors: William H. Meeks III, Giuseppe Tinaglia

Venue/year: Journal für die reine und angewandte Mathematik 660 (2011), 181–190.

DOI/arXiv/URL: 10.1515/CRELLE.2011.073; arXiv:0801.3409, <https://arxiv.org/abs/0801.3409>

Result: Theorem 1.2, the Finite Genus Isometry Extension Theorem, states that if $M \subset \mathbb{R}^3$ is complete embedded CMC of finite genus and is minimal, then every intrinsic isometry $\sigma : M \rightarrow M$ extends to an isometry of $\mathbb{R}^3$. This covers all finite-genus minimal cases but not infinite-genus one-ended surfaces.

6. The classical theory of minimal surfaces

Authors: William H. Meeks III, Joaquín Pérez

Venue/year: Bulletin of the American Mathematical Society 48 (2011), no. 3, 325–407.

DOI/URL: 10.1090/S0273-0979-2011-01334-9, <https://www.ams.org/bull/2011-48-03/>

Result/status summary: Conjecture 11.10 is exactly the Isometry Conjecture: if $M \in \mathcal{C}$, every intrinsic isometry of M extends to an ambient isometry of $\mathbb{R}^3$. The survey says it is known for more-than-one-end surfaces, doubly periodic surfaces, periodic finite-topology quotients, and finite genus. It also states the reduction to checking one-ended infinite-genus properly embedded examples with a plane intersection containing a simple closed curve. This confirms the full statement was not then resolved.

7. A Survey on Classical Minimal Surface Theory

Authors: William H. Meeks III, Joaquín Pérez

Venue/year: American Mathematical Society, University Lecture Series 60, 2012.

DOI/URL: 10.1090/ulect/060, <https://www.ams.org/books/ulect/060/>

Result/status summary: Repeats the Isometry Conjecture as an outstanding problem and lists the same known cases. It further notes reductions using the Dynamics Theorem and Local Picture Theorem, but does not claim a full proof.

8. The local picture theorem on the scale of topology

Authors: William H. Meeks III, Joaquín Pérez, Antonio Ros

Venue/year: Journal of Differential Geometry 109 (2018), no. 3, 509–565.

DOI/arXiv/URL: 10.4310/jdg/1531188195; arXiv:1505.06761, <https://arxiv.org/abs/1505.06761>

Result/status summary: Gives structural restrictions on potential counterexamples. Remark 5.6 says a complete embedded minimal surface with an intrinsic isometry not extending to an ambient isometry would have to have zero flux and constrained local pictures; the authors describe this as potentially useful for proving the classical conjecture. This is diagnostic progress, not a proof of the full conjecture.

3.16.3.4. What remains unresolved

The published theorems cover properly embedded minimal surfaces with more than one end and all finite-genus cases, among other periodic/special cases. I found no citable theorem proving the full statement for every properly embedded minimal surface. The unresolved part is the possible one-ended, infinite-genus properly embedded minimal surface case, especially the zero-flux configurations singled out by the later reduction and local-picture work. Therefore the correct verdict for the problem is **partially resolved**.

3.16.4. Remaining open problems recorded by the database

3.16.4.1. Open problem 1

Let $M \subset \mathbb{R}^3$ be a properly embedded minimal surface with exactly one end and infinite genus. Prove that every intrinsic isometry of M extends to an ambient Euclidean isometry of $\mathbb{R}^3$.

Reference supplied by the database. William H. Meeks III and Giuseppe Tinaglia, “The rigidity of embedded constant mean curvature surfaces,” *Journal für die reine und angewandte Mathematik* **660** (2011), 181–190, especially the outstanding isometry conjecture in the introduction and Theorem 1.2 (the finite-genus case), DOI: <https://doi.org/10.1515/CRELLE.2011.073>. Together with Hyeon In Choi, William H. Meeks III, and Brian White, “A

rigidity theorem for properly embedded minimal surfaces in $\mathbb{R}^3$," *Journal of Differential Geometry* **32** (1990), no. 1, 65–76 (the case of at least two ends), DOI: <https://doi.org/10.4310/jdg/1214445037>, these results isolate the stated one-ended infinite-genus case as the remaining open case.

3.16.5. Primary-source pointer

The originating ICM record is W. H., *Recent progress on the geometry of surfaces in R and on the use of computer graphics as a research tool* (1986). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1986.1/ICM1986.1.ocr.pdf#page=653>.

3.17. 130. W. H., Recent progress on the geometry of surfaces in $\mathbb{R}$ and on the use of computer... [1986-geom-42]

Record: 1986-geom-42

Subfield: Classical and local differential geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1986: W. H., Recent progress on the geometry of surfaces in $\mathbb{R}$ and on the use of computer graphics as a research tool

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1986.1/ICM1986.1.ocr.pdf#page=653>

3.17.1. Problem statement

Prove that for every surface M that is homeomorphic to a closed surface with at least three points removed, there exists a proper embedding $f : M \rightarrow \mathbb{R}^3$ such that the embedded surface $f(M)$ has constant mean curvature 1.

3.17.2. Historical background, literature, and current progress

3.17.3. Verdict: partially resolved

3.17.3.1. Precise problem

The problem asks for an existence theorem for complete finite-topology constant-mean-curvature surfaces in Euclidean three-space: for every orientable surface of type (g, k) , i.e. a closed genus g surface with $k \geq 3$ punctures, does there exist a proper embedding in $\mathbb{R}^3$ with constant mean curvature $H = 1$?

The normalization $H = 1$ is harmless up to Euclidean dilation. The word “embedding” is essential here: results for properly immersed or merely Alexandrov/almost embedded CMC surfaces do not by themselves settle the problem statement.

3.17.3.2. Current progress

1. Complete constant mean curvature surfaces in Euclidean three-space

- Authors: Nicolaos Kapouleas
- Venue: Annals of Mathematics (2), 131(2), 239-330
- Year: 1990
- DOI/URL: DOI 10.2307/1971494; JSTOR <https://www.jstor.org/stable/1971494>
- Relevance: Kapouleas constructed many complete finite-topology CMC surfaces by gluing Delaunay pieces and spheres. The 1987 announcement states the construction gives properly immersed complete CMC surfaces of genus g with m ends in the

relevant ranges, and later CMC moduli papers cite Kapouleas for nonemptiness of the Alexandrov-embedded moduli spaces $M_{g,k}$ for $k > 2$.

- Limitation: this is not a direct proof of the exact problem because the problem asks for proper embeddings. The citable statements I found distinguish properly immersed/Alexandrov embedded surfaces from strict embedded surfaces.

2. Triunduloids: Embedded constant mean curvature surfaces with three ends and genus zero

- Authors: Karsten Grosse-Brauckmann, Robert B. Kusner, John M. Sullivan
- Venue: Journal für die reine und angewandte Mathematik, 564, 35-61
- Year: 2003
- DOI/arXiv/URL: DOI 10.1515/crll.2003.093; arXiv:math/0102183
- Relevance: this constructs/classifies the complete embedded CMC surfaces with exactly three ends and genus zero. Thus the problem is resolved for the sphere minus three points.
- Limitation: it does not cover three-ended surfaces of positive genus.

3. Embedded constant mean curvature surfaces in Euclidean three-space

- Authors: Christine Breiner, Nikolaos Kapouleas
- Venue: Mathematische Annalen, 360(3-4), 1041-1108
- Year: 2014
- DOI/arXiv/URL: arXiv:1210.3394
- Relevance: this refines Kapouleas' construction and proves embeddedness for a large class of finite-topology CMC examples arising from flexible pre-embedded graphs.
- Limitation: the theorem is graph-hypothesis based and is presented as producing a large class of embedded examples, not every pair (g, k) with $k \geq 3$. It therefore does not by itself imply the full problem.

4. Constant Mean Curvature surfaces with prescribed finite topologies

- Author: Stephen J. Kleene
- Venue: arXiv preprint
- Year: 2023
- DOI/arXiv/URL: arXiv:2309.08344
- Relevance: Theorem 1.1 states that for every $g \geq 0$ and every $k \geq 4$, there is a family of complete embedded CMC surfaces in $\mathbb{R}^3$ with genus g and k ends. This fully resolves the problem statement for all surfaces with at least four punctures.
- Limitation: the theorem assumes $k \geq 4$. The same preprint lists "Complete embedded CMC surfaces with three ends and any genus" as a work "in preparation", not as a published or arXiv theorem.

5. The structure of complete embedded surfaces with constant mean curvature

- Authors: Nicholas J. Korevaar, Robert Kusner, Bruce Solomon
- Venue: Journal of Differential Geometry, 30(2), 465-503
- Year: 1989
- DOI/URL: DOI 10.4310/jdg/1214443598
- Relevance: this proves the Delaunay asymptotic structure of annular ends and the classification of the two-ended embedded finite-topology case as Delaunay. It explains why the original question starts at three punctures.
- Limitation: it is structural/classificatory, not an existence theorem for all $k \geq 3$.

3.17.3.3. What remains unresolved

The citable literature found fully covers all genera with $k \geq 4$ ends and covers the genus-zero case with $k = 3$ ends. I did not find a published or arXiv result proving the missing case: proper embedded CMC $H = 1$ surfaces of arbitrary positive genus with exactly three ends.

Therefore the exact statement is not fully resolved. The correct status is **partially resolved**.

3.17.4. Remaining open problems recorded by the database

3.17.4.1. Open problem 1

For every integer $g \geq 1$, construct a complete properly embedded constant-mean-curvature-one surface in $\mathbb{R}^3$ having genus g and exactly three ends.

Reference supplied by the database. Stephen J. Kleene, “Constant Mean Curvature surfaces with prescribed finite topologies,” arXiv preprint (2023), Theorem 1.1 and bibliography item “Complete embedded CMC surfaces with three ends and any genus, in preparation,” [arXiv:2309.08344](https://arxiv.org/abs/2309.08344).

3.17.5. Primary-source pointer

The originating ICM record is W. H., *Recent progress on the geometry of surfaces in $\mathbb{R}^3$ and on the use of computer graphics as a research tool* (1986). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1986.1/ICM1986.1.ocr.pdf#page=653>.

3.18. 131. W. H., Recent progress on the geometry of surfaces in $\mathbb{R}$ and on the use of computer... [1986-geom-43]

Record: 1986-geom-43

Subfield: Classical and local differential geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1986: W. H., Recent progress on the geometry of surfaces in $\mathbb{R}$ and on the use of computer graphics as a research tool

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1986.1/ICM1986.1.ocr.pdf#page=653>

3.18.1. Problem statement

Prove that every properly embedded surface $M \subset \mathbb{R}^3$ with constant mean curvature that is contained in a half-space $H \subset \mathbb{R}^3$ is invariant under reflection across some plane parallel to the boundary plane ∂H of the half-space.

3.18.2. Historical background, literature, and current progress

3.18.3. Verdict: partially resolved

3.18.3.1. Precise problem

The problem asks the half-space Alexandrov-symmetry problem for constant-mean-curvature surfaces in Euclidean 3-space: if a properly embedded CMC surface $M \subset \mathbb{R}^3$ is contained in a half-space H , must M be invariant under reflection in some plane parallel to ∂H ?

This is stronger than the known finite-topology/finite-ends statements. If such a reflecting plane exists for a surface lying on one side of ∂H , then the surface is also trapped in a slab parallel to ∂H . Thus unresolved slab-symmetry questions are directly relevant to this problem.

3.18.3.2. Literature found

1. **Recent Progress on the Geometry of Surfaces in $\mathbb{R}^3$ and on the Use of Computer Graphics as a Research Tool**

Authors: William H. Meeks III.

Venue/year: Proceedings of the International Congress of Mathematicians, Berkeley, 1986, published 1987.

DOI/arXiv/URL: no DOI found; IMU proceedings OCR: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1986.1/ICM1986.1.ocr.pdf>; author bibliography: <https://profmeeks.wordpress.com/my-research-papers-2/>

Relevance: this is the source family for the problem. It places the question in the Alexandrov reflection theory of properly embedded CMC surfaces.

2. The strong halfspace theorem for minimal surfaces

Authors: David Hoffman and William H. Meeks III.

Venue/year: *Inventiones Mathematicae* 101, 373-377, 1990.

DOI/arXiv/URL: DOI 10.1007/BF01231507; bibliographic listing at https://www.bimsa.cn/research_detail/mincmcr3.html

Result: a connected proper nonplanar minimal surface in $\mathbb{R}^3$ is not contained in a half-space.

Applicability: this resolves the zero-mean-curvature subcase of the problem. A minimal properly embedded surface in a half-space is therefore a plane parallel to the boundary plane, hence has the required reflection symmetry. This is only one subcase.

3. The topology and geometry of embedded surfaces of constant mean curvature

Authors: William H. Meeks III.

Venue/year: *Journal of Differential Geometry* 27, 539-552, 1988.

DOI/arXiv/URL: DOI 10.4310/jdg/1214442008; Project Euclid page <https://projecteuclid.org/journals/journal-of-differential-geometry/volume-27/issue-3/The-topology-and-geometry-of-embedded-surfaces-of-constant-mean/10.4310/jdg/1214442008.full>

Result/progress: proves major topology restrictions for properly embedded nonzero CMC surfaces, including the no-one-end phenomenon cited in later surveys.

Applicability: important background, but it does not prove the full half-space reflection conclusion.

4. The structure of complete embedded surfaces with constant mean curvature

Authors: Nicholas J. Korevaar, Rob Kusner, and Bruce Solomon.

Venue/year: *Journal of Differential Geometry* 30, 465-503, 1989.

DOI/arXiv/URL: DOI 10.4310/jdg/1214443598; bibliographic listing at https://www.bimsa.cn/research_detail/mincmcr3.html

Result: the finite-ended version is resolved. Expositions of the KKS theorem state that if a properly embedded CMC surface has finitely many ends and is contained in a half-space bounded by a plane π , then it has a plane of symmetry parallel to π , and hence lies in a slab. See also Jesse Ratzkin, *An End-to-End Gluing Construction for Surfaces of Constant Mean Curvature*, thesis, 2001, Appendix C, Theorem 17, which explicitly quotes this KKS consequence: <https://sites.math.washington.edu/~pollack/students/Ratzkin-thesis-2001.pdf>

Applicability: this is the closest positive result, but the extra hypothesis “finitely many ends” or finite topology is not present in the problem statement.

5. Properly embedded surfaces with constant mean curvature

Authors: Antonio Ros and Harold Rosenberg.

Venue/year: *American Journal of Mathematics* 132(6), 1429-1443, 2010.

DOI/arXiv/URL: DOI 10.1353/ajm.2010.0015; JSTOR stable page <https://www.jstor.org/stable/40931045>; author PDF via ResearchGate https://www.researchgate.net/publication/228395641_Properly_embedded_surfaces_with_constant_mean_curvature

Result/progress: proves a maximum principle at infinity for properly embedded H -surfaces. The paper explicitly identifies as unsolved whether a properly embedded CMC surface in a horizontal slab admits a horizontal plane of symmetry.

Applicability: because the stated half-space statement would imply the corresponding slab symmetry statement, this is evidence that the unrestricted problem was not resolved by the Ros-Rosenberg maximum principle.

6. The Dynamics Theorem for CMC surfaces in $\mathbb{R}^3$

Authors: William H. Meeks III and Giuseppe Tinaglia.

Venue/year: Journal of Differential Geometry 85(1), 141-173, 2010.

DOI/arXiv/URL: DOI 10.4310/jdg/1284557928; arXiv:0805.1427, <https://arxiv.org/abs/0805.1427>

Result/progress: for a properly embedded nonzero CMC surface with bounded second fundamental form, the paper studies translational limits $T(M)$. Its abstract states that if M has infinitely many ends, there is a nonempty minimal T -invariant set in $T(M)$ consisting entirely of surfaces with planes of Alexandrov symmetry. Searchable text of the paper also states that if M lies in a half-space, a minimal element of $T(M)$ has a horizontal Alexandrov symmetry plane.

Applicability: this is substantial asymptotic progress, but it proves symmetry for certain translational limit surfaces, not for the original M . It also assumes bounded second fundamental form, which is not in the problem statement.

7. Geometry of constant mean curvature surfaces in $\mathbb{R}^3$

Authors: William H. Meeks III and Giuseppe Tinaglia.

Venue/year: Journal of the European Mathematical Society 27(6), 2467-2476, 2025.

DOI/arXiv/URL: DOI 10.4171/JEMS/1416; EMS Press PDF <https://ems.press/content/serial-article-files/50378>

Result/progress: proves major global structure results for nonzero CMC surfaces, including that complete simply connected embedded nonzero CMC surfaces in $\mathbb{R}^3$ are round spheres and further finite-topology consequences.

Applicability: these results do not remove the finite-topology or asymptotic restrictions needed to prove the exact half-space reflection statement for every properly embedded CMC surface.

3.18.3.3. Conclusion

The problem is not fully resolved as stated. The minimal case is settled by Hoffman-Meeks, compact embedded CMC surfaces are classical Alexandrov spheres, and the nonzero CMC finite-ended/finite-topology half-space case is settled by Korevaar-Kusner-Solomon. Later Ros-Rosenberg and Meeks-Tinaglia papers provide maximum-principle and dynamical/asymptotic progress, but they do not prove that an arbitrary properly embedded CMC surface contained in a half-space is itself symmetric about a plane parallel to the boundary plane.

What remains unresolved is the unrestricted case with no finite-topology, finite-end, bounded-curvature, or asymptotic-limit replacement hypothesis.

3.18.4. Remaining open problems recorded by the database

3.18.4.1. Open problem 1

Prove that every properly embedded constant-mean-curvature surface in $\mathbb{R}^3$ that is contained between two parallel planes has a plane of reflection symmetry parallel to those planes.

Reference supplied by the database. Antonio Ros and Harold Rosenberg, "Properly embedded surfaces with constant mean curvature," *American Journal of Mathematics* **132** (2010),

no. 6, 1429–1443, explicit unsolved slab-symmetry problem in the introduction, DOI: <https://doi.org/10.1353/ajm.2010.0015>.

3.18.5. Primary-source pointer

The originating ICM record is W. H., *Recent progress on the geometry of surfaces in R and on the use of computer graphics as a research tool* (1986). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1986.1/ICM1986.1.ocr.pdf#page=653>.

3.19. 132. W. H., Recent progress on the geometry of surfaces in $\mathbb{R}$ and on the use of computer... [1986-geom-44]

Record: 1986-geom-44

Subfield: Classical and local differential geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1986: W. H., Recent progress on the geometry of surfaces in $\mathbb{R}$ and on the use of computer graphics as a research tool

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1986.1/ICM1986.1.ocr.pdf#page=653>

3.19.1. Problem statement

Prove that every smooth embedded closed surface $M \subset \mathbb{R}^3$ of genus 2 whose total Gaussian curvature

$$K(M) = \int_M |K_G| dA$$

is less than or equal to the total Gaussian curvature of every smooth embedded closed surface ambiently isotopic to M in $\mathbb{R}^3$ satisfies

$$K(M) = 4\pi(2 + 1) = 12\pi.$$

3.19.2. Historical background, literature, and current progress

3.19.3. Verdict: partially resolved

3.19.3.1. Precise problem

Let $M \subset \mathbb{R}^3$ be a smooth embedded closed orientable surface of genus 2, and let

$$\mathcal{K}(M) = \int_M |K_G| dA.$$

The problem asks whether every genus-2 embedding that minimizes $\mathcal{K}$ among all smooth embedded closed surfaces ambiently isotopic to it must be absolutely tight, i.e. must satisfy

$$\mathcal{K}(M) = 4\pi(2 + 1) = 12\pi.$$

Equivalently, in the normalized convention $\tau(M) = \mathcal{K}(M)/(2\pi)$, the question is whether every genus-2 isotopy-tight embedding has $\tau = 6$, and whether a non-tight genus-2 isotopy-tight embedding can exist.

3.19.3.2. Current status

I did not find a published or otherwise citable result that fully resolves this exact genus-2 question. The literature gives strong partial results: the Chern-Lashof lower bound gives $\mathcal{K}(M) \geq 12\pi$; surfaces of genus 2 below the next threshold $\mathcal{K} < 20\pi$ are unknotted; Kuiper

and Meeks rule out the first possible non-tight genus-2 isotopy-tight level $\mathcal{K} = 20\pi$; the torus analogue is fully solved; and knotted isotopy-tight examples exist in every genus $g \geq 3$. What remains unresolved is whether a genus-2 isotopy-tight embedding with higher total absolute curvature, necessarily beyond the excluded 20π level, exists.

3.19.3.3. Key literature and applicability

1. On the total curvature of immersed manifolds

- Authors: S.-S. Chern and R. K. Lashof
- Venue/year: American Journal of Mathematics 79 (1957), 306-318
- DOI/arXiv/URL: no arXiv; bibliographic listing at <https://www.ams.org/tran/1965-115-00/S0002-9947-1965-0202094-9/>
- Contribution: Establishes the total absolute curvature lower bound in terms of Betti numbers. For a closed orientable genus-2 surface, this gives $\mathcal{K}(M) \geq 2\pi(2 + 2g) = 12\pi$. This proves the absolute lower bound but not that every isotopy-class minimizer attains it.

2. On the total curvature of immersed manifolds. II

- Authors: S.-S. Chern and R. K. Lashof
- Venue/year: Michigan Mathematical Journal 5 (1958), 5-12
- DOI/arXiv/URL: DOI 10.1307/mmj/1028998005; <https://projecteuclid.org/journals/michigan-mathematical-journal/volume-5/issue-1/On-the-total-curvature-of-immersed-manifolds-II/10.1307/mmj/1028998005.full>
- Contribution: Continuation of the Chern-Lashof total curvature theory. It supports the tightness framework but does not address ambient isotopy classes of knotted genus-2 surfaces.

3. A criterion for an embedded surface in R^3 to be unknotted

- Author: H. R. Morton
- Venue/year: Lecture Notes in Mathematics 722, Springer, 1979, pp. 92-98
- DOI/arXiv/URL: cited in Springer reference list at <https://link.springer.com/article/10.1007/BF01389134>
- Contribution: One of the sources, together with work of Meeks and Langevin-Rosenberg, for the unknottedness threshold for embedded surfaces. In the later Banchoff-Kuhnel survey this appears as: if $(1/2\pi) \int_M |K| < 8 - \chi(M)$, then M is unknotted. For genus 2, $8 - \chi(M) = 10$, so this covers $\mathcal{K} < 20\pi$, not the full isotopy-tight question.

4. On curvature integrals and knots

- Authors: R. Langevin and H. Rosenberg
- Venue/year: Topology 15 (1976), 405-416
- DOI/arXiv/URL: no arXiv found; cited at <https://link.springer.com/article/10.1007/BF01871945>
- Contribution: Early curvature-integral work relating total curvature and knotting, including the torus direction. It is part of the background for the unknottedness threshold but does not settle genus-2 isotopy tightness.

5. Total curvature for knotted surfaces

- Authors: N. H. Kuiper and W. H. Meeks III

- Venue/year: *Inventiones Mathematicae* 77:1 (1984), 25-69
- DOI/arXiv/URL: DOI 10.1007/BF01389134; <https://link.springer.com/article/10.1007/BF01389134>; IHES PDF https://repo-archives.ihes.fr/FONDS_IHES/I_Prepublications/KUIPER/1972-1991/M_83_37/M_83_37_web.pdf
- Contribution: Defines isotopy-tight embedded surfaces and proves the key structural theorem: a relatively/isotopy-tight genus g surface has normalized curvature in the discrete form $2 + 2g + 4(k - 1)$. It proves that if a genus-2 surface had the first non-tight value $\tau = 10$, equivalently $\mathcal{K} = 20\pi$, then it is unknotted. It also constructs knotted isotopy-tight examples for every $g > 2$ with $\tau = 2g + 6$. This is major partial progress but does not rule out genus-2 isotopy-tight embeddings with $\tau \geq 14$.

6. Minimal total curvature, a look at old and new results

- Author: N. H. Kuiper
- Venue/year: *Russian Mathematical Surveys* 40:4 (1985), 49-55
- DOI/arXiv/URL: https://www.mathnet.ru/php/getFT.phtml?jrn_id=rm&option_lang=rus&paperid=2703&what=fullteng
- Contribution: Survey of the total-absolute-curvature theory. It summarizes the unknottedness threshold, Kuiper-Meeks examples, the nonattainment of the $\beta + 4$ level for genera 1 and 2, and the fundamental lemma for isotopy-tight surfaces. It explicitly leaves many related problems open and does not provide a full genus-2 resolution.

7. The total curvature of a knotted torus

- Authors: N. H. Kuiper and W. H. Meeks III
- Venue/year: *Journal of Differential Geometry* 26:3 (1987), 371-384
- DOI/arXiv/URL: DOI 10.4310/jdg/1214441483; <https://projecteuclid.org/journals/journal-of-differential-geometry/volume-26/issue-3/The-total-curvature-of-a-knotted-torus/10.4310/jdg/1214441483.pdf>
- Contribution: Fully resolves the genus-1 analogue by proving there is no isotopy-tight knotted torus. The introduction states that the corresponding genus-2 non-tight isotopy-tight existence question was still unknown. This confirms that the 1984 results had not resolved the exact problem.

8. Recent progress on the geometry of surfaces in R^3 and on the use of computer graphics as a research tool

- Author: W. H. Meeks III
- Venue/year: *Proceedings of the International Congress of Mathematicians, Berkeley 1986*, Vol. 1, 551-560, published 1987
- DOI/arXiv/URL: no arXiv found; IMU proceedings OCR PDF <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1986.1/ICM1986.1.ocr.pdf>
- Contribution: Original source context for the problem. It points to the total-curvature/isotopy-tight surface problem area but does not supply a theorem resolving the genus-2 case.

9. Tight Submanifolds, Smooth and Polyhedral

- Authors: Thomas F. Banchoff and Wolfgang Kuhnel

- Venue/year: in *Tight and Taut Submanifolds*, MSRI Publications 32, Cambridge University Press, 1997, pp. 51-118
- DOI/arXiv/URL: DOI 10.1017/9781009701815.003; <https://www.cambridge.org/core/books/abs/tight-and-taut-submanifolds/tight-submanifolds-smooth-and-polyhedral/D0929532D954165E5470391EF8A5EF42>
- Contribution: Later survey restating the relevant known theorem: a knotted torus is never isotopy tight, and knotted isotopy-tight surfaces exist for genus $g \geq 3$ with $(1/2\pi) \int |K| = 8 - \chi(M)$. It does not state a solution for genus 2, which remains the missing case between the torus theorem and the genus $g \geq 3$ examples.

10. Regular homotopy and total curvature II: sphere immersions into 3-space

- Author: Tobias Ekholm
- Venue/year: Algebraic & Geometric Topology 6 (2006), 493-512
- DOI/arXiv/URL: DOI 10.2140/agt.2006.6.493; <https://msp.org/agt/2006/6-1/p17.xhtml>; arXiv version <https://arxiv.org/pdf/math/0310266>
- Contribution: Later work on relatively isotopy-tight immersions and local minima of the total curvature functional. It cites the Kuiper-Meeks structure theorem and proves non-injectivity results for certain RIT sphere immersions. Its hypotheses concern sphere immersions/local minima, not embedded genus-2 ambient isotopy-class minimizers, so it does not resolve this problem.

3.19.3.4. What remains unresolved

The exact missing theorem would have to prove that no non-tight genus-2 isotopy-tight embedding exists, or equivalently that every genus-2 ambient-isotopy-class minimizer has $\mathcal{K} = 12\pi$. Existing results cover the absolute lower bound, the unknotted region below 20π , the first non-tight level 20π , the torus analogue, and the existence of higher-genus counterexamples. They do not eliminate possible genus-2 isotopy-tight embeddings at higher normalized curvature levels such as $\tau \geq 14$. Therefore the literature only partially resolves the problem.

3.19.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

3.19.5. Primary-source pointer

The originating ICM record is W. H., *Recent progress on the geometry of surfaces in R and on the use of computer graphics as a research tool* (1986). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1986.1/ICM1986.1.ocr.pdf#page=653>.

3.20. 133. Isometric Embeddings of Riemannian Manifolds [1990-geom-306]

Record: 1990-geom-306

Subfield: Classical and local differential geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1990: Isometric Embeddings of Riemannian Manifolds

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1990.2/ICM1990.2.ocr.pdf#page=383>

3.20.1. Problem statement

Determine whether, for every integer $n \geq 1$, every smooth n -dimensional Riemannian manifold (M, g) , and every point $p \in M$, there exist an open neighborhood $U \subseteq M$ of p and a smooth immersion $u : U \rightarrow \mathbb{R}^{n(n+1)/2}$ such that

$$u^*\langle \cdot, \cdot \rangle_{\mathbb{R}^{n(n+1)/2}} = g|_U.$$

3.20.2. Historical background, literature, and current progress

Verdict: unresolved. This is the smooth local Schläfli problem in the formally determined dimension $s_n = n(n+1)/2$. The literature does not prove the assertion for arbitrary smooth metrics, and it does not provide an accepted smooth counterexample. The gap already remains for $n = 2$, where the question is whether every smooth surface metric is locally smoothly isometrically immersible in $\mathbb{R}^3$.

Established progress.

- *Isometric Embedding of Riemannian Manifolds in Euclidean Spaces*, Qing Han and Jia-Xing Hong, Mathematical Surveys and Monographs 130, American Mathematical Society, 2006, DOI 10.1090/surv/130. This monograph records the Janet-Cartan-Burstin theorem: every real-analytic n -metric has a local real-analytic isometric embedding in the exact dimension s_n . It also records the independent smooth local theorems of Gromov-Rokhlin and Greene in the larger target $\mathbb{R}^{s_n+n}$. Neither result settles the stated smooth problem in $\mathbb{R}^{s_n}$.
- *Local Isometric Embedding of Surfaces with Nonpositive Gaussian Curvature*, Qing Han, Jia-Xing Hong, and Chang-Shou Lin, Journal of Differential Geometry 63(3) (2003), 475-520, DOI 10.4310/jdg/1090426772; *On the Isometric Embedding of Surfaces with Gauss Curvature Changing Sign Cleanly*, Qing Han, Communications on Pure and Applied Mathematics 58(2) (2005), 285-295, DOI 10.1002/cpa.20054; and *On the Local Isometric Embedding in $\mathbb{R}^3$ of Surfaces with Gaussian Curvature of Mixed Sign*, Qing Han and Marcus Khuri, Communications in Analysis and Geometry 18(4) (2010), 649-704, DOI 10.4310/CAG.2010.v18.n4.a2. These prove local existence for important two-dimensional curvature regimes, including specified nonpositive and finite-order mixed-sign degeneracies. Their additional curvature and finite-order hypotheses do not cover an arbitrary smooth surface metric, whose Gaussian curvature may vanish to infinite order with

unrestricted zero set.

- *Isometric Embedding via Strongly Symmetric Positive Systems*, Gui-Qiang Chen, Jeanne Clelland, Marshall Slemrod, Dehua Wang, and Deane Yang, *Asian Journal of Mathematics* 22(1) (2018), 1-40, DOI 10.4310/AJM.2018.v22.n1.a1. Theorem 2 gives a smooth local embedding in $\mathbb{R}^{s_n}$ for $n = 2, 3$ wherever the Riemann curvature tensor is nonzero. The paper also surveys further three-dimensional results when curvature vanishes cleanly. Points with more degenerate curvature, and general higher dimensions, remain outside these theorems.
- *On the Isometric Version of Whitney's Strong Embedding Theorem*, Wentao Cao and Laszlo Szekelyhidi Jr., *Advances in Mathematics* 460 (2025), Article 110040, DOI 10.1016/j.aim.2024.110040. Its introduction describes the full smooth analogue of Janet's local theorem as still outstanding. Its own theorem constructs global $C^{1,\theta}$ isometric embeddings in $\mathbb{R}^{2n}$, not smooth embeddings in $\mathbb{R}^{s_n}$.

Counterexample caveat. *Counterexamples for Local Isometric Embedding*, Nikolai Nadirashvili and Yu Yuan, 2002, [arXiv:math/0208127](https://arxiv.org/abs/math/0208127), claims in Theorem 1.1 a smooth surface metric having no local C^3 isometric embedding in $\mathbb{R}^3$, which would disprove the statement. That claim appeared only as the first arXiv version and is not treated as a resolution by the subsequent literature. In particular, the same authors' peer-reviewed paper *Improving Pogorelov's Isometric Embedding Counterexample*, *Calculus of Variations and Partial Differential Equations* 32 (2008), 319-323, DOI 10.1007/s00526-007-0140-7, proves a counterexample only for a $C^{2,1}$ metric and explicitly states that local C^∞ , or even C^2 , embeddability of arbitrary C^∞ metrics remains open. Thus the published counterexample does not satisfy the smooth-metric hypothesis, and the unrefereed 2002 claim cannot presently be counted as a verified resolution.

Consequently, neither an affirmative theorem nor a verified counterexample covers the exact quantifiers and regularity in the statement.

3.20.3. Remaining open problems recorded by the database

3.20.3.1. Open problem 1

Determine whether, for every smooth two-dimensional Riemannian manifold (M, g) and every point $p \in M$, there exist an open neighborhood $U \subseteq M$ of p and a smooth immersion $u : U \rightarrow \mathbb{R}^3$ such that

$$u^*\langle \cdot, \cdot \rangle_{\mathbb{R}^3} = g|_U.$$

Reference supplied by the database. Qing Han, *Local Isometric Embedding of Surfaces in $\mathbb{R}^3$* , Max Planck Institute for Mathematics in the Sciences Preprint 18/2003 (Lecture Note), 2003, conjecture reposed by Shing-Tung Yau in the abstract and introduction, https://www.mis.mpg.de/de/publications/preprint-repository/lecture_note/2003/issue-18.

3.20.3.2. Open problem 2

Determine whether, for every smooth two-dimensional Riemannian manifold (M, g) and every point $p \in M$, there exist an open neighborhood $U \subseteq M$ of p and a C^2 immersion $u : U \rightarrow \mathbb{R}^3$ such that

$$u^*\langle \cdot, \cdot \rangle_{\mathbb{R}^3} = g|_U.$$

Reference supplied by the database. Nikolai Nadirashvili and Yu Yuan, *Improving Pogorelov's Isometric Embedding Counterexample*, *Calculus of Variations and Partial Differential Equations* 32 (2008), 319-323, explicitly identified open question in the introduction, <https://doi.org/10.1007/s00526-007-0140-7>.

3.20.4. Primary-source pointer

The originating ICM record is *Isometric Embeddings of Riemannian Manifolds* (1990). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1990.2/ICM1990.2.ocr.pdf#page=383>.

3.21. 134. Geometry and Analytic Theory of Frobenius Manifolds [1998-geom-339]

Record: 1998-geom-339

Subfield: Classical and local differential geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1998: Geometry and Analytic Theory of Frobenius Manifolds

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.2/ICM1998.2.ocr.pdf#page=317>

3.21.1. Problem statement

Prove or disprove that every connected polynomial semisimple complex Frobenius manifold is equivalent to a finite direct product of the canonical polynomial Frobenius manifolds associated with finite irreducible Coxeter groups, in the following sense.

Let M be a connected complex manifold whose holomorphic tangent bundle is equipped with a commutative, associative multiplication $\circ$, a unit vector field e , a flat nondegenerate symmetric metric η , and an Euler vector field E . Let ∇ be the Levi-Civita connection of η , and suppose that

$$\eta(X \circ Y, Z) = \eta(X, Y \circ Z), \quad \nabla e = 0,$$

that ∇c is totally symmetric for $c(X, Y, Z) = \eta(X \circ Y, Z)$, and that

$$\nabla \nabla E = 0, \quad \mathcal{L}_E(\circ) = \circ, \quad \mathcal{L}_E \eta = (2 - d)\eta$$

for some charge d . Assume that M admits global flat coordinates $t^1, \dots, t^n$ in which its Frobenius potential F , defined up to a polynomial of degree at most two by

$$\frac{\partial^3 F}{\partial t^\alpha \partial t^\beta \partial t^\gamma} = \eta\left(\frac{\partial}{\partial t^\alpha} \circ \frac{\partial}{\partial t^\beta}, \frac{\partial}{\partial t^\gamma}\right),$$

is a polynomial. Assume also that the Frobenius algebra $(T_t M, \circ)$ is semisimple at some point $t \in M$, and hence at a generic point.

For a finite irreducible Coxeter group $W \subset O(\mathbb{R}^r)$, acting on $\mathbb{C}^r$ through the complexification of its reflection representation, let

$$M_W = \mathbb{C}^r / W$$

denote the orbit space equipped with the canonical polynomial Dubrovin–Saito Frobenius structure constructed from a system of basic W -invariant polynomials. Prove or disprove that there exist finite irreducible Coxeter groups $W_1, \dots, W_k$ such that M is biholomorphic to $M_{W_1} \times \dots \times M_{W_k}$ by a map preserving the multiplication, unit, metric, and Euler vector field, where the product carries the componentwise product Frobenius structure.

3.21.2. Historical background, literature, and current progress

The exact statement is **unresolved**. The literature contains a deep classification theorem under an additional positivity hypothesis, but no verified result removes that hypothesis and proves the global classification stated here; nor did the search locate a counterexample satisfying all of the stated axioms.

Boris Dubrovin, *Geometry and analytic theory of Frobenius manifolds*, **Documenta Mathematica, Extra Volume ICM 1998, Vol. II** (1998), 315–326, [arXiv:math/9807034](https://arxiv.org/abs/math/9807034). Section 5.1 states the conjecture that polynomial semisimple Frobenius manifolds are the finite-Coxeter orbit-space examples and their direct sums. Dubrovin’s earlier *Differential geometry of the space of orbits of a Coxeter group*, **Surveys in Differential Geometry** 4 (1998), 181–211, <https://doi.org/10.4310/SDG.1998.v4.n1.a4>, constructs the canonical polynomial semisimple Frobenius structures and formulates the classification with positive invariant degrees.

Claus Hertling, *Multiplication on the tangent bundle* (1999), Theorem 19.3, [arXiv:math/9910116](https://arxiv.org/abs/math/9910116); incorporated as Theorem 5.26 in *Frobenius Manifolds and Moduli Spaces for Singularities*, **Cambridge Tracts in Mathematics** 151, Cambridge University Press (2002), <https://doi.org/10.1017/CBO9780511543104>. Hertling proves that a germ of a generically semisimple Frobenius manifold whose Euler field has strictly positive spectrum decomposes into Coxeter-orbit Frobenius germs (with the factors constrained to have the same Coxeter number). This proves the positive-spectrum version in every dimension, but positivity is not a hypothesis of the problem in `statement.md`, and the theorem is local rather than the asserted global equivalence.

Misha Feigin, Daniele Valeri, and Johan Wright, *Flat coordinates of algebraic Frobenius manifolds in small dimensions*, **Journal of Geometry and Physics** 200 (2024), 105151, <https://doi.org/10.1016/j.geomphys.2024.105151>, [arXiv:2309.01577](https://arxiv.org/abs/2309.01577). Its introduction accurately records the present scope of Hertling’s result: Dubrovin’s polynomial classification was proved “with the added assumption that the Euler vector field has positive degrees.” It studies algebraic, rather than polynomial, extensions and does not close the missing nonpositive-spectrum or global cases.

Thus the strongest established progress is a complete local classification under positive Euler degrees. What remains for the exact problem is to prove the classification without any positivity restriction and upgrade the applicable local conclusion to the stated global Frobenius-manifold equivalence, or else exhibit a polynomial semisimple counterexample under those unrestricted hypotheses.

3.21.3. Remaining open problems recorded by the database

No separate descendant problem is listed in the website’s “Some remaining open problems” field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

3.21.4. Primary-source pointer

The originating ICM record is *Geometry and Analytic Theory of Frobenius Manifolds* (1998). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.2/ICM1998.2.ocr.pdf#page=317>.

3.22. 135. Some New Developments of Realization of Surfaces into $\mathbb{R}^3$ [2002-geom-252]

Record: 2002-geom-252

Subfield: Classical and local differential geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Some New Developments of Realization of Surfaces into $\mathbb{R}^3$

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=154>

3.22.1. Problem statement

Prove that for every smooth two-dimensional Riemannian manifold (M, g) and every point $p \in M$, there exist an open neighborhood $U \subset M$ of p and a smooth embedding $f : U \rightarrow \mathbb{R}^3$ such that $f^*g_{\text{Euc}} = g|_U$, where g_{Euc} is the standard Euclidean metric on $\mathbb{R}^3$.

3.22.2. Historical background, literature, and current progress

The exact problem is **unresolved**. No accepted theorem proves local smooth embeddability into $\mathbb{R}^3$ for every smooth two-dimensional Riemannian metric, and no accepted smooth counterexample is known.

The strongest progress covers substantial special cases. Chang-Shou Lin proved finite-regularity local embeddings for sufficiently differentiable metrics with nonnegative Gaussian curvature, and Gen Nakamura and Yoshihiro Maeda obtained smooth embeddings for smooth metrics whose Gaussian curvature changes sign cleanly. Qing Han, Jia-Xing Hong, and Chang-Shou Lin proved smooth local existence for nonpositive curvature under specified nondegeneracy/stability conditions. Qing Han and Marcus Khuri later treated finite-order vanishing with transverse zero curves, while Tsung-Yin Lin treated tangent zero curves satisfying cusp hypotheses. None of these hypotheses exhausts arbitrary smooth metrics, particularly unrestricted infinitely degenerate zero sets of Gaussian curvature.

Known counterexamples do not settle the stated smooth problem. Nikolai Nadirashvili and Yu Yuan constructed a $C^{2,1}$ nonnegatively curved metric having no local C^2 isometric embedding. Their earlier arXiv v1, *Counterexamples for Local Isometric Embedding* (arXiv:math/0208127, 2002), claims a smooth metric with no local C^3 embedding, but that claim was not published or incorporated as an established resolution: the same authors' 2008 paper explicitly says that local C^∞ (or even C^2) embeddability of arbitrary C^∞ metrics remains open, and later peer-reviewed papers likewise identify smooth counterexamples as open while citing only the $C^{2,1}$ result. I found no later correction or validated publication resolving this discrepancy.

Key references:

- Chang-Shou Lin, *The Local Isometric Embedding in $\mathbb{R}^3$ of 2-Dimensional Riemannian Manifolds with Nonnegative Curvature*, Journal of Differential Geometry 21 (1985), 213–230, direct URL: <https://projecteuclid.org/journals/journal-of-differential-geometry/volume-21/issue-2/The-local-isometric>

-embedding-in- $\mathbb{R}^3$ -of-2-dimensional-Riemannian/jdg/1214439563.full.

- Gen Nakamura and Yoshihiro Maeda, *Local Smooth Isometric Embeddings of Low-Dimensional Riemannian Manifolds into Euclidean Spaces*, Transactions of the American Mathematical Society 313 (1989), 1–51, direct URL: <https://www.ams.org/journals/tran/1989-313-01/S0002-9947-1989-0980068-6/>.
- Qing Han, Jia-Xing Hong, and Chang-Shou Lin, *Local Isometric Embedding of Surfaces with Nonpositive Gaussian Curvature*, Journal of Differential Geometry 63 (2003), 475–520, direct URL: <https://projecteuclid.org/journals/journal-of-differential-geometry/volume-63/issue-3/Local-isometric-embedding-of-surfaces-with-nonpositive-Gaussian-curvature/jdg/1090426989.full>.
- Nikolai Nadirashvili and Yu Yuan, *Improving Pogorelov's Isometric Embedding Counterexample*, Calculus of Variations and Partial Differential Equations 32 (2008), 319–323, DOI: <https://doi.org/10.1007/s00526-007-0140-7>.
- Qing Han and Marcus A. Khuri, *On the Local Isometric Embedding in $\mathbb{R}^3$ of Surfaces with Gaussian Curvature of Mixed Sign*, Communications in Analysis and Geometry 18 (2010), DOI: <https://doi.org/10.4310/CAG.2010.v18.n4.a2>.
- Tsung-Yin Lin, *On the Local Isometric Embedding in $\mathbb{R}^3$ of Surfaces with Zero Sets of Gaussian Curvature Forming Cusp Domains*, Communications in Partial Differential Equations 41 (2016), DOI: <https://doi.org/10.1080/03605302.2016.1227336>.

3.22.3. Remaining open problems recorded by the database

3.22.3.1. Open problem 1

Determine whether, for every smooth two-dimensional Riemannian manifold (M, g) with nonnegative Gaussian curvature and every point $p \in M$, there exist an open neighborhood $U \subset M$ of p and a smooth embedding $f : U \rightarrow \mathbb{R}^3$ such that $f^*g_{\text{Euc}} = g|_U$, where g_{Euc} is the standard Euclidean metric on $\mathbb{R}^3$.

Reference supplied by the database. Nikolai Nadirashvili and Yu Yuan, *Improving Pogorelov's Isometric Embedding Counterexample*, Calculus of Variations and Partial Differential Equations 32 (2008), no. 3, 319–323, open problem stated on p. 320, DOI: <https://doi.org/10.1007/s00526-007-0140-7>.

3.22.3.2. Open problem 2

Determine whether, for every smooth two-dimensional Riemannian manifold (M, g) and every point $p \in M$, there exist an open neighborhood $U \subset M$ of p and a C^2 embedding $f : U \rightarrow \mathbb{R}^3$ such that $f^*g_{\text{Euc}} = g|_U$, where g_{Euc} is the standard Euclidean metric on $\mathbb{R}^3$.

Reference supplied by the database. Nikolai Nadirashvili and Yu Yuan, *Improving Pogorelov's Isometric Embedding Counterexample*, Calculus of Variations and Partial

Differential Equations 32 (2008), no. 3, 319–323, open problem stated on p. 320, DOI: <https://doi.org/10.1007/s00526-007-0140-7>.

3.22.4. Primary-source pointer

The originating ICM record is *Some New Developments of Realization of Surfaces into R^3* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=154>.

3.23. 136. Properly embedded minimal surfaces with finite topology [2006-ag-115]

Record: 2006-ag-115

Subfield: Classical and local differential geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2006: Properly embedded minimal surfaces with finite topology

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.2/ICM2006.2.ocr.pdf#page=917>

3.23.1. Problem statement

For integers $k \geq 0$ and $r \geq 1$, let $\mathcal{M}(k, r)$ denote the moduli space of properly embedded minimal surfaces $\Sigma \subset \mathbb{R}^3$ of finite topology, genus k , and exactly r horizontal ends. Prove that $\mathcal{M}(k, r)$ is empty whenever $r > k + 2$.

3.23.2. Historical background, literature, and current progress

3.23.3. Status: partially resolved

3.23.3.1. Precise problem

For integers $k \geq 0$ and $r \geq 1$, let $\mathcal{M}(k, r)$ be the moduli space of properly embedded minimal surfaces in $\mathbb{R}^3$ with finite topology, genus k , and exactly r horizontal ends. The problem asks for the nonexistence statement

$$\mathcal{M}(k, r) = \emptyset \quad \text{whenever } r > k + 2.$$

This is the nonexistence half of the Hoffman-Meeks finite topology conjecture, in the formulation for finite-topology properly embedded minimal surfaces with horizontal ends.

3.23.3.2. Literature checked

1. Embedded minimal surfaces of finite topology

Authors: David Hoffman and William H. Meeks III

Venue/year: Annals of Mathematics (2) 131, no. 1, 1-34, 1990

DOI/URL: <https://doi.org/10.2307/1971506>; <https://annals.math.princeton.edu/1990/131-1/p01>

Relevance: Original Hoffman-Meeks finite-topology context and examples. Later papers and surveys attribute the finite-topology genus/ends conjecture to Hoffman and Meeks.

2. Properly embedded minimal surfaces with finite topology

Author: Antonio Ros

Venue/year: Proceedings of the International Congress of Mathematicians, Madrid 2006,

Vol. II, 907-926, European Mathematical Society, published 2007

DOI/URL: <https://doi.org/10.4171/022-2/43>; <https://ems.press/books/standalone/21>

Relevance: States the exact $M(k, r)$ formulation: $M(k, r)$ should be empty for $r > k + 2$. It records that $k = 0$ is known and that for higher genus only a sufficiently-large- r nonexistence result was known at that time.

3. Topologie et courbure des surfaces minimales proprement plongées de $\mathbb{R}^3$

Author: Pascal Collin

Venue/year: Annals of Mathematics (2) 145, no. 1, 1-31, 1997

DOI/URL: <https://doi.org/10.2307/2951822>; <https://annals.math.princeton.edu/articles/13049>

Relevance: Proves that a properly embedded minimal surface in $\mathbb{R}^3$ with finite topology and at least two ends has finite total curvature. Since $r > k + 2$ implies $r > 1$, this reduces the stated setting to the finite-total-curvature/plane-or-catenoid-end setting used in the later literature, but does not prove the $r \leq k + 2$ bound.

4. On embedded complete minimal surfaces of genus zero

Authors: Francisco J. López and Antonio Ros

Venue/year: Journal of Differential Geometry 33, no. 1, 293-300, 1991

DOI/URL: <https://doi.org/10.4310/jdg/1214446040>; <https://projecteuclid.org/journals/journal-of-differential-geometry/volume-33/issue-1/On-embedded-complete-minimal-surfaces-of-genus-zero/10.4310/jdg/1214446040.full>

Relevance: Proves the genus-zero case: the only embedded complete genus-zero finite-total-curvature examples are the plane and catenoid. Therefore $M(0, r) = \emptyset$ for $r > 2$. This is only the $k = 0$ case of the problem.

5. Bounds on the topology and index of minimal surfaces

Authors: William H. Meeks III, Joaquín Pérez, Antonio Ros

Venue/year: Acta Mathematica 223, no. 1, 113-149, 2019

DOI/URL: <https://doi.org/10.4310/ACTA.2019.v223.n1.a2>; <https://www.intlpress.com/api/bgcloud-front/resource/pdf/volume/1805558284297904130-1805558284297904130-fb9a5257cf584f50b9d843f7768fe5a9.pdf>

Relevance: Theorem 1.4 proves that, for fixed genus, a properly embedded finite-topology minimal surface has a bound on the number of ends depending only on the genus. This gives $M(k, r) = \emptyset$ for all sufficiently large r depending on k , but it does not identify the sharp bound $k + 2$.

6. Minimal Surfaces of Finite Genus: Classification, Dynamics and Laminations

Author: Joaquín Pérez

Venue/year: arXiv preprint/survey, 2025

arXiv/URL: arXiv:2510.12319; <https://arxiv.org/abs/2510.12319>; <https://arxiv.org/html/2510.12319v1>

Relevance: Section 5 restates the Hoffman-Meeks finite topology conjecture and says that even the genus $g = 1$ case remains open. It identifies the Meeks-Pérez-Ros genus-dependent end bound as the best partial answer up to date.

3.23.3.3. Conclusion

The literature does **not** fully resolve the exact statement in the problem statement. Known results prove the genus-zero case and prove that, for each fixed genus k , there is some finite

upper bound on the number of ends. However, the sharp bound $r \leq k + 2$ remains unproved in positive genus; in particular the first case $k = 1, r = 4$ is still not covered by a published or citable theorem. Therefore the correct verdict is **partially resolved**.

3.23.4. Remaining open problems recorded by the database

3.23.4.1. Open problem 1

Prove the genus-one case of the Hoffman–Meeks end bound: a complete properly embedded minimal surface in $\mathbb{R}^3$ with finite total curvature and genus one has at most three ends; equivalently, $\mathcal{M}(1, r) = \emptyset$ for every $r \geq 4$.

Reference supplied by the database. Matthias Weber, “Recent Progress on Minimal Surfaces with Low Genus,” in Olivier Biquard, Bernhard Leeb, and Gang Tian (organizers), “Differentialgeometrie im Großen,” *Oberwolfach Reports* 6 (2009), 1943–1972, contribution at pp. 1966–1969, especially Conjectures 4.1 and 4.3 and the intervening genus-one classification discussion; DOI: <https://doi.org/10.4171/OWR/2009/35>.

3.23.5. Primary-source pointer

The originating ICM record is *Properly embedded minimal surfaces with finite topology* (2006). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.2/ICM2006.2.ocr.pdf#page=917>.

3.24. 137. Properly embedded minimal surfaces with finite topology [2006-ag-116]

Record: 2006-ag-116

Subfield: Classical and local differential geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2006: Properly embedded minimal surfaces with finite topology

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.2/ICM2006.2.ocr.pdf#page=917>

3.24.1. Problem statement

For integers $k \geq 0$ and $r \geq 1$, let $M(k, r)$ denote the moduli space of properly embedded minimal surfaces $\Sigma \subset \mathbb{R}^3$ of genus k with r horizontal ends. Prove that $M(k, r)$ is either empty or noncompact; equivalently, if it is nonempty, it is not compact in the sense that not every sequence in $M(k, r)$ has a subsequence converging to a single limit surface still lying in $M(k, r)$.

3.24.2. Historical background, literature, and current progress

Verdict: partially resolved.

Precise problem checked: for each genus $k \geq 0$ and number $r \geq 1$ of horizontal ends, decide whether the moduli space $M(k, r)$ of properly embedded minimal surfaces in $\mathbb{R}^3$ is either empty or noncompact. In the source literature this is the Ros compactness/noncompactness question for finite-topology embedded minimal surfaces, most explicitly cited for $r \geq 4$; the low-end cases $r < 4$ are treated separately.

Key source:

- Title: Properly embedded minimal surfaces with finite topology Authors: Antonio Ros Venue: Proceedings of the International Congress of Mathematicians, Madrid 2006, Vol. II, pp. 907-926, European Mathematical Society Year: 2006/2007 DOI/URL: DOI 10.4171/022-2/43, <https://ems.press/books/standalone/21/427> Role: This is the source of the moduli-space question. Ros defines $M(k, r)$ for properly embedded finite-topology minimal surfaces with horizontal ends and records the state of compactness and classification questions. Later authors cite Ros's conjecture for $r \geq 4$: if $M(g, r)$ is nonempty, then it should be noncompact.

Global partial progress:

- Title: Bounds on the topology and index of minimal surfaces Authors: William H. Meeks III, Joaquin Perez, Antonio Ros Venue: Acta Mathematica 223(1), 113-149 Year: 2019 DOI/URL: <https://intlpress.com/api/bgcloud-front/resource/pdf/volume/1805558284297904130-1805558284297904130-fb9a5257cf584f50b9d843f7768fe5a9.pdf> Result used: Theorem 1.4 proves that a properly embedded minimal surface in $\mathbb{R}^3$ with finite topology has a bound on

the number of ends depending only on its genus. Why only partial: this gives a major finiteness/topological obstruction toward the Hoffman-Meeks finite topology conjecture, but it does not prove that every nonempty $\mathcal{M}(k, r)$ is noncompact, nor that the compact cases predicted by Ros are empty.

- Title: Minimal Surfaces of Finite Genus: Classification, Dynamics and Laminations Author: Joaquin Perez Venue: arXiv preprint / survey Year: 2025 arXiv/URL: arXiv:2510.12319, <https://arxiv.org/html/2510.12319v1> Result/status used: The survey states that the Hoffman-Meeks finite topology conjecture remains open, even in the simplest positive-genus finite-topology cases, and lists open problems including sharper bounds on the number of ends. Why only partial: it documents the current state after the Meeks-Perez-Ros program; it does not report a theorem resolving Ros's full noncompactness question.

Special cases and related progress:

- Title: On embedded complete minimal surfaces of genus zero Authors: Francisco J. Lopez, Antonio Ros Venue: Journal of Differential Geometry 33(1), 293-300 Year: 1991 DOI/arXiv/URL: bibliographic reference in the current surveys; no arXiv found Result used: the only complete embedded genus-zero finite-total-curvature examples are the plane and catenoid. Why only partial: this resolves the genus-zero finite-total-curvature cases but not positive genus and $r \geq 4$.
- Title: Uniqueness, symmetry, and embeddedness of minimal surfaces Author: Richard M. Schoen Venue: Journal of Differential Geometry 18, 791-809 Year: 1983 DOI/arXiv/URL: bibliographic reference in the current surveys Result used: the complete embedded finite-total-curvature two-ended case is the catenoid. Why only partial: this resolves $r = 2$, but not higher-ended moduli spaces.
- Title: Properly embedded minimal planar domains Authors: William H. Meeks III, Joaquin Perez, Antonio Ros Venue: Annals of Mathematics 181(2), 473-546 Year: 2015 DOI/arXiv/URL: cited in the 2025 survey, <https://arxiv.org/html/2510.12319v1> Result used: classification of properly embedded minimal planar domains, including the Riemann minimal examples in the infinite-topology genus-zero setting. Why only partial: it is a genus-zero classification theorem and does not settle the finite-topology positive-genus compactness question.
- Title: Non-compact families of complete, properly immersed minimal surfaces with finite topology Authors: Stephen J. Kleene, Niels Martin Moller Venue: arXiv preprint Year: 2014, revised 2024 arXiv/URL: arXiv:1409.8381, <https://arxiv.org/pdf/1409.8381> Result used: Theorem 1.1 constructs continuous families of complete minimal surfaces of genus g with four catenoidal ends for large g , giving evidence for noncompactness phenomena. Why only partial: the authors explicitly state that their constructed surfaces are a priori non-embedded and explain that the embedded analogue is expected but not proved there; hence this does not resolve $\mathcal{M}(k, r)$ for properly embedded surfaces.

Conclusion: I found no published or otherwise citable theorem that proves the full statement for all $k \geq 0$ and $r \geq 1$. The literature resolves several low-topology or genus-zero cases and proves strong global bounds on the number of ends in fixed genus, but the exact Ros noncompactness/emptiness statement for the embedded moduli spaces remains open in the remaining higher-genus, higher-end cases.

3.24.3. Remaining open problems recorded by the database

3.24.3.1. Open problem 1

For every integer $G \geq 2$, prove that every complete, properly embedded minimal surface in $\mathbb{R}^3$ with finite total curvature, genus G , and exactly three horizontal ends is a Costa–Hoffman–Meeks surface of genus G ; equivalently, prove that $\mathcal{M}(G, 3)$ consists precisely of the Costa–Hoffman–Meeks surfaces.

Reference supplied by the database. Martin Traizet, “A balancing condition for weak limits of families of minimal surfaces,” *Commentarii Mathematici Helvetici* **79** (2004), no. 4, 798–825, Conjecture 2, DOI: <https://doi.org/10.1007/s00014-004-0805-1>.

3.24.4. Primary-source pointer

The originating ICM record is *Properly embedded minimal surfaces with finite topology* (2006). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.2/ICM2006.2.ocr.pdf#page=917>.

3.25. 138. Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries [2010-geom-104]

Record: 2010-geom-104

Subfield: Classical and local differential geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2010: Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=836>

3.25.1. Problem statement

Prove or disprove that every compact immersed surface S in the product space $\mathbb{H}^2 \times \mathbb{R}$ with constant mean curvature and nonnegative second variation for all compactly supported volume-preserving variations, equivalently for all smooth variational functions f on S with $\int_S f = 0$, is a rotational constant-mean-curvature sphere, meaning a sphere invariant under rotations about a vertical axis in $\mathbb{H}^2 \times \mathbb{R}$.

3.25.2. Historical background, literature, and current progress

Verdict: partially resolved.

Precise problem checked: classify compact immersed constant-mean-curvature surfaces in $\mathbb{H}^2 \times \mathbb{R}$ that are stable for all compactly supported volume-preserving variations, equivalently whose CMC stability quadratic form is nonnegative on all smooth functions with zero integral. The proposed conclusion is that every such surface must be a rotational CMC sphere.

Current progress:

1. Title: Global properties of constant mean curvature surfaces in $\mathbb{H}^2 \times \mathbb{R}$ Authors: Barbara Nelli and Harold Rosenberg Venue: Pacific Journal of Mathematics 226(1), 137-152 Year: 2006 DOI/URL: <https://doi.org/10.2140/pjm.2006.226.137> ; <https://msp.org/pjm/2006/226-1/p07.xhtml> Relevance: Establishes foundational global restrictions for CMC surfaces in $\mathbb{H}^2 \times \mathbb{R}$. In particular, later papers cite it for the fact that compact immersed CMC surfaces in $\mathbb{H}^2 \times \mathbb{R}$ must have $H > 1/2$. This narrows the possible range but does not classify stable compact immersed examples.
2. Title: A Hopf differential for constant mean curvature surfaces in $S^2 \times \mathbb{R}$ and $\mathbb{H}^2 \times \mathbb{R}$ Authors: Uwe Abresch and Harold Rosenberg Venue: Acta Mathematica 193, 141-174 Year: 2004 DOI/URL: <https://doi.org/10.1007/BF02392562> Relevance: Introduces the Abresch-Rosenberg holomorphic quadratic differential and gives the Hopf-type classification of immersed CMC spheres in these product spaces. Thus, if stability could be used to prove that the surface is topologically a sphere, this result would identify it as rotational. It does not by itself prove that every stable compact immersed CMC surface is a sphere.
3. Title: On stable constant mean curvature surfaces in $S^2 \times \mathbb{R}$ and $\mathbb{H}^2 \times \mathbb{R}$ Author: Rabah

Souam Venue: Transactions of the American Mathematical Society 362(6), 2845-2857
 Year: 2010 DOI/arXiv/URL: <https://doi.org/10.1090/S0002-9947-10-04826-9> ; arXiv:0709.4204 ; <https://arxiv.org/abs/0709.4204> Relevance: This is the main partial result for the exact stability question. It proves that all rotational CMC spheres in $\mathbb{H}^2 \times \mathbb{R}$ are stable, and proves the desired converse only under the additional hypothesis $H \geq 1/\sqrt{2}$: a closed stable immersed CMC surface with $H \geq 1/\sqrt{2}$ is a rotational sphere. For lower H , it gives genus bounds, for example genus at most one when $1/\sqrt{3} < H < 1/\sqrt{2}$ and genus at most two at $H = 1/\sqrt{3}$, but does not rule out stable nonspherical compact immersed surfaces throughout the full remaining range.

4. Title: On the uniqueness of isoperimetric solutions and imbedded soap bubbles in non-compact symmetric spaces, I Authors: Wu-Teh Hsiang and Wu-Yi Hsiang Venue: Inventiones Mathematicae 98, 39-58 Year: 1989 DOI/URL: <https://doi.org/10.1007/BF01388843> Relevance: Gives uniqueness/isoperimetric and embedded soap-bubble results for the rotational CMC spheres in the relevant product spaces. This supports the rotational-sphere picture for embedded or isoperimetric hypersurfaces, but the problem asks about compact immersed weakly stable CMC surfaces, so this does not cover the full statement.
5. Title: Constant mean curvature surfaces in 3-dimensional Thurston geometries Authors: Isabel Fernandez and Pablo Mira Venue: Invited contribution to the Proceedings of the International Congress of Mathematicians 2010; Proceedings of the ICM, Volume II, Hindustan Book Agency, pp. 830-861 Year: 2010 arXiv/URL: arXiv:1004.4752 ; <https://arxiv.org/abs/1004.4752> Relevance: Survey of compact CMC surfaces in Thurston geometries. It explicitly records the Nelli-Rosenberg conjectural status: compact weakly stable CMC surfaces in $\mathbb{H}^2 \times \mathbb{R}$ are expected to be rotational CMC spheres. This survey source treats the statement as conjectural/open rather than settled.
6. Title: On Stability and Isoperimetry of Constant Mean Curvature Spheres of $\mathbb{H}^n \times \mathbb{R}$ and $\mathbb{S}^n \times \mathbb{R}$ Authors: Ronaldo F. de Lima, Maria F. Elbert, and Barbara Nelli Venue: To appear in Mathematische Zeitschrift; preprint Year: 2023 preprint, revised 2024 arXiv/URL: arXiv:2301.11038 ; <https://arxiv.org/abs/2301.11038> Relevance: Extends and refines stability/isoperimetry properties of the known rotational CMC spheres, proving in particular that all rotational CMC spheres in $\mathbb{H}^n \times \mathbb{R}$ are stable. It does not prove the converse classification of all compact immersed weakly stable CMC surfaces in $\mathbb{H}^2 \times \mathbb{R}$.

What remains unresolved: the exact statement still lacks a citable theorem covering every compact immersed weakly stable CMC surface in $\mathbb{H}^2 \times \mathbb{R}$. The literature proves the desired conclusion for $H \geq 1/\sqrt{2}$, proves the known rotational spheres are stable, and proves related embedded/isoperimetric uniqueness results. I did not find a published or citable result ruling out all possible stable nonspherical compact immersed CMC surfaces in the remaining mean-curvature regime $1/2 < H < 1/\sqrt{2}$. Therefore the problem is partially resolved, not fully resolved and not fully open.

3.25.3. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

3.25.4. Primary-source pointer

The originating ICM record is *Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries* (2010). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=836>.

3.26. 139. Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries [2010-geom-105]

Record: 2010-geom-105

Subfield: Classical and local differential geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2010: Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=836>

3.26.1. Problem statement

Prove or disprove that, in the Heisenberg space $\text{Nil}_3 = \mathbb{E}^3(0, \tau)$ with $\tau \neq 0$, the simply connected three-dimensional Heisenberg group endowed with its standard left-invariant homogeneous metric, the isoperimetric regions, namely regions of prescribed volume minimizing boundary area, are precisely the regions bounded by the canonical rotational constant-mean-curvature spheres; that is, the embedded CMC spheres invariant under rotations about the vertical axis of the fibration $\mathbb{E}^3(0, \tau) \rightarrow \mathbb{R}^2$, unique up to ambient isometry for their mean curvature.

3.26.2. Historical background, literature, and current progress

3.26.3. Verdict: partially resolved

3.26.3.1. Precise problem

The problem asks for the full isoperimetric classification in the Riemannian Heisenberg space $\text{Nil}_3 = \mathbb{E}^3(0, \tau)$, $\tau \neq 0$: for every prescribed volume, are the perimeter-minimizing regions exactly the regions enclosed by the canonical rotational constant-mean-curvature spheres, up to ambient isometry?

3.26.3.2. Current literature status

No citable published result found proves the full statement. The strongest current results classify the candidate CMC spheres and prove several necessary, conditional, asymptotic, or small-volume facts, but they do not prove that every isoperimetric region in Nil_3 has spherical topology, nor that every canonical rotational CMC sphere is globally isoperimetric for its enclosed volume.

Key literature:

- **Constant mean curvature surfaces in 3-dimensional Thurston geometries;** Isabel Fernández and Pablo Mira; *Proceedings of the International Congress of Mathematicians, Volume II (Invited Lectures)*, Hindustan Book Agency, 2010, pp. 830-861; arXiv:1004.4752,

<https://arxiv.org/abs/1004.4752>. This is the source survey. It states that for the remaining $E^3(\kappa, \tau)$ spaces, including Nil_3 , the isoperimetric problem is open; it records only the small-volume consequence that isoperimetric solutions are canonical rotational CMC spheres with large mean curvature.

- **CMC spheres in the Heisenberg group**; Valentina Franceschi, Francescopaolo Montefalcone, Roberto Monti; *Analysis and Geometry in Metric Spaces* 7(1), 109-129, 2019; DOI: 10.1515/agms-2019-0006; arXiv:1611.09215, <https://arxiv.org/abs/1611.09215>. This paper studies the explicit Riemannian Heisenberg CMC spheres and says they are conjectured to be the isoperimetric sets. It proves supporting results, including classification of topological CMC spheres as translations of the model spheres and a restricted quantitative stability inequality, but explicitly does not prove the isoperimetric property.
- **Isoperimetric domains of large volume in homogeneous three-manifolds**; William H. Meeks III, Pablo Mira, Joaquín Pérez, Antonio Ros; *Advances in Mathematics* 264, 546-592, 2014; DOI: 10.1016/j.aim.2014.07.025; arXiv:1303.4222, <https://arxiv.org/abs/1303.4222>. This gives asymptotic information for large-volume isoperimetric domains in homogeneous three-manifolds. It does not classify Nil_3 isoperimetric domains as canonical rotational CMC spheres.
- **Constant mean curvature spheres in homogeneous three-manifolds**; William H. Meeks III, Pablo Mira, Joaquín Pérez, Antonio Ros; *Inventiones Mathematicae* 224, 147-244, 2021; DOI: 10.1007/s00222-020-01008-y; <https://link.springer.com/article/10.1007/s00222-020-01008-y>. This fully classifies immersed CMC spheres in homogeneous three-manifolds. In particular, combined with the Abresch-Rosenberg theory, CMC spheres in Nil_3 are the rotational ones and are unique up to isometry for their mean curvature. This is not enough to solve the problem statement because an isoperimetric boundary is only known a priori to be a stable embedded CMC surface, not necessarily a sphere.
- **Uniqueness of Constant Mean Curvature Spheres**; Pablo Mira and Joaquín Pérez; in *New Trends in Geometric Analysis*, RSME Springer Series, vol. 10, Springer, 2023, pp. 365-393; DOI: 10.1007/978-3-031-39916-9_13. This recent survey explicitly says that in Nil_3 , although all CMC spheres are rotational and weakly stable, it is not known whether isoperimetric regions are topological balls; the obstruction is the lack of reflections/Alexandrov reflection methods in Nil_3 .

Important non-resolution:

- **Isoperimetric domains in homogeneous three-manifolds and the isoperimetric constant of the Heisenberg group H^1** ; Jih-Hsin Cheng, Andrea Malchiodi, Paul Yang; arXiv preprint, 2015; arXiv:1501.07357, <https://arxiv.org/abs/1501.07357>. This preprint claimed a broad topological-ball theorem that would be relevant, but arXiv marks it as withdrawn due to a problem in Section 2. It should not be treated as resolving literature.

3.26.3.3. What remains unresolved

The missing full step is to exclude non-spherical isoperimetric boundaries in Nil_3 , or equivalently to prove a theorem strong enough to show that every isoperimetric region is a topological ball/sphere-bounded region. Once that topology is known, the CMC-sphere classification would identify the boundary with a canonical rotational CMC sphere. Conversely, the global

minimizing property of the entire canonical family is still only conjectural outside the known small-volume/conditional regimes.

3.26.4. Remaining open problems recorded by the database

3.26.4.1. Open problem 1

Prove or disprove that every isoperimetric region in the Riemannian Heisenberg space Nil_3 is a topological three-ball (equivalently, that its boundary is a topological two-sphere).

Reference supplied by the database. Pablo Mira and Joaquín Pérez, “Uniqueness of Constant Mean Curvature Spheres,” in *New Trends in Geometric Analysis*, RSME Springer Series, vol. 10, Springer, Cham, 2023, pp. 365–393; discussion of the open isoperimetric-topology step for Nil_3 (no numbered problem). DOI: 10.1007/978-3-031-39916-9_13. https://doi.org/10.1007/978-3-031-39916-9_13

3.26.5. Primary-source pointer

The originating ICM record is *Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries* (2010). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=836>.

3.27. 140. Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries [2010-geom-106]

Record: 2010-geom-106

Subfield: Classical and local differential geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2010: Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=836>

3.27.1. Problem statement

Let $\mathbb{E}^3(\kappa, \tau)$ be a fixed homogeneous space with $\kappa < 0$ and $\tau \neq 0$, equivalently the universal cover of $\mathrm{PSL}(2, \mathbb{R})$ endowed with one of its standard left-invariant homogeneous metrics. Prove or disprove that the isoperimetric regions in $\mathbb{E}^3(\kappa, \tau)$ are exactly the compact regions bounded by the canonical rotational constant-mean-curvature spheres. Here an isoperimetric region means a region whose boundary has least area among regions with the same enclosed volume. The canonical rotational constant-mean-curvature spheres are the embedded CMC H -spheres in $\mathbb{E}^3(\kappa, \tau)$ that are invariant under rotations about a vertical fiber; for $\kappa - 4\tau^2 < 0$, such a sphere exists uniquely up to ambient isometry precisely when $H^2 > -\kappa/4$.

3.27.2. Historical background, literature, and current progress

3.27.3. Verdict: partially resolved

3.27.3.1. Precise problem checked

For a fixed simply connected homogeneous space $\mathbb{E}^3(\kappa, \tau)$ with $\kappa < 0$ and $\tau \neq 0$, i.e. the universal cover of $\mathrm{PSL}(2, \mathbb{R})$ with one of its standard 4-dimensional-isometry-group metrics, determine whether every isoperimetric region of every prescribed volume is exactly the compact region bounded by the canonical rotational embedded CMC sphere, and conversely whether every such spherical region is isoperimetric.

3.27.3.2. Literature and current progress

1. Constant mean curvature surfaces in 3-dimensional Thurston geometries

Authors: Isabel Fernández and Pablo Mira.

Venue/year: Proceedings of the International Congress of Mathematicians 2010; arXiv preprint, 2010.

DOI/arXiv/URL: arXiv:1004.4752, <https://arxiv.org/abs/1004.4752>.

Contribution: This survey is the apparent source of the problem. In Section 4.4–4.5 it explicitly says that the isoperimetric problem is open in Nil_3 , the universal cover of

$\mathrm{PSL}(2, \mathbb{R})$, and Berger spheres outside the Torralbo–Urbano range. It also records the partial fact that for small volumes the general isoperimetric theory plus the Abresch–Rosenberg CMC-sphere uniqueness theorem gives canonical rotational CMC spheres with large H as the isoperimetric solutions.

2. Rotationally invariant constant mean curvature surfaces in homogeneous 3-manifolds

Author: Francisco Torralbo.

Venue/year: *Differential Geometry and its Applications* 28 (2010), no. 5, 593–607.

DOI/arXiv/URL: DOI 10.1016/j.difgeo.2010.04.007; arXiv:0911.5128, <https://arxiv.org/abs/0911.5128>.

Contribution: Classifies rotational CMC surfaces in Berger spheres and in $\mathrm{SL}(2, \mathbb{R})$ -type homogeneous spaces, including the canonical rotational CMC spheres used in the conjecture. This supplies the candidate spheres and their ODE geometry, but it does not prove that their enclosed domains minimize area for all prescribed volumes.

3. Compact stable constant mean curvature surfaces in homogeneous 3-manifolds

Authors: Francisco Torralbo and Francisco Urbano.

Venue/year: *Indiana University Mathematics Journal* 61 (2012), no. 3, 1129–1156.

DOI/arXiv/URL: DOI 10.1512/iumj.2012.61.4667; <https://iumj.org/article/5606/>.

Contribution: Proves that the constant-mean-curvature spheres in the special linear group case and in Nil_3 are stable, and gives partial classification results for compact stable CMC surfaces under additional hypotheses. Stability is necessary for isoperimetry, but not sufficient; the paper does not prove the full isoperimetric classification in $\mathrm{PSL}(2, \mathbb{R})$.

4. Isoperimetric domains of large volume in homogeneous three-manifolds

Authors: William H. Meeks III, Pablo Mira, Joaquín Pérez, and Antonio Ros.

Venue/year: *Advances in Mathematics* 264 (2014), 546–592.

DOI/arXiv/URL: DOI 10.1016/j.aim.2014.07.025; arXiv:1303.4222, <https://arxiv.org/abs/1303.4222>.

Contribution: Gives large-volume asymptotics for isoperimetric domains in noncompact simply connected homogeneous three-manifolds: radii diverge, area/volume ratios converge to the Cheeger constant, boundary mean curvatures converge to $\mathrm{Ch}(X)/2$, and large boundaries are approximated by a CMC foliation. This is substantial asymptotic progress, but it does not identify the exact minimizers for every finite volume as the canonical rotational CMC spheres.

5. Constant mean curvature spheres in homogeneous three-manifolds

Authors: William H. Meeks III, Pablo Mira, Joaquín Pérez, and Antonio Ros.

Venue/year: *Inventiones Mathematicae* 224 (2021), 147–244.

DOI/arXiv/URL: DOI 10.1007/s00222-020-01008-y; arXiv:1706.09394, <https://arxiv.org/abs/1706.09394>.

Contribution: Completely classifies immersed CMC spheres in homogeneous three-manifolds. For X diffeomorphic to $\mathbb{R}^3$, CMC spheres exist exactly for $|H| > \mathrm{Ch}(X)/2$, are unique up to isometry for fixed H , are maximally symmetric, and in rotationally symmetric $\mathbb{E}^3(\kappa, \tau)$ spaces are rotational. This proves the uniqueness/classification of the candidate boundaries, but it does not prove that every isoperimetric boundary has genus zero, nor that each candidate sphere is globally area-minimizing for its enclosed volume.

6. Isoperimetric domains in homogeneous three-manifolds and the isoperimetric constant of the Heisenberg group H^1

Authors: Jih-Hsin Cheng, Andrea Malchiodi, and Paul Yang.

Venue/year: arXiv preprint, 2015.

DOI/arXiv/URL: arXiv:1501.07357, <https://arxiv.org/abs/1501.07357>.

Contribution: The preprint claimed that isoperimetric sets in homogeneous three-spaces diffeomorphic to $\mathbb{R}^3$ are topological balls, which together with the CMC-sphere classification would have been close to resolving this problem. However, the arXiv record states that the paper was withdrawn due to a problem in Section 2, so it cannot be counted as a valid resolution.

3.27.3.3. What remains unresolved

No valid published or otherwise citable result found proves the full biconditional in the problem statement: for every volume in every $\mathbb{E}^3(\kappa, \tau)$ with $\kappa < 0$, $\tau \neq 0$, the isoperimetric regions are exactly the compact regions bounded by canonical rotational CMC spheres. The remaining gap is global isoperimetric minimality and exclusion of all other possible isoperimetric boundaries for all volumes. The literature gives strong partial progress: small-volume solutions, stability of the candidate spheres, large-volume asymptotics, and complete uniqueness/classification of CMC spheres.

3.27.4. Remaining open problems recorded by the database

3.27.4.1. Open problem 1

Prove or disprove that every isoperimetric region in a fixed homogeneous space $\mathbb{E}^3(\kappa, \tau)$ with $\kappa < 0$ and $\tau \neq 0$ is a topological three-ball (equivalently, that every isoperimetric boundary is a topological two-sphere).

Reference supplied by the database. William H. Meeks III, Joaquín Pérez, and Giuseppe Tinaglia, “Constant Mean Curvature Surfaces,” *Surveys in Differential Geometry* 21 (2016), 179–287, Conjecture 13.8(1), “Isoperimetric Domains Conjecture” (numbered Conjecture 15.18(1) in the author-manuscript pagination). DOI: 10.4310/SDG.2016.v21.n1.a6. arXiv:1605.02512. <https://doi.org/10.4310/SDG.2016.v21.n1.a6>

3.27.4.2. Open problem 2

Prove or disprove that every canonical rotational constant-mean-curvature sphere in a fixed homogeneous space $\mathbb{E}^3(\kappa, \tau)$ with $\kappa < 0$ and $\tau \neq 0$ bounds an isoperimetric region.

Reference supplied by the database. William H. Meeks III, Joaquín Pérez, and Giuseppe Tinaglia, “Constant Mean Curvature Surfaces,” *Surveys in Differential Geometry* 21 (2016), 179–287, Conjecture 13.8(2), “Isoperimetric Domains Conjecture” (numbered Conjecture 15.18(2) in the author-manuscript pagination). DOI: 10.4310/SDG.2016.v21.n1.a6. arXiv:1605.02512. <https://doi.org/10.4310/SDG.2016.v21.n1.a6>

3.27.5. Primary-source pointer

The originating ICM record is *Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries* (2010). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=836>.

3.28. 141. Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries [2010-geom-107]

Record: 2010-geom-107

Subfield: Classical and local differential geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2010: Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=836>

3.28.1. Problem statement

Prove or disprove that the following Alexandrov classification holds in the non-product homogeneous spaces $\mathbb{E}^3(\kappa, \tau)$ with $\tau \neq 0$: in the Heisenberg space Nil_3 , in the universal covering of $\text{PSL}(2, \mathbb{R})$ with its left-invariant homogeneous metric, and in hemispheres of Berger 3-spheres, every compact embedded surface with constant mean curvature is, up to an ambient isometry, the canonical rotational constant-mean-curvature sphere with the same mean curvature; that is, the rotational CMC 2-sphere invariant under rotations about a fiber (the vertical axis) of the canonical fibration of the corresponding $\mathbb{E}^3(\kappa, \tau)$ space over a 2-dimensional space form, unique among rotational CMC spheres for that mean curvature whenever such a sphere exists.

3.28.2. Historical background, literature, and current progress

3.28.3. Verdict: partially resolved

3.28.3.1. Precise problem

The problem asks for the Alexandrov classification of compact embedded constant-mean-curvature surfaces in the non-product homogeneous spaces $\mathbb{E}^3(\kappa, \tau)$, $\tau \neq 0$: Nil_3 , $\widetilde{\text{PSL}}(2, \mathbb{R})$ / $\widetilde{\text{SL}}_2(\mathbb{R})$, and Berger hemispheres. The desired conclusion is that every such surface is congruent to the canonical rotational CMC sphere with the same mean curvature, whenever that sphere exists. This is stronger than the Hopf problem, because no genus-zero hypothesis is allowed.

3.28.3.2. Literature checked and current progress

1. Isabel Fernández and Pablo Mira, "Constant mean curvature surfaces in 3-dimensional Thurston geometries", *Proceedings of the International Congress of Mathematicians, Hyderabad 2010*, Vol. II, Hindustan Book Agency, 2010, pp. 830-861. arXiv:1004.4752. URL: <https://arxiv.org/abs/1004.4752>.

This is the source of the problem. Section 4.5 states that the Alexandrov problem for $\tau \neq 0$, namely in Nil_3 , $\widetilde{\text{PSL}}(2, \mathbb{R})$, and Berger hemispheres, is a major open problem and

conjectures that the canonical rotational spheres are the only compact embedded CMC surfaces. The same survey records the known partial results: existence/uniqueness of rotational CMC spheres, Hopf uniqueness for immersed CMC spheres, and some isoperimetric/stability results.

2. Uwe Abresch and Harold Rosenberg, "A Hopf differential for constant mean curvature surfaces in $S^2 \times \mathbb{R}$ and $H^2 \times \mathbb{R}$ ", *Acta Mathematica* 193(2), 141-174, 2004. DOI: 10.1007/BF02392562. URL: <https://doi.org/10.1007/BF02392562>. Uwe Abresch and Harold Rosenberg, "Generalized Hopf differentials", *Matemática Contemporânea* 28, 1-28, 2005. MR 2195187.

These papers give the Abresch-Rosenberg holomorphic differential and solve the Hopf-type genus-zero problem in the four-dimensional-isometry homogeneous spaces: any immersed CMC sphere is rotational. This is important partial progress but does not imply the Alexandrov statement, because an arbitrary compact embedded CMC surface need not be known to be a sphere.

3. Francisco Torralbo, "Rotationally invariant constant mean curvature surfaces in homogeneous 3-manifolds", *Differential Geometry and its Applications* 28(5), 593-607, 2010. DOI: 10.1016/j.difgeo.2010.04.007. arXiv:0911.5128. URL: <https://arxiv.org/abs/0911.5128>.

This classifies rotational CMC surfaces in Berger spheres and in the special linear group setting, including the rotational CMC spheres and Delaunay-type examples. It verifies the rotational model family, but only within the rotational class, so it does not rule out non-rotational compact embedded CMC surfaces.

4. Francisco Torralbo and Francisco Urbano, "Compact stable constant mean curvature surfaces in homogeneous 3-manifolds", *Indiana University Mathematics Journal* 61(3), 1129-1156, 2012. DOI: 10.1512/iumj.2012.61.4667. arXiv:0906.1439. URL: <https://arxiv.org/abs/0906.1439>.

This classifies stable CMC spheres in Berger spheres, Nil_3 , and $\widetilde{\text{SL}}_2(\mathbb{R})$, and classifies stable compact orientable CMC surfaces in certain Berger-sphere parameter ranges, with isoperimetric consequences. The hypotheses are stability and/or restricted Berger parameters, so this is not the unrestricted Alexandrov classification.

5. William H. Meeks III, Pablo Mira, Joaquín Pérez and Antonio Ros, "Constant mean curvature spheres in homogeneous three-manifolds", *Inventiones Mathematicae* 224, 147-244, 2021. DOI: 10.1007/s00222-020-01008-y. URL: <https://doi.org/10.1007/s00222-020-01008-y>.

This gives a complete classification of immersed CMC spheres in arbitrary homogeneous three-manifolds. It subsumes and extends Hopf-type uniqueness, but its theorem is explicitly about genus-zero immersed spheres. It therefore does not prove that every compact embedded CMC surface in Nil_3 or $\widetilde{\text{SL}}_2(\mathbb{R})$ has genus zero.

6. William H. Meeks III, Pablo Mira, Joaquín Pérez and Antonio Ros, "Constant mean curvature spheres in homogeneous three-spheres", *Journal of Differential Geometry* 120(2), 307-343, 2022. DOI: 10.4310/jdg/1645207520. arXiv:1308.2612. URL: <https://doi.org/10.4310/jdg/1645207520>.

This resolves uniqueness/existence of CMC spheres in homogeneous three-spheres, including Berger-sphere settings. Again, it is a CMC-sphere theorem rather than a classification of all compact embedded CMC surfaces in the Berger-hemisphere Alexandrov sense.

7. Antonio Bueno, "Delaunay surfaces of prescribed mean curvature in Nil_3 and $\widetilde{\text{SL}}_2(\mathbb{R})$ ", arXiv preprint, submitted 2020, revised 2021. arXiv:2003.13018. DOI:

10.48550/arXiv.2003.13018. URL: <https://arxiv.org/abs/2003.13018>.

The introduction explicitly states that, for CMC surfaces, the Alexandrov problem in Nil_3 and $\widetilde{\text{SL}}_2(\mathbb{R})$ is still an outstanding major open problem. The paper studies rotational prescribed-mean-curvature analogues and does not resolve the CMC Alexandrov classification.

3.28.3.3. What remains unresolved

No published or otherwise citable theorem found covers the full statement. The literature proves uniqueness of CMC spheres and gives strong rotational, stable, and isoperimetric special cases, but the decisive missing step for the full Alexandrov classification is to show that an arbitrary compact embedded CMC surface in Nil_3 and in $\widetilde{\text{PSL}}(2, \mathbb{R})$ must be genus zero/rotational. Because those two cases are still described in recent literature as open, the problem is not fully resolved.

3.28.4. Remaining open problems recorded by the database

3.28.4.1. Open problem 1

Prove or disprove that every compact embedded constant-mean-curvature surface in the Heisenberg space Nil_3 is a canonical rotational constant-mean-curvature sphere.

Reference supplied by the database. Pablo Mira and Joaquín Pérez, “Uniqueness of Constant Mean Curvature Spheres,” in *New Trends in Geometric Analysis*, RSME Springer Series, vol. 10, Springer, Cham, 2023, pp. 365–393, Conjecture 9.2. DOI: 10.1007/978-3-031-39916-9_13. https://doi.org/10.1007/978-3-031-39916-9_13

3.28.4.2. Open problem 2

Prove or disprove that every compact embedded constant-mean-curvature surface in the universal cover of $\text{PSL}(2, \mathbb{R})$, equipped with a standard left-invariant homogeneous metric $\mathbb{E}^3(\kappa, \tau)$ with $\kappa < 0$ and $\tau \neq 0$, is a canonical rotational constant-mean-curvature sphere.

Reference supplied by the database. Isabel Fernández and Pablo Mira, “Constant Mean Curvature Surfaces in 3-Dimensional Thurston Geometries,” in *Proceedings of the International Congress of Mathematicians, Hyderabad 2010*, vol. II, Hindustan Book Agency, New Delhi, 2010, pp. 830–861, §4.5 “Open problems” (the $\widetilde{\text{PSL}}(2, \mathbb{R})$ special case of the Alexandrov problem; no numbered conjecture). arXiv:1004.4752. <https://arxiv.org/abs/1004.4752>

3.28.4.3. Open problem 3

Prove or disprove that every compact embedded constant-mean-curvature surface contained in a hemisphere of a Berger three-sphere is a canonical rotational constant-mean-curvature sphere.

Reference supplied by the database. Isabel Fernández and Pablo Mira, “Constant Mean Curvature Surfaces in 3-Dimensional Thurston Geometries,” in *Proceedings of the International Congress of Mathematicians, Hyderabad 2010*, vol. II, Hindustan Book Agency, New Delhi, 2010, pp. 830–861, §4.5 “Open problems” (the Berger-hemisphere special case of the Alexandrov problem; no numbered conjecture). arXiv:1004.4752. <https://arxiv.org/abs/1004.4752>

3.28.5. Primary-source pointer

The originating ICM record is *Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries* (2010). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=836>.

3.29. 142. Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries [2010-geom-108]

Record: 2010-geom-108

Subfield: Classical and local differential geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2010: Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=836>

3.29.1. Problem statement

Prove that every immersed two-sphere S of constant mean curvature in the Thurston geometry Sol_3 , realized as the Lie group $\mathbb{R}^3$ with left-invariant metric

$$ds^2 = e^{2x_3} dx_1^2 + e^{-2x_3} dx_2^2 + dx_3^2,$$

is weakly stable: if η is a unit normal to S , $\mathcal{B}$ is its second fundamental form, and $\mathcal{L} = \Delta + \|\mathcal{B}\|^2 + \text{Ric}(\eta)$ is its Jacobi operator, then

$$-\int_S f \mathcal{L}(f) dA \geq 0$$

for every smooth function f on S satisfying $\int_S f dA = 0$.

3.29.2. Historical background, literature, and current progress

3.29.3. Verdict: partially resolved

3.29.3.1. Problem checked

The problem asks whether every immersed constant-mean-curvature two-sphere in the standard Thurston geometry Sol_3 ,

$$ds^2 = e^{2x_3} dx_1^2 + e^{-2x_3} dx_2^2 + dx_3^2,$$

is weakly stable for volume-preserving variations, i.e. whether

$$-\int_S f (\Delta f + \|\mathcal{B}\|^2 f + \text{Ric}(\eta)f) dA \geq 0$$

for every smooth zero-mean function f on S .

3.29.3.2. Status

I did not find a published or otherwise citable theorem proving this full weak-stability statement for all CMC spheres in Sol_3 . Recent citable literature still treats the exact assertion as

open. The problem is not fully open, however, because the CMC spheres in Sol3 have been classified and are known to be embedded and of index one.

3.29.3.3. Progress in the literature

1. Existence and uniqueness of constant mean curvature spheres in Sol3

Authors: Benoit Daniel and Pablo Mira.

Venue: Journal für die reine und angewandte Mathematik 685 (2013), 1-32.

DOI/arXiv/URL: DOI <https://doi.org/10.1515/crelle-2012-0016>; arXiv 0812.3059; <https://arxiv.org/abs/0812.3059>.

Result: Theorems 1.1 and 1.2 prove existence/uniqueness and embeddedness for $H > 1/\sqrt{3}$, with index one, and prove a conditional uniqueness theorem when a CMC H -sphere satisfies one of several hypotheses including weak stability, index one, or Gauss-map diffeomorphism. This gives major partial progress but does not prove weak stability for every CMC sphere.

2. Constant mean curvature spheres in Sol3

Author: William H. Meeks III.

Venue: American Journal of Mathematics 135(3) (2013), 763-775.

DOI/arXiv/URL: DOI <https://doi.org/10.1353/ajm.2013.0025>; JSTOR page <https://www.jstor.org/stable/23525906>.

Result: Completes the Sol3 CMC-sphere classification for every $H > 0$: existence, uniqueness up to left translation, embeddedness, and index one. This still does not imply the volume-preserving weak-stability inequality, since index one is weaker than weak stability.

3. Constant mean curvature spheres in homogeneous three-manifolds

Authors: William H. Meeks III, Pablo Mira, Joaquin Perez, Antonio Ros.

Venue: Inventiones Mathematicae 224 (2021), 147-244.

DOI/arXiv/URL: DOI <https://doi.org/10.1007/s00222-020-01008-y>; arXiv 1706.09394; <https://link.springer.com/article/10.1007/s00222-020-01008-y>.

Result: Gives the general homogeneous-three-manifold classification. For Sol3 it confirms the known picture: unique embedded CMC spheres with index one and nullity three. The theorem's conclusions do not include weak stability.

4. Uniqueness of Constant Mean Curvature Spheres

Authors: Pablo Mira and Joaquin Perez.

Venue: New Trends in Geometric Analysis, RSME Springer Series 10 (2023), 365-393.

DOI/URL: DOI https://doi.org/10.1007/978-3-031-39916-9_13; https://link.springer.com/chapter/10.1007/978-3-031-39916-9_13.

Result: This recent survey explicitly states that the Sol3 CMC spheres classified by Daniel-Mira and Meeks are embedded and have index one, but that it is not known whether they are weakly stable; it records the conjecture "Any CMC sphere in Sol3 is weakly stable."

3.29.3.4. What remains unresolved

The missing step is a citable proof that the known index-one CMC spheres in Sol3 satisfy the full zero-mean Jacobi quadratic-form inequality. The available classification and index-one theorems are necessary background, but they do not by themselves prove weak stability.

I also checked the arXiv preprint **Isoperimetric domains in homogeneous three-manifolds and the isoperimetric constant of the Heisenberg group H_1** by Jih-Hsin Cheng, Andrea Malchiodi, and Paul Yang, arXiv 1501.07357; arXiv marks it withdrawn because of a problem in Section 2, so I did not count it as resolving literature.

3.29.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

3.29.5. Primary-source pointer

The originating ICM record is *Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries* (2010). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=836>.

3.30. 143. Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries [2010-geom-109]

Record: 2010-geom-109

Subfield: Classical and local differential geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2010: Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=836>

3.30.1. Problem statement

Prove that every constant mean curvature sphere in the Thurston geometry Sol_3 , realized as $\mathbb{R}^3$ with the standard left-invariant metric $ds^2 = e^{2x_3} dx_1^2 + e^{-2x_3} dx_2^2 + dx_3^2$, is the boundary of an isoperimetric region in Sol_3 , that is, of a region whose boundary has least area among regions with the same enclosed volume.

3.30.2. Historical background, literature, and current progress

3.30.3. Verdict: partially resolved

3.30.3.1. Precise problem

The problem asks whether, in the standard Thurston geometry $\text{Sol}_3 = (\mathbb{R}^3, e^{2x_3} dx_1^2 + e^{-2x_3} dx_2^2 + dx_3^2)$, every compact constant mean curvature two-sphere S is the boundary of an isoperimetric region: equivalently, whether the domain enclosed by S minimizes boundary area among all regions in Sol_3 with the same enclosed volume.

3.30.3.2. Current status

No published or otherwise reliable citable result found proves this for all CMC spheres in Sol_3 . The literature fully classifies the CMC spheres in Sol_3 , and proves the desired isoperimetric conclusion in the small-volume/large-mean-curvature regime, but the all- $H > 0$ isoperimetric assertion remains open. In particular, a 2023 survey by Mira and Perez still records that it is not known whether the Sol_3 CMC spheres are weakly stable, and says that a positive answer would imply that all CMC spheres in Sol_3 bound isoperimetric regions.

3.30.3.3. Key literature

1. "Constant mean curvature surfaces in 3-dimensional Thurston geometries"

Authors: Isabel Fernandez and Pablo Mira.

Venue/year: Proceedings of the International Congress of Mathematicians, Volume II, Hindustan Book Agency, 2010, pp. 830-861.

arXiv/DOI/URL: arXiv:1004.4752; DOI 10.1142/9789814324359_0076; <https://arxiv.org/abs/1004.4752>.

Role: This is the original survey source matching the problem statement. It states the Sol_3 classification theorem of Daniel-Mira and Meeks, but then lists as open: whether CMC spheres in Sol_3 are weakly stable and whether they all bound isoperimetric regions.

2. **“Existence and uniqueness of constant mean curvature spheres in Sol_3 ”**

Authors: Benoit Daniel and Pablo Mira.

Venue/year: Journal fuer die reine und angewandte Mathematik (Crelle’s Journal) 685 (2013), 1-32.

arXiv/DOI/URL: arXiv:0812.3059; DOI 10.1515/crelle-2012-0016; <https://arxiv.org/abs/0812.3059>.

Role: Proves existence and uniqueness of embedded CMC H -spheres for $H > 1/\sqrt{3}$, and proves a conditional uniqueness theorem for spheres satisfying properties such as being isoperimetric, weakly stable, index one, or having diffeomorphic Gauss map. This is major progress but does not prove that every CMC sphere is isoperimetric.

3. **“Constant mean curvature spheres in Sol_3 ”**

Author: William H. Meeks III.

Venue/year: American Journal of Mathematics 135 no. 3 (2013), 763-775.

DOI/URL: DOI 10.1353/ajm.2013.0025; JSTOR <https://www.jstor.org/stable/23525906>.

Role: Completes the Sol_3 classification by proving existence and uniqueness of an immersed CMC H -sphere for each $H > 0$. The result gives uniqueness/embeddedness/index-one information, not the global area-minimizing isoperimetric property.

4. **“Constant mean curvature spheres in homogeneous three-manifolds”**

Authors: William H. Meeks III, Pablo Mira, Joaquin Perez, Antonio Ros.

Venue/year: Inventiones Mathematicae 224 (2021), 147-244.

arXiv/DOI/URL: arXiv:1706.09394; DOI 10.1007/s00222-020-01008-y; <https://arxiv.org/abs/1706.09394>.

Role: Gives a broad classification of CMC spheres in homogeneous three-manifolds. It records, in the proof discussion around Proposition 3.4, that for H sufficiently large the spheres S_H bound small isoperimetric regions. This is only a large- H special case and does not cover all CMC spheres in Sol_3 .

5. **“Uniqueness of Constant Mean Curvature Spheres”**

Authors: Pablo Mira and Joaquin Perez.

Venue/year: RSME Springer Series chapter, Springer, 2023.

DOI/URL: DOI 10.1007/978-3-031-39916-9_13; author PDF <https://wpd.ugr.es/~jperez/wordpress/wp-content/uploads/authorsample-MiraPerez.pdf>.

Role: A recent survey of the classification theory. In its open-problems section it states that, although the Sol_3 CMC spheres have index one, it is not known whether they are weakly stable; it records Daniel-Mira’s conjecture that every Sol_3 CMC sphere is weakly stable and notes that this would imply that all such spheres bound isoperimetric regions. This is strong evidence that the exact question is not fully resolved as of the current literature.

6. **“Isoperimetric domains in homogeneous three-manifolds and the isoperimetric constant of the Heisenberg group H^1 ”**

Authors: Jih-Hsin Cheng, Andrea Malchiodi, Paul Yang.

Venue/year: arXiv preprint, 2015.

arXiv/URL: arXiv:1501.07357; <https://arxiv.org/abs/1501.07357>.

Role: This appeared in searches because it claimed structural results for isoperimetric sets in homogeneous spaces diffeomorphic to $\mathbb{R}^3$, but arXiv marks it withdrawn due to a problem in Section 2. I therefore did not treat it as resolving citable evidence.

3.30.3.4. What remains unresolved

The missing step is to prove, for every $H > 0$, that the unique CMC H -sphere $S_H \subset \text{Sol}_3$ is the boundary of a globally area-minimizing region for its enclosed volume. Known index-one and uniqueness results do not imply this, and the available isoperimetric result only covers sufficiently large H (small enclosed volumes). Thus the exact all-cases statement is not fully resolved.

3.30.4. Remaining open problems recorded by the database

3.30.4.1. Open problem 1

Prove or disprove that every constant-mean-curvature sphere in Sol_3 is weakly stable under volume-preserving variations.

Reference supplied by the database. Pablo Mira and Joaquín Pérez, “Uniqueness of Constant Mean Curvature Spheres,” in *New Trends in Geometric Analysis*, RSME Springer Series, vol. 10, Springer, Cham, 2023, pp. 365–393, Conjecture 9.3 (attributed there to Benoît Daniel and Pablo Mira). DOI: 10.1007/978-3-031-39916-9_13. https://doi.org/10.1007/978-3-031-39916-9_13

3.30.5. Primary-source pointer

The originating ICM record is *Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries* (2010). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=836>.

3.31. 144. Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries [2010-geom-110]

Record: 2010-geom-110

Subfield: Classical and local differential geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2010: Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=836>

3.31.1. Problem statement

Prove that there is no complete stable immersed surface in $\mathbb{H}^2 \times \mathbb{R}$ with constant mean curvature H satisfying $H > 1/2$.

3.31.2. Historical background, literature, and current progress

3.31.3. Verdict: partially resolved

3.31.3.1. Precise problem

The problem asks for the nonexistence of complete stable immersed constant-mean-curvature surfaces in the product space $\mathbb{H}^2 \times \mathbb{R}$ when the constant mean curvature satisfies $H > 1/2$. In the original CMC-surface context, “stable” is the strong/two-sided stability condition for compactly supported variations, not merely weak volume-preserving stability of closed CMC spheres.

3.31.3.2. Current progress

Original/open-problem source Title: Constant mean curvature surfaces in 3-dimensional Thurston geometries

Authors: Isabel Fernández and Pablo Mira

Venue/year: Proceedings of the International Congress of Mathematicians 2010, Vol. II, Hindustan Book Agency/World Scientific, pp. 830-861, 2010/2011

DOI/arXiv/URL: arXiv:1004.4752; DOI: 10.1142/9789814324359_0076; <https://arxiv.org/abs/1004.4752>

Status contribution: This survey is the source matching the dataset entry. It records that there are no complete stable CMC surfaces in $\mathbb{H}^2 \times \mathbb{R}$ for $H > 1/\sqrt{3}$, and says the expected optimal statement is for $H > 1/2$. It also records the parabolic-conformal-type case as known. Thus the survey itself presents the exact $H > 1/2$ problem as not fully proved.

Stronger mean-curvature range Title: Global properties of constant mean curvature surfaces in $\mathbb{H}^2 \times \mathbb{R}$

Authors: Barbara Nelli and Harold Rosenberg

Venue/year: Pacific Journal of Mathematics 226(1), 137-152, 2006

DOI/arXiv/URL: DOI: 10.2140/pjm.2006.226.137; <https://msp.org/pjm/2006/226-1/pjm-v226-n1-p07-p.pdf>

Key result: Theorem B proves that in $\mathbb{H}^2 \times \mathbb{R}$ there is no complete noncompact stable H -surface with $H > 1/\sqrt{3}$, with compact boundary or without boundary.

Why partial: Since $1/\sqrt{3} \approx 0.577 > 1/2$, this leaves the interval $1/2 < H \leq 1/\sqrt{3}$ open for arbitrary complete stable immersed surfaces.

Parabolic conformal type Title: Parabolic stable surfaces with constant mean curvature

Authors: José M. Manzano, Joaquín Pérez, and M. Magdalena Rodríguez

Venue/year: Calculus of Variations and Partial Differential Equations 42(1-2), 137-152, 2011

DOI/arXiv/URL: arXiv:0910.5373; DOI: 10.1007/s00526-010-0383-6; <https://arxiv.org/abs/0910.5373>

Key result: For an orientable, parabolic, complete immersed stable surface with constant mean curvature H in $\mathbb{H}^2 \times \mathbb{R}$, one has $|H| \leq 1/2$; if equality holds, the surface is an entire graph or a vertical horocylinder.

Why partial: This proves the desired nonexistence only under the additional assumption that the surface is parabolic as a Riemannian/conformal surface. It does not rule out possible hyperbolic conformal type examples with $1/2 < H \leq 1/\sqrt{3}$.

Curvature decay/integrability hypotheses Title: Finite Index Operators on Surfaces

Authors: José M. Espinar

Venue/year: Journal of Geometric Analysis 23, 415-437, 2013

DOI/arXiv/URL: DOI: 10.1007/s12220-011-9259-z; <https://link.springer.com/article/10.1007/s12220-011-9259-z>

Key result: Among other applications, the paper proves that there are no complete stable H -surfaces $\Sigma \subset \mathbb{H}^2 \times \mathbb{R}$ with $H > 1/2$ when either $K_e^+ = \max\{0, K_e\} \in L^1(\Sigma)$ or the extrinsic curvature satisfies a decay bound $|K_e(q)| \leq M/d(p_0, q)$.

Why partial: The threshold is exactly $H > 1/2$, but the theorem imposes extra integrability/decay assumptions on the extrinsic curvature. These hypotheses are not present in the problem statement.

Related compact/weak-stability result Title: On stable constant mean curvature surfaces in $\mathbb{S}^2 \times \mathbb{R}$ and $\mathbb{H}^2 \times \mathbb{R}$

Authors: Rabah Souam

Venue/year: Transactions of the American Mathematical Society 362(6), 2845-2857, 2010

DOI/arXiv/URL: arXiv:0709.4204; DOI: 10.48550/arXiv.0709.4204; <https://arxiv.org/abs/0709.4204>

Status contribution: This paper studies compact weakly stable CMC surfaces, proving for example stability/classification results for rotational CMC spheres and genus bounds. This is useful context but does not resolve the strong complete-stability nonexistence question in the problem statement.

3.31.3.3. What remains unresolved

I did not find published or otherwise citable literature proving the unrestricted statement that every complete strongly stable immersed CMC surface in $\mathbb{H}^2 \times \mathbb{R}$ must satisfy $|H| \leq 1/2$. The unresolved part is the general case, especially possible complete stable surfaces of hyperbolic conformal type with $1/2 < H \leq 1/\sqrt{3}$, without the parabolicity or extrinsic-curvature hypotheses used in the known partial results.

3.31.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

3.31.5. Primary-source pointer

The originating ICM record is *Constant Mean Curvature Surfaces in 3-dimensional Thurston Geometries* (2010). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=836>.

3.32. 145. New applications of Min-max Theory [2014-geom-127]

Record: 2014-geom-127

Subfield: Classical and local differential geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: New applications of Min-max Theory

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=957>

3.32.1. Problem statement

Prove or disprove that, for every integer $g \geq 2$, the Lawson embedded minimal surface $\xi_{1,g} \subset S^3$ of genus g , in the round unit 3-sphere, minimizes the Willmore energy among all compact genus- g surfaces in S^3 ; equivalently, every compact surface $\Sigma \subset S^3$ of genus g satisfies

$$\mathcal{W}(\Sigma) \geq \mathcal{W}(\xi_{1,g}),$$

where

$$\mathcal{W}(\Sigma) = \int_{\Sigma} \left(1 + \left(\frac{k_1 + k_2}{2} \right)^2 \right) d\mu$$

and k_1, k_2 are the principal curvatures of Σ with respect to the standard metric on S^3 .

3.32.2. Historical background, literature, and current progress

3.32.3. Verdict: partially resolved

3.32.3.1. Precise problem

For every integer $g \geq 2$, decide whether the Lawson embedded minimal surface $\xi_{1,g} \subset S^3$ of genus g minimizes the Willmore energy

$$\mathcal{W}(\Sigma) = \int_{\Sigma} \left(1 + \left(\frac{k_1 + k_2}{2} \right)^2 \right) d\mu$$

among all compact genus- g surfaces in the round unit S^3 . The literature often writes the same Lawson genus- g family as $\xi_{g,1}$; the Lawson indices are interchangeable up to the usual symmetry of the construction.

3.32.3.2. Verdict

This is not fully resolved. The best available literature gives important special cases, existence/asymptotic results, numerical evidence, and a recent symmetry-restricted theorem, but no published or citable result proves

$$\mathcal{W}(\Sigma) \geq \mathcal{W}(\xi_{1,g})$$

for every compact genus- g surface in S^3 for every $g \geq 2$.

3.32.3.3. Key literature and current progress

1. Complete minimal surfaces in S^3

Authors: H. Blaine Lawson, Jr.

Venue/year: Annals of Mathematics (2) 92, 335-374, 1970.

DOI/URL: [doi:10.2307/1970625](https://doi.org/10.2307/1970625); JSTOR stable URL <http://www.jstor.org/stable/1970625>.

Role: Constructs the Lawson minimal surfaces $\xi_{m,k}$ in S^3 . This supplies the candidate surfaces but does not address Willmore minimization.

2. Comparison surfaces for the Willmore problem

Author: Robert Kusner.

Venue/year: Pacific Journal of Mathematics 138(2), 317-345, 1989.

DOI/URL: [doi:10.2140/pjm.1989.138.317](https://doi.org/10.2140/pjm.1989.138.317); <https://msp.org/pjm/1989/138-2/p08.xhtml>.

Role: Original source of Kusner's higher-genus Lawson minimizer conjecture (Conjecture 8.4 in later citations) and comparison-surface estimates such as $\mathcal{W}(\xi_{g,1}) < 8\pi$. It proposes the problem; it does not prove the full conjecture.

3. Minimizing the squared mean curvature integral for surfaces in space forms

Authors: Lucas Hsu, Rob Kusner, John Sullivan.

Venue/year: Experimental Mathematics 1(3), 191-207, 1992.

DOI/URL: [doi:10.1080/10586458.1992.10504258](https://doi.org/10.1080/10586458.1992.10504258); Project Euclid record <https://projecteuclid.org/euclid.em/1048622023>.

Role: Numerical and experimental evidence for Lawson-type minimizers in low genus. This is supporting evidence, not a proof of the full inequality.

4. Existence of surfaces minimizing the Willmore functional and Existence of minimizing Willmore surfaces of prescribed genus

Authors: Leon Simon; Matthias Bauer and Ernst Kuwert.

Venues/years: Communications in Analysis and Geometry 1(2), 281-326, 1993, [doi:10.4310/CAG.1993.v1.n2.a4](https://doi.org/10.4310/CAG.1993.v1.n2.a4); International Mathematics Research Notices 2003(10), 553-576, 2003, [doi:10.1155/S1073792803208072](https://doi.org/10.1155/S1073792803208072).

Role: These works establish existence of Willmore minimizers for prescribed genus under the necessary compactness/Douglas-type framework. They do not identify the minimizer as $\xi_{1,g}$.

5. The large genus limit of the infimum of the Willmore energy

Authors: Ernst Kuwert, Yuxiang Li, Reiner Schaetzle.

Venue/year: American Journal of Mathematics 132(1), 37-51, 2010.

DOI/URL: [doi:10.1353/ajm.0.0100](https://doi.org/10.1353/ajm.0.0100).

Role: Shows the large-genus limit of the infimum is 8π , consistent with the Lawson-surface picture. This is asymptotic information about the infimum, not identification of the minimizers for each g .

6. Min-Max theory and the Willmore conjecture

Authors: Fernando C. Marques, Andre Neves.

Venue/year: Annals of Mathematics 179(2), 683-782, 2014.

DOI/URL: [doi:10.4007/annals.2014.179.2.6](https://doi.org/10.4007/annals.2014.179.2.6).

Role: Fully resolves the genus-1 Willmore conjecture, identifying the Clifford torus. It gives a general lower bound $\mathcal{W} \geq 2\pi^2$ for positive genus, but this is weaker than the

genus $g \geq 2$ Lawson-minimizer claim and does not prove the problem statement.

7. Area estimates for high genus Lawson surfaces via DPW

Authors: Lynn Heller, Sebastian Heller, Martin Traizet.

Venue/year: Journal of Differential Geometry 124(1), 1-35, 2023; preprint arXiv:1907.07139.

DOI/URL: [doi:10.4310/jdg/1685121318](https://doi.org/10.4310/jdg/1685121318); <https://arxiv.org/abs/1907.07139>.

Role: Gives high-genus asymptotic area estimates for Lawson surfaces, for example $\text{Area}(\zeta_{1,g}) = 8\pi(1 - \frac{\log 2}{2g} + O(g^{-2}))$ in the relevant case. Since a minimal surface in S^3 has Willmore energy equal to area, this estimates the conjectured minimum, but it does not prove minimality among all genus- g surfaces.

8. The Lawson surfaces are determined by their symmetries and topology

Authors: Nikolaos Kapouleas, David Wiygul.

Venue/year: Journal fuer die reine und angewandte Mathematik 786, 155-173, 2022.

DOI/URL: [doi:10.1515/crelle-2021-0087](https://doi.org/10.1515/crelle-2021-0087).

Role: Proves that a closed embedded minimal surface in the round S^3 with the symmetries and genus of a Lawson surface is congruent to that Lawson surface. This is a minimal-surface characterization with symmetry assumptions, not a global Willmore minimization theorem.

9. The Willmore problem for surfaces with symmetry

Authors: Rob Kusner, Ying Lu, Peng Wang.

Venue/year: arXiv preprint, submitted 2024, revised 2025.

DOI/arXiv/URL: arXiv:2410.12582v2; [doi:10.48550/arXiv.2410.12582](https://doi.org/10.48550/arXiv.2410.12582); <https://arxiv.org/abs/2410.12582>.

Key result: Theorem 1.1 proves that if $M \subset S^3$ is a closed embedded genus mk surface invariant under specified finite ambient symmetry groups associated to $\zeta_{m,k}$, then

$$\mathcal{W}(M) \geq \mathcal{W}(\zeta_{m,k}) = \text{Area}(\zeta_{m,k}) \geq \mathcal{W}(\zeta_{mk,1}),$$

with equality conditions identifying the Lawson surface, and equality in the second inequality only for $k = 1$.

Applicability check: Taking $k = 1$ gives the desired Lawson lower bound for the symmetric embedded competitors of genus m . the problem has no symmetry hypothesis and asks for all compact genus- g competitors, so this is only partial progress. The paper's abstract explicitly states that the general genus $g > 1$ Lawson minimizer conjecture remains open.

10. Large topology asymptotics for spectrally extremal minimal surfaces in $\mathbb{B}^3$ and $\mathbb{S}^3$

Authors: Mikhail Karpukhin, Peter McGrath, Daniel Stern.

Venue/year: Calculus of Variations and Partial Differential Equations 65, article 119, 2026; preprint arXiv:2502.10225.

DOI/arXiv/URL: [doi:10.1007/s00526-026-03288-3](https://doi.org/10.1007/s00526-026-03288-3); <https://arxiv.org/abs/2502.10225>.

Role: Confirms, in a class of spectrally extremal low-area minimal examples, the special role of Lawson $\zeta_{\gamma,1}$ surfaces and compares large-topology area behavior. This is strong contextual evidence but does not cover all Willmore competitors.

11. The Analysis of Willmore Surfaces and its Generalizations in Higher Dimensions

Authors: Tian Lan, Dorian Martino, Tristan Riviere.

Venue/year: arXiv survey/preprint, 2025, revised 2026.

DOI/arXiv/URL: arXiv:2511.01777v3; [doi:10.48550/arXiv.2511.01777](https://doi.org/10.48550/arXiv.2511.01777); <https://arxiv.org/abs/2511.01777>.

Role: A current survey of Willmore-surface analysis. It states that after Marques-Neves the genus-1 case is solved, but the higher-genus Lawson candidate problem remains largely open. This agrees with the theorem-by-theorem applicability check above.

3.32.3.4. What remains unresolved

The open part is the full, unrestricted global minimization statement for every $g \geq 2$: proving that every compact genus- g surface in round S^3 , without imposed Lawson symmetries, fixed conformal structure, minimality, or membership in a special constructed family, satisfies $\mathcal{W}(\Sigma) \geq \mathcal{W}(\xi_{1,g})$. Existing results do not remove these restrictions.

3.32.4. Remaining open problems recorded by the database

3.32.4.1. Open problem 1

Prove that the genus-two Lawson surface $\xi_{1,2} \subset S^3$ minimizes the Willmore energy among all compact genus-two surfaces in S^3 , with equality only for Möbius images of $\xi_{1,2}$.

Reference supplied by the database. Robert Kusner, “Comparison surfaces for the Willmore problem,” *Pacific Journal of Mathematics* **138**(2) (1989), 317–345; the genus-two case of Kusner’s proposed higher-genus Lawson-surface minimization conjecture. DOI: <https://doi.org/10.2140/pjm.1989.138.317>. Its open status, including for $g = 2$, is reaffirmed in Rob Kusner, Ying Lü, and Peng Wang, “The Willmore problem for surfaces with symmetry,” preprint (2024), introduction, [arXiv:2410.12582](https://arxiv.org/abs/2410.12582).

3.32.5. Primary-source pointer

The originating ICM record is *New applications of Min-max Theory* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=957>.

3.33. 146. Evolution of form and shape [2022-ag-217]

Record: 2022-ag-217

Subfield: Classical and local differential geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2022: Evolution of form and shape

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-560-5-volume-2.pdf#page=267>

3.33.1. Problem statement

Prove or disprove the Calabi-Yau conjectures for embedded minimal surfaces: let $\Sigma \subset \mathbb{R}^3$ be a complete embedded minimal surface, meaning a smooth embedded surface that is complete for the Riemannian metric induced from Euclidean space and whose mean curvature vector is identically zero. Prove or disprove that Σ is unbounded in $\mathbb{R}^3$. More ambitiously, prove or disprove that if Σ is non-flat (equivalently, not a plane), then for every line $L \subset \mathbb{R}^3$, the orthogonal projection of Σ onto L is an unbounded subset of L .

3.33.2. Historical background, literature, and current progress

3.33.3. Verdict: partially resolved

3.33.3.1. Precise problem

The problem asks whether every complete embedded minimal surface Σ in $\mathbb{R}^3$ is unbounded, and whether the stronger Calabi-Yau projection statement holds: if Σ is not a plane, then for every line L in $\mathbb{R}^3$, the orthogonal projection of Σ to L is unbounded.

Equivalently, the stronger part asks for a halfspace/slab exclusion with no topology, genus, curvature, injectivity-radius, or end-space assumptions: a nonflat complete embedded minimal surface should not be contained in any halfspace perpendicular to a chosen line.

3.33.3.2. Status

This is not fully resolved as stated. The literature proves the statement in important embedded cases, especially finite topology and finite genus with countably many ends. However, the unrestricted embedded case remains open in the infinite-genus or uncountably-many-ends regime. Recent literature explicitly notes that it is still open whether a complete bounded embedded minimal surface exists in $\mathbb{R}^3$; such a surface, if it exists, would necessarily lie outside the finite-topology and finite-genus/countable-end cases.

The correct verdict for this problem is therefore *partially resolved*.

3.33.3.3. Progress and key literature

1. Original source

- Title: “Problems in Differential Geometry”
- Author: E. Calabi
- Venue: Proceedings of the United States-Japan Seminar in Differential Geometry, Kyoto, Japan, 1965, ed. S. Kobayashi and J. Eells, Jr.; Nippon Hyoronsha, Tokyo
- Year: 1966
- DOI/arXiv/URL: no DOI found; cited in Colding-Minicozzi as p. 170
- Relevance: This is the source of the Calabi conjectures that a complete minimal hypersurface in Euclidean space is unbounded, and that a complete nonflat minimal hypersurface has unbounded projection in every relevant linear direction.

2. Immersed counterexamples, not embedded

- Title: “A complete minimal surface in R^3 between two parallel planes”
- Authors: Luquesio P. de M. Jorge, Frederico Xavier
- Venue: Annals of Mathematics (2), 112(1), 203-206
- Year: 1980
- DOI/URL: <https://doi.org/10.2307/1971325> ; Annals page <https://annals.math.princeton.edu/1980/112-1/p08>
- Relevance: Gives a complete nonflat minimal immersion contained between two parallel planes, so the projection conjecture is false for immersed surfaces. It does not resolve the problem statement because the statement requires embeddedness.
- Title: “Hadamard’s and Calabi-Yau’s conjectures on negatively curved and minimal surfaces”
- Author: Nikolai Nadirashvili
- Venue: Inventiones Mathematicae, 126, 457-465
- Year: 1996
- DOI/URL: <https://doi.org/10.1007/s002220050106>
- Relevance: Constructs a complete minimal immersion in the unit ball, disproving the unboundedness conjecture for immersed surfaces. It does not resolve the embedded problem.

3. Proper surfaces imply the projection statement

- Title: “The strong halfspace theorem for minimal surfaces”
- Authors: David Hoffman, William H. Meeks III
- Venue: Inventiones Mathematicae, 101, 373-377
- Year: 1990
- DOI/URL: <https://doi.org/10.1007/BF01231506>
- Relevance: A connected proper minimal surface in R^3 contained in a halfspace must be a plane. Therefore, once properness is known for a class of complete embedded minimal surfaces, the stated stronger projection claim follows for that class: bounded projection to a line would put the surface in a halfspace perpendicular to that line.

4. Finite topology: resolved

- Title: “The Calabi-Yau conjectures for embedded surfaces”
- Authors: Tobias H. Colding, William P. Minicozzi II
- Venue: *Annals of Mathematics* (2), 167(1), 211-243
- Year: 2008
- DOI/arXiv/URL: <https://doi.org/10.4007/annals.2008.167.211> ; arXiv:math/0404197
- Key result: Corollary 0.12 says the plane is the only complete embedded minimal surface with finite topology in a halfspace of R^3 ; Corollary 0.13 says a complete embedded minimal surface with finite topology in R^3 is proper.
- Relevance: This proves both parts of the problem for complete embedded minimal surfaces of finite topology, but it has an extra finite-topology hypothesis and therefore does not cover the full statement.

5. Positive injectivity radius: resolved

- Title: “The minimal lamination closure theorem”
- Authors: William H. Meeks III, Harold Rosenberg
- Venue: *Duke Mathematical Journal*, 133(3), 467-497
- Year: 2006
- DOI/URL: <https://doi.org/10.1215/S0012-7094-06-13332-X>
- Relevance: The closure theorem implies properness in R^3 for complete embedded minimal surfaces with positive injectivity radius. This is another major partial result, but the statement does not assume positive injectivity radius.

6. Finite genus with countably many ends: resolved

- Title: “The embedded Calabi-Yau conjecture for finite genus”
- Authors: William H. Meeks III, Joaquin Perez, Antonio Ros
- Venue: *Duke Mathematical Journal*, 170(13), 2891-2956
- Year: 2021
- DOI/arXiv/URL: <https://doi.org/10.1215/00127094-2020-0087> ; arXiv:1806.03104
- Key result: For a complete embedded minimal surface in R^3 of finite genus, properness is obtained under countability of the end space; the arXiv abstract states the finite-genus equivalences involving properness, at most two limit ends, and countability conditions.
- Relevance: This extends the finite-topology progress substantially. It still has extra hypotheses and does not prove the unrestricted statement.

7. Recent survey/current status

- Title: “Complete minimal surfaces with Cantor ends in minimally convex domains”
- Author: Antonio Alarcon
- Venue: arXiv preprint
- Year: 2024
- DOI/arXiv/URL: arXiv:2410.13687, <https://arxiv.org/abs/2410.13687>
- Relevance: This recent survey/preprint explicitly records the current boundary: finite topology and finite genus with countably many ends are covered by Colding-Minicozzi and Meeks-Perez-Ros, while it remains open whether there exists a complete bounded embedded minimal surface in R^3 of infinite genus or with uncount-

ably many ends.

- Title: “Minimal Surfaces of Finite Genus: Classification, Dynamics and Laminations”
- Author: Joaquín Pérez
- Venue: arXiv/preprint survey
- Year: 2025
- DOI/arXiv/URL: arXiv:2510.12319, <https://arxiv.org/html/2510.12319v1>
- Relevance: Surveys the finite-genus program and describes the embedded Calabi-Yau problem as resolved for finite genus with countably many ends, while emphasizing remaining properness questions beyond the known hypotheses.

3.33.3.4. What remains unresolved

No citable theorem found proves the full unrestricted statement for every complete embedded minimal surface in $\mathbb{R}^3$. The unresolved cases are complete embedded minimal surfaces without finite topology, without finite genus/countable-end hypotheses, and without positive injectivity-radius or curvature hypotheses. In particular, the possible infinite-genus or uncountably-many-ends cases are not covered by the resolving theorems above.

3.33.4. Remaining open problems recorded by the database

3.33.4.1. Open problem 1

Prove that every complete embedded minimal surface in $\mathbb{R}^3$ of finite genus is properly embedded in $\mathbb{R}^3$.

Reference supplied by the database. William H. Meeks III, Joaquín Pérez, and Antonio Ros, “The embedded Calabi–Yau conjecture for finite genus,” *Duke Mathematical Journal* 170(13) (2021), 2891–2956, Conjecture 1.1. DOI: <https://doi.org/10.1215/00127094-2020-0087>; arXiv: [1806.03104](https://arxiv.org/abs/1806.03104). Its still-open status and the best known countable-ends case are recorded in Joaquín Pérez, “Minimal Surfaces of Finite Genus: Classification, Dynamics and Laminations,” 2025, Section 6 and Theorem 6.1, arXiv: [2510.12319](https://arxiv.org/abs/2510.12319).

3.33.4.2. Open problem 2

Let M be an orientable open surface every end of which has infinite genus. Prove that M admits a complete proper minimal embedding into every smooth bounded domain $D \subset \mathbb{R}^3$.

Reference supplied by the database. Leonor Ferrer, Francisco Martín, and William H. Meeks III, “Existence of proper minimal surfaces of arbitrary topological type,” *Advances in*

Mathematics 231(1) (2012), 378–413, Embedded Calabi–Yau Conjecture 1(1) (attributed there to Martín, Meeks, Nadirashvili, Pérez, and Ros). DOI: <https://doi.org/10.1016/j.aim.2012.05.007>; arXiv: 0903.4194. The underlying existence of any nonproper complete embedded minimal surface in $\mathbb{R}^3$ is still listed as open in Joaquín Pérez, “Minimal Surfaces of Finite Genus: Classification, Dynamics and Laminations,” 2025, Section 7, arXiv: 2510.12319.

3.33.5. Primary-source pointer

The originating ICM record is *Evolution of form and shape* (2022). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-560-5-volume-2.pdf#page=267>.

4. Symplectic and contact geometry

The symplectic and contact records are strongly shaped by Floer theory, Gromov–Witten theory, mirror symmetry, Hamiltonian dynamics, and contact topology. Many statements began as broad programs in ICM lectures rather than single yes–no conjectures. Consequently, the literature often proves one direction, a monotone or exact case, or a categorical enhancement while leaving foundational and general-case comparisons unresolved.

Coverage: 31 website records; **source years:** 1994–2022; **separately listed descendant open problems:** 56.

4.1. 147. Homological algebra of mirror symmetry [1994-ag-53]

Record: 1994–ag–53

Subfield: Symplectic and contact geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1994: Homological algebra of mirror symmetry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.1/ICM1994.1.ocr.pdf#page=192>

4.1.1. Problem statement

Let (V, ω) be a closed symplectic manifold. For $\beta \in H_2(V, \mathbf{Z})$ and $n \geq 3$, let

$$I_{0,n;\beta} \in H_*\left(\overline{M}_{0,n}(\mathbf{C}) \times V^n; \mathbf{Q}\right)$$

denote the genus-zero Gromov–Witten homology class, where $\overline{M}_{0,n}$ is the Deligne–Mumford compactification of the moduli stack of stable genus-zero curves with n marked points. Let

$$\mathcal{H} := \bigoplus_k H^k(V, \mathbf{C})$$

be the total cohomology of V , regarded as a $\mathbf{Z}$ -graded super vector space. For an even element $\gamma \in \mathcal{H} \otimes \Lambda$, where Λ is an auxiliary supercommutative algebra, define the genus-zero Gromov–Witten pre-potential

$$\Phi(\gamma) := \sum_{\beta \in H_2(V, \mathbf{Z})} e^{-\int_{\beta} \omega} \sum_{n \geq 3} \frac{1}{n!} \int_{I_{0,n;\beta}} 1_{\overline{M}_{0,n}} \otimes \gamma \otimes \cdots \otimes \gamma,$$

where $1_{\overline{M}_{0,n}}$ is the identity cohomology class on $\overline{M}_{0,n}$.

Prove that if the symplectic cohomology class $[\omega] \in H^2(V, \mathbf{R})$ is sufficiently positive, in the sense that its pairings $\int_{\beta} \omega$ with the pseudo-holomorphic curve classes counted by the Gromov-Witten classes are taken large enough for the exponential weights $e^{-\int_{\beta} \omega}$ to decay, then the series defining Φ is absolutely convergent for γ in some neighborhood $\mathcal{U}$ of 0 in $\mathcal{H}$.

4.1.2. Historical background, literature, and current progress

4.1.3. Verdict: partially resolved

4.1.3.1. Precise problem

The problem asks for Kontsevich's convergence conjecture for the genus-zero Gromov-Witten prepotential: for an arbitrary closed symplectic manifold (V, ω) , does making the symplectic class sufficiently positive make

$$\sum_{\beta} \exp(-\int_{\beta} \omega) \sum_{n \geq 3} (1/n!) < I_{0,n;\beta}, 1 \times \gamma^n >$$

absolutely convergent for γ in a neighborhood of zero in total cohomology?

This is a quantitative growth problem for genus-zero Gromov-Witten invariants as the curve class/energy increases. The full statement is not just formal GW theory over a Novikov ring; it asks for honest analytic convergence after choosing sufficiently decaying symplectic weights.

4.1.3.2. Original source

- **Title:** Homological Algebra of Mirror Symmetry
- **Author:** Maxim Kontsevich
- **Venue/year:** Proceedings of the International Congress of Mathematicians, Zurich 1994, Vol. I, Birkhauser, 1995, pp. 120-139
- **DOI/arXiv/URL:** DOI https://doi.org/10.1007/978-3-0348-9078-6_11; arXiv alg-geom/9411018; <https://arxiv.org/abs/alg-geom/9411018>
- **Relevance:** This is the source of the exact conjecture. Kontsevich states that the series Φ should be absolutely convergent near zero when $[\omega]$ is sufficiently positive, and immediately notes that without this conjecture one can instead work formally over a Novikov-type coefficient ring.

4.1.3.3. Literature giving partial progress

- **Title:** Convergence of quantum cohomology by quantum Lefschetz
- **Author:** Hiroshi Iritani
- **Venue/year:** Journal fuer die reine und angewandte Mathematik 610 (2007), 29-69
- **DOI/arXiv/URL:** DOI <https://doi.org/10.1515/CRELLE.2007.067>; arXiv math/0506236; <https://arxiv.org/abs/math/0506236>
- **Progress:** Proves convergence of twisted genus-zero theory under the assumption that the original genus-zero theory converges; as an application it proves convergence/semisimplicity results for projective toric manifolds. This is significant but

conditional or restricted to special algebraic targets, so it does not cover arbitrary closed symplectic manifolds.

- **Title:** On the Convergence of Gromov-Witten Potentials and Givental's Formula
- **Authors:** Tom Coates and Hiroshi Iritani
- **Venue/year:** Michigan Mathematical Journal 64 no. 3 (2015), 587-631
- **DOI/arXiv/URL:** DOI <https://doi.org/10.1307/mmj/1441116660>; arXiv 1203.4193; <https://arxiv.org/abs/1203.4193>
- **Progress:** Theorem 1.1 proves NF-convergence of the total descendant potential for smooth projective varieties satisfying Formal Semisimplicity, Genus-Zero Convergence, and Analytic Semisimplicity. Corollary 1.2 gives compact toric varieties and complete flag varieties; Corollary 1.3 gives certain total spaces of negative line bundles over compact toric varieties. This paper explicitly treats genus-zero convergence as an assumption in the main theorem and says convergence of quantum cohomology is not known in general, so it cannot resolve Kontsevich's full conjecture.
- **Title:** The genus 0 Gromov-Witten invariants of projective complete intersections
- **Author:** Aleksey Zinger
- **Venue/year:** Geometry & Topology 18 no. 2 (2014), 1035-1114
- **DOI/arXiv/URL:** DOI <https://doi.org/10.2140/gt.2014.18.1035>; arXiv 1106.1633; <https://arxiv.org/abs/1106.1633>
- **Progress:** Establishes mirror-formula structure and growth bounds for genus-zero Gromov-Witten invariants of projective complete intersections in the stated setting. These bounds imply positive-radius convergence for corresponding generating functions in these special cases, but only for projective complete intersections and specified insertions/classes, not the full closed symplectic case.
- **Title:** Some conjectures on the asymptotic behavior of Gromov-Witten invariants
- **Author:** Aleksey Zinger
- **Venue/year:** Handbook for Mirror Symmetry of Calabi-Yau and Fano Manifolds, Advanced Lectures in Mathematics 47, Higher Education Press/International Press, 2019, pp. 523-550; arXiv version 2016/2017
- **DOI/arXiv/URL:** arXiv 1610.02971; <https://arxiv.org/abs/1610.02971>
- **Progress:** Surveys and formulates energy-growth conjectures for GW invariants. It states that these asymptotic/energy-bound problems are outside current techniques in full generality and explains their connection to positive radii of convergence of GW generating functions. This is evidence that the general convergence problem remains unresolved.
- **Title:** Energy Bounds and Vanishing Results for the Gromov-Witten Invariants of the Projective Space
- **Author:** Aleksey Zinger
- **Venue/year:** Journal of Geometry and Physics 145 (2019), Article 103479
- **DOI/arXiv/URL:** DOI <https://doi.org/10.1016/j.geomphys.2019.103479>; arXiv 1801.07242; <https://arxiv.org/abs/1801.07242>
- **Progress:** Proves uniform energy bounds and vanishing results for arbitrary-genus GW

invariants of projective space, supporting the convergence philosophy in a very special family. It does not address arbitrary closed symplectic manifolds.

4.1.3.4. Why the full problem is not resolved

The exact problem is about every closed symplectic manifold after making $[\omega]$ sufficiently positive. The strongest convergence theorems found either:

- assume the target is a smooth projective variety and impose analytic/genus-zero convergence and semisimplicity hypotheses,
- prove convergence for special classes such as compact toric varieties, complete flag varieties, projective spaces, or projective complete intersections,
- or formulate the required energy-growth estimates as conjectures.

None of the citable literature found proves an unconditional analytic convergence theorem for the genus-zero Gromov-Witten prepotential for all closed symplectic manifolds. Therefore the correct status is **partially resolved**, not fully resolved.

4.1.4. Remaining open problems recorded by the database

4.1.4.1. Open problem 1

Let X be a smooth complex projective variety. Prove that its genus-zero primary Gromov–Witten potential has a nonempty domain of absolute convergence after the Novikov variables are specialized sufficiently far inside the ample cone.

Reference supplied by the database. David A. Cox and Sheldon Katz, *Mirror Symmetry and Algebraic Geometry*, Mathematical Surveys and Monographs **68**, American Mathematical Society, Providence, RI, 1999, Conjecture 8.1.2, DOI: <https://doi.org/10.1090/surv/068>. See also Maxim Kontsevich and Yuri Manin, “Gromov-Witten Classes, Quantum Cohomology, and Enumerative Geometry,” *Communications in Mathematical Physics* **164** (1994), no. 3, 525–562, §4.3, DOI: <https://doi.org/10.1007/BF02101490>. Its still-open status is explicitly confirmed in Mark McLean, “Bounds for Genus Zero Gromov-Witten Invariants,” 2026, introduction following Theorem 1.1, arXiv: [2606.16997](https://arxiv.org/abs/2606.16997).

4.1.4.2. Open problem 2

Let (X, ω) be a compact Calabi–Yau threefold. Prove that there is a constant $C_X > 0$ such that every genus-zero, unmarked Gromov–Witten invariant satisfies

$$|\langle \rangle_{0,\beta}^X| \leq C_X^{\langle \omega, \beta \rangle} \quad \text{for all } \beta \in H_2(X; \mathbf{Z}).$$

Reference supplied by the database. Aleksey Zinger, “The Genus 0 Gromov–Witten Invariants of Projective Complete Intersections,” *Geometry & Topology* **18** (2014), no. 2, 1035–1114, Conjecture 1 and the Calabi–Yau threefold specialization discussed immediately afterward,

DOI: <https://doi.org/10.2140/gt.2014.18.1035>, arXiv: [1106.1633](https://arxiv.org/abs/1106.1633). Re-stated as Conjecture 2.5 in Aleksey Zinger, “Some Conjectures on the Asymptotic Behavior of Gromov-Witten Invariants,” in *Handbook for Mirror Symmetry of Calabi-Yau and Fano Manifolds*, Advanced Lectures in Mathematics **47**, Higher Education Press and International Press, 2019, pp. 523–550, arXiv: [1610.02971](https://arxiv.org/abs/1610.02971). The exponential bound remains conjectural in Mark McLean, “Bounds for Genus Zero Gromov-Witten Invariants,” 2026, arXiv: [2606.16997](https://arxiv.org/abs/2606.16997).

4.1.5. Primary-source pointer

The originating ICM record is *Homological algebra of mirror symmetry* (1994). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.1/ICM1994.1.ocr.pdf#page=192>.

4.2. 148. Homological algebra of mirror symmetry [1994-ag-54]

Record: 1994-ag-54

Subfield: Symplectic and contact geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1994: Homological algebra of mirror symmetry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.1/ICM1994.1.occr.pdf#page=192>

4.2.1. Problem statement

Let (V, ω) be a closed $2n$ -dimensional symplectic manifold with $c_1(T_V) = 0$, and let W be a mirror-dual n -dimensional complex algebraic manifold. Let

$$LV = \{(x, L) : x \in V, L \subset T_x V \text{ is Lagrangian}\},$$

and fix the $\mathbb{Z}$ -covering $\overline{LV} \rightarrow LV$ that induces the universal cover on each Lagrangian-Grassmannian fiber. Let $F(V)$ denote Fukaya's A_∞ -category whose basic objects are Lagrangian submanifolds $\mathcal{L} \subset V$ equipped with a lift $\mathcal{L} \rightarrow \overline{LV}$ of the natural map $\mathcal{L} \rightarrow LV$; for transversely intersecting objects, the morphism space is the complex vector space generated by their intersection points, graded by the Maslov index, with differential and higher compositions defined by signed counts of pseudo-holomorphic discs weighted by $\exp(-\int \phi^* \omega)$. Allow, if necessary, a suitably enlarged Fukaya A_∞ -category, for example one whose objects include pairs $(\mathcal{L}, \mathcal{E})$ with $\mathcal{E}$ a unitary local system on $\mathcal{L}$. For an A_∞ -category C , let $\mathcal{D}^b(C)$ be the triangulated category $H(\text{Tw}(C))$ obtained from one-sided twisted complexes over C , and let $\text{Coh}(W)$ be the abelian category of coherent sheaves on W .

Prove or disprove that $\mathcal{D}^b(F(V))$, or the corresponding derived category of a suitable enlargement of $F(V)$, embeds as a full triangulated subcategory of $\mathcal{D}^b(\text{Coh}(W))$. In the strongest expected form, prove or disprove that this derived Fukaya category is equivalent to the bounded derived category of coherent sheaves on W .

4.2.2. Historical background, literature, and current progress

4.2.3. Verdict: partially resolved

4.2.3.1. Precise problem checked

The problem asks for Kontsevich's homological mirror symmetry conjecture in the original compact Calabi-Yau setting: for every closed symplectic manifold (V, ω) with $c_1(T_V) = 0$ and a mirror-dual complex algebraic manifold W , the derived Fukaya category, possibly after the standard enlargement by local systems/bounding data, should fully embed in $\mathcal{D}^b(\text{Coh}(W))$; in the strongest expected form it should be equivalent to $\mathcal{D}^b(\text{Coh}(W))$.

This is a universal statement over all such mirror pairs. A theorem for elliptic curves, K3 surfaces, torus fibrations, hypersurfaces, or Batyrev/Greene-Plesser families is therefore only par-

tial progress unless it covers all closed symplectic Calabi-Yau manifolds and all mirror-dual algebraic manifolds.

4.2.3.2. Status summary

The problem is not fully resolved. Live arXiv/web searches found many citable proofs for important special cases and broad families, but no theorem proving the statement for arbitrary closed symplectic Calabi-Yau manifolds with $c_1=0$ and arbitrary mirrors. Recent broad-family papers themselves describe the compact Calabi-Yau HMS theorem as known only in a limited list of cases before their new classes.

The correct status is therefore **partially resolved**, not fully open and not fully resolved.

4.2.3.3. Key literature and progress

1. Homological Algebra of Mirror Symmetry

- Authors: Maxim Kontsevich
- Venue/year: Proceedings of the International Congress of Mathematicians, Zurich 1994, Vol. I, Birkhauser, 1995, pp. 120-139
- arXiv/DOI/URL: arXiv:alg-geom/9411018, <https://arxiv.org/abs/alg-geom/9411018>, DOI <https://doi.org/10.48550/arXiv.alg-geom/9411018>
- Role: Original source of the homological mirror symmetry formulation. It formulates the conjecture; it does not prove the universal statement.

2. Categorical Mirror Symmetry: The Elliptic Curve

- Authors: Alexander Polishchuk, Eric Zaslow
- Venue/year: Advances in Theoretical and Mathematical Physics 2 (1998), 443-470
- arXiv/DOI/URL: arXiv:math/9801119, <https://arxiv.org/abs/math/9801119>, DOI <https://doi.org/10.4310/ATMP.1998.v2.n2.a9>
- Result/progress: Proves the HMS equivalence for elliptic curves, matching a suitable Fukaya category of the mirror two-torus with the derived category of coherent sheaves on the elliptic curve. This resolves the one-dimensional elliptic-curve case only.

3. Homological mirror symmetry for the quintic 3-fold

- Authors: Yuichi Nohara, Kazushi Ueda
- Venue/year: Geometry & Topology 16 (2012), 1967-2001
- arXiv/DOI/URL: arXiv:1103.4956, <https://arxiv.org/abs/1103.4956>, DOI <https://doi.org/10.2140/gt.2012.16.1967>
- Result/progress: Proves HMS for the quintic Calabi-Yau threefold. This is a major compact Calabi-Yau threefold example, but only one family/example, not the full conjecture.

4. Homological Mirror Symmetry for the Quartic Surface

- Authors: Paul Seidel
- Venue/year: Memoirs of the American Mathematical Society 236, no. 1116 (2015)
- arXiv/DOI/URL: arXiv:math/0310414, <https://arxiv.org/abs/math/0310414>, DOI <https://doi.org/10.1090/memo/1116>

- Result/progress: Proves Kontsevich’s mirror conjecture for the quartic surface on the symplectic side. This is a K3/quartic-surface case, not a general theorem for all Calabi-Yau symplectic manifolds.

5. Homological Mirror Symmetry for Calabi-Yau hypersurfaces in projective space

- Author: Nicholas Sheridan
- Venue/year: *Inventiones Mathematicae* 199 (2015), no. 1, 1-186
- arXiv/DOI/URL: arXiv:1111.0632, <https://arxiv.org/abs/1111.0632>, DOI <https://doi.org/10.1007/s00222-014-0507-2>
- Result/progress: Proves HMS for smooth d -dimensional Calabi-Yau hypersurfaces in projective space for $d > 2$ (including the quintic threefold). This is a broad projective-hypersurface family, but it is still far narrower than arbitrary closed symplectic Calabi-Yau manifolds.

6. Homological mirror symmetry without correction

- Author: Mohammed Abouzaid
- Venue/year: *Journal of the American Mathematical Society* 34 (2021), no. 4, 1059-1173
- arXiv/DOI/URL: arXiv:1703.07898, <https://arxiv.org/abs/1703.07898>, DOI <https://doi.org/10.1090/jams/973>
- Result/progress: For closed symplectic manifolds equipped with a Lagrangian torus fibration over a base $\mathbb{Q}$, proves a fully faithful embedding of the Fukaya category of tautologically unobstructed graded Lagrangians into the derived category of twisted coherent sheaves on a rigid analytic mirror, under the technical hypothesis $\pi_2(Q) = 0$. This resembles the embedding form of the problem, but has strong extra hypotheses and does not cover arbitrary $c_1=0$ symplectic manifolds or all Fukaya objects.

7. Homological mirror symmetry for generalized Greene-Plesser mirrors

- Authors: Nick Sheridan, Ivan Smith
- Venue/year: *Inventiones Mathematicae* 224 (2021), 627-682
- arXiv/DOI/URL: arXiv:1709.08937, <https://arxiv.org/abs/1709.08937>, DOI <https://doi.org/10.1007/s00222-020-01018-w>
- Result/progress: Proves HMS for all Greene-Plesser mirror pairs and certain Batyrev-Borisov mirror pairs/Calabi-Yau complete intersections. These are important compact Calabi-Yau mirror families, but still special classes.

8. Homological mirror symmetry for Batyrev mirror pairs

- Authors: Sheel Ganatra, Andrew Hanlon, Jeff Hicks, Daniel Pomerleano, Nick Sheridan
- Venue/year: arXiv preprint, 2024, revised 2024
- arXiv/DOI/URL: arXiv:2406.05272, <https://arxiv.org/abs/2406.05272>, DOI <https://doi.org/10.48550/arXiv.2406.05272>
- Result/progress: Proves HMS for a large new class of Batyrev mirror pairs of Calabi-Yau hypersurfaces in toric varieties. The theorem has extra hypotheses, including dimension at least 3, characteristic conditions, smoothness/MPCS/connectedness conditions, and for the equivalence form assumes standard structural hypotheses on the relative Fukaya category. This is substantial progress but not a theorem for all closed symplectic Calabi-Yau manifolds.

9. Homological mirror symmetry for projective K3 surfaces

- Authors: Paul Hacking, Ailsa Keating
- Venue/year: arXiv preprint, 2025
- arXiv/DOI/URL: arXiv:2503.05680, <https://arxiv.org/abs/2503.05680>, DOI <https://doi.org/10.48550/arXiv.2503.05680>
- Result/progress: Proves HMS for projective K3 surfaces with integral Kahler class, with mirror a Picard-rank 19 K3 surface over $\mathbb{C}((q))$. This is a major K3 result, but it covers complex dimension two and integral Kahler classes, not the full arbitrary-dimensional statement.

10. A Gentle Introduction to Homological Mirror Symmetry

- Author: Raf Bocklandt
- Venue/year: London Mathematical Society Student Texts 99, Cambridge University Press, 2021
- DOI/URL: <https://doi.org/10.1017/9781108692458>
- Role: Survey/textbook background. It confirms HMS is a broad active program with many examples and variants, but it is not a resolving theorem for the universal compact Calabi-Yau statement.

4.2.3.4. Why no cited result fully resolves the problem statement

The problem is not restricted to elliptic curves, abelian varieties, K3 surfaces, hypersurfaces in projective or toric varieties, Batyrev/Greene-Plesser mirrors, or manifolds admitting a suitable torus fibration. The strongest recent broad-family results still impose geometric hypotheses and, in some cases, structural assumptions on the Fukaya category. Therefore their hypotheses do not imply the full statement for every closed symplectic $2n$ -manifold with $c_1(TV) = 0$ and every mirror-dual algebraic W .

What remains unresolved is a general construction/proof of the required full embedding or equivalence for arbitrary compact Calabi-Yau mirror pairs, including arbitrary closed symplectic Calabi-Yau manifolds, the correct universally enlarged Fukaya category, and the verification that it matches $D^b(\text{Coh}(W))$ without the special geometric hypotheses used in known cases.

4.2.4. Remaining open problems recorded by the database

4.2.4.1. Open problem 1

Let

$$V_\lambda = Z(Q_{1,\lambda}, Q_{2,\lambda}) \subset \mathbf{P}^5$$

be a smooth Calabi-Yau threefold given by the intersection of the two cubics

$$Q_{1,\lambda} = x_0^3 + x_1^3 + x_2^3 - 3\lambda x_3 x_4 x_5, \quad Q_{2,\lambda} = x_3^3 + x_4^3 + x_5^3 - 3\lambda x_0 x_1 x_2.$$

Prove homological mirror symmetry for V_λ : its split-closed derived Fukaya category is equivalent to the bounded derived category of coherent sheaves on either the Libgober-Teitelbaum mirror stack $\mathcal{U}_{LT,\lambda}$ or the derived-equivalent Batyrev-Borisov mirror stack $\mathcal{Z}_\lambda$.

Reference supplied by the database. Aimeric Malter, “A Derived Equivalence of the Libgober–Teitelbaum and the Batyrev–Borisov Mirror Constructions,” *International Mathematics Research Notices* **2024** (2024), no. 3, 2099–2137, introduction after Theorem 1.1 (explicit expectation that both mirror derived categories are equivalent to the Fukaya category of the intersection of two cubics), DOI: <https://doi.org/10.1093/imrn/rnad081>.

4.2.5. Primary-source pointer

The originating ICM record is *Homological algebra of mirror symmetry* (1994). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.1/ICM1994.1.ocr.pdf#page=192>.

4.3. 149. Fibrations in Symplectic Topology [1998-geom-62]

Record: 1998-geom-62

Subfield: Symplectic and contact geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1998: Fibrations in Symplectic Topology

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.1/ICM1998.1.ocr.pdf#page=341>

4.3.1. Problem statement

Determine whether there is an intrinsic C^0 notion of closeness for symplectic structures ω and ω' , at least in dimension 4, not merely ordinary C^0 -closeness of cohomologous symplectic forms, such that sufficiently close symplectic structures are isotopic through symplectic structures.

In particular, determine whether the following type of intrinsic C^0 notion, or any analogous one, yields such a stability result: declare two symplectic structures ω and ω' to be ε -close on a compact domain K if

$$|w_G(U, \omega) - w_G(U, \omega')| \leq \varepsilon$$

for every open subset $U \subset K$, where $w_G(U, \omega)$ is the Gromov width of U with respect to ω , namely the supremum of the numbers πr^2 such that the standard ball $B^{2n}(r)$ symplectically embeds in (U, ω) .

4.3.2. Historical background, literature, and current progress

4.3.3. Status: partially resolved

4.3.3.1. Precise problem

The problem asks for a symplectic analogue of Colin's C^0 -stability theorem for contact structures: define an intrinsic C^0 notion of closeness for symplectic structures, preferably not just ordinary C^0 closeness of cohomologous closed 2-forms, such that sufficiently close symplectic structures on a fixed manifold are isotopic through symplectic structures. The question is asked at least in dimension 4. A proposed candidate is closeness measured by the Gromov widths of all open subsets in a compact domain:

$$|w_G(U, \omega) - w_G(U, \omega')| \leq \varepsilon$$

for every open $U \subset K$.

4.3.3.2. Verdict

I found no published or otherwise citable result that fully resolves this exact question. In particular, I found no theorem that defines an intrinsic C^0 topology on the space of symplectic

structures and proves the requested local isotopy stability, even in dimension 4, and no theorem proving that the all-open-subsets Gromov-width candidate implies such stability.

The problem is not completely untouched: there is substantial partial and adjacent progress in C^0 symplectic topology, capacity-based characterizations of symplectic maps, ordinary C^0 perturbations of symplectic forms in non-squeezing problems, and uniqueness/classification questions for symplectic forms. These results do not imply the full statement because their hypotheses concern maps rather than the topology on symplectic structures, use extrinsic C^0 norms on forms, add special deformation/nondegeneracy assumptions, or prove rigidity of capacities rather than isotopy of nearby structures.

4.3.3.3. Literature and progress

1. Dusa McDuff, “Fibrations in Symplectic Topology”, Documenta Mathematica, Extra Volume ICM 1998, Vol. I, 339-357, 1998. URL: <https://ems.press/content/book-chapter-files/27111>.

This is the original source of the question. In Question 3.2 McDuff asks for a symplectic analogue of Colin’s C^0 -stability result and explicitly says that the appropriate notion of C^0 -closeness is unclear. She also suggests the Gromov-width condition over all open subsets and states that the consequences of such a definition are not known.

2. Vincent Colin, “Stabilité topologique des structures de contact en dimension 3”, Duke Mathematical Journal 99(2), 329-351, 1999. DOI: 10.1215/S0012-7094-99-09912-X. URL: <https://projecteuclid.org/journals/duke-mathematical-journal/volume-99/issue-2/Stabilit%C3%A9-topologique-des-structures-de-contact-en-dimension-3/10.1215/S0012-7094-99-09912-X.full>.

This is the contact-stability theorem used as the model for McDuff’s question. It does not address symplectic structures.

3. Yong-Geun Oh and Stefan Muller, “The group of Hamiltonian homeomorphisms and C^0 -symplectic topology”, Journal of Symplectic Geometry 5(2), 167-219, 2007. arXiv:math/0402210. URL: <https://projecteuclid.org/journals/journal-of-symplectic-geometry/volume-5/issue-2/The-group-of-Hamiltonian-homeomorphisms-and-C0-symplectic-topology/jsg/1202004455.pdf>.

This develops C^0 symplectic topology for Hamiltonian homeomorphisms and continuous Hamiltonian flows. It gives a rigorous C^0 theory for maps on a fixed symplectic manifold, but it does not define an intrinsic C^0 topology on symplectic structures themselves or prove isotopy stability of nearby structures.

4. Vincent Humilière, Remi Leclercq, and Sobhan Seyfaddini, “Coisotropic rigidity and C^0 -symplectic geometry”, Duke Mathematical Journal 164(4), 767-799, 2015. DOI: 10.1215/00127094-2881701. arXiv:1305.1287. URL: <https://projecteuclid.org/journals/duke-mathematical-journal/volume-164/issue-4/----Custom-HTML----Coisotropic/10.1215/00127094-2881701.full>.

This proves C^0 -rigidity of coisotropic submanifolds and characteristic foliations under symplectic homeomorphisms, generalizing the Gromov-Eliashberg map-rigidity viewpoint. It is important C^0 rigidity, but its object is symplectic homeomorphisms/submanifolds, not closeness or isotopy of symplectic forms.

5. Lev Buhovsky and Emmanuel Opshtein, “Some quantitative results in C^0 symplectic geometry”, *Inventiones Mathematicae* 205(1), 1-56, 2016. DOI: 10.1007/s00222-015-0626-4. arXiv:1404.0875. URL: <https://hal.science/hal-03432689/>.

The paper studies behavior of symplectic homeomorphisms with respect to numerical invariants such as capacities and proves rigidity/flexibility results for submanifolds. This is relevant to the capacity side of the question, but it does not prove that Gromov-width closeness of symplectic structures forces isotopy.

6. Stefan Muller, “ C^0 -characterization of symplectic and contact embeddings and Lagrangian rigidity”, *International Journal of Mathematics* 30(9), 1950035, 48 pp., 2019. DOI: 10.1142/S0129167X19500356. arXiv:1607.03135. URL: <https://www.worldscientific.com/doi/10.1142/S0129167X19500356>.

This gives C^0 characterizations of symplectic embeddings/diffeomorphisms using shape and Lagrangian rigidity. It supports the idea that intrinsic symplectic data can characterize symplectic maps, but it is not a stability theorem for nearby symplectic structures.

7. Yasha Savelyev, “A remark on deformation of Gromov non-squeezing”, *Differential Geometry and its Applications*, 2025, article 102262. DOI: 10.1016/j.difgeo.2025.102262. arXiv:2302.13985. URL: <https://arxiv.org/abs/2302.13985>.

Savelyev proves that Gromov non-squeezing persists under certain ordinary C^0 perturbations of the symplectic form on the target, including special dimension-4 cases. This is the closest progress I found toward the Gromov-width/deformation direction. It still uses an extrinsic C^0 norm on forms and special non-squeezing hypotheses, and it does not produce an intrinsic topology with local isotopy stability.

8. Spencer Cattalani, “On Gromov Width under C^0 Deformations”, arXiv:2507.11466, 2025. DOI: 10.48550/arXiv.2507.11466. URL: <https://arxiv.org/abs/2507.11466>.

Cattalani constructs a bounded symplectic structure on $S^2 \times \mathbb{R}^4$ with infinite Gromov width, giving a higher-dimensional counterexample to a conjecture of Savelyev, and proves positive results for special classes. This clarifies the limits of ordinary C^0 deformation statements about Gromov width. It does not settle McDuff’s intrinsic C^0 stability problem, and the counterexample is not a theorem about the proposed all-open-subsets width topology on compact domains.

9. Dietmar Salamon, “Uniqueness of Symplectic Structures”, *Acta Mathematica Vietnamica* 38(1), 123-144, 2013. DOI: 10.1007/s40306-012-0004-x. arXiv:1211.2940. URL: <https://arxiv.org/abs/1211.2940>.

This survey records what is known about uniqueness, isotopy, homotopy, and deformation equivalence of symplectic forms in many examples. It gives useful background and many partial uniqueness results, especially in dimension 4, but not an intrinsic C^0 local stability theorem.

10. Shengzhen Ning, “Cohomologous symplectic forms with different Gromov widths”, arXiv:2505.09550, 2025. DOI: 10.48550/arXiv.2505.09550. URL: <https://arxiv.org/abs/2505.09550>.

This shows that in dimensions at least 6 there are closed manifolds with cohomologous symplectic forms of different Gromov widths. It is related to the role of Gromov width in distinguishing symplectic structures, but it is higher-dimensional and does not address the dimension-4 intrinsic C^0 stability question.

4.3.3.4. What remains unresolved

The full problem remains unresolved: find, or rule out, an intrinsic C^0 notion of closeness for symplectic structures, especially in dimension 4, for which sufficiently close symplectic structures are isotopic through symplectic structures. In particular, the literature found does not determine whether the proposed topology using $w_G(U, \omega)$ for all open $U \subset K$ has the desired stability property.

4.3.4. Remaining open problems recorded by the database

4.3.4.1. Open problem 1

Let ω_0 be the product symplectic form on $S^2 \times \mathbb{R}^2$, normalized so that $\int_{S^2} \omega_0 = 1$, and use its standard product metric to define the comass norm. If ω is any symplectic form with $\|\omega - \omega_0\| < \infty$, must its Gromov width satisfy $c_G(\omega) \leq 1 + \|\omega - \omega_0\|$?

Reference supplied by the database. Yasha Savelyev, “A Remark on Deformation of Gromov Non-squeezing,” *Differential Geometry and its Applications* **100** (2025), Article 102262, Conjecture 1 (specialized to the still-unresolved four-dimensional case), DOI: 10.1016/j.difgeo.2025.102262, arXiv:2302.13985, <https://doi.org/10.1016/j.difgeo.2025.102262>. Current status is recorded in Spencer Cattalani, “On Gromov Width under C^0 Deformations,” accepted in *Journal of Symplectic Geometry*, 2026 preprint, Conjecture 1, Theorem 2, and Remark 8 (counterexample in dimensions at least six, while the construction does not settle the four-dimensional case), arXiv:2507.11466, <https://arxiv.org/abs/2507.11466>.

4.3.5. Primary-source pointer

The originating ICM record is *Fibrations in Symplectic Topology* (1998). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.1/ICM1998.1.ocr.pdf#page=341>.

4.4. 150. Some Open Problems in Dynamical Systems [1998-geom-69]

Record: 1998-geom-69

Subfield: Symplectic and contact geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1998: Some Open Problems in Dynamical Systems

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.2/ICM1998.2.ocr.pdf#page=799>

4.4.1. Problem statement

Let (M^{2n}, ω) be a compact C^∞ symplectic manifold with Euler-Poincaré characteristic $\chi(M^{2n}) = 0$, and suppose that M^{2n} does not admit a fibration over S^1 . Prove that every symplectic diffeomorphism $f : (M^{2n}, \omega) \rightarrow (M^{2n}, \omega)$ that is homotopic to the identity map has a fixed point. In fact, prove that the number of fixed points of f is at least

$$\min_X \#\{p \in M^{2n} : X(p) = 0\},$$

where the minimum is taken over all smooth symplectic vector fields X on (M^{2n}, ω) , equivalently over all vector fields X for which $i_X \omega = v$ for some closed 1-form v on M^{2n} .

4.4.2. Historical background, literature, and current progress

4.4.3. Verdict: partially resolved

4.4.3.1. Precise problem

The problem is Michael Herman's Conjecture 5.8 from "Some Open Problems in Dynamical Systems" (Documenta Mathematica, Extra Volume ICM 1998, Vol. II, 797-808). It asks whether a compact smooth symplectic manifold (M^{2n}, ω) with $\chi(M) = 0$ and no fibration over S^1 has the following fixed point property: every symplectic diffeomorphism $f : (M, \omega) \rightarrow (M, \omega)$ homotopic to the identity has a fixed point, and in fact

$$\#\text{Fix}(f) \geq \min_X \#\{p : X(p) = 0\},$$

where X ranges over symplectic vector fields, equivalently $i_X \omega$ ranges over closed 1-forms.

Tischler's theorem explains the fibration condition: a closed manifold carrying a nowhere-zero closed 1-form fibers over S^1 .

4.4.3.2. Current literature status

No published or otherwise citable result I found proves Herman's full statement in this generality. Existing results prove important special cases and weaker lower bounds, mainly for

symplectomorphisms lying in the identity component through symplectic isotopies, usually under nondegeneracy hypotheses, and with lower bounds given by Novikov homology rather than by the actual minimum number of zeros of closed 1-forms. This does not cover all symplectic diffeomorphisms merely homotopic to the identity, and it does not prove the stated zero-count lower bound in all cases.

4.4.3.3. Key sources and what they prove

1. Some Open Problems in Dynamical Systems

Michael R. Herman. Documenta Mathematica, Extra Volume ICM 1998, Vol. II, 797-808, 1998. URL: <https://ems.press/content/book-chapter-files/27206>

This is the original source. Section 5.8 states the conjecture exactly: if $\chi(M^{2n}) = 0$ and M^{2n} does not fiber over S^1 , then every symplectic diffeomorphism homotopic to the identity has a fixed point, with the stronger lower bound by the minimal number of zeros of a symplectic vector field. Herman also notes that the conjecture is easy for f C^1 -near the identity.

2. On fibering certain foliated manifolds over S^1

David Tischler. Topology 9, 153-154, 1970. DOI: 10.1016/0040-9383(70)90037-6.

Gives the standard implication used in Herman's formulation: a nowhere-zero closed 1-form on a compact manifold yields a fibration over S^1 . This identifies why the no-fibration hypothesis forces zeros for every closed 1-form, but it is not a fixed-point theorem for symplectomorphisms.

3. Symplectic fixed points, the Calabi invariant and Novikov homology

Hông Vân Lê and Kaoru Ono. Topology 34(1), 155-176, 1995. DOI: 10.1016/0040-9383(94)E0015-C. URL: https://archive.mpim-bonn.mpg.de/4017/1/preprint_1993_27.pdf

Proves a Floer-Novikov lower bound for nondegenerate symplectomorphisms isotopic to the identity through symplectomorphisms, under monotonicity-type assumptions. The lower bound is the sum of Novikov Betti numbers associated to the flux/Calabi invariant. This is substantial partial progress, but it has extra hypotheses and does not cover arbitrary homotopic-to-identity symplectomorphisms or Herman's stated minimum-zero bound.

4. Fixed points of non-Hamiltonian symplectomorphisms

Luisa D. Stelling. Journal of Geometric Analysis 11, 693-716, 2001. DOI: 10.1007/BF02930763.

URL: <https://link.springer.com/article/10.1007/BF02930763>

Studies an Arnold-type fixed point problem for non-Hamiltonian symplectic maps and obtains lower bounds for maps with sufficiently small Calabi invariant. This is a small-flux/small-Calabi special case, not the full Herman conjecture.

5. Floer-Novikov cohomology and symplectic fixed points

Kaoru Ono. Journal of Symplectic Geometry 3, 545-563, 2005. DOI: 10.4310/JSG.2005.v3.n4.a3.

Project Euclid URL: <http://projecteuclid.org/euclid.jsg/1154467629>

Extends Floer-Novikov methods and gives a lower bound for nondegenerate time-one maps of symplectic isotopies on compact symplectic manifolds in terms of minimal Novikov numbers. This improves the generality of the symplectic-isotopy case, but the lower bound can be weaker than the minimum number of zeros of closed 1-forms and still does not address all homotopic-to-identity symplectomorphisms.

6. **The Calabi invariant and the least number of periodic solutions of locally Hamiltonian equations**
 Hồng Vân Lê. arXiv:1511.00638, 2015. DOI: 10.48550/arXiv.1511.00638. URL: <https://arxiv.org/abs/1511.00638>
 Proves lower bounds for nondegenerate locally Hamiltonian equations on compact symplectic manifolds in terms of Novikov homology associated to the Calabi invariant, generalizing the homological Arnold conjecture. This is again a Floer-Novikov/Novikov-Betti bound, not Herman's full zero-count statement for all homotopic-to-identity symplectic diffeomorphisms.
7. **Floer-Novikov cohomology and symplectic fixed points, revisited**
 Kaoru Ono and Hồng Vân Lê. Proceedings of the International Geometry Center 13(4), 89-115, 2020. DOI: 10.15673/tmgc.v13i4.1809. URL: <https://journals.ontu.edu.ua/index.php/geometry/article/view/1809>
 States sharper modern versions: for a smooth symplectic isotopy $\{\phi_t\}$ from the identity with nondegenerate ϕ_1 , the number of fixed points is bounded below by Novikov-Betti numbers of the actual flux. The authors explicitly frame this as a result for symplectomorphisms symplectically isotopic to the identity. It is strong partial progress but does not cover maps merely homotopic to the identity.
8. **A Symplectic Fixed Point Theorem and Floer-Novikov Theory in the Presence of an Infinite Fundamental Group**
 Hồng Vân Lê and Kaoru Ono. Vietnam Journal of Mathematics 54, 441-450, 2026; published online 2025. DOI: 10.1007/s10013-025-00772-9. URL: <https://link.springer.com/article/10.1007/s10013-025-00772-9>
 Gives a newer Floer-Novikov existence result for 1-periodic orbits on certain product symplectic manifolds, including situations where a relevant Floer-Novikov homology group vanishes in a degree. This is additional partial progress, but it is explicitly for certain product manifolds and does not resolve Herman's all-manifolds/all-homotopic-symplectomorphisms statement.
9. **Arnold conjecture and Gromov-Witten invariant**
 Kenji Fukaya and Kaoru Ono. Topology 38(5), 933-1048, 1999. DOI: 10.1016/S0040-9383(98)00042-1.
 Together with the next item, this proves the homological Arnold conjecture for Hamiltonian diffeomorphisms on general compact symplectic manifolds. It covers the Hamiltonian subcase of Herman's problem, but Herman's conjecture is about all symplectic diffeomorphisms homotopic to the identity and uses closed 1-forms/symplectic vector fields, so this is only partial progress.
10. **Floer homology and Arnold conjecture**
 Gang Liu and Gang Tian. Journal of Differential Geometry 49(1), 1-74, 1998. DOI: 10.4310/jdg/1214460936. URL: <https://projecteuclid.org/euclid.jdg/1214460936>
 Another proof of the homological Arnold conjecture for Hamiltonian diffeomorphisms. It does not extend to arbitrary non-Hamiltonian symplectic diffeomorphisms homotopic to the identity.
11. **Closed 1-forms with at most one zero**
 Michael Farber and Dirk Schütz. Topology 45(3), 465-473, 2006. DOI: 10.1016/j.top.2005.06.006. URL: <https://maths.dur.ac.uk/users/dirk.schuetz/d10.pdf>
 Shows that every nonzero real cohomology class on a closed connected smooth manifold has a closed 1-form with at most one zero. This clarifies Herman's lower-bound term: when $H^1(M; \mathbb{R}) \neq 0$ and M does not fiber over S^1 , the minimum over nonzero closed

1-forms is often just one. This topological result does not supply the missing symplectic fixed point theorem.

4.4.3.4. What remains unresolved

The unresolved part is the full Herman conjecture: proving, for every compact symplectic (M, ω) with $\chi(M) = 0$ and no fibration over S^1 , that every symplectic diffeomorphism homotopic to the identity has a fixed point, and proving the stated lower bound by the minimum number of zeros of symplectic vector fields.

The main gaps in the existing literature are:

- results generally require f to be the time-one map of a symplectic isotopy from the identity, not merely homotopic to the identity;
- results generally assume nondegenerate fixed points or prove homological/Novikov-Betti lower bounds;
- Novikov-Betti bounds are weaker than the exact minimum-zero bound appearing in the problem statement;
- recent Floer-Novikov refinements still cover special classes or product situations rather than all compact symplectic manifolds in Herman's statement.

Therefore the correct status is **partially resolved**, not fully resolved.

4.4.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

4.4.5. Primary-source pointer

The originating ICM record is *Some Open Problems in Dynamical Systems* (1998). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.2/ICM1998.2.ocr.pdf#page=799>.

4.5. 151. Geometry on the Group of Hamiltonian Diffeomorphisms [1998-geom-73]

Record: 1998-geom-73

Subfield: Symplectic and contact geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1998: Geometry on the Group of Hamiltonian Diffeomorphisms

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.2/ICM1998.2.ocr.pdf#page=403>

4.5.1. Problem statement

For every connected symplectic manifold (M, Ω) without boundary, let $\text{Ham}(M, \Omega)$ be the group of Hamiltonian diffeomorphisms. For a smooth Hamiltonian path $\{f_t\}_{t \in [0,1]}$ from ϕ to ψ , generated by a normalized Hamiltonian $F(x, t)$ (compactly supported in M if M is noncompact, and of zero mean with respect to the symplectic volume if M is closed), set

$$\text{length}\{f_t\} = \int_0^1 \left(\max_M F(\cdot, t) - \min_M F(\cdot, t) \right) dt,$$

and let $\rho(\phi, \psi)$ be the infimum of this length over all such paths. Prove that $(\text{Ham}(M, \Omega), \rho)$ has infinite diameter; equivalently, the distances $\rho(\text{id}, \phi)$ are unbounded as ϕ ranges over $\text{Ham}(M, \Omega)$.

4.5.2. Historical background, literature, and current progress

4.5.3. Verdict: partially resolved

4.5.3.1. Precise problem

The problem asks for the general Hofer diameter conjecture: for every connected symplectic manifold without boundary, with compactly supported Hamiltonians in the noncompact case and normalized Hamiltonians in the closed case, is the Hofer metric on $\text{Ham}(M, \Omega)$ unbounded, equivalently does $\text{Ham}(M, \Omega)$ have infinite Hofer diameter?

This matches Polterovich's ICM formulation:

- Title: Geometry on the Group of Hamiltonian Diffeomorphisms
- Author: Leonid Polterovich
- Venue/year: Documenta Mathematica, Extra Volume ICM 1998, Vol. II, pp. 401-410, 1998
- DOI/arXiv/URL: <https://ems.press/content/book-chapter-files/27169>
- Relevant item: Conjecture 6.A states that the diameter of $\text{Ham}(M, \Omega)$ with respect to Hofer's metric is infinite.
- Status relevance: This is the original all-manifolds problem, not a theorem.

4.5.3.2. Evidence that the full problem is not resolved

Recent literature still states the all-symplectic-manifolds assertion as open in general.

- Title: Hofer geometry of A3-configurations
- Author: Adrian Dawid
- Venue/year: arXiv preprint, 2025 version
- DOI/arXiv/URL: arXiv:2402.16773, <https://arxiv.org/abs/2402.16773>
- Relevant statement: The introduction says that the Hofer diameter conjecture asserts infinite Hofer diameter for all symplectic manifolds and that this question is still open in general.
- Applicability check: The paper proves quasi-flats for Liouville domains with an A3-configuration of exact Lagrangian spheres. This is a substantial partial result, but it has strong Liouville/exact/Lagrangian-configuration hypotheses and does not imply the problem statement for every connected boundaryless symplectic manifold.
- Title: Spectral diameter of negatively monotone manifolds
- Author: Yuhan Sun
- Venue/year: arXiv preprint 2024; listed by the author's page as accepted for Journal of Symplectic Geometry
- DOI/arXiv/URL: arXiv:2410.21416, <https://arxiv.org/abs/2410.21416>
- Relevant statement: The introduction describes the Hofer diameter conjecture as the question whether the Hofer norm is unbounded for any closed symplectic manifold, and proves quasi-isometric embeddings under negative-monotonicity and heavy-Lagrangian hypotheses.
- Applicability check: The theorem assumes a closed negatively monotone symplectic manifold of dimension at least six containing an incompressible heavy Lagrangian. These hypotheses are far narrower than the stated all connected symplectic manifolds.

4.5.3.3. Main partial progress

- Title: Hofer's diameter and Lagrangian intersections
- Author: Leonid Polterovich
- Venue/year: International Mathematics Research Notices, 1998, no. 4, pp. 217-223
- DOI/arXiv/URL: DOI 10.1155/S1073792898000178; arXiv:math/9801054, <https://arxiv.org/abs/math/9801054>
- Result: Proves that the Hamiltonian diffeomorphism group of the 2-sphere has infinite Hofer diameter.
- Applicability check: This resolves the special case $M = S^2$, not arbitrary symplectic manifolds.
- Title: Hofer's L-infinity-geometry: energy and stability of Hamiltonian flows, parts I and II
- Authors: Francois Lalonde and Dusa McDuff

- Venue/year: *Inventiones Mathematicae* 122 (1995), pp. 1-33 and 35-69; erratum *Inventiones Mathematicae* 123 (1996), p. 613
- DOI/arXiv/URL: DOI 10.1007/BF01231437 and 10.1007/BF01231438; arXiv:math/9503227 and arXiv:math/9503228
- Result: Early Hofer-geometry results used in Polterovich's ICM survey to establish infinite diameter for closed oriented surfaces of positive genus and related cases.
- Applicability check: Important surface and stability/geodesic results, but not a theorem for all symplectic manifolds.
- Title: On the action spectrum for closed symplectically aspherical manifolds
- Author: Matthias Schwarz
- Venue/year: *Pacific Journal of Mathematics* 193 (2000), no. 2, pp. 419-461
- DOI/arXiv/URL: <https://msp.org/pjm/2000/193-2/pjm-v193-n2-p10-p.pdf>
- Result: Develops spectral invariants for closed symplectically aspherical manifolds; Section 5.1 gives examples and methods yielding infinite Hofer diameter in symplectically aspherical settings.
- Applicability check: Requires symplectic asphericity or related hypotheses, so it does not cover rational/monotone/non-aspherical closed manifolds in general.
- Title: A comparison of Hofer's metrics on Hamiltonian diffeomorphisms and Lagrangian submanifolds
- Author: Yaron Ostrover
- Venue/year: *Communications in Contemporary Mathematics* 5 (2003), no. 5, pp. 803-811
- DOI/arXiv/URL: DOI 10.1142/S0219199703001154; arXiv:math/0207070, <https://arxiv.org/abs/math/0207070>
- Result: As a corollary, proves infinite Hofer diameter for closed symplectic manifolds with $\pi_2(M) = 0$.
- Applicability check: The hypothesis $\pi_2(M) = 0$ excludes many closed symplectic manifolds, so it is partial.
- Title: Monodromy in Hamiltonian Floer theory
- Author: Dusa McDuff
- Venue/year: *Commentarii Mathematici Helvetici* 85 (2010), no. 1, pp. 95-133
- DOI/arXiv/URL: DOI 10.4171/CMH/189; arXiv:0801.1328, <https://arxiv.org/abs/0801.1328>
- Result: Gives conditions under which asymptotic spectral invariants descend from the universal cover to $\text{Ham}(M, \omega)$; when they descend, $\text{Ham}(M, \omega)$ has infinite Hofer diameter. The paper states this establishes infinite diameter in many new cases.
- Applicability check: The result depends on extra conditions involving Chern class, quantum homology, minimal Chern number, or Gromov-Witten behavior; it is not a universal theorem.
- Title: Hofer's metrics and boundary depth

- Author: Michael Usher
- Venue/year: Annales scientifiques de l'Ecole Normale Supérieure (4) 46 (2013), no. 1, pp. 57-129
- DOI/arXiv/URL: DOI 10.24033/asens.2185; arXiv:1107.4599, <https://arxiv.org/abs/1107.4599>
- Result: Theorem 1.1 proves infinite Hofer diameter, indeed infinite-dimensional quasi-isometric embeddings, for closed symplectic manifolds admitting a nonconstant autonomous Hamiltonian whose contractible closed orbits are all constant.
- Applicability check: This dynamical hypothesis covers many new examples but fails for other manifolds; Usher explicitly notes that it is not yet known whether Hofer's metric is always unbounded.
- Title: Hofer geometry via toric degeneration
- Author: Yusuke Kawamoto
- Venue/year: Mathematische Annalen 390 (2024), pp. 721-762
- DOI/arXiv/URL: DOI 10.1007/s00208-023-02775-8; arXiv:2210.14726, <https://arxiv.org/abs/2210.14726>
- Result: Uses toric degeneration to produce homogeneous quasimorphisms and Hofer-geometry consequences for quadric hypersurfaces and del Pezzo surfaces.
- Applicability check: This gives further important closed examples, not all symplectic manifolds.

4.5.3.4. What remains unresolved

No citable theorem found proves the full statement for every connected symplectic manifold without boundary. The available literature proves infinite Hofer diameter in many classes, including closed surfaces, S^2 , closed symplectically aspherical manifolds, closed manifolds satisfying McDuff's spectral-invariant descent criteria, manifolds satisfying Usher's autonomous-Hamiltonian dynamical condition, and several more recent Liouville, monotone, negatively monotone, toric, and heavy-Lagrangian settings.

The gap is the absence of a theorem covering all closed symplectic manifolds and all non-compact connected symplectic manifolds in the exact formulation of the problem, without extra asphericity, monotonicity, Floer-theoretic descent, dynamical, exactness, or Lagrangian-configuration hypotheses. Therefore the correct status is partially resolved.

4.5.4. Remaining open problems recorded by the database

4.5.4.1. Open problem 1

Does the Hamiltonian diffeomorphism group of the monotone one-point blow-up of $\mathbb{CP}^2$ have infinite diameter with respect to Hofer's metric?

Reference supplied by the database. Dusa McDuff, "Monodromy in Hamiltonian Floer theory," *Commentarii Mathematici Helvetici* **85** (2010), no. 1, 95–133, discussion following Corol-

lary 1.5 (the paper explicitly states that infinite Hofer diameter is unknown for the monotone one-point blow-up of $\mathbb{CP}^2$); DOI: <https://doi.org/10.4171/CMH/189>; arXiv: 0801.1328.

4.5.5. Primary-source pointer

The originating ICM record is *Geometry on the Group of Hamiltonian Diffeomorphisms* (1998). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.2/ICM1998.2.ocr.pdf#page=403>.

4.6. 152. Geometric Physics [1998-ag-74]

Record: 1998-ag-74

Subfield: Symplectic and contact geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1998: Geometric Physics

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.1/ICM1998.1.ocr.pdf#page=539>

4.6.1. Problem statement

Let M be a Calabi-Yau manifold of complex dimension d , let W be its mirror Calabi-Yau manifold, and let V be a holomorphic vector bundle on M with total Chern class $c(V) \in \bigoplus_p H^{p,p}(M)$. Represent the components of $c(V)$ by Poincaré-dual holomorphic cycles in M , equivalently by the corresponding collection of D-branes wrapped on those cycles.

Prove or disprove that the mirror of this collection representing V is a real d -dimensional submanifold $C \subset W$ such that holomorphicity of V corresponds to C being Lagrangian with respect to the Kähler form on W , and stability of V corresponds to C being a minimal-area Lagrangian cycle.

4.6.2. Historical background, literature, and current progress

4.6.3. Verdict: partially resolved

4.6.3.1. Precise problem

The problem asks for a general mirror-symmetry theorem for Calabi-Yau mirror pairs: given a Calabi-Yau manifold M of complex dimension d , its mirror W , and a stable holomorphic vector bundle V on M , does the D-brane charge represented by the Chern-class components of V correspond to a real d -dimensional submanifold $C \subset W$, with holomorphicity of V equivalent to C being Lagrangian and stability of V equivalent to C being minimal-area, more precisely a supersymmetric or special Lagrangian cycle?

This is stronger than ordinary homological mirror symmetry: it asks for an object-level geometric representative and for the stability/minimal-special-Lagrangian correspondence in all Calabi-Yau cases.

4.6.3.2. Original source

- Title: Extending Mirror Conjecture to Calabi-Yau with Bundles
- Author: Cumrun Vafa
- Venue: Communications in Contemporary Mathematics 1(1), 65-70
- Year: 1999

- DOI/arXiv/URL: DOI 10.1142/S0219199799000043; arXiv:hep-th/9804131; <https://arxiv.org/abs/hep-th/9804131>
- Relevance: This is the original conjectural source. It defines mirror symmetry for a Calabi-Yau with a stable bundle in terms of supersymmetric cycles on the mirror. It is a physics proposal, not a proof for arbitrary Calabi-Yau manifolds or arbitrary vector bundles.
- Title: Geometric Physics
- Author: Cumrun Vafa
- Venue: Proceedings of the International Congress of Mathematicians, Vol. I (Berlin, 1998), 537-556; Documenta Mathematica
- Year: 1998
- DOI/arXiv/URL: arXiv:hep-th/9810149; <https://arxiv.org/abs/hep-th/9810149>
- Relevance: This ICM survey contains the distinctive “extension to bundles” discussion underlying the problem statement wording and explicitly treats checks as ongoing rather than as a completed theorem.

4.6.3.3. Partial progress

- Title: Mirror Symmetry is T-Duality
- Authors: Andrew Strominger, Shing-Tung Yau, Eric Zaslow
- Venue: Nuclear Physics B 479, 243-259
- Year: 1996
- DOI/arXiv/URL: DOI 10.1016/0550-3213(96)00434-8; arXiv:hep-th/9606040; <https://arxiv.org/abs/hep-th/9606040>
- Progress: Gives the SYZ framework in which mirrors arise from dual special Lagrangian torus fibrations, with the mirror as a moduli space of such cycles with flat connections. This is foundational support for the statement, but it is not a proof for all Calabi-Yau manifolds or all stable bundles.
- Title: Categorical Mirror Symmetry: The Elliptic Curve
- Authors: Alexander Polishchuk, Eric Zaslow
- Venue: Advances in Theoretical and Mathematical Physics 2(2), 443-470
- Year: 1998
- DOI/arXiv/URL: DOI 10.4310/ATMP.1998.v2.n2.a9; arXiv:math/9801119; <https://arxiv.org/abs/math/9801119>
- Progress: Proves Kontsevich’s categorical mirror equivalence for elliptic curves and gives a detailed dictionary between coherent sheaves and Lagrangian objects. This is a genuine one-dimensional special case, not a general theorem for higher-dimensional Calabi-Yau manifolds or arbitrary stable bundles.
- Title: From Special Lagrangian to Hermitian-Yang-Mills via Fourier-Mukai Transform
- Authors: Naichung Conan Leung, Shing-Tung Yau, Eric Zaslow

- Venue: Advances in Theoretical and Mathematical Physics 4(6), 1319-1341; also AMS/IP Studies in Advanced Mathematics 23, 209-225
- Year: 2000/2001
- DOI/arXiv/URL: DOI 10.4310/ATMP.2000.v4.n6.a5; arXiv:math/0005118; <https://arxiv.org/abs/math/0005118>
- Progress: Verifies Vafa's mirror conjecture with bundles in a semi-flat special case. The hypotheses are restrictive: the mirror pair is assumed to be given by dual flat torus fibrations, singular fibers are avoided, and the main construction treats A-cycles that are sections, with multisections discussed separately. Therefore it does not cover the full statement.
- Title: Special Lagrangians, stable bundles and mean curvature flow
- Authors: R. P. Thomas, S.-T. Yau
- Venue: Communications in Analysis and Geometry 10(5), 1075-1113
- Year: 2002
- DOI/arXiv/URL: DOI 10.4310/CAG.2002.v10.n5.a8; arXiv:math/0104197; <https://arxiv.org/abs/math/0104197>
- Progress: Develops the Thomas-Yau stability and Lagrangian mean-curvature-flow conjectural framework. It proves uniqueness of special Lagrangians in Hamiltonian deformation classes under mild conditions and proves the conjecture in some symmetric examples, but it does not prove the general stability/existence correspondence.
- Title: Conjectures on Bridgeland stability for Fukaya categories of Calabi-Yau manifolds, special Lagrangians, and Lagrangian mean curvature flow
- Author: Dominic Joyce
- Venue: EMS Surveys in Mathematical Sciences 2(1), 1-62
- Year: 2015
- DOI/arXiv/URL: DOI 10.4171/EMSS/8; arXiv:1401.4949; <https://arxiv.org/abs/1401.4949>
- Progress: Gives a modern conjectural formulation using Bridgeland stability on Fukaya categories and Lagrangian mean curvature flow with surgeries. Its role is to refine the conjecture; it is not a resolving theorem.
- Title: Moment Maps, Nonlinear PDE and Stability in Mirror Symmetry, I: Geodesics
- Authors: Tristan C. Collins, Shing-Tung Yau
- Venue: Annals of PDE 7, article 11
- Year: 2021
- DOI/arXiv/URL: DOI 10.1007/s40818-021-00100-7; arXiv:1811.04824; <https://arxiv.org/abs/1811.04824>
- Progress: Studies the deformed Hermitian-Yang-Mills equation, mirror to the special Lagrangian equation, from a variational and stability viewpoint. It proves geodesic and obstruction results in hypercritical/line-bundle settings and discusses toric Landau-Ginzburg mirrors. It also makes clear that the full Bridgeland-stability/PDE correspondence and higher-rank bundle case remain open.

4.6.3.4. Why this is not fully resolved

No located theorem proves the exact statement for arbitrary Calabi-Yau mirror pairs and arbitrary stable holomorphic vector bundles. The strongest direct verification, Leung-Yau-Zaslow, assumes a semi-flat dual torus-fibration model and mainly section-type A-cycles. The modern stability side is still formulated through Thomas-Yau/Joyce/Bridgeland conjectures and through the deformed Hermitian-Yang-Mills program, where general Bridgeland stability conditions in Calabi-Yau dimension greater than 2 and higher-rank non-abelian dHYM remain major unresolved issues.

Thus the literature contains substantial special cases and conceptual progress, but not a published or otherwise citable result covering the full statement/all cases asked in this problem.

4.6.4. Remaining open problems recorded by the database

4.6.4.1. Open problem 1

Let L be a graded Lagrangian submanifold of a Calabi-Yau manifold X . Normalize the phase of the holomorphic volume form so that the cohomological phase of L is zero. Suppose that either (i) for every graded Lagrangian connect-sum decomposition $L_1 \# L_2 \approx L$, the interval $[\phi(L_1), \phi(L_2)]$ is not contained in $(\inf_L \theta_L, \sup_L \theta_L)$, or (ii) for every such decomposition, the volume of L is at most the sum of the cohomological volumes of L_1 and L_2 . Prove that Lagrangian mean-curvature flow starting from L exists for all time and converges to the unique special Lagrangian in the Hamiltonian deformation class of L .

Reference supplied by the database. Richard P. Thomas and Shing-Tung Yau, “Special Lagrangians, stable bundles and mean curvature flow,” *Communications in Analysis and Geometry* **10** (2002), no. 5, 1075–1113, Conjecture 7.3 (conditions (7.1) and (7.2)). DOI: <https://doi.org/10.4310/CAG.2002.v10.n5.a8>. arXiv: [math/0104197](https://arxiv.org/abs/math/0104197).

4.6.4.2. Open problem 2

For a compact or suitably convex-at-infinity Calabi-Yau manifold (M, J, g, Ω) , construct a natural Bridgeland stability condition $(Z, \mathcal{P})$ on its derived Fukaya category such that the central charge is $Z([L, E, b]) = \int_L \Omega$ and every brane supported on a special Lagrangian of phase $e^{i\pi\phi}$ is semistable of phase ϕ .

Reference supplied by the database. Dominic Joyce, “Conjectures on Bridgeland stability for Fukaya categories of Calabi-Yau manifolds, special Lagrangians, and Lagrangian mean curvature flow,” *EMS Surveys in Mathematical Sciences* **2** (2015), no. 1, 1–62, Conjecture 3.2(a)–(b). DOI: <https://doi.org/10.4171/EMSS/8>. arXiv: [1401.4949](https://arxiv.org/abs/1401.4949).

4.6.4.3. Open problem 3

Let A_α be the category of objects of the derived Fukaya category represented by unobstructed graded Lagrangian branes whose phase functions take values in $(\pi\alpha, \pi(\alpha+1))$, and let $\overline{A}_\alpha$ be its isomorphism closure. Prove that $\overline{A}_\alpha$ is the heart of a bounded t -structure and that slope semistability in this heart, with central charge $Z([L, E, b]) = \int_L \Omega$, gives the expected Bridgeland stability condition.

Reference supplied by the database. Dominic Joyce, “Conjectures on Bridgeland stability for Fukaya categories of Calabi–Yau manifolds, special Lagrangians, and Lagrangian mean curvature flow,” *EMS Surveys in Mathematical Sciences* **2** (2015), no. 1, 1–62, Conjecture 3.5. DOI: <https://doi.org/10.4171/EMSS/8>. arXiv: 1401.4949.

4.6.4.4. Open problem 4

Prove Joyce’s neck-pinch conjecture for Lagrangian mean-curvature flow with surgeries in Calabi–Yau m -folds, $m \geq 2$: a Lawlor neck may shrink at a finite singular time to two transversely intersecting special-Lagrangian planes and continue as an immersed Lagrangian brane with the prescribed local system and bounding cochain, and such neck pinches are generic under sufficiently small Hamiltonian perturbations of the initial Lagrangian.

Reference supplied by the database. Dominic Joyce, “Conjectures on Bridgeland stability for Fukaya categories of Calabi–Yau manifolds, special Lagrangians, and Lagrangian mean curvature flow,” *EMS Surveys in Mathematical Sciences* **2** (2015), no. 1, 1–62, Conjecture 3.16. DOI: <https://doi.org/10.4171/EMSS/8>. arXiv: 1401.4949.

4.6.4.5. Open problem 5

For a generic unobstructed compact graded Lagrangian brane (L, E, b) in a compact or suitably convex-at-infinity Calabi–Yau manifold, construct the unique all-time Lagrangian mean-curvature flow with finitely many prescribed surgeries that stays in the isomorphism class of (L, E, b) in an enlarged derived Fukaya category and converges, as a graded Lagrangian integral current, to a finite ordered sum of special Lagrangians whose phases give the semistable factors of the brane.

Reference supplied by the database. Dominic Joyce, “Conjectures on Bridgeland stability for Fukaya categories of Calabi–Yau manifolds, special Lagrangians, and Lagrangian mean curvature flow,” *EMS Surveys in Mathematical Sciences* **2** (2015), no. 1, 1–62, Conjecture 3.34. DOI: <https://doi.org/10.4171/EMSS/8>. arXiv: 1401.4949.

4.6.5. Primary-source pointer

The originating ICM record is *Geometric Physics* (1998). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.1/ICM1998.1.ocr.pdf#page=539>.

4.7. 153. Geometric Physics [1998-ag-80]

Record: 1998-ag-80

Subfield: Symplectic and contact geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1998: Geometric Physics

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.1/ICM1998.1.ocr.pdf#page=539>

4.7.1. Problem statement

Let M be a Calabi-Yau manifold of complex dimension d , let $Q \in H_d(M, \mathbb{Z})$ be a middle-dimensional homology class, and let Ω denote a nowhere-zero holomorphic d -form on M . For a complex structure on M , define

$$V(Q, \Omega) = \frac{|\Omega(Q)|}{\sqrt{\int_M \Omega \wedge \bar{\Omega}}},$$

where $\Omega(Q)$ denotes the period of Ω over Q . Minimize $V(Q, \Omega)$ over the moduli space of complex structures of the Calabi-Yau, and suppose that the minimum exists and is attained at a non-degenerate Calabi-Yau; denote the minimum by $V_{\min}$.

Prove or disprove that the growth S of the cohomology of the moduli space of pairs (L, ∇) , where $L \subset M$ is a real d -dimensional minimal Lagrangian submanifold representing Q and ∇ is a flat connection on L , satisfies

$$S = \exp(c(d) V_{\min}^{d-1}),$$

where $c(d)$ is a universal constant depending only on d .

4.7.2. Historical background, literature, and current progress

4.7.3. Verdict: partially resolved

4.7.3.1. Precise problem

The problem asks for Vafa's proposed asymptotic count/entropy for A-branes on a Calabi-Yau d -fold: for a middle-dimensional class $Q \in H_d(M, \mathbb{Z})$, minimize the normalized period

$$V(Q, \Omega) = \frac{|\Omega(Q)|}{\sqrt{\int_M \Omega \wedge \bar{\Omega}}}$$

over complex structures, assume the minimum is attained at a nondegenerate Calabi-Yau, and prove that the growth of the cohomology of the moduli space of pairs (L, ∇) , with L a real d -dimensional minimal/special Lagrangian representative of Q and ∇ a flat connection on L , is

$$S = \exp(c(d) V_{\min}^{d-1})$$

with $c(d)$ depending only on d . This is a global enumerative/cohomological growth statement, not just a local deformation statement for special Lagrangians.

4.7.3.2. Source and related progress

- **Geometric Physics.** Cumrun Vafa. Proceedings of the International Congress of Mathematicians, Berlin 1998, Vol. I, Documenta Mathematica Extra Volume ICM 1998, pp. 537-556, 1998. DOI: <https://doi.org/10.4171/DMS/1-1/19>. arXiv: <https://arxiv.org/abs/hep-th/9810149>.
This is the exact source of the question. Section 6 frames the $d = 3$ case as a black-hole prediction and then conjectures the general d -dimensional formula $S = \exp(c(d)V_{\min}^{d-1})$. Vafa says the formula is vacuous for $d = 1$, verifiable for $d = 2$, predicted for $d = 3$, and asks whether it is true in all dimensions. The paper does not prove the full conjecture.
- **Minimal Cycles, Black Holes and QFT's.** Cumrun Vafa. Preprint, 1998. arXiv: <https://arxiv.org/abs/hep-th/9805213>. DOI: <https://doi.org/10.48550/arXiv.hep-th/9805213>.
Gives physical context for minimal cycles and black-hole entropy. It supports the motivation for the conjecture but does not provide a mathematical proof of the cohomology-growth formula for moduli of minimal Lagrangians.
- **Black Holes and Calabi-Yau Threefolds.** Cumrun Vafa. Advances in Theoretical and Mathematical Physics 2 (1998), 207-218. arXiv: <https://arxiv.org/abs/hep-th/9711067>. DOI: <https://doi.org/10.48550/arXiv.hep-th/9711067>.
Computes/compares black-hole entropy predictions in Calabi-Yau threefold settings and verifies a related surprising growth for cohomology of moduli of holomorphic curves in elliptic threefolds. This is related by the physics and mirror-symmetry motivation, but it is not the problem as stated, about all Calabi-Yau d -folds and moduli of minimal Lagrangian d -folds with flat connections.
- **Deformations of Calibrated Submanifolds.** Robert C. McLean. Communications in Analysis and Geometry 6 (1998), no. 4, 705-747. DOI: <https://doi.org/10.4310/CAG.1998.v6.n4.a4>. URL: <https://www.intlpress.com/site/pub/pages/journals/items/cag/content/vols/0006/0004/a004/>
Proves the fundamental local deformation theorem: the moduli space of nearby compact special Lagrangian deformations is smooth of dimension $b^1(L)$. This is necessary background for understanding the moduli spaces, but it is purely local and does not count global cohomology or prove Vafa's asymptotic formula.
- **The moduli space of special Lagrangian submanifolds.** Nigel J. Hitchin. Annali della Scuola Normale Superiore di Pisa, Classe di Scienze 25 (1997), no. 3-4, 503-515. arXiv: <https://arxiv.org/abs/dg-ga/9711002>. URL: <https://eudml.org/doc/84303>.
Develops the local geometry of the moduli space of deformations of a compact special Lagrangian, building on McLean. This is structural progress, not an enumerative theorem or an asymptotic cohomology-growth result.
- **On counting special Lagrangian homology 3-spheres.** Dominic Joyce. Contemporary Mathematics 314 (2002), 125-151. arXiv: <https://arxiv.org/abs/hep-th/9907013>. DOI: <https://doi.org/10.48550/arXiv.hep-th/9907013>.
Attempts to define an invariant of almost Calabi-Yau 3-folds by weighted counts of special Lagrangian rational homology 3-spheres. Joyce explicitly treats the singular behav-

ior needed for such invariants as not fully understood and formulates conjectural wall-crossing behavior. This is partial progress in dimension 3, but it is much narrower than Vafa's all- d , all-classes cohomology-growth formula.

- **Special Lagrangian submanifolds with isolated conical singularities. V. Survey and applications.** Dominic Joyce. *Journal of Differential Geometry* 63 (2003), 279-348. arXiv: <https://arxiv.org/abs/math/0303272>. DOI: <https://doi.org/10.48550/arXiv.math/0303272>.

Surveys and proves major results on singular special Lagrangians, desingularization, connected sums, and boundaries of moduli spaces. These results address important compactification/singularity issues but do not yield Vafa's global asymptotic cohomology count.

- **Sigma Models for Bundles on Calabi-Yau: A Proposal for Matrix String Compactifications.** C. Hofman and J.-S. Park. *Nuclear Physics B* 561 (1999), 125-156. DOI: [https://doi.org/10.1016/S0550-3213\(99\)00548-9](https://doi.org/10.1016/S0550-3213(99)00548-9). arXiv: <https://arxiv.org/abs/hep-th/9904150>.

Discusses Vafa's extension of mirror symmetry involving stable bundles on one side and minimal Lagrangian submanifolds on the mirror side. The authors propose sigma-model structures whose A-model and elliptic genus relate to moduli of stable bundles, but they do not compute the Vafa asymptotic growth formula.

- **Gauge Theory in Higher Dimensions, II.** Simon K. Donaldson and Ed Segal. *Surveys in Differential Geometry* 16, *Geometry of Special Holonomy and Related Topics*, International Press, 2011, pp. 1-41. arXiv: <https://arxiv.org/abs/0902.3239>.

Develops a formal Floer/gauge-theoretic framework connected to counting calibrated submanifolds and says that, for counting special Lagrangian submanifolds, a final theory is still lacking. This is strong evidence that the general enumerative foundation needed for Vafa's conjecture was not available.

- **Transversality for perturbed special Lagrangian submanifolds.** Emily Autumn Windes. *Annals of Global Analysis and Geometry* 68, article 6, 2025. DOI: <https://doi.org/10.1007/s10455-025-10006-3>. arXiv: <https://arxiv.org/abs/2406.17948>.

Proves that the moduli space of perturbed special Lagrangian submanifolds in the Donaldson-Segal setup is generically a set of isolated points. This is recent progress toward a counting theory in real dimension 3/Calabi-Yau dimension 3, but it does not define the full invariant, prove compactness/wall-crossing enough for Vafa's formula, or handle all dimensions.

- **Tropical Quantum Field Theory, Mirror Polyvector Fields, and Multiplicities of Tropical Curves.** Travis Mandel and Helge Ruddat. *International Mathematics Research Notices* 2023, no. 4, 3249-3304. DOI: <https://doi.org/10.1093/imrn/rnab332>. URL: <https://academic.oup.com/imrn/article/2023/4/3249/6454977>.

Relates tropical multiplicities to Joyce-type weights in settings where tropical curves produce Lagrangian submanifolds in torus fibrations. This is useful special-geometry progress, but it does not settle the global cohomology-growth formula in the problem statement.

4.7.3.3. What remains unresolved

No published or otherwise citable result found proves the full statement as written. The missing pieces include:

- a general, rigorous compactified moduli theory for minimal/special Lagrangian d -folds with flat connections in arbitrary compact Calabi-Yau d -folds;
- a theorem identifying the relevant cohomology growth of those moduli spaces for all classes Q , or even for all large multiples of Q ;
- a proof that the exponent is universally $c(d)V_{\min}^{d-1}$ with $c(d)$ independent of the Calabi-Yau and with exactly Vafa's period-minimization hypotheses;
- a verification that related stable-bundle or BPS/black-hole results imply the Lagrangian moduli statement without extra hypotheses. Current literature provides motivation, local deformation theory, proposed invariants, singularity/desingularization analysis, and some special-case or perturbed transversality results, but not a theorem covering the full problem.

Therefore the correct status is **partially resolved**, not fully resolved.

4.7.4. Remaining open problems recorded by the database

4.7.4.1. Open problem 1

For a generic almost Calabi-Yau threefold M , construct a rational-valued invariant $I : H_3(M, \mathbb{Z}) \rightarrow \mathbb{Q}$ by a finite weighted count of compact immersed, possibly singular, special Lagrangian rational homology 3-spheres in each class. For a nonsingular immersed N , the weight should be $|H_1(N, \mathbb{Z})|/|G(N)|$, and the invariant should be unchanged when the Kähler class varies while the period class of the holomorphic volume form is fixed up to nonzero complex scaling.

Reference supplied by the database. Dominic Joyce, “On counting special Lagrangian homology 3-spheres,” in *Topology and Geometry: Commemorating SISTAG*, Contemporary Mathematics **314**, American Mathematical Society, Providence, RI, 2002, 125–151, Conjecture 7.3(a)–(c). arXiv: [hep-th/9907013](https://arxiv.org/abs/hep-th/9907013). MR1941627.

4.7.4.2. Open problem 2

Prove the basic wall-crossing formula for Joyce's special-Lagrangian counting invariant. If primitive classes $\chi^+, \chi^- \in H_3(M, \mathbb{Z})$ align in phase and $I(\chi^+ + \chi^-)^\pm$ denote the values on the two sides of the corresponding wall, then

$$I(\chi^+ + \chi^-)^+ - I(\chi^+ + \chi^-)^- = (\chi^+ \cap \chi^-) I(\chi^+) I(\chi^-).$$

Reference supplied by the database. Dominic Joyce, “On counting special Lagrangian homology 3-spheres,” in *Topology and Geometry: Commemorating SISTAG*, Contemporary Mathematics **314**, American Mathematical Society, Providence, RI, 2002, 125–151, Conjecture 7.3(d), equation (7.9). arXiv: [hep-th/9907013](https://arxiv.org/abs/hep-th/9907013). MR1941627.

4.7.4.3. Open problem 3

Determine conditions under which the zero-dimensional moduli space of perturbed special Lagrangian 3-submanifolds in a fixed homology class and with fixed topology, for a generic tamed closed G_2 -pair on a closed 6-manifold, is compact; more generally, construct the anticipated compactification by rectifiable currents and describe the singular objects in its boundary well enough to define a count.

Reference supplied by the database. Emily Autumn Windes, “Transversality for perturbed special Lagrangian submanifolds,” *Annals of Global Analysis and Geometry* **68** (2025), article 6, Introduction, unnumbered compactness/compactification problem following the volume-bounds proposition. DOI: <https://doi.org/10.1007/s10455-025-10006-3>. arXiv: 2406.17948.

4.7.4.4. Open problem 4

Construct a Floer theory for special Lagrangian 3-submanifolds of a Calabi–Yau threefold using the critical points and flow theory of the Donaldson–Segal Lagrange-multiplier functional, including the perturbation, compactness, and singularity analysis needed for the theory to be well defined.

Reference supplied by the database. Emily Autumn Windes, “Transversality for perturbed special Lagrangian submanifolds,” *Annals of Global Analysis and Geometry* **68** (2025), article 6, Introduction, unnumbered question on constructing a Floer theory for special Lagrangian submanifolds. DOI: <https://doi.org/10.1007/s10455-025-10006-3>. arXiv: 2406.17948.

4.7.5. Primary-source pointer

The originating ICM record is *Geometric Physics* (1998). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.1/ICM1998.1.ocr.pdf#page=539>.

4.8. 154. Fibrations in Symplectic Topology [1998-geom-275]

Record: 1998-geom-275

Subfield: Symplectic and contact geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1998: Fibrations in Symplectic Topology

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.1/ICM1998.1.ocr.pdf#page=341>

4.8.1. Problem statement

Let (M, ω) be a closed symplectic manifold, and let

$$p: P \rightarrow S^2$$

be a fiber bundle with fiber (M, ω) whose structure group reduces to the group $\text{Ham}(M, \omega)$ of Hamiltonian symplectomorphisms of (M, ω) . Suppose that the quantum product on $H^*(M; \mathbb{Q})$ is trivial, meaning that it equals the ordinary cup product. Prove that there is an isomorphism of graded $\mathbb{Q}$ -algebras

$$H^*(P; \mathbb{Q}) \cong H^*(S^2; \mathbb{Q}) \otimes_{\mathbb{Q}} H^*(M; \mathbb{Q}),$$

where the algebra structure on the right is the graded tensor-product structure induced by the ordinary cup products on the two factors.

4.8.2. Historical background, literature, and current progress

4.8.3. Verdict: unresolved

The exact assertion is not known in full. The literature proves additive cohomological splitting for every Hamiltonian fibration over S^2 and proves multiplicative splitting under hypotheses stronger than the triviality of the small quantum product, but no located result proves that the trivial quantum product alone forces the stated graded $\mathbb{Q}$ -algebra isomorphism. No counterexample to the exact assertion was found either.

- Dusa McDuff, “Fibrations in symplectic topology,” *Documenta Mathematica*, Extra Volume ICM 1998, I, 339–357 (1998). DOI: <https://doi.org/10.4171/DMS/1-1/10>. Claim 4.5 is the source of the exact problem. Its surrounding text calls the statement plausible and says that the full proof details had not been worked out.
- Dusa McDuff, “Quantum homology of fibrations over S^2 ,” *International Journal of Mathematics* 11(5), 665–721 (2000). DOI: <https://doi.org/10.1142/S0129167X00000337>; arXiv: [math/9905092](https://arxiv.org/abs/math/9905092). Theorem 1.1 proves the unconditional additive splitting $H^*(P; \mathbb{Q}) \cong H^*(M; \mathbb{Q}) \otimes H^*(S^2; \mathbb{Q})$. Theorem 1.2 proves the required ring splitting if all nonconstant vertical Gromov–Witten invariants of the total space with at most four insertions vanish. Proposition 3.23 proves it when the relevant Seidel element has a pure scalar-monomial form together with vanishing nonconstant three-point invariants

of M . These are substantive special cases, including the case in which the fibers admit no nonconstant J -holomorphic spheres. Crucially, the paper explicitly states that it is still unclear whether ring splitting follows from the sole assumption that the quantum product of M is trivial.

- **Mohammed Abouzaid, Mark McLean, and Ivan Smith, “Complex cobordism, Hamiltonian loops and global Kuranishi charts” (2021).** arXiv: [2110.14320](https://arxiv.org/abs/2110.14320). Theorems 1.1 and 1.3 strengthen additive splitting to integral cohomology and to every complex-oriented generalized cohomology theory. The result is expressly additive, and the paper notes that Hamiltonian fibrations need not split multiplicatively in general. It therefore does not settle the additional multiplicative conclusion under the problem’s trivial-quantum-product hypothesis.

What remains open is precisely the removal of McDuff’s additional vertical Gromov–Witten or Seidel-element hypotheses: one must prove, or disprove by a counterexample, that equality of the small quantum product of M with its ordinary cup product is by itself enough to make the ordinary rational cohomology ring of every associated Hamiltonian fibration over S^2 a product ring.

4.8.4. Remaining open problems recorded by the database

4.8.4.1. Open problem 1

Let (M, ω) be a closed symplectic manifold, let $\phi: S^1 \rightarrow \text{Ham}(M, \omega)$ be a loop based at the identity, and let

$$p: P_\phi = (D_+^2 \times M) \cup_\phi (D_-^2 \times M) \rightarrow S^2$$

be the Hamiltonian fibration obtained from ϕ by clutching along the equator. Let

$$\rho: \pi_1(\text{Ham}(M, \omega)) \rightarrow QH_{\text{ev}}(M; \Lambda_{\mathbb{R}})^\times$$

be the Seidel representation, where $\Lambda_{\mathbb{R}}$ is the real Novikov ring and the unit of quantum homology is $[M]$. Suppose that $\rho([\phi]) = [M]$. Determine whether there must be an isomorphism of graded $\mathbb{Q}$ -algebras

$$H^*(P_\phi; \mathbb{Q}) \cong H^*(S^2; \mathbb{Q}) \otimes_{\mathbb{Q}} H^*(M; \mathbb{Q}),$$

with the graded tensor-product multiplication on the right.

Reference supplied by the database. Dusa McDuff, “Quantum homology of fibrations over S^2 ,” *International Journal of Mathematics* 11(5), 665–721 (2000), Section 1.2, discussion following Theorem 1.5. DOI: 10.1142/S0129167X00000337; arXiv:math/9905092, <https://arxiv.org/abs/math/9905092>.

4.8.5. Primary-source pointer

The originating ICM record is *Fibrations in Symplectic Topology* (1998). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.1/ICM1998.1.ocr.pdf#page=341>.

4.9. 155. Geometry on the Group of Hamiltonian Diffeomorphisms [1998-geom-310]

Record: 1998-geom-310

Subfield: Symplectic and contact geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1998: Geometry on the Group of Hamiltonian Diffeomorphisms

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.2/ICM1998.2.ocr.pdf#page=403>

4.9.1. Problem statement

Let (M^{2n}, ω) be a connected symplectic manifold without boundary, and let $U \subset M$ be open. For $\phi \in \text{Ham}_c(M, \omega)$, define its Hofer norm by

$$\|\phi\|_H = \inf_H \int_0^1 \left(\max_{x \in M} H_t(x) - \min_{x \in M} H_t(x) \right) dt,$$

where the infimum is over all compactly supported smooth time-dependent Hamiltonians $H: M \times [0, 1] \rightarrow \mathbb{R}$ whose time-one Hamiltonian flow is ϕ . Define the displacement energy of U by

$$e(U) = \inf\{\|\phi\|_H : \phi \in \text{Ham}_c(M, \omega), \phi(U) \cap U = \emptyset\},$$

with $e(U) = +\infty$ if no such ϕ exists. Define the Gromov width of U by

$$c_G(U) = \sup\{\pi r^2 : \text{there is a symplectic embedding } (B^{2n}(r), \omega_0) \hookrightarrow (U, \omega|_U)\},$$

where $B^{2n}(r) \subset \mathbb{R}^{2n}$ is the open radius- r ball, $\omega_0 = \sum_{j=1}^n dx_j \wedge dy_j$, and $c_G(\emptyset) = 0$. Prove that $e(U) \geq c_G(U)$ for every such (M, ω) and U .

4.9.2. Historical background, literature, and current progress

Verdict: unresolved.

The exact assertion is the sharp energy–Gromov-width inequality for every connected symplectic manifold, including arbitrary noncompact and nonconvex ambient manifolds. No proof or counterexample covering that full class was found.

Leonid Polterovich, *Geometry on the Group of Hamiltonian Diffeomorphisms*, Documenta Mathematica, Extra Volume ICM 1998 II (1998), 401–410, [direct URL](#), states this as the conjectural removal of the factor $1/2$ from the general inequality.

Francois Lalonde and Dusa McDuff, *The Geometry of Symplectic Energy*, Annals of Mathematics (2) 141 (1995), 349–371, <https://doi.org/10.2307/2118524>, Theorem 1.1(ii), prove the unconditional bound $e(A) \geq \frac{1}{2}c_G(A)$ for compact subsets of an arbitrary symplectic manifold. Applied to compact subballs of an open set, this yields $e(U) \geq \frac{1}{2}c_G(U)$ in the setting of the problem, but it does not give the requested sharp constant.

Urs Frauenfelder, Viktor Ginzburg, and Felix Schlenk, *Energy Capacity Inequalities via an Action Selector*, in *Geometry, Spectral Theory, Groups, and Dynamics*, Contemporary Mathematics 387, American Mathematical Society (2005), 129–152, [arXiv:math/0402404](https://arxiv.org/abs/math/0402404), Theorem 1 and Corollary 1, prove $c_G(A) \leq c_{HZ}(A) \leq c_{HZ}^\circ(A, M) \leq e(A, M)$ whenever the ambient manifold has an exhaustion admitting action selectors. This proves the desired inequality for the broad classes for which such selectors are constructed, not for every symplectic manifold.

Michael Usher, *The Sharp Energy-Capacity Inequality*, Communications in Contemporary Mathematics 12 (2010), 457–473, <https://doi.org/10.1142/S0219199710003889>, [arXiv:0808.1592](https://arxiv.org/abs/0808.1592), Theorem 1.1, proves $c_{HZ}^\circ(A, M) \leq e(A, M)$ for every subset of every closed symplectic manifold. Since $c_G \leq c_{HZ} \leq c_{HZ}^\circ$, this resolves the exact assertion for closed M . The same paper also obtains $c_{HZ}(A) \leq e(A, M)$ for manifolds of Type (C), including additional noncompact examples.

Yoshihiro Sugimoto, *The Sharp Energy-Capacity Inequality on Convex Symplectic Manifolds*, Journal of Fixed Point Theory and Applications 21 (2019), article 13, <https://doi.org/10.1007/s11784-018-0650-2>, [arXiv:1903.07051](https://arxiv.org/abs/1903.07051), Theorem 2, proves $c_{HZ}^\circ(A, M) \leq e(A, M)$ for every subset of every convex symplectic manifold, with no asphericity assumption. Thus it proves the desired assertion for all convex ambient manifolds.

What remains is the sharp inequality for noncompact symplectic manifolds outside these and the other action-selector/exhaustion classes. The cited results neither exhaust all connected symplectic manifolds without boundary nor supply a counterexample, so they do not completely resolve the statement as written.

4.9.3. Remaining open problems recorded by the database

4.9.3.1. Open problem 1

Let (M^{2n}, ω) be a connected noncompact symplectic manifold without boundary. For $\phi \in \text{Ham}_c(M, \omega)$, define

$$\|\phi\|_H = \inf_H \int_0^1 \left(\max_{x \in M} H_t(x) - \min_{x \in M} H_t(x) \right) dt,$$

where the infimum is over all compactly supported smooth time-dependent Hamiltonians $H: M \times [0, 1] \rightarrow \mathbb{R}$ whose time-one Hamiltonian flow is ϕ . For a subset $A \subset M$, define

$$e(A, M) = \inf\{\|\phi\|_H : \phi \in \text{Ham}_c(M, \omega), \phi(A) \cap A = \emptyset\},$$

with value $+\infty$ if no such ϕ exists.

Let $\mathcal{H}(A, M)$ consist of all autonomous Hamiltonians $F \in C_c^\infty(M)$ such that $\text{supp } F \subset A$, $F \geq 0$, and both $F^{-1}(0)$ and $F^{-1}(\max F)$ contain nonempty open subsets. Call $F \in \mathcal{H}(A, M)$ HZ° -admissible if its Hamiltonian flow has no nonconstant periodic orbit of period less than one that is contractible in M , and define the π_1 -sensitive Hofer–Zehnder capacity by

$$c_{HZ}^\circ(A, M) = \sup\{\max F : F \in \mathcal{H}(A, M) \text{ is } HZ^\circ\text{-admissible}\},$$

with the supremum equal to 0 when the set is empty.

Determine whether

$$c_{HZ}^\circ(A, M) \leq e(A, M)$$

for every subset A of every such noncompact symplectic manifold (M, ω) .

Reference supplied by the database. Yoshihiro Sugimoto, “The Sharp Energy-Capacity Inequality on Convex Symplectic Manifolds,” *Journal of Fixed Point Theory and Applications* 21 (2019), Article 13, Question 1. DOI: <https://doi.org/10.1007/s11784-018-0650-2>; arXiv: 1903.07051.

4.9.4. Primary-source pointer

The originating ICM record is *Geometry on the Group of Hamiltonian Diffeomorphisms* (1998). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.2/ICM1998.2.ocr.pdf#page=403>.

4.10. 156. Geometry of Symplectic Intersections [2002-geom-82]

Record: 2002-geom-82

Subfield: Symplectic and contact geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Geometry of Symplectic Intersections

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.2/ICM2002.2.ocr.pdf#page=234>

4.10.1. Problem statement

Let $L \subset (\mathbb{C}^n, \omega_{\text{std}})$ be a closed Lagrangian submanifold, and let $\mu_L : \pi_2(\mathbb{C}^n, L) \rightarrow \mathbb{Z}$ denote its Maslov homomorphism, equivalently the Maslov class of L in $H^1(L; \mathbb{Z})$. Determine the possible classes μ_L for Lagrangian submanifolds of $\mathbb{C}^n$, subject to the Floer-theoretic vanishing principle that, since every compact subset of $\mathbb{C}^n$ can be Hamiltonianly displaced from itself, every such L whose self-Floer homology is well defined must satisfy $HF(L, L) = 0$.

4.10.2. Historical background, literature, and current progress

4.10.3. Verdict: partially resolved

4.10.3.1. Precise problem checked

The problem is broader than the original torus version of Audin's conjecture. In the cited ICM discussion, the question is not merely whether a Lagrangian torus has minimal Maslov number 2, but what restrictions Floer-theoretic vanishing imposes on the Maslov homomorphism/class of an arbitrary closed Lagrangian submanifold

$$L \subset (\mathbb{C}^n, \omega_{\text{std}}).$$

Since every compact subset of $\mathbb{C}^n$ is Hamiltonianly displaceable, any version of self-Floer homology that is defined for such an L must vanish. The intended problem is therefore the Maslov-class restriction/realization problem for closed Lagrangians in $\mathbb{C}^n$, with the original Audin torus conjecture as a distinguished special case.

4.10.3.2. Literature and current progress

1. Michele Audin, "Fibres normaux d'immersions en dimension double, points doubles d'immersions lagrangiennes et plongements totalement reels", *Commentarii Mathematici Helvetici* 63 (1988), 593-623. DOI: 10.1007/BF02566781. This is the original source traditionally cited for Audin's Maslov-class questions. Later papers identify the original concrete conjecture as the statement that every Lagrangian torus $T^n \subset \mathbb{C}^n$ has Maslov number 2, equivalently bounds a disk of Maslov index 2.

2. **Paul Biran**, “Geometry of Symplectic Intersections”, *Proceedings of the International Congress of Mathematicians, Beijing 2002, Vol. II*, Higher Education Press, 2002, 241-255; [arXiv:math/0304260](#). This is the source corresponding to the dataset entry. Section 4 states the displaceability/Floer-vanishing principle: for Lagrangians in manifolds with the non-intersection property, $HF(L, L) = 0$ whenever it is well-defined, and applying this to $\mathbb{C}^n$ gives restrictions on possible Maslov classes. Biran explicitly treats this as a source of restrictions, not as a complete classification.
3. **Yong-Geun Oh**, “Floer cohomology, spectral sequences, and the Maslov class of Lagrangian embeddings”, *International Mathematics Research Notices* 1996, no. 7, 305-346. DOI: [10.1155/S1073792896000219](#). Oh gives the Floer-theoretic spectral-sequence interpretation behind the Maslov-class restrictions. The results apply under hypotheses such as monotonicity/minimal Maslov-number assumptions and provide important restrictions, but not a complete answer for arbitrary closed Lagrangians in $\mathbb{C}^n$.
4. **Claude Viterbo**, “A new obstruction to embedding Lagrangian tori”, *Inventiones Mathematicae* 100 (1990), 301-320. DOI: [10.1007/BF01231188](#). Viterbo proved strong restrictions for Lagrangian tori, including early cases of the Audin torus problem. This is key partial progress but is restricted to tori.
5. **Leonid Polterovich**, “Monotone Lagrange submanifolds of linear spaces and the Maslov class in cotangent bundles”, *Mathematische Zeitschrift* 207 (1991), 217-222. DOI: [10.1007/BF02571385](#). Polterovich proved Maslov-class rigidity results for monotone Lagrangians and low-dimensional/special cases. These results are again partial: they do not classify possible Maslov classes for all closed Lagrangians in $\mathbb{C}^n$.
6. **Lev Buhovsky**, “The Maslov class of Lagrangian tori and quantum products in Floer cohomology”, *Journal of Topology and Analysis* 2 (2010), no. 1, 57-75. DOI: [10.1142/S1793525310000240](#); [arXiv:math/0608063](#). Buhovsky proved the monotone version of Audin’s torus conjecture in all dimensions: a monotone Lagrangian torus in $\mathbb{R}^{2n} \cong \mathbb{C}^n$ has minimal Maslov number 2. This resolves the monotone torus case only.
7. **Mihai Damian**, “Floer homology on the universal cover, Audin’s conjecture and other constraints on Lagrangian submanifolds”, *Commentarii Mathematici Helvetici* 87 (2012), 433-462/463. DOI: [10.4171/CMH/259](#); [arXiv:1006.3398](#). Damian proves generalized Audin-type statements for monotone displaceable Lagrangians, including that if L is aspherical, or more generally its universal cover has no odd homology, then its Maslov number is 2. The monotonicity and topological assumptions are extra hypotheses, so this is partial progress for the broader problem stated here.
8. **Kai Cieliebak and Klaus Mohnke**, “Punctured holomorphic curves and Lagrangian embeddings”, *Inventiones Mathematicae* 212 (2018), no. 1, 213-295. DOI: [10.1007/s00222-017-0767-8](#); [arXiv:1411.1870](#). Theorem 1.2(b) proves Audin’s original torus conjecture: every Lagrangian torus in $\mathbb{C}^n$ admits a disk of Maslov index 2. This fully resolves the torus special case, including non-monotone tori. It does not cover arbitrary closed Lagrangian submanifolds.
9. **Kei Irie**, “Chain level loop bracket and pseudo-holomorphic disks”, *Journal of Topology* 13 (2020), no. 2, 870-938. DOI: [10.1112/topo.12140](#); [arXiv:1801.04633](#). Irie’s work gives another proof of Audin’s conjecture in the torus case and proves/generalizes Fukaya’s string-topology approach for closed, oriented, spin $K(\pi, 1)$ Lagrangians, producing Maslov-index-2 holomorphic disks. These are important generalizations but still require strong topological hypotheses.
10. **Tobias Ekholm, Yakov Eliashberg, Emmy Murphy, and Ivan Smith**, “Constructing ex-

act Lagrangian immersions with few double points”, *Geometric and Functional Analysis* 23 (2013), 1772–1803. DOI: 10.1007/s00039-013-0243-6; arXiv:1303.0588. Their construction yields a Lagrangian embedding $S^1 \times S^2 \rightarrow \mathbb{R}_{\text{st}}^6 \cong \mathbb{C}^3$ with vanishing Maslov class. This rules out any unrestricted theorem saying that every closed Lagrangian in $\mathbb{C}^n$ must have nonzero Maslov class.

11. Yin Li, “Aspherical Lagrangian submanifolds, Audin’s conjecture and cyclic dilations”, arXiv:2308.05086, 2023; listed by the author as to appear in *Selecta Mathematica* (N.S.). This proves a general Audin-type Maslov-index-2 disk result for closed oriented $K(\pi, 1)$ Lagrangians in Liouville domains with finite first Gutt-Hutchings capacity, extending the $\mathbb{C}^n$ aspherical/ $K(\pi, 1)$ story. It remains conditional on asphericity-type assumptions and is not a classification for all closed Lagrangians.
12. Shuhao Li, “The topology of Lagrangian submanifolds via open-closed string topology”, arXiv:2604.08481, 2026 preprint. The paper proves a new partial Maslov-class rigidity result: for closed, oriented, spin $L \subset \mathbb{C}^n$, if $\pi_2(L) = 0$, then the Maslov class is nonzero. Its introduction also surveys the current state and explicitly notes the Ekholm-Eliashberg-Murphy-Smith example $S^2 \times S^1 \subset \mathbb{C}^3$ with vanishing Maslov class. Thus even the latest literature gives further partial restrictions, not a complete solution of the full problem.

4.10.3.3. Why this is not fully resolved

The original Audin torus conjecture is fully resolved by Cieliebak-Mohnke, with alternative/generalized proofs by Fukaya-Irie and later work. However, the dataset entry is stated for arbitrary closed Lagrangian submanifolds $L \subset \mathbb{C}^n$ and asks about possible Maslov classes/restrictions coming from Floer-theoretic vanishing. No cited theorem gives a complete characterization of all possible Maslov homomorphisms/classes for arbitrary closed Lagrangians in $\mathbb{C}^n$.

Moreover, the broad nonzero-Maslov formulation is not true without hypotheses: $S^1 \times S^2$ admits a Maslov-zero Lagrangian embedding in $\mathbb{C}^3$. Current positive results require extra assumptions such as torus, monotone, aspherical/ $K(\pi, 1)$, spin, $\pi_2(L) = 0$, nonpositive curvature, or vanishing H^2 . Therefore the correct status of the full problem is **partially resolved**, not fully resolved.

4.10.4. Remaining open problems recorded by the database

4.10.4.1. Open problem 1

Does the connected sum $T^3 \# T^3$ admit a Lagrangian embedding into $(\mathbb{C}^3, \omega_{\text{std}})$?

Reference supplied by the database. Shuhao Li, “The topology of Lagrangian submanifolds via open-closed string topology,” arXiv preprint (2026), Remark following Theorem 1.1, item (3), where this existence problem is explicitly identified as wide open; arXiv: 2604.08481, DOI: <https://doi.org/10.48550/arXiv.2604.08481>.

4.10.4.2. Open problem 2

If M_1 and M_2 are closed hyperbolic 3-manifolds, does their connected sum $M_1 \# M_2$ admit a Lagrangian embedding into $(\mathbb{C}^3, \omega_{\text{std}})$?

Reference supplied by the database. Shuhao Li, “The topology of Lagrangian submanifolds via open-closed string topology,” arXiv preprint (2026), Remark following Theorem 1.1, item (3), where the existence or nonexistence of Lagrangian embeddings of connected sums of hyperbolic 3-manifolds into $\mathbb{C}^3$ is explicitly identified as wide open; arXiv: [2604.08481](https://arxiv.org/abs/2604.08481), DOI: <https://doi.org/10.48550/arXiv.2604.08481>.

4.10.5. Primary-source pointer

The originating ICM record is *Geometry of Symplectic Intersections* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.2/ICM2002.2.ocr.pdf#page=234>.

4.11. 157. Géométrie de Contact de la Dimension Trois vers les Dimensions Supérieures [2002-geom-83]

Record: 2002-geom-83

Subfield: Symplectic and contact geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Géométrie de Contact de la Dimension Trois vers les Dimensions Supérieures

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.2/ICM2002.2.ocr.pdf#page=389>

4.11.1. Problem statement

Prouver ou réfuter que, sur toute variété de contact close de dimension trois (V, ξ) , tout flot de Reeb, c'est-à-dire le flot du champ de Reeb R_α associé à une forme de contact α telle que $\xi = \ker \alpha$ et caractérisé par $\alpha(R_\alpha) = 1$ et $i_{R_\alpha} d\alpha = 0$, admet une section de Poincaré-Birkhoff : une surface compacte qui rencontre toutes les orbites du flot, dont l'intérieur est transversal au flot et dont chaque composante du bord est une orbite périodique.

4.11.2. Historical background, literature, and current progress

4.11.3. Verdict: partially resolved

4.11.3.1. Precise problem

The problem asks whether every Reeb flow of every contact form α defining an arbitrary closed three-dimensional contact manifold (V, α) admits a Poincaré-Birkhoff/Birkhoff global surface of section: a compact surface meeting every orbit, with interior transverse to the Reeb flow, and with every boundary component a periodic Reeb orbit.

This is stronger than Giroux's ICM corollary, which asserts only that every closed contact 3-manifold admits some adapted Reeb flow with such a section.

4.11.3.2. Original source and status of the question

- Title: Geometrie de contact: de la dimension trois vers les dimensions superieures
- Author: Emmanuel Giroux
- Venue/year: Proceedings of the International Congress of Mathematicians, Vol. II, Beijing 2002, Higher Education Press, 2002, pp. 405-414
- arXiv/URL: arXiv:math/0305129, <https://arxiv.org/abs/math/0305129>
- Relevant point: Corollaire 6 states that on every closed contact 3-manifold there exists a Reeb flow admitting a Poincaré-Birkhoff section. Immediately afterward Giroux says that it is not excluded that every Reeb flow admits such a section, which would imply the Weinstein conjecture. Thus the exact all-Reeb-flows statement in this problem was not proved there.

4.11.3.3. Main partial progress

- Title: The dynamics on three-dimensional strictly convex energy surfaces
- Authors: Helmut Hofer, Krzysztof Wysocki, Eduard Zehnder
- Venue/year: Annals of Mathematics 148 (1998), no. 1, 197-289
- DOI/URL: 10.2307/120994, <https://annals.math.princeton.edu/articles/12987>
- Result: proves disk-like global surfaces of section for Hamiltonian/Reeb dynamics on strictly convex energy surfaces in $\mathbb{R}^4$. This is a foundational special case, but it assumes a convex/dynamically convex S^3 -type setting and does not cover arbitrary contact 3-manifolds or arbitrary Reeb flows.
- Title: On the existence of supporting broken book decompositions for contact forms in dimension 3
- Authors: Vincent Colin, Pierre Dehornoy, Ana Rechtman
- Venue/year: Inventiones mathematicae 231 (2023), no. 3, 1489-1539
- DOI/arXiv/URL: 10.1007/s00222-022-01160-7, arXiv:2001.01448, <https://arxiv.org/abs/2001.01448>
- Result: every nondegenerate contact form in dimension 3 is carried by a broken book decomposition. This gives a system of transverse surfaces and strong dynamical consequences, but a broken book is weaker than a Birkhoff/global surface of section. The paper formulates “Every Reeb vector field has a Birkhoff section” as a conjecture, so it does not resolve the full statement.
- Title: Existence of Birkhoff sections for Kupka-Smale Reeb flows of closed contact 3-manifolds
- Authors: Gonzalo Contreras, Marco Mazzucchelli
- Venue/year: Geometric and Functional Analysis 32 (2022), no. 5, 951-979
- DOI/arXiv/URL: 10.1007/s00039-022-00616-5, arXiv:2110.07491, <https://arxiv.org/abs/2110.07491>
- Result: any Reeb vector field on an orientable closed 3-manifold satisfying the Kupka-Smale condition admits a Birkhoff section. Consequently C^∞ -generic contact forms have Reeb flows with Birkhoff sections. This is strong generic progress, but the Kupka-Smale/nondegeneracy-transversality hypothesis is an extra assumption absent from the problem statement.
- Title: Generic properties of 3-dimensional Reeb flows: Birkhoff sections and entropy
- Authors: Vincent Colin, Pierre Dehornoy, Umberto Hryniewicz, Ana Rechtman
- Venue/year: Commentarii Mathematici Helvetici 99 (2024), no. 3, 557-611
- DOI/arXiv/URL: 10.4171/CMH/573, arXiv:2202.01506, <https://arxiv.org/abs/2202.01506>
- Result: if the Liouville measure of a nondegenerate contact form on a closed 3-manifold can be approximated by periodic Reeb orbits, then the Reeb flow admits a partial-strong Birkhoff section; using Irie’s equidistribution theorem this gives an open dense/generic existence result. This again does not cover arbitrary, possibly degenerate Reeb flows.

- Title: Homoclinic orbits, Reeb chords and nice Birkhoff sections for Reeb flows in 3D
- Authors: Vincent Colin, Umberto Hryniewicz, Ana Rechtman
- Venue/year: arXiv preprint, 2025
- arXiv/URL: arXiv:2501.11725, <https://arxiv.org/abs/2501.11725>
- Result: for C-infinity-generic contact forms defining a fixed co-oriented contact structure on a closed 3-manifold, the authors obtain embedded Birkhoff/global surfaces of section with additional prescribed features. This improves the generic picture but remains generic rather than universal.
- Title: From curve shortening to flat link stability and Birkhoff sections of geodesic flows
- Authors: Marcelo R. R. Alves, Marco Mazzucchelli
- Venue/year: Journal de l'Ecole polytechnique - Mathematiques 12 (2025), 801-851
- DOI/URL: 10.5802/jep.301, <https://jep.centre-mersenne.org/articles/10.5802/jep.301/>
- Result: proves Birkhoff sections for geodesic flows of any closed connected orientable Riemannian surface. These are Reeb flows on unit tangent bundles, so this is an important subclass, but not the full class of Reeb flows on arbitrary closed contact 3-manifolds.

4.11.3.4. What remains unresolved

I found no published or otherwise citable theorem proving that every Reeb flow, without genericity, nondegeneracy, Kupka-Smale, equidistribution, dynamical convexity, geodesic-flow, or other extra hypotheses, admits a compact Poincare-Birkhoff/global surface of section on every closed contact 3-manifold. the exact statement is therefore not fully resolved; the literature gives substantial partial progress only.

4.11.4. Remaining open problems recorded by the database

4.11.4.1. Open problem 1

Does every nondegenerate Reeb vector field on a closed contact three-manifold admit a Birkhoff section?

Reference supplied by the database. Vincent Colin, Pierre Dehornoy, and Ana Rechtman, “On the existence of supporting broken book decompositions for contact forms in dimension 3,” *Inventiones Mathematicae* 231, no. 3 (2023), 1489–1539, §4.6 (discussion preceding Conjecture 4.14). DOI: <https://doi.org/10.1007/s00222-022-01160-7>. Preprint: [arXiv:2001.01448](https://arxiv.org/abs/2001.01448).

4.11.5. Primary-source pointer

The originating ICM record is *Géométrie de Contact de la Dimension Trois vers les Dimensions Supérieures* (2002). The source link retained by the dataset is <https://www.mathunion.org>.

org/fileadmin/ICM/Proceedings/ICM2002.2/ICM2002.2.ocr.pdf#page=389.

4.12. 158. Géométrie de Contact de la Dimension Trois vers les Dimensions Supérieures [2002-geom-84]

Record: 2002-geom-84

Subfield: Symplectic and contact geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Géométrie de Contact de la Dimension Trois vers les Dimensions Supérieures

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.2/ICM2002.2.ocr.pdf#page=389>

4.12.1. Problem statement

Prove that, for every closed cooriented contact manifold (V, ξ) and every contact form α defining ξ , the Reeb vector field R_α characterized by $\alpha(R_\alpha) = 1$ and $\iota_{R_\alpha} d\alpha = 0$ has at least one periodic orbit.

4.12.2. Historical background, literature, and current progress

4.12.3. Status check: conjecture de Weinstein

4.12.3.1. Verdict

Partially resolved.

4.12.3.2. Precise problem

The problem asks for the full modern Weinstein conjecture for contact manifolds: for every closed cooriented contact manifold (V, ξ) , in every odd dimension, and for every contact form α with $\ker \alpha = \xi$, the Reeb vector field R_α has at least one periodic orbit.

This is not just the three-dimensional conjecture, nor only the hypersurface-of-contact-type version in a specified symplectic manifold. It quantifies over all closed cooriented contact manifolds and all defining contact forms.

4.12.3.3. Current status

No published or otherwise citable result found resolves this exact statement in all dimensions and for all closed cooriented contact manifolds. The literature proves many important special cases, including all closed contact 3-manifolds, but in dimensions at least five the general case remains only partially understood.

4.12.3.4. Key literature and what it proves

1. Alan Weinstein, “On the hypotheses of Rabinowitz’ periodic orbit theorems”, *Journal of Differential Equations* 33(3), 353-358, 1979. DOI: 10.1016/0022-0396(79)90070-6.
 - Original source associated with the Weinstein conjecture. It reformulates periodic-orbit existence in contact-type terms, following Rabinowitz’s variational results, but does not prove the full arbitrary closed contact manifold statement.
2. Claude Viterbo, “A proof of Weinstein’s conjecture in $\mathbb{R}^{2n}$ ”, *Annales de l’Institut Henri Poincaré. C, Analyse non lineaire* 4(4), 337-356, 1987. URL: https://www.numdam.org/item/AIHPC_1987__4_4_337_0/
 - Proves the conjecture for contact-type hypersurfaces in $\mathbb{R}^{2n}$. This is a major high-dimensional result, but it requires the contact manifold to occur as such a hypersurface and therefore does not cover arbitrary closed cooriented contact manifolds.
3. Helmut Hofer, “Pseudoholomorphic curves in symplectizations with applications to the Weinstein conjecture in dimension three”, *Inventiones Mathematicae* 114, 515-563, 1993. DOI: 10.1007/BF01232679.
 - Establishes important three-dimensional cases using pseudoholomorphic curves, including overtwisted contact 3-manifolds and other cases, but not all dimensions and not all closed contact manifolds.
4. Clifford Henry Taubes, “The Seiberg-Witten equations and the Weinstein conjecture”, *Geometry & Topology* 11, 2117-2202, 2007. arXiv: math/0611007. DOI: 10.2140/gt.2007.11.2117.
 - Proves that if M is a compact oriented 3-manifold and α is a contact 1-form, then the Reeb vector field has a closed integral curve. This fully resolves the problem statement in dimension three only. The proof uses dimension-specific Seiberg-Witten/Floer input, so it does not imply the arbitrary-dimensional statement.
5. Michael Hutchings, “Taubes’s proof of the Weinstein conjecture in dimension three”, *Bulletin of the American Mathematical Society* 47(1), 73-125, 2010. arXiv: 0906.2444. DOI: 10.1090/S0273-0979-09-01282-8.
 - Survey of Taubes’s proof and surrounding literature. It states the 3-dimensional theorem and notes that in higher dimensions, although the conjecture is known for compact hypersurfaces of contact type in $\mathbb{R}^{2n}$ and other cases, the general problem is wide open. This confirms that Taubes’s theorem does not settle the full all-dimensional statement.
6. Peter Albers and Helmut Hofer, “On the Weinstein conjecture in higher dimensions”, *Commentarii Mathematici Helvetici* 84(2), 429-436, 2009. arXiv: 0705.3953.
 - Proves the Weinstein conjecture for closed PS-overtwisted contact manifolds, giving a contractible periodic orbit for every supporting contact form. This is a genuine higher-dimensional family but assumes the presence of a plastikstufe/PS-overtwisted structure, so it is only partial progress.
7. Peter Albers, Urs Fuchs, and Will J. Merry, “Orderability and the Weinstein conjecture”, *Compositio Mathematica* 151(12), 2251-2272, 2015. DOI: 10.1112/S0010437X15007642.
 - Proves the Weinstein conjecture for contact manifolds whose identity component of the contactomorphism group is non-orderable, linking orderability and hypertightness. This gives another significant class but does not cover arbitrary closed contact manifolds.

8. Hansjorg Geiges and Kai Zehmisch, “The Weinstein conjecture for connected sums”, *International Mathematics Research Notices* 2016(2), 325-342, 2016. DOI: 10.1093/imrn/rnv124. arXiv: 1407.4307.
 - Proves the conjecture for non-trivial contact connected sums under topological assumptions such as non-trivial fundamental group or torsion-free homology. These hypotheses exclude many contact manifolds and therefore do not resolve the full statement.
9. Matthew Strom Borman, Yakov Eliashberg, and Emmy Murphy, “Existence and classification of overtwisted contact structures in all dimensions”, *Acta Mathematica* 215(2), 281-361, 2015. DOI: 10.1007/s11511-016-0134-4. arXiv: 1404.6157.
 - Provides the modern all-dimensional theory of overtwisted contact structures. Together with results such as Albers-Hofer for PS-overtwisted/overtwisted-type structures, it enlarges the class of high-dimensional contact manifolds where Weinstein-type conclusions are known. It does not prove the conjecture for tight or arbitrary high-dimensional contact manifolds.
10. Yanqiao Ding, “A survey on the Weinstein conjecture”, *Frontiers of Mathematics in China* 20(1), 1-15, 2025. DOI: 10.3868/s140-DDD-025-0003-x.
 - Recent survey article on variational and pseudoholomorphic-curve approaches. It lists the major known special cases and later progress, but does not report a theorem resolving the arbitrary-dimensional closed contact manifold case.
11. Lina Deschamps, Levin Maier, and Tom Stalljohann, “On the contact type conjecture for exact magnetic systems”, arXiv: 2508.01113, 2025 preprint.
 - States the Weinstein conjecture for closed contact manifolds and explicitly notes that in dimension three it was resolved by Hofer and Taubes, while in higher dimensions it remains only partially understood. It then uses the conjecture only for classes where it is known, again confirming that no general higher-dimensional resolution is available.

4.12.3.5. Why the problem is not fully resolved

Taubes’s theorem proves the exact Reeb-periodic-orbit conclusion for every contact form on every closed oriented contact 3-manifold, so the conjecture is fully solved in dimension three. However, the statement has no dimension restriction. The higher-dimensional literature found in the search proves the result under extra hypotheses: contact-type hypersurfaces in special symplectic manifolds, fillability/cobordism assumptions, PS-overtwisted or overtwisted-type structures, non-orderability, connected-sum hypotheses, or other special geometric/topological conditions.

Thus the unresolved part is: closed cooriented contact manifolds of dimension at least five with arbitrary contact structure and arbitrary supporting contact form, without the additional hypotheses appearing in the known theorems.

4.12.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website’s “Some remaining open problems” field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

4.12.5. Primary-source pointer

The originating ICM record is *Géométrie de Contact de la Dimension Trois vers les Dimensions Supérieures* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.2/ICM2002.2.ocr.pdf#page=389>.

4.13. 159. Geometry and Nonlinear Analysis [2002-geom-85]

Record: 2002-geom-85

Subfield: Symplectic and contact geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Geometry and Nonlinear Analysis

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.1/ICM2002.1.ocr.pdf#page=474>

4.13.1. Problem statement

Prove that if (M, g, ω) is a compact Kähler-Einstein complex surface with Kähler form ω , then every smooth embedded real surface $S_0 \subset M$ that is symplectic with respect to ω is isotopic in M to a smooth embedded real surface $S \subset M$ that is still symplectic with respect to ω and has zero mean curvature with respect to the Kähler-Einstein metric g .

4.13.2. Historical background, literature, and current progress

4.13.3. Verdict: partially resolved

4.13.3.1. Precise problem

The problem asks whether Tian's symplectic isotopy/minimal-surface problem holds in full:

Given any compact Kähler-Einstein complex surface (M, g, ω) and any smooth embedded real surface S_0 such that $\omega|_{S_0}$ is a positive area form, must S_0 be isotopic in M to another smooth embedded ω -symplectic surface S whose mean curvature vector with respect to g is zero?

Equivalently, every embedded symplectic surface in every compact Kähler-Einstein complex surface should have a symplectic minimal representative in its smooth isotopy class.

4.13.3.2. Original source

- Title: *Symplectic isotopy in four dimension*
- Author: Gang Tian
- Venue: *First International Congress of Chinese Mathematicians (Beijing, 1998)*, AMS/IP Studies in Advanced Mathematics 20, American Mathematical Society/International Press
- Year: 2001
- Pages: 143-147
- DOI/URL: <https://doi.org/10.1090/amsip/020/09>
- Role: This is the cited source for Tian's problem. Later papers quote it as the conjecture that every embedded orientable closed symplectic surface in a compact Kähler-Einstein surface is isotopic to a symplectic minimal surface.

4.13.3.3. Main partial progress

- Title: *Moving Symplectic Curves in Kähler-Einstein Surfaces*
- Authors: Jingyi Chen, Gang Tian
- Venue: *Acta Mathematica Sinica, English Series* 16(4), 541-548
- Year: 2000
- DOI/URL: <https://doi.org/10.1007/s101140000075>
- Result: Derives the parabolic equation for the Kähler angle along mean curvature flow in a Kähler-Einstein surface and proves that a symplectic curve remains symplectic as long as the smooth mean curvature flow exists.
- Limitation: This is preservation along an existing smooth flow, not global existence/convergence for all initial symplectic surfaces.
- Title: *Mean Curvature Flow of Surfaces in 4-Manifolds*
- Authors: Jingyi Chen, Jiayu Li
- Venue: *Advances in Mathematics* 163, 287-309
- Year: 2001
- DOI/URL: <https://doi.org/10.1006/aima.2001.2008>
- Result: Develops codimension-two mean-curvature-flow estimates used in the symplectic Kähler-Einstein setting, including preservation/monotonicity tools cited by later work.
- Limitation: Does not prove the full isotopy-to-minimal statement.
- Title: *Two-Dimensional Graphs Moving by Mean Curvature Flow*
- Authors: Jingyi Chen, Jiayu Li, Gang Tian
- Venue: *Acta Mathematica Sinica, English Series* 18(2), 209-224
- Year: 2002
- DOI/URL: <https://doi.org/10.1007/s101140200163>
- Result: Gives long-time existence and convergence in graphical cases.
- Limitation: the problem asks for arbitrary embedded symplectic surfaces, not only graphical initial data.
- Title: *Mean Curvature Flow of Surfaces in Einstein Four-Manifolds*
- Author: Mu-Tao Wang
- Venue: *Journal of Differential Geometry* 57, 301-338
- Year: 2001
- DOI/URL: <https://projecteuclid.org/euclid.jdg/1090348113>
- Result: Studies mean curvature flow of surfaces in Einstein four-manifolds; in the Kähler-Einstein/calibrated setting it is part of the standard evidence that the symplectic condition is preserved under the flow.
- Limitation: It does not establish global smooth flow and convergence for every symplectic surface.

- Title: *Singularities of Codimension Two Mean Curvature Flow of Symplectic Surfaces*
- Authors: Jingyi Chen, Jiayu Li
- Venue: arXiv preprint
- Year: 2002
- arXiv/URL: <https://arxiv.org/abs/math/0208227>
- Result: For a compact symplectic surface in a compact Kähler-Einstein surface, analyzes first singular-time tangent cones; the theorem gives tangent cones as finite unions of complex 2-planes and records no Type I singularities for symplectic mean curvature flow.
- Limitation: Type II singularities remain a central obstruction; tangent-cone control is not a proof that the flow always exists globally or reaches a smooth minimal representative.
- Title: *The Mean Curvature Flow Approach to the Symplectic Isotopy Problem*
- Authors: Xiaoli Han, Jiayu Li
- Venue: *International Mathematics Research Notices* 2005(26), 1611-1620
- Year: 2005
- DOI/arXiv/URL: <https://doi.org/10.1155/IMRN.2005.1611>; <https://arxiv.org/abs/math/0411284>
- Result: If the Kähler-Einstein surface has positive scalar curvature and the initial surface is sufficiently close to a holomorphic curve, the mean curvature flow exists globally and converges to a holomorphic curve.
- Limitation: Requires positive scalar curvature and a smallness/closeness hypothesis on the initial surface.
- Title: *Symplectic critical surfaces in Kähler surfaces*
- Authors: Xiaoli Han, Jiayu Li
- Venue: *Journal of the European Mathematical Society* 12(2), 505-527
- Year: 2010
- DOI/arXiv/URL: <https://doi.org/10.4171/JEMS/207>; <https://arxiv.org/abs/0711.2211>
- Result: Introduces a variational functional on symplectic surfaces, computes its Euler-Lagrange equation, proves ellipticity, and shows that compact symplectic critical surfaces satisfy strong minimal/constant-angle conclusions; for nonnegative scalar curvature the critical surfaces are holomorphic.
- Limitation: This is a variational framework and conditional structure theorem; it does not prove existence of such a critical/minimal surface in every isotopy class. The paper itself states the natural conjecture that each homotopy class of symplectic surfaces should contain a symplectic critical surface.
- Title: *Minimal surfaces and deformations of holomorphic curves in Kähler-Einstein manifolds*
- Author: Claudio Arezzo
- Venue: *Annali della Scuola Normale Superiore di Pisa, Classe di Scienze* 29(2), 473-481
- Year: 2000
- DOI/arXiv/URL: https://www.numdam.org/item/ASNSP_2000_4_29_

[2_473_0/](#)

- Result: Constructs stable nonholomorphic minimal surfaces in negative-scalar-curvature Kähler-Einstein four-manifolds.
- Limitation: Shows that in negative scalar curvature the target minimal surfaces need not be holomorphic, but does not prove existence in every symplectic isotopy class.
- Title: *Long time existence of the symplectic mean curvature flow*
- Authors: Xiaoli Han, Jiayu Li
- Venue: arXiv preprint
- Year: 2011
- arXiv/URL: <https://arxiv.org/abs/1107.0829>
- Result: In Kähler surfaces of constant positive holomorphic sectional curvature, proves long-time existence and convergence to a holomorphic curve under explicit curvature/Kähler-angle pinching hypotheses.
- Limitation: Applies only under strong pinching assumptions, not to all embedded symplectic surfaces.
- Title: *Symplectic mean curvature flow in CP²*
- Authors: Xiaoli Han, Jiayu Li, Liuqing Yang
- Venue: *Calculus of Variations and Partial Differential Equations* 48, 111-129
- Year: 2013
- DOI/URL: <https://doi.org/10.1007/s00526-012-0544-x>
- Result: Proves global existence and convergence for symplectic mean curvature flow in CP² under suitable curvature pinching conditions.
- Limitation: Covers a pinched subclass of initial surfaces in CP², not arbitrary symplectic surfaces.
- Title: *Gradient flow for beta-symplectic critical surfaces*
- Authors: Xiaoli Han, Jiayu Li, Jun Sun
- Venue: *Annales de l'Institut Henri Poincaré C, Analyse Non Linéaire* 41, 1083-1116
- Year: 2024
- DOI/URL: <https://doi.org/10.4171/AIHPC/100>
- Result: Studies a gradient flow for the beta-symplectic-critical-surface functional, proving preservation of the symplectic condition, a monotonicity formula, epsilon-regularity, and finite flat-plane tangent cones for the flow.
- Limitation: Still a regularity/flow framework; it does not prove existence of smooth minimal representatives for every isotopy class.
- Title: *Tangent Flows of Symplectic Mean Curvature Flows*
- Authors: Jingyi Chen, Xiaoli Han, Jiayu Li, Jun Sun
- Venue: *Journal of Mathematical Study* 59(1), 40-59
- Year: 2026
- DOI/URL: <https://doi.org/10.4208/jms.v59n1.26.03>

- Result: Recent tangent-flow progress: the tangent cone at the first blow-up time of mean curvature flow of a closed symplectic surface in a compact Kähler-Einstein surface is a finite union of planes in R^4 , and Type I* tangent cones are holomorphic cones.
- Limitation: This improves singularity analysis but still does not establish the required global smooth flow/convergence or an alternative all-class existence theorem.

4.13.3.4. Related symplectic-isotopy evidence

- Title: *On the holomorphicity of genus two Lefschetz fibrations*
- Authors: Bernd Siebert, Gang Tian
- Venue: *Annals of Mathematics* 161(2), 959-1020
- Year: 2005
- DOI/URL: <https://doi.org/10.4007/annals.2005.161.959>
- Result: Includes the strongest classical smooth CP^2 symplectic-isotopy progress: smooth symplectic plane curves are isotopic to complex curves through degree 17.
- Limitation: This is only a low-degree result. Since CP^2 with the Fubini-Study metric is Kähler-Einstein, the problem statement would in particular imply the unresolved high-degree smooth symplectic isotopy problem in CP^2 .
- Title: *A new approach to the symplectic isotopy problem*
- Author: Laura Starkston
- Venue: *Journal of Topology* 13(4), 1649-1689
- Year: 2020
- DOI/arXiv/URL: <https://doi.org/10.1112/topo.12158>; <https://arxiv.org/abs/1709.02544>
- Result: Recasts the smooth symplectic isotopy problem in CP^2 in terms of Lagrangian disk data and explicitly notes that progress after Siebert-Tian's degree-17 result had stalled.
- Limitation: Provides an equivalent/reductive approach, not a full solution.

4.13.3.5. Why the full problem is not resolved

No citable result found proves the full statement for every compact Kähler-Einstein complex surface and every embedded symplectic surface. The literature proves preservation of symplecticity along smooth mean curvature flow, analyzes singularities, and proves convergence under additional assumptions such as graphical initial data, closeness to a holomorphic curve, positive scalar curvature, CP^2 , constant holomorphic sectional curvature, or curvature/Kähler-angle pinching.

The unrestricted case remains open. In particular, the Fubini-Study CP^2 case is already a necessary subcase; for positive scalar curvature, known results relate symplectic minimal representatives to holomorphic curves, so a full proof would imply the still-open high-degree smooth symplectic isotopy problem in CP^2 . Thus the correct status is partially resolved, not fully resolved.

4.13.4. Remaining open problems recorded by the database

4.13.4.1. Open problem 1

Let $\Sigma \subset \mathbb{CP}^2$ be a smooth symplectically embedded surface representing $d[\mathbb{CP}^1]$. Must there exist $d(d-1)/2$ pairwise disjoint embedded Lagrangian disks whose interiors are disjoint from Σ and whose boundary curves form a d -splitting system on Σ , meaning that cutting Σ along them gives d genus-zero components, each with $d-1$ boundary components, with each boundary curve separating a distinct pair of components?

Reference supplied by the database. Laura Starkston, “A new approach to the symplectic isotopy problem,” *Journal of Symplectic Geometry* 18, no. 3 (2020), 939–960, Definition 1.1, Theorem 1.2, and the discussion immediately following Theorem 1.2. DOI: <https://doi.org/10.4310/JSG.2020.v18.n3.a11>. Preprint: [arXiv:1709.02544](https://arxiv.org/abs/1709.02544).

4.13.5. Primary-source pointer

The originating ICM record is *Geometry and Nonlinear Analysis* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.1/ICM2002.1.ocr.pdf#page=474>.

4.14. 160. Mirror Symmetry and Quantum Geometry (2) [2002-ag-94]

Record: 2002-ag-94

Subfield: Symplectic and contact geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Mirror Symmetry and Quantum Geometry (2)

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=422>

4.14.1. Problem statement

Prove or disprove that for any Calabi-Yau manifold X admitting a special Lagrangian torus fibration $\pi : X \rightarrow B$, the mirror of X is obtained, up to the quantum corrections required in the presence of singular fibers, by applying T-duality fiber by fiber: each smooth torus fiber $T^n = \pi^{-1}(b)$ is replaced by its dual torus $\widehat{T}^n \cong H^1(T^n, U(1))$, equivalently the moduli space of flat $U(1)$ connections, or holonomies, on T^n , and under this dualization a D-brane wrapped on a torus fiber is transformed into a D0-brane at the corresponding point of the dual fiber.

4.14.2. Historical background, literature, and current progress

4.14.3. Status check: Strominger-Yau-Zaslow (SYZ) conjecture

Verdict: partially resolved.

4.14.3.1. Precise problem checked

The problem asks for the full SYZ mirror construction: given a Calabi-Yau manifold X with a special Lagrangian torus fibration $\pi : X \rightarrow B$, the mirror should be obtained by dualizing each smooth torus fiber T^n to $\widehat{T}^n \cong H^1(T^n, U(1))$, with a mathematically precise treatment of singular fibers by quantum/instanton corrections, and with the corresponding brane transform identifying a fiber brane with a point/D0-brane on the dual fiber. This is stronger than proving only the existence of special Lagrangian tori on a large open set, or proving a topological/logarithmic analogue, or proving the construction for a particular class of examples.

4.14.3.2. Current status

No published or otherwise standard citable result proves the full statement in this generality. The literature gives major partial results: semi-flat/no-singular-fiber constructions, topological compactifications in special examples, the Gross-Siebert logarithmic/tropical reconstruction program, wall-crossing and open Gromov-Witten corrections for toric and related examples, family-Floer constructions in the absence of singular fibers and some local singular models, and recent metric SYZ results for maximal degenerations. These do not together imply the full

statement for arbitrary Calabi-Yau manifolds admitting special Lagrangian torus fibrations with singular fibers and quantum corrections.

4.14.3.3. Key literature and what it proves

- **Mirror Symmetry is T-Duality**; Andrew Strominger, Shing-Tung Yau, Eric Zaslow; *Nuclear Physics B* 479 (1996), 243-259; DOI: 10.1016/0550-3213(96)00434-8; arXiv: hep-th/9606040, <https://arxiv.org/abs/hep-th/9606040>. This is the original proposal. It argues that a mirror is the moduli space of special Lagrangian torus fibers with flat $U(1)$ -connections and that mirror symmetry is fiberwise T-duality. It is not a complete mathematical proof, especially around singular fibers.
- **Singularities of special Lagrangian fibrations and the SYZ Conjecture**; Dominic Joyce; *Communications in Analysis and Geometry* 11 (2003), no. 5, 859-907; arXiv: math/0011179, <https://arxiv.org/abs/math/0011179>. Joyce constructs and studies local examples of special Lagrangian fibrations and emphasizes that singularities are poorly understood; he argues that generic discriminants can be codimension one and that stronger/literal versions of SYZ are unlikely to be true as stated. This is evidence against treating the problem statement as fully resolved.
- **Topological Mirror Symmetry**; Mark Gross; *Inventiones Mathematicae* 144 (2001), 75-137; arXiv: math/9909015, <https://arxiv.org/abs/math/9909015>. Gross proves a topological version for certain T^3 -fibrations, including the quintic/mirror quintic at the topological level. This supports the SYZ picture but omits the special Lagrangian metric condition, analytic singular-fiber corrections, and full brane/Floer-theoretic mirror equivalence.
- **Mirror symmetry and the Strominger-Yau-Zaslow conjecture**; Mark Gross; *Current Developments in Mathematics 2012* (International Press, 2013), 133-191; DOI: 10.4310/CDM.2012.v2012.n1.a3; arXiv: 1212.4220, <https://arxiv.org/abs/1212.4220>. This survey records that the original differential-geometric conjecture had successes but that Joyce's work made the literal form unlikely; it describes the Gross-Siebert program as an algebro-geometric route inspired by SYZ, not as a proof of the full original statement.
- **From real affine geometry to complex geometry**; Mark Gross, Bernd Siebert; *Annals of Mathematics* 174 (2011), 1301-1428; DOI: 10.4007/annals.2011.174.3.1; arXiv: math/0703822, <https://arxiv.org/abs/math/0703822>. This gives a reconstruction theorem producing degenerations of Calabi-Yau manifolds from affine manifolds with singularities and scattering/tropical data. It is a major logarithmic/tropical realization of SYZ-style quantum corrections, but it does not prove that every special Lagrangian torus fibration produces the mirror by literal fiberwise T-duality across singular fibers.
- **Mirror symmetry and T-duality in the complement of an anticanonical divisor**; Denis Auroux; *Journal of Gokova Geometry Topology* 1 (2007), 51-91; arXiv: 0706.3207, <https://arxiv.org/abs/0706.3207>. Auroux studies concrete wall-crossing and quantum corrections in SYZ-type constructions. This is important special-case progress, not a general resolution.
- **SYZ mirror symmetry for toric Calabi-Yau manifolds**; Kwokwai Chan, Siu-Cheong Lau, Naichung Conan Leung; *Journal of Differential Geometry* 90 (2012), no. 2, 177-250; arXiv: 1006.3830, <https://arxiv.org/abs/1006.3830>. The authors construct SYZ mirrors of toric Calabi-Yau manifolds using T-duality corrected by open Gromov-Witten

invariants. This covers a rich special class, but not arbitrary Calabi-Yau fibrations or the full singular-fiber problem.

- **Family Floer cohomology and mirror symmetry**; Mohammed Abouzaid; arXiv preprint (2014), arXiv: 1404.2659, <https://arxiv.org/abs/1404.2659>. Abouzaid explains a family-Floer construction of the mirror from a Lagrangian torus fibration in the absence of singular fibers. The no-singular-fiber hypothesis is a substantial restriction relative to this problem.
- **Family Floer mirror space for local SYZ singularities**; Hang Yuan; *Forum of Mathematics, Sigma* 12 (2024), e119; DOI: 10.1017/fms.2024.107; arXiv: 2206.04652, <https://arxiv.org/abs/2206.04652>. Yuan gives a precise family-Floer/non-archimedean SYZ statement and proves it for toric Calabi-Yau manifolds with Gross Lagrangian fibrations, including local singular-fiber behavior. This is significant progress with singularities, but still special-case/local rather than the full conjecture.
- **Valuative independence and metric SYZ conjecture**; Yang Li; arXiv preprint (2026), arXiv: 2605.00516, <https://arxiv.org/abs/2605.00516>, together with **Valuative independence for Calabi-Yau varieties**; Harold Blum, Yuchen Liu; arXiv preprint (2026), arXiv: 2604.27890, <https://arxiv.org/abs/2604.27890>. These recent works address the metric SYZ conjecture for polarized maximal degenerations: roughly, the existence of a special Lagrangian T^n -fibration on an open subset of X_t of measure at least $1 - \delta$. Even if counted as resolving this metric formulation, it is not the same as the full mirror construction: it does not provide a global dual mirror with singular-fiber quantum corrections and the D-brane/D0-brane transform in all cases.

4.14.3.4. Claimed full-resolution result not counted

The search also found **The SYZ Conjecture through Computational Equivalence**; Logan Nye; *Advances in Pure Mathematics* 15 (2025), no. 2, 145-179; DOI: 10.4236/apm.2025.152007. Although it claims to prove SYZ, its conclusion is formulated through a quantum-complexity/computational-equivalence framework and does not supply a standard theorem in differential, symplectic, or algebraic geometry proving the full fiberwise dual torus construction with singular-fiber corrections. It therefore does not resolve the stated mathematical question.

4.14.3.5. What remains unresolved

The unresolved part is the general, precise, global SYZ theorem: starting from a special Lagrangian torus fibration with singular fibers on an arbitrary relevant Calabi-Yau manifold, construct the mirror by fiberwise dualization plus quantum corrections, prove that the corrected compactified dual is the mirror in the full mathematical sense intended by mirror symmetry, and establish the corresponding brane/Fourier-Mukai/family-Floer transform across singular fibers. Current results prove this package only in restricted settings or in weakened/reformulated versions.

4.14.4. Remaining open problems recorded by the database

4.14.4.1. Open problem 1

Prove that the two piecewise-smooth special Lagrangian fibrations F and F' constructed by Joyce are the generic local models for codimension-one singularities of special Lagrangian fibrations of almost Calabi–Yau threefolds: locally the fibrations should have the same topology, their fibres should approach those of F or F' , and the singular fibres should have conical singularities modeled on Joyce’s special Lagrangian T^2 -cone $C \subset \mathbb{C}^3$.

Reference supplied by the database. Dominic Joyce, “Singularities of Special Lagrangian Fibrations and the SYZ Conjecture,” *Communications in Analysis and Geometry* **11** (2003), no. 5, 859–907, Conjecture 5.5. arXiv: [math/0011179](https://arxiv.org/abs/math/0011179); journal PDF: [International Press](#).

4.14.4.2. Open problem 2

Prove that Joyce’s piecewise-smooth special Lagrangian fibration $\hat{F}$ is a generic local model for a codimension-two singularity of a special Lagrangian fibration of an almost Calabi–Yau threefold, including the transition where two multiplicity-one conical singular points merge into a multiplicity-two nonconical singularity.

Reference supplied by the database. Dominic Joyce, “Singularities of Special Lagrangian Fibrations and the SYZ Conjecture,” *Communications in Analysis and Geometry* **11** (2003), no. 5, 859–907, Conjecture 6.7. arXiv: [math/0011179](https://arxiv.org/abs/math/0011179); journal PDF: [International Press](#).

4.14.4.3. Open problem 3

Prove that a generic special Lagrangian fibration of an almost Calabi–Yau threefold which contains singular fibres is never a smooth map, but only a piecewise-smooth map; in particular, under a small generic deformation, the codimension-two graph-like discriminant of a smooth fibration should thicken into codimension-one ribbons.

Reference supplied by the database. Dominic Joyce, “Singularities of Special Lagrangian Fibrations and the SYZ Conjecture,” *Communications in Analysis and Geometry* **11** (2003), no. 5, 859–907, Conjecture 7.15 and the discussion immediately following it. arXiv: [math/0011179](https://arxiv.org/abs/math/0011179); journal PDF: [International Press](#).

4.14.4.4. Open problem 4

Prove or disprove that for generic mirror Calabi–Yau threefolds M and $\widehat{M}$, even when special Lagrangian fibrations $f : M \rightarrow B$ and $\widehat{f} : \widehat{M} \rightarrow \widehat{B}$ exist, in general there is no homeomorphism $B \cong \widehat{B}$ which simultaneously identifies their discriminant loci and makes their nonsingular 3-torus fibres have dual homology.

Reference supplied by the database. Dominic Joyce, “Singularities of Special Lagrangian Fibrations and the SYZ Conjecture,” *Communications in Analysis and Geometry* **11** (2003), no. 5, 859–907, Conjecture 8.1. arXiv: [math/0011179](https://arxiv.org/abs/math/0011179); journal PDF: [International Press](#).

4.14.4.5. Open problem 5

Construct the family-Floer mirror space for a higher-dimensional Lagrangian torus fibration whose singular fibres are not Lagrangian immersions, by defining and controlling the Floer theory of those singular fibres and gluing it compatibly to the dual-torus charts over the smooth locus.

Reference supplied by the database. Mohammed Abouzaid, “Family Floer Cohomology and Mirror Symmetry,” in *Proceedings of the International Congress of Mathematicians—Seoul 2014*, Vol. II, Kyung Moon Sa, Seoul, 2014, pp. 813–836, especially Introduction and §1.2, item (2), which identifies nonimmersed higher-dimensional singular fibres as beyond existing family-Floer techniques. arXiv: [1404.2659](https://arxiv.org/abs/1404.2659).

4.14.4.6. Open problem 6

Let $\mathcal{X} \rightarrow D$ be a polarized toric degeneration with intersection complex $(\check{B}, \check{D})$, and let ω_t represent the first Chern class of the polarization on $\mathcal{X}_t$. Prove that there is an open set $\check{U} \subseteq \check{B}$, whose complement retracts onto the discriminant locus of $\check{B}$, such that every smooth fibre $\mathcal{X}_t$ is a symplectic compactification of the semi-flat torus bundle $\check{X}(\check{U})$.

Reference supplied by the database. Mark Gross, “Mirror Symmetry and the Strominger–Yau–Zaslow Conjecture,” in *Current Developments in Mathematics 2012*, International Press, 2013, pp. 133–191, conjecture in §4 following the construction of the intersection complex (the polarized-toric-degeneration symplectic compactification conjecture). DOI: <https://doi.org/10.4310/CDM.2012.v2012.n1.a3>; arXiv: [1212.4220](https://arxiv.org/abs/1212.4220).

4.14.5. Primary-source pointer

The originating ICM record is *Mirror Symmetry and Quantum Geometry* (2) (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=422>.

4.15. 161. Fukaya Categories and Deformations [2002-geom-278]

Record: 2002-geom-278

Subfield: Symplectic and contact geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Fukaya Categories and Deformations

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.2/ICM2002.2.ocr.pdf#page=338>

4.15.1. Problem statement

Let Y be a smooth complex projective variety, and let $\{X_z\}_{z \in \mathbb{P}^1}$ be a Lefschetz pencil of hyperplane sections of Y . Assume that $X = X_0$ is smooth, that its divisor class satisfies $K_Y \cong \mathcal{O}_Y(-X)$, and that the base locus $D = X_0 \cap X_\infty$ is a smooth hyperplane section of X . Let $U \subset X$ be a sufficiently small open neighborhood of D , and equip $M = X \setminus U$ with the restriction of the Fubini–Study Kähler form. Then M is a compact exact symplectic manifold with contact-type boundary and $c_1(M) = 0$.

Let $\mathcal{F}(M)$ be the A_∞ Fukaya category over $\mathbb{Q}$ whose objects are graded exact Lagrangian submanifolds of M , let $SH^*(M)$ be the symplectic cohomology of M , and let $HH^*(\mathcal{F}(M), \mathcal{F}(M))$ be the Hochschild cohomology of $\mathcal{F}(M)$. Consider the canonical open–closed string map

$$SH^*(M) \longrightarrow HH^*(\mathcal{F}(M), \mathcal{F}(M)),$$

defined by counting inhomogeneous pseudoholomorphic maps from disks with one interior puncture and an arbitrary number of boundary punctures, using near the interior puncture the Hamiltonian perturbation that defines $SH^*(M)$ and imposing along the boundary the Lagrangian boundary conditions that define $\mathcal{F}(M)$. Prove that this map is an isomorphism.

4.15.2. Historical background, literature, and current progress

The exact assertion is **unresolved**. Seidel formulated it as Conjecture 4 for the compact exact Fukaya category whose objects are closed graded exact Lagrangians. No result located proves that canonical map is an isomorphism for every complement arising from an anticanonical Lefschetz pencil as in the statement.

The strongest general progress concerns a different category. Ganatra proves that, for a non-degenerate Liouville manifold, the canonical closed–open map

$$SH^*(M) \longrightarrow HH^*(\mathcal{W}(M), \mathcal{W}(M))$$

is an isomorphism for the **wrapped** Fukaya category (Theorem 1.1). Chantraine–Dimitroglou Rizell–Ghiggini–Golovko prove that cocores generate the wrapped Fukaya category of every finite-type Weinstein manifold and that its open–closed map to symplectic cohomology is an isomorphism (Theorems 1.1 and 1.4). Together these results establish the wrapped-category analogue for the Weinstein completions occurring here. They do not identify $HH^*(\mathcal{W}(M))$

with $HH^*(\mathcal{F}(M))$, and the compact and wrapped Fukaya categories are not interchangeable in this assertion.

There is substantive compact-category progress in special cases. Lekili–Ueda prove that for the Milnor fiber $\check{V}$ of a weighted-homogeneous isolated hypersurface singularity with grading defect $\check{d}_0 = \check{h} - \sum_i \check{d}_i \neq 0$, one has $HH^*(\mathcal{W}(\check{V})) \cong HH^*(\mathcal{F}(\check{V}))$ (Theorem 6.2 and Corollary 6.7); combining this with the wrapped result yields $SH^*(\check{V}) \cong HH^*(\mathcal{F}(\check{V}))$ (Corollary 6.4), which they explicitly identify as a proof of Seidel’s Conjecture 4 in that case. This does not cover arbitrary Y , pencil, and smooth base locus in the statement. The remaining gap is a general argument for the compact exact Fukaya category, or a comparison theorem strong enough to transfer the wrapped isomorphism to it, under precisely the stated hypotheses.

Key references:

- Paul Seidel, “Fukaya Categories and Deformations,” *Proceedings of the International Congress of Mathematicians, Beijing 2002*, Vol. II, Higher Education Press, 2002, pp. 351–360, Conjecture 4; arXiv:math/0206155. This is the source of the exact compact-category assertion.
- Sheel Ganatra, “Symplectic Cohomology and Duality for the Wrapped Fukaya Category,” Ph.D. thesis, Massachusetts Institute of Technology, 2012; arXiv:1304.7312 (2013), Theorem 1.1. This proves both string maps are isomorphisms under non-degeneracy, but for $\mathcal{W}(M)$.
- Baptiste Chantraine, Georgios Dimitroglou Rizell, Paolo Ghiggini, and Roman Golovko, “Geometric Generation of the Wrapped Fukaya Category of Weinstein Manifolds and Sectors,” *Annales Scientifiques de l’École Normale Supérieure* 57 (2024), no. 1, 1–85, Theorems 1.1 and 1.4; DOI: 10.24033/asens.2570; arXiv:1712.09126. This removes the non-degeneracy gap for finite-type Weinstein manifolds, again in the wrapped setting.
- Yankı Lekili and Kazushi Ueda, “Homological Mirror Symmetry for Milnor Fibers via Moduli of A_∞ -Structures,” *Journal of Topology* 15 (2022), no. 3, 1058–1106, Theorem 6.2 and Corollaries 6.4 and 6.7; DOI: 10.1112/topo.12248; arXiv:1806.04345. This proves the compact-category assertion for the stated class of weighted-homogeneous Milnor fibers.

4.15.3. Remaining open problems recorded by the database

4.15.3.1. Open problem 1

Let

$$w(x_1, \dots, x_n) = \sum_{i=1}^n \prod_{j=1}^n x_j^{a_{ij}} \in \mathbb{C}[x_1, \dots, x_n]$$

be an invertible polynomial: the exponent matrix $A = (a_{ij})$ has nonnegative integer entries and nonzero determinant, and w is weighted homogeneous with an isolated critical point at the origin. Define its Berglund–Hübsch transpose by

$$\check{w}(x_1, \dots, x_n) = \sum_{i=1}^n \prod_{j=1}^n x_j^{a_{ji}},$$

and let $\check{V}_{\check{w}} = \{\check{w} = 1\} \subset \mathbb{C}^n$, equipped with its standard exact symplectic structure and

Liouville completion. Let

$$\Gamma_w = \left\{ (t_0, t_1, \dots, t_n) \in (\mathbb{C}^*)^{n+1} \mid \prod_{j=1}^n t_j^{a_{1j}} = \dots = \prod_{j=1}^n t_j^{a_{nj}} = t_0 t_1 \dots t_n \right\}.$$

Let $\text{mf}([\mathbb{A}^{n+1}/\Gamma_w], w + x_0 x_1 \dots x_n)$ denote the idempotent completion of the dg category of Γ_w -equivariant matrix factorizations of $w + x_0 x_1 \dots x_n$, and let $\mathcal{W}(\check{V}_{\check{w}})$ denote the wrapped Fukaya category of $\check{V}_{\check{w}}$. Prove that there is a quasi-equivalence

$$\text{mf}([\mathbb{A}^{n+1}/\Gamma_w], w + x_0 x_1 \dots x_n) \simeq \mathcal{W}(\check{V}_{\check{w}}).$$

\end{openproblem}

Reference supplied by the database. Yankı Lekili and Kazushi Ueda, “Homological Mirror Symmetry for Milnor Fibers via Moduli of A_∞ -Structures,” *Journal of Topology* 15 (2022), no. 3, 1058–1106, Conjecture 1.4, DOI: 10.1112/topo.12248; arXiv:1806.04345.

4.15.4. Primary-source pointer

The originating ICM record is *Fukaya Categories and Deformations* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.2/ICM2002.2.ocr.pdf#page=338>.

4.16. 162. Fukaya Categories and Deformations [2002-geom-279]

Record: 2002-geom-279

Subfield: Symplectic and contact geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Fukaya Categories and Deformations

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.2/ICM2002.2.ocr.pdf#page=338>

4.16.1. Problem statement

Let Y be a smooth complex projective variety and let $X = X_0 \subset Y$ be a smooth hyperplane section such that $K_Y \cong \mathcal{O}_Y(-X)$. Suppose that X_0 belongs to a Lefschetz pencil $\{X_z\}_{z \in \mathbb{P}^1}$ of such hyperplane sections whose smooth base locus is $D = X_0 \cap X_\infty$. Thus K_X is trivial. Equip X with the restriction of the Fubini–Study Kähler form, choose a sufficiently small tubular neighbourhood U of D , and put $M = X \setminus U$, with the restricted symplectic form; then M is a compact exact symplectic manifold with contact-type boundary and $c_1(M) = 0$.

Let $\mathcal{F}(M \subset X)$ be the curved A_∞ -deformation over $\mathbb{Q}[[t]]$ of the exact Fukaya category $\mathcal{F}(M)$ obtained by using the same graded Spin exact Lagrangian submanifolds of the interior of M as objects and by counting pseudoholomorphic polygons in X , with a polygon having intersection multiplicity k with D weighted by t^k . Its special fibre at $t = 0$ is $\mathcal{F}(M)$. Define its generic fibre $\mathcal{F}(M \subset X)_{\text{gen}}$ by adjoining to each object a degree-one element α of positive t -adic order satisfying the Maurer–Cartan equation

$$\mu^0 + \mu^1(\alpha) + \mu^2(\alpha, \alpha) + \dots = 0,$$

and then inverting t ; this is an A_∞ -category over $\mathbb{Q}((t)) = \mathbb{Q}[t^{-1}][[t]]$. Let Λ_t be the Novikov field over $\mathbb{Q}$ consisting of formal sums $\sum_i a_i t^{\lambda_i}$ with $a_i \in \mathbb{Q}$, $\lambda_i \in \mathbb{R}$, and $\lambda_i \rightarrow +\infty$, and let $\mathcal{F}(X)$ be the Fukaya A_∞ -category of the closed symplectic manifold X over Λ_t . For an A_∞ -category $\mathcal{A}$, write

$$D^\pi(\mathcal{A}) = H^0(\text{Tw}^\pi(\mathcal{A}))$$

for the triangulated category obtained from $\mathcal{A}$ by forming twisted complexes and then taking the idempotent completion.

Prove that there is a canonical equivalence of Λ_t -linear triangulated categories

$$D^\pi(\mathcal{F}(M \subset X)_{\text{gen}} \otimes_{\mathbb{Q}((t))} \Lambda_t) \cong D^\pi(\mathcal{F}(X)).$$

4.16.2. Historical background, literature, and current progress

The assertion is unresolved in the stated generality. No source found proves that, for every smooth anticanonical hyperplane section in every smooth projective Y satisfying the Lefschetz-pencil hypotheses, the generic fibre of Seidel’s relative Fukaya category is canonically equivalent after split-closure to the entire closed Fukaya category. The missing general step is

not merely the construction of the relative category: one must also compare its coefficient-specialized generic fibre with the closed Fukaya category and prove that the Lagrangians lying in the divisor complement split-generate all unobstructed objects of the latter.

Key references and progress:

- Paul Seidel, “Fukaya categories and deformations,” *Proceedings of the International Congress of Mathematicians (Beijing)*, Vol. II, 2002, pp. 351–360, arXiv:math/0206155, <https://arxiv.org/abs/math/0206155>. Conjecture 5 is the exact assertion at issue. Seidel identifies split-generation of the closed Fukaya category by the pencil’s vanishing cycles as the intended mechanism, but gives no proof in general.
- Paul Seidel, “Homological Mirror Symmetry for the Quartic Surface,” *Memoirs of the American Mathematical Society* 236, no. 1116 (2015), DOI: 10.1090/memo/1116, arXiv:math/0310414. This proves the quartic-K3 case of homological mirror symmetry using a relative Fukaya category. In particular, Corollary 9.6 proves that the relevant vanishing cycles split-generate the affine, relative, and closed split-closed derived Fukaya categories in the surface setup considered there. It realizes the proposed mechanism in a major special case, not for arbitrary Y or arbitrary dimension.
- Yuichi Nohara and Kazushi Ueda, “Homological mirror symmetry for the quintic 3-fold,” *Geometry & Topology* 16 (2012), 1967–2001, DOI: 10.2140/gt.2012.16.1967, arXiv:1103.4956. Theorem 1.1 proves HMS for the quintic threefold, and the proof establishes split-generation of its closed Fukaya category by a distinguished collection of vanishing cycles. This is another special case of the generation strategy, restricted to the quintic-pencil geometry.
- Nicholas Sheridan, “Homological Mirror Symmetry for Calabi–Yau Hypersurfaces in Projective Space,” *Inventiones Mathematicae* 199 (2015), 1–186, DOI: 10.1007/s00222-014-0507-2, arXiv:1111.0632. This develops and computes the relative Fukaya category for Calabi–Yau hypersurfaces in projective space and proves HMS in complex dimension at least three. Together with later automatic-generation results, it supplies broad projective-hypersurface special cases, but it does not cover anticanonical sections of an arbitrary smooth projective Y .
- Timothy Perutz and Nick Sheridan, “Automatic split-generation for the Fukaya category,” *International Mathematics Research Notices* 2023, no. 19, 16708–16747, DOI: 10.1093/imrn/rnac363, arXiv:1510.03848. Theorems 1.1, 1.2, and 3.4 give open–closed/closed–open criteria which force a full subcategory to split-generate a compact Calabi–Yau Fukaya category. These are conditional criteria: the required nonvanishing or injectivity is not proved for the vanishing cycles of every pencil in the present statement.
- Timothy Perutz and Nick Sheridan, “Constructing the relative Fukaya category,” *Journal of Symplectic Geometry* 21, no. 5 (2023), 997–1076, arXiv:2203.15482, <https://arxiv.org/abs/2203.15482>. This rigorously constructs Seidel’s relative Fukaya category for semipositive smooth projective varieties, including the Calabi–Yau case, and proves that its special fibre is the exact Fukaya category of the complement. It does not prove the generic-fibre equivalence with the full closed Fukaya category; its introduction states that further properties needed for applications remain for future work.
- Nick Sheridan, “Constructing the big relative Fukaya category, and its open–closed maps,” preprint, 2025, arXiv:2511.01656, <https://arxiv.org/abs/2511.01656>. This constructs the big relative category and its open–closed and closed–open maps and proves Abouzaid’s split-generation criterion in that setting. The introduction ex-

PLICITLY says that it does not yet verify all assumptions required by the cited automatic-generation and HMS applications. Thus it supplies important foundational progress but no unconditional proof of the present assertion.

- **Sheel Ganatra, Andrew Hanlon, Jeff Hicks, Daniel Pomerleano, and Nick Sheridan, “Homological mirror symmetry for Batyrev mirror pairs,” preprint, 2024, arXiv:2406.05272, <https://arxiv.org/abs/2406.05272>.** Theorem B constructs a fully faithful HMS embedding for a large class of Calabi–Yau hypersurfaces in toric varieties. Its upgrade to full HMS is explicitly conditional on additional relative-Fukaya-category hypotheses needed for split-generation. Even after those hypotheses are supplied, the theorem concerns a restricted toric class and cannot exhaust the varieties allowed in the present statement.

Accordingly, the literature establishes the relative category, general split-generation criteria, and substantial hypersurface and toric special cases. It does not establish the required canonical equivalence for all data covered by the statement.

4.16.3. Remaining open problems recorded by the database

4.16.3.1. Open problem 1

Let Y be a smooth complex projective variety and let $X = X_0 \subset Y$ be a smooth hyperplane section such that $K_Y \cong \mathcal{O}_Y(-X)$. Suppose that X_0 belongs to a Lefschetz pencil $\{X_z\}_{z \in \mathbb{P}^1}$ with smooth base locus $D = X_0 \cap X_\infty$. Equip X with the restricted Fubini–Study Kähler form, choose a sufficiently small tubular neighbourhood U of D , and set $M = X \setminus U$. Let $\mathcal{F}(M \subset X)_{\text{gen}}$ be the generic fibre over $\mathbb{Q}((t))$ of the relative Fukaya category whose objects are graded Spin exact Lagrangian submanifolds of M equipped with positive-order Maurer–Cartan elements and whose operations count pseudoholomorphic polygons in X , weighted by t to their intersection multiplicity with D . Let Λ_t be the universal Novikov field over $\mathbb{Q}$, and let $\mathcal{F}(X)$ be the Fukaya A_∞ -category of X over Λ_t . Construct a cohomologically full and faithful Λ_t -linear A_∞ -functor

$$\mathcal{F}(M \subset X)_{\text{gen}} \otimes_{\mathbb{Q}((t))} \Lambda_t \hookrightarrow \mathcal{F}(X)$$

which sends each Lagrangian brane in M with its Maurer–Cartan element to the corresponding Lagrangian brane in X with the induced bounding cochain, and prove that this functor is compatible with the open–closed string maps.

Reference supplied by the database. Timothy Perutz and Nick Sheridan, “Automatic split-generation for the Fukaya category,” *International Mathematics Research Notices* 2023, no. 19 (2023), 16708–16747, Conjecture 1.8, DOI: 10.1093/imrn/rnac363; arXiv:1510.03848, <https://arxiv.org/abs/1510.03848>.

4.16.3.2. Open problem 2

Let Y be a smooth complex projective variety and let $X = X_0 \subset Y$ be a smooth hyperplane section such that $K_Y \cong \mathcal{O}_Y(-X)$. Suppose that X_0 belongs to a Lefschetz pencil $\{X_z\}_{z \in \mathbb{P}^1}$ with smooth base locus $D = X_0 \cap X_\infty$. Equip X with the restricted Fubini–Study Kähler form, choose a sufficiently small tubular neighbourhood U of D , and set $M = X \setminus U$. Choose a distinguished basis of vanishing paths for the pencil, and let $S_1, \dots, S_m \subset M$ be the associated graded Spin Lagrangian vanishing spheres, equipped with the brane and bounding-cochain data that make them objects of the closed Fukaya A_∞ -category $\mathcal{F}(X)$ over the universal Novikov field. Prove that $S_1, \dots, S_m$ split-generate

$$D^\pi(\mathcal{F}(X)) = H^0(\mathrm{Tw}^\pi(\mathcal{F}(X)));$$

that is, every object of this category belongs to the smallest full triangulated idempotent-closed subcategory containing $S_1, \dots, S_m$.

Reference supplied by the database. Paul Seidel, “Fukaya categories and deformations,” Proceedings of the International Congress of Mathematicians (Beijing), Vol. II (2002), 351–360, discussion following Conjecture 5, arXiv:math/0206155, <https://arxiv.org/abs/math/0206155>.

4.16.4. Primary-source pointer

The originating ICM record is *Fukaya Categories and Deformations* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.2/ICM2002.2.ocr.pdf#page=338>.

4.17. 163. Symplectic field theory and its applications [2006-geom-102]

Record: 2006-geom-102

Subfield: Symplectic and contact geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2006: Symplectic field theory and its applications

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.1/ICM2006.1.ocr.pdf#page=227>

4.17.1. Problem statement

Prove that for a Stein fillable contact manifold (V, ξ) , the cylindrical contact homology $CH_*(V, \xi)$ is well defined; that is, it is independent of the auxiliary choices of contact form λ , almost complex structure J , and augmentation used to form the linearized contact-homology complex. Determine whether the same independence holds more generally for symplectically fillable contact manifolds (V, ξ) .

4.17.2. Historical background, literature, and current progress

4.17.3. Verdict: partially resolved

4.17.3.1. Precise problem

The problem asks whether the cylindrical/linearized contact homology $CH_*(V, \xi)$ of a contact manifold is a well-defined contact invariant, independent of the contact form λ , compatible almost complex structure J , and the augmentation used to linearize the contact homology DGA. The asserted target range is at least all Stein fillable contact manifolds, and possibly all symplectically fillable contact manifolds.

This is stronger than the now-standard statement that the full contact homology algebra is well defined: it asks for a single cylindrical/linearized homology independent of augmentation, and for a theorem covering all Stein fillable or all symplectically fillable cases.

4.17.3.2. Original source

- Title: "Symplectic field theory and its applications"
- Author: Yakov Eliashberg
- Venue/year: International Congress of Mathematicians, Vol. I, European Mathematical Society, Zurich, 2007, pp. 217-246; ICM Madrid 2006 lecture
- DOI/URL: <https://doi.org/10.4171/022-1/10> ; PDF source found at <https://www.staff.science.uu.nl/~crain101/Floer/symp13.pdf>
- Relevant content: Eliashberg states that if the contact homology algebra has a unique augmentation, then the linearized homology is independent of λ, J, ϵ , and says it seems likely that cylindrical contact homology is well defined at least for all Stein fillable, per-

haps symplectically fillable, contact manifolds. The same passage lists only examples then known, including subcritical Stein-fillable manifolds with $c_1(\xi) = 0$.

4.17.3.3. Current progress

- Title: “Cylindrical contact homology of subcritical Stein-fillable contact manifolds”
- Author: Mei-Lin Yau
- Venue/year: Geometry & Topology 8 (2004), no. 3, 1243-1280
- DOI/arXiv/URL: <https://doi.org/10.2140/gt.2004.8.1243> ; arXiv:math/0409542
- Result/status: Computes and establishes cylindrical contact homology for subcritical Stein-fillable contact manifolds under the additional hypothesis $c_1(\xi) = 0$. This is a significant special case, but it does not cover all Stein fillable manifolds, nor symplectically fillable manifolds in general.
- Title: “ S^1 -equivariant symplectic homology and linearized contact homology”
- Authors: Frederic Bourgeois and Alexandru Oancea
- Venue/year: International Mathematics Research Notices 2017, no. 13, 3849-3937
- DOI/arXiv/URL: <https://doi.org/10.1093/imrn/rnw029> ; arXiv:1212.3731
- Result/status: Shows that, over rational coefficients, positive S^1 -equivariant symplectic homology is isomorphic to linearized contact homology when the latter is defined, and gives a rigorous S^1 -equivariant substitute. This supplies a powerful replacement and comparison theorem, but not a theorem that cylindrical/linearized contact homology is independent of augmentation for all Stein or symplectically fillable contact manifolds.
- Title: “Contact homology and virtual fundamental cycles”
- Author: John Pardon
- Venue/year: Journal of the American Mathematical Society 32 (2019), 825-919
- DOI/arXiv/URL: <https://doi.org/10.1090/jams/924> ; arXiv:1508.03873
- Result/status: Constructs contact homology in the Eliashberg-Givental-Hofer sense using coherent virtual fundamental cycles. This resolves the basic foundations of the full contact homology DGA/algebra as an invariant, but it does not by itself prove that the linearized homology for an augmentation is independent of the augmentation in the sense asked in the problem statement.
- Title: “Semi-global Kuranishi charts and the definition of contact homology”
- Authors: Erkao Bao and Ko Honda
- Venue/year: Advances in Mathematics 414 (2023), article 108864
- DOI/arXiv/URL: <https://doi.org/10.1016/j.aim.2023.108864> ; arXiv:1512.00580
- Result/status: Defines the contact homology algebra for any contact manifold via semi-global Kuranishi charts and proves invariance under auxiliary choices. Like Pardon, this addresses full contact homology, not the full augmentation-independence claim for cylindrical/linearized contact homology in all Stein/symplectically fillable cases.

- Title: “Contact Homology and Linearization Without DGA Homotopies”
- Author: Julian Chaidez
- Venue/year: International Mathematics Research Notices 2025, no. 21, article rnaf325
- DOI/arXiv/URL: <https://doi.org/10.1093/imrn/rnaf325> ; arXiv:2412.20676
- Result/status: Proves that the set of weak-equivalence classes of augmentations of a closed contact manifold is well defined up to canonical bijection, and that the linearized contact homology associated to an augmentation class is well defined up to non-canonical isomorphism. It also proves augmentation-independence for strongly asymptotically dynamically convex (SADC) contact manifolds, and notes that subcritical Stein-fillable contact manifolds with zero first Chern class are SADC. However, it explicitly does not supply a single augmentation-independent linearized contact homology for arbitrary Stein fillable or symplectically fillable contact manifolds.
- Title: “Contact manifolds with flexible fillings”
- Author: Oleg Lazarev
- Venue/year: Geometric and Functional Analysis 30 (2020), no. 1, 188-254
- DOI/arXiv/URL: <https://doi.org/10.1007/s00039-020-00524-6> ; arXiv:1610.04837
- Result/status: Develops asymptotically dynamically convex contact structures and proves filling-independence results for positive symplectic homology in that setting, including flexible-fillable cases under hypotheses. This is important partial progress related to the linearized contact-homology substitute, but it is not all Stein fillable or all symplectically fillable contact manifolds.
- Title: “Contact homology of contact manifolds and its applications”
- Author: Frederic Bourgeois
- Venue/year: arXiv preprint/survey arXiv:2504.16540, 2025; listed by the author as for a *Celebratio Mathematica* volume on Yakov Eliashberg
- DOI/arXiv/URL: <https://arxiv.org/abs/2504.16540>
- Result/status: A current survey explicitly distinguishes full contact homology, cylindrical contact homology, and linearized contact homology. It states that cylindrical contact homology is well defined under extra Reeb-index hypotheses excluding problematic rigid planes, and that linearized contact homology depends on the augmentation, while the collection over all augmentations is a contact invariant. This matches the conclusion that the full statement is not presently proved.

4.17.3.4. What remains unresolved

No citable theorem found proves the exact statement for all Stein fillable contact manifolds, let alone all symplectically fillable contact manifolds. The literature now gives rigorous full contact homology for general contact manifolds and rigorous invariance of the collection of linearized homologies over augmentation classes. It also gives augmentation uniqueness/independence for special classes such as SADC contact manifolds, including subcritical Stein-fillable cases with $c_1 = 0$, and related filling-independence results for positive symplectic homology under ADC/flexible hypotheses.

The missing part is the full augmentation-independence and cylindrical/linearized invariant claimed in the problem statement for every Stein fillable contact manifold, and a fortiori for every symplectically fillable contact manifold. Therefore the problem is partially resolved, not fully resolved.

4.17.4. Remaining open problems recorded by the database

4.17.4.1. Open problem 1

Is cylindrical contact homology well defined for every subcritical Stein-fillable contact manifold without assuming that the first Chern class $c_1(\xi)$ vanishes?

Reference supplied by the database. Yakov Eliashberg, “Symplectic field theory and its applications,” in *Proceedings of the International Congress of Mathematicians, Madrid 2006*, Vol. I, European Mathematical Society, Zürich, 2007, pp. 217–246, §3.2, example (a) following formula (43). DOI: <https://doi.org/10.4171/022-1/10>.

4.17.5. Primary-source pointer

The originating ICM record is *Symplectic field theory and its applications* (2006). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.1/ICM2006.1.ocr.pdf#page=227>.

4.18. 164. Gromov-Witten invariants and moduli spaces of curves [2006-ag-108]

Record: 2006-ag-108

Subfield: Symplectic and contact geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2006: Gromov-Witten invariants and moduli spaces of curves

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.2/ICM2006.2.ocr.pdf#page=801>

4.18.1. Problem statement

Let V^{2d} be a compact symplectic manifold with a compatible almost-complex structure such that $H^{\text{odd}}(V; \mathbb{C}) = 0$, and fix a basis $\{\gamma_1, \dots, \gamma_N\}$ of $H^*(V; \mathbb{C})$ with γ_1 the identity. For $g \geq 0$, let $F_g(t)$ be the genus- g Gromov-Witten potential, viewed as a formal power series in the big phase-space variables t_n^α with coefficients in the Novikov ring:

$$F_g(t) = \sum_{k \geq 0} \frac{1}{k!} \sum_{\substack{\alpha_1, \dots, \alpha_k \in \{1, \dots, N\} \\ n_1, \dots, n_k \geq 0}} t_{n_1}^{\alpha_1} \cdots t_{n_k}^{\alpha_k} \sum_{A \in H_2(V; \mathbb{Z})} q^A \langle \tau_{n_1}(\gamma_{\alpha_1}) \cdots \tau_{n_k}(\gamma_{\alpha_k}) \rangle_{g, A}.$$

Assume that V has semisimple quantum cohomology in the sense that, for the big quantum product

$$\omega_1 \circ \omega_2 = \llbracket \omega_1 \omega_2 \gamma^\alpha \rrbracket_0 \gamma_\alpha,$$

the endomorphism $W \mapsto \mathcal{X} \circ W$ of the span of the primary fields γ_α has distinct eigenvalues at generic points of the big phase space; here $\llbracket \cdots \rrbracket_0$ denotes genus-0 correlation functions, γ^α is dual to γ_α under the Poincaré pairing, and $\mathcal{X}$ is the Euler vector field determined by the grading and by cup product with $c_1(V)$.

Prove that, for every integer $g \geq 2$, F_g is explicitly recoverable from the universal equations, namely the partial differential equations for $F_0, \dots, F_g$ obtained by pulling tautological relations among powers of ψ -classes and boundary classes on the Deligne-Mumford spaces $\overline{M}_{h,k}$ of stable curves back to the stable-map moduli spaces $\overline{M}_{h,k}(V, A)$.

4.18.2. Historical background, literature, and current progress

4.18.3. Verdict: partially resolved

4.18.3.1. Precise problem

The problem asks whether, for a compact symplectic target V with no odd cohomology and generically semisimple big quantum cohomology, every higher-genus descendant potential F_g for $g \geq 2$ is explicitly recoverable from the universal differential equations obtained by pulling tautological/topological-recursion relations on $\overline{M}_{h,k}$ back to stable-map moduli spaces.

This is not just the broader Givental-Teleman semisimple reconstruction statement. The ex-

act issue is whether the universal equations coming from tautological relations are known to determine F_g in all genera.

4.18.3.2. Current status

No published theorem found proves the exact all-genus statement via universal equations/topological recursion relations. The literature gives substantial partial progress:

- genus 1 is known by Dubrovin-Zhang;
- genus 2 is known by Xiaobo Liu, using universal equations/topological recursion methods;
- Givental-Teleman proves a closely related but different all-genus semisimple reconstruction theorem for CohFT/Gromov-Witten theory;
- recent papers continue to produce new universal equations and still describe the all-universal-equations determination route as conjectural.

Therefore the correct status for this problem is partially resolved.

4.18.3.3. Key literature

1. Xiaobo Liu, “Gromov-Witten invariants and moduli spaces of curves”, in *Proceedings of the International Congress of Mathematicians, Madrid 2006, Vol. II*, European Mathematical Society, 2006, pp. 791-812. DOI: 10.4171/022-2/38. URL: <https://ems.press/books/standalone/21/422>.

This is the source matching the problem statement. Its abstract states the expectation that, for semisimple quantum cohomology, information from the geometry of moduli spaces of curves might determine all higher-genus Gromov-Witten invariants.

2. Boris Dubrovin and Youjin Zhang, “Bi-Hamiltonian hierarchies in 2D topological field theory at one-loop approximation”, *Communications in Mathematical Physics* 198(2), 311-361, 1998. DOI: 10.1007/s002200050480. arXiv/hep-th:9712232.

This is cited in later universal-equation literature as proving the genus-1 semisimple case.

3. Xiaobo Liu, “Genus-2 Gromov-Witten invariants for manifolds with semisimple quantum cohomology”, *American Journal of Mathematics* 129(2), 463-498, 2007. DOI: 10.1353/ajm.2007.0012. arXiv:math/0310410.

Gives a formula for the genus-2 generating function for semisimple quantum cohomology and proves the genus-2 Virasoro conjecture. This resolves the $g = 2$ case but not all $g \geq 2$.

4. Constantin Teleman, “The structure of 2D semi-simple field theories”, *Inventiones Mathematicae* 188, 525-588, 2012. DOI: 10.1007/s00222-011-0352-5. arXiv:0712.0160.

Teleman’s Theorem 8.15 says that Gromov-Witten classes of a compact symplectic manifold are uniquely determined by c_1 and the quantum multiplication law at a semisimple point. This confirms Givental’s higher-genus reconstruction conjecture, but it is not a proof that the tautological universal equations themselves explicitly determine every F_g .

5. Y.-P. Lee, “Invariance of tautological equations II: Gromov-Witten theory”, *Journal of the American Mathematical Society* 22(2), 331-352, 2009. DOI: 10.1090/S0894-0347-08-00616-4. arXiv:math/0605708.

Proves strong invariance properties of tautological equations under Givental loop-group actions. This supports the tautological-equation program but does not by itself supply an all-genus determination theorem for the F_g .

6. Emily Clader, Felix Janda, Xin Wang, and Dmitry Zakharov, “Topological recursion relations from Pixton’s formula”, *Michigan Mathematical Journal* 73(2), 227–241, 2023. DOI: 10.1307/mmj/20195795. URL: <https://projecteuclid.org/journals/michigan-mathematical-journal/volume-73/issue-2/Topological-Recursion-Relations-from-Pixtons-Formula/10.1307/mmj/20195795.full>.

The introduction explicitly states that, for semisimple targets, universal equations are known to determine all Gromov-Witten invariants in genus 1 or 2, but that the genus-3 analogue does not follow from the currently known universal equations.

7. Felix Janda and Xin Wang, “Universal Equations for Higher Genus Gromov-Witten Invariants from Hodge Integrals”, *Communications in Mathematical Physics* 405, Article 47, 2024. DOI: 10.1007/s00220-024-04931-w.

Establishes new all-genus universal equations, showing ongoing progress, but not a full reconstruction theorem from all such equations.

8. Xin Wang, “New Push-Forward Relation and Universal Equation for Higher Genus Gromov-Witten invariants”, *Communications in Mathematical Physics* 407, Article 118, 2026. DOI: 10.1007/s00220-026-05634-0.

Gives another new universal equation and push-forward relation. This is further partial progress rather than a complete solution.

9. Felix Janda and Xin Wang, “Structures in topological recursion relations”, arXiv:2601.20673, 2026.

This recent preprint still phrases the statement that the collection of all universal equations determines all higher-genus Gromov-Witten invariants from genus-zero data in the semisimple case as a conjecture, distinct from the Givental or Dubrovin-Zhang approaches.

4.18.3.4. What remains unresolved

What remains open is a citable all-genus theorem proving that, for every compact symplectic V satisfying the stated semisimplicity hypothesis and every $g \geq 2$, F_g is explicitly recoverable from the universal PDEs obtained from tautological relations on moduli spaces of curves.

Teleman’s theorem means the broader semisimple all-genus reconstruction problem is solved by CohFT classification/Givental reconstruction, but the exact universal-equations formulation in this problem is not fully resolved.

4.18.4. Remaining open problems recorded by the database

4.18.4.1. Open problem 1

For a target with semisimple quantum cohomology, prove that all genus-3 Gromov–Witten invariants are explicitly determined by genus-0 data through the universal equations obtained from tautological relations on moduli spaces of stable curves.

Reference supplied by the database. Emily Clader, Felix Janda, Xin Wang, and Dmitry Zakharov, “Topological recursion relations from Pixton’s formula,” *Michigan Mathematical Journal* **73** (2023), no. 2, 227–241, Introduction (explicitly observing that the genus-3 analogue of the known genus-1 and genus-2 reconstruction does not follow from the currently known universal equations). DOI: <https://doi.org/10.1307/mmj/20195795>; arXiv: [1704.02011](https://arxiv.org/abs/1704.02011).

4.18.4.2. Open problem 2

For every $g \geq 2$, prove that the universal pure-descendant Gromov–Witten equation in Janda and Wang’s Corollary 2 (their equation arising from the $r = 0$ extension of the Liu–Pandharipande boundary push-forward relation) is a consequence of the degree- g topological recursion relations of Clader–Janda–Wang–Zakharov.

Reference supplied by the database. Felix Janda and Xin Wang, “Universal equations for higher genus Gromov–Witten invariants from Hodge integrals,” *Communications in Mathematical Physics* **405** (2024), no. 2, Article 47, Conjecture 11 in §3.3 (verified there only for $g = 2$). DOI: <https://doi.org/10.1007/s00220-024-04931-w>; arXiv: [2401.00899](https://arxiv.org/abs/2401.00899).

4.18.4.3. Open problem 3

Let a degree- g topological recursion relation contain a term $c(\xi_{\Gamma_{0,g,[n]}})_*(1)$ for the bouquet graph with one genus-0 vertex, g loops, and all n markings. Assume every other graph has at least two vertices and that there are no κ -classes and no genus- h vertex decorated by a ψ -monomial of degree at least $h + \delta_h^0$. Prove that $c = 0$.

Reference supplied by the database. Felix Janda and Xin Wang, “Structures in topological recursion relations,” *Transactions of the American Mathematical Society, Series B* **12** (2025), 1223–1263, Conjecture 3. DOI: <https://doi.org/10.1090/btran/238>; arXiv: [2601.20673](https://arxiv.org/abs/2601.20673).

4.18.5. Primary-source pointer

The originating ICM record is *Gromov-Witten invariants and moduli spaces of curves* (2006). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/>

[ICM/Proceedings/ICM2006.2/ICM2006.2.ocr.pdf#page=801.](#)

4.19. 165. Gromov-Witten invariants and moduli spaces of curves [2006-ag-109]

Record: 2006-ag-109

Subfield: Symplectic and contact geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2006: Gromov-Witten invariants and moduli spaces of curves

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.2/ICM2006.2.ocr.pdf#page=801>

4.19.1. Problem statement

Let V be a compact symplectic manifold, and let $F_g(t)$ be the genus- g Gromov-Witten descendant potential of V on the big phase space with coordinates t_n^α corresponding to descendants $\tau_n(\gamma_\alpha)$ of a homogeneous basis $\{\gamma_\alpha\}$ of $H^*(V; \mathbb{C})$. For vector fields $\mathcal{W}_i$ on the big phase space, write

$$\langle\langle \mathcal{W}_1 \cdots \mathcal{W}_k \rangle\rangle_g$$

for the k -th covariant derivative of F_g ; in particular $\langle\langle \mathcal{W} \rangle\rangle_g = \mathcal{W}(F_g)$.

Let $\mathcal{L}_n$, $n \geq -1$, be the Virasoro vector fields used in the vector-field formulation of the Gromov-Witten Virasoro constraints: they are defined recursively by $\mathcal{L}_n = -R^{n+1}(\mathcal{S})$, where $\mathcal{S} = -\sum_{m,\alpha} (t_m^\alpha - \delta_{m,1} \delta_{\alpha,1}) \tau_{m-1}(\gamma_\alpha)$ is the string vector field, with $\tau_{-1}(\gamma_\alpha) = 0$, and $R = GT + C$,

$$T(\mathcal{W}) = \tau_+(\mathcal{W}) - \langle\langle \mathcal{W} \gamma^\alpha \rangle\rangle_0 \gamma_\alpha.$$

Here τ_+ raises descendant level, $G(\tau_m(\gamma_\alpha)) = (m + b_\alpha) \tau_m(\gamma_\alpha)$ is the grading operator with $b_\alpha = \frac{1}{2}(\deg \gamma_\alpha - d + 1)$ when $\dim_{\mathbb{R}} V = 2d$ and, for a smooth projective d -fold with $\gamma_\alpha \in H^{p_\alpha, q_\alpha}(V)$, with the Virasoro weight $b_\alpha = p_\alpha - \frac{1}{2}(d - 1)$, C is induced by cup product with $c_1(V)$, and repeated Greek indices are summed using the Poincaré pairing $\eta_{\alpha\beta} = \int_V \gamma_\alpha \cup \gamma_\beta$, so that $\gamma^\alpha = \eta^{\alpha\beta} \gamma_\beta$. Universal equations mean the partial differential equations for the potentials F_h obtained by translating tautological relations among ψ -classes and boundary strata on the Deligne-Mumford moduli spaces $\overline{\mathcal{M}}_{h,k}$ through the stabilization maps from moduli spaces of stable maps to V .

Prove that, for every compact symplectic manifold V , every $g \geq 1$, and every $n \geq 2$, the genus- g one-point Virasoro correlation $\langle\langle \mathcal{L}_n \rangle\rangle_g$ can be computed from $\langle\langle \mathcal{L}_1 \rangle\rangle_g$ together with lower-genus Gromov-Witten data, meaning the potentials F_h and their correlation functions for $h < g$. Prove also that, if V is a smooth projective variety, then the full Gromov-Witten Virasoro conjecture, namely the assertion that the total descendant potential $Z_V = \exp(\sum_{g \geq 0} \lambda^{2g-2} F_g)$ is annihilated by the half-Virasoro sequence of Eguchi-Hori-Xiong/Katz differential operators, follows from the universal equations together with the L_1 -constraint, i.e. the $n = 1$ Virasoro constraint.

4.19.2. Historical background, literature, and current progress

4.19.3. Verdict: partially resolved

4.19.3.1. Precise problem

The problem asks for an all-genus reduction statement in Gromov-Witten theory: for every compact symplectic target V , every $g \geq 1$, and every $n \geq 2$, the genus- g one-point Virasoro correlation $\langle\langle \mathcal{L}_n \rangle\rangle_g$ should be computable from $\langle\langle \mathcal{L}_1 \rangle\rangle_g$ and lower-genus data via universal equations coming from tautological relations on $\overline{M}_{g,k}$. For smooth projective V , this would imply that the full Eguchi-Hori-Xiong/Katz Virasoro conjecture follows from universal equations plus the L_1 -constraint.

4.19.3.2. Status

I found substantial low-genus and special-target progress, but no published or otherwise citable result proving the exact all-genus, all-compact-symplectic/smooth-projective statement in the problem statement. Recent literature still treats the corresponding all-genus determination by universal equations as conjectural, and describes the non-semisimple Virasoro problem as open. The correct status is therefore **partially resolved**.

4.19.3.3. Key literature and what it proves

1. Xiaobo Liu, "Gromov-Witten invariants and moduli spaces of curves," *Proceedings of the International Congress of Mathematicians, Vol. II, European Mathematical Society, 2006*, pp. 791-812; published 2007. DOI: 10.4171/022-2/38. URL: <https://ems.press/books/standalone/21/422>.

This is the source most closely matching the problem statement. Its abstract says the paper explains how much Gromov-Witten information is determined by moduli spaces of curves and says that, in the semisimple case, this information is believed to determine all higher genus invariants and help study the Virasoro conjecture. This is not a proof of the all-genus statement.

2. Xiaobo Liu, "Quantum product on the big phase space and the Virasoro conjecture," *Advances in Mathematics* 169(2), 313-375, 2002. DOI: 10.1006/aima.2001.2062; arXiv:math/0104030. URL: <https://arxiv.org/abs/math/0104030>.

This paper introduces the big-phase-space quantum product and the recursive Virasoro vector-field formalism used in the problem statement. It proves the genus-2 reduction: for any manifold satisfying the genus-1 L_1 -constraint, the genus-2 L_1 -constraint implies the genus-2 Virasoro conjecture, and for $n \geq 2$ the genus-2 L_n -constraint can be recursively computed from the genus-2 L_1 -constraint plus lower-genus terms. This is exactly analogous to this problem but only in genus 2, not all g .

3. Xiaobo Liu, "Elliptic Gromov-Witten invariants and Virasoro conjecture," *Communications in Mathematical Physics* 216, 705-728, 2001. DOI: 10.1007/s002200000357; arXiv:math/9907113. URL: <https://arxiv.org/abs/math/9907113>.

Gives necessary and sufficient conditions for the genus-1 Virasoro conjecture. This is low-genus progress and supports the $L_1/SL(2)$ -reduction pattern, but it does not prove the all-genus claim.

4. Xiaobo Liu and Gang Tian, “Virasoro constraints for quantum cohomology,” *Journal of Differential Geometry* 50(3), 537-590, 1998. DOI: 10.4310/jdg/1214424970; arXiv:math/9806028. URL: <https://arxiv.org/abs/math/9806028>.

Proves the genus-0 Virasoro constraints for quantum cohomology. This is foundational progress, but the problem concerns every $g \geq 1$.

5. Xiaobo Liu, “Genus-2 Gromov-Witten invariants for manifolds with semisimple quantum cohomology,” arXiv:math/0310410, 2004. URL: <https://arxiv.org/abs/math/0310410>.

Proves the genus-2 Virasoro conjecture for manifolds with semisimple quantum cohomology by giving a formula for F_2 . This is a special semisimple/low-genus result, not the general all-genus reduction in the problem statement.

6. Constantin Teleman, “The structure of 2D semi-simple field theories,” *Inventiones Mathematicae* 188, 525-588, 2012. DOI: 10.1007/s00222-011-0352-5. URL: <https://link.springer.com/article/10.1007/s00222-011-0352-5>.

Teleman’s classification confirms Givental’s higher-genus reconstruction conjecture and implies the Virasoro conjecture for targets with semisimple quantum cohomology. This is a powerful full-genus theorem under a semisimplicity hypothesis. It does not cover arbitrary smooth projective varieties or arbitrary compact symplectic manifolds, and it is not the claimed universal-equations-plus- L_1 reduction.

7. Andrei Okounkov and Rahul Pandharipande, “Virasoro constraints for target curves,” *Inventiones Mathematicae* 163, 47-108, 2006. DOI: 10.1007/s00222-005-0455-y; arXiv:math/0308097. URL: <https://arxiv.org/abs/math/0308097>.

Proves generalized Virasoro constraints for the relative Gromov-Witten theories of all nonsingular target curves. This is another major special-target result, not a proof for all smooth projective targets.

8. Tom Coates, Alexander Givental, and Hsian-Hua Tseng, “Virasoro constraints for toric bundles,” *Forum of Mathematics, Pi* 12, e4, 2024. DOI: 10.1017/fmp.2024.2. URL: <https://www.cambridge.org/core/journals/forum-of-mathematics-pi/article/virasoro-constraints-for-toric-bundles/8BE7221F9F98EBE37DEDAFB62F4ACBC5>.

Proves the Virasoro conjecture for toric bundles exactly when it holds for the base, with corollaries including bases with generically semisimple quantum cohomology, compact Riemann surfaces, K3 surfaces, and Calabi-Yau manifolds of dimension at least 3. This substantially enlarges the known cases, but remains a special-family theorem and does not imply the all-target statement.

9. Felix Janda and Xin Wang, “Universal Equations for Higher Genus Gromov-Witten Invariants from Hodge Integrals,” *Communications in Mathematical Physics* 405, article 47, 2024. DOI: 10.1007/s00220-024-04931-w. URL: <https://link.springer.com/article/10.1007/s00220-024-04931-w>.

Establishes new universal equations for higher-genus Gromov-Witten invariants from Hodge integrals. This is directly relevant progress on the universal-equation side, but it does not prove that universal equations plus L_1 imply all Virasoro constraints in every genus.

10. Xin Wang, “New Push-Forward Relation and Universal Equation for Higher Genus Gromov-Witten invariants,” *Communications in Mathematical Physics* 407, article 118, 2026. DOI: 10.1007/s00220-026-05634-0. URL: <https://link.springer.co>

[m/article/10.1007/s00220-026-05634-0](https://arxiv.org/abs/2601.20673).

Proves new push-forward tautological relations and obtains a new partial differential equation for higher-genus descendant Gromov-Witten invariants of any smooth projective variety. This is further progress, but still only supplies additional universal equations, not the full all-genus reduction asserted in the problem statement.

11. Felix Janda and Xin Wang, “Structures in topological recursion relations,” arXiv:2601.20673, 2026. URL: <https://arxiv.org/pdf/2601.20673>.

This preprint gives structural results and applications to Gromov-Witten universal equations. It explicitly states that it is still conjectured that all such equations determine all higher-genus Gromov-Witten invariants in terms of genus-0 invariants in the semisimple case. That language is inconsistent with a known proof of the stated all-genus claim.

12. Xin Wang, “Virasoro constraints for Hodge integrals,” arXiv:2506.10033, 2025. URL: <https://arxiv.org/pdf/2506.10033>.

A recent preprint on Virasoro constraints for Hodge integrals states that the Virasoro conjecture for targets with non-semisimple quantum cohomology is widely open. This supports the conclusion that the stated all-target smooth-projective implication is not known.

4.19.3.4. What remains unresolved

No source found proves that for every compact symplectic V , every $g \geq 1$, and every $n \geq 2$, $\langle\langle \mathcal{L}_n \rangle\rangle_g$ is computable from $\langle\langle \mathcal{L}_1 \rangle\rangle_g$ and lower-genus data via universal equations. Likewise, no source proves for all smooth projective varieties that universal equations together with the L_1 -constraint imply the full Eguchi-Hori-Xiong/Katz Virasoro constraints. The known results cover genus 0, genus 1/2 reductions, semisimple quantum cohomology, target curves, toric-bundle families, and many new universal equations, but not the full statement in this problem.

4.19.4. Remaining open problems recorded by the database

4.19.4.1. Open problem 1

Let X be an arbitrary smooth projective variety. Prove the genus-one L_1 Virasoro constraint for the descendant Gromov-Witten theory of X , equivalently $\Psi_{1,1;\emptyset} = 0$, without assuming that the quantum cohomology of X is semisimple.

Reference supplied by the database. Xin Wang, “Virasoro constraints for Hodge integrals,” *Communications in Number Theory and Physics* **19** (2025), no. 3, 631–682, Introduction (the vanishing of $\Psi_{1,n;\emptyset}$ for general targets is explicitly identified as open and reduced to $\Psi_{1,1;\emptyset}$); DOI: <https://doi.org/10.4310/CNTP.250829234004>; arXiv: 2506.10033.

4.19.4.2. Open problem 2

Let X and Y be smooth projective varieties. Prove that the Virasoro conjecture holds for the product $X \times Y$ if and only if it holds for both X and Y , without a semisimplicity hypothesis on either factor.

Reference supplied by the database. Hsian-Hua Tseng, “A note on Virasoro constraints for products,” preprint (2026), Problem 1.1 (with the case in which one factor has semisimple quantum cohomology proved as Theorem 1.2); arXiv: [2603.22486](https://arxiv.org/abs/2603.22486).

4.19.5. Primary-source pointer

The originating ICM record is *Gromov-Witten invariants and moduli spaces of curves* (2006). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.2/ICM2006.2.ocr.pdf#page=801>.

4.20. 166. Symplectic field theory and its applications [2006-geom-282]

Record: 2006-geom-282

Subfield: Symplectic and contact geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2006: Symplectic field theory and its applications

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.1/ICM2006.1.ocr.pdf#page=227>

4.20.1. Problem statement

Let (V^{2n-1}, ξ) be a closed cooriented contact manifold that is symplectically fillable: there is a compact symplectic manifold (W^{2n}, ω) with $\partial W = V$ such that V is a convex boundary of contact type and its induced contact structure is ξ . Choose a nondegenerate contact form λ with $\ker \lambda = \xi$, let R_λ be its Reeb vector field, and choose an almost complex structure J on ξ compatible with $d\lambda|_\xi$. Extend J to the standard cylindrical almost complex structure on the symplectization $\mathbb{R} \times V$ by requiring it to be translation invariant and to satisfy $J(\partial_s) = R_\lambda$, where s is the $\mathbb{R}$ -coordinate.

Let $(\text{fock}_{\lambda,J}, \partial_{\lambda,J})$ denote the genus-zero rational symplectic-field-theory differential graded algebra with no marked-point variables. Thus $\text{fock}_{\lambda,J}$ is the graded supercommutative algebra over $\mathbb{C}$ generated by variables associated with the closed Reeb orbits of λ , and $\partial_{\lambda,J}$ is the derivation defined by counts of rigid finite-energy J -holomorphic curves in $\mathbb{R} \times V$ having one positive puncture and arbitrarily many negative punctures. An augmentation is a degree-preserving unital algebra homomorphism

$$\varepsilon : (\text{fock}_{\lambda,J}, \partial_{\lambda,J}) \rightarrow (\mathbb{C}, 0)$$

such that $\varepsilon \circ \partial_{\lambda,J} = 0$.

Prove that $(\text{fock}_{\lambda,J}, \partial_{\lambda,J})$ admits at least one augmentation for every symplectically fillable (V, ξ) . Determine whether, for every symplectically fillable (V, ξ) , the contact form λ and compatible almost complex structure J can be chosen so that this augmentation is unique.

4.20.2. Historical background, literature, and current progress

Verdict: unresolved.

The existence half is now established, but the uniqueness half is not. Modern foundations rigorously define the contact-homology DGA. Exact fillings directly induce augmentations. For a general convex contact-type (strong) filling, the rational-SFT filling construction produces an augmentation after Novikov completion and implies that the ordinary contact homology cannot vanish. Over the algebraically closed field $\mathbb{C}$, nonvanishing of contact homology is the only algebraic obstruction to a degree-preserving augmentation; hence every strongly symplectically fillable contact manifold has at least one $\mathbb{C}$ -valued augmentation as required in the first sentence.

The second sentence remains open. Recent work makes weak-equivalence classes of augmentations and the associated linearized contact homologies invariant without assuming DGA homotopies and proves uniqueness of an augmentation class for special asymptotically dynamically convex classes. It does not prove that every symplectically fillable contact manifold admits some choice of contact form and almost complex structure for which the resulting DGA has literally one augmentation. No counterexample to that universal assertion was found either.

Established progress and key references.

- John Pardon, *Contact homology and virtual fundamental cycles*, Journal of the American Mathematical Society **32** (2019), 825-919, DOI <https://doi.org/10.1090/jams/924>, arXiv:1508.03873. Pardon constructs contact homology in arbitrary dimensions and unital DGA maps for exact symplectic cobordisms. Applied to exact fillings, this gives augmentations, but the exactness hypothesis does not cover every convex contact-type filling in the problem.
- Erkao Bao and Ko Honda, *Semi-global Kuranishi charts and the definition of contact homology*, Advances in Mathematics **414** (2023), 108864, DOI <https://doi.org/10.1016/j.aim.2023.108864>, arXiv:1512.00580. They construct the contact-homology DGA and maps for Liouville cobordisms. Their Warning 1.0.5 explicitly notes that they do not prove homotopy invariance of the cobordism maps or invariance of the collection of linearized contact homologies, so this does not settle augmentation uniqueness.
- Agustin Moreno and Zhengyi Zhou, *A landscape of contact manifolds via rational SFT*, Geometry & Topology **29** (2025), 3465-3565, DOI <https://doi.org/10.2140/gt.2025.29.3465>, arXiv:2012.04182. Proposition 3.14 constructs the exact-filling augmentation, and Corollary 3.15 constructs the strong-filling augmentation over the Novikov field and proves that finite algebraic planar torsion obstructs strong fillability. Since vanishing contact homology is algebraic planar torsion zero, a strong filling forces ordinary contact homology to be nonzero.
- Chi Li and Zhengyi Zhou, *Kahler compactification of $\mathbb{C}^n$ and Reeb dynamics*, Inventiones Mathematicae **243** (2026), 181-208, DOI <https://doi.org/10.1007/s00222-025-01377-2>, arXiv:2409.10275. Remark 3.9 records the algebraic implication used here: over an algebraically closed field, vanishing contact homology is the only obstruction to an algebraic augmentation. Combined with Moreno-Zhou, this proves the $\mathbb{C}$ -valued existence assertion for every strong filling, although the augmentation need not be the direct geometric count associated to that filling.
- Julian Chaidez, *Contact Homology and Linearization Without DGA Homotopies*, International Mathematics Research Notices **2025** (2025), no. 21, rnaf325, DOI <https://doi.org/10.1093/imrn/rnaf325>, arXiv:2412.20676. The paper proves that weak-equivalence classes of augmentations and the resulting isomorphism classes of linearized contact homologies are contact invariants. Corollaries 5.9-5.10 establish at-most-one or unique graded augmentation classes for SADC and 1-SADC contact manifolds, respectively; these are special cases and concern equivalence classes rather than literal uniqueness of an augmentation of a chosen DGA.
- Frederic Bourgeois, *Contact homology of contact manifolds and its applications*, Celebratio Mathematica (2025), [direct URL](#). The survey explains the direct geometric filling augmentation by counting holomorphic planes under the additional assumption that a strong filling is symplectically aspherical. This complements the algebraic nonvanishing argument used for general strong fillings.

4.20.3. Remaining open problems recorded by the database

4.20.3.1. Open problem 1

Let (V^{2n-1}, ξ) be a closed cooriented contact manifold that is Stein fillable, meaning that (V, ξ) is the strictly pseudoconvex contact boundary of a compact Stein domain. Determine whether cylindrical contact homology is defined for every such (V, ξ) : namely, whether suitable nondegenerate contact forms and compatible cylindrical almost complex structures give a chain complex generated by good closed Reeb orbits, with differential defined by counts of rigid finite-energy holomorphic cylinders, such that the differential squares to zero and the resulting homology is independent of all auxiliary choices.

Reference supplied by the database. Michael Hutchings, with Yasha Eliashberg and John Etnyre, *Open problems*, in *Holomorphic Curves in Contact Geometry*, American Institute of Mathematics workshop, 4-9 August 2003, Question 1.9, <https://aimath.org/WWN/contactgeom2/articles/html/28a/>.

4.20.4. Primary-source pointer

The originating ICM record is *Symplectic field theory and its applications* (2006). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.1/ICM2006.1.ocr.pdf#page=227>.

4.21. 167. Lagrangian submanifolds from the local model to the cluster complex [2006-geom-285]

Record: 2006-geom-285

Subfield: Symplectic and contact geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2006: Lagrangian submanifolds from the local model to the cluster complex

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.2/ICM2006.2.ocr.pdf#page=779>

4.21.1. Problem statement

Prove that for every integer $n \geq 1$, every compact Lagrangian submanifold $L \subset \mathbb{C}^n$ with respect to the standard symplectic form $\omega_0 = \sum_{j=1}^n dx_j \wedge dy_j$, and every point $x \in L$, there exists a nonconstant map $u: D \rightarrow \mathbb{C}^n$ that is continuous on the closed unit disc $D = \{z \in \mathbb{C} : |z| \leq 1\}$ and holomorphic on its interior with respect to the standard complex structure on $\mathbb{C}^n$, such that $u(\partial D) \subseteq L$ and $x \in u(\partial D)$.

4.21.2. Historical background, literature, and current progress

The exact problem is unresolved. Gromov proved that every compact Lagrangian submanifold of $\mathbb{C}^n$ bounds at least one nonconstant standard holomorphic disc, but his theorem does not put the boundary of such a disc through an arbitrarily prescribed point of L . Later uniruledness results impose additional hypotheses (such as monotonicity, orientability, regularity, or homological restrictions), or establish existence in prescribed boundary homotopy classes without prescribing a boundary point. No proof or counterexample covering every compact Lagrangian $L \subset \mathbb{C}^n$, every n , and every $x \in L$ was found.

Key literature progress:

- “Pseudo holomorphic curves in symplectic manifolds,” Mikhail Gromov, *Inventiones Mathematicae* 82 (1985), 307–347, DOI <https://doi.org/10.1007/BF01388806>. Theorem 0.4.A2 (and Section 2.3.B) proves the existence of a nonconstant holomorphic disc with boundary on an arbitrary closed Lagrangian submanifold of $\mathbb{C}^n$. It supplies no prescribed-point condition and hence proves only the weaker existence statement.
- “Cluster homology: an overview of the construction and results,” Octav Cornea and François Lalonde, *Electronic Research Announcements of the American Mathematical Society* 12 (2006), 1–12, DOI <https://doi.org/10.1090/S1079-6762-06-00154-5>. Section 4.1 explicitly calls the statement the Gromov–Sikorav problem. Corollary 4.4 gives, for compact orientable relatively spin displaceable Lagrangians, a dichotomy between a disc through every point and the existence of a disc of sufficiently low Maslov index; Corollary 4.2 removes the obstruction only under an additional rational-homology restriction and initially yields discs through a dense set for generic J . These hypotheses and alternatives do not cover arbitrary compact Lagrangians in $\mathbb{C}^n$.

- “A geometric refinement of a theorem of Chekanov,” François Charette, *Journal of Symplectic Geometry* 10 (2012), 475–491, DOI <https://doi.org/10.4310/JSG.2012.v10.n3.a6>, arXiv:1011.0392. Theorem 1.2 proves the prescribed-point conclusion in the monotone setting: after taking a Hamiltonian-disjoint copy of a monotone $L \subset \mathbb{C}^n$, the alternative curves are forced to be discs with boundary on L . This resolves an important special class but not nonmonotone Lagrangians.
- “Gromov width and uniruling for orientable Lagrangian surfaces,” François Charette, *Algebraic & Geometric Topology* 15 (2015), 1439–1451, DOI <https://doi.org/10.2140/agt.2015.15.1439>. Theorem 3.1 establishes prescribed-point holomorphic-curve existence for closed orientable Lagrangian surfaces and regular compatible almost complex structures, without monotonicity. It does not cover higher dimensions or nonorientable Lagrangians, and its stated regularity condition is not part of the exact problem.
- “Holomorphic disks with boundary on compact Lagrangian surface,” Jingyi Chen, preprint (2025), arXiv:2505.19407. Theorem 1.1 proves that, for a compact oriented Lagrangian surface in a homogeneously regular Kähler surface, generators of the kernel of $\pi_1(L) \rightarrow \pi_1(M)$ are represented by boundaries of holomorphic discs. In particular this gives multiple standard holomorphic discs for Lagrangian tori in $\mathbb{C}^2$, but it does not require their boundaries to pass through a prescribed point.
- “Exact Lagrangian caps and non-uniruled Lagrangian submanifolds,” Georgios Dimitroglou Rizell, *Arkiv för Matematik* 53 (2015), 37–64, DOI <https://doi.org/10.1007/s11512-014-0202-y>, arXiv:1306.4667. Proposition 1.4 and Corollary 1.5 give compact Lagrangians in $\mathbb{C}^n$ ($n \geq 3$) that are not uniruled in the stronger sense requiring a uniform area bound for every compatible almost complex structure. This does not disprove the present problem, which fixes the standard complex structure and asks for no uniform area bound.

What remains open is precisely whether, without monotonicity, orientability, spin, homological, regularity, or area-bound assumptions, the standard complex structure on $\mathbb{C}^n$ always admits a nonconstant attached holomorphic disc through each prescribed point of every compact Lagrangian submanifold.

4.21.3. Remaining open problems recorded by the database

No separate descendant problem is listed in the website’s “Some remaining open problems” field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

4.21.4. Primary-source pointer

The originating ICM record is *Lagrangian submanifolds from the local model to the cluster complex* (2006). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.2/ICM2006.2.ocr.pdf#page=779>.

4.22. 168. Lagrangian submanifolds from the local model to the cluster complex [2006-geom-297]

Record: 2006-geom-297

Subfield: Symplectic and contact geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2006: Lagrangian submanifolds from the local model to the cluster complex

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.2/ICM2006.2.ocr.pdf#page=779>

4.22.1. Problem statement

Let V be a connected smooth manifold, let $\pi : T^*V \rightarrow V$ be its cotangent-bundle projection, and let λ be the canonical Liouville 1-form on T^*V , defined by

$$\lambda_{(q,p)}(\xi) = p(d\pi_{(q,p)}(\xi)).$$

Equip T^*V with the symplectic form $\omega = d\lambda$. Prove that every nonempty compact embedded submanifold $L \subset T^*V$ without boundary satisfying

$$\dim L = \dim V, \quad \omega|_L = 0, \quad \text{and} \quad \lambda|_L = df$$

for some smooth function $f : L \rightarrow \mathbb{R}$ is Hamiltonian isotopic to the zero section. In other words, prove that there exists a Hamiltonian isotopy $(\phi_t)_{t \in [0,1]}$ of T^*V such that $\phi_0 = \text{id}_{T^*V}$ and

$$\phi_1(L) = \{(q, 0) : q \in V\}.$$

4.22.2. Historical background, literature, and current progress

The statement is **unresolved**. It is the nearby Lagrangian conjecture. For a noncompact connected base V , the assertion is vacuous: Lalonde and Sikorav prove that the cotangent projection of a compact exact Lagrangian is surjective, which would force V to be compact. For closed V , however, the claimed Hamiltonian isotopy is still explicitly described in recent literature as a major open problem, and no collection of known results covers all closed bases.

The strongest general result currently available is topological rather than symplectic: Abouzaid and Kragh prove that the projection $L \rightarrow V$ is a simple homotopy equivalence. Abouzaid, Courte, Guillermou, and Kragh subsequently prove strong restrictions on the stable Lagrangian Gauss map, and Abouzaid, Álvarez-Gavela, Courte, and Kragh prove that the normal invariant of $L \rightarrow V$ is 2-torsion. None of these results implies that L is diffeomorphic to V , much less Hamiltonian isotopic to the zero section. The full conclusion is known only in special cases, including $V = S^2$ and $V = \mathbb{T}^2$; for example, the case of closed orientable surfaces of genus at least 2 remains open.

Key references:

- François Lalonde and Jean-Claude Sikorav, “Sous-variétés lagrangiennes et lagrangiennes exactes des fibrés cotangents,” *Commentarii Mathematici Helvetici* **66** (1991), 18–33, DOI <https://doi.org/10.1007/BF02566634>. They prove the surjectivity of $L \rightarrow V$, excluding compact exact Lagrangians over noncompact connected bases.
- Mohammed Abouzaid and Thomas Kragh, “Simple Homotopy Equivalence of Nearby Lagrangians,” *Acta Mathematica* **220** (2018), 207–237, DOI <https://doi.org/10.4310/ACTA.2018.v220.n2.a1>, arXiv:1603.05431. Their Theorem 1 proves that $L \rightarrow V$ is a simple homotopy equivalence for every closed exact Lagrangian, but does not produce a Hamiltonian isotopy.
- Mohammed Abouzaid, Sylvain Courte, Stéphane Guillermou, and Thomas Kragh, “Twisted Generating Functions and the Nearby Lagrangian Conjecture,” *Duke Mathematical Journal* **174** (2025), 949–1011, DOI <https://doi.org/10.1215/00127094-2024-0052>, arXiv:2011.13178. They prove that the stable Lagrangian Gauss map induces the zero map on homotopy groups (and is null-homotopic for spheres), a further obstruction result short of isotopy.
- Mohammed Abouzaid, Daniel Álvarez-Gavela, Sylvain Courte, and Thomas Kragh, “Normal Invariant of Nearby Lagrangians via Twisted Derivative,” preprint (2025; revised 2026), arXiv:2505.12515. They prove that the normal invariant of the homotopy equivalence $L \rightarrow V$ is 2-torsion, leaving the diffeomorphism and Hamiltonian-isotopy questions open.
- Richard Hind, “Lagrangian Unknottedness in Stein Surfaces,” *Asian Journal of Mathematics* **16** (2012), 1–36, DOI <https://doi.org/10.4310/AJM.2012.v16.n1.a1>. He proves Hamiltonian uniqueness for Lagrangian spheres in T^*S^2 , giving the resolved S^2 special case.
- Georgios Dimitroglou Rizell, Elizabeth Goodman, and Alexander Ivrii, “Lagrangian Isotopy of Tori in $S^2 \times S^2$ and $\mathbb{CP}^2$,” *Geometric and Functional Analysis* **26** (2016), 1297–1358, DOI <https://doi.org/10.1007/s00039-016-0388-1>, arXiv:1602.08821. Theorem B proves the full nearby Lagrangian conjecture for $T^*\mathbb{T}^2$, only a special base.
- Roger Casals, *A Microlocal Introduction to Legendrian Submanifolds*, CBMS Summer School lecture notes (draft), 2025, [direct PDF](#). Section 1.3 explicitly records the nearby Lagrangian conjecture for every closed orientable surface of genus at least 2 as open.
- Noah Porcelli, “ C^0 Lagrangian Monodromy,” *Bulletin of the London Mathematical Society* (2025), DOI <https://doi.org/10.1112/blms.70162>. Conjecture 2.1 records the exact Hamiltonian-isotopy assertion as one of the main open problems in symplectic topology.

4.22.3. Remaining open problems recorded by the database

4.22.3.1. Open problem 1

Let V be a closed connected smooth manifold, let λ be the canonical Liouville 1-form on T^*V , and equip T^*V with the symplectic form $d\lambda$. Prove that every closed smooth manifold L admitting an exact Lagrangian embedding $L \hookrightarrow T^*V$ is diffeomorphic to V .

Reference supplied by the database. François Lalonde, “Lagrangian Submanifolds: From the Local Model to the Cluster Complex,” *Proceedings of the International Congress of Mathematicians, Madrid 2006*, Volume II, European Mathematical Society, 2006, pp. 779–798, Section 2, consequence (I), DOI: <https://doi.org/10.4171/022-2/37>.

4.22.3.2. Open problem 2

Let Σ be a closed connected orientable smooth surface of genus at least 2, let λ be the canonical Liouville 1-form on $T^*\Sigma$, and equip $T^*\Sigma$ with the symplectic form $d\lambda$. Prove that every compact embedded exact Lagrangian submanifold $L \subset T^*\Sigma$ without boundary is Hamiltonian isotopic to the zero section.

Reference supplied by the database. Roger Casals, *A Microlocal Introduction to Legendrian Submanifolds*, draft lecture notes for the CBMS Summer School “Legendrian Links and the Microlocal Theory of Sheaves,” 2025, Section 1.3, comment 3 after Conjecture 1.3.1, https://www.math.ucdavis.edu/~casals/SheavesCBMS_Notes_RCasals.pdf.

4.22.3.3. Open problem 3

Let $n \geq 2$, let Σ be a closed smooth n -manifold homotopy equivalent to S^n , and let λ be the canonical Liouville 1-form on T^*S^n . Prove that if Σ admits an exact Lagrangian embedding into $(T^*S^n, d\lambda)$, then Σ is diffeomorphic to S^n .

Reference supplied by the database. Daniel Álvarez-Gavela, “Framed Bordism of Lagrangian Homotopy Spheres via Generating Functions,” preprint, 2026, Section 1.1, arXiv:2606.02409, <https://arxiv.org/abs/2606.02409>.

4.22.4. Primary-source pointer

The originating ICM record is *Lagrangian submanifolds from the local model to the cluster complex* (2006). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.2/ICM2006.2.ocr.pdf#page=779>.

4.23. 169. Knots and dynamics [2006-geom-316]

Record: 2006-geom-316

Subfield: Symplectic and contact geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2006: Knots and dynamics

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.1/ICM2006.1.ocr.pdf#page=257>

4.23.1. Problem statement

Let (S^2, ω) be the two-sphere equipped with an area form. For every open disk $D \subset S^2$ satisfying

$$\int_D \omega < \frac{1}{2} \int_{S^2} \omega,$$

identify the group $\text{Ham}_c(D, \omega|_D)$ of compactly supported Hamiltonian diffeomorphisms of D with a subgroup of $\text{Ham}(S^2, \omega)$ by extending its elements by the identity outside D . Its Calabi homomorphism is

$$\text{Cal}_D(\phi_H^1) = \int_0^1 \int_D H_t \omega dt,$$

where the compactly supported time-dependent Hamiltonian $H_t: D \rightarrow \mathbb{R}$ generates an isotopy with time-one map ϕ_H^1 .

Determine whether there is exactly one homogeneous quasimorphism $\mu: \text{Ham}(S^2, \omega) \rightarrow \mathbb{R}$ such that

$$\mu|_{\text{Ham}_c(D, \omega|_D)} = \text{Cal}_D$$

for every such disk D . Here homogeneous means that $\mu(\phi^n) = n\mu(\phi)$ for all $\phi \in \text{Ham}(S^2, \omega)$ and $n \in \mathbb{Z}$, and quasimorphism means that

$$\sup_{\phi, \psi \in \text{Ham}(S^2, \omega)} |\mu(\phi\psi) - \mu(\phi) - \mu(\psi)| < \infty.$$

4.23.2. Historical background, literature, and current progress

The exact uniqueness problem remains unresolved. Entov and Polterovich constructed a homogeneous Calabi quasimorphism on $\text{Ham}(S^2, \omega)$, so existence is known, but no cited result proves that every homogeneous quasimorphism with the stated restrictions equals it or constructs a second quasimorphism satisfying those restrictions for every disk of area below one half.

Michael Entov and Leonid Polterovich, “Calabi quasimorphism and quantum homology,” *International Mathematics Research Notices* 2003, no. 30 (2003), 1635–1676, DOI <https://doi.org/10.1155/S1073792803210011>, arXiv:math/0205247. Theorem 1.3 proves existence on S^2 . Theorem 5.2 proves an important partial uniqueness result: all Calabi quasi-

morphisms continuous in the paper's Hamiltonian C^0 sense have the same median formula on time-one maps of autonomous Hamiltonians. It does not prove equality on arbitrary Hamiltonian diffeomorphisms, and the problem here assumes no continuity.

Étienne Ghys, "Knots and dynamics," *Proceedings of the International Congress of Mathematicians, Madrid 2006*, Volume I, European Mathematical Society, 2007, 247–277, DOI <https://doi.org/10.4171/022-1/11>. Section 2.8 records precisely the uniqueness of the Calabi quasimorphism on $\text{Ham}(S^2)$ as an open question.

Michael Entov, Leonid Polterovich, Pierre Py, and Michael Khanevsky, "On continuity of quasimorphisms for symplectic maps," in *Perspectives in Analysis, Geometry, and Topology*, Progress in Mathematics 296, Birkhäuser/Springer, 2012, 169–197, DOI https://doi.org/10.1007/978-0-8176-8277-4_8. Theorem 3 characterizes C^0 -continuous homogeneous quasimorphisms on surface Hamiltonian groups by vanishing on all sufficiently small disk-supported subgroups. Consequently, the difference of any two quasimorphisms satisfying the present Calabi condition would be a C^0 -continuous homogeneous quasimorphism vanishing on every disk subgroup of area below one half. This is a useful reduction, not a uniqueness theorem.

Daniel Cristofaro-Gardiner, Vincent Humilière, Cheuk Yu Mak, Sobhan Seyfaddini, and Ivan Smith, "Quantitative Heegaard Floer cohomology and the Calabi invariant," *Forum of Mathematics, Pi* 10 (2022), e27, DOI <https://doi.org/10.1017/fmp.2022.18>, arXiv:2105.11026. Theorems 7.6–7.7 construct linearly independent link-spectral homogeneous quasimorphisms and give their support-control property. Proposition 7.12 shows that for each fixed $A < \frac{1}{2} \int_{S^2} \omega$ there is a nonzero homogeneous quasimorphism vanishing on every Hamiltonian diffeomorphism supported in a disk of area at most A . Thus one obtains nonunique extensions of the Calabi rule up to any prescribed subcritical threshold A , arbitrarily close to one half. The construction does not produce one nonzero difference that vanishes for all disks with area strictly below one half, which is exactly the remaining gap in the stated problem.

4.23.3. Remaining open problems recorded by the database

4.23.3.1. Open problem 1

Let (S^2, ω) be the two-sphere equipped with an area form. For every open disk $D \subset S^2$ with

$$\int_D \omega < \frac{1}{2} \int_{S^2} \omega,$$

extend each element of $\text{Ham}_c(D, \omega|_D)$ by the identity outside D and regard the resulting map as an element of $\text{Ham}(S^2, \omega)$. Define

$$\text{Cal}_D(\phi_H^1) = \int_0^1 \int_D H_t \omega dt$$

for every compactly supported Hamiltonian $H: [0, 1] \times D \rightarrow \mathbb{R}$ with time-one map ϕ_H^1 . Determine whether there is exactly one homogeneous quasimorphism $\mu: \text{Ham}(S^2, \omega) \rightarrow \mathbb{R}$ such that

$$\mu|_{\text{Ham}_c(D, \omega|_D)} = \text{Cal}_D$$

for every such disk D and such that μ is continuous in the following sense: whenever smooth time-dependent Hamiltonians $H^{(j)}: [0, 1] \times S^2 \rightarrow \mathbb{R}$ converge uniformly to a

smooth Hamiltonian H , one has

$$\mu(\phi_{H^{(j)}}^1) \rightarrow \mu(\phi_H^1).$$

Here homogeneous means that $\mu(\phi^n) = n\mu(\phi)$ for every $\phi \in \text{Ham}(S^2, \omega)$ and $n \in \mathbb{Z}$, and quasimorphism means that

$$\sup_{\phi, \psi \in \text{Ham}(S^2, \omega)} |\mu(\phi\psi) - \mu(\phi) - \mu(\psi)| < \infty.$$

Reference supplied by the database. Michael Entov and Leonid Polterovich, “Calabi quasimorphism and quantum homology,” *International Mathematics Research Notices* 2003, no. 30 (2003), 1635–1676, Remark 1.4(1), DOI: 10.1155/S1073792803210011, arXiv:math/0205247, <https://arxiv.org/abs/math/0205247>.

4.23.4. Primary-source pointer

The originating ICM record is *Knots and dynamics* (2006). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.1/ICM2006.1.ocr.pdf#page=257>.

4.24. 170. Gauge theory and mirror symmetry [2014-geom-124]

Record: 2014-geom-124

Subfield: Symplectic and contact geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: Gauge theory and mirror symmetry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=1329>

4.24.1. Problem statement

Let T be a compact torus with Lie algebra $\mathfrak{t}$, let $T_{\mathbb{C}}^{\vee}$ be its complexified dual torus, and let X be a symplectic manifold with Hamiltonian T -action and moment map $\mu : X \rightarrow \mathfrak{t}^*$. Using the unitary mirror/spectral decomposition of the Fukaya category $\mathfrak{F}(X)$ over $T_{\mathbb{C}}^{\vee}$, write $\mathfrak{Vect}_{\tau}$ for the irreducible topological T -representation classified by $\tau \in T_{\mathbb{C}}^{\vee}$, i.e. the category of complex vector spaces with that topological T -action. If $q = \operatorname{Re} \log \tau \in \mathfrak{t}^*$ is a regular value of μ , prove that the multiplicity category, or τ -spectral component, of $\mathfrak{Vect}_{\tau}$ in $\mathfrak{F}(X)$ is equivalent to the Fukaya category of the symplectic quotient

$$X//_q T = \mu^{-1}(q)/T,$$

with imaginary curving, equivalently B -field, $\operatorname{Im} \log \tau$.

4.24.2. Historical background, literature, and current progress

4.24.3. Verdict: partially resolved

4.24.3.1. Problem identified

The problem asks for Teleman's torus symplectic-quotient conjecture: for every compact torus T , every Hamiltonian T -symplectic manifold X , and every $\tau \in T_{\mathbb{C}}^{\vee}$ whose real logarithm $q = \operatorname{Re} \log \tau$ is a regular moment-map value, the τ -spectral/multiplicity component of the Fukaya category $\mathfrak{F}(X)$ should be equivalent to the Fukaya category of the Marsden-Weinstein quotient $X//_q T = \mu^{-1}(q)/T$, with B -field $\operatorname{Im} \log \tau$.

This is Conjecture 4.2, "Torus symplectic quotients", in Teleman's ICM article.

4.24.3.2. Current literature status

The problem is not fully resolved in the exact generality of the problem. There is substantial partial progress: several papers prove Floer-cohomology-level, local mirror, exact/Liouville, or special-geometric versions of the expected relationship, and one recent preprint states a quasi-isomorphism under a detailed setup. However, I did not find a published or otherwise citable result proving the full spectral-component equivalence for arbitrary Hamiltonian compact-torus actions and arbitrary Fukaya categories with the stated B -field.

4.24.3.3. Key literature and applicability

Original source

- Title: “Gauge theory and mirror symmetry”
- Author: Constantin Teleman
- Venue/year: Proceedings of the International Congress of Mathematicians, Seoul 2014, Vol. II, Kyung Moon Sa, 2014, pp. 1309-1332
- arXiv/DOI/URL: arXiv:1404.6305, <https://arxiv.org/abs/1404.6305>, DOI <https://doi.org/10.48550/arXiv.1404.6305>
- Relevant result: Conjecture 4.2 states that the multiplicity of Vect_τ in $\mathfrak{F}(X)$ is the Fukaya category of the symplectic reduction at $\text{Re log } \tau$, with imaginary curving/B-field $\text{Im log } \tau$. Teleman explicitly frames it as a conjecture and notes singular values are not covered.

Regular-value and spectral-component progress

- Title: “Equivariant Fukaya categories at singular values”
- Authors: Yanki Lekili and Ed Segal
- Venue/year: Glasgow Mathematical Journal 68(2), May 2026, pp. 197-221; published online 2025
- DOI/arXiv/URL: DOI <https://doi.org/10.1017/S0017089525100487>; arXiv:2304.10969, <https://arxiv.org/abs/2304.10969>
- Relevant content: The paper states that Teleman and Fukaya proposed equivalence between Fukaya categories of symplectic quotients and equivariant/spectral Fukaya categories. It explains that for regular values the corresponding component should be the Fukaya category of the quotient, possibly with bulk deformation, and formulates extensions to singular values.
- Applicability check: This is progress and evidence, not a full proof of the problem. The paper repeatedly formulates conjectures, discusses examples and mirror heuristics, and points out possible sign ambiguities and bulk-deformation issues.
- Title: “Equivariant Lagrangian correspondence and a conjecture of Teleman”
- Authors: Siu-Cheong Lau, Nai-Chung Conan Leung, Yan-Lung Leon Li
- Venue/year: arXiv preprint, 2023; revised 2025
- DOI/arXiv/URL: arXiv:2312.13926, <https://arxiv.org/abs/2312.13926>, DOI <https://doi.org/10.48550/arXiv.2312.13926>
- Relevant content: The abstract says the equivariant Lagrangian-correspondence functor induces a quasi-isomorphism between the equivariant derived Fukaya category at a regular moment-map level and the derived Fukaya category of the corresponding quotient. The paper also proves precise relations between equivariant and reduced Lagrangian Floer cohomology and local mirror potentials.
- Applicability check: This is the closest result found, but it still does not fully settle the problem statement. Its main theorem is under Setup 4.14, assumes a free action on the relevant moment level and weak unobstructedness of L , L_π , and $\bar{L}$ (possibly after bulk deformation). Its Theorem 4.26 is explicitly a localized abelian mirror statement, with globalization deferred to forthcoming work. The problem asks for the entire τ -spectral Fukaya-category equivalence for all Hamiltonian compact-torus manifolds at all regular values, with the stated B -field and no extra unobstructedness/topological/toric/localization hy-

potheses.

Floer-homology-level and foundational partial results

- Title: “Equivariant Lagrangian Floer homology via cotangent bundles of EG_N ”
- Author: Guillem Cazassus
- Venue/year: Journal of Topology 17(1), 2024, article e12328
- DOI/arXiv/URL: DOI <https://doi.org/10.1112/topo.12328>; arXiv:2202.10097, <https://arxiv.org/abs/2202.10097>
- Relevant content: Constructs equivariant Lagrangian Floer homology for Hamiltonian compact-group actions and constructs Kirwan morphisms relating equivariant Floer complexes to Floer complexes on smooth reductions.
- Applicability check: This supplies important Floer-theoretic machinery, but it is not a Fukaya-category equivalence of spectral components.
- Title: “Equivariant Floer homology is isomorphic to reduced Floer homology”
- Author: Julio Sampietro Christ
- Venue/year: arXiv preprint, 2025
- DOI/arXiv/URL: arXiv:2505.02647, <https://arxiv.org/abs/2505.02647>, DOI <https://doi.org/10.48550/arXiv.2505.02647>
- Relevant content: Proves Cazassus’ conjecture that, under free action on the zero moment level, equivariant Lagrangian Floer homology agrees with Floer homology of the Marsden-Weinstein reduction.
- Applicability check: The theorem is for pairs of G -invariant Lagrangians contained in the zero level, with relative exactness and compactness/convexity assumptions, and gives a homology isomorphism rather than the full Fukaya-category spectral-component equivalence.
- Title: “Moment Lagrangian correspondences are unobstructed after bulk deformation”
- Author: Yao Xiao
- Venue/year: arXiv preprint, 2024
- DOI/arXiv/URL: arXiv:2405.11169, <https://arxiv.org/abs/2405.11169>, DOI <https://doi.org/10.48550/arXiv.2405.11169>
- Relevant content: Proves unobstructedness after bulk deformation for moment Lagrangian correspondences induced by symplectic reduction maps at free zero level sets, assuming equivariant Kuranishi structures and compatible perturbations.
- Applicability check: This addresses a key obstruction in the correspondence approach, but it is conditional on technical structures and does not itself prove the stated equivalence.

Exact/Liouville-sector and special-geometric results

- Title: “Equivariant Partially Wrapped Fukaya Categories on Liouville Sectors”
- Authors: Dongwook Choa, Jiawei Hu, Siu-Cheong Lau, Yan-Lung Leon Li

- Venue/year: arXiv preprint, 2025
- DOI/arXiv/URL: arXiv:2512.24382, <https://arxiv.org/abs/2512.24382>, DOI <https://doi.org/10.48550/arXiv.2512.24382>
- Relevant content: Develops equivariant wrapped/partially wrapped Fukaya categories for Liouville sectors and a correspondence theory. In the free-action case it produces a fully faithful A_∞ -functor from a full subcategory of the equivariant wrapped Fukaya category to the wrapped Fukaya category of the exact quotient.
- Applicability check: This is an exact/Liouville-sector result and is fully faithful from a full subcategory, not the full general equivalence for arbitrary Hamiltonian compact-torus symplectic manifolds and all spectral components.
- Title: “The Hamiltonian reduction of hypertoric mirror symmetry”
- Authors: Michael McBreen, Vivek Shende, Peng Zhou
- Venue/year: Advances in Mathematics 493, 2026, article 110881
- DOI/arXiv/URL: DOI <https://doi.org/10.1016/j.aim.2026.110881>; arXiv:2405.07955, <https://arxiv.org/abs/2405.07955>
- Relevant content: Proves a “Fukaya category commutes with reduction” theorem for the Hamiltonian torus action on a multiplicative hypertoric variety.
- Applicability check: This is a strong special-case theorem, but it is restricted to multiplicative hypertoric geometry and does not cover arbitrary Hamiltonian torus actions.

4.24.3.4. What remains unresolved

The unresolved part is the full, global categorical statement in Teleman’s Conjecture 4.2 exactly as in the problem statement: an equivalence between the τ -spectral/multiplicity category of $\mathfrak{F}(X)$ and the Fukaya category of $X//_q T$ with B -field $\text{Im } \log \tau$, for general compact torus T , general Hamiltonian T -symplectic manifolds X , and all regular values $q = \text{Re } \log \tau$, without imposing exact/Liouville, toric/hypertoric, relative exactness, weak unobstructedness, free-action-on-selected-Lagrangians, local mirror-chart, or bulk-deformation hypotheses beyond those in the problem statement.

Therefore the correct status is: partially resolved.

4.24.4. Remaining open problems recorded by the database

4.24.4.1. Open problem 1

Let

$$X = \{(x, y, z, w) \in \mathbb{C}^4 : xy + zw = 1\} \cong T^*S^3$$

and let S^1 act with weights $((1, -1, 1, -1))$. Prove that the spectral components of its S^1 -equivariant wrapped Fukaya category satisfy

$$\mathcal{W}^{S^1}(X)_1 \simeq \mathcal{W}(T^*S^2), \quad \mathcal{W}^{S^1}(X)_\lambda \simeq 0 \quad (\lambda \neq 1),$$

where the zero-level Hamiltonian reduction of (X) is T^*S^2 .

Reference supplied by the database. Yanki Lekili and Ed Segal, “Equivariant Fukaya Categories at Singular Values,” *Glasgow Mathematical Journal* **68** (2026), no. 2, 197–221, prediction in Example 2.8, especially equation (2.12), DOI: <https://doi.org/10.1017/S0017089525100487>, arXiv: 2304.10969.

4.24.5. Primary-source pointer

The originating ICM record is *Gauge theory and mirror symmetry* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=1329>.

4.25. 171. Quasi-morphisms and quasi-states in symplectic topology [2014-geom-132]

Record: 2014-geom-132

Subfield: Symplectic and contact geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: Quasi-morphisms and quasi-states in symplectic topology

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=1167>

4.25.1. Problem statement

Let (M, ω) be a closed connected symplectic manifold, and let $C^\infty(M)$ and $C(M)$ denote the real-valued smooth and continuous functions on M . Let $\mathbb{A} \subset C^\infty(M)$ be a finite-dimensional Poisson-commutative vector subspace, meaning that $\{F, G\} = 0$ for every $F, G \in \mathbb{A}$. Define the moment map of $\mathbb{A}$ by

$$\Phi_{\mathbb{A}} : M \rightarrow \mathbb{A}^*, \quad \langle \Phi_{\mathbb{A}}(x), F \rangle = F(x).$$

Is it true that, for every such $\mathbb{A}$, there exist $p \in \mathbb{A}^*$ with $\Phi_{\mathbb{A}}^{-1}(p) \neq \emptyset$ and a symplectic quasi-state $\zeta : C(M) \rightarrow \mathbb{R}$ such that the fiber $\Phi_{\mathbb{A}}^{-1}(p)$ is heavy with respect to ζ ? Here a symplectic quasi-state is a normalized, monotone, homogeneous functional on $C(M)$ whose restriction to $C^\infty(M)$ is additive on Poisson-commuting functions, vanishes on functions supported in stably displaceable open subsets, and is invariant under $\text{Symp}_0(M)$, the identity component of the symplectomorphism group; a closed set $X \subset M$ is heavy with respect to ζ if $\zeta(H) \geq \inf_X H$ for every $H \in C(M)$. Equivalently, the desired fiber should satisfy

$$\zeta(H) \geq \inf_{\Phi_{\mathbb{A}}^{-1}(p)} H$$

for every $H \in C(M)$.

4.25.2. Historical background, literature, and current progress

4.25.3. Status: partially resolved

4.25.3.1. Problem identified

The problem asks whether every finite-dimensional Poisson-commutative subspace $\mathbb{A} \subset C^\infty(M)$ on an arbitrary closed connected symplectic manifold has a nonempty moment-map fiber $\Phi_{\mathbb{A}}^{-1}(p)$ that is heavy with respect to some genuine symplectic quasi-state $\zeta : C(M) \rightarrow \mathbb{R}$, where “genuine” is understood in the sense stated in the problem: normalized, monotone, homogeneous, additive on Poisson-commuting smooth functions, vanishing on stably displaceable supports, and $\text{Symp}_0(M)$ -invariant.

4.25.3.2. Current literature status

Original source

- Title: Quasi-morphisms and quasi-states in symplectic topology
- Author: Michael Entov
- Venue/year: Proceedings of the International Congress of Mathematicians 2014, Seoul; arXiv preprint 2014
- DOI/arXiv/URL: arXiv:1404.6408, <https://arxiv.org/abs/1404.6408>
- Relevant item: Question 4.9 asks whether any finite-dimensional Poisson-commutative subspace has a heavy fiber, at least with respect to some symplectic quasi-state. The preceding Corollary 4.8 gives only a non-displaceable fiber.
- Hypothesis/conclusion check: This is the source of the problem. The survey distinguishes genuine symplectic quasi-states from partial symplectic quasi-states; for arbitrary idempotents, including the fundamental class, one generally gets only the weaker partial version.

Earlier full result for non-displaceability, not heaviness

- Title: Quasi-states and symplectic intersections
- Authors: Michael Entov and Leonid Polterovich
- Venue/year: Commentarii Mathematici Helvetici 81 (2006), 75-99
- DOI/arXiv/URL: arXiv:math/0410338, <https://arxiv.org/abs/math/0410338>
- Relevant result: Theorem 2.1 proves that every finite-dimensional Poisson-commutative subspace of $C^\infty(M)$ has at least one non-displaceable fiber.
- Hypothesis/conclusion check: This covers all closed symplectic manifolds and all such A , but the conclusion is non-displaceability, not heaviness with respect to a genuine symplectic quasi-state.

Cases where genuine quasi-states are available

- Title: Symplectic quasi-states and semi-simplicity of quantum homology
- Authors: Michael Entov and Leonid Polterovich
- Venue/year: Toric Topology, Contemporary Mathematics 460, American Mathematical Society, 2008, 47-70
- DOI/arXiv/URL: DOI 10.1090/conm/460/09010; arXiv:0705.3735, <https://arxiv.org/abs/0705.3735>
- Relevant result: Constructs Calabi quasi-morphisms and genuine symplectic quasi-states for manifolds whose quantum homology has a suitable field direct summand.
- Hypothesis/conclusion check: Combined with later heavy-fiber theorems, this gives the literal conclusion for these field-split/field-factor cases. The field-factor hypothesis is extra and is not known for all closed symplectic manifolds.

Pseudoheavy fibers and simple partial quasi-states

- Title: Existence of pseudo-heavy fibers of moment maps
- Authors: Morimichi Kawasaki and Ryuma Orita
- Venue/year: Communications in Contemporary Mathematics 23 (2021), no. 5, 2050047
- DOI/arXiv/URL: DOI 10.1142/S0219199720500479; arXiv:1901.09395, <https://arxiv.org/abs/1901.09395>

- Relevant results: Theorem 1.7 proves that for any closed symplectic manifold, any finite-dimensional Poisson-commutative A , and any partial symplectic quasi-state, some fiber is pseudoheavy. Corollary 1.9 gives a heavy fiber when the partial symplectic quasi-state is simple.
- Hypothesis/conclusion check: Pseudoheaviness is weaker than heaviness, and simplicity is an extra hypothesis. The paper also gives examples of pseudoheavy fibers that are not heavy, so this does not resolve the problem statement as stated.

Strong recent heavy-fiber theorem in the spectral-invariant sense

- Title: Symplectic topology and ideal-valued measures
- Authors: Adi Dickstein, Yaniv Ganor, Leonid Polterovich, and Frol Zapolsky
- Venue/year: arXiv preprint, 2021; revised 2024
- DOI/arXiv/URL: arXiv:2107.10012, <https://arxiv.org/abs/2107.10012>
- Relevant contribution: Develops quantum-cohomology ideal-valued quasi-measures and SH-heavy sets, providing the framework used in later relative symplectic cohomology results.
- Hypothesis/conclusion check: This is major progress toward heavy-type rigidity and big-fiber theorems, but it does not itself supply a genuine scalar symplectic quasi-state for every closed symplectic manifold.
- Title: A characterization of heaviness in terms of relative symplectic cohomology
- Authors: Cheuk Yu Mak, Yuhan Sun, and Umut Varolgunes
- Venue/year: Journal of Topology 17 (2024), no. 1, e12327
- DOI/arXiv/URL: DOI 10.1112/topo.12327; arXiv:2301.12625, <https://arxiv.org/abs/2301.12625>
- Relevant results: Theorem 1.7 proves that a compact set is heavy if and only if its relative symplectic cohomology over the Novikov field is nonzero. Corollary 1.8 proves that the union of two Poisson-commuting non-heavy compact sets is non-heavy. Corollary 1.9 states that any involutive map $\Phi : M \rightarrow \mathbb{R}^N$ has at least one heavy fiber, and the paper explicitly says this gives a positive answer to Entov's Question 4.9.
- Hypothesis/conclusion check: Choosing a basis of A identifies Φ_A with an involutive map $M \rightarrow \mathbb{R}^N$, so Corollary 1.9 covers the same fibers. However, the heaviness in this paper is the standard spectral-invariant/idempotent notion, with the unit idempotent omitted in the introduction. This is the heavy-fiber conclusion for the homogenized spectral invariant/partial quasi-state framework. It implies the literal genuine-quasi-state version in cases where the chosen idempotent gives a genuine Entov-Polterovich symplectic quasi-state, for example field-factor cases. It does not by itself prove that every closed connected symplectic manifold admits a genuine symplectic quasi-state satisfying the stated axioms.

4.25.3.3. What remains unresolved

The original heavy-fiber question has very strong progress, and in the standard spectral-invariant/partial-quasi-state sense Mak-Sun-Varolgunes prove the expected heavy fiber for every involutive map. But the problem statement as written asks for a genuine symplectic

quasi-state on every closed connected symplectic manifold. I found no published or otherwise citable theorem proving that all closed connected symplectic manifolds admit such a genuine quasi-state, or that the Mak-Sun-Varolgunes heavy fiber can always be promoted to heaviness with respect to a genuine quasi-state satisfying the full additivity and homogeneity axioms.

Therefore the exact statement is partially resolved, not fully resolved.

4.25.4. Remaining open problems recorded by the database

4.25.4.1. Open problem 1

Construct a symplectic quasi-state on the standard symplectic four-torus $(T^4, \omega_{\text{std}})$, or prove that none exists.

Reference supplied by the database. Michael Entov, “Quasi-morphisms and quasi-states in symplectic topology,” in *Proceedings of the International Congress of Mathematicians—Seoul 2014*, Vol. II, Kyung Moon Sa, Seoul, 2014, 1147–1171, §3 (explicitly identifies T^4 as the basic open case for existence of a symplectic quasi-state). [arXiv:1404.6408](https://arxiv.org/abs/1404.6408). Current evidence and status: Adi Dickstein and Frol Zapolsky, “Constraints on symplectic quasi-states,” *International Journal of Mathematics* **37**(7) (2026), DOI: <https://doi.org/10.1142/S0129167X26500291>, [arXiv:2407.08014](https://arxiv.org/abs/2407.08014).

4.25.5. Primary-source pointer

The originating ICM record is *Quasi-morphisms and quasi-states in symplectic topology* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=1167>.

4.26. 172. Gauge theory and mirror symmetry [2014-ag-152]

Record: 2014-ag-152

Subfield: Symplectic and contact geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: Gauge theory and mirror symmetry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=1329>

4.26.1. Problem statement

Let G be a compact Lie group, let $L \subset G$ be a Levi subgroup, and let $G^\vee$ and $L^\vee$ denote the Langlands dual groups. For a group H equal to $G^\vee$ or $L^\vee$, choose a Borel subgroup of $H_{\mathbb{C}}$ with unipotent radical N_H , Lie algebra $\mathfrak{n}_H$, and regular character $\chi_H \in \mathfrak{n}_H^*$. Define the Toda space

$$T(H) := (N_H, \chi_H) \backslash T^*H_{\mathbb{C}} // (N_H, \chi_H),$$

the holomorphic symplectic quotient of $T^*H_{\mathbb{C}}$ by the left and right N_H -actions, reduced at (χ_H, χ_H) . The Toda isomorphism identifies $T(H)$ with the BFM holomorphic symplectic manifold

$$BFM(H) = T_{\text{reg}}^*H_{\mathbb{C}} //_{\text{Ad}} H_{\mathbb{C}}.$$

Assume $N_{L^\vee} = N_{G^\vee} \cap L_{\mathbb{C}}^\vee$, let w_0 be a representative of the longest Weyl element of $G^\vee$, and let $w_{L^\vee}$ be a representative of the longest Weyl element of $L^\vee$. Let

$$N_{G^\vee} \times_{N_{L^\vee}} w_0 w_{L^\vee}^{-1} \cdot L_{\mathbb{C}}^\vee \times_{N_{L^\vee}} N_{G^\vee} \subset G_{\mathbb{C}}^\vee$$

be the inclusion that determines an open holomorphic symplectic embedding $T(L^\vee) \hookrightarrow T(G^\vee)$.

Prove or disprove that, via the Toda isomorphism, this embedding is the mirror of the symplectic induction functor SInd_L^G , which sends a categorical representation $\mathcal{C}$ of L to the Fukaya category of the flag variety G/L with coefficients in the associated local system of categories $G \times_L \mathcal{C} \rightarrow G/L$. Equivalently, prove or disprove that the flag variety G/L , viewed as the Hamiltonian G -manifold used for this induction and as a domain wall between pure three-dimensional L - and G -gauge theories, has as its mirror the above open embedding between the Toda/BFM spaces of $L^\vee$ and $G^\vee$.

4.26.2. Historical background, literature, and current progress

4.26.3. Verdict: partially resolved

4.26.3.1. Precise problem

The problem asks for the status of Teleman's Levi/Toda mirror-symmetry conjecture: for every compact Lie group G and Levi $L \subset G$, the open holomorphic symplectic embedding

$$T(L^\vee) \hookrightarrow T(G^\vee)$$

coming from the $N_{G^\vee} \times_{N_{L^\vee}} w_0 w_{L^\vee}^{-1} L_C^\vee \times_{N_{L^\vee}} N_{G^\vee}$ open piece should, after the Toda/BFM identification, be the mirror of the symplectic induction domain wall SInd_L^G implemented by the flag variety G/L , including the categorical Fukaya/KRS interpretation.

4.26.3.2. Status

This is not fully resolved. The holomorphic symplectic geometry on the BFM/Toda side is substantially established, and there is strong decategorified and special-case evidence from quantum cohomology and Rietsch mirrors. However, I found no citable theorem proving the full categorical statement in all compact Lie group/Levi cases. The exact mirror identification with symplectic induction still depends on foundations that the literature itself describes as conjectural, especially equivariant Fukaya categories with coefficients and the Kapustin-Rozansky-Saulina/Rozansky-Witten 2-category framework.

4.26.3.3. Key literature and progress

1. Gauge theory and mirror symmetry

Author: Constantin Teleman.

Venue/year: Proceedings of the International Congress of Mathematicians–Seoul 2014, Vol. II, Kyung Moon Sa, 1309–1332, 2014.

arXiv/URL: arXiv:1404.6305, <https://arxiv.org/abs/1404.6305>; PDF at <https://math.berkeley.edu/~teleman/math/icmtalk.pdf>.

Relevant results: This is the original source of the problem. Teleman proves the Toda/BFM holomorphic symplectic isomorphism in Theorem 6.3 and states that the Levi open cell inclusion determines an open embedding $T(L) \subset T(G)$. The exact mirror statement is Conjecture 6.5, not a theorem. Remark 6.7 explicitly says the conjecture is difficult to prove without precise definitions of equivariant Fukaya categories with coefficients and the KRS 2-category. Theorem 6.8 proves a smooth foliation of $\mathrm{BFM}(G^\vee)$ by leaves indexed by Levi/cuspidal data, but the categorical interpretation of those leaves is still conjectural.

2. Equivariant homology and K-theory of affine Grassmannians and Toda lattices

Authors: Roman Bezrukavnikov, Michael Finkelberg, Ivan Mirkovic.

Venue/year: Compositio Mathematica 141 no. 3, 746–768, 2005.

DOI/URL: <https://doi.org/10.1112/S0010437X04001228>; Cambridge page <https://www.cambridge.org/core/journals/compositio-mathematica/article/equivariant-homology-and-ktheory-of-affine-grassmannians-and-toda-lattices/EDBE1754F7F5B202B9998A90A9D3CD8D>.

Relevant results: Foundational BFM/Toda result identifying equivariant homology/K-theory of affine Grassmannians with universal centralizer/Toda structures. It supports the BFM side used by Teleman, but it does not prove the mirror-domain-wall assertion for symplectic induction.

3. A mirror symmetric construction of $qH_T^*(G/P)_{(q)}$

Author: Konstanze Rietsch.

Venue/year: Advances in Mathematics 217 no. 6, 2401–2442, 2008.

DOI/arXiv: <https://doi.org/10.1016/j.aim.2007.08.010>; arXiv:math/0511124.

Relevant results: Gives Lie-theoretic Landau-Ginzburg mirrors for G/P and recovers Peterson presentations of equivariant quantum cohomology after localizing quantum

parameters. This is key evidence for Teleman's final-section recovery of Rietsch mirrors, but it is a quantum-cohomology/mirror-family result, not a proof that the Toda open embedding is the KRS/Fukaya mirror of SInd_L^G .

4. Crystals and Mirror Constructions for Quotients

Author: George W. Melvin.

Venue/year: PhD thesis, University of California, Berkeley, 2017.

URL: <https://escholarship.org/uc/item/1xm099m7>; PDF also at https://personal.colby.edu/personal/g/gwmelvin/thesis_final.pdf.

Relevant results: Surveys Teleman's construction and explicitly distinguishes the proved foliation from the conjectural Fukaya-category statement. Melvin notes that the BFM foliation is proved in Teleman's Theorem 6.8, while the Fukaya-category interpretation depends on unproven homological mirror symmetry equivalences and the conjectural KRS 2-category. The thesis verifies a quantum-cohomology consequence in low-rank polygon-space examples, so it is partial progress only.

5. The role of Coulomb branches in 2D gauge theory

Author: Constantin Teleman.

Venue/year: Journal of the European Mathematical Society 23 no. 11, 3497–3520, 2021.

DOI/arXiv: <https://doi.org/10.4171/JEMS/1071>; arXiv:1801.10124.

Relevant results: Develops the Coulomb-branch/BFM interpretation and actions on equivariant quantum cohomology. This strengthens the rigorous BFM and quantum-cohomology framework, but it does not prove the full categorical Levi-domain-wall mirror statement.

6. Affine nil-Hecke algebras and quantum cohomology

Authors: Eduardo Gonzalez, Cheuk Yu Mak, Dan Pomerleano.

Venue/year: Advances in Mathematics 415, Article 108867, 2023.

DOI/arXiv: <https://doi.org/10.1016/j.aim.2023.108867>; arXiv:2202.05785.

Relevant results: Constructs nonabelian shift-operator actions and proves that $QH_G^*(M)$ defines a canonical holomorphic Lagrangian subvariety in $BFM(G_{\mathbb{C}}^V)$, confirming part of Teleman's expectation at the level of Lagrangian support. The authors still state that the relevant Rozansky-Witten 2-category is not yet rigorously defined, and their flag-variety example is described as conjectural pending comparison with the module structure. Thus it is important partial progress, not a full resolution.

7. Homological Mirror Symmetry for the universal centralizers

Author: Xin Jin.

Venue/year: arXiv preprint, 2022 revision.

arXiv/URL: arXiv:2206.09035, <https://arxiv.org/abs/2206.09035>. Earlier adjoint-case version: arXiv:2107.13395.

Relevant results: Proves homological mirror symmetry for universal centralizers/Toda spaces J_G , with a partially wrapped Fukaya category on J_G on the A-side and coherent sheaves on a quotient of a dual maximal torus on the B-side. This is substantial progress around Toda/BFM spaces, but it is not the same as Teleman's domain-wall statement: it does not identify the Levi open embedding $T(L^V) \hookrightarrow T(G^V)$ with the symplectic induction functor G/L between L - and G -gauge theories.

8. Equivariant Lagrangian correspondence and a conjecture of Teleman

Authors: Siu-Cheong Lau, Nai-Chung Conan Leung, Yan-Lung Leon Li.

Venue/year: arXiv preprint, 2023, revised later.

arXiv/URL: arXiv:2312.13926, <https://arxiv.org/abs/2312.13926>.

Relevant results: Addresses a Teleman conjecture about equivariant Lagrangian correspondences and the relation between disc potentials upstairs and downstairs under symplectic quotient. This is adjacent progress on Teleman’s quotient-mirror program, but it concerns a different quotient/potential statement and does not settle the Levi/Toda symplectic-induction domain wall in this problem.

4.26.3.4. What remains unresolved

The remaining missing theorem would have to give a rigorous all-cases categorical mirror equivalence showing that the open symplectic embedding $T(L^\vee) \hookrightarrow T(G^\vee)$, under the Toda/BFM isomorphism, represents the domain wall/functor SInd_L^G on categorical representations, with Fukaya categories of G/L with coefficient systems matching the appropriate KRS/Rozansky-Witten 2-categorical morphism. Existing results prove the geometric embedding, the Toda/BFM identification, related foliation statements, quantum-cohomology shadows, and some special or adjacent Teleman conjectures, but not this full categorical identification.

4.26.4. Remaining open problems recorded by the database

4.26.4.1. Open problem 1

Let G be a compact Lie group with maximal torus T , and let $q \in T^\vee$ specify a one-dimensional representation of T . Prove the torus-Levi case of symplectic-induction mirror symmetry: under the Toda isomorphism, the cotangent fibre at q maps to a Lagrangian leaf $\Lambda(q) \subset \mathrm{BFM}(G^\vee)$ representing the symplectically induced representation, equivalently the G -equivariant Fukaya category of G/T with quantum parameter q .

Reference supplied by the database. Constantin Teleman, “Gauge theory and mirror symmetry,” in *Proceedings of the International Congress of Mathematicians—Seoul 2014*, Vol. II, Kyung Moon Sa, Seoul, 2014, pp. 1309–1332, Conjecture 6.5 and its torus-Levi specialization in Example 6.6. arXiv: [1404.6305](https://arxiv.org/abs/1404.6305); ICM proceedings PDF: [ICM 2014, Vol. II](#).

4.26.4.2. Open problem 2

Let $L \subset G$ be a Levi subgroup, let T be a maximal torus, and let ν be a regular value of the moment map for the Hamiltonian T -action on G/L . If $t \in Z(L_\mathbb{C})$ denotes the exponential of the symplectic structure on G/L and $\Lambda(t)$ is the Lagrangian support of its T -equivariant Fukaya category in $\mathrm{BFM}(T^\vee) = T^*T_\mathbb{C}^\vee$, prove that the Fukaya category of the symplectic reduction $(G/L)//_\nu T$ is the category $\mathrm{Hom}(S_\nu, \Lambda(t))$, where S_ν is the cotangent fibre over $\exp(\nu) \in T^\vee$.

Reference supplied by the database. George W. Melvin, *Crystals and Mirror Constructions for Quotients*, Ph.D. thesis, University of California, Berkeley, 2017, Conjecture 3.2.6 (attributed there to Constantin Teleman). eScholarship permalink: <https://escholarship.org/uc/item/1xm099m7>.

4.26.4.3. Open problem 3

Under the hypotheses of the preceding symplectic-reduction problem, assume that $(G/L)//_{\nu}T$ is Fano. Prove that its quantum cohomology is the Jacobian ring of the restriction of the T -equivariant superpotential to the fibre over $\exp(2\pi i\nu)$ of the equivariant-structure map on the mirror of G/L , with the quantum deformation obtained by varying $t \in Z(L_{\mathbb{C}})$; if G has finite nontrivial centre Z , account for every critical point with multiplicity $|Z|$.

Reference supplied by the database. George W. Melvin, *Crystals and Mirror Constructions for Quotients*, Ph.D. thesis, University of California, Berkeley, 2017, Conjecture 3.2.7. eScholarship permalink: <https://escholarship.org/uc/item/1xm099m7>.

4.26.5. Primary-source pointer

The originating ICM record is *Gauge theory and mirror symmetry* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=1329>.

4.27. 173. The structure of algebraic varieties [2014-ag-173]

Record: 2014-ag-173

Subfield: Symplectic and contact geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: The structure of algebraic varieties

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.1/ICM2014.1.pdf#page=411>

4.27.1. Problem statement

Prove that rational connectedness is determined by the underlying symplectic structure: if X and Y are smooth projective complex varieties whose underlying smooth manifolds, with their natural symplectic structures, are symplectomorphic, then X is rationally connected if and only if Y is rationally connected.

Here a projective variety Z is rationally connected if, for every finite set of points $z_1, \dots, z_r \in Z$, there exists a morphism $\phi : \mathbb{CP}^1 \rightarrow Z$ whose image contains $z_1, \dots, z_r$.

4.27.2. Historical background, literature, and current progress

4.27.3. Verdict: partially resolved

4.27.3.1. Precise problem

The problem asks whether rational connectedness is determined by the natural symplectic structure of a smooth projective complex variety: if smooth projective complex varieties X and Y are symplectomorphic as smooth manifolds with their natural symplectic forms, must X be rationally connected if and only if Y is rationally connected?

The literature usually formulates this as invariance under symplectic deformation equivalence. That is slightly broader than a single symplectomorphism of chosen natural symplectic forms, so positive results for symplectic deformation equivalence apply to the stated symplectomorphism formulation. No citable result found covers all smooth projective varieties in all dimensions.

4.27.3.2. Original source and restatement

- **Title:** Low degree polynomial equations: arithmetic, geometry and topology
Author: Janos Kollar
Venue/year: European Congress of Mathematics, Vol. I (Budapest, 1996), Progress in Mathematics 168, Birkhauser, 1998, pp. 255-288
arXiv/URL: arXiv:alg-geom/9607016, <https://arxiv.org/abs/alg-geom/9607016>
Relevant result: Conjecture 4.2.7 states that for smooth projective varieties over $\mathbb{C}$, rational connectedness should be invariant under symplectic deformation equivalence. The-

orem 4.2.10 proves the analogous invariance for uniruledness, and Kollar notes that the rational-connectedness conjecture follows in the case $\dim H^2(X, \mathbb{Q}) = 1$. This is evidence and a special case, not a full solution.

- **Title:** The structure of algebraic varieties
Author: Janos Kollar
Venue/year: Proceedings of the International Congress of Mathematicians, Seoul 2014, Vol. I, Kyung Moon Sa, 2014, pp. 395-419/420
arXiv/URL: arXiv:1407.7478, <https://arxiv.org/abs/1407.7478>
Relevant result: Conjecture 7.4 restates that being rationally connected is a property of the underlying symplectic structure and cites only partial results.

4.27.3.3. Partial progress

- **Title:** Rationally connected 3-folds and symplectic geometry
Author: Claire Voisin
Venue/year: Asterisque 322, 2008, pp. 1-21
arXiv/URL: arXiv:0801.1396, <https://arxiv.org/abs/0801.1396>; Numdam: https://www.numdam.org/item/AST_2008__322__1_0/
Relevant result: Theorem 0.8 proves the conjecture for compact Kahler threefolds when X is Fano or when X is rationally connected and $b_2(X) \leq 2$. The paper also observes the surface case. These are restricted cases only.
- **Title:** Symplectic geometry of rationally connected threefolds
Author: Zhiyu Tian
Venue/year: Duke Mathematical Journal 161, no. 5, 2012, pp. 803-843
DOI/arXiv: <https://doi.org/10.1215/00127094-1548398>; arXiv:1006.2486, <https://arxiv.org/abs/1006.2486>
Relevant result: Theorem 1.7 proves Kollar's conjecture in complex dimension 3. This fully resolves the threefold case, but not higher dimensions.
- **Title:** Symplectic geometry and rationally connected 4-folds
Author: Zhiyu Tian
Venue/year: Journal fur die reine und angewandte Mathematik 698, 2015, pp. 221-244
DOI/arXiv: <https://doi.org/10.1515/crelle-2013-0005>; arXiv:1208.4343, <https://arxiv.org/abs/1208.4343>
Relevant result: Theorem 1.2 proves rational connectedness for smooth projective fourfolds symplectic deformation equivalent to either a Fano fourfold of pseudo-index at least 2 or a rationally connected fourfold with second Betti number 2. This is partial fourfold progress and does not cover arbitrary fourfolds.
- **Title:** Symplectic invariance of rational surfaces on Kahler manifolds
Author: Jason Michael Starr
Venue/year: arXiv preprint, 2019 version
DOI/arXiv: <https://doi.org/10.48550/arXiv.1803.06412>; arXiv:1803.06412, <https://arxiv.org/abs/1803.06412>
Relevant result: The paper proves Kollar's conjecture for Kahler manifolds symplectically deformation equivalent to G/P and for certain low-degree complete intersections in such spaces. This is higher-dimensional evidence, but it has strong geometric hypotheses and does not imply the full all-varieties statement.
- **Title:** Towards the Symplectic Graber-Harris-Starr Theorems
Author: Zhiyu Tian

Venue/year: Acta Mathematica Sinica, English Series 42, 2026, pp. 791-805

DOI/arXiv: <https://doi.org/10.1007/s10114-026-5298-z>;
arXiv:1208.4340, <https://arxiv.org/abs/1208.4340>

Relevant result: This develops related questions about symplectic rational connectedness and Gromov-Witten invariants. It explicitly treats Kollar's conjecture as known for threefolds and then studies related inductive questions; it is not a proof of the full conjecture.

- **Title:** Gromov-Witten invariants and rationally connectedness

Author: Jianxun Hu

Venue/year: In *Gromov-Witten theory, gauge theory and dualities*, Proceedings of the Centre for Mathematics and its Applications, Australian National University, vol. 48, 2019, 23 pp.

URL: <https://maths.anu.edu.au/research/cma-proceedings/gromov-witten-theory-gauge-theory-and-dualities-anukioloa-6-16-january>

Relevant result: Survey-style account of symplectic rational connectedness and Gromov-Witten approaches. It records Tian's threefold theorem and Tian's fourfold partial results; it does not claim a full solution.

- **Title:** Cohomological splitting over rationally connected bases

Authors: Shaoyun Bai, Daniel Pomerleano, Guangbo Xu

Venue/year: arXiv preprint, 2024

DOI/arXiv: <https://doi.org/10.48550/arXiv.2406.00931>;
arXiv:2406.00931, <https://arxiv.org/abs/2406.00931>

Relevant result: Introduces and uses enumerative/symplectic rational connectedness for cohomological splitting of Hamiltonian fibrations. It cites Kollar's conjecture and the Voisin/Tian work as motivation, but its theorems concern Hamiltonian fibrations over enumeratively rationally connected bases, not invariance of algebraic rational connectedness for arbitrary symplectomorphic projective varieties.

4.27.3.4. What remains unresolved

The complete statement remains open in dimensions at least 4 for arbitrary smooth projective complex varieties. Existing results cover all dimensions up to 3 and several special fourfold or higher-dimensional families, but I found no published or otherwise citable theorem proving that every pair of symplectomorphic smooth projective complex varieties has the same rational-connectedness status.

4.27.4. Remaining open problems recorded by the database

4.27.4.1. Open problem 1

Is every smooth complex projective rationally connected variety symplectically rationally connected, in the sense that it has a nonzero genus-zero Gromov-Witten invariant with two point-class insertions (and possibly further insertions)?

Reference supplied by the database. Jianxun Hu, "Gromov-Witten Invariants and Rationally Connectedness," in Peter Bouwknegt, Brett Parker, and Bai-Ling Wang (eds.), *Gromov-Witten Theory, Gauge Theory and Dualities: Proceedings of the Centre for Mathematics and its Appli-*

cations, *Australian National University*, vol. 48, Australian National University, Canberra, 2019, pp. 74–96, Conjecture 3.4. ISBN 978-0-6481056-2-6. https://maths.anu.edu.au/files/4gromov-witten-invariants_and_rationally_connectedness.pdf

4.27.4.2. Open problem 2

For smooth complex projective fourfolds, is rational connectedness invariant under symplectic deformation equivalence (and hence under symplectomorphism)?

Reference supplied by the database. Zhiyu Tian, “Symplectic Geometry and Rationally Connected 4-Folds,” *Journal für die reine und angewandte Mathematik* 698 (2015), 221–244, Conjecture 1.1 and the fourfold problem studied in the paper. DOI: 10.1515/crelle-2013-0005. arXiv:1208.4343. <https://doi.org/10.1515/crelle-2013-0005>

4.27.4.3. Open problem 3

Let X be a smooth Fano fourfold of Picard number one, and let C be a general embedded very free rational curve obtained by the Kollár–Miyaoka–Mori quotient construction from a free rational curve of minimal anticanonical degree. Is the normal bundle $N_{C/X}$ balanced?

Reference supplied by the database. Zhiyu Tian, “Symplectic Geometry and Rationally Connected 4-Folds,” *Journal für die reine und angewandte Mathematik* 698 (2015), 221–244, Remark 1.5 (the explicit question on whether the normal bundle of a general such curve is balanced). DOI: 10.1515/crelle-2013-0005. arXiv:1208.4343. <https://doi.org/10.1515/crelle-2013-0005>

4.27.4.4. Open problem 4

Does every extremal ray of the Mori cone of a smooth complex projective fourfold contain a curve class that gives a nonzero genus-zero Gromov–Witten invariant, so that the ray remains effective under symplectic deformation?

Reference supplied by the database. Zhiyu Tian, “Symplectic Geometry and Rationally Connected 4-Folds,” *Journal für die reine und angewandte Mathematik* 698 (2015), 221–244, §3 (the explicitly identified problem of showing that all extremal rays support curve classes giving nonzero Gromov–Witten invariants). DOI: 10.1515/crelle-2013-0005. arXiv:1208.4343. <https://doi.org/10.1515/crelle-2013-0005>

4.27.5. Primary-source pointer

The originating ICM record is *The structure of algebraic varieties* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.1/ICM2014.1.pdf#page=411>.

4.28. 174. Gauge theory and mirror symmetry [2014-geom-283]

Record: 2014-geom-283

Subfield: Symplectic and contact geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: Gauge theory and mirror symmetry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=1329>

4.28.1. Problem statement

Work in a Fukaya-categorical framework in which unitary B -field twists, Fukaya categories with local coefficients in categories, topological actions of compact Lie groups, and equivariant functor categories are defined. Let G be a compact connected Lie group acting in a Hamiltonian fashion on a compact symplectic manifold (X, ω) , and let $\Phi : X \rightarrow \mathfrak{g}^*$ be an equivariant moment map. Let $\mu \in \mathfrak{g}^*$ be a regular value of Φ , let

$$G_\mu := \{g \in G : \text{Ad}_g^* \mu = \mu\}$$

be its coadjoint stabilizer, and let ν be a compatible parameter specifying a unitary B -field twist $i\nu$. Define the symplectic reduction at the coadjoint orbit $G \cdot \mu$ by

$$X //_{G \cdot \mu} G := \Phi^{-1}(G \cdot \mu) / G,$$

equivalently $\Phi^{-1}(\mu) / G_\mu$.

A topological categorical representation of a compact Lie group is a linear category endowed with an action that factors through the topology of the group. Let $\text{Vect}_{\mu+i\nu}$ be the one-dimensional topological categorical representation of G_μ determined by the complexified parameter $\mu + i\nu$. For a topological categorical representation $\mathcal{C}$ of G_μ , define its symplectic induction to G by

$$\text{SInd}_{G_\mu}^G(\mathcal{C}) := \mathcal{F}(G/G_\mu; G \times_{G_\mu} \mathcal{C}),$$

the Fukaya category of the coadjoint orbit G/G_μ , with its Kirillov–Kostant–Souriau symplectic form determined by μ , and with coefficients in the associated local system of categories $G \times_{G_\mu} \mathcal{C} \rightarrow G/G_\mu$. For topological categorical G -representations $\mathcal{A}$ and $\mathcal{B}$, let $\text{Hom}_G(\mathcal{A}, \mathcal{B})$ denote the category of G -equivariant functors from $\mathcal{A}$ to $\mathcal{B}$, called the multiplicity category of $\mathcal{A}$ in $\mathcal{B}$. Equip $\mathcal{F}(X)$ with the topological G -action induced by the Hamiltonian action on X .

Prove that there is an equivalence of categories

$$\mathcal{F}(X //_{G \cdot \mu} G, i\nu) \simeq \text{Hom}_G(\text{SInd}_{G_\mu}^G(\text{Vect}_{\mu+i\nu}), \mathcal{F}(X)).$$

4.28.2. Historical background, literature, and current progress

The exact assertion is **unresolved**. No citable source found proves the stated equivalence for every compact symplectic X , compact connected G , regular value μ , and unitary B -field parameter ν . The most recent sources continue to treat even closely related regular-value categorical equivalences as conjectural, or prove results only after imposing substantially different hypotheses.

- Constantin Teleman, “Gauge theory and mirror symmetry,” *Proceedings of the International Congress of Mathematicians, Seoul 2014*, Vol. II, 1309–1332, 2014, arXiv:1404.6305. Conjecture 6.2 is the original nonabelian symplectic-induction/multiplicity statement. Teleman explicitly presents the general Fukaya-categorical framework as partly experimental and without proofs in full generality.
- Yanki Lekili and Ed Segal, “Equivariant Fukaya categories at singular values,” *Glasgow Mathematical Journal* 68(2), 197–221, 2026 (published online 2025), DOI: 10.1017/S0017089525100487; arXiv:2304.10969. Equation (1.7) still calls the regular-value S^1 spectral-component equivalence a Teleman conjecture. The paper supplies mirror-symmetry evidence and examples, while identifying spin-sign, unobstructedness, bounding-cochain, and possible bulk-deformation issues. It does not prove the arbitrary compact, nonabelian statement.
- Daniel Pomerleano and Constantin Teleman, “Quantization commutes with reduction again: the quantum GIT conjecture I,” preprint, 2024 (revised 2026), arXiv:2405.20301. Theorem 1 gives a quantum-cohomology/gauged- A -model reduction result for compact G -monotone manifolds with smooth free reduction. The paper separately states the Fukaya-category equivalence as Conjecture 1 and says it is not addressed there. Thus this is strong decategorified progress under additional monotonicity and linearization assumptions, not a proof of the asserted categorical equivalence.
- Siu-Cheong Lau, Nai-Chung Conan Leung, and Yan-Lung Leon Li, “Equivariant Lagrangian correspondence and a conjecture of Teleman,” preprint, 2023, arXiv:2312.13926. This constructs equivariant Lagrangian-correspondence machinery and proves an open-string, local relation between disc potentials; the equivariant obstruction is solved in the compact semi-Fano toric setting. It does not establish an equivalence of the full categories in the statement.
- Dongwook Choa, Jiawei Hu, Siu-Cheong Lau, and Yan-Lung Leon Li, “Equivariant Partially Wrapped Fukaya Categories on Liouville Sectors,” preprint, 2025 (revised 2026), arXiv:2512.24382. Theorem 1.3/6.15 constructs, for an exact Liouville manifold with a free compact-group action, a fully faithful A_∞ -functor from the full subcategory generated by invariant Lagrangians in the zero moment level to the wrapped Fukaya category of the exact quotient, and identifies those two full subcategories. This is a rigorous categorical reduction result in the noncompact exact setting, but it neither covers compact X nor identifies Teleman’s full multiplicity category with the quotient category for all parameters.
- Michael McBreen, Vivek Shende, and Peng Zhou, “The Hamiltonian reduction of hypertoric mirror symmetry,” *Advances in Mathematics* 493, article 110881, 2026, DOI: 10.1016/j.aim.2026.110881; arXiv:2405.07955. Theorems 1.4–1.5 prove a “Fukaya category commutes with reduction” result for the Hamiltonian torus action on multiplicative hypertoric varieties using wrapped Fukaya categories and mirror symmetry. This is a special noncompact hypertoric case and does not cover the compact nonabelian conjecture.

Accordingly, established work supplies categorical foundations, functors, local open-string identities, a quantum-cohomology analogue, and exact/hypertoric special cases. What remains is a construction and proof of the precise symplectic-induction multiplicity equivalence in the full compact nonabelian setting, including the stated B -field dependence and all regular values.

4.28.3. Remaining open problems recorded by the database

4.28.3.1. Open problem 1

Work in a Fukaya-categorical framework in which unitary B -field twists, topological torus actions, and equivariant functor categories are defined. Let T be a compact torus acting in a Hamiltonian fashion on a compact symplectic manifold (X, ω) , with moment map $\Phi : X \rightarrow \mathfrak{t}^*$. Let $\tau \in T_{\mathbb{C}}^{\vee}$ be a complex character parameter for which

$$\mu := \operatorname{Re} \log \tau \in \mathfrak{t}^*$$

is a regular value of Φ , and write $\nu := \operatorname{Im} \log \tau$ for the associated unitary B -field parameter. Let $\operatorname{Vect}_{\tau}$ be the one-dimensional topological categorical representation of T classified by τ , equip $\mathcal{F}(X)$ with the topological T -action induced by the Hamiltonian action, and let $\operatorname{Hom}_T(\operatorname{Vect}_{\tau}, \mathcal{F}(X))$ be the category of T -equivariant functors. Prove that

$$\mathcal{F}(\Phi^{-1}(\mu)/T, i\nu) \simeq \operatorname{Hom}_T(\operatorname{Vect}_{\tau}, \mathcal{F}(X)).$$

Reference supplied by the database. Constantin Teleman, “Gauge theory and mirror symmetry,” Proceedings of the International Congress of Mathematicians, Seoul 2014, Vol. II, Kyung Moon Sa, Seoul, 2014, pp. 1309-1332, Conjecture 4.2 (“Torus symplectic quotients”), arXiv:1404.6305, <https://arxiv.org/abs/1404.6305>.

4.28.3.2. Open problem 2

Let G be a compact connected Lie group acting in a Hamiltonian fashion on a compact symplectic manifold (X, ω) , with moment map $\mu : X \rightarrow \mathfrak{g}^*$. Assume that the action is anticanonically linearized, in the sense that

$$[\omega - \mu] = c_1(TX) \in H_G^2(X),$$

and assume that G acts freely on $\mu^{-1}(0)$, so that

$$X // G := \mu^{-1}(0)/G$$

is a smooth symplectic manifold. Let $\mathcal{F}(X)^G$ denote the homotopy fixed-point category of the topological G -action on the monotone Fukaya category $\mathcal{F}(X)$. Prove that the moment-level Lagrangian correspondence

$$\Gamma_\mu := \{(x, [x]) \mid \mu(x) = 0\} \subset X^- \times (X // G)$$

induces an equivalence of A_∞ -categories

$$\mathcal{F}(X)^G \simeq \mathcal{F}(X // G).$$

Reference supplied by the database. Daniel Pomerleano and Constantin Teleman, “Quantization commutes with reduction again: the quantum GIT conjecture I,” preprint, 2024, revised 2026, Section 1.3, Conjecture 1, arXiv:2405.20301, <https://arxiv.org/abs/2405.20301>.

4.28.4. Primary-source pointer

The originating ICM record is *Gauge theory and mirror symmetry* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=1329>.

4.29. 175. Stability conditions in symplectic topology [2018-geom-299]

Record: 2018-geom-299

Subfield: Symplectic and contact geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Stability conditions in symplectic topology

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/uploads/ICM-2018-vol2-ver1-eb.pdf#page=1078>

4.29.1. Problem statement

Let (X, ω) be a symplectic manifold of real dimension $2n$, either closed or convex at infinity, and assume that $c_1(X) = 0$. Let $D^\pi \mathcal{F}(X, \omega)$ denote the split-closed derived Fukaya category of (X, ω) , and let $K(D^\pi \mathcal{F}(X, \omega))$ be its numerical Grothendieck group. Let $\mathcal{T}_X$ be the Teichmüller space of marked pairs (J, Ω) on the fixed smooth manifold X , where J is an integrable complex structure compatible with ω and $\Omega \in \Omega^{n,0}(X; J)$ is a nowhere-vanishing J -holomorphic volume form.

Construct an injective map

$$\mathcal{T}_X \rightarrow \text{Stab}(D^\pi \mathcal{F}(X, \omega)), \quad (J, \Omega) \mapsto \sigma_{J, \Omega} = (Z_{J, \Omega}, \mathcal{P}_{J, \Omega}),$$

to the space of locally finite numerical Bridgeland stability conditions satisfying the support property. Thus $Z_{J, \Omega} : K(D^\pi \mathcal{F}(X, \omega)) \rightarrow \mathbb{C}$ is the central charge and, for each $\phi \in \mathbb{R}$, $\mathcal{P}_{J, \Omega}(\phi)$ is the full additive subcategory of semistable objects of phase ϕ . Require that every Lagrangian brane L defining an object of $D^\pi \mathcal{F}(X, \omega)$ satisfy

$$Z_{J, \Omega}([L]) = \int_L \Omega,$$

where $[L] \in K(D^\pi \mathcal{F}(X, \omega))$ is its numerical Grothendieck class, and require every special Lagrangian brane L of phase ϕ , meaning that $\Omega|_L$ has constant argument $\pi\phi$, to lie in $\mathcal{P}_{J, \Omega}(\phi)$.

In addition, construct a version of Lagrangian mean curvature flow for Lagrangian branes that permits surgeries at finite-time singularities and whose output, for every object $E \in D^\pi \mathcal{F}(X, \omega)$, realizes the Harder–Narasimhan filtration of E with respect to $\sigma_{J, \Omega}$: a finite sequence of exact triangles whose successive factors lie in $\mathcal{P}_{J, \Omega}(\phi_1), \dots, \mathcal{P}_{J, \Omega}(\phi_k)$ for strictly decreasing phases $\phi_1 > \dots > \phi_k$.

4.29.2. Historical background, literature, and current progress

The problem is **unresolved**. It asks for one construction valid for every closed or convex-at-infinity symplectic Calabi–Yau manifold, together with a global Lagrangian mean-curvature flow with surgery realizing every Harder–Narasimhan filtration. No cited result supplies either construction in this generality, let alone both. The current literature still explicitly describes the basic passage from a holomorphic volume form to a stability condition on the derived Fukaya category as open.

- Ivan Smith, *Stability conditions in symplectic topology*, in *Proceedings of the International Congress of Mathematicians–Rio de Janeiro 2018*, Vol. II, World Scientific, 2018, pp. 969–991, DOI 10.1142/9789813272880_0084, arXiv:1711.04263. Section 3.1 presents the injection, period central charge, special-Lagrangian semistability, and mean-curvature-flow realization of Harder–Narasimhan filtrations as conjectural.
- Dominic Joyce, *Conjectures on Bridgeland stability for Fukaya categories of Calabi–Yau manifolds, special Lagrangians, and Lagrangian mean curvature flow*, *EMS Surveys in Mathematical Sciences* 2 (2015), no. 1, 1–62, DOI 10.4171/EMSS/8, arXiv:1401.4949. Joyce formulates the stability condition and a conjectural Lagrangian mean-curvature flow with surgeries whose limiting special-Lagrangian currents are the semistable factors; this is a conjectural framework, not a proof.
- Fabian Haiden, Ludmil Katzarkov, and Maxim Kontsevich, *Flat surfaces and stability structures*, *Publications Mathématiques de l’IHES* 126 (2017), 247–318, DOI 10.1007/s10240-017-0095-y, arXiv:1409.8611. Theorems 5.2 and 5.3 construct stability conditions from finite-type flat surfaces, with central charge given by periods and semistable objects represented by constant-phase geodesics, and identify the marked-flat-surface space bianalytically with a union of components of the relevant surface Fukaya-category stability space. This is substantive progress in real dimension two, but it does not cover arbitrary (X, ω) or construct the requested mean-curvature flow with surgery.
- Jason D. Lotay and Goncalo Oliveira, *Special Lagrangians, Lagrangian mean curvature flow and the Gibbons–Hawking ansatz*, *Journal of Differential Geometry* 126 (2024), no. 3, 1121–1184, DOI 10.4310/jdg/1717348872, arXiv:2002.10391. They prove versions of special-Lagrangian existence and long-time Lagrangian mean-curvature flow for circle-invariant Lagrangians in Gibbons–Hawking Calabi–Yau two-folds. These symmetry-restricted results neither construct the general Bridgeland map nor realize Harder–Narasimhan filtrations for all Fukaya objects.
- Yang Li, *Thomas–Yau conjecture and holomorphic curves*, *EMS Surveys in Mathematical Sciences* 12 (2025), 323–475, DOI 10.4171/EMSS/96, arXiv:2203.01467. Li explicitly states that constructing a stability condition on the derived Fukaya category from a holomorphic volume form remains open. The paper proves Floer-theoretic obstructions under extra hypotheses and develops a variational program for exact, quantitatively almost-calibrated Lagrangians in Calabi–Yau Stein manifolds, but explicitly does not complete the Thomas–Yau conjecture.
- Yu-Wei Fan, *Special Lagrangians and Bridgeland stable objects beyond geometric stability conditions: the product case*, preprint, 2026, arXiv:2602.03041. Theorem 1.3 constructs a finite-dimensional family of stability conditions on certain stopped wrapped Fukaya categories associated with $(\mathbb{C}^\times)^n$, with period central charges, and proves that stable objects admit special-Lagrangian representatives. This is a restricted product case and does not prove the converse special-Lagrangian semistability statement or any flow-with-surgery realization.

Thus the literature establishes important low-dimensional, symmetric, exact, stopped, and product-type cases, as well as obstruction and uniqueness results, but the universal injective map and the global surgery flow required by the problem remain unconstructed.

4.29.3. Remaining open problems recorded by the database

4.29.3.1. Open problem 1

Let (M, J, g, Ω) be a Calabi–Yau manifold of complex dimension m , and let $L \subset M$ be a compact graded Lagrangian with finitely many isolated conical singularities. Assume that, near each singular point, L is asymptotic to a stable special Lagrangian cone $C \subset \mathbb{C}^m$, where stability means that the obstruction space in the deformation theory of special Lagrangians with that conical singularity vanishes. Extend Lagrangian Floer cohomology, its obstruction and bounding-cochain theory, and the derived Fukaya category $D^b \mathcal{F}(M)$ so that such singular Lagrangians are objects, in a manner compatible with the usual theories for smooth Lagrangians and with compactification and gluing for moduli spaces of J -holomorphic discs whose boundaries lie on L and may pass through its singular points.

Reference supplied by the database. Dominic Joyce, *Conjectures on Bridgeland stability for Fukaya categories of Calabi–Yau manifolds, special Lagrangians, and Lagrangian mean curvature flow*, EMS Surveys in Mathematical Sciences 2 (2015), no. 1, 1–62, Problem 3.20. DOI: 10.4171/EMSS/8; arXiv:1401.4949.

4.29.3.2. Open problem 2

For an integer $n \geq 2$, equip $(\mathbb{C}^\times)^n$ with the split exact symplectic form

$$\omega_n = \pi_1^* \omega_{\mathbb{C}^\times} + \cdots + \pi_n^* \omega_{\mathbb{C}^\times},$$

where $\mathbb{C}^\times \cong T^*S^1$ has its canonical exact symplectic form, and let $\mathfrak{s}_n$ be the product stop. Let $\mathcal{S}_n \subset \mathbb{C}^n$ be the connected n -dimensional parameter locus for the product-type Bridgeland stability conditions

$$\sigma_{\mathbf{a},c} \in \text{Stab}(H^0 \text{Tw } \mathcal{W}((\mathbb{C}^\times)^n, \mathfrak{s}_n)), \quad (\mathbf{a}, c) = (a_1, \dots, a_n, c) \in \mathcal{S}_n \times \mathbb{C},$$

whose central charges are

$$Z_{\mathbf{a},c}(\mathbb{L}) = \int_{[\mathbb{L}]} \Omega_{\mathbf{a},c}, \quad \Omega_{\mathbf{a},c} = \exp\left(z_1 + \cdots + z_n + c + \frac{e^{a_1}}{z_1} + \cdots + \frac{e^{a_n}}{z_n}\right) \frac{dz_1}{z_1} \cdots \frac{dz_n}{z_n}.$$

Determine whether every special Lagrangian brane in $((\mathbb{C}^\times)^n, \omega_n)$ with respect to $\Omega_{\mathbf{a},c}$ that defines an object of $H^0 \text{Tw } \mathcal{W}((\mathbb{C}^\times)^n, \mathfrak{s}_n)$ is $\sigma_{\mathbf{a},c}$ -stable.

Reference supplied by the database. Yu-Wei Fan, *Special Lagrangians and Bridgeland stable objects beyond geometric stability conditions: the product case*, arXiv preprint (2026), Remark 1.8(i). arXiv:2602.03041; <https://arxiv.org/abs/2602.03041>.

4.29.4. Primary-source pointer

The originating ICM record is *Stability conditions in symplectic topology* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/uploads/ICM-2018-vol2-ver1-eb.pdf#page=1078>.

4.30. 176. Gamma classes and quantum cohomology [2022-geom-288]

Record: 2022-geom-288

Subfield: Symplectic and contact geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2022: Gamma classes and quantum cohomology

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-562-9-volume-4.pdf#page=195>

4.30.1. Problem statement

Let X be a smooth projective complex variety, or a compact symplectic manifold, of complex dimension n . Assume that its genus-zero quantum product converges on a domain $U \subset H^{\text{even}}(X; \mathbb{C})$. Write $\star_\tau$ for the quantum product at $\tau \in U$, write $(\cdot, \cdot)$ for the Poincaré pairing, and let $N = \dim_{\mathbb{C}} H^{\text{even}}(X; \mathbb{C})$. Fix a homogeneous basis $\{\phi_a\}$ of $H^{\text{even}}(X; \mathbb{C})$ and the corresponding linear coordinates $\{\tau^a\}$ on U . Define the quantum connection on the trivial vector bundle $H^{\text{even}}(X; \mathbb{C}) \times (U \times \mathbb{C})$ by

$$\nabla_{\partial/\partial\tau^a} = \frac{\partial}{\partial\tau^a} + \frac{1}{z}(\phi_a \star_\tau), \quad \nabla_{z\partial_z} = z\frac{\partial}{\partial z} - \frac{1}{z}(E \star_\tau) + \mu,$$

where

$$E = c_1(X) + \sum_a \left(1 - \frac{\deg(\phi_a)}{2}\right) \tau^a \phi_a, \quad \mu(\phi_a) = \left(\frac{\deg(\phi_a)}{2} - \frac{n}{2}\right) \phi_a.$$

Let $K(X) = K_{\text{top}}^0(X)$. If $\delta_1, \dots, \delta_n$ are the Chern roots of TX , define the Gamma class

$$\hat{\Gamma}_X = \prod_{j=1}^n \Gamma(1 + \delta_j).$$

Let $L(\tau, z)z^{-\mu}z^{c_1(X)}$ be the fundamental solution of the quantum connection normalized at the large-radius limit. For $V \in K(X)$, define the associated Gamma-integral flat section by

$$\mathfrak{s}(V)(\tau, z) = L(\tau, z)z^{-\mu}z^{c_1(X)} \left((2\pi)^{-n/2} \hat{\Gamma}_X (2\pi i)^{\deg/2} \text{ch}(V) \right),$$

where $(2\pi i)^{\deg/2}$ acts on $H^{2p}(X; \mathbb{C})$ by multiplication by $(2\pi i)^p$.

Assume that the Frobenius algebra $(H^{\text{even}}(X; \mathbb{C}), \star_{\tau_0})$ is semisimple at some $\tau_0 \in U$. After restricting to a neighborhood of τ_0 , choose analytic idempotents $\psi_1, \dots, \psi_N$ and analytic functions $u_1, \dots, u_N$ such that

$$\psi_i \star_\tau \psi_j = \delta_{ij} \psi_i, \quad E \star_\tau \psi_i = u_i \psi_i,$$

and define normalized idempotents $\Psi_i = (\psi_i, \psi_i)^{-1/2} \psi_i$, with a choice of square root for each i . A phase $\phi \in \mathbb{R}$ is admissible at τ_0 if

$$e^{i\phi} \notin \mathbb{R}_{>0}(u_i(\tau_0) - u_j(\tau_0)) \quad \text{for all } i \neq j.$$

For an admissible phase ϕ , let $y_1^\phi, \dots, y_N^\phi$ be the uniquely determined basis of ∇ -flat sections near $\tau = \tau_0$ and the ray $\arg z = \phi$ such that, for some $\varepsilon > 0$,

$$e^{u_i(\tau)/z} y_i^\phi(\tau, z) \longrightarrow \Psi_i(\tau) \quad \text{as } z \rightarrow 0 \text{ in } |\arg z - \phi| < \pi + \varepsilon.$$

Prove that there exist classes $\mathcal{E}_1^\phi, \dots, \mathcal{E}_N^\phi \in K(X)$ such that

$$y_i^\phi(\tau, z) = \mathfrak{s}(\mathcal{E}_i^\phi)(\tau, z), \quad i = 1, \dots, N,$$

in a neighborhood of $\tau = \tau_0$ and the ray $\arg z = \phi$.

4.30.2. Historical background, literature, and current progress

The exact problem is **unresolved**. It is the topological form of Gamma Conjecture II: at every semisimple point, every sectorial asymptotic basis should be contained in the Gamma-integral lattice coming from topological K -theory. The literature proves this in several important geometric families, usually in the stronger form in which the K -classes lift to a full exceptional collection, but no result covers all smooth projective varieties or compact symplectic manifolds satisfying the stated convergence and semisimplicity hypotheses. No counterexample to this exact topological assertion was found.

Hiroshi Iritani, “Gamma Classes and Quantum Cohomology,” *Proceedings of the International Congress of Mathematicians 2022*, Vol. 4, EMS Press, 2023, pp. 2552–2574, DOI <https://doi.org/10.4171/ICM2022/156>, arXiv:2307.15938. Conjecture 2.7 states exactly the topological K -theory assertion in the problem, without a Fano hypothesis or an exceptional-collection hypothesis. The article records proofs only for type-A Grassmannians, Fano toric manifolds, and quadrics, and therefore leaves the universal claim open.

Sergey Galkin, Vasily Golyshev, and Hiroshi Iritani, “Gamma Classes and Quantum Cohomology of Fano Manifolds: Gamma Conjectures,” *Duke Mathematical Journal* **165** (2016), no. 11, 2005–2077, DOI <https://doi.org/10.1215/00127094-3476593>, arXiv:1404.6407. Conjecture 4.6.1 formulates the stronger Fano/exceptional-collection version. Theorems and constructions in Sections 5–6 prove it for projective spaces and all type-A Grassmannians; these cases imply the topological statement here but do not address arbitrary semisimple quantum cohomology.

Bohan Fang and Peng Zhou, “Gamma II for Toric Varieties from Integrals on T-Dual Branes and Homological Mirror Symmetry,” arXiv preprint (2019), arXiv:1903.05300. Theorem 6.4 proves the stronger Gamma II conjecture for complete smooth toric Fano varieties at the semisimple points considered there, using oscillatory integrals and homological mirror symmetry. This supplies a broad special family, not a general theorem under the hypotheses of the problem.

Xiaowen Hu and Hua-Zhong Ke, “Gamma Conjecture II for Quadrics,” *Mathematische Annalen* **387** (2023), 927–983, DOI <https://doi.org/10.1007/s00208-022-02477-7>, arXiv:2103.15143. Theorem 6.1 proves Gamma Conjecture II for every smooth quadric hypersurface and also establishes convergence of its full quantum cohomology. Their Theorem 3.8 gives a useful sufficient criterion for general Fano manifolds, but that criterion is not automatic from convergence and semisimplicity.

Jianxun Hu, Hua-Zhong Ke, Changzheng Li, and Zhitong Su, “Gamma Conjecture II via Global Gamma-I,” arXiv preprint (2026), arXiv:2606.07418. Theorem 1.4 proves Gamma Conjecture II for all del Pezzo surfaces, and Corollary 1.6 extends it to blow-ups of $\mathbb{P}^2$ at at

most eight points. Its strategy theorem requires additional hypotheses involving a global Gamma-I property, the small quantum spectrum, and exceptional collections. The paper explicitly presents the known proof landscape as a small collection of geometric families rather than a resolution of the universal conjecture.

Thus the established progress consists of family-specific proofs and conditional criteria. What remains is the assertion for every convergent semisimple quantum cohomology allowed by the statement, including non-Fano projective varieties outside the known families and the full compact-symplectic setting.

4.30.3. Remaining open problems recorded by the database

4.30.3.1. Open problem 1

Let G be a connected complex semisimple algebraic group, let $P \subset G$ be a parabolic subgroup, and let $X = G/P$, of complex dimension n . Assume that the genus-zero quantum product of X converges on a domain $U \subset H^{\text{even}}(X; \mathbb{C})$. Write $\star_\tau$ for the quantum product at $\tau \in U$, write $(\cdot, \cdot)$ for the Poincaré pairing, and let $N = \dim_{\mathbb{C}} H^{\text{even}}(X; \mathbb{C})$. Fix a homogeneous basis $\{\phi_a\}$ and corresponding coordinates $\{\tau^a\}$, and define the quantum connection by

$$\nabla_{\partial/\partial\tau^a} = \frac{\partial}{\partial\tau^a} + \frac{1}{z}(\phi_a \star_\tau), \quad \nabla_{z\partial_z} = z\frac{\partial}{\partial z} - \frac{1}{z}(E \star_\tau) + \mu,$$

where

$$E = c_1(X) + \sum_a \left(1 - \frac{\deg(\phi_a)}{2}\right) \tau^a \phi_a, \quad \mu(\phi_a) = \left(\frac{\deg(\phi_a)}{2} - \frac{n}{2}\right) \phi_a.$$

Let $K(X) = K_{\text{top}}^0(X)$ and

$$\hat{\Gamma}_X = \prod_{j=1}^n \Gamma(1 + \delta_j),$$

where $\delta_1, \dots, \delta_n$ are the Chern roots of TX . Let $L(\tau, z)z^{-\mu}z^{c_1(X)}$ be the fundamental solution normalized at the large-radius limit, and for $V \in K(X)$ define

$$\mathfrak{s}(V)(\tau, z) = L(\tau, z)z^{-\mu}z^{c_1(X)} \left((2\pi)^{-n/2} \hat{\Gamma}_X(2\pi i)^{\deg/2} \text{ch}(V) \right),$$

where $(2\pi i)^{\deg/2}$ acts on $H^{2p}(X; \mathbb{C})$ by multiplication by $(2\pi i)^p$.

Suppose that $(H^{\text{even}}(X; \mathbb{C}), \star_{\tau_0})$ is semisimple at some $\tau_0 \in U$. Near τ_0 , choose analytic idempotents $\psi_1, \dots, \psi_N$ and analytic eigenvalues $u_1, \dots, u_N$ satisfying

$$\psi_i \star_\tau \psi_j = \delta_{ij} \psi_i, \quad E \star_\tau \psi_i = u_i \psi_i,$$

and set $\Psi_i = (\psi_i, \psi_i)^{-1/2} \psi_i$. For a phase $\phi \in \mathbb{R}$ satisfying

$$e^{i\phi} \notin \mathbb{R}_{>0}(u_i(\tau_0) - u_j(\tau_0)) \quad \text{for all } i \neq j,$$

let $y_1^\phi, \dots, y_N^\phi$ be the uniquely determined basis of ∇ -flat sections such that, for some $\varepsilon > 0$,

$$e^{u_i(\tau)/z} y_i^\phi(\tau, z) \longrightarrow \Psi_i(\tau) \quad \text{as } z \rightarrow 0 \text{ in } |\arg z - \phi| < \pi + \varepsilon.$$

Prove that there exist classes $\mathcal{E}_1^\phi, \dots, \mathcal{E}_N^\phi \in K(X)$ such that

$$y_i^\phi(\tau, z) = \mathfrak{s}(\mathcal{E}_i^\phi)(\tau, z), \quad i = 1, \dots, N,$$

near $\tau = \tau_0$ and the ray $\arg z = \phi$.

Reference supplied by the database. Daewoong Cheong and Changzheng Li, “On the Conjecture O of GGI for G/P ,” *Advances in Mathematics* **306** (2017), 704–721. DOI: <https://doi.org/10.1016/j.aim.2016.10.033>.

4.30.4. Primary-source pointer

The originating ICM record is *Gamma classes and quantum cohomology* (2022). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-562-9-volume-4.pdf#page=195>.

4.31. 177. Gamma classes and quantum cohomology [2022-geom-332]

Record: 2022-geom-332

Subfield: Symplectic and contact geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2022: Gamma classes and quantum cohomology

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-562-9-volume-4.pdf#page=195>

4.31.1. Problem statement

Let X be a compact symplectic manifold of real dimension $2n$, equipped with a compatible almost-complex structure. Work on the even cohomology $H^{\text{even}}(X; \mathbb{C})$. Suppose that the genus-zero quantum product $\star_\tau$ converges at a parameter $\tau \in H^{\text{even}}(X; \mathbb{C})$. For a homogeneous basis $\{\phi_i\}$, with $\tau = \sum_i \tau^i \phi_i$, set

$$E = c_1(X) + \sum_i \left(1 - \frac{\deg \phi_i}{2}\right) \tau^i \phi_i, \quad \mu(\alpha) = \left(\frac{\deg \alpha}{2} - \frac{n}{2}\right) \alpha$$

for homogeneous α . Let $\text{QDM}(X)_\tau$ be the restriction to $\{\tau\} \times \mathbb{C}_z$ of the quantum cohomology D -module, namely the trivial bundle with fiber $H^{\text{even}}(X; \mathbb{C})$ and quantum connection whose z -direction is

$$\nabla_{z\partial_z} = z\partial_z - \frac{1}{z}(E\star_\tau) + \mu.$$

Let $C \subset \mathbb{C}$ be the set of distinct eigenvalues of $E\star_\tau$.

Assume that the germ of $\text{QDM}(X)_\tau$ at $z = 0$ is of exponential type: its formal completion over $\mathbb{C}[[z]]$ admits a decomposition of meromorphic connections

$$\widehat{\Phi} : \widehat{\text{QDM}}(X)_\tau \cong \bigoplus_{u \in C} (e^{u/z} \otimes \widehat{\mathcal{R}}_u),$$

where $e^{u/z}$ is the rank-one connection $(\mathbb{C}\{z\}, d + d(u/z))$ and each $\mathcal{R}_u$ is a free $\mathbb{C}\{z\}$ -module equipped with a regular-singular connection. Here $\widehat{\mathcal{R}}_u = \mathcal{R}_u \otimes_{\mathbb{C}\{z\}} \mathbb{C}[[z]]$.

Choose an admissible phase $\phi \in \mathbb{R}$, meaning that

$$e^{i\phi} \notin \mathbb{R}_{>0}(u - u')$$

for all distinct $u, u' \in C$. For some $\varepsilon > 0$, let

$$I = \left\{ z \in \mathbb{C}^\times : |\arg z - \phi| < \frac{\pi}{2} + \varepsilon \right\},$$

where the choice of the real lift ϕ fixes a lift of I to the universal cover of $\mathbb{C}^\times$. The Hukuhara–Turrittin theorem gives the sectorial analytic lift

$$\Phi_I : \text{QDM}(X)_\tau|_I \cong \bigoplus_{u \in C} e^{u/z} \otimes \mathcal{R}_u|_I$$

whose asymptotic expansion at $z = 0$ is $\widehat{\Phi}$. Consequently, the vector space S_I of ∇ -flat sections on I decomposes as

$$S_I = \bigoplus_{u \in C} \mathcal{U}_u,$$

where $\mathcal{U}_u$ is the flat-section space of the summand $e^{u/z} \otimes \mathcal{R}_u$. Use the chosen lift of I to identify S_I with the space $S(X)$ of multivalued flat sections of $\text{QDM}(X)_\tau$ on the universal cover of $\mathbb{C}^\times$.

Let

$$K(X) = K_{\text{top}}^0(X) / \text{torsion}.$$

For the Chern roots $\delta_1, \dots, \delta_n$ of TX , define the Gamma class

$$\widehat{\Gamma}_X = \prod_{j=1}^n \Gamma(1 + \delta_j),$$

with the product expanded as a symmetric power series in the Chern roots. Let

$$\mathfrak{s} : K(X) \longrightarrow S(X)$$

be the Gamma-integral-structure map

$$\mathfrak{s}(V)(\tau, z) = L(\tau, z) z^{-\mu} z^{c_1(X)} \left((2\pi)^{-n/2} \widehat{\Gamma}_X (2\pi i)^{\deg/2} \text{ch}(V) \right).$$

Here $(2\pi i)^{\deg/2}$ acts on $H^{2k}(X; \mathbb{C})$ by multiplication by $(2\pi i)^k$, and $L(\tau, z) z^{-\mu} z^{c_1(X)}$ is the fundamental solution of the quantum connection normalized by

$$L(\tau, z) = (\text{id} + O(e^\sigma, \tau')) e^{-\sigma/z}$$

at the large-radius limit, where $\tau = \sigma + \tau'$ with $\sigma \in H^2(X; \mathbb{C})$ and $\tau' \in \bigoplus_{k \neq 2} H^k(X; \mathbb{C})$. Prove that there is a direct-sum decomposition of lattices

$$K(X) = \bigoplus_{u \in C} V_u^\phi$$

such that, for every $u \in C$,

$$\mathcal{U}_u = \mathfrak{s}(V_u^\phi) \otimes_{\mathbb{Z}} \mathbb{C}.$$

4.31.2. Historical background, literature, and current progress

Verdict: unresolved.

The statement is the topological lattice form of the Sanda–Shamoto/Galkin conjecture, recorded as Conjecture 2.13 in Hiroshi Iritani, *Gamma classes and quantum cohomology*, Proceedings of the International Congress of Mathematicians 2022, Vol. 4 (EMS Press, 2023), 2552–2574, DOI <https://doi.org/10.4171/ICM2022/156>. Iritani states precisely that the Stokes decomposition of an exponential-type quantum connection should be induced by a direct-sum decomposition of the Gamma-integral topological K -lattice. The article presents it as a conjecture for compact symplectic manifolds, not as a theorem.

The strongest directly relevant non-semisimple result is Fumihiko Sanda and Yota Shamoto, *An analogue of Dubrovin's conjecture*, Annales de l'Institut Fourier 70 (2020), no. 2, 621–682, DOI <https://doi.org/10.5802/aif.3321>, arXiv:1705.05989. Their Theorem 7.9

proves the stronger mutation-system/categorical form for smooth Fano complete intersections in projective space of Fano index greater than one. It therefore yields the required lattice compatibility for that special class and its small quantum connection, but not for arbitrary compact symplectic X , arbitrary convergent τ , or all exponential-type connections.

In the semisimple setting, Gamma Conjecture II implies the asserted rank-one lattice decomposition. Established cases include:

- Sergey Galkin, Vasily Golyshev, and Hiroshi Iritani, *Gamma classes and quantum cohomology of Fano manifolds: Gamma conjectures*, Duke Mathematical Journal 165 (2016), no. 11, 2005–2077, DOI <https://doi.org/10.1215/00127094-3476593>, arXiv:1404.6407. The paper proves Gamma Conjecture II for projective spaces and type-A Grassmannians.
- Bohan Fang and Peng Zhou, *Gamma II for toric varieties from integrals on T-dual branes and homological mirror symmetry*, preprint (2019), arXiv:1903.05300. Theorem 6.4 proves Gamma Conjecture II for smooth toric Fano manifolds.
- Xiaowen Hu and Hua-Zhong Ke, *Gamma conjecture II for quadrics*, Mathematische Annalen 387 (2023), 927–983, DOI <https://doi.org/10.1007/s00208-022-02477-7>, arXiv:2103.15143. Theorem 6.1 proves convergence of the full quantum cohomology and Gamma Conjecture II for smooth quadrics.
- Jianxun Hu, Hua-Zhong Ke, Changzheng Li, and Zhitong Su, *Gamma conjecture II via global Gamma-I*, preprint (2026), arXiv:2606.07418. Theorem 1.4 proves Gamma Conjecture II for all del Pezzo surfaces; the paper also describes Grassmannians, toric Fano manifolds, and quadrics as the previously known general families.

Related recent progress does not close the scope gap. Yefeng Shen and Mark Shoemaker, *Quantum spectrum and Gamma structure for standard flips*, preprint (2025), arXiv:2502.08762, proves Gamma-class decompositions compatible with standard semiorthogonal decompositions for projective bundles, blow-ups, and standard flips after restriction to an extremal curve direction. This is not a theorem about the full quantum connection in the statement.

Thus there is substantial progress for special Fano geometries, including genuinely non-semisimple complete intersections, but no cited result or compatible collection of results covers every compact symplectic manifold and every convergent exponential-type quantum connection required by the exact problem.

4.31.3. Remaining open problems recorded by the database

4.31.3.1. Open problem 1

Let X be a smooth complex Fano manifold of complex dimension n . Suppose that its genus-zero quantum product converges at a Kahler parameter $q = e^\tau$, with $\tau \in H^2(X; \mathbb{C})$, and that the operator

$$c_1(X) \star_q : H^*(X; \mathbb{C}) \longrightarrow H^*(X; \mathbb{C})$$

has a simple rightmost eigenvalue u , meaning that u is simple and $\operatorname{Re}(u) > \operatorname{Re}(v)$ for every other eigenvalue v . Let $S(\tau, z)z^{-\mu}z^{c_1(X)}$ be the fundamental solution of the quantum connection normalized at the large-radius limit. Define the principal asymptotic class $A_X(\tau) \in H^*(X; \mathbb{C})$, up to multiplication by a nonzero scalar, by requiring that for some $m \geq 0$,

$$\|e^{\mu/z} S(\tau, z) z^{-\mu} z^{c_1(X)} A_X(\tau)\| = O(z^{-m}) \quad (z \rightarrow 0^+).$$

Let

$$\hat{\Gamma}_X = \prod_{j=1}^n \Gamma(1 + \delta_j), \quad \text{Ch}(V) = \sum_{k \geq 0} (2\pi i)^k \text{ch}_k(V),$$

where $\delta_1, \dots, \delta_n$ are the Chern roots of TX . Determine whether there always exists a class $V \in K_{\text{top}}^0(X)$ such that

$$[A_X(\tau)] = [\hat{\Gamma}_X \text{Ch}(V)]$$

in the projective space $\mathbb{P}(H^*(X; \mathbb{C}))$.

Reference supplied by the database. Sergey Galkin, Jianxun Hu, Hiroshi Iritani, Huazhong Ke, Changzheng Li, and Zhitong Su, “Revisiting Gamma conjecture I: counterexamples and modifications,” preprint, 2024 (revised 2025), Question 6.13, arXiv:2405.16979, <https://arxiv.org/abs/2405.16979>.

4.31.3.2. Open problem 2

Let X be a smooth complex Fano manifold of complex dimension n . Assume that its big genus-zero quantum product is convergent and semisimple at some $\tau_0 \in H^{\text{even}}(X; \mathbb{C})$ and that the bounded derived category $D_{\text{coh}}^b(X)$ admits a full exceptional collection. Let $\psi_1, \dots, \psi_N$ be an idempotent basis for $(H^{\text{even}}(X; \mathbb{C}), \star_{\tau_0})$, let

$$E \star_{\tau_0} \psi_i = u_i \psi_i, \quad \Psi_i = (\psi_i, \psi_i)^{-1/2} \psi_i,$$

where $(\cdot, \cdot)$ is the Poincaré pairing, and choose a phase $\phi \in \mathbb{R}$ such that $e^{i\phi}$ is not a positive real multiple of $u_i - u_j$ whenever $u_i \neq u_j$. Let $y_i^\phi(\tau, z)$ be the asymptotically exponential flat section of the quantum connection characterized near $\tau = \tau_0$ by

$$e^{u_i/z} y_i^\phi(\tau, z) \rightarrow \Psi_i$$

as $z \rightarrow 0$ in a sector $|\arg z - \phi| < \frac{\pi}{2} + \varepsilon$ for some $\varepsilon > 0$. Define

$$\hat{\Gamma}_X = \prod_{j=1}^n \Gamma(1 + \delta_j)$$

for the Chern roots δ_j of TX , and define the Gamma-integral flat section associated with $V \in K_{\text{top}}^0(X)$ by

$$\mathfrak{s}(V)(\tau, z) = L(\tau, z) z^{-\mu} z^{c_1(X)} \left((2\pi)^{-n/2} \hat{\Gamma}_X (2\pi i)^{\deg/2} \text{ch}(V) \right),$$

where $L(\tau, z) z^{-\mu} z^{c_1(X)}$ is the fundamental solution normalized at the large-radius limit. Prove that, for every admissible phase ϕ , there is a full exceptional collection $(\mathcal{E}_1^\phi, \dots, \mathcal{E}_N^\phi)$ in $D_{\text{coh}}^b(X)$ such that

$$y_i^\phi(\tau, z) = \mathfrak{s}(\mathcal{E}_i^\phi)(\tau, z)$$

near $\tau = \tau_0$ and $\arg z = \phi$ for every i .

Reference supplied by the database. Sergey Galkin, Vasily Golyshev, and Hiroshi Iritani, “Gamma classes and quantum cohomology of Fano manifolds: Gamma conjectures,” *Duke Mathematical Journal* 165 (2016), no. 11, 2005–2077, Conjecture 4.6.1 (Gamma Conjecture II), DOI: 10.1215/00127094-3476593, arXiv:1404.6407.

4.31.4. Primary-source pointer

The originating ICM record is *Gamma classes and quantum cohomology* (2022). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-562-9-volume-4.pdf#page=195>.

5. Metric and topological geometry

These records trace a broad arc from classical embedding and transformation questions to coarse geometry, the Ribe program, nonlinear dimension reduction, and the geometry of singular metric spaces. The central objects range from finite metric spaces and Banach spaces to Gromov–Hausdorff limits and large-scale assembly maps. The common theme is the extraction of topological, linear, or algorithmic structure from metric information, often with a sharp dependence on dimension or distortion as the unresolved point.

Coverage: 28 website records; **source years:** 1932–2022; **separately listed descendant open problems:** 30.

5.1. 178. Un problème de M. Bricard [1932-geom-226]

Record: 1932-geom-226

Subfield: Metric and topological geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1932: Un problème de M. Bricard

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1932.2/ICM1932.2.ocr.pdf#page=185>

5.1.1. Problem statement

Let S^2 be the unit sphere with its geodesic distance d_{S^2} , and fix $0 < \alpha < \pi/2$. Determine the minimum length of a simple rectifiable curve $\gamma : [0, 1] \rightarrow S^2$ such that

$$S^2 = \bigcup_{t \in [0, 1]} \{x \in S^2 : d_{S^2}(x, \gamma(t)) \leq \alpha\},$$

and construct a curve attaining this minimum. In the aircraft interpretation, if the spherical Earth has radius R and the aircraft flies at constant altitude $h > 0$, then $\alpha = \arccos(R/(R + h))$, γ is the radial projection of the flight path onto the unit sphere, and the flight-path length is $(R + h)$ times the length of γ .

5.1.2. Historical background, literature, and current progress

The problem is **unresolved** for arbitrary $0 < \alpha < \pi/2$. Errera proved that if an open simple rectifiable curve of length L sweeps out area Δ with spherical caps of radius α , then

$$\Delta \leq 2L \sin \alpha + 2\pi(1 - \cos \alpha).$$

Consequently every curve in the stated problem satisfies the sharp lower bound

$$L \geq \pi \frac{1 + \cos \alpha}{\sin \alpha} = \pi \cot(\alpha/2).$$

He constructed a two-pole spherical spiral attaining equality when $\pi/(2\alpha)$ is an integer, but did not treat general α .

Gerlach and von der Mosel provide further exact special cases. In Section 5, Lemmas 5.1 and 5.2 construct simple $C^{1,1}$, sphere-filling open curves for every $\alpha = \pi/m$, $m \geq 2$, and Theorem 5.4 classifies the sphere-filling open curves within their positive-thickness class. Their open tubular neighborhood includes the two endpoint caps, so for $m \geq 3$ these curves are admissible for the present problem and, together with Errera's lower bound, are genuine minimizers of length $\pi \cot(\alpha/2)$. Theorem 5.4 also shows why their no-overlap construction cannot extend to an intermediate radius, but it does not determine an optimizer that covers with overlap and it imposes thickness and endpoint-separation hypotheses absent from the present problem.

Thus the known results do not determine the minimum or a minimizing curve when $\alpha \notin \{\pi/m : m = 3, 4, \dots\}$. O'Rourke's equivalent formulation asks for the shortest path whose radius- d disks cover S^2 ; his 2022 status update says the questions remain unsolved and singles out $d = 5\pi/12$ as an unclear case. No later paper located in searches through August 2026 resolves these missing radii.

Key references:

- Alfred Errera, "Un problème de M. Bricard," *Verhandlungen des Internationalen Mathematiker-Kongresses Zürich 1932*, vol. II, 1932, pp. 165–166. Direct source: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1932.2/ICM1932.2.ocr.pdf>.
- Henryk Gerlach and Heiko von der Mosel, "What Are the Longest Ropes on the Unit Sphere?", *Archive for Rational Mechanics and Analysis* 201(1), 2011, pp. 303–342. DOI: <https://doi.org/10.1007/s00205-010-0390-y>.
- Henryk Gerlach and Heiko von der Mosel, "On Sphere-Filling Ropes," *The American Mathematical Monthly* 118(10), 2011, pp. 863–876. DOI: <https://doi.org/10.4169/amer.math.monthly.118.10.863>; arXiv:1005.4609.
- Joseph O'Rourke, "Optimal Inspection Path on a Sphere," *MathOverflow*, Question 130255, 2013, updated 2022. <https://mathoverflow.net/questions/130255/optimal-inspection-path-on-a-sphere>.

5.1.3. Remaining open problems recorded by the database

5.1.3.1. Open problem 1

Let S^2 be the unit sphere with geodesic distance d_{S^2} . Determine the minimum length of

a simple rectifiable curve $\gamma : [0, 1] \rightarrow S^2$ such that

$$S^2 = \bigcup_{t \in [0, 1]} \left\{ x \in S^2 : d_{S^2}(x, \gamma(t)) \leq \frac{5\pi}{12} \right\},$$

and construct a curve attaining this minimum.

Reference supplied by the database. Joseph O'Rourke, "Optimal Inspection Path on a Sphere," MathOverflow, Question 130255, 2013 (updated 9 August 2022), Q1 and the explicitly discussed case $d = 5\pi/12$. <https://mathoverflow.net/questions/130255/optimal-inspection-path-on-a-sphere>.

5.1.4. Primary-source pointer

The originating ICM record is *Un problème de M. Bricard* (1932). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1932.2/ICM1932.2.ocr.pdf#page=185>.

5.2. 179. H., La topologie dans les fondements de la géométrie [1954-geom-228]

Record: 1954-geom-228

Subfield: Metric and topological geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1954: H., La topologie dans les fondements de la géométrie

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1954.3/ICM1954.3.ocr.pdf#page=178>

5.2.1. Problem statement

Let $\mathcal{D} = (P, \mathcal{L}, I)$ be a topological projective plane: any two distinct points of P are incident with a unique line of $\mathcal{L}$, any two distinct lines are incident with a unique point, and there exist four points no three of which are collinear. Assume that P and $\mathcal{L}$ are Hausdorff topological spaces, that P is compact, that every line, regarded as a subset of P , is closed and connected, and that the join and intersection maps

$$(P \times P) \setminus \Delta_P \rightarrow \mathcal{L}, \quad (p, q) \mapsto p \vee q,$$

and

$$(\mathcal{L} \times \mathcal{L}) \setminus \Delta_{\mathcal{L}} \rightarrow P, \quad (L, M) \mapsto L \wedge M,$$

are continuous. Prove or disprove that P must be a finite-dimensional topological manifold, that every line must be a topological submanifold of P , and that $\dim P = 2^k$ for some positive integer k .

5.2.2. Historical background, literature, and current progress

The conjecture is unresolved. No result found proves from the stated weak compactness, connectedness, incidence, and continuity hypotheses that the point space is finite-dimensional, that it is a topological manifold, or that its lines are topological submanifolds.

The strongest general progress is conditional on finite dimension. Rainer Löwen, “Topology and dimension of stable planes: On a conjecture of H. Freudenthal,” *Journal für die reine und angewandte Mathematik* **343** (1983), 108–122, DOI <https://doi.org/10.1515/crll.1983.343.108>, proves that a finite-dimensional compact connected topological projective plane has point-space dimension 2, 4, 8, or 16; its lines are homotopy equivalent to spheres of dimension 1, 2, 4, or 8, respectively, and the relevant spaces are integral ENR homology manifolds. Thus the power-of-two conclusion is known after adding finite-dimensionality, but a homology manifold need not automatically be a topological manifold.

Helmut Salzmann, “Topological planes,” *Advances in Mathematics* **2** (1967), 1–60, DOI [https://doi.org/10.1016/S0001-8708\(67\)80002-1](https://doi.org/10.1016/S0001-8708(67)80002-1), proves the complementary conditional result: if a point row (line) is already a manifold, then all point rows and line pencils are spheres S^l with l one of 1, 2, 4, 8, and the point, line, and flag spaces are manifolds of dimensions $2l$, $2l$, and $3l$. This does not derive the manifold hypothesis from the assumptions in

the problem.

Linus Kramer, “Projective Planes and their Look-Alikes,” *Journal of Differential Geometry* 64 (2003), 1–55, DOI <https://doi.org/10.4310/jdg/1090426887>, explicitly isolates the remaining steps: prove finite-dimensionality; in finite dimension, upgrade the integral ENR homology manifold to a genuine manifold; and, once it is a manifold, determine its homeomorphism type. It also records that dimensions 2 and 4 are understood, while the higher-dimensional manifold-recognition issue remains.

Later literature continues to retain these qualifications. Helmut R. Salzmann, “Compact planes, mostly 8-dimensional. A retrospect,” arXiv:1402.0304 (2014), [arXiv:1402.0304](https://arxiv.org/abs/1402.0304), states that every line in an 8-dimensional plane is homotopy equivalent to S^4 , but that whether all such lines are manifolds (hence homeomorphic to S^4) is very difficult and unresolved; it reports the analogous known homotopy result for 16-dimensional planes. Rainer Löwen, “Locally classical stable planes,” arXiv:2504.19265 (2025), [arXiv:2504.19265](https://arxiv.org/abs/2504.19265), proves a classification only under the additional hypothesis of local classicality and explicitly notes that point spaces of stable planes are not known to be manifolds in general.

Consequently, neither the possible infinite-dimensional case nor manifold and submanifold recognition in all finite-dimensional cases is settled, so the exact assertion in the statement is not completely resolved.

5.2.3. Remaining open problems recorded by the database

5.2.3.1. Open problem 1

Let $\mathcal{P} = (P, \mathcal{L}, I)$ be a compact connected topological projective plane: P and $\mathcal{L}$ are compact connected Hausdorff spaces, every two distinct points are incident with a unique line, every two distinct lines are incident with a unique point, there exist four points no three of which are collinear, and the join and intersection maps are continuous. Prove that the covering dimension of P is finite.

Reference supplied by the database. Linus Kramer, “Projective Planes and their Look-Alikes,” *Journal of Differential Geometry* 64 (2003), no. 1, 1–55, open problem (*) and step (1) on pp. 3–4. DOI: <https://doi.org/10.4310/jdg/1090426887>.

5.2.3.2. Open problem 2

Let $\mathcal{P} = (P, \mathcal{L}, I)$ be a compact connected topological projective plane whose point space P has finite positive covering dimension. Prove that P is a topological manifold.

Reference supplied by the database. Linus Kramer, “Projective Planes and their Look-Alikes,” *Journal of Differential Geometry* 64 (2003), no. 1, 1–55, open problem (*) and step (2) on pp. 3–4. DOI: <https://doi.org/10.4310/jdg/1090426887>.

5.2.3.3. Open problem 3

Let $\mathcal{D} = (P, \mathcal{L}, I)$ be a compact connected topological projective plane whose point space P has covering dimension 8. Prove or disprove that every line of $\mathcal{D}$, with its subspace topology from P , is homeomorphic to the 4-sphere S^4 .

Reference supplied by the database. Helmut R. Salzmann, “Compact planes, mostly 8-dimensional. A retrospect,” arXiv preprint (2014), Section 1.1, p. 2. arXiv:1402.0304, <https://arxiv.org/abs/1402.0304>.

5.2.4. Primary-source pointer

The originating ICM record is *H., La topologie dans les fondements de la géométrie* (1954). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1954.3/ICM1954.3.ocr.pdf#page=178>.

5.3. 180. MODERN DEVELOPMENT OF SURFACE THEORY [1958-geom-8]

Record: 1958-geom-8

Subfield: Metric and topological geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1958: MODERN DEVELOPMENT OF SURFACE THEORY

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1958/ICM1958.ocr.pdf#page=67>

5.3.1. Problem statement

Determine whether every abstract M.B.C. (manifold of bounded curvature) admits a local isometric realization in ordinary Euclidean space E^3 as a surface of bounded positive external curvature.

Here an abstract M.B.C. is a two-dimensional metric manifold with an intrinsic metric such that each point has a neighbourhood U for which the sums of the triangle excesses

$$\delta(T) = \bar{\alpha} + \bar{\beta} + \bar{\gamma} - \pi,$$

taken over arbitrary finite families of pairwise non-overlapping geodesic triangles $T \subset U$, are bounded above by a constant depending only on U ; $\bar{\alpha}, \bar{\beta}, \bar{\gamma}$ are the upper angles at the vertices, defined as the limiting upper values of the corresponding Euclidean comparison angles determined by the metric side lengths. A surface $S \subset E^3$ has bounded positive external curvature if, for every subdivision of S into pieces S_i , $\sigma^+(S_i)$ denotes the area on the unit sphere of the normal directions to supporting planes at the non-boundary points of S_i , and the sums $\sum_i \sigma^+(S_i)$ are bounded by a constant independent of the subdivision.

5.3.2. Historical background, literature, and current progress

5.3.3. Verdict: partially resolved

5.3.3.1. Precise problem checked

The problem asks whether every abstract two-dimensional M.B.C. (manifold of bounded curvature, in Alexandrov's intrinsic sense of locally bounded sums of positive triangle excesses) admits local isometric realization in ordinary Euclidean 3-space as a surface of bounded positive external/extrinsic curvature. The target class is the class Alexandrov describes via uniformly bounded sums of spherical areas of supporting-plane normal directions over arbitrary subdivisions.

This is stronger than merely producing a Nash-Kuiper-type C^1 isometric embedding: Alexandrov explicitly wanted a "Gaussian" realization class where intrinsic and external curvature remain connected.

5.3.3.2. Current status

I found substantial partial progress, but no published or otherwise citable theorem that covers the full statement for all abstract two-dimensional M.B.C.s. The full local realization in E^3 by surfaces of bounded positive external curvature therefore remains unresolved.

5.3.3.3. Key literature and progress

Original source

- Title: “Modern development of surface theory”
- Author: A. D. Alexandrov
- Venue: Proceedings of the International Congress of Mathematicians, 14-21 August 1958, edited by J. A. Todd, Cambridge University Press
- Year: 1960
- DOI/arXiv/URL: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1958/ICM1958.ocr.pdf>
- Relevant content: Alexandrov defines M.B.C.s intrinsically using bounded sums of triangle excesses, then asks which Euclidean surfaces are M.B.C.s intrinsically and whether any M.B.C. can be embedded locally in E^3 . He says the desired embeddings are “Gaussian embeddings.” He identifies surfaces of bounded positive external curvature as the expected general class, but states that at that time their curvature/Gauss theorem theory was not yet available and that general embedding results were known only for convex surfaces.

Convex and nonnegative-curvature cases

- Title: “Intrinsic Geometry of Convex Surfaces” / related Alexandrov convex-surface embedding theorem
- Author: A. D. Alexandrov; uniqueness developed by A. V. Pogorelov
- Venue: Russian original, Gostekhizdat; later references and modern accounts in Alexandrov geometry texts
- Year: 1948 and later
- DOI/arXiv/URL: modern accessible account: <https://anton-petrinin.github.io/alexandrov/alexandrov.pdf>
- Progress: The Alexandrov embedding theorem realizes geodesic CBB(0) metrics on S^2 as closed convex surfaces in E^3 , with uniqueness due to Pogorelov. This fully resolves important positive/nonnegative convex cases, but it does not cover arbitrary local M.B.C.s of mixed or negative curvature.

Foundations of M.B.C. theory

- Title: “Two-dimensional manifolds of bounded curvature (Foundations of the intrinsic geometry of surfaces)”
- Authors: A. D. Aleksandrov and V. A. Zalgaller
- Venue: Trudy Mat. Inst. Steklov 63; English translation as Intrinsic Geometry of Surfaces, Translations of Mathematical Monographs 15, AMS
- Year: 1962; English translation 1967
- DOI/arXiv/URL: <https://www.mathnet.ru/eng/tm1580>
- Progress: This is the foundational treatment of two-dimensional M.B.C.s. It develops the intrinsic class but does not supply the missing general local realization theorem in E^3

by surfaces of bounded positive external curvature.

General C^1 realization, but not in the bounded-positive-external-curvature class

- Title: “Realization of a metrized two-dimensional manifold by a surface in E^3 ”
- Author: Yu. D. Burago
- Venue: Dokl. Akad. Nauk SSSR 135:6, 1301-1302; English translation: Soviet Math. Dokl. 1, 1364-1365
- Year: 1960
- DOI/arXiv/URL: <https://www.mathnet.ru/eng/dan24409>
- Progress: Burago proves that if an orientable two-dimensional metric manifold admits proportional approximation by polyhedral metrics, then it is realized by a non-self-intersecting surface in E^3 with a tangent plane at every point. A corollary covers orientable M.B.C.s under extra exclusions such as no point of curvature 2π and suitable boundary/no boundary assumptions.
- Why this is not full resolution: The surface class in the theorem is a C^1 /tangent-plane class, not the class of surfaces of bounded positive external curvature. The proof is in the Nash-Kuiper flexibility direction and does not establish the Gaussian curvature connection Alexandrov required.

Bounded extrinsic/positive extrinsic curvature theory and structural results

- Title: “Surfaces of Bounded Extrinsic Curvature”
- Author: A. V. Pogorelov
- Venue: Kharkov University Press; later related material in Extrinsic Geometry of Convex Surfaces, AMS Translations of Mathematical Monographs 35
- Year: 1956; English book translation 1973
- DOI/arXiv/URL: Google Books record: https://books.google.com/books/about/Extrinsic_Geometry_of_Convex_Surfaces.html?id=PI4mo2yX8HoC
- Progress: Pogorelov developed the theory of surfaces of bounded extrinsic curvature and Gauss-type relations in that setting. This is central background, but it treats a more regular/specific extrinsic class and does not prove that every abstract M.B.C. has the desired local realization.
- Title: “ C^1 -smooth isometric imbeddings”
- Author: S. Z. Shefel’
- Venue: Sibirsk. Mat. Zh. 15:6, 1372-1393; English translation in Siberian Mathematical Journal
- Year: 1974
- DOI/arXiv/URL: <https://doi.org/10.1007/BF00966565>; <https://link.springer.com/article/10.1007/BF00966565>
- Progress: Gives structural theorems for C^1 isometric immersions under sign and affine-stability hypotheses. In particular, C^1, α immersions with $\alpha > 2/3$ of metrics in the classes A^+ , A_0 , A^- have corresponding normal nonnegative/developable/saddle structure and bounded extrinsic curvature.
- Why this is partial: It assumes regularity/sign/stability hypotheses and is not an exis-

tence theorem for arbitrary M.B.C.s of mixed curvature.

- Title: “ C^1 -smooth surfaces of bounded positive extrinsic curvature”
- Author: S. Z. Shefel’
- Venue: Sibirsk. Mat. Zh. 16:5, 1122-1123; English translation: Siberian Math. J. 16:5, 863-864
- Year: 1975
- DOI/arXiv/URL: <https://doi.org/10.1007/BF00967119>; <https://www.mathnet.ru/eng/smj4203>
- Progress: Studies C^1 surfaces of bounded positive extrinsic curvature and proves that, if such a surface is an isometric immersion of an M.B.C. of class A^+ , A_0 , or A^- , then it has the corresponding normal nonnegative-curvature, developable, or saddle extrinsic structure.
- Why this is partial: The theorem assumes the desired bounded-positive-extrinsic-curvature surface already exists and assumes intrinsic sign classes. It does not construct such a realization for every abstract M.B.C.
- Title: “On the extrinsic curvature and the extrinsic structure of normal developable C^1 surfaces”
- Author: I. Kh. Sabitov
- Venue: Mathematical Notes 87, 874-879
- Year: 2010
- DOI/arXiv/URL: <https://doi.org/10.1134/S0001434610050275>; <https://link.springer.com/article/10.1134/S0001434610050275>
- Progress: Modern continuation of the low-regularity extrinsic-curvature structural theory, proving that normal C^1 developable surfaces in Shefel’s sense have zero extrinsic curvature in Pogorelov’s sense.
- Why this is partial: It addresses a structural special case, not the general local realization of M.B.C.s.

Later surveys/background

- Title: “The Geometry of Surfaces in Euclidean Spaces”
- Authors: Yu. D. Burago and S. Z. Shefel’
- Venue: Geometry III, Encyclopaedia of Mathematical Sciences 48, Springer, pp. 1-85; Russian original in Itogi Nauki i Tekhniki, Sovremennye Problemy Matematiki, Fundamental’nye Napravleniya 48
- Year: 1992 English version; 1989 Russian version
- DOI/arXiv/URL: https://link.springer.com/chapter/10.1007/978-3-662-02751-6_1
- Progress: Survey of the connection between metric classes and surface classes in Euclidean spaces, including bounded extrinsic curvature, C^1 isometric immersions, convex/saddle/developable surfaces, and open directions. It does not give a theorem resolving the full Alexandrov expectation for arbitrary M.B.C.s.

- Title: “Two-Dimensional Manifolds of Bounded Curvature”
- Author: Yu. G. Reshetnyak
- Venue: Geometry IV, Encyclopaedia of Mathematical Sciences 70, Springer, pp. 3-163
- Year: 1993
- DOI/arXiv/URL: https://doi.org/10.1007/978-3-662-02897-1_1 ; https://link.springer.com/chapter/10.1007/978-3-662-02897-1_1
- Progress: A major survey of the intrinsic M.B.C. theory. Its bibliography records Alexandrov-Zalgaller, Burago, Shefel, Pogorelov, and related work, but not a full local E^3 realization theorem in the bounded-positive-external-curvature class.

Smooth local embedding literature: evidence of only special-case progress

- Title: “The local isometric embedding in R^3 of two-dimensional Riemannian manifolds with Gaussian curvature changing sign to finite order on a curve”
- Author: Marcus A. Khuri
- Venue: Journal of Differential Geometry 76, 249-291
- Year: 2007
- DOI/arXiv/URL: https://www.math.stonybrook.edu/~khuri/Khuri_JDG.pdf
- Progress: Khuri states that the local realization problem for two-dimensional Riemannian manifolds in R^3 “has only been partially answered” and proves existence under a finite-order sign-changing-curve hypothesis.
- Why this matters: Smooth Riemannian metrics are a subclass of M.B.C.s. The smooth literature still treats nonzero curvature, sign, finite-order vanishing, or other hypotheses rather than a general all-cases theorem. This supports the conclusion that no stronger published resolution of Alexandrov’s arbitrary M.B.C. realization problem is available.

5.3.3.4. What remains unresolved

No located theorem proves that every abstract two-dimensional M.B.C., with no sign, smoothness, orientability/boundary exclusions, or curvature-zero exclusions beyond the definition itself, is locally isometric to a surface in E^3 whose positive external curvature is bounded in Alexandrov’s supporting-plane normal-image sense.

The closest positive results either:

- realize only convex/nonnegative-curvature global cases;
- produce a C^1 /tangent-plane realization without bounded positive external curvature;
- assume the target surface already has bounded positive extrinsic curvature and then prove structure;
- impose sign, Holder regularity, affine-stability, finite-order vanishing, or other additional hypotheses.

Therefore the problem should be classified as partially resolved, not fully resolved and not fully open.

5.3.4. Remaining open problems recorded by the database

5.3.4.1. Open problem 1

Does every smooth two-dimensional Riemannian manifold admit a smooth local isometric embedding into $\mathbb{R}^3$? Equivalently, are there any smooth (or sufficiently smooth) counterexamples to the local isometric-embedding problem for surfaces?

Reference supplied by the database. Qing Han and Marcus Khuri, “On the local isometric embedding in $\mathbb{R}^3$ of surfaces with Gaussian curvature of mixed sign,” *Communications in Analysis and Geometry* **18** (2010), no. 4, 649–704, opening question and Remark 1.1 (explicitly stating that the existence of smooth counterexamples remained open), DOI: <https://doi.org/10.4310/CAG.2010.v18.n4.a2>. The paper proves only the finite-order, transverse-zero-set case; subsequent special-case progress includes Siyuan Lu, “On the local isometric embedding in $\mathbb{R}^3$ of surfaces with zero sets of Gaussian curvature forming cusp domains,” *Communications in Partial Differential Equations* **41** (2016), no. 10, 1541–1570, DOI: <https://doi.org/10.1080/03605302.2016.1227336>, and does not settle the general smooth question.

5.3.5. Primary-source pointer

The originating ICM record is *MODERN DEVELOPMENT OF SURFACE THEORY* (1958). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1958/ICM1958.ocr.pdf#page=67>.

5.4. 181. Spaces with Curvature Bounded Below [1994-geom-61]

Record: 1994-geom-61

Subfield: Metric and topological geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1994: Spaces with Curvature Bounded Below

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.1/ICM1994.1.ocr.pdf#page=589>

5.4.1. Problem statement

Formulate an Alexandrov-space analogue of the rational ellipticity conjecture for complete, locally compact, finite-dimensional Alexandrov spaces M with curvature bounded below by 0. The formulation should generalize the conjecture that every compact simply connected Riemannian manifold with nonnegative sectional curvature is rationally elliptic, meaning that the rational Betti numbers of its loop space grow at most polynomially, and it should incorporate not only the underlying topology of M but also the extremal subsets of M .

Here an extremal subset of an Alexandrov space M is a closed subset $F \subset M$ such that, for every distance function $f = \text{dist}_p$ with $p \notin F$, if $q \in F$ realizes the distance from p to F , then

$$\sup_{\xi \in \Sigma_q} f'_q(\xi) \leq 0,$$

where Σ_q is the space of directions at q . The formulation should also account for conjectures about positively curved manifolds, since spherical suspensions and joins of spaces with curvature at least 1 are Alexandrov spaces with curvature at least 1.

5.4.2. Historical background, literature, and current progress

5.4.3. Verdict: partially resolved

5.4.3.1. Precise problem checked

The statement asks for Perelman's speculative Alexandrov-space analogue of the rational ellipticity, or Bott-Grove-Halperin, conjecture. In its classical manifold form, the conjecture says that every compact simply connected Riemannian manifold with nonnegative sectional curvature is rationally elliptic, equivalently that the rational Betti numbers of its based loop space grow at most polynomially. The problem asks for an extension to complete, locally compact, finite-dimensional Alexandrov spaces with curvature bounded below by 0, and further says that the correct formulation should incorporate extremal subsets, not just the topology of the underlying space.

Because every compact nonnegatively curved Riemannian manifold is an Alexandrov space with curvature bounded below by 0, any theorem resolving the full Alexandrov statement would in particular resolve the classical Bott-Grove-Halperin conjecture for manifolds. The literature I found does not do this.

5.4.3.2. Original and status-setting sources

- Title:** Spaces with Curvature Bounded Below
Author: G. Perelman
Venue: Proceedings of the International Congress of Mathematicians, Zurich 1994, Birkhauser, Basel, pp. 517-525
Year: 1995
DOI/URL: DOI https://doi.org/10.1007/978-3-0348-9078-6_45, https://link.springer.com/chapter/10.1007/978-3-0348-9078-6_45
Relevance: This is the Perelman ICM source for Alexandrov spaces with curvature bounded below, including the extremal-subset viewpoint. It is the source context for the stated speculative extension, not a proof of rational ellipticity.
- Title:** Open Problems in Non-negative Sectional Curvature
Author/Compiler: Martin Kerin, compiled from the 2007 AIM workshop on nonnegative sectional curvature
Venue: American Institute of Mathematics workshop problem list
Year: 2007
URL: <https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf>
Relevance: Problem 12 explicitly asks for an Alexandrov analogue of rational ellipticity and, in particular, asks whether almost non-negatively curved manifolds are rationally elliptic. This is direct evidence that the Alexandrov analogue was open after Perelman's proposal.

5.4.3.3. Partial progress

- Title:** Nilpotency, almost nonnegative curvature, and the gradient flow on Alexandrov spaces
Authors: Vitali Kapovitch, Anton Petrunin, Wilderich Tuschmann
Venue: Annals of Mathematics, 171(1), 343-373
Year: 2010
DOI/URL: DOI <https://doi.org/10.4007/annals.2010.171.343>, <https://annals.math.princeton.edu/2010/171-1/p06>
Progress: The paper proves strong structural results for almost nonnegatively curved manifolds, including virtual nilpotency of fundamental groups and a fibration theorem. It formulates a rational-ellipticity conjecture for closed almost nonnegatively curved manifolds and reduces part of the problem to the simply connected case. This is substantial progress around the conjecture, but it does not prove rational ellipticity for all nonnegatively curved manifolds or any full Alexandrov/extremal-subset analogue.
- Title:** Almost non-negative curvature and rational ellipticity in cohomogeneity two
Authors: Karsten Grove, Burkhard Wilking, Joseph Yeager; appendix by Steve Halperin
Venue: Annales de l'Institut Fourier, 69(7), 2921-2939
Year: 2019
DOI/URL: DOI <https://doi.org/10.5802/aif.3304>, https://www.numdam.org/article/AIF_2019__69_7_2921_0/
Progress: The main theorem proves that simply connected almost non-negatively curved manifolds with an isometric action of cohomogeneity at most two are rationally elliptic. This verifies an important symmetry-restricted special case of the manifold conjecture,

not the full manifold conjecture and not the full Alexandrov/extremal-subset problem.

- **Title:** Nonnegatively curved quotient spaces with boundary
Author: Wolfgang Spindeler
Venue: Manuscripta Mathematica, 153, 55-67
Year: 2017
DOI/arXiv/URL: DOI <https://doi.org/10.1007/s00229-016-0875-y>, arXiv 1510.01908, <https://arxiv.org/abs/1510.01908>
Progress: The paper proves that simply connected nonnegatively curved torus manifolds with nonempty boundary quotient are rationally elliptic and obtains related classification results. This is another special case under a strong torus-action hypothesis.
- **Title:** Torus manifolds and non-negative curvature
Author: Michael Wiemeler
Venue: Journal of the London Mathematical Society, 91(3), 667-692
Year: 2015
DOI/arXiv/URL: DOI <https://doi.org/10.1112/jlms/jdv017>, arXiv 1401.0403, <https://arxiv.org/abs/1401.0403>
Progress: This gives structure and rational-homotopy information for nonnegatively curved torus manifolds. It supports rational ellipticity in a highly symmetric manifold setting, but does not cover general nonnegatively curved manifolds or Alexandrov spaces with extremal subsets.
- **Title:** Rational ellipticity of G-manifolds from their quotients
Authors: Elahe Khalili Samani, Marco Radeschi
Venue: Compositio Mathematica, 161(2), 244-256
Year: 2025
DOI/arXiv/URL: DOI <https://doi.org/10.1112/S0010437X24007656>, arXiv 2312.08202, <https://arxiv.org/abs/2312.08202>
Progress: The paper proves rational ellipticity criteria for certain compact simply connected smooth manifolds from the topology of their metric quotients under isometric group actions, including a cohomogeneity-three infinitesimally polar case. Although metric quotients are Alexandrov spaces when the original manifold has lower curvature bounds, this is a quotient/submetry theorem for special group actions, not a full rational-ellipticity theorem for arbitrary Alexandrov spaces or arbitrary extremal-subset data.

5.4.3.4. What remains unresolved

No citable theorem found proves that every complete, locally compact, finite-dimensional Alexandrov space with curvature bounded below by 0 satisfies the proposed rational-ellipticity analogue, especially not in a formulation incorporating all extremal subsets. In fact, the full Alexandrov statement would imply the classical Bott-Grove-Halperin conjecture for compact simply connected nonnegatively curved Riemannian manifolds, and the cited recent literature continues to treat that manifold conjecture as open except in special cases.

Therefore the exact problem considered here is not fully resolved. The correct status is **partially resolved**: there is substantial progress in special cases and related structural results, but the full Alexandrov/extremal-subset analogue remains open.

5.4.4. Remaining open problems recorded by the database

5.4.4.1. Open problem 1

Is every closed simply connected almost nonnegatively curved manifold rationally elliptic? Here “almost nonnegatively curved” means that the manifold admits metrics g_i with $\sec(M, g_i) > -1/i$ and $\text{diam}(M, g_i) \leq 1/i$.

Reference supplied by the database. Vitali Kapovitch, Anton Petrunin, and Wilderich Tuschmann, “Nilpotency, Almost Nonnegative Curvature, and the Gradient Flow on Alexandrov Spaces,” *Annals of Mathematics* (2) **171** (2010), no. 1, 343–373, §6.3, Conjecture 6.3.1 (reduced there to the simply connected case), DOI: 10.4007/annals.2010.171.343, arXiv:math/0506273, <https://doi.org/10.4007/annals.2010.171.343>. See also Martin Kerin (compiler), “Open Problems in Non-negative Sectional Curvature,” American Institute of Mathematics workshop problem list, 2007, Problem 12, <https://aimath.org/WWN/nnsectcurvature/nnsectcurvature.pdf>.

5.4.5. Primary-source pointer

The originating ICM record is *Spaces with Curvature Bounded Below* (1994). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.1/ICM1994.1.ocr.pdf#page=589>.

5.5. 182. Combinatorics and convexity [1994-geom-240]

Record: 1994-geom-240

Subfield: Metric and topological geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1994: Combinatorics and convexity

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.2/ICM1994.2.ocr.pdf#page=659>

5.5.1. Problem statement

For every integer $d \geq 2$, let $S^{d-1} = \{x \in \mathbb{R}^d : \|x\|_2 = 1\}$, equipped with its $(d-1)$ -dimensional spherical surface measure vol_{d-1} . For $0 \leq r \leq \pi$, let v_r denote the spherical surface measure of a geodesic spherical cap of radius r in S^{d-1} . Prove that if $A \subseteq S^{d-1}$ is measurable and contains no pair of orthogonal vectors, meaning that $\langle x, y \rangle \neq 0$ for all distinct $x, y \in A$, then

$$\text{vol}_{d-1}(A) \leq 2v_{\pi/4}.$$

5.5.2. Historical background, literature, and current progress

The problem is **unresolved**. Let α_d be the supremum of $\text{vol}_{d-1}(A) / \text{vol}_{d-1}(S^{d-1})$ over measurable orthogonal-pair-free sets. The two open antipodal caps of radius $\pi/4$ show that α_d is at least the normalized value of $2v_{\pi/4}$, so the stated inequality is precisely the assertion that this construction is optimal. It is known for $d = 2$, but remains open for every $d \geq 3$. In particular, for $d = 3$ the current bounds are

$$1 - \frac{1}{\sqrt{2}} = 0.292893 \dots \leq \alpha_3 \leq 0.2953,$$

and the strict gap means that even this first open dimension does not satisfy a proved version of the requested bound.

Key literature and progress:

- H. S. Witsenhausen, "Spherical Sets Without Orthogonal Point Pairs," *The American Mathematical Monthly* 81(10) (1974), 1101–1102, <https://doi.org/10.1080/00029890.1974.11993744>. Witsenhausen formulated the extremal-measure problem, exhibited the double-cap lower bound, and proved $\alpha_d \leq 1/d$. This settles $d = 2$ because the two bounds coincide there, but not any $d \geq 3$.
- Peter Frankl and Richard M. Wilson, "Intersection Theorems with Geometric Consequences," *Combinatorica* 1(4) (1981), 357–368, <https://doi.org/10.1007/BF02579457>; and A. M. Raigorodskii, "On a Bound in Borsuk's Problem," *Russian Mathematical Surveys* 54(2) (1999), 453–454, <https://doi.org/10.1070/RM1999v054n02ABEH000146>. Their forbidden-intersection constructions imply exponentially decreasing asymptotic upper bounds for α_d ; Raigorodskii's refinement gives

$\alpha_d \leq (1+o(1))(1.1547\dots)^{-d}$. This is still exponentially larger than the double-cap density, whose exponential base is $\sqrt{2}$.

- Evan DeCorte and Oleg Pikhurko, “Spherical Sets Avoiding a Prescribed Set of Angles,” *International Mathematics Research Notices* 2016(20) (2016), 6095–6117, <https://doi.org/10.1093/imrn/rnv319>. They proved $\alpha_3 \leq 0.313$ and proved that the supremum defining α_d is attained for every $d \geq 3$; neither result identifies the maximizer or proves the double-cap value.
- Evan DeCorte, Fernando Mário de Oliveira Filho, and Frank Vallentin, “Complete Positivity and Distance-Avoiding Sets,” *Mathematical Programming* 191 (2022), 487–558, <https://doi.org/10.1007/s10107-020-01562-6>. Their completely positive optimization bounds improved the three-dimensional upper bound to $\alpha_3 \leq 0.30153$ and also improved bounds in several other low dimensions, without reaching the double-cap values.
- Bram Bekker, Olga Kuryatnikova, Fernando Mário de Oliveira Filho, and Juan C. Vera, “Optimization Hierarchies for Distance-Avoiding Sets in Compact Spaces,” *Transactions of the American Mathematical Society* 379(1) (2026), 33–70, <https://doi.org/10.1090/tran/9578>. They proved convergent optimization hierarchies for this problem and rigorous bounds for $3 \leq d \leq 8$, including $\alpha_3 \leq 0.297742$, still above the double-cap lower bound.
- Domonkos Czifra, Ákos Dúcz, Máté Matolcsi, Dániel Varga, and Pál Zsámboki, “Improved Bounds for the Double Cap Conjecture,” preprint (2026), [arXiv:2605.28709](https://arxiv.org/abs/2605.28709). Theorem 1.1 proves the current strongest three-dimensional upper bound $\alpha_3 \leq 0.2953$ using harmonic analysis and a finite orthogonality graph. The paper explicitly records that the conjecture remains open for all $d \geq 3$.

Thus the established results give exactness only in dimension 2, quantitative upper bounds in fixed low dimensions, and exponential asymptotic upper bounds. What remains is to close the gap to the double-cap measure in every dimension $d \geq 3$.

5.5.3. Remaining open problems recorded by the database

5.5.3.1. Open problem 1

Let $S^2 = \{x \in \mathbb{R}^3 : \|x\|_2 = 1\}$, equipped with spherical surface area. Prove that every measurable set $A \subseteq S^2$ satisfying $\langle x, y \rangle \neq 0$ for all distinct $x, y \in A$ has

$$\text{area}(A) \leq 4\pi \left(1 - \frac{1}{\sqrt{2}}\right).$$

Reference supplied by the database. Domonkos Czifra, Ákos Dúcz, Máté Matolcsi, Dániel Varga, and Pál Zsámboki, “Improved Bounds for the Double Cap Conjecture,” arXiv preprint (2026), Introduction and Theorem 1.1, [arXiv:2605.28709](https://arxiv.org/abs/2605.28709), <https://arxiv.org/abs/2605.28709>.

5.5.4. Primary-source pointer

The originating ICM record is *Combinatorics and convexity* (1994). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/>

[ICM1994.2/ICM1994.2.ocr.pdf#page=659.](#)

5.6. 183. Dimension Theory and Large Riemannian Manifolds [1998-geom-65]

Record: 1998-geom-65

Subfield: Metric and topological geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1998: Dimension Theory and Large Riemannian Manifolds

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.2/ICM1998.2.ocr.pdf#page=425>

5.6.1. Problem statement

Prove or disprove that for every metric space X that is uniformly contractible and has bounded geometry, the coarse Baum-Connes assembly map

$$A : K_*(X) \rightarrow K_*(C^*(X))$$

is a monomorphism. Determine whether, in the stronger isomorphism version of the conjecture, this assembly map is an isomorphism.

Here X is uniformly contractible if for every $R > 0$ there is $S > 0$ such that every R -ball $B_R(x)$ centered at x can be contracted to a point inside $B_S(x)$, for every $x \in X$. The space X has bounded geometry if for every $\epsilon > 0$ and every $R > 0$ there is an integer m such that every R -ball in X contains an ϵ -net with fewer than m points. The notation $C^*(X)$ denotes the C^* -algebra associated to X in the coarse Baum-Connes assembly construction, and $K_*(C^*(X))$ denotes its K -theory.

5.6.2. Historical background, literature, and current progress

5.6.3. Verdict: partially resolved

5.6.3.1. Precise problem checked

The problem asks whether, for every uniformly contractible bounded-geometry metric space X , the coarse Baum-Connes assembly map

$$A : K_*(X) \rightarrow K_*(C^*(X))$$

is injective, and whether the stronger isomorphism form is true. In modern terminology, the injective form is the coarse Novikov conjecture for this class, while the isomorphism form is the coarse Baum-Connes conjecture for this class. The uniform contractibility and bounded-geometry hypotheses are part of the exact question.

5.6.3.2. Current status

I found no published or otherwise citable theorem that proves injectivity for all uniformly contractible bounded-geometry metric spaces, nor a citable counterexample satisfying both uni-

form contractibility and bounded geometry that disproves the injective statement. Therefore the exact problem is not fully resolved.

There is substantial partial progress. The assembly map is an isomorphism for many important subclasses, including finite-asymptotic-dimension spaces, spaces uniformly/coarsely embeddable into Hilbert space, Busemann non-positively curved spaces, coarsely convex spaces, and certain relative expanders. There are also counterexamples to the broader coarse Baum-Connes isomorphism conjecture for general bounded-geometry spaces, especially expanders and related warped-cone/asymptotic-expander constructions, but those do not prove failure of the exact uniformly contractible bounded-geometry injectivity statement in this problem.

5.6.3.3. Key literature and relevance

- **A. N. Dranishnikov**, “Dimension theory and large Riemannian manifolds”, *Documenta Mathematica*, Extra Volume ICM II, 1998, pp. 423-432. URL: <https://eudml.org/doc/231645> and <https://www2.math.ethz.ch/EMIS/journals/DMJDMV/xvol-icm/06/Dranishnikov.MAN.ps.gz>.
This is the dataset source. It states the coarse Baum-Connes/Novikov problem for uniformly contractible bounded-geometry metric spaces and notes the role of the bounded-geometry hypothesis.
- **John Roe**, “Coarse cohomology and index theory on complete Riemannian manifolds”, *Memoirs of the American Mathematical Society* 104, no. 497, 1993. URL: <https://doi.org/10.1090/memo/0497>.
Foundational source for coarse cohomology, Roe algebras, and the coarse assembly framework on complete Riemannian manifolds.
- **Nigel Higson and John Roe**, “On the coarse Baum-Connes conjecture”, in *Novikov Conjectures, Index Theorems and Rigidity*, Vol. 2, London Mathematical Society Lecture Note Series 227, Cambridge University Press, 1995, pp. 227-254.
Foundational formulation and analysis of the coarse Baum-Connes assembly map.
- **Guoliang Yu**, “The Novikov conjecture for groups with finite asymptotic dimension”, *Annals of Mathematics* (2) 147, no. 2, 1998, pp. 325-355. DOI: 10.2307/121011.
Proves the relevant Novikov/coarse injectivity consequences for finite-asymptotic-dimension settings. This is a major positive special case, but finite asymptotic dimension is an extra hypothesis not present in the problem statement.
- **Guoliang Yu**, “The coarse Baum-Connes conjecture for spaces which admit a uniform embedding into Hilbert space”, *Inventiones Mathematicae* 139, 2000, pp. 201-240. DOI: 10.1007/s002229900032.
Proves the coarse Baum-Connes assembly map is an isomorphism for bounded-geometry spaces admitting a uniform embedding into Hilbert space. This covers many uniformly contractible bounded-geometry examples but not all such spaces.
- **Gennadi Kasparov and Guoliang Yu**, “The coarse geometric Novikov conjecture and uniform convexity”, *Advances in Mathematics* 206, no. 1, 2006, pp. 1-56; arXiv:math/0507599.
Proves injectivity for bounded-geometry spaces uniformly embeddable into uniformly convex Banach spaces. Again, this is a strong special case with an additional embedding hypothesis.
- **Georges Skandalis, Jean-Louis Tu, and Guoliang Yu**, “The coarse Baum-Connes conjecture and groupoids”, *Topology* 41, no. 4, 2002, pp. 807-834. DOI: 10.1016/S0040-

9383(01)00004-0.

Identifies the coarse assembly map for bounded-geometry discrete metric spaces with a groupoid Baum-Connes map and gives another route to the Hilbert-space embedding theorem. It does not settle the full uniformly contractible bounded-geometry class.

- **Nigel Higson, Vincent Lafforgue, and Georges Skandalis, “Counterexamples to the Baum-Connes conjecture”, *Geometric and Functional Analysis* 12, 2002, pp. 330-354. DOI: 10.1007/s00039-002-8249-5.**

Gives counterexamples related to expanders and shows failure of surjectivity for coarse Baum-Connes maps in bounded-geometry settings. These are counterexamples to broader isomorphism formulations, not a resolution of the stated uniform-contractibility injectivity problem.

- **Rufus Willett and Guoliang Yu, “Higher index theory for certain expanders and Gromov monster groups I”, *Advances in Mathematics* 229, no. 3, 2012, pp. 1380-1416. DOI: 10.1016/j.aim.2011.10.024.**

Shows that for expander sequences with girth tending to infinity, the coarse Baum-Connes assembly map is injective but not surjective. This is important negative progress for the isomorphism form in general bounded-geometry spaces, but the expander coarse disjoint unions are not the uniformly contractible spaces of the problem.

- **Tomohiro Fukaya and Shin-ichi Oguni, “The coarse Baum-Connes conjecture for Busemann non-positively curved spaces”, *Kyoto Journal of Mathematics* 56, no. 1, 2016, pp. 1-12. DOI: 10.1215/21562261-3445129; arXiv:1304.3224.**

Proves isomorphism for proper Busemann non-positively curved spaces. This is a substantial positive subclass, not the full class.

- **Tomohiro Fukaya and Shin-ichi Oguni, “A coarse Cartan-Hadamard theorem with application to the coarse Baum-Connes conjecture”, *Journal of Topology and Analysis* 12, no. 3, 2020, pp. 857-895. DOI: 10.1142/S1793525319500675; arXiv:1705.05588.**

Proves the conjecture for proper coarsely convex spaces, including further geometric examples such as systolic complexes. This is still an extra coarse convexity hypothesis.

- **Jintao Deng, Qin Wang, and Guoliang Yu, “The coarse Baum-Connes conjecture for certain relative expanders”, *Advances in Mathematics* 424, 2023, Article 109047; arXiv:2102.10617.**

Proves the conjecture for a class of relative expanders and special box spaces that need not coarsely embed into Hilbert space. This enlarges the positive known class but remains a special construction.

- **Christos Kitsios, Thomas Schick, and Federico Vigolo, “Coarse Baum-Connes and warped cones: failure of surjectivity in odd degree”, arXiv:2504.21811, 2025. URL: <https://arxiv.org/abs/2504.21811>.**

Gives recent warped-cone counterexamples to surjectivity of the coarse Baum-Connes map. This further shows that broad isomorphism formulations fail, but it does not supply a full resolution of the stated injectivity claim for all uniformly contractible bounded-geometry spaces.

- **Jianguo Zhang, “On the quantitative coarse Baum-Connes conjecture with coefficients”, *Science China Mathematics* 69, 2026, pp. 1537-1564. DOI: 10.1007/s11425-024-2484-5.**

Recent work showing active development of quantitative/coefficient variants and listing the current landscape of positive classes and counterexamples. It does not resolve the exact question.

5.6.3.4. What remains unresolved

The unresolved core is the all-cases injectivity statement:

For every uniformly contractible bounded-geometry metric space X ,
 $A : K_*(X) \rightarrow K_*(C^*(X))$ is injective.

The stronger isomorphism statement is known in many large subclasses and false in broader bounded-geometry settings, but I did not find a theorem whose hypotheses exactly include all uniformly contractible bounded-geometry spaces and whose conclusion settles either injectivity or isomorphism in all cases. Thus the problem should be classified as **partially resolved**, not fully resolved and not fully open.

5.6.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

5.6.5. Primary-source pointer

The originating ICM record is *Dimension Theory and Large Riemannian Manifolds* (1998). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.2/ICM1998.2.ocr.pdf#page=425>.

5.7. 184. Basis Problems in Combinatorial Set Theory [1998-geom-72]

Record: 1998-geom-72

Subfield: Metric and topological geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1998: Basis Problems in Combinatorial Set Theory

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.2/ICM1998.2.ocr.pdf#page=45>

5.7.1. Problem statement

Prove or disprove that for every compact topological space X that does not contain a subspace homeomorphic to the discrete topological space on the first uncountable ordinal ω_1 , there exist a compact metric space M and a continuous surjection $f : X \rightarrow M$ such that $|f^{-1}(m)| \leq 2$ for every $m \in M$.

5.7.2. Historical background, literature, and current progress

5.7.3. Verdict: partially resolved

5.7.3.1. Precise problem

The problem asks whether every compact space K with no subspace homeomorphic to the discrete space $D(\omega_1)$ admits a continuous surjection $f : K \rightarrow M$ onto a compact metric space such that every fiber has size at most 2. Equivalently, must every compactum of countable spread be a premetric compactum of degree at most 2?

The original ICM source phrases this as Todorćević's Conjecture 6 with a forcing-axiom tag: under PFA, every compact space not containing $D(\omega_1)$ admits an at most 2-to-1 map onto a compact metric space. the problem statement omits the PFA tag, but the literature treats the general compact-space problem as a Fremlin/Todorćević topological basis problem rather than as a settled ZFC theorem.

5.7.3.2. Current status

I did not find a published or otherwise citable result proving the full statement for all compact spaces. Later sources continue to state the full compact-space version as an open problem, while proving substantial special cases. Therefore the problem should be treated as partially resolved, not fully resolved.

5.7.3.3. Key literature and progress

1. Basis Problems in Combinatorial Set Theory

Authors: Stevo Todorćević.

Venue: Documenta Mathematica, Extra Volume ICM 1998, Vol. II, pp. 43-52, 1998.

DOI/arXiv/URL: <https://emis.de/ft/10334> ; also in ICM 1998 proceedings.

Result/status: Conjecture 6 states, with the tag [PFA], that every compact space not containing $D(\omega_1)$ admits an at most 2-to-1 continuous map onto a compact metric space. The article says this is related to Fremlin's problem and then explicitly restricts to test cases because the needed Ramsey-theoretic principles were not available. It records a special-case theorem for pointwise compact subsets of the first Baire class, not a proof for arbitrary compact spaces.

2. Compact subsets of the first Baire class

Author: Stevo Todorćević.

Venue: Journal of the American Mathematical Society 12, no. 4, pp. 1179-1212, 1999.

DOI/arXiv/URL: DOI 10.1090/S0894-0347-99-00312-4; <https://www.ams.org/jams/1999-12-04/S0894-0347-99-00312-4/>

Result/status: Proves the relevant 2-to-1 conclusion for compact subsets of Baire class 1 functions on a Polish space, often called Rosenthal compacta: if such a compactum contains no uncountable discrete subspace, then it admits an at most 2-to-1 map onto a compact metric space. This verifies the conjecture for an important concrete subclass, but it assumes the compact space is Rosenthal/Baire-1 and so does not cover arbitrary compact spaces.

3. The Proper Forcing Axiom

Author: Justin Tatch Moore.

Venue: Proceedings of the International Congress of Mathematicians 2010, Vol. II, pp. 3-29, 2010.

DOI/arXiv/URL: <https://pi.math.cornell.edu/~justin/Ftp/ICM.pdf>

Result/status: Theorem 8.7 summarizes Todorćević's Baire-1/Rosenthal compacta result, including the at most 2-to-1 conclusion when there is no uncountable discrete subspace. Immediately afterward, Problem 8.9 asks whether, assuming PFA, every compact K with no uncountable discrete subspace admits an at most 2-to-1 map onto a metric space. This confirms that the full PFA compact-space question was still open in this survey.

4. Perfect compacta and basis problems in topology

Authors: Gary Gruenhage and Justin Tatch Moore.

Venue: Open Problems in Topology II, Elsevier, 2007, pp. 151-157; preprint dated 2005.

DOI/arXiv/URL: <https://pi.math.cornell.edu/~justin/Ftp/basisprob.pdf>

Result/status: Discusses Fremlin's central question asking whether it is consistent that every perfect compactum admits a continuous at most two-to-one map onto a metric space, and calls such spaces premetric of order 2. It records partial results: under Martin's Maximum, Fremlin obtained a map with fibers of size at most 2 on a comeager subset of the range; under PFA/OCA there are positive results for cometrizable spaces; and Todorćević proved the Rosenthal compacta case. These results are explicitly presented as partial approaches, and the paper notes that reductions such as the zero-dimensional case were still unknown.

5. Analysis of a topological basis problem

Authors: Yinhe Peng and Stevo Todorćević.

Venue: Acta Mathematica Hungarica 167, pp. 419-475, 2022.

DOI/arXiv/URL: DOI 10.1007/s10474-022-01253-y; <https://link.springer.com/article/10.1007/s10474-022-01253-y>

Result/status: Gives recent PFA analysis of the associated topological basis problem via inner/outer topologies and finite-to-one perfect maps. In accompanying 2022 slides, the Fremlin question “Is every compact space in $D(\aleph_1)^\perp$ an at most 2-to-1 continuous preimage of a compact metric space?” is still displayed as a question, while the Baire-class-one theorem is listed as a special-case theorem. The 2022 results provide structural progress but do not prove the full compact-space statement.

6. One-dimensional definable topological spaces in o-minimal structures

Authors: Chris Miller and Sergei Starchenko.

Venue: arXiv preprint, 2023.

DOI/arXiv/URL: arXiv:2310.04510; <https://arxiv.org/abs/2310.04510>

Result/status: Section 9.1 reviews Fremlin’s conjecture for perfectly normal compact Hausdorff spaces and states that it asks for consistency of every such space admitting an at most 2-to-1 map onto a compact metric space. The paper proves definable/o-minimal one-dimensional analogues, not the unrestricted compact-space conjecture.

5.7.3.4. What remains unresolved

No cited theorem covers all compact spaces K with no copy of $D(\omega_1)$. The existing positive results require extra hypotheses such as being a Rosenthal compactum/compact subset of Baire class 1 functions, metrizable, monotonically normal, definable in an o-minimal setting, or they provide only weaker fiber conclusions on a comeager subset. Thus the full statement in the problem statement remains unresolved, with substantial partial progress.

5.7.4. Remaining open problems recorded by the database

5.7.4.1. Open problem 1

Does every perfectly normal compact Hausdorff space admit a continuous map into a metric space all of whose fibers are metrizable?

Reference supplied by the database. Gary Gruenhage and Justin Tatch Moore, “Perfect Compacta and Basis Problems in Topology,” in Elliott Pearl (ed.), *Open Problems in Topology II*, Elsevier, 2007, pp. 151–157, Question 7, preprint dated June 16, 2006, <https://pi.math.cornell.edu/~justin/Ftp/basisprob.pdf>.

5.7.4.2. Open problem 2

If a perfectly normal compact Hausdorff space admits a continuous map into a metric space with metrizable fibers, must it admit an at-most-two-to-one continuous map into a metric space?

Reference supplied by the database. Gary Gruenhage and Justin Tatch Moore, “Perfect Compacta and Basis Problems in Topology,” in Elliott Pearl (ed.), *Open Problems in Topology II*, Elsevier, 2007, pp. 151–157, Question 8, preprint dated June 16, 2006, <https://pi.math.cornell.edu/~justin/Ftp/basisprob.pdf>.

5.7.5. Primary-source pointer

The originating ICM record is *Basis Problems in Combinatorial Set Theory* (1998). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.2/ICM1998.2.ocr.pdf#page=45>.

5.8. 185. Finite Metric Spaces - Combinatorics, Geometry and Algorithms [2002-geom-79]

Record: 2002-geom-79

Subfield: Metric and topological geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Finite Metric Spaces - Combinatorics, Geometry and Algorithms

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=554>

5.8.1. Problem statement

Prove that there exists a positive constant C such that for every finite graph G , equipped with its shortest-path graph metric, with girth g and minimum degree at least 3, the least distortion $c_2(G)$ of an embedding of this metric into a Euclidean ℓ_2 space satisfies

$$c_2(G) \geq Cg.$$

5.8.2. Historical background, literature, and current progress

5.8.3. Verdict: partially resolved

5.8.3.1. Precise problem

The problem asks whether there is a universal constant $C > 0$ such that every finite graph G , equipped with its shortest-path metric, whose vertices all have degree at least 3 and whose girth is g , satisfies

$$c_2(G) \geq Cg,$$

where $c_2(G)$ is the least bilipschitz distortion of an embedding of the graph metric into Hilbert/Euclidean ℓ_2 space.

5.8.3.2. Original source

Title: The geometry of graphs and some of its algorithmic applications

Authors: Nathan Linial, Eran London, Yuri Rabinovich

Venue/year: Combinatorica 15(2), 215-245, 1995; preliminary FOCS 1994 version

DOI/URL: <https://doi.org/10.1007/BF01200757>; <https://www.cs.huji.ac.il/~nati/PAPERS/llr.pdf>

Status contribution: Problem 8.7 asks exactly whether, if all vertices in G have degree at least 3, every embedding of G into Euclidean space has distortion $\Omega(\text{girth}(G))$.

5.8.3.3. Known partial progress

Title: Girth and Euclidean distortion

Authors: Nathan Linial, Avner Magen, Assaf Naor

Venue/year: Geometric and Functional Analysis 12(2), 380-394, 2002; STOC 2002 version

DOI/arXiv/URL: <https://doi.org/10.1007/s00039-002-8251-y>; STOC DOI <https://doi.org/10.1145/509907.510009>; https://www.cs.huji.ac.il/~nati/PAPERS/euc_girth.pdf

Progress: Theorem 1.3 proves that for every k -regular graph G , $k > 2$, with girth g , one has $c_2(G) \geq C\sqrt{g}$. The paper explicitly describes this as only a partial proof of the Linial-London-Rabinovich conjecture and states that the original $\Omega(g)$ conjecture remains open. This covers the regular subcase only and gives $\sqrt{g}$, not g .

Title: On metric Ramsey-type phenomena

Authors: Yair Bartal, Nathan Linial, Manor Mendel, Assaf Naor

Venue/year: Annals of Mathematics 162(2), 643-709, 2005

DOI/arXiv/URL: <https://doi.org/10.4007/annals.2005.162.643>; arXiv:math/0406353; <https://annals.math.princeton.edu/2005/162-2/p02>

Progress: Theorem 6.1 strengthens the $\sqrt{g}$ lower bound beyond regular graphs: if G has girth g and average degree δ , then for the full metric $c_2(G) \geq ((\delta - 2)/\delta)\sqrt{\lfloor g/2 \rfloor} - 1$. Since minimum degree at least 3 implies average degree $\delta \geq 3$, this gives $c_2(G) = \Omega(\sqrt{g})$ for the full class. It still falls short of the conjectured linear $\Omega(g)$ bound.

Title: Finite metric spaces - combinatorics, geometry and algorithms

Author: Nathan Linial

Venue/year: Proceedings of the International Congress of Mathematicians, Vol. III, Beijing 2002, Higher Ed. Press, 573-586, 2002

arXiv/URL: arXiv:math/0304466; <https://arxiv.org/abs/math/0304466>

Status contribution: Section 3.3 restates the problem for graphs of girth g with all degrees at least 3, records the Linial-Magen-Naor $\Omega(\sqrt{g})$ theorem, and lists the state of knowledge as between $\Omega(\sqrt{g})$ and $O(g)$.

Title: An introduction to the Ribe program

Author: Assaf Naor

Venue/year: Japanese Journal of Mathematics 7(2), 167-233, 2012

DOI/arXiv/URL: <https://doi.org/10.1007/s11537-012-1222-7>; arXiv:1205.5993; <https://arxiv.org/abs/1205.5993>

Status contribution: Section 9.4 records the average-degree $\Omega(\sqrt{g})$ lemma and then asks, even for finite 3-regular connected graphs of girth g , what the growth rate of $c_2(g)$ is and in particular whether it grows asymptotically faster than $\sqrt{g}$. Since 3-regular graphs are a subclass of the stated graphs, this later survey confirms that the Euclidean problem is not known even beyond the $\sqrt{g}$ scale, let alone at the conjectured linear scale.

5.8.3.4. Nearby but non-resolving work

Title: Low-distortion embeddings of graphs with large girth

Author: Mikhail I. Ostrovskii

Venue/year: Journal of Functional Analysis 262(8), 3548-3555, 2012

DOI/arXiv/URL: <https://doi.org/10.1016/j.jfa.2012.01.020> ;
arXiv:1108.2542 ; <https://arxiv.org/abs/1108.2542>

Relevance: This constructs bounded-degree graphs with growing girth and uniformly bounded ℓ_1 distortion, resolving a related ℓ_1 question. It does not prove or disprove the stated bilipschitz ℓ_2 lower bound $c_2(G) \geq \Omega(g)$; coarse embeddability or ℓ_1 -embeddability is not the same as uniformly bounded Euclidean distortion.

5.8.3.5. What remains unresolved

I found no published or citable theorem proving $c_2(G) \geq Cg$ for all finite graphs of girth g and minimum degree at least 3, nor a citable counterexample settling the exact conjecture negatively. The best directly applicable lower bound located is $\Omega(\sqrt{g})$, including the full minimum-degree-at-least-3 class via the average-degree theorem of Bartal-Linial-Mendel-Naor. Thus the problem is partially resolved, not fully resolved.

5.8.4. Remaining open problems recorded by the database

5.8.4.1. Open problem 1

For each g , let $c_2^{(3)}(g)$ be the infimum of the least Euclidean distortions $c_2(G)$ over all finite connected 3-regular graphs G of girth g . Does

$$\frac{c_2^{(3)}(g)}{\sqrt{g}} \rightarrow \infty \quad \text{as } g \rightarrow \infty?$$

Reference supplied by the database. Assaf Naor, “An introduction to the Ribe program,” *Japanese Journal of Mathematics* 7 (2012), no. 2, 167–233, Question 7; DOI: <https://doi.org/10.1007/s11537-012-1222-7>; arXiv: 1205.5993.

5.8.5. Primary-source pointer

The originating ICM record is *Finite Metric Spaces - Combinatorics, Geometry and Algorithms* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=554>.

5.9. 186. Finite Metric Spaces - Combinatorics, Geometry and Algorithms [2002-geom-89]

Record: 2002-geom-89

Subfield: Metric and topological geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Finite Metric Spaces - Combinatorics, Geometry and Algorithms

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=554>

5.9.1. Problem statement

For given integers $n \geq 1$ and real numbers $\epsilon > 0$, what is the smallest integer $k = k(n, \epsilon)$ such that every finite metric space (X, d) with $|X| = n$ that embeds isometrically into some ℓ_1 space admits an embedding $f : X \rightarrow \ell_1^k$ with distortion less than $1 + \epsilon$, i.e. after rescaling, $d(x, y) \leq \|f(x) - f(y)\|_1 < (1 + \epsilon)d(x, y)$ for all distinct $x, y \in X$?

5.9.2. Historical background, literature, and current progress

5.9.3. Status: partially resolved

5.9.3.1. Precise problem

For each integer $n \geq 1$ and each $\epsilon > 0$, determine the least target dimension $k(n, \epsilon)$ such that every n -point metric space that is an ℓ_1 metric, equivalently every finite subset of some L_1 or ℓ_1 space, has a bi-Lipschitz embedding into ℓ_1^k with distortion less than $1 + \epsilon$. In cut-metric language this asks how many weighted cut coordinates are always enough, and sometimes necessary, to multiplicatively approximate every n -point ℓ_1 metric within factor $1 + \epsilon$.

5.9.3.2. Verdict

The problem is not fully resolved. The literature proves strong general upper and lower bounds, including a general upper bound $k(n, \epsilon) = O(n/\epsilon^2)$ and near-linear lower bounds for small ϵ , but these do not match and do not identify the smallest integer $k(n, \epsilon)$ for all n, ϵ .

5.9.3.3. Original source

- Title: Finite metric spaces–combinatorics, geometry and algorithms
- Author: Nathan Linial
- Venue/year: Proceedings of the International Congress of Mathematicians, Beijing 2002, vol. 3, pp. 573-586; arXiv version submitted 2003
- DOI/arXiv/URL: arXiv:math/0304466, <https://arxiv.org/abs/math/0304466>

- **Relevance:** This ICM survey states Open Problem 5 in essentially the same form: determine the smallest $k = k(n, \epsilon)$ such that every n -point metric in ℓ_1 embeds into ℓ_1^k with distortion $< 1 + \epsilon$. At that time the recorded bounds were only roughly $\Omega(\log n) \leq k \leq O(n \log n)$.

5.9.3.4. Progress in the literature

- **Title:** More on embedding subspaces of L_p in l_r^n
- **Author:** Gideon Schechtman
- **Venue/year:** *Compositio Mathematica* 61(2), pp. 159-169, 1987
- **DOI/arXiv/URL:** <https://eudml.org/doc/89819>
- **Result and relevance:** Provides finite-dimensional embedding results for subspaces of L_p , including the historical L_1 subspace upper-bound line. For an n -point subset of L_1 , the span of the point differences has dimension at most $n - 1$, so these subspace results gave general finite-dimensional ℓ_1 upper bounds, but not the optimal $k(n, \epsilon)$.
- **Title:** Embedding subspaces of L_1 into l_1^N
- **Author:** Michel Talagrand
- **Venue/year:** *Proceedings of the American Mathematical Society* 108(2), pp. 363-369, 1990
- **DOI/arXiv/URL:** DOI 10.1090/S0002-9939-1990-0994792-4
- **Result and relevance:** Improved the finite-dimensional L_1 subspace embedding estimates. This is part of the upper-bound background cited in later dimension-reduction papers, but it does not settle the finite metric problem exactly.
- **Title:** Dimension Reduction in the ℓ_1 Norm
- **Authors:** Moses Charikar and Amit Sahai
- **Venue/year:** 43rd IEEE Symposium on Foundations of Computer Science (FOCS 2002), pp. 551-560, IEEE, 2002
- **DOI/arXiv/URL:** DOI 10.1109/SFCS.2002.1181979
- **Result and relevance:** Shows that linear dimension reduction is impossible for general ℓ_1 point sets and gives positive results for special ℓ_1 -embeddable metric classes. This is important partial progress but does not answer the nonlinear finite-metric question in the problem statement.
- **Title:** On the impossibility of dimension reduction in ℓ_1
- **Authors:** Bo Brinkman and Moses Charikar
- **Venue/year:** *Journal of the ACM* 52(5), pp. 766-788, 2005; preliminary version FOCS 2003
- **DOI/arXiv/URL:** DOI 10.1145/1089023.1089026; preliminary DOI 10.1109/SFCS.2003.1238224
- **Result and relevance:** Constructs n -point subsets of ℓ_1 for which every embedding into ℓ_1^d with distortion D requires $d \geq n^{\Omega(1/D^2)}$. For distortion $1 + \epsilon$, it also gives a lower bound $d \geq n^{1/2 - O(\epsilon \log(1/\epsilon))}$. This disproves a Johnson-Lindenstrauss-style logarithmic dimension theorem for ℓ_1 , but it is still only a lower bound.
- **Title:** Embedding the diamond graph in L_p and dimension reduction in L_1

- Authors: James R. Lee and Assaf Naor
- Venue/year: Geometric and Functional Analysis 14(4), pp. 745-747, 2004
- DOI/arXiv/URL: DOI 10.1007/s00039-004-0473-8; arXiv:math/0407520
- Result and relevance: Gives a short geometric proof using diamond graphs that there are n -point subsets of L_1 whose distortion- D embeddings into ℓ_1^d need $d \geq n^{\Omega(1/D^2)}$. This supports the lower-bound side but does not determine the optimal $k(n, \epsilon)$.
- Title: On Multiplicative λ -Approximations and Some Geometric Applications
- Authors: Ilan Newman and Yuri Rabinovich
- Venue/year: SIAM Journal on Computing 42(3), pp. 855-883, 2013; arXiv preprint "Finite Volume Spaces and Sparsification", 2010
- DOI/arXiv/URL: DOI 10.1137/100801809; arXiv:1002.3541, <https://arxiv.org/abs/1002.3541>
- Result and relevance: Proves that every ℓ_1 metric on n points can be embedded into $\ell_1^{O(n/\epsilon^2)}$ with distortion $(1 + \epsilon)/(1 - \epsilon)$. Reparameterizing ϵ gives a $1 + \epsilon$ distortion upper bound $k(n, \epsilon) = O(n/\epsilon^2)$ for exactly the full class in the problem statement. This is the best general upper-bound type result found, but it is not matched by known lower bounds.
- Title: Near Linear Lower Bound for Dimension Reduction in L_1
- Authors: Alexandr Andoni, Moses S. Charikar, Ofer Neiman, and Huy L. Nguyen
- Venue/year: 52nd IEEE Symposium on Foundations of Computer Science (FOCS 2011), pp. 315-323, IEEE, 2011
- DOI/arXiv/URL: DOI 10.1109/FOCS.2011.87; author PDF <https://www.cs.columbia.edu/~andoni/papers/dim2-focs.pdf>
- Result and relevance: Improves the $1 + \epsilon$ lower bound substantially: for small ϵ , some n -point subsets of ℓ_1 require dimension at least $n^{1-O(1/\log(1/\epsilon))}$. This is near-linear as $\epsilon \rightarrow 0$, but still falls short of matching the $O(n/\epsilon^2)$ upper bound and does not yield the exact smallest $k(n, \epsilon)$.
- Title: Entropy-based Bounds on Dimension Reduction in L_1
- Author: Oded Regev
- Venue/year: Israel Journal of Mathematics 195, pp. 825-832, 2013; arXiv version 2011
- DOI/arXiv/URL: DOI 10.1007/s11856-012-0137-6; arXiv:1108.1283, <https://arxiv.org/abs/1108.1283>
- Result and relevance: Gives an alternative entropy proof of the Brinkman-Charikar and Andoni-Charikar-Neiman-Nguyen lower bounds: $d \geq N^{\Omega(1/D^2)}$ for distortion $D > 1$, and $d \geq N^{1-O(1/\log(1/\epsilon))}$ for distortion $1 + \epsilon$. This clarifies the lower-bound mechanism but does not close the gap.
- Title: Dimension Reduction for Finite Trees in ℓ_1
- Authors: James R. Lee, Arnaud de Mesmay, and Mohammad Moharrami
- Venue/year: Discrete & Computational Geometry 50, pp. 977-1032, 2013
- DOI/arXiv/URL: DOI 10.1007/s00454-013-9536-7

- Result and relevance: Shows that every n -point tree metric embeds into $\ell_1^{C(\epsilon) \log n}$ with distortion $1 + \epsilon$, with related lower bounds for star metrics. This is significant for a special subclass of ℓ_1 metrics, but the problem asks for all n -point ℓ_1 metrics.
- Title: Metric dimension reduction: A snapshot of the Ribe program
- Author: Assaf Naor
- Venue/year: Proceedings of the International Congress of Mathematicians, Rio de Janeiro 2018, vol. I, pp. 759-837; arXiv:1809.02376
- DOI/arXiv/URL: arXiv:1809.02376, <https://arxiv.org/abs/1809.02376>
- Result and relevance: Survey context for metric dimension reduction and its open challenges. Later papers continue to cite the Brinkman-Charikar and Andoni-Charikar-Neiman-Nguyen bounds as the relevant ℓ_1 lower-bound state rather than a complete determination of $k(n, \epsilon)$.

5.9.3.5. What remains unresolved

The exact function $k(n, \epsilon)$, even up to matching asymptotic order in the main parameter ranges, remains unknown. The current general results imply

$$n^{1-O(1/\log(1/\epsilon))} \lesssim k(n, \epsilon) \lesssim O(n/\epsilon^2)$$

for $1 + \epsilon$ distortion, with earlier and complementary bounds for fixed larger distortion. Because no cited theorem gives matching upper and lower bounds, or an exact smallest integer, for all n and ϵ , the problem should be classified as partially resolved rather than fully resolved.

5.9.4. Remaining open problems recorded by the database

5.9.4.1. Open problem 1

Determine the asymptotic growth of $c_{10}(n)$, the least integer d such that every n -point subset of L_1 embeds into ℓ_1^d with distortion at most 10; in particular, close the gap between the known polynomial lower bound and linear upper bound.

Reference supplied by the database. James R. Lee, “Open problem: Dimension reduction in L_1 ,” *tcs math* (research blog), February 19, 2010, explicit fixed-distortion instance $c_{10}(n)$. [Post](#).

5.9.5. Primary-source pointer

The originating ICM record is *Finite Metric Spaces - Combinatorics, Geometry and Algorithms* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=554>.

5.10. 187. Finite Metric Spaces - Combinatorics, Geometry and Algorithms [2002-geom-90]

Record: 2002-geom-90

Subfield: Metric and topological geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Finite Metric Spaces - Combinatorics, Geometry and Algorithms

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=554>

5.10.1. Problem statement

For integers $n \geq 1$ and real numbers $t \geq 1$, what is the largest value of $f(n, t)$ such that every metric space (X, d) with $|X| = n$ contains a subset $Y \subseteq X$ with $|Y| \geq f(n, t)$ whose restricted metric $d|_{Y \times Y}$ has Euclidean distortion at most t ?

Equivalently, if $c_2(Y)$ denotes the least distortion of an embedding of the metric space $(Y, d|_{Y \times Y})$ into some finite-dimensional ℓ_2 space, where distortion at most t means that for some embedding $\varphi : Y \rightarrow \ell_2^m$ and constants $\alpha \geq \beta > 0$ one has

$$\beta d(y, y') \leq \|\varphi(y) - \varphi(y')\|_2 \leq \alpha d(y, y')$$

for all $y, y' \in Y$ and $\alpha/\beta \leq t$, determine the largest $f(n, t)$ for which every n -point metric space (X, d) has such a subset Y with $c_2(Y) \leq t$.

5.10.2. Historical background, literature, and current progress

5.10.3. Verdict: partially resolved

5.10.3.1. Problem identified

The problem asks for the finite metric Ramsey function for Hilbert/Euclidean targets: for each integer $n \geq 1$ and distortion $t \geq 1$, determine the largest integer $f(n, t)$ such that every n -point metric space contains a subset Y of size at least $f(n, t)$ with Euclidean distortion $c_2(Y) \leq t$. In the literature this is usually written as the Euclidean metric Ramsey function $R_2(t, n)$.

The original source found is:

- Title: *Finite Metric Spaces - Combinatorics, Geometry and Algorithms*
- Author: Nathan Linial
- Venue/year: Proceedings of the International Congress of Mathematicians, Beijing 2002; arXiv version, 2003
- DOI/arXiv/URL: arXiv:math/0304466, <https://arxiv.org/abs/math/0304466>
- Relevance: Open Problem 8 asks exactly for the largest $f(\cdot, \cdot)$ such that every n -point

metric (X, d) has a subset Y with $|Y| \geq f(n, t)$ and $c_2(Y) \leq t$. It notes that for t close to 1 the answer is $\Theta(\log n)$, while larger t behaves differently.

5.10.3.2. Main progress

- Title: *On Hilbertian subsets of finite metric spaces*
- Authors: Jean Bourgain, Tadeusz Figiel, Vitali D. Milman
- Venue/year: Israel Journal of Mathematics 55(2), 147-152, 1986
- DOI/arXiv/URL: DOI 10.1007/BF02801990
- Result: For every $\epsilon > 0$, every finite metric space has a subset of size at least $C(\epsilon) \log |X|$ whose restricted metric embeds into Hilbert space with distortion $1+\epsilon$. They also prove logarithmic sharpness for some distortion $\alpha_0 > 1$. This establishes the original small-distortion nonlinear Dvoretzky/Ramsey phenomenon, but it does not determine $f(n, t)$ for all t .
- Title: *On metric Ramsey-type phenomena*
- Authors: Yair Bartal, Nathan Linial, Manor Mendel, Assaf Naor
- Venue/year: Annals of Mathematics 162(2), 643-709, 2005
- DOI/arXiv/URL: DOI 10.4007/annals.2005.162.643; arXiv:math/0406353
- Result: This paper explicitly studies $R_2(\alpha, n)$. It proves nearly tight upper and lower bounds, including a phase transition: for every fixed $1 < \alpha < 2$, $R_2(\alpha, n) = \Theta_\alpha(\log n)$, while for every fixed $\alpha > 2$, $R_2(\alpha, n)$ is polynomial in n with exponent strictly between 0 and 1. For large α , their bounds include $R_2(\alpha, n) \geq n^{1 - C \log(2\alpha)/\alpha}$ and $R_2(\alpha, n) \leq Cn^{1 - c/\alpha}$. This is major partial progress, but not an exact determination of $f(n, t)$, and it leaves a logarithmic gap in the large-distortion lower bound.
- Title: *Some low distortion metric Ramsey problems*
- Authors: Yair Bartal, Nathan Linial, Manor Mendel, Assaf Naor
- Venue/year: Discrete & Computational Geometry 33(1), 27-41, 2005
- DOI/arXiv/URL: DOI 10.1007/s00454-004-1100-z; arXiv:math/0406358
- Result: This follow-up records and uses the ℓ_2 phase transition at distortion 2, and extends related low-distortion threshold questions to ℓ_p spaces. For this problem it is supporting evidence about the ℓ_2 low-distortion regime rather than a full determination of $f(n, t)$.
- Title: *Ramsey partitions and proximity data structures*
- Authors: Manor Mendel, Assaf Naor
- Venue/year: Journal of the European Mathematical Society 9(2), 253-275, 2007
- DOI/arXiv/URL: DOI 10.4171/JEMS/79; arXiv:cs/0511084
- Result: For every ϵ in $(0, 1)$, every n -point metric space has a subset of size at least $n^{1-\epsilon}$ that embeds into Hilbert space, indeed into an ultrametric and hence into Hilbert space, with distortion $O(1/\epsilon)$. The paper states this result is best possible, matching the known lower examples up to universal constants. Equivalently, for large distortion t , this gives $R_2(t, n) \geq n^{1 - O(1/t)}$ and the known upper examples

give $R_2(t, n) \leq n^{1-\Omega(1/t)}$. This resolves the large-distortion order of the exponent up to constants, but still not the exact largest value requested in the problem statement.

- Title: *Scale-oblivious metric fragmentation and the nonlinear Dvoretzky theorem*
- Authors: Assaf Naor, Terence Tao
- Venue/year: Israel Journal of Mathematics 192, 489-504, 2012
- DOI/arXiv/URL: DOI 10.1007/s11856-012-0039-7; arXiv:1003.4013
- Result: Gives a simpler randomized proof for every distortion $D > 2$, producing a polynomial-size subset that embeds into an ultrametric with distortion D , with an explicit exponent from an optimization equation. The paper explicitly says the behavior at distortion $D = 2$ is unknown and asks related sharp-exponent questions near 2.
- Title: *Ultrametric subsets with large Hausdorff dimension*
- Authors: Manor Mendel, Assaf Naor
- Venue/year: Inventiones Mathematicae 192(1), 1-54, 2013
- DOI/arXiv/URL: DOI 10.1007/s00222-012-0402-7; arXiv:1106.0879
- Result: Extends the nonlinear Dvoretzky/metric Ramsey phenomenon to compact metric spaces and summarizes the finite results: for $D \in (1, 2)$ the finite Ramsey function is logarithmic in n , while for $D \in (2, \infty)$ it is polynomial in n ; for large D , the $O(1/\epsilon)$ distortion theorem and the matching expander-based obstruction are sharp up to constants.
- Title: *Advances in Metric Ramsey Theory and its Applications*
- Author: Yair Bartal
- Venue/year: arXiv preprint, 2021
- DOI/arXiv/URL: arXiv:2104.03484, <https://arxiv.org/abs/2104.03484>
- Result: Gives deterministic constructions and strengthened algorithmic versions of metric Ramsey theorems with tight asymptotic tradeoffs. This is later progress but does not claim an exact formula for $R_2(t, n)$ in all cases.

5.10.3.3. Why this is not fully resolved

The literature does not give a citable theorem determining the exact largest value of $f(n, t)$ for all integers $n \geq 1$ and all real $t \geq 1$. What is known is a strong collection of asymptotic and regime results:

- For fixed $1 < t < 2$, $f(n, t) = \Theta_t(\log n)$.
- For fixed $t > 2$, $f(n, t)$ is polynomial in n .
- For large t , $f(n, t)$ has exponent $1 - \Theta(1/t)$ in the sense that $n^{1-C/t} \leq f(n, t) \leq n^{1-c/t}$ up to universal constants.

This does not cover the exact question. In particular, the exact threshold value $t = 2$ is not determined by these theorems, and the exact constants/exponents for general t are not known. Therefore the correct status is partially resolved, not fully resolved.

5.10.4. Remaining open problems recorded by the database

5.10.4.1. Open problem 1

Determine the asymptotic order of $R_2(2, n)$, the largest integer m such that every n -point metric space contains an m -point subspace that embeds into Hilbert space with distortion at most 2.

Reference supplied by the database. Yair Bartal, Nathan Linial, Manor Mendel, and Assaf Naor, “On metric Ramsey-type phenomena,” *Annals of Mathematics* (2) 162, no. 2 (2005), 643–709, §7 (“Open Problems”), explicit question of determining $R_2(2, n)$. DOI: <https://doi.org/10.4007/annals.2005.162.643>. Preprint: [arXiv:math/0406353](https://arxiv.org/abs/math/0406353).

5.10.5. Primary-source pointer

The originating ICM record is *Finite Metric Spaces - Combinatorics, Geometry and Algorithms* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=554>.

5.11. 188. Higher index theory of elliptic operators and geometry of groups [2006-geom-95]

Record: 2006-geom-95

Subfield: Metric and topological geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2006: Higher index theory of elliptic operators and geometry of groups

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.2/ICM2006.2.ocr.pdf#page=1633>

5.11.1. Problem statement

Prove that for every discrete metric space Γ with bounded geometry, meaning that for every $r > 0$ there is a number $N(r) > 0$ such that every ball of radius r in Γ has at most $N(r)$ elements, the coarse Baum-Connes assembly map

$$\mu : \lim_{d \rightarrow \infty} K_*(P_d(\Gamma)) \longrightarrow K_*(C^*(\Gamma))$$

is injective. Here $P_d(\Gamma)$ is the Rips complex with vertex set Γ , in which a finite subset spans a simplex exactly when all pairwise distances are at most d , and $C^*(\Gamma)$ is the Roe algebra of Γ , the operator-norm closure of the locally compact finite-propagation operators on $\ell^2(\Gamma) \otimes H$ for a separable infinite-dimensional Hilbert space H .

5.11.2. Historical background, literature, and current progress

5.11.3. Status check: Coarse Strong Novikov Conjecture

Verdict: partially resolved.

5.11.3.1. Precise problem

The problem asks whether, for every discrete metric space X of bounded geometry, the reduced coarse Baum-Connes assembly map

$$\mu : \lim_{d \rightarrow \infty} K_*(P_d(X)) \longrightarrow K_*(C^*(X))$$

is injective. This is the injectivity part of the coarse Baum-Connes conjecture, usually called the coarse Novikov conjecture or coarse strong Novikov conjecture.

The problem is not the maximal coarse assembly map, not an equivariant/twisted/rational variant, and not the full isomorphism statement. A result resolves this problem only if it proves the above injectivity for all bounded-geometry discrete metric spaces, or gives a counterexample to that exact injectivity statement.

5.11.3.2. Current status

I found no published or citable theorem proving injectivity for all bounded-geometry discrete metric spaces, and no counterexample to this exact bounded-geometry injectivity statement. The literature gives many important special cases. Counterexamples to the full coarse Baum-Connes conjecture for bounded-geometry spaces are known, but the standard ones obstruct surjectivity, not the injectivity asked here.

Thus the full statement in the problem statement remains unresolved, while substantial special cases are known.

5.11.3.3. Original and background sources

1. Higher index theory of elliptic operators and geometry of groups

Authors: Guoliang Yu.

Venue/year: Proceedings of the International Congress of Mathematicians, Madrid 2006, Vol. II, European Mathematical Society, Zurich, 2006, pp. 1623-1639.

DOI/URL: DOI <https://doi.org/10.4171/022-2/77>; EMS page <https://ems.press/books/standalone/21>; IMU proceedings PDF <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.2/ICM2006.2.ocr.pdf>.

Relevance: This is the dataset's source context and discusses higher index theory, the coarse Baum-Connes framework, and the coarse Novikov/injectivity problem.

2. On the Coarse Baum-Connes Conjecture

Authors: Nigel Higson and John Roe.

Venue/year: In *Novikov Conjectures, Index Theorems and Rigidity*, Vol. 2 (Oberwolfach, 1993), London Mathematical Society Lecture Note Series 227, Cambridge University Press, 1995, pp. 227-254.

DOI/arXiv/URL: no DOI found in the search results; bibliographic reference appears in later papers such as <https://ems.press/journals/jncg/articles/9871882>.

Relevance: Standard early formulation/reference for the coarse Baum-Connes assembly map and its injectivity/surjectivity problems.

5.11.3.4. Positive partial results

1. The Novikov conjecture for groups with finite asymptotic dimension

Authors: Guoliang Yu.

Venue/year: *Annals of Mathematics* 147(2), 325-355, 1998.

DOI/URL: DOI <https://doi.org/10.2307/121011>; Annals page <https://annals.math.princeton.edu/articles/12915>.

Result: Theorem 7.1 proves the coarse Baum-Connes conjecture for proper metric spaces with finite asymptotic dimension.

Applicability check: This implies the stated injectivity for bounded-geometry spaces of finite asymptotic dimension, but finite asymptotic dimension is an extra hypothesis and is not automatic for all bounded-geometry spaces.

2. The coarse Baum-Connes conjecture for spaces which admit a uniform embedding into Hilbert space

Authors: Guoliang Yu.

Venue/year: *Inventiones Mathematicae* 139, 201-240, 2000.

DOI/URL: DOI <https://doi.org/10.1007/s002229900032>; Springer page <https://link.springer.com/article/10.1007/s002229900032>.

Result: Theorem 1.1 proves the coarse Baum-Connes conjecture for discrete bounded-geometry spaces admitting a uniform/coarse embedding into Hilbert space.

Applicability check: This is stronger than injectivity on that subclass, but the embedding hypothesis is additional and fails for important bounded-geometry examples such as many expanders.

3. The coarse Baum-Connes conjecture and groupoids

Authors: Georges Skandalis, Jean-Louis Tu, Guoliang Yu.

Venue/year: *Topology* 41(4), 807-834, 2002.

DOI/URL: DOI [https://doi.org/10.1016/S0040-9383\(01\)00004-0](https://doi.org/10.1016/S0040-9383(01)00004-0); author PDF <https://webusers.imj-prg.fr/~georges.skandalis/Publications/Publi32.pdf>.

Result: Gives a groupoid approach and a new proof that bounded-geometry spaces uniformly embeddable into Hilbert space satisfy the coarse assembly isomorphism.

Applicability check: Again resolves the problem statement only for the Hilbert-embeddable subclass.

4. The coarse geometric Novikov conjecture and uniform convexity

Authors: Gennadi Kasparov and Guoliang Yu.

Venue/year: *Advances in Mathematics* 206(1), 1-56, 2006.

DOI/arXiv/URL: DOI <https://doi.org/10.1016/j.aim.2005.08.004>; arXiv math/0507599, <https://arxiv.org/abs/math/0507599>.

Result: Proves the coarse geometric Novikov conjecture for bounded-geometry spaces uniformly/coarsely embeddable into uniformly convex Banach spaces.

Applicability check: This proves injectivity under a Banach-space embeddability hypothesis, not for arbitrary bounded-geometry spaces.

5. The coarse Novikov conjecture and Banach spaces with Property (H)

Authors: Xiaoman Chen, Qin Wang, Guoliang Yu.

Venue/year: *Journal of Functional Analysis* 268(9), 2754-2786, 2015.

DOI/URL: DOI <https://doi.org/10.1016/j.jfa.2015.02.001>; bibliographic page https://www.researchgate.net/publication/272753709_The_coarse_Novikov_conjecture_and_Banach_spaces_with_Property_H.

Result: Proves the coarse Novikov conjecture for bounded-geometry metric spaces coarsely embeddable into Banach spaces with Property (H).

Applicability check: This includes spaces coarsely embeddable into many classical Banach spaces, but still assumes a coarse embedding into a target with Property (H).

6. Higher index theory for certain expanders and Gromov monster groups, I

Authors: Rufus Willett and Guoliang Yu.

Venue/year: *Advances in Mathematics* 229(3), 1380-1416, 2012.

DOI/arXiv/URL: DOI <https://doi.org/10.1016/j.aim.2011.10.024>; arXiv 1012.4150, <https://arxiv.org/abs/1012.4150>.

Result: For sequences of expander graphs with girth tending to infinity, the coarse assembly map is injective but not surjective.

Applicability check: This is important because it proves injectivity for some non-Hilbert-embeddable bounded-geometry spaces, but it is still a special family.

7. A Bott periodicity theorem for ℓ^p -spaces and the coarse Novikov conjecture at infinity

Authors: Liang Guo, Zheng Luo, Qin Wang, Yazhou Zhang.

Venue/year: *Journal of Functional Analysis* 286(2), article 110215, 2024.

DOI/arXiv/URL: DOI <https://doi.org/10.1016/j.jfa.2023.110215>; arXiv 2207.04193, <https://arxiv.org/abs/2207.04193>.

Result: Proves the coarse Novikov conjecture for bounded-geometry metric spaces admitting a fibred coarse embedding into an ell^p -space.

Applicability check: The fibred coarse embedding hypothesis is substantial and does not cover all bounded-geometry metric spaces.

8. ell^p -coarse Baum-Connes conjecture for ell^q -coarse embeddable spaces

Authors: Jinmin Wang, Zhizhang Xie, Guoliang Yu, Bo Zhu.

Venue/year: arXiv preprint, 2024; version 2 dated 2025.

DOI/arXiv/URL: DOI <https://doi.org/10.48550/arXiv.2411.15070>; arXiv 2411.15070, <https://arxiv.org/abs/2411.15070>.

Result: Theorem 1.1 proves the ordinary coarse Baum-Connes conjecture for discrete bounded-geometry spaces coarsely embedding into ell^q for some finite q .

Applicability check: This implies injectivity for that subclass and strengthens earlier Hilbert-space results, but still assumes coarse embeddability into ell^q . It does not prove the universal bounded-geometry statement.

5.11.3.5. Negative results and why they do not settle this problem

1. Counterexamples to the Baum-Connes conjecture

Authors: Nigel Higson, Vincent Lafforgue, Georges Skandalis.

Venue/year: *Geometric and Functional Analysis* 12(2), 330-354, 2002.

DOI/URL: DOI <https://doi.org/10.1007/s00039-002-8249-5>; Penn State bibliographic page <https://pure.psu.edu/en/publications/counterexamples-to-the-baum-connes-conjecture/>; author PDF <https://vlaafforg.perso.math.cnrs.fr/files/counterex.pdf>.

Result: Gives counterexamples to several Baum-Connes-type statements, including surjectivity failures for coarse geometric spaces.

Applicability check: the problem asks only for injectivity of the coarse assembly map. These bounded-geometry expander-type counterexamples do not give a failure of injectivity of the stated map.

2. Expanders are counterexamples to the ell^p coarse Baum-Connes conjecture

Authors: Yeong Chyuan Chung and Piotr W. Nowak.

Venue/year: *Journal of Noncommutative Geometry* 17(1), 305-331, 2023.

DOI/arXiv/URL: DOI <https://doi.org/10.4171/JNCG/498>; arXiv 1811.10457, <https://arxiv.org/abs/1811.10457>; EMS page <https://ems.press/journals/jncg/articles/9871882>.

Result: Shows non-surjectivity for an ell^p analogue of the coarse Baum-Connes assembly map for certain expanders; the paper's introduction also summarizes that classical expander counterexamples are surjectivity obstructions and that injectivity is known for certain classes of expanders.

Applicability check: This is not a counterexample to the original C^* -Roe-algebra injectivity statement in the problem statement.

3. Non-bounded-geometry counterexamples

Several sources note that Yu constructed a counterexample to coarse Novikov-type injectivity when the bounded-geometry hypothesis is omitted. For example, Chung-Nowak's 2023 paper explicitly states that such a counterexample exists without bounded geome-

try.

Applicability check: The statement includes bounded geometry as a central hypothesis, so these examples do not resolve the problem statement.

5.11.3.6. Status indicators from recent literature

1. The strongly quasi-local coarse Novikov conjecture and Banach spaces with Property (H)

Authors: Xiaoman Chen, Kun Gao, Jiawen Zhang.

Venue/year: *Kyoto Journal of Mathematics* 64(1), 31-74, 2024.

DOI/arXiv/URL: DOI <https://doi.org/10.1215/21562261-2023-0010>; arXiv 2109.11118, <https://arxiv.org/abs/2109.11118>; Project Euclid page <https://projecteuclid.org/journals/kyoto-journal-of-mathematics/volume-64/issue-1/The-strongly-quasi-local-coarse-Novikov-conjecture-and-Banach-spaces/10.1215/21562261-2023-0010.full>.

Relevance: Its introduction says that, despite many positive results, no counterexample to the coarse Novikov conjecture had been discovered. The paper proves a strongly quasi-local variant under Property (H) embeddability, not the universal statement in this problem.

5.11.3.7. What remains unresolved

It remains unknown whether the reduced coarse assembly map

$$\mu : \lim_{d \rightarrow \infty} K_*(P_d(X)) \rightarrow K_*(C^*(X))$$

is injective for every discrete bounded-geometry metric space X . Existing theorems prove injectivity, or even full isomorphism, for large classes such as finite asymptotic dimension spaces, Hilbert-embeddable spaces, uniformly convex Banach-embeddable spaces, Property (H)-embeddable spaces, ℓ^q -embeddable spaces, and some fibred-coarse-embeddable or expander families. None of these hypotheses is part of the problem statement, and no theorem found removes them for all bounded-geometry spaces.

5.11.4. Remaining open problems recorded by the database

5.11.4.1. Open problem 1

Does there exist a sequence $(p_n)_{n \in \mathbb{N}}$ with $p_n > 1$ and $p_n \rightarrow \infty$ such that the Hilbertian direct sum $(\bigoplus_{n=1}^{\infty} \ell_{p_n})_{\ell_2}$ has Kasparov–Yu Property (H)?

Reference supplied by the database. Qingjin Cheng, “Sphere equivalence, Property H, and Banach expanders,” *Studia Mathematica* **233** (2016), no. 1, 67–83, Question 3.7. DOI: <https://doi.org/10.4064/sm8396-4-2016>.

5.11.4.2. Open problem 2

Does the Banach sequence space c_0 have Kasparov–Yu Property (H)?

Reference supplied by the database. Qingjin Cheng, “Sphere equivalence, Property H, and Banach expanders,” *Studia Mathematica* **233** (2016), no. 1, 67–83, Remark 3.8(ii). DOI: <https://doi.org/10.4064/sm8396-4-2016>.

5.11.5. Primary-source pointer

The originating ICM record is *Higher index theory of elliptic operators and geometry of groups* (2006). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.2/ICM2006.2.ocr.pdf#page=1633>.

5.12. 189. Quasiconformal geometry of fractals [2006-geom-101]

Record: 2006-geom-101

Subfield: Metric and topological geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2006: Quasiconformal geometry of fractals

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.2/ICM2006.2.ocr.pdf#page=1359>

5.12.1. Problem statement

Let X be the metric two-sphere obtained from a subdivision rule on $\mathbb{S}^2$ by the associated tile metric d_λ , in which level- n tiles have diameter comparable to λ^n for some $\lambda < 1$ sufficiently close to 1, and let $f : X \rightarrow X$ be the postcritically finite topologically holomorphic expanding map associated with the subdivision rule. Let P_f be the finite postcritical set of f . The Ahlfors regular conformal dimension of X is

$$\dim_c X = \inf\{q : \text{there exists an Ahlfors } q\text{-regular metric space} \\ \text{quasisymmetrically equivalent to } X\}.$$

Here Ahlfors q -regular means that, with respect to q -dimensional Hausdorff measure, the measure of each ball of radius r not exceeding the diameter of the space is bounded above and below by constant multiples of r^q , and quasisymmetrically equivalent means related by a quasisymmetric homeomorphism.

For an f -stable multicurve $\Gamma = \{\gamma_1, \dots, \gamma_n\}$, meaning a finite collection of pairwise disjoint, pairwise non-homotopic, nonperipheral Jordan curves in $\mathbb{S}^2 \setminus P_f$ such that every component of $f^{-1}(\gamma_j)$ is either peripheral or homotopic in $\mathbb{S}^2 \setminus P_f$ to one of the curves γ_i , where nonperipheral means that each complementary component of the curve contains at least two points of P_f , define for each $Q \geq 2$ the modified Thurston matrix $A(\Gamma, Q) = (m_{ij}^Q)$ by

$$m_{ij}^Q = \sum_{\alpha} \frac{1}{d_{i,j,\alpha}^{Q-1}},$$

where the sum is over the components $\gamma_{i,j,\alpha}$ of $f^{-1}(\gamma_j)$ homotopic to γ_i in $\mathbb{S}^2 \setminus P_f$, and $d_{i,j,\alpha}$ is the degree of $f : \gamma_{i,j,\alpha} \rightarrow \gamma_j$. Let $\lambda(f, \Gamma, Q)$ be the largest nonnegative eigenvalue of $A(\Gamma, Q)$.

Prove that

$$\dim_c X = \inf\{Q \geq 2 : \lambda(f, \Gamma, Q) < 1 \text{ for every } f\text{-stable multicurve } \Gamma\}.$$

5.12.2. Historical background, literature, and current progress

5.12.3. Verdict: partially resolved

5.12.3.1. Precise problem checked

The problem asks for the Bonk-Geyer-Pilgrim conformal-dimension formula for visual/tile-metric two-spheres coming from finite subdivision rules, equivalently expanding postcritically finite Thurston-type maps on the sphere: whether

$$\text{ARCdim}(X) = \inf\{Q \geq 2 : \lambda(f, \Gamma, Q) < 1 \text{ for every } f\text{-stable multicurve } \Gamma\}.$$

This is an exact equality statement. It is not just asking for one inequality, for the rational/unobstructed case, for examples, or for a different critical exponent defined by all curve families or graph energies.

5.12.3.2. Literature found

1. Mario Bonk, “Quasiconformal geometry of fractals”, *Proceedings of the International Congress of Mathematicians, Madrid 2006, Vol. II, European Mathematical Society, 2007*, pp. 1349-1373. DOI: 10.4171/022-2/64. URL: <https://ems.press/books/standalone/21/448>

This is the source context for the conjectural formula. Bonk describes metric two-spheres associated with subdivision rules and, citing discussions with Lukas Geyer and Kevin Pilgrim, states the conjecture that the Ahlfors regular conformal dimension is the infimum of the parameters $Q \geq 2$ for which all stable multicurve modified Thurston matrices have spectral radius < 1 . This is the exact problem statement.

2. Peter Haissinsky and Kevin M. Pilgrim, “Thurston obstructions and Ahlfors regular conformal dimension”, *Journal de Mathématiques Pures et Appliquées* (9) 90 (2008), no. 3, 229-241. DOI: 10.1016/j.matpur.2008.04.006. arXiv:0706.1123. URL: <https://arxiv.org/abs/0706.1123>

This is the main partial resolution. For an expanding branched covering $f : S^2 \rightarrow S^2$ with finite postcritical set, the paper associates a canonical quasisymmetry class of Ahlfors regular metrics and proves a lower bound for the Ahlfors regular conformal dimension in terms of the same multicurve spectral data. In the notation of the problem, it proves the obstruction direction: if some stable multicurve has $\lambda(f, \Gamma, Q) \geq 1$, then the conformal dimension cannot be below the corresponding Q . This gives $\text{ARCdim}(X) \geq$ the multicurve critical exponent. It does not prove the reverse inequality, i.e. it does not construct Ahlfors regular metrics of every dimension above the multicurve threshold.

3. John M. Mackay and Jeremy T. Tyson, “Conformal Dimension: Theory and Application”, *University Lecture Series 54, American Mathematical Society, 2010*. DOI: 10.1090/ulect/054.

Section 7.4 formulates the same statement as Conjecture 7.4.6, attributed to Bonk-Geyer-Pilgrim, in the form $\text{ARCdim}(\mathcal{G}_f) = p_f$, where p_f is the supremum of the critical exponents of irreducible stable multicurves. The book explicitly records that one direction is known and cites Haissinsky-Pilgrim for $\text{ARCdim} \mathcal{G}_f \geq p_f$. This confirms that, as of that survey, the equality was not known.

4. **Peter Haissinsky and Kevin M. Pilgrim, “Coarse expanding conformal dynamics”, *Asterisque* 325 (2009), viii+139 pp. arXiv:math/0612617. URL: <https://arxiv.org/abs/math/0612617>**

This monograph develops the canonical conformal gauge/coarse expanding conformal framework that includes finite subdivision rule examples, and records how finite subdivision rules with mesh going to zero and bounded valence fit into topologically coarse expanding conformal dynamics. It supplies the ambient setting for the Haissinsky-Pilgrim lower-bound theorem, but it does not prove the full Bonk-Geyer-Pilgrim equality.

5. **Peter Haissinsky and Kevin M. Pilgrim, “Examples of coarse expanding conformal maps”, *Discrete and Continuous Dynamical Systems* 32 (2012), no. 7, 2403-2416. DOI: 10.3934/dcds.2012.32.2403. URL: <https://www.aims sciences.org/article/doi/10.3934/dcds.2012.32.2403>**

This paper extends the class of examples and discusses implications for computing conformal dimension. It gives useful examples and applications of the coarse expanding conformal framework, but not a theorem covering all subdivision-rule spheres and all stable multicurves in the stated formula.

6. **Peter Haissinsky and Kevin M. Pilgrim, “Minimal Ahlfors regular conformal dimension of coarse expanding conformal dynamics on the sphere”, *Duke Mathematical Journal* 163 (2014), no. 13, 2517-2559. DOI: 10.1215/00127094-2819408. arXiv:1103.4019. URL: <https://arxiv.org/abs/1103.4019>**

This proves strong results about when the Ahlfors regular conformal dimension of a topologically coarse expanding conformal sphere is attained. It is relevant to the “moreover” part often stated with the conjecture, but it does not identify the conformal dimension with the multicurve critical exponent in all cases.

7. **Mario Bonk and Daniel Meyer, “Expanding Thurston Maps”, *Mathematical Surveys and Monographs* 225, American Mathematical Society, 2017. DOI: 10.1090/surv/225. arXiv:1009.3647. URL: <https://arxiv.org/abs/1009.3647>**

This monograph gives the modern theory of expanding Thurston maps and visual metrics, including that expanding Thurston maps produce visual spheres and that rationality corresponds to quasisymmetric equivalence with the Riemann sphere in the relevant setting. It supports the interpretation of the stated d_λ tile metrics as visual metrics, but it does not resolve the Bonk-Geyer-Pilgrim conformal-dimension formula.

8. **Matias Carrasco Piaggio, “On the conformal gauge of a compact metric space”, *Annales scientifiques de l’Ecole Normale Supérieure, Serie 4*, 46 (2013), no. 3, 495-548. DOI: 10.24033/asens.2195. URL: <https://www.numdam.org/articles/10.24033/asens.2195/>**

Carrasco gives a general combinatorial-modulus description of Ahlfors regular conformal gauges and conformal dimension for compact metric spaces. This is important background and provides a broad critical-exponent framework, but it does not specialize the critical exponent for all subdivision-rule visual spheres to the stable-multicurve Thurston-matrix quantity appearing in the problem statement.

9. **Kevin M. Pilgrim and Dylan P. Thurston, “Ahlfors-regular conformal dimension and energies of graph maps”, *Groups, Geometry, and Dynamics* 20 (2026), no. 3, 971-1050. DOI: 10.4171/GGD/914. arXiv:2112.09041. URL: <https://ems.press/journals/ggd/articles/14299346>**

This recent work gives upper and lower bounds for the Ahlfors regular conformal di-

mension of connected Julia sets of hyperbolic rational maps using asymptotic conformal energies of graph maps. It is substantial later progress in the same area, but its hypotheses and conclusion differ from the problem statement: it concerns hyperbolic rational-map Julia sets and graph-energy bounds, not the exact all-stable-multicurve formula for arbitrary subdivision-rule visual two-spheres.

10. Insung Park, “Julia sets with Ahlfors-regular conformal dimension one”, arXiv:2209.13384, version 2, 2026. URL: <https://arxiv.org/abs/2209.13384>

Park characterizes postcritically finite hyperbolic rational maps whose Julia sets have Ahlfors regular conformal dimension one and uses finite subdivision rules/graph virtual endomorphisms to estimate conformal dimensions. This is later progress for a different extremal problem involving Julia sets, not a proof of the stated dimension formula for metric two-spheres.

5.12.3.3. Why the problem is not fully resolved

The exact formula requires both inequalities. The literature proves the obstruction/lower-bound direction via Haissinsky-Pilgrim: multicurves with large modified Thurston spectral radius force the Ahlfors regular conformal dimension to be at least the associated critical exponent.

I found no published or otherwise citable result proving the converse in the full stated generality: for every subdivision-rule visual/tile sphere and every Q above the multicurve threshold, there exists an Ahlfors Q -regular metric quasisymmetrically equivalent to X . Later work supplies general combinatorial-modulus frameworks, attainment results, examples, and graph-energy estimates in related rational-map settings, but none of these hypotheses and conclusions imply the exact Bonk-Geyer-Pilgrim equality for all stable multicurves and all subdivision-rule visual spheres.

Therefore the current status is **partially resolved**.

5.12.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website’s “Some remaining open problems” field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

5.12.5. Primary-source pointer

The originating ICM record is *Quasiconformal geometry of fractals* (2006). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.2/ICM2006.2.ocr.pdf#page=1359>.

5.13. 190. Higher index theory of elliptic operators and geometry of groups [2006-geom-281]

Record: 2006-geom-281

Subfield: Metric and topological geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2006: Higher index theory of elliptic operators and geometry of groups

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.2/ICM2006.2.ocr.pdf#page=1633>

5.13.1. Problem statement

Prove that for every integer $n \geq 0$ there exists an exponent $p_n \in (1, \infty)$ with the following property. If M is a compact smooth manifold of dimension n and Γ is a countable subgroup of the diffeomorphism group $\text{Diff}(M)$, equip Γ with any proper left-invariant metric d . Then there is a map

$$f : \Gamma \rightarrow C_{p_n}(H)$$

into the Schatten- p_n class on a separable infinite-dimensional Hilbert space H ,

$$C_{p_n}(H) = \{T \in \mathcal{B}(H) : \text{Tr}((T^*T)^{p_n/2}) < \infty\},$$

endowed with the norm

$$\|T\|_{p_n} = \text{Tr}((T^*T)^{p_n/2})^{1/p_n},$$

and there are nondecreasing functions $\rho_-, \rho_+ : [0, \infty) \rightarrow [0, \infty)$ satisfying

$$\lim_{r \rightarrow \infty} \rho_-(r) = \lim_{r \rightarrow \infty} \rho_+(r) = \infty$$

such that

$$\rho_-(d(g, h)) \leq \|f(g) - f(h)\|_{p_n} \leq \rho_+(d(g, h))$$

for all $g, h \in \Gamma$.

5.13.2. Historical background, literature, and current progress

The exact assertion is **unresolved**. In particular, no source found proves that an exponent can be chosen solely from the manifold dimension and then works for every compact (n)-manifold and every countable subgroup of its diffeomorphism group. The literature continues to state even weaker forms as open: allowing (p) merely to exist for each subgroup is weaker than the uniform p_n required here.

The conjecture was explicitly posed as Conjecture 4.9 in Guoliang Yu, "Higher index theory of elliptic operators and geometry of groups," *Proceedings of the International Congress of Mathematicians, Madrid 2006*, Vol. II, European Mathematical Society, 2006, pp. 1623–1639, DOI <https://doi.org/10.4171/022-2/77>. Yu specifies that (p) is to depend on (dim M), and notes that the conjecture was open even for the circle.

Gennadi Kasparov and Guoliang Yu, “The Novikov conjecture and geometry of Banach spaces,” *Geometry & Topology* 16 (2012), 1859–1880, DOI <https://doi.org/10.2140/gt.2012.16.1859>, restate the Schatten-class coarse-embedding conjecture. Their Theorems 1.2 and 1.3 prove strong Novikov consequences **conditional** on a coarse embedding into a Banach space with Property (H); they do not produce the embedding asserted here.

Guoliang Yu, “The Novikov conjecture,” *Russian Mathematical Surveys* 74:3 (2019), 525–541, DOI <https://doi.org/10.1070/RM9882>, gives particularly clear later status evidence. Open Question 3.15 asks whether every countable subgroup of the diffeomorphism group of a compact smooth manifold coarsely embeds into C_p for some ($p > 1$), without the present statement’s uniform dimension-only choice. Open Question 3.9 asks whether every countable subgroup of the circle diffeomorphism group coarsely embeds into Hilbert space. Thus neither this special direction nor the weaker non-uniform Schatten formulation was settled there.

There is substantial progress for restricted classes. Erik Guentner, Nigel Higson, and Shmuel Weinberger, “The Novikov conjecture for linear groups,” *Publications Mathématiques de l’IHÉS* 101 (2005), 243–268, DOI <https://doi.org/10.1007/s10240-005-0030-5>, prove that every countable linear group coarsely embeds into Hilbert space, hence into C_2 . More generally, Nathaniel P. Brown and Erik Guentner, “Uniform embeddings of bounded geometry spaces into reflexive Banach space,” *Proceedings of the American Mathematical Society* 133 (2005), 2045–2050, DOI <https://doi.org/10.1090/S0002-9939-05-07721-X>, prove that every bounded-geometry metric space embeds into an explicit reflexive Banach space, but that target is not a fixed Schatten class and so does not settle the problem.

Recent diffeomorphism-group work reaches the intended Novikov application by different geometry but still does not prove the stated embedding. Sherry Gong, Jianchao Wu, and Guoliang Yu, “The Novikov conjecture, the group of volume preserving diffeomorphisms and Hilbert-Hadamard spaces,” *Geometric and Functional Analysis* 31 (2021), 206–267, DOI <https://doi.org/10.1007/s00039-021-00563-7>, proves the rational strong Novikov conjecture for geometrically discrete subgroups of volume-preserving diffeomorphisms via proper actions on admissible Hilbert-Hadamard spaces. Sherry Gong, Jianchao Wu, Zhizhang Xie, and Guoliang Yu, “The Novikov conjecture, the group of diffeomorphisms and continuous fields of Hilbert-Hadamard spaces,” arXiv preprint (2023; revised 2025), [arXiv:2310.01219](https://arxiv.org/abs/2310.01219), removes volume preservation for subgroups satisfying the additional μ -discreteness hypothesis. Neither result supplies a coarse embedding into C_{p_n} , and neither covers all countable subgroups under the hypotheses of the present statement.

The zero-dimensional case is elementary because a compact (0)-manifold has finite diffeomorphism group. Hilbert-coarsely-embeddable subgroups (including the linear examples above) also satisfy the desired conclusion with ($p=2$). What remains open is the universal assertion for every positive dimension, already including the circle case, together with the dimension-uniform exponent required by the statement.

5.13.3. Remaining open problems recorded by the database

5.13.3.1. Open problem 1

Determine whether the following assertion holds. Let Γ be any countable subgroup of the diffeomorphism group $\text{Diff}(S^1)$ of the circle, equipped with any proper left-invariant metric d . Then there exist a separable Hilbert space H , a map $f : \Gamma \rightarrow H$, and nondecreas-

ing functions $\rho_-, \rho_+ : [0, \infty) \rightarrow [0, \infty)$ satisfying

$$\lim_{r \rightarrow \infty} \rho_-(r) = \lim_{r \rightarrow \infty} \rho_+(r) = \infty$$

such that

$$\rho_-(d(g, h)) \leq \|f(g) - f(h)\|_H \leq \rho_+(d(g, h))$$

for all $g, h \in \Gamma$.

Reference supplied by the database. Guoliang Yu, “The Novikov conjecture,” *Russian Mathematical Surveys* 74(3) (2019), 525–541, Open Question 3.9, DOI: <https://doi.org/10.1070/RM9882>.

5.13.4. Primary-source pointer

The originating ICM record is *Higher index theory of elliptic operators and geometry of groups* (2006). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.2/ICM2006.2.ocr.pdf#page=1633>.

5.14. 191. Volume Comparison via Boundary Distances [2010-geom-121]

Record: 2010-geom-121

Subfield: Metric and topological geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2010: Volume Comparison via Boundary Distances

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=775>

5.14.1. Problem statement

Let S^1 be the circle with its intrinsic length metric and total length 2π . Let M be a compact orientable Riemannian surface whose boundary is identified with S^1 , and suppose that for every $x, y \in \partial M = S^1$, the distance $d_M(x, y)$ in M is at least the intrinsic circle distance between x and y . Prove that $\text{area}(M) \geq 2\pi$. Equivalently, prove that the filling volume of S^1 , defined as the infimum of $\text{area}(M)$ over all such M , is 2π , with equality realized by the standard round hemisphere of radius 1.

5.14.2. Historical background, literature, and current progress

5.14.3. Verdict: partially resolved

5.14.3.1. Precise problem

The problem asks for Gromov's circle filling conjecture for the Riemannian circle of length 2π : if M is any compact orientable Riemannian surface with ∂M identified with that circle and with no boundary shortcuts, then $\text{area}(M) \geq 2\pi$, with the round unit hemisphere realizing equality. Since the boundary itself contains arcs of length equal to the intrinsic circle distance, the hypothesis $d_M(x, y) \geq d_{S^1}(x, y)$ is equivalent here to equality of the boundary distance function $d_M|_{\partial M \times \partial M} = d_{S^1}$.

The literature I found does not prove this for all compact orientable filling surfaces. It proves important special cases and lower bounds, but the higher-genus case remains open.

5.14.3.2. Main sources and progress

1. M. Gromov, "Filling Riemannian manifolds", *Journal of Differential Geometry* **18** (1983), 1-147. DOI/URL: <https://doi.org/10.4310/jdg/1214509283>. This is the original source for filling volume and the circle filling problem. Later papers consistently cite it as the source of the conjecture. The disk/simply connected filling case is known from the Pu inequality argument: identify antipodal boundary points to get a projective-plane metric with systole at least π , giving area at least 2π . This does not cover fillings of arbitrary genus.

2. Victor Bangert, Christopher Croke, Sergei V. Ivanov, and Mikhail G. Katz, “Filling area conjecture and ovalless real hyperelliptic surfaces”, *Geometric and Functional Analysis* 15 (2005), 577-597. arXiv:math/0405583, URL: <https://arxiv.org/abs/math/0405583>. Theorem 1.7 proves a relative Pu-type inequality for ovalless real hyperelliptic surfaces, and Corollary 1.8 states that all orientable genus-one fillings of the circle satisfy the filling area conjecture. The hypotheses/conclusion are not enough for the full statement, because the statement quantifies over all compact orientable surfaces, including arbitrary higher genus.
3. Neil N. Katz, “A modulus for curves from distance”, *Annales Academiae Scientiarum Fennicae Mathematica* 35 (2010), 3-21. DOI: <https://doi.org/10.5186/aasfm.2010.3501>. The introduction summarizes the status: Gromov proved the disk case, Bangert-Croke-Ivanov-M. Katz extended it to a torus with a disk removed, and the higher-genus question remains open. This is useful because it directly addresses the same filling-volume formulation and confirms that the known genus-one result is only partial.
4. Sergei Ivanov, “Volume comparison via boundary distances”, *Proceedings of the International Congress of Mathematicians 2010*, Vol. II, 769-784; arXiv:1004.2505, URL: <https://arxiv.org/abs/1004.2505>. Section 2.2 states Gromov’s circle filling conjecture in the same form and says the disk case is well known via Pu’s inequality, the genus-one case follows from Bangert-Croke-Ivanov-Katz, and higher genus remains open. This is a survey/status source, not a full solution.
5. Gregory R. Chambers, “On minimal higher genus fillings”, *Journal of Topology and Analysis*, online ready, DOI: <https://doi.org/10.1142/S1793525326500184>; arXiv:2202.01342, URL: <https://arxiv.org/abs/2202.01342>. The main theorem proves that if M has genus G and is compared with a disk D with unique geodesics and dominating boundary distances, then

$$\left(1 + \frac{2G}{\pi}\right) \text{Area}(M) \geq \text{Area}(D).$$

Taking D to be the hemisphere would only give $\text{Area}(M) \geq 2\pi/(1 + 2G/\pi)$, which is weaker than 2π for $G > 0$. Thus it is genuine higher-genus progress but not a resolution of Gromov’s conjecture.

6. Roger Züst, “The Riemannian hemisphere is almost calibrated in the injective hull of its boundary”, *Calculus of Variations and Partial Differential Equations* 64, Article 297 (2025). DOI: <https://doi.org/10.1007/s00526-025-03133-z>; arXiv:2104.04498, URL: <https://arxiv.org/abs/2104.04498>. This paper explicitly states that the higher-genus question is widely open. It proves an almost-calibration/stability lower bound for metric-current or oriented Lipschitz-surface fillings near the hemisphere in the injective hull. The result supplies positive evidence and local/near-hemisphere control, but not the global 2π area lower bound for every orientable Riemannian filling.
7. Joseph Briggs and Chris Wells, “A discrete view of Gromov’s filling area conjecture”, arXiv:2602.17859v1 (2026), URL: <https://arxiv.org/abs/2602.17859>. This recent preprint states that the full conjecture remains unresolved and proves Theorem 1.2: any compact Riemannian surface isometrically filling the Riemannian circle of length 2π has area at least $(\sqrt{3}/4)\pi^2$, approximately 1.36π . This is an all-topology quantitative lower bound, even without orientability, but it is strictly below the conjectured 2π .

5.14.3.3. What remains unresolved

the exact statement remains open for compact orientable Riemannian fillings of genus at least 2. No located published or citable preprint proves that every such filling has area at least 2π . The known results cover genus 0, genus 1, certain stability/near-hemisphere settings, and weaker universal lower bounds, so the correct status is **partially resolved**.

5.14.4. Remaining open problems recorded by the database

5.14.4.1. Open problem 1

Prove that every compact orientable Riemannian surface (M) of genus (2) with one boundary component identified with the Riemannian circle S^1 of length 2π , and satisfying

$$d_M(x, y) \geq d_{S^1}(x, y) \quad \text{for all } x, y \in \partial M,$$

has area at least 2π .

Reference supplied by the database. Mikhail G. Katz, “Filling Area Conjecture,” *Website for Systolic Geometry and Topology*, section explicitly offering a prize for the genus-(2) case, accessed July 21, 2026, <https://u.cs.biu.ac.il/~katzmik/sgtdirectory/filling.html>. Current open-status verification: Joseph Briggs and Chris Wells, “A Discrete View of Gromov’s Filling Area Conjecture,” arXiv preprint (2026), Conjecture 1.1 and the discussion following it, arXiv: [2602.17859](https://arxiv.org/abs/2602.17859).

5.14.5. Primary-source pointer

The originating ICM record is *Volume Comparison via Boundary Distances* (2010). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=775>.

5.15. 192. Isoperimetric inequalities and asymptotic geometry [2014-geom-126]

Record: 2014-geom-126

Subfield: Metric and topological geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: Isoperimetric inequalities and asymptotic geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=1067>

5.15.1. Problem statement

Let X be a proper cocompact $\text{CAT}(0)$ -space, and let its Euclidean rank be k , meaning that k is the maximal dimension of an isometrically embedded copy of Euclidean space in X . For a metric space X and $r \geq 0$, define $\text{FV}_{m+1}(X, r)$ to be the supremum of $\text{Fillvol}_X(z)$ over all Lipschitz m -cycles z in X with mass $\mathbf{M}(z) \leq r$, where $\text{Fillvol}_X(z)$ is the least mass of a Lipschitz $(m+1)$ -chain c in X satisfying $\partial c = z$, and is $+\infty$ if no such c exists. Prove that for every integer $m \geq k$, there exists a constant C such that

$$\text{FV}_{m+1}(X, r) \leq Cr$$

for all $r \geq 0$.

5.15.2. Historical background, literature, and current progress

5.15.3. Verdict: partially resolved

5.15.3.1. Precise problem

The problem asks for Gromov's isoperimetric gap/rank conjecture for proper cocompact $\text{CAT}(0)$ spaces: if X has Euclidean rank k , then every Lipschitz m -cycle in X , for every $m \geq k$, should admit a Lipschitz $(m+1)$ -chain filling with mass bounded linearly by the mass of the cycle. Equivalently, since the asymptotic rank equals the Euclidean rank for proper cocompact $\text{CAT}(0)$ spaces, it asks whether $\text{CAT}(0)$ spaces of asymptotic rank at most m satisfy a linear m -dimensional filling inequality for all m -cycles.

5.15.3.2. Status

I did not find a published or citable theorem proving the full statement for all proper cocompact $\text{CAT}(0)$ spaces and all $m \geq k$. The literature gives substantial progress and several important special cases, but recent papers still state the general $\text{CAT}(0)$ rank ≥ 2 case as incomplete. Therefore the correct verdict is **partially resolved**.

5.15.3.3. Original source and formulation

- **Title:** *Asymptotic invariants of infinite groups*
- **Author:** Mikhail Gromov
- **Venue/year:** In *Geometric Group Theory, Vol. 2 (Sussex, 1991)*, London Mathematical Society Lecture Note Series 182, Cambridge University Press, 1993, pp. 1–295
- **URL:** <https://www.ihes.fr/~gromov/wp-content/uploads/2018/08/492.pdf>
- **Role:** The conjectural linear filling statement is traced in later papers to Gromov, section 6.B2, p. 128(b).
- **Title:** *Guido's Book of Conjectures*
- **Collector/editor:** Indira Chatterji
- **Venue/year:** L'Enseignement Mathématique 54 (2008)
- **URL:** <https://chatterji.perso.math.cnrs.fr/papers/GMFinal.pdf>
- **Role:** Conjecture 45.1 records the statement in essentially the same form: every proper cocompact CAT(0) space of Euclidean rank r should admit a linear isoperimetric inequality for k -cycles for every $k \geq r$.

5.15.3.4. Progress and related results

- **Title:** *Higher-dimensional linear isoperimetric inequalities in hyperbolic groups*
- **Author:** Urs Lang
- **Venue/year:** International Mathematics Research Notices 2000, no. 13, 709–717
- **DOI:** 10.1155/S1073792800000384
- **Progress:** Proves higher-dimensional linear filling inequalities in the hyperbolic setting. This covers the rank-one/hyperbolic part of the conjectural picture under the standard geometric hypotheses, but not general higher-rank CAT(0) spaces.
- **Title:** *The asymptotic rank of metric spaces*
- **Author:** Stefan Wenger
- **Venue/year:** Commentarii Mathematici Helvetici 86 (2011), 247–275
- **DOI:** 10.4171/CMH/223
- **arXiv:** math/0701212
- **Progress:** Shows that for complete metric spaces with cone-type inequalities, including CAT(0) spaces, cycles in dimensions at least the asymptotic rank satisfy a sub-Euclidean filling inequality: $FV_{m+1}(r) = o(r^{(m+1)/m})$. It also records that for proper cocompact CAT(0) spaces the asymptotic rank equals Euclidean rank. This is strong evidence/progress but is weaker than the required linear bound $O(r)$.
- **Title:** *Characterizations of higher rank hyperbolicity*
- **Authors:** Tommaso Goldhirsch and Urs Lang
- **Venue/year:** Mathematische Zeitschrift 305 (2023), Article 13

- **DOI:** 10.1007/s00209-023-03339-x
- **arXiv:** 2112.15565
- **Progress:** Establishes a linear filling inequality for n -cycles with controlled density in spaces of asymptotic rank at most n . This is a restricted class of cycles, so it does not imply the stated supremum over all Lipschitz m -cycles.
- **Title:** *Linear isoperimetric inequality for homogeneous Hadamard manifolds*
- **Author:** Hjalti Isleifsson
- **Venue/year:** Journal of Topology and Analysis, online 2023 / issue 2025, 8 pp.
- **DOI:** 10.1142/S1793525323500334
- **arXiv:** 2203.09166
- **Progress:** Proves the expected linear inequality for homogeneous Hadamard manifolds, extending the known symmetric-space case. This is a significant special class, not all proper cocompact CAT(0) spaces.
- **Title:** *Minimal tetrahedra and an isoperimetric gap theorem in non-positive curvature*
- **Authors:** Cornelia Drutu, Urs Lang, Panos Papasoglu, and Stephan Stadler
- **Venue/year:** arXiv preprint, 2025
- **arXiv/DOI:** arXiv:2502.03389, <https://doi.org/10.48550/arXiv.2502.03389>
- **Progress:** For proper CAT(0) spaces of asymptotic rank at most 2, proves a homotopical δ -isoperimetric inequality for Lipschitz 2-spheres, with exponent $1 + \delta$, and related results for bounded-genus surfaces. The paper explicitly notes that this is weaker than the conjectured linear bound and does not handle homological fillings of general 2-cycles.
- **Title:** *Isoperimetric inequalities in Hadamard spaces of asymptotic rank two*
- **Authors:** Urs Lang, Stephan Stadler, and David Urech
- **Venue/year:** arXiv preprint, 2025
- **arXiv/DOI:** arXiv:2506.04882, <https://doi.org/10.48550/arXiv.2506.04882>
- **Progress:** Proves a homological δ -isoperimetric inequality $M(V) \leq CM(T)^{1+\delta}$ for general k -cycles, $k \geq 2$, in CAT(0) spaces of asymptotic rank at most 2 and finite asymptotic Nagata dimension; it covers examples such as Hadamard 3-manifolds and finite-dimensional CAT(0) cube complexes. The authors explicitly leave open whether a linear filling inequality holds even under their assumptions and whether the finite asymptotic Nagata dimension assumption can be removed.

5.15.3.5. What remains unresolved

The full problem asks for a uniform linear filling bound for **all** Lipschitz m -cycles in **every** proper cocompact CAT(0) space, for **every** $m \geq$ Euclidean rank. Current literature proves the conjectured linear bound only in special settings such as hyperbolic/rank-one cases, symmetric or homogeneous Hadamard manifolds, and restricted cycle classes such as controlled-density cycles. For general proper cocompact CAT(0) spaces of rank at least 2, the best general theorem I found is sub-Euclidean or $1 + \delta$ -type, not linear. Thus the exact statement is not fully resolved.

5.15.4. Remaining open problems recorded by the database

5.15.4.1. Open problem 1

Let X be a CAT(0) space of asymptotic rank at most 2 and finite asymptotic Nagata dimension. Prove that, for every integer $k \geq 2$, there is a constant $C = C(X, k)$ such that every compactly supported integral k -cycle T in X has a compactly supported integral filling V satisfying

$$\partial V = T, \quad \mathbf{M}(V) \leq C \mathbf{M}(T).$$

Reference supplied by the database. Urs Lang, Stephan Stadler, and David Urech, “Isoperimetric inequalities in Hadamard spaces of asymptotic rank two,” *Advances in Mathematics* **502** (2026), Article 111123. Open question immediately following Theorem 1.1 (whether the theorem’s $\mathbf{M}(T)^{1+\delta}$ estimate can be replaced by a linear isoperimetric filling inequality). DOI: <https://doi.org/10.1016/j.aim.2026.111123>. Preprint: [arXiv:2506.04882](https://arxiv.org/abs/2506.04882).

5.15.5. Primary-source pointer

The originating ICM record is *Isoperimetric inequalities and asymptotic geometry* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=1067>.

5.16. 193. Metric dimension reduction A snapshot of the Ribe program [2018-geom-144]

Record: 2018-geom-144

Subfield: Metric and topological geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Metric dimension reduction A snapshot of the Ribe program

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>

5.16.1. Problem statement

Prove or disprove that for every sequence $(X_n)_{n=1}^\infty$ such that each X_n is a linear subspace of $\ell_\infty^n = (\mathbb{R}^n, \|\cdot\|_\infty)$ and

$$\limsup_{n \rightarrow \infty} \frac{\dim(X_n)}{\log n} = \infty,$$

the Pythagorean direct sum

$$X = \left(\bigoplus_{n=1}^\infty X_n \right)_2 = \left\{ (x_n)_{n=1}^\infty : x_n \in X_n, \sum_{n=1}^\infty \|x_n\|_\infty^2 < \infty \right\}, \quad \|(x_n)\|_X = \left(\sum_{n=1}^\infty \|x_n\|_\infty^2 \right)^{1/2},$$

is finitarily universal: there exists a constant $\alpha \geq 1$ such that every finite metric space $(\mathcal{F}, d_{\mathcal{F}})$ admits a bi-Lipschitz embedding into X with distortion at most α , i.e., there are a map $f : \mathcal{F} \rightarrow X$ and a scaling factor $\tau > 0$ such that for all $u, v \in \mathcal{F}$,

$$\tau d_{\mathcal{F}}(u, v) \leq \|f(u) - f(v)\|_X \leq \alpha \tau d_{\mathcal{F}}(u, v).$$

5.16.2. Historical background, literature, and current progress

5.16.3. Verdict: partially resolved

5.16.3.1. Precise problem

The problem asks whether every sequence of linear subspaces $X_n \subseteq \ell_\infty^n$ with

$$\limsup_{n \rightarrow \infty} \frac{\dim X_n}{\log n} = \infty$$

has the property that the Hilbert/Pythagorean sum $(\oplus_n X_n)_2$ is finitarily universal, i.e. embeds every finite metric space with a distortion bound independent of the finite metric space.

Equivalently in the formulation used in the literature, this is the universality version of Pisier's dichotomy problem. A standard sufficient route is to prove that, along such a sequence, the subspaces X_n contain uniformly isomorphic copies of $\ell_\infty^{m_n}$ with $m_n \rightarrow \infty$. That would imply finitary universality, since every finite metric space embeds isometrically into some finite-dimensional ℓ_∞^m . The unresolved point is whether the hypothesis $\dim X_n / \log n \rightarrow \infty$ along a subsequence is already enough for this.

5.16.3.2. Literature checked

Original source

- **Title:** *Remarques sur un résultat non publié de B. Maurey*
- **Author:** Gilles Pisier
- **Venue/year:** Séminaire d'analyse fonctionnelle 1980-1981, Exposé V, École Polytechnique, Palaiseau, 1981
- **DOI/arXiv/URL:** no DOI found; Numdam URL: https://www.numdam.org/item/SAF_1980-1981____A5_0/
- **Status relevance:** Pisier formulates the subspace analogue of Maurey's quotient theorem and the associated threshold question for subspaces of ℓ_∞^m . The note identifies the logarithmic threshold as the natural boundary: if $\dim E$ is only of order $\log m$, no conclusion of this kind can be forced, while the conjectural statement asks for ℓ_∞^k -subspaces when $\dim E$ is asymptotically larger than $\log m$. It is the source of the problem, not a resolution.

Survey formulation of the exact problem

- **Title:** *Metric dimension reduction: A snapshot of the Ribe program*
- **Author:** Assaf Naor
- **Venue/year:** Proceedings of the International Congress of Mathematicians, Rio de Janeiro 2018, Vol. I, pp. 759-837; World Scientific, 2018/2019
- **DOI/arXiv/URL:** DOI 10.1142/9789813272880_0029; arXiv:1809.02376
- **Status relevance:** Naor states exactly the stated metric-universality question as Question 6, "Pisier's dichotomy problem": for arbitrary $X_n \subseteq \ell_\infty^n$ with $\limsup \dim(X_n)/\log n = \infty$, does $(X_1 \oplus X_2 \oplus \dots)_2$ become universal? The survey explicitly lists this among concrete universality questions whose answer is unknown, and records only stronger-growth cases as known.

Partial progress: proportional/large-dimensional cases

- **Title:** *Large subspaces of ℓ_∞^n and estimates of the Gordon-Lewis constant*
- **Authors:** Tadeusz Figiel and William B. Johnson
- **Venue/year:** Israel Journal of Mathematics 37 (1980), no. 1-2, pp. 92-112
- **DOI/arXiv/URL:** DOI 10.1007/BF02762871
- **Status relevance:** This gives sharp information for very large/proportional-dimensional subspaces of ℓ_∞^N . It is far from the logarithmic-threshold regime in the problem statement.
- **Title:** *Subspaces of L_∞^N , arithmetical diameter and Sidon sets*
- **Author:** Jean Bourgain
- **Venue/year:** In *Probability in Banach Spaces V*, Lecture Notes in Mathematics 1153, Springer, 1985, pp. 96-127
- **DOI/arXiv/URL:** DOI 10.1007/BFb0074947; IHES archive PDF: https://repositaires.ihses.fr/FONDS_IHES/I_Prepublications/BOURGAIN/1984-1989/M_85_11/M_85_11_web.pdf
- **Status relevance:** Bourgain proves a partial form of the desired conclusion for substantially larger subspaces. As summarized by Naor and by Schechtman-Tomczak-Jaegermann, Bourgain's method yields the desired uniform ℓ_∞^k -subspaces, and hence

the stated finitary universality conclusion, under a stronger hypothesis such as $\dim X_N$ exceeding about $(\log N)^4$. This does not cover arbitrary sequences with $\dim X_N / \log N \rightarrow \infty$.

Partial progress: current best general quantitative bound found

- **Title:** *Polylog Dimensional Subspaces of ℓ_∞^N*
- **Authors:** Gideon Schechtman and Nicole Tomczak-Jaegermann
- **Venue/year:** In *Geometric Aspects of Functional Analysis: Israel Seminar (GAFA) 2017-2019, Volume II*, Lecture Notes in Mathematics 2266, Springer, 2020, pp. 325-334
- **DOI/arXiv/URL:** DOI 10.1007/978-3-030-46762-3_15; arXiv:1812.03678
- **Status relevance:** The paper states Pisier's dichotomy problem in the equivalent Banach-space form: whether for every sequence $n_N \leq N$ with $n_N / \log N \rightarrow \infty$, every n_N -dimensional subspace of ℓ_∞^N contains uniformly isomorphic $\ell_\infty^{m_N}$ with $m_N \rightarrow \infty$. Its Theorem 1 proves the stronger-hypothesis result: if $n > (\log N \log \log N)^2$, then every n -dimensional subspace of ℓ_∞^N contains a subspace of dimension

$$m = c(\eta) \frac{\sqrt{n}}{\log N \log \log N}$$

whose distance from ℓ_∞^m is at most $1 + \eta$. Thus the conclusion follows when

$$\limsup_{N \rightarrow \infty} \frac{\dim X_N}{(\log N)^2 (\log \log N)^2} = \infty,$$

but this is much stronger than the stated assumption $\limsup \dim X_N / \log N = \infty$.

Generic-case progress

- **Title:** *Random Banach spaces: The limitations of the method*
- **Authors:** Piotr Mankiewicz and Stanislaw J. Szarek
- **Venue/year:** *Mathematika* 41 (1994), no. 2, pp. 239-250; corrigendum *Mathematika* 42 (1995), pp. 220-221
- **DOI/arXiv/URL:** DOI 10.1112/S002557930000735X; arXiv:math/9305203
- **Status relevance:** The authors prove a version of "Pisier's dichotomy conjecture" for generic subspaces of ℓ_∞^N in the Haar-measure sense. This is not a resolution of the problem because the statement quantifies over every possible sequence of subspaces.

5.16.3.3. Why the full statement is not resolved

The strongest general result found is Schechtman-Tomczak-Jaegermann's 2020 theorem, which requires roughly $(\log N)^2 (\log \log N)^2$ -scale dimension growth. Bourgain's earlier result requires an even stronger polylogarithmic condition. Generic/random-subspace results do not address arbitrary subspaces. I found no published or citable theorem proving the exact hypothesis, and no published or citable counterexample disproving it.

What remains open is the full logarithmic-threshold case: arbitrary $X_N \subseteq \ell_\infty^N$ with $\dim X_N / \log N$ unbounded but possibly much smaller than $(\log N)^2 (\log \log N)^2$. The current literature therefore gives partial progress, not a full resolution.

5.16.4. Remaining open problems recorded by the database

5.16.4.1. Open problem 1

Construct sequences of finite-dimensional normed spaces E_n, F_n with $\dim E_n = \dim F_n = n$ and subspaces $Z_n \subseteq \mathcal{L}(E_n, F_n)$ such that $\dim Z_n/n \rightarrow \infty$, while there is a universal integer k_0 for which no Z_n contains a subspace that is 2-isomorphic to ℓ_∞^k for any $k > k_0$.

Reference supplied by the database. Gideon Schechtman and Nicole Tomczak-Jaegermann, “Polylog Dimensional Subspaces of ℓ_∞^N ,” in Bo’az Klartag and Emanuel Milman (eds.), *Geometric Aspects of Functional Analysis: Israel Seminar (GAFA) 2017–2019, Volume II*, Lecture Notes in Mathematics **2266**, Springer, 2020, pp. 325–334, final Remark 1 (the proposed operator-space route to a negative solution of Pisier’s dichotomy problem), https://doi.org/10.1007/978-3-030-46762-3_15; arXiv:1812.03678.

5.16.5. Primary-source pointer

The originating ICM record is *Metric dimension reduction A snapshot of the Ribe program* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>.

5.17. 194. Metric dimension reduction A snapshot of the Ribe program [2018-geom-145]

Record: 2018-geom-145

Subfield: Metric and topological geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Metric dimension reduction A snapshot of the Ribe program

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>

5.17.1. Problem statement

For each positive integer k , let $P_1(\mathbb{R}^k)$ denote the set of Borel probability measures μ on Euclidean $\mathbb{R}^k$ with finite first moment, meaning that $\int_{\mathbb{R}^k} \|x - y\|_2 d\mu(y) < \infty$ for every $x \in \mathbb{R}^k$, equipped with the Wasserstein-1 metric

$$W_1(\mu, \nu) = \inf_{\pi} \int_{\mathbb{R}^k \times \mathbb{R}^k} \|x - y\|_2 d\pi(x, y),$$

where the infimum is over all Borel probability measures π on $\mathbb{R}^k \times \mathbb{R}^k$ whose marginals are μ and ν . Is $(P_1(\mathbb{R}^2), W_1)$ universal in the finitary bi-Lipschitz sense: does there exist a constant $\alpha \geq 1$ such that every finite metric space $(\mathcal{F}, d_{\mathcal{F}})$ admits a map $f : \mathcal{F} \rightarrow P_1(\mathbb{R}^2)$ and a scaling factor $\tau > 0$ satisfying

$$\tau d_{\mathcal{F}}(x, y) \leq W_1(f(x), f(y)) \leq \alpha \tau d_{\mathcal{F}}(x, y)$$

for all $x, y \in \mathcal{F}$? More generally, does there exist an integer $k \geq 2$ for which $(P_1(\mathbb{R}^k), W_1)$ has this universality property?

5.17.2. Historical background, literature, and current progress

this problem is fully open

5.17.3. Remaining open problems recorded by the database

5.17.3.1. Open problem 1

Is the Wasserstein space $(P_1(\ell_2), W_1)$ universal in the finitary bi-Lipschitz sense?

Reference supplied by the database. Assaf Naor, “Metric Dimension Reduction: A Snapshot of the Ribe Program,” in *Proceedings of the International Congress of Mathematicians—Rio de Janeiro 2018*, Vol. I: Plenary Lectures, eds. Boyan Sirakov, Paulo Ney de Souza, and Marcelo Viana, World Scientific, 2018, pp. 759–837, §1.2.2, immediately following “Bourgain’s

universality problem” (explicitly stating that it is unknown whether $P_1(\ell_2)$ is universal). arXiv:1809.02376, <https://arxiv.org/abs/1809.02376>.

5.17.4. Primary-source pointer

The originating ICM record is *Metric dimension reduction A snapshot of the Ribe program* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>.

5.18. 195. Metric dimension reduction A snapshot of the Ribe program [2018-geom-146]

Record: 2018-geom-146

Subfield: Metric and topological geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Metric dimension reduction A snapshot of the Ribe program

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>

5.18.1. Problem statement

Let $p \in (1, 2)$. Let $\ell_p \widehat{\otimes} \ell_2$ denote the projective tensor product of the Banach spaces ℓ_p and ℓ_2 , i.e. the completion of their algebraic tensor product under the norm whose unit ball is the convex hull of $\{x \otimes y : \|x\|_{\ell_p} \leq 1, \|y\|_{\ell_2} \leq 1\}$. Determine whether $\ell_p \widehat{\otimes} \ell_2$, viewed as a metric space with its norm metric, is universal in the following finitary sense: does there exist a constant $\alpha \geq 1$ such that every finite metric space $(\mathcal{F}, d_{\mathcal{F}})$ admits a map $f : \mathcal{F} \rightarrow \ell_p \widehat{\otimes} \ell_2$ and a scaling factor $\tau > 0$ satisfying

$$\tau d_{\mathcal{F}}(x, y) \leq \|f(x) - f(y)\|_{\ell_p \widehat{\otimes} \ell_2} \leq \alpha \tau d_{\mathcal{F}}(x, y)$$

for all $x, y \in \mathcal{F}$.

5.18.2. Historical background, literature, and current progress

5.18.3. Verdict: partially resolved

5.18.3.1. Precise problem

For each fixed $p \in (1, 2)$, decide whether the Banach space

$$X_p = \ell_p \widehat{\otimes} \pi \ell_2$$

is finitarily universal as a metric space, i.e. whether there is a constant α independent of the finite metric space such that every finite metric embeds into X_p with distortion at most α , after rescaling.

For Banach targets this is equivalent to asking whether X_p fails to have finite cotype. Indeed, failure of finite cotype is equivalent, by Maurey-Pisier, to uniform finite representability of ℓ_∞^n , and every n -point metric embeds isometrically into ℓ_∞^n . Conversely, finite cotype gives the standard metric-cotype obstruction to finite-metric universality.

5.18.3.2. Literature and progress

1. Gilles Pisier, "Random series of trace class operators", in *Proceedings Cuarto CLAPEM Mexico 1990. Contribuciones en probabilidad y estadística matemática*,

- pp. 29-42, 1992. arXiv:1103.2090, URL: <https://arxiv.org/abs/1103.2090>. Pisier formulates the relevant open finite-cotype problem for projective tensor products. In the final remarks he states that $L_p \widehat{\otimes}_\pi L_q$ has cotype $\max(p, q)$ when $2 \leq p, q < \infty$, but that it remains open whether $L_p \widehat{\otimes}_\pi L_q$ has finite cotype for $1 < p < 2$ and $1 < q \leq 2$, explicitly including $L_p \widehat{\otimes}_\pi L_2$. Since ℓ_p is an L_p -space, this is the original source-level open problem covering the stated case.
2. Gilles Pisier, “Factorization of operator valued analytic functions”, *Advances in Mathematics* 93(1):61-125, 1992. DOI: 10.1016/0001-8708(92)90025-G. This gives the positive finite-cotype/nonuniversality result for $L_p \widehat{\otimes}_\pi L_q$ in the range $p, q \in [2, \infty)$. It verifies nonuniversality at and beyond the Hilbert endpoint, but it does not cover the stated range $1 < p < 2, q = 2$.
 3. Nicole Tomczak-Jaegermann, “The moduli of smoothness and convexity and the Rademacher averages of the trace classes S_p ($1 \leq p < \infty$)”, *Studia Mathematica* 50(2):163-182, 1974. URL: <https://eudml.org/doc/217886>. This includes the finite-cotype behavior of trace classes; in particular $\ell_2 \widehat{\otimes}_\pi \ell_2 \cong S_1$ has cotype 2, so the endpoint $p = 2$ is not universal. This is an endpoint result, not a resolution for $1 < p < 2$.
 4. Jop Briët, Assaf Naor, and Oded Regev, “Locally decodable codes and the failure of cotype for projective tensor products”, *Electronic Research Announcements in Mathematical Sciences* 19:120-130, 2012. DOI: 10.3934/era.2012.19.120; arXiv:1208.0539, URL: <https://arxiv.org/abs/1208.0539>. Theorem 1.1 proves that if $p_1, p_2, p_3 \in (1, \infty)$ and $1/p_1 + 1/p_2 + 1/p_3 \leq 1$, then $\ell_{p_1} \widehat{\otimes}_\pi \ell_{p_2} \widehat{\otimes}_\pi \ell_{p_3}$ does not have finite cotype. It also implies that for every $p \in (1, \infty)$ there exists a Banach space X of finite cotype such that $\ell_p \widehat{\otimes}_\pi X$ fails finite cotype. This is substantial partial progress, but X is not ℓ_2 , and the theorem is three-fold rather than the two-fold tensor product in the problem statement.
 5. Assaf Naor, “Metric dimension reduction: A snapshot of the Ribe program”, in *Proceedings of the International Congress of Mathematicians - Rio de Janeiro 2018, Vol. I. Plenary Lectures*, pp. 759-837, World Scientific, 2018/2019. DOI: 10.1142/9789813272880_0029; arXiv:1809.02376, URL: <https://arxiv.org/abs/1809.02376>. Question 8 in this survey is exactly the question: for $p \in (1, 2)$, is $\ell_p \widehat{\otimes}_\pi \ell_2$ universal? The survey records the known endpoint results and identifies Briët-Naor-Regev’s three-fold theorem as the best available result in this direction, not as a solution of the two-fold problem.
 6. Qingying Bu and Patrick N. Dowling, “On type and cotype of tensor products of Banach spaces”, *Journal of Mathematical Analysis and Applications* 553(2), Article 129913, 2026. DOI: 10.1016/j.jmaa.2025.129913, URL: <https://www.sciencedirect.com/science/article/pii/S0022247X25006948>. This recent adjacent paper studies type/cotype phenomena for tensor products, but the available article metadata and abstract do not state a theorem resolving whether $\ell_p \widehat{\otimes}_\pi \ell_2$ has finite cotype for $1 < p < 2$. Searches for the exact phrase, the Pisier/Naor question, and post-2018 finite-cotype tensor-product results found no citable full resolution.

5.18.3.3. What remains unresolved

No published or otherwise citable theorem found in the literature proves either:

- $\ell_p \widehat{\otimes}_\pi \ell_2$ fails finite cotype for every $1 < p < 2$, which would imply metric universality; or
- $\ell_p \widehat{\otimes}_\pi \ell_2$ has finite cotype for every $1 < p < 2$, which would imply nonuniversality.

Thus the exact problem remains open, while related endpoint cases and higher-fold/projective-

tensor-product variants constitute partial progress.

5.18.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

5.18.5. Primary-source pointer

The originating ICM record is *Metric dimension reduction A snapshot of the Ribe program* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>.

5.19. 196. Metric dimension reduction A snapshot of the Ribe program [2018-geom-147]

Record: 2018-geom-147

Subfield: Metric and topological geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Metric dimension reduction A snapshot of the Ribe program

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>

5.19.1. Problem statement

For metric spaces (F, d_F) and (X, d_X) , let $C_X(F)$ denote the infimum of all $\alpha \geq 1$ for which there exist a map $f : F \rightarrow X$ and a scaling factor $\tau > 0$ such that

$$\tau d_F(u, v) \leq d_X(f(u), f(v)) \leq \alpha \tau d_F(u, v)$$

for every $u, v \in F$. Say that a metric space X is finitarily universal if there is a constant $\alpha < \infty$ such that every finite metric space F satisfies $C_X(F) \leq \alpha$.

Is there a universal constant $\theta_0 > 0$ such that every metric space X that is not finitarily universal admits an exponent $\theta_X > \theta_0$ and a constant $c_X > 0$ with the following property: for arbitrarily large integers n , there is an n -point metric space G_n such that

$$C_X(G_n) \geq c_X (\log n)^{\theta_X}?$$

5.19.2. Historical background, literature, and current progress

5.19.3. Verdict: partially resolved

5.19.3.1. Precise problem

The problem asks for the uniform-exponent version of the metric cotype dichotomy. Mendel and Naor's metric cotype dichotomy says that if a metric space X is not finitarily universal, then X admits arbitrarily large finite metric spaces whose least distortion into X grows at least like a positive power of $\log n$. The open part in this problem is whether that positive exponent can always be chosen larger than one universal constant $\theta_0 > 0$, independent of X .

5.19.3.2. Current literature

1. Metric cotype

- Title: *Metric Cotype*
- Authors: Manor Mendel and Assaf Naor
- Venue: *Annals of Mathematics* 168(1), 247-298
- Year: 2008

- DOI/arXiv/URL: DOI <https://doi.org/10.4007/annals.2008.168.247>; arXiv math/0506201; <https://arxiv.org/abs/math/0506201>
- Status relevance: The paper proves the metric cotype dichotomy in an X -dependent form. In the notation of the problem, it gives: if X is not universal/finitarily universal, then there is some exponent $\alpha(X) > 0$ and arbitrarily large finite metric spaces M_n with $C_X(M_n) \gtrsim_X (\log |M_n|)^{\alpha(X)}$. This is substantial partial progress, but it does not give a lower bound on $\alpha(X)$ that is uniform over all non-finitarily universal metric spaces.

2. Metric dimension reduction: A snapshot of the Ribe program

- Title: *Metric dimension reduction: A snapshot of the Ribe program*
- Author: Assaf Naor
- Venue: *Proceedings of the International Congress of Mathematicians - Rio de Janeiro 2018*, Vol. I, World Scientific, 759-837
- Year: 2018
- DOI/arXiv/URL: arXiv 1809.02376; <https://arxiv.org/abs/1809.02376>
- Status relevance: The survey restates the Mendel-Naor theorem as Theorem 9 and then asks the exact uniform-exponent problem as Question 10. It explicitly identifies the unknown point as whether the logarithmic power $\theta(X)$ in the dichotomy can be bounded below by a universal positive constant. It also notes that known computed examples at that time were consistent even with exponent at least 1, while a proposed Wasserstein scenario could give a negative answer.

3. Snowflake universality of Wasserstein spaces

- Title: *Snowflake universality of Wasserstein spaces*
- Authors: Alexandr Andoni, Assaf Naor, and Ofer Neiman
- Venue: *Annales Scientifiques de l'Ecole Normale Supérieure* 51(3), 657-700
- Year: 2018
- DOI/arXiv/URL: DOI <https://doi.org/10.24033/asens.2363>; arXiv 1509.08677; <https://numdam.org/articles/10.24033/asens.2363/>
- Status relevance: Section 1.5.1 formulates the same metric cotype dichotomy problem as Question 13. The paper gives Wasserstein snowflake universality results and asks Question 14 about whether every n -point metric space embeds into $P_p(\mathbb{R}^3)$ with distortion $\lesssim_p (\log n)^{1-1/p}$. The authors explain that a positive answer to that Wasserstein question would imply exponents tending to 0 as $p \rightarrow 1^+$, hence would disprove the uniform-exponent problem. However, Question 14 is left open and therefore does not resolve the problem.

5.19.3.3. What remains unresolved

I found no published or citable preprint result proving either of the two conclusions needed for a full resolution:

- a positive theorem giving a universal $\theta_0 > 0$ for every non-finitarily universal metric space X ; or
- a negative construction of non-finitarily universal spaces X_j for which every admissible logarithmic obstruction exponent tends to 0.

Thus the exact problem is not fully resolved. The correct status is **partially resolved**: the X -dependent metric cotype dichotomy is known, and Wasserstein-space work identifies a plausible route to a negative answer, but the universal-exponent question remains open.

5.19.4. Remaining open problems recorded by the database

5.19.4.1. Open problem 1

For $p \in (1, 2]$, is there a constant $C_p < \infty$ such that every n -point metric space X embeds into $(\mathcal{D}_p(\mathbb{R}^3), W_p)$ with distortion at most $C_p(\log n)^{1-1/p}$?

Reference supplied by the database. Alexandr Andoni, Assaf Naor, and Ofer Neiman, “Snowflake Universality of Wasserstein Spaces,” *Annales Scientifiques de l’École Normale Supérieure*, 4e série, 51 (2018), no. 3, 657–700, Question 14 (§1.5.1). DOI: 10.24033/asens.2363. arXiv:1509.08677. <https://doi.org/10.24033/asens.2363>.

5.19.4.2. Open problem 2

Let G be an n -vertex 3-regular graph whose normalized adjacency matrix has second-largest eigenvalue $\lambda_2(G) \leq 99/100$, and let $\Sigma_{\lceil \log n \rceil}(G)$ be obtained by replacing every edge of G by a path of $\lceil \log n \rceil$ edges. What is the asymptotic growth rate of the least distortion $c_{\mathcal{D}_2(\mathbb{R}^3)}(\Sigma_{\lceil \log n \rceil}(G))$? In particular, is it $O(\sqrt{\log n})$, or even $o(\log n)$?

Reference supplied by the database. Alexandr Andoni, Assaf Naor, and Ofer Neiman, “Snowflake Universality of Wasserstein Spaces,” *Annales Scientifiques de l’École Normale Supérieure*, 4e série, 51 (2018), no. 3, 657–700, Question 36 (§6.3, p. 694). DOI: 10.24033/asens.2363. arXiv:1509.08677. <https://doi.org/10.24033/asens.2363>.

5.19.5. Primary-source pointer

The originating ICM record is *Metric dimension reduction A snapshot of the Ribe program* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>.

5.20. 197. Metric dimension reduction A snapshot of the Ribe program [2018-geom-148]

Record: 2018-geom-148

Subfield: Metric and topological geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Metric dimension reduction A snapshot of the Ribe program

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>

5.20.1. Problem statement

Suppose that $p \in [1, \infty) \setminus \{2\}$. Do there exist constants $\alpha = \alpha(p)$ and $\beta = \beta(p)$ in $[1, \infty)$, depending only on p , such that for every $n \in \mathbb{N}$ and every subset $C \subseteq \ell_p$ with $|C| = n$, there is a finite-dimensional normed space $(Y, \|\cdot\|_Y)$ with $\dim Y \leq \beta \log n$ and a map $f : C \rightarrow Y$ for which there exists a scaling factor $\tau > 0$ satisfying

$$\tau \|x - y\|_{\ell_p} \leq \|f(x) - f(y)\|_Y \leq \alpha \tau \|x - y\|_{\ell_p}$$

for all $x, y \in C$?

5.20.2. Historical background, literature, and current progress

5.20.3. Verdict: partially resolved

5.20.3.1. Precise problem

For a fixed $p \in [1, \infty) \setminus \{2\}$, decide whether there are constants $\alpha(p), \beta(p) < \infty$ such that every n -point subset $C \subseteq \ell_p$ embeds with bi-Lipschitz distortion at most $\alpha(p)$ into some normed space Y of dimension at most $\beta(p) \log n$. The target space is allowed to depend on C and is not required to be an ℓ_p space or a subspace of ℓ_p .

5.20.3.2. Status

I found no published or citable result that proves either the asserted $O(\log n)$ constant-distortion reduction for all finite subsets of ℓ_p , $p \neq 2$, or a counterexample to it when the target is an arbitrary finite-dimensional normed space. The problem is therefore not fully resolved.

It is not best described as fully open, because there is substantial partial progress on nearby formulations: growing-distortion Hilbert embeddings for subsets of L_p , strong lower bounds when the target is restricted to ℓ_1^k or ℓ_p^k , and recent arbitrary-target lower bounds for Hamming metrics. None of these results has the hypotheses and conclusion needed to settle the exact question above.

5.20.3.3. Key literature and applicability checks

1. Metric dimension reduction: A snapshot of the Ribe program

Assaf Naor. Proceedings of the International Congress of Mathematicians–Rio de Janeiro 2018, Vol. I, World Scientific, 2018, pp. 759–837. DOI: https://doi.org/10.1142/9789813272880_0029; arXiv: 1809.02376; URL: <https://arxiv.org/abs/1809.02376>.

This is the source of the question. It explicitly asks whether every n -point subset of ℓ_p , $p \neq 2$, embeds with constant distortion into some $O(\log n)$ -dimensional normed space, and notes that even an $o(n)$ dimensional bound was open at the time. This source does not resolve the problem.

2. Euclidean distortion and the sparsest cut

Sanjeev Arora, James R. Lee, Assaf Naor. Journal of the American Mathematical Society 21(1):1–21, 2008. DOI: <https://doi.org/10.1090/S0894-0347-07-00573-5>; arXiv: math/0508154; URL: <https://arxiv.org/abs/math/0508154>.

This proves that every n -point metric of negative type, in particular every n -point subset of L_1 , embeds into Hilbert space with distortion $O(\sqrt{\log n \log \log n})$. As summarized by Naor’s ICM article, this gives nonconstant Hilbert distortion for $p \in [1, 2]$. After Johnson-Lindenstrauss, the target can be made $O(\log n)$ -dimensional, but the distortion still grows with n . Hence it is only partial progress, not a solution to the constant distortion question.

3. Euclidean embedding, randomized clustering, and Lipschitz extension for finite and doubling subsets of L_p when $p > 2$

Assaf Naor, Kevin Ren. Ars Inveniendi Analytica, 2026, Paper No. 2, 60 pp. DOI: <https://doi.org/10.15781/d72xfr59>; arXiv: 2502.10543; URL: <https://arxiv.org/abs/2502.10543>.

This proves that for fixed $p > 2$, every n -point subset of L_p embeds into Hilbert space with distortion $p^3(\log n)^{1/2+o(1)}$, improving Bourgain’s general $O(\log n)$ guarantee. Since $\ell_p \subset L_p$, this applies to the stated spaces for $p > 2$. Applying Johnson-Lindenstrauss gives an $O(\log n)$ -dimensional Hilbert target, but the distortion is still unbounded in n , so it does not imply the requested constants $\alpha(p), \beta(p)$.

4. The Power of Recursive Embeddings for ℓ_p Metrics

Robert Krauthgamer, Nir Petruschka, Shay Sapir. arXiv preprint, 2025. arXiv: 2503.18508; URL: <https://arxiv.org/abs/2503.18508>.

This gives additional state-of-the-art recursive embedding and decomposition results for ℓ_p metrics with $p > 2$, including improved Hilbert distortion bounds in some regimes. These are again growing-distortion results and therefore do not settle the constant-distortion $O(\log n)$ -dimension problem.

5. On the impossibility of dimension reduction in ℓ_1

Bo Brinkman, Moses Charikar. Journal of the ACM 52(5):766–788, 2005. DOI: <https://doi.org/10.1145/1089023.1089026>; URL: <https://dl.acm.org/doi/10.1145/1089023.1089026>.

This gives n -point subsets of ℓ_1 requiring polynomial dimension for constant-distortion embeddings into ℓ_1^k . It is strong evidence against a Johnson-Lindenstrauss analogue with the same target norm, but it does not rule out embeddings into an arbitrary k -dimensional normed space as allowed in the question.

6. Dimensionality of Hamming Metrics and Rademacher Type

Alexandros Eskenazis. 40th International Symposium on Computational Geometry (SoCG 2024), LIPIcs 293, Art. 55, 2024. DOI: <https://doi.org/10.4230/LIPIcs.SocG.2024.55>; URL: <https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.SocG.2024.55>.

This proves arbitrary-target dimension lower bounds for embeddings of Hamming cubes in terms of Rademacher type constants. Since Hamming cubes are subsets of ℓ_1 , this is relevant to the $p = 1$ case, but the lower bounds are compatible with dimension proportional to $\log|C|$ for arbitrary targets and hence do not disprove the desired statement.

5.20.3.4. What remains unresolved

The remaining gap is exactly the question: for any fixed $p \in [1, \infty) \setminus \{2\}$, no cited theorem gives a constant $\alpha(p)$ and logarithmic dimension constant $\beta(p)$ for embeddings of all finite subsets of ℓ_p into arbitrary finite-dimensional normed spaces, and no cited theorem proves that such constants cannot exist.

5.20.4. Remaining open problems recorded by the database

5.20.4.1. Open problem 1

Fix $p \in [1, \infty) \setminus \{2\}$. Do there exist a constant $\alpha_p < \infty$ and a function $k_p(n) = o(n)$ such that every n -point subset of ℓ_p embeds with distortion at most α_p into some normed space of dimension at most $k_p(n)$?

Reference supplied by the database. Assaf Naor, “Metric Dimension Reduction: A Snapshot of the Ribe Program,” in *Proceedings of the International Congress of Mathematicians—Rio de Janeiro 2018*, Vol. I: Plenary Lectures, eds. Boyan Sirakov, Paulo Ney de Souza, and Marcelo Viana, World Scientific, 2018, pp. 759–837, immediately after Question 13 (§1.3, p. 767), explicitly stating that even the bound $k = o(n)$ is open for every fixed $p \neq 2$. arXiv:1809.02376. <https://arxiv.org/abs/1809.02376>.

5.20.5. Primary-source pointer

The originating ICM record is *Metric dimension reduction A snapshot of the Ribe program* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>.

5.21. 198. Metric dimension reduction A snapshot of the Ribe program [2018-geom-150]

Record: 2018-geom-150

Subfield: Metric and topological geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Metric dimension reduction A snapshot of the Ribe program

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>

5.21.1. Problem statement

Does every subset S of a Hilbert space H , equipped with the metric d_S induced by H , that is doubling admit a bi-Lipschitz embedding into $\mathbb{R}^k$ for some $k \in \mathbb{N}$? Here “doubling” means that there exists $K \in [2, \infty)$ such that every closed ball $B_S(x, r)$ in S can be covered by at most K closed balls in S of radius $r/2$, and “bi-Lipschitz embedding into $\mathbb{R}^k$ ” means that there are $k \in \mathbb{N}$, a map $f : S \rightarrow \mathbb{R}^k$, and constants $0 < c \leq C < \infty$ such that

$$c d_S(x, y) \leq \|f(x) - f(y)\| \leq C d_S(x, y)$$

for all $x, y \in S$.

5.21.2. Historical background, literature, and current progress

5.21.3. Verdict: partially resolved

5.21.3.1. Precise problem

The problem asks the Lang-Plaut problem: if S is an arbitrary doubling subset of a Hilbert space H , with the metric inherited from H , must S admit a bi-Lipschitz embedding into some finite-dimensional Euclidean space $\mathbb{R}^k$?

The exact problem remains open in full generality. I found no published or citable preprint result proving either that every doubling Hilbert subset embeds bi-Lipschitzly into some $\mathbb{R}^k$, or constructing a doubling subset of Hilbert space that fails to do so.

5.21.3.2. Original source

- Title: “Bilipschitz Embeddings of Metric Spaces into Space Forms”
- Authors: Urs Lang and Conrad Plaut
- Venue: *Geometriae Dedicata* 87, 285-307
- Year: 2001
- DOI/URL: <https://doi.org/10.1023/A:1012093209450>
- Relevance: Question 2.4 asks whether every doubling subset of a Hilbert space admits a bilipschitz embedding into some Euclidean space. The paper proves positive embedding

results under additional geodesic/convex-bicombing hypotheses, but those hypotheses do not cover arbitrary subsets of Hilbert space.

5.21.3.3. Main partial progress and status evidence

- Title: “Assouad’s theorem with dimension independent of the snowflaking”
- Authors: Assaf Naor and Ofer Neiman
- Venue: Revista Matematica Iberoamericana 28(4), 1123-1142
- Year: 2012
- DOI/arXiv/URL: <https://doi.org/10.4171/RMI/706> ; arXiv:1012.2307
- Result/progress: For every doubling metric space (X, d) , each snowflake $(X, d^{1-\epsilon})$, $0 < \epsilon < 1/2$, embeds bi-Lipschitzly into $\mathbb{R}^{N(K)}$, where the dimension is independent of ϵ . Section 1.1 states the Lang-Plaut problem and explains that it remains open; the paper also proposes possible counterexample routes via snowflakes of the Heisenberg group. This is partial because it embeds snowflaked metrics, not the original Hilbert-subset metric.
- Title: “A doubling subset of L_p for $p > 2$ that is inherently infinite dimensional”
- Authors: Vincent Lafforgue and Assaf Naor
- Venue: Geometriae Dedicata 172, 387-398
- Year: 2014
- DOI/arXiv/URL: <https://doi.org/10.1007/s10711-013-9924-4> ; arXiv:1308.4554
- Result/progress: Theorem 1.1 constructs, for every $p > 2$, a doubling subset of L_p that does not admit a bi-Lipschitz embedding into any $\mathbb{R}^k$. The introduction explicitly says the Hilbert case and the Lang-Plaut question remain open. This is strong evidence against naive positive approaches, but it does not resolve the Hilbert case $p = 2$.
- Title: “On the Impossibility of Dimension Reduction for Doubling Subsets of ℓ_p ”
- Authors: Yair Bartal, Lee-Ad Gottlieb, and Ofer Neiman
- Venue: SIAM Journal on Discrete Mathematics 29(3), 1207-1222
- Year: 2015
- DOI/arXiv/URL: <https://doi.org/10.1137/140977655> ; arXiv:1308.4996
- Result/progress: Gives finite doubling subsets of ℓ_p , $p > 2$, for which embeddings into low-dimensional ℓ_p^d require large distortion, yielding another negative answer to the $p > 2$ analogue. It does not apply to ℓ_2 /Hilbert.
- Title: “No dimension reduction for doubling subsets of ℓ_q when $q > 2$ revisited”
- Authors: Florent P. Baudier, Krzysztof Swiecicki, and Andrew Swift
- Venue: Journal of Mathematical Analysis and Applications 504(2), Paper 125407
- Year: 2021
- DOI/arXiv/URL: <https://doi.org/10.1016/j.jmaa.2021.125407> ; arXiv:2103.05080

- Result/progress: Reproves and extends the $q > 2$ impossibility results, including obstructions for broader target classes. The paper formulates the Hilbert/ell₂ version as the original open problem and treats the $q > 2$ results as analogues, not as a solution.
- Title: “Bi-Lipschitz embeddings revisited”
- Authors: H. Movahedi-Lankarani and R. Wells
- Venue: arXiv preprint
- Year: 2025
- DOI/arXiv/URL: arXiv:2501.07648, <https://arxiv.org/abs/2501.07648>
- Result/progress: Gives sufficient conditions for compact metric-measure spaces, including compact subsets of Hilbert space satisfying additional hypotheses involving the canonical distance map and a measure. The paper explicitly states that the problem of whether every doubling subset of Hilbert space embeds into Euclidean space is still open. These hypotheses are not implied by doubling alone.
- Title: “Coarse and pointwise tangent fields”
- Authors: Guy C. David, Sylvester Eriksson-Bique, and Raanan Schul
- Venue: arXiv preprint
- Year: 2025; version checked: arXiv v2 dated 17 Apr 2026
- DOI/arXiv/URL: arXiv:2508.04653, <https://arxiv.org/abs/2508.04653>
- Result/progress: Shows that doubling subsets of Hilbert space admit pointwise and coarse tangent fields with dimension bounded by Nagata/Assouad dimension. This gives direct structural information for the exact class in the Lang-Plaut problem, but it is not a bi-Lipschitz embedding theorem into finite-dimensional Euclidean space.

5.21.3.4. What remains unresolved

The full arbitrary-Hilbert-subset case remains unresolved. The available positive results either require snowflaking, geodesic/convex structure, compact metric-measure hypotheses, or give structural tangent-field information rather than an embedding. The available negative results solve analogues in L_p or ℓ_p for $p > 2$, not in Hilbert space $p = 2$. Therefore no cited theorem’s hypotheses and conclusion cover all doubling subsets of Hilbert space with the original induced metric.

5.21.4. Remaining open problems recorded by the database

5.21.4.1. Open problem 1

For every $K > 1$, do there exist constants $a(K), b(K) > 0$ such that, for every metric space (X, d) with doubling constant at most K and every $\varepsilon \in (0, 1/2)$, there is a map $f : X \rightarrow \ell_2$ satisfying

$$a(K) \sqrt{\varepsilon} d(x, y)^{1-\varepsilon} \leq \|f(x) - f(y)\|_2 \leq d(x, y)^{1-\varepsilon} \quad (x, y \in X),$$

whose image $f(X)$ has doubling constant at most $b(K)$?

Reference supplied by the database. Assaf Naor and Ofer Neiman, “Assouad’s Theorem with Dimension Independent of the Snowflaking,” *Revista Matemática Iberoamericana* 28 (2012), no. 4, 1123–1142, Question 2 (§1.1). DOI: 10.4171/RMI/706. arXiv:1012.2307. <https://doi.org/10.4171/RMI/706>.

5.21.5. Primary-source pointer

The originating ICM record is *Metric dimension reduction A snapshot of the Ribe program* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>.

5.22. 199. Metric dimension reduction A snapshot of the Ribe program [2018-geom-151]

Record: 2018-geom-151

Subfield: Metric and topological geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Metric dimension reduction A snapshot of the Ribe program

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>

5.22.1. Problem statement

Does ℓ_1 admit average metric dimension reduction, in the following precise sense: is there a constant $\alpha \in [1, \infty)$ such that for every $n \in \mathbb{N}$ there is an integer k_n with

$$\lim_{n \rightarrow \infty} \frac{\log k_n}{\log n} = 0$$

and such that, for every n -point subset $C \subseteq \ell_1$, there are a linear subspace $F \subseteq \ell_1$ with $\dim(F) = k_n$ and a map $f : C \rightarrow F$ satisfying

$$\|f(x) - f(y)\|_1 \leq \alpha \|x - y\|_1 \quad \text{for all } x, y \in C$$

and

$$\frac{1}{n^2} \sum_{x \in C} \sum_{y \in C} \|f(x) - f(y)\|_1 \geq \frac{1}{n^2} \sum_{x \in C} \sum_{y \in C} \|x - y\|_1?$$

More generally, for $p \in [1, 2)$, does ℓ_p have the same property, with the ℓ_p norm in place of the ℓ_1 norm and with the constant α and the sequence $k_n = n^{o(1)}$ allowed to depend on p ?

5.22.2. Historical background, literature, and current progress

this problem is fully open

5.22.3. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

5.22.4. Primary-source pointer

The originating ICM record is *Metric dimension reduction A snapshot of the Ribe program* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>

[n/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758.](#)

5.23. 200. Metric dimension reduction A snapshot of the Ribe program [2018-geom-152]

Record: 2018-geom-152

Subfield: Metric and topological geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Metric dimension reduction A snapshot of the Ribe program

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>

5.23.1. Problem statement

Let $X = \ell_3 \widehat{\otimes} \ell_3$, where ℓ_3 is the Banach space of scalar sequences $a = (a_i)$ with $\sum_i |a_i|^3 < \infty$, equipped with the norm $\|a\|_3 = (\sum_i |a_i|^3)^{1/3}$, and $\widehat{\otimes}$ denotes the projective tensor product, i.e. the completion of $\ell_3 \otimes \ell_3$ under the norm whose unit ball is the convex hull of $\{u \otimes v : \|u\|_3 \leq 1, \|v\|_3 \leq 1\}$. Does there exist a constant $\alpha \in [1, \infty)$ and a sequence of integers k_n with $\lim_{n \rightarrow \infty} (\log k_n) / \log n = 0$ such that, for every $n \in \mathbb{N}$ and every n -point subset $C \subseteq X$, there are a linear subspace $F \subseteq X$ with $\dim(F) = k_n$ and a map $f : C \rightarrow F$ satisfying

$$\|f(x) - f(y)\|_X \leq \alpha \|x - y\|_X \quad \text{for all } x, y \in C,$$

and

$$\frac{1}{n^2} \sum_{x \in C} \sum_{y \in C} \|f(x) - f(y)\|_X \geq \frac{1}{n^2} \sum_{x \in C} \sum_{y \in C} \|x - y\|_X?$$

5.23.2. Historical background, literature, and current progress

this problem is fully open

5.23.3. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

5.23.4. Primary-source pointer

The originating ICM record is *Metric dimension reduction A snapshot of the Ribe program* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>.

5.24. 201. Metric dimension reduction A snapshot of the Ribe program [2018-geom-153]

Record: 2018-geom-153

Subfield: Metric and topological geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Metric dimension reduction A snapshot of the Ribe program

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>

5.24.1. Problem statement

Determine whether there exists an infinite-dimensional Banach space Y such that Y admits average metric dimension reduction, but the projective tensor product $\ell_2 \widehat{\otimes} Y$ fails to admit average metric dimension reduction.

Here, for Banach spaces X and Z , $X \widehat{\otimes} Z$ is the completion of $X \otimes Z$ under the projective tensor norm whose unit ball is the convex hull of

$$\{x \otimes z \in X \otimes Z : \|x\|_X \leq 1, \|z\|_Z \leq 1\}.$$

An infinite-dimensional Banach space X admits average metric dimension reduction if there is a constant $\alpha_X \in [1, \infty)$ and a sequence of integers k_n with

$$\lim_{n \rightarrow \infty} \frac{\log k_n}{\log n} = 0$$

such that for every $n \in \mathbb{N}$ and every n -point subset $C \subseteq X$, there is a linear subspace $F \subseteq X$ with $\dim(F) = k_n$ and a map $f : C \rightarrow F$ satisfying

$$\|f(x) - f(y)\|_X \leq \alpha_X \|x - y\|_X \quad \text{for all } x, y \in C,$$

and

$$\frac{1}{n^2} \sum_{x \in C} \sum_{y \in C} \|f(x) - f(y)\|_X \geq \frac{1}{n^2} \sum_{x \in C} \sum_{y \in C} \|x - y\|_X.$$

5.24.2. Historical background, literature, and current progress

5.24.3. Verdict: partially resolved

5.24.3.1. Precise problem

The problem asks whether there is an infinite-dimensional Banach space Y such that Y admits average metric dimension reduction, but the projective tensor product $\ell_2 \widehat{\otimes} Y$ fails to admit average metric dimension reduction.

5.24.3.2. Current literature status

I did not find a published or citable result that resolves the exact ℓ_2 -endpoint question. The closest available theorem proves the analogous non-stability phenomenon when the first tensor factor is ℓ_p for every $p > 2$, and the same source explicitly formulates the $p = 2$ endpoint as an open question.

5.24.3.3. Relevant literature and progress

1. Metric dimension reduction: A snapshot of the Ribe program

- Authors: Assaf Naor
- Venue: Proceedings of the International Congress of Mathematicians–Rio de Janeiro 2018, Vol. I, World Scientific, 2018, pp. 759–837
- arXiv: 1809.02376
- URL: <https://arxiv.org/abs/1809.02376>
- Relevant result: Theorem 29 proves that there exist Banach spaces X, Y that admit average metric dimension reduction while $X \widehat{\otimes} Y$ does not; moreover, for every $p \in (2, \infty)$, one can take $X = \ell_p$.
- Why this is only partial: Immediately after Theorem 29, Naor asks Question 31: whether the same conclusion holds for $p = 2$, namely whether there is Y admitting average metric dimension reduction such that $\ell_2 \widehat{\otimes} Y$ does not. Thus Theorem 29 does not cover the exact statement.

2. Locally decodable codes and the failure of cotype for projective tensor products

- Authors: Jop Briët, Assaf Naor, Oded Regev
- Venue: Electronic Research Announcements in Mathematical Sciences, 19, 2012, pp. 120–138
- DOI: 10.3934/era.2012.19.120
- arXiv: 1208.0539
- URL: <https://arxiv.org/abs/1208.0539>
- Relevant result: Theorem 1.1 proves that if $p_1, p_2, p_3 \in (1, \infty)$ and $1/p_1 + 1/p_2 + 1/p_3 \leq 1$, then $\ell_{p_1} \widehat{\otimes} \ell_{p_2} \widehat{\otimes} \ell_{p_3}$ does not have finite cotype. This is the tensor-product obstruction used in Naor's proof of Theorem 29.
- Why this is only partial: It gives finite-cotype/universality input for triple tensor products, but it does not itself prove failure of average metric dimension reduction for any space of the form $\ell_2 \widehat{\otimes} Y$ with Y admitting average metric dimension reduction.

3. Impossibility of dimension reduction in the nuclear norm

- Authors: Assaf Naor, Gilles Pisier, Gideon Schechtman
- Venue: Discrete & Computational Geometry, 63(2), 2020, pp. 319–345
- DOI: 10.1007/s00454-019-00162-2
- arXiv: 1710.08896
- URL: <https://arxiv.org/abs/1710.08896>
- Relevant result: Theorem 1 proves strong lower bounds for ordinary bi-Lipschitz metric dimension reduction in S_1 , where $S_1 \cong \ell_2 \widehat{\otimes} \ell_2$.
- Why this is only partial: the problem concerns average metric dimension reduction. Naor's ICM survey explicitly notes that it is unknown whether S_1 admits average metric dimension reduction, so this result does not resolve the problem statement by taking $Y = \ell_2$.

5.24.3.4. What remains unresolved

The exact endpoint question remains unresolved: no located result proves either the existence or nonexistence of an infinite-dimensional Banach space Y that admits average metric dimension reduction while $\ell_2 \widehat{\otimes} Y$ fails to admit average metric dimension reduction. The $p > 2$ projective tensor result and the non-average S_1 dimension-reduction obstruction are genuine partial progress, but they do not cover all hypotheses and conclusions of the problem statement.

5.24.4. Remaining open problems recorded by the database

5.24.4.1. Open problem 1

Does the Schatten–von Neumann trace class $S_1 \cong \ell_2 \widehat{\otimes} \ell_2$ admit average metric dimension reduction? Explicitly, is there a constant $D < \infty$ and a sequence $k_n = n^{o(1)}$ such that every n -point subset $C \subseteq S_1$ has a D -Lipschitz map into a k_n -dimensional subspace of S_1 whose average pairwise distance is at least that of C ?

Reference supplied by the database. Assaf Naor, “Metric Dimension Reduction: A Snapshot of the Ribe Program,” in *Proceedings of the International Congress of Mathematicians—Rio de Janeiro 2018*, Vol. I: Plenary Lectures, eds. Boyan Sirakov, Paulo Ney de Souza, and Marcelo Viana, World Scientific, 2018, pp. 759–837, §1.4.1, discussion between Corollary 28 and Theorem 29 (explicitly stating that it is unknown whether S_1 admits average metric dimension reduction). arXiv:1809.02376. <https://arxiv.org/abs/1809.02376>.

5.24.5. Primary-source pointer

The originating ICM record is *Metric dimension reduction A snapshot of the Ribe program* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>.

5.25. 202. Metric dimension reduction A snapshot of the Ribe program [2018-geom-154]

Record: 2018-geom-154

Subfield: Metric and topological geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Metric dimension reduction A snapshot of the Ribe program

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>

5.25.1. Problem statement

Let (W, d_W) be a metric space and let $p \in [1, \infty)$. For a metric space (X, d_X) , let $P_p(X)$ denote the set of Borel probability measures μ on X with finite p -th moment, and equip $P_p(X)$ with the Wasserstein distance

$$W_p(\mu, \nu) = \inf_{\pi} \left(\int_{X \times X} d_X(x, y)^p d\pi(x, y) \right)^{1/p},$$

where the infimum is over all couplings π of μ and ν . Fix $\alpha > 1$, $n \in \mathbb{N}$, and $\mu_1, \dots, \mu_n \in P_p(W)$. What is the asymptotic behavior of the smallest $k \in \mathbb{N}$ for which there exists a subset $S \subseteq W$, equipped with the metric inherited from W , with $|S| \leq k$, and a map $f : \{\mu_1, \dots, \mu_n\} \rightarrow P_p(S)$ such that, for some scaling factor $\tau > 0$,

$$\tau W_p(\mu_i, \mu_j) \leq W_p(f(\mu_i), f(\mu_j)) \leq \alpha \tau W_p(\mu_i, \mu_j)$$

for all $i, j \in \{1, \dots, n\}$?

5.25.2. Historical background, literature, and current progress

5.25.3. Verdict: partially resolved

5.25.3.1. Precise problem

The problem is Naor's Wasserstein version of metric dimension reduction: given an n -point subset $\mu_1, \dots, \mu_n$ of a Wasserstein space $P_p(W)$, determine the asymptotic size of the smallest subset S of the original ground metric space W such that the finite metric space induced by Wasserstein distances on the μ_i embeds with distortion α into $P_p(S)$. Equivalently, it asks how well one can replace arbitrary measures by measures supported on a small subset of the same ambient metric space while preserving all pairwise Wasserstein distances up to a fixed bi-Lipschitz distortion.

5.25.3.2. Status

I found no published or otherwise citable theorem that determines this minimal support size in the full generality of the problem: arbitrary metric ground space W , all p in $[1, \infty)$, all $\alpha > 1$, and all finite subsets of $P_p(W)$. The literature gives important obstructions and special/adjacent results, but not the full asymptotic answer.

5.25.3.3. Partial progress and related literature

1. Title: Metric dimension reduction: A snapshot of the Ribe program Authors: Assaf Naor Venue/year: Proceedings of the International Congress of Mathematicians–Rio de Janeiro 2018, Vol. I, World Scientific, 2018, pp. 759-837 DOI/arXiv/URL: DOI 10.1142/9789813272880_0029; arXiv:1809.02376; <https://arxiv.org/abs/1809.02376> Relevance: This is the original source. The problem appears as Question 32 in the Wasserstein-spaces subsection. Naor records that the bi-Lipschitz structure of Wasserstein spaces is poorly understood in this dimension-reduction setting. He also gives obstructions: for unrestricted W , one cannot hope for $k = n^{o(1)}$ with constant distortion, since point masses already contain arbitrary hard finite metrics; and even for $W = \ell_1$, universality of $P_1(\ell_1)$ gives hard examples obstructing subpolynomial support-size reduction.
2. Title: Similarity estimation techniques from rounding algorithms Authors: Moses S. Charikar Venue/year: Proceedings of the 34th Annual ACM Symposium on Theory of Computing (STOC 2002), pp. 380-388 DOI/arXiv/URL: DOI 10.1145/509907.509965; <https://dl.acm.org/doi/10.1145/509907.509965> Relevance: Together with Indyk-Thaper below, this gives logarithmic-distortion embeddings of finite earthmover/Wasserstein-1 metrics into ℓ_1 , used in Naor's obstruction discussion. It does not solve the support-size question.
3. Title: Fast Image Retrieval via Embeddings Authors: Piotr Indyk and Nitin Thaper Venue/year: 3rd International Workshop on Statistical and Computational Theories of Vision, 2003 DOI/arXiv/URL: <https://people.csail.mit.edu/indyk/emd.pdf> Relevance: Provides the standard embedding approach for earthmover distance used in the same obstruction framework. It concerns embeddings into ℓ_1 , not embeddings into $P_p(S)$ with $S \subset W$.
4. Title: The metrical interpretation of superreflexivity in Banach spaces Authors: Jean Bourgain Venue/year: Israel Journal of Mathematics 56, 222-230, 1986 DOI/arXiv/URL: DOI 10.1007/BF02766125; <https://link.springer.com/article/10.1007/BF02766125> Relevance: Cited in the Wasserstein literature for the universality of $P_1(\ell_1)$. This supplies negative evidence against small-support reduction even when the ground space itself is ℓ_1 , but it is not a full answer to the question.
5. Title: Snowflake universality of Wasserstein spaces Authors: Alexandr Andoni, Assaf Naor, Ofer Neiman Venue/year: Annales scientifiques de l'Ecole Normale Supérieure, Serie 4, 51(3), 657-700, 2018 DOI/arXiv/URL: DOI 10.24033/asens.2363; arXiv:1509.08677; <https://www.numdam.org/articles/10.24033/asens.2363/> Relevance: Proves that for $p > 1$, every finite metric space has its $1/p$ -snowflake embedded almost isometrically into $P_p(R^3)$, with sharpness results for p in $(1, 2]$. Naor uses this as partial information for Question 32 when $W = R^3$ and $p > 1$. It still does not answer the stated support-size problem because it does not show how small $S \subset R^3$ can be, nor does it determine whether $P_p(S)$ for small S has the necessary low-dimensional

normed embedding behavior.

6. Title: Impossibility of Sketching of the 3D Transportation Metric with Quadratic Cost
Authors: Alexandr Andoni, Assaf Naor, Ofer Neiman Venue/year: 43rd International Colloquium on Automata, Languages, and Programming (ICALP 2016), LIPIcs 55, article 83, pp. 83:1-83:14 DOI/arXiv/URL: DOI 10.4230/LIPIcs.ICALP.2016.83; <https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.ICALP.2016.83> Relevance: Conference/CS version of the 3D transportation-metric obstruction. It supports the same partial picture for $P_p(\mathbb{R}^3)$ but is not a full support-size theorem.
7. Title: Wasserstein distance and metric trees Authors: Maxime Mathey-Prevot and Alain Valette Venue/year: L'Enseignement Mathématique 69(3/4), 315-333, 2023 DOI/arXiv/URL: DOI 10.4171/LEM/1052; arXiv:2110.02115; <https://ems.press/journals/lem/articles/11033098> Relevance: Shows that if a finite metric space X stochastically embeds with distortion D into finite metric trees, then $P_1(X)$ embeds bi-Lipschitzly into ℓ_1 with distortion D , and gives tree formulas for Wasserstein distance. This is useful structural progress for tree-like ground spaces, but it does not determine the minimal $|S|$ in Naor's question.
8. Title: Coarse Embeddability of Wasserstein Space and the Space of Persistence Diagrams Authors: Neil Pritchard and Thomas Weighill Venue/year: Discrete & Computational Geometry 74, 358-373, 2025 DOI/arXiv/URL: DOI 10.1007/s00454-024-00674-6; arXiv:2307.12884; <https://link.springer.com/article/10.1007/s00454-024-00674-6> Relevance: Recent work showing equivalences between embeddability questions for Wasserstein space on $\mathbb{R}^2$ and persistence diagrams, and explicitly noting that important embeddability cases for Wasserstein space on the plane remain open. This reinforces that the low-dimensional Wasserstein embedding landscape is still unresolved, but it does not answer Question 32.

5.25.3.4. What remains unresolved

The full asymptotic behavior of the smallest support size k remains unknown. In particular, the literature does not give a theorem covering all metric spaces W and all p , nor does it settle the main “nice ground space” cases highlighted by Naor, such as $p=1$ over low-dimensional Euclidean spaces. Existing results rule out some overly optimistic answers and give related embedding theorems, but they do not determine the support-size reduction modulus asked for in the problem statement.

5.25.4. Remaining open problems recorded by the database

5.25.4.1. Open problem 1

Fix $\alpha > 1$. Given arbitrary measures $\mu_1, \dots, \mu_n \in P_1(\mathbb{R}^2)$, determine the asymptotic order of the smallest $k = k(n, \alpha)$ for which there is a set $S \subseteq \mathbb{R}^2$ with $|S| \leq k$ such that the finite metric space $(\{\mu_1, \dots, \mu_n\}, W_1)$ embeds with distortion at most α into $(P_1(S), W_1)$.

Reference supplied by the database. Assaf Naor, “Metric Dimension Reduction: A Snapshot of the Ribe Program,” in *Proceedings of the International Congress of Mathematicians—Rio de*

Janeiro 2018, Vol. I: Plenary Lectures, eds. Boyan Sirakov, Paulo Ney de Souza, and Marcelo Viana, World Scientific, 2018, pp. 759–837, Question 32 and the ensuing discussion in §1.4.2 (explicitly singling out low-dimensional ground spaces and the image-retrieval case $p = 1$, $M \subset \mathbb{R}^2$). arXiv:1809.02376. <https://arxiv.org/abs/1809.02376>.

5.25.5. Primary-source pointer

The originating ICM record is *Metric dimension reduction A snapshot of the Ribe program* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol1-ver1-eb.pdf#page=758>.

5.26. 203. Dimension theory of self-similar sets and measures [2018-geom-261]

Record: 2018-geom-261

Subfield: Metric and topological geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Dimension theory of self-similar sets and measures

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol3-ver1-eb.pdf#page=540>

5.26.1. Problem statement

Let Λ be a finite set, and let $\Phi = \{\varphi_i\}_{i \in \Lambda}$ be an iterated function system of contracting similarities of $\mathbb{R}$, where the contraction ratio of φ_i is $r_i \in (0, 1)$. Let X_Φ be the unique nonempty compact set satisfying

$$X_\Phi = \bigcup_{i \in \Lambda} \varphi_i(X_\Phi).$$

Given a probability vector $p = (p_i)_{i \in \Lambda}$ with every $p_i > 0$, let $\mu_{\Phi, p}$ be the unique Borel probability measure satisfying

$$\mu_{\Phi, p} = \sum_{i \in \Lambda} p_i (\varphi_i)_* \mu_{\Phi, p}.$$

Define the similarity dimension $s(\Phi)$ by

$$\sum_{i \in \Lambda} r_i^{s(\Phi)} = 1,$$

and define the Lyapunov dimension by

$$s(\Phi, p) = \frac{-\sum_{i \in \Lambda} p_i \log p_i}{-\sum_{i \in \Lambda} p_i \log r_i}.$$

For a Borel probability measure μ , write

$$\dim_H \mu = \inf\{\dim_H E : E \subseteq \mathbb{R} \text{ is Borel and } \mu(E) > 0\}.$$

For a word $\mathbf{i} = i_1 \cdots i_n \in \Lambda^n$, set $\varphi_{\mathbf{i}} = \varphi_{i_1} \circ \cdots \circ \varphi_{i_n}$. An exact overlap means that $\varphi_{\mathbf{i}} = \varphi_{\mathbf{j}}$ for two distinct words $\mathbf{i}, \mathbf{j} \in \Lambda^n$ of some common length $n \geq 1$.

Prove or disprove that a dimension drop can occur only if there is an exact overlap; equivalently, prove or disprove both implications

$$\dim_H X_\Phi < \min\{1, s(\Phi)\} \implies \Phi \text{ has an exact overlap}$$

and

$$\dim_H \mu_{\Phi, p} < \min\{1, s(\Phi, p)\} \implies \Phi \text{ has an exact overlap}.$$

5.26.2. Historical background, literature, and current progress

The problem is **unresolved**. No proof or counterexample is known for arbitrary finite systems of contracting similarities of $\mathbb{R}$. In particular, the available theorems do not cover every system having nonalgebraic, nonhomogeneous contraction ratios and unrestricted translations.

The main established progress is as follows.

- **“On self-similar sets with overlaps and inverse theorems for entropy,”** Michael Hochman, *Annals of Mathematics* 180 (2014), 773–822, <https://doi.org/10.4007/annals.2014.180.2.7>. Theorem 1.1 proves, for self-similar measures on the line, that dimension drop forces super-exponential concentration of distinct cylinders; the set statement follows by choosing the natural weights. Consequently, exponential separation rules out dimension drop. Theorem 1.5 proves the exact-overlap conclusion when all coefficients of the similarities are algebraic. This is a reduction and a large special case, not a proof that super-exponential concentration always produces an exact overlap.
- **“Iterated function systems with super-exponentially close cylinders,”** Simon Baker, *Advances in Mathematics* 379 (2021), Article 107548, <https://doi.org/10.1016/j.aim.2020.107548>, and **“Super-exponential condensation without exact overlaps,”** Balázs Bárány and Antti Käenmäki, *Advances in Mathematics* 379 (2021), Article 107549, <https://doi.org/10.1016/j.aim.2020.107549>. These papers construct systems with super-exponentially close cylinders but no exact overlaps. They show that Hochman’s necessary condition cannot by itself be replaced by “exact overlap.” They do not exhibit dimension drop, so they are not counterexamples to the problem.
- **“Proof of the exact overlaps conjecture for systems with algebraic contractions,”** Ariel Rapaport, *Annales scientifiques de l’École Normale Supérieure* 55 (2022), 1357–1377, <https://doi.org/10.24033/asens.2518>. Theorem 2 proves the measure implication for arbitrary translations when every signed contraction factor is algebraic, and Corollary 3 proves the corresponding attractor statement. Thus all weights and translations are covered in this class, but arbitrary nonalgebraic contraction factors are not.
- **“Self-similar measures associated to a homogeneous system of three maps,”** Ariel Rapaport and Péter P. Varjú, *Duke Mathematical Journal* 173 (2024), 513–602, <https://doi.org/10.1215/00127094-2023-0019>. Appendix A proves the measure conclusion for homogeneous systems with rational translations and arbitrary positive weights. Theorem 1.8 proves it for the equal-weight three-map family $\{x \mapsto \lambda x, x \mapsto \lambda x + 1, x \mapsto \lambda x + \tau\}$ when $\lambda > 2^{-2/3}$, and Corollary 1.4 shows that the exceptional parameter set in the full two-parameter family has Hausdorff dimension zero. These results leave other homogeneous three-map parameters open.
- **“Dimension of homogeneous iterated function systems with algebraic translations,”** De-Jun Feng and Zhou Feng, *Journal of the London Mathematical Society* (2025), <https://doi.org/10.1112/jlms.70222>. Theorem 1.2 proves the measure implication for every homogeneous IFS $\{x \mapsto \lambda x + t_j\}$ with arbitrary nonzero $|\lambda| < 1$, provided all translations t_j are algebraic. The attractor conclusion follows by applying the measure result with the natural similarity-dimension weights. Transcendental translation configurations, and general nonhomogeneous systems, remain outside this theorem.

Hence the strongest results cover exponential separation, all algebraic contraction factors, and several important homogeneous families, but they do not exhaust the arbitrary IFSs quantified over in the statement. The unresolved gap is whether a dimension drop in one of the re-

maining systems must force an actual equality between two finite cylinder maps, or whether a counterexample exists.

5.26.3. Remaining open problems recorded by the database

5.26.3.1. Open problem 1

Let $0 < \lambda < 1$ and $\tau \in \mathbb{R}$, and consider the three contracting similarities

$$f_1(x) = \lambda x, \quad f_2(x) = \lambda x + 1, \quad f_3(x) = \lambda x + \tau.$$

Let $\mu_{\lambda, \tau}$ be the unique Borel probability measure satisfying

$$\mu_{\lambda, \tau} = \frac{1}{3} \sum_{j=1}^3 (f_j)_* \mu_{\lambda, \tau},$$

and define $\dim_H \mu_{\lambda, \tau} = \inf\{\dim_H E : E \subseteq \mathbb{R} \text{ is Borel and } \mu_{\lambda, \tau}(E) > 0\}$. For a word $\mathbf{i} = i_1 \cdots i_n \in \{1, 2, 3\}^n$, write $f_{\mathbf{i}} = f_{i_1} \circ \cdots \circ f_{i_n}$. Prove or disprove that, if $f_{\mathbf{i}} \neq f_{\mathbf{j}}$ for every $n \geq 1$ and every pair of distinct words $\mathbf{i}, \mathbf{j} \in \{1, 2, 3\}^n$, then

$$\dim_H \mu_{\lambda, \tau} = \min \left\{ 1, \frac{\log 3}{\log(1/\lambda)} \right\}.$$

Reference supplied by the database. “Self-similar measures associated to a homogeneous system of three maps,” Ariel Rapaport and Péter P. Varjú, *Duke Mathematical Journal* 173 (2024), no. 3, 513–602, Conjecture 1.2 specialized to the IFS (1.2), <https://doi.org/10.1215/00127094-2023-0019>.

5.26.3.2. Open problem 2

For $0 < \lambda < 1$ and $\tau \in \mathbb{R}$, let

$$\Phi_{\lambda, \tau} = \{f_{a,b}^{\lambda, \tau}(x) = \lambda x + a + b\tau : (a, b) \in \{-1, 1\}^2\}.$$

For each $N \geq 1$, choose N maps independently and uniformly from $\Phi_{\lambda, \tau}$, compose them in order, and denote the resulting random similarity by $Y_N(\lambda, \tau)$. If $A \subseteq (0, 1) \times \mathbb{R}$, regard $Y_N|_A$ as the finite-valued random function that maps each $(\lambda, \tau) \in A$ to $Y_N(\lambda, \tau)$, and define

$$h(A) = \lim_{N \rightarrow \infty} \frac{1}{N} H(Y_N|_A), \quad H(Z) = - \sum_z \mathbb{P}(Z = z) \log \mathbb{P}(Z = z).$$

Write $h(\lambda, \tau) = h(\{(\lambda, \tau)\})$. An exact-overlap curve is a one-dimensional component $\gamma \subseteq (0, 1) \times \mathbb{R}$ of the solution set of an identity

$$f_{a_1, b_1}^{\lambda, \tau} \circ \cdots \circ f_{a_n, b_n}^{\lambda, \tau} = f_{a'_1, b'_1}^{\lambda, \tau} \circ \cdots \circ f_{a'_n, b'_n}^{\lambda, \tau}$$

between two distinct words $((a_1, b_1), \dots, (a_n, b_n))$ and $((a'_1, b'_1), \dots, (a'_n, b'_n))$ in $(\{-1, 1\}^2)^n$, for some $n \geq 1$; its entropy rate is $h(\gamma)$.

For an algebraic number λ whose primitive minimal polynomial factors as $P(x) = a_d \prod_{j=1}^d (x - \alpha_j)$, define its Mahler measure by

$$M(\lambda) = |a_d| \prod_{j=1}^d \max\{1, |\alpha_j|\},$$

and set $M(\lambda) = \infty$ when λ is transcendental.

Determine whether, for every $\varepsilon > 0$, there exist $\delta > 0$ and $C < \infty$ such that every $(\lambda, \tau) \in (\varepsilon, 1 - \varepsilon) \times \mathbb{R}$ satisfying

$$h(\lambda, \tau) \leq \min\{\log 4, \log(1/\lambda)\} - \varepsilon$$

also satisfies at least one of the following alternatives:

$$M(\lambda) < C,$$

or (λ, τ) lies on an exact-overlap curve γ such that

$$h(\gamma) \leq \min\{\log 4, \log(1/\lambda)\} - \delta.$$

Reference supplied by the database. “Entropy rates in the dimension theory of self-similar measures,” Péter P. Varjú, 2025, Question 11, arXiv:2509.22042 [math.DS], <https://arxiv.org/abs/2509.22042>.

5.26.4. Primary-source pointer

The originating ICM record is *Dimension theory of self-similar sets and measures* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol3-ver1-eb.pdf#page=540>.

5.27. 204. On some properties of sparse sets a survey [2022-geom-267]

Record: 2022-geom-267

Subfield: Metric and topological geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2022: On some properties of sparse sets a survey

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-562-9-volume-4.pdf#page=867>

5.27.1. Problem statement

For every integer $d \geq 2$ and every compact set $E \subseteq \mathbb{R}^d$, define its Euclidean distance set by

$$\Delta(E) = \{\|x - y\|_2 : x, y \in E\}.$$

Prove that if the Hausdorff dimension of E satisfies $\dim_{\text{H}}(E) > d/2$, then $\Delta(E)$ has positive one-dimensional Lebesgue measure.

5.27.2. Historical background, literature, and current progress

The conjecture is **unresolved**. No result currently covers every compact $E \subseteq \mathbb{R}^d$ with $\dim_{\text{H}}(E) > d/2$, and recent literature continues to identify the conjecture as open in every dimension $d \geq 2$.

The threshold $d/2$ is sharp in the sense motivating the conjecture: Falconer constructed, for every $s \leq d/2$, compact sets of Hausdorff dimension s whose distance sets have zero Lebesgue measure, while proving positivity when $\dim_{\text{H}}(E) > (d + 1)/2$. See K. J. Falconer, “On the Hausdorff dimensions of distance sets,” *Mathematika* 32(2) (1985), 206–212, DOI: <https://doi.org/10.1112/S0025579300010998>.

For $d = 2$, the strongest general positive-measure theorem remains the $5/4$ threshold: if $\dim_{\text{H}}(E) > 5/4$, there is an $x \in E$ for which the pinned set $\{|x - y| : y \in E\}$ has positive measure, hence $\Delta(E)$ has positive measure. See Larry Guth, Alex Iosevich, Yumeng Ou, and Hong Wang, “On Falconer’s distance set problem in the plane,” *Inventiones Mathematicae* 219 (2020), 779–830, DOI: <https://doi.org/10.1007/s00222-019-00917-x>, arXiv:1808.09346.

For every $d \geq 3$, the strongest general theorem found gives positivity when

$$\dim_{\text{H}}(E) > \frac{d}{2} + \frac{1}{4} - \frac{1}{8d + 4}.$$

It likewise proves the stronger existence of a pin $x \in E$ with a positive-measure pinned distance set. See Xiumin Du, Yumeng Ou, Kevin Ren, and Ruixiang Zhang, “New improvement to Falconer distance set problem in higher dimensions,” preprint (2023, revised 2024), arXiv:2309.04103.

There is sharper progress for structured sets, but it does not resolve the stated universal claim. In the plane, Bochen Liu proves the conjectured threshold when the set of pins has equal Hausdorff and packing dimensions; taking the same set for points and pins settles compact planar E with $\dim_H(E) = \dim_P(E) > 1$. See Bochen Liu, “Lebesgue measure of distance sets with regular pins and multi-scale Mizohata–Takeuchi-type estimates,” preprint (2026), arXiv:2603.15328. Minh-Quy Pham proves the conjecture for product sets $A = A_1 \times A_2$ when the one-dimensional factor A_2 is Salem, under the stated threshold $\dim_H(A) > d/2$; see “On Falconer type functions and the distance set problem,” *Mathematische Annalen* (2026), DOI: <https://doi.org/10.1007/s00208-026-03427-3>, arXiv:2510.15118.

Thus the remaining gap is the full range $1 < \dim_H(E) \leq 5/4$ for arbitrary planar compact sets and, for each $d \geq 3$, the range

$$\frac{d}{2} < \dim_H(E) \leq \frac{d}{2} + \frac{1}{4} - \frac{1}{8d+4}$$

for arbitrary compact sets. Results requiring regularity, product structure, or Fourier-dimension hypotheses do not cover these cases.

5.27.3. Remaining open problems recorded by the database

5.27.3.1. Open problem 1

For every compact set $E \subseteq \mathbb{R}^2$, define

$$\Delta(E) = \{\|x - y\|_2 : x, y \in E\}.$$

Prove that if $\dim_H(E) > 1$, then $\Delta(E)$ has positive one-dimensional Lebesgue measure.

Reference supplied by the database. Larry Guth, Alex Iosevich, Yumeng Ou, and Hong Wang, “On Falconer’s distance set problem in the plane,” *Inventiones Mathematicae* 219(3) (2020), 779–830; Falconer Distance Conjecture and planar case in the Introduction; DOI: <https://doi.org/10.1007/s00222-019-00917-x>; arXiv:1808.09346.

5.27.3.2. Open problem 2

For every integer $d \geq 3$ and every compact set $E \subseteq \mathbb{R}^d$, define

$$\Delta(E) = \{\|x - y\|_2 : x, y \in E\}.$$

Prove that if $\dim_H(E) > d/2$, then $\Delta(E)$ has positive one-dimensional Lebesgue measure.

Reference supplied by the database. Xiumin Du, Yumeng Ou, Kevin Ren, and Ruixiang Zhang, “New improvement to Falconer distance set problem in higher dimensions,” preprint (2023), Falconer Conjecture and Theorem 1.2 in Section 1; arXiv:2309.04103.

5.27.4. Primary-source pointer

The originating ICM record is *On some properties of sparse sets a survey* (2022). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-562-9-volume-4.pdf#page=867>.

5.28. 205. Slices and distances on two problems of Furstenberg and Falconer [2022-geom-268]

Record: 2022-geom-268

Subfield: Metric and topological geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2022: Slices and distances on two problems of Furstenberg and Falconer

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-562-9-volume-4.pdf#page=909>

5.28.1. Problem statement

Prove that for every integer $d \geq 2$ and every Borel set $X \subset \mathbb{R}^d$ with Hausdorff dimension $\dim_{\text{H}}(X) \geq d/2$, the Euclidean distance set

$$\Delta(X) := \{\|x - y\|_2 : x, y \in X\}$$

has Hausdorff dimension $\dim_{\text{H}}(\Delta(X)) = 1$.

5.28.2. Historical background, literature, and current progress

Verdict: unresolved. The exact assertion remains open in every ambient dimension $d \geq 2$. In particular, no known theorem gives $\dim_{\text{H}} \Delta(X) = 1$ for every Borel set at the critical hypothesis $\dim_{\text{H}} X = d/2$. A current paper explicitly records this same dimension conjecture as open for all $d \geq 2$: Jonathan M. Fraser and Thang Pham, *On Fourier decay and the distance set problem*, preprint, 2026, arXiv:2604.19486. Its full-dimension conclusions require additional Fourier-decay or Fourier-spectrum hypotheses and therefore do not resolve the stated problem.

The strongest endpoint progress for unrestricted planar sets is a pinned lower bound. Peter Cholak, Marianna Csörnyei, Neil Lutz, Patrick Lutz, Elvira Mayordomo, and D. M. Stull, *Algorithmic Information Bounds for Distances and Orthogonal Projections*, preprint, 2025, arXiv:2509.05211, Theorem 12, prove that every analytic $E \subset \mathbb{R}^2$ with $s = \dim_{\text{H}} E \leq 1$ satisfies

$$\sup_{x \in E} \dim_{\text{H}} \Delta_x(E) \geq \frac{3s}{4}.$$

Thus at $s = 1$, $\dim_{\text{H}} \Delta(E) \geq 3/4$, which is still short of the required value 1.

In dimensions $d \geq 3$, Xiumin Du, Yumeng Ou, Kevin Ren, and Ruixiang Zhang, *New improvement to Falconer distance set problem in higher dimensions*, preprint, 2024, arXiv:2309.04103v2, Theorem 1.3, prove for compact endpoint sets that

$$\sup_{x \in E} \dim_{\text{H}} \Delta_x(E) \geq \frac{d+2}{2(d+1)}.$$

This is strictly greater than the classical $1/2$ bound but strictly less than 1. The same paper proves positive Lebesgue measure only under the stronger threshold $\dim_{\text{H}} E > d/2 + 1/4 - 1/(8d+4)$, so it does not cover the hypotheses in the statement.

Pablo Shmerkin and Hong Wang, *On the distance sets spanned by sets of dimension $d/2$ in $\mathbb{R}^d$* , *Geometric and Functional Analysis* **35** (2025), 283–358, DOI: <https://doi.org/10.1007/s00039-024-00696-5>, Theorem 1.4, prove the desired full Hausdorff dimension in every $d \geq 2$ under the additional assumption $\dim_H E = \dim_P E = d/2$. This settles the critical case for that regular subclass, including Ahlfors-regular sets, but the packing-dimension hypothesis is absent from the original problem. The remaining gap is precisely the full-dimensional conclusion for arbitrary Borel sets allowed by the statement.

5.28.3. Remaining open problems recorded by the database

5.28.3.1. Open problem 1

Prove that for every Borel set $X \subset \mathbb{R}^2$ with Hausdorff dimension $\dim_H(X) = 1$, the Euclidean distance set

$$\Delta(X) := \{\|x - y\|_2 : x, y \in X\}$$

has Hausdorff dimension $\dim_H(\Delta(X)) = 1$.

Reference supplied by the database. Jonathan M. Fraser and Thang Pham, *On Fourier decay and the distance set problem*, preprint, 2026, Section 1.1, equation (1.1), arXiv:2604.19486, <https://arxiv.org/abs/2604.19486>.

5.28.4. Primary-source pointer

The originating ICM record is *Slices and distances on two problems of Furstenberg and Falconer* (2022). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-562-9-volume-4.pdf#page=909>.

6. Manifolds and global analysis

The problems here connect the topology of manifolds with elliptic operators, index theory, moduli spaces, gauge theory, and global nonlinear analysis. Their histories often pass through a succession of increasingly flexible categories: smooth, topological, coarse, equivariant, or derived. Partial results typically isolate hypotheses under which compactness, transversality, or an index obstruction can be controlled.

Coverage: 21 website records; **source years:** 1974–2022; **separately listed descendant open problems:** 37.

6.1. 206. Eigenvalues of the Laplacian and Invariants of Manifolds [1974-geom-301]

Record: 1974-geom-301
Subfield: Manifolds and global analysis
Normalized status: Open / open-ended
Dataset review state: unreviewed (reviewed flag: no)
Original source: ICM 1974: Eigenvalues of the Laplacian and Invariants of Manifolds
Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1974.1/ICM1974.1.ocr.pdf#page=243>

6.1.1. Problem statement

Let $q \geq 1$, and let M be a closed oriented Riemannian manifold of dimension $4q - 1$ whose sectional curvatures are strictly negative. Let $\Gamma = \pi_1(M)$, let $\chi : \Gamma \rightarrow O(m)$ be an orthogonal representation, and let $E_\chi \rightarrow M$ be the associated flat real vector bundle. Write d_χ for the exterior differential on E_χ -valued forms and $*$ for the Hodge-star operator, acting trivially on the E_χ -factor. Define the self-adjoint odd signature operator A_χ on even-degree E_χ -valued differential forms by

$$A_\chi|_{\Omega^{2p}(M; E_\chi)} = (-1)^{q+p+1}(*d_\chi - d_\chi*).$$

For $\text{Re } s$ sufficiently large, define its eta function by

$$\eta(A_\chi, s) = \sum_{\lambda \in \text{Spec}(A_\chi) \setminus \{0\}} \text{sign}(\lambda) |\lambda|^{-s},$$

where eigenvalues are counted with multiplicity, and let $\eta(A_\chi, 0)$ denote its value at $s = 0$ obtained by meromorphic continuation.

Every nontrivial primitive conjugacy class in Γ , together with an orientation, determines a primitive oriented closed geodesic γ . Let ℓ_γ be its length and let P_γ be the linear Poincaré return map of the geodesic flow on the vector space transverse to the corresponding periodic orbit in the unit tangent bundle. Negative curvature makes P_γ hyperbolic. Let $\tau_1(\gamma), \dots, \tau_{4q-2}(\gamma) > 1$ be the moduli of its expanding eigenvalues, counted with algebraic multiplicity. The oriented normal bundle N_γ has rank $4q - 2$; on the complexification of its middle exterior power $\Lambda^{2q-1} N_\gamma^*$, the normal Hodge-star operator has eigenvalues $+i$ and $-i$. Let $\varphi^\pm(\gamma)$ denote parallel transport around γ , restricted to the corresponding $\pm i$ -eigenbundle. For $\operatorname{Re} s$ sufficiently large, define

$$Z_\chi^\pm(s) = \prod_{[\gamma]_{\text{prim}}} \prod_{n_1, \dots, n_{4q-2} \geq 0} \det \left(I - \varphi^\pm(\gamma) \otimes \chi(\gamma) \exp \left[-s\ell_\gamma - \sum_{j=1}^{4q-2} \left(n_j + \frac{1}{2} \right) \log \tau_j(\gamma) \right] \right),$$

where the outer product is over primitive oriented closed geodesics.

Prove that the locally integrable remainder terms in the closed-geodesic wave-trace expansions make no contribution after the $+$ and $-$ terms are divided. More precisely, prove that $Z_\chi^+(s)/Z_\chi^-(s)$ admits a meromorphic continuation to the complex plane, is holomorphic and nonzero at $s = 0$, and satisfies

$$\exp(\pi i \eta(A_\chi, 0)) = \frac{Z_\chi^+(0)}{Z_\chi^-(0)}.$$

Assume in addition that the twisted de Rham cohomology $H^*(M; E_\chi)$ vanishes. For each p , let

$$\Delta_p^\chi = d_\chi d_\chi^* + d_\chi^* d_\chi$$

be the Hodge Laplacian on E_χ -valued p -forms, and define

$$\log \det' \Delta_p^\chi = - \left. \frac{d}{ds} \right|_{s=0} \sum_{\lambda \in \operatorname{Spec}(\Delta_p^\chi) \setminus \{0\}} \lambda^{-s}, \quad \log T(M, \chi) = \frac{1}{2} \sum_{p=0}^{4q-1} (-1)^p p \log \det' \Delta_p^\chi.$$

Prove that the locally integrable remainder terms in the closed-geodesic wave-trace expansions of the operators Δ_p^χ cancel in this alternating weighted sum. Equivalently, prove that the Ray–Singer analytic torsion $T(M, \chi)$ is determined entirely by the principal closed-geodesic terms, or corresponding closed-geodesic zeta factors, in those wave-trace expansions.

6.1.2. Historical background, literature, and current progress

The exact problem is **unresolved**. The constant-curvature and locally symmetric cases are known, and modern microlocal methods establish meromorphic continuation for several related dynamical zeta functions in variable negative curvature. However, the literature does not prove that the locally integrable wave-trace remainders cancel for every closed strictly negatively curved $(4q - 1)$ -manifold. In particular, no result located proves both the stated eta identity for Z_χ^+/Z_χ^- and the stated analytic-torsion cancellation for arbitrary variable negative curvature and every orthogonal representation χ .

Key established results and limitations:

- John J. Millson, “Closed geodesics and the η -invariant,” *Annals of Mathematics* (2)

- 108 (1978), 1–39. DOI: <https://doi.org/10.2307/1970928>. Millson proves the closed-geodesic Selberg-zeta formula for the eta invariant of the odd signature operator on compact manifolds of constant negative curvature. This proves the eta component in the real-hyperbolic special case, but not for a general variable negatively curved metric.
- **Henri Moscovici and Robert J. Stanton**, “Eta invariants of Dirac operators on locally symmetric manifolds,” *Inventiones Mathematicae* 95 (1989), 629–666. DOI: <https://doi.org/10.1007/BF01393895>. This extends closed-geodesic eta formulas to compact locally symmetric spaces using the Selberg trace formula. Local symmetry is an additional hypothesis absent from the problem.
 - **David Fried**, “Analytic torsion and closed geodesics on hyperbolic manifolds,” *Inventiones Mathematicae* 84 (1986), 523–540. DOI: <https://doi.org/10.1007/BF01388745>. For an acyclic orthogonal representation on a closed oriented odd-dimensional hyperbolic manifold, Fried proves that the Ruelle zeta function is regular at zero and that its value there equals the square of the Ray–Singer torsion. This settles the torsion relation only in constant negative curvature.
 - **Henri Moscovici and Robert J. Stanton**, “R-torsion and zeta functions for locally symmetric manifolds,” *Inventiones Mathematicae* 105 (1991), 185–216. DOI: <https://doi.org/10.1007/BF01232263>; **Shu Shen**, “Analytic torsion, dynamical zeta functions, and the Fried conjecture,” *Analysis & PDE* 11 (2018), 1–74. DOI: <https://doi.org/10.2140/apde.2018.11.1>, arXiv:1602.00664. These works establish the analytic-torsion/Ruelle-zeta identity for closed locally symmetric reductive manifolds, with Shen filling a gap in the earlier argument. They do not cover arbitrary negatively curved metrics.
 - **Paolo Giulietti, Carlangelo Liverani, and Mark Pollicott**, “Anosov flows and dynamical zeta functions,” *Annals of Mathematics* (2) 178 (2013), 687–773. DOI: <https://doi.org/10.4007/annals.2013.178.2.6>; **Frederic Faure and Masato Tsujii**, “The semiclassical zeta function for geodesic flows on negatively curved manifolds,” *Inventiones Mathematicae* 208 (2017), 851–998. DOI: <https://doi.org/10.1007/s00222-016-0701-5>, arXiv:1311.4932. These papers prove meromorphic continuation for broad Anosov dynamical zeta functions, including the semiclassical/Gutzwiller–Voros zeta associated with contact Anosov geodesic flows. They do not identify the value at zero with the eta invariant or prove the wave-trace remainder cancellation asserted here.
 - **Nguyen Viet Dang, Colin Guillarmou, Gabriel Riviere, and Shu Shen**, “The Fried conjecture in small dimensions,” *Inventiones Mathematicae* 220 (2020), 525–579. DOI: <https://doi.org/10.1007/s00222-019-00935-9>, arXiv:1807.01189. The authors prove deformation and low-dimensional results and obtain the Fried formula for geodesic flows sufficiently close to the flow of a hyperbolic three-manifold. Their theorem covers a neighborhood of the hyperbolic case, not every negatively curved metric.
 - **Yann Chaubet and Nguyen Viet Dang**, “Dynamical torsion for contact Anosov flows,” *Analysis & PDE* 17 (2024), 2619–2681. arXiv:1911.09931, [publisher URL](#). They construct a deformation-invariant dynamical torsion and, for contact Anosov flows connected to hyperbolic geodesic flows, relate the leading Laurent coefficient of the Ruelle zeta function to Reidemeister torsion and a finite-dimensional zero-resonance torsion. The extra resonant factor is precisely a remaining obstruction to the unrestricted torsion identity; the paper does not address the eta quotient in the problem.
 - **Jean-Michel Bismut and Shu Shen**, “Anosov Vector Fields and Fried Sections,” *Communications in Mathematical Physics* 406, article 240 (2025). DOI: <https://doi.org/10.1007/s00220-025-05000-0>.

[i.org/10.1007/s00220-025-05400-8](https://doi.org/10.1007/s00220-025-05400-8). They construct a canonical nonzero Fried section of the determinant-of-cohomology line for a general Anosov vector field and prove its flatness in families. Their Section 9 explicitly formulates, rather than proves, the required comparison with the Reidemeister/Ray–Singer metric as the modified Fried conjecture $\|\tau_\nu(i_Z)\|_\nu = 1$. Thus this latest general framework leaves the torsion assertion open and does not prove the eta identity.

Thus the strongest results establish the desired phenomena for constant-curvature or locally symmetric manifolds and supply substantial analytic continuation and deformation theory in variable curvature. The missing part is the claimed exact cancellation of the wave-trace remainders, and hence the two stated spectral identities, over the full class of strictly negatively curved manifolds.

6.1.3. Remaining open problems recorded by the database

6.1.3.1. Open problem 1

Let $q \geq 1$, and let M be a closed oriented Riemannian manifold of dimension $4q - 1$ with strictly negative sectional curvature. Let $\Gamma = \pi_1(M)$, let $\chi : \Gamma \rightarrow O(m)$ be an orthogonal representation, and let $E_\chi \rightarrow M$ be the associated flat bundle. On even-degree E_χ -valued differential forms define the odd signature operator

$$A_\chi|_{\Omega^{2p}(M; E_\chi)} = (-1)^{q+p+1}(*d_\chi - d_\chi*),$$

and define $\eta(A_\chi, 0)$ by meromorphically continuing

$$\eta(A_\chi, s) = \sum_{\lambda \in \text{Spec}(A_\chi) \setminus \{0\}} \text{sign}(\lambda) |\lambda|^{-s}$$

from the half-plane where the series converges.

For each primitive oriented closed geodesic γ , let ℓ_γ be its length, let P_γ be the linear Poincaré return map transverse to the corresponding orbit of the geodesic flow, and let $\tau_1(\gamma), \dots, \tau_{4q-2}(\gamma) > 1$ be the moduli of its expanding eigenvalues, counted with multiplicity. On the complexified middle exterior power $\Lambda^{2q-1} N_\gamma^*$ of the oriented normal bundle of γ , let $\varphi^\pm(\gamma)$ be parallel transport around γ restricted to the $\pm i$ -eigenspace of the normal Hodge-star operator. For $\text{Re } s$ sufficiently large, set

$$Z_\chi^\pm(s) = \prod_{[\gamma]_{\text{prim}}} \prod_{n_1, \dots, n_{4q-2} \geq 0} \det \left(I - \varphi^\pm(\gamma) \otimes \chi(\gamma) \exp \left[-s\ell_\gamma - \sum_{j=1}^{4q-2} \left(n_j + \frac{1}{2} \right) \log \tau_j(\gamma) \right] \right).$$

Prove that the locally integrable remainder terms in the closed-geodesic wave-trace expansions cancel in the quotient $Z_\chi^+(s)/Z_\chi^-(s)$, that this quotient extends meromorphically to the complex plane and is holomorphic and nonzero at $s = 0$, and that

$$\exp(\pi i \eta(A_\chi, 0)) = \frac{Z_\chi^+(0)}{Z_\chi^-(0)}.$$

Reference supplied by the database. I. M. Singer, “Eigenvalues of the Laplacian and invariants of manifolds,” in R. D. James (ed.), *Proceedings of the International Congress of*

Mathematicians, Vancouver 1974, Vol. 1, Canadian Mathematical Congress, Montreal, 1975, pp. 187–200, conjecture at the end of Section V, <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1974.1/ICM1974.1.ocr.pdf>.

6.1.3.2. Open problem 2

Let $q \geq 1$, and let M be a closed oriented Riemannian manifold of dimension $4q - 1$ with strictly negative sectional curvature. Let $\Gamma = \pi_1(M)$, and let $\chi : \Gamma \rightarrow O(m)$ be an orthogonal representation whose associated flat bundle E_χ is acyclic, so that $H^*(M; E_\chi) = 0$. For every primitive oriented closed geodesic γ , let ℓ_γ be its length and let $\varepsilon_\gamma \in \{1, -1\}$ be its orientation index, equal to 1 when the unstable bundle of the geodesic flow restricted to γ is orientable and to -1 otherwise. Define the twisted Ruelle zeta function for $\operatorname{Re} s$ sufficiently large by

$$R_\chi(s) = \prod_{[\gamma]_{\text{prim}}} \det(I - \varepsilon_\gamma \chi(\gamma) e^{-s\ell_\gamma}).$$

For the twisted Hodge Laplacians Δ_p^χ , define the Ray–Singer analytic torsion by

$$\log T(M, \chi) = \frac{1}{2} \sum_{p=0}^{\dim M} (-1)^p p \log \det' \Delta_p^\chi, \quad \log \det' \Delta_p^\chi = - \left. \frac{d}{dz} \right|_{z=0} \sum_{\lambda \in \operatorname{Spec}(\Delta_p^\chi) \setminus \{0\}} \lambda^{-z}.$$

Prove that $R_\chi(s)$ has a meromorphic continuation to the complex plane, is holomorphic and nonzero at $s = 0$, and satisfies

$$R_\chi(0) = T(M, \chi)^2.$$

Reference supplied by the database. David Fried, “Meromorphic zeta functions for analytic flows,” *Communications in Mathematical Physics* 174 (1995), 161–190, conjecture on p. 181, DOI: 10.1007/BF02099469.

6.1.4. Primary-source pointer

The originating ICM record is *Eigenvalues of the Laplacian and Invariants of Manifolds* (1974). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1974.1/ICM1974.1.ocr.pdf#page=243>.

6.2. 207. Algebraic Curves of Infinite Genus Arising in the Theory of Nonlinear Waves [1978-ag-18]

Record: 1978-ag-18

Subfield: Manifolds and global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1978: Algebraic Curves of Infinite Genus Arising in the Theory of Nonlinear Waves

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1978.2/ICM1978.2.ocr.pdf#page=273>

6.2.1. Problem statement

Let $Q = -D^2 + q(x)$ be Hill's operator with periodic potential, and let $\Delta(\lambda)$ be one-half the trace of the monodromy matrix of $Qf = \lambda f$. Assume the exceptional double spectra are absent, so the multiplier curve

$$\Omega = \{(\lambda, m) : m^2 - 2\Delta(\lambda)m + 1 = 0\}$$

is a nonsingular hyperelliptic curve with infinitely many branch points accumulating at the ramified point ∞ . Its genus is $g = \infty$. The roots of $\Delta = +1$ and $\Delta = -1$ give the periodic and antiperiodic spectra: the ground state is λ_0 , and the remaining spectral points occur in pairs $\lambda_{2j-1} \leq \lambda_{2j}$ tending to $+\infty$. The j -th lacuna is the interval $[\lambda_{2j-1}, \lambda_{2j}]$. Let DFK be the class of differentials of the first kind consisting of differentials

$$\omega = \frac{\varphi(\lambda) d\lambda}{\sqrt{\Delta(\lambda)^2 - 1}},$$

where φ is entire of order $1/2$ and type at most 1, satisfies $\varphi(\lambda^*) = \varphi(\lambda)^*$, satisfies

$$\int_0^\infty |\varphi(\lambda)|^2 \lambda^{3/2} d\lambda < \infty,$$

and has values on the lacunae that vanish rapidly as $j \rightarrow \infty$.

Formulate and prove a Riemann-Roch theorem for Ω and the class DFK that gives a useful replacement for the classical dimension and codimension terms in this infinite-genus setting, where such dimensions and codimensions may be infinite, and that is compatible with the following zero-counting behavior: every differential of the first kind on Ω with a zero of finite multiplicity $2m$ at ∞ has $2g - 2 - 2m$ additional finite zeros on Ω , counted in the special Hill-surface sense used for these curves; equivalently, it has $2g - 2$ zeros in all. Account also for the fact that a differential ω in DFK can have a zero of infinite multiplicity at ∞ without vanishing identically, but this cannot occur if, for example, the lacuna lengths decay exponentially: there exist constants $a, b > 0$ such that

$$\lambda_{2j} - \lambda_{2j-1} \leq a \exp(-bj)$$

as $j \rightarrow \infty$.

6.2.2. Historical background, literature, and current progress

6.2.3. Verdict: partially resolved

6.2.3.1. Precise problem checked

The problem asks for the status of a Riemann-Roch-type theorem for the infinite-genus hyperelliptic multiplier curve of a periodic Hill operator in the KdV case, using the McKean-Trubowitz/DFK class of holomorphic differentials

$$\omega = \frac{\varphi(\lambda)d\lambda}{\sqrt{\Delta(\lambda)^2 - 1}},$$

where φ has the stated entire-function, L^2 , reality, and lacuna-decay conditions. The unresolved part is not the construction of the Hill surface itself, but whether there is a useful Riemann-Roch theorem for this infinite-genus curve and this natural class of differentials/divisors: a theorem determining the relevant dimensions/codimensions and replacing the finite-genus formula without imposing extra hypotheses not present in the problem statement.

6.2.3.2. Literature found

1. H. P. McKean, "Algebraic curves of infinite genus arising in the theory of nonlinear waves," in *Proceedings of the International Congress of Mathematicians (Helsinki, 1978)*, vol. 2, Acad. Sci. Fennica, 1980, pp. 777-783. URL: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1978.2/ICM1978.2.ocr.pdf>. This is the apparent source for the problem. Live searches of the proceedings locate the explicit statement that the Riemann-Roch theorem is open for these Hill surfaces, while the zero-counting aspect and the possibility of infinite multiplicity at infinity are known.
2. H. P. McKean and E. Trubowitz, "Hill's surfaces and their theta functions," *Bulletin of the American Mathematical Society* 84, no. 6 (1978), 1042-1085. DOI: 10.1090/S0002-9904-1978-14542-X. URL: <https://projecteuclid.org/journals/bulletin-of-the-american-mathematical-society/volume-84/issue-6/Hills-surfaces-and-their-theta-functions/bams/1183541462.pdf>. This is the main Hill-surface/theta-function reference. It develops the infinite-genus Hill surface, DFK-type differentials, Abel sums, and theta functions, but does not give a full Riemann-Roch theorem for the DFK class.
3. H. P. McKean and E. Trubowitz, "Hill's operator and hyperelliptic function theory in the presence of infinitely many branch points," *Communications on Pure and Applied Mathematics* 29 (1976), 143-226. DOI: 10.1002/cpa.3160290203. This gives the foundational hyperelliptic function theory for Hill operators with infinitely many branch points. It is essential background but not a full Riemann-Roch resolution.
4. M. U. Schmidt, "Integrable systems and Riemann surfaces of infinite genus," *Memoirs of the American Mathematical Society* 122, no. 581 (1996), viii+111 pp.; preprint arXiv:solv-int/9412006. URL: <https://arxiv.org/pdf/solv-int/9412006>. Schmidt proves a Riemann-Roch theorem for divisors of "Riemann-Roch type" using modified cohomology. The theorem is important partial progress, but its own formulation restricts to divisors satisfying special asymptotic/trivialization conditions; the paper notes that identifying enough such divisors is a separate main problem. Thus it does not cover all DFK differentials/divisors on arbitrary Hill curves in the problem statement.

5. Franz Merkl, “A Riemann Roch Theorem for infinite genus Riemann surfaces,” *Inventiones Mathematicae* 139, no. 2 (2000), 391-437. DOI: 10.1007/s002229900031. URL: <http://research.tue.nl/en/publications/a-riemann-roch-theorem-for-infinite-genus-riemann-surfaces>. Merkl proves a Riemann-Roch theorem for a class of holomorphic line bundles over infinite-genus Riemann surfaces, with pointwise asymptotic growth conditions and asymptotic bounds on gluing and transition functions. This is a genuine infinite-genus Riemann-Roch theorem, but the hypotheses and section spaces are not shown to coincide with the McKean-Trubowitz DFK class for every nonsingular Hill multiplier curve in the problem statement.
6. Joel Feldman, Horst Knorrer, and Eugene Trubowitz, *Riemann Surfaces of Infinite Genus*, CRM Monograph Series 20, American Mathematical Society, Providence, 2003. DOI: 10.1090/crmm/020. URL: <https://bookstore.ams.org/CRMM/20> and author PDF <https://www.math.ubc.ca/~feldman/papers/allriem.pdf>. The monograph develops a broad theory for marked infinite-genus surfaces satisfying explicit geometric hypotheses, including Royden-Riemann-Roch for finite divisors on parabolic surfaces and theta-function results. This is not the requested full Hill/DFK Riemann-Roch theorem: it works with different hypotheses and does not settle the dimensions/codimensions for the DFK class in the problem statement.
7. Ilya Zakharevich, “Curves of infinite genus I. Riemann–Roch theorem for small degree,” arXiv:math/0202057, 2002. URL: <https://arxiv.org/abs/math/0202057>; and “Quasi-algebraic geometry of curves I. Riemann-Roch theorem and Jacobian,” arXiv:alg-geom/9710013, 1997. URL: <https://arxiv.org/abs/alg-geom/9710013>. These preprints develop Riemann-Roch analogues for infinite-genus curves with growth-at-infinity conditions and handle-length restrictions. They are related partial progress, but they are not a published theorem covering the full Hill/DFK problem as stated.
8. Thomas Kappeler and Peter Topalov, “On normalized differentials on hyperelliptic curves of infinite genus,” *Journal of Differential Geometry* 105, no. 2 (2017), 209-248. DOI: 10.4310/jdg/1486522814. arXiv:1408.7107, URL: <https://arxiv.org/abs/1408.7107>. This constructs normalized differentials on infinite-genus hyperelliptic curves and gives zero-distribution estimates. It is strong later progress on differentials but not a Riemann-Roch theorem for the DFK class.
9. Kenneth T-R. McLaughlin and Patrik V. Nabelek, “A Riemann-Hilbert problem approach to infinite gap Hill’s operators and the Korteweg-de Vries equation,” *International Mathematics Research Notices* 2021, no. 2, 1288-1348. DOI: 10.1093/imrn/rnz156. arXiv:1810.07818, URL: <https://arxiv.org/abs/1810.07818>. This gives a Riemann-Hilbert formulation and uniqueness results for periodic infinite-gap Hill operators. It is current Hill/KdV infinite-gap literature, but it does not prove the missing Riemann-Roch theorem.

6.2.3.3. Why this is not fully resolved

The later literature does contain several Riemann-Roch-type theorems for infinite-genus surfaces, but none found in the search proves a theorem covering exactly the full setting: arbitrary nonsingular KdV/Hill multiplier curves with the McKean-Trubowitz DFK class and no extra exponential/asymptotic/trivialization hypotheses beyond those in the statement. The available results either:

- treat finite divisors or different function spaces;

- impose explicit geometric, gluing, handle-length, or growth conditions;
- restrict to divisors of “Riemann-Roch type” or admissible line bundles;
- prove normalized differential or inverse spectral/Riemann-Hilbert results rather than Riemann-Roch dimensions for DFK.

Therefore the correct current status is **partially resolved**: the original zero-counting and special-hypothesis results remain established, and substantial related infinite-genus Riemann-Roch theory exists, but the full Hill/DFK Riemann-Roch problem is not fully resolved by citable literature.

6.2.4. Remaining open problems recorded by the database

6.2.4.1. Open problem 1

For a completed infinite-genus spectral curve Y in Schmidt’s framework, find and geometrically characterize sufficiently many divisors of Riemann–Roch type, beyond the classes covered by the known generic separation and admissibility criteria, so that the infinite-genus Riemann–Roch theorem applies to the spectral line bundles of interest.

Reference supplied by the database. Martin U. Schmidt, “Integrable Systems and Riemann Surfaces of Infinite Genus,” *Memoirs of the American Mathematical Society* 122, no. 581 (1996), viii+111 pp., §5, immediately after Serre Duality Theorem 5.3 (explicitly identified as “the main problem”). ISBN: 978-0-8218-0445-2. DOI: 10.1090/memo/0581. arXiv: solv-int/9412006. <https://arxiv.org/abs/solv-int/9412006>

6.2.5. Primary-source pointer

The originating ICM record is *Algebraic Curves of Infinite Genus Arising in the Theory of Nonlinear Waves* (1978). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1978.2/ICM1978.2.ocr.pdf#page=273>.

6.3. 208. Some Problems in Complex Analytic Geometry with Growth Conditions [1978-geom-27]

Record: 1978-geom-27

Subfield: Manifolds and global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1978: Some Problems in Complex Analytic Geometry with Growth Conditions

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1978.2/ICM1978.2.ocr.pdf#page=141>

6.3.1. Problem statement

Let $X \subset \mathbb{C}^n$ be a pure d -dimensional entire analytic set, let

$$X[r] = X \cap \{z : \|z\| < r\},$$

let φ be the standard Kahler form on $\mathbb{C}^n$, and let Ω_X be the curvature matrix of X at its smooth points. Define the Chern forms $c_i(\Omega_X)$ by

$$\det \left(tI + \frac{\sqrt{-1}}{2\pi} \Omega_X \right) = \sum_{i=0}^d (-1)^i t^{d-i} c_i(\Omega_X),$$

and, for $0 \leq i \leq d$, define

$$\mu_i(X, r) = \frac{1}{(d-i)! r^{2(d-i)}} \int_{X[r]} c_i(\Omega_X) \wedge \varphi^{d-i}.$$

Set

$$\mu(X, r) = \sum_{i=0}^d \mu_i(X, r).$$

Determine whether there is a Bezout-type estimate bounding the refined growth indicator $\mu(V \cap W, r)$ of the intersection of entire analytic subvarieties $V, W \subset \mathbb{C}^n$ in terms of the refined growth indicators $\mu(V, r)$ and $\mu(W, r)$.

6.3.2. Historical background, literature, and current progress

6.3.3. Verdict: partially resolved

6.3.3.1. Precise problem

The problem asks whether Griffiths's refined curvature-integral growth invariant for a pure-dimensional entire analytic set $X \subset \mathbb{C}^n$,

$$\mu(X, r) = \sum_i \mu_i(X, r), \quad \mu_i(X, r) = \frac{1}{(d-i)! r^{2(d-i)}} \int_{X[r]} c_i(\Omega_X) \wedge \varphi^{d-i},$$

satisfies a Bezout-type inequality for intersections: namely, whether the refined growth indicator of $V \cap W$ can be bounded in terms of the refined growth indicators of the entire analytic subvarieties V and W themselves, without restricting to the ordinary Euclidean volume growth term.

6.3.3.2. Literature checked

1. Phillip A. Griffiths, "Some Problems in Complex Analytic Geometry with Growth Conditions," Proceedings of the International Congress of Mathematicians, Helsinki 1978, pp. 645-651, Acad. Sci. Fennica, 1980. URL: <https://publications.ias.edu/sites/default/files/someproblems.pdf>. This is the original source of the question. Griffiths states the ordinary transcendental Bezout problem, recalls that the ordinary estimate holds in some hypersurface/linear-section settings but fails in higher codimension, then proposes the curvature-integral refinement and asks whether there is a Bezout estimate for $\mu(V, r) = \sum_i \mu_i(V, r)$.
2. Phillip A. Griffiths, "Complex differential and integral geometry and curvature integrals associated to singularities of complex analytic varieties," Duke Mathematical Journal 45 no. 3 (1978), 427-512. DOI: 10.1215/S0012-7094-78-04522-2. URL: <https://publications.ias.edu/sites/default/files/complexdiff.pdf>. This paper develops the Chern-form curvature integrals, monotonicity, tube-volume interpretation, and complex Crofton/kinematic formulas underlying the refined indicator. It supplies the geometric framework for the question, but it does not prove a global Bezout inequality for $\mu(V \cap W, r)$ in terms of $\mu(V, r)$ and $\mu(W, r)$.
3. Wilhelm Stoll, "A Bezout estimate for complete intersections," Annals of Mathematics (2) 96 no. 2 (1972), 361-401. DOI: 10.2307/1970793. URL: <https://annals.math.princeton.edu/1972/96-2/p07>. Stoll proves a Bezout estimate in the complete-intersection/value-distribution setting. This is important partial progress on the classical transcendental Bezout problem, but it is for the ordinary growth framework and special complete-intersection hypotheses, not the full curvature-integral refined indicator for arbitrary analytic subvariety intersections.
4. Maurizio Cornalba and Bernard Shiffman, "A counterexample to the 'Transcendental Bezout problem'," Annals of Mathematics (2) 96 no. 2 (1972), 402-406. DOI: 10.2307/1970794. URL: <https://annals.math.princeton.edu/1972/96-2/p08>. This disproves the naive ordinary maximum-modulus/volume version in dimension at least two. It motivates Griffiths's refined curvature question but does not settle the refined question.
5. Phillip A. Griffiths, "On the Bezout problem for entire analytic sets," Annals of Mathematics (2) 100 no. 3 (1974), 533-552. DOI: 10.2307/1970957. URL: <https://annals.math.princeton.edu/1974/100-3/p02>. This gives partial positive results for the ordinary entire-analytic-set Bezout problem, especially around curves and linear sections. It does not prove the later 1978 refined Chern-form growth estimate for all V, W .
6. James A. Carlson, "A moving lemma for the transcendental Bezout problem," Annals of Mathematics (2) 103 no. 2 (1976), 305-330. DOI: 10.2307/1971008. URL: <https://annals.math.princeton.edu/1976/103-2/p06>. Carlson proves moving/generic-intersection results for the transcendental Bezout problem. These are partial results under movement/genericity and ordinary growth notions; they do not imply a bound for the refined curvature indicator of arbitrary intersections.

7. James R. King, “Refined residues, Chern forms, and intersections,” in *Value Distribution Theory, Part A, Pure and Applied Mathematics 25*, Marcel Dekker, New York, 1974, pp. 169-190. MR50 #7565; Zbl 0307.32002. King’s work is relevant to Chern forms and intersection theory, but the available citations and later use concern residue/intersection formulas rather than a full transcendental Bezout growth estimate for Griffiths’s $\sum_i \mu_i$.
8. Shanyu Ji, “Bezout estimate for entire holomorphic maps and their Jacobians,” *American Journal of Mathematics* 117 no. 2 (1995), 395-403. DOI: 10.2307/2374920. URL: <https://www.jstor.org/stable/2374920>. Ji obtains Bezout-type estimates for entire maps under hypotheses involving the Jacobian. This is a substantial special/conditional result for maps and zeros, but the hypotheses and conclusion do not cover arbitrary analytic subvarieties V, W with Griffiths’s refined curvature growth.
9. B. Q. Li and B. A. Taylor, “On the Bezout problem and area of interpolating varieties in $\mathbb{C}^n$,” *American Journal of Mathematics* 118 no. 5 (1996), 989-1010; and B. Q. Li, “On the Bezout problem and area of interpolating varieties in $\mathbb{C}^n$, II,” *American Journal of Mathematics* 120 no. 6 (1998), 1191-1198. These papers give further partial results for area/interpolation formulations of the Bezout problem. They do not address the full refined Chern-form growth indicator for arbitrary intersections.
10. Lev Buhovsky, Iosif Polterovich, Leonid Polterovich, Egor Shelukhin, and Vukasin Stojisavljevic, “Persistent transcendental Bezout theorems,” *Forum of Mathematics, Sigma* 12 (2024), e72, 1-28. DOI: 10.1017/fms.2024.49. arXiv:2307.02937. URL: <https://www.cambridge.org/core/journals/forum-of-mathematics-sigma/article/persistent-transcendental-bezout-theorems/C973B8F28E9FAD19237DEE5DD2AE0426>. This recent paper proves bounds for modified coarse/persistent zero counts of entire maps. It explicitly changes the counting invariant to a persistence-based coarse count and uses maximum-modulus growth, so it is not a resolution of Griffiths’s curvature-integral refined Bezout problem.

6.3.3.3. What remains unresolved

I found no published or otherwise citable theorem whose hypotheses and conclusion imply the full statement in the problem statement: a Bezout-type estimate bounding the refined curvature-integral growth indicator $\mu(V \cap W, r)$ of arbitrary intersections of pure-dimensional entire analytic subvarieties in $\mathbb{C}^n$ in terms of the corresponding refined indicators of V and W .

The literature gives important partial progress and related variants, including complete-intersection estimates, moving/generic-intersection results, estimates with Jacobian or interpolation hypotheses, average/area estimates, and recent persistent/coarse zero-count bounds. Each changes the invariant, adds extra hypotheses, or treats a special setting. Thus the exact Griffiths refined curvature-integral Bezout question remains unresolved in full.

6.3.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website’s “Some remaining open problems” field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

6.3.5. Primary-source pointer

The originating ICM record is *Some Problems in Complex Analytic Geometry with Growth Conditions* (1978). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1978.2/ICM1978.2.ocr.pdf#page=141>.

6.4. 209. Spectral geometry and non-compact Riemannian manifolds [1983-geom-34]

Record: 1983-geom-34

Subfield: Manifolds and global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1983: Spectral geometry and non-compact Riemannian manifolds

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1983.1/ICM1983.1.ocr.pdf#page=627>

6.4.1. Problem statement

Prove that for every discrete torsion-free subgroup $\Gamma \subset \mathrm{SL}(2, \mathbb{R})$ for which the hyperbolic quotient $\Gamma \backslash \mathbb{H}$ has finite volume, where $\mathbb{H}$ is the upper half-plane with the Poincaré metric, the self-adjoint Laplace-Beltrami operator on L^2 functions on $\Gamma \backslash \mathbb{H}$ has infinitely many eigenvalues in its point spectrum.

6.4.2. Historical background, literature, and current progress

6.4.3. Verdict: partially resolved

6.4.3.1. Precise problem

The statement asks whether every torsion-free cofinite Fuchsian group $\Gamma \subset \mathrm{SL}(2, \mathbb{R})$ has the property that the Laplace-Beltrami operator on $L^2(\Gamma \backslash \mathbb{H})$ has infinitely many eigenvalues in its point spectrum.

For cocompact Γ , this is a standard consequence of compactness and Weyl's law. The real Roelcke-Selberg problem is the noncompact finite-area case, where the continuous spectrum is $[1/4, \infty)$ and only the discrete part of the spectrum is counted. Since the residual spectrum and the eigenvalues below $1/4$ are finite in the usual finite-area setting, the question is essentially whether every torsion-free nonuniform lattice has infinitely many cuspidal L^2 eigenfunctions.

6.4.3.2. Verdict

Partially resolved. The full statement for all torsion-free cofinite noncompact Fuchsian groups is not resolved by citable literature found in the search. It is true in important special cases, especially congruence/arithmetic groups, but no valid theorem found covers all cofinite torsion-free groups. There is also strong contrary/generic evidence, from Phillips-Sarnak type deformation theory and later work, that the full universal assertion may fail.

6.4.3.3. Original and background sources

- Title: Über die Wellengleichung bei Grenzkreisgruppen erster Art Authors: Walter Roelcke Venue: Sitzungsberichte der Heidelberger Akademie der Wissenschaften, Mathematisch-naturwissenschaftliche Klasse 1953/1955, 159-267 Year: 1956 DOI/arXiv/URL: MR18,476d Relevance: One of Roelcke's original spectral-analysis sources for finite area Fuchsian quotients. Later literature identifies the infinite-discrete-spectrum expectation as the Roelcke-Selberg conjecture or hypothesis, though Petridis-Risager note that it is unclear how formally Roelcke and Selberg stated it as a conjecture.
- Title: Das Eigenwertproblem der automorphen Formen in der hyperbolischen Ebene. I, II Authors: Walter Roelcke Venue: Mathematische Annalen Year: 1966-1967 DOI/arXiv/URL: Part I, Math. Ann. 167 (1966), 292-337; Part II, Math. Ann. 168 (1967), 261-324 Relevance: Foundational source on the automorphic Laplace eigenvalue problem. It does not settle the modern all-cofinite-groups question.
- Title: Harmonic analysis and discontinuous groups in weakly symmetric Riemannian spaces with applications to Dirichlet series Authors: Atle Selberg Venue: Journal of the Indian Mathematical Society (N.S.) 20, 47-87 Year: 1956 DOI/arXiv/URL: MR0088511 Relevance: Selberg's trace formula proves abundance/Weyl-law results for congruence groups. This gives infinitely many cuspidal eigenvalues for the relevant torsion-free congruence subgroups, but it is an arithmetic special case, not the full statement.

6.4.3.4. Later progress and limitations

- Title: Artin-Takagi formula for the Selberg zeta function and the Roelcke hypothesis Authors: Alexei B. Venkov Venue: Doklady Akademii Nauk SSSR 247, no. 3, 540-543 Year: 1979 DOI/arXiv/URL: Soviet Math. Dokl. translation 20, no. 4, 745-748 Relevance: Gives further arithmetic/commensurability-type progress around the Roelcke hypothesis. It is not a proof for arbitrary torsion-free cofinite Fuchsian groups.
- Title: On cusp forms for co-finite subgroups of $\mathrm{PSL}(2, \mathbb{R})$ Authors: Ralph Phillips; Peter Sarnak Venue: Inventiones Mathematicae 80, no. 2, 339-364 Year: 1985 DOI/arXiv/URL: <https://doi.org/10.1007/BF01388610> Relevance: Developed the deformation/instability theory of cusp forms for cofinite groups. This work is central to why the original universal Roelcke-Selberg expectation is no longer generally believed, but it does not itself give a full counterexample to the torsion-free quotient statement.
- Title: The Weyl theorem and the deformation of discrete groups Authors: Ralph Phillips; Peter Sarnak Venue: Communications on Pure and Applied Mathematics 38, 853-866 Year: 1985 DOI/arXiv/URL: MR87f:11035 Relevance: Shows how Weyl-law behavior for discrete spectrum can be destroyed under deformation. This is strong partial/generic evidence, not a complete resolution of all cases.
- Title: Spectra of hyperbolic surfaces Authors: Peter Sarnak Venue: Bulletin of the American Mathematical Society (N.S.) 40, no. 4, 441-478 Year: 2003 DOI/arXiv/URL: <https://doi.org/10.1090/S0273-0979-03-00991-1> Relevance: Survey source. It records that modular/congruence surfaces have abundant Maass cusp forms, while for general finite-area hyperbolic surfaces the existence problem remains subtle. It states Phillips-Sarnak type conjectures that generic finite-area surfaces are not essentially cuspidal and, except for the once-punctured torus Teichmüller space, generically have only finitely many discrete eigenvalues. These conjectures are incompatible with a naive universal Weyl-law version and show the full statement is not settled by the known arith-

metric results.

- Title: Hyperbolic triangles without embedded eigenvalues Authors: Luc Hillairet; Chris Judge Venue: *Annals of Mathematics* (2) 187, no. 2, 301-377 Year: 2018 DOI/arXiv/URL: <https://doi.org/10.4007/annals.2018.187.2.1> ; arXiv:1402.4533 Relevance: Proves that generic one-cusp hyperbolic triangles with Neumann boundary have no nonconstant square-integrable eigenfunctions. The paper also explicitly presents the corresponding finite-area hyperbolic-surface question as a basic unresolved problem. This is important negative model evidence but not a counterexample to the problem statement, because the result concerns triangle Neumann problems/model orbifold-type settings, not every torsion-free quotient $\Gamma \backslash \mathbb{H}$.
- Title: Higher Order Deformations of Hyperbolic Spectra Authors: Yiannis N. Petridis; Morten S. Røisager Venue: *Schrödinger Operators, Spectral Analysis and Number Theory*, Springer Proceedings in Mathematics & Statistics 348, Springer, Cham, 223-239 Year: 2021 DOI/arXiv/URL: https://doi.org/10.1007/978-3-030-68490-7_11 ; arXiv:1907.00800 Relevance: Modern review of the Roelcke-Selberg conjecture and deformation theory. It states that for a finite-volume hyperbolic quotient the discrete spectrum may be finite or infinite, that Selberg proved Weyl law for congruence groups, and that the belief $N_d(\lambda) \rightarrow \infty$ for general cofinite groups is called the Roelcke-Selberg conjecture. It also explains that the difficulty is counting embedded eigenvalues in the continuous spectrum and that Phillips-Sarnak's work weakened general belief in the conjecture.

6.4.3.5. Claimed full resolution not accepted

- Title: Selberg Formula for Cofinite Groups and the Roelke Conjecture Authors: Dmitry A. Popov Venue: arXiv preprints Year: 2015/2017, withdrawn 2019 DOI/arXiv/URL: arXiv:1506.08886 and arXiv:1705.05612 Relevance: These preprints claimed to prove the Roelke conjecture for arbitrary cofinite Fuchsian groups, but arXiv records the later version arXiv:1705.05612 as withdrawn and unavailable. The earlier arXiv record states that a reviewer pointed out an unproved assertion and that the preprint was withdrawn. Therefore this cannot be treated as published or citable resolving literature.

6.4.3.6. What remains unresolved

No located published or otherwise reliable citable theorem proves that every torsion-free non-compact cofinite $\Gamma \subset \mathrm{SL}(2, \mathbb{R})$ has infinitely many L^2 point-spectrum eigenvalues. The known positive theorems cover compact quotients and arithmetic/congruence families; the general nonarithmetic finite-area torsion-free case remains unresolved.

6.4.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

6.4.5. Primary-source pointer

The originating ICM record is *Spectral geometry and non-compact Riemannian manifolds* (1983). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1983.1/ICM1983.1.ocr.pdf#page=627>.

6.5. 210. Nishio) Trace Formulae in Spectral Geometry [1990-geom-47]

Record: 1990-geom-47

Subfield: Manifolds and global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1990: Nishio) Trace Formulae in Spectral Geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1990.1/ICM1990.1.ocr.pdf#page=665>

6.5.1. Problem statement

Let M be a closed nonpositively curved Riemannian manifold, let $X = \widetilde{M}$ be its universal covering with deck transformation group $\Gamma = \pi_1(M)$, and let Δ_X be the Laplacian on X . For the identity conjugacy class $[1] \subset \Gamma$, equivalently the null-homotopic free homotopy class of closed curves in M , define the even wave-trace distribution $\Theta_{[1]}$ on $\mathbb{R}$ by

$$\langle \Theta_{[1]}, f \rangle = \int_{\mathcal{D}} U_f(x, x) dx, \quad f \in C_c^\infty(\mathbb{R}),$$

where $\mathcal{D}$ is a fundamental domain for the Γ -action on X , and $U_f(x, y)$ is the kernel of

$$\hat{f}(\sqrt{\Delta_X}) = \int_{-\infty}^{\infty} f(t) \cos(t\sqrt{\Delta_X}) dt, \quad \hat{f}(s) = \int_{-\infty}^{\infty} f(t) \cos(st) dt.$$

Prove that $\Theta_{[1]}$, regarded as a function away from its possible singularity at $t = 0$, satisfies

$$\Theta_{[1]}(t) = O(|t|^{-\alpha}) \quad \text{as } t \rightarrow +\infty \text{ and as } t \rightarrow -\infty$$

for some $\alpha > 0$.

6.5.2. Historical background, literature, and current progress

6.5.3. Status: partially resolved

6.5.3.1. Precise problem checked

The problem asks whether, for every closed nonpositively curved Riemannian manifold M , the identity-conjugacy-class term in the Nishio-Sunada wave trace formula

$$\Theta_{[1]}(t) = \text{Tr}_\Gamma (\cos(t\sqrt{\Delta_{\widetilde{M}}}))$$

is a smooth function for $t \neq 0$ whose long-time size satisfies $\Theta_{[1]}(t) = O(|t|^{-\alpha})$ as $t \rightarrow \pm\infty$ for some $\alpha > 0$.

6.5.3.2. Verdict

I did not find a published or citable result proving this full statement for all closed nonpositively curved Riemannian manifolds. The literature gives the trace-formula framework, temperedness of the identity term, related heat-kernel/density-of-states results, and special homogeneous cases, but not the stated pointwise polynomial decay of the wave trace in the full variable-curvature nonpositive-curvature setting.

6.5.3.3. Relevant literature and what it proves

1. Trace formulae in spectral geometry

Authors: Toshikazu Sunada and Manabu Nishio.

Venue/year: Proceedings of the International Congress of Mathematicians, Kyoto 1990, Vol. I, Springer-Verlag, Tokyo, 1991, pp. 577-585.

DOI/arXiv/URL: no DOI found; full text/source page: https://www.researchgate.net/publication/236210645_Trace_formulae_in_spectral_geometry; ICM proceedings OCR: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1990.2/ICM1990.2.ocr.pdf>

Relevance: This is the original source for the conjugacy-class distributions $\Theta_{[\sigma]}$. It proves the generalized trace formula, support/singular-support statements, and that the identity-class distribution is tempered. Temperedness is much weaker than $\Theta_{[1]}(t) = O(|t|^{-\alpha})$ as a function at infinity.

2. An overview of Sunada's work up to age 60

Authors: Atsushi Katsuda and Polly W. Sy.

Venue/year: Contemporary Mathematics 484, *Spectral Analysis in Geometry and Number Theory*, 2009, pp. 7-42.

DOI/arXiv/URL: DOI 10.1090/conm/484/09462; PDF: <https://www.mims.meiji.ac.jp/publications/2008/abst00013.pdf>

Relevance: The survey restates the Nishio-Sunada generalized trace formula and the explicit formula for $\Theta_{[\sigma]}$. It confirms the framework and known properties but does not state a resolution of the polynomial decay question.

3. On the wave equation on a compact Riemannian manifold without conjugate points

Author: Pierre H. Bérard.

Venue/year: Mathematische Zeitschrift 155, 1977, pp. 249-276.

DOI/arXiv/URL: DOI 10.1007/BF02028444.

Relevance: Bérard's no-conjugate-points wave parametrix is foundational for later spectral estimates on such manifolds. It gives strong control useful for logarithmic Weyl remainders, but it does not prove long-time polynomial decay of the von Neumann wave trace on the universal cover.

4. Density of states in spectral geometry

Authors: Toshiaki Adachi and Toshikazu Sunada.

Venue/year: Commentarii Mathematici Helvetici 68, 1993, pp. 480-493.

DOI/arXiv/URL: DOI 10.1007/BF02565831; Springer page: <https://link.springer.com/article/10.1007/BF02565831>

Relevance: Develops density-of-states methods in spectral geometry related to the von Neumann trace. This is relevant background for $\Theta_{[1]}$, but it concerns density-of-states/heat-type spectral information rather than the required pointwise large- $|t|$ decay of the wave trace.

5. Von Neumann spectra near the spectral gap

Authors: Alan L. Carey, Thierry Coulhon, Varghese Mathai, and John Phillips.

Venue/year: Bulletin des Sciences Mathématiques 122, 1998, pp. 203-242.

DOI/arXiv/URL: DOI 10.1016/S0007-4497(98)80088-2; arXiv:dg-ga/9610019, <https://arxiv.org/abs/dg-ga/9610019>

Relevance: Studies von Neumann heat-theta functions on universal covers and introduces invariants $\beta_j(M, g)$. The paper notes that Sunada's positivity conjecture for the function case $\beta_0(M, g) > 0$ was proved by Terry Lyons. This gives related polynomial heat-trace information, but a heat-trace estimate does not by itself imply the stated oscillatory wave-trace decay.

6. Asymptotic behaviour of spectra of compact quotients of certain symmetric spaces

Author: Ramesh Gangolli.

Venue/year: Acta Mathematica 121, 1968, pp. 151-192.

DOI/arXiv/URL: DOI 10.1007/BF02391912.

Relevance: In locally symmetric/nonpositive-curvature settings, the identity contribution can be analyzed through the Selberg trace formula and Harish-Chandra Plancherel theory. These homogeneous cases are important supporting examples, but they impose much stronger hypotheses than an arbitrary closed nonpositively curved manifold.

6.5.3.4. What remains unresolved

The missing result is a theorem covering every closed nonpositively curved Riemannian manifold, with no local-symmetry, rank-one, pinching, analyticity, or other extra assumptions, proving that the identity-class wave trace is represented away from 0 by a function with some polynomial decay at both infinities. Existing results establish temperedness and related heat/density-of-states estimates, but I found no citable theorem converting those into the exact wave-trace bound required in the problem statement.

6.5.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

6.5.5. Primary-source pointer

The originating ICM record is *Nishio* *Trace Formulae in Spectral Geometry* (1990). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1990.1/ICM1990.1.ocr.pdf#page=665>.

6.6. 211. Static and Moving Defects in Liquid Crystals [1990-geom-239]

Record: 1990-geom-239

Subfield: Manifolds and global analysis

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1990: Static and Moving Defects in Liquid Crystals

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1990.2/ICM1990.2.ocr.pdf#page=411>

6.6.1. Problem statement

Let $E_g(u; \Omega) = \int_{\Omega} |Du|_g^2 dx$ be the Dirichlet energy of a map into the two-sphere S^2 equipped with a smooth Riemannian metric g . An energy-minimizing tangent map is a nonconstant map $\Phi \in W_{\text{loc}}^{1,2}(\mathbb{R}^m, S^2)$ such that $\Phi(\lambda x) = \Phi(x)$ for every $\lambda > 0$ and almost every $x \in \mathbb{R}^m$, and such that $E_g(\Phi; U) \leq E_g(v; U)$ for every bounded open set $U \subseteq \mathbb{R}^m$ and every $v \in W^{1,2}(U, S^2)$ having the same trace as Φ on ∂U . Classify all energy-minimizing tangent maps (i) when $m \geq 4$ and g is the standard round metric, and (ii) when $m = 3$ and g is an arbitrary smooth non-round metric on S^2 .

6.6.2. Historical background, literature, and current progress

Status: unresolved.

No citable result located gives an exhaustive classification in either requested branch. The literature classifies the three-dimensional round-target benchmark and constructs or constrains important higher-dimensional and non-round examples, but it does not cover every $m \geq 4$ tangent map or every smooth metric on S^2 .

- Haïm Brezis, Jean-Michel Coron, and Elliott H. Lieb, “Harmonic Maps with Defects,” *Communications in Mathematical Physics* 107 (1986), 649–705, DOI [10.1007/BF01205490](https://doi.org/10.1007/BF01205490). Their classification proves that, for $\mathbb{R}^3$ and the round S^2 , the only nonconstant homogeneous minimizers are $x \mapsto \pm Rx/|x|$. This is the solved base case, not either of the two cases requested here.
- Jean-Michel Coron and Robert Gulliver, “Minimizing p-Harmonic Maps into Spheres,” *Journal für die reine und angewandte Mathematik* 401 (1989), 82–100, DOI [10.1515/crll.1989.401.82](https://doi.org/10.1515/crll.1989.401.82). In particular, they prove that the degree-zero homogeneous extension of the complex Hopf fibration $S^3 \rightarrow S^2$ is energy minimizing on B^4 . Thus the $m = 4$ branch already contains a genuinely non-cylindrical example, but the paper does not classify all minimizers $\mathbb{R}^4 \rightarrow S^2$, much less all $m \geq 4$.
- Robert Hardt and Fang-Hua Lin, “The Singular Set of an Energy Minimizing Map from B^4 to S^2 ,” *Manuscripta Mathematica* 69 (1990), 275–289, DOI [10.1007/BF02567926](https://doi.org/10.1007/BF02567926). They obtain strong structural and compactness information for four-dimensional minimizers and their tangent maps, including the local finite-curve description of the singular set, but not an exhaustive list of four-dimensional tangent maps.

- Heayong Shin, “Degree 1 Singularities of Energy Minimizing Maps to a Bumpy Sphere,” *Mathematische Annalen* 293 (1992), 509–521, DOI 10.1007/BF01444732. For every smooth metric on S^2 , Shin proves existence and uniqueness up to domain rotation of a degree-one minimizing tangent map. This classifies the degree-one sector only; it does not rule out minimizing tangent maps of larger absolute degree for an arbitrary metric.
- Fang-Hua Lin, “Singular Sets of Energy Minimizing Maps,” *RIMS Kôkyûroku* 770 (1991), 161–170, [direct PDF](#). Lin proves that if the target metric is sufficiently C^1 -close to the round metric, every three-dimensional minimizing tangent map has absolute degree one. Section 4 explicitly leaves removal of that closeness assumption open. Consequently, arbitrary smooth non-round metrics remain uncovered.
- Michał Miśkiewicz, “On Hölder Regularity of the Singular Set of Energy Minimizing Harmonic Maps into Closed Manifolds,” *Calculus of Variations and Partial Differential Equations* 59 (2020), article 36, DOI 10.1007/s00526-019-1685-y. This later work uses the Brezis–Coron–Lieb classification to identify the maximally cylindrical tangent maps in every domain dimension and derives regularity of the top-dimensional singular stratum. It expressly treats only that symmetry class, leaving lower-symmetry higher-dimensional tangent maps unclassified.

Therefore the strongest verified progress consists of the complete $m = 3$, round-metric classification; the Hopf and cylindrical higher-dimensional families plus structural bounds; uniqueness within the degree-one non-round sector; and a full non-round classification only for metrics sufficiently close to round. The remaining gaps are an exhaustive classification of lower-symmetry tangent maps for $m \geq 4$ and exclusion or description of higher-degree tangent maps for arbitrary smooth metrics on S^2 .

6.6.3. Remaining open problems recorded by the database

6.6.3.1. Open problem 1

Let g be an arbitrary smooth Riemannian metric on S^2 . Prove or disprove that every nonconstant map $\Phi \in W_{\text{loc}}^{1,2}(\mathbb{R}^3, S^2)$ which is homogeneous of degree zero and locally minimizes the Dirichlet energy $\int |D\Phi|_g^2$ has $|\deg(\Phi|_{S^2})| = 1$.

Reference supplied by the database. Fang Hua Lin, “Singular Sets of Energy Minimizing Maps,” *RIMS Kôkyûroku* 770 (1991), 161–170, Section 4 (“Open problems”), <https://www.kurims.kyoto-u.ac.jp/~kyodo/kokyuroku/contents/pdf/0770-13.pdf>.

6.6.4. Primary-source pointer

The originating ICM record is *Static and Moving Defects in Liquid Crystals* (1990). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1990.2/ICM1990.2.ocr.pdf#page=411>.

6.7. 212. Positive scalar curvature metrics - Existence and classification questions [1994-geom-60]

Record: 1994-geom-60

Subfield: Manifolds and global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1994: Positive scalar curvature metrics - Existence and classification questions

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.1/ICM1994.1.ocr.pdf#page=697>

6.7.1. Problem statement

For every discrete group π and integer n , let $B\pi$ denote the classifying space of π , and let $C_r^*\pi$ be the reduced real C^* -algebra of π . Let $R_n(B\pi)$ be Stolz's relative spin bordism group: its elements are represented by triples (N, f, h) , where N is a compact n -dimensional spin manifold, possibly with boundary, $f : N \rightarrow B\pi$ is a map, and h is a positive scalar curvature metric on ∂N ; two such triples are bordant if their boundary triples are joined by a positive-scalar-curvature bordism and the closed spin manifold obtained by gluing the two representatives to that boundary bordism bounds over $B\pi$. The index homomorphism

$$\theta : R_n(B\pi) \longrightarrow KO_n(C_r^*\pi)$$

sends the class of (N, f, h) to the real K -theoretic index of the Dirac operator on N determined by f and the boundary metric h . Fix a Bott manifold $\mathcal{B}$, meaning a simply connected 8-dimensional spin manifold with $\hat{A}(\mathcal{B}) = 1$. Taking products with $\mathcal{B}$ gives homomorphisms $R_{n+8k}(B\pi) \rightarrow R_{n+8(k+1)}(B\pi)$, and

$$R_n(B\pi)[\mathcal{B}^{-1}] := \varinjlim_{k \geq 0} R_{n+8k}(B\pi).$$

Prove that, under the Bott-periodicity identifications $KO_{n+8k}(C_r^*\pi) \cong KO_n(C_r^*\pi)$, the homomorphisms θ induce an isomorphism

$$R_n(B\pi)[\mathcal{B}^{-1}] \xrightarrow{\cong} KO_n(C_r^*\pi).$$

6.7.2. Historical background, literature, and current progress

6.7.3. Verdict: partially resolved

6.7.3.1. Precise problem

The statement is the Bott-stabilized Stolz relative index conjecture for positive scalar curvature. In spin notation, it asks whether, for every discrete group Γ and every integer n , the relative

higher index map

$$\theta : R_n^{\text{spin}}(B\Gamma)[B^{-1}] \longrightarrow KO_n(C_r^*(\Gamma; \mathbb{R}))$$

is an isomorphism, where $R_n^{\text{spin}}(B\Gamma)[B^{-1}]$ is obtained from Stolz's R -group by taking the direct limit under products with an 8-dimensional Bott manifold.

6.7.3.2. Verdict

This problem is not fully resolved. It has substantial partial progress, mainly on surjectivity/rational surjectivity under Baum-Connes-type hypotheses, but the full integral isomorphism for all discrete groups is not known. In particular, recent citable literature explicitly formulates exactly this map as the Stolz conjecture and states that injectivity of the Stolz map is still wide open, even for the trivial group.

6.7.3.3. Literature and progress

1. **Stephan Stolz, "Positive Scalar Curvature Metrics – Existence and Classification Questions",** *Proceedings of the International Congress of Mathematicians, Zurich 1994*, Vol. 1, Birkhauser, 1995, pp. 625–636. DOI/URL: https://doi.org/10.1007/978-3-0348-9078-6_56. This is the original ICM source for the existence/classification framework around positive scalar curvature metrics and Stolz's R -groups.
2. **Jonathan Rosenberg and Stephan Stolz, "Metrics of positive scalar curvature and connections with surgery",** *Surveys on Surgery Theory*, Vol. 2, *Annals of Mathematics Studies* 149, Princeton University Press, 2001, pp. 353–386. URL: <https://www.math.umd.edu/~jmr/psc.pdf>. Section 5 defines the index homomorphism $\theta : R_n(\gamma) \rightarrow KO_n(C_\gamma^*)$, observes that it is not generally injective before stabilization, defines $R_n(\gamma)[Bt^{-1}]$, and states Conjecture 5.7: the stable homomorphism $\theta^{st} : R_n(\gamma)[Bt^{-1}] \rightarrow KO_n(C_\gamma^*)$ is an isomorphism. The spin case is the problem statement. The same discussion notes that surjectivity is related to Baum-Connes and that injectivity appears hard.
3. **Jonathan Rosenberg, "Manifolds of Positive Scalar Curvature: A Progress Report",** *Surveys in Differential Geometry* 11, 2006, pp. 259–294. URL: <https://www.math.umd.edu/~jmr/psc2006.pdf>. Conjecture 2.4 restates Stolz's problem: the index map $R_{n+1}(\pi) \rightarrow KO_{n+1}(C_R^*(\pi))$ should be stably an isomorphism, with stability given by the direct limit under product with the Bott manifold. This matches the problem.
4. **Paolo Piazza and Thomas Schick, "Rho-classes, index theory and Stolz' positive scalar curvature sequence",** *Journal of Topology* 7 (2014), no. 4, 965–1004. DOI/arXiv: <https://doi.org/10.1112/jtopol/jtt048>, arXiv:1210.6892. They construct a map from Stolz's positive scalar curvature exact sequence to the Higson-Roe analytic surgery exact sequence and identify the analytic targets of the R -group index and rho classes. This is important structural progress, but it does not prove the Bott-stabilized index map is an isomorphism.
5. **Zhizhang Xie and Guoliang Yu, "Positive scalar curvature, higher rho invariants and localization algebras",** *Advances in Mathematics* 262 (2014), 823–866. DOI/arXiv: <https://doi.org/10.1016/j.aim.2014.06.001>, arXiv:1302.4418. They give localization-algebra constructions relating higher rho invariants and relative higher

indices, including a map from Stolz's sequence to analytic surgery. Again, this supplies comparison machinery but not the full isomorphism in the conjecture.

6. Zhizhang Xie, Guoliang Yu and Rudolf Zeidler, "On the range of the relative higher index and the higher rho-invariant for positive scalar curvature", *Advances in Mathematics* 390 (2021), Article 107897. DOI/arXiv: <https://doi.org/10.1016/j.aim.2021.107897>, arXiv:1712.03722. Corollary 1.6/3.16 proves that if the real Baum-Connes assembly map for Γ is rationally surjective, then the Bott-stabilized relative index map

$$R_n^{\text{spin}}(B\Gamma)[Bt^{-1}] \rightarrow KO_n(C_r^*\Gamma)$$

is rationally surjective. This is strong partial progress on the range of the map, but it is only rational surjectivity under an additional hypothesis and does not address full integral injectivity.

7. Jinmin Wang, Zhizhang Xie and Guoliang Yu, "Approximations of delocalized eta invariants by their finite analogues", *L'Enseignement Mathématique* 69 (2023), 51–91. DOI/URL: <https://doi.org/10.4171/LEM/1048>, <https://ems.press/content/serial-article-files/44474>. Appendix A formulates exactly the assertion as Conjecture A.3, "Stolz conjecture": the map

$$\theta : R_n^{\text{spin}}(B\Gamma)[B^{-1}] \rightarrow KO_n(C_r^*(\Gamma; \mathbb{R}))$$

is an isomorphism. The authors then state that the injectivity of the Stolz map, and of the generalized Stolz map, is "wide open" and not even known for Γ trivial. This directly rules out a fully resolved verdict.

6.7.3.4. What remains unresolved

The unresolved part is the full isomorphism assertion in all cases. Known results do not prove integral surjectivity and injectivity for every discrete group. The principal missing point is injectivity of the Bott-stabilized Stolz map; current literature records it as open even in the simplest case of the trivial group. Therefore the correct status is **partially resolved**, not fully resolved.

6.7.4. Remaining open problems recorded by the database

6.7.4.1. Open problem 1

Prove, already for the trivial group $\Gamma = \{e\}$, that the Bott-stabilized Stolz relative index map $\theta : R_n^{\text{spin}}(\text{pt})[B^{-1}] \rightarrow KO_n(\mathbb{R})$ is injective.

Reference supplied by the database. Jinmin Wang, Zhizhang Xie, and Guoliang Yu, "Approximations of Delocalized Eta Invariants by Their Finite Analogues," *L'Enseignement Mathématique* 69 (2023), no. 1–2, 51–91, Appendix A, Conjecture A.3 and the paragraph following Conjecture A.4 (explicitly stating that injectivity is not known even for the trivial group), DOI: 10.4171/LEM/1048, arXiv:2003.03401, <https://doi.org/10.4171/LEM/1048>.

6.7.5. Primary-source pointer

The originating ICM record is *Positive scalar curvature metrics - Existence and classification questions* (1994). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.1/ICM1994.1.ocr.pdf#page=697>.

6.8. 213. Finite Dimensional Approximations in Geometry [2002-ag-88]

Record: 2002-ag-88

Subfield: Manifolds and global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Finite Dimensional Approximations in Geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.2/ICM2002.2.ocr.pdf#page=380>

6.8.1. Problem statement

Let $\overline{M}$ be a compactified moduli space, as in symplectic geometry or Donaldson theory and in the Kotschick-Morgan conjecture, whose local structure is encoded by Fukaya-Ono Kuranishi data or Ruan virtual-neighborhood data. Determine whether there exists a finite-dimensional approximation A of the ambient infinite-dimensional configuration space such that $\overline{M} \subset A$, the Kuranishi or virtual-neighborhood structure is retained sufficiently that changes of auxiliary choices are represented by cobordisms, and the resulting cobordism class can be interpreted by a stable Pontrjagin-Thom construction.

6.8.2. Historical background, literature, and current progress

6.8.3. Verdict: partially resolved

6.8.3.1. Precise problem

The problem is not just the classical Kotschick-Morgan wall-crossing conjecture. The source problem is Furuta's broader "global approximation" question: given a compactified moduli space described locally by Fukaya-Ono Kuranishi data or Ruan virtual-neighborhood data, construct a finite-dimensional ambient approximation of the original infinite-dimensional configuration space that contains the compactified moduli space, keeps enough of the local Kuranishi/virtual-neighborhood data to make changes of choices into cobordisms, and places the resulting cobordism class in a stable Pontrjagin-Thom/stable homotopy framework.

6.8.3.2. Original source

- Title: *Finite dimensional approximations in geometry*
- Author: Mikio Furuta
- Venue/year: Proceedings of the International Congress of Mathematicians, Beijing 2002, vol. 2, pp. 395-404; arXiv version submitted 2003
- DOI/arXiv/URL: arXiv:math/0304295; <https://doi.org/10.48550/arXiv.math/0304295>; <https://arxiv.org/abs/math/0304295>
- Relevance: Section 4 states the problem in essentially this form. Furuta observes that for symplectic geometry or Donaldson theory this stable Pontrjagin-Thom/global finite-

dimensional approximation construction had not been done, and that the same difficulty occurs for the Kotschick-Morgan conjecture. The paper contrasts this with the Seiberg-Witten case, where compactness allows a global finite-dimensional approximation leading to the Bauer-Furuta stable cohomotopy invariant.

6.8.3.3. Progress found

- Title: *Global Kuranishi charts and a product formula in symplectic Gromov-Witten theory*
- Authors: Amanda Hirschi and Mohan Swaminathan
- Venue/year: *Selecta Mathematica*, New Series, 30, article 87, 2024
- DOI/arXiv/URL: <https://doi.org/10.1007/s00029-024-00982-y>; arXiv:2212.11797; <https://arxiv.org/abs/2212.11797>
- Result/progress: Constructs global Kuranishi charts for moduli spaces of stable pseudo-holomorphic maps to closed symplectic manifolds in all genera and proves a product formula for symplectic Gromov-Witten invariants. This is strong progress on the symplectic Gromov-Witten part of Furuta's question, including global finite-dimensional chart technology and cobordism/equivalence control for choices, but it is not a full solution for arbitrary Fukaya-Ono/Ruan moduli problems or Donaldson/Kotschick-Morgan theory.
- Title: *Gromov-Witten Invariants in Complex and Morava-Local K-Theories*
- Authors: Mohammed Abouzaid, Mark McLean, and Ivan Smith
- Venue/year: *Geometric and Functional Analysis* 34, pp. 1647-1733, 2024
- DOI/arXiv/URL: <https://doi.org/10.1007/s00039-024-00697-4>; arXiv:2307.01883; <https://arxiv.org/abs/2307.01883>
- Result/progress: Constructs Gromov-Witten-type invariants valued in complex K-theory and Morava-local complex-oriented cohomology theories. The key geometric input is global Kuranishi charts for stable maps of arbitrary genus. This advances the generalized-cohomology/stable-homotopy direction for symplectic Gromov-Witten theory, but it remains a special symplectic GW framework rather than the full ambient approximation problem for all Kuranishi/virtual-neighborhood compactifications.
- Title: *Complex cobordism, Hamiltonian loops and global Kuranishi charts*
- Authors: Mohammed Abouzaid, Mark McLean, and Ivan Smith
- Venue/year: Preprint, 2021
- DOI/arXiv/URL: arXiv:2110.14320; <https://doi.org/10.48550/arXiv.2110.14320>; <https://arxiv.org/abs/2110.14320>
- Result/progress: Builds global Kuranishi charts for moduli spaces of pseudoholomorphic spheres and uses them in generalized cohomology arguments. This is an important genus-zero symplectic case, but it does not address Donaldson theory or the full generality of Furuta's problem.
- Title: *An integral Euler cycle in normally complex orbifolds and $\mathbb{Z}$ -valued Gromov-Witten type invariants*
- Authors: Shaoyun Bai and Guangbo Xu
- Venue/year: Preprint, 2022

- DOI/arXiv/URL: arXiv:2201.02688; <https://arxiv.org/abs/2201.02688>
- Result/progress: Provides derived orbifold chart/bordism technology for genus-zero Gromov-Witten-type invariants, including invariance under choices in a bordism group. This is relevant partial progress toward Pontrjagin-Thom-type refinements, but it is not a complete solution of the general ambient finite-dimensional approximation problem.
- Title: *An $SO(3)$ -Monopole Cobordism Formula Relating Donaldson and Seiberg-Witten Invariants*
- Authors: Paul M. N. Feehan and Thomas G. Leness
- Venue/year: *Memoirs of the American Mathematical Society* 256, no. 1226, 2018
- DOI/arXiv/URL: <https://doi.org/10.1090/memo/1226>; arXiv:math/0203047; <https://arxiv.org/abs/math/0203047>
- Result/progress: Proves an $SO(3)$ -monopole analogue of the Kotschick-Morgan wall-crossing phenomenon and reduces the classical Kotschick-Morgan conjecture to further anti-self-dual gluing input. This is related gauge-theoretic progress but does not give the finite-dimensional ambient approximation and stable Pontrjagin-Thom construction asked for in the problem statement.

6.8.3.4. Claimed resolution checked

- Title: *A Smooth Compactification of the Moduli Space of Instantons and Its Application*
- Author: Bohui Chen
- Venue/year: Preprint, 2002
- DOI/arXiv/URL: arXiv:math/0204287; <https://arxiv.org/abs/math/0204287>
- Claim and assessment: The abstract claims a proof of the Kotschick-Morgan conjecture via a smooth compactification of Donaldson moduli spaces. Even if read as addressing the classical Donaldson wall-crossing conjecture, this is narrower than Furuta's global finite-dimensional ambient approximation/stable Pontrjagin-Thom problem. Later Feehan-Leness discussions describe Chen's approach as an attempt requiring difficult transition-map/gluing analysis, and their published monograph treats related results as analogues/reductions rather than a full resolution of Furuta's stated problem. This item therefore does not fully resolve the problem.

6.8.3.5. What remains unresolved

No published or otherwise citable theorem found gives a construction covering the full statement of the problem: a general finite-dimensional ambient approximation for compactified moduli spaces with Fukaya-Ono Kuranishi or Ruan virtual-neighborhood structures, including Donaldson theory and the Kotschick-Morgan setting, with changes of auxiliary choices represented by cobordisms and with the resulting class interpreted by a stable Pontrjagin-Thom construction.

The symplectic Gromov-Witten case has substantial modern solutions via global Kuranishi charts and generalized cohomology-valued invariants, but these are special cases with extra structure and do not settle the full cross-context problem Furuta asked.

6.8.4. Remaining open problems recorded by the database

6.8.4.1. Open problem 1

Let $X = M/\mathbb{Z}_\alpha$ be a four-orbifold of the type considered by Qiao, with $b_2^+(X) = 1$, and let $D(e)$ be the link of the reducible pseudo-connections associated with an orbifold wall W_e . Prove the orbifold Kotschick–Morgan conjecture: if q is the intersection form and $r = (e^2 - p_1(P))/4$, then

$$\langle (2\mu)^d, D(e) \rangle = \sum_{i=0}^r a_i(r, O, X) q^{r-i} e^{d-2r-2i},$$

where the coefficients $a_i(r, O, X)$ depend only on r , the equivariant bundle data O , and the homotopy type of X .

Reference supplied by the database. Shuaige Qiao, *Bubble Tree Compactification of Instanton Moduli Spaces*, Ph.D. thesis, Australian National University, 2021, Chapter 7, Section 7.1, “The Kotschick–Morgan conjecture on orbifolds,” displayed “Conjecture (Kotschick–Morgan conjecture for orbifolds),” pp. 133–135. DOI: <https://doi.org/10.25911/B3M9-WX61>; repository record: <http://hdl.handle.net/1885/233020>.

6.8.4.2. Open problem 2

Construct the proposed bubble-tree compactification of the universal instanton bundle over a general smooth four-manifold, with compactified total space $\mathcal{D}_{k,1}^0$, compactified marked-point base $\mathcal{D}_{k,1}$, and maps $\mathcal{D}_{k,1}^0 \rightarrow \mathcal{D}_{k,1} \rightarrow \overline{M}_k$ and $\mathcal{D}_{k,1} \rightarrow X$, so that the family index class extends over the lower strata and gives a gauge-theoretic definition of K-theoretic Donaldson invariants beyond nonsingular projective surfaces.

Reference supplied by the database. Shuaige Qiao, *Bubble Tree Compactification of Instanton Moduli Spaces*, Ph.D. thesis, Australian National University, 2021, Chapter 7, Section 7.2, “K-theoretic Donaldson invariant,” especially the proposed compactification diagram and the statement “Details of this construction need to be clarified,” pp. 135–137. DOI: <https://doi.org/10.25911/B3M9-WX61>; repository record: <http://hdl.handle.net/1885/233020>.

6.8.4.3. Open problem 3

Construct a bubble-tree compactification of the $U(2)$ -monopole moduli space that incorporates virtual-neighborhood data around the nonregular reducible monopoles in every bubbling stratum, and use it to complete the $PU(2)$ -monopole cobordism approach to Witten’s conjectural formula relating Donaldson and Seiberg–Witten invariants.

Reference supplied by the database. Shuaige Qiao, *Bubble Tree Compactification of Instanton Moduli Spaces*, Ph.D. thesis, Australian National University, 2021, Chapter 7, Section 7.3, “Witten’s conjecture and bubble tree for non-abelian monopole moduli space,” pp. 137–138. DOI: <https://doi.org/10.25911/B3M9-WX61>; repository record: <http://hdl.handle.net/1885/233020>.

6.8.4.4. Open problem 4

Associate to general Floer data a well-defined stable homotopy type whose ordinary homology recovers the corresponding Floer homology; in particular, construct a true model in the case of nontrivial polarization and disc bubbling by combining the required twisted-spectrum structure with a compatible Kuranishi- or polyfold-type compactification of the flow category.

Reference supplied by the database. Alice Hedenlund and Tasos Moulinos, “Twisted Spectra Revisited,” *Algebraic & Geometric Topology* **25** (2025), no. 9, 5693–5754, Question 0.1, Question 0.2, and Remark 0.3, pp. 5694–5695. DOI: <https://doi.org/10.2140/agt.2025.25.5693>; arXiv: [2312.07403](https://arxiv.org/abs/2312.07403).

6.8.5. Primary-source pointer

The originating ICM record is *Finite Dimensional Approximations in Geometry* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.2/ICM2002.2.ocr.pdf#page=380>.

6.9. 214. Kähler manifolds and transcendental techniques in algebraic geometry [2006-geom-97]

Record: 2006-geom-97

Subfield: Manifolds and global analysis

Normalized status: Partially resolved

Dataset review state: under_review (reviewed flag: no)

Original source: ICM 2006: Kähler manifolds and transcendental techniques in algebraic geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.1/ICM2006.1.ocr.pdf#page=163>

6.9.1. Problem statement

Prove or disprove that if $\pi : \mathcal{X} \rightarrow S$ is a deformation of compact complex manifolds over an irreducible analytic base space S , that is, π is a proper holomorphic map whose fibers are smooth compact complex manifolds, and locally near every point of every fiber $\mathcal{X}$ is isomorphic to a product over S , with π identified with the projection to the base, and if X_{t_0} is Kähler for some $t_0 \in S$, where $X_t = \pi^{-1}(t)$ for $t \in S$, then there exists a finite, or possibly countable, union $\bigcup_v S_v$ of proper analytic strata $S_v \subsetneq S$ such that X_t is Kähler for every $t \in S \setminus \bigcup_v S_v$, and X_t is bimeromorphic to a Kähler manifold for every $t \in \bigcup_v S_v$.

6.9.2. Historical background, literature, and current progress

6.9.3. Verdict: partially resolved

6.9.3.1. Precise problem

The problem is Demailly's/Demailly–Paun's deformation-stability question for the Kahler property. In modern terms it asks:

Given a locally trivial proper holomorphic family of compact complex manifolds over an irreducible analytic base, with at least one Kahler fiber, is the Kahler locus the complement of a finite or countable union of proper analytic strata, and are all remaining fibers at least in Fujiki class C, i.e. bimeromorphic to compact Kahler manifolds?

Equivalently, it combines two assertions:

1. global/countable-Zariski openness of the Kahler property from one Kahler fiber;
2. closedness of the Fujiki-class property for the exceptional/non-Kahler fibers, already including the disk-limit problem: if all punctured fibers are Kahler, is the central fiber in Fujiki class C?

6.9.3.2. Current literature status

The exact statement is not fully resolved. The most difficult remaining point is the Fujiki-class conclusion for limits of Kahler manifolds; even the one-parameter disk version is still treated

in recent papers as a conjecture, with only equivalent criteria or extra hypotheses known. The generic/countable-Zariski openness part from a single Kahler fiber is also not known in the full generality of the problem.

6.9.3.3. Key literature and progress

Original source and foundational results

- Title: Kähler manifolds and transcendental techniques in algebraic geometry
 Authors: Jean-Pierre Demailly
 Venue: Proceedings of the International Congress of Mathematicians, Madrid 2006, Vol. I, EMS, pp. 153-186
 Year: 2007
 DOI/URL: <https://doi.org/10.4171/022-1/8>; https://www-fourier.univ-grenoble-alpes.fr/~demailly/manuscripts/madrid_icm2006_print.pdf
 Relevance: This is the source context for the problem. It states the conjectural countable analytic stratification and the expected Fujiki-class conclusion for the exceptional fibers.
- Title: Numerical characterization of the Kähler cone of a compact Kähler manifold
 Authors: Jean-Pierre Demailly and Mihai Păun
 Venue: Annals of Mathematics (2) 159, no. 3, 1247-1274
 Year: 2004
 DOI/arXiv/URL: <https://doi.org/10.4007/annals.2004.159.1247>; [arXiv:math/0105176](https://arxiv.org/abs/math/0105176); <https://annals.math.princeton.edu/2004/159-3/p05>
 Relevance: Theorem 0.9 proves invariance of the Kahler cone for very general fibers in a deformation whose fibers are already compact Kahler manifolds. Section 5 formulates Conjecture 5.1: if one fiber is Kahler, then outside a countable union of proper analytic subsets the fibers should be Kahler. The paper explicitly presents this as a conjecture, so it does not prove the full hypothesis.
- Title: On deformations of complex analytic structures. III. Stability theorems for complex structures
 Authors: Kunihiko Kodaira and Donald C. Spencer
 Venue: Annals of Mathematics (2) 71, 43-76
 Year: 1960
 DOI/URL: <https://doi.org/10.2307/1969879>
 Relevance: Proves local openness of the Kahler property: nearby fibers of a Kahler fiber are Kahler. This is much weaker than the global countable-stratification assertion in the problem statement.
- Title: An example of a non-Kählerian complex-analytic deformation of Kählerian complex structures
 Authors: Heisuke Hironaka
 Venue: Annals of Mathematics (2) 75, 190-208
 Year: 1962
 URL: <https://www.jstor.org/stable/1970426>
 Relevance: Shows that a limit of Kahler/projective manifolds need not itself be Kahler in dimension 3. Hironaka's central fiber is Moishezon, hence in Fujiki class C, so it supports the expected weaker conclusion but does not prove it generally.

Partial results for Moishezon/projective analogues

- Title: Deformation limits of projective manifolds: Hodge numbers and strongly Gauduchon metrics
 Authors: Dan Popovici
 Venue: *Inventiones Mathematicae* 194, 515-534
 Year: 2013
 DOI/URL: <https://doi.org/10.1007/s00222-013-0449-0>
 Relevance: Proves that limits of projective fibers are Moishezon under additional hypotheses such as local constancy of certain Hodge numbers or the existence of a strongly Gauduchon metric on the limit fiber. This is a projective/Moishezon analogue with extra assumptions, not the Kahler-to-Fujiki statement.
- Title: Deformation limit and bimeromorphic embedding of Moishezon manifolds
 Authors: Sheng Rao and I-Hsun Tsai
 Venue: *Communications in Contemporary Mathematics* 23, no. 8, Paper No. 2050087
 Year: 2021
 DOI/arXiv/URL: <https://doi.org/10.1142/S021919972050087X> ; arXiv:1901.10627
 Relevance: If uncountably many fibers are Moishezon and the reference fiber satisfies deformation invariance of the Hodge number of type (0,1) or admits a strongly Gauduchon metric, then the reference fiber is Moishezon. Still conditional and about Moishezon fibers.
- Title: Smooth deformation limit of Moishezon manifolds is Moishezon
 Authors: Mu-Lin Li, Sheng Rao, Kai Wang, and Meng-jiao Wang
 Venue: arXiv preprint
 Year: 2024/2025
 DOI/arXiv/URL: <https://doi.org/10.48550/arXiv.2407.02022> ; arXiv:2407.02022
 Relevance: Proves the Moishezon disk-limit conjecture and even that uncountably many Moishezon fibers force all fibers to be Moishezon. This is important progress for the algebraic/bimeromorphic-projective analogue, but it does not apply to general Kahler fibers, which need not be Moishezon.

Partial results for Kahler/Fujiki limits

- Title: Compact Complex Manifolds with Small Gauduchon Cone
 Authors: Dan Popovici and Luis Ugarte
 Venue: *Proceedings of the London Mathematical Society*
 Year: 2018
 DOI/arXiv/URL: <https://doi.org/10.1112/plms.12110> ; arXiv:1407.5070
 Relevance: Surveys and develops a program toward closedness of Fujiki class C under holomorphic deformations. It states the class-C limit conjecture and gives stability results for related Gauduchon-type conditions. It is explicitly programmatic, not a proof of the problem.
- Title: The limits of Kahler manifolds under holomorphic deformations
 Authors: Mu-Lin Li and Wanmin Liu
 Venue: arXiv preprint
 Year: 2024, revised 2026
 DOI/arXiv/URL: <https://doi.org/10.48550/arXiv.2406.14076> ; arXiv:2406.14076

Relevance: Treats the disk conjecture as longstanding. It proves that a Kahler limit is actually Kahler under extra metric and cohomological assumptions on the central fiber, and recovers the surface case. These hypotheses are not present in the problem statement.

- Title: The holomorphic limit of Kahler manifolds

Authors: Mu-Lin Li

Venue: arXiv preprint

Year: 2024

DOI/arXiv/URL: <https://doi.org/10.48550/arXiv.2409.19957> ; arXiv:2409.19957

Relevance: For a smooth family over a disk with all punctured fibers Kahler, Theorem 1.2 gives an equivalence: the central fiber is in Fujiki class C if and only if it has finite upper volume. The same paper says it is an important open problem whether this upper volume is always finite. Therefore this is a strong criterion, not a full resolution of the problem.

- Title: Volumes of Bott-Chern Classes

Authors: Sébastien Boucksom, Vincent Guedj, and Chinh H. Lu

Venue: Peking Mathematical Journal

Year: 2025

DOI/URL: <https://doi.org/10.1007/s42543-025-00105-2>

Relevance: Gives a characterization of Fujiki class C via the bounded mass property plus existence of a closed Bott-Chern class with positive volume. This supplies major analytic context for recent work on Kahler limits, but it still leaves the needed bounded-mass/positive-volume conditions to be proved for arbitrary Kahler deformation limits.

6.9.3.4. Why the full problem is not resolved

No cited theorem found proves the full implication:

if one fiber in an arbitrary locally trivial proper family over an irreducible analytic base is Kahler, then all very general fibers are Kahler and every exceptional fiber is in Fujiki class C.

The closest general disk statement,

if all punctured fibers are Kahler, then the central fiber is in Fujiki class C,

is still presented as a conjecture in 2024-2026 work and is only known under additional hypotheses or through equivalent criteria such as finite upper volume/bounded mass. Since the statement requires all bases, all dimensions, no projectivity/Moishezon assumption, and no extra metric or cohomological condition on exceptional fibers, the literature gives partial progress only.

6.9.4. Remaining open problems recorded by the database

6.9.4.1. Open problem 1

Let $\pi : \mathcal{X} \rightarrow \Delta$ be a proper holomorphic submersion over the unit disc, with compact complex fibers X_t . If X_t is Kähler for every $t \neq 0$, must the central fiber X_0 belong to Fujiki class C, i.e. be bimeromorphic to a compact Kähler manifold?

Reference supplied by the database. Mu-Lin Li and Wanmin Liu, “The limits of Kähler manifolds under holomorphic deformations,” arXiv preprint (2024; revised 2026), Conjecture 1.1, [arXiv:2406.14076](#).

6.9.4.2. Open problem 2

If a compact complex manifold X carries a closed positive $(1, 1)$ -current T whose Bott–Chern class has positive volume $\text{vol}_n(\{T\}) > 0$, must X belong to Fujiki class C ?

Reference supplied by the database. Mu-Lin Li, “The holomorphic limit of Kähler manifolds,” arXiv preprint (2024; revised 2026), Conjecture 1.3 (the Boucksom–Demailly–Păun conjecture), [arXiv:2409.19957](#).

6.9.5. Primary-source pointer

The originating ICM record is *Kähler manifolds and transcendental techniques in algebraic geometry* (2006). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.1/ICM2006.1.ocr.pdf#page=163>.

6.10. 215. Kähler manifolds and transcendental techniques in algebraic geometry [2006-geom-98]

Record: 2006-geom-98

Subfield: Manifolds and global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2006: Kähler manifolds and transcendental techniques in algebraic geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.1/ICM2006.1.ocr.pdf#page=163>

6.10.1. Problem statement

Prove or disprove that for every compact Kähler manifold X of complex dimension n , the Kodaira dimension $\kappa(X)$ equals the numerical dimension of the canonical class:

$$\kappa(X) = \nu(X) := \nu(c_1(K_X)),$$

where $K_X = \Lambda^n T_X^*$ is the canonical bundle of X , $c_1(K_X) \in H^{1,1}(X, \mathbb{R})$ is its first Chern class, and the numerical dimension of a class $\alpha \in H^{1,1}(X, \mathbb{R})$ is defined by

$$\nu(\alpha) = -\infty \quad \text{if } \alpha \text{ is not pseudo-effective,}$$

while, if α is pseudo-effective, meaning that it is represented by a closed positive $(1,1)$ -current,

$$\nu(\alpha) = \max\{p \in \{0, 1, \dots, n\} : \langle \alpha^p \rangle \neq 0\},$$

with $\langle \alpha^p \rangle$ denoting the p -fold movable intersection product of α .

6.10.2. Historical background, literature, and current progress

6.10.3. Verdict: partially resolved

6.10.3.1. Precise problem

The problem asks for the generalized abundance conjecture for compact Kähler manifolds: for every compact Kähler manifold X , in every dimension, the Kodaira dimension of X should equal the numerical dimension of the canonical class,

$$\kappa(X) = \nu(c_1(K_X)),$$

where ν is defined using the movable/non-pluripolar intersection product and is $-\infty$ for non-pseudo-effective classes.

This is an all-dimensional compact Kähler statement, not just the projective case, not just minimal models, and not just the case where K_X is nef.

6.10.3.2. Status check

I found no citable literature proving this full all-dimensional compact Kahler statement. Recent literature still treats the corresponding Kahler abundance problem as open in higher dimension. Since projective manifolds are compact Kahler manifolds, the problem statement would in particular cover the projective generalized abundance problem in all dimensions; the projective/Kahler abundance literature still has open cases in dimension at least 4.

The correct status is therefore partially resolved: substantial cases are known, especially the extreme numerical-dimension cases and compact Kahler threefolds, but the full statement remains open.

6.10.3.3. Key literature and progress

1. Original/source formulation

- **Title:** “Kahler manifolds and transcendental techniques in algebraic geometry”
- **Author:** Jean-Pierre Demailly
- **Venue:** Proceedings of the International Congress of Mathematicians, Madrid 2006, Vol. I, EMS Press, pp. 153-186
- **Year:** 2007
- **DOI/URL:** DOI <https://doi.org/10.4171/022-1/8>; <https://ems.press/books/standalone/20/363>
- **Relevant result:** Conjecture 5.9 states the generalized abundance conjecture for arbitrary compact Kahler manifolds: $\kappa(X) = \nu(c_1(K_X))$. Theorem 5.10 records known extreme cases $\nu(X) = -\infty, 0, n$.
- **Scope:** This is the source/problem statement and partial progress only; it does not resolve the intermediate cases in arbitrary dimension.

2. Movable intersection/numerical dimension framework and early partial cases

- **Title:** “The pseudo-effective cone of a compact Kahler manifold and varieties of negative Kodaira dimension”
- **Authors:** Sebastien Boucksom, Jean-Pierre Demailly, Mihai Paun, Thomas Peternell
- **Venue:** Journal of Algebraic Geometry 22, no. 2, pp. 201-248
- **Year:** 2013
- **DOI/arXiv/URL:** DOI <https://doi.org/10.1090/S1056-3911-2012-00574-8>; arXiv math/0405285; <https://arxiv.org/abs/math/0405285>
- **Relevant result:** Section 3.8 formulates the generalized abundance conjecture using the numerical dimension; remarks record $\kappa(X) \leq \nu(X)$, the easy cases $\nu = -\infty, n$, and partial information for $\nu = 0$, including projective dimension ≤ 4 under the hypotheses stated there.
- **Scope:** Gives foundational tools and partial cases, not the full conjecture.

3. Relation with the usual projective abundance conjecture

- **Title:** “Numerical analogues of the Kodaira dimension and the Abundance Conjecture”
- **Author:** Thomas Eckl
- **Venue:** Manuscripta Mathematica 150, pp. 337-356
- **Year:** 2016
- **DOI/arXiv/URL:** DOI <https://doi.org/10.1007/s00229-015-0815-x>; arXiv 1505.01262; <https://arxiv.org/abs/1505.01262>

- **Relevant result:** Shows that, for non-uniruled smooth complex projective varieties, the usual MMP abundance conjecture and generalized abundance formulated with numerical analogues of Kodaira dimension are equivalent.
- **Scope:** This confirms that the conjecture contains deep unresolved projective abundance cases; it is not a full compact Kahler proof.

4. Compact Kahler threefold minimal models

- **Title:** “Minimal models for Kahler threefolds”
- **Authors:** Andreas Hoering, Thomas Peternell
- **Venue:** *Inventiones Mathematicae* 203, no. 1, pp. 217-264
- **Year:** 2016
- **DOI/arXiv/URL:** DOI <https://doi.org/10.1007/s00222-015-0592-x>; arXiv 1304.4013; <https://arxiv.org/abs/1304.4013>
- **Relevant result:** Proves that a non-uniruled compact Kahler threefold has a minimal model.
- **Scope:** Supplies the threefold MMP input; it is dimension 3 only.

5. Abundance for compact Kahler threefolds

- **Title:** “Abundance for Kahler threefolds”
- **Authors:** Frederic Campana, Andreas Hoering, Thomas Peternell
- **Venue:** *Annales scientifiques de l’Ecole Normale Supérieure*, Serie 4, 49, no. 4, pp. 971-1025
- **Year:** 2016
- **DOI/URL:** DOI <https://doi.org/10.24033/asens.2301>; <https://www.numdam.org/articles/10.24033/asens.2301/>
- **Relevant result:** Theorem 1.1 proves that if X is a normal $\mathbb{Q}$ -factorial compact Kahler threefold with terminal singularities and K_X nef, then K_X is semi-ample. The paper explains that for nef K_X , equality $\kappa(X) = \nu(X)$ is sufficient for abundance in this setting.
- **Scope:** Resolves the minimal threefold case, but not arbitrary dimension.

6. Log abundance for compact Kahler threefolds, first part

- **Title:** “On the log abundance for compact Kahler 3-folds”
- **Authors:** Omprokash Das, Wenhao Ou
- **Venue:** *Manuscripta Mathematica* 173, no. 1-2, pp. 341-404
- **Year:** 2024
- **DOI/arXiv/URL:** DOI <https://doi.org/10.1007/s00229-023-01467-6>; arXiv 2201.01202; <https://arxiv.org/abs/2201.01202>
- **Relevant result:** Proves log abundance for log canonical compact Kahler threefold pairs when $K_X + \Delta$ is nef and $\nu(K_X + \Delta) \neq 2$, including non-vanishing and semi-ameness in the stated cases.
- **Scope:** Major dimension-3/log progress; excludes $\nu = 2$, later handled separately.

7. Log abundance for compact Kahler threefolds, completion of the $\nu = 2$ case

- **Title:** “On the Log Abundance for Compact Kahler threefolds II”
- **Authors:** Omprokash Das, Wenhao Ou
- **Venue:** *Proceedings of the London Mathematical Society* 132, no. 3, article e70141
- **Year:** 2026
- **DOI/arXiv/URL:** DOI <https://doi.org/10.1112/plms.70141>; arXiv 2306.00671; <https://arxiv.org/abs/2306.00671>

- **Relevant result:** Proves that for a log canonical compact Kahler threefold pair with $K_X + \Delta$ nef and $\nu(X, K_X + \Delta) = 2$, $K_X + \Delta$ is semi-ample. Combined with the previous Das-Ou paper, this gives log abundance for log canonical compact Kahler threefold pairs.
- **Scope:** Completes the threefold log case, but still only in dimension 3.

8. Current higher-dimensional status

- **Title:** “An approach to the abundance conjecture for Kahler varieties via algebraic reduction”
- **Author:** Zhiyuan Jiang
- **Venue:** arXiv preprint
- **Year:** 2026
- **arXiv/URL:** arXiv 2604.03879; <https://arxiv.org/abs/2604.03879>
- **Relevant result/status statement:** The introduction states that abundance is open in dimension ≥ 4 , that there are major open cases even in dimension 4, and that abundance for Kahler varieties of arbitrary dimension is a major open problem. The paper proves an inductive criterion and some special Kahler fourfold cases, e.g. $a(X) = 3$, under its hypotheses.
- **Scope:** Confirms that the full statement is not resolved; it gives only conditional or special higher-dimensional progress.

6.10.3.4. What remains unresolved

The full conjecture remains open for compact Kahler manifolds in arbitrary dimension, especially intermediate numerical dimensions $0 < \nu(c_1(K_X)) < \dim X$ in dimension at least 4. Known threefold and selected fourfold/projective results do not cover all compact Kahler manifolds and do not imply the full equality $\kappa(X) = \nu(c_1(K_X))$ in every dimension.

6.10.4. Remaining open problems recorded by the database

6.10.4.1. Open problem 1

Let X be a compact Kähler fourfold whose canonical bundle K_X is nef. Must K_X be semiample, i.e. must some positive tensor power $K_X^{\otimes m}$ be generated by global sections?

Reference supplied by the database. Zhiyuan Jiang, “An approach to the abundance conjecture for Kähler varieties via algebraic reduction,” arXiv preprint (2026), Conjecture 1.1 and the discussion identifying dimension four as open, [arXiv:2604.03879](https://arxiv.org/abs/2604.03879).

6.10.4.2. Open problem 2

If the canonical class K_X of a compact Kähler manifold X is pseudo-effective, must X have nonnegative Kodaira dimension, $\kappa(X) \geq 0$?

Reference supplied by the database. Jean-Pierre Demailly, “Kähler manifolds and transcendental techniques in algebraic geometry,” in *Proceedings of the International Congress of Mathematicians, Madrid 2006*, Vol. I, European Mathematical Society, Zürich, 2007, pp. 153–186, §5.5 (the non-vanishing consequence of Conjecture 5.9). DOI: <https://doi.org/10.4171/022-1/8>.

6.10.4.3. Open problem 3

Is the algebraic reduction map of every compact Kähler variety almost holomorphic?

Reference supplied by the database. Zhiyuan Jiang, “An approach to the abundance conjecture for Kähler varieties via algebraic reduction,” arXiv preprint (2026), Conjecture 1.2, [arXiv:2604.03879](https://arxiv.org/abs/2604.03879).

6.10.5. Primary-source pointer

The originating ICM record is *Kähler manifolds and transcendental techniques in algebraic geometry* (2006). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.1/ICM2006.1.ocr.pdf#page=163>.

6.11. 216. Kähler manifolds and transcendental techniques in algebraic geometry [2006-geom-99]

Record: 2006-geom-99

Subfield: Manifolds and global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2006: Kähler manifolds and transcendental techniques in algebraic geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.1/ICM2006.1.ocr.pdf#page=163>

6.11.1. Problem statement

Prove that for every compact Kähler manifold X of complex dimension n , if $\mathcal{E} \subset H^{1,1}(X, \mathbb{R})$ is the pseudo-effective cone, namely the closed convex cone of cohomology classes $\{T\}$ of closed positive currents T of type $(1,1)$ on X , and $\mathcal{M} \subset H_{\mathbb{R}}^{n-1,n-1}(X)$ is the cone of movable classes, namely the closure of the convex cone generated by cohomology classes of currents of the form $\mu_*(\tilde{\omega}_1 \wedge \cdots \wedge \tilde{\omega}_{n-1})$, where $\mu : \tilde{X} \rightarrow X$ ranges over modifications and $\tilde{\omega}_1, \dots, \tilde{\omega}_{n-1}$ range over Kähler forms on $\tilde{X}$, then $\mathcal{E}$ and $\mathcal{M}$ are dual under the Poincaré duality pairing $H^{1,1}(X, \mathbb{R}) \times H_{\mathbb{R}}^{n-1,n-1}(X) \rightarrow \mathbb{R}$, $(\alpha, \beta) \mapsto \int_X \alpha \wedge \beta$; equivalently, $\mathcal{E} = \mathcal{M}^\vee$ and $\mathcal{M} = \mathcal{E}^\vee$ with respect to this pairing.

6.11.2. Historical background, literature, and current progress

6.11.3. Verdict: partially resolved

6.11.3.1. Precise problem

The problem asks for the Boucksom-Demailly-Paun-Peternell cone-duality conjecture in its full compact Kähler form: for every compact Kähler manifold X , the pseudo-effective cone $\mathcal{E} \subset H^{1,1}(X, \mathbb{R})$ of closed positive $(1,1)$ -current classes should be dual, under Poincaré duality, to the movable cone $\mathcal{M} \subset H_{\mathbb{R}}^{n-1,n-1}(X, \mathbb{R})$ generated by pushforwards $\mu_*(\tilde{\omega}_1 \wedge \cdots \wedge \tilde{\omega}_{n-1})$ from modifications.

6.11.3.2. Literature status

1. Jean-Pierre Demailly, "Kähler manifolds and transcendental techniques in algebraic geometry", Proceedings of the International Congress of Mathematicians, Madrid 2006, Vol. I, European Mathematical Society, 2007, pp. 153-186. DOI: <https://doi.org/10.4171/022-1/8>. URL: <https://ems.press/books/standalone/20> This ICM source states the Kähler analogue as a conjecture: for arbitrary compact Kähler X , $\mathcal{E}$ and $\mathcal{M}$ should be dual cones.
2. Sébastien Boucksom, Jean-Pierre Demailly, Mihai Păun, Thomas Peternell, "The pseudo-effective cone of a compact Kähler manifold and varieties of negative Kodaira dimension"

- sion”, *Journal of Algebraic Geometry* 22 (2013), no. 2, 201-248. DOI: <https://doi.org/10.1090/S1056-3911-2012-00574-8>; arXiv: math/0405285; URL: <https://arxiv.org/abs/math/0405285> This is the original BDPP source for the cone-duality conjecture. It formulates the arbitrary compact Kähler statement as Conjecture 2.3, and proves projective/algebraic consequences, including the projective movable-curve duality in the Néron-Severi setting. This does not prove the full problem because the statement allows all compact Kähler manifolds and all real $(1, 1)$ -classes.
3. David Witt Nyström, “Duality between the pseudoeffective and the movable cone on a projective manifold”, with an appendix by Sébastien Boucksom, *Journal of the American Mathematical Society* 32 (2019), no. 3, 675-689. DOI: <https://doi.org/10.1090/jams/922>; arXiv: 1602.03778; URL: <https://arxiv.org/abs/1602.03778> The main theorem proves exactly the $\mathcal{E}$ - $\mathcal{M}$ duality in full real Hodge cohomology when X is projective. This is substantial progress, but projectivity is an extra hypothesis not present in the problem statement.
 4. Jian Xiao, “Weak transcendental holomorphic Morse inequalities on compact Kähler manifolds”, *Annales de l’Institut Fourier* 65 (2015), no. 3, 1367-1379. DOI: <https://doi.org/10.5802/aif.2959>; arXiv: 1308.2878; URL: <https://aif.centre-mersenne.org/articles/10.5802/aif.2959/> This proves weak transcendental Morse inequalities and is explicitly described as a partial improvement of BDPP. It supplies related partial progress, not the full cone duality.
 5. Nicholas McCleerey, “Plurisupported Currents on Compact Kähler Manifolds”, arXiv preprint, 2021. DOI: <https://doi.org/10.48550/arXiv.2106.12017>; arXiv: 2106.12017; URL: <https://arxiv.org/abs/2106.12017> Theorem 1.6 / Theorem 3.12 proves the BDPP cone-duality conjecture for compact Kähler manifolds Y admitting a proper modification $\mu : X \rightarrow Y$ where X carries a plurisupported current whose class is Kähler. This is conditional and does not cover arbitrary compact Kähler manifolds.
 6. Hsueh-Yung Lin, “On the dual positive cones and the algebraicity of a compact Kähler manifold”, *Épjournal de Géométrie Algébrique*, special volume in honour of Claire Voisin, 2024, epiga:10771. DOI: <https://doi.org/10.46298/epiga.2024.10771>; arXiv: 2001.10654; URL: <https://arxiv.org/abs/2001.10654> This recent work still refers to the equality of the movable cone with the dual of the pseudo-effective cone as conjectural in the compact Kähler setting, and studies related algebraicity consequences instead of proving the cone duality.
 7. Junyu Cao and Valentino Tosatti, “Regularity of the volume function”, *Bulletin of the London Mathematical Society*, 2026. DOI: <https://doi.org/10.1112/blms.70212>; arXiv: 2505.14495; URL: <https://arxiv.org/abs/2505.14495> The results are stated for projective manifolds, or more generally compact Kähler manifolds satisfying the BDPP conjecture, and the paper notes that proving the corresponding result for all compact Kähler manifolds without BDPP remains interesting. This is further evidence that the full compact Kähler cone-duality statement is not presently settled.

6.11.3.3. Hypothesis check

The strongest citable theorem matching the conclusion is Witt Nyström’s theorem, but it assumes X is projective. McCleerey’s theorem covers additional compact Kähler manifolds only under the existence of a suitable plurisupported Kähler class after modification. These hypotheses are not automatic for an arbitrary compact Kähler manifold.

Therefore the exact problem is not fully resolved. It is also not fully open, since the projective case and some conditional compact Kähler cases are proved. The correct status is partially resolved.

6.11.4. Remaining open problems recorded by the database

6.11.4.1. Open problem 1

Let X be a compact Kähler n -fold and let α be a pseudo-effective real $(1,1)$ -class. Does the orthogonality identity

$$\langle \alpha^{n-1} \rangle \cdot \alpha = \text{Vol}(\alpha)$$

always hold, where $\langle \alpha^{n-1} \rangle$ is the movable positive product?

Reference supplied by the database. Valentino Tosatti, “Orthogonality of divisorial Zariski decompositions for classes with volume zero,” *Tohoku Mathematical Journal* (2) **71** (2019), no. 1, 1–8, Conjecture 1.1. DOI: <https://doi.org/10.2748/tmj/1552100439>.

6.11.4.2. Open problem 2

Let X be a compact Kähler n -fold and let $\alpha, \beta \in H^{1,1}(X, \mathbb{R})$ be nef classes. Is the sharp transcendental Morse inequality

$$\text{Vol}(\alpha - \beta) \geq \alpha^n - n \alpha^{n-1} \cdot \beta$$

always valid?

Reference supplied by the database. Jian Xiao, “Weak transcendental holomorphic Morse inequalities on compact Kähler manifolds,” *Annales de l’Institut Fourier* **65** (2015), no. 3, 1367–1379, Conjecture 1.1(ii) (originally Boucksom–Demailly–Păun–Peternell, Conjecture 10.1(ii)). DOI: <https://doi.org/10.5802/aif.2959>.

6.11.5. Primary-source pointer

The originating ICM record is *Kähler manifolds and transcendental techniques in algebraic geometry* (2006). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.1/ICM2006.1.ocr.pdf#page=163>.

6.12. 217. A Hyperbolic Dispersion Estimate, with Applications to the Linear Schrödinger Equation [2010-geom-103]

Record: 2010-geom-103

Subfield: Manifolds and global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2010: A Hyperbolic Dispersion Estimate, with Applications to the Linear Schrödinger Equation

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.3/ICM2010.3.pdf#page=663>

6.12.1. Problem statement

Let X be a compact d -dimensional Riemannian manifold with negative sectional curvatures, and let Φ^t be the geodesic flow on the unit cotangent bundle

$$S^*X = \{(x, \xi) \in T^*X : \|\xi\|_x = 1\}.$$

For any family of L^2 -normalized Laplace-Beltrami eigenfunctions ϕ_λ satisfying $-\Delta\phi_\lambda = \lambda\phi_\lambda$ with $\lambda \rightarrow +\infty$, let μ be any associated semiclassical measure, i.e. any weak limit of the microlocal distributions

$$\langle \mu_\lambda, a \rangle = \langle \phi_\lambda, \text{Op}_{\lambda^{-1/2}}(a)\phi_\lambda \rangle_{L^2(X)}, \quad a \in C_c^\infty(T^*X).$$

Prove that

$$h_{KS}(\mu; \Phi^t) \geq \frac{1}{2} \int_{S^*X} \sum_{j=1}^{d-1} \lambda_j^+(\rho) d\mu(\rho),$$

where $h_{KS}(\mu; \Phi^t)$ is the Kolmogorov-Sinai entropy of the Φ^t -invariant probability measure μ on S^*X , and $\lambda_j^+(\rho)$ are the nonnegative Lyapunov exponents of the geodesic flow at $\rho = (x, \xi) \in S^*X$.

6.12.2. Historical background, literature, and current progress

6.12.3. Status verdict: partially resolved

6.12.3.1. Precise problem

The problem asks whether every semiclassical measure of Laplace-Beltrami eigenfunctions on an arbitrary compact negatively curved d -dimensional Riemannian manifold satisfies the sharp lower bound

$$h_{KS}(\mu; \Phi^t) \geq \frac{1}{2} \int_{S^*X} \sum_{j=1}^{d-1} \lambda_j^+(\rho) d\mu(\rho).$$

Equivalently, this is the conjectural lower bound by one half of the Ruelle-Pesin upper bound, for all compact negatively curved manifolds and all dimensions.

6.12.3.2. Literature checked

Original source / formulation

- Title: A Hyperbolic Dispersion Estimate, with Applications to the Linear Schrödinger Equation
- Author: Nalini Anantharaman
- Venue: Proceedings of the International Congress of Mathematicians 2010, Volume III, Hindustan Book Agency, pp. 1839-1861
- Year: 2010
- DOI/URL: https://doi.org/10.1142/9789814324359_0124 ; PDF copy: https://www.math.u-bordeaux.fr/~pthieull/Recherche/KamFaible/Publications/Anantharaman2010_01.pdf
- Relevance: Section 3.2 states the conjectured bound exactly for variable negative curvature. The same discussion says the existing method proves positivity of entropy in variable negative curvature and the half-maximal entropy bound in constant curvature, but not the stated variable-curvature formula. It also records that the inequality had been proved in dimension $d = 2$ by Rivière and that, at that time, it was not known how to prove it when the stable/unstable bundles have dimension $d - 1 > 1$.

General positive entropy and constant-curvature progress

- Title: Entropy and the localization of eigenfunctions
- Author: Nalini Anantharaman
- Venue: Annals of Mathematics, Second Series 168(2), 435-475
- Year: 2008
- DOI/URL: <https://doi.org/10.4007/annals.2008.168.435> ; <https://annals.math.princeton.edu/2008/168-2/p02>
- Relevance: Proves that semiclassical measures for Anosov geodesic flows have positive metric entropy. This rules out complete concentration on closed geodesics, but it does not prove the explicit half-Ruelle lower bound in the problem statement.
- Title: Half-delocalization of eigenfunctions for the Laplacian on an Anosov manifold
- Authors: Nalini Anantharaman and Stéphane Nonnenmacher
- Venue: Annales de l'Institut Fourier 57(7), 2465-2523
- Year: 2007
- DOI/arXiv/URL: <https://doi.org/10.5802/aif.2340> ; arXiv:math-ph/0610019 ; <https://numdam.org/articles/10.5802/aif.2340/>
- Relevance: Gives an entropy lower bound for Anosov manifolds that equals half the maximal entropy in constant sectional curvature. For variable curvature, the bound is weaker and is not the integral of half the sum of positive Lyapunov exponents.
- Title: Entropy of eigenfunctions
- Authors: Nalini Anantharaman, Herbert Koch, and Stéphane Nonnenmacher
- Venue: New Trends in Mathematical Physics, Proceedings of the XVth International Congress on Mathematical Physics, Springer, pp. 1-22
- Year: 2009

- DOI/arXiv/URL: https://doi.org/10.1007/978-90-481-2810-5_1 ; arXiv:0704.1564
- Relevance: Improves the explicit lower bound in variable negative curvature, but the theorem still has a maximal-expansion error term and does not imply the sharp half-Ruelle bound in general.

Cases where the requested bound is proved

- Title: Entropy of semiclassical measures in dimension 2
- Author: Gabriel Rivière
- Venue: Duke Mathematical Journal 155(2), 271-335
- Year: 2010
- DOI/arXiv/URL: <https://doi.org/10.1215/00127094-2010-056> ; arXiv:0809.0230 ; <https://projecteuclid.org/journals/duke-mathematical-journal/volume-155/issue-2/Entropy-of-semiclassical-measures-in-dimension-2/10.1215/00127094-2010-056.full>
- Relevance: Theorem 1.2 proves that on compact Riemannian surfaces with Anosov geodesic flow, every semiclassical measure satisfies the lower bound by one half of the Ruelle upper bound. This resolves the problem statement when $d = 2$, but not all dimensions.
- Title: Entropy of Semiclassical Measures for Nonpositively Curved Surfaces
- Author: Gabriel Rivière
- Venue: Annales Henri Poincaré 11(6), 1085-1116
- Year: 2010
- DOI/arXiv/URL: <https://doi.org/10.1007/s00023-010-0055-2> ; arXiv:0911.1840 ; <https://link.springer.com/article/10.1007/s00023-010-0055-2>
- Relevance: Extends the surface entropy lower bound to compact nonpositively curved surfaces with the appropriate Lyapunov exponent. This is stronger in curvature assumptions for surfaces, but still only two-dimensional.

Later surveys and newer related progress

- Title: Semiclassical behaviour of quantum eigenstates
- Author: Gabriel Rivière
- Venue: arXiv preprint / lecture notes
- Year: 2019
- DOI/arXiv/URL: <https://doi.org/10.48550/arXiv.1905.12303> ; arXiv:1905.12303
- Relevance: States the general Anosov conjecture as Conjecture 6.17 and then says that the theorem answers the question in dimension 2, while in higher dimensions the conjecture remains open. It lists only partial explicit bounds for locally symmetric spaces and

multidimensional torus maps beyond the surface case.

- Title: Quantum ergodicity and semiclassical measures: mathematical results
- Author: Stéphane Nonnenmacher
- Venue: arXiv preprint, chapter for Comprehensive Quantum Mechanics
- Year: 2026
- DOI/arXiv/URL: <https://doi.org/10.48550/arXiv.2606.12098> ; arXiv:2606.12098
- Relevance: A current survey of entropy and delocalization constraints. It cites the classical entropy-bound works and describes recent higher-dimensional results as delocalization/support results in locally symmetric settings, not as a proof of the full sharp half-Ruelle entropy lower bound for arbitrary compact variable negatively curved manifolds.
- Title: Semiclassical Measures on Hyperbolic Manifolds
- Authors: Elena Kim and Nicholas Miller
- Venue: arXiv preprint
- Year: 2025
- DOI/arXiv/URL: <https://doi.org/10.48550/arXiv.2503.01528> ; arXiv:2503.01528
- Relevance: Proves support restrictions for semiclassical measures on compact real hyperbolic manifolds. These are constant-curvature locally symmetric manifolds and the result is about support/delocalization, not the full variable-curvature entropy bound.
- Title: Semiclassical measures for complex hyperbolic quotients
- Authors: Jayadev Athreya, Semyon Dyatlov, and Nicholas Miller
- Venue: Geometric and Functional Analysis 35(4), 979-1050
- Year: 2025
- DOI/arXiv/URL: arXiv:2402.06477 ; <https://link.springer.com/journal/139/articles>
- Relevance: Gives support restrictions for compact complex hyperbolic quotients, a locally symmetric negative-curvature class with different Lyapunov exponents. This is important related progress but neither covers arbitrary negatively curved manifolds nor proves the requested entropy inequality.

6.12.3.3. Hypothesis/conclusion check

The statement quantifies over every compact d -dimensional Riemannian manifold with negative sectional curvatures, including variable curvature and dimensions $d \geq 3$. The published theorems above cover:

- positive entropy in the general Anosov/negative-curvature setting;
- the sharp half-Ruelle bound in dimension 2;
- the sharp half-maximal entropy bound in constant real hyperbolic curvature and isotropic-expansion cases;
- weaker explicit entropy bounds in variable curvature;
- newer support/delocalization results for surfaces and locally symmetric higher-

dimensional spaces.

None of these hypotheses and conclusions imply the full statement for arbitrary variable negatively curved manifolds in dimensions $d \geq 3$. The obstruction repeatedly identified in the literature is the presence of several distinct Lyapunov exponents and higher-dimensional stable/unstable directions.

6.12.3.4. What remains unresolved

The exact inequality

$$h_{KS}(\mu; \Phi^t) \geq \frac{1}{2} \int_{S^*X} \sum_{j=1}^{d-1} \lambda_j^+(\rho) d\mu(\rho)$$

remains open for general compact variable negatively curved manifolds in dimensions $d \geq 3$. Therefore the problem is not fully resolved, but it is not fully open because it is proved for surfaces and for constant-curvature/isotropic-expansion special cases, with additional weaker general entropy bounds.

6.12.4. Remaining open problems recorded by the database

6.12.4.1. Open problem 1

Let X be a compact three-dimensional Riemannian manifold of variable negative sectional curvature. Does every semiclassical measure μ of Laplace eigenfunctions satisfy

$$h_{KS}(\mu; \Phi^t) \geq \frac{1}{2} \int_{S^*X} (\lambda_1^+(\rho) + \lambda_2^+(\rho)) d\mu(\rho)?$$

Reference supplied by the database. Gabriel Rivi re, “Semiclassical behaviour of quantum eigenstates,” lecture notes, 2019, Conjecture 6.17 and the subsequent discussion identifying only the two-dimensional case as proved, [arXiv:1905.12303](https://arxiv.org/abs/1905.12303).

6.12.5. Primary-source pointer

The originating ICM record is *A Hyperbolic Dispersion Estimate, with Applications to the Linear Schr dinger Equation* (2010). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.3/ICM2010.3.pdf#page=663>.

6.13. 218. Equidistribution in Homogeneous Spaces and Number Theory [2010-geom-111]

Record: 2010-geom-111

Subfield: Manifolds and global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2010: Equidistribution in Homogeneous Spaces and Number Theory

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.1/ICM2010.1.pdf#page=522>

6.13.1. Problem statement

Let M be a compact Riemannian manifold with negative sectional curvature, let $d \text{ vol}$ be its Riemannian volume measure, and let

$$\text{vol}(M) = \int_M d \text{ vol}.$$

Let $(\phi_i)_{i \geq 1}$ be an $L^2(M, d \text{ vol})$ -orthonormal sequence of eigenfunctions of the Laplacian on M , so that $\Delta \phi_i = \lambda_i \phi_i$ for some eigenvalue λ_i . Prove that, for every $f \in C^\infty(M)$,

$$\lim_{i \rightarrow \infty} \int_M f(x) |\phi_i(x)|^2 d \text{ vol}(x) = \frac{1}{\text{vol}(M)} \int_M f(x) d \text{ vol}(x).$$

6.13.2. Historical background, literature, and current progress

6.13.3. Verdict: partially resolved

6.13.3.1. Precise problem

The statement is the physical-space form of the Rudnick-Sarnak Quantum Unique Ergodicity (QUE) conjecture. It asks whether, for every compact Riemannian manifold M of negative sectional curvature and every orthonormal sequence of Laplace eigenfunctions ϕ_i with eigenvalues tending to infinity, the probability measures

$$|\phi_i(x)|^2 d \text{ vol}(x)$$

converge weakly to the normalized Riemannian volume measure $d \text{ vol} / \text{vol}(M)$. Equivalently, it asks that there be no exceptional subsequence of eigenfunctions whose mass has a different macroscopic limit.

6.13.3.2. Verdict

Partially resolved. The exact full statement for arbitrary compact negatively curved manifolds and arbitrary orthonormal eigenfunction sequences is not known. The literature proves

density-one equidistribution and several strong restrictions on possible exceptional limits, and it proves QUE in special arithmetic/Hecke settings, but all of these have extra hypotheses or weaker conclusions.

6.13.3.3. Literature and status

1. Original source and formulation:

- Title: The behaviour of eigenstates of arithmetic hyperbolic manifolds
- Authors: Zeev Rudnick and Peter Sarnak
- Venue: Communications in Mathematical Physics 161, no. 1, 195-213
- Year: 1994
- DOI/URL: <https://doi.org/10.1007/BF02099418>; <https://projecteuclid.org/journals/communications-in-mathematical-physics/volume-161/issue-1/The-behaviour-of-eigenstates-of-arithmetic-hyperbolic-manifolds/cmp/1104269797.full>
- Relevance: This is the Rudnick-Sarnak source associated with QUE. It studies arithmetic hyperbolic manifolds and proves no strong localization (“scarring”) onto totally geodesic submanifolds in that arithmetic setting. It does not prove the full conjecture for all compact negatively curved manifolds.

2. problem source / exact Conjecture 8:

- Title: Equidistribution in homogeneous spaces and number theory
- Author: Elon Lindenstrauss
- Venue: Proceedings of the International Congress of Mathematicians 2010, Vol. I, Hindustan Book Agency, pp. 531-557
- Year: 2010
- URL: <https://math.huji.ac.il/~elon/Publications/ICM2010.pdf>
- Relevance: Section 4.8 states Conjecture 8 in the same form as this problem and notes that both the physical-space and stronger phase-space versions are open. It also records the arithmetic Hecke-eigenfunction theorem as a special case, not a solution of the general conjecture.

3. Quantum ergodicity, a density-one result:

- Title: Ergodic properties of eigenfunctions
- Author: A. I. Shnirelman
- Venue: Uspekhi Mat. Nauk 29, no. 6(180), 181-182
- Year: 1974
- URL: <https://www.mathnet.ru/eng/rm4463>
- Title: Ergodicite et fonctions propres du laplacien
- Author: Yves Colin de Verdiere
- Venue: Communications in Mathematical Physics 102, 497-502
- Year: 1985
- DOI/URL: <https://doi.org/10.1007/BF01209296>
- Title: Uniform distribution of eigenfunctions on compact hyperbolic surfaces

- Author: Steven Zelditch
- Venue: Duke Mathematical Journal 55, no. 4, 919-941
- Year: 1987
- DOI/URL: <https://doi.org/10.1215/S0012-7094-87-05546-3>
- Relevance: These works prove the Quantum Ergodicity theorem: if the geodesic flow is ergodic, then a density-one subsequence of any eigenbasis equidistributes. Negative sectional curvature implies ergodicity/Anosov dynamics, so this applies here. It is only a density-one statement and does not exclude exceptional subsequences, so it does not resolve QUE.

4. Arithmetic QUE special cases:

- Title: Invariant measures and arithmetic quantum unique ergodicity
- Author: Elon Lindenstrauss
- Venue: Annals of Mathematics 163, no. 1, 165-219
- Year: 2006
- DOI/URL: <https://doi.org/10.4007/annals.2006.163.165>
- Relevance: Proves arithmetic QUE for compact arithmetic hyperbolic surfaces, using Hecke symmetry and measure rigidity. The hypotheses require arithmetic structure and Hecke-Laplace eigenfunctions, so this does not cover arbitrary compact negatively curved manifolds or arbitrary eigenbases.
- Title: Quantum unique ergodicity for $SL_2(\mathbb{Z}) \backslash \mathbb{H}$
- Author: Kannan Soundararajan
- Venue: Annals of Mathematics 172, no. 2, 1529-1538
- Year: 2010
- DOI/arXiv/URL: <https://doi.org/10.4007/annals.2010.172.1529> ; arXiv:0901.4060
- Relevance: Eliminates escape of mass for Hecke-Maass forms on the modular surface; combined with Lindenstrauss this establishes QUE for the noncompact modular surface. This is not the compact general statement and still uses Hecke eigenfunctions.
- Title: Joint quasimodes, positive entropy, and quantum unique ergodicity
- Authors: Shimon Brooks and Elon Lindenstrauss
- Venue: Inventiones Mathematicae 198, no. 1, 219-259
- Year: 2014
- DOI/arXiv/URL: <https://doi.org/10.1007/s00222-014-0502-7> ; arXiv:1112.5311
- Relevance: Gives QUE for certain joint quasimodes/eigenfunctions involving Hecke operators on compact congruence surfaces. This again depends on arithmetic/Hecke structure and is not the unrestricted Rudnick-Sarnak conjecture.

5. Restrictions on exceptional quantum limits:

- Title: Entropy and the localization of eigenfunctions

- Author: Nalini Anantharaman
 - Venue: *Annals of Mathematics* 168, no. 2, 435-475
 - Year: 2008
 - DOI/URL: <https://doi.org/10.4007/annals.2008.168.435>
 - Relevance: For compact negatively curved, more generally Anosov, manifolds, Wigner measures of eigenfunctions have positive metric entropy. This rules out total concentration on a closed geodesic but does not force the Liouville measure.
 - Title: Half-delocalization of eigenfunctions for the Laplacian on an Anosov manifold
 - Authors: Nalini Anantharaman and Stephane Nonnenmacher
 - Venue: *Annales de l'Institut Fourier* 57, no. 7, 2465-2523
 - Year: 2007
 - DOI/arXiv/URL: <https://doi.org/10.5802/aif.2340> ; arXiv:math-ph/0610019
 - Relevance: Gives stronger entropy lower bounds, equal to half the maximal entropy in constant negative curvature. This is a major partial result but still permits many non-Liouville invariant measures.
 - Title: Semiclassical measures on hyperbolic surfaces have full support
 - Authors: Semyon Dyatlov and Long Jin
 - Venue: *Acta Mathematica* 220, no. 2, 297-339
 - Year: 2018
 - DOI/arXiv/URL: <https://doi.org/10.4310/ACTA.2018.v220.n2.a3> ; arXiv:1705.05019
 - Relevance: Shows every semiclassical measure on a compact hyperbolic surface has full support. Full support is weaker than being Liouville, so this is partial progress only.
 - Title: Control of eigenfunctions on surfaces of variable curvature
 - Authors: Semyon Dyatlov, Long Jin, and Stephane Nonnenmacher
 - Venue: *Journal of the American Mathematical Society* 35, no. 2, 361-465
 - Year: 2022
 - DOI/arXiv/URL: <https://doi.org/10.1090/jams/979> ; arXiv:1906.08923
 - Relevance: Extends the full-support theorem to compact negatively curved surfaces and more generally surfaces with Anosov geodesic flow. Again, full support does not imply equidistribution to normalized volume.
6. Recent surveys confirming the current status:
- Title: Recent progress on the quantum unique ergodicity conjecture
 - Author: Peter Sarnak
 - Venue: *Bulletin of the American Mathematical Society* 48, no. 2, 211-228
 - Year: 2011
 - DOI/URL: <https://doi.org/10.1090/S0273-0979-2011-01323-4>

- Relevance: Surveys arithmetic QUE breakthroughs and general microlocal progress; it treats the general Rudnick-Sarnak conjecture as unresolved.
- Title: Macroscopic limits of chaotic eigenfunctions
- Author: Semyon Dyatlov
- Venue: Proceedings of the International Congress of Mathematicians 2022, Vol. 5, EMS Press, pp. 3704-3723
- Year: 2023
- DOI/arXiv/URL: <https://doi.org/10.4171/ICM2022/22> ; arXiv:2109.09053
- Relevance: States the QUE conjecture for Anosov systems and says it is known in arithmetic hyperbolic surface cases with Hecke symmetry, but open in general despite entropy and full-support progress.
- Title: Quantum ergodicity and semiclassical measures: mathematical results
- Author: Stephane Nonnenmacher
- Venue: arXiv preprint
- Year: 2026
- arXiv/URL: arXiv:2606.12098 ; <https://arxiv.org/abs/2606.12098>
- Relevance: A current survey that formulates QUE for compact manifolds of negative curvature and states that the conjecture remains open, even for constant negative curvature, outside the arithmetic Hecke-eigenfunction setting.

6.13.3.4. What remains unresolved

No published or citable result found proves that every orthonormal sequence of Laplace eigenfunctions on every compact negatively curved Riemannian manifold satisfies the stated equidistribution limit. The unresolved part is exactly the removal of the density-one restriction and the removal of all extra arithmetic/Hecke/special-surface hypotheses. In particular, the literature still does not prove that the normalized Liouville/volume measure is the only possible quantum limit for arbitrary compact negatively curved manifolds.

6.13.4. Remaining open problems recorded by the database

6.13.4.1. Open problem 1

Let M be a compact hyperbolic surface. Prove that every sequence of L^2 -normalized Laplace eigenfunctions whose eigenvalues tend to infinity equidistributes in physical space, without assuming that the eigenfunctions are simultaneous Hecke eigenfunctions.

Reference supplied by the database. Ankit Bisain, Peter Humphries, Andrei Mandelsham, Noah Walsh, and Xun Wang, “Subconvexity Implies Effective Quantum Unique Ergodicity for Hecke–Maaß Cusp Forms on $\mathrm{SL}_2(\mathbb{Z}) \backslash \mathrm{SL}_2(\mathbb{R})$,” *Essential Number Theory* 3, no. 1 (2024), 101–148, §1A (explicitly identifies QUE for compact hyperbolic surfaces, without the arith-

metic Hecke restriction, as the Rudnick–Sarnak conjecture and as far from proved; no separate conjecture number). DOI: 10.2140/ent.2024.3.101. <https://doi.org/10.2140/ent.2024.3.101>

6.13.5. Primary-source pointer

The originating ICM record is *Equidistribution in Homogeneous Spaces and Number Theory* (2010). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.1/ICM2010.1.pdf#page=522>.

6.14. 219. Moduli spaces of manifolds [2014-ag-161]

Record: 2014-ag-161

Subfield: Manifolds and global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: Moduli spaces of manifolds

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=1217>

6.14.1. Problem statement

For $n = 1$, let $W_g = \#^g(S^1 \times S^1)$ be the closed oriented surface of genus g , and fix an embedded disk $D^2 \subset W_g$. For a smooth oriented W_g -bundle $\pi : E \rightarrow B$, with vertical tangent bundle $T_\pi E$ and Euler class $e(T_\pi E)$, define

$$\kappa_{e^k}(\pi) = \pi_! (e(T_\pi E)^k) \in H^{2k-2}(B; \mathbb{Q}), \quad k \geq 2.$$

Determine, for each genus g , all universal relations among the Miller-Morita-Mumford classes κ_{e^k} in the tautological subrings, equivalently determine the kernels of the natural homomorphisms

$$\mathbb{Q}[\kappa_{e^k} \mid k \geq 2] \rightarrow H^*(B \operatorname{Diff}(W_g, D^2); \mathbb{Q})$$

and

$$\mathbb{Q}[\kappa_{e^k} \mid k \geq 2] \rightarrow H^*(B \operatorname{Diff}^+(W_g); \mathbb{Q}),$$

where $\operatorname{Diff}(W_g, D^2)$ denotes diffeomorphisms restricting to the identity near D^2 .

6.14.2. Historical background, literature, and current progress

6.14.3. Status check

Verdict: partially resolved.

6.14.3.1. Precise problem

For the genus g oriented surface $W_g = \#^g(S^1 \times S^1)$, determine the full kernel of the natural map

$$\mathbb{Q}[\kappa_{e^k} \mid k \geq 2] \rightarrow H^*(B \operatorname{Diff}(W_g, D^2); \mathbb{Q})$$

and similarly for

$$\mathbb{Q}[\kappa_{e^k} \mid k \geq 2] \rightarrow H^*(B \operatorname{Diff}^+(W_g); \mathbb{Q}).$$

Equivalently, after identifying $B \operatorname{Diff}^+(W_g)$ with the orbifold moduli space $\mathcal{M}_g$ and κ_{e^k} with the usual surface κ_{k-1} classes, the question asks for all universal relations among the κ -classes in the tautological κ -subring, in every genus, including the disk-fixed/marked-disk version.

6.14.3.2. Literature and progress

1. Carel Faber, “A conjectural description of the tautological ring of the moduli space of curves”, in *Moduli of Curves and Abelian Varieties, Aspects of Mathematics E33*, Vieweg, 1999, pp. 109-129. arXiv:math/9711218. URL: <https://arxiv.org/abs/math/9711218> This is the original source of the all-genus conjectural description of $R^*(M_g)$. The arXiv abstract states that the paper formulates conjectures giving a rather complete description of the tautological ring of M_g . This is the source of the conjectural all-genus answer, not a proof of the full all-genus relations.
2. Eduard Looijenga, “On the tautological ring of M_g ”, *Inventiones Mathematicae* 121 (1995), 411-419. arXiv:alg-geom/9501010. URL: <https://arxiv.org/abs/alg-geom/9501010> Looijenga proved the key vanishing/socle result: tautological products in degree $d > g - 2$ vanish and degree $g - 2$ is at most one-dimensional. This is major structural progress, but it does not determine all relations among the κ -classes.
3. Carel Faber, “Tautological algebras of moduli spaces of curves”, in *Moduli Spaces of Riemann Surfaces*, IAS/Park City Mathematics Series 20, AMS/IAS, 2013, pp. 197-219. URL: <https://www.staff.science.uu.nl/~faber010/faber.jcpB.pdf> Faber’s survey records the state of the conjecture: large parts are proved, the complete conjecture was proved computationally for $g \leq 23$, and the relations used for $g \leq 23$ appear insufficient to prove the Gorenstein prediction in genus 24. This supports the stated “known up to $g \leq 23$ ” statement but also confirms that the all-genus problem was not settled there.
4. Qizheng Yin, “On the tautological rings of $M_{g,1}$ and its universal Jacobian”, arXiv:1206.3783, 2012. URL: <https://arxiv.org/abs/1206.3783> Yin gives additional relations for the one-pointed case and proves Faber-type conjectures for $M_{g,1}$ up to $g = 19$, and by pushing forward obtains another proof for M_g up to $g = 23$. This is partial evidence for the disk-fixed/marked version, not an all-genus solution.
5. R. Pandharipande and A. Pixton, “Relations in the tautological ring of the moduli space of curves”, *Pure and Applied Mathematics Quarterly* 17 (2021), no. 2, 717-771. arXiv:1301.4561. URL: <https://arxiv.org/abs/1301.4561> DOI: <https://doi.org/10.48550/arXiv.1301.4561> This paper proves the Faber-Zagier relations in the Chow tautological ring of M_g . It explicitly defines the kernel problem for $\mathbb{Q}[\kappa_1, \kappa_2, \dots] \rightarrow R^*(M_g)$, proves that the Faber-Zagier expressions are genuine relations, and states as Conjecture 2 that the Faber-Zagier ideal is the entire kernel for all g . The same paper says that whether additional cohomological relations exist is not settled and that no non-Faber-Zagier relation has been found. Therefore it proves a large, central family of relations but not completeness.
6. R. Pandharipande, A. Pixton, and D. Zvonkine, “Relations on $\overline{M}_{g,n}$ via 3-spin structures”, *Journal of the American Mathematical Society* 28 (2015), 279-309. arXiv:1303.1043. URL: <https://arxiv.org/abs/1303.1043> This proves Pixton’s generalized relations in cohomology for stable pointed curves. It includes the Faber-Zagier relations on M_g as a special/open-interior shadow, but it does not prove that these relations are complete.
7. Felix Janda, “Relations on $\overline{M}_{g,n}$ via equivariant Gromov-Witten theory of $\mathbb{P}^1$ ”, *Algebraic Geometry* 4 (2017), 311-336. URL: <https://algebraicgeometry.nl/2017-4/2017-4-006.pdf> DOI: 10.14231/AG-2017-006 Janda proves Pixton’s relations in Chow using stable maps to $\mathbb{P}^1$. This strengthens validity of the known relations but still does not establish that they generate all relations.

8. Samir Canning, “The tautological ring of $\overline{M}_{g,n}$ is rarely Gorenstein”, *Geometry & Topology* 29 (2025), no. 7, 3905–3919; arXiv:2406.10516. URL: <https://arxiv.org/abs/2406.10516> DOI: 10.2140/gt.2025.29.3905 This recent work concerns the compactified spaces $\overline{M}_{g,n}$, proving non-Gorenstein results when $g \geq 2$ and $2g + n \geq 24$, and explicitly states that finding a complete set of relations among tautological generators remains a fundamental open problem. Its hypotheses and conclusion are not the same as the smooth κ -subring problem considered here.

6.14.3.3. Why this is not fully resolved

The strongest directly relevant all-genus theorem proves that the Faber-Zagier relations are true relations. The missing step is completeness: no published or otherwise citable result found proves that the Faber-Zagier ideal, or any other explicit ideal, is exactly the full kernel of the κ -class map for all genera in rational cohomology of $B\text{Diff}^+(W_g)$ and $B\text{Diff}(W_g, D^2)$.

The literature repeatedly formulates completeness as conjectural: Pandharipande-Pixton state the all-genus kernel equality as a conjecture, and later work on Pixton relations proves validity rather than completeness. The known complete computations for small genus, especially $g \leq 23$ for M_g , and the proven Faber-Zagier/Pixton relations are substantial progress. The all-genus universal relation problem remains unresolved.

6.14.4. Remaining open problems recorded by the database

6.14.4.1. Open problem 1

For every genus $g \geq 2$ and every $0 \leq d \leq g - 2$, is the multiplication pairing

$$R^d(M_g) \times R^{g-2-d}(M_g) \longrightarrow R^{g-2}(M_g) \cong \mathbb{Q}$$

perfect (equivalently, is $R^*(M_g)$ Gorenstein with socle in degree $g - 2$)?

Reference supplied by the database. Carel Faber, “A Conjectural Description of the Tautological Ring of the Moduli Space of Curves,” in Carel Faber and Eduard Looijenga (eds.), *Moduli of Curves and Abelian Varieties*, Aspects of Mathematics E33, Vieweg, 1999, pp. 109–129, Conjecture 1 (perfect-pairing/Gorenstein part). DOI: 10.1007/978-3-322-90172-9_6. https://doi.org/10.1007/978-3-322-90172-9_6; arXiv:math/9711218, <https://arxiv.org/abs/math/9711218>

6.14.4.2. Open problem 2

For every genus $g \geq 2$ and every $0 \leq d \leq g - 1$, is the multiplication pairing

$$R^d(M_{g,1}) \times R^{g-1-d}(M_{g,1}) \longrightarrow R^{g-1}(M_{g,1}) \cong \mathbb{Q}$$

perfect (equivalently, is $R^*(M_{g,1})$ Gorenstein with socle in degree $g - 1$)?

Reference supplied by the database. Qizheng Yin, “On the Tautological Rings of $\mathcal{M}_{g,1}$ and Its Universal Jacobian,” preprint, first posted 2012 and revised 2026, Conjecture 3.1(i) (Faber’s conjecture for $\mathcal{M}_{g,1}$); arXiv:1206.3783. <https://arxiv.org/abs/1206.3783>

6.14.4.3. Open problem 3

Is the tautological ring $R^*(\mathcal{M}_{24})$ not Gorenstein?

Reference supplied by the database. Aaron Pixton, *The Tautological Ring of the Moduli Space of Curves*, Ph.D. dissertation, Princeton University, 2013, Appendix A, Conjecture A.1 (the $\mathcal{M}_{24}$ case), ProQuest LLC, MR 3153424. <https://math.mit.edu/~apixton/papers/tautological.pdf>

6.14.5. Primary-source pointer

The originating ICM record is *Moduli spaces of manifolds* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=1217>.

6.15. 220. Algebraic surfaces with minimal Betti numbers [2018-ag-177]

Record: 2018-ag-177

Subfield: Manifolds and global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Algebraic surfaces with minimal Betti numbers

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/uploads/ICM-2018-vol2-ver1-eb.pdf#page=784>

6.15.1. Problem statement

Every faithful smooth action of the circle group S^1 on the 5-sphere S^5 that is pseudofree, meaning that it is free except for finitely many orbits whose isotropy groups are nontrivial finite cyclic groups, has at most three exceptional orbits, where the exceptional orbits are precisely those non-free orbits with finite cyclic isotropy.

6.15.2. Historical background, literature, and current progress

6.15.3. Verdict: partially resolved

6.15.3.1. Precise problem checked

The problem asks whether every faithful smooth pseudofree circle action

$$S^1 \times S^5 \rightarrow S^5$$

has at most three exceptional/non-free orbits, where pseudofree means free away from finitely many orbits with nontrivial finite cyclic isotropy. Equivalently, in the standard orbit-space formulation, it asks whether the associated simply connected smooth compact 4-manifold with $H_2 \cong \mathbb{Z}$ can have boundary a connected sum of more than three lens spaces with pairwise coprime isotropy orders.

6.15.3.2. Status

I did not find any published or otherwise citable theorem proving the full unrestricted smooth statement. The literature contains substantial partial progress, especially for algebraic orbit spaces and for additional geometric/topological hypotheses, but the full smooth pseudofree S^1 -action problem on S^5 remains unresolved.

The verdict is therefore **partially resolved**, not fully resolved: there are citable special-case and obstruction results, but they do not cover all smooth pseudofree circle actions on S^5 .

6.15.3.3. Key literature and what it proves

1. Deane Montgomery and C. T. Yang, "Differentiable pseudo-free circle actions on homotopy seven spheres," *Proceedings of the Second Conference on Compact Transformation Groups, Lecture Notes in Mathematics 298*, Springer, 1972, pp. 41-101. DOI: <https://doi.org/10.1007/BFb0070033>.

This is one of the sources motivating the problem: in dimension 7 and higher odd dimensions there are pseudofree circle actions with arbitrarily many exceptional orbits. It does not resolve the S^5 case.

2. Ronald Fintushel and Ronald J. Stern, "Pseudofree orbifolds," *Annals of Mathematics* (2) 122(2), 1985, pp. 335-364. DOI: <https://doi.org/10.2307/1971306>.

Develops the pseudofree orbifold/4-manifold viewpoint used in later formulations. It supplies background and tools, not a full proof of the Montgomery-Yang bound.

3. Ronald Fintushel and Ronald J. Stern, "O(2) actions on the 5-sphere," *Inventiones Mathematicae* 87(3), 1987, pp. 457-476. DOI: <https://doi.org/10.1007/BF01389237>.

This is the standard cited source for the Montgomery-Yang problem in this form. Kollár later notes that the proposed partial solution in this paper is incorrect, citing Math Reviews MR0874031; therefore it cannot be counted as a full resolution.

4. János Kollár, "Is there a topological Bogomolov-Miyaoka-Yau inequality?", *Pure and Applied Mathematics Quarterly* 4(2), 2008, pp. 203-236. DOI: <https://doi.org/10.4310/PAMQ.2008.v4.n2.a1>. arXiv: math/0602562.

Kollár states the Montgomery-Yang problem for circle actions on S^5 , gives the 4-manifold/lens-space correspondence, and formulates algebraic analogues. He explicitly records that the Fintushel-Stern proposed partial solution is incorrect. This paper frames the problem and related conjectures but does not prove the full statement.

5. Jin Hong Kim, "The number of exceptional orbits of a pseudofree circle action on S^5 ," *Journal of Fixed Point Theory and Applications* 9, 2011, pp. 55-62. DOI: <https://doi.org/10.1007/s11784-010-0039-3>.

Proves bounds under additional Riemannian hypotheses: for pseudofree isometric S^1 -actions on nonnegatively curved 5-manifolds, and in particular on S^5 , the number of exceptional circle orbits is at most five under the paper's hypotheses. This is related progress but weaker than "at most three" and assumes extra geometry.

6. DongSeon Hwang and JongHae Keum, "The maximum number of singular points on rational homology projective planes," *Journal of Algebraic Geometry* 20(3), 2011, pp. 495-523. DOI: <https://doi.org/10.1090/S1056-3911-10-00532-1>.

Gives algebraic bounds for quotient singularities on rational homology projective planes. This supports the algebraic Montgomery-Yang program but does not cover arbitrary smooth orbit spaces.

7. DongSeon Hwang and JongHae Keum, "Algebraic Montgomery-Yang problem: the non-cyclic case," *Mathematische Annalen* 350(3), 2011, pp. 721-754. DOI: <https://doi.org/10.1007/s00208-010-0565-8>. arXiv: 0904.2975.

Proves the algebraic Montgomery-Yang conjecture when the rational homology projective plane has at least one non-cyclic singularity. This is a genuine special case of the

algebraic analogue, not a solution of the full smooth S^5 action problem.

8. **DongSeon Hwang and JongHae Keum, "Algebraic Montgomery-Yang problem: the nonrational surface case,"** *Michigan Mathematical Journal* 62(1), 2013, pp. 3-37. DOI: <https://doi.org/10.1307/mmj/1363958239>.

Solves another algebraic special case, namely nonrational surfaces. It leaves the rational cyclic ample-canonical case outside its scope.

9. **DongSeon Hwang and JongHae Keum, "Algebraic Montgomery-Yang problem: the log del Pezzo surface case,"** *Journal of the Mathematical Society of Japan* 66(4), 2014, pp. 1073-1089. DOI: <https://doi.org/10.2969/jmsj/06641073>.

Proves the algebraic Montgomery-Yang statement for log del Pezzo surfaces of Picard number one. Again, this is a special algebraic setting.

10. **DongSeon Hwang, "Algebraic Montgomery-Yang problem and cascade conjecture,"** *arXiv preprint, 2020/2021*. arXiv: 2012.13355.

States that the algebraic Montgomery-Yang problem remains open for rational $\mathbb{Q}$ -homology projective planes with cyclic quotient singularities and ample canonical divisor. It proves the algebraic conjecture assuming the additional cascade conjecture, so the result is conditional.

11. **Woohyeok Jo, Jongil Park, and Kyungbae Park, "Algebraic Montgomery-Yang problem and smooth obstructions,"** *Transactions of the American Mathematical Society* 378(4), 2025, pp. 2969-3003. DOI: <https://doi.org/10.1090/tran/9364>. arXiv: 2402.04569.

This is the strongest recent progress found. It states that the algebraic Montgomery-Yang problem has not been completely resolved, summarizes known partial results, formulates a smooth 4-manifold obstruction problem, and rules out many four-singularity types using Donaldson theory, Heegaard Floer correction terms, and related conditions. The paper explicitly says that infinitely many singularity types are still not ruled out by its conditions, so it does not resolve the full problem.

12. **Jihao Liu and Wenfei Liu, "The minimal volume of stable surfaces of rank one,"** *arXiv preprint, 2026*. arXiv: 2605.05641.

This very recent preprint applies a plurigenus filter to Kollár's algebraic Montgomery-Yang problem after the Jo-Park-Park reductions. It reports that Jo-Park-Park leave 16 values of $p \leq 50000$ to be excluded in a remaining algebraic family and that the new filter excludes 8 of those 16. This is additional partial progress in the algebraic analogue, not a full resolution of either the algebraic problem or the original unrestricted smooth S^5 problem.

6.15.3.4. What remains unresolved

No cited result proves that every faithful smooth pseudofree S^1 -action on the standard S^5 has at most three exceptional orbits. Existing results either:

- treat algebraic/projective orbit spaces rather than arbitrary smooth orbit spaces;
- impose extra assumptions such as non-cyclic singularities, nonrationality, log del Pezzo structure, cascade existence, bounded/computational singularity families, curvature, or isometric action;
- prove a weaker numerical bound such as at most five exceptional orbits; or
- rule out many candidate boundary/singularity types without excluding all possible cases.

Thus the exact statement is not fully resolved by the current literature.

6.15.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

6.15.5. Primary-source pointer

The originating ICM record is *Algebraic surfaces with minimal Betti numbers* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/uploads/ICM-2018-vol2-ver1-eb.pdf#page=784>.

6.16. 221. Lattice subgroups acting on manifolds [2022-geom-165]

Record: 2022-geom-165

Subfield: Manifolds and global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2022: Lattice subgroups acting on manifolds

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-563-6-volume-5.pdf#page=33>

6.16.1. Problem statement

Let G be a connected noncompact simple Lie group of real rank at least 2, meaning that the dimension of a maximal $\mathbb{R}$ -split abelian subalgebra of the Lie algebra of G is at least 2. Let $\Gamma \subset G$ be a lattice, meaning a discrete subgroup such that G/Γ has finite invariant volume. Let M be a compact manifold, and let vol be a smooth volume form on M . Fix a differentiability regularity class, write $\text{Diff}(M)$ for the diffeomorphism group of M in that regularity class, and write $\text{Diff}_{\text{vol}}(M)$ for the subgroup of diffeomorphisms preserving vol .

Define

$$v(G) = \min\{\dim(G/H) : H \subset G \text{ is a proper closed subgroup}\},$$

let $n(G)$ be the minimal dimension of a nontrivial real linear representation of the Lie algebra of G , and define

$$d(G) = \min\{\dim(G_{\text{cpt}}/H) : H \subset G_{\text{cpt}} \text{ is a proper closed subgroup}\},$$

where G_{cpt} denotes the connected compact Lie group whose Lie algebra is a compact real form of the complexification of the Lie algebra of G .

Prove that the following assertions hold:

1. If $\dim(M) < \min\{n(G), d(G), v(G)\}$, then every homomorphism $\alpha : \Gamma \rightarrow \text{Diff}(M)$ has finite image.
2. If $\dim(M) < \min\{n(G), d(G)\}$, then every homomorphism $\alpha : \Gamma \rightarrow \text{Diff}_{\text{vol}}(M)$ has finite image.
3. If $\dim(M) < \min\{v(G), n(G)\}$, then for every homomorphism $\alpha : \Gamma \rightarrow \text{Diff}(M)$, there exists an $\alpha(\Gamma)$ -invariant Riemannian metric on M , at least a C^0 Riemannian metric.
4. If $\dim(M) < n(G)$, then for every homomorphism $\alpha : \Gamma \rightarrow \text{Diff}_{\text{vol}}(M)$, there exists an $\alpha(\Gamma)$ -invariant Riemannian metric on M , at least a C^0 Riemannian metric.

6.16.2. Historical background, literature, and current progress

6.16.3. Verdict: partially resolved

6.16.3.1. Problem identified

The problem asks for the general Zimmer/Farb-Shalen low-dimensional action conjecture for a lattice Γ in a connected noncompact simple Lie group G of real rank at least 2. In the notation of the problem, it asks whether actions of Γ on compact manifolds below the critical dimensions defined by $n(G)$, $d(G)$, and $v(G)$ must either have finite image or preserve a Riemannian metric, with separate volume-preserving and non-volume-preserving versions.

6.16.3.2. Status summary

This exact general statement is not fully resolved. There is major progress, including sharp results for real-split simple groups and many non-split families, but I found no published or standard arXiv-citable theorem covering all connected noncompact simple real-rank-at-least-2 Lie groups and all four clauses at the stated bounds. Recent non-split work still states results for specified families and weaker bounds in some cases.

6.16.3.3. Key literature and progress

1. **Robert J. Zimmer, “Actions of semisimple groups and discrete subgroups.”**
 Venue/year: Proceedings of the International Congress of Mathematicians, Berkeley, 1986, American Mathematical Society, 1987, pp. 1247-1258.
 DOI/arXiv/URL: no DOI found; bibliographic URL: <https://mathscinet.ams.org/mathscinet-getitem?mr=934298>
 Role: original Zimmer-program source for rigidity questions about actions of higher-rank lattices on manifolds.
2. **Aaron Brown, “Lattice subgroups acting on manifolds.”**
 Venue/year: Proceedings of the International Congress of Mathematicians 2022, Vol. 5, EMS Press, 2023, pp. 3388-3411.
 DOI/arXiv/URL: DOI 10.4171/ICM2022/163; <https://ems.press/books/standalone/277/5519>
 Progress: survey/source for the formulation and current Zimmer-program landscape. It discusses low-dimensional existence questions under the name Zimmer’s conjecture, but it is not itself a full proof of the general statement.
3. **Aaron Brown, David Fisher, Sebastian Hurtado, “Zimmer’s conjecture: Subexponential growth, measure rigidity, and strong property (T).”**
 Venue/year: Annals of Mathematics (2) 196 (2022), no. 3, 891-940.
 DOI/arXiv/URL: DOI 10.4007/annals.2022.196.3.1; arXiv:1608.04995; <https://annals.math.princeton.edu/2022/196-3/p01>
 Progress: states the full Zimmer/Farb-Shalen conjecture and proves several major cocompact-lattice cases. It proves the non-volume-preserving case for cocompact lattices in higher-rank split simple Lie groups and certain volume-preserving cases, but explicitly gives only partial results for non-split groups.
4. **Aaron Brown, David Fisher, Sebastian Hurtado, “Zimmer’s conjecture for actions of $SL(m, \mathbb{Z})$.”**

Venue/year: *Inventiones Mathematicae* 221 (2020), no. 3, 1001-1060.

DOI/arXiv/URL: DOI 10.1007/s00222-020-00962-x; arXiv:1710.02735.

Progress: proves Zimmer's conjecture for C^2 actions by finite-index subgroups of $SL(m, \mathbb{Z})$ for $m > 3$. This is a sharp and important nonuniform arithmetic case, but it does not cover arbitrary simple G .

5. **Aaron Brown, David Fisher, Sebastian Hurtado, "Zimmer's conjecture for non-uniform lattices: escape of mass and growth of cocycles."**

Venue/year: arXiv preprint, 2021; current version v3, 2024.

DOI/arXiv/URL: arXiv:2105.14541; DOI 10.48550/arXiv.2105.14541; <https://arxiv.org/abs/2105.14541>

Progress: extends the BFH method to nonuniform lattices. Theorem A proves the expected bounds for lattices in connected $\mathbb{R}$ -split simple Lie groups of real rank at least 2. Theorem B gives broader bounds using the minimal resonant codimension $r(G)$. For non- $\mathbb{R}$ -split groups, $r(G)$ can be strictly smaller than $v(G)$, so this is partial for the full statement.

6. **Aaron Brown, Danijela Damjanović, Zhiyuan Zhang, " C^1 actions on manifolds by lattices in Lie groups."**

Venue/year: *Compositio Mathematica* 158 (2022), no. 3, 529-549.

DOI/arXiv/URL: DOI 10.1112/S0010437X22007278; arXiv:1801.04009; <https://www.cambridge.org/core/journals/compositio-mathematica/article/c1-actions-on-manifolds-by-lattices-in-lie-groups/05C8DE5533527E5AEE7764FDE4D40897>

Progress: obtains C^1 -regularity results when the rank of a uniform lattice is larger than the manifold dimension, with sharp bounds for lattices in $SL(n, \mathbb{R})$. These are important low-regularity results but not a full resolution of the general $n(G), d(G), v(G)$ statement.

7. **Jinpeng An, Aaron Brown, Zhiyuan Zhang, "Zimmer's conjecture for non-split semisimple Lie groups."**

Venue/year: arXiv preprint, 2024.

DOI/arXiv/URL: arXiv:2411.13858; DOI 10.48550/arXiv.2411.13858; <https://arxiv.org/abs/2411.13858>

Progress: proves many new non- $\mathbb{R}$ -split cases. Theorem 1.3 covers connected complex semisimple groups without rank-one simple factors and specified families including $SL(n, \mathbb{H})$, $SO^+(m, n)$, $SU(m, n)$, $SO^*(2n)$, and EII , with stated parameter ranges; it also gives weaker $v(G) - 2$ bounds for some $Sp(m, n)$. The paper's own formulation is "many new cases", not all cases, so it does not fully resolve the problem statement.

6.16.3.4. What remains unresolved

The unresolved part is the full non- $\mathbb{R}$ -split generality: all connected noncompact simple Lie groups of real rank at least 2, all lattices, and all four finite-image/invariant-metric conclusions at the exact critical dimensions involving $n(G)$, $d(G)$, and $v(G)$. Current literature covers the real-split case and many non-split families, but not every simple real form and not every volume-preserving/invariant-metric clause at the stated sharp bounds.

6.16.4. Remaining open problems recorded by the database

6.16.4.1. Open problem 1

Let Γ be a lattice in a connected Lie group whose Lie algebra is $\mathfrak{so}(n, n+1)$ for $n \geq 3$, or $\mathfrak{so}(n, n)$ for $n \geq 4$. Prove that every sufficiently smooth volume-preserving action of Γ on a compact manifold of dimension $n(\mathfrak{g}) - 1$ has finite image (thus, respectively, in dimensions $2n$ and $2n - 1$).

Reference supplied by the database. Aaron Brown, Federico Rodriguez Hertz, and Zhiren Wang, “Boundary Actions by Higher-Rank Lattices: Classification and Embedding in Low Dimensions, Local Rigidity, Smooth Factors,” 2024, §2.8, especially the paragraph following Theorem 2.17 (explicitly identifying this as the missing dimension in the volume-preserving Zimmer conjecture and reducing it to ruling out an embedded projective subaction), arXiv: [2405.16202](https://arxiv.org/abs/2405.16202).

6.16.4.2. Open problem 2

Let $G = \mathrm{Sp}(m, n)$ with $6 \leq n \leq m \leq (n^2 - 3n + 6)/4$, and let $\Gamma < G$ be a lattice. If M is a compact manifold of either of the two still-uncovered dimensions

$$\dim M = v(G) - 2 = 4m + 4n - 7 \quad \text{or} \quad \dim M = v(G) - 1 = 4m + 4n - 6,$$

must every $C^{1+\varepsilon}$ action $\Gamma \rightarrow \mathrm{Diff}(M)$ have finite image?

Reference supplied by the database. Jinpeng An, Aaron Brown, and Zhiyuan Zhang, “Zimmer’s Conjecture for Non-Split Semisimple Lie Groups,” 2024, Conjecture 1.1 (non-volume-preserving Zimmer conjecture), the paragraph preceding Conjecture 1.2, and Theorem 3.12 (which proves only the weaker range $\dim M < v(G) - 2$ for the displayed $\mathrm{Sp}(m, n)$ family). arXiv: [2411.13858](https://arxiv.org/abs/2411.13858).

6.16.5. Primary-source pointer

The originating ICM record is *Lattice subgroups acting on manifolds* (2022). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-563-6-volume-5.pdf#page=33>.

6.17. 222. Rigidity, lattices, and invariant measures beyond homogeneous dynamics [2022-geom-168]

Record: 2022-geom-168

Subfield: Manifolds and global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2022: Rigidity, lattices, and invariant measures beyond homogeneous dynamics

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-563-6-volume-5.pdf#page=129>

6.17.1. Problem statement

Let G be a simple Lie group of real rank at least 2, let $\Gamma < G$ be a lattice, let M be a compact manifold, and let $\rho : \Gamma \rightarrow \text{Diff}(M)$ be an action. Suppose that either $\rho(\Gamma)$ preserves a rigid geometric structure on M , or there exists $\gamma \in \Gamma$ such that the diffeomorphism $f = \rho(\gamma)$ admits a dominated splitting of TM , meaning a nontrivial continuous Df -invariant splitting $TM = E \oplus F$ for which there are constants $K > 0$ and $0 < \lambda < 1$ such that

$$\|Df^n|_{E_x}\| \|(Df^n|_{F_x})^{-1}\| \leq K\lambda^n$$

for every $x \in M$ and every $n \geq 0$. Prove that the action ρ is generalized quasiaffine. Here generalized quasiaffine means that, after identifying M equivariantly with a skew-product space, the Γ -action has the following form. There is a collection of data consisting of a topological group A , a closed subgroup $B < A$, a compact group C of affine diffeomorphisms of A/B , an affine action of Γ on A/B commuting with the action of C , a compact Riemannian manifold F , and a C^1 cocycle

$$\alpha : \Gamma \times C \backslash A/B \rightarrow \text{Isom}(F),$$

such that Γ acts on $C \backslash A/B \times F$ by

$$\gamma \cdot (C[a], y) = (C(\gamma \cdot [a]), \alpha(\gamma, C[a])y).$$

An affine diffeomorphism of A/B means that each acting element d sends $[a]$ to $[t_d L_d(a)]$, where $t_d \in A$ and L_d is a continuous automorphism of A .

6.17.2. Historical background, literature, and current progress

6.17.3. Verdict: partially resolved

6.17.3.1. Precise problem

The problem is Fisher's Conjecture 2.10 from the Zimmer-program literature. It asks whether every action $\rho : \Gamma \rightarrow \text{Diff}(M)$ of a lattice Γ in a simple Lie group G of real rank at least 2 on a compact manifold M is generalized quasiaffine, assuming either:

1. $\rho(\Gamma)$ preserves a rigid geometric structure on M , or
2. some element $\rho(\gamma)$ has a nontrivial dominated splitting of TM .

Here generalized quas affine is the skew-product/compact-quotient affine model described in the statement.

6.17.3.2. Status

I found no published or citable theorem proving this full statement for arbitrary higher-rank simple G , arbitrary lattice $\Gamma < G$, arbitrary compact M , and either of the two stated hypotheses. The problem is not fully open, because there is substantial special-case progress, but the exact conjecture remains unresolved.

6.17.3.3. Original source

Title: Rigidity, lattices, and invariant measures beyond homogeneous dynamics

Author: David Fisher

Venue: Proceedings of the International Congress of Mathematicians 2022, EMS Press, 2023; also arXiv preprint

Year: 2022/2023

arXiv/URL: arXiv:2111.14922, <https://arxiv.org/abs/2111.14922>; EMS URL: <https://ems.press/content/book-chapter-files/33258>

Relevant item: Conjecture 2.10 states exactly this rigid-structure/dominated-splitting-to-generalized-quas affine problem. Immediately after the conjecture, Fisher says the best evidence for the dynamical form is Homin Lee's work in dimension n , while the geometric form has more evidence going back to Zimmer but "the question remains mostly open." This source therefore does not resolve the problem statement; it identifies it as an open conjecture with partial evidence.

6.17.3.4. Partial progress

Title: Global rigidity of higher rank lattice actions with dominated splitting

Author: Homin Lee

Venue: Ergodic Theory and Dynamical Systems, 44(3), 799-828

Year: 2024

DOI/arXiv/URL: DOI 10.1017/etds.2023.31; arXiv:2010.11874, <https://arxiv.org/abs/2010.11874>

What it proves: For volume-preserving smooth actions of lattices in $SL(n, \mathbb{R})$, $n \geq 3$, on closed n -manifolds, if one element admits a dominated splitting, then the manifold is the flat n -torus and the action is smoothly conjugate to an affine action. Analogous results are given for lattices in $Sp(2n, \mathbb{R})$ and $SO(n, n)$ on closed $2n$ -manifolds.

Why it is only partial: It assumes volume preservation, restricts the ambient groups and dimensions, and concludes standard affine behavior in those cases. It does not cover arbitrary simple real-rank-at-least-2 G , arbitrary compact M , or the rigid-geometric-structure alternative.

Title: Positive entropy actions by higher-rank lattices

Authors: Aaron Brown and Homin Lee

Venue: arXiv preprint

Year: 2024, revised 2025

DOI/arXiv/URL: arXiv:2409.05991, <https://arxiv.org/abs/2409.05991>

What it proves: For smooth actions by lattices in higher-rank simple Lie groups, assuming one element has positive topological entropy, the paper proves rigidity results. In particular, for lattices in $SL(n, \mathbb{R})$ acting on n -manifolds, positive entropy forces the lattice to be commensurable with $SL(n, \mathbb{Z})$, and such actions admit an absolutely continuous probability measure with positive metric entropy and are measurably conjugate to affine actions on infra-tori.

Why it is only partial: Positive entropy is not the same as the stated dominated-splitting or rigid-structure hypotheses, the main stated n -manifold conclusion is for $SL(n, \mathbb{R})$, and the conclusion is measurable/structural rather than the full generalized quasiaffine classification in all cases.

Title: Partially hyperbolic lattice actions on 2-step nilmanifolds

Authors: Homin Lee and Sven Sandfeldt

Venue: arXiv preprint

Year: 2024

DOI/arXiv/URL: arXiv:2410.00784, <https://arxiv.org/abs/2410.00784>

What it proves: Global rigidity for higher-rank lattice actions on tori or 2-step nilmanifolds containing a partially hyperbolic element with one-dimensional center, under a technical assumption on the partially hyperbolic element; the action is forced to be affine in that setting.

Why it is only partial: It is restricted to tori/2-step nilmanifolds, partially hyperbolic elements with one-dimensional center, and an extra technical assumption. It does not cover general compact manifolds or the rigid-geometric-structure alternative.

Title: Global smooth and topological rigidity of hyperbolic lattice actions

Authors: Aaron Brown, Federico Rodriguez Hertz, and Zhiren Wang

Venue: Annals of Mathematics, 186(3), 913-972

Year: 2017

DOI/arXiv/URL: DOI 10.4007/annals.2017.186.3.3; arXiv:1512.06720, <https://arxiv.org/abs/1512.06720>

What it proves: For smooth actions of higher-rank lattices on compact nilmanifolds, if the action lifts to the universal cover and its linear data contains a hyperbolic element, there is a semiconjugacy to the linear data; with an Anosov element, the semiconjugacy becomes a smooth conjugacy.

Why it is only partial: It assumes nilmanifold topology, lifting/linear-data hypotheses, and hyperbolicity/Anosov-type behavior; it is evidence for the dominated-splitting branch but not a proof of the general conjecture.

Title: Rigidity of weakly hyperbolic actions of higher real rank semisimple Lie groups and their lattices

Authors: Gregory A. Margulis and Nantian Qian

Venue: Ergodic Theory and Dynamical Systems, 21(1), 121-164

Year: 2001

DOI/URL: DOI 10.1017/S0143385701001109

What it proves: Local rigidity for standard algebraic actions under weak hyperbolicity conditions, and C^0 global rigidity for volume-preserving higher-rank lattice Anosov actions on nilmanifolds with a finite orbit.

Why it is only partial: It treats standard/nilmanifold/Anosov settings with additional hypotheses and does not classify arbitrary compact-manifold actions with a dominated splitting as generalized quasiaffine.

Title: Invariant Rigid Geometric Structures and Smooth Projective Factors

Authors: Amos Nevo and Robert J. Zimmer

Venue: Geometric and Functional Analysis, 19, 520-535

Year: 2009

DOI/arXiv/URL: DOI 10.1007/s00039-009-0005-7; arXiv:0708.0766, <https://arxiv.org/abs/0708.0766>

What it proves: For actions of noncompact simple Lie groups preserving analytic rigid geometric structures of algebraic type on compact manifolds, under a full-support stationary-measure hypothesis, either there is a smooth equivariant map to a homogeneous projective variety on an open dense conull set or a Gromov fundamental-group representation has Zariski closure containing the acting Lie algebra.

Why it is only partial: It concerns connected Lie group actions rather than the full lattice-action conjecture as stated, assumes analyticity/algebraic type and full-support stationary measures, and gives a projective-factor/fundamental-group alternative rather than generalized quasiaffine classification.

Title: Rigid Geometric Structures and Representations of Fundamental Groups

Author: David Fisher

Venue: Rigidity in Dynamics and Geometry, Springer, pp. 135-147

Year: 2002

DOI/URL: DOI 10.1007/978-3-662-04743-9_6; https://link.springer.com/chapter/10.1007/978-3-662-04743-9_6

What it proves: Under analytic, ergodic, volume-preserving, rigid-geometric-structure hypotheses, a higher-rank lattice action is either isometric with finite $\pi_1(M)$, or $\pi_1(M)$ has a large linear representation; related results describe zero-entropy extensions of standard arithmetic examples.

Why it is only partial: These are strong restrictions under additional hypotheses, not a classification of all rigid-structure-preserving lattice actions as generalized quasiaffine.

Title: Rigid geometric structures and actions of semisimple Lie groups

Author: Renato Feres

Venue: Rigidite, groupe fondamental et dynamique, Panoramas et Syntheses 13, Societe Mathematique de France, pp. 121-167

Year: 2002

URL: <https://www.math.wustl.edu/~feres/strassbourg.pdf>

What it provides: A survey and framework for rigid geometric structures, including results of Zimmer/Gromov giving algebraic-hull, local-symmetry, and fundamental-group restrictions for semisimple actions preserving rigid structures.

Why it is only partial: It supplies restrictions and background machinery, not the full generalized quasiaffine conclusion for all higher-rank lattice actions in the problem statement.

Title: Lattices in semisimple groups and distal geometric structures

Author: Robert J. Zimmer

Venue: Inventiones Mathematicae, 80, 123-137

Year: 1985

DOI/URL: DOI 10.1007/BF01388551

What it proves: Early Zimmer-program rigidity results for lattice actions preserving distal geometric structures.

Why it is only partial: It covers a special class of geometric structures/actions and does not establish the full generalized quasiaffine classification.

Title: On the automorphism group of a compact Lorentz manifold and other geometric manifolds

Author: Robert J. Zimmer

Venue: Inventiones Mathematicae, 83, 411-424

Year: 1986

DOI/URL: DOI 10.1007/BF01394415

What it proves: Restrictions on automorphism groups of compact Lorentz and related geometric manifolds.

Why it is only partial: It is important evidence for the rigid-geometric-structure branch, but it is not a full classification theorem for arbitrary higher-rank lattice actions as generalized quas affine.

6.17.3.5. What remains unresolved

The unresolved part is exactly the broad statement in the problem statement: without dimension restrictions, volume preservation, nilmanifold/topological assumptions, Anosov/partial-hyperbolicity strengthening, analytic/full-support stationary-measure assumptions, or restrictions to special ambient groups, prove that every higher-rank lattice action preserving a rigid geometric structure or containing an element with a dominated splitting is generalized quas affine.

I found no post-2022 paper claiming a theorem with hypotheses broad enough to cover both alternatives in this exact form. The correct current verdict is therefore partially resolved.

6.17.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

6.17.5. Primary-source pointer

The originating ICM record is *Rigidity, lattices, and invariant measures beyond homogeneous dynamics* (2022). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-563-6-volume-5.pdf#page=129>.

6.18. 223. The regularity theory for the area functional (in geometric measure theory) [2022-geom-170]

Record: 2022-geom-170

Subfield: Manifolds and global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2022: The regularity theory for the area functional (in geometric measure theory)

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-560-5-volume-2.pdf#page=313>

6.18.1. Problem statement

Prove that if Σ is an m -dimensional stationary integral varifold in an open set $U \subset \mathbb{R}^{m+n}$, and $\text{Sing}(\Sigma)$ denotes the set of points of the support of Σ at which Σ is not locally a smooth m -dimensional minimal submanifold, then

$$\dim_{\mathcal{H}} \text{Sing}(\Sigma) \leq m - 1.$$

6.18.2. Historical background, literature, and current progress

6.18.3. Status check

6.18.3.1. Verdict

Partially resolved. The exact statement remains open for arbitrary stationary integral varifolds, but there is substantial progress in special classes and on related structural problems.

6.18.3.2. Precise problem

The problem asks whether every stationary integral m -varifold V in an open set $U \subset \mathbb{R}^{m+n}$ has interior singular set

$\text{Sing}(V) = \{\text{points of } \text{spt}(V) \text{ at which } \text{spt}(V) \text{ is not locally a smooth } m\text{-dimensional minimal submanifold}\}$

with Hausdorff dimension at most $m - 1$. Equivalently, the interior singular set should have codimension at least 1 in the support. The statement has no extra assumptions such as area minimizing, stability, codimension one, graphicality, multiplicity one, absence of classical singularities, or an epsilon-regularity hypothesis.

6.18.3.3. Why the full statement is not resolved

The best current general theory does not prove this all-cases bound. The obstruction is not the usual Almgren stratification S^k : for stationary integral varifolds, S^k has dimension at

most k and is rectifiable, but singular points with a flat tangent cone of multiplicity greater than one can lie in the top stratum $S^m \setminus S^{m-1}$. Controlling these flat higher-multiplicity or branch-type singular points is precisely the unresolved part.

Current literature even records the weaker question “is the singular set H^m -null?” as open for general stationary integral varifolds. Since the stated bound $\dim_H \text{Sing}(V) \leq m - 1$ would imply $H^m(\text{Sing}(V)) = 0$, these statements confirm that the problem is not fully resolved.

6.18.3.4. Main status sources

1. Title: The regularity theory for the area functional (in geometric measure theory) Authors: Camillo De Lellis Venue: Proceedings of the International Congress of Mathematicians 2022, Vol. 2, Plenary Lectures, EMS Press, Berlin, 2023, pp. 872-913 DOI/arXiv/URL: DOI 10.4171/ICM2022/122; arXiv:2110.11324; <https://ems.press/books/standalone/274/5418> Relevance: This ICM survey is the source context for the problem. It states Almgren’s theorem that the stratum S_k of a stationary integral varifold has Hausdorff dimension at most k and Naber-Valtorta’s rectifiability improvement. It then explains that Allard’s multiplicity-one epsilon-regularity does not control flat tangent cones of higher multiplicity, and says that one could reasonably conjecture that the singular set of a stationary integral varifold has dimension at most $m - 1$. The stated best general consequence is only that the singular set is closed with empty relative interior. Thus it does not resolve the problem statement.
2. Title: Remarks and conjectures on stationary varifolds dedicated to Enrico Bombieri Authors: Camillo Brena, Stefano Decio, Camillo De Lellis Venue: Atti Accad. Naz. Lincei Cl. Sci. Fis. Mat. Natur. 36 (2025), no. 3, pp. 409-443; published online by EMS Press in 2026 DOI/arXiv/URL: DOI 10.4171/RLM/1077; arXiv:2503.00651; <https://ems.press/journals/rlm/articles/14299484> Relevance: This paper states the exact codimension-one assertion as Conjecture 1.1: the interior singular set of a stationary integer rectifiable m -dimensional varifold has codimension not smaller than 1. It also states the weaker Conjecture 1.2 that the interior singular set is H^m -null and explicitly says this remains open even decades after Allard. The paper proves and reproves height, Lipschitz approximation, and excess-decay tools, but these are partial results and not a proof of the problem.
3. Title: A Branch Set Stratification for Stationary Varifolds with Epsilon-Regularity Authors: Brian Krummel, Paul Minter, Neshan Wickramasekera Venue: arXiv preprint, 2026 DOI/arXiv/URL: arXiv:2606.01511; <https://arxiv.org/abs/2606.01511> Relevance: This very recent preprint says that for general stationary integral n -varifolds it is still open whether the singular set, equivalently the branch set in the relevant context, can have positive H^n measure, and that in full generality essentially only density of the regular set is known. Its main theorem proves that branch points of density $\leq Q$ have Hausdorff dimension $\leq n - 2$ only for classes satisfying an epsilon-regularity property near planes of integer multiplicity $\leq Q$. The paper then obtains consequences for two-valued Lipschitz graphs and stable codimension-one classes. These extra hypotheses do not cover arbitrary stationary integral varifolds.

6.18.3.5. Important partial progress

1. Title: On the First Variation of a Varifold Authors: William K. Allard Venue: *Annals of Mathematics* (2) 95 (1972), no. 3, pp. 417-491 DOI/arXiv/URL: DOI 10.2307/1970868; <https://www.jstor.org/stable/1970868> Progress: Allard's interior regularity theorem gives smooth graphical regularity near multiplicity-one flat points and implies that the regular set of a stationary integral varifold is relatively open and dense. It does not bound the Hausdorff dimension of the entire singular set because higher-multiplicity flat tangent cones are outside the multiplicity-one hypothesis.
2. Title: The singular structure and regularity of stationary varifolds Authors: Aaron Naber, Daniele Valtorta Venue: *Journal of the European Mathematical Society* 22 (2020), no. 10, pp. 3305-3382 DOI/arXiv/URL: DOI 10.4171/JEMS/987; arXiv:1505.03428; <https://ems.press/content/serial-article-files/32493> Progress: For integral varifolds with bounded mean curvature, including stationary varifolds, the classical strata S^k are countably k -rectifiable, with quantitative Minkowski estimates for the quantitative strata. This is a major structural theorem, but it does not imply $\text{Sing}(V) \subset S^{m-1}$; flat higher-multiplicity singular points can be top-stratum points.
3. Title: C^∞ rectifiability of stationary varifolds Authors: Camillo Brena, Camillo De Lellis, Federico Franceschini Venue: arXiv preprint, 2025 DOI/arXiv/URL: arXiv:2503.00649; <https://arxiv.org/abs/2503.00649> Progress: Proves that the support of any stationary m -dimensional integer rectifiable varifold in $U \subset \mathbb{R}^{m+n}$ is C^∞ rectifiable up to an H^m -null set. This is general and important, but it is not the requested singular-set dimension bound: being covered a.e. by smooth graphs does not rule out an H^m -large set of points where the support is not locally a single smooth minimal submanifold.
4. Title: Dimension of the singular set for 2-valued stationary Lipschitz graphs Authors: Jonas Hirsch, Luca Spolaor Venue: *Duke Mathematical Journal* 175 (2026), no. 8, pp. 1453-1518 DOI/arXiv/URL: DOI 10.1215/00127094-2025-0060; arXiv:2401.00279; <https://projecteuclid.org/journals/duke-mathematical-journal/volume-175/issue-8/Dimension-of-the-singular-set-for-2-valued-stationary-Lipschitz/10.1215/00127094-2025-0060.full> Progress: Proves the optimal codimension-one bound for the singular set of a two-valued Lipschitz graph whose associated graph varifold is stationary. This is a genuine stationary result, but it assumes multiplicity two and Lipschitz graphicality, so it is a special case.
5. Title: On the Nature of Stationary Integral Varifolds near Multiplicity 2 Planes Authors: Spencer Becker-Kahn, Paul Minter, Neshan Wickramasekera Venue: arXiv preprint, 2025 DOI/arXiv/URL: arXiv:2507.13148; <https://arxiv.org/abs/2507.13148> Progress: Proves an epsilon-regularity theorem near multiplicity-two planes under a topological structural condition on the lower-density part of the support, and obtains consequences for stationary two-valued Lipschitz graphs. The theorem is local and conditional, not a general theorem for all stationary integral varifolds.
6. Title: A general regularity theory for stable codimension 1 integral varifolds Authors: Neshan Wickramasekera Venue: *Annals of Mathematics* (2) 179 (2014), no. 3, pp. 843-1007 DOI/arXiv/URL: DOI 10.4007/annals.2014.179.3.2; arXiv:0911.4883; <https://annals.math.princeton.edu/2014/179-3/p02> Progress: Gives optimal regularity for stable codimension-one integral varifolds satisfying the alpha-structural hypothesis. The stability, codimension-one, and structural assumptions are not present in the problem statement.

7. Title: A structure theory for stable codimension 1 integral varifolds with applications to area minimising hypersurfaces mod p Authors: Paul Minter, Neshan Wickramasekera Venue: Journal of the American Mathematical Society 37 (2024), pp. 861-927 DOI/arXiv/URL: DOI 10.1090/jams/1032; arXiv:2111.11202; <https://www.ams.org/journals/jams/2024-37-03/S0894-0347-2023-01032-X/> Progress: Develops a structure theory for stable codimension-one stationary integral varifolds admitting no low-density classical singularities, with applications to area-minimizing hypersurfaces mod p . Again this has extra stability and structural hypotheses.
8. Title: Regularity of area minimizing currents mod p Authors: Camillo De Lellis, Jonas Hirsch, Andrea Marchese, Salvatore Stuvard Venue: Geometric and Functional Analysis 30 (2020), no. 5, pp. 1224-1336 DOI/arXiv/URL: DOI 10.1007/s00039-020-00546-0; arXiv:1909.05172; <https://link.springer.com/article/10.1007/s00039-020-00546-0> Progress: Proves that the interior singular set of an m -dimensional area-minimizing current mod p has Hausdorff dimension at most $m - 1$. This matches the numerical bound but only for area-minimizing currents mod p , a much stronger variational assumption than stationarity of an arbitrary integral varifold.

6.18.3.6. What remains unresolved

What remains open is the flat higher-multiplicity/branch-point part of the singular set for a completely arbitrary stationary integral varifold. Existing theorems either handle multiplicity-one points, stratify non-flat tangent-cone points, prove rectifiability of the support up to H^m -null sets, or impose extra assumptions such as minimizing, stability, codimension one, graphicality, multiplicity two, no classical singularities, or an epsilon-regularity class condition. None of these hypotheses is available in the problem statement.

Therefore the literature does not fully resolve the problem as stated, but the problem is not “fully open” because the sources above give substantial partial progress and solve important special cases.

6.18.4. Remaining open problems recorded by the database

6.18.4.1. Open problem 1

Let V be a stationary integer rectifiable m -dimensional varifold in an open set $U \subset \mathbb{R}^{m+n}$. Is its interior singular set $\mathcal{H}^m$ -null?

Reference supplied by the database. Camillo Brena, Stefano Decio, and Camillo De Lellis, “Remarks and conjectures on stationary varifolds dedicated to Enrico Bombieri,” *Atti della Accademia Nazionale dei Lincei, Classe di Scienze Fisiche, Matematiche e Naturali. Rendiconti Lincei. Matematica e Applicazioni* **36** (2025), no. 3, 409–443 (published online 3 February 2026), Conjecture 1.2. DOI: <https://doi.org/10.4171/RLM/1077>; arXiv: 2503.00651.

6.18.4.2. Open problem 2

Let V be a stationary integer rectifiable m -dimensional varifold in an open set $U \subset \mathbb{R}^{m+n}$, and let $x_0 \in \text{spt}(V) \cap U$. Suppose that V has a unique flat tangent cone at x_0 and

$$\lim_{r \downarrow 0} \frac{\|V\|(\{x : \Theta(V, x) \neq \Theta(V, x_0)\} \cap B_r(x_0))}{r^m} = 0.$$

Must x_0 be a regular point of V ?

Reference supplied by the database. Camillo Brena, Stefano Decio, and Camillo De Lellis, “Remarks and conjectures on stationary varifolds dedicated to Enrico Bombieri,” *Atti della Accademia Nazionale dei Lincei, Classe di Scienze Fisiche, Matematiche e Naturali. Rendiconti Lincei. Matematica e Applicazioni* **36** (2025), no. 3, 409–443 (published online 3 February 2026), Conjecture 1.3. DOI: <https://doi.org/10.4171/RLM/1077>; arXiv: 2503.00651.

6.18.4.3. Open problem 3

Let V and x_0 satisfy the following conditions: V is a stationary integer rectifiable m -dimensional varifold in an open set $U \subset \mathbb{R}^{m+n}$, $x_0 \in \text{spt}(V) \cap U$, V has a unique flat tangent cone at x_0 , and

$$\lim_{r \downarrow 0} \frac{\|V\|(\{x : \Theta(V, x) \neq \Theta(V, x_0)\} \cap B_r(x_0))}{r^m} = 0.$$

Does there exist a smooth classical minimal m -dimensional graph M in a neighborhood of x_0 such that

$$\int_{B_r(x_0)} \text{dist}(x, M)^2 d\|V\|(x) = o(r^N)$$

for every $N \in \mathbb{N}$?

Reference supplied by the database. Camillo Brena, Stefano Decio, and Camillo De Lellis, “Remarks and conjectures on stationary varifolds dedicated to Enrico Bombieri,” *Atti della Accademia Nazionale dei Lincei, Classe di Scienze Fisiche, Matematiche e Naturali. Rendiconti Lincei. Matematica e Applicazioni* **36** (2025), no. 3, 409–443 (published online 3 February 2026), Conjecture 1.4. DOI: <https://doi.org/10.4171/RLM/1077>; arXiv: 2503.00651.

6.18.4.4. Open problem 4

Let V be a stationary m -dimensional varifold in an open set $U \subset \mathbb{R}^{m+n}$, let $x_0 \in \text{spt}(V) \cap U$, and let M be a classical smooth minimal m -dimensional surface. Suppose that

$$\int_{B_r(x_0)} \text{dist}(x, M)^2 d\|V\|(x) = o(r^N)$$

for every $N \in \mathbb{N}$. Must $\text{spt}(V)$ be contained in M in some neighborhood of x_0 ?

Reference supplied by the database. Camillo Brena, Stefano Decio, and Camillo De Lellis, “Remarks and conjectures on stationary varifolds dedicated to Enrico Bombieri,” *Atti della Accademia Nazionale dei Lincei, Classe di Scienze Fisiche, Matematiche e Naturali. Rendiconti Lincei. Matematica e Applicazioni* **36** (2025), no. 3, 409–443 (published online 3 February 2026), Conjecture 1.5. DOI: <https://doi.org/10.4171/RLM/1077>; arXiv: 2503.00651.

6.18.4.5. Open problem 5

Let V be a stationary m -dimensional varifold in an open set $U \subset \mathbb{R}^{m+n}$, and let $h \in C^1(U)$ be harmonic on V , meaning that

$$\int \nabla_{T_V} h \cdot \nabla_{T_V} \varphi \, d\|V\| = 0$$

for every $\varphi \in C_c^1(U)$. If $x_0 \in \text{spt}(V) \cap U$ and

$$\int_{B_r(x_0)} h^2 \, d\|V\| = o(r^N)$$

for every $N \in \mathbb{N}$, must h vanish identically on $\text{spt}(V) \cap B_\rho(x_0)$ for some $0 < \rho < \text{dist}(x_0, \partial U)$?

Reference supplied by the database. Camillo Brena, Stefano Decio, and Camillo De Lellis, “Remarks and conjectures on stationary varifolds dedicated to Enrico Bombieri,” *Atti della Accademia Nazionale dei Lincei, Classe di Scienze Fisiche, Matematiche e Naturali. Rendiconti Lincei. Matematica e Applicazioni* **36** (2025), no. 3, 409–443 (published online 3 February 2026), Conjecture 1.7. DOI: <https://doi.org/10.4171/RLM/1077>; arXiv: 2503.00651.

6.18.5. Primary-source pointer

The originating ICM record is *The regularity theory for the area functional (in geometric measure theory)* (2022). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-560-5-volume-2.pdf#page=313>.

6.19. 224. The regularity theory for the area functional (in geometric measure theory) [2022-geom-172]

Record: 2022-geom-172

Subfield: Manifolds and global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2022: The regularity theory for the area functional (in geometric measure theory)

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-560-5-volume-2.pdf#page=313>

6.19.1. Problem statement

Let $m, n \geq 1$, let $\Gamma \subset \mathbb{R}^{m+n}$ be a smooth closed oriented $(m-1)$ -dimensional submanifold, and let Σ be an m -dimensional area-minimizing integral current in $\mathbb{R}^{m+n}$ whose boundary is Γ counted with multiplicity one. Call a point $p \in \Gamma$ a boundary regular point of Σ if, in some neighborhood U of p , either $\Sigma \cap U$ is a smooth minimal m -dimensional submanifold with boundary $\Gamma \cap U$, or there is a smooth minimal m -dimensional submanifold $\Lambda \subset U$ without boundary such that $\Gamma \cap U \subset \Lambda$, $\Gamma \cap U$ separates Λ into two regions Λ^+ and Λ^- , and $\Sigma \cap U = (k+1)\Lambda^+ + k\Lambda^-$ as currents for some positive integer k . Prove that the set of boundary singular points, namely the points of Γ that are not boundary regular points, has zero $(m-1)$ -dimensional Hausdorff measure.

6.19.2. Historical background, literature, and current progress

Verdict: partially resolved.

6.19.2.1. Problem identified

The statement asks whether, for every $m, n \geq 1$, every smooth closed oriented $(m-1)$ -submanifold $\Gamma \subset \mathbb{R}^{m+n}$, and every m -dimensional area-minimizing integral current Σ with $\partial\Sigma = [\Gamma]$, the boundary singular set

$$\Gamma \setminus \text{Reg}_b(\Sigma)$$

has zero $\mathcal{H}^{m-1}$ -measure, where boundary regularity includes both the ordinary one-sided smooth boundary case and the collapsed/two-sided local model $(k+1)\Lambda^+ + k\Lambda^-$.

6.19.2.2. Current status

The exact full statement is not known. The literature proves substantial special cases and density/rectifiability results, but I found no published or citable result proving $\mathcal{H}^{m-1}(\text{Sing}_b(\Sigma)) = 0$ for arbitrary codimension and general smooth multiplicity-one boundary. Recent work explicitly lists this as open, even for $Q = 1$.

6.19.2.3. Key literature and progress

1. Boundary regularity and embedded solutions for the oriented Plateau problem

Authors: Robert Hardt, Leon Simon.

Venue/year: Annals of Mathematics (2) 110, no. 3, 439-486, 1979.

DOI/URL: <https://doi.org/10.2307/1971233>

Relevance: Resolves the codimension-one hypersurface case by proving boundary regularity for oriented area-minimizing hypersurfaces. This gives an empty boundary singular set in that special case, but does not cover arbitrary codimension.

2. On the boundary behavior of mass-minimizing integral currents

Authors: Camillo De Lellis, Guido De Philippis, Jonas Hirsch, Annalisa Massaccesi.

Venue/year: Memoirs of the American Mathematical Society 291, no. 1446, 2023.

DOI/arXiv/URL: <https://doi.org/10.1090/memo/1446> ; arXiv:1809.09457, <https://arxiv.org/abs/1809.09457>

Relevance: Theorem 1.6 proves that the boundary regular set is dense in Γ under the general higher-codimension multiplicity-one hypotheses. The same work states Conjecture 2.8: $\mathcal{H}^{m-1}(\text{Sing}_b(T)) = 0$. Thus it is partial progress, not a resolution.

3. The regularity theory for the area functional (in geometric measure theory)

Author: Camillo De Lellis.

Venue/year: ICM 2022, Vol. 2 Plenary Lectures, EMS Press, 872-913, 2023.

DOI/arXiv/URL: <https://doi.org/10.4171/ICM2022/122> ; arXiv:2110.11324, <https://arxiv.org/abs/2110.11324>

Relevance: Theorem 13.2 states the dense relatively open boundary regular set result using exactly the two regularity alternatives in the problem statement. Section 16.5 asks exactly whether this can be improved to zero $(m - 1)$ -dimensional Hausdorff measure for the boundary singular set.

4. Nonclassical minimizing surfaces with smooth boundary

Authors: Camillo De Lellis, Guido De Philippis, Jonas Hirsch.

Venue/year: Journal of Differential Geometry 122, no. 2, 205-222, 2022.

DOI/arXiv/URL: <https://doi.org/10.4310/jdg/1669998183> ; arXiv:1906.09488, <https://arxiv.org/abs/1906.09488>

Relevance: Constructs smooth boundary examples with large boundary singular sets; the Memoirs paper records an example with boundary singular set of Hausdorff dimension 1 but $\mathcal{H}^1$ -null. This rules out a naive dimension estimate in full smooth generality, but does not disprove or prove the measure-null conjecture.

5. An Allard-type boundary regularity theorem for $2d$ minimizing currents at smooth curves with arbitrary multiplicity

Authors: Camillo De Lellis, Stefano Nardulli, Simone Steinbrüchel.

Venue/year: Publications mathématiques de l'IHÉS 140, 37-154, 2024.

DOI/arXiv/URL: <https://doi.org/10.1007/s10240-024-00144-y> ; arXiv:2111.02991, <https://arxiv.org/abs/2111.02991>

Relevance: Gives a sharp Allard-type boundary regularity theorem for two-dimensional currents under density and convex-barrier hypotheses. This is important special-case progress, but its hypotheses do not cover the full statement.

6. Density of the boundary regular set of $2d$ area minimizing currents with arbitrary codimension and multiplicity

Authors: Stefano Nardulli, Reinaldo Resende.

Venue/year: Advances in Mathematics 455, article 109889, 2024.

DOI/arXiv/URL: <https://doi.org/10.1016/j.aim.2024.109889> ;

arXiv:2204.11947, <https://arxiv.org/abs/2204.11947>

Relevance: Extends density of the boundary regular set in dimension $m = 2$ and arbitrary codimension/multiplicity. It remains a density result, not a proof of $\mathcal{H}^{m-1}$ -nullity of all boundary singular points.

7. On the regularity of area minimizing currents at boundaries with arbitrary multiplicity

Authors: Ian Fleschler, Reinaldo Resende.

Venue/year: arXiv preprint, 2024; revised version 2025.

DOI/arXiv/URL: arXiv:2410.04566, <https://arxiv.org/abs/2410.04566>

Relevance: Proves $\mathcal{H}^{m-3}$ -rectifiability for density $Q/2$ boundary singular points and an open dense boundary regular set under broad arbitrary-multiplicity assumptions. It explicitly states as Open question 2 whether $\text{Sing}_b(T)$ is $\mathcal{H}^{m-1}$ -null, says this was originally Conjecture 2.8, and says it remains widely open even when $Q = 1$. This directly prevents a “fully resolved” verdict for the stated multiplicity-one problem.

8. An essential one sided boundary singularity for a 3-dimensional area minimizing current in $\mathbb{R}^5$

Author: Ian Fleschler.

Venue/year: arXiv preprint, 2025.

DOI/arXiv/URL: arXiv:2502.15047, <https://arxiv.org/abs/2502.15047>

Relevance: Constructs essential one-sided singularities in a higher-multiplicity setting and shows sharpness of related rectifiability theory. It is not a counterexample to the stated $Q = 1$ null-measure statement, but it clarifies why boundary regularity remains difficult.

6.19.2.4. What remains unresolved

The unresolved part is exactly the full statement: for arbitrary codimension n , arbitrary dimension m , and a general smooth multiplicity-one boundary Γ , no known theorem proves that all non-regular boundary points form an $\mathcal{H}^{m-1}$ -null subset of Γ . Existing results give full regularity in codimension one, open/dense regularity in the general multiplicity-one case, and stronger rectifiability under density, dimension-two, convex-barrier, or arbitrary-multiplicity subcase hypotheses.

6.19.3. Remaining open problems recorded by the database

6.19.3.1. Open problem 1

Let T be an m -dimensional area-minimizing integral current with smooth multiplicity-one boundary Γ in a smooth ambient manifold Σ . A boundary singular point is called *genuine* if it is not of crossing type, i.e. it cannot locally be separated as $T = T_1 + T_2$ with $\partial T_1 = 0$ and the point boundary-regular for T_2 . Is the Hausdorff dimension of the set of genuine boundary singularities at most $m - 2$?

Reference supplied by the database. Camillo De Lellis, Guido De Philippis, Jonas Hirsch, and Annalisa Massaccesi, *On the Boundary Behavior of Mass-Minimizing Integral Currents*, *Mem-oirs of the American Mathematical Society* **291** (2023), no. 1446, Conjecture 2.7 (also stated as Conjecture 1.7). DOI: <https://doi.org/10.1090/memo/1446>; arXiv: 1809.09457.

6.19.3.2. Open problem 2

Let (g^+, g^-) be a $(Q - \frac{1}{2})$ -valued function with a C^3 interface (γ, φ) over a domain $\Omega \subset \mathbb{R}^m$, and suppose that (g^+, g^-) minimizes the Dirichlet energy. Is the Hausdorff dimension of its set of genuine boundary singularities at most $m - 2$?

Reference supplied by the database. Camillo De Lellis, Guido De Philippis, Jonas Hirsch, and Annalisa Massaccesi, *On the Boundary Behavior of Mass-Minimizing Integral Currents*, *Mem-oirs of the American Mathematical Society* **291** (2023), no. 1446, Conjecture 2.11. DOI: <https://doi.org/10.1090/memo/1446>; arXiv: 1809.09457.

6.19.3.3. Open problem 3

Let (g^+, g^-) be a $(Q - \frac{1}{2})$ -valued function with a C^3 interface (γ, φ) over a domain $\Omega \subset \mathbb{R}^m$, and suppose that (g^+, g^-) minimizes the Dirichlet energy. Is its boundary singular set $\mathcal{H}^{m-1}$ -null?

Reference supplied by the database. Camillo De Lellis, Guido De Philippis, Jonas Hirsch, and Annalisa Massaccesi, *On the Boundary Behavior of Mass-Minimizing Integral Currents*, *Mem-oirs of the American Mathematical Society* **291** (2023), no. 1446, Conjecture 2.12. DOI: <https://doi.org/10.1090/memo/1446>; arXiv: 1809.09457.

6.19.3.4. Open problem 4

Let T be a two-dimensional area-minimizing integral current with smooth multiplicity-one boundary Γ in a smooth ambient manifold Σ , and assume that both Σ and Γ are real analytic. Is the union of the boundary singular set and the interior singular set of T discrete?

Reference supplied by the database. Camillo De Lellis, Guido De Philippis, Jonas Hirsch, and Annalisa Massaccesi, *On the Boundary Behavior of Mass-Minimizing Integral Currents*, *Mem-oirs of the American Mathematical Society* **291** (2023), no. 1446, Conjecture 2.14 (the White conjecture). DOI: <https://doi.org/10.1090/memo/1446>; arXiv: 1809.09457.

6.19.3.5. Open problem 5

Does there exist a two-dimensional area-minimizing integral current in Euclidean space, with smooth multiplicity-one boundary, having a boundary singular point that is an accumulation point of boundary branch points?

Reference supplied by the database. Camillo De Lellis, Guido De Philippis, Jonas Hirsch, and Annalisa Massaccesi, *On the Boundary Behavior of Mass-Minimizing Integral Currents*, *Memoirs of the American Mathematical Society* **291** (2023), no. 1446, Question 2.15. DOI: <https://doi.org/10.1090/memo/1446>; arXiv: [1809.09457](https://arxiv.org/abs/1809.09457).

6.19.4. Primary-source pointer

The originating ICM record is *The regularity theory for the area functional (in geometric measure theory)* (2022). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-560-5-volume-2.pdf#page=313>.

6.20. 225. The regularity theory for the area functional (in geometric measure theory) [2022-geom-174]

Record: 2022-geom-174

Subfield: Manifolds and global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2022: The regularity theory for the area functional (in geometric measure theory)

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-560-5-volume-2.pdf#page=313>

6.20.1. Problem statement

Let Σ be an m -dimensional area-minimizing integral current in $\mathbb{R}^{m+n}$, with $m \geq 3$ and $n \geq 2$, and let p be an interior singular point of Σ . Suppose that, for some sequence of scales $r_k \downarrow 0$, the rescaled currents $(\Sigma - p)/r_k$ converge to a tangent cone Σ_0 at p , and that Σ_0 is the union of two distinct m -dimensional linear planes in $\mathbb{R}^{m+n}$, each counted with multiplicity 1. Must Σ_0 be independent of the choice of blow-up sequence, equivalently the unique tangent cone to Σ at p ?

6.20.2. Historical background, literature, and current progress

6.20.3. Verdict: partially resolved

6.20.3.1. Precise problem

The problem asks whether an arbitrary interior singular point of an m -dimensional area-minimizing integral current $T \subset \mathbb{R}^{m+n}$, $m \geq 3$, $n \geq 2$, has a unique tangent cone once one blow-up cone is

$$T_0 = \llbracket P_1 \rrbracket + \llbracket P_2 \rrbracket,$$

where $P_1 \neq P_2$ are m -dimensional linear planes, each with multiplicity one. Equivalently, it asks whether every sequence of blow-up scales at that same point converges to the same two-plane cone.

6.20.3.2. Status

No citable theorem found resolves this exact pointwise question in all cases. The literature gives important partial results: conditional uniqueness for cylindrical cones under extra hypotheses such as Simon's no-hole condition, uniqueness in low-dimensional or smooth-link settings, and recent $\mathcal{H}^{m-2}$ -a.e. uniqueness and rectifiability theorems for higher-codimension area-minimizing currents. These do not imply uniqueness at the particular point p in the problem statement, because the statement has no no-hole assumption and is not restricted to almost every singular point.

6.20.3.3. Key literature and comparison

1. **The regularity theory for the area functional (in geometric measure theory)**
Author: Camillo De Lellis
Venue/year: International Congress of Mathematicians 2022, Vol. 2, Plenary lectures, EMS Press, pp. 872-913, 2023 publication of proceedings
DOI/arXiv/URL: DOI 10.4171/ICM2022/122; arXiv:2110.11324; <https://ems.press/content/book-chapter-files/33147>
Relevance: This is the apparent source of the problem. It explicitly states this exact two-plane tangent-cone uniqueness question and says it is still open, even in the basic case $m = 3$ in $\mathbb{R}^5$. It also explains that the issue is tied to whether Simon-type cylindrical tangent-cone results can be used without the no-hole condition.
2. **Cylindrical tangent cones and the singular set of minimal submanifolds**
Author: Leon Simon
Venue/year: Journal of Differential Geometry 38(3), 585-652, 1993
DOI/arXiv/URL: DOI 10.4310/jdg/1214454484; <https://projecteuclid.org/journals/journal-of-differential-geometry/volume-38/issue-3/Cylindrical-tangent-cones-and-the-singular-set-of-minimal-submanifolds/10.4310/jdg/1214454484.full>
Relevance: Gives uniqueness for cylindrical tangent cones under structural hypotheses, including integrability and a no-hole condition. A two-plane cone with a common spine is a natural cylindrical-cone case, but the statement does not assume the no-hole condition, so this is only conditional progress.
3. **On the radial behavior of minimal surfaces and the uniqueness of their tangent cones**
Authors: William K. Allard and Frederick J. Almgren, Jr.
Venue/year: Annals of Mathematics 113(2), 215-265, 1981
DOI/arXiv/URL: DOI 10.2307/2006984; <https://annals.math.princeton.edu/198/113-2/p01>
Relevance: Foundational uniqueness theorem for regular/smooth-link tangent-cone settings. It does not cover the singular-link two-plane cone in the general common-spine case posed here.
4. **Analysis of singularities of area minimizing currents: planar frequency, branch points of rapid decay, and weak locally uniform approximation**
Authors: Brian Krummel and Neshan Wickramasekera
Venue/year: arXiv preprint, 2023
DOI/arXiv/URL: arXiv:2304.10653; DOI 10.48550/arXiv.2304.10653; <https://arxiv.org/abs/2304.10653>
Relevance: Develops the planar-frequency and decomposition framework used in the later uniqueness/rectifiability results. It provides structural alternatives but not the pointwise uniqueness asserted in the problem statement.
5. **Analysis of singularities of area minimizing currents: a uniform height bound, estimates away from branch points of rapid decay, and uniqueness of tangent cones**
Authors: Brian Krummel and Neshan Wickramasekera
Venue/year: arXiv preprint, 2023
DOI/arXiv/URL: arXiv:2304.10272; DOI 10.48550/arXiv.2304.10272; <https://arxiv.org/abs/2304.10272>
Relevance: Proves, together with the companion work, that locally area-minimizing currents of codimension at least two have unique tangent cones at $\mathcal{H}^{n-2}$ -almost every point, and proves rectifiability of the relevant singular set. This is substantial partial progress,

but it is an a.e. statement and does not force uniqueness at the arbitrary point p in the problem statement.

6. The Fine Structure of the Singular Set of Area-Minimizing Integral Currents III: Frequency 1 Flat Singular Points and $\mathcal{H}^{m-2}$ -a.e. Uniqueness of Tangent Cones

Authors: Camillo De Lellis, Paul Minter, Anna Skorobogatova

Venue/year: arXiv preprint, 2023; revised 2024

DOI/arXiv/URL: arXiv:2304.11553; DOI 10.48550/arXiv.2304.11553; <https://arxiv.org/abs/2304.11553>

Relevance: Independently proves that an area-minimizing integral current in codimension greater than one has a unique tangent cone, a superposition of planes, at $\mathcal{H}^{m-2}$ -almost every point, and contributes to countable $(m-2)$ -rectifiability of the singular set. Again, this is not a pointwise theorem for every singular point with a two-plane tangent cone.

7. Singularities and tangent cones for area-minimizing and semicalibrated currents

Author: Anna Skorobogatova

Venue/year: Séminaire Laurent Schwartz – EDP et applications (2024-2025), Talk no. 5, published online 2025

DOI/arXiv/URL: DOI 10.5802/slsedp.175; <https://proceedings.centre-mersenne.org/articles/10.5802/slsedp.175/>

Relevance: Survey of the recent breakthroughs. It states the current higher-dimensional result as $\mathcal{H}^{m-2}$ -a.e. uniqueness and rectifiability, confirming that the available broad theorem is almost-everywhere rather than pointwise for every singular point.

6.20.3.4. What remains unresolved

The missing theorem would need to prove that, for every interior singular point p satisfying the stated two-plane tangent-cone hypothesis, all blow-up sequences converge to the same cone, without adding a no-hole condition, integrability condition, genericity/a.e. restriction, boundary hypothesis, or low-dimensional assumption. I found no published or arXiv result proving that full statement.

6.20.4. Remaining open problems recorded by the database

6.20.4.1. Open problem 1

Let T be a three-dimensional area-minimizing integral current in $\mathbb{R}^5$, and let p be an interior singular point. Suppose that one tangent cone to T at p is the union of two distinct three-dimensional linear planes, each counted with multiplicity one. Must this cone be the unique tangent cone to T at p ?

Reference supplied by the database. Camillo De Lellis, “The regularity theory for the area functional (in geometric measure theory),” in *Proceedings of the International Congress of Mathematicians 2022*, Vol. 2, Plenary Lectures, EMS Press, Berlin, 2023, pp. 872–913, §16.7 (“Uniqueness of tangent cones”), the explicitly identified special case $m = 3, n = 2$ of the final open question. DOI: <https://doi.org/10.4171/ICM2022/122>; arXiv: 2110.11324.

6.20.5. Primary-source pointer

The originating ICM record is *The regularity theory for the area functional (in geometric measure theory)* (2022). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-560-5-volume-2.pdf#page=313>.

6.21. 226. Emergent complex geometry [2022-ag-215]

Record: 2022-ag-215

Subfield: Manifolds and global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2022: Emergent complex geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-562-9-volume-4.pdf#page=99>

6.21.1. Problem statement

Let X be an n -dimensional Fano manifold. For each positive integer k , set

$$N_k := \dim_{\mathbb{C}} H^0(X, -kK_X).$$

Choose a basis $s_1^{(k)}, \dots, s_{N_k}^{(k)}$ of $H^0(X, -kK_X)$, and let $\det S^{(k)}$ be the Slater determinant section of $-kK_{X^{N_k}}$ given locally by

$$\det S^{(k)}(x_1, \dots, x_{N_k}) := \det(s_i^{(k)}(x_j)).$$

Let

$$D_{N_k} := \frac{1}{k} \operatorname{div}(\det S^{(k)})$$

be the anticanonical $\mathbb{Q}$ -divisor on X^{N_k} cut out by the multivalued section $(\det S^{(k)})^{1/k}$ of $-K_{X^{N_k}}$, and let $\operatorname{lct}(D_{N_k})$ denote its log canonical threshold.

Let $\mathcal{P}(X)$ be the space of probability measures on X . Fix a smooth Kähler metric on X , with volume form dV , and let E be the associated pluricomplex energy on $\mathcal{P}(X)$ in the anticanonical class: on smooth probability volume forms it is characterized, up to an additive constant, by the first variation $dE(\mu) = -u_\mu$, where u_μ solves

$$\frac{(\omega_0 + \frac{i}{2\pi} \partial \bar{\partial} u_\mu)^n}{\int_X \omega_0^n} = \mu$$

for the curvature form ω_0 of the fixed Hermitian metric on $-K_X$, and E is extended to $\mathcal{P}(X)$ as the greatest lower-semicontinuous extension. Let $\operatorname{Ent}_{dV}(\mu) = \int_X \log(d\mu/dV) d\mu$ when μ is absolutely continuous with respect to dV , and let $\operatorname{Ent}_{dV}(\mu) = +\infty$ otherwise. For $\beta < 0$, define

$$F_\beta(\mu) = \begin{cases} \beta E(\mu) + \operatorname{Ent}_{dV}(\mu), & E(\mu) < +\infty, \\ +\infty, & E(\mu) = +\infty. \end{cases}$$

Prove or disprove that, with N ranging through the integers N_k as $k \rightarrow \infty$,

$$\lim_{k \rightarrow \infty} \operatorname{lct}(D_{N_k}) = \Gamma(X), \quad \Gamma(X) := \sup_{\beta < 0} \left\{ -\beta : \inf_{\mu \in \mathcal{P}(X)} F_\beta(\mu) > -\infty \right\}.$$

6.21.2. Historical background, literature, and current progress

Verdict: partially resolved.

Precise problem checked: for every Fano manifold X , prove the full asymptotic identity

$$\lim_{k \rightarrow \infty} \text{lct}(D_{N_k}) = \Gamma(X),$$

where D_{N_k} is the N_k -point Slater-determinant anticanonical $\mathbb{Q}$ -divisor on X^{N_k} , and $\Gamma(X)$ is the macroscopic free-energy boundedness threshold defined from the pluricomplex energy plus entropy.

Current literature does not prove this identity in full generality.

Key literature and progress:

1. Title: “Emergent complex geometry” Authors: Robert J. Berman Venue/year: Proceedings of the International Congress of Mathematicians 2022, Vol. 4, EMS Press, 2022 DOI/arXiv/URL: DOI 10.4171/ICM2022/59; <https://ems.press/content/book-chapter-files/33209>; arXiv:2109.00307 Relevance: This is the source for this problem. Conjecture 2.7 states the same limiting identity between the Slater-determinant lct threshold and the free-energy threshold $\Gamma(X)$ for Fano manifolds. It is presented as a conjecture suggested by extending large-deviation/partition-function asymptotics to negative inverse temperature.
2. Title: “An invitation to Kähler-Einstein metrics and random point processes” Authors: Robert J. Berman Venue/year: arXiv preprint, 2020 DOI/arXiv/URL: arXiv:2003.11358; <https://arxiv.org/abs/2003.11358> Relevance: Section 3.2.2 formulates the same problem as Conjecture 3.10, $\gamma(X) = \Gamma(X)$, with $\gamma_k(X) = \text{lct}(X^{N_k}, D_k/k)$. The paper records one-sided and conditional progress and notes that the converse implication and convergence problem remain open in general.
3. Title: “On the K-stability of Fano varieties and anticanonical divisors” Authors: Kento Fujita and Yuji Odaka Venue/year: Tohoku Mathematical Journal (2) 70, no. 4, 511–521, 2018 DOI/arXiv/URL: DOI 10.2748/tmj/1546570823; arXiv:1602.01305 Relevance: Proves the algebro-geometric one-sided comparison underlying partial progress: the Berman-Gibbs/microscopic threshold is bounded above by the Fujita-Odaka delta thresholds, hence Berman-Gibbs stability implies K-stability. This does not give the missing reverse inequality needed for the full limit identity.
4. Title: “The probabilistic vs the quantization approach to Kähler-Einstein geometry” Authors: Robert J. Berman Venue/year: Mathematische Annalen 388, 4383–4404, 2024; published online 2023 DOI/arXiv/URL: DOI 10.1007/s00208-023-02627-5; arXiv:2109.06575 Relevance: Gives a quantitative bound on the partition functions and proves $\gamma(X) \leq$ the analytic/coercivity threshold; combined with known results this gives the direction “uniform Gibbs stability implies existence of a unique Kähler-Einstein metric.” The paper explicitly states that the missing lower bound on $\gamma(X)$, equivalently upgrading $\gamma(X) \leq \delta(X)$ to equality, remains to be established.
5. Title: “Kähler-Einstein metrics and Archimedean zeta functions” Authors: Robert J. Berman Venue/year: Proceedings of the 8th European Congress of Mathematics, EMS Press, 2023 DOI/arXiv/URL: <https://ems.press/content/book-chapter-files/28296>; arXiv:2112.04791 Relevance: Develops conditional/analytic approaches via Archimedean zeta functions and proves convergence/equivalence results for log Fano curves. It states that the converse to uniform Gibbs stability implying K-stability is widely open in general.

6. Title: “Microscopic stability thresholds and constant scalar curvature Kähler metrics” Authors: Takahiro Aoi Venue/year: arXiv preprint, 2024 DOI/arXiv/URL: arXiv:2410.22090; <https://arxiv.org/abs/2410.22090> Relevance: Extends related microscopic-threshold techniques to polarized manifolds and cscK metrics. It explicitly states that the limiting equality $\delta(L) = \gamma(L)$ is still open, even in the Fano case.
7. Title: “Gibbs polystability of Fano manifolds, stability thresholds and symmetry breaking” Authors: Rolf Andreasson, Robert J. Berman and Ludvig Svensson Venue/year: arXiv preprint, 2026 DOI/arXiv/URL: arXiv:2511.16173; <https://arxiv.org/abs/2511.16173> Relevance: Gives newer progress for log Fano manifolds with automorphisms. It proves the relevant microscopic/analytic threshold conjectures for log Fano curves and gives higher-dimensional examples and one-sided estimates, but it explicitly leaves the general log Fano/Fano conjectures for future work.

What remains unresolved:

The full reverse inequality/lower bound needed to prove

$$\Gamma(X) \leq \liminf_{k \rightarrow \infty} \text{lct}(D_{N_k})$$

for every Fano manifold is not known. Existing results prove one-sided bounds, implications for uniform Gibbs stability, conditional large-deviation/partition-function results, and complete curve/log-curve cases, but no citable theorem found covers all Fano manifolds and proves the exact equality in the problem statement.

6.21.3. Remaining open problems recorded by the database

6.21.3.1. Open problem 1

With $\gamma(X) := \liminf_{k \rightarrow \infty} \text{lct}(D_{N_k})$ and $\Gamma(X)$ the macroscopic stability threshold defined from F_β , prove the still-missing reverse inequality $\Gamma(X) \leq \gamma(X)$ for every Fano manifold X .

Reference supplied by the database. Robert J. Berman, “The probabilistic vs the quantization approach to Kähler–Einstein geometry,” *Mathematische Annalen* 388 (2024), 4383–4404, Conjecture 1.1 and Corollary 1.3 (which proves only $\gamma(X) \leq \Gamma(X)$, leaving the reverse implication). DOI: <https://doi.org/10.1007/s00208-023-02627-5>.

6.21.3.2. Open problem 2

Let X be a uniformly Gibbs-stable Fano manifold. For the canonical N_k -particle Gibbs measures $\mu_{-1}^{(N_k)}$, let $\Gamma_{N_k, -1}$ be the law of the empirical measure and let Γ_{-1} be any weak limit point. Prove the “upper bound property of the mean energy”

$$\limsup_{k \rightarrow \infty} E^{(N_k)}(\mu_{-1}^{(N_k)}) \leq \int_{\mathcal{P}(X)} E(\mu) d\Gamma_{-1}(\mu).$$

Reference supplied by the database. Robert J. Berman, “An invitation to Kähler–Einstein metrics and random point processes,” *Surveys in Differential Geometry* 23 (2020), 35–87, equation (7.4), the discussion “Towards the missing Step 2,” and Theorem 7.6. arXiv: [2003.11358](https://arxiv.org/abs/2003.11358).

6.21.3.3. Open problem 3

For a uniformly Gibbs-stable Fano manifold X , prove that the meromorphic continuations in β of the canonical Archimedean partition functions $Z_{N_k, \beta}$ are holomorphic and nonvanishing on one complex neighborhood of the interval $[-1, 0]$ that is independent of k .

Reference supplied by the database. Robert J. Berman, “An invitation to Kähler–Einstein metrics and random point processes,” *Surveys in Differential Geometry* 23 (2020), 35–87, Section 7.1, especially the paragraph following Theorem 7.9 (the proposed uniform zero-free-neighborhood approach, explicitly described as completely open in general). arXiv: [2003.11358](https://arxiv.org/abs/2003.11358).

6.21.4. Primary-source pointer

The originating ICM record is *Emergent complex geometry* (2022). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-562-9-volume-4.pdf#page=99>.

7. Geometric flows and geometric PDE

The records in this chapter concern the use of evolution equations and nonlinear PDE to construct or classify geometric structures. Ricci-type flows, elliptic equations, and canonical-metric problems illustrate the characteristic division between short-time or conditional theory and global existence, convergence, and singularity control.

Coverage: 19 website records; **source years:** 1994–2022; **separately listed descendant open problems:** 30.

7.1. 227. Interface dynamics in phase transitions [1994-geom-243]

Record: 1994-geom-243

Subfield: Geometric flows and geometric PDE

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1994: Interface dynamics in phase transitions

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.2/ICM1994.2.ocr.pdf#page=429>

7.1.1. Problem statement

Construct a mathematically well-posed generalized weak evolution of interfaces that is defined past singularities and agrees with the corresponding classical smooth evolution for as long as the latter exists, both for Hele–Shaw free-boundary motion and for multiphase interface evolution, including interfaces meeting at triple junctions, where the comparison or avoidance principle used for single-front generalized propagation is unavailable.

7.1.2. Historical background, literature, and current progress

The problem is **unresolved**. Modern work supplies robust post-singularity theories for important Hele–Shaw models and global weak existence for multiphase mean-curvature flow, but there is still no single mathematically well-posed generalized evolution covering multiphase interfaces through arbitrary singularities. In particular, the strongest multiphase uniqueness results are weak–strong uniqueness results: they identify the weak flow only while a classical network or cluster remains smooth. At topology changes, distinct physically meaningful

planar network continuations can occur, so a satisfactory general continuation or selection framework remains missing.

For the one-phase, zero-surface-tension Hele–Shaw problem, Inwon C. Kim, *Uniqueness and Existence Results on the Hele–Shaw and the Stefan Problems*, Archive for Rational Mechanics and Analysis 168 (2003), 299–328, DOI <https://doi.org/10.1007/s00205-003-0251-z>, introduced viscosity solutions and proved existence and uniqueness. More recently, Thomas Alazard and Herbert Koch, *The Hele–Shaw Semi-flow* (2025 revision), arXiv:2312.13678, Theorems 2.3 and 2.5, constructed a unique global semiflow that agrees with smooth solutions, permits topology changes for connected fluid domains lying between two horizontal hyperplanes, and becomes analytic after finite time. These are substantive solutions of major Hele–Shaw variants, but their hypotheses and physical laws do not cover every evolution encompassed by the unqualified phrase “Hele–Shaw free-boundary motion” in the problem.

For multiphase mean-curvature flow, Salvatore Stuvard and Yoshihiro Tonegawa, *On the Existence of Canonical Multi-phase Brakke Flows*, Advances in Calculus of Variations 17 (2024), 33–78, DOI <https://doi.org/10.1515/acv-2021-0093>, Theorem 2.11, proved global existence from arbitrary closed rectifiable multiphase data of a Brakke flow coupled to evolving grains, with their distributional velocity equal to generalized mean curvature. This prevents the artificial instantaneous disappearance allowed by unrestricted Brakke flow, but does not prove unconditional uniqueness through singularities.

Julian Fischer, Sebastian Hensel, Tim Laux, and Theresa M. Simon, *The Local Structure of the Energy Landscape in Multiphase Mean Curvature Flow: Weak-Strong Uniqueness and Stability of Evolutions*, Journal of the European Mathematical Society (online first, 2025), DOI <https://doi.org/10.4171/JEMS/1577>, proved weak–strong uniqueness for planar multiphase BV and varifold-BV solutions, including regular networks with triple junctions, only up to the first topology change. Sebastian Hensel and Tim Laux, *Weak-Strong Uniqueness for the Mean Curvature Flow of Double Bubbles*, Interfaces and Free Boundaries 25 (2023), 37–107, DOI <https://doi.org/10.4171/IFB/484>, established the analogous result for three-dimensional double bubbles meeting along triple lines. Neither theorem selects a unique evolution after singularities.

Carlo Mantegazza, Matteo Novaga, Alessandra Pluda, and Felix Schulze, *Evolution of Networks with Multiple Junctions*, Astérisque 452 (2024), DOI <https://doi.org/10.24033/ast.1225>, develops classical planar network flow, short-time restart constructions from nonregular networks, and continuation across several controlled singularities. Its Chapter 15 still lists as open the multiplicity-one conjecture, uniqueness or selection of restarted flows for generic nonregular networks, and whether singular times can accumulate. Thus the global existence and partial consistency results do not yet amount to the well-posed post-singularity multiphase evolution requested here.

7.1.3. Remaining open problems recorded by the database

7.1.3.1. Open problem 1

Let $\{S_t\}_{0 \leq t < T}$ be the curvature flow of an embedded regular network in a smooth strictly convex planar domain: at each regular time, S_t is a finite union of smooth embedded curves meeting only at triple junctions with angles 120° , its boundary endpoints are fixed, and every curve has normal velocity equal to its curvature vector. Prove or disprove that every self-similarly shrinking network obtained either as a subsequential limit

of Huisken's rescaling at a singular point or as a subsequential limit of parabolic rescalings $\lambda_j(S_{T+\lambda_j^{-2}t} - x_0)$, with $\lambda_j \rightarrow \infty$, is embedded and has multiplicity one.

Reference supplied by the database. Carlo Mantegazza, Matteo Novaga, Alessandra Pluda, and Felix Schulze, *Evolution of Networks with Multiple Junctions*, Astérisque 452, Société Mathématique de France, 2024, Open Problem 10.1 (Multiplicity-One Conjecture), DOI: <https://doi.org/10.24033/ast.1225>.

7.1.3.2. Open problem 2

Let a planar network evolve so that, between singular times, it is a finite union of smooth embedded curves moving with normal velocity equal to curvature and meeting only at 120° triple junctions, and at each singular time it is continued by a smooth network flow starting from the resulting nonregular limit network. Assume that no curve incident to a fixed boundary endpoint collapses. Determine whether the set of singular times must have no finite accumulation point.

Reference supplied by the database. Carlo Mantegazza, Matteo Novaga, Alessandra Pluda, and Felix Schulze, *Evolution of Networks with Multiple Junctions*, Astérisque 452, Société Mathématique de France, 2024, Open Problem 13.1 (second question), DOI: <https://doi.org/10.24033/ast.1225>.

7.1.4. Primary-source pointer

The originating ICM record is *Interface dynamics in phase transitions* (1994). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.2/ICM1994.2.ocr.pdf#page=429>.

7.2. 228. Lagrangian intersections, 3-manifolds with boundary, and the Atiyah-Floer conjecture [1994-geom-309]

Record: 1994-geom-309

Subfield: Geometric flows and geometric PDE

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1994: Lagrangian intersections, 3-manifolds with boundary, and the Atiyah-Floer conjecture

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.1/ICM1994.1.ocr.pdf#page=598>

7.2.1. Problem statement

Let

$$\mathbb{H}^4 = \{(x_0, x_1, x_2, x_3) \in \mathbb{R}^4 : x_0 \geq 0\}$$

with boundary $\partial\mathbb{H}^4 = \{x_0 = 0\}$, and let $\mathfrak{g} = \mathfrak{su}(2)$ be equipped with the inner product $\langle \zeta, \eta \rangle = \text{tr}(\zeta^* \eta)$ and its induced norm. Let

$$A = \sum_{j=0}^3 A_j dx_j \in \Omega^1(\mathbb{H}^4, \mathfrak{g})$$

be a smooth connection on the trivial $\text{SU}(2)$ -bundle over $\mathbb{H}^4$, and write its curvature components as

$$F_{ij} = \partial_i A_j - \partial_j A_i + [A_i, A_j].$$

Assume that

$$F_{01} = F_{23}, \quad F_{02} = F_{31}, \quad F_{03} = F_{12}$$

on $\mathbb{H}^4$, and that

$$A_0|_{\partial\mathbb{H}^4} = 0, \quad F_{23}|_{\partial\mathbb{H}^4} = 0.$$

Prove that either $F_{ij} = 0$ identically for every i, j , or

$$\mathcal{YM}(A) := \frac{1}{2} \int_{\mathbb{H}^4} \sum_{i < j} |F_{ij}|^2 dx_0 dx_1 dx_2 dx_3 \geq 8\pi^2.$$

7.2.2. Historical background, literature, and current progress

The exact problem is **unresolved**. No proof or counterexample was found for the sharp assertion that every nonflat smooth self-dual $\text{SU}(2)$ connection on $\mathbb{H}^4$ satisfying only the stated temporal-gauge and slice-flat boundary conditions has Yang–Mills energy at least $8\pi^2$.

Dietmar Salamon, *Lagrangian Intersections, 3-Manifolds with Boundary, and the Atiyah–Floer Conjecture*, in *Proceedings of the International Congress of Mathematicians, Zurich 1994*, Birkhauser, 1995, pp. 526–536, Conjecture 3.2, DOI 10.1007/978-3-0348-9078-6_46. This is the proposing source. It states the exact half-space energy gap and says that a proof should follow the pat-

tern of Uhlenbeck's removable-singularity theorem, but that the details had not been carried out.

Katrin Wehrheim, *Anti-Self-Dual Instantons with Lagrangian Boundary Conditions II: Bubbling*, *Communications in Mathematical Physics* **258** (2005), 275–315, DOI 10.1007/s00220-005-1349-y, arXiv:math/0401377. The introduction explicitly identifies Salamon's conjectured quantization for half-space limit instantons, then explains that local rescaling loses the global part of the Lagrangian boundary condition. Theorem 1.2 proves concentration of some constant $\varepsilon_0 > 0$ for sequences on $\mathbb{H}^2 \times \Sigma$ satisfying a stronger global handlebody-induced Lagrangian boundary condition. Lemma 2.8 obtains partial boundary regularity under slice-wise flatness alone. These results do not determine the energy of a half-space instanton and do not give the sharp constant $8\pi^2$.

Katrin Wehrheim, *Energy Identity for Anti-Self-Dual Instantons on $\mathbb{C} \times \Sigma$* , *Mathematical Research Letters* **13** (2006), no. 1, 161–166, DOI 10.4310/MRL.2006.v13.n1.a12, arXiv:math/0503341. Theorem 1.1 proves that a finite-energy $SU(2)$ ASD instanton on the boundaryless product $\mathbb{C} \times \Sigma$ has energy in $4\pi^2\mathbb{N}_0$. This is a genuine exact energy-quantization result in a related bubbling model, but its compact surface factor and absence of a boundary mean that it does not cover the problem here.

Tomasz S. Mrowka and Katrin Wehrheim, *L^2 -Topology and Lagrangians in the Space of Connections over a Riemann Surface*, *Geometric and Functional Analysis* **20** (2010), 1278–1305, DOI 10.1007/s00039-010-0086-3, arXiv:1005.0731. Theorem 6.1 extends Wehrheim's positive- ε_0 bubbling and removable-singularity results to a class of general gauge-invariant Lagrangian boundary conditions. It still assumes a global Lagrangian condition and yields no sharp $8\pi^2$ classification for connections satisfying only the local boundary condition in the statement.

Thus the established progress supplies elliptic regularity, removable-singularity machinery, and non-sharp energy concentration under stronger or different hypotheses. What remains open is precisely the asserted $8\pi^2$ gap for every connection satisfying the original half-space hypotheses.

7.2.3. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

7.2.4. Primary-source pointer

The originating ICM record is *Lagrangian intersections, 3-manifolds with boundary, and the Atiyah-Floer conjecture* (1994). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.1/ICM1994.1.ocr.pdf#page=598>.

7.3. 229. Fourier Analysis of Null Forms and Non-linear Wave Equations [1998-geom-71]

Record: 1998-geom-71

Subfield: Geometric flows and geometric PDE

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1998: Fourier Analysis of Null Forms and Non-linear Wave Equations

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.3/ICM1998.3.ocr.pdf#page=51>

7.3.1. Problem statement

Let n be the number of spatial variables, let M be a Riemannian manifold, and let $\phi : \mathbb{R}^{1+n} \rightarrow M$ be a wave map satisfying, in local coordinates,

$$\square \phi^i + \Gamma_{jk}^i(\phi) \partial_\alpha \phi^j \partial^\alpha \phi^k = 0,$$

where indices are raised with the Minkowski metric $(-1, 1, \dots, 1)$.

Prove that the Cauchy problem for this equation is locally well posed in time for initial data

$$(\phi(0, \cdot), \partial_t \phi(0, \cdot)) = (\phi_0, \phi_1)$$

of Sobolev regularity

$$H^{n/2}(\mathbb{R}^n) \times H^{n/2-1}(\mathbb{R}^n),$$

using both the null-form structure

$$Q_0(u, v) = \partial_\alpha u \partial^\alpha v$$

occurring in the nonlinearity and the geometric information coming from the target manifold. In the model case $M = S^{k-1} \subset \mathbb{R}^k$, this geometric structure is expressed by the equivalent constrained equation

$$\square \phi + \phi(\partial_\alpha \phi \cdot \partial^\alpha \phi) = 0, \quad |\phi| = 1.$$

7.3.2. Historical background, literature, and current progress

7.3.3. Verdict: partially resolved

7.3.3.1. Precise problem checked

The problem asks for the sharp endpoint local well-posedness theory for wave maps

$$\square \phi^i + \Gamma_{jk}^i(\phi) \partial_\alpha \phi^j \partial^\alpha \phi^k = 0$$

from $\mathbb{R}^{1+n}$ to a Riemannian target M , with arbitrary initial data in the scaling-critical Sobolev class

$$H^{n/2}(\mathbb{R}^n) \times H^{n/2-1}(\mathbb{R}^n).$$

The issue is not merely the scalar null form Q_0 ; the endpoint theory must also use the target geometry, e.g. for $M = S^{k-1}$ the constrained equation

$$\square \phi + \phi(\partial_\alpha \phi \cdot \partial^\alpha \phi) = 0, \quad |\phi| = 1.$$

No published or otherwise standard citable result was found that proves this full endpoint local well-posedness statement for arbitrary large critical Sobolev data, all spatial dimensions, and all Riemannian targets. The problem is therefore not fully resolved.

7.3.3.2. Main partial progress

1. Local and global results for wave maps. I

Daniel Tataru. *Communications in Partial Differential Equations* 23 (1998), no. 9-10, 1781-1793. DOI: <https://doi.org/10.1080/03605309808821400>. URL: <https://math.berkeley.edu/~tataru/papers/wm.pdf>

This established scale-invariant small-data theory in critical Besov-type spaces in high dimensions. It was a critical-regularity result, but the data space is smaller than $H^{n/2} \times H^{n/2-1}$, and the result is small-data, not arbitrary large endpoint-data local well-posedness.

2. On global existence and scattering for the wave maps equation

Daniel Tataru. *American Journal of Mathematics* 123 (2001), no. 1, 37-77. DOI: <https://doi.org/10.1353/ajm.2001.0005>.

This extended the critical small-data wave map theory, including low-dimensional aspects, in spaces adapted to the null form. It does not settle arbitrary critical Sobolev data for general targets.

3. Global regularity of wave maps I. Small critical Sobolev norm in high dimension

Terence Tao. *International Mathematics Research Notices* 2001, no. 6, 299-328. URL: <https://doi.org/10.1155/S1073792801000128>

Tao proved small-data global regularity at the critical Sobolev norm in high dimensions for sphere targets, using a microlocal renormalization/gauge mechanism. This is genuine $\dot{H}^{n/2} \times \dot{H}^{n/2-1}$ progress, but only for small data and special targets/dimensions.

4. Global regularity of wave maps II. Small energy in two dimensions

Terence Tao. *Communications in Mathematical Physics* 224 (2001), no. 2, 443-544. DOI: <https://doi.org/10.1007/PL00005588>.

This gives small-energy regularity in the two-dimensional critical case. It is important endpoint progress, but again small-data and not a general large-data local theory for all targets.

5. On the global regularity of wave maps in the critical Sobolev norm

Sergiu Klainerman and Igor Rodnianski. *International Mathematics Research Notices* 2001, no. 13, 655-677.

This is another critical Sobolev small-data result, using geometric/gauge structure. Its hypotheses do not cover arbitrary large critical data for arbitrary Riemannian targets.

6. On the well-posedness of the wave map problem in high dimensions

Andrea Nahmod, Atanas Stefanov, and Karen Uhlenbeck. *Communications in Analysis and Geometry* 11 (2003), no. 1, 49-83. URL: <https://www.intlpress.com/si>

[te/pub/pages/journals/items/cag/content/vols/0011/0001/a004/](https://pub/pages/journals/items/cag/content/vols/0011/0001/a004/)

This treats high-dimensional wave maps into compact Lie groups and Riemannian symmetric spaces, obtaining small critical-data well-posedness in that geometric setting. It is special-target and small-data progress.

7. Rough solutions for the wave maps equation

Daniel Tataru. *American Journal of Mathematics* 127 (2005), no. 2, 293-377. DOI: <https://doi.org/10.1353/ajm.2005.0014>. PDF: <https://math.berkeley.edu/~tataru/papers/wavemaps.pdf>

This is the strongest general-target critical Sobolev result found. Tataru proves global well-posedness for initial data small in the critical homogeneous Sobolev space for wave maps into a uniformly isometrically embedded Riemannian manifold. The paper explicitly says the analogous large-data problem remains largely open. Thus it does not imply the full arbitrary-data local well-posedness statement.

8. Energy Dispersed Large Data Wave Maps in 2 + 1 Dimensions

Jacob Sterbenz and Daniel Tataru. *Communications in Mathematical Physics* 298 (2010), no. 1, 139-230. DOI: <https://doi.org/10.1007/s00220-010-1061-4>.

This develops a large-data theory in the energy-critical 2 + 1-dimensional case under energy-dispersion assumptions. It is a major large-data endpoint result, but it is dimension-specific and assumes additional dispersion/threshold-type hypotheses.

9. Regularity of Wave-Maps in Dimension 2 + 1

Jacob Sterbenz and Daniel Tataru. *Communications in Mathematical Physics* 298 (2010), no. 1, 231-264. DOI: <https://doi.org/10.1007/s00220-010-1062-3>.

For compact targets in 2+1 dimensions, this proves that regularity and dispersive bounds persist unless a soliton-like concentration occurs; together with the companion paper it yields threshold-type global regularity below harmonic-map obstructions. This is not the full arbitrary-target/all-dimensions endpoint local theory asked in the problem statement.

10. Global regularity of wave maps V. Large data local wellposedness and perturbation theory in the energy class

Terence Tao. arXiv preprint, 2008. arXiv: 0808.0368. URL: <https://arxiv.org/abs/0808.0368>

This gives large-data local well-posedness in the energy class for wave maps from $\mathbb{R}^{1+2}$ to hyperbolic space $\mathbb{H}^m$. It is strong evidence and a major special case, but it is restricted to dimension 2 and hyperbolic targets, and it remains an unpublished preprint.

11. Unconditional well-posedness for wave maps

Nader Masmoudi and Fabrice Planchon. *Journal of Hyperbolic Differential Equations* 9 (2012), no. 2, 223-237. DOI: <https://doi.org/10.1142/S0219891612500075>. arXiv: 1111.4374. URL: <https://arxiv.org/abs/1111.4374>

This proves uniqueness in the natural critical class $C([0, T]; \dot{H}^{d/2}) \times C^1([0, T]; \dot{H}^{d/2-1})$ for $d \geq 4$, under bounded-curvature hypotheses. It is a uniqueness theorem, not a complete existence and continuous-dependence theorem for arbitrary endpoint data.

12. Ill-posedness for one-dimensional wave maps at the critical regularity

Terence Tao. *American Journal of Mathematics* 122 (2000), no. 3, 451-463. arXiv: math/9811169. URL: <https://arxiv.org/abs/math/9811169>

This shows that in $\mathbb{R}^{1+1}$ the wave map equation is in general ill-posed in the critical space $\dot{H}^{1/2}$. Therefore, if the problem statement is read literally as including $n = 1$, the endpoint well-posedness assertion is not true in full generality. Most positive critical

wave-map theory is consequently interpreted for $n \geq 2$ or with additional structure.

7.3.3.3. What remains unresolved

the exact statement would require a citable theorem giving local well-posedness for arbitrary data in $H^{n/2} \times H^{n/2-1}$, using both null form and geometry, for general Riemannian targets and all intended spatial dimensions. Existing literature proves:

- small-data global well-posedness at critical Sobolev regularity for broad classes of targets, culminating in Tataru's general-target small-data theorem;
- sharper special-target results such as spheres, compact Lie groups, symmetric spaces, and hyperbolic spaces;
- substantial $2 + 1$ -dimensional energy-critical large-data/threshold theory;
- uniqueness in critical classes in some higher-dimensional settings.

These do not cover arbitrary large endpoint Sobolev data for all general Riemannian targets. The current status of the problem is therefore **partially resolved**, not fully resolved.

7.3.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

7.3.5. Primary-source pointer

The originating ICM record is *Fourier Analysis of Null Forms and Non-linear Wave Equations* (1998). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.3/ICM1998.3.ocr.pdf#page=51>.

7.4. 230. Bubbling and Regularity Issues in Geometric Non-linear Analysis [2002-geom-256]

Record: 2002-geom-256

Subfield: Geometric flows and geometric PDE

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Bubbling and Regularity Issues in Geometric Non-linear Analysis

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=194>

7.4.1. Problem statement

Let (V^6, J, ω) be a compact almost Hermitian manifold obtained by a generic almost-complex perturbation of a Calabi–Yau threefold, and let $E \rightarrow V$ be a Hermitian vector bundle with structure group $SU(2)$. Consider the almost-complex vortex equations for a unitary connection A on E and an $\text{End}(E)$ -valued $(0, 3)$ -form f :

$$F_A^{0,2} = \bar{\partial}_A^* f, \quad F_A^{1,1} \wedge \omega^2 = [f, \bar{f}].$$

By dimensional reduction, regard each pair (A, f) as a torus-invariant almost-complex anti-self-dual connection $\tilde{A}$ on the pullback of E to $T^2 \times V$. Let (A_i, f_i) be a sequence of smooth solutions on the fixed bundle E such that the Yang–Mills energies $\int_{T^2 \times V} |F_{\tilde{A}_i}|^2$ are uniformly bounded and the pairs converge weakly modulo unitary gauge transformations to a weak solution (A_∞, f_∞) represented by smooth fields away from a closed set. Define the blow-up set $\Sigma \subset V$ by requiring $T^2 \times \Sigma$ to be the support of the defect measure in

$$|F_{\tilde{A}_i}|^2 d \text{vol}_{T^2 \times V} \rightarrow |F_{\tilde{A}_\infty}|^2 d \text{vol}_{T^2 \times V} + \nu,$$

and assume that Σ , with its multiplicity, is a pseudoholomorphic two-dimensional rectifiable cycle, so its approximate tangent planes are J -invariant almost everywhere. Let $S(A_\infty, f_\infty)$ be the set of points of V at which no local unitary gauge makes the weak limit smooth. Prove that $S(A_\infty, f_\infty) \subseteq \Sigma$ and that $S(A_\infty, f_\infty)$ is discrete.

7.4.2. Historical background, literature, and current progress

The problem is unresolved. The literature establishes weak compactness, rectifiability and pseudoholomorphic structure of the concentration cycle, and several removable-singularity results, but it does not prove the exact assertion for the almost-complex vortex equations: no result found shows, for a generic nonintegrable almost-complex perturbation, both that every intrinsic singularity of the weak limit belongs to the defect-measure blow-up cycle and that the entire intrinsic singular set is discrete.

- “Gauge theory and calibrated geometry, I,” Gang Tian, *Annals of Mathematics* 151 (2000), 193–268. DOI: <https://doi.org/10.2307/121116>; arXiv: [math/0010015](https://arxiv.org/abs/math/0010015). Tian derives the six-dimensional vortex system as equation (6.2.2),

proves that the blow-up locus of higher-dimensional anti-self-dual instantons is a calibrated integral current, and obtains removable-singularity and tangent-cone results. However, Conjecture 2 only predicts that the intrinsic singular set of an admissible calibrated anti-self-dual instanton has Hausdorff codimension at least six; this is the dimension bound that would reduce to isolated points in the present six-dimensional setting, and it is not proved there.

- **“Bubbling and regularity issues in geometric non-linear analysis,”** Tristan Rivière, *Proceedings of the International Congress of Mathematicians, Beijing 2002*, Vol. III, Higher Education Press (2002), 197–208. arXiv: [math/0304396](https://arxiv.org/abs/math/0304396). The exact assertion appears as open problem (i) in the discussion of torus-invariant instantons and the vortex equations on a Calabi–Yau threefold after perturbation to a nonintegrable almost-complex structure.
- **“A singularity removal theorem for Yang–Mills fields in higher dimensions,”** Terence Tao and Gang Tian, *Journal of the American Mathematical Society* 17 (2004), 557–593. DOI: <https://doi.org/10.1090/S0894-0347-04-00457-6>; arXiv: [math/0209352](https://arxiv.org/abs/math/0209352). This proves smooth removability for admissible stationary Yang–Mills fields under a sufficiently small scale-invariant curvature bound. It supplies an important local regularity tool but does not establish the global codimension-six/discreteness conclusion or the required containment in the defect cycle for the coupled almost-complex vortex system.
- **“The singular set of 1-1 integral currents,”** Tristan Rivière and Gang Tian, *Annals of Mathematics* 169 (2009), 741–794. DOI: <https://doi.org/10.4007/annals.2009.169.741>; arXiv: [math/0310052](https://arxiv.org/abs/math/0310052). This proves that a two-dimensional integral almost-complex cycle, under the stated local compatible-symplectic hypothesis, is a smooth pseudoholomorphic curve away from isolated points. It resolves the separate regularity question for the blow-up cycle itself, not the assertion that singularities of the limiting vortex fields are isolated and lie on that cycle.
- **“A weak compactness theorem of the Donaldson–Thomas instantons on compact Kähler threefolds,”** Yuuji Tanaka, *Journal of Mathematical Analysis and Applications* 408 (2013), 27–34. DOI: <https://doi.org/10.1016/j.jmaa.2013.05.059>; arXiv: [0805.2195](https://arxiv.org/abs/0805.2195). Under the additional assumptions that the threefold is Kähler and the Higgs fields have uniformly bounded L^2 -norm, Theorem 5.1 gives smooth subsequential convergence away from a closed set of finite real two-dimensional Hausdorff measure. This is a special-case weak compactness theorem, not discreteness in the almost-complex setting.
- **“A removal singularity theorem of the Donaldson–Thomas instanton on compact Kähler threefolds,”** Yuuji Tanaka, *Journal of Mathematical Analysis and Applications* 411 (2014), 422–428. DOI: <https://doi.org/10.1016/j.jmaa.2013.09.053>; arXiv: [1305.2998](https://arxiv.org/abs/1305.2998). Theorem 1.3 and Corollary 3.2 remove the two-dimensional top stratum of the Kähler-case singular set. Corollary 2.6 leaves a residual set $S^{(0)}$ with $\mathcal{H}^2(S^{(0)}) = 0$, and the paper explicitly states that finiteness of $\mathcal{H}^0(S^{(0)})$ and discreteness are still expected. Thus even this strongest direct treatment does not prove the exact almost-complex statement.
- **“Compactness for Ω -Yang–Mills connections,”** Xuemiao Chen and Richard A. Wentworth, *Calculus of Variations and Partial Differential Equations* 61 (2022), Article 58. DOI: <https://doi.org/10.1007/s00526-021-02178-0>; arXiv: [2106.09131](https://arxiv.org/abs/2106.09131). This extends weak compactness, rectifiability, tangent-cone, calibrated-cycle, and small-energy removal results to Ω -Yang–Mills connections with Ω not necessarily

closed. It still records the assertion that essential singularities of finite-energy Ω -anti-self-dual instantons have codimension at least six as conjectural, and it does not prove the requested discreteness for the dimensionally reduced vortex fields.

Accordingly, the established results control the two-dimensional concentration set and remove substantial classes of singularities, especially in the integrable Kähler case. What remains is the full exclusion of nonisolated intrinsic singularities and the proof that every surviving singular point lies on the pseudoholomorphic defect-measure cycle for the generic almost-complex equations.

7.4.3. Remaining open problems recorded by the database

7.4.3.1. Open problem 1

Let (Z, ω) be a compact Kähler threefold and let $E \rightarrow Z$ be a Hermitian vector bundle of rank r . A Donaldson–Thomas instanton is a unitary connection A on E together with a $u(E)$ -valued $(0, 3)$ -form u satisfying

$$F_A^{0,2} = 0, \quad \bar{\partial}_A^* u = 0, \quad F_A^{1,1} \wedge \omega^2 + [u, \bar{u}] + \frac{i\lambda(E)}{3} \text{Id}_E \omega^3 = 0,$$

where $\lambda(E) = 6\pi(c_1(E) \smile [\omega]^2)/(r[\omega]^3)$. Let (A_n, u_n) be a sequence of Donaldson–Thomas instantons such that $\sup_n \int_Z |u_n|^2 dV_g < \infty$, and, after passing to a subsequence and applying unitary gauge transformations, let (A, u) be its weak limit. Set

$$m(A, u) = |F_A^\perp|^2 - |[u, \bar{u}]|^2,$$

where $F_A^\perp$ is the component of F_A orthogonal to the Kähler form, and suppose that

$$m(A_n, u_n) dV_g \rightharpoonup \mu$$

as Radon measures. For $z \in Z$, define

$$\Theta(\mu, z) = \lim_{\rho \downarrow 0} \rho^{-2} \mu(B_\rho(z)),$$

let $\varepsilon > 0$ be the regularity threshold in the weak compactness theorem, and set

$$S = \{z \in Z : \Theta(\mu, z) \geq \varepsilon\}, \quad S^{(0)} = \left\{z \in S : \lim_{\rho \downarrow 0} \rho^{-2} \int_{B_\rho(z)} |F_A|^2 dV_g \geq \varepsilon\right\}.$$

Prove that $\mathcal{H}^0(S^{(0)}) < \infty$ and that $S^{(0)}$ is discrete.

Reference supplied by the database. Yuuji Tanaka, “A removal singularity theorem of the Donaldson–Thomas instanton on compact Kähler threefolds,” *Journal of Mathematical Analysis and Applications* 411 (2014), no. 1, 422–428, Corollary 2.6 and the open expectation immediately following it. DOI: <https://doi.org/10.1016/j.jmaa.2013.09.053>; arXiv:1305.2998.

7.4.4. Primary-source pointer

The originating ICM record is *Bubbling and Regularity Issues in Geometric Non-linear Analysis* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=194>.

7.5. 231. Evolution of Curves and Surfaces by Mean Curvature [2002-geom-335]

Record: 2002-geom-335

Subfield: Geometric flows and geometric PDE

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Evolution of Curves and Surfaces by Mean Curvature

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.1/ICM2002.1.ocr.pdf#page=522>

7.5.1. Problem statement

Let $M_0 \subset \mathbb{R}^3$ be a smooth, compact, connected, embedded surface without boundary that is diffeomorphic to either the sphere S^2 or the torus T^2 . Prove that there is exactly one mean-curvature evolution of M_0 in the class of Ilmanen enhanced Brakke flows of varifolds starting from the unit-multiplicity varifold associated with M_0 .

Equivalently, let $u : \mathbb{R}^3 \times [0, \infty) \rightarrow \mathbb{R}$ be the viscosity solution of the level-set mean-curvature equation

$$\partial_t u = |\nabla u| \operatorname{div} \left(\frac{\nabla u}{|\nabla u|} \right),$$

with initial data equal to the signed distance from M_0 , and let $M_t = \{x \in \mathbb{R}^3 : u(x, t) = 0\}$. Prove that M_t has empty interior for every $t > 0$; that is, the level-set flow starting from M_0 never fattens.

7.5.2. Historical background, literature, and current progress

The exact problem is **unresolved**. The assertion is now known for every smooth embedded 2-sphere, but no citable result was found that proves nonfattening for every smooth embedded torus in $\mathbb{R}^3$, nor is there a proved fattening counterexample of genus one. Thus the torus case prevents a complete resolution of the stated sphere-or-torus assertion.

The spherical case is complete. Bamler and Kleiner prove the multiplicity-one conjecture for flows of embedded surfaces in $\mathbb{R}^3$ and explicitly deduce, by combining their theorem with Brendle's genus-zero shrinker classification and the mean-convex-neighborhood/nonfattening theory, that the level-set flow of every embedded 2-sphere does not fatten; they also obtain uniqueness in their class of almost-regular flows. Since the problem states level-set nonfattening as the equivalent formulation, this covers the entire S^2 alternative.

For a torus, the strongest general criteria remain conditional. Hershkovits and White prove nonfattening when every singularity is of mean-convex or mean-concave type. Choi, Haslhofer, and Hershkovits prove the required mean-convex neighborhoods at multiplicity-one spherical or cylindrical singularities, and hence nonfattening while all singularities are spherical or cylindrical. Generic-flow results, together with Bamler–Kleiner, consequently give nonfattening after arbitrarily small generic perturbations of a torus, but this does not cover each prescribed torus. Genus-one singularities genuinely occur in suitable one-parameter families of initial tori, so the genus-zero classification cannot simply be reused. Conversely, the rigor-

ous smooth fattening examples of Ilmanen and White have genus g only for sufficiently large g (in particular, $g \geq 3$); they do not furnish a genus-one counterexample. White's 2026 genus-monotonicity theorem also applies only as long as singularities are spherical or cylindrical and therefore does not close the remaining positive-genus singularity case.

Key references:

- **Evolution of Curves and Surfaces by Mean Curvature**, Brian White, *Proceedings of the International Congress of Mathematicians, Beijing 2002*, Vol. I, 525–538 (2002), [arXiv:math/0212407](https://arxiv.org/abs/math/0212407). This is the source that explicitly conjectures uniqueness for smooth embedded spheres and tori.
- **Embedded self-similar shrinkers of genus 0**, Simon Brendle, *Annals of Mathematics* 183 (2016), 715–728, <https://doi.org/10.4007/annals.2016.183.2.6>. It classifies the properly embedded genus-zero shrinkers relevant to spherical initial data as the plane, round sphere, and round cylinder.
- **Nonfattening of Mean Curvature Flow at Singularities of Mean Convex Type**, Or Herskovits and Brian White, *Communications on Pure and Applied Mathematics* 73 (2020), 558–580, <https://doi.org/10.1002/cpa.21852>. It proves nonfattening under the stated local mean-convex/mean-concave singularity hypothesis.
- **Ancient low-entropy flows, mean-convex neighborhoods, and uniqueness**, Kyeongsu Choi, Robert Haslhofer, and Or Herskovits, *Acta Mathematica* 228 (2022), 217–301, <https://doi.org/10.4310/ACTA.2022.v228.n2.a1>. Theorems 1.7 and 1.9 establish mean-convex neighborhoods and nonfattening when all singularities are multiplicity-one spheres or cylinders; Theorem 1.12 gave the sphere conclusion conditional on multiplicity one.
- **On the Multiplicity One Conjecture for Mean Curvature Flows of surfaces**, Richard H. Bamler and Bruce Kleiner, arXiv preprint (2023; revised 2024), [arXiv:2312.02106](https://arxiv.org/abs/2312.02106). It proves multiplicity one for embedded surface flows and unconditionally deduces nonfattening and well-posedness for embedded 2-spheres.
- **Revisiting generic mean curvature flow in $\mathbb{R}^3$** , Otis Chodosh, Kyeongsu Choi, Christos Mantoulidis, and Felix Schulze, *Journal für die reine und angewandte Mathematik* (2026 special issue; published online 2025), <https://doi.org/10.1515/crelle-2025-0064>. Together with Bamler–Kleiner, it resolves Huisken's genericity conjecture, yielding only spherical and cylindrical singularities for generic embedded initial surfaces and hence nonfattening for generic, but not all, tori.
- **Genus-one singularities in mean curvature flow**, Adrian Chun-Pong Chu and Ao Sun, *Geometry & Topology* 29 (2025), 4299–4340, <https://doi.org/10.2140/gt.2025.29.4299>. It forces a genus-one singularity in certain one-parameter families of initial tori and constructs a new embedded genus-one self-shrinker; it does not decide whether those torus flows fatten.
- **Fattening in mean curvature flow**, Tom Ilmanen and Brian White, *Ars Inveniendi Analytica* (2025), Paper No. 4, <https://doi.org/10.15781/cybg-mf32>, [arXiv:2406.18703](https://arxiv.org/abs/2406.18703). It proves fattening from smooth compact embedded initial surfaces of every sufficiently large genus (necessarily at least 3), leaving genus one untouched.
- **The Genus-Decreasing Property of Mean Curvature Flow, I**, Brian White, arXiv preprint (2026), [arXiv:2601.13787](https://arxiv.org/abs/2601.13787). It proves monotonicity of the genus of the regular set while all singularities are multiplicity-one planes, spheres, or cylinders, but does not handle the possible genus-one singularities that form the remaining obstruction here.

7.5.3. Remaining open problems recorded by the database

7.5.3.1. Open problem 1

Let $M_0 \subset \mathbb{R}^3$ be a smooth, compact, connected, embedded surface without boundary that is diffeomorphic to the torus T^2 . Prove that there is exactly one mean-curvature evolution of M_0 in the class of Ilmanen enhanced Brakke flows of varifolds starting from the unit-multiplicity varifold associated with M_0 . Equivalently, if $u : \mathbb{R}^3 \times [0, \infty) \rightarrow \mathbb{R}$ is the viscosity solution of $\partial_t u = |\nabla u| \operatorname{div}(\nabla u / |\nabla u|)$ with initial data equal to the signed distance from M_0 , prove that the zero set $\{x \in \mathbb{R}^3 : u(x, t) = 0\}$ has empty interior for every $t > 0$.

Reference supplied by the database. Brian White, “Evolution of Curves and Surfaces by Mean Curvature,” *Proceedings of the International Congress of Mathematicians, Beijing 2002*, Vol. I, 2002, pp. 525–538, Section 4 (unnumbered conjecture), arXiv:math/0212407, <https://arxiv.org/abs/math/0212407>.

7.5.4. Primary-source pointer

The originating ICM record is *Evolution of Curves and Surfaces by Mean Curvature* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.1/ICM2002.1.ocr.pdf#page=522>.

7.6. 232. On non-Kähler Calabi-Yau Threefolds with Balanced Metrics [2010-ag-139]

Record: 2010-ag-139

Subfield: Geometric flows and geometric PDE

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2010: On non-Kähler Calabi-Yau Threefolds with Balanced Metrics

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=711>

7.6.1. Problem statement

For an integer $k \geq 2$, let X be a non-Kähler Calabi-Yau threefold obtained by the Clemens-Friedman construction whose underlying smooth manifold is the connected sum $\#_k(S^3 \times S^3)$ of k copies of $S^3 \times S^3$, and let Ω be a nowhere-vanishing holomorphic three-form on X . Does there exist a Hermitian metric ω on X and a Hermitian metric h on the holomorphic tangent bundle $T'X$ such that, writing F_h for the curvature of h , R_ω for the curvature of the Hermitian connection determined by ω , and α' for the string parameter, the Strominger system

$$d(\|\Omega\|_\omega \omega^2) = 0, \quad F_h^{2,0} = F_h^{0,2} = 0, \quad F_h \wedge \omega^2 = 0,$$

$$\sqrt{-1} \partial \bar{\partial} \omega = \frac{\alpha'}{4} (\text{tr}(R_\omega \wedge R_\omega) - \text{tr}(F_h \wedge F_h))$$

is satisfied?

7.6.2. Historical background, literature, and current progress

7.6.3. Verdict: partially resolved

7.6.3.1. Precise problem

For each integer $k \geq 2$, and for a non-Kähler Calabi-Yau threefold X obtained by the Clemens-Friedman conifold-transition construction with underlying smooth manifold $\#_k(S^3 \times S^3)$, decide whether there are Hermitian metrics ω on X and h on $T^{1,0}X$ solving the full Strominger/Hull-Strominger system:

- the conformally balanced equation $d(\|\Omega\|_\omega \omega^2) = 0$;
- the Hermitian-Yang-Mills equations for the Chern curvature F_h on $T^{1,0}X$;
- the anomaly cancellation/Bianchi identity $\partial \bar{\partial} \omega = (\alpha'/4)(\text{tr}(R_\omega \wedge R_\omega) - \text{tr}(F_h \wedge F_h))$.

7.6.3.2. Verdict

The problem is partially resolved, not fully resolved. The literature proves the conformally balanced part and the Hermitian-Yang-Mills tangent-bundle part for the relevant conifold-transition/Clemens setting, including the $\#_k(S^3 \times S^3)$ examples. I found no citable result proving the final anomaly cancellation equation for these Clemens-Friedman threefolds, and recent survey/lecture literature explicitly states that this additional equation has not yet been solved through conifold transitions.

7.6.3.3. Progress and relevant literature

1. C. Herbert Clemens, “Double solids”, *Advances in Mathematics* 47 no. 2 (1983), 107-230. DOI: [https://doi.org/10.1016/0001-8708\(83\)90025-7](https://doi.org/10.1016/0001-8708(83)90025-7). This is part of the original conifold-transition/Clemens background.
2. Robert Friedman, “Simultaneous resolution of threefold double points”, *Mathematische Annalen* 274 (1986), 671-689. DOI: <https://doi.org/10.1007/BF01458602>; and “On threefolds with trivial canonical bundle”, in *Complex Geometry and Lie Theory*, Proc. Sympos. Pure Math. 53, AMS, 1991, 103-134. DOI: <https://doi.org/10.1090/pspum/053/1141199>. These give the Friedman side of the Clemens-Friedman construction and smoothing criteria for nodal Calabi-Yau threefolds.
3. P. Lu and G. Tian, “The complex structure on a connected sum of $S^3 \times S^3$ with trivial canonical bundle”, *Mathematische Annalen* 298 (1994), 761-764. DOI: <https://doi.org/10.1007/BF01459760>; and “Complex structures on connected sums of $S^3 \times S^3$ ”, in *Manifolds and Geometry*, Symposia Mathematica XXXVI, Cambridge Univ. Press, 1996, 284-293. These are standard sources for the non-Kähler Calabi-Yau complex structures on connected sums of copies of $S^3 \times S^3$.
4. Jixiang Fu, “On non-Kähler Calabi-Yau threefolds with balanced metrics”, *Proceedings of the International Congress of Mathematicians 2010*, Vol. II, Hindustan Book Agency, 2010, 705-716. DOI/URL: https://doi.org/10.1142/9789814324359_0070. This ICM article identifies the Strominger system as the intended canonical-geometry problem for these balanced non-Kähler Calabi-Yau threefolds.
5. Jixiang Fu, Jun Li, and Shing-Tung Yau, “Balanced metrics on non-Kähler Calabi-Yau threefolds”, *Journal of Differential Geometry* 90 no. 1 (2012), 81-129. DOI: <https://doi.org/10.4310/jdg/1335209490>; arXiv: 0809.4748. The main theorem constructs balanced metrics on non-Kähler Calabi-Yau threefolds obtained by smoothing after contracting $(-1, -1)$ curves. Corollary 1.3 applies to the complex structures on $\#_k(S^3 \times S^3)$ for any $k \geq 2$. The paper also explains that this is meant as a step toward the Strominger system; it does not solve the anomaly cancellation equation.
6. Yuri D. Bozhkov, “The specific Hermitian geometry of certain three-folds”, *Rivista di Matematica della Università di Parma* (5) 4 (1995), 61-68. MR1395324; Zbl 0865.53059. This proves stability properties of the tangent bundle for the $\#_k(S^3 \times S^3)$ complex threefolds. Via Li-Yau’s non-Kähler Hermitian-Einstein theorem, this gives Hermitian-Yang-Mills metrics in the relevant stable setting, but it does not address the anomaly equation.
7. Tristan C. Collins, Sebastien Picard, and Shing-Tung Yau, “Stability of the tangent bundle through conifold transitions”, *Communications on Pure and Applied Mathematics* 77 (2024), 284-371. DOI: <https://doi.org/10.1002/cpa.22135>; arXiv: 2102.11170. Their main theorem proves that for small smoothings in a conifold transition, $T^{1,0}X_t$ admits a Hermitian-Yang-Mills metric with respect to the Fu-Li-Yau con-

formally balanced metrics, and gives quantitative behavior near vanishing cycles. This supplies the second equation in the stated system for the tangent bundle, but again not the anomaly equation.

8. Sebastien Picard, “Calabi-Yau threefolds across quadratic singularities”, in 2024 MATRIX Annals, Part II, MATRIX Book Series 8, Springer, 2026. DOI: https://doi.org/10.1007/978-3-032-16206-9_17; arXiv: 2501.19313. Theorem 3.2 summarizes the current conifold-transition results: for small $\mathfrak{t}$, the non-Kähler smoothing has $(\Omega_{\mathfrak{t}}, g_{\mathfrak{t}}, h_{\mathfrak{t}})$ satisfying $d(|\Omega_{\mathfrak{t}}|_{\Omega_{\mathfrak{t}}} \Omega_{\mathfrak{t}}^2) = 0$ and $F_{h_{\mathfrak{t}}} \wedge \Omega_{\mathfrak{t}}^2 = 0$ on the tangent bundle. Immediately afterward, the text says that the heterotic Bianchi identity/anomaly cancellation equation is an additional equation and “has not yet been solved through conifold transitions”; it is stated as Yau’s conjecture.
9. Anna Fino, Gueo Grantcharov, and Luigi Vezzoni, “Solutions to the Hull-Strominger System with Torus Symmetry”, Communications in Mathematical Physics 388 (2021), 947-967. DOI: <https://doi.org/10.1007/s00220-021-04223-7>; arXiv: 1901.10322. This gives full Hull-Strominger solutions on other non-Kähler Calabi-Yau threefolds, including certain manifolds with $S^3 \times S^3$ summands, but the topology and construction are different from the pure Clemens-Friedman $\#_k(S^3 \times S^3)$ problem considered here. It is related progress, not a resolution.

7.6.3.4. What remains unresolved

No located paper proves that, for every $k \geq 2$ and every Clemens-Friedman non-Kähler Calabi-Yau threefold diffeomorphic to $\#_k(S^3 \times S^3)$, one can choose the conformally balanced metric ω and tangent-bundle Hermitian-Yang-Mills metric h so that the coupled anomaly cancellation equation also holds. The missing step is not just a technical detail: recent accounts isolate it as the remaining Bianchi/anomaly equation in Yau’s conjectural Strominger-system picture through conifold transitions.

7.6.4. Remaining open problems recorded by the database

7.6.4.1. Open problem 1

Given a Clemens–Friedman complex threefold $X \cong \#_{k \geq 2}(S^3 \times S^3)$ and a holomorphic vector bundle $V \rightarrow X$, find a positive Hermitian form ω and a Hermitian metric h on V satisfying the Bianchi identity

$$dd^c \omega = -\operatorname{tr}(F_h \wedge F_h).$$

Reference supplied by the database. Mario García-Fernández, “Pluriclosed flow and the Hull–Strominger system,” Herts Mathematical Physics Seminar, University of Hertfordshire, 30 October 2024, slide 43, Problem, <https://hertsmathphys.github.io/2024-10-30-Garc%C3%ADa.pdf>.

7.6.5. Primary-source pointer

The originating ICM record is *On non-Kähler Calabi-Yau Threefolds with Balanced Metrics* (2010). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2010.2/ICM2010.2.pdf#page=711>.

7.7. 233. Extremal Kähler metrics [2014-geom-123]

Record: 2014-geom-123

Subfield: Geometric flows and geometric PDE

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: Extremal Kähler metrics

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=1035>

7.7.1. Problem statement

Let M be a Fano manifold, and let $m > 0$ be such that there is a smooth divisor $D \in |mK_M^{-1}|$. Consider the conical continuity method with initial equation

$$\text{Ric}(\omega_t) = t\omega_t + \frac{1-t}{m}[D],$$

where $[D]$ denotes the current of integration over D .

Determine whether the following assertion holds. If, as t approaches a singular time $T < 1$, the solutions converge to a pair (W, Δ) , where W is a $\mathbf{Q}$ -Fano variety and Δ is the limiting Weil divisor corresponding to D , then the process can be continued on the current pair by solving the weak conical Kähler-Ricci soliton equation

$$\text{Ric}(\omega_t) + L_{X_t}\omega_t = t\omega_t + \frac{1-t}{m}[\Delta],$$

where X_t is a holomorphic vector field preserving Δ and $[\Delta]$ is the current of integration over Δ . Moreover, after passing through only finitely many such singular times and replacing the pair (W, Δ) by the successive limiting pairs $(W_1, \Delta_1), \dots, (W_k, \Delta_k)$, one can solve this equation all the way up to $t = 1$. At $t = 1$, the final variety W_k is a $\mathbf{Q}$ -Fano variety admitting a weak Kähler-Ricci soliton, and there is a test configuration, equivalently a $\mathbf{C}^*$ -equivariant flat degeneration, for the polarized Fano manifold (M, K_M^{-1}) whose central fiber is W_k .

7.7.2. Historical background, literature, and current progress

7.7.3. Verdict: partially resolved

7.7.3.1. Precise problem checked

The statement is Székelyhidi's proposed continuation picture for the conical continuity method on a Fano manifold. Starting with a smooth anticanonical multiple divisor $D \in |mK_M^{-1}|$, one follows

$$\text{Ric}(\omega_t) = t\omega_t + \frac{1-t}{m}[D].$$

If the path reaches a singular time $T < 1$ and converges to a $\mathbf{Q}$ -Fano pair (W, Δ) , the conjectural claim is that one can continue on the limiting current pair by solving a weak conical Kähler-

Ricci soliton equation with a holomorphic vector field preserving Δ . After only finitely many such jumps, the process should reach $t = 1$, producing a $\mathbf{Q}$ -Fano variety W_k with a weak Kähler-Ricci soliton and a $\mathbf{C}^*$ -equivariant flat degeneration/test configuration of $(M, -K_M)$ with central fiber W_k .

The exact problem is therefore not merely the existence of some Kähler-Ricci soliton degeneration of M , but the existence and finite termination of this specific conical-continuity continuation through singular pairs.

7.7.3.2. Original source

- Title: *Extremal Kähler metrics*
- Author: Gábor Székelyhidi
- Venue: Proceedings of the International Congress of Mathematicians, Seoul 2014; arXiv preprint
- Year: 2014
- DOI/arXiv/URL: arXiv:1405.4836, <https://arxiv.org/abs/1405.4836>
- Relevance: Conjecture 11 in this article is the source of the problem. It explicitly describes the conical continuity method with divisor D , continuation through limiting pairs by weak conical Kähler-Ricci soliton equations, finite singular times, and the final test-configuration degeneration. This is stated as a conjectural picture, not as a theorem.

7.7.3.3. Later progress

- Title: *Kähler-Ricci flow, Kähler-Einstein metric, and K-stability*
- Authors: Xiuxiong Chen, Song Sun, Bing Wang
- Venue: Geometry & Topology
- Year: 2018
- DOI/arXiv/URL: DOI 10.2140/gt.2018.22.3145; arXiv:1508.04397, <https://arxiv.org/abs/1508.04397>
- Relevance: This proves major compactness and degeneration results for the Kähler-Ricci flow on Fano manifolds, including algebraic degeneration phenomena tied to K-stability/Kähler-Einstein existence. It supports the broader expectation that analytic limits are $\mathbf{Q}$ -Fano degenerations, but it is about the Kähler-Ricci flow/Kähler-Einstein setting and does not prove the finite conical-continuity soliton-jump procedure in the problem statement.
- Title: *The Kähler-Ricci flow and optimal degenerations*
- Authors: Ruadhaí Dervan, Gábor Székelyhidi
- Venue: Journal of Differential Geometry
- Year: 2020
- DOI/arXiv/URL: DOI 10.4310/jdg/1594266881; arXiv:1612.07299, <https://arxiv.org/abs/1612.07299>
- Relevance: This paper proves that, when the Kähler-Ricci flow does not converge to a Kähler-Einstein metric, the limiting behavior is related to an optimal degeneration and a Kähler-Ricci soliton type central fiber. This is close in spirit to the final degeneration predicted by the conjecture. However, it does not establish Székelyhidi's conical conti-

nunity path with divisors, continuation across singular times, or finite termination of that process.

- Title: *Algebraic uniqueness of Kähler-Ricci flow limits and optimal degenerations of Fano varieties*
- Author: Han Li
- Venue: Geometry & Topology
- Year: 2023
- DOI/arXiv/URL: DOI 10.2140/gt.2023.27.1465; arXiv:2009.01098, <https://arxiv.org/abs/2009.01098>
- Relevance: This proves algebraic uniqueness results for Kähler-Ricci flow limits and optimal degenerations of Fano varieties. It strengthens the algebraic understanding of the limiting soliton/degeneration picture, but it is still not a proof of the finite conical-continuity continuation through the pairs (W_i, Δ_i) .
- Title: *The existence of the Kähler-Ricci soliton degeneration*
- Authors: Chenyang Xu, Ziquan Zhuang
- Venue: Forum of Mathematics, Pi
- Year: 2024
- DOI/arXiv/URL: DOI 10.1017/fmp.2024.11; arXiv:2406.04590, <https://arxiv.org/abs/2406.04590>
- Relevance: The authors prove existence of a Kähler-Ricci soliton degeneration for any log Fano pair. In particular, the theorem gives an algebraic degeneration whose central fiber admits the relevant soliton structure in a broad log Fano setting. This resolves a major algebraic/existence part of the picture predicted by Székelyhidi. It still does not prove that the particular conical continuity method described in the problem can be continued through singular limiting pairs, nor that only finitely many such analytic singular times occur.
- Title: *Twisted conical Kähler-Ricci solitons on Fano manifolds*
- Authors: Zhiwei Wang, Xiaohua Zhu
- Venue: Science China Mathematics
- Year: 2023
- DOI/arXiv/URL: DOI 10.1007/s11425-022-2125-3; <https://link.springer.com/article/10.1007/s11425-022-2125-3>
- Relevance: This studies existence of twisted conical Kähler-Ricci solitons under stability/properness hypotheses. It is relevant to the weak conical soliton equation appearing in the conjectural continuation, but it supplies conditional existence results rather than the global finite-step continuation asserted in the problem statement.

7.7.3.4. Why the problem is not fully resolved

The literature now contains strong results proving existence and uniqueness properties of Kähler-Ricci soliton degenerations, especially via Kähler-Ricci flow and non-Archimedean/algebraic methods. These results cover important pieces of the predicted final output: a $\mathbf{Q}$ -Fano/log Fano central fiber, soliton structure, and test-configuration style

degeneration.

However, I found no published or otherwise citable theorem proving the full statement as written: starting from the conical continuity method with a smooth divisor $D \in |mK_M^{-1}|$, every singular time $T < 1$ can be crossed by solving the weak conical Kähler-Ricci soliton equation on the limiting pair, only finitely many such crossings occur, and this analytic procedure reaches $t = 1$. The closest later theorems establish related flow or algebraic degeneration results, often without the divisor path and without the finite sequence of conical-continuity jumps.

Thus the correct current status of the problem is **partially resolved**: major algebraic and flow-theoretic components of the expected endpoint are known, but the full conical-continuity finite-continuation conjecture remains unresolved.

7.7.4. Remaining open problems recorded by the database

7.7.4.1. Open problem 1

Prove that the continuation of the conical continuity method through limiting pairs (W_i, Δ_i) cannot encounter infinitely many singular parameters: only finitely many changes of the pair are required before the parameter reaches $(t=1)$.

Reference supplied by the database. Gábor Székelyhidi, “Extremal Kähler Metrics,” in *Proceedings of the International Congress of Mathematicians—Seoul 2014*, Vol. II, Kyung Moon Sa, Seoul, 2014, pp. 1017–1032, Conjecture 11, arXiv: [1405.4836](https://arxiv.org/abs/1405.4836).

7.7.5. Primary-source pointer

The originating ICM record is *Extremal Kähler metrics* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=1035>.

7.8. 234. Geometric approaches to semilinear elliptic equations [2014-geom-125]

Record: 2014-geom-125

Subfield: Geometric flows and geometric PDE

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: Geometric approaches to semilinear elliptic equations

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.3/ICM2014.3.pdf#page=957>

7.8.1. Problem statement

Let $u : \mathbb{R}^3 \rightarrow \mathbb{R}$ be a bounded entire solution of the Allen-Cahn equation

$$\Delta u + u - u^3 = 0 \quad \text{in } \mathbb{R}^3.$$

Define the Morse index $m(u)$ to be the maximal dimension of a vector space $E \subset C_c^\infty(\mathbb{R}^3)$ such that

$$\mathcal{Q}(\psi, \psi) := \int_{\mathbb{R}^3} (|\nabla \psi|^2 + (3u^2 - 1)\psi^2) < 0$$

for every nonzero $\psi \in E$. Prove that if $m(u) = 1$ and $\nabla u \neq 0$ outside some bounded subset of $\mathbb{R}^3$, then u is axially symmetric: equivalently, there is a line in $\mathbb{R}^3$ such that u is invariant under all rotations about that line, or, after a rigid motion, $u(x_1, x_2, x_3) = U(\sqrt{x_1^2 + x_2^2}, x_3)$ for some function U .

7.8.2. Historical background, literature, and current progress

7.8.3. Status check

Verdict: partially resolved.

7.8.3.1. Precise problem

The problem asks whether every bounded entire solution

$$\Delta u + u - u^3 = 0 \quad \text{in } \mathbb{R}^3$$

with Morse index $m(u) = 1$ and with $\nabla u \neq 0$ outside a compact set must be axially symmetric, i.e. invariant under rotations about some line.

7.8.3.2. Current literature status

I found no published or citable preprint theorem that proves this exact implication. The closest results either assume axial symmetry from the start, assume detailed two-end asymptotics and

extra rate conditions, work only in dimension two, or prove a finite-end/asymptotic structure under an added bounded-energy-density hypothesis without proving axial symmetry.

7.8.3.3. Key literature and hypothesis checks

1. Manuel del Pino, Michal Kowalczyk, Juncheng Wei, “Entire solutions of the Allen-Cahn equation and complete embedded minimal surfaces of finite total curvature in $\mathbb{R}^3$ ”, *Journal of Differential Geometry* 93 (2013), 67-131. arXiv:0902.2047, URL: <https://arxiv.org/abs/0902.2047>.

This is the original source for the classification question. In Section 12, the authors state as an open expected classification that a bounded solution in $\mathbb{R}^3$ with finite Morse index and $\nabla u \neq 0$ outside a bounded set should have finitely many planar/catenoidal ends, and separately that if the Morse index is one then the solution should be axially symmetric with two catenoidal ends. The paper constructs Allen-Cahn solutions from nondegenerate finite-total-curvature minimal surfaces and proves Morse-index correspondence for those constructed examples, but it does not classify arbitrary Morse-index-one solutions.

2. Juncheng Wei, “Geometric approaches to semilinear elliptic equations”, in *Proceedings of the International Congress of Mathematicians, Seoul 2014*, Vol. III, Kyung Moon SA, 2014, pp. 941-964. URL: <https://www.mathunion.org/icm/proceedings>.

This survey/source context records the geometric Allen-Cahn program and points back to the finite-index and axial-symmetry classification problems of del Pino-Kowalczyk-Wei. It is not a resolving theorem.

3. Changfeng Gui, Yong Liu, Juncheng Wei, “Two-end solutions to the Allen-Cahn equation in $\mathbb{R}^3$ ”, *Advances in Mathematics* 320 (2017), 926-992. DOI: 10.1016/j.aim.2017.09.013. arXiv:1502.05963, URL: <https://arxiv.org/abs/1502.05963>.

This paper constructs and studies a complete family of axially symmetric two-end solutions with logarithmic growth rates. It is important progress toward the catenoid-type picture, but the solutions are axially symmetric by construction/definition in the analysis. It does not prove that an arbitrary Morse-index-one solution satisfying the problem statement hypotheses is axially symmetric.

4. Kelei Wang, Juncheng Wei, “Finite Morse index implies finite ends”, *Communications on Pure and Applied Mathematics* 72 (2019), 1044-1119. DOI: 10.1002/cpa.21812. arXiv:1705.06831, URL: <https://arxiv.org/abs/1705.06831>.

This proves finite Morse index implies finitely many ends and linear energy growth for Allen-Cahn solutions in $\mathbb{R}^2$, and in particular that Morse-index-one solutions in $\mathbb{R}^2$ have four ends. It is strong lower-dimensional progress, but it does not apply to the stated $\mathbb{R}^3$ problem.

5. Changfeng Gui, Kelei Wang, Juncheng Wei, “Axially symmetric solutions of Allen-Cahn equation with finite Morse index”, *Transactions of the American Mathematical Society* 373 (2020), 3649-3668. DOI: 10.1090/tran/8035. arXiv:1901.11263, URL: <https://arxiv.org/abs/1901.11263>.

The main result says that axially symmetric finite-index solutions in dimension three have finitely many ends, and that an axially symmetric solution with Morse index one has exactly two ends. This is close but conditional on the desired symmetry. Since axial symmetry is a hypothesis rather than a conclusion, the theorem does not resolve the problem statement.

6. Weizhao Liang, Jianmin Yang, “Qualitative properties of two-end solutions to the Allen-Cahn equation in $\mathbb{R}^3$ ”, *Advanced Nonlinear Studies* 25 (2025), no. 3, 871–901. DOI: 10.1515/ans-2023-0186. URL: <https://doi.org/10.1515/ans-2023-0186>.

This paper proves axial symmetry and monotonicity for two-end solutions whose nodal set has prescribed logarithmic asymptotics $z = k_i \log |x'| + c_i$ and whose rates satisfy $k_1 - k_2 > 2\sqrt{2}$, plus nonexistence in another rate range. This is genuine partial progress toward axial symmetry of two-end solutions, but the statement does not assume two-end asymptotics or the rate gap, and current literature does not derive those hypotheses from $m(u) = 1$ and $\nabla u \neq 0$ outside a compact set.

7. Enric Florit-Simon, “Phase transitions with bounded index: Parallels to De Giorgi’s conjecture”, arXiv preprint, submitted 2026. arXiv:2602.03136, DOI: 10.48550/arXiv.2602.03136, URL: <https://arxiv.org/abs/2602.03136>.

Theorem 1.4 proves a strong three-dimensional structure theorem for finite-index Allen-Cahn solutions with bounded energy density/quadratic energy growth: such solutions have finitely many parallel ends, planar or catenoidal. This confirms a substantial part of the del Pino-Kowalczyk-Wei finite-end picture under an extra energy-growth hypothesis. However, the paper explicitly says it does not address the related question of proving that solutions in $\mathbb{R}^3$ with two regular ends, or with index one, are axially symmetric. Thus it does not resolve the problem statement.

7.8.3.4. What remains unresolved

The missing step is a theorem that starts only from the stated hypotheses—bounded entire solution in $\mathbb{R}^3$, Morse index one, and $\nabla u \neq 0$ outside a compact set—and proves axial symmetry. Existing results either assume the axial symmetry, assume the two-end asymptotic model plus extra rate restrictions, work in $\mathbb{R}^2$, or add bounded energy density while still leaving the index-one axial-symmetry question open. Therefore the exact problem is not fully resolved.

7.8.4. Remaining open problems recorded by the database

7.8.4.1. Open problem 1

Prove that every bounded entire two-ended solution of the Allen–Cahn equation

$$\Delta u + u - u^3 = 0 \quad \text{in } \mathbb{R}^3$$

is axially symmetric.

Reference supplied by the database. Changfeng Gui, Kelei Wang, and Juncheng Wei, “Axially symmetric solutions of the Allen–Cahn equation with finite Morse index,” *Transactions of the American Mathematical Society* 373(5) (2020), 3649–3668. Explicit open conjecture stated in the introduction following Theorem 1.6 (“all two-ended solutions ... must be axially symmetric”). DOI: <https://doi.org/10.1090/tran/8035>. Preprint: [arXiv:1901.05186](https://arxiv.org/abs/1901.05186).

7.8.5. Primary-source pointer

The originating ICM record is *Geometric approaches to semilinear elliptic equations* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.3/ICM2014.3.pdf#page=957>.

7.9. 235. Ricci flows with unbounded curvature [2014-geom-133]

Record: 2014-geom-133

Subfield: Geometric flows and geometric PDE

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: Ricci flows with unbounded curvature

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=1051>

7.9.1. Problem statement

Let (M^3, g_0) be a smooth, complete three-dimensional Riemannian manifold with non-negative Ricci curvature, $\text{Ric}_{g_0} \geq 0$, with no assumption that the full curvature tensor of g_0 is bounded.

Prove that there exist a time $T > 0$ and a smooth Ricci flow $g(t)$ on M , defined for $t \in [0, T)$, such that

$$\frac{\partial}{\partial t} g(t) = -2 \text{Ric}_{g(t)}, \quad g(0) = g_0.$$

Moreover, prove that this flow preserves nonnegative Ricci curvature, $\text{Ric}_{g(t)} \geq 0$ for all $t \in [0, T)$, and is unique among smooth Ricci flows starting from g_0 and satisfying these properties.

7.9.2. Historical background, literature, and current progress

7.9.3. Verdict: partially resolved

7.9.3.1. Precise problem checked

The problem asks whether every smooth complete three-dimensional Riemannian manifold (M^3, g_0) with $\text{Ric}_{g_0} \geq 0$, allowing unbounded full curvature tensor, admits a short-time smooth Ricci flow $g(t)$ with $g(0) = g_0$, preserving nonnegative Ricci curvature, and unique among smooth flows with these properties. The surrounding literature usually states the source conjecture in the stronger form that the resulting flow should also be complete for positive times.

7.9.3.2. Current progress

1. Peter M. Topping, "Ricci flows with unbounded curvature", Proceedings of the International Congress of Mathematicians, Seoul 2014, Vol. II, 1033-1048, Kyung Moon Sa, 2014. arXiv:1408.6866; DOI: 10.48550/arXiv.1408.6866. This is the original survey/source context. It explains why unbounded curvature creates nonuniqueness and completeness issues, and formulates the relevant program of starting Ricci flow without bounded curvature assumptions.

2. Yi Lai, “Producing 3D Ricci flows with non-negative Ricci curvature via singular Ricci flows”, *Geometry & Topology* 25 (2021), no. 7, 3629–3690. DOI: 10.2140/gt.2021.25.3629; arXiv:2004.05291. Theorem 1.3 proves that if (M, g) is a complete 3-dimensional Riemannian manifold with $Ric \geq 0$, then there is $T > 0$ and a smooth Ricci flow $(M, g(t))$ on $[0, T)$ with $g(0) = g$ and $Ric(g(t)) \geq 0$. The paper explicitly says that completeness of this flow is not guaranteed. It also does not prove the uniqueness asserted in the problem statement.
3. Bing-Long Chen, Guoyi Xu, and Zhuhong Zhang, “Local pinching estimates in 3-dim Ricci flow”, *Mathematical Research Letters* 20 (2013), no. 5, 845–855. DOI: 10.4310/MRL.2013.v20.n5.a3; arXiv:1206.1814. This proves preservation results for curvature cones in complete 3-dimensional Ricci flows without assuming bounded curvature, including nonnegative Ricci curvature. It supports the preservation part once an appropriate complete flow exists, but it is not an existence or uniqueness theorem for the full initial-data problem.
4. Raphael Hochard, “Short-time existence of the Ricci flow on complete, non-collapsed 3-manifolds with Ricci curvature bounded from below”, preprint, 2016. arXiv:1603.08726. This gives short-time complete existence under a Ricci lower bound plus a uniform positive lower bound on volumes of unit balls. Combined with preservation results, it covers a noncollapsed special case, but the statement allows collapse at infinity.
5. Esther Cabezas-Rivas and Burkhard Wilking, “How to produce a Ricci flow via Cheeger-Gromoll exhaustion”, *Journal of the European Mathematical Society* 17 (2015), no. 12, 3153–3194. DOI: 10.4171/JEMS/578. This constructs complete Ricci flows under non-negative complex sectional curvature. In dimension three this is nonnegative sectional curvature, which is strictly stronger than nonnegative Ricci curvature.
6. Andrew D. McLeod and Peter M. Topping, “Pyramid Ricci flow in higher dimensions”, *Mathematische Zeitschrift* 296 (2020), 511–523. DOI: 10.1007/s00209-020-02472-1. Conjecture 1.1 states the WPIC1 version; in dimension three this is the complete nonnegative-Ricci Ricci-flow existence problem. The paper constructs pyramid/local-in-time-space flows, not a single complete flow on $M \times [0, T)$.
7. Albert Chau and Adam Martens, “Pseudolocality and completeness for nonnegative Ricci curvature limits of 3D singular Ricci flows”, *Mathematische Zeitschrift* 307 (2024), no. 4, Paper No. 63. DOI: 10.1007/s00209-024-03524-6; arXiv:2307.08088. Theorem 1.1 proves completeness of Lai’s flow provided the initial metric satisfies a volume-ratio lower bound tending to zero at infinity; Theorem 1.2 also covers compactly supported perturbations of complete nonnegative-sectional-curvature metrics. These are significant partial results but add hypotheses not present in the problem statement.
8. Man-Chun Lee and Peter M. Topping, “Three-manifolds with non-negatively pinched Ricci curvature”, preprint, 2022; arXiv:2204.00504. It proves complete Ricci-flow existence for complete noncompact 3-manifolds with the stronger pinching condition $Ric \geq \epsilon Rg \geq 0$, and is used in work on Hamilton’s pinching conjecture. This does not cover arbitrary nonnegative Ricci curvature.
9. Man-Chun Lee, “Uniqueness of Ricci flow with scaling invariant estimates”, preprint, submitted 2025, revised 2026; arXiv:2503.20292; DOI: 10.48550/arXiv.2503.20292. Theorem 1.1 proves uniqueness for complete noncompact Ricci flows satisfying $|Rm| \leq \alpha/t$; Corollary 1.1 gives uniqueness in a uniformly noncollapsed 3D $Ric \geq 0$ setting. This does not prove uniqueness for arbitrary smooth flows from arbitrary complete $Ric \geq 0$ initial data, nor for Lai’s general flow without an a priori completeness/curvature-decay hypothesis.

7.9.3.3. What remains unresolved

No published or otherwise citable result found covers the full problem exactly. Lai resolves the general existence and Ricci-preservation part as stated without positive-time completeness, but not uniqueness. The stronger source conjecture asking for a complete Ricci flow remains open in full generality: Chau-Martens, Hochard, Cabezas-Rivas-Wilking, and Lee-Topping cover important cases with extra volume, noncollapsing, sectional/complex-sectional curvature, or pinching assumptions. Existing uniqueness theorems also impose extra hypotheses such as completeness plus scaling-invariant curvature bounds or noncollapsing. Therefore the correct status is partially resolved.

7.9.4. Remaining open problems recorded by the database

7.9.4.1. Open problem 1

Let (M^3, g_0) be smooth and complete with $\text{Ric}_{g_0} \geq 0$, allowing unbounded curvature and collapse. Prove the short-time existence of a smooth complete Ricci flow $g(t)$ with $g(0) = g_0$, without requiring uniqueness.

Reference supplied by the database. Peter M. Topping, “Ricci flows with unbounded curvature,” in *Proceedings of the International Congress of Mathematicians—Seoul 2014*, Vol. II, Kyung Moon Sa, Seoul, 2014, 1033–1048, §4 (explicitly states that short-time existence from complete three-manifolds with nonnegative Ricci curvature and unbounded curvature remains open). [arXiv:1408.6866](https://arxiv.org/abs/1408.6866).

7.9.4.2. Open problem 2

Let (M^3, g_0) be a smooth complete, possibly collapsed manifold with nonnegative sectional curvature and no upper curvature bound. Prove that the Cabezas-Rivas–Wilking Ricci flow from g_0 is unique among smooth complete Ricci flows preserving nonnegative sectional curvature and having initial metric g_0 .

Reference supplied by the database. Peter M. Topping, “Ricci flows with unbounded curvature,” in *Proceedings of the International Congress of Mathematicians—Seoul 2014*, Vol. II, Kyung Moon Sa, Seoul, 2014, 1033–1048, Question following Theorem 4.1 (explicitly asks for uniqueness of the Cabezas-Rivas–Wilking solution), [arXiv:1408.6866](https://arxiv.org/abs/1408.6866). Man-Chun Lee, “Uniqueness of Ricci flow with scaling invariant estimates,” preprint (2025), proves the three-dimensional case under uniform noncollapse, leaving the collapsed case above open, [arXiv:2503.20292](https://arxiv.org/abs/2503.20292).

7.9.5. Primary-source pointer

The originating ICM record is *Ricci flows with unbounded curvature* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=1051>.

7.10. 236. Ricci flows with unbounded curvature [2014-geom-134]

Record: 2014-geom-134

Subfield: Geometric flows and geometric PDE

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: Ricci flows with unbounded curvature

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=1051>

7.10.1. Problem statement

Let (M, g_0) be a complete Riemannian manifold with nonnegative complex sectional curvature. Suppose that $g(t)$ is the short-time Ricci flow on $[0, \varepsilon)$ constructed by Cabezas-Rivas and Wilking, with $g(0) = g_0$ and with nonnegative complex sectional curvature for every $t \in [0, \varepsilon)$. Determine whether this Ricci flow is unique: that is, whether every Ricci flow $\tilde{g}(t)$ with $\tilde{g}(0) = g_0$ agrees with $g(t)$ on their common time interval.

7.10.2. Historical background, literature, and current progress

7.10.3. Verdict: partially resolved

7.10.3.1. Precise problem

The problem asks whether the short-time Ricci flow constructed by Cabezas-Rivas and Wilking from any complete Riemannian manifold with nonnegative complex sectional curvature is unique. Equivalently: if another Ricci flow starts from the same initial metric, must it agree with the Cabezas-Rivas–Wilking flow on the common time interval? The question is meant in the unbounded-curvature, complete noncompact setting and without adding a global curvature bound such as $|Rm| \leq C/t$ to every competing flow.

7.10.3.2. Literature status

1. Peter M. Topping, “Ricci flows with unbounded curvature”, *Proceedings of the International Congress of Mathematicians, Seoul 2014*, pp. 1033–1048, arXiv:1408.6866, DOI 10.48550/arXiv.1408.6866. Topping states this exact question after quoting the Cabezas-Rivas–Wilking existence theorem: proving uniqueness of the solution is an open question, and a priori there could be many branches with the same initial metric. This is the source formulation of the problem, not a resolution.
2. Esther Cabezas-Rivas and Burkhard Wilking, “How to produce a Ricci flow via Cheeger-Gromoll exhaustion”, *Journal of the European Mathematical Society* 17 (2015), no. 12, 3153–3194, DOI 10.4171/JEMS/582, arXiv:1107.0606. Theorem 1.1 proves short-time existence for complete open manifolds with nonnegative complex sectional curvature, with no upper curvature bound. Corollary 1.3 gives a $c(n, v_0)/t$ curvature bound under a uniform lower volume bound on unit balls. Theorem 1.4

constructs complete flows with nonnegative/positive curvature conditions whose curvature is unbounded for all times or becomes unbounded later. This paper supplies the flow and shows why bounded-curvature uniqueness is not enough, but it does not prove uniqueness of the constructed flow in full generality.

3. **Bing-Long Chen and Xi-Ping Zhu**, “Uniqueness of the Ricci flow on complete noncompact manifolds”, *Journal of Differential Geometry* 74 (2006), no. 1, 119-154, DOI 10.4310/jdg/1175266184, arXiv:math/0505447. This proves uniqueness for complete noncompact Ricci flows with bounded curvature. It applies to the classical Hamilton-Shi setting and to any positive-time interval on which the relevant flows have uniformly bounded curvature, but it does not cover the unbounded-curvature Cabezas-Rivas–Wilking cases.
4. **Brett Kotschwar**, “An energy approach to the problem of uniqueness for the Ricci flow”, *Communications in Analysis and Geometry* 22 (2014), no. 1, 149-176, DOI 10.4310/CAG.2014.v22.n1.a3, arXiv:1206.3225. Kotschwar gives energy-method uniqueness results, including extensions beyond uniformly bounded curvature under additional growth/integrability hypotheses. These results are important partial progress but do not give unconditional uniqueness for every complete flow from an arbitrary complete initial metric with nonnegative complex sectional curvature.
5. **Li Sheng and Xiaojie Wang**, “On Uniqueness of Complete Ricci Flow Solution with Curvature Bounded from Below”, preprint, 2013, arXiv:1310.1611, DOI 10.48550/arXiv.1310.1611. The abstract states uniqueness for complete noncompact noncollapsing initial data whose complex sectional curvature is bounded below and scalar curvature is bounded above, in the class of complete metrics with complex sectional curvature bounded below. This has extra hypotheses: noncollapsing, scalar curvature upper bound, and a restricted competitor class. It therefore does not settle the all-cases question.
6. **Man-Chun Lee**, “On the uniqueness of Ricci flow”, preprint, revised 2018, arXiv:1706.06848, DOI 10.48550/arXiv.1706.06848. This studies complete noncompact flows with curvature bounded above by C/t and proves uniqueness under further hypotheses such as polynomial growth of the initial curvature and a smallness condition on the flow’s Ricci curvature. It is partial progress only.
7. **Man-Chun Lee**, “Uniqueness of Ricci flow with scaling invariant estimates”, preprint, arXiv:2503.20292v2, 2026, DOI 10.48550/arXiv.2503.20292. Theorem 1.1 proves that if two complete noncompact Ricci flows start from the same complete metric and both satisfy $|Rm(g(t))| + |Rm(\tilde{g}(t))| \leq \alpha/t$ on $(0, T]$, then they are equal. Corollary 1.2 gives a strong uniqueness consequence in dimension three for uniformly noncollapsed initial manifolds with $Ric \geq 0$. This is the closest later result found. It still assumes the scaling-invariant curvature bound for both flows, or derives it only in special settings such as the three-dimensional uniformly noncollapsed case. Cabezas-Rivas–Wilking explicitly exhibit higher-dimensional/collapsed phenomena with unbounded curvature not controlled by a global C/t bound, so this theorem does not imply uniqueness for all initial metrics in the problem statement.
8. **Richard H. Bamler, Esther Cabezas-Rivas and Burkhard Wilking**, “The Ricci flow under almost non-negative curvature conditions”, *Inventiones Mathematicae* 217 (2019), 95-126, DOI 10.1007/s00222-019-00864-7, arXiv:1707.03002. This broadens existence and smoothing results under almost nonnegative curvature hypotheses, including short-time existence without upper curvature bounds in related settings. It does not provide a full uniqueness theorem for the Cabezas-Rivas–Wilking flow.

7.10.3.3. Conclusion

No published or citable result found proves uniqueness for the full Cabezas-Rivas–Willing short-time Ricci flow from every complete initial metric with nonnegative complex sectional curvature against all competing Ricci flows from the same metric. The literature resolves several special classes, especially bounded-curvature flows, flows with $|Rm| \leq C/t$, and some noncollapsed low-dimensional settings. The collapsed/unbounded-curvature general case remains unresolved.

7.10.4. Remaining open problems recorded by the database

7.10.4.1. Open problem 1

Prove uniqueness of the Cabezas-Rivas–Willing Ricci flow when the initial manifold is a complete, possibly collapsed three-manifold of nonnegative sectional curvature and unbounded curvature.

Reference supplied by the database. Peter M. Topping, “Ricci flows with unbounded curvature,” in *Proceedings of the International Congress of Mathematicians—Seoul 2014*, Vol. II, Kyung Moon Sa, Seoul, 2014, 1033–1048, explicit open question following Theorem 4.1. [arXiv:1408.6866](https://arxiv.org/abs/1408.6866). Man-Chun Lee, “Uniqueness of Ricci flow with scaling invariant estimates,” preprint (2025), Theorem 1.1 and Corollary 1.1, establishes uniqueness under a C/t curvature bound and, in dimension three, under uniform noncollapse; it does not cover the collapsed case stated above. [arXiv:2503.20292](https://arxiv.org/abs/2503.20292).

7.10.5. Primary-source pointer

The originating ICM record is *Ricci flows with unbounded curvature* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.2/ICM2014.2.pdf#page=1051>.

7.11. 237. Geometric approaches to semilinear elliptic equations [2014-ag-153]

Record: 2014-ag-153

Subfield: Geometric flows and geometric PDE

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: Geometric approaches to semilinear elliptic equations

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.3/ICM2014.3.pdf#page=957>

7.11.1. Problem statement

Let $w : \mathbb{R} \rightarrow (-1, 1)$ be the one-dimensional Allen-Cahn heteroclinic profile, defined by

$$w'' + w - w^3 = 0, \quad w(0) = 0, \quad \lim_{\zeta \rightarrow \pm\infty} w(\zeta) = \pm 1,$$

equivalently $w(\zeta) = \tanh(\zeta/\sqrt{2})$. In $\mathbb{R}^3$, write $r = \sqrt{x_1^2 + x_2^2}$ and $z = x_3$, and consider bounded axially symmetric solutions $u = u(r, z)$ of the Allen-Cahn equation

$$u_{zz} + u_{rr} + r^{-1}u_r + u - u^3 = 0, \quad r \in [0, \infty), \quad z \in \mathbb{R},$$

which are regular on the axis and satisfy

$$u(r, z) = u(r, -z), \quad u_r(0, z) = 0, \quad u_z > 0 \text{ for } z > 0, \quad u_r < 0 \text{ for } r > 0.$$

Prove or disprove that for each $\kappa \in (\sqrt{2}, \infty)$, there is a unique solution u_κ in this class for which there exists a constant c_κ , depending on κ , such that in the region $z > 0$,

$$u_\kappa(r, z) = w(z - \kappa \ln r + c_\kappa) + o(1) \quad \text{as } r \rightarrow \infty,$$

and that for $\kappa \in (0, \sqrt{2}]$, no solution in this class has such an asymptotic behavior.

7.11.2. Historical background, literature, and current progress

7.11.3. Verdict: partially resolved

7.11.3.1. Precise problem checked

The problem asks for the full classification of bounded axially symmetric Allen-Cahn solutions in $\mathbb{R}^3$, even in z , regular on the axis, satisfying $u_z > 0$ for $z > 0$ and $u_r < 0$ for $r > 0$, whose upper interface has logarithmic asymptotic

$$u(r, z) = \tanh\left(\frac{z - \kappa \log r + c_\kappa}{\sqrt{2}}\right) + o(1) \quad (r \rightarrow \infty, z > 0).$$

The asserted full answer is: for every $\kappa > \sqrt{2}$, exactly one such solution exists; for $0 < \kappa \leq \sqrt{2}$, no such solution exists.

7.11.3.2. Literature found

1. Two-end solutions to the Allen-Cahn equation in $\mathbb{R}^3$

Authors: Changfeng Gui, Yong Liu, Juncheng Wei.

Venue/year: *Advances in Mathematics* 320 (2017), 926-992.

DOI/arXiv/URL: DOI <https://doi.org/10.1016/j.aim.2017.09.013>;

arXiv 1502.05963, <https://arxiv.org/abs/1502.05963>.

Relevant result: The paper defines the same axial two-end growth rate by

$$\|u(r, \cdot) - H(\cdot - k \log r + c)\|_{L^\infty(0, \infty)} \rightarrow 0$$

with $H(t) = \tanh(t/\sqrt{2})$. Its Theorem 1 proves existence for every $k \in (\sqrt{2}, \infty)$. It also proves monotonicity of two-end solutions with $k > \sqrt{2}$, and Theorem 2 proves nonexistence only for $k \in (0, \sqrt{2}/2]$. The introduction explicitly says that uniqueness for $k > \sqrt{2}$ and nonexistence for $k \in (\sqrt{2}/2, \sqrt{2}]$ are expected, not proved there.

2. Qualitative properties of two-end solutions to the Allen-Cahn equation in $\mathbb{R}^3$

Authors: Weizhao Liang, Jianmin Yang.

Venue/year: *Advanced Nonlinear Studies* 25(3) (2025), 871-901.

DOI/URL: DOI <https://doi.org/10.1515/ans-2023-0186>, <https://doi.org/10.1515/ans-2023-0186>.

Relevant result: The paper treats general two-end solutions with ends $z = k_i \log |x'| + c_i$. It proves axial symmetry and monotonicity when $k_1 - k_2 > 2\sqrt{2}$. In the symmetric situation of this problem, $k_1 = \kappa$, $k_2 = -\kappa$, so this condition is exactly $\kappa > \sqrt{2}$. It also proves nonexistence for $-\sqrt{2}/2 < k_2 < k_1 < \sqrt{2}/2$ and for $k_1 = -k_2 = \sqrt{2}/2$, which in the symmetric case covers $\kappa \leq \sqrt{2}/2$, but not $\sqrt{2}/2 < \kappa \leq \sqrt{2}$. It does not prove uniqueness for fixed $\kappa > \sqrt{2}$.

3. Solutions with multiple catenoidal ends to the Allen-Cahn equation in $\mathbb{R}^3$

Authors: Oscar Agudelo, Manuel del Pino, Juncheng Wei.

Venue/year: *Journal de Mathématiques Pures et Appliquées* 103(1) (2015), 142-218.

DOI: <https://doi.org/10.1016/j.matpur.2014.03.010>.

Relevance: This is earlier Lyapunov-Schmidt progress constructing axially symmetric Allen-Cahn solutions with catenoidal/logarithmic ends in special regimes. It is cited by Gui-Liu-Wei as partial prior progress and does not by itself give the full one-parameter classification or the missing nonexistence range.

4. Entire solutions of the Allen-Cahn equation and complete embedded minimal surfaces of finite total curvature in $\mathbb{R}^3$

Authors: Manuel del Pino, Michał Kowalczyk, Juncheng Wei.

Venue/year: *Journal of Differential Geometry* 93(1) (2013), 67-131.

URL: <https://projecteuclid.org/journals/journal-of-differential-geometry/volume-93/issue-1/Entire-solutions-of-the-Allen-Cahn-equation-and-complete-embedded-minimal/10.4310/jdg/1361800878.full>.

Relevance: Earlier gluing construction connecting Allen-Cahn interfaces with large dilations of finite-total-curvature minimal surfaces, including catenoidal behavior. It supplies special large-growth examples, not the full threshold/uniqueness statement.

7.11.3.3. What remains unresolved

the exact statement is not fully resolved. The literature proves existence for all $\kappa > \sqrt{2}$, but I found no citable theorem proving uniqueness for each fixed $\kappa > \sqrt{2}$ in the stated class. The nonexistence side is also incomplete: current theorems rule out the symmetric small-rate range only up to $\kappa \leq \sqrt{2}/2$, while the interval

$$\sqrt{2}/2 < \kappa \leq \sqrt{2}$$

remains not covered by the cited nonexistence results. Therefore the status is **partially resolved**, not fully resolved.

7.11.4. Remaining open problems recorded by the database

7.11.4.1. Open problem 1

Prove that for every fixed growth rate $\kappa > \sqrt{2}$ there is, up to the normalizations imposed in the definition of an axially symmetric two-end solution, a unique two-end solution of the Allen–Cahn equation in $\mathbb{R}^3$ with asymptotic profile $H(z - \kappa \log r + c_\kappa)$.

Reference supplied by the database. Changfeng Gui, Yong Liu, and Juncheng Wei, “Two-end solutions to the Allen–Cahn equation in $\mathbb{R}^3$,” *Advances in Mathematics* **320** (2017), 926–992, unnumbered expected uniqueness statement immediately following Theorem 2. DOI: <https://doi.org/10.1016/j.aim.2017.09.013>; arXiv: 1502.05963.

7.11.4.2. Open problem 2

Prove that no axially symmetric two-end solution of the Allen–Cahn equation in $\mathbb{R}^3$ has growth rate

$$\frac{\sqrt{2}}{2} < \kappa \leq \sqrt{2}.$$

Reference supplied by the database. Changfeng Gui, Yong Liu, and Juncheng Wei, “Two-end solutions to the Allen–Cahn equation in $\mathbb{R}^3$,” *Advances in Mathematics* **320** (2017), 926–992, unnumbered expected nonexistence statement immediately following Theorem 2. DOI: <https://doi.org/10.1016/j.aim.2017.09.013>; arXiv: 1502.05963.

7.11.4.3. Open problem 3

Prove that every two-end solution of the Allen–Cahn equation in $\mathbb{R}^3$ has Morse index one.

Reference supplied by the database. Changfeng Gui, Yong Liu, and Juncheng Wei, “Two-end solutions to the Allen–Cahn equation in $\mathbb{R}^3$,” *Advances in Mathematics* **320** (2017), 926–992, unnumbered open expectation in the introduction following Theorem 2. DOI: <https://doi.org/10.1016/j.aim.2017.09.013>; arXiv: 1502.05963.

7.11.5. Primary-source pointer

The originating ICM record is *Geometric approaches to semilinear elliptic equations* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.3/ICM2014.3.pdf#page=957>.

7.12. 238. Liouville equations from a variational point of view [2014-geom-259]

Record: 2014-geom-259

Subfield: Geometric flows and geometric PDE

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: Liouville equations from a variational point of view

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.3/ICM2014.3.pdf#page=357>

7.12.1. Problem statement

Let (M, g) be a smooth closed four-dimensional Riemannian manifold. Define its Q -curvature and Paneitz operator by

$$Q_g = -\frac{1}{12} (\Delta_g R_g - R_g^2 + 3|\text{Ric}_g|^2)$$

and

$$P_g \varphi = \Delta_g^2 \varphi + \text{div}_g \left(\left(\frac{2}{3} R_g g - 2 \text{Ric}_g \right) d\varphi \right),$$

where R_g and Ric_g are the scalar and Ricci curvatures of g . Assume that $\ker P_g$ consists precisely of the constant functions and that the total Q -curvature satisfies

$$q_P := \int_M Q_g dV_g = 8k\pi^2$$

for some positive integer k . Determine whether there always exists a smooth function u on M , normalized by $\int_M e^{4u} dV_g = 1$, such that

$$P_g u + 2Q_g = 2q_P e^{4u} \quad \text{on } M.$$

Equivalently, determine whether every such conformal class contains a metric of constant Q -curvature.

7.12.2. Historical background, literature, and current progress

The problem is **unresolved**. The resonant values $q_P = 8k\pi^2$ are precisely the values at which bubbling can destroy compactness. The literature gives substantial sufficient conditions for existence, but no result found proves existence for every closed four-manifold satisfying only $\ker P_g = \mathbb{R}$, for every $k \geq 1$; nor does it provide a counterexample to that universal assertion.

- Zindine Djadli and Andrea Malchiodi, *Existence of conformal metrics with constant Q -curvature*, **Annals of Mathematics** 168 (2008), 813-858, <https://doi.org/10.4007/annals.2008.168.813>. Their main theorem proves existence when $\ker P_g = \mathbb{R}$ and $q_P \notin 8\pi^2 \mathbb{N}_{>0}$. Thus it identifies, rather than resolves, exactly the resonant cases in the present problem.

- Jiayu Li, Yuxiang Li, and Pan Liu, *The Q-curvature on a 4-dimensional Riemannian manifold (M, g) with $\int_M Q dV_g = 8\pi^2$* , **Advances in Mathematics** 231 (2012), 2194–2223, <https://doi.org/10.1016/j.aim.2012.06.002>, arXiv:math/0608543. For the first resonant level they assume that P_g is nonnegative with kernel the constants and prove existence under an additional explicit positive-mass-type inequality at a minimizing point of the blow-up energy. This is genuine progress for $k = 1$, but the extra positivity and mass hypotheses are not consequences of the assumptions in the problem.
- Mohameden Ould Ahmedou and Cheikh Birahim Ndiaye, *Morse theory and the resonant Q-curvature problem*, preprint (2014), arXiv:1409.7919. They treat all four-dimensional resonant levels $q_p = 8m\pi^2$, describe critical points at infinity, and derive Morse- and Poincaré-Hopf-type sufficient criteria for existence. Their theorems require nondegeneracy of an associated finite-dimensional Hamiltonian and additional index conditions, so they do not give unconditional existence.
- Cheikh Birahim Ndiaye, *Topological methods for the resonant Q-curvature problem in arbitrary even dimension*, **Journal of Geometry and Physics** 140 (2019), 178–213, <https://doi.org/10.1016/j.geomphys.2019.02.007>. This develops resonant Morse theory and algebraic-topological existence criteria in every even dimension. In dimension four it strengthens the conditional resonant theory, but still does not remove the Hamiltonian nondegeneracy/topological hypotheses needed to conclude existence.
- Cheikh Birahim Ndiaye, *Leray-Schauder degree for the resonant Q-curvature problem in even dimensions*, preprint (2022), arXiv:2206.12964. The paper proves an exact bubbling-rate formula, conditional compactness, and degree/existence results under nondegeneracy, Morse, and sign or positive-mass-type assumptions. These results do not exhaust metrics for which the associated mass quantity vanishes or has mixed sign.
- Yuchen Bi and Jiayu Li, *The prescribed Q-curvature flow for arbitrary even dimension in a critical case*, **Advances in Mathematics** 438 (2024), Article 109478, <https://doi.org/10.1016/j.aim.2023.109478>, arXiv:2106.03137. At the first critical level, and with the GJMS/Paneitz operator weakly positive, they prove flow convergence under additional geometric inequalities and recover the Li-Li-Liu existence theorem. This neither treats general $k \geq 2$ nor arbitrary negative spectrum.

Consequently, established progress covers important special and generic cases, including all resonant levels under suitable finite-dimensional nondegeneracy and index assumptions. What remains open is the full unconditional existence question stated here, especially degenerate bubbling configurations and conformal classes with indefinite Paneitz operator.

7.12.3. Remaining open problems recorded by the database

7.12.3.1. Open problem 1

Let (M, g) be a smooth closed locally conformally flat four-dimensional Riemannian manifold with

$$\int_M Q_g dV_g = 8\pi^2,$$

and assume that its Paneitz operator P_g is nonnegative and has kernel consisting exactly of the constant functions. For each $p \in M$, let G_p be the normalized Green function

determined by

$$P_g G_p + 2Q_g = 16\pi^2 \delta_p, \quad \int_M G_p dV_g = 0,$$

and write

$$G_p(x) = -2 \log |x| + S_0(p) + O(|x|)$$

in locally conformally flat coordinates centered at p . Let p' minimize

$$-16\pi^2 S_0(p) + 2 \int_M Q_g G_p dV_g$$

over $p \in M$. Choose locally conformally flat coordinates $x = (x^1, \dots, x^4)$ centered at p' in which

$$g = e^{2f(x)} \sum_{i=1}^4 (dx^i)^2, \quad f(x) = \sum_i c_i x^i + \frac{1}{2} \sum_{i,j} c_{ij} x^i x^j + O(|x|^3),$$

and

$$G_{p'}(x) = -2 \log |x| + S_0(p') + \sum_i a_i x^i + \frac{1}{2} \sum_{i,j} a_{ij} x^i x^j + O(|x|^3).$$

Prove that

$$\sum_{i=1}^4 \left(\frac{a_{ii} + c_{ii}}{2} + 2(a_i + c_i)^2 \right) \geq 0,$$

with equality if and only if (M, g) is conformally equivalent to the standard round four-sphere.

Reference supplied by the database. Jiayu Li, Yuxiang Li, and Pan Liu, "The Q-curvature on a 4-dimensional Riemannian manifold (M, g) with $\int_M Q dV_g = 8\pi^2$," *Advances in Mathematics* 231 (2012), 2194-2223, Conjecture at the end of Section 5, <https://doi.org/10.1016/j.aim.2012.06.002>; arXiv:math/0608543.

7.12.4. Primary-source pointer

The originating ICM record is *Liouville equations from a variational point of view* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.3/ICM2014.3.pdf#page=357>.

7.13. 239. On large time behavior of growth by birth and spread [2018-geom-262]

Record: 2018-geom-262

Subfield: Geometric flows and geometric PDE

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: On large time behavior of growth by birth and spread

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol3-ver1-eb.pdf#page=916>

7.13.1. Problem statement

Let $N \geq 1$, let $r : \mathbb{R}^N \rightarrow [0, \infty)$ be Lipschitz continuous and compactly supported, and let $u_0 \in \text{BUC}(\mathbb{R}^N)$. Let u be the viscosity solution of

$$u_t - \left(\operatorname{div} \frac{\nabla u}{|\nabla u|} + 1 \right) |\nabla u| = r(x) \quad \text{in } \mathbb{R}^N \times (0, \infty), \quad u(\cdot, 0) = u_0,$$

where the level-set mean-curvature operator is interpreted in the viscosity sense at points where $\nabla u = 0$. It is known that the locally uniform limit

$$R = \lim_{t \rightarrow \infty} \frac{u(x, t)}{t}$$

exists, is constant in x , and is independent of u_0 .

Prove that there is a viscosity solution $w : \mathbb{R}^N \rightarrow \mathbb{R}$ of the stationary equation

$$R - \left(\operatorname{div} \frac{\nabla w}{|\nabla w|} + 1 \right) |\nabla w| = r(x) \quad \text{in } \mathbb{R}^N$$

such that

$$u(\cdot, t) - Rt \rightarrow w \quad \text{locally uniformly in } \mathbb{R}^N \quad \text{as } t \rightarrow \infty.$$

Characterize the uniqueness sets for this stationary equation, meaning the subsets $A \subseteq \mathbb{R}^N$ for which any two viscosity solutions that agree on A agree everywhere (up to any unavoidable additive constant), and determine how the limiting profile w depends on the initial datum u_0 .

7.13.2. Historical background, literature, and current progress

The problem is **unresolved**. The asymptotic speed is known to exist, and full convergence, uniqueness sets, and the dependence of the profile on the initial datum have been obtained in important special cases, but no result found covers every nonnegative compactly supported Lipschitz source r and every bounded uniformly continuous u_0 in the nonradial whole-space problem.

Yoshikazu Giga, Hiroyoshi Mitake, Takeshi Ohtsuka, and Hung V. Tran, "Existence of asymptotic speed of solutions to birth-and-spread type nonlinear partial differential equations," *In-*

diana University Mathematics Journal **70**(1) (2021), 121–156, <https://doi.org/10.1512/iumj.2021.70.8305>, proves the locally uniform existence of the large-time average $u(x, t)/t \rightarrow R$ for this forced level-set mean-curvature equation (as an application of its general framework). This establishes the constant R , not convergence of $u - Rt$.

Yoshikazu Giga, Hiroyoshi Mitake, and Hung V. Tran, “Remarks on large time behavior of level-set mean curvature flow equations with driving and source terms,” *Discrete and Continuous Dynamical Systems - B* **25**(10) (2020), 3983–3999, <https://doi.org/10.3934/dcdsb.2019228>, gives the strongest direct partial progress. Theorem 1.1 proves full convergence when both the C_c^1 source and the initial datum are radial; Theorem 1.3 gives an optimal-control representation of the limiting profile and identifies the radial set

$$\{\rho \geq n - 1 : \tilde{r}(\rho) = R\}$$

as a uniqueness set. Theorem 1.2 proves convergence for zero initial data in a restricted nonradial case where the maximum set of the source contains a ball of radius $n - 1$ and decomposes into smooth sets satisfying a curvature inequality; Proposition 3.2 makes that maximum set a uniqueness set within the growth class relevant to the limit. The paper explicitly leaves convergence unknown without those additional nonradial hypotheses. Consequently, these results neither prove full convergence for arbitrary r, u_0 , nor characterize all uniqueness sets, nor determine the general dependence on u_0 .

Adrian D. Calderon, “Large-time behavior for a cutoff level-set mean curvature G-equation with source term,” *Nonlinear Analysis* **268** (2026), Article 114085, <https://doi.org/10.1016/j.na.2026.114085>, proves convergence and uniqueness-set results in periodic and radial settings for a related equation containing the cutoff $(-\operatorname{div}(Du/|Du|) + 1)_+$ (and possibly a wind term). Because that cutoff changes the operator and the paper retains symmetry/periodicity assumptions, it does not resolve the equation stated here.

The original open problem was posed in Yoshikazu Giga, “On Large Time Behavior of Growth by Birth and Spread,” *Proceedings of the International Congress of Mathematicians 2018*, Vol. 3, World Scientific (2018), https://doi.org/10.1142/9789813272880_0139. The remaining gap is the unrestricted nonradial whole-space problem, including a general uniqueness-set theory and a formula or characterization of the profile’s dependence on arbitrary bounded uniformly continuous initial data.

7.13.3. Remaining open problems recorded by the database

7.13.3.1. Open problem 1

Let $n \geq 2$, let $f \in C_c^1(\mathbb{R}^n)$ be nonnegative and nonzero, set $c = \max_{\mathbb{R}^n} f$, and suppose that the maximum set $U = \{x \in \mathbb{R}^n : f(x) = c\}$ contains a closed ball of radius $n - 1$. For arbitrary $u_0 \in BUC(\mathbb{R}^n)$, let u be the viscosity solution of

$$u_t - \left(\operatorname{div} \frac{\nabla u}{|\nabla u|} + 1 \right) |\nabla u| = f(x) \quad \text{in } \mathbb{R}^n \times (0, \infty), \quad u(\cdot, 0) = u_0.$$

Prove that there exists a viscosity solution $v : \mathbb{R}^n \rightarrow \mathbb{R}$ of

$$- \left(\operatorname{div} \frac{\nabla v}{|\nabla v|} + 1 \right) |\nabla v| = f(x) - c \quad \text{in } \mathbb{R}^n$$

such that $u(\cdot, t) - ct \rightarrow v$ locally uniformly in $\mathbb{R}^n$ as $t \rightarrow \infty$, without imposing a curvature decomposition on U or restricting u_0 to be zero.

Reference supplied by the database. Yoshikazu Giga, Hiroyoshi Mitake, and Hung V. Tran, “Remarks on large time behavior of level-set mean curvature flow equations with driving and source terms,” *Discrete and Continuous Dynamical Systems - B* 25(10) (2020), 3983–3999, unnumbered open question in the Introduction following assumptions (A3)–(A5), <https://doi.org/10.3934/dcdsb.2019228>.

7.13.3.2. Open problem 2

Fix $a > 0$ and let

$$U_a = \{x \in \mathbb{R}^2 : \operatorname{dist}(x, [-a, a] \times \{0\}) \leq 1\}$$

be the closed planar stadium of radius 1. Classify all functions $v \in C(U_a)$ that are viscosity solutions of

$$- \left(\operatorname{div} \frac{\nabla v}{|\nabla v|} + 1 \right) |\nabla v| = 0 \quad \text{in } \operatorname{int}(U_a).$$

Reference supplied by the database. Yoshikazu Giga, Hiroyoshi Mitake, and Hung V. Tran, “Remarks on large time behavior of level-set mean curvature flow equations with driving and source terms,” *Discrete and Continuous Dynamical Systems - B* 25(10) (2020), 3983–3999, Section 4 and its concluding discussion, <https://doi.org/10.3934/dcdsb.2019228>.

7.13.4. Primary-source pointer

The originating ICM record is *On large time behavior of growth by birth and spread* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol3-ver1-eb.pdf#page=916>.

7.14. 240. On large time behavior of growth by birth and spread [2018-geom-263]

Record: 2018-geom-263

Subfield: Geometric flows and geometric PDE

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: On large time behavior of growth by birth and spread

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol3-ver1-eb.pdf#page=916>

7.14.1. Problem statement

Let $N \geq 1$, $a \geq 0$, and $\sigma \in \mathbb{R}$. Let $M : \mathbb{R}^N \rightarrow [0, \infty)$ be continuous and positively homogeneous of degree one, let $r : \mathbb{R}^N \rightarrow \mathbb{R}$ be continuous and compactly supported, and let $\gamma : \mathbb{R}^N \rightarrow [0, \infty)$ be a convex, positively homogeneous, piecewise-linear crystalline anisotropy satisfying $\gamma(p) > 0$ for $p \neq 0$. Consider the forced crystalline level-set equation

$$u_t - M(-\nabla u) (a \operatorname{div}(-z) + \sigma) = r(x), \quad z \in \partial\gamma(-\nabla u),$$

on $\mathbb{R}^N \times (0, \infty)$, where $\partial\gamma$ is the convex subdifferential and $\operatorname{div}(-z)$ is interpreted as crystalline anisotropic curvature through Cahn–Hoffman vector fields. Prove that the class of continuous crystalline viscosity solutions of this equation is closed under locally uniform convergence: if $(u_j)_{j \geq 1}$ is a sequence of such solutions and $u_j \rightarrow u$ locally uniformly on $\mathbb{R}^N \times (0, \infty)$, then u is also a crystalline viscosity solution of the same equation.

7.14.2. Historical background, literature, and current progress

The problem is **unresolved**. The exact closure-under-locally-uniform-limits assertion is known in the planar case, but no result was found that proves it for a fixed crystalline anisotropy in every dimension covered by the statement. The principal higher-dimensional papers prove comparison, well-posedness, and stability when *regularized anisotropies* converge to a crystalline anisotropy; that is not the requested stability of an arbitrary locally uniformly convergent sequence of solutions of one fixed crystalline equation.

- Mi-Ho Giga and Yoshikazu Giga, “Generalized Motion by Nonlocal Curvature in the Plane,” *Archive for Rational Mechanics and Analysis* 159 (2001), 295–333, DOI <https://doi.org/10.1007/s002050100154>. The paper establishes the planar crystalline viscosity theory, including the stability result that resolves the assertion for $N = 2$.
- Yoshikazu Giga and Norbert Požár, “A Level Set Crystalline Mean Curvature Flow of Surfaces,” *Advances in Differential Equations* 21 (2016), 631–698, [arXiv:1601.01802](https://arxiv.org/abs/1601.01802). It introduces the higher-dimensional faceted-test-function viscosity theory and proves comparison and stability under regularization, but not closure under locally uniform limits of solutions for the fixed crystalline operator.
- Yoshikazu Giga and Norbert Požár, “Approximation of General Facets by Regular Facets with Respect to Anisotropic Total Variation Energies and Its Application to the Crys-

talline Mean Curvature Flow,” *Communications on Pure and Applied Mathematics* 71 (2018), 1461–1491, DOI <https://doi.org/10.1002/cpa.21752>. It extends well-posedness to arbitrary dimension by approximating facets and supplies the test functions needed for comparison, but does not supply the limit-closure property asked here.

- Yoshikazu Giga, “On Large Time Behavior of Growth by Birth and Spread,” *Proceedings of the International Congress of Mathematicians 2018*, Vol. II (2018), 2305–2328, Problem 4, DOI https://doi.org/10.1142/9789813272880_0139. This is the proposing source; it states that only $N = 2$ was known and identifies instability of the higher-dimensional viscosity definition under the required limiting procedure as the gap.
- Yoshikazu Giga and Norbert Požár, “Viscosity Solutions for the Crystalline Mean Curvature Flow with a Nonuniform Driving Force Term,” *SN Partial Differential Equations and Applications* 1 (2020), Article 39, DOI <https://doi.org/10.1007/s42985-020-00040-0>. Theorems 4.1 and 4.4 prove relaxed-limit stability for solutions with smooth anisotropies converging to the crystalline one. They explicitly require that the operator not depend directly on x , so they do not prove the exact fixed-anisotropy statement with the additive source $r(x)$.
- Yoshikazu Giga and Norbert Požár, “Motion by Crystalline-Like Mean Curvature: A Survey,” *Bulletin of Mathematical Sciences* 12 (2022), Article 2230004, DOI <https://doi.org/10.1142/S1664360722300043>. Section 6.3 explains that the needed Perron/limit stability for crystalline viscosity solutions remains unclear in higher dimensions because crystalline curvature can jump when a facet bends or breaks; its Theorems 6.9–6.10 summarize only regularized-anisotropy stability.

Thus the established progress covers $N = 2$ and several approximation regimes, while the fixed-crystalline closure assertion in higher dimensions, and in particular for the source-driven equation stated here, remains open.

7.14.3. Remaining open problems recorded by the database

7.14.3.1. Open problem 1

Let $a \geq 0$ and $\sigma \in \mathbb{R}$. Let $M : \mathbb{R}^3 \rightarrow [0, \infty)$ be continuous and positively homogeneous of degree one, let $r : \mathbb{R}^3 \rightarrow \mathbb{R}$ be continuous and compactly supported, and let $\gamma : \mathbb{R}^3 \rightarrow [0, \infty)$ be a convex, positively homogeneous, piecewise-linear crystalline anisotropy satisfying $\gamma(p) > 0$ for $p \neq 0$. Consider

$$u_t - M(-\nabla u) (a \operatorname{div}(-z) + \sigma) = r(x), \quad z \in \partial\gamma(-\nabla u),$$

on $\mathbb{R}^3 \times (0, \infty)$, with $\operatorname{div}(-z)$ interpreted as crystalline anisotropic curvature through Cahn–Hoffman vector fields. Prove that if $(u_j)_{j \geq 1}$ is a sequence of continuous crystalline viscosity solutions and $u_j \rightarrow u$ locally uniformly on $\mathbb{R}^3 \times (0, \infty)$, then u is a crystalline viscosity solution of the same equation.

Reference supplied by the database. Yoshikazu Giga, “On Large Time Behavior of Growth by Birth and Spread,” *Proceedings of the International Congress of Mathematicians 2018*, Vol. II, 2018, pp. 2305–2328, Problem 4 and the paragraph following it. DOI: https://doi.org/10.1142/9789813272880_0139.

7.14.4. Primary-source pointer

The originating ICM record is *On large time behavior of growth by birth and spread* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol3-ver1-eb.pdf#page=916>.

7.15. 241. On large time behavior of growth by birth and spread [2018-geom-284]

Record: 2018-geom-284

Subfield: Geometric flows and geometric PDE

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: On large time behavior of growth by birth and spread

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol3-ver1-eb.pdf#page=916>

7.15.1. Problem statement

Let $N \geq 2$, $a \geq 0$, and $\sigma > 0$. Let $M : \mathbb{S}^{N-1} \rightarrow (0, \infty)$ be continuous, and extend it to a positively one-homogeneous function on $\mathbb{R}^N$ by

$$M(p) = |p|M\left(\frac{p}{|p|}\right) \quad \text{for } p \neq 0, \quad M(0) = 0.$$

Let $\gamma : \mathbb{R}^N \rightarrow [0, \infty)$ be convex, positively one-homogeneous, positive away from the origin, and crystalline, meaning that γ is piecewise linear. Let $r : \mathbb{R}^N \rightarrow [0, \infty)$ be Lipschitz continuous and compactly supported. Let $u : \mathbb{R}^N \times [0, \infty) \rightarrow \mathbb{R}$ be the unique continuous crystalline level-set viscosity solution of

$$u_t - M(-\nabla u)(aH_\gamma[u] + \sigma) = r(x) \quad \text{in } \mathbb{R}^N \times (0, \infty),$$

with initial condition $u(x, 0) = 0$ for every $x \in \mathbb{R}^N$. Here ∇u is the spatial gradient and $H_\gamma[u]$ is the crystalline anisotropic mean-curvature operator associated with the interfacial energy γ , interpreted in the crystalline level-set viscosity sense; when γ is differentiable away from the origin, this operator is

$$H_\gamma[u] = \operatorname{div}(-\nabla \gamma(-\nabla u)).$$

Prove that u is spatially Lipschitz continuous with a bound uniform in time: there exists a constant $K < \infty$, depending only on N, a, σ, M, γ , and r , such that

$$|u(x, t) - u(y, t)| \leq K|x - y|$$

for all $x, y \in \mathbb{R}^N$ and all $t > 0$.

7.15.2. Historical background, literature, and current progress

The exact problem remains **unresolved**. The original source explicitly poses it as Problem 5, after proving the corresponding time-uniform spatial Lipschitz estimate only for smooth anisotropies. I found no later proof or counterexample covering a crystalline γ in arbitrary dimension with the additive compactly supported source $r(x)$ and the full range of mobilities in the statement.

Yoshikazu Giga, “On Large Time Behavior of Growth by Birth and Spread,” *Proceedings of the International Congress of Mathematicians (ICM 2018)*, World Scientific, 2019, pp. 2287–2310, DOI: [10.1142/9789813272880_0139](https://doi.org/10.1142/9789813272880_0139). Lemma 3.4 proves the asserted uniform spatial Lipschitz bound when $\gamma \in C^2(\mathbb{R}^N \setminus \{0\})$; Problem 5 asks for its extension to crystalline γ . Thus the smooth-anisotropy case is established, but the hypothesis central to the present problem is excluded.

Yoshikazu Giga, Hiroyoshi Mitake, Takeshi Ohtsuka, and Hung V. Tran, “Existence of Asymptotic Speed of Solutions to Birth-and-Spread Type Nonlinear Partial Differential Equations,” *Indiana University Mathematics Journal* 70 (2021), 121–156, DOI: [10.1512/iumj.2021.70.8305](https://doi.org/10.1512/iumj.2021.70.8305), arXiv:1808.06312. Proposition 3.4 gives only the general estimate $|u(x + y, t) - u(x, t)| \leq \|Dr\|_\infty |y|t$, which grows with time. Its concrete second-order application is the smooth isotropic forced mean-curvature equation, not the crystalline equation in the statement, so it does not supply the required uniform bound.

Yoshikazu Giga and Norbert Pořár, “Viscosity Solutions for the Crystalline Mean Curvature Flow with a Nonuniform Driving Force Term,” *SN Partial Differential Equations and Applications* 1 (2020), Article 39, DOI: [10.1007/s42985-020-00040-0](https://doi.org/10.1007/s42985-020-00040-0), arXiv:2006.04375. Theorems 5.1 and 5.3 establish spatial Lipschitz continuity and approximation-based existence for crystalline **geometric** level-set equations whose spatially varying force occurs inside the normal-velocity law. Their estimate is a finite-time propagation estimate, and their equation has no additive birth term $r(x)$ of the form in the present problem. Consequently these results do not imply the claimed time-uniform estimate.

Antonin Chambolle, Massimiliano Morini, Matteo Novaga, and Marcello Ponsiglione, “Existence and Uniqueness for Anisotropic and Crystalline Mean Curvature Flows,” *Journal of the American Mathematical Society* 32 (2019), 779–824, DOI: [10.1090/jams/919](https://doi.org/10.1090/jams/919), arXiv:1702.03094. This proves well-posedness and approximation stability for anisotropic and crystalline geometric flows with spatially Lipschitz forcing and arbitrary convex mobilities. It advances the crystalline level-set theory needed around the problem, but does not treat the non-geometric additive source or prove the uniform spatial Lipschitz estimate requested here.

Thus the known literature supplies the estimate for smooth anisotropy, standard first-order subcases such as $a = 0$, and substantial well-posedness and finite-time regularity theory for related crystalline geometric flows. What remains open is precisely a spatial Lipschitz constant independent of t for the additive-source crystalline birth-and-spread equation under all hypotheses in the statement.

7.15.3. Remaining open problems recorded by the database

No separate descendant problem is listed in the website’s “Some remaining open problems” field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

7.15.4. Primary-source pointer

The originating ICM record is *On large time behavior of growth by birth and spread* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol3-ver1-eb.pdf#page=916>.

7.16. 242. Evolution of form and shape [2022-geom-266]

Record: 2022-geom-266

Subfield: Geometric flows and geometric PDE

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2022: Evolution of form and shape

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-560-5-volume-2.pdf#page=267>

7.16.1. Problem statement

Let $\{M_t\}_{t \geq 0}$ be a mean curvature flow starting from a smooth, closed, embedded hyper-surface in $\mathbb{R}^{n+1}$, with normal velocity $\partial_t x = -H\nu$. Define its space-time singular set by

$$\mathcal{S} = \{(x, t) \in \mathbb{R}^{n+1} \times (0, \infty) : \{M_t\} \text{ is not smooth in any space-time neighborhood of } (x, t)\}.$$

Assume that every point of $\mathcal{S}$ is cylindrical, meaning that at least one tangent flow there is a multiplicity-one shrinking round cylinder $\mathbb{S}^k \times \mathbb{R}^{n-k}$ for some $1 \leq k \leq n$. Prove that $\mathcal{S}$ has only finitely many connected components.

7.16.2. Historical background, literature, and current progress

Verdict: unresolved (as of 2026-08-03).

No citable paper or preprint was found that proves the assertion for every mean curvature flow in the statement. The main unresolved issue is to exclude infinitely many connected components in the lower-dimensional parts of the cylindrical singular strata; existing dimension, rectifiability, and local regularity results do not imply this global finiteness.

Key literature and partial progress.

- *The singular set of mean curvature flow with generic singularities*, Tobias Holck Colding and William P. Minicozzi II, **Inventiones Mathematicae** 204 (2016), 443-471, <https://doi.org/10.1007/s00222-015-0617-5>, arXiv:1405.5187. Theorem 0.1 places the space-time singular set in finitely many compact embedded Lipschitz submanifolds of dimension at most $n - 1$, together with a set of dimension at most $n - 2$; Theorem 0.3 gives additional time-slice structure. The paper states the exact finiteness assertion as Conjecture 0.8, and its lower-dimensional remainder can still have infinitely many components.
- *Regularity of cylindrical singular sets of mean curvature flow*, Ao Sun, Zhihan Wang, and Jinxin Xue, arXiv preprint (2025), [arXiv:2509.01707](https://arxiv.org/abs/2509.01707). Theorem 1.1 refines the cylindrical stratification: after a lower-dimensional exceptional part, each k -cylindrical stratum is locally contained in a k -dimensional $C^{2,\alpha}$ submanifold, while the remaining nondegenerate part has parabolic Hausdorff dimension at most $k - 1$ (and consists of isolated points when $k = 1$). These local and dimensional conclusions do not establish finitely many global components.

- *Stability and singular set of the two-convex level set flow*, Siao-Hao Guo, arXiv preprint (2024), [arXiv:2412.08977](https://arxiv.org/abs/2412.08977). Theorem 1.2 proves that a two-convex level-set flow has a singular set that is a finite disjoint union of points and compact C^1 curves **under the additional assumption that the flow has finitely many singular times**. Thus it proves a restricted conditional case, not the statement here.
- *Mean curvature flow through singularities for surfaces of rotation*, Steven Altschuler, Sigurd B. Angenent, and Yoshikazu Giga, **Journal of Geometric Analysis** 5 (1995), 293-358, <https://doi.org/10.1007/BF02921800>. For the rotationally symmetric graphical class studied there, the weak flow is regular except at finitely many isolated space-time singularities, giving a genuine special case only.

A 2026 Hubei University seminar announcement, *Finiteness of Connected Components of the Singular Set in Mean Curvature Flow with Cylindrical Singularities* (speaker: Lingzhong Zeng, 18 April 2026, [direct URL](#)), reports a claimed resolution of the exact conjecture. However, the announcement contains neither a proof nor a reference to a paper or preprint, and exhaustive title-, author-, phrase-, DOI-, and arXiv searches found no citable manuscript whose hypotheses and argument could be verified. It therefore does not supply a literature resolution under the stated standard.

7.16.3. Remaining open problems recorded by the database

7.16.3.1. Open problem 1

Let $\{M_t\}_{t \geq 0}$ be the level-set mean curvature flow starting from a smooth, closed, embedded, mean-convex hypersurface $M_0 \subset \mathbb{R}^{n+1}$, where mean convexity means that the mean-curvature vector of M_0 points weakly into the compact region bounded by M_0 . Prove that the space-time singular set

$$\mathcal{S} = \{(x, t) \in \mathbb{R}^{n+1} \times (0, \infty) : \{M_t\} \text{ is not smooth in any space-time neighborhood of } (x, t)\}$$

has only finitely many connected components.

Reference supplied by the database. Tobias Holck Colding, William P. Minicozzi II, and Erik Kjaer Pedersen, *Mean curvature flow*, Bulletin of the American Mathematical Society 52 (2015), no. 2, 297-333, Conjecture 6.5 and the following statement that its mean-convex special case is completely open. DOI: <https://doi.org/10.1090/S0273-0979-2015-01468-0>.

7.16.3.2. Open problem 2

Let $\{M_t\}_{t \geq 0}$ be the level-set mean curvature flow starting from a smooth, closed, embedded, mean-convex surface $M_0 \subset \mathbb{R}^3$. Prove that there are only finitely many singular times; that is, only finitely many $t > 0$ for which M_t is not a smooth surface.

Reference supplied by the database. Tobias Holck Colding and William P. Minicozzi II, *Differentiability of the arrival time*, Communications on Pure and Applied Mathematics 69 (2016),

no. 12, 2349-2363, Conjecture 1.2. DOI: <https://doi.org/10.1002/cpa.21635>.

7.16.3.3. Open problem 3

Let $\mathcal{M}$ be an n -dimensional unit-regular, cyclic mod 2, integral Brakke flow in $\mathbb{R}^{n+1}$, starting from a smooth, closed, embedded hypersurface, and assume that every singularity has multiplicity one and is either spherical or a neck pinch. Here a spherical singularity has the shrinking round sphere $\mathbb{S}^n(\sqrt{-2nt})$, $t < 0$, as its tangent flow, while a neck-pinch singularity has the shrinking round cylinder $\mathbb{S}^{n-1}(\sqrt{-2(n-1)t}) \times \mathbb{R}$, $t < 0$, as its tangent flow. Prove or disprove that no neck-pinch singularity is a space-time accumulation point of distinct spherical singularities.

Reference supplied by the database. Joshua M. Daniels-Holgate, *Approximation of mean curvature flow with generic singularities by smooth flows with surgery*, *Advances in Mathematics* 410 (2022), Part A, Article 108715, explicit open question in the introduction concerning accumulation of spherical singularities at a neck pinch. DOI: <https://doi.org/10.1016/j.aim.2022.108715>; arXiv:2104.11647.

7.16.4. Primary-source pointer

The originating ICM record is *Evolution of form and shape* (2022). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-560-5-volume-2.pdf#page=267>.

7.17. 243. Formation of singularities in nonlinear dispersive PDEs [2022-geom-331]

Record: 2022-geom-331

Subfield: Geometric flows and geometric PDE

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2022: Formation of singularities in nonlinear dispersive PDEs

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-563-6-volume-5.pdf#page=499>

7.17.1. Problem statement

Let

$$\mathbb{S}^2 = \{y \in \mathbb{R}^3 : |y| = 1\}, \quad \dot{H}^1(\mathbb{R}^2; \mathbb{S}^2) = \{u : \mathbb{R}^2 \rightarrow \mathbb{S}^2 : \nabla u \in L^2(\mathbb{R}^2)\}.$$

For $u \in \dot{H}^1(\mathbb{R}^2; \mathbb{S}^2)$, define

$$\mathcal{E}(u) = \frac{1}{2} \int_{\mathbb{R}^2} |\nabla u(x)|^2 dx, \quad \deg(u) = \frac{1}{4\pi} \int_{\mathbb{R}^2} u \cdot (\partial_1 u \times \partial_2 u) dx.$$

Consider the energy-critical Schrödinger map Cauchy problem

$$\partial_t u = u \times \Delta u, \quad u(0, x) = u_0(x), \quad u : \mathbb{R}_t \times \mathbb{R}_x^2 \rightarrow \mathbb{S}^2.$$

Here, scattering forward or backward in time means the following. Choose along u an oriented orthonormal frame (u, v, w) satisfying the Coulomb condition

$$\partial_1 A_1 + \partial_2 A_2 = 0, \quad A_j = w \cdot \partial_j v,$$

and define the complex differentiated fields

$$\psi_j = v \cdot \partial_j u + i w \cdot \partial_j u, \quad j = 1, 2.$$

The solution scatters as $t \rightarrow \pm\infty$ if there exist $\psi_{1,\pm}, \psi_{2,\pm} \in L^2(\mathbb{R}^2; \mathbb{C})$ such that

$$\lim_{t \rightarrow \pm\infty} \|\psi_j(t) - e^{it\Delta} \psi_{j,\pm}\|_{L^2(\mathbb{R}^2)} = 0, \quad j = 1, 2.$$

Prove that for every $u_0 \in \dot{H}^1(\mathbb{R}^2; \mathbb{S}^2)$ with

$$\deg(u_0) = 0 \quad \text{and} \quad \mathcal{E}(u_0) < 8\pi,$$

the corresponding maximal energy-class solution is defined for all $t \in \mathbb{R}$ and scatters as $t \rightarrow -\infty$ and as $t \rightarrow +\infty$.

7.17.2. Historical background, literature, and current progress

The problem is **unresolved**. The 8π assertion is the strong threshold conjecture for two-dimensional Schrödinger maps into $\mathbb{S}^2$. Perelman explicitly records it as open in full generality, and the later literature located through August 2026 does not supply a theorem covering arbitrary degree-zero energy-class data with energy below 8π .

Established partial progress is as follows.

- Ioan Bejenaru, Alexandru D. Ionescu, Carlos E. Kenig, and Daniel Tataru, “Global Schrödinger maps in dimensions $d \geq 2$: Small data in the critical Sobolev spaces,” *Annals of Mathematics* 173 (2011), 1443–1506, [arXiv:0807.0265](#). This proves global well-posedness (with the corresponding dispersive bounds/scattering formulation for differentiated fields) for data small in the critical norm. It does not treat all data below either 4π or 8π .
- Paul Smith, “Global regularity of critical Schrödinger maps: subthreshold dispersed energy,” arXiv preprint (2011; revised 2012), [arXiv:1112.0251](#). This proves global smooth regularity for smooth data of energy below 4π under an additional frequency-by-frequency energy-dispersion hypothesis. Thus it does not cover arbitrary subthreshold data, and its theorem does not provide the full 8π statement.
- Stephen Gustafson and Eva Koo, “Global well-posedness for 2D radial Schrödinger maps into the sphere,” arXiv preprint (2011), [arXiv:1105.5659](#). This proves global well-posedness for radial H^2 maps with no size restriction, with scattering for the associated differentiated nonlocal Schrödinger variable. Radial maps form only a special symmetry class.
- Ioan Bejenaru, Alexandru D. Ionescu, Carlos E. Kenig, and Daniel Tataru, “Equivariant Schrödinger maps in two spatial dimensions,” *Duke Mathematical Journal* 162 (2013), 1967–2025, DOI: [10.1215/00127094-2293611](#), [arXiv:1112.6122](#). This proves global existence and scattering for equivariant data with energy below 4π . The paper expressly notes that extending the degree-zero argument to 8π requires a new idea because the relevant virial term loses its sign.
- Benjamin Dodson and Paul Smith, “A controlling norm for energy-critical Schrödinger maps,” *Transactions of the American Mathematical Society* 367 (2015), 7193–7220, DOI: [10.1090/S0002-9947-2015-06417-4](#), [arXiv:1302.3879](#). This gives continuation while a critical spacetime L^4 norm stays bounded, reducing large-data global well-posedness to controlling that norm rather than proving the required bound for every datum in the conjectured range.
- Benjamin Dodson and Jeremy L. Marzuola, “Schrödinger maps to a Kähler manifold in two dimensions,” arXiv preprint (2025), [arXiv:2512.19559](#). This proves global well-posedness and scattering for small data in a Besov space, including the spherical target, but does not address arbitrary large data below 8π .

The source that identifies the exact remaining conjecture is Galina Perelman, “Formation of singularities in nonlinear dispersive PDEs,” *Proceedings of the International Congress of Mathematicians 2022*, vol. 5, EMS Press (2022), 3854–3879, DOI: [10.4171/ICM2022/123](#). What remains open is removal of all auxiliary smallness, dispersion, regularity, and symmetry assumptions and proof of global energy-class existence plus scattering for every degree-zero datum throughout the full range $\mathcal{E}(u_0) < 8\pi$.

7.17.3. Remaining open problems recorded by the database

7.17.3.1. Open problem 1

Let

$$\mathbb{S}^2 = \{y \in \mathbb{R}^3 : |y| = 1\}, \quad \dot{H}^1(\mathbb{R}^2; \mathbb{S}^2) = \{u : \mathbb{R}^2 \rightarrow \mathbb{S}^2 : \nabla u \in L^2(\mathbb{R}^2)\},$$

and define

$$\mathcal{E}(u) = \frac{1}{2} \int_{\mathbb{R}^2} |\nabla u(x)|^2 dx.$$

Consider the Schrödinger map Cauchy problem

$$\partial_t u = u \times \Delta u, \quad u(0, x) = u_0(x), \quad u : \mathbb{R}_t \times \mathbb{R}_x^2 \rightarrow \mathbb{S}^2.$$

Choose along u an oriented orthonormal frame (u, v, w) satisfying the Coulomb condition

$$\partial_1 A_1 + \partial_2 A_2 = 0, \quad A_j = w \cdot \partial_j v,$$

and set

$$\psi_j = v \cdot \partial_j u + i w \cdot \partial_j u, \quad j = 1, 2.$$

Say that u scatters as $t \rightarrow \pm\infty$ if there exist $\psi_{1,\pm}, \psi_{2,\pm} \in L^2(\mathbb{R}^2; \mathbb{C})$ such that

$$\lim_{t \rightarrow \pm\infty} \|\psi_j(t) - e^{it\Delta} \psi_{j,\pm}\|_{L^2(\mathbb{R}^2)} = 0, \quad j = 1, 2.$$

Prove that for every $u_0 \in \dot{H}^1(\mathbb{R}^2; \mathbb{S}^2)$ with $\mathcal{E}(u_0) < 4\pi$, the corresponding maximal energy-class solution is defined for all $t \in \mathbb{R}$ and scatters as $t \rightarrow -\infty$ and as $t \rightarrow +\infty$.

Reference supplied by the database. Ioan Bejenaru, Alexandru D. Ionescu, Carlos E. Kenig, and Daniel Tataru, “Equivariant Schrödinger Maps in Two Spatial Dimensions: The $\mathbb{H}^2$ Target,” *Kyoto Journal of Mathematics* 56 (2016), no. 2, 283–323, Conjecture 1.2(b), DOI: <https://doi.org/10.1215/21562261-3478889>, arXiv:1212.2566.

7.17.3.2. Open problem 2

Let

$$\mathbb{S}^2 = \{y \in \mathbb{R}^3 : |y| = 1\}, \quad \dot{H}^1(\mathbb{R}^2; \mathbb{S}^2) = \{u : \mathbb{R}^2 \rightarrow \mathbb{S}^2 : \nabla u \in L^2(\mathbb{R}^2)\}.$$

For $u \in \dot{H}^1(\mathbb{R}^2; \mathbb{S}^2)$, define

$$\mathcal{E}(u) = \frac{1}{2} \int_{\mathbb{R}^2} |\nabla u(x)|^2 dx, \quad \deg(u) = \frac{1}{4\pi} \int_{\mathbb{R}^2} u \cdot (\partial_1 u \times \partial_2 u) dx.$$

Let

$$R = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

and call a map 1-equivariant if, in polar coordinates $x = (r \cos \theta, r \sin \theta)$, it has the form

$$u(r, \theta) = e^{\theta R} \bar{u}(r)$$

for some $\bar{u} : [0, \infty) \rightarrow \mathbb{S}^2$.

Consider the Schrödinger map Cauchy problem

$$\partial_t u = u \times \Delta u, \quad u(0, x) = u_0(x), \quad u : \mathbb{R}_t \times \mathbb{R}_x^2 \rightarrow \mathbb{S}^2.$$

Choose along u an oriented orthonormal frame (u, v, w) satisfying the Coulomb condition

$$\partial_1 A_1 + \partial_2 A_2 = 0, \quad A_j = w \cdot \partial_j v,$$

and set

$$\psi_j = v \cdot \partial_j u + i w \cdot \partial_j u, \quad j = 1, 2.$$

Say that u scatters as $t \rightarrow \pm\infty$ if there exist $\psi_{1,\pm}, \psi_{2,\pm} \in L^2(\mathbb{R}^2; \mathbb{C})$ such that

$$\lim_{t \rightarrow \pm\infty} \|\psi_j(t) - e^{it\Delta} \psi_{j,\pm}\|_{L^2(\mathbb{R}^2)} = 0, \quad j = 1, 2.$$

Prove that for every 1-equivariant $u_0 \in \dot{H}^1(\mathbb{R}^2; \mathbb{S}^2)$ with

$$\deg(u_0) = 0 \quad \text{and} \quad \mathcal{E}(u_0) < 8\pi,$$

the corresponding maximal energy-class solution is defined for all $t \in \mathbb{R}$ and scatters as $t \rightarrow -\infty$ and as $t \rightarrow +\infty$.

Reference supplied by the database. Ioan Bejenaru, Alexandru D. Ionescu, Carlos E. Kenig, and Daniel Tataru, “Equivariant Schrödinger Maps in Two Spatial Dimensions,” *Duke Mathematical Journal* 162 (2013), no. 11, 1967–2025, open conjecture in the Introduction following the main theorem, DOI: <https://doi.org/10.1215/00127094-2293611>, arXiv:1112.6122.

7.17.4. Primary-source pointer

The originating ICM record is *Formation of singularities in nonlinear dispersive PDEs* (2022). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-563-6-volume-5.pdf#page=499>.

7.18. 244. Some recent developments in Ricci flow [2022-geom-334]

Record: 2022-geom-334

Subfield: Geometric flows and geometric PDE

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2022: Some recent developments in Ricci flow

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-562-9-volume-4.pdf#page=75>

7.18.1. Problem statement

Develop a rigorous notion of a generalized Ricci-flow spacetime in dimension four and prove that, for every closed smooth four-dimensional manifold M and every smooth Riemannian metric g on M , there exists such a spacetime for all forward times with initial time-slice isometric to (M, g) , satisfying the following requirements: its regular spacetime region is smooth and its time-dependent metric satisfies

$$\partial_t g(t) = -2 \operatorname{Ric}_{g(t)};$$

the generalized flow continues past every finite singular time; time-slices may become disconnected, with points in distinct components regarded as being at infinite distance; the only nonsmooth points permitted in a time-slice are isolated orbifold singularities; and every change in the topology of the time-slices occurs in a region whose parabolically rescaled local singularity model is cylindrical or conical.

7.18.2. Historical background, literature, and current progress

Verdict: unresolved. No citable work found through 4 August 2026 constructs the stated all-forward generalized flow for every smooth metric on every closed four-manifold. The literature supplies a weak-flow framework, control of a first singular time, several restricted existence theorems, and local models for resolving many cones, but it does not assemble these ingredients into the required arbitrary-data continuation through every singular time.

- Richard H. Bamler, “Some recent developments in Ricci flow,” *Proceedings of the International Congress of Mathematicians 2022*, Vol. 4, 2022, pp. 2432-2455, DOI [10.4171/ICM2022/58](https://doi.org/10.4171/ICM2022/58). Conjecture 5.8 is the exact four-dimensional problem. The accompanying discussion leaves the notion of flow through singularities open and suggests metric-flow partial regularity, possibly disconnected slices, and isolated orbifold points.
- Richard H. Bamler, “Structure theory of non-collapsed limits of Ricci flows,” preprint, 2020, [arXiv:2009.03243](https://arxiv.org/abs/2009.03243), together with “Compactness theory of the space of super Ricci flows,” *Inventiones Mathematicae* 233 (2023), 1121-1277, DOI [10.1007/s00222-023-01196-3](https://doi.org/10.1007/s00222-023-01196-3). These works construct metric-flow limits with a smooth Ricci-flow regular part and parabolic codimension-four singular set; for a compact classical flow they construct and

analyze its first singular time-slice and its shrinking-soliton or static-cone tangent models. They do not construct a forward continuation from that slice.

- Beomjun Choi and Robert Haslhofer, “Ricci limit flows and weak solutions,” *Journal of the European Mathematical Society* 27 (2025), no. 12, 5155-5171, DOI [10.4171/JEMS/1565](https://doi.org/10.4171/JEMS/1565). This proves that Bamler’s noncollapsed Ricci-limit flows are weak solutions in the Haslhofer-Naber sense, reconciling two candidate frameworks. It is not an arbitrary-data, all-forward existence theorem.
- Robert Haslhofer, “Uniqueness and stability of singular Ricci flows in higher dimensions,” *Proceedings of the American Mathematical Society* 150 (2022), 5433-5437, DOI [10.1090/proc/16108](https://doi.org/10.1090/proc/16108). Corollary 1.2 gives a unique singular Ricci flow for compact four-manifolds whose initial metric has positive isotropic curvature. This extra curvature hypothesis excludes general (M, g) .
- Timothy Buttsworth, Maximilien Hallgren, and Yongjia Zhang, “Canonical surgeries in rotationally invariant Ricci flow,” *Transactions of the American Mathematical Society* 377 (2024), 7877-7944, DOI [10.1090/tran/9220](https://doi.org/10.1090/tran/9220). Theorem 1.1 constructs an all-forward surgery flow and a limiting Ricci-flow spacetime for closed rotationally invariant metrics in every dimension greater than three. Rotational invariance is a substantial additional hypothesis, and the result treats the cylindrical surgery regime rather than arbitrary conical and orbifold singularities.
- Richard H. Bamler and Eric Chen, “Degree theory for 4-dimensional asymptotically conical gradient expanding solitons,” *Communications on Pure and Applied Mathematics* 79 (2026), DOI [10.1002/cpa.70024](https://doi.org/10.1002/cpa.70024), [arXiv:2305.03154](https://arxiv.org/abs/2305.03154). They prove that every nonnegative-scalar-curvature cone over S^3 has a smooth asymptotically conical gradient expander, and obtain an orbifold expander for links S^3/Γ . This provides local outgoing models for many conical singularities, but not a gluing/surgery theorem or a global flow for arbitrary initial data; Conjecture 1.17 leaves general orientable links open.
- Robert Haslhofer, “On κ -solutions and canonical neighborhoods in 4d Ricci flow,” preprint, 2024, [arXiv:2308.01448v2](https://arxiv.org/abs/2308.01448v2). The canonical-neighborhood theorem near cylindrical singularities is conditional on Conjecture 1.7, the still-open classification of ancient asymptotically cylindrical four-dimensional metric Ricci flows. Thus even the cylindrical local structure needed for a general construction is not yet unconditional.

The remaining gap is global: one must control all singularities arising from an arbitrary closed four-dimensional initial metric, choose and glue suitable outgoing evolutions at both cylindrical and conical models, allow the prescribed orbifold slices and disconnected components, and iterate the construction without finite-time breakdown. None of the cited special-case or local results proves all of these requirements.

7.18.3. Remaining open problems recorded by the database

7.18.3.1. Open problem 1

Classify, up to parabolic scaling, isometry, and finite isometric quotient, all four-dimensional κ -solutions: ancient Ricci flows $(N^4, g(t))_{t \in (-\infty, 0]}$ whose time-slices are complete, whose curvature is bounded on every compact time interval, whose curvature operator is nonnegative and scalar curvature is positive, and which are κ -noncollapsed at every scale for some $\kappa > 0$.

Reference supplied by the database. Robert Haslhofer, “On κ -solutions and canonical neighborhoods in 4d Ricci flow,” arXiv preprint, 2024, Conjecture 1.3, arXiv:2308.01448v2, <https://arxiv.org/abs/2308.01448v2>.

7.18.3.2. Open problem 2

Let $(N^4, g(t))_{t \in (-\infty, 0]}$ be a noncompact four-dimensional κ -solution whose tangent flow at $t = -\infty$, obtained as a pointed parabolic blow-down limit, is the round shrinking product $\mathbb{R}^2 \times S^2$. Prove that, for every $t \leq 0$, the Tits cone of the complete nonnegatively curved manifold $(N, g(t))$, namely the Euclidean cone over its ideal boundary equipped with the Tits metric, is either a ray or splits isometrically as a product $\mathbb{R} \times C$ with a metric cone C .

Reference supplied by the database. Robert Haslhofer, “On κ -solutions and canonical neighborhoods in 4d Ricci flow,” arXiv preprint, 2024, discussion immediately following Conjecture 1.3, arXiv:2308.01448v2, <https://arxiv.org/abs/2308.01448v2>.

7.18.3.3. Open problem 3

Let (N^3, h) be a closed orientable Riemannian three-manifold such that the cone metric $\gamma = dr^2 + r^2h$ on $(0, \infty) \times N$ has nonnegative scalar curvature. Prove that there exist integers $k, \ell \geq 0$ with $k + \ell \geq 1$ and finite subgroups $\Gamma_1, \dots, \Gamma_k \subset SO(4)$, each acting freely on S^3 , for which N is diffeomorphic to

$$(S^3/\Gamma_1) \# \dots \# (S^3/\Gamma_k) \# \underbrace{(S^1 \times S^2) \# \dots \# (S^1 \times S^2)}_{\ell \text{ factors}},$$

and there is a complete four-dimensional gradient expanding Ricci soliton (X, g, f) with nonnegative scalar curvature,

$$\text{Ric}_g + \nabla_g^2 f + \frac{1}{2}g = 0,$$

that is asymptotic to γ and whose underlying orbifold is diffeomorphic to the interior of the boundary connected sum

$$(D^4/\Gamma_1) \#_{\partial} \dots \#_{\partial} (D^4/\Gamma_k) \#_{\partial} \underbrace{(S^1 \times D^3) \#_{\partial} \dots \#_{\partial} (S^1 \times D^3)}_{\ell \text{ factors}}.$$

Reference supplied by the database. Richard H. Bamler and Eric Chen, “Degree theory for 4-dimensional asymptotically conical gradient expanding solitons,” Communications on Pure and Applied Mathematics 79 (2026), 1151-1298, Conjecture 1.17, DOI 10.1002/cpa.70024, arXiv:2305.03154.

7.18.3.4. Open problem 4

For every integer $k \geq 3$, construct a Ricci flow on $S^1 \times (S^3/\mathbb{Z}_k)$, where the generator of $\mathbb{Z}_k$ acts freely on $S^3 \subset \mathbb{C}^2$ by $(z_1, z_2) \mapsto (e^{2\pi i/k} z_1, e^{2\pi i/k} z_2)$, that develops a quotient-neck singularity with tangent flow $\mathbb{R} \times (S^3/\mathbb{Z}_k)$ and admits at least two distinct continuous forward Ricci-flow continuations through that singularity: one locally modeled on the four-dimensional Bryant steady soliton modulo $\mathbb{Z}_k$, and another locally modeled on the complete $U(2)$ -invariant Appleton steady gradient Ricci soliton on the complex line bundle $O_{\mathbb{CP}^1}(-k)$.

Reference supplied by the database. Robert Haslhofer, “On κ -solutions and canonical neighborhoods in 4d Ricci flow,” arXiv preprint, 2024, Conjecture 1.9, arXiv:2308.01448v2, <https://arxiv.org/abs/2308.01448v2>.

7.18.4. Primary-source pointer

The originating ICM record is *Some recent developments in Ricci flow* (2022). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-562-9-volume-4.pdf#page=75>.

7.19. 245. Entropy in mean curvature flow [2022-geom-337]

Record: 2022-geom-337

Subfield: Geometric flows and geometric PDE

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2022: Entropy in mean curvature flow

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-562-9-volume-4.pdf#page=299>

7.19.1. Problem statement

Prove that, for every integer n with $1 \leq n \leq 6$, there exists a constant $\varepsilon_0(n) > 0$ such that the following holds. Let $\Sigma \subset \mathbb{R}^{n+1}$ be a smooth, complete, embedded hypersurface without boundary satisfying the self-shrinker equation

$$\mathbf{H}_\Sigma + \frac{x^\perp}{2} = 0,$$

where $\mathbf{H}_\Sigma$ is the mean-curvature vector of Σ and $x^\perp$ is the orthogonal projection of the position vector x onto the normal space of Σ at x . Suppose that Σ is neither a hyperplane through the origin nor the round self-shrinking sphere

$$\mathbb{S}^n = \{x \in \mathbb{R}^{n+1} : |x| = \sqrt{2n}\}.$$

Then

$$\lambda[\Sigma] \geq \lambda[\mathbb{S}^n] + \varepsilon_0(n),$$

where the entropy of an n -dimensional hypersurface $\Gamma \subset \mathbb{R}^{n+1}$ is

$$\lambda[\Gamma] = \sup_{\substack{x_0 \in \mathbb{R}^{n+1} \\ t_0 > 0}} (4\pi t_0)^{-n/2} \int_{\Gamma} \exp\left(-\frac{|x - x_0|^2}{4t_0}\right) d\mathcal{H}^n(x),$$

and $\mathcal{H}^n$ denotes n -dimensional Hausdorff measure.

7.19.2. Historical background, literature, and current progress

The problem is **unresolved**. It is a single assertion quantified over every $1 \leq n \leq 6$, and the noncompact cases in dimensions $n = 3, 4, 5, 6$ have not been proved or disproved. No counterexample is known.

Established cases and partial progress:

- Uwe Abresch and Joel Langer, “The normalized curve shortening flow and homothetic solutions,” *Journal of Differential Geometry* **23** (1986), 175–196, <https://doi.org/10.4310/jdg/1214440025>. Their classification of self-shrinking planar curves gives the assertion for $n = 1$: among smooth embedded examples, the only nonflat shrinker is the round circle, so the stated implication is vacuous after the circle and line are excluded.

- Tobias Holck Colding, Tom Ilmanen, William P. Minicozzi II, and Brian White, “The round sphere minimizes entropy among closed self-shrinkers,” *Journal of Differential Geometry* **95** (2013), 53-69, <https://doi.org/10.4310/jdg/1375124609>, [arXiv:1205.2043](https://arxiv.org/abs/1205.2043). Their main theorem proves, for every dimension, a positive entropy gap above the round sphere for **closed** self-shrinkers. Thus every compact case in the problem is settled, but the theorem does not cover noncompact shrinkers.
- Jacob Bernstein and Lu Wang, “A topological property of asymptotically conical self-shrinkers of small entropy,” *Duke Mathematical Journal* **166** (2017), 403-435, <https://doi.org/10.1215/00127094-3715082>, [arXiv:1504.01996](https://arxiv.org/abs/1504.01996). Combining their low-entropy topology theorem with Brendle’s genus-zero classification, they prove that the round sphere has least entropy among all nonflat two-dimensional self-shrinkers and that the round cylinder has second least entropy, with a positive gap to every other shrinker. Consequently the exact assertion holds for $n = 2$, for example with $\varepsilon_0(2) = \lambda[\mathbb{S}^1] - \lambda[\mathbb{S}^2] > 0$.
- Or Hershkovits and Brian White, “Sharp entropy bounds for self-shrinkers in mean curvature flow,” *Geometry & Topology* **23** (2019), 1611-1619, <https://doi.org/10.2140/gt.2019.23.1611>, [arXiv:1803.00637](https://arxiv.org/abs/1803.00637). They obtain sharp lower entropy bounds for smooth **closed** self-shrinkers from nontrivial homology. These strengthen the compact theory under topological information but do not address the missing noncompact cases.
- Lu Wang, “Entropy in mean curvature flow,” *Proceedings of the International Congress of Mathematicians 2022*, Vol. 4 (EMS Press, 2022), 2656-2676, <https://doi.org/10.4171/ICM2022/116>. Conjecture 2.2 is the exact problem; the survey proves/reviews the $n = 2$ result and states that the conjecture is wide open in general, identifying the lack of a classification of self-shrinkers in dimensions at least three as a central obstruction.
- Fagui Li and Yuhang Zhao, “Gaussian-Weighted Curvature Gaps for Self-Shrinkers,” preprint (2026), [arXiv:2606.14360](https://arxiv.org/abs/2606.14360). This recent work derives curvature-gap consequences from the Colding-Ilmanen-Minicozzi-White entropy gap for closed shrinkers and from the Bernstein-Wang theorem only in dimension two. It does not establish the required entropy gap for complete noncompact shrinkers in dimensions 3 through 6.

Therefore the unresolved portion is exactly the existence, for each $n = 3, 4, 5, 6$, of a uniform positive gap above $\lambda[\mathbb{S}^n]$ for all smooth complete embedded noncompact self-shrinkers other than the hyperplane; the known compact and low-dimensional results do not exhaust those cases.

7.19.3. Remaining open problems recorded by the database

7.19.3.1. Open problem 1

Prove that there exists a constant $\varepsilon_0 > 0$ such that the following holds. Let $\Sigma \subset \mathbb{R}^4$ be a smooth, complete, embedded three-dimensional hypersurface without boundary satisfying the self-shrinker equation

$$\mathbf{H}_\Sigma + \frac{x^\perp}{2} = 0,$$

where $\mathbf{H}_\Sigma$ is the mean-curvature vector of Σ and $x^\perp$ is the orthogonal projection of the position vector x onto the normal space of Σ at x . Suppose that Σ is neither a hyperplane

through the origin nor the round self-shrinking sphere

$$\mathbb{S}^3 = \{x \in \mathbb{R}^4 : |x| = \sqrt{6}\}.$$

Then

$$\lambda[\Sigma] \geq \lambda[\mathbb{S}^3] + \varepsilon_0,$$

where the entropy of a three-dimensional hypersurface $\Gamma \subset \mathbb{R}^4$ is

$$\lambda[\Gamma] = \sup_{\substack{x_0 \in \mathbb{R}^4 \\ t_0 > 0}} (4\pi t_0)^{-3/2} \int_{\Gamma} \exp\left(-\frac{|x - x_0|^2}{4t_0}\right) d\mathcal{H}^3(x),$$

and $\mathcal{H}^3$ denotes three-dimensional Hausdorff measure.

Reference supplied by the database. Tobias Holck Colding, Tom Ilmanen, William P. Minicozzi II, and Brian White, “The round sphere minimizes entropy among closed self-shrinkers,” *Journal of Differential Geometry* **95** (2013), no. 1, 53-69, Conjecture 0.10. <https://doi.org/10.4310/jdg/1375124609>

7.19.3.2. Open problem 2

Prove that there exists a constant $\varepsilon_0 > 0$ such that the following holds. Let $\Sigma \subset \mathbb{R}^5$ be a smooth, complete, embedded four-dimensional hypersurface without boundary satisfying the self-shrinker equation

$$\mathbf{H}_{\Sigma} + \frac{x^{\perp}}{2} = 0,$$

where $\mathbf{H}_{\Sigma}$ is the mean-curvature vector of Σ and $x^{\perp}$ is the orthogonal projection of the position vector x onto the normal space of Σ at x . Suppose that Σ is neither a hyperplane through the origin nor the round self-shrinking sphere

$$\mathbb{S}^4 = \{x \in \mathbb{R}^5 : |x| = \sqrt{8}\}.$$

Then

$$\lambda[\Sigma] \geq \lambda[\mathbb{S}^4] + \varepsilon_0,$$

where the entropy of a four-dimensional hypersurface $\Gamma \subset \mathbb{R}^5$ is

$$\lambda[\Gamma] = \sup_{\substack{x_0 \in \mathbb{R}^5 \\ t_0 > 0}} (4\pi t_0)^{-2} \int_{\Gamma} \exp\left(-\frac{|x - x_0|^2}{4t_0}\right) d\mathcal{H}^4(x),$$

and $\mathcal{H}^4$ denotes four-dimensional Hausdorff measure.

Reference supplied by the database. Tobias Holck Colding, Tom Ilmanen, William P. Minicozzi II, and Brian White, “The round sphere minimizes entropy among closed self-shrinkers,” *Journal of Differential Geometry* **95** (2013), no. 1, 53-69, Conjecture 0.10. <https://doi.org/10.4310/jdg/1375124609>

[g/10.4310/jdg/1375124609](https://doi.org/10.4310/jdg/1375124609)

7.19.3.3. Open problem 3

Prove that there exists a constant $\varepsilon_0 > 0$ such that the following holds. Let $\Sigma \subset \mathbb{R}^6$ be a smooth, complete, embedded five-dimensional hypersurface without boundary satisfying the self-shrinker equation

$$\mathbf{H}_\Sigma + \frac{x^\perp}{2} = 0,$$

where $\mathbf{H}_\Sigma$ is the mean-curvature vector of Σ and $x^\perp$ is the orthogonal projection of the position vector x onto the normal space of Σ at x . Suppose that Σ is neither a hyperplane through the origin nor the round self-shrinking sphere

$$\mathbb{S}^5 = \{x \in \mathbb{R}^6 : |x| = \sqrt{10}\}.$$

Then

$$\lambda[\Sigma] \geq \lambda[\mathbb{S}^5] + \varepsilon_0,$$

where the entropy of a five-dimensional hypersurface $\Gamma \subset \mathbb{R}^6$ is

$$\lambda[\Gamma] = \sup_{\substack{x_0 \in \mathbb{R}^6 \\ t_0 > 0}} (4\pi t_0)^{-5/2} \int_{\Gamma} \exp\left(-\frac{|x - x_0|^2}{4t_0}\right) d\mathcal{H}^5(x),$$

and $\mathcal{H}^5$ denotes five-dimensional Hausdorff measure.

Reference supplied by the database. Tobias Holck Colding, Tom Ilmanen, William P. Minicozzi II, and Brian White, “The round sphere minimizes entropy among closed self-shrinkers,” *Journal of Differential Geometry* **95** (2013), no. 1, 53-69, Conjecture 0.10. <https://doi.org/10.4310/jdg/1375124609>

7.19.3.4. Open problem 4

Prove that there exists a constant $\varepsilon_0 > 0$ such that the following holds. Let $\Sigma \subset \mathbb{R}^7$ be a smooth, complete, embedded six-dimensional hypersurface without boundary satisfying the self-shrinker equation

$$\mathbf{H}_\Sigma + \frac{x^\perp}{2} = 0,$$

where $\mathbf{H}_\Sigma$ is the mean-curvature vector of Σ and $x^\perp$ is the orthogonal projection of the position vector x onto the normal space of Σ at x . Suppose that Σ is neither a hyperplane through the origin nor the round self-shrinking sphere

$$\mathbb{S}^6 = \{x \in \mathbb{R}^7 : |x| = \sqrt{12}\}.$$

Then

$$\lambda[\Sigma] \geq \lambda[\mathbb{S}^6] + \varepsilon_0,$$

where the entropy of a six-dimensional hypersurface $\Gamma \subset \mathbb{R}^7$ is

$$\lambda[\Gamma] = \sup_{\substack{x_0 \in \mathbb{R}^7 \\ t_0 > 0}} (4\pi t_0)^{-3} \int_{\Gamma} \exp\left(-\frac{|x - x_0|^2}{4t_0}\right) d\mathcal{H}^6(x),$$

and $\mathcal{H}^6$ denotes six-dimensional Hausdorff measure.

Reference supplied by the database. Tobias Holck Colding, Tom Ilmanen, William P. Minicozzi II, and Brian White, “The round sphere minimizes entropy among closed self-shrinkers,” *Journal of Differential Geometry* **95** (2013), no. 1, 53-69, Conjecture 0.10. <https://doi.org/10.4310/jdg/1375124609>

7.19.4. Primary-source pointer

The originating ICM record is *Entropy in mean curvature flow* (2022). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-562-9-volume-4.pdf#page=299>.

8. Convex geometry

The convex-geometric records focus on extremal approximation and dimension- dependent structure. They exemplify a persistent divide between existence theorems and effective or sharp versions: algorithms and asymptotic bounds may be available even when optimal constants or exact generality remain unknown.

Coverage: 12 website records; **source years:** 1974–2022; **separately listed descendant open problems:** 13.

8.1. 246. Some Open Questions on Convex Surfaces [1974-geom-233]

Record: 1974-geom-233
Subfield: Convex geometry
Normalized status: Open / open-ended
Dataset review state: unreviewed (reviewed flag: no)
Original source: ICM 1974: Some Open Questions on Convex Surfaces
Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1974.1/ICM1974.1.ocr.pdf#page=535>

8.1.1. Problem statement

Let $n \geq 4$, let p be an integer with $1 < p < n - 1$, and let μ be a finite nonnegative Borel measure on the unit sphere $S^{n-1} \subset \mathbb{R}^n$ such that every open hemisphere has positive μ -measure. Characterize, by necessary and sufficient conditions on μ , when there exists a compact convex body $K \subset \mathbb{R}^n$ with nonempty interior whose p th area measure satisfies

$$S_p(K, \cdot) = \mu.$$

Here the area measures $S_j(K, \cdot)$, $0 \leq j \leq n - 1$, are defined by the local Steiner formula

$$S_{n-1}(K + \lambda B^n, \omega) = \sum_{j=0}^{n-1} \binom{n-1}{j} S_j(K, \omega) \lambda^{n-1-j}$$

for every Borel set $\omega \subset S^{n-1}$ and every $\lambda \geq 0$, where B^n is the Euclidean unit ball and S_{n-1} is the usual surface-area measure. In the smooth strictly convex case, if H is the positively homogeneous support function of K and $R_1(u), \dots, R_{n-1}(u)$ are the eigenvalues of $D^2H(u)$ on $u^\perp$ (the principal radii of curvature at outer normal u), then

$$dS_p(K, u) = \binom{n-1}{p}^{-1} e_p(R_1(u), \dots, R_{n-1}(u)) du,$$

where e_p is the p th elementary symmetric polynomial and du is spherical area measure.

8.1.2. Historical background, literature, and current progress

The problem is **unresolved**. No necessary-and-sufficient characterization is known for arbitrary finite Borel measures in any of the unrestricted intermediate orders $1 < p < n - 1$. The centroid condition $\int_{S^{n-1}} u d\mu(u) = 0$ and known restrictions on concentration or support do not give a complete criterion. The main unresolved point is to distinguish those data for which the associated intermediate Hessian equation has a genuinely convex support-function solution; existence of a merely p -admissible solution is not enough.

Pengfei Guan and Xi-Nan Ma, “The Christoffel-Minkowski Problem I: Convexity of Solutions of a Hessian Equation,” *Inventiones Mathematicae* 151 (2003), 553–577, DOI: <https://doi.org/10.1007/s00222-002-0259-2>. Theorem 1.3 proves existence and uniqueness up to translation for smooth positive densities φ when $\varphi^{-1/p}$ has convex degree-one homogeneous extension (originally with a path condition). This is a substantial sufficient condition, not a necessary-and-sufficient characterization.

Pengfei Guan, Xi-Nan Ma, and Feng Zhou, “The Christoffel-Minkowski Problem III: Existence and Convexity of Admissible Solutions,” *Communications on Pure and Applied Mathematics* 59 (2006), 1352–1376, DOI: <https://doi.org/10.1002/cpa.20118>. Theorem 1.3 shows that every positive $C^{1,1}$ density satisfying the centroid condition yields a solution in the Gårding p -admissible class. Such a solution need not have positive-semidefinite spherical Hessian and therefore need not be the support function of a convex body; the paper’s full-rank theorem gives convexity only under an additional structural condition.

Paul Bryan, Mohammad N. Ivaki, and Julian Scheuer, “Christoffel-Minkowski Flows,” *Transactions of the American Mathematical Society* 376 (2023), 2373–2393, DOI: <https://doi.org/10.1090/tran/8683>; arXiv:2005.14680. Corollary 1.2 gives a curvature-flow proof of smooth strictly convex solvability when the density is centered and its negative $1/p$ power is spherically convex. The authors explicitly note that this convexity hypothesis is sufficient but not necessary, so the regular problem is not classified.

Yong Huang, Deane Yang, and Gaoyong Zhang, “Minkowski Problems for Geometric Measures,” *Bulletin of the American Mathematical Society* 62 (2025), 359–425, DOI: <https://doi.org/10.1090/bull/1858>; arXiv:2502.05427. Section 5.3.2 states that the Christoffel-Minkowski problem for measures remains unsolved for every intermediate order and summarizes the smooth sufficient-condition and local symmetric results.

Leo Brauner, Georg C. Hofstätter, and Oscar Ortega-Moreno, “Mixed Christoffel-Minkowski Problems for Bodies of Revolution,” preprint (2025), arXiv:2508.09794. Corollary A gives necessary and sufficient conditions for centered zonal Borel measures, with no regularity assumption, to be intermediate area measures of bodies of revolution. The introduction explicitly says that the unrestricted intermediate problem remains widely open.

Fabian Mussnig and Jacopo Ulivelli, “Explicit Solutions to Christoffel-Minkowski Problems and Hessian Equations under Rotational Symmetries,” preprint (2025; revised 2026), arXiv:2508.11600. Theorem 1.1 independently characterizes rotation-invariant Borel measures that are area measures of bodies of revolution and reconstructs the support function explicitly.

Its axial-symmetry hypothesis leaves the general measures in the stated problem untreated.

8.1.3. Remaining open problems recorded by the database

8.1.3.1. Open problem 1

Let $n \geq 4$, let p be an integer with $1 < p < n - 1$, and let $\varphi \in C^\infty(S^{n-1})$ be positive and satisfy

$$\int_{S^{n-1}} u \varphi(u) du = 0.$$

Characterize, by necessary and sufficient conditions on φ , when there exists a function $H \in C^\infty(S^{n-1})$ such that the symmetric bilinear form

$$Q_H = \nabla_{S^{n-1}}^2 H + H g_{S^{n-1}}$$

is positive definite everywhere and

$$e_p(Q_H) = \varphi$$

on S^{n-1} , where $g_{S^{n-1}}$ is the round metric, $\nabla_{S^{n-1}}^2$ is its covariant Hessian, and $e_p(Q_H)$ is the p th elementary symmetric polynomial of the eigenvalues of Q_H with respect to $g_{S^{n-1}}$.

Reference supplied by the database. Pengfei Guan and Xi-Nan Ma, “The Christoffel-Minkowski Problem I: Convexity of Solutions of a Hessian Equation,” *Inventiones Mathematicae* 151 (2003), 553–577, the explicit “fundamental question” in the introduction, DOI: <https://doi.org/10.1007/s00222-002-0259-2>.

8.1.3.2. Open problem 2

Let $n \geq 4$ and let μ be a finite nonnegative Borel measure on S^{n-1} such that every open hemisphere has positive μ -measure. Characterize, by necessary and sufficient conditions on μ , when there exists a compact convex body $K \subset \mathbb{R}^n$ with nonempty interior such that

$$S_{n-2}(K, \cdot) = \mu.$$

Here $S_{n-2}(K, \cdot)$ is the $(n-2)$ nd area measure defined, together with the other area measures, by the local Steiner formula

$$S_{n-1}(K + \lambda B^n, \omega) = \sum_{j=0}^{n-1} \binom{n-1}{j} S_j(K, \omega) \lambda^{n-1-j}$$

for every Borel set $\omega \subset S^{n-1}$ and every $\lambda \geq 0$, where B^n is the Euclidean unit ball and S_{n-1} is surface-area measure.

Reference supplied by the database. Erwin Lutwak, Dongmeng Xi, Deane Yang, and Gaoyong Zhang, “Chord Measures in Integral Geometry and Their Minkowski Problems,” *Communications on Pure and Applied Mathematics* 77 (2024), no. 7, 3277–3330, the “Chord Minkowski Problem” discussion following equation (1.21), DOI: <https://doi.org/10.1002/cpa.22190>; arXiv:2502.08082.

8.1.3.3. Open problem 3

Let $n \geq 4$ and let $K \subset \mathbb{R}^n$ be a compact convex body with nonempty interior. For $q > 0$, define the chord integral

$$I_q(K) = \int_{\mathcal{L}^n} |K \cap \ell|^q d\ell,$$

where $\mathcal{L}^n$ is the space of affine lines, $|K \cap \ell|$ is the length of the chord $K \cap \ell$, and the rigid-motion-invariant measure $d\ell$ is normalized to be probability measure on rotations and $(n-1)$ -dimensional Lebesgue measure on parallel translations. Define the q th chord measure $F_q(K, \cdot)$ on S^{n-1} by the first-variation identity

$$\left. \frac{d}{dt} \right|_{t=0+} I_q(K + tL) = \int_{S^{n-1}} h_L(u) dF_q(K, u)$$

for every compact convex body $L \subset \mathbb{R}^n$, where $h_L(u) = \max_{x \in L} \langle x, u \rangle$. Determine whether, for every Borel set $\eta \subset S^{n-1}$,

$$\lim_{q \rightarrow 0+} F_q(K, \eta) = \frac{(n-1)\omega_{n-1}}{n\omega_n} S_{n-2}(K, \eta),$$

where ω_m is the volume of the Euclidean unit ball in $\mathbb{R}^m$ and $S_{n-2}(K, \cdot)$ is the $(n-2)$ nd area measure characterized by the local Steiner formula

$$S_{n-1}(K + \lambda B^n, \omega) = \sum_{j=0}^{n-1} \binom{n-1}{j} S_j(K, \omega) \lambda^{n-1-j}.$$

Reference supplied by the database. Erwin Lutwak, Dongmeng Xi, Deane Yang, and Gaoyong Zhang, “Chord Measures in Integral Geometry and Their Minkowski Problems,” *Communications on Pure and Applied Mathematics* 77 (2024), no. 7, 3277–3330, the explicit continuity question following equation (1.18), DOI: <https://doi.org/10.1002/cpa.22190>; arXiv:2502.08082.

8.1.4. Primary-source pointer

The originating ICM record is *Some Open Questions on Convex Surfaces* (1974). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1974.1/ICM1974.1.ocr.pdf#page=535>.

8.2. 247. Some Open Questions on Convex Surfaces [1974-geom-234]

Record: 1974-geom-234

Subfield: Convex geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1974: Some Open Questions on Convex Surfaces

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1974.1/ICM1974.1.ocr.pdf#page=535>

8.2.1. Problem statement

Let $n \geq 3$ and let $1 \leq p < q \leq n - 1$. For a convex body $K \subset \mathbb{R}^n$ (a compact convex set with nonempty interior), let $S_j(K, \cdot)$, $0 \leq j \leq n - 1$, denote its j -th area measure on the unit sphere S^{n-1} , defined by the local Steiner formula

$$S_{n-1}(K + \lambda B^n, \omega) = \sum_{j=0}^{n-1} \binom{n-1}{j} S_j(K, \omega) \lambda^{n-1-j}$$

for every Borel set $\omega \subseteq S^{n-1}$ and every $\lambda \geq 0$, where B^n is the Euclidean unit ball and $S_{n-1}(K + \lambda B^n, \cdot)$ is the surface area measure of $K + \lambda B^n$.

Determine whether, for every convex body $K \subset \mathbb{R}^n$, there necessarily exists a convex body $L \subset \mathbb{R}^n$, possibly different from K , such that

$$S_q(L, \cdot) = S_p(K, \cdot).$$

8.2.2. Historical background, literature, and current progress

Verdict: unresolved.

The question is affirmative when $q = n - 1$. Every area measure $S_p(K, \cdot)$ is centered and, for a full-dimensional convex body, is not concentrated on a great subsphere; the Aleksandrov-Fenchel-Jessen solution of the Minkowski problem therefore realizes it as the surface area measure $S_{n-1}(L, \cdot)$ of another convex body. Firey already records this endpoint argument in the source below. Consequently the full question is affirmative in dimension 3, where $(p, q) = (1, 2)$ is the only possible pair.

For intermediate target order $q \leq n - 2$, no complete answer for arbitrary convex bodies is known. The 2025 survey by Huang, Yang, and Zhang states that prescribing a general Borel measure as $S_i(K, \cdot)$ remains unsolved for every $1 < i < n - 1$; its strongest general smooth existence result is the Guan-Ma theorem, which requires the additional condition that the homogeneous extension of $f^{-1/i}$ be convex. There is no verified result showing that $f = dS_p(K)/d\omega$, or a general possibly singular measure $S_p(K, \cdot)$, automatically satisfies such a sufficient condition for target order q . Thus these results do not close the cases $n \geq 4$ and $p < q \leq n - 2$.

There is substantial special-case progress. Firey classified sufficiently regular area measures

of bodies of revolution, which implies the desired passage from a lower to a higher order in that smooth rotationally symmetric setting. Mussnig and Ulivelli's Theorem 1.1 removes the regularity restriction for bodies of revolution, and Section 5.5 explicitly deduces that if $\mu = S_j(K, \cdot)$ for such a body, then $\mu = S_k(L, \cdot)$ for every $k > j$. This does not cover convex bodies without rotational symmetry.

Key references:

- William J. Firey, "Some Open Questions on Convex Surfaces," *Proceedings of the International Congress of Mathematicians, Vancouver 1974*, Vol. 1, Canadian Mathematical Congress, 1975, pp. 479–484. Direct source: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1974.1/ICM1974.1.ocr.pdf>. This poses the exact question and proves the endpoint $q = n - 1$.
- William J. Firey, "Intermediate Christoffel-Minkowski Problems for Figures of Revolution," *Israel Journal of Mathematics* 8 (1970), 384–390. DOI: <https://doi.org/10.1007/BF02798684>. This gives the smooth body-of-revolution classification underlying the affirmative special case.
- Pengfei Guan and Xi-Nan Ma, "The Christoffel-Minkowski Problem I: Convexity of Solutions of a Hessian Equation," *Inventiones Mathematicae* 151 (2003), 553–577. DOI: <https://doi.org/10.1007/s00222-002-0259-2>. This proves existence and regularity for intermediate area measures under an additional convexity hypothesis on the prescribed density, not for every $S_p(K, \cdot)$.
- Yong Huang, Deane Yang, and Gaoyong Zhang, "Minkowski Problems for Geometric Measures," *Bulletin of the American Mathematical Society* 62 (2025), 359–425. DOI: <https://doi.org/10.1090/bull/1858>; arXiv:2502.05427. Section 5.3 surveys the exact area-measure prescription problem and states that the intermediate cases remain unsolved for general measures.
- Fabian Mussnig and Jacopo Ulivelli, "Explicit Solutions to Christoffel-Minkowski Problems and Hessian Equations under Rotational Symmetries," arXiv preprint, version 2, 2026. arXiv:2508.11600, <https://arxiv.org/abs/2508.11600>. Theorem 1.1 solves the body-of-revolution prescription problem without regularity assumptions, and Section 5.5 explicitly derives the lower-order-to-higher-order realization in that class.

8.2.3. Remaining open problems recorded by the database

8.2.3.1. Open problem 1

Let $n \geq 4$ and let $1 \leq p < q \leq n - 2$. For a convex body $K \subset \mathbb{R}^n$ (a compact convex set with nonempty interior), let $S_j(K, \cdot)$, $0 \leq j \leq n - 1$, denote its j -th area measure on the unit sphere S^{n-1} , defined by the local Steiner formula

$$S_{n-1}(K + \lambda B^n, \omega) = \sum_{j=0}^{n-1} \binom{n-1}{j} S_j(K, \omega) \lambda^{n-1-j}$$

for every Borel set $\omega \subseteq S^{n-1}$ and every $\lambda \geq 0$, where B^n is the Euclidean unit ball and $S_{n-1}(K + \lambda B^n, \cdot)$ is the surface area measure of $K + \lambda B^n$.

Determine whether, for every convex body $K \subset \mathbb{R}^n$, there exists a convex body $L \subset \mathbb{R}^n$, possibly different from K , such that

$$S_q(L, \cdot) = S_p(K, \cdot).$$

Reference supplied by the database. William J. Firey, "Some Open Questions on Convex Surfaces," in *Proceedings of the International Congress of Mathematicians, Vancouver 1974*, Vol. 1, edited by Ralph D. James, Canadian Mathematical Congress, Montreal, 1975, pp. 479–484 (un-numbered question on p -th and q -th area functions), <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1974.1/ICM1974.1.ocr.pdf>.

8.2.4. Primary-source pointer

The originating ICM record is *Some Open Questions on Convex Surfaces* (1974). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1974.1/ICM1974.1.ocr.pdf#page=535>.

8.3. 248. Some Open Questions on Convex Surfaces [1974-geom-235]

Record: 1974-geom-235

Subfield: Convex geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1974: Some Open Questions on Convex Surfaces

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1974.1/ICM1974.1.ocr.pdf#page=535>

8.3.1. Problem statement

Let $n \geq 3$, let $1 \leq p < n - 1$, and let $f : S^{n-1} \rightarrow (0, \infty)$ be a prescribed smooth function, where S^{n-1} is the unit sphere in $\mathbb{R}^n$. Determine whether there exists a smooth strictly convex embedding $X : S^{n-1} \rightarrow \mathbb{R}^n$ such that, at every $u \in S^{n-1}$,

$$f(u) = \sigma_p(R_1(X(u)), \dots, R_{n-1}(X(u))) = \sum_{1 \leq i_1 < \dots < i_p \leq n-1} R_{i_1}(X(u)) \cdots R_{i_p}(X(u)),$$

where $R_1, \dots, R_{n-1}$ are the principal radii of curvature of the convex hypersurface $X(S^{n-1})$. The data are free in the sense that X is not required to be the inverse Gauss map, so the parameter u need not be the outer unit normal at $X(u)$.

8.3.2. Historical background, literature, and current progress

The problem remains unresolved. Firey posed precisely the free-data question for the intermediate elementary symmetric functions of the principal radii, after distinguishing it from the normal-data Christoffel–Minkowski problem. I found no later theorem or counterexample that treats every positive smooth datum, every $n \geq 3$, and every $1 \leq p < n - 1$.

The relevant endpoint does not close this gap. Herman Gluck, “**The generalized Minkowski problem in differential geometry in the large,**” *Annals of Mathematics* 96 (1972), 245–276, DOI <https://doi.org/10.2307/1970788>, proves the free-data existence theorem for Gauss–Kronecker curvature. Equivalently, this settles the product $R_1 \cdots R_{n-1}$, the excluded endpoint $p = n - 1$, but not any intermediate p .

For $p = 1$, William J. Firey, “**Christoffel’s problem for general convex bodies,**” *Mathematika* 15 (1968), 7–21, DOI <https://doi.org/10.1112/S0025579300002321>, gives necessary and sufficient conditions when the sum of the principal radii is prescribed as a function of the outer normal. This is normal data: the unresolved free-data step is to find a diffeomorphism of the sphere that turns an arbitrary prescribed f into an admissible normal datum.

For intermediate orders, Pengfei Guan and Xi-Nan Ma, “**The Christoffel–Minkowski problem. I. Convexity of solutions of a Hessian equation,**” *Inventiones Mathematicae* 151 (2003), 553–577, DOI <https://doi.org/10.1007/s00222-002-0259-2>, prove normal-data existence and uniqueness when the prescribed function satisfies the centroid condition and an additional spherical-convexity condition on its negative p th root. Paul Bryan, Moham-

mad N. Ivaki, and Julian Scheuer, “Christoffel–Minkowski flows,” *Transactions of the American Mathematical Society* 376 (2023), 2373–2393, DOI <https://doi.org/10.1090/tran/8683>, reprove this conditional result by curvature flow and explicitly formulate the regular problem as finding necessary and sufficient conditions. These hypotheses are sufficient but not necessary, and neither paper proves that an arbitrary free datum can be rearranged by a diffeomorphism to satisfy them. Thus the verified progress covers endpoints and restricted normal-data classes, while arbitrary free data remain open.

The proposing source is William J. Firey, “Some Open Questions on Convex Surfaces,” in *Proceedings of the International Congress of Mathematicians, Vancouver 1974*, vol. 1, Canadian Mathematical Congress, 1975, 479–484, [official proceedings PDF](#).

8.3.3. Remaining open problems recorded by the database

8.3.3.1. Open problem 1

Let $n \geq 3$ and let $g : S^{n-1} \rightarrow (0, \infty)$ be a prescribed smooth function. Determine whether there exists a smooth strictly convex embedding $X : S^{n-1} \rightarrow \mathbb{R}^n$ such that

$$g(u) = R_1(X(u)) + \cdots + R_{n-1}(X(u))$$

for every $u \in S^{n-1}$, where $R_1, \dots, R_{n-1}$ are the principal radii of curvature of $X(S^{n-1})$ and the parameter u is not required to be the outer unit normal at $X(u)$.

Reference supplied by the database. William J. Firey, “Some Open Questions on Convex Surfaces,” *Proceedings of the International Congress of Mathematicians, Vancouver 1974*, vol. 1, Canadian Mathematical Congress, 1975, pp. 479–484 (the “In particular” free-data question), <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1974.1/ICM1974.1.ocr.pdf>.

8.3.3.2. Open problem 2

Let $n \geq 4$ and $1 < p < n-1$. Characterize the positive smooth functions $\varphi : S^{n-1} \rightarrow (0, \infty)$ satisfying

$$\int_{S^{n-1}} u \varphi(u) d\omega(u) = 0$$

for which there exists a smooth strictly convex body $K \subset \mathbb{R}^n$ whose support function $s : S^{n-1} \rightarrow \mathbb{R}$ satisfies

$$\sigma_p(\nabla_{S^{n-1}}^2 s + s g_{S^{n-1}}) = \varphi \quad \text{on } S^{n-1},$$

where $g_{S^{n-1}}$ is the round metric and the eigenvalues of $\nabla_{S^{n-1}}^2 s + s g_{S^{n-1}}$ are the principal radii of curvature of ∂K at the point whose outer unit normal is u .

Reference supplied by the database. Paul Bryan, Mohammad N. Ivaki, and Julian Scheuer, “Christoffel–Minkowski Flows,” *Transactions of the American Mathematical Society* 376 (2023), no. 4, 2373–2393 (regular Christoffel–Minkowski problem in the introduction), DOI: 10.1090/tran/8683; arXiv:2005.14680.

8.3.4. Primary-source pointer

The originating ICM record is *Some Open Questions on Convex Surfaces* (1974). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1974.1/ICM1974.1.ocr.pdf#page=535>.

8.4. 249. Some Open Questions on Convex Surfaces [1974-geom-236]

Record: 1974-geom-236

Subfield: Convex geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1974: Some Open Questions on Convex Surfaces

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1974.1/ICM1974.1.ocr.pdf#page=535>

8.4.1. Problem statement

Let $n \geq 3$, let $\Omega = S^{n-1} \subset \mathbb{R}^n$, and fix an integer p with $1 \leq p \leq n-1$. Let $B \in \mathcal{Q}(\Omega)$ be a prescribed C^1 quadratic differential form on Ω ; thus $B(u, du) = \langle du B(u), du \rangle$ for a symmetric matrix field on tangent vectors to Ω . Determine whether there exist a smooth convex hypersurface Σ , meaning the boundary of a compact convex body with nonempty interior in $\mathbb{R}^n$, and a smooth embedding $F : \Omega \rightarrow \Sigma$ such that $F^* I_p^\Sigma = B$. Here, if $x : \Omega \rightarrow \Sigma$ is the inverse Gauss map of Σ , H is its support function extended positively homogeneously to $\mathbb{R}^n$, and $\mathcal{H}(u) = D^2 H(u)$, then the p -th fundamental form is defined in Gauss-map coordinates by

$$(x^* I_p^\Sigma)_u(du, du) = \langle du \mathcal{H}(u)^p, du \rangle.$$

The embedding F is not required to parametrize Σ by its outer unit normals.

8.4.2. Historical background, literature, and current progress

The problem is **unresolved**. No citable result located gives a global existence criterion or construction for every prescribed form B , every $n \geq 3$, and every $1 \leq p \leq n-1$ in the free-data sense of the statement.

William J. Firey, “Some Open Questions on Convex Surfaces,” *Proceedings of the International Congress of Mathematicians, Vancouver 1974*, Vol. 1 (1975), 479–484, [direct PDF](#), defines I_p by $\langle du \mathcal{H}^p, du \rangle$, proves a necessary-and-sufficient condition for the different problem in which the tensor is prescribed in Gauss-map coordinates (“normal data”), and explicitly leaves the corresponding free-data problem open.

There is a complete special case when $n = 3$ and $p = 2$, where I_2 is the ordinary induced metric. Hans Lewy, “On the Existence of a Closed Convex Surface Realizing a Given Riemannian Metric,” *Proceedings of the National Academy of Sciences* 24 (1938), 104–106, <https://doi.org/10.1073/pnas.24.2.104>, proves analytic realization under positive Gauss curvature. Louis Nirenberg, “The Weyl and Minkowski Problems in Differential Geometry in the Large,” *Communications on Pure and Applied Mathematics* 6 (1953), 337–394, <https://doi.org/10.1002/cpa.3160060303>, solves the sufficiently smooth positive-curvature Weyl problem. These results realize only the classical first-fundamental-form case and do not address the other values of p or arbitrary C^1 forms.

For the narrower $n = 3, p = 1$ case, Elie Cartan proved local solvability for real-analytic non-degenerate prescribed second fundamental forms. Robert L. Bryant, “On Surfaces with Prescribed Shape Operator,” *Results in Mathematics* 40 (2001), 88–121, <https://doi.org/10.1007/BF03322701>, [arXiv:math/0107083](https://arxiv.org/abs/math/0107083), summarizes Cartan’s theorem and states that little is known in the smooth category or global setting. Bryant’s paper studies prescribed shape operators, which are not prescribed quadratic forms, so its classification does not resolve Firey’s problem.

Thus the literature establishes local analytic progress and one classical global special case, while the global smooth free-data realization problem for the remaining fundamental forms, and a unified criterion covering the exact statement, remain open.

8.4.3. Remaining open problems recorded by the database

8.4.3.1. Open problem 1

Let B be a smooth positive-definite quadratic differential form on S^2 . Determine whether there exists a smooth strictly convex embedding $F : S^2 \rightarrow \mathbb{R}^3$ whose second fundamental form, computed with respect to the outward unit normal and pulled back to S^2 by F , is equal to B .

Reference supplied by the database. Robert L. Bryant, “On Surfaces with Prescribed Shape Operator,” *Results in Mathematics* 40 (2001), 88–121, Section 1.4 (“Prescribed second fundamental form”), <https://doi.org/10.1007/BF03322701>; [arXiv:math/0107083](https://arxiv.org/abs/math/0107083), <https://arxiv.org/abs/math/0107083>.

8.4.4. Primary-source pointer

The originating ICM record is *Some Open Questions on Convex Surfaces* (1974). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1974.1/ICM1974.1.ocr.pdf#page=535>.

8.5. 250. Some Open Questions on Convex Surfaces [1974-geom-237]

Record: 1974-geom-237

Subfield: Convex geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1974: Some Open Questions on Convex Surfaces

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1974.1/ICM1974.1.ocr.pdf#page=535>

8.5.1. Problem statement

Let $n \geq 2$, let $\Omega = S^{n-1} \subset \mathbb{R}^n$, and let $\Sigma = \partial K$ and $\Sigma' = \partial K'$ be the boundaries of compact convex bodies with nonempty interiors. For $0 \leq p \leq n-1$, let $S_p(\Sigma; \cdot)$ denote the p -th area measure of Σ , characterized by the local Steiner formula

$$S_{n-1}(\partial(K + \lambda B); \omega) = \sum_{p=0}^{n-1} \binom{n-1}{p} S_p(\Sigma; \omega) \lambda^{n-p-1}$$

for every $\lambda \geq 0$ and every Borel set $\omega \subseteq \Omega$, where B is the Euclidean unit ball and $S_{n-1}(\partial(K + \lambda B); \omega)$ is the $(n-1)$ -dimensional area of the inverse spherical image of ω on $\partial(K + \lambda B)$.

Fix Σ' , and define a Borel measure $\mu_{\Sigma, \Sigma'}$ on Ω by

$$\mu_{\Sigma, \Sigma'}(\omega) = \sum_{p=0}^{n-1} \binom{n-1}{p} S_p(\Sigma; \omega) S_{n-p-1}(\Sigma'; \Omega).$$

Determine whether $\mu_{\Sigma, \Sigma'}$ determines Σ uniquely up to translation; equivalently, determine whether

$$\mu_{\Sigma_1, \Sigma'} = \mu_{\Sigma_2, \Sigma'}$$

implies that Σ_1 and Σ_2 are translates for all convex surfaces $\Sigma_1, \Sigma_2 \subset \mathbb{R}^n$.

8.5.2. Historical background, literature, and current progress

The problem is **unresolved**. For a smooth strictly convex body, the density of the measure in the statement is a positive linear combination of the elementary symmetric functions of the principal radii of curvature. Equality of two such measures is therefore the uniqueness problem for a non-homogeneous mixed Christoffel–Minkowski equation. The problem is solved in ambient dimensions two and three, but no theorem or counterexample covering arbitrary convex bodies in ambient dimension at least four was found. In fact, the recent literature explicitly continues to list higher-dimensional uniqueness as open, even in the smooth PDE setting.

Key references and established progress:

- William J. Firey, “Kinematic Measures for Sets of Support Figures,” *Mathematika* **21**

(1974), 270–281, DOI <https://doi.org/10.1112/S0025579300008664>. Firey derives the kinematic measure as the stated linear combination of area measures and poses its uniqueness problem.

- Rolf Schneider, “Bestimmung eines konvexen Körpers durch gewisse Beruhrmasse,” *Archiv der Mathematik* **27** (1976), 99–105, DOI <https://doi.org/10.1007/BF01224647>. Schneider proves uniqueness up to translation for general measures in ambient dimension three, removing the smoothness assumption from Firey’s three-dimensional result. Ambient dimension two follows directly because the order-zero area measure is fixed and the remaining term is the ordinary surface-area measure.
- Pengfei Guan, Zhizhang Wang, and Xiangwen Zhang, “A Proof of Alexandrov’s Uniqueness Theorem for Convex Surfaces in $\mathbb{R}^3$,” *Annales de l’Institut Henri Poincaré C, Analyse non lineaire* **33** (2016), 329–336, DOI <https://doi.org/10.1016/j.anihpc.2014.09.011>. Their Theorem 1 gives the smooth three-dimensional rigidity theorem for any C^1 function of the two principal curvatures whose two partial derivatives have product greater than zero; the kinematic density satisfies this monotonicity condition.
- Pengfei Guan and Xiangwen Zhang, “A Class of Curvature Type Equation,” *Pure and Applied Mathematics Quarterly* **17** (2021), 865–907, DOI <https://doi.org/10.4310/PAMQ.2021.v17.n3.a2>. Equation (1.2) is precisely the smooth equation obtained by prescribing a positive linear combination of area measures. The authors state that existence and uniqueness of convex solutions remain open when a lower-order coefficient is nonzero. Their Theorem 1.1 proves existence of an admissible solution under a symmetry-group hypothesis, but does not prove the uniqueness required here and does not by itself ensure that the solution is a convex hypersurface.
- Paul Bryan, Mohammad N. Ivaki, and Julian Scheuer, “Constant Rank Theorems for Curvature Problems via a Viscosity Approach,” *Calculus of Variations and Partial Differential Equations* **62** (2023), article 98, DOI <https://doi.org/10.1007/s00526-023-02442-5>. Theorem 2.4 supplies the missing full-rank/strict-convexity conclusion for a class of non-homogeneous sums of elementary symmetric functions under explicit conditions. It is regularity progress, not a uniqueness theorem.
- Mohammad N. Ivaki, “Uniqueness of Solutions to a Class of Non-Homogeneous Curvature Problems,” 2023, arXiv:2307.06252. The introduction explicitly states that kinematic-measure uniqueness is still open in sphere dimension $n \geq 3$, equivalently ambient dimension at least four. Theorem 1.1 proves a genuine special case: for constant data, every even C^2 convex solution in the stated class is a centered sphere. This does not compare arbitrary pairs of convex bodies with an arbitrary common measure, so it does not resolve the problem in the statement.

Thus the remaining gap is higher-dimensional global injectivity, up to translations, of the fixed positive linear combination $\Sigma \mapsto \mu_{\Sigma, \Sigma'}$ for arbitrary Borel area measures (and already for general smooth strictly convex surfaces). None of the cited partial results exhausts that case.

8.5.3. Remaining open problems recorded by the database

8.5.3.1. Open problem 1

Let $n \geq 4$ and $\Omega = S^{n-1}$. For a C^2 strictly convex surface $\Sigma \subset \mathbb{R}^n$ with positive principal radii of curvature $R_1(u), \dots, R_{n-1}(u)$ at outer normal $u \in \Omega$, define its p -th area measure,

$0 \leq p \leq n-1$, by

$$dS_p(\Sigma; u) = \frac{e_p(R_1(u), \dots, R_{n-1}(u))}{\binom{n-1}{p}} du,$$

where e_p is the p -th elementary symmetric polynomial and du is spherical area measure. Fix another such surface Σ' , and define

$$\mu_{\Sigma, \Sigma'}(\omega) = \sum_{p=0}^{n-1} \binom{n-1}{p} S_p(\Sigma; \omega) S_{n-p-1}(\Sigma'; \Omega)$$

for every Borel set $\omega \subseteq \Omega$. Determine whether, for every fixed Σ' , equality $\mu_{\Sigma_1, \Sigma'} = \mu_{\Sigma_2, \Sigma'}$ for two C^2 strictly convex surfaces $\Sigma_1, \Sigma_2 \subset \mathbb{R}^n$ with positive principal radii of curvature implies that Σ_1 and Σ_2 are translates.

Reference supplied by the database. Mohammad N. Ivaki, “Uniqueness of Solutions to a Class of Non-Homogeneous Curvature Problems,” 2023, Introduction (the uniqueness question for kinematic measures in sphere dimension $n \geq 3$), arXiv:2307.06252, <https://arxiv.org/abs/2307.06252>.

8.5.4. Primary-source pointer

The originating ICM record is *Some Open Questions on Convex Surfaces* (1974). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1974.1/ICM1974.1.ocr.pdf#page=535>.

8.6. 251. Some Open Questions on Convex Surfaces [1974-geom-273]

Record: 1974-geom-273

Subfield: Convex geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1974: Some Open Questions on Convex Surfaces

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1974.1/ICM1974.1.ocr.pdf#page=535>

8.6.1. Problem statement

Let $n \geq 3$, let $\mathbb{S}^{n-1} \subset \mathbb{R}^n$ be the unit sphere, and let $g : \mathbb{S}^{n-1} \rightarrow (0, \infty)$ be continuous. Determine whether there always exist a closed strictly convex C^2 hypersurface $\Sigma \subset \mathbb{R}^n$ with positive principal curvatures and a C^∞ diffeomorphism $h : \mathbb{S}^{n-1} \rightarrow \mathbb{S}^{n-1}$ isotopic to the identity such that

$$g(h(u)) = R_1(u) + \cdots + R_{n-1}(u) \quad \text{for every } u \in \mathbb{S}^{n-1},$$

where $R_1(u), \dots, R_{n-1}(u)$ are the principal radii of curvature of Σ at the point whose outer unit normal is u . Equivalently, determine whether every positive continuous normal vector field $g(u)u$ can be deformed, by reparametrizing its scalar factor through such a diffeomorphism h , into the normal vector field whose scalar factor is the sum of the principal radii of curvature of a strictly convex hypersurface.

8.6.2. Historical background, literature, and current progress

The exact problem remains **unresolved**. The literature resolves the Christoffel problem once a function of the outer normal has been fixed, and it separately shows that a positive field can be rearranged to satisfy the Christoffel equation's linear moment condition. It does not show that these two requirements can always be met by the same rearrangement while preserving the positive-definite radius tensor needed for a strictly convex C^2 hypersurface.

- William J. Firey, "Some Open Questions on Convex Surfaces," *Proceedings of the International Congress of Mathematicians, Vancouver 1974*, vol. 1, Canadian Mathematical Congress, 1975, pp. 479–484, [direct PDF](#). Firey explicitly proposes the present free-data problem after reviewing Gluck's deformation theorem and the ordinary Christoffel problem.
- Herman Gluck, "The Generalized Minkowski Problem in Differential Geometry in the Large," *Annals of Mathematics* 96 (1972), 245–276, [DOI 10.2307/1970788](https://doi.org/10.2307/1970788). Gluck proves the free-data Minkowski theorem by deforming a positive continuous normal field through a sphere diffeomorphism isotopic to the identity. In the present setting this supplies a rearrangement $f = g \circ h$ satisfying the necessary balance condition $\int_{\mathbb{S}^{n-1}} uf(u) d\omega(u) = 0$, but it does not ensure that the solution of the Christoffel equation has positive-definite spherical Hessian.

- William J. Firey, “Christoffel’s Problem for General Convex Bodies,” *Mathematika* 15 (1968), 7–21, DOI [10.1112/S0025579300002321](https://doi.org/10.1112/S0025579300002321). Firey gives necessary and sufficient conditions for a fixed function or measure to be the first area measure, equivalently the sum of principal radii in the smooth case. These conditions include a global convexity condition stronger than the balance condition, and the paper does not prove that an arbitrary g has a diffeomorphic rearrangement satisfying it.
- Qi-Rui Li, Dongrui Wan, and Xu-Jia Wang, “The Christoffel Problem by Fundamental Solution of the Laplace Equation,” preprint, 2019, [arXiv:1911.10507](https://arxiv.org/abs/1911.10507). For a fixed positive Lipschitz function f , Theorem 1.2 reformulates the exact solvability criterion as the balance condition together with an explicit condition guaranteeing convexity of the solution of $\Delta_{\mathbb{S}^{n-1}}H + (n - 1)H = f$; Theorems 3.1–3.4 give additional sufficient convexity criteria. No result there constructs, for every positive continuous g , a diffeomorphism h for which $g \circ h$ satisfies both requirements.

Thus the established progress reduces the free-data question to finding a single isotopically trivial diffeomorphism that enforces both balance and the Firey–Berg convexity criterion. No complete construction or counterexample covering every $n \geq 3$ and every positive continuous g was found; in particular, the gap remains even before addressing the full continuous-data/ C_+^2 regularity demanded in the statement.

8.6.3. Remaining open problems recorded by the database

No separate descendant problem is listed in the website’s “Some remaining open problems” field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

8.6.4. Primary-source pointer

The originating ICM record is *Some Open Questions on Convex Surfaces* (1974). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1974.1/ICM1974.1.ocr.pdf#page=535>.

8.7. 252. Geometric Algorithms and Algorithmic Geometry [1990-geom-45]

Record: 1990-geom-45

Subfield: Convex geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1990: Geometric Algorithms and Algorithmic Geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1990.1/ICM1990.1.ocr.pdf#page=227>

8.7.1. Problem statement

Determine whether the following computational problem is NP-hard: given a convex polytope $K \subset \mathbb{R}^n$ as the solution set of an explicit finite system of rational linear inequalities, and given an error tolerance $\varepsilon > 0$, compute an ellipsoid that approximates, with error at most ε , the Löwner-John ellipsoid of K , namely the unique minimum-volume ellipsoid containing K .

8.7.2. Historical background, literature, and current progress

Verdict: partially resolved.

Precise problem checked: given a bounded convex polytope K in $\mathbb{R}^n$ by an explicit rational H-description $Ax \leq b$, and a tolerance $\varepsilon > 0$, compute an ellipsoid approximating the exact Loewner-John ellipsoid of K , i.e. the unique minimum-volume ellipsoid containing K , to arbitrary prescribed accuracy. This is the H-presentation outer/minimum-volume enclosing ellipsoid problem, not the V-presentation point-set MVEE problem and not the polar maximum-volume inscribed ellipsoid problem.

The literature does not appear to contain a theorem fully resolving the Khachiyan-Todd conjecture as stated, namely NP-hardness of arbitrary-accuracy approximation of the true minimum-volume enclosing ellipsoid for all rational H-polytopes. There is substantial partial progress and closely related hardness/relaxation work, but the available citable results either cover different input presentations, the polar inscribed problem, exact/containment hardness claims without a precise arbitrary-accuracy approximation theorem for this problem, or coarse/relaxation-based outer ellipsoids.

Key literature and status:

1. Title: Geometric algorithms and algorithmic geometry Authors: Laszlo Lovasz Venue/year: Proceedings of the International Congress of Mathematicians, Kyoto 1990, Vol. 1, Springer-Verlag, 1991, pp. 139-154 DOI/arXiv/URL: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1990.1/ICM1990.1.ocr.pdf> Relevance: Section 3.1 states the original presentation-dependent contrast: for a polytope given by vertices, the Loewner-John ellipsoid can be approximated to any ε in polynomial time; for a polytope given by inequalities, the polar maximum-volume inscribed ellipsoid can be approximated in polynomial

time; Khachiyan and Todd conjectured that approximating the outer Loewner-John ellipsoid from a linear-inequality presentation is NP-hard.

2. Title: On the complexity of approximating the maximal inscribed ellipsoid for a polytope Authors: Leonid G. Khachiyan and Michael J. Todd Venue/year: Mathematical Programming 61, 137-159, 1993 DOI/arXiv/URL: <https://doi.org/10.1007/BF01582144> Relevance: This gives polynomial-time approximation results for the maximum-volume ellipsoid inscribed in an H-polytope. It resolves the polar/inscribed side mentioned by Lovasz, but it does not solve the minimum-volume circumscribed ellipsoid problem for H-polytopes.
3. Title: On Khachiyan's algorithm for the computation of minimum-volume enclosing ellipsoids Authors: Michael J. Todd and E. Alper Yildirim Venue/year: Discrete Applied Mathematics 155(13), 1731-1744, 2007 DOI/arXiv/URL: <https://doi.org/10.1016/j.dam.2007.02.013> Relevance: Provides polynomial approximation algorithms for the MVEE of a finite point set, equivalently a V-presented polytope. This is an important special case, but it does not cover an H-polytope with exponentially many vertices.
4. Title: Minimal Ellipsoid Circumscribing a Polytope Defined by a System of Linear Inequalities Authors: Jun-Ya Gotoh and Hiroshi Konno Venue/year: Journal of Global Optimization 34, 1-14, 2006 DOI/arXiv/URL: <https://doi.org/10.1007/s10898-005-3883-8> Relevance: Gives constraint-generation algorithms for the H-polytope circumscribed ellipsoid problem. The paper emphasizes that vertex-based MVEE algorithms may fail to be efficient when the inequality description has many vertices, and reports numerical success in low dimensions. It does not prove a polynomial-time arbitrary-accuracy algorithm or NP-hardness theorem resolving the conjecture.
5. Title: Sharpening geometric inequalities using computable symmetry measures Authors: Rene Brandenberg and Stefan Konig Venue/year: Mathematika 61(3), 559-580, 2015 DOI/arXiv/URL: <https://doi.org/10.1112/S0025579314000291> ; <https://arxiv.org/abs/1310.4368> Relevance: Explicitly records the state of the H-presented MVEE problem after the earlier work: it says arbitrary-accuracy approximation of the maximum-volume inscribed ellipsoid is polynomial-time, while it is not known whether the same is true for the minimum-volume enclosing ellipsoid; it cites the Khachiyan-Todd conjecture that the latter is NP-hard. It also notes a coarse multiplicative outer approximation from the inscribed ellipsoid and John's theorem, at about $(1+\epsilon)^d$, with possible symmetry-based improvements. These are approximation-factor results, not arbitrary-accuracy computation of the true Loewner-John ellipsoid.
6. Title: On the complexity of four polyhedral set containment problems Authors: Robert M. Freund and James B. Orlin Venue/year: Mathematical Programming 33, 139-145, 1985 DOI/arXiv/URL: <https://doi.org/10.1007/BF01582241> Relevance: A foundational containment-complexity reference used in later MVEP papers to justify hardness of containment/feasibility subproblems. It supports why H-description outer ellipsoid constraints are algorithmically hard, but it is not itself a theorem proving arbitrary-accuracy hardness for computing the Loewner-John ellipsoid of every rational H-polytope.
7. Title: Finding Minimum Volume Circumscribing Ellipsoids Using Generalized Copositive Programming Authors: Areesh Mittal and Grani A. Hanasusanto Venue/year: Operations Research 70(5), 2867-2882, 2022; published online 2021 DOI/arXiv/URL: <https://doi.org/10.1287/opre.2021.2156> ; <https://arxiv.org/ab>

[s/1807.07507](#) Relevance: Gives an exact generalized copositive reformulation of the minimum-volume ellipsoid problem for compact convex sets and SDP restrictions for sets described by affine and convex quadratic inequalities, including polytopes. The paper states that finding the MVEE of a polytope described by affine inequalities is NP-hard, citing Boyd-Vandenberghe and Freund-Orlin, but its main positive results are exact copositive reformulation and tractable conservative SDP approximations. These approximations return feasible outer ellipsoids and are compared to scaled MVIE methods, but they are not a full arbitrary-accuracy algorithm or a self-contained approximation-hardness theorem matching the problem statement.

8. Title: Robust Optimization for Models with Uncertain Second-Order Cone and Semidefinite Programming Constraints Authors: Jianzhe Zhen, Frans J. C. T. de Ruiter, Ernst Roos, and Dick den Hertog Venue/year: INFORMS Journal on Computing 34(1), 196-210, 2022 DOI/arXiv/URL: <https://doi.org/10.1287/ijoc.2020.1025> Relevance: Develops robust-optimization reformulations and approximation/lower-bound techniques, with numerical experiments including minimum-volume circumscribing ellipsoids of polytopes. This is further partial progress on tractable approximations and bounds, not a full resolution of the Khachiyan-Todd arbitrary-accuracy H-polytope question.

Conclusion: the exact question remains unresolved in the sense required here. The strongest current literature I found provides polynomial algorithms for V-presented MVEE and H-presented maximal inscribed ellipsoids, coarse John-type outer approximations, constraint-generation methods, exact copositive reformulations, and SDP/decision-rule approximations. I did not find a published or otherwise citable theorem whose hypotheses and conclusion fully prove NP-hardness of computing an epsilon-accurate Loewner-John ellipsoid for every rational H-polytope, nor a polynomial-time algorithm for that exact task.

8.7.3. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

8.7.4. Primary-source pointer

The originating ICM record is *Geometric Algorithms and Algorithmic Geometry* (1990). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1990.1/ICM1990.1.ocr.pdf#page=227>.

8.8. 253. On Some Inequalities for Gaussian Measures [2002-geom-251]

Record: 2002-geom-251

Subfield: Convex geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: On Some Inequalities for Gaussian Measures

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.2/ICM2002.2.ocr.pdf#page=782>

8.8.1. Problem statement

For every integer $n \geq 2$, let $S^{n-1} = \{x \in \mathbb{R}^n : \|x\|_2 = 1\}$, and let σ_{n-1} be the normalized rotation-invariant measure on S^{n-1} . Prove or disprove that if $A \subseteq S^{n-1}$ is a measurable antipodally symmetric set (that is, $A = -A$) and, for some $t \in [0, 1]$,

$$P = \{x = (x_1, \dots, x_n) \in S^{n-1} : |x_1| \leq t\}$$

satisfies $\sigma_{n-1}(A) = \sigma_{n-1}(P)$, then

$$\text{vol}_n(\text{conv}(A)) \geq \text{vol}_n(\text{conv}(P)),$$

where conv denotes the Euclidean convex hull and vol_n denotes n -dimensional Lebesgue measure.

8.8.2. Historical background, literature, and current progress

The problem is **unresolved**. The asserted extremal inequality is proved in dimensions $n = 2$ and $n = 3$, but no proof or counterexample covering any general dimension $n \geq 4$ was found. Thus the quantified statement for every $n \geq 2$ is not completely resolved.

Rafał Łatała and Marek Pycia, “On the Volume of Convex Hulls of Sets on Spheres,” *Geometriae Dedicata* **63** (1996), 153–157, DOI: <https://doi.org/10.1007/BF00148215>. They prove that a spherical cap minimizes convex-hull volume among arbitrary measurable spherical sets of fixed Haar measure. More importantly for the exact problem, they prove that among centrally symmetric sets the equatorial strip is a minimizer for $n = 2, 3$. Their higher-dimensional cap theorem does not resolve the antipodally symmetric strip problem; the reviewed proof of the symmetric result uses a cylindrical projection that ceases to be measure-preserving or measure-increasing in higher dimensions.

V. A. Zalgaller and V. N. Sudakov, “Some Problems Concerning Centrally Symmetric Convex Bodies,” *Journal of Soviet Mathematics* **10** (1978), no. 3, 454–460 (translation of *Zapiski Nauchnykh Seminarov LOMI* **45** (1974), 75–82), DOI: <https://doi.org/10.1007/BF01476853>. They prove, for $n = 2, 3$, the associated extremal problem for Gaussian integrals over origin-symmetric convex bodies, with a slab between two parallel hyperplanes as extremizer. This supplies the low-dimensional result used in later accounts but gives no theorem for $n \geq 4$.

Rafał Łatała, “On Some Inequalities for Gaussian Measures,” *Proceedings of the International Congress of Mathematicians, Beijing 2002*, Vol. II, Higher Education Press, 2002, 813–822; arXiv: [math/0304343](https://arxiv.org/abs/math/0304343). Conjecture 5.2 is the exact spherical convex-hull statement here; the survey records only $n \leq 3$ as known and explains that it would imply a broader radial-measure S -inequality. Searches of later work on spherical convex-hull inequalities, radial-measure and product-measure S -inequalities, citation databases, and equivalent formulations found no subsequent result closing the $n \geq 4$ gap.

8.8.3. Remaining open problems recorded by the database

8.8.3.1. Open problem 1

For every integer $n \geq 4$, let ν be a rotation-invariant Borel measure on $\mathbb{R}^n$ that is absolutely continuous with respect to Lebesgue measure and has density $x \mapsto f(\|x\|_2)$ for some nonincreasing function $f : [0, \infty) \rightarrow [0, \infty)$. Prove or disprove that if $K \subseteq \mathbb{R}^n$ is an origin-symmetric convex set and

$$P = \{x \in \mathbb{R}^n : |\langle x, u \rangle| \leq a\}$$

is a symmetric strip for some unit vector $u \in \mathbb{R}^n$ and $a \geq 0$, with $\nu(K) = \nu(P)$, then

$$\nu(\lambda K) \geq \nu(\lambda P)$$

for every $\lambda \geq 1$.

Reference supplied by the database. Piotr Nayar and Tomasz Tkocz, “ S -Inequality for Certain Product Measures,” *Mathematische Nachrichten* **287** (2014), no. 4, 398–404, identification of Łatała’s radial-measure conjecture as open in the Introduction, DOI: 10.1002/mana.201200294, <https://doi.org/10.1002/mana.201200294>.

8.8.4. Primary-source pointer

The originating ICM record is *On Some Inequalities for Gaussian Measures* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.2/ICM2002.2.ocr.pdf#page=782>.

8.9. 254. Isomorphic and almost-isometric problems in high-dimensional convex geometry [2006-geom-96]

Record: 2006-geom-96

Subfield: Convex geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2006: Isomorphic and almost-isometric problems in high-dimensional convex geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.2/ICM2006.2.ocr.pdf#page=1557>

8.9.1. Problem statement

Fix a positive integer k . Determine whether there exist constants $c(k), c'(k) > 0$, depending only on k , such that for every $0 < \varepsilon < 1$, if

$$N = \left\lceil c'(k) \left(\frac{1}{\varepsilon} \right)^{c(k)} \right\rceil,$$

then for every centrally symmetric convex body $K \subset \mathbb{R}^N$ (that is, $K = -K$, with K compact, convex, and with nonempty interior), there exist $r > 0$ and a k -dimensional linear subspace $E \subset \mathbb{R}^N$ such that

$$(1 - \varepsilon)rD_E \subset K \cap E \subset (1 + \varepsilon)rD_E,$$

where $D_E = \{x \in E : |x| \leq 1\}$ is the unit ball of E for the standard Euclidean norm inherited from $\mathbb{R}^N$. Determine in particular whether one may take $c(k) = (k - 1)/2$.

8.9.2. Historical background, literature, and current progress

8.9.3. Verdict: partially resolved

8.9.3.1. Precise problem

This is the fixed-dimensional, almost-isometric Dvoretzky problem: for each fixed positive integer k , whether every sufficiently high-dimensional centrally symmetric convex body has a k -dimensional central section whose Banach-Mazur distance to the Euclidean ball is $1 + O(\varepsilon)$, with the ambient dimension N bounded by a polynomial in $1/\varepsilon$. The conjectured sharp exponent is $c(k) = (k - 1)/2$.

8.9.3.2. Status

The exact problem is not fully resolved. Current literature still treats the polynomial dependence on $1/\varepsilon$ for arbitrary centrally symmetric convex bodies as the conjectural almost-isometric variant of Dvoretzky's theorem. The full result is known in dimension $k = 2$, and

polynomial/near-optimal estimates are known under strong symmetry assumptions on the norm/body, but these do not cover arbitrary centrally symmetric convex bodies for all $k \geq 3$.

8.9.3.3. Key literature and progress

1. Title: Isomorphic and almost-isometric problems in high-dimensional convex geometry
 Authors: Bo'az Klartag
 Venue: Proceedings of the International Congress of Mathematicians, Madrid 2006, Vol. II, pp. 1547-1562
 Year: 2006
 DOI/arXiv/URL: <https://www.weizmann.ac.il/math/klartag/sites/math.klartag/files/uploads/congress.pdf>
 Role: Original source for this problem's Question 1.2. Klartag states the fixed- k polynomial question, records the conjectured exponent $(k - 1)/2$, notes the special $k = 2$ result, and says that the problem was unresolved even for $k = 3$ at that time.
2. Title: A weak version of the epsilon-Dvoretzky conjecture for normed spaces
 Authors: Bo'az Klartag and Tomer Novikov
 Venue: Israel Journal of Mathematics
 Year: 2026
 DOI/arXiv/URL: DOI 10.1007/s11856-026-2911-x; arXiv:2206.13273; <https://link.springer.com/article/10.1007/s11856-026-2911-x>
 Role: Current citable status update. It states the best general Dvoretzky-type bound as $n \geq \exp(Ck|\log \varepsilon|/\varepsilon)$, which is still exponential in $1/\varepsilon$. It explicitly describes polynomial dependence as the conjectural almost-isometric variant, known for $k = 2$ or for 1-symmetric norms. Its main theorem gives a polynomial-dimensional subspace whose norm is close to an unconditional norm, not necessarily close to a Euclidean norm, so it does not imply the conclusion.
3. Title: Dichotomies, structure, and concentration in normed spaces
 Authors: Grigoris Paouris and Petros Valettas
 Venue: Advances in Mathematics 332, pp. 438-464
 Year: 2018
 DOI/arXiv/URL: DOI 10.1016/j.aim.2018.05.022; arXiv:1708.05149; <https://arxiv.org/abs/1708.05149>
 Role: Gives the best general quantitative Dvoretzky estimate cited in the recent literature: subspace dimension at least $c\varepsilon \log n / \log(1/\varepsilon)$. Inverting this for fixed k gives n exponential, not polynomial, in $1/\varepsilon$.
4. Title: A few observations on the connections between local theory and some other fields
 Authors: Vitali D. Milman
 Venue: Geometric Aspects of Functional Analysis (1986/87), Lecture Notes in Mathematics 1317, Springer, pp. 283-289
 Year: 1988
 DOI/arXiv/URL: cited in Klartag-Novikov 2026 and Klartag 2006
 Role: Source cited for the conjectural polynomial dependence and for the $k = 2$ case, whose proof is attributed to Gromov. This is a genuine special case only; it does not settle $k \geq 3$.
5. Title: Almost Euclidean sections in spaces with a symmetric basis
 Authors: Jean Bourgain and Joram Lindenstrauss
 Venue: Geometric Aspects of Functional Analysis (1987-88), Lecture Notes in Mathematics 1376, Springer, pp. 278-288

Year: 1989

DOI/arXiv/URL: cited in Klartag 2006 and Klartag-Novikov 2026

Role: Establishes polynomial/near-optimal dependence for spaces with a symmetric basis, a strong extra symmetry assumption. This is partial progress only because arbitrary centrally symmetric convex bodies need not have such symmetries.

6. Title: Almost Euclidean sections in symmetric spaces and concentration of order statistics
 Authors: Konstantin E. Tikhomirov
 Venue: Journal of Functional Analysis 265, pp. 2074-2088
 Year: 2013
 DOI/arXiv/URL: cited at <https://link.springer.com/article/10.1007/s11856-026-2911-x>
 Role: Further progress in symmetric spaces, again under symmetry assumptions not present in the problem.
7. Title: Explicit Euclidean embeddings in permutation invariant normed spaces
 Authors: Daniel Fresen
 Venue: Advances in Mathematics 266, pp. 1-16
 Year: 2014
 DOI/arXiv/URL: DOI 10.1016/j.aim.2014.07.017; arXiv:1401.0203; <https://arxiv.org/abs/1401.0203>
 Role: Gives explicit Euclidean embeddings in permutation-invariant normed spaces with $n \lesssim \log N / \log(1/\varepsilon)$. This is another special-symmetry result and does not cover arbitrary centrally symmetric convex bodies.
8. Title: Topological aspects of the Dvoretzky theorem
 Authors: Dmitri Burago, Sergei Ivanov, and Serge Tabachnikov
 Venue: Journal of Topology and Analysis 2, pp. 453-467
 Year: 2010
 DOI/arXiv/URL: DOI 10.1142/S1793525310000410; arXiv:0907.5041; <https://arxiv.org/abs/0907.5041>
 Role: Studies topological approaches and their limitations. It does not prove the full polynomial Euclidean-section statement.

8.9.3.4. What remains unresolved

No located published or citable theorem proves that for every fixed $k \geq 3$ and every centrally symmetric convex body $K \subset \mathbb{R}^N$, a $(1 + \varepsilon)/(1 - \varepsilon)$ -Euclidean k -dimensional section exists once N is merely polynomial in $1/\varepsilon$. The known full arbitrary-body estimates remain exponential in $1/\varepsilon$, while the polynomial results require extra symmetry or prove a weaker non-Euclidean conclusion.

8.9.4. Remaining open problems recorded by the database

8.9.4.1. Open problem 1

Does the almost-isometric Dvoretzky conjecture hold already for $k = 3$? More precisely, do constants $C, c > 0$ exist such that, for every $0 < \varepsilon < 1$, every centrally symmetric convex body $K \subset \mathbb{R}^N$ with $N = \lfloor c\varepsilon^{-C} \rfloor$ has a three-dimensional subspace E and an $r > 0$

satisfying

$$(1 - \varepsilon)rD_E \subset K \cap E \subset (1 + \varepsilon)rD_E?$$

Can one in particular take the conjecturally sharp exponent $C = 1$?

Reference supplied by the database. Bo'az Klartag, "Isomorphic and almost-isometric problems in high-dimensional convex geometry," in *Proceedings of the International Congress of Mathematicians, Madrid 2006*, Vol. II, European Mathematical Society, Zürich, 2006, pp. 1547–1562, discussion following Question 1.2. [EMS Press article](#).

8.9.5. Primary-source pointer

The originating ICM record is *Isomorphic and almost-isometric problems in high-dimensional convex geometry* (2006). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.2/ICM2006.2.ocr.pdf#page=1557>.

8.10. 255. Convexity, complexity, and high dimensions [2006-geom-289]

Record: 2006-geom-289

Subfield: Convex geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2006: Convexity, complexity, and high dimensions

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.2/ICM2006.2.ocr.pdf#page=1609>

8.10.1. Problem statement

Prove that there exist universal constants $a, b \geq 1$ such that, for every positive integer n and every pair of origin-symmetric convex bodies $K, B \subset \mathbb{R}^n$,

$$b^{-1} \log_2 N(B^\circ, aK^\circ) \leq \log_2 N(K, B) \leq b \log_2 N(B^\circ, a^{-1}K^\circ).$$

Here a convex body is a compact convex set with nonempty interior; origin-symmetric means $A = -A$; the polar body of a convex body A containing the origin in its interior is

$$A^\circ = \left\{ u \in \mathbb{R}^n : \sup_{x \in A} \langle x, u \rangle \leq 1 \right\};$$

and $N(A, C)$ is the least number of translates of C whose union covers A .

8.10.2. Historical background, literature, and current progress

The problem is **unresolved**. It is the full, dimension-free metric-entropy duality conjecture for two arbitrary origin-symmetric convex bodies. The literature proves important special cases and dimension-dependent substitutes, but no verified result supplies universal constants a, b for every pair K, B as required here. The two displayed inequalities are equivalent by interchanging K and B and using the bipolar theorem, so the remaining gap is already present in either one-sided inequality.

- “Duality of Metric Entropy,” S. Artstein, V. D. Milman, and S. J. Szarek, *Annals of Mathematics* 159 (2004), 1313–1328. DOI: <https://doi.org/10.4007/annals.2004.159.1313>; arXiv:math/0407236. Theorem 1 proves the conjectured dimension-free comparison when one of the two bodies is a Euclidean ball and, by affine invariance, when one body is an ellipsoid. It does not cover two arbitrary symmetric convex bodies.
- “On Convexified Packing and Entropy Duality,” S. Artstein, V. D. Milman, S. J. Szarek, and N. Tomczak-Jaegermann, *Geometric and Functional Analysis* 14 (2004), 1134–1141. DOI: <https://doi.org/10.1007/s00039-004-0486-3>; arXiv:math/0407238. This paper proves a duality theorem for convexified packing and derives the covering-number conjecture when one of the associated Banach spaces is K -convex, with constants depending on its K -convexity constant. Arbitrary finite-

dimensional normed spaces do not have a uniform bound on that parameter, so this does not yield the universal constants demanded by the problem.

- “A Remark on Two Duality Relations,” Emanuel Milman, *Integral Equations and Operator Theory* 57 (2007), 217–228. DOI: <https://doi.org/10.1007/s00020-006-1479-4>; arXiv:math/0603461. For arbitrary symmetric convex bodies it proves the general estimate

$$N(K, B) \leq N\left(B^\circ, \frac{1}{C \log(1+n)} K^\circ\right)^{C \log(1+n) \log \log(2+n)}$$

with a universal C . This is the strongest general form located here, but its dilation and exponent depend on n , precisely the loss excluded by the stated problem.

- “Metric Entropy Duality and the Sample Complexity of Outcome Indistinguishability,” Lunjia Hu, Charlotte Peale, and Omer Reingold, *Proceedings of the 33rd International Conference on Algorithmic Learning Theory, Proceedings of Machine Learning Research* 167 (2022), 515–552. arXiv:2203.04536; <https://proceedings.mlr.press/v167/hu22a.html>. Section 4 explicitly states that Pietsch’s full metric-entropy duality conjecture remains open, distinguishes the ellipsoidal result above, and proves a different variant with an additional quadratic scale-dependent factor rather than the universal-constant comparison requested here.

Thus the established progress covers ellipsoids, more generally uniformly K -convex settings, and all bodies with logarithmic dimension losses. What remains open is removal of every geometric restriction and every dependence on n , simultaneously for all symmetric convex bodies.

8.10.3. Remaining open problems recorded by the database

8.10.3.1. Open problem 1

Prove that, for every $r \in (0, 1]$, there exists a constant $c = c(r) > 0$ such that for every positive integer n and every polytope $P = \text{conv}(V) \subset \mathbb{R}^n$, where V is the set of vertices of P , if

$$rB_2^n \subset P \quad \text{and} \quad N(P, B_2^n) < e^{cn},$$

then $|V| > e^{cn}$. Here $B_2^n = \{x \in \mathbb{R}^n : \|x\|_2 \leq 1\}$, and $N(A, C)$ is the least number of translates of C whose union covers A .

Reference supplied by the database. V. D. Milman and S. J. Szarek, “A Geometric Lemma and Duality of Entropy Numbers,” in *Geometric Aspects of Functional Analysis: Israel Seminar 1996–2000*, Lecture Notes in Mathematics 1745, Springer, 2000, pp. 191–222, Conjecture 2.2. <https://doi.org/10.1007/BFb0107215>

8.10.4. Primary-source pointer

The originating ICM record is *Convexity, complexity, and high dimensions* (2006). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2006.2/ICM2006.2.ocr.pdf#page=1609>.

8.11. 256. Some questions related to the reverse Minkowski theorem [2022-geom-169]

Record: 2022-geom-169

Subfield: Convex geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2022: Some questions related to the reverse Minkowski theorem

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-564-3-volume-6.pdf#page=747>

8.11.1. Problem statement

Determine whether there are universal constants $c, C > 0$ and a fixed polynomial P such that, for every integer $n \geq 1$ and every centrally symmetric convex body $K \subseteq \mathbb{R}^n$ (that is, $K = -K$) with induced Minkowski norm

$$\|x\|_K = \inf\{s \geq 0 : x \in sK\},$$

if $X \sim N(0, I_n)$ is a standard Gaussian vector in $\mathbb{R}^n$, then

$$c \leq \mathbb{E}[\|X\|_K] \min_{\substack{W \subseteq \mathbb{R}^n \text{ linear subspace} \\ 1 \leq \dim(W) \leq n}} \text{vol}_d(\pi_W(K))^{1/d} \leq CP(\log n),$$

where π_W is orthogonal projection onto W and vol_d is d -dimensional Lebesgue volume on W .

8.11.2. Historical background, literature, and current progress

8.11.3. Verdict: partially resolved

8.11.3.1. Precise problem

The problem asks whether there are universal constants $c, C > 0$ and a fixed polynomial P such that every centrally symmetric convex body $K \subseteq \mathbb{R}^n$ satisfies

$$c \leq \mathbb{E}\|X\|_K \min_{1 \leq d = \dim W \leq n} \text{vol}_d(\pi_W K)^{1/d} \leq CP(\log n),$$

where $X \sim N(0, I_n)$. Equivalently, a large Gaussian norm expectation should always be explained, up to a polylogarithmic factor, by some low-volume orthogonal projection of K .

8.11.3.2. Current literature status

Original source

- Title: Some questions related to the reverse Minkowski theorem
- Author: Oded Regev

- Venue: Proceedings of the International Congress of Mathematicians 2022, Vol. 6, EMS Press, pp. 4898-4912
- Year: 2022; EMS online publication 2023
- DOI/URL: <https://doi.org/10.4171/ICM2022/134>, <https://ems.press/books/standalone/278/5582>
- Relevance: Section 5 formulates exactly this projection question as equation (5.3). Regev states that the analogous section bound follows from Milman-Pisier, but that the projection bound is not known for arbitrary symmetric convex bodies. The paper proves the projection bound only for Voronoi cells of lattices, using the reverse Minkowski theorem.

Earlier formulation and partial progress

- Title: Towards Strong Reverse Minkowski-type Inequalities for Lattices
- Authors: Daniel Dadush and Oded Regev
- Venue: 57th IEEE Symposium on Foundations of Computer Science (FOCS 2016), pp. 447-456
- Year: 2016
- DOI/arXiv/URL: <https://doi.org/10.1109/FOCS.2016.55>, arXiv:1606.06913, <https://arxiv.org/abs/1606.06913>
- Relevance: Section 10 introduces the same continuous convex-geometric relaxation. It records that the lower bound is standard and that the upper bound with sections $K \cap W$ follows from Milman-Pisier, but explicitly says the corresponding upper bound with projections $\pi_W(K)$ is not known for all symmetric convex bodies.
- Title: Gaussian processes and mixed volumes
- Authors: V. D. Milman and G. Pisier
- Venue: Annals of Probability 15, no. 1, pp. 292-304
- Year: 1987
- DOI/URL: <https://doi.org/10.1214/aop/1176992271>, <https://projecteuclid.org/journals/annals-of-probability/volume-15/issue-1/Gaussian-Processes-and-Mixed-Volumes/10.1214/aop/1176992271.full>
- Relevance: The volume-number theorem gives the analogous upper bound, up to $O(\log^2(n+1))$, when $\text{vol}_d(\pi_W K)$ is replaced by $\text{vol}_d(K \cap W)$. This does not imply the stated projection statement because $\pi_W K$ can be larger than $K \cap W$.
- Title: A reverse Minkowski theorem
- Authors: Oded Regev and Noah Stephens-Davidowitz
- Venue: Annals of Mathematics (2) 199, no. 1, pp. 1-49
- Year: 2024
- DOI/arXiv/URL: <https://doi.org/10.4007/annals.2024.199.1.1>, arXiv:1611.05979
- Relevance: This proves Dadush's lattice reverse Minkowski conjecture for stable lattices and implies the desired projection-type estimate for Voronoi cells of lattices. A Voronoi cell is a special centrally symmetric convex body, so this is not a proof for all K .

- Title: The Subspace Flatness Conjecture and Faster Integer Programming
- Authors: Victor Reis and Thomas Rothvoss
- Venue: arXiv preprint; conference version in FOCS 2023
- Year: 2023; updated versions through 2026
- arXiv/URL: arXiv:2303.14605, <https://arxiv.org/abs/2303.14605>
- Relevance: This settles a related Kannan-Lovasz/subspace-flatness problem up to $O(\log^3 n)$, using the reverse Minkowski theorem. Its conclusion concerns the covering radius $\mu(\Lambda, K)$ and the best volume-based lattice lower bound $\mu_{KL}(\Lambda, K)$, not the Gaussian norm expectation times the minimum projection volume for every symmetric convex body. It therefore does not resolve this problem's exact question.

8.11.3.3. What remains unresolved

The exact upper bound

$$\mathbb{E}\|X\|_K \min_{1 \leq d = \dim W \leq n} \text{vol}_d(\pi_W K)^{1/d} \leq \text{polylog}(n)$$

for every centrally symmetric convex body $K \subseteq \mathbb{R}^n$ remains unresolved in the literature I found.

Known results cover:

- the universal lower bound;
- the section analogue with $K \cap W$, by Milman-Pisier;
- the projection statement for lattice Voronoi cells, via Regev-Stephens-Davidowitz;
- related lattice covering-radius/subspace-flatness problems.

These do not cover all centrally symmetric convex bodies and therefore do not fully resolve the problem.

8.11.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

8.11.5. Primary-source pointer

The originating ICM record is *Some questions related to the reverse Minkowski theorem* (2022). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-564-3-volume-6.pdf#page=747>.

8.12. 257. Analysis of high-dimensional distributions using pathwise methods [2022-geom-265]

Record: 2022-geom-265

Subfield: Convex geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2022: Analysis of high-dimensional distributions using pathwise methods

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-564-3-volume-6.pdf#page=95>

8.12.1. Problem statement

Prove that there exists a universal constant $c > 0$ such that, for every integer $n \geq 1$ and every isotropic log-concave probability measure ν on $\mathbb{R}^n$, its Cheeger constant satisfies

$$\psi_\nu := \inf_{\substack{A \subseteq \mathbb{R}^n \text{ Borel} \\ 0 < \nu(A) < 1}} \frac{\nu^+(\partial A)}{\nu(A)(1 - \nu(A))} \geq c,$$

where isotropic means that $\int_{\mathbb{R}^n} x d\nu(x) = 0$ and $\int_{\mathbb{R}^n} xx^\top d\nu(x) = I_n$, and

$$\nu^+(\partial A) := \limsup_{\varepsilon \rightarrow 0^+} \frac{\nu(A_\varepsilon \setminus A)}{\varepsilon}, \quad A_\varepsilon := \{x \in \mathbb{R}^n : \inf_{y \in A} \|x - y\|_2 \leq \varepsilon\}.$$

8.12.2. Historical background, literature, and current progress

The problem is **unresolved**. The conjectured lower bound must be independent of both the dimension and the measure, but every currently verified general estimate still depends on the dimension. (Some references use the reciprocal “KLS constant” and use $\min\{\nu(A), 1 - \nu(A)\}$ rather than $\nu(A)(1 - \nu(A))$; these conventions differ only by universal factors.)

- Ravi Kannan, László Lovász, and Miklós Simonovits, “Isoperimetric Problems for Convex Bodies and a Localization Lemma,” *Discrete & Computational Geometry* 13 (1995), 541–559, <https://doi.org/10.1007/BF02574061>. This paper introduced the conjecture and proved the earlier dimension-dependent isoperimetric estimate of order $n^{-1/2}$ in isotropic position.
- Yin Tat Lee and Santosh S. Vempala, “Eldan’s Stochastic Localization and the KLS Hyperplane Conjecture: An Improved Lower Bound for Expansion,” *58th IEEE Symposium on Foundations of Computer Science (FOCS)* (2017), 998–1007, <https://doi.org/10.1109/FOCS.2017.96>. This established the general lower bound $\psi_\nu \gtrsim n^{-1/4}$.
- Yuansi Chen, “An Almost Constant Lower Bound of the Isoperimetric Coefficient in the KLS Conjecture,” *Geometric and Functional Analysis* 31 (2021), 34–61, <https://doi.org/10.1007/s00039-021-00558-4>; arXiv:2011.13661. Chen improved the general lower bound to $\psi_\nu \geq \exp[-O(\sqrt{\log n \log \log n})] = n^{-o(1)}$, still short of a universal positive constant.

- Bo'az Klartag, "Logarithmic Bounds for Isoperimetry and Slices of Convex Sets," *Ars Inveniendi Analytica* (2023), Paper No. 4, <https://doi.org/10.15781/jsjy-0b06>; arXiv:2303.14938. Klartag proved the KLS conjecture up to a factor $\sqrt{\log n}$, giving $\psi_\nu \gtrsim (\log n)^{-1/2}$.
- Brayden Letwin, "The KLS Constant Is $O(\log^{1/4} n)$," preprint (2026), arXiv:2607.24164. Theorem 1.1 improves the reciprocal KLS constant to $O((\log n)^{1/4})$; equivalently, up to universal convention factors, it gives the strongest located general bound $\psi_\nu \gtrsim (\log n)^{-1/4}$. The same paper explicitly continues to call the dimension-free assertion a central open problem.

Thus substantial progress has reduced the remaining loss to a fourth root of a logarithm, and the Poincaré/KLS inequality has also been proved there with a universal constant for quadratic test functions, but neither result supplies the required dimension-free Cheeger lower bound for all isotropic log-concave measures.

8.12.3. Remaining open problems recorded by the database

8.12.3.1. Open problem 1

Prove that there exists a universal constant $c > 0$ such that the following holds. For every integer $n \geq 1$, let $K \subset \mathbb{R}^n$ be a convex body that is unconditional, meaning that $(x_1, \dots, x_n) \in K$ implies $(\varepsilon_1 x_1, \dots, \varepsilon_n x_n) \in K$ for every $(\varepsilon_1, \dots, \varepsilon_n) \in \{-1, 1\}^n$, and assume that its normalized uniform probability measure μ_K is isotropic, meaning that $\int_K x d\mu_K(x) = 0$ and $\int_K xx^T d\mu_K(x) = I_n$. Then

$$\inf_{\substack{A \subseteq \mathbb{R}^n \text{ Borel} \\ 0 < \mu_K(A) < 1}} \frac{\mu_K^+(\partial A)}{\mu_K(A)(1 - \mu_K(A))} \geq c,$$

where

$$\mu_K^+(\partial A) := \limsup_{\varepsilon \rightarrow 0^+} \frac{\mu_K(A_\varepsilon \setminus A)}{\varepsilon}, \quad A_\varepsilon := \{x \in \mathbb{R}^n : \inf_{y \in A} \|x - y\|_2 \leq \varepsilon\}.$$

Reference supplied by the database. Bo'az Klartag and Joseph Lehec, "Isoperimetric Inequalities in High-Dimensional Convex Sets," *Bulletin of the American Mathematical Society* 62(4) (2025), 575–642, Section 4 after Corollary 36, <https://doi.org/10.1090/bull/1869>; arXiv:2406.01324.

8.12.3.2. Open problem 2

Prove that there exists a universal constant $c > 0$ with the following property. Let $N \geq 1$, $p \geq 2$, and $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$, and let $E_N^{\mathbb{F}}$ be the real Euclidean space of $N \times N$ self-adjoint matrices over $\mathbb{F}$, equipped with the Hilbert–Schmidt norm. Define the self-adjoint Schatten

p -ball

$$B_{p,N}^{\mathbb{F}} := \left\{ M \in E_N^{\mathbb{F}} : \left(\sum_{i=1}^N |\lambda_i(M)|^p \right)^{1/p} \leq 1 \right\},$$

where $\lambda_1(M), \dots, \lambda_N(M)$ are the eigenvalues of M . Let μ be normalized Lebesgue measure on $B_{p,N}^{\mathbb{F}}$, let Σ be its covariance operator on $E_N^{\mathbb{F}}$, and let $\tilde{\mu}$ be the pushforward of μ under $M \mapsto \Sigma^{-1/2}(M - \int M d\mu(M))$. Then

$$\inf_{\substack{A \subseteq E_N^{\mathbb{F}} \text{ Borel} \\ 0 < \tilde{\mu}(A) < 1}} \frac{\tilde{\mu}^+(\partial A)}{\tilde{\mu}(A)(1 - \tilde{\mu}(A))} \geq c,$$

where

$$\tilde{\mu}^+(\partial A) := \limsup_{\varepsilon \rightarrow 0^+} \frac{\tilde{\mu}(A_\varepsilon \setminus A)}{\varepsilon}, \quad A_\varepsilon := \left\{ X \in E_N^{\mathbb{F}} : \inf_{Y \in A} \|X - Y\|_{\text{HS}} \leq \varepsilon \right\}.$$

Reference supplied by the database. Charlie Dworaczek Guera, Ronan Memin, and Michel Pain, “CLT for β -Ensembles with Freud Weights, Application to the KLS Conjecture in Schatten Balls,” preprint (2026), Section 1.2.2 and Conjecture 1.7, arXiv:2511.05386v3, <https://arxiv.org/abs/2511.05386>.

8.12.4. Primary-source pointer

The originating ICM record is *Analysis of high-dimensional distributions using pathwise methods* (2022). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-564-3-volume-6.pdf#page=95>.

9. Applications of geometry / global analysis

These problems arise where geometric structures organize questions from mathematical physics, representation theory, topology, and nonlinear analysis. They include conjectural correspondences as well as concrete vanishing, rigidity, and enumeration statements. The research record therefore mixes theorem-driven progress with the gradual construction of a broader framework.

Coverage: 8 website records; **source years:** 1986–2018; **separately listed descendant open problems:** 15.

9.1. 258. E., Physics and geometry [1986-geom-38]

Record: 1986-geom-38

Subfield: Applications of geometry / global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1986: E., Physics and geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1986.1/ICM1986.1.ocr.pdf#page=369>

9.1.1. Problem statement

Determine the proper mathematical and logical setting underlying string theory, understood as a proposed framework for reconciling gravitation with quantum mechanics and known perturbatively through constructions involving Riemann surfaces, modular forms, and representations of infinite-dimensional Lie algebras, and formulate the fundamental generalization of the usual geometric concepts that in general relativity are expressed through Riemannian or pseudo-Riemannian geometry, metrics, connections, and curvature.

9.1.2. Historical background, literature, and current progress

9.1.3. Verdict: partially resolved

9.1.3.1. Precise problem identified

The problem is not a sharply formalized theorem. It is a broad programmatic question from Edward Witten's 1986 ICM address: find the proper mathematical and logical setting for string

theory as a framework reconciling gravitation and quantum mechanics, in which the usual Riemannian/pseudo-Riemannian language of spacetime, metrics, connections, and curvature is replaced or enlarged by a genuinely “stringy” geometry. The exact question to check is therefore whether the literature now supplies a complete mathematical framework for full string/M-theory, not only for special perturbative sectors or particular backgrounds.

9.1.3.2. Status assessment

The literature does not fully resolve this problem. There is extensive partial progress: perturbative string theory, two-dimensional conformal field theory, rational CFT, topological strings, mirror symmetry, D-branes, noncommutative geometry, derived/categorical geometry, string field theory, AdS/CFT, and matrix models all supply pieces of the new geometry Witten anticipated. However, recent surveys still treat the full definition of string theory/M-theory, especially a background-independent and generally non-perturbative formulation, as unknown. The best-developed frameworks are tied to sectors, backgrounds, extra hypotheses, or toy/simplified theories, so none covers the full statement.

9.1.3.3. Original source

- Title: “Physics and geometry”
- Author: Edward Witten
- Venue/year: Proceedings of the International Congress of Mathematicians, Berkeley 1986, Vol. 1, American Mathematical Society, 1987, pp. 267-303
- DOI/arXiv/URL: no DOI found; CERN scan: <https://cds.cern.ch/record/181783/files/er-000093203.pdf> ; IMU proceedings scan: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1986.1/ICM1986.1.ocr.pdf>
- Relevance: This is the source of the problem statement. Witten describes string theory as involving Riemann surfaces, modular forms, and infinite-dimensional Lie algebra representation theory, and says the proper logical setting is not known but should involve a fundamental generalization of usual geometry.

9.1.3.4. Partial progress and why it is not a full resolution

Mathematical foundations of perturbative string theory and QFT

- Title: “Survey of mathematical foundations of QFT and perturbative string theory”
- Authors: Hisham Sati and Urs Schreiber
- Venue/year: arXiv preprint, 2011; introduction to the AMS volume below
- DOI/arXiv/URL: arXiv:1109.0955, <https://arxiv.org/abs/1109.0955> ; DOI: <https://doi.org/10.48550/arXiv.1109.0955>
- Progress: Surveys progress on mathematical formulations of quantum field theory and perturbative string theory.
- Limitation: The scope is explicitly perturbative. It does not define full non-perturbative, background-independent string/M-theory.
- Title: “Mathematical Foundations of Quantum Field Theory and Perturbative String Theory”

- Editors: Hisham Sati and Urs Schreiber
- Venue/year: Proceedings of Symposia in Pure Mathematics, Vol. 83, American Mathematical Society, 2011, 354 pp.
- DOI/arXiv/URL: AMS page <https://bookstore.ams.org/pspum-83/>
- Progress: Collects rigorous work on higher categories, quantization, defects, orientifolds, rational CFT, conformal nets, supersymmetric field theories, and generalized cohomology.
- Limitation: The title and contents are about QFT and perturbative string theory. This is foundational progress but not a complete setting for full string theory.

Current broad surveys

- Title: “A Panorama Of Physical Mathematics c. 2022”
- Authors: Ibrahima Bah, Daniel S. Freed, Gregory W. Moore, Nikita Nekrasov, Shlomo S. Razamat, Sakura Schafer-Nameki
- Venue/year: arXiv preprint, 2022; version 3 in 2024
- DOI/arXiv/URL: arXiv:2211.04467, <https://arxiv.org/abs/2211.04467> ; DOI: <https://doi.org/10.48550/arXiv.2211.04467>
- Progress: Gives an up-to-date survey of physical mathematics, including string theory/M-theory, topological strings, perturbation theory, anomalies, noncommutative spacetime, and holography.
- Limitation: Section 3.1 asks for the definition of string theory and M-theory and answers that it is not known. It explicitly treats matrix theory and AdS/CFT as profound but partial, background-dependent answers, not satisfactory definitions of full string/M-theory.
- Title: “Les Houches lectures on non-perturbative topological strings”
- Author: Marcos Marino
- Venue/year: arXiv preprint, 2024/2025 lecture notes for the Les Houches School on Quantum Geometry
- DOI/arXiv/URL: arXiv:2411.16211, <https://arxiv.org/abs/2411.16211> ; DOI: <https://doi.org/10.48550/arXiv.2411.16211>
- Progress: Reviews non-perturbative definitions for topological strings in important toric Calabi-Yau settings via spectral theory.
- Limitation: This is for topological string theory, a simplified subsector. The introduction emphasizes that non-perturbative definitions of string theories do not exist in general.
- Title: “String Theory and Math: Why This Marriage May Last. Mathematics and dualities of quantum physics”
- Author: Mina Aganagic
- Venue/year: Bulletin of the American Mathematical Society 53 (2016), no. 1, 93-115
- DOI/arXiv/URL: arXiv:1508.06642, <https://arxiv.org/abs/1508.06642> ; DOI: <https://doi.org/10.1090/bull/1517>
- Progress: Surveys how string dualities have generated major mathematical structures and conjectures.

- Limitation: It documents rich mathematical output from string theory, not a theorem giving the full mathematical/logical setting of string theory.

String field theory

- Title: “Non-commutative geometry and string field theory”
- Author: Edward Witten
- Venue/year: Nuclear Physics B 268 (1986), no. 2, 253-294
- DOI/arXiv/URL: DOI: [https://doi.org/10.1016/0550-3213\(86\)90155-0](https://doi.org/10.1016/0550-3213(86)90155-0)
- Progress: Early string field theory paper showing algebraic/noncommutative structures in string interactions.
- Limitation: It initiated an important framework but does not give a complete definition of all string/M-theory.
- Title: “String Field Theory: A Review”
- Authors: Ashoke Sen and Barton Zwiebach
- Venue/year: arXiv preprint, 2024; expanded review for the Handbook of Quantum Gravity
- DOI/arXiv/URL: arXiv:2405.19421, <https://arxiv.org/abs/2405.19421> ; DOI: <https://doi.org/10.48550/arXiv.2405.19421>
- Progress: Reviews consistent gauge-invariant string field theories for bosonic open/closed strings, open superstrings, heterotic strings, and type II strings, using A-infinity/L-infinity structures, moduli of Riemann surfaces, CFT/BCFT, and BV quantization.
- Limitation: This is major progress for string field theory, perturbation theory, and certain non-perturbative phenomena such as tachyon condensation, but it is not a generally accepted complete mathematical/logical foundation for full background-independent string/M-theory.

Mirror symmetry and stringy geometry

- Title: “A pair of Calabi-Yau manifolds as an exactly soluble superconformal theory”
- Authors: Philip Candelas, Xenia C. de la Ossa, Paul S. Green, Linda Parkes
- Venue/year: Nuclear Physics B 359 (1991), 21-74
- DOI/arXiv/URL: DOI: [https://doi.org/10.1016/0550-3213\(91\)90292-6](https://doi.org/10.1016/0550-3213(91)90292-6)
- Progress: Found landmark mirror-symmetry computations linking Calabi-Yau geometry, CFT, and enumerative predictions.
- Limitation: Resolves important special enumerative and compactification questions, not the full string-theoretic foundation.
- Title: “Homological Algebra of Mirror Symmetry”
- Author: Maxim Kontsevich

- Venue/year: Proceedings of the International Congress of Mathematicians, Zurich 1994, Birkhauser, 1995
- DOI/arXiv/URL: arXiv:alg-geom/9411018, <https://arxiv.org/abs/alg-geom/9411018>
- Progress: Formulates homological mirror symmetry via equivalences of Fukaya categories and derived categories, a major categorical form of stringy geometry.
- Limitation: It is a mirror-symmetry program/conjectural framework, proven in many special cases, not a complete foundation for all string theory.
- Title: “Mirror Symmetry is T-Duality”
- Authors: Andrew Strominger, Shing-Tung Yau, Eric Zaslow
- Venue/year: Nuclear Physics B 479 (1996), 243-259
- DOI/arXiv/URL: arXiv:hep-th/9606040, <https://arxiv.org/abs/hep-th/9606040> ; DOI: [https://doi.org/10.1016/0550-3213\(96\)00434-8](https://doi.org/10.1016/0550-3213(96)00434-8)
- Progress: Proposes the SYZ geometric mechanism for mirror symmetry via dual special Lagrangian torus fibrations.
- Limitation: It explains a central duality in favorable Calabi-Yau settings, but not the full mathematical setting of string theory.
- Title: “Open Strings and Extended Mirror Symmetry”
- Author: Johannes Walcher
- Venue/year: Fields Institute Communications 54, “Modular Forms and String Duality”, American Mathematical Society, 2008, pp. 279-297
- DOI/arXiv/URL: AMS page <https://pubs.ams.org/view?ProductCode=FIC%2F54> ; PDF mirror <https://inspirehep.net/files/524d5ed62a8b4c90a2ac42a3709b2c2b>
- Progress: Reviews open-string extensions of mirror symmetry and D-brane geometry.
- Limitation: It explicitly says there is no complete definition of string theory and no precise idea of the final generalized geometry, so the work is partial progress.

Noncommutative geometry and matrix theory

- Title: “Noncommutative Geometry and Matrix Theory: Compactification on Tori”
- Authors: Alain Connes, Michael R. Douglas, Albert Schwarz
- Venue/year: Journal of High Energy Physics 1998, no. 02, 003
- DOI/arXiv/URL: arXiv:hep-th/9711162, <https://arxiv.org/abs/hep-th/9711162> ; DOI: <https://doi.org/10.1088/1126-6708/1998/02/003>
- Progress: Applies noncommutative geometry to Matrix-theory compactifications and noncommutative tori.
- Limitation: Covers toroidal/noncommutative backgrounds in matrix theory, not a full universal geometry of string theory.
- Title: “String Theory and Noncommutative Geometry”
- Authors: Nathan Seiberg and Edward Witten

- Venue/year: Journal of High Energy Physics 1999, no. 09, 032
- DOI/arXiv/URL: arXiv:hep-th/9908142, <https://arxiv.org/abs/hep-th/9908142> ; DOI: <https://doi.org/10.1088/1126-6708/1999/09/032>
- Progress: Shows how noncommutative gauge theory arises from open strings in a background B-field and gives the Seiberg-Witten map.
- Limitation: It identifies a precise stringy noncommutative regime, not the full mathematical/logical setting of string theory.

Conformal field theory and categorical geometry

- Title: “Conformal field theory and a new geometry”
- Author: Liang Kong
- Venue/year: Proceedings of Symposia in Pure Mathematics 83, American Mathematical Society, 2011, pp. 199-242
- DOI/arXiv/URL: arXiv:1107.3649, <https://arxiv.org/abs/1107.3649> ; DOI: <https://doi.org/10.1090/pspum/083/2742430>
- Progress: Reviews open-closed rational conformal field theory via vertex operator algebras and proposes a geometry based on CFTs and D-branes.
- Limitation: The results concern rational/open-closed CFT and a proposed new geometry. They do not supply a complete formulation of arbitrary string backgrounds or non-perturbative string/M-theory.

9.1.3.5. Conclusion

the full question remains unresolved, but it is not “fully open” in the sense of having no progress. The correct status is partially resolved: many important sectors and mathematical structures have been developed and in some cases rigorously proved, but no citable theorem or framework fully covers the entire claimed setting for string theory in all cases.

9.1.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website’s “Some remaining open problems” field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

9.1.5. Primary-source pointer

The originating ICM record is *E., Physics and geometry* (1986). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1986.1/ICM1986.1.ocr.pdf#page=369>.

9.2. 259. Mirror symmetry and moduli spaces of superconformal field theories [1994-geom-58]

Record: 1994-geom-58

Subfield: Applications of geometry / global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1994: Mirror symmetry and moduli spaces of superconformal field theories

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.2/ICM1994.2.ocr.pdf#page=600>

9.2.1. Problem statement

Determine whether the following assertion holds: Let X be a compact manifold, let g be a Ricci-flat Kähler metric on X with respect to a fixed complex structure $\mathcal{J}$, let ω_g be its Kähler form, and let $B \in H^2(X, \mathbb{R}/\mathbb{Z})$ be the B -field in the corresponding $N = (2, 2)$ supersymmetric nonlinear σ -model. If the Kähler class $[\omega_g]$ lies sufficiently deep in the Kähler cone of the complex manifold $X_{\mathcal{J}}$, then the renormalization-flow trajectory starting from the sigma-model data (g, B) determines a unique conformally invariant theory, equivalently a unique $N = (2, 2)$ superconformal field theory fixed point in the same universality class.

9.2.2. Historical background, literature, and current progress

9.2.3. Status: partially resolved

9.2.3.1. Precise problem

The problem asks for a nonperturbative existence and uniqueness statement for Calabi-Yau nonlinear sigma models: for every compact manifold X , fixed complex structure $\mathcal{J}$, Ricci-flat Kahler metric g , and B -field $B \in H^2(X, \mathbb{R}/\mathbb{Z})$, with $[\omega_g]$ sufficiently deep in the Kahler cone, the renormalization-group trajectory of the $N = (2, 2)$ nonlinear sigma model should determine a unique conformally invariant theory, equivalently a unique $N = (2, 2)$ SCFT fixed point in the same universality class.

This is stronger than Yau's theorem on Ricci-flat Kahler metrics and stronger than perturbative beta-function cancellation. It requires a mathematically or physically citable result constructing the full IR SCFT and proving uniqueness of the RG-flow endpoint for the whole class of large-radius Ricci-flat Kahler sigma-model data.

9.2.3.2. Literature checked and progress

1. David R. Morrison, "Mirror symmetry and moduli spaces of superconformal field theories."

- Venue: Proceedings of the International Congress of Mathematicians, Zurich 1994, vol. 2, Birkhauser, 1995, pp. 1304-1314.
 - arXiv: [alg-geom/9411019](https://arxiv.org/abs/alg-geom/9411019).
 - DOI/URL: <https://doi.org/10.48550/arXiv.alg-geom/9411019> and https://doi.org/10.1007/978-3-0348-9078-6_125.
 - Relevance: This is the original source of the problem statement. Morrison explicitly formulates the assertion as an expectation: Ricci-flat Kahler metrics deep in the Kahler cone are expected to determine a unique conformally invariant theory in the same universality class. The article also warns that the one-loop semiclassical sigma-model moduli space is only an approximation to the true SCFT moduli space. It is not a proof of the full claim.
2. **Luis Alvarez-Gaume, Sidney R. Coleman, and Paul H. Ginsparg, “Finiteness of Ricci Flat $N = 2$ Supersymmetric σ -Models.”**
- Venue: Communications in Mathematical Physics 103 (1986), 423.
 - URL: <https://projecteuclid.org/journals/communication-s-in-mathematical-physics/volume-103/issue-3/Finiteness-of-Ricci-flat-N2-supersymmetric-sigma-models/cmp/1104114791.pdf>.
 - Relevance: Early perturbative finiteness result for Ricci-flat $N = 2$ supersymmetric sigma models. Later four-loop computations showed that naive Ricci-flat metrics are not finite without scheme/counterterm corrections, so this does not by itself prove the full conformal fixed-point statement.
3. **Marcus T. Grisaru, A. E. M. van de Ven, and Daniela Zanon, “Four Loop beta Function for the $N = 1$ and $N = 2$ Supersymmetric Nonlinear Sigma Model in Two-Dimensions.”**
- Venue: Physics Letters B 173 (1986), 423-428.
 - DOI: [https://doi.org/10.1016/0370-2693\(86\)90408-9](https://doi.org/10.1016/0370-2693(86)90408-9).
 - Relevance: Computes higher-loop beta-function contributions and shows why naive Ricci-flatness is not enough as an uncorrected all-order statement. This is important negative/diagnostic progress, not a resolution.
4. **Marcus T. Grisaru, A. E. M. van de Ven, and Daniela Zanon, “Two-Dimensional Supersymmetric Sigma Models on Ricci Flat Kahler Manifolds Are Not Finite.”**
- Venue: Nuclear Physics B 277 (1986), 388-408.
 - DOI: [https://doi.org/10.1016/0550-3213\(86\)90448-7](https://doi.org/10.1016/0550-3213(86)90448-7).
 - Relevance: Shows a nonzero four-loop obstruction for uncorrected Ricci-flat Kahler sigma models. This confirms that the full problem cannot be reduced to “Ricci-flat implies conformal” in a naive perturbative scheme.
5. **Dennis Nemeschansky and Ashoke Sen, “Conformal Invariance of Supersymmetric Sigma Models on Calabi-Yau Manifolds.”**
- Venue: Physics Letters B 178 (1986), 365-369.
 - DOI: [https://doi.org/10.1016/0370-2693\(86\)91394-8](https://doi.org/10.1016/0370-2693(86)91394-8).
 - Relevance: This is the main positive perturbative result: by using finite local counterterms or equivalently corrections to the Ricci-flat metric order by order, the sigma-model beta functions can be made to vanish to all orders in perturbation theory. This is substantial progress toward conformal invariance, but it is still an order-by-order perturbative statement. It does not prove convergence, construct the full nonperturbative SCFT, or prove uniqueness of the RG-flow endpoint for all large-radius data.

6. Ian T. Jardine and Callum Quigley, “Conformal Invariance of (0,2) Sigma Models on Calabi-Yau Manifolds.”

- Venue: Journal of High Energy Physics 03 (2018), 090.
- arXiv: 1801.04336.
- DOI: [https://doi.org/10.1007/JHEP03\(2018\)090](https://doi.org/10.1007/JHEP03(2018)090).
- Relevance: The abstract and introduction summarize the Nemeschansky-Sen result as showing that the Ricci-flat metric can be corrected order by order in perturbation theory to produce a conformally invariant (2,2) nonlinear sigma model, and extend analogous methods to (0,2) models under additional assumptions. This confirms the scope of the known result: all-order perturbative beta-function cancellation, not a full proof of RG-flow existence and uniqueness.

7. Philip Candelas, Xenia C. de la Ossa, Paul S. Green, and Linda Parkes, “A pair of Calabi-Yau manifolds as an exactly soluble superconformal theory.”

- Venue: Nuclear Physics B 359 (1991), 21-74.
- DOI: [https://doi.org/10.1016/0550-3213\(91\)90292-6](https://doi.org/10.1016/0550-3213(91)90292-6).
- Relevance: Provides exact and highly influential SCFT/mirror-symmetry evidence in special Calabi-Yau examples, especially the quintic mirror story and massless-sector computations. It does not cover arbitrary compact Ricci-flat Kahler targets or prove uniqueness of RG-flow endpoints from the stated sigma-model data.

8. Werner Nahm and Katrin Wendland, “A hiker’s guide to K3: Aspects of $N = (4,4)$ superconformal field theory with central charge $c = 6$.”

- Venue: Communications in Mathematical Physics 216 (2001), 85-138.
- DOI: <https://doi.org/10.1007/PL00005548>.
- Relevance: Gives a detailed description of the moduli space of $N = (4,4)$, $c = 6$ SCFTs related to K3 surfaces and locates many known K3 models. This is special to the hyperkahler/K3 setting and does not imply the full $N = (2,2)$ Calabi-Yau statement for all compact targets.

9. Yi-Zhi Huang, “Some open problems in mathematical two-dimensional conformal field theory.”

- Venue: Lie Algebras, Vertex Operator Algebras, and Related Topics, Contemporary Mathematics 695, American Mathematical Society, 2017, pp. 123-138.
- arXiv: 1606.04493.
- DOI: <https://doi.org/10.1090/conm/695/13999>.
- Relevance: Describes open problems in rigorous mathematical construction of two-dimensional CFTs. The discussion reinforces that mathematically constructing interacting nonlinear sigma-model CFTs remains difficult; it does not provide a theorem resolving the claim.

9.2.3.3. Why the full problem is not resolved

The strongest directly relevant results establish perturbative, order-by-order vanishing of beta functions after correcting the sigma-model couplings. They do not prove convergence of the perturbation series, existence of the full nonperturbative RG flow, existence of an actual SCFT fixed point for every large-radius Ricci-flat Kahler input, or uniqueness of that fixed point in the same universality class.

Known exact SCFTs, toroidal/Narain models, Gepner models, quintic mirror computations, and K3 moduli-space analyses are important special cases or evidence. They have extra structure or cover restricted families and therefore do not imply the full statement for arbitrary compact Ricci-flat Kahler manifolds with B -field deep in the Kahler cone.

9.2.3.4. Verdict

The problem is **partially resolved**. The literature gives substantial perturbative and special-case evidence, but I found no published or otherwise citable theorem proving the full existence-and-uniqueness statement in all cases covered by the problem statement.

9.2.4. Remaining open problems recorded by the database

9.2.4.1. Open problem 1

Construct a rigorous correspondence under which the Fermat quintic Calabi–Yau threefold is assigned the ($N=2$) Gepner superconformal field theory built from five level-(3) ($N=2$) minimal models (with the usual projection/orbifold), and prove that this theory is the quantum nonlinear sigma model of the Fermat quintic at the Gepner point.

Reference supplied by the database. Yi-Zhi Huang, “Some Open Problems in Mathematical Two-Dimensional Conformal Field Theory,” in *Lie Algebras, Vertex Operator Algebras, and Related Topics*, Contemporary Mathematics **695**, American Mathematical Society, 2017, pp. 123–138, §4, Problem (CY-funct) (including the Fermat-quintic/Gepner-model requirement), DOI: 10.1090/conm/695/13999, arXiv:1606.04493, <https://arxiv.org/abs/1606.04493>. The special model was originally proposed in Doron Gepner, “Exactly Solvable String Compactifications on Manifolds of $SU(N)$ Holonomy,” *Physics Letters B* **199** (1987), no. 3, 380–388, DOI: 10.1016/0370-2693(87)90938-5.

9.2.5. Primary-source pointer

The originating ICM record is *Mirror symmetry and moduli spaces of superconformal field theories* (1994). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1994.2/ICM1994.2.ocr.pdf#page=600>.

9.3. 260. String Theory and Duality [1998-ag-82]

Record: 1998-ag-82

Subfield: Applications of geometry / global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1998: String Theory and Duality

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.2/ICM1998.2.ocr.pdf#page=231>

9.3.1. Problem statement

Let X and Y be Calabi-Yau threefolds used as the compact factors in ten-dimensional type II string compactifications to four-dimensional Minkowski space $M \cong \mathbb{R}^{3,1}$. For the type IIA compactification on X , suppose the local string-theoretic data consist of a Ricci-flat metric on X , a B -field in $H^2(X, \mathbb{R}/\mathbb{Z})$, a Ramond-Ramond field in $H^{\text{odd}}(X, \mathbb{R}/\mathbb{Z})$, and a dilaton-axion $\Phi \in \mathbb{C}$. For the type IIB compactification on Y , suppose the corresponding data consist of a Ricci-flat metric on Y , a B -field in $H^2(Y, \mathbb{R}/\mathbb{Z})$, a Ramond-Ramond field in $H^{\text{even}}(Y, \mathbb{R}/\mathbb{Z})$, and a dilaton-axion $\Phi \in \mathbb{C}$. Determine whether (X, Y) is a mirror pair, that is, whether these two compactified string theories are physically equivalent in the sense that the physics they determine in the uncompactified space M is isomorphic.

9.3.2. Historical background, literature, and current progress

9.3.3. Status check: mirror symmetry

Verdict: partially resolved.

9.3.3.1. Precise problem

The problem is not asking for a single enumerative mirror theorem or for Hodge-number symmetry. It uses the strongest physics formulation of Calabi-Yau threefold mirror symmetry: for Calabi-Yau threefolds X and Y , equipped with the type-II compactification data listed in the problem statement (Ricci-flat metric, B -field, Ramond-Ramond fields, and dilaton-axion), X and Y form a mirror pair exactly when type IIA string theory compactified on X and type IIB string theory compactified on Y give isomorphic four-dimensional physics on $M \cong \mathbb{R}^{3,1}$. Thus a full resolution would require a mathematically precise and proven equivalence of the full compactified string theories, including the quantum and nonperturbative data, for all cases covered by the statement.

9.3.3.2. Original/source formulation

- Title: "String Theory and Duality" Authors: Paul S. Aspinwall Venue/year: Proceedings of the International Congress of Mathematicians, Vol. II, Berlin 1998, pp. 229-238; arXiv

preprint, 1998 DOI/arXiv/URL: arXiv:math/9809004, <https://arxiv.org/abs/math/9809004> Relevance: This is the source matching the problem statement. It defines a mirror pair of Calabi-Yau threefolds as one for which type IIA on X is physically equivalent to type IIB on Y , and explicitly notes that the definition is mathematically unsatisfying because it depends on physics. It also treats the moduli-space identifications and curve-counting consequences as approximate or sectorial rather than a rigorous theorem proving full physical equivalence.

- Title: “Mirror Symmetry and the Type II String” Authors: David R. Morrison Venue/year: Nuclear Physics B Proceedings Supplements 46, 146-155, 1996 DOI/arXiv/URL: arXiv:hep-th/9512016, <https://arxiv.org/abs/hep-th/9512016> Relevance: Morrison formulates the same full-strength expectation: for a mirror pair, mirror symmetry should extend to an isomorphism between type IIA compactified on X and type IIB compactified on Y , with nonperturbative effects included. The paper studies implications for semiclassical moduli spaces; it is not a proof of the full equivalence.
- Title: “The geometry underlying mirror symmetry” Authors: David R. Morrison Venue/year: In New Trends in Algebraic Geometry, Cambridge University Press, 1999, pp. 283-310 DOI/arXiv/URL: DOI 10.1017/CBO9780511721540.011, <https://doi.org/10.1017/CBO9780511721540.011> Relevance: This survey explicitly says that precise mathematical formulations of string-theoretic mirror symmetry had been elusive, because the traditional formulation predicts equivalence of physical theories without specifying a geometric relationship. It supports treating the full physical-equivalence statement as unresolved, not as a theorem.

9.3.3.3. Partial progress and why it does not fully resolve the problem statement

- Title: “Duality in Calabi-Yau Moduli Space” Authors: Brian R. Greene and M. Ronen Plesser Venue/year: Nuclear Physics B 338, 15-37, 1990 DOI/arXiv/URL: DOI 10.1016/0550-3213(90)90622-K Progress: Gives foundational examples of mirror duality in Calabi-Yau/Gepner-orbifold moduli spaces and helped identify topologically distinct Calabi-Yau geometries yielding the same conformal field theory data. Limitation: This is a construction and analysis of important examples of world-sheet conformal field theories, not a theorem covering all Calabi-Yau threefold compactifications with the full type-II data in the problem statement.
- Title: “A Pair of Calabi-Yau Manifolds as an Exactly Soluble Superconformal Theory” Authors: Philip Candelas, Xenia C. de la Ossa, Paul S. Green, Linda Parkes Venue/year: Nuclear Physics B 359, 21-74, 1991 DOI/arXiv/URL: DOI 10.1016/0550-3213(91)90292-6 Progress: Produced the celebrated quintic mirror example and the mirror-symmetry predictions for rational curves, solving a key massless/topological sector of a particular mirror pair. Limitation: It concerns a specific family and the superconformal/topological data used for enumerative predictions. It does not prove full type IIA/type IIB equivalence for arbitrary Calabi-Yau threefolds, nor the RR/dilaton/nonperturbative parts of the stated compactification data.
- Title: “Dual Polyhedra and Mirror Symmetry for Calabi-Yau Hypersurfaces in Toric Varieties” Authors: Victor V. Batyrev Venue/year: Journal of Algebraic Geometry 3, no. 3, 493-535, 1994 DOI/arXiv/URL: arXiv:alg-geom/9310003, <https://arxiv.org/abs/alg-geom/9310003> Progress: Gives a broad mathematical construction of mirror candidate families for Calabi-Yau hypersurfaces in toric varieties via reflexive polytopes. Limitation: The result constructs mirror candidates and verifies Hodge-theoretic/combinatorial predictions in a large class. It does not prove the full physical

equivalence of type-II compactifications.

- Title: “Homological Algebra of Mirror Symmetry” Authors: Maxim Kontsevich Venue/year: Proceedings of the International Congress of Mathematicians, Zurich 1994, Birkhauser, 1995, pp. 120-139 DOI/arXiv/URL: DOI 10.1007/978-3-0348-9078-6_11; arXiv:alg-geom/9411018, <https://arxiv.org/abs/alg-geom/9411018> Progress: Formulates homological mirror symmetry as an equivalence between the derived category of coherent sheaves on one side and the Fukaya category on the mirror side. Limitation: This is a rigorous mathematical shadow of mirror symmetry for topological/open-string sectors. Even where proved, it is not the same as the full four-dimensional type-II physical equivalence with all compactification fields.
- Title: “A Mirror Theorem for Toric Complete Intersections” Authors: Alexander Givental Venue/year: Topological Field Theory, Primitive Forms and Related Topics, Progress in Mathematics 160, Birkhauser, 1998, pp. 141-175 DOI/arXiv/URL: arXiv:alg-geom/9701016, <https://arxiv.org/abs/alg-geom/9701016> Progress: Proves mirror formulas for genus-zero Gromov-Witten invariants of non-negative complete intersections in symplectic toric manifolds, a rigorous version of a major enumerative prediction of mirror symmetry. Limitation: The theorem covers a genus-zero A-model/B-model enumerative sector under toric complete-intersection hypotheses. It does not imply full equivalence of type IIA and type IIB compactified string theories.
- Title: “Mirror Principle I” and “The Candelas-de la Ossa-Green-Parkes Formula” Authors: Bong H. Lian, Kefeng Liu, Shing-Tung Yau Venue/year: Asian Journal of Mathematics 1, 729-763, 1997; Nuclear Physics B Proceedings Supplements 67, 106-114, 1998 DOI/arXiv/URL: arXiv:alg-geom/9712011, <https://arxiv.org/abs/alg-geom/9712011>; DOI 10.1016/S0920-5632(98)00126-1 Progress: Gives rigorous mirror-principle proofs including the Candelas-de la Ossa-Green-Parkes quintic formula. Limitation: These are rigorous enumerative mirror theorems, not full physical equivalence theorems for arbitrary Calabi-Yau compactifications.
- Title: “Mirror Symmetry is T-Duality” Authors: Andrew Strominger, Shing-Tung Yau, Eric Zaslow Venue/year: Nuclear Physics B 479, 243-259, 1996 DOI/arXiv/URL: DOI 10.1016/0550-3213(96)00434-8; arXiv:hep-th/9606040, <https://arxiv.org/abs/hep-th/9606040> Progress: Proposes the SYZ geometric mechanism: mirror symmetry should arise from dual special Lagrangian torus fibrations. Limitation: The paper argues for a geometric explanation and states a conjectural mechanism. The existence, compactification, singular-fiber behavior, and full connection to type-II physical equivalence are not proved in the all-case form needed by the problem statement.
- Title: “From Real Affine Geometry to Complex Geometry” Authors: Mark Gross and Bernd Siebert Venue/year: Annals of Mathematics 174, 1301-1428, 2011 DOI/arXiv/URL: DOI 10.4007/annals.2011.174.3.1; arXiv:math/0703822, <https://arxiv.org/abs/math/0703822> Progress: Constructs degenerations of Calabi-Yau manifolds from affine/tropical data and is a central result in the Gross-Siebert program, an algebro-geometric incarnation of SYZ ideas. Limitation: It constructs and controls important mirror families under degeneration/tropical hypotheses and supplies a program toward mirror symmetry. It does not prove full type-II string compactification equivalence in the stated sense.
- Title: “Dirichlet Branes and Mirror Symmetry” Authors: Paul S. Aspinwall, Tom Bridgeland, Alastair Craw, Michael R. Douglas, Mark Gross, Anton Kapustin, Gregory W. Moore, Graeme Segal, Balazs Szendroi, P. M. H. Wilson Venue/year: Clay Mathematics Monographs, Vol. 4, American Mathematical Society/Clay Mathematics Institute, 2009

DOI/arXiv/URL: URL <https://www.claymath.org/library/monographs/cmim04.pdf> Progress: Gives an extensive mathematical-physics account of D-branes and mirror symmetry. It states as Conjecture 1.3 that type IIA on a Calabi-Yau threefold X is dual to type IIB on a mirror Calabi-Yau threefold Y . Limitation: The relevant statement is explicitly presented as a conjecture. The book develops consequences and mathematical models of sectors of the conjecture rather than proving the full physical equivalence.

- Title: “Homological Mirror Symmetry for Calabi-Yau Hypersurfaces in Projective Space” Authors: Nick Sheridan Venue/year: *Inventiones Mathematicae* 199, no. 1, 1-186, 2015 DOI/arXiv/URL: DOI 10.1007/s00222-014-0507-2; arXiv:1111.0632, <https://arxiv.org/abs/1111.0632> Progress: Proves homological mirror symmetry for smooth d -dimensional Calabi-Yau hypersurfaces in projective space for $d \geq 3$, including the quintic threefold case. Limitation: This is a major categorical theorem for a specific class of mirror pairs and a specific mathematical formulation. It does not cover all Calabi-Yau threefolds and does not prove equivalence of the full type-II compactified physical theories including RR fields and dilaton-axion.
- Title: “Homological Mirror Symmetry for Generalized Greene-Plesser Mirrors” Authors: Nick Sheridan and Ivan Smith Venue/year: *Inventiones Mathematicae* 224, 627-682, 2021 DOI/arXiv/URL: DOI 10.1007/s00222-020-01018-w; arXiv:1709.08937, <https://arxiv.org/abs/1709.08937> Progress: Proves Kontsevich’s homological mirror symmetry conjecture for broad classes including Greene-Plesser mirror pairs and certain Batyrev-Borisov mirror constructions. Limitation: The hypotheses are restricted classes of mirror constructions, and the conclusion is a homological/categorical equivalence. It still does not prove the all-case, full string-theoretic physical equivalence.

9.3.3.4. Conclusion

No citable theorem found in the literature resolves exactly the full statement. The literature contains extensive and deep partial resolutions: explicit mirror constructions, rigorous enumerative mirror theorems, SYZ/Gross-Siebert geometric programs, and homological mirror symmetry proofs for important classes including the quintic and many Greene-Plesser/Batyrev-Borisov mirrors. However, these results either apply only to special classes, prove only topological/enumerative/categorical sectors, or require extra hypotheses. They do not establish a mathematically precise equivalence of the full type IIA and type IIB compactified string theories for all Calabi-Yau threefold mirror pairs with the complete field data listed in the problem statement. Therefore the status is partially resolved.

9.3.4. Remaining open problems recorded by the database

9.3.4.1. Open problem 1

For a mirror pair of Calabi-Yau threefolds $((X, Y))$, is the split-closed derived Fukaya category of (X) equivalent to the bounded derived category of coherent sheaves on (Y) ,

$$D^{\pi} \text{Fuk}(X) \simeq D^b \text{Coh}(Y),$$

with the analogous equivalence after exchanging (X) and (Y) ?

Reference supplied by the database. Maxim Kontsevich, “Homological Algebra of Mirror Symmetry,” in *Proceedings of the International Congress of Mathematicians, Zürich, 1994*, Vol. I, Birkhäuser, Basel, 1995, pp. 120–139, “Homological Mirror Conjecture,” arXiv: [alg-geom/9411018](https://arxiv.org/abs/alg-geom/9411018).

9.3.4.2. Open problem 2

Do mirror Calabi–Yau threefolds (X) and (Y), near the appropriate large-complex-structure limits, admit special Lagrangian (3)-torus fibrations over a common base whose nonsingular fibers are fiberwise dual, so that the mirror is obtained as the moduli space of the fibers together with flat (U(1))-connections (with the necessary corrections at singular fibers)?

Reference supplied by the database. Andrew Strominger, Shing-Tung Yau, and Eric Zaslow, “Mirror Symmetry Is T-Duality,” *Nuclear Physics B* 479, no. 1–2 (1996), 243–259, Strominger–Yau–Zaslow conjecture, DOI: [https://doi.org/10.1016/0550-3213\(96\)00434-8](https://doi.org/10.1016/0550-3213(96)00434-8); arXiv: [hep-th/9606040](https://arxiv.org/abs/hep-th/9606040).

9.3.4.3. Open problem 3

For every positive integer (d), does a general smooth quintic threefold in $\mathbb{P}^4$ contain only finitely many rational curves of degree (d) (and, in the strong form, are these curves smooth, isolated, and equipped with normal bundle $\mathcal{O}_{\mathbb{P}^1}(-1) \oplus \mathcal{O}_{\mathbb{P}^1}(-1)$)?

Reference supplied by the database. C. Herbert Clemens, “Some Results on Abel–Jacobi Mappings,” in Phillip A. Griffiths (ed.), *Topics in Transcendental Algebraic Geometry*, Annals of Mathematics Studies 106, Princeton University Press, Princeton, 1984, pp. 289–304, p. 300 (Clemens conjecture).

9.3.5. Primary-source pointer

The originating ICM record is *String Theory and Duality* (1998). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.2/ICM1998.2.ocr.pdf#page=231>.

9.4. 261. Mathematical Results Inspired by Physics [2002-ag-95]

Record: 2002-ag-95

Subfield: Applications of geometry / global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2002: Mathematical Results Inspired by Physics

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=446>

9.4.1. Problem statement

After fixing the underlying four-manifold and gauge data, let M_n denote the moduli space of self-dual connections in the n -th topological sector, and let $\chi(M_n)$ denote its Euler number. Prove that the Euler-number generating series

$$Z(\tau) = \sum_n \chi(M_n) q^n, \quad q = e^{2\pi i \tau},$$

is modular, in the sense that it transforms as a modular form in the modular parameter τ .

9.4.2. Historical background, literature, and current progress

9.4.3. Verdict: partially resolved

9.4.3.1. Precise problem

The problem asks for the general Vafa-Witten S-duality prediction: after fixing an oriented four-manifold and gauge/topological data, let M_n be the instanton/self-dual connection moduli space in the n -th sector. Is the Euler-number partition function

$$Z(\tau) = \sum_n \chi(M_n) q^n, \quad q = e^{2\pi i \tau},$$

modular in τ in the sense predicted by S-duality?

This is a broad all-cases question. It is not just the K3 case, the projective-plane case, rank 2, or the modern algebraic-geometry version for polarized projective surfaces.

9.4.3.2. Verdict

Partially resolved. There are rigorous results proving the predicted modularity or corrected modular/mock-modular behavior in important special cases, especially for K3 surfaces, rational/projective surfaces, and some algebraic Vafa-Witten invariants. I did not find a published or otherwise citable theorem that proves the full statement for arbitrary four-manifolds and all gauge data/topological sectors.

9.4.3.3. Key literature and status

1. Title: "A Strong Coupling Test of S-Duality"
 Authors: Cumrun Vafa and Edward Witten
 Venue: Nuclear Physics B 431, pp. 3-77
 Year: 1994
 DOI/arXiv/URL: DOI 10.1016/0550-3213(94)90097-3; arXiv:hep-th/9408074; <https://arxiv.org/abs/hep-th/9408074>
 Role: Original source of the S-duality prediction. It studies topologically twisted $N = 4$ supersymmetric Yang-Mills theory on four-manifolds and gives exact physical tests and formulas in examples. It is not a mathematical proof of the full all-four-manifold statement.
2. Title: "Euler Characteristics of $SU(2)$ Instanton Moduli Spaces on Rational Elliptic Surfaces"
 Author: Kota Yoshioka
 Venue: Communications in Mathematical Physics 205, pp. 501-517
 Year: 1999
 DOI/arXiv/URL: DOI 10.1007/s002200050687; arXiv:math/9805003; <https://arxiv.org/abs/math/9805003>
 Role: Computes Euler characteristics of $SU(2)$ instanton/Gieseker moduli spaces on rational elliptic surfaces and checks the modular property of $Z_{SU(2)}$ and $Z_{SO(3)}$ conjectured by Vafa and Witten. This is a genuine special-case verification, not a proof for all four-manifolds or all gauge groups.
3. Title: "On blowup formulae for the S-duality conjecture of Vafa and Witten"
 Authors: Wei-Ping Li and Zhenbo Qin
 Venue: Inventiones Mathematicae 136, pp. 451-482
 Year: 1999
 DOI/arXiv/URL: DOI 10.1007/s002220050316; arXiv:math/9805054; <https://arxiv.org/abs/math/9805054>
 Role: Proves blow-up formulae for virtual Hodge polynomials and Euler numbers of rank-2 sheaf/Uhlenbeck compactification spaces over algebraic surfaces, verifying the blow-up formulae predicted by the Vafa-Witten S-duality conjecture. This supports and propagates special cases but does not prove the base modularity statement in all cases.
4. Title: "From sheaves on P^2 to a generalization of the Rademacher expansion"
 Authors: Kathrin Bringmann and Jan Manschot
 Venue: American Journal of Mathematics 135(4), pp. 1039-1065
 Year: 2013
 DOI/arXiv/URL: DOI 10.1353/ajm.2013.0031; arXiv:1006.0915; <https://arxiv.org/abs/1006.0915>
 Role: Proves a Vafa-Witten conjecture for rank-2 sheaves on $\mathbb{P}^2$, expressing Euler-number generating functions as mixed mock modular forms. This is an important exact result, but it is for a specific surface and rank and also shows that the naive phrase "is a modular form" must be interpreted carefully in cases with modular anomaly/mock modularity.
5. Title: "Vafa-Witten Theory and Iterated Integrals of Modular Forms"
 Author: Jan Manschot
 Venue: Communications in Mathematical Physics 371, pp. 787-831
 Year: 2019
 DOI/arXiv/URL: DOI 10.1007/s00220-019-03389-5; arXiv:1709.10098; <https://arxiv.org/abs/1709.10098>

- Role: Analyzes modular anomaly for Vafa-Witten partition functions, including $SU(3)$ on $\mathbb{P}^2$, and identifies higher-depth mock modular behavior. This is strong progress for projective-plane/higher-rank cases, not a full all-manifold proof.
6. Title: “Vafa-Witten invariants for projective surfaces I: stable case”
 Authors: Yuuji Tanaka and Richard P. Thomas
 Venue: Journal of Algebraic Geometry 29, pp. 603-668
 Year: 2020
 DOI/arXiv/URL: DOI 10.1090/jag/738; arXiv:1702.08487; <https://arxiv.org/abs/1702.08487>
 Role: Gives a mathematical definition of Vafa-Witten invariants for polarized projective surfaces in the stable case via symmetric obstruction theories and virtual localization. When the Vafa-Witten vanishing theorem applies, the invariant is the signed Euler characteristic of the instanton moduli space; in general there are additional rational contributions. This is foundational but is limited to projective surfaces and does not prove the full statement.
 7. Title: “Vafa-Witten invariants for projective surfaces II: semistable case”
 Authors: Yuuji Tanaka and Richard P. Thomas
 Venue: Pure and Applied Mathematics Quarterly 13(3), pp. 517-562
 Year: 2017/2018
 DOI/arXiv/URL: DOI 10.4310/PAMQ.2017.v13.n3.a6; arXiv:1702.08488; <https://arxiv.org/abs/1702.08488>
 Role: Extends the definition to semistable Higgs pairs and calculates K3-surface invariants in terms of modular forms generalizing and proving Vafa-Witten predictions in that setting. This is a major special-case result for projective surfaces, not a proof for arbitrary four-manifolds and all gauge data.
 8. Title: “Sheaf counting on local K3 surfaces”
 Authors: Daves Maulik and Richard P. Thomas
 Venue: Pure and Applied Mathematics Quarterly 14(3-4), pp. 419-441
 Year: 2018/2019
 DOI/arXiv/URL: DOI 10.4310/PAMQ.2018.v14.n3.a1; arXiv:1806.02657; <https://arxiv.org/abs/1806.02657>
 Role: Proves a conjecture of Tanaka-Thomas used in the semistable K3-surface Vafa-Witten theory. This strengthens the K3 case but does not cover general four-manifolds.
 9. Title: “A proof of all ranks S-duality conjecture for K3 surfaces”
 Authors: Yunfeng Jiang and Hsian-Hua Tseng
 Venue: Asian Journal of Mathematics 28(2), pp. 317-344
 Year: 2024
 DOI/arXiv/URL: arXiv:2003.09562; <https://arxiv.org/abs/2003.09562>; journal listing <https://intlpress.com/JDetail/1829551722763767809>
 Role: Proves the all-ranks S-duality conjecture for K3 surfaces in the $SU(r)/Z_r$ Vafa-Witten framework, using Tanaka-Thomas $SU(r)$ invariants and twisted-sheaf counts. This is one of the strongest resolving-looking results, but its hypothesis is specifically K3 surfaces, so it does not imply the full statement.
 10. Title: “Virtual Refinements of the Vafa-Witten Formula”
 Authors: Lothar Goettsche and Martijn Kool
 Venue: Communications in Mathematical Physics 376, pp. 1-49
 Year: 2020
 DOI/arXiv/URL: DOI 10.1007/s00220-020-03748-7; arXiv:1703.07196; <https://arxiv.org/abs/1703.07196>

[v.org/abs/1703.07196](https://arxiv.org/abs/1703.07196)

Role: Formulates and checks refined conjectural formulas for virtual χ_y -genera and virtual Euler characteristics of rank-2 sheaf moduli spaces on surfaces with holomorphic 2-form, recovering part of Vafa-Witten's formula for minimal general type surfaces. It is explicitly conjectural in its broad form and thus is partial progress, not a full resolution.

11. Title: "SU(r) Vafa-Witten Invariants, Ramanujan's Continued Fractions, and Cosmic Strings"
 Authors: Lothar Goettsche, Martijn Kool, and Ties Laarakker
 Venue: Michigan Mathematical Journal 75(1), pp. 3-63
 Year: 2025
 DOI/arXiv/URL: DOI 10.1307/mmj/20226202; arXiv:2108.13413; <https://arxiv.org/abs/2108.13413>
 Role: Gives conjectural structure formulas for SU(r) Vafa-Witten partition functions for surfaces with holomorphic 2-form and proves/checks vertical parts in low virtual dimensions. This documents active progress but also confirms that broad higher-rank/projective-surface formulas remain conjectural beyond proved pieces.
12. Title: analytic work on Vafa-Witten equations on closed four-manifolds
 Authors/items: Yuuji Tanaka, "Some boundedness properties of solutions to the Vafa-Witten equations on closed four-manifolds," Q. J. Math. 68 (2017), pp. 1203-1225, DOI 10.1093/qmath/hax015, arXiv:1308.0862; Clifford H. Taubes, "The behavior of sequences of solutions to the Vafa-Witten equations," arXiv:1702.04610
 Role: Develops analytic compactness/boundedness behavior for the equations over closed four-manifolds. These papers are foundational for moduli-space theory, but they do not prove modularity of Euler-number generating series.

9.4.3.4. What remains unresolved

No cited theorem found proves that, for every fixed underlying four-manifold and all relevant gauge/topological data, the Euler-number generating series of self-dual connection/instanton moduli spaces transforms as the predicted modular form.

The strongest rigorous results have additional hypotheses: K3 surfaces; rational elliptic surfaces; $\mathbb{P}^2$; smooth projective/polarized algebraic surfaces; rank restrictions or specialized SU(r)/SU(r)/Z_r frameworks; stable or semistable sheaf/Higgs-pair compactifications; virtual rather than ordinary Euler characteristics; or modular completions/mock-modular forms rather than ordinary holomorphic modular forms.

Therefore the exact broad problem is not fully resolved. The current literature supports the prediction in many important cases and has a robust projective-surface theory, but the general four-manifold/gauge-data modularity statement remains open beyond those cases.

9.4.4. Remaining open problems recorded by the database

9.4.4.1. Open problem 1

Let S be a smooth projective surface with $b_1(S) = 0$, $p_g(S) > 0$, and with Seiberg–Witten basic classes 0 and $K_S \neq 0$. If $M_S^H(2, c_1, c_2)$ contains no strictly Gieseker H -semistable

sheaves, prove that its normalized virtual χ_y -genus is the coefficient of $x^{4c_2-c_1^2-3\chi(O_S)}$ in

$$8 \left(\frac{1}{2} \prod_{n \geq 1} \frac{1}{(1-x^{2n})^{10}(1-x^{2n}y)(1-x^{2n}y^{-1})} \right)^{\chi(O_S)} \left(\frac{2\bar{\eta}(x^4)^2}{\theta_3(x, y^{1/2})} \right)^{K_S^2}.$$

Reference supplied by the database. Lothar Göttsche and Martijn Kool, “Virtual Refinements of the Vafa–Witten Formula,” *Communications in Mathematical Physics* **376** (2020), 1–49, Conjecture 1.1. DOI: <https://doi.org/10.1007/s00220-020-03748-7>; arXiv: 1703.07196.

9.4.4.2. Open problem 2

For a polarized projective surface and fixed charge $\alpha = (r, c_1, c_2)$, prove the Tanaka–Thomas universal wall-crossing formula expressing the virtual-localization invariants of stable Joyce–Song Higgs pairs $P_\alpha^\perp(N)$, for all sufficiently large auxiliary twists N , in terms of rational Vafa–Witten invariants VW_{α_i} . In particular, prove in full generality that the resulting Vafa–Witten invariant is independent of N ; when $H^{0,1}(S) \neq 0$ or $H^{0,2}(S) \neq 0$, prove the predicted one-term formula

$$P_{r,L,c_2}^\perp(N) = (-1)^{\chi(\alpha(N))-1} \chi(\alpha(N)) \text{VW}_{r,L,c_2}(S).$$

Reference supplied by the database. Yuuji Tanaka and Richard P. Thomas, “Vafa–Witten Invariants for Projective Surfaces II: Semistable Case,” *Pure and Applied Mathematics Quarterly* **13** (2017), no. 3, 517–562, Conjecture 1.2 (equivalently Conjecture 6.5 in the detailed development). DOI: <https://doi.org/10.4310/PAMQ.2017.v13.n3.a6>; arXiv: 1702.08488.

9.4.4.3. Open problem 3

Let (S, H) be a smooth polarized projective surface with $H_1(S, \mathbb{Z}) = 0$ and $p_g(S) > 0$, let r be prime, and let $c_1 \in H^2(S, \mathbb{Z})$ be algebraic. Prove that the mathematically defined $\text{SU}(r)$ and $\text{SU}(r)/\mathbb{Z}_r$ Vafa–Witten partition functions are meromorphic modular functions on the upper half-plane satisfying

$$Z_{c_1}^{\text{SU}(r)}(-1/\tau) = (-1)^{(r-1)\chi(O_S)} \left(\frac{r\tau}{i} \right)^{-e(S)/2} Z_{c_1}^{\text{SU}(r)/\mathbb{Z}_r}(\tau).$$

Reference supplied by the database. Yunfeng Jiang and Martijn Kool, “Twisted Sheaves and $\text{SU}(r)/\mathbb{Z}_r$ Vafa–Witten Theory,” *Mathematische Annalen* **382** (2022), nos. 1–2, 719–743, Conjecture 1.2. DOI: <https://doi.org/10.1007/s00208-021-02303-6>; arXiv: 2006.10368.

9.4.4.4. Open problem 4

For every rank $r > 1$, prove the universal Seiberg–Witten structure formula for the virtual Euler characteristics of Gieseker–Maruyama moduli spaces $M_S^H(r, c_1, c_2)$ on smooth polarized surfaces with $b_1(S) = 0$ and $p_g(S) > 0$: namely, prove the existence of universal power series D_0 and D_{ij} , depending only on r , such that all coefficients $e^{\text{vir}}(M_S^H(r, c_1, c_2))$ are given by the formula of Göttsche–Kool–Laarakker involving only $\chi(O_S)$, K_S^2 , the Seiberg–Witten invariants of S , and the intersection products of their basic classes.

Reference supplied by the database. Lothar Göttsche, Martijn Kool, and Ties Laarakker, “SU(r) Vafa–Witten Invariants, Ramanujan’s Continued Fractions, and Cosmic Strings,” *Michigan Mathematical Journal* **75** (2025), no. 1, 3–63, Conjecture 1.10 (restated as Conjecture 4.1). DOI: <https://doi.org/10.1307/mmj/20226202>; arXiv: [2108.13413](https://arxiv.org/abs/2108.13413).

9.4.4.5. Open problem 5

For every rank $r > 1$, prove that the universal series C_0, C_{ij} governing the vertical Vafa–Witten contribution and the universal series D_0, D_{ij} governing the horizontal contribution are Fourier expansions of meromorphic functions on the upper half-plane and are exchanged by the modular S -transformation:

$$D_0(\tau) = C_0(-1/\tau), \quad D_{ij}(\tau) = C_{ij}(-1/\tau).$$

Reference supplied by the database. Lothar Göttsche, Martijn Kool, and Ties Laarakker, “SU(r) Vafa–Witten Invariants, Ramanujan’s Continued Fractions, and Cosmic Strings,” *Michigan Mathematical Journal* **75** (2025), no. 1, 3–63, Conjecture 1.13 (restated as Conjecture 4.2). DOI: <https://doi.org/10.1307/mmj/20226202>; arXiv: [2108.13413](https://arxiv.org/abs/2108.13413).

9.4.5. Primary-source pointer

The originating ICM record is *Mathematical Results Inspired by Physics* (2002). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2002.3/ICM2002.3.ocr.pdf#page=446>.

9.5. 262. Inverse problems for linear and non-linear hyperbolic equations [2018-geom-140]

Record: 2018-geom-140

Subfield: Applications of geometry / global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Inverse problems for linear and non-linear hyperbolic equations

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol4-ver1-eb.pdf#page=1116>

9.5.1. Problem statement

Let (M, g) be a 4-dimensional globally hyperbolic time-oriented Lorentzian manifold of signature $(-, +, +, +)$, written using the product representation $M = \mathbb{R} \times N$, and let $t_0 > 0$. Let $\mu \subset (0, t_0) \times N$ be a timelike path, let $V \subset M$ be an open neighborhood of μ , and let $p^-, p^+ \in \mu$ with $p^- \ll p^+$, where $p \ll q$ means that p and q can be joined by a future-pointing timelike curve. Write

$$I^+(p) = \{q \in M : p \ll q\}, \quad I^-(p) = \{q \in M : q \ll p\},$$

and define the chronological double cone

$$I(p^-, p^+) = I^+(p^-) \cap I^-(p^+).$$

For a nowhere-vanishing smooth coefficient a and a first-order differential operator $B(x, D)$ with smooth x -dependent coefficients, consider the nonlinear wave equation

$$\square_g u(x) + B(x, D)u(x) + a(x)u(x)^2 = f(x), \quad x \in (-\infty, t_0) \times N,$$

with $u = 0$ and $f = 0$ in $(-\infty, 0) \times N$. For sufficiently small C^k sources f supported in V , where k is sufficiently large, let u be the corresponding solution and let

$$L_V : f \mapsto u|_V$$

be the source-to-solution measurement operator on V . Given the timelike path μ , its neighborhood V , and the operator L_V , is it possible to construct the metric tensor g and the first-order operator $B(x, D)$ on $I(p^-, p^+)$, up to the natural local gauge ambiguity for this inverse problem, namely local changes of variables and the induced transformation of the differential operator that leave the observed source-to-solution data on V unchanged?

9.5.2. Historical background, literature, and current progress

9.5.3. Verdict: partially resolved

9.5.3.1. Precise problem checked

The problem asks whether the local active source-to-solution map near a timelike path for

$$\Box_g u + B(x, D)u + a(x)u^2 = f$$

on an arbitrary 4-dimensional globally hyperbolic Lorentzian manifold determines, on the chronological diamond $I(p^-, p^+)$, both the full Lorentzian metric tensor g and the arbitrary smooth first-order operator $B(x, D)$, up to the natural local gauge by changes of variables and induced operator transformations.

I did not find a published or citable theorem that proves this exact simultaneous recovery result for an arbitrary first-order term $B(x, D)$ in this local source-to-solution setting.

9.5.3.2. Main partial progress

1. Inverse problems for Lorentzian manifolds and non-linear hyperbolic equations

Authors: Yaroslav Kurylev, Matti Lassas, Gunther Uhlmann.

Venue/year: *Inventiones Mathematicae* 212 (2018), 781-857.

DOI/arXiv/URL: DOI <https://doi.org/10.1007/s00222-017-0780-y>; arXiv 1405.3386, <https://arxiv.org/abs/1405.3386>.

Result checked: Theorem 1.5 treats the local source-to-solution map for $\Box_g u + a(x)u^2 = f$ with nonvanishing a on a 3+1 dimensional globally hyperbolic Lorentzian manifold. It reconstructs the differentiable structure and the conformal class of g on $I(p^-, p^+)$, with the conformal factor only in additional cases such as Ricci-flat regions. It does not include an unknown first-order operator $B(x, D)$ and does not generally recover the full metric tensor.

2. Inverse problems for semilinear wave equations on Lorentzian manifolds

Authors: Matti Lassas, Gunther Uhlmann, Yiran Wang.

Venue/year: *Communications in Mathematical Physics* 360 (2018), 555-609.

DOI/arXiv/URL: DOI <https://doi.org/10.1007/s00220-018-3133-7>; arXiv 1606.06261, <https://arxiv.org/abs/1606.06261>.

Result checked: Theorems 1.2-1.5 and 6.1-6.2 extend the local source-to-solution method to general semilinear terms and scalar zeroth-order linear terms. The paper explicitly leaves determination of the cubic, quadratic, and linear Taylor terms to future work in the fixed-metric coefficient theorem, and its lower-order linear term is scalar $Q(x)u$, not an arbitrary first-order differential operator $B(x, D)u$.

3. Inverse problems for quadratic derivative nonlinear wave equations

Authors: Yiran Wang, Ting Zhou.

Venue/year: *Communications in Partial Differential Equations* 44 (2019), 1140-1178.

DOI/arXiv/URL: DOI <https://doi.org/10.1080/03605302.2019.1607895>; arXiv 1612.04437, <https://arxiv.org/abs/1612.04437>.

Result checked: Theorem 1.1 has nearly the same local timelike-path/source-to-solution geometry and recovers the conformal class of the metric, and in special cases the metric itself, for a quadratic derivative nonlinear equation. This is a different nonlinear model and does not recover a general first-order linear operator $B(x, D)$.

4. Inverse problems for non-linear hyperbolic equations with disjoint sources and receivers

Authors: Ali Feizmohammadi, Matti Lassas, Lauri Oksanen.

Venue/year: Forum of Mathematics, Pi 9 (2021), e10.

DOI/arXiv/URL: DOI <https://doi.org/10.1017/fmp.2021.8>; arXiv 2006.12158, <https://arxiv.org/abs/2006.12158>.

Result checked: Theorem 1.1 recovers the Lorentzian metric, not just its conformal class, for semilinear equations $\Box_g u + u^m = f$ with $m \geq 3$ except for the exceptional $(n, m) = (3, 3)$ case. It does not cover the quadratic nonlinearity $a(x)u^2$ in 3+1 dimensions with arbitrary a , and it has no unknown first-order $B(x, D)$.

5. The Dirichlet-to-Neumann map for a semilinear wave equation on Lorentzian manifolds

Authors: Peter Hintz, Gunther Uhlmann, Jian Zhai.

Venue/year: Communications in Partial Differential Equations 47 (2022), 2363-2400.

DOI/arXiv/URL: DOI <https://doi.org/10.1080/03605302.2022.2131866>; arXiv 2103.08110, <https://arxiv.org/abs/2103.08110>.

Result checked: Theorem 1.3 recovers the Lorentzian metric and a quartic coefficient a from boundary Dirichlet-to-Neumann data up to diffeomorphism and a conformal obstruction. This is boundary data, assumes no conjugate points in the relevant region, uses a quartic nonlinearity, and does not contain an arbitrary first-order B .

6. An inverse boundary value problem for a semilinear wave equation on Lorentzian manifolds

Authors: Peter Hintz, Gunther Uhlmann, Jian Zhai.

Venue/year: International Mathematics Research Notices 2022, no. 17, 13181-13211.

DOI/arXiv/URL: DOI <https://doi.org/10.1093/imrn/rnab088>; arXiv 2005.10447, <https://arxiv.org/abs/2005.10447>.

Result checked: The main theorem recovers nonlinear Taylor coefficients from a Neumann-to-Dirichlet map when the metric is fixed and null geodesics have no cut points. It is coefficient recovery for a different boundary problem, not simultaneous recovery of g and B .

7. Detection of Hermitian connections in wave equations with cubic non-linearity

Authors: Xi Chen, Matti Lassas, Lauri Oksanen, Gabriel P. Paternain.

Venue/year: Journal of the European Mathematical Society 24 (2022), 2191-2232.

DOI/arXiv/URL: DOI <https://doi.org/10.4171/JEMS/1136>; <https://ems.press/journals/jems/articles/8965734>.

Result checked: Theorem 1 recovers a Hermitian connection, a first-order lower-order structure, from a source-to-solution map for a cubic connection wave equation in Minkowski space. The Lorentzian metric is fixed and flat, the equation is vector-valued with a connection structure, and it is not the arbitrary scalar $B(x, D)$ problem.

8. Inverse problem for connections in semi-linear wave equations

Authors: Lauri Oksanen, Ruochong Zhang.

Venue/year: arXiv preprint, 2025.

DOI/arXiv/URL: arXiv 2509.25971, <https://arxiv.org/abs/2509.25971>.

Result checked: Theorem 1.1 extends connection recovery from source-to-solution data to arbitrary globally hyperbolic Lorentzian manifolds, but the metric g is given and the operator is a Hermitian connection wave operator with cubic nonlinearity. It does not simultaneously reconstruct an unknown metric and a completely general first-order scalar operator.

9. Inverse problems for real principal type operators

Authors: Lauri Oksanen, Mikko Salo, Plamen Stefanov, Gunther Uhlmann.

Venue/year: American Journal of Mathematics 146 (2024), 161-240.

DOI/arXiv/URL: DOI <https://doi.org/10.1353/ajm.2024.a922595>; arXiv 2001.07599, <https://arxiv.org/abs/2001.07599>.

Result checked: The nonlinear theorem fixes the real-principal-type operator P and recovers the nonlinearity under analytic and cubic-or-higher assumptions; other theorems recover boundary/scattering data or lower-order ray transforms under separate hypotheses. It does not solve the local Lorentzian source-to-solution problem with unknown g and arbitrary first-order B .

9.5.3.3. Why the exact problem remains unresolved

The literature fully covers important pieces: local nonlinear interactions recover the conformal Lorentzian geometry for $\Box_g u + au^2$; later work recovers the full metric in several different nonlinear or boundary settings; and recent work recovers connection-type first-order data when the metric and operator structure are fixed. However, I found no theorem whose hypotheses match the full setting and whose conclusion reconstructs both:

- the full metric tensor g on $I(p^-, p^+)$, and
- an arbitrary first-order differential operator $B(x, D)$,

from the same local source-to-solution operator for $\Box_g u + B(x, D)u + a(x)u^2 = f$ on arbitrary 3+1 dimensional globally hyperbolic Lorentzian manifolds, modulo only the natural local gauge.

Therefore the correct status is **partially resolved**, not fully resolved.

9.5.4. Remaining open problems recorded by the database

9.5.4.1. Open problem 1

Let (M, g) be a known globally hyperbolic Lorentzian manifold, let $\Omega \subset M$ be an open observation region, and set $\mathbb{D} = J^+(\Omega) \cap J^-(\Omega)$. For an unknown scalar potential V , let $\mathcal{L}_V f = u|_{\Omega}$, where $(\Box_g + V)u = f$ and u vanishes in the past. Does the linearized source-to-solution map $\mathcal{L}_V$ determine $V|_{\mathbb{D}}$? This is open even in the corresponding Minkowski-space formulation without additional geometric assumptions.

Reference supplied by the database. Ali Feizmohammadi and Lauri Oksanen, “Recovery of zeroth order coefficients in non-linear wave equations,” Journal of the Institute of Mathematics of Jussieu 21 (2022), no. 2, 367–393, explicit open question in Section 2 immediately after the definition of the linearized source-to-solution map $\mathcal{L}_V$, DOI: 10.1017/S1474748020000122, <https://doi.org/10.1017/S1474748020000122>; arXiv:1903.12636, <https://arxiv.org/abs/1903.12636>.

9.5.5. Primary-source pointer

The originating ICM record is *Inverse problems for linear and non-linear hyperbolic equations* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol4-ver1-eb.pdf#page=1116>.

9.6. 263. Inverse problems for linear and non-linear hyperbolic equations [2018-geom-141]

Record: 2018-geom-141

Subfield: Applications of geometry / global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Inverse problems for linear and non-linear hyperbolic equations

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol4-ver1-eb.pdf#page=1116>

9.6.1. Problem statement

Let (M, g) be a 4-dimensional globally hyperbolic time-oriented Lorentzian manifold of signature $(-, +, +, +)$. For $p, q \in M$, write $p \ll q$ if p and q can be joined by a future-pointing timelike curve, set

$$I^+(p) = \{q \in M : p \ll q\}, \quad I^-(p) = \{q \in M : q \ll p\},$$

and define the double cone

$$I(p, q) = I^+(p) \cap I^-(q).$$

Using the product representation $M = \mathbb{R} \times N$ of a globally hyperbolic spacetime, let $t_0 > 0$, let $\mu \subset (0, t_0) \times N$ be a timelike curve, and let $V \subset M$ be an open neighborhood of μ . For a nowhere-vanishing smooth coefficient a , consider the semilinear wave equation

$$\square_g u(x) + a(x)u(x)^2 = f(x), \quad x \in (-\infty, t_0) \times N,$$

with $u = 0$ and $f = 0$ on $(-\infty, 0) \times N$. For sufficiently small C^k sources f supported in V , let L_V be the active measurement operator

$$L_V : f \mapsto u|_V,$$

where u is the corresponding solution. Suppose that the given data consist of the time-like curve μ , its neighborhood $V \subset M$, an approximation to the measurement operator L_V with some error, and points $p^-, p^+ \in \mu$ with $p^- \ll p^+$. Is it possible to reconstruct the double cone $I(p^-, p^+)$ and the metric g on $I(p^-, p^+)$ with an error estimate controlled in terms of a priori geometric bounds for M and the error in the given data?

9.6.2. Historical background, literature, and current progress

9.6.3. Verdict: partially resolved

9.6.3.1. Precise problem

The problem asks for a quantitative stability/reconstruction theorem for the local active inverse problem for

$$\square_g u + a(x)u^2 = f$$

on a 4-dimensional globally hyperbolic Lorentzian spacetime. The requested result would start from approximate local source-to-solution data $L_V : f \mapsto u|_V$, a timelike observation curve μ , a neighborhood V , and points $p^- \ll p^+$ on μ , and would reconstruct both the causal diamond $I(p^-, p^+)$ and the Lorentzian metric g on it with an explicit error estimate controlled by a priori geometry and measurement error.

9.6.3.2. Partial progress found

1. Inverse problems for Lorentzian manifolds and non-linear hyperbolic equations

Authors: Yaroslav Kurylev, Matti Lassas, Gunther Uhlmann.

Venue/year: *Inventiones Mathematicae* 212(3), 781-857, 2018.

DOI/arXiv/URL: DOI <https://doi.org/10.1007/s00222-017-0780-y>; arXiv 1405.3386, <https://arxiv.org/abs/1405.3386>.

Relevant result: Theorem 1.5 treats the exact local active source-to-solution map for the quadratic equation with nowhere-vanishing a in dimension $1+3$. It determines $I(p^-, p^+)$ as a differentiable manifold and determines the conformal class of g there, from exact equality of the maps. This is the closest match to this problem, but it is not a stability theorem for approximate data and its general conclusion is conformal, not full metric recovery. Full metric recovery is only indicated in additional special situations.

2. Inverse problems for linear and non-linear hyperbolic equations

Author: Matti Lassas.

Venue/year: *Proceedings of the International Congress of Mathematicians 2018, Rio de Janeiro*, Vol. III, pp. 3739-3760, 2018.

DOI/URL: World Scientific chapter URL https://www.worldscientific.com/doi/10.1142/9789813272880_0199.

Relevant result: Survey/source discussing inverse problems for hyperbolic equations and the nonlinear Lorentzian active measurement method. It records the uniqueness technology but does not provide the requested quantitative stability/error estimate.

3. Inverse Problems for Semilinear Wave Equations on Lorentzian Manifolds

Authors: Matti Lassas, Gunther Uhlmann, Yiran Wang.

Venue/year: *Communications in Mathematical Physics* 360, 555-609, 2018.

DOI/arXiv/URL: DOI <https://doi.org/10.1007/s00220-018-3135-7>; arXiv 1606.06261, <https://arxiv.org/abs/1606.06261>.

Relevant result: Theorems 1.2 and 1.3 extend the exact-data source-to-solution uniqueness theory to more general semilinear nonlinearities $H(x, u)$. They again give conformal recovery in general, with full metric recovery only under extra assumptions such as H independent of x or Ricci-flat metrics. The paper explicitly uses exact equality of measurement maps and gives no error estimate for approximate active data.

4. Inverse problems for quadratic derivative nonlinear wave equations

Authors: Yiran Wang, Ting Zhou.

Venue/year: Communications in Partial Differential Equations 44(11), 1140-1158, 2019.

DOI/arXiv/URL: DOI <https://doi.org/10.1080/03605302.2019.1612908>; arXiv 1612.04437, <https://arxiv.org/abs/1612.04437>.

Relevant result: Gives exact-data recovery results for a different class of quadratic derivative nonlinear wave equations, sometimes recovering the metric up to diffeomorphism under structural hypotheses. It does not cover the scalar quadratic nonlinearity $a(x)u^2$ in the problem statement and gives no approximate-data stability theorem.

5. Reconstruction of Lorentzian Manifolds from Boundary Light Observation Sets

Authors: Peter Hintz, Gunther Uhlmann.

Venue/year: International Mathematics Research Notices 2019(22), 6949-6987, 2019.

DOI/arXiv/URL: DOI <https://doi.org/10.1093/imrn/rnx320>; arXiv 1705.01215, <https://arxiv.org/abs/1705.01215>.

Relevant result: Reconstructs topological, differentiable, and conformal structures from boundary light observation sets under boundary/convexity-type hypotheses. This is related to the geometric reconstruction stage, but it is not the local active semilinear measurement problem and is not a quantitative stability theorem for L_V .

6. The Dirichlet-to-Neumann map for a semilinear wave equation on Lorentzian manifolds

Authors: Peter Hintz, Gunther Uhlmann, Jian Zhai.

Venue/year: Communications in Partial Differential Equations 47(12), 2363-2400, 2022.

DOI/arXiv/URL: DOI <https://doi.org/10.1080/03605302.2022.2122837>; arXiv 2103.08110, <https://arxiv.org/abs/2103.08110>.

Relevant result: Recovers the metric and coefficient, up to natural obstructions, from a boundary Dirichlet-to-Neumann map for a quartic semilinear wave equation. This is a boundary inverse problem with exact data and a different nonlinearity/data type, so it does not resolve the local active approximate-data question in the problem statement.

7. Stability estimates for inverse problems for semi-linear wave equations on Lorentzian manifolds

Authors: Matti Lassas, Tony Liimatainen, Leyter Potenciano-Machado, Teemu Tyni.

Venue/year: Analysis & PDE 18, 1065-1118, 2025.

DOI/arXiv/URL: DOI <https://doi.org/10.2140/apde.2025.18.1065>; arXiv 2106.12257, <https://arxiv.org/abs/2106.12257>.

Relevant result: Proves a Hölder stability estimate for recovering a potential q in $\square_g u + qu^m = 0$, $m \geq 4$, from boundary Dirichlet-to-Neumann data on a fixed Lorentzian background. This is the closest stability result found, but it recovers a coefficient rather than the metric/double cone, uses boundary DN data rather than local active data, and does not cover the quadratic case $m = 2$.

9.6.3.3. What remains unresolved

No published or otherwise citable result found during the live search proves the full statement in the problem statement. In particular, the literature does not currently provide a theorem that, for arbitrary globally hyperbolic 4-dimensional (M, g) and nowhere-vanishing $a(x)$, reconstructs the double cone and the full Lorentzian metric from an approximate local active map L_V with an error estimate depending only on a priori geometric bounds and the data error.

The exact active-measurement problem is substantially understood at the level of conformal reconstruction, and several related boundary or coefficient-recovery problems have unique-

ness or stability results. The missing pieces for the problem are the quantitative stability of the nonlinear active reconstruction pipeline and full metric recovery in the general quadratic $a(x)u^2$ setting without extra hypotheses.

9.6.4. Remaining open problems recorded by the database

9.6.4.1. Open problem 1

Let (M, g) be a fixed, known globally hyperbolic Lorentzian manifold with timelike boundary, and consider $\square_g u + q(x)u^2 = 0$. Establish a quantitative (for example, Hölder) stability estimate for recovering the unknown coefficient q on the causally observable region from the nonlinear Dirichlet-to-Neumann map, without assuming boundary convexity or that two lightlike geodesics intersect at most once.

Reference supplied by the database. Matti Lassas, Tony Liimatainen, Leyter Potenciano-Machado, and Teemu Tyni, “Stability and Lorentzian geometry for an inverse problem of a semilinear wave equation,” *Analysis & PDE* 18 (2025), no. 5, 1065–1118, discussion immediately after Corollary 1.2 (the cases $m = 2$ and $m = 3$ are left for future work, with $m = 2$ requiring different techniques), DOI: 10.2140/apde.2025.18.1065, <https://doi.org/10.2140/apde.2025.18.1065>; arXiv:2106.12257, <https://arxiv.org/abs/2106.12257>.

9.6.5. Primary-source pointer

The originating ICM record is *Inverse problems for linear and non-linear hyperbolic equations* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol4-ver1-eb.pdf#page=1116>.

9.7. 264. Inverse problems for linear and non-linear hyperbolic equations [2018-geom-142]

Record: 2018-geom-142

Subfield: Applications of geometry / global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Inverse problems for linear and non-linear hyperbolic equations

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol4-ver1-eb.pdf#page=1116>

9.7.1. Problem statement

Let (M, g) be a four-dimensional globally hyperbolic time-oriented Lorentzian manifold, written as $M = \mathbb{R} \times N$, and let $t_0 > 0$. Let $\mu \subset (0, t_0) \times N$ be a timelike curve, let $V \subset M$ be an open neighborhood of μ , and let $p^-, p^+ \in \mu$ with $p^- \ll p^+$. For a sufficiently small smooth source f supported in V , let u be the solution of the semilinear wave equation

$$\square_g u(x) + a(x)u(x)^2 = f(x), \quad x \in (-\infty, t_0) \times N,$$

with $u = 0$ and $f = 0$ in $(-\infty, 0) \times N$, where $a \in C^\infty(M)$ is nowhere vanishing, and let

$$L_V f := u|_V$$

be the observed response in V . Can one construct a single such source f so that the data consisting of the set V and the single measurement $L_V f$ uniquely determine the chronological diamond

$$I(p^-, p^+) = I^+(p^-) \cap I^-(p^+),$$

where $I^+(p)$ and $I^-(p)$ are the chronological future and past of p , and uniquely determine the metric g on $I(p^-, p^+)$?

9.7.2. Historical background, literature, and current progress

Verdict: partially resolved.

9.7.2.1. Precise problem checked

The problem asks whether, for the four-dimensional globally hyperbolic Lorentzian semilinear wave equation

$$\square_g u + a(x)u^2 = f, \quad a \in C^\infty(M), a \neq 0,$$

one can choose a single sufficiently small smooth source f supported in an observer neighborhood V of a timelike curve so that the single observed field $u|_V = L_V f$, together with V ,

uniquely determines both the chronological diamond

$$I(p^-, p^+) = I^+(p^-) \cap I^-(p^+)$$

and the Lorentzian metric g on that diamond. The distinctive point is the one-source/one-measurement requirement; the data are not the full local source-to-solution map for all small sources.

9.7.2.2. Literature and progress

1. Original source / survey context.

- Title: “Inverse problems for linear and non-linear hyperbolic equations”
- Author: Matti Lassas
- Venue/year: Proceedings of the International Congress of Mathematicians, Rio de Janeiro 2018; World Scientific, 2018/2019
- DOI/URL: [doi:10.1142/9789813272880_0199](https://doi.org/10.1142/9789813272880_0199)
- Relevance: This survey discusses the local inverse problems for nonlinear hyperbolic equations that underlie the question. It does not give a theorem proving the one-fixed-source statement.

2. Main full-map Lorentzian nonlinear result.

- Title: “Inverse problems for Lorentzian manifolds and non-linear hyperbolic equations”
- Authors: Yaroslav Kurylev, Matti Lassas, Gunther Uhlmann
- Venue/year: *Inventiones Mathematicae* 212, 781-857, 2018
- DOI/arXiv/URL: [doi:10.1007/s00222-017-0780-y](https://doi.org/10.1007/s00222-017-0780-y); arXiv:1405.3386; <https://arxiv.org/abs/1405.3386>
- Result checked: The paper proves recovery results for a four-dimensional Lorentzian manifold from local source-to-solution data for nonlinear wave equations, using nonlinear interaction of waves. The data are a local source-to-solution map for many small sources supported in an observation region, not a single fixed source. The recovery is formulated in terms of the causal geometry/conformal structure, with metric recovery requiring additional information or hypotheses in the surrounding theory. Thus this is major partial progress but does not resolve the exact question.

3. General semilinear source-to-solution map result.

- Title: “Inverse Problems for Semilinear Wave Equations on Lorentzian Manifolds”
- Authors: Matti Lassas, Gunther Uhlmann, Yiran Wang
- Venue/year: *Communications in Mathematical Physics* 360, 555-609, 2018
- DOI/arXiv/URL: [doi:10.1007/s00220-018-3135-7](https://doi.org/10.1007/s00220-018-3135-7); arXiv:1606.06261; <https://arxiv.org/abs/1606.06261>
- Result checked: The theorem determines the conformal class of the Lorentzian metric from the full source-to-solution map for a semilinear wave equation, and obtains metric determination only under extra hypotheses such as special nonlinearity/geometry assumptions. It does not treat data consisting of one single source response $L_V f$, and its hypotheses/conclusions do not imply the full one-source metric recovery asked in the problem statement.

4. Disjoint source/receiver full-map progress.

- Title: “Inverse problems for nonlinear hyperbolic equations with disjoint sources and receivers”
- Authors: Ali Feizmohammadi, Matti Lassas, Lauri Oksanen
- Venue/year: Forum of Mathematics, Pi 9:e10, 2021
- DOI/arXiv/URL: [doi:10.1017/fmp.2021.11](https://doi.org/10.1017/fmp.2021.11); arXiv:2006.12158; <https://arxiv.org/abs/2006.12158>
- Result checked: This paper proves metric recovery in a causal diamond from full source-to-solution data in a disjoint source/receiver setting for equations of the form $\square_g u + u^m = f$, under restrictions including $m \geq 3$ and excluding the conformal exceptional case $(n, m) = (3, 3)$. The stated equation is quadratic, $m = 2$, with a general nowhere-vanishing coefficient $a(x)$, and the problem asks for one fixed source rather than a full map. Hence this is partial progress only.

5. Single-observer but still full-map progress.

- Title: “Determining Lorentzian manifold from non-linear wave observation at a single point”
- Authors: Medet Nursultanov, Lauri Oksanen, Leo Tzou
- Venue/year: arXiv preprint, 2023; Journal of Differential Equations 444, Article 113563, 2025
- DOI/arXiv/URL: arXiv:2310.06925; [doi:10.1016/j.jde.2025.113563](https://doi.org/10.1016/j.jde.2025.113563); <https://arxiv.org/abs/2310.06925>
- Result checked: This work reduces the observation geometry to a timelike curve/single-point-type observation for a nonlinear wave model, but it still uses a source-to-solution map for all suitable sources and concerns a cubic nonlinearity with conformal recovery. It does not prove that a single smooth source response determines the metric for the quadratic problem stated here.

6. Related one-measurement inverse-problem ideas.

- Title: “An inverse problem for the wave equation with one measurement and the pseudorandom source”
- Authors: Tapio Helin, Matti Lassas, Lauri Oksanen
- Venue/year: Analysis & PDE 5(5), 887-912, 2012
- DOI/URL: [doi:10.2140/apde.2012.5.887](https://doi.org/10.2140/apde.2012.5.887)
- Relevance: This shows that one carefully chosen pseudorandom source can determine boundary spectral data in a linear wave inverse problem. It is conceptually relevant to the one-measurement theme, but it is not the Lorentzian semilinear metric-recovery problem in the problem statement.
- Title: “Global recovery of a time-dependent coefficient for the wave equation from a single measurement”
- Authors: Ali Feizmohammadi, Yavar Kian
- Venue/year: Asymptotic Analysis 131(3-4), 513-539, 2023
- DOI/arXiv/URL: [doi:10.3233/ASY-221779](https://doi.org/10.3233/ASY-221779); arXiv:2002.11598; <https://arxiv.org/abs/2002.11598>
- Relevance: This is a genuine single-measurement theorem, but it recovers a coefficient for a wave equation on a known background geometry, not an unknown Lorentzian metric in the nonlinear four-dimensional setting of the problem.

9.7.2.3. What remains unresolved

the exact question remains unresolved in the literature I found. Existing nonlinear Lorentzian recovery theorems require the full source-to-solution map, or recover only

conformal/geometric structure without the full metric unless extra assumptions are imposed. Later results reduce receiver geometry or prove one-measurement recovery for different unknowns, but they do not cover all hypotheses and the full conclusion of the problem: a single smooth source $f \in V$ for $\square_g u + a(x)u^2 = f$ in four spacetime dimensions determining both $I(p^-, p^+)$ and the metric g on that diamond.

9.7.3. Remaining open problems recorded by the database

9.7.3.1. Open problem 1

Let (M, g) be a compact non-trapping Riemannian manifold with nonempty boundary. Prove that its full scattering relation (including the travel time of every boundary-to-boundary geodesic) determines (M, g) up to a boundary-fixing isometry.

Reference supplied by the database. Günther Uhlmann, “The Cauchy Data and the Scattering Relation,” in *Geometric Methods in Inverse Problems and PDE Control*, IMA Volumes in Mathematics and its Applications 137, Springer, 2004, pp. 263–287, Conjecture 4.1, <https://sites.math.washington.edu/~gunther/Papers/imafinal.pdf>. The non-trapping formulation and its explicit role as the remaining metric-recovery step after one pseudorandom-source measurement are stated in Tapio Helin, Matti Lassas, and Lauri Oksanen, “An Inverse Problem for the Wave Equation with One Measurement and the Pseudorandom Source,” *Analysis & PDE* 5 (2012), no. 5, 887–912, introduction after Corollary 1, <https://doi.org/10.2140/apde.2012.5.887>; arXiv:1011.2527.

9.7.4. Primary-source pointer

The originating ICM record is *Inverse problems for linear and non-linear hyperbolic equations* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol4-ver1-eb.pdf#page=1116>.

9.8. 265. Liouville quantum gravity as a metric space and a scaling limit [2018-geom-143]

Record: 2018-geom-143

Subfield: Applications of geometry / global analysis

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Liouville quantum gravity as a metric space and a scaling limit

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol4-ver1-eb.pdf#page=226>

9.8.1. Problem statement

Let Q_n be a uniformly random planar quadrangulation with n faces. Regard Q_n as a topological surface by replacing each quadrilateral face with a Euclidean unit square $[0, 1]^2$ and gluing square edges according to the adjacency relations of Q_n , identifying matching boundary lengths. Choose a conformal map from this random surface to the Riemann sphere $\mathbb{S}^2$, and in a local conformal coordinate $z = x + iy$ write the induced metric as

$$e^{\lambda_n(z)}(dx^2 + dy^2),$$

thereby defining the random conformal factor λ_n .

Prove or disprove that, as $n \rightarrow \infty$, the fields λ_n converge in law as random distributions on $\mathbb{S}^2$ to $\sqrt{8/3}h$, where h is a Gaussian free field on $\mathbb{S}^2$, i.e. a Gaussian random distribution whose covariance is locally given by the Green's function of the Laplacian. Equivalently, prove or disprove that the associated area measures

$$e^{\lambda_n(z)} dx dy$$

converge in law weakly as random measures to the $\sqrt{8/3}$ -Liouville quantum gravity area measure associated with h , namely the Gaussian multiplicative chaos measure

$$\mu_h^{\sqrt{8/3}} = \lim_{\varepsilon \rightarrow 0} \varepsilon^{(\sqrt{8/3})^2/2} e^{\sqrt{8/3} h_\varepsilon(z)} dx dy = \lim_{\varepsilon \rightarrow 0} \varepsilon^{4/3} e^{\sqrt{8/3} h_\varepsilon(z)} dx dy,$$

where $h_\varepsilon(z)$ denotes the circle-average regularization of h .

9.8.2. Historical background, literature, and current progress

9.8.3. Verdict: partially resolved

9.8.3.1. Precise problem checked

The problem asks for a theorem proving that if Q_n is a uniformly random spherical planar quadrangulation with n faces, made into a polyhedral Riemann surface by replacing each face by a Euclidean unit square and then conformally uniformizing to $\mathbb{S}^2$, then the resulting conformal factor λ_n , or equivalently the pushed-forward Riemannian area measure $e^{\lambda_n(z)} dx dy$,

converges to the $\gamma = \sqrt{8/3}$ Liouville quantum gravity sphere measure associated with a Gaussian free field.

I found no published or otherwise citable theorem that proves this exact finite-sphere, uniform-quadrangulation, unit-square/Riemann-uniformization statement. The literature proves the metric-measure scaling limit of uniform quadrangulations and identifies the Brownian map with the $\sqrt{8/3}$ -LQG metric sphere, and it proves conformal-embedding convergence for some related random planar map models/embeddings, but these results do not imply convergence of the specific conformal factors or area measures in the problem statement.

9.8.3.2. Relevant literature and current progress

1. **Grégory Miermont**, “The Brownian map is the scaling limit of uniform random plane quadrangulations”, *Acta Mathematica* 210 (2013), 319-401. DOI: 10.1007/s11511-013-0096-8. arXiv:1104.1606.

This proves that uniform random plane quadrangulations, with graph distance rescaled by $n^{-1/4}$, converge to the Brownian map. This is a central result for the metric scaling limit of the same discrete objects appearing in the problem statement. It does not prove convergence of the Riemann uniformization of the unit-square surface, nor convergence of the conformal factor λ_n or of the pushed-forward Euclidean area measure in a fixed conformal coordinate.

2. **Jean-François Le Gall**, “Uniqueness and universality of the Brownian map”, *Annals of Probability* 41 (2013), 2880-2960. DOI: 10.1214/12-AOP792. arXiv:1105.4842.

This gives Brownian-map universality results for broad classes of random planar maps. It supports the metric-measure side of the $\sqrt{8/3}$ -LQG picture but, like Miermont’s theorem, is not a theorem about the conformal embedding obtained by gluing unit squares and uniformizing to the sphere.

3. **Bertrand Duplantier and Scott Sheffield**, “Liouville quantum gravity and KPZ”, *Inventiones Mathematicae* 185 (2011), 333-393. DOI: 10.1007/s00222-010-0308-1. arXiv:0808.1560.

This constructs and analyzes Liouville quantum gravity measures via Gaussian multiplicative chaos and explains the $\gamma = \sqrt{8/3}$ connection with random planar maps through KPZ. It supplies the rigorous LQG-measure framework used in the statement, but it does not prove the convergence of uniform quadrangulations under the unit-square conformal uniformization.

4. **François David, Antti Kupiainen, Rémi Rhodes, and Vincent Vargas**, “Liouville quantum gravity on the Riemann sphere”, *Communications in Mathematical Physics* 342 (2016), 869-907. DOI: 10.1007/s00220-016-2579-6. arXiv:1410.7318.

This gives a rigorous construction of Liouville quantum gravity on the sphere. It establishes the continuum target object, not the convergence of the discrete quadrangulation surfaces in the problem statement.

5. **Jason Miller and Scott Sheffield**, “Liouville quantum gravity and the Brownian map I: The QLE(8/3,0) metric”, *Inventiones Mathematicae* 219 (2020), 75-152. DOI: 10.1007/s00222-019-00905-1. arXiv:1507.00719; “II: Geodesics and continuity of the embedding”, *Annals of Probability* 49 (2021), 2732-2829. DOI: 10.1214/21-AOP1506. arXiv:1605.03563; “III: The conformal structure is determined”, *Probability Theory and Related Fields* 179 (2021), 1183-1211. DOI: 10.1007/s00440-021-01026-8.

arXiv:1608.05391.

This trilogy identifies the Brownian map with the $\sqrt{8/3}$ -Liouville quantum gravity sphere equipped with its metric and shows that the conformal structure is determined in the continuum. Together with the quadrangulation-to-Brownian-map convergence, it is very strong evidence for the assertion. However, the results do not state or imply, without additional work, that the conformal maps obtained by uniformizing the discrete unit-square quadrangulation surfaces have conformal factors or pushed-forward area measures converging to the LQG measure.

6. **Nicolas Curien, “A glimpse of the conformal structure of random planar maps”, *Communications in Mathematical Physics* 333 (2015), 1417-1463. DOI: 10.1007/s00220-014-2196-5. arXiv:1308.1807.**

This studies conformal structure for random planar maps using circle packing and Riemann uniformization in a related setting. It gives partial evidence and techniques for conformal convergence questions, but it is not a proof of the finite uniform quadrangulation/unit-square sphere statement in this problem.

7. **Ewain Gwynne, Nina Holden, and Xin Sun, “Convergence of uniform triangulations under the Cardy embedding”, *Acta Mathematica* 230 (2023), 93-203. DOI: 10.4310/ACTA.2023.v230.n1.a2. arXiv:1905.13207.**

This proves convergence to $\sqrt{8/3}$ -LQG for uniform triangulations under the Cardy embedding, a major conformal-embedding result for a closely related random planar map model. It is not the same model and not the same embedding: the problem concerns uniform quadrangulations made into unit-square Riemann surfaces and then uniformized to the sphere.

8. **Federico Bertacco, Ewain Gwynne, and Scott Sheffield, “Scaling limits of planar maps under the Smith embedding”, *Annals of Probability* 53 (2025). DOI: 10.1214/24-AOP1731. arXiv:2306.02988.**

This proves γ -LQG scaling limits for mated-CRT maps under the Smith embedding. It is another strong conformal-embedding theorem, but its hypotheses and embedding are different from the uniform quadrangulation/unit-square Riemann uniformization in the problem statement.

9. **Nina Holden and Pu Yu, “Circle packing and Riemann uniformization of random planar maps in an ergodic scale-free environment”, preprint (2026). arXiv:2603.06528.**

This recent preprint studies circle packing and Riemann uniformization for infinite random planar maps in a scale-free environment. It gives relevant progress on Riemann-uniformization methods but does not prove the specific convergence of finite uniform quadrangulations with unit-square faces to the $\sqrt{8/3}$ -LQG sphere measure.

10. **Jason Miller, “Liouville quantum gravity as a metric space and a scaling limit”, *Proceedings of the International Congress of Mathematicians 2018, Vol. IV, World Scientific* (2018), 2945-2971. arXiv:1712.01571.**

This survey explains the Brownian-map/ $\sqrt{8/3}$ -LQG metric equivalence and its role as the scaling limit of random planar maps. It is useful background and likely close to the source of the formulation, but it is a survey of metric scaling-limit results and does not contain a theorem proving the exact unit-square conformal-factor convergence asserted here.

9.8.3.3. What remains unresolved

The missing theorem is a direct convergence result for the conformal uniformization of the discrete polyhedral surfaces obtained from uniform spherical quadrangulations by gluing Euclidean unit squares. In particular, the literature I found does not prove that the random measures $e^{\lambda_n(z)} dx dy$, after a specified normalization of the Möbius freedom, converge to the $\sqrt{8/3}$ -LQG sphere area measure, nor that the fields λ_n converge as distributions to $\sqrt{8/3}$ times a Gaussian free field. Existing Brownian-map convergence and Brownian-map/LQG identification resolve the metric-measure scaling-limit side, while Cardy/Smith/circle-packing/Riemann-uniformization results resolve related models or embeddings, so the correct status is **partially resolved**, not fully resolved.

9.8.4. Remaining open problems recorded by the database

9.8.4.1. Open problem 1

Let $T_{\infty,\infty}$ be the uniform infinite half-plane triangulation, conformally uniformized by $\mathbb{H}$, and explore it by an independent chordal SLE_6 . If $\mathcal{H}^+(i)$ is the right horodistance after the i th peeling step and $\eta_i^+ := \mathcal{H}^+(i+1) - \mathcal{H}^+(i + \frac{1}{2})$ is the discrepancy between the next peeling edge and the half-step prediction, prove Curien's condition (*):

$$\sup_{t \in [0,1]} \frac{\eta_1^+ + \cdots + \eta_{[tn]}^+}{n^{2/3}} \xrightarrow[n \rightarrow \infty]{\mathbb{P}} 0.$$

Reference supplied by the database. Nicolas Curien, "A Glimpse of the Conformal Structure of Random Planar Maps," *Communications in Mathematical Physics* **333** (2015), no. 3, 1417–1463, Section 2.5, condition (*), <https://doi.org/10.1007/s00220-014-2196-5>; arXiv:1308.1807.

9.8.4.2. Open problem 2

Uniformize the uniform infinite half-plane triangulation in $\mathbb{H}$ so that the root-edge endpoints map to $-\frac{1}{2}$ and $\frac{1}{2}$ and the infinite end maps to ∞ . If $\mathcal{X}_n$ is the image of the n th boundary vertex to the right of the root, prove

$$\frac{\log \mathcal{X}_n}{\log n} \xrightarrow[n \rightarrow \infty]{\mathbb{P}} 3,$$

and determine whether the family $(n^{-3} \mathcal{X}_n)_{n \geq 1}$ is tight.

Reference supplied by the database. Nicolas Curien, "A Glimpse of the Conformal Structure of Random Planar Maps," *Communications in Mathematical Physics* **333** (2015), no. 3, 1417–1463, Section 5, first question ("location of the n th vertex"), <https://doi.org/10.1007/s00220-014-2196-5>; arXiv:1308.1807.

9.8.4.3. Open problem 3

Let ν be a random locally finite measure on $\mathbb{R}$, independent of a chordal SLE_6 curve γ in $\mathbb{H}$ from 0 to ∞ , and set

$$\Gamma_\nu^+ = \{(t, \nu[0, \gamma_t]) : \gamma_t \in \mathbb{R}_+\}, \quad \Gamma_\nu^- = \{(t, \nu[\gamma_t, 0]) : \gamma_t \in \mathbb{R}_-\}.$$

Let S^+ and S^- be independent spectrally negative 3/2-stable Lévy processes and let

$$\mathcal{M}^\pm = \{(t, -S_t^\pm) : S_t^\pm = \inf_{0 \leq s \leq t} S_s^\pm\}.$$

Does the identity in law $(\Gamma_\nu^+, \Gamma_\nu^-) \stackrel{d}{=} (\mathcal{M}^+, \mathcal{M}^-)$, up to a reparametrization of the first coordinate, uniquely characterize the law of ν ?

Reference supplied by the database. Nicolas Curien, “A Glimpse of the Conformal Structure of Random Planar Maps,” *Communications in Mathematical Physics* **333** (2015), no. 3, 1417–1463, Section 5.2, Question “Characterization,” <https://doi.org/10.1007/s00220-014-2196-5>; arXiv:1308.1807.

9.8.5. Primary-source pointer

The originating ICM record is *Liouville quantum gravity as a metric space and a scaling limit* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2018/ICM-2018-vol4-ver1-eb.pdf#page=226>.

10. Incidence, axiomatic, and classical geometry

These records return to foundational geometric configurations and incidence structures. Their statements are often elementary to formulate, but their modern treatments draw on combinatorics, topology, algebra, and computational search. The surviving problems tend to be sharp universal claims for which many structured or low-dimensional cases are known.

Coverage: 8 website records; **source years:** 1893–2022; **separately listed descendant open problems:** 13.

10.1. 266. CONSIDÉRATIONS GÉNÉRALES SUR LA MESURE DE LA SIMPLICITÉ DANS LES SCIENCES MATHÉMATIQUES ET... [1893-geom-1]

Record: 1893-geom-1

Subfield: Incidence, axiomatic, and classical geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1893: CONSIDÉRATIONS GÉNÉRALES SUR LA MESURE DE LA SIMPLICITÉ DANS LES SCIENCES MATHÉMATIQUES ET APPLICATION L'ÉVALUATION THÉORIQUE DE LA SIMPLICITÉ DES TRACÉS GÉOMÉTRIQUES, OU GÉOMÉTROGRAPHIE

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1893/ICM1893.ocr.pdf#page=159>

10.1.1. Problem statement

In straightedge-and-compass geometry, where the simplicity of a construction is the total number of elementary operations R_1, R_2, C_1, C_2, C_3 , with R_1 = passing the edge of the straightedge through a given point, R_2 = drawing a line, C_1 = placing one point of the compass at a given point, C_2 = placing one point of the compass at an undetermined point of a drawn line, and C_3 = drawing a circle, determine, for a given length AB and a given positive integer n , a construction of a length equal to $n \cdot AB$ that minimizes this total number of elementary operations.

10.1.2. Historical background, literature, and current progress

10.1.3. Verdict: partially resolved

10.1.3.1. Precise problem checked

The problem asks for the exact geometrographic optimization problem posed by Emile Lemoine: in ruler-and-compass geometry, with simplicity equal to the total number of elementary operations $R1, R2, C1, C2, C3$, determine, for every given segment AB and every positive integer n , a construction of a segment of length $n \cdot AB$ whose total operation count is minimal.

This is not merely the constructibility of $n \cdot AB$; repeated transfer of AB gives a construction. The unresolved part is proving optimality, uniformly in n , in Lemoine's specific operation-count model.

10.1.3.2. Literature checked

Original source Title: "Considerations generales sur la mesure de la simplicité dans les sciences mathématiques et application à l'évaluation théorique de la simplicité des traces géométriques, ou géométrographie"

Author: Emile Lemoine

Venue: Mathematical Papers Read at the International Mathematical Congress Held in Connection with the World's Columbian Exposition, Chicago, 1893, ed. E. H. Moore, O. Bolza, H. Maschke, H. S. White; Macmillan, New York

Year: 1896, pp. 143-154

DOI/arXiv/URL: https://archive.org/download/mathematicalp00socigoog/mathematicalp00socigoog_djvu.txt

Lemoine defines the five elementary operations and the "simplicity" coefficient as their total count. He then states the exact problem: given a length AB , find as simply as possible a length equal to n times AB . He explicitly says he does not know a general solution, and that the problem is of the same kind as, but more complicated than, Dellac's minimum-multiplication problem for computing A^n .

Geometrography surveys and later work Title: "Two Approaches to Geometrography"

Authors: Jorma K. Merikoski and Timo Tossavainen

Venue: Journal of Geometry and Graphics 14(1), pp. 15-28

Year: 2010

DOI/arXiv/URL: <https://www.heldermann-verlag.de/jgg/jgg14/j14h1toss.pdf>

This paper surveys Lemoine's operation-count model, gives the five basic operations, and describes later statistical approaches to construction accuracy. It studies example constructions such as perpendiculars and regular pentagons. It does not give a theorem determining the optimal construction of $n \cdot AB$ for all n , nor does it give lower bounds proving optimality in Lemoine's model.

Title: Computational Geometry: An Introduction

Authors: Franco P. Preparata and Michael Ian Shamos

Venue: Springer-Verlag, Texts and Monographs in Computer Science

Year: 1985

DOI/arXiv/URL: <https://www.cs.kent.edu/~dragan/CG/CG-Book.pdf>

The historical discussion treats Lemoine's geometrography as an early time-complexity measure for Euclidean constructions, but emphasizes that Lemoine did not prove necessary lower bounds for operation counts. This supports the conclusion that known geometrographic constructions are not, by themselves, resolutions of the optimality problem.

Title: "The Complexity of Geometric Constructions"

Author: Francois Labelle

Venue: Web project, McGill University

Year: 1997; updated 2014

DOI/arXiv/URL: <https://www.cs.mcgill.ca/~sqrt/cons/constructions.html>

Labelle's page collects efficient or "best known" constructions under a related line/circle drawing-count model and states that, once constructibility is known, little is available on the minimum number of required steps. This is useful evidence about the state of the construction-complexity literature, but it uses a different complexity measure and does not resolve Lemoine's all- n problem.

Related number-theoretic problem: addition chains Title: "Efficient Generation of Minimal Length Addition Chains"

Author: Edward G. Thurber

Venue: SIAM Journal on Computing 28(4), pp. 1247-1263

Year: 1999

DOI/arXiv/URL: <https://doi.org/10.1137/S0097539795295663> ; PDF: https://wwwhomes.uni-bielefeld.de/achim/siam_thurber.pdf

Thurber identifies Dellac's 1894 question on the minimum number of multiplications needed to compute A^m as the source of addition chains, defines the minimal addition-chain length $l(n)$, and gives exact branch-and-bound methods for generating minimal chains. This is substantial progress on the analogous arithmetic problem Lemoine mentioned. It does not imply an optimal ruler-and-compass construction for $n \leq AB$, because Lemoine's geometric problem has additional operation costs and geometric reuse issues.

Title: "Calculating Optimal Addition Chains"

Author: Neill Michael Clift

Venue: Computing 91, pp. 265-284

Year: 2011

DOI/arXiv/URL: <https://doi.org/10.1007/s00607-010-0118-8> ; article page: <https://link.springer.com/article/10.1007/s00607-010-0118-8>

Clift gives faster exact computation of optimal addition chains, including computation of all optimal addition-chain lengths for $n \leq 2^{32}$. This is finite-range and arithmetic-chain progress, not a proof resolving the geometrographic optimum for all positive integers n .

Title: "Computing Sequences with Addition Chains"

Authors: Peter Downey, Benton Leong, and Ravi Sethi

Venue: SIAM Journal on Computing 10(3), pp. 638-646

Year: 1981

DOI/arXiv/URL: <https://doi.org/10.1137/0210047>

This paper proves complexity results for generalized addition-chain sequence problems. These results concern arithmetic addition chains, not Lemoine's straightedge-and-compass operation

model, and therefore do not resolve the exact geometric problem.

10.1.3.3. Applicability check

the exact problem requires both:

1. for every positive integer n , an explicit construction of a segment of length $n \cdot AB$ in Lemoine's $R1$, $R2$, $C1$, $C2$, $C3$ model; and
2. a proof that no construction with a smaller total operation count exists in that same model.

The addition-chain literature addresses an analogous arithmetic optimization: minimizing multiplications for A^n , equivalently minimizing additions in a chain for n . Even a shortest addition chain does not automatically minimize Lemoine's geometric cost, because a geometric realization must pay for drawing/placing lines and circles, reusing already drawn loci, transferring lengths, and choosing starting points. Lemoine explicitly described his geometric problem as more complicated than the Dellac/addition-chain problem.

The geometrography literature gives definitions, examples, comparisons, best-known constructions, and statistical accuracy models. It does not provide a complete lower-bound theory or a theorem whose hypotheses and conclusion imply optimal $n \cdot AB$ constructions for all n .

10.1.3.4. What remains unresolved

No citable theorem was found that determines, for every positive integer n , the minimum possible total number of Lemoine elementary operations needed to construct a segment of length $n \cdot AB$, together with an optimal construction attaining that number. The problem is therefore not fully resolved. The literature gives related and partial progress through addition-chain theory and through studies of geometrographic complexity for selected constructions, so the appropriate verdict is partially resolved rather than fully resolved.

10.1.4. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

10.1.5. Primary-source pointer

The originating ICM record is *CONSIDÉRATIONS GÉNÉRALES SUR LA MESURE DE LA SIMPLICITÉ DANS LES SCIENCES MATHÉMATIQUES ET APPLICATION L'ÉVALUATION THÉORIQUE DE LA SIMPLICITÉ DES TRACÉS GÉOMÉTRIQUES, OU GÉOMÉTROGRAPHIE* (1893). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1893/ICM1893.ocr.pdf#page=159>.

10.2. 267. V. Theory of Functions and Geometry [1893-geom-2]

Record: 1893-geom-2

Subfield: Incidence, axiomatic, and classical geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1893: V. Theory of Functions and Geometry

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1893.2/ICM1893.2.ocr.pdf#page=43>

10.2.1. Problem statement

Determine the possible forms of the circular or spherical quadrangles and, more generally, polygons that arise in the conformal-geometric treatment of functions defined by second-order linear differential equations with more than three singular points, generalizing the hypergeometric case in which three singular points correspond to the vertices of a circular or spherical triangle.

10.2.2. Historical background, literature, and current progress

10.2.3. Verdict: partially resolved

10.2.3.1. Precise problem identified

The problem is not asking for a single numerical theorem about one equation. It is Klein's Schwarz-map program for second-order linear differential equations with more than three regular singular points. In the hypergeometric case, the ratio of two independent solutions maps the upper half-plane (or a disk) to a circular/spherical triangle. For equations with four or more singular points, the analogous Schwarz images are circular or spherical quadrilaterals and, in general, circular/spherical polygons. The problem is to obtain a general classification/overview of the possible forms of these polygons, equivalently to understand the corresponding Schwarzian/Fuchsian equations and accessory parameters. In the modern positive-curvature real/unitary formulation, this becomes the classification of spherical n -gons or of second-order Fuchsian equations with real parameters and unitary monodromy; for $n=4$ this is the Heun-equation case.

10.2.3.2. Original source

- Title: The Evanston Colloquium: Lectures on Mathematics
- Author: Felix Klein
- Venue/year: Macmillan, New York, 1894; lectures delivered at Northwestern University after the 1893 Chicago Congress of Mathematics; AMS reprints later issued
- DOI/arXiv/URL: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1893.2/ICM1893.2.ocr.pdf>
- Relevance: Lecture V says that for differential equations with more than three singular points, triangles must be replaced by quadrangles and other polygons, and that the dif-

ficuity is the geometrical problem of obtaining a general view of the possible polygon forms. This is exactly the problem represented by the problem statement.

10.2.3.3. Progress found

- Title: Ueber Kreisbogenpolygone. Erste Abhandlung
- Author: A. Schoenflies
- Venue/year: Mathematische Annalen 42 (1893), 377-408
- DOI/arXiv/URL: <https://doi.org/10.1007/BF01444164>
- Progress: Early work on circular-arc polygons, directly cited in Klein's discussion. It gives important constructive/reduction analysis for circular polygons, but later literature treats it as only a partial extension of Klein's triangle classification.
- Title: Ueber Kreisbogendreiecke und Kreisbogenvierecke
- Author: A. Schoenflies
- Venue/year: Mathematische Annalen 44 (1894), 105-124
- DOI/arXiv/URL: <https://doi.org/10.1007/BF01446976>
- Progress: Early special treatment of circular triangles and quadrilaterals. It does not give a modern complete all-cases classification of the polygons arising from arbitrary higher-singularity second-order equations.
- Title: Ueber die gestaltlichen Verhaeltnisse der Kreisbogenvierecke; Ueber die geometrischen Eigenschaften der Kreisbogenvierecke
- Author: W. Ihlenburg
- Venue/year: Goettinger Nachrichten (1908), 225-230; Nova Acta Leopoldina 91 (1909), 1-79 plus tables
- DOI/arXiv/URL: no DOI found; bibliographic data cited in Eremenko-Gabrielov-Tarasov, Conformal Geometry and Dynamics 20 (2016), 128-175
- Progress: Further analysis of circular quadrilaterals. Modern authors state that the Schoenflies-Ihlenburg algorithm constructs all circular quadrilaterals, but still "falls short of a complete classification": prescribed angles and sides need not determine a unique quadrilateral, and the geodesic/spherical cases are hard to isolate.
- Title: Metrics with conic singularities and spherical polygons
- Authors: Alexandre Eremenko, Andrei Gabrielov, Vitaly Tarasov
- Venue/year: Illinois Journal of Mathematics 58(3) (2014), 739-755
- DOI/arXiv/URL: <https://doi.org/10.1215/ijm/1441790388> ; arXiv:1405.1738, <https://arxiv.org/abs/1405.1738>
- Progress: Gives a modern formulation of spherical n-gons and states that the problem is equivalent to classification of some second-order linear differential equations with regular singularities, real parameters, and unitary monodromy. It enumerates/classifies the case in which exactly two corner angles are not multiples of π , with the remaining angles integral. This is substantial progress for arbitrary n , but it is a restricted case.
- Title: Metrics with four conic singularities and spherical quadrilaterals

- Authors: Alexandre Eremenko, Andrei Gabrielov, Vitaly Tarasov
- Venue/year: Conformal Geometry and Dynamics 20 (2016), 128-175
- DOI/arXiv/URL: <https://doi.org/10.1090/ecgd/295> ; arXiv:1409.1529, <https://arxiv.org/abs/1409.1529>
- Progress: Classifies spherical quadrilaterals up to isometry in cases with integer-angle hypotheses and identifies the equivalent problem as classification of Heun equations with real parameters and unitary monodromy. This resolves important quadrilateral subcases, but not all quadrilaterals/polygons asked for by Klein's general problem.
- Title: Spherical quadrilaterals with three non-integer angles
- Authors: Alexandre Eremenko, Andrei Gabrielov, Vitaly Tarasov
- Venue/year: Journal of Mathematical Physics, Analysis, Geometry 12 (2016), 134-167
- DOI/arXiv/URL: <https://doi.org/10.15407/mag12.02.134> ; arXiv:1504.02928, <https://arxiv.org/abs/1504.02928>
- Progress: Treats the quadrilateral case in which one angle is an integer multiple of π and the other three are not. It is another major Heun/unitary-monodromy subcase, but still assumes a special angle pattern.
- Title: Circular Pentagons and Real Solutions of Painleve VI Equations
- Authors: Alexandre Eremenko, Andrei Gabrielov
- Venue/year: Communications in Mathematical Physics 355 (2017), 51-95
- DOI/arXiv/URL: <https://doi.org/10.1007/s00220-017-2921-y> ; arXiv:1611.01356, <https://arxiv.org/abs/1611.01356>
- Progress: Associates a class of real Painleve VI solutions to one-parameter families of circular pentagons and gives an algorithm for special-point counts and order. This shows active progress beyond quadrilaterals, but it is a special family of pentagons, not a full classification of all possible polygon forms.
- Title: Fuchsian Equations with Three Non-Apparent Singularities
- Authors: Alexandre Eremenko, Vitaly Tarasov
- Venue/year: SIGMA 14 (2018), article 058, 12 pages
- DOI/arXiv/URL: <https://doi.org/10.3842/SIGMA.2018.058> ; arXiv:1801.08529, <https://arxiv.org/abs/1801.08529>
- Progress: Proves that a second-order Fuchsian equation with n singularities of which $n-3$ are apparent is related to a hypergeometric equation by a polynomial-coefficient differential operator, generically surjectively, and applies this to curvature-1 metrics with all but three angles integral. This is a strong special case, but it assumes $n-3$ apparent singularities/all but three integer angles.
- Title: Metrics of constant positive curvature with four conic singularities on the sphere
- Author: Alexandre Eremenko
- Venue/year: Proceedings of the American Mathematical Society 148(9) (2020), 3957-3965
- DOI/arXiv/URL: arXiv:1905.02537, <https://arxiv.org/abs/1905.02537>
- Progress: Proves finiteness of metrics for four prescribed points and prescribed non- 2π -

multiple angles. Finiteness is important, but it is not a classification or “general overview” of the possible quadrilateral forms.

- Title: Classification of Generic Spherical Quadrilaterals
- Author: Andrei Gabrielov
- Venue/year: Arnold Mathematical Journal 9 (2023), 151-203
- DOI/arXiv/URL: <https://doi.org/10.1007/s40598-022-00204-3>; arXiv:2008.07999, <https://arxiv.org/abs/2008.07999>
- Progress: Classifies generic spherical quadrilaterals up to isometry, where “generic” means the side images lie on four distinct circles with no triple intersections. This is a modern classification theorem, but the genericity hypothesis excludes non-generic four-circle configurations and does not address higher polygons.
- Title: Metrics of constant positive curvature with conic singularities. A survey
- Author: Alexandre Eremenko
- Venue/year: survey/preprint, 2021; updated online PDF lists later 2023 references
- DOI/arXiv/URL: arXiv:2103.13364, <https://arxiv.org/abs/2103.13364>; updated PDF: <https://www.math.purdue.edu/~eremenko/dvi/survey1.pdf>
- Progress/status: The survey states that for curvature $\kappa \leq 0$ the conic-singularity metric problem has a complete solution, but for $\kappa > 0$ the problem is wide open. It also states that classification of metrics is equivalent to classification of developing maps, and lists many partial results, including quadrilateral and polygon special cases. This is strong evidence against a fully resolved verdict.

10.2.3.4. Why this is not fully resolved

The literature gives complete answers for several important subfamilies: classical triangles, all-integer-angle/rational-function cases, n -gons with exactly two non-integer angles, quadrilaterals with one or more integer angles, generic quadrilaterals, and some apparent-singularity/Fuchsian reductions. These results all impose extra hypotheses on angles, monodromy, genericity, number of sides, or apparent singularities.

the problem asks for the general overview/classification of the possible quadrangles and more general polygons that replace Schwarz triangles for second-order equations with more than three singular points. No citable source found proves a theorem covering all such polygons or all associated Fuchsian equations/accessory parameters. Current surveys still describe the positive-curvature/spherical-polygon version as wide open in general.

Therefore the correct status is: partially resolved.

10.2.4. Remaining open problems recorded by the database

10.2.4.1. Open problem 1

Characterize all quadruples $(\alpha_0, \alpha_1, \alpha_2, \alpha_3) \in \mathbb{R}_{>0}^4$ for which there exists a spherical quadrilateral with geodesic sides and interior angles $\pi\alpha_0, \pi\alpha_1, \pi\alpha_2, \pi\alpha_3$. In particular, determine the additional necessary and sufficient conditions beyond the Gauss–Bonnet inequality $\sum_{j=0}^3 \alpha_j > 2$ when angles larger than π are allowed.

Reference supplied by the database. Alexandre Eremenko, “Possible shapes of spherical quadrilaterals,” problem note in the author’s *Unsolved Problems in Function Theory*, originally dated February 16, 2014, revised July 21, 2025, no problem number and no journal publication; the opening problem explicitly asks which prescribed angle quadruples are realizable. [PDF](#) and [problem-list page](#).

10.2.5. Primary-source pointer

The originating ICM record is *V. Theory of Functions and Geometry* (1893). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1893.2/ICM1893.2.ocr.pdf#page=43>.

10.3. 268. ON GALOIS GEOMETRIES [1958-geom-272]

Record: 1958-geom-272

Subfield: Incidence, axiomatic, and classical geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1958: ON GALOIS GEOMETRIES

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1958/ICM1958.ocr.pdf#page=552>

10.3.1. Problem statement

For every integer $h \geq 1$, let $q = 2^h$ and let $\text{PG}(2, q)$ be the Desarguesian projective plane whose points are the one-dimensional subspaces of $\mathbb{F}_q^3$ and whose lines are the two-dimensional subspaces of $\mathbb{F}_q^3$. Classify all subsets H of the points of $\text{PG}(2, q)$ such that $|H| = q + 2$ and every line contains at most two points of H . Such a set H is called a hyperoval (or, in the terminology of the source, an oval).

10.3.2. Historical background, literature, and current progress

The problem is **unresolved**. Complete classifications are known only for bounded orders, and the available structural theorems cover restricted classes of hyperovals rather than all hyperovals in every $\text{PG}(2, 2^h)$.

The small-order classification is complete through $q = 64$. Segre proved that for $q = 2, 4, 8$ every hyperoval is a conic together with its nucleus. Hall proved that $\text{PG}(2, 16)$ has exactly two equivalence classes, the regular and Lunelli-Sce classes. Penttila and Royle proved that $\text{PG}(2, 32)$ has exactly six classes. Vandendriessche's Main Theorem proves that $\text{PG}(2, 64)$ has exactly four classes: regular, Adelaide, Subiaco I, and Subiaco II. His conclusion explicitly identifies $\text{PG}(2, 128)$ as the next open case and estimates that the same exhaustive method would require more than 10^{25} CPU-years. Thus these results exhaust only $h \leq 6$, not the quantifier over every h .

There is substantial algebraic partial progress. After a projective change of coordinates, hyperovals correspond to o -polynomials over $\mathbb{F}_q$. Caullery and Schmidt classify every o -polynomial of degree less than $\frac{1}{2}q^{1/4}$, up to their stated equivalence, as x^6 or x^{2^k} , and consequently classify exceptional o -polynomials. Ding and Zieve strengthen the degree range to $\deg f \leq q^{1/4}$. These bounds do not include arbitrary o -polynomials, whose normalized degree may grow to nearly q . Even the narrower classification of monomial o -polynomials remains a stated hard open problem: Kölsch and Kyureghyan record exhaustive computation for $h \leq 30$, the classification of fixed exponents occurring for infinitely many extensions, and low-degree classifications, but no general classification. Known infinite families and sporadic examples therefore do not constitute an exhaustive answer for all even prime powers.

Key references:

- Beniamino Segre, "On Galois Geometries," *Proceedings of the International Congress of Mathematicians, Edinburgh 1958*, Cambridge University Press, 1960, pp. 488-499. [Official](#)

[proceedings PDF](#). This is the source that states the complete even-order classification problem and gives the initial $q \leq 8$ result.

- Marshall Hall Jr., “Ovals in the Desarguesian Plane of Order 16,” *Annali di Matematica Pura ed Applicata* 102 (1975), 159-176. DOI: [10.1007/BF02410604](https://doi.org/10.1007/BF02410604). This proves the two-class classification for $q = 16$.
- Tim Penttila and Gordon F. Royle, “Classification of Hyperovals in $PG(2, 32)$,” *Journal of Geometry* 50 (1994), 151-158. DOI: [10.1007/BF01222672](https://doi.org/10.1007/BF01222672). This exhaustive search proves that there are exactly six classes for $q = 32$.
- Peter Vandendriessche, “Classification of the Hyperovals in $PG(2, 64)$,” *The Electronic Journal of Combinatorics* 26(2) (2019), #P2.35. DOI: [10.37236/7589](https://doi.org/10.37236/7589). Its Main Theorem gives all four classes for $q = 64$ and its conclusion explains why $q = 128$ remains out of computational reach.
- Florian Caullery and Kai-Uwe Schmidt, “On the Classification of Hyperovals,” *Advances in Mathematics* 283 (2015), 195-203. DOI: [10.1016/j.aim.2015.07.016](https://doi.org/10.1016/j.aim.2015.07.016); [arXiv:1403.2880](https://arxiv.org/abs/1403.2880). Theorem 1.2 classifies low-degree o -polynomials, and Corollary 1.3 classifies exceptional o -polynomials.
- Zhiguo Ding and Michael E. Zieve, “Determination of Hyperovals by Lines Through a Few Points,” preprint, revised 2026. [arXiv:2309.10866](https://arxiv.org/abs/2309.10866). Theorem 1.2 strengthens the low-degree classification to $\deg f \leq q^{1/4}$, still far short of unrestricted degree.
- Lukas Kölsch and Gohar Kyureghyan, “The Classifications of o -Monomials and of 2-to-1 Binomials Are Equivalent,” *Designs, Codes and Cryptography* 93 (2025), 961-970. DOI: [10.1007/s10623-024-01463-1](https://doi.org/10.1007/s10623-024-01463-1). The paper explicitly describes general o -monomial classification as open and records the known computational and low-degree coverage.

10.3.3. Remaining open problems recorded by the database

10.3.3.1. Open problem 1

Let $q = 128$, and let $PG(2, q)$ be the projective plane whose points are the one-dimensional subspaces of $\mathbb{F}_q^3$ and whose lines are the two-dimensional subspaces of $\mathbb{F}_q^3$. Classify, up to the action of the projective semilinear group $P\Gamma L(3, q)$, all sets H of $q + 2 = 130$ points such that every line contains at most two points of H .

Reference supplied by the database. Peter Vandendriessche, “Classification of the Hyperovals in $PG(2, 64)$,” *The Electronic Journal of Combinatorics* 26(2) (2019), #P2.35, open question in the conclusion following Theorem 25. DOI: <https://doi.org/10.37236/7589>.

10.3.3.2. Open problem 2

For every integer $n \geq 2$, let $q = 2^n$. Classify all integers d with $1 \leq d \leq q - 2$ for which

$$H_{n,d} = \{(1 : t : t^d) : t \in \mathbb{F}_q\} \cup \{(0 : 1 : 0), (0 : 0 : 1)\}$$

is a hyperoval in $PG(2, q)$; that is, $H_{n,d}$ has $q + 2$ points and no three of them are collinear. Here $(x : y : z)$ denotes the one-dimensional subspace of $\mathbb{F}_q^3$ spanned by a nonzero vector (x, y, z) .

Reference supplied by the database. Lukas Kölsch and Gohar Kyureghyan, “The Classifications of o -Monomials and of 2-to-1 Binomials Are Equivalent,” *Designs, Codes and Cryptography* 93 (2025), 961-970, open classification problem stated after Theorem 3.2. DOI: <https://doi.org/10.1007/s10623-024-01463-1>.

10.3.4. Primary-source pointer

The originating ICM record is *ON GALOIS GEOMETRIES* (1958). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1958/ICM1958.ocr.pdf#page=552>.

10.4. 269. Combinatorial theory [1970-geom-11]

Record: 1970-geom-11

Subfield: Incidence, axiomatic, and classical geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1970: Combinatorial theory

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1970.3/ICM1970.3.ocr.pdf#page=237>

10.4.1. Problem statement

Let $G = G(S)$ be a finite combinatorial geometry on a finite set S , with matroid closure operation, rank function r , geometric lattice $L(G)$ of flats, and characteristic polynomial p_G . Let $G^\perp(S)$ be the orthogonal geometry whose bases are exactly the complements in S of bases of G . Fix a finite field $GF(q)$, and define the critical exponent of G to be the least integer c such that $p_G(q^c) > 0$.

Prove or disprove Rota's form of Hadwiger's conjecture: the critical exponent of G is characterized by the absence of the corresponding forbidden segments, equivalently obstruction intervals $[A, B]$ of flats of length $r(B) - r(A)$, in $L(G)$ or in $L(G^\perp)$.

10.4.2. Historical background, literature, and current progress

10.4.3. Verdict: partially resolved

10.4.3.1. Precise problem checked

The problem is not about Hadwiger's theorem in convex geometry. It is the graph-minor Hadwiger conjecture expressed in Rota/Crapo critical-problem language for finite combinatorial geometries/matroids. In ordinary graph language the underlying conjecture is:

For every integer $t \geq 1$, every graph with no K_t minor is $(t - 1)$ -colorable; equivalently, every graph with chromatic number at least t has a K_t minor.

Rota's formulation says that for a finite combinatorial geometry $G(S)$, with flats forming the geometric lattice $L(G)$, dual/orthogonal geometry $G^\perp(S)$, and critical exponent defined by characteristic-polynomial values $p_G(q^c)$, Hadwiger's conjecture should be viewed as an obstruction criterion: critical exponent should be controlled by the absence of certain forbidden segments/obstruction intervals in $L(G)$ or $L(G^\perp)$.

10.4.3.2. Status

The full problem is not resolved. It is partially resolved: the exact Hadwiger conjecture is proved only in small cases, while the first exact open case is $t = 7$. Since the Rota critical-problem statement is a reformulation of the same graph-minor conjecture, the unresolved $t = 7$ graph case prevents a full resolution of the question.

10.4.3.3. Key literature

1. Hugo Hadwiger, “Über eine Klassifikation der Streckenkomplexe”, *Vierteljahrsschrift der Naturforschenden Gesellschaft in Zürich* 88 (1943), 133-142/143. URL: https://ngzh.ch/wp-content/uploads/2024/08/88_17.pdf

This is the original graph-theoretic source. The modern form is that every graph with chromatic number at least t contains a K_t minor.

2. Gian-Carlo Rota, “Combinatorial theory, old and new”, in *Actes du Congrès International des Mathématiciens* (Nice, 1970), Tome 3, Gauthier-Villars, Paris, 1971, pp. 229-233. URL: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1970.3/ICM1970.3.ocr.pdf>

This is the source of the stated language. Rota defines the critical exponent via $p_G(q^c) > 0$, describes the critical problem for combinatorial geometries, and states that the critical exponent is conjecturally related to absence of “forbidden segments” or “obstructions” in the geometric lattice of $G(S)$ or $G^\perp(S)$; he then says Hadwiger’s conjecture can be restated in these terms. This is a reformulation/program, not a proof.

3. Geoff Whittle, “Dowling group geometries and the critical problem”, *Journal of Combinatorial Theory, Series B* 47(1) (1989), 80-92. DOI: 10.1016/0095-8956(89)90067-1. URL: https://homepages.ecs.vuw.ac.nz/~whittle/pubs/dowling_geometries_and_critical_problem.pdf

Whittle extends the critical-problem framework using Dowling group geometries. He proves that for suitable matroids embeddable as minors of $Q_r(A)$, a critical exponent over a finite group A is well-defined and determined by characteristic-polynomial evaluation. He explicitly observes that this lets every case of Hadwiger’s conjecture be stated as a critical problem. This verifies the reformulation but does not resolve Hadwiger’s conjecture.

4. G. A. Dirac, “A Property of 4-Chromatic Graphs and some Remarks on Critical Graphs”, *Journal of the London Mathematical Society* s1-27(1) (1952), 85-92. DOI: 10.1112/jlms/s1-27.1.85.

Together with Hadwiger’s original work, this covers the small cases up to $t \leq 4$.

5. Klaus Wagner, “Über eine Eigenschaft der ebenen Komplexe”, *Mathematische Annalen* 114 (1937), 570-590. DOI: 10.1007/BF01594196. URL: <https://eudml.org/doc/159935>

Wagner proved the relevant K_5 -minor-free structure/equivalence with the Four Color Theorem. In combination with the Four Color Theorem, this gives Hadwiger’s conjecture for $t = 5$.

6. Kenneth Appel and Wolfgang Haken, “Every planar map is four colorable. Part I: Discharging”, *Illinois Journal of Mathematics* 21(3) (1977), 429-490. DOI: 10.1215/ijm/1256049011; Kenneth Appel, Wolfgang Haken, and John Koch, “Every planar map is four colorable. Part II: Reducibility”, *Illinois Journal of Mathematics* 21(3) (1977), 491-567. DOI: 10.1215/ijm/1256049012.

These prove the Four Color Theorem, which, through Wagner’s equivalence, supplies the $t = 5$ case.

7. Neil Robertson, Paul Seymour, and Robin Thomas, “Hadwiger’s conjecture for K_6 -free graphs”, *Combinatorica* 13(3) (1993), 279-361. DOI: 10.1007/BF01202354.

This proves the $t = 6$ case: every graph with no K_6 minor is 5-colorable. It is the strongest fully settled exact case.

8. Paul Seymour, “Hadwiger’s conjecture”, in *Open Problems in Mathematics*, Springer, 2016, pp. 417-437. DOI: 10.1007/978-3-319-32162-2_13.

Seymour’s survey records the conjecture as a major open problem and identifies K_7 -minor-free 6-colorability as the first unsettled exact case.

9. Sergey Norin, “Recent progress towards Hadwiger’s conjecture”, *Proceedings of the International Congress of Mathematicians 2022*, EMS Press, 2023, pp. 4606-4620. URL: <https://ems.press/content/book-chapter-files/33307>

Norin states Hadwiger’s conjecture, summarizes that it is known for $t \leq 6$, and says settling the exact conjecture for $t \geq 7$ remains extremely challenging. The paper surveys weakenings and asymptotic progress, not a complete solution.

10. Michelle Delcourt and Luke Postle, “Reducing linear Hadwiger’s conjecture to coloring small graphs”, *Journal of the American Mathematical Society* 38(2) (2025), 481-507. DOI: 10.1090/jams/1047; arXiv:2108.01633.

This gives the current type of asymptotic progress: every K_t -minor-free graph is $O(t \log \log t)$ -colorable and linear Hadwiger reduces to coloring small graphs. This is far weaker than the exact $(t - 1)$ -colorability required by Hadwiger’s conjecture.

11. Sergey Norin and Agnes Totschnig, “Every graph with no $K_7^\vee$ -minor is 6-colorable”, arXiv:2507.03244, 2025. URL: <https://arxiv.org/abs/2507.03244>

This recent preprint works near the $t = 7$ frontier and explicitly states that the next open case is $t = 7$, that every K_7 -minor-free graph is known to be 8-colorable, and that even 7-colorability for K_7 -minor-free graphs is open. Its theorem excludes the weaker graph $K_7^\vee$, not K_7 itself, so it is partial progress only.

10.4.3.4. Why no full-resolution theorem applies

The strongest exact published theorem is the $t = 6$ case of Robertson-Seymour-Thomas. The modern surveys and 2025 frontier work still identify $t = 7$ as open. Results such as $O(t\sqrt{\log t})$, $O(t(\log t)^\beta)$, and $O(t \log \log t)$ -colorability are asymptotic weakenings and do not imply $(t - 1)$ -colorability. Likewise, results excluding $K_7^\vee$, K_7^- , or extra minors have stronger hypotheses than the full obstruction criterion and do not cover all K_7 -minor-free graphs.

Therefore the exact problem is not fully resolved, but it is not fully open either because the cases $t \leq 6$ and many weakenings/special cases are resolved.

10.4.4. Remaining open problems recorded by the database

10.4.4.1. Open problem 1

Prove the first unresolved exact case of Hadwiger’s conjecture: every graph with no K_7 minor is 6-colorable.

Reference supplied by the database. Paul Seymour, “Hadwiger’s conjecture,” in Michael Th. Rassias (ed.), *Open Problems in Mathematics*, Springer, Cham, 2016, pp. 417–437 (the first open case $t = 7$), DOI: https://doi.org/10.1007/978-3-319-32162-2_13. Current-status confirmation: Sergey Norin and Agnes Totschnig, “Every graph with no $K_7^\vee$ -minor is 6-colorable,” arXiv:2507.03244 (2025), [arXiv:2507.03244](https://arxiv.org/abs/2507.03244); their theorem concerns the strictly stronger exclusion of $K_7^\vee$, not exclusion of K_7 alone.

10.4.4.2. Open problem 2

Prove the still-open weakening of the $t = 7$ case: every graph with no K_7 minor is 7-colorable.

Reference supplied by the database. Sergey Norin and Agnes Totschnig, “Every graph with no $K_7^\vee$ -minor is 6-colorable,” arXiv:2507.03244 (2025), introduction (identifying even 7-colorability of all K_7 -minor-free graphs as open), [arXiv:2507.03244](https://arxiv.org/abs/2507.03244). See also Sergey Norin, “Recent progress towards Hadwiger’s conjecture,” in *Proceedings of the International Congress of Mathematicians 2022*, Vol. 6, EMS Press, Berlin, 2023, pp. 4606–4620, DOI: <https://doi.org/10.4171/ICM2022/143>.

10.4.4.3. Open problem 3

Prove Hadwiger’s conjecture for graphs of independence number at most two: every n -vertex graph G with $\alpha(G) \leq 2$ contains a $K_{\lfloor n/2 \rfloor}$ minor.

Reference supplied by the database. Zoltán Füredi, András Gyárfás, and Gábor Simonyi, “Connected matchings and Hadwiger’s conjecture,” *Combinatorics, Probability and Computing* **14** (2005), no. 3, 435–438, Conjecture 1.1 and its $\alpha(G) = 2$ specialization, DOI: <https://doi.org/10.1017/S0963548305006759>. Current-status confirmation: Sergey Norin and Paul Seymour, “Dense minors of graphs with independence number two,” *Journal of Combinatorial Theory, Series B* **176** (2026), 101–110, introduction, DOI: <https://doi.org/10.1016/j.jctb.2025.08.005>.

10.4.4.4. Open problem 4

Find a constant-factor improvement over the Duchet–Meyniel bound for graphs of independence number two: prove that there is an absolute $\varepsilon > 0$ such that every n -vertex graph G with $\alpha(G) = 2$ contains a complete minor of order at least $(1/3 + \varepsilon)n$.

Reference supplied by the database. Paul Seymour, “Hadwiger’s conjecture,” in Michael Th. Rassias (ed.), *Open Problems in Mathematics*, Springer, Cham, 2016, pp. 417–437, Question 4.8, DOI: https://doi.org/10.1007/978-3-319-32162-2_13. Current-status confirmation: Sergey Norin and Paul Seymour, “Dense minors of graphs with independence number two,” *Journal of Combinatorial Theory, Series B* **176** (2026), 101–110, introduction (calling

any constant-factor improvement a well-known open problem), DOI: <https://doi.org/10.1016/j.jctb.2025.08.005>.

10.4.4.5. Open problem 5

Prove the Füredi–Gyárfás–Simonyi connected-matching conjecture: for every positive integer t , every graph G on $4t - 1$ vertices with $\alpha(G) = 2$ contains a connected matching of size at least t .

Reference supplied by the database. Zoltán Füredi, András Gyárfás, and Gábor Simonyi, “Connected matchings and Hadwiger’s conjecture,” *Combinatorics, Probability and Computing* **14** (2005), no. 3, 435–438, connected-matching conjecture, DOI: <https://doi.org/10.1017/S0963548305006759>. Current-status confirmation: Rong Chen and Zijian Deng, “Connected matching in graphs with independence number two,” arXiv:2409.05920 (2024), proving the conjecture only for $t \leq 22$, [arXiv:2409.05920](https://arxiv.org/abs/2409.05920).

10.4.5. Primary-source pointer

The originating ICM record is *Combinatorial theory* (1970). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1970.3/ICM1970.3.ocr.pdf#page=237>.

10.5. 270. Intersection Theorems for Finite Sets and Geometric Applications [1986-geom-39]

Record: 1986-geom-39

Subfield: Incidence, axiomatic, and classical geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1986: Intersection Theorems for Finite Sets and Geometric Applications

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1986.2/ICM1986.2.ocr.pdf#page=555>

10.5.1. Problem statement

Let X be an n -element set, and for an integer r write

$$\binom{X}{r} = \{A \subseteq X : |A| = r\}.$$

Let k and l be integers with $k > l \geq 0$ and $k \leq 2l + 1$. Prove that if $\mathcal{F} \subseteq \binom{X}{k}$ is a family of k -element subsets of X such that no two distinct members $F, F' \in \mathcal{F}$ satisfy $|F \cap F'| = l$, then

$$|\mathcal{F}| \leq \frac{\binom{2k-l-1}{k} \binom{n}{l}}{\binom{2k-l-1}{l}}.$$

Moreover, prove that equality holds if and only if there is an $(l, 2k - l - 1, n)$ -Steiner system $\mathcal{S} \subseteq \binom{X}{2k-l-1}$, meaning that every l -element subset of X is contained in exactly one member of $\mathcal{S}$, such that

$$\mathcal{F} = \Delta_k(\mathcal{S}),$$

where

$$\Delta_k(\mathcal{S}) = \{H \in \binom{X}{k} : H \subseteq S \text{ for some } S \in \mathcal{S}\}$$

is the k -shadow of $\mathcal{S}$.

10.5.2. Historical background, literature, and current progress

10.5.3. Verdict: partially resolved

10.5.3.1. Precise problem checked

The problem asks for the exact maximum size of a k -uniform family $\mathcal{F} \subseteq \binom{X}{k}$ with no two distinct members satisfying $|F \cap F'| = l$, in the range $k > l \geq 0$ and $k \leq 2l + 1$. The proposed answer is

$$|\mathcal{F}| \leq \frac{\binom{2k-l-1}{k} \binom{n}{l}}{\binom{2k-l-1}{l}},$$

with equality exactly when $\mathcal{F}$ is the k -shadow of an $(l, 2k - l - 1, n)$ -Steiner system.

Equivalently, setting $t = l + 1$, this is the $k < 2t$ Steiner-system regime of the Erdős-Sós forbidden intersection problem for $g(n, k, t)$, where one forbids intersections of size $t - 1$.

10.5.3.2. Literature status

Original/source context

- Title: *Intersection theorems for finite sets and geometric applications*
- Author: Peter Frankl
- Venue: Proceedings of the International Congress of Mathematicians, Vol. 1, 2, Berkeley, California, 1986, American Mathematical Society, Providence, RI, pp. 1419-1430
- Year: 1987
- DOI/arXiv/URL: no DOI found; bibliographic listing at <https://www.renyi.hu/~pfrankl/FPpubl.html>
- Relevance: This is the cited ICM source for the conjectural forbidden-intersection/Steiner-shadow extremal statement.

Early formulation and partial verification

- Title: *Erdős-Ko-Rado theorem—22 years later*
- Authors: Michel Deza and Peter Frankl
- Venue: SIAM Journal on Algebraic and Discrete Methods 4(4), 419-431
- Year: 1983
- DOI/arXiv/URL: <https://doi.org/10.1137/0604042>
- Relevance: Conjecture 3.8 states the same dichotomy for $m(n, k, \{0, \dots, k\} \setminus \{l\})$ for sufficiently large n . It records that the inequality in the $k \leq 2l + 1$ case was known when $k - l$ is a prime power, not in all cases.
- Title: *An extremal set theoretical characterization of some steiner systems*
- Author: Peter Frankl
- Venue: Combinatorica 3, 193-199
- Year: 1983
- DOI/arXiv/URL: <https://doi.org/10.1007/BF02579293>
- Relevance: This proves the exact upper bound and equality characterization under the extra arithmetic hypothesis that $k - l$ is prime. The Springer abstract states: if $k \leq 2t + 1$ and $k - t$ is prime, then the same bound holds and equality is characterized by the corresponding Steiner system. With $t = l$, this is precisely a prime-difference subcase of the problem, so it is only partial progress.
- Title: *Intersection theorems with geometric consequences*
- Authors: Peter Frankl and Richard M. Wilson
- Venue: Combinatorica 1, 357-368
- Year: 1981
- DOI/arXiv/URL: <https://doi.org/10.1007/BF02579457>
- Relevance: Frankl-Wilson polynomial-method bounds imply the inequality part in the

prime-power cases. Later literature explicitly summarizes this as covering $k - l$ a prime power. This still leaves composite non-prime-power values of $k - l$, and it does not by itself provide the full equality classification in every problem case.

- Title: *Forbidding just one intersection*
- Authors: Peter Frankl and Zoltán Füredi
- Venue: Journal of Combinatorial Theory, Series A 39(2), 160-176
- Year: 1985
- DOI/arXiv/URL: [https://doi.org/10.1016/0097-3165\(85\)90035-4](https://doi.org/10.1016/0097-3165(85)90035-4) ; PDF <https://www.renyi.hu/~pfrankl/j26.pdf>
- Relevance: This is a central paper on the Erdős-Sós forbidden-intersection problem. It solves the $k \geq 2t$ umvirate regime for large n and gives asymptotic/conditional progress in the $k < 2t$ regime, but it does not remove the arithmetic restrictions needed for the exact Steiner-system statement in this problem.

Current status source

- Title: *Stability for the complete intersection theorem, and the forbidden intersection problem of Erdős and Sós*
- Authors: David Ellis, Nathan Keller, and Noam Lifshitz
- Venue: Journal of the European Mathematical Society 26, 1611-1654
- Year: 2024
- DOI/arXiv/URL: <https://doi.org/10.4171/JEMS/1441> ; PDF <https://ems.press/content/serial-article-files/47528>
- Relevance: The introduction defines the forbidden intersection problem $g(n, k, t)$. It states that Frankl proved the $k < 2t$ Steiner-system bound when $k - t + 1$ is prime and that Frankl-Wilson proved the inequality when $k - t + 1$ is a prime power, but that removing the prime hypothesis remains open. Translating $t = l + 1$, the unresolved range $k < 2t$ is exactly the stated $k \leq 2l + 1$ range. Their main theorem resolves broad large- k ranges where the extremal families are Ahlswede-Khachatrian/umvirate-type families, not the full Steiner-shadow regime asked here.

10.5.3.3. Conclusion

The exact statement is not fully resolved in the literature. It is proved in important special cases, especially when $k - l$ is prime, and the inequality is known in prime-power cases. Recent literature still identifies the general $k < 2t$ case, equivalently $k \leq 2l + 1$, as a remaining open part of the Erdős-Sós forbidden intersection problem. What remains unresolved is the full bound and equality characterization for all integers $k > l \geq 0$ with $k \leq 2l + 1$, especially composite non-prime-power values of $k - l$.

10.5.4. Remaining open problems recorded by the database

10.5.4.1. Open problem 1

Prove the forbidden-intersection bound and its equality characterization in the remaining arithmetic regime where $k - l$ is not a prime power: if $k > l \geq 0$, $k \leq 2l + 1$, and $\mathcal{F} \subseteq \binom{[n]}{k}$ has $|A \cap B| \neq l$ for distinct $A, B \in \mathcal{F}$, then

$$|\mathcal{F}| \leq \frac{\binom{2k-l-1}{k} \binom{n}{l}}{\binom{2k-l-1}{l}},$$

with equality only for the k -shadow of an $(l, 2k - l - 1, n)$ -Steiner system.

Reference supplied by the database. David Ellis, Nathan Keller, and Noam Lifshitz, “Stability for the complete intersection theorem, and the forbidden intersection problem of Erdős and Sós,” *Journal of the European Mathematical Society* 26, no. 5 (2024), 1611–1654, Problem 1.3 and the discussion after Theorem 1.4 (Frankl’s conjecture; the prime-power cases follow from Frankl–Wilson, while removal of that hypothesis is stated to remain open), DOI: 10.4171/JEMS/1441. <https://doi.org/10.4171/JEMS/1441>

10.5.5. Primary-source pointer

The originating ICM record is *Intersection Theorems for Finite Sets and Geometric Applications* (1986). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1986.2/ICM1986.2.ocr.pdf#page=555>.

10.6. 271. Finite Geometries, Varieties and Codes [1998-geom-249]

Record: 1998-geom-249

Subfield: Incidence, axiomatic, and classical geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1998: Finite Geometries, Varieties and Codes

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.3/ICM1998.3.ocr.pdf#page=399>

10.6.1. Problem statement

Let q be a prime power and let $n \geq 2$ satisfy $q \geq n + 1$. A k -arc in the n -dimensional projective space $\text{PG}(n, q)$ over $\text{GF}(q)$ is a set K of $k \geq n + 1$ points such that no $n + 1$ points of K lie in a hyperplane. Prove that every such k -arc satisfies

$$k \leq \begin{cases} q + 2, & \text{if } q \text{ is even and } n \in \{2, q - 2\}, \\ q + 1, & \text{otherwise.} \end{cases}$$

10.6.2. Historical background, literature, and current progress

Verdict: unresolved.

Writing $r = n + 1$, the problem is exactly the linear MDS conjecture for an r -dimensional code over $\mathbb{F}_q$. The exceptional bounds for $r = 3$ and, by duality, $r = q - 1$ in even characteristic are known, and many nonexceptional ranges are proved, but no theorem or compatible collection of theorems covers every nonprime q and every middle dimension. No counterexample to the stated bounds is known.

Key progress:

- Simeon Ball, "On sets of vectors of a finite vector space in which every subset of basis size is a basis," *Journal of the European Mathematical Society* 14 (2012), 733-748, DOI <https://doi.org/10.4171/JEMS/316>. Ball proves the bound $q + 1$ when $r \leq p = \text{char}(\mathbb{F}_q)$; together with duality this proves the full conjecture when q is prime.
- Simeon Ball and Jan De Beule, "On sets of vectors of a finite vector space in which every subset of basis size is a basis II," *Designs, Codes and Cryptography* 65 (2012), 5-14, DOI <https://doi.org/10.1007/s10623-012-9658-6>. For nonprime $q = p^h$, they prove the MDS bound for $r \leq 2p - 2$, with the dual result covering the corresponding high-dimensional range.
- Simeon Ball and Michel Lavrauw, "Planar arcs," *Journal of Combinatorial Theory, Series A* 160 (2018), 261-287, DOI <https://doi.org/10.1016/j.jcta.2018.06.015>. Bounds forcing sufficiently large planar arcs into conics yield further MDS-conjecture ranges by projection, notably for odd square and nonsquare fields, but still leave unrestricted middle dimensions untreated.
- Simeon Ball and Michel Lavrauw, "Arcs in finite projective spaces," *EMS Surveys in Math-*

emational Sciences 6 (2019), 133-172, DOI <https://doi.org/10.4171/EMSS/33>. Conjecture 21 is the nonexceptional core of the present statement; Section 19 records proofs for $r = 4, 5$, additional small-dimensional and parity-specific cases, the ranges above, and asymptotic ranges, while presenting the general assertion as open.

- Yansheng Wu, Cunsheng Ding, and Tingfang Chen, “When Does the Extended Code of an MDS Code Remain MDS?,” *IEEE Transactions on Information Theory* 71 (2025), 263-272, DOI <https://doi.org/10.1109/TIT.2024.3494813>. Section V again states the same MDS conjecture, explicitly says that a proof remains open, and lists $r \leq 2p - 2$ and further odd-field ranges as partial results.

Thus the remaining gap is the asserted $q+1$ upper bound for general extension fields $q = p^h$, $h > 1$, in the middle dimensions not covered by the characteristic bounds, duality, low-dimensional theorems, or the known geometric/asymptotic ranges.

10.6.3. Remaining open problems recorded by the database

10.6.3.1. Open problem 1

Let $q \neq 9$ be an odd square prime power. An arc in $\text{PG}(2, q)$ is a set of points no three of which are collinear, and it is complete if it is not properly contained in a larger arc. Prove that the second-largest possible cardinality of a complete arc in $\text{PG}(2, q)$ is $q - \sqrt{q} + 1$; equivalently, prove that every complete arc having fewer than $q + 1$ points has at most $q - \sqrt{q} + 1$ points and that this bound is attained.

Reference supplied by the database. Simeon Ball and Michel Lavrauw, “Planar arcs,” *Journal of Combinatorial Theory, Series A* 160 (2018), 261-287, Conjecture 4, <https://doi.org/10.1016/j.jcta.2018.06.015>.

10.6.3.2. Open problem 2

Let q be a prime power and let r satisfy $6 \leq r \leq q - 5$. Prove that every set of $q + 1$ points in $\text{PG}(r - 1, q)$ such that every r of them span $\text{PG}(r - 1, q)$ is projectively equivalent to the normal rational curve

$$\{(1, t, t^2, \dots, t^{r-1}) : t \in \text{GF}(q)\} \cup \{(0, \dots, 0, 1)\}.$$

Reference supplied by the database. Simeon Ball and Michel Lavrauw, “Arcs in finite projective spaces,” *EMS Surveys in Mathematical Sciences* 6 (2019), no. 1-2, 133-172, Conjecture 22, <https://doi.org/10.4171/EMSS/33>.

10.6.3.3. Open problem 3

Let q be a prime power of characteristic p , let $0 \leq s \leq q$, and let

$$2 \leq r \leq \min \left\{ p + s(p - 2), \frac{q + 4 - s}{2} \right\}.$$

Let $G = (g_1, \dots, g_{2r-3+s})$ be an ordered family of nonzero vectors in $\text{GF}(q)^r$ such that every r members of G form a basis. Fix a basis B of $\text{GF}(q)^r$. Define a matrix $M_G^{\uparrow s}$ whose rows are indexed by the $(r-1)$ -subsets C of G , whose columns are indexed by ordered pairs (U, A) with U an s -subset of G and A an $(r-2)$ -subset of $G \setminus U$, and whose entries are

$$M_G^{\uparrow s}(C, (U, A)) = \begin{cases} \prod_{u \in U} \det_B(u, C), & \text{if } A \subset C \subset G \setminus U, \\ 0, & \text{otherwise.} \end{cases}$$

Here $\det_B(u, C)$ is the determinant of the matrix whose first row is u and whose remaining rows are the members of C in the order induced by G , with all vectors written in the basis B . Prove that $M_G^{\uparrow s}$ has full row rank over $\text{GF}(q)$.

Reference supplied by the database. Ameera Chowdhury, "Inclusion Matrices and the MDS Conjecture," The Electronic Journal of Combinatorics 23 (2016), no. 4, Paper P4.29, Conjecture 1.9, <https://doi.org/10.37236/5713>.

10.6.4. Primary-source pointer

The originating ICM record is *Finite Geometries, Varieties and Codes* (1998). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1998.3/ICM1998.3.ocr.pdf#page=399>.

10.7. 272. Approximate algebraic structure [2014-geom-258]

Record: 2014-geom-258

Subfield: Incidence, axiomatic, and classical geometry

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2014: Approximate algebraic structure

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.1/ICM2014.1.pdf#page=357>

10.7.1. Problem statement

Prove that, for every fixed real constant $K > 0$, the following holds. Let p tend to infinity through the odd primes, and for each such p let A_p be a subset of the projective plane $\text{PG}(2, p)$ over $\mathbb{F}_p$ such that $|A_p| \geq p/K$ and no projective line contains three distinct points of A_p . Then there is a projective plane curve C_p over $\mathbb{F}_p$, defined by a nonzero homogeneous polynomial of degree at most 3, such that

$$\frac{|A_p \setminus C_p(\mathbb{F}_p)|}{p} \rightarrow 0 \quad \text{as } p \rightarrow \infty.$$

10.7.2. Historical background, literature, and current progress

The conjecture is unresolved. For prime-order planes, the strongest general asymptotic result still applies only to arcs whose density is very close to 1: Voloch proved that a complete arc in $\text{PG}(2, p)$ that is not a conic has fewer than $44p/45 + 2$ points. Since every arc extends to a complete arc, this implies that every arc with more than $44p/45 + 2$ points is contained in a conic, and hence proves the stated conclusion in that high-density range. It gives no conclusion for a general fixed lower density $1/K$.

Later work of Ball and Lavrauw proves that an arc in $\text{PG}(2, p)$ of size at least $p - \sqrt{p} + 7/2$ is contained in a conic and classifies the arcs of sizes $p - 1$ and $p - 2$. These are refined near-maximal results, but for sufficiently large primes their density range is narrower than Voloch's and they do not address the remaining fixed-positive-density regime. Cubic constructions of size about $p/2$ also show why degree 3, rather than degree 2, is necessary in the conjectured conclusion.

Ben Green's problem list, most recently updated in December 2025, continues to state the exact conjecture as unsolved Problem 68 and cites Voloch's $44p/45$ conic-containment result as the known structural progress. Thus no verified result covers arcs of every size $|A| \geq p/K$, even allowing $o(p)$ exceptional points, and no counterexample is known.

Key references:

- **Approximate algebraic structure**, Ben Green, *Proceedings of the International Congress of Mathematicians, Seoul 2014*, Vol. I, Kyung Moon Sa, 2014, pp. 355-379, arXiv:1404.0093. Section 4.4 explicitly formulates the present conjecture and records the cubic examples.
- **Arcs in projective planes over prime fields**, José Felipe Voloch, *Journal of Geometry* 38 (1990), 198-200, DOI: <https://doi.org/10.1007/BF01222904>. The theorem

proves that a complete nonconic arc has fewer than $44p/45 + 2$ points.

- **Planar arcs**, Simeon Ball and Michel Lavrauw, *Journal of Combinatorial Theory, Series A* 160 (2018), 261-287, DOI: <https://doi.org/10.1016/j.jcta.2018.06.015>, arXiv:1705.10940. Theorem 3 gives conic containment at size $p - \sqrt{p} + 7/2$, and the paper classifies arcs of sizes $p - 1$ and $p - 2$.
- **100 Open Problems**, Ben Green, online problem collection, most recent update December 2025, Problem 68, [direct PDF](#). It records the exact conjecture as open and identifies the same remaining gap.

10.7.3. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

10.7.4. Primary-source pointer

The originating ICM record is *Approximate algebraic structure* (2014). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2014.1/ICM2014.1.pdf#page=357>.

10.8. 273. Combinatorial geometry takes the lead [2022-geom-160]

Record: 2022-geom-160

Subfield: Incidence, axiomatic, and classical geometry

Normalized status: Partially resolved

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2022: Combinatorial geometry takes the lead

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-559-9-volume-1.pdf#page=433>

10.8.1. Problem statement

Prove that for every finite configuration of points $P_1, \dots, P_n$ in $\mathbb{R}^d$, if $\mathcal{F}_r$ denotes the set of distinct rank r flats determined by these points, where a rank r flat means an $(r-1)$ -dimensional affine flat and a flat determined by the points means the minimal affine flat spanned by some subset of $\{P_1, \dots, P_n\}$, then the finite sequence $(|\mathcal{F}_r|)_r$, as r ranges over the possible ranks, is unimodal; that is, there exists an index m such that

$$|\mathcal{F}_1| \leq \dots \leq |\mathcal{F}_m| \geq |\mathcal{F}_{m+1}| \geq \dots.$$

10.8.2. Historical background, literature, and current progress

10.8.3. Status: partially resolved

10.8.3.1. Precise problem checked

The problem asks for Rota's unimodality conjecture for the Whitney numbers of the second kind of a real-realizable matroid: given a finite point configuration in $\mathbb{R}^d$, count the distinct affine flats of each rank determined by subsets of the points, and prove that the sequence of these counts is unimodal. Equivalently, after homogenizing the affine points to vectors $(P_i, 1)$, this asks whether the numbers of rank- r flats in every matroid realizable over $\mathbb{R}$ form a unimodal sequence.

10.8.3.2. Literature and progress

1. Gian-Carlo Rota, "Combinatorial theory, old and new," in *Actes du Congres International des Mathematiciens (Nice, 1970)*, Tome 3, Gauthier-Villars, Paris, 1971, pp. 229-233.
 - URL/source record: https://www.openproblemgarden.org/op/rotas_unimodal_conjecture
 - This is the original source cited in later papers for the conjecture that the rank-level sizes of a geometric lattice, equivalently Whitney numbers of the second kind of a simple matroid, are unimodal.

2. **June Huh and Botong Wang**, “Enumeration of points, lines, planes, etc.,” *Acta Mathematica* 218 (2017), no. 2, 297-317.
 - DOI: <https://doi.org/10.4310/ACTA.2017.v218.n2.a2>
 - arXiv: <https://arxiv.org/abs/1609.05484>
 - Theorem/result: for a finite subset E of the projectivization of an r -dimensional vector space over a field, with L_p the rank- p subspaces spanned by subsets of E , they prove top-heavy inequalities and lower-half monotonicity. In particular, for $p \leq q$ and $p+q \leq r$, $|L_p| \leq |L_{r-q}|$, with an order-preserving injection, and for $p < r/2$, $|L_p| \leq |L_{p+1}|$.
 - Applicability: this includes the real affine point configurations after homogenization/restricting to the affine span, so it gives major partial progress for exactly the stated setting.
 - Limitation: it does not prove the full unimodality statement. The paper explicitly states that Rota conjectured unimodality of the rank-level sizes and that “the unimodality for the upper half remains as an outstanding open problem.” Thus Huh-Wang proves the lower-half/top-heavy part, not the required eventual decrease for all upper ranks.
3. **Tom Braden, June Huh, Jacob P. Matherne, Nicholas Proudfoot, and Botong Wang**, “Singular Hodge theory for combinatorial geometries,” arXiv preprint, 2020; version 4, 2023.
 - arXiv: <https://arxiv.org/abs/2010.06088>
 - DOI: <https://doi.org/10.48550/arXiv.2010.06088>
 - Theorem/result: Theorem 1.1 proves the Dowling-Wilson top-heavy conjecture for all matroids: for any $k \leq j \leq \text{rk } M - k$, $|L_k(M)| \leq |L_j(M)|$, with compatible injections.
 - Applicability: this strengthens Huh-Wang by removing representability hypotheses, so it also covers the real-realizable configurations in the problem statement.
 - Limitation: this still proves top-heaviness/order-matchings, not full unimodality of the entire sequence of Whitney numbers of the second kind.
4. **Luis Ferroni and Benjamin Schroeter**, “Face enumeration for split matroid polytopes,” *Combinatorics, Probability and Computing*, published online 2025.
 - DOI: <https://doi.org/10.1017/S0963548325000021>
 - arXiv: <https://arxiv.org/abs/2310.05487>
 - Relevance: Section 3.2 summarizes the status after the top-heavy theorem and says that the vector counting flats of each rank is the Whitney-number sequence of the second kind; after noting that the Top-Heavy Conjecture was settled, it states that it remains open to prove or disprove unimodality of the Whitney numbers of the second kind.
5. **Paul D. Seymour**, “On the points-lines-planes conjecture,” *Journal of Combinatorial Theory, Series B* 33 (1982), no. 1, 17-26.
 - DOI: [https://doi.org/10.1016/0095-8956\(82\)90054-5](https://doi.org/10.1016/0095-8956(82)90054-5)
 - Relevance: proves a special stronger log-concavity-type points-lines-planes inequality under additional hypotheses, such as no line containing five or more points. This is special-case progress and does not cover arbitrary real point configurations or the full rank sequence.

10.8.3.3. Verdict

The problem is **partially resolved**, not fully resolved. The exact problem's setting is covered by the Huh-Wang representable-matroid theorem and by the later all-matroids top-heavy theorem, but those results prove only lower-half monotonicity/top-heaviness. They do not imply the full unimodality sequence required by Rota's conjecture, because they do not control the possible behavior within the upper half after the middle ranks. No citable result found in the current literature proves full unimodality for every real point configuration.

10.8.4. Remaining open problems recorded by the database

10.8.4.1. Open problem 1

Let M be a simple matroid representable over $\mathbb{R}$, of rank r , and let $W_k(M)$ be its number of rank- k flats. Is the upper-half sequence

$$W_{\lfloor r/2 \rfloor}(M), W_{\lfloor r/2 \rfloor + 1}(M), \dots, W_r(M)$$

unimodal? Equivalently, can the already proved lower-half monotonicity/top-heavy inequalities be completed by ruling out a decrease followed by a later increase among the upper rank levels?

Reference supplied by the database. June Huh and Botong Wang, "Enumeration of Points, Lines, Planes, Etc.," *Acta Mathematica* **218** (2017), no. 2, 297–317, discussion immediately after Conjecture 1 explicitly identifying unimodality of the "upper half" as the outstanding open problem, DOI: <https://doi.org/10.4310/ACTA.2017.v218.n2.a2>, arXiv: [1609.05484](https://arxiv.org/abs/1609.05484). The full second-kind Whitney-number unimodality problem remains open after the top-heavy theorem; see Luis Ferroni and Benjamin Schröter, "Valuative Invariants for Large Classes of Matroids," *Journal of the London Mathematical Society* (2) **110** (2024), no. 3, article e12984, §8, DOI: <https://doi.org/10.1112/jlms.12984>, arXiv: [2208.04893](https://arxiv.org/abs/2208.04893).

10.8.5. Primary-source pointer

The originating ICM record is *Combinatorial geometry takes the lead* (2022). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-559-9-volume-1.pdf#page=433>.

11. Singularity theory

These records study the local and global structure of singular maps, critical points, monodromy, and invariants of surface singularities. They connect differential topology and complex analytic geometry with deformation theory, resolution graphs, and low-dimensional topology. The surviving questions ask which algebraic or topological data determine an analytic singularity and how local models fit into global classification and realization problems.

Coverage: 5 website records; **source years:** 1970–2018; **separately listed descendant open problems:** 4.

11.1. 274. Structure locale des morphismes analytiques [1970-geom-229]

Record: 1970-geom-229

Subfield: Singularity theory

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1970: Structure locale des morphismes analytiques

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1970.2/ICM1970.2.ocr.pdf#page=265>

11.1.1. Problem statement

Determine whether every germ of a morphism between real analytic spaces (respectively, between complex analytic spaces) has a polynomial local model that is locally isotopic to the original morphism as a stratified morphism and has the same local incidence scheme. Here the incidence scheme is the finite directed graph of the local strata, labelled by their dimensions, in which there is an arrow from a stratum to every lower-dimensional stratum incident to its closure.

11.1.2. Historical background, literature, and current progress

The exact problem remains unresolved. The literature gives complete algebraization results for analytic set germs and for scalar-valued analytic function germs, over both $\mathbb{R}$ and $\mathbb{C}$, but no verified theorem or counterexample settles arbitrary analytic morphism germs while preserving their full local stratified incidence scheme up to stratified isotopy.

René Thom, “Structure locale des morphismes analytiques,” *Actes du Congrès International des Mathématiciens, Nice 1970*, vol. 2, Gauthier-Villars, 1971, pp. 257–265 ([IMU proceedings](#)). Thom formulates the problem for real and complex analytic morphisms and notes that a positive answer would imply local topological algebraicity of analytic sets.

Tadeusz Mostowski, “Topological equivalence between analytic and algebraic sets,” *Bulletin of the Polish Academy of Sciences. Mathematics* 32 (1984), no. 7–8, 393–400 ([zbMATH record](#)). Mostowski proves the induced set-germ consequence in the real case; the proof also works over $\mathbb{C}$, as checked explicitly in the next reference. This does not algebraize a general morphism or preserve all morphism incidence data.

Marcin Bilski, Adam Parusiński, and Guillaume Rond, “Local topological algebraicity of analytic function germs,” *Journal of Algebraic Geometry* 26 (2017), 177–197, <https://doi.org/10.1090/jag/667>, [arXiv:1401.5771](#). Theorem 1.2 proves that every real or complex analytic function germ $(\mathbb{K}^n, 0) \rightarrow (\mathbb{K}, 0)$ is right-topologically equivalent to a polynomial germ; Corollary 1.4 allows a singular analytic source and replaces it by an affine algebraic variety. The proof constructs a Thom stratification and uses the second Thom–Mather isotopy lemma. Thus the original problem is settled for a one-dimensional target. Example 6.3 shows that their source-homeomorphism formulation cannot simply be extended to maps into $\mathbb{K}^m$, $m > 1$, but it is not a counterexample to Thom’s more general stratified-isotopy question.

Guillaume Rond, “Local topological algebraicity with algebraic coefficients of analytic sets or functions,” *Algebra & Number Theory* 12 (2018), 1215–1231, <https://doi.org/10.2140/ant.2018.12.1215>, [arXiv:1706.05938](#). Theorem 1 strengthens the set/function result by obtaining polynomial models over the algebraic numbers. Remark 5 again explains why the scalar theorem does not extend by the same source-only equivalence to arbitrary vector-valued map germs.

Adam Parusiński and Guillaume Rond, “Płoski approximation theorem,” *Annales Polonici Mathematici* 135 (2025), 281–294, <https://doi.org/10.4064/ap241218-2-6>, [arXiv:2505.12356](#). This recent survey reviews the equisingular-deformation method and states the current algebraization theorems for analytic set germs (Theorem 3.4) and scalar analytic function germs (Theorem 3.7), without a corresponding theorem for general morphisms. The remaining gap is therefore the case of arbitrary positive-dimensional targets together with preservation of the complete local incidence structure and stratified isotopy.

11.1.3. Remaining open problems recorded by the database

11.1.3.1. Open problem 1

Let $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. Determine whether, for every analytic set germ $(V, 0) \subset (\mathbb{K}^n, 0)$, there exists a germ of a bi-Lipschitz homeomorphism $h : (\mathbb{K}^n, 0) \rightarrow (\mathbb{K}^n, 0)$ such that $h(V)$ is the germ of an algebraic subset of $\mathbb{K}^n$.

Reference supplied by the database. Marcin Bilski, Adam Parusiński, and Guillaume Rond, “Local topological algebraicity of analytic function germs,” *Journal of Algebraic Geometry* 26 (2017), 177–197, open question following Corollary 1.4, <https://doi.org/10.1090/jag/667>; [arXiv:1401.5771](#).

11.1.3.2. Open problem 2

Let $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. Determine whether every analytic function germ $g : (\mathbb{K}^n, 0) \rightarrow (\mathbb{K}, 0)$ is bi-Lipschitz right-equivalent to a polynomial germ; that is, whether there exists a germ of a bi-Lipschitz homeomorphism $h : (\mathbb{K}^n, 0) \rightarrow (\mathbb{K}^n, 0)$ such that $g \circ h$ is a polynomial germ.

Reference supplied by the database. Marcin Bilski, Adam Parusiński, and Guillaume Rond, “Local topological algebraicity of analytic function germs,” *Journal of Algebraic Geometry* 26 (2017), 177–197, open question following Corollary 1.4, <https://doi.org/10.1090/jag/667>; arXiv:1401.5771.

11.1.4. Primary-source pointer

The originating ICM record is *Structure locale des morphismes analytiques* (1970). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1970.2/ICM1970.2.ocr.pdf#page=265>.

11.2. 275. Critical Points of Smooth Functions [1974-geom-230]

Record: 1974-geom-230

Subfield: Singularity theory

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1974: Critical Points of Smooth Functions

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1974.1/ICM1974.1.ocr.pdf#page=75>

11.2.1. Problem statement

Let $f : (\mathbb{R}^n, 0) \rightarrow (\mathbb{R}, 0)$ be a C^∞ function germ with an isolated critical point at 0 and finite Milnor number

$$\mu(f) = \dim_{\mathbb{R}} \frac{\mathbb{R}[[x_1, \dots, x_n]]}{(\partial f / \partial x_1, \dots, \partial f / \partial x_n)} < \infty.$$

For $\phi \in C_c^\infty(\mathbb{R}^n)$ supported in a sufficiently small neighborhood of 0, set

$$I_{f,\phi}(h) = \int_{\mathbb{R}^n} e^{if(x)/h} \phi(x) dx, \quad h \rightarrow 0^+,$$

and define the degree of singularity of f by

$$\beta(f) = \inf \left\{ \gamma \in \mathbb{R} : I_{f,\phi}(h) = O_\phi(h^{n/2-\gamma}) \text{ as } h \rightarrow 0^+ \text{ for every such } \phi \right\}.$$

Because $\mu(f) < \infty$, the germ f is smoothly right-equivalent to a polynomial germ. Define its stable complex type to be the stable right-equivalence class of the complexification of such a polynomial representative, where stable right-equivalence allows biholomorphic changes of source coordinates and the addition of nondegenerate quadratic forms in new variables.

Call the critical point simple if its stable complex type is one of

$$A_k : x^{k+1} \ (k \geq 1), \quad D_k : x^2 y + y^{k-1} \ (k \geq 4), \quad E_6 : x^3 + y^4, \quad E_7 : x^3 + xy^3, \quad E_8 : x^3 + y^5.$$

Call it parabolic if its stable complex type is one of

$$\begin{aligned} T_{3,3,3} : x^3 + y^3 + z^3 + axyz, \quad a^3 + 27 &\neq 0, \\ T_{2,4,4} : x^4 + y^4 + z^2 + ax^2 y^2, \quad a^2 &\neq 4, \\ T_{2,3,6} : x^3 + y^6 + z^2 + ax^2 y^2, \quad 4a^3 + 27 &\neq 0, \end{aligned}$$

and call it hyperbolic if its stable complex type is

$$T_{p,q,r} : x^p + y^q + z^r + axyz, \quad p, q, r \geq 2, \quad \frac{1}{p} + \frac{1}{q} + \frac{1}{r} < 1, \quad a \neq 0.$$

Prove that every finite-multiplicity isolated critical point that is neither simple, parabolic, nor hyperbolic satisfies

$$\beta(f) > \frac{1}{2}.$$

11.2.2. Historical background, literature, and current progress

The problem is **unresolved**. I found neither a proof of the strict inequality for every finite-multiplicity real critical-point germ outside the three stated classes nor a counterexample. The literature gives substantial computations and several broad conditional results, but it does not cover arbitrary higher-modality germs.

V. I. Arnol'd, "Remarks on the Stationary Phase Method and Coxeter Numbers," *Russian Mathematical Surveys* **28**(5) (1973), 19–48, DOI <https://doi.org/10.1070/RM1973v028n05ABEH001609>. Definition 2.1.1 gives the estimate-based invariant used in the statement. Proposition 4.3.1 computes it for quasihomogeneous germs; Proposition 4.3.2 gives values strictly greater than 1/2 for the fourteen exceptional unimodal families; and Corollary 4.4.4 gives value 1/2 for every hyperbolic germ. The paper's proposed general lower-bound argu-

ment uses its then-unproved semicontinuity hypothesis, so it does not prove the strict assertion in the problem.

V. I. Arnol'd, "Critical Points of Smooth Functions and Their Normal Forms," *Russian Mathematical Surveys* **30**(5) (1975), 1–75, DOI <https://doi.org/10.1070/RM1975v030n05ABEH001521>. Theorem XIX records the exact values for simple, parabolic, and hyperbolic singularities and immediately states the present strict inequality for all remaining singularities as an expectation. The same article classifies all unimodal singularities and lists all bimodal singularities, thereby delimiting the first finite collection of cases in which the invariant can be checked explicitly.

A. N. Varchenko, "Newton Polyhedra and Estimation of Oscillating Integrals," *Functional Analysis and Its Applications* **10**(3) (1976), 175–196, DOI <https://doi.org/10.1007/BF01075524>. Varchenko determines the leading exponent from the Newton polyhedron when the real principal part is nondegenerate under the paper's hypotheses, proves the adapted-coordinate formula for analytic phases in two variables, and calculates the indices of the singularities in Arnol'd's classified lists. These results verify many cases of the desired inequality. The same paper gives counterexamples to Arnol'd's separate semicontinuity conjecture and shows that the real oscillation index can depend on the real form, preventing that conjectural reduction from resolving the present problem.

David Chillingworth and Carmen Romero-Fuster, "Remarks on the Singularity Indices of Arnol'd and Berry," *Proceedings of the Royal Society of Edinburgh, Section A: Mathematics* **94**(3–4) (1983), 339–350, DOI <https://doi.org/10.1017/S0308210500015705>. This gives an invariant algebraic interpretation and formulas for quasihomogeneous functions and their unfoldings, while explicitly noting the difficulties for nonquasihomogeneous functions. It therefore extends the computable part of the problem but not the unrestricted assertion.

V. I. Arnold, S. M. Gusein-Zade, and A. N. Varchenko, *Singularities of Differentiable Maps, Volume 2: Monodromy and Asymptotics of Integrals*, Birkhäuser, 1988; reprint 2012, DOI <https://doi.org/10.1007/978-0-8176-8343-6>. Sections 6.1.9–6.1.10 and Chapter 9 calculate the singular index for all simple, unimodal, and bimodal critical points, all listed critical points of Milnor number below 16, and all listed classes of codimension below 10. The tabulated non-simple, non-parabolic, non-hyperbolic cases have index greater than $1/2$. The book also exhibits complex-equivalent real germs with different real indices, confirming that results about the complex spectrum alone do not exhaust the real problem.

What remains is the strict inequality for finite-multiplicity germs outside those classifications and Newton-polyhedron hypotheses, particularly higher-modality, Newton-degenerate real germs. No source found proves that these remaining germs cannot have $\beta = 1/2$ (or smaller), and no source found constructs such a germ.

11.2.3. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

11.2.4. Primary-source pointer

The originating ICM record is *Critical Points of Smooth Functions* (1974). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proc>

[eedings/ICM1974.1/ICM1974.1.ocr.pdf#page=75.](#)

11.3. 276. Le Groupe de Monodromie du Déploiement des Singularités Isolées de Courbes Planes. II [1974-geom-231]

Record: 1974-geom-231

Subfield: Singularity theory

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1974: Le Groupe de Monodromie du Déploiement des Singularités Isolées de Courbes Planes. II

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1974.1/ICM1974.1.ocr.pdf#page=451>

11.3.1. Problem statement

Classify the polynomial germs $f : (\mathbb{C}^{n+1}, 0) \rightarrow (\mathbb{C}, 0)$ having an isolated critical point at the origin for which there exist a Milnor ball $B_\varepsilon \subset \mathbb{C}^{n+1}$, a one-parameter polynomial deformation $F : \mathbb{C}^{n+1} \times \mathbb{C} \rightarrow \mathbb{C}$ with $F(x, 0) = f(x)$, a number $\eta > 0$, and a parameter $t_0 \in \mathbb{C}$ with $|t_0| \leq \eta$, such that $\{x \in \mathbb{C}^{n+1} : F(x, t) = 0\}$ meets ∂B_ε transversely for every $|t| \leq \eta$, while the polynomial $\tilde{f}(x) = F(x, t_0)$ has real coefficients and has exactly $\mu(f)$ critical points in B_ε , all of which lie in $B_\varepsilon \cap \mathbb{R}^{n+1}$. Here

$$\mu(f) = \dim_{\mathbb{C}} \frac{\mathbb{C}\{x_0, \dots, x_n\}}{(\partial f / \partial x_0, \dots, \partial f / \partial x_n)}$$

is the Milnor number. (The $\mu(f)$ critical points of $\tilde{f}$ are then nondegenerate.)

11.3.2. Historical background, literature, and current progress

The problem is **unresolved**. No complete criterion is known for an arbitrary isolated hypersurface singularity, and the literature still treats existence of a deformation with all μ Morse critical points real (an M -morsification, or strong real morsification) only for special classes. In particular, even a stronger existence question restricted to real plane-curve singularities remains conjectural, so the available results do not classify all germs in the statement.

Norbert A'Campo, "Le groupe de monodromie du déploiement des singularités isolées de courbes planes. II," *Proceedings of the International Congress of Mathematicians (Vancouver, 1974)*, vol. 1, Canadian Mathematical Congress, 1975, pp. 395–404, [direct PDF](#). Section 1 poses the present question. Theorem 1 proves a substantive special case: if a real plane-curve germ factors into analytically irreducible real factors (so all local branches are real), it has a real Morse confluence, indeed one with at most three critical values. The same article also gives the Chebyshev construction for every one-variable isolated singularity. These results do not cover plane curves with non-real conjugate branches or arbitrary higher-dimensional singularities.

Peter Leviant and Eugenio Shustin, "Morsifications of real plane curve singularities," *Journal of Singularities* **18** (2018), 307–328, DOI <https://doi.org/10.5427/jsing.2018.18p>, arXiv:1703.05510. Their Theorem 1 proves existence for real plane-curve singularities satisfying an explicit admissibility condition at every real infinitely near point; this includes all to-

tally real and all real quasihomogeneous plane-curve singularities, as well as further cases with conjugate branches. Their Conjecture 1 states that every real plane-curve singularity has a real morsification. Their convention additionally requires every saddle to have critical value zero, so their theorem implies the all-critical-points-real condition in the present problem, while the conjecture highlights a remaining gap in a restricted class.

Lars Andersen, “On Isolated Real Singularities I,” preprint (2021), arXiv:2110.04407. Definition 1 distinguishes weak real morsifications from strong ones, the latter having exactly the complex Milnor number of real critical points. Proposition 2.0.1 gives a real Morse perturbation for every real isolated hypersurface singularity but does not ensure that all μ critical points are real. Proposition 2.0.2 proves the strong property when the complexification is a simple ADE singularity, in any dimension. Thus generic real morsifiability is not the condition asked here, and the paper supplies only ADE coverage for the exact maximal-reality requirement.

Sergey Fomin, Pavlo Pylyavskyy, Eugenii Shustin, and Dylan Thurston, “Morsifications and mutations,” *Journal of the London Mathematical Society* **105** (2022), 2478–2554, DOI <https://doi.org/10.1112/jlms.12566>, arXiv:1711.10598. Definition 1.10 again requires all critical points to be real (and all saddles to lie at level zero), and Conjecture 1.12 repeats that every real plane-curve singularity should possess such a morsification. The authors record only the totally real case and the Leviant–Shustin class as established. This later explicit conjectural status rules out a complete resolution of the classification in the statement.

V. A. Vassiliev, “A Few Problems on Monodromy and Discriminants,” *Arnold Mathematical Journal* **1** (2015), 201–209, DOI <https://doi.org/10.1007/s40598-015-0011-9>, arXiv:1504.01997. Section 6 formulates open problems about the number and distribution of real critical points in real morsifications; Problem 6C asks whether every chamber of the discriminant complement of a versal deformation contains a morsification with all $\mu(f)$ critical points real, noting this only for simple singularities. This does not itself equal the present existence/classification question, but it confirms that maximal real morsification beyond the simple classes remains incompletely understood.

Consequently, the established special cases leave both the general existence question and the requested classification open; no counterexample or exhaustive necessary-and-sufficient criterion covering all dimensions and all isolated polynomial germs was found.

11.3.3. Remaining open problems recorded by the database

11.3.3.1. Open problem 1

Prove that every reduced real analytic plane-curve germ $C = \{f(x, y) = 0\} \subset (\mathbb{C}^2, 0)$, where $f \in \mathbb{R}\{x, y\}$ has an isolated critical point at the origin, admits a real analytic one-parameter deformation $f_t \in \mathbb{R}\{x, y\}$, $f_0 = f$, such that, for every sufficiently small $t > 0$, the function f_t has exactly $\mu(f)$ critical points in a fixed Milnor ball, all of them are real and nondegenerate, and every saddle point of $f_t|_{\mathbb{R}^2}$ has critical value zero. Here $\mu(f) = \dim_{\mathbb{C}} \mathbb{C}\{x, y\} / (\partial f / \partial x, \partial f / \partial y)$.

Reference supplied by the database. Sergey Fomin, Pavlo Pylyavskyy, Eugenii Shustin, and Dylan Thurston, “Morsifications and mutations,” *Journal of the London Mathematical Society* **105**(4) (2022), 2478–2554, Conjecture 1.12. DOI: <https://doi.org/10.1112/jlms.12566>; arXiv:1711.10598.

11.3.4. Primary-source pointer

The originating ICM record is *Le Groupe de Monodromie du Déploiement des Singularités Isolées de Courbes Planes. II* (1974). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1974.1/ICM1974.1.ocr.pdf#page=451>.

11.4. 277. Le Groupe de Monodromie du Déploiement des Singularités Isolées de Courbes Planes. II [1974-geom-232]

Record: 1974-geom-232

Subfield: Singularity theory

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 1974: Le Groupe de Monodromie du Déploiement des Singularités Isolées de Courbes Planes. II

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1974.1/ICM1974.1.ocr.pdf#page=451>

11.4.1. Problem statement

Let $f : (\mathbb{C}^3, 0) \rightarrow (\mathbb{C}, 0)$ be a holomorphic function germ with an isolated critical point at 0, and let

$$\mu(f) = \dim_{\mathbb{C}} \frac{\mathbb{C}\{x, y, z\}}{(\partial f / \partial x, \partial f / \partial y, \partial f / \partial z)}$$

be its Milnor number. Say that f admits a *Morse confluence with k critical values* if there exist a sufficiently small Milnor ball $B \subset \mathbb{C}^3$, a one-parameter holomorphic deformation $F : B \times D \rightarrow \mathbb{C}$ with $F(\cdot, 0) = f$, and $t_0 \in D$ such that $F(\cdot, t_0)$ has exactly $\mu(f)$ critical points in B , all nondegenerate, and the images of these critical points comprise exactly k distinct values.

It is known that every suspension $f(x, y, z) = g(x, y) + z^a$, where $a \geq 2$ and $g : (\mathbb{C}^2, 0) \rightarrow (\mathbb{C}, 0)$ is a simple isolated plane hypersurface singularity, admits such a Morse confluence for some k with $1 \leq k \leq 3$. Classify all isolated hypersurface singularities $f : (\mathbb{C}^3, 0) \rightarrow (\mathbb{C}, 0)$ that admit a Morse confluence with k critical values for some $1 \leq k \leq 3$.

11.4.2. Historical background, literature, and current progress

The problem is **unresolved**. No citable source located gives a necessary-and-sufficient classification of all isolated germs $f : (\mathbb{C}^3, 0) \rightarrow (\mathbb{C}, 0)$ having a full Morse perturbation on at most three critical levels.

The strongest verified progress is as follows.

- Norbert A'Campo, *Le groupe de monodromie du déploiement des singularités isolées de courbes planes. II*, Proceedings of the International Congress of Mathematicians (Vancouver, 1974), Vol. 1, Canadian Mathematical Congress, 1975, pp. 395-404, [direct proceedings PDF](#). Theorem 2 proves that an isolated hypersurface singularity has a Morse confluence with at most two critical values if and only if it is simple (ADE). The same paper constructs the stated three-value family $g(x, y) + z^a$, with g simple, and then explicitly poses the classification for germs in three variables. Thus it completely settles the subcase $k \leq 2$ and supplies a substantial family for $k = 3$, but not the requested classification.
- S. M. Gusein-Zade, *Singularities admitting perturbations with a small number of critical values*,

Functional Analysis and Its Applications 21(1) (1987), 66-68, <https://doi.org/10.1007/BF01077991>. Assertion 1 proves that, for a singularity of modality m , almost every point of its μ -constant stratum admits no perturbation into μ Morse critical points with fewer than $m + 2$ critical values. In particular, for modality at least two, the desired three-value property is generically impossible. The paper also gives a bimodal $J_{k,0}$ -type family whose generic members have no such three-value perturbation. These are exclusion results for generic members, not a classification of all exceptional members; after stabilization by a nondegenerate square they apply to germs in three variables.

- Vladimir I. Arnold (ed.), *Arnold's Problems*, Springer/PHASIS, 2004-2005, Problem 1975-13, <https://doi.org/10.1007/b138219>. This records the broader problem of determining the minimum number of critical values in a full morsification and states the dimension-based value only conjecturally. It supplies no classification that closes A'Campo's three-variable question.

Consequently, the established results characterize the one- and two-level cases, construct important three-level examples, and show that three-level morsifications are nongeneric in modality at least two. What remains open is a necessary-and-sufficient description of every exceptional three-variable isolated hypersurface singularity admitting exactly three critical values.

11.4.3. Remaining open problems recorded by the database

No separate descendant problem is listed in the website's "Some remaining open problems" field. This does not imply that the main statement has been resolved; the status assessment above governs the record.

11.4.4. Primary-source pointer

The originating ICM record is *Le Groupe de Monodromie du Déploiement des Singularités Isolées de Courbes Planes. II* (1974). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1974.1/ICM1974.1.ocr.pdf#page=451>.

11.5. 278. Pairs of invariants of surface singularities [2018-geom-330]

Record: 2018-geom-330

Subfield: Singularity theory

Normalized status: Open / open-ended

Dataset review state: unreviewed (reviewed flag: no)

Original source: ICM 2018: Pairs of invariants of surface singularities

Source link: <https://www.mathunion.org/fileadmin/ICM/Proceedings/uploads/ICM-2018-vol2-ver1-eb.pdf#page=840>

11.5.1. Problem statement

Let

$$f = f_d + f_{d+1} + f_{d+2} + \cdots \in \mathbb{C}\{x, y, z\}$$

be a convergent power series defining an isolated hypersurface surface singularity

$$(X, 0) = (\{f = 0\}, 0) \subset (\mathbb{C}^3, 0),$$

where f_j is homogeneous of degree j and $f_d \neq 0$. Assume that the projective plane curve

$$C = \{f_d = 0\} \subset \mathbb{CP}^2$$

is reduced, irreducible, rational, and cuspidal, and that

$$\{f_{d+1} = 0\} \cap \text{Sing}(C) = \emptyset.$$

Thus $(X, 0)$ is the superisolated surface singularity associated with C .

Let $p_1, \dots, p_v$ be the cusps of C . For each i , let $\mathcal{K}_i = C \cap S_\varepsilon^3(p_i) \subset S_\varepsilon^3(p_i) \cong S^3$ be the local algebraic knot of the plane-curve singularity (C, p_i) , and let $\Delta_i(t) \in \mathbb{Z}[t]$ be its Alexander polynomial in the normalization $\Delta_i(0) = \Delta_i(1) = 1$. Set

$$\Delta(t) = \prod_{i=1}^v \Delta_i(t).$$

Define

$$R(t) = \frac{1}{d} \sum_{\xi^d=1} \frac{\Delta(\xi t^{1/d})}{(1 - \xi t^{1/d})^2} - \frac{1 - t^d}{(1 - t)^3},$$

where the sum is over all complex d -th roots of unity and the root-of-unity average is interpreted formally (it is independent of the choice of $t^{1/d}$). For these data, the resulting expression $R(t)$ is a polynomial.

Prove or disprove that every such superisolated surface singularity satisfies

$$R(1) \leq 0.$$

11.5.2. Historical background, literature, and current progress

The problem is unresolved. It is the “Weak Conjecture” $R(1) \leq 0$ introduced by Bodnár and Némethi. No general proof or counterexample was found in the subsequent literature.

The strongest complete case result is for curves with at most two cusps. Bodnár and Némethi prove the lattice-cohomological equivalent

$$\mathrm{eu} \, \mathbb{H}_{\mathrm{can}}^*(S_{-d}^3(\#\mathcal{K}_i)) \leq \mathrm{eu} \, \mathbb{H}_{\mathrm{can}}^0(S_{-d}^3(\#\mathcal{K}_i))$$

when $v \leq 2$, using the surgery computation of Némethi and Román and the semigroup-distribution theorem of Borodzik and Livingston. They also verify $R(1) \leq 0$ for every multicuspidal family then known: three infinite tricuspidal series, the exceptional tricuspidal quintic, and the four-cuspidal quintic. Their degree-eight tricuspidal example disproves only the stronger coefficientwise conjecture; it satisfies the weaker inequality in this problem.

Later, Koras and Palka proved that a rational cuspidal plane curve has at most four cusps and that the four-cusp curve is the unique four-cuspidal quintic. Combined with the preceding checks, this covers $v = 1, 2, 4$. The remaining gap is the general $v = 3$ case: the known tricuspidal families satisfy the inequality, but tricuspidal rational cuspidal curves have not been classified in full and no argument covers every possible tricuspidal curve. Palka and Peřka’s classification covers curves satisfying the additional Negativity Conjecture, so it does not exhaust the hypotheses of the present problem.

Key references:

- József Bodnár and András Némethi, “Lattice cohomology and rational cuspidal curves,” *Mathematical Research Letters* **23** (2016), 339–375. DOI: <https://doi.org/10.4310/MRL.2016.v23.n2.a3>; arXiv: [1405.0437](https://arxiv.org/abs/1405.0437). Conjectures 1.2.10/3.2.2 are the exact inequality and its lattice-cohomology form; Sections 3.3 and 4 prove $v \leq 2$ and check all listed $v \geq 3$ examples.
- Maciej Borodzik and Charles Livingston, “Heegaard Floer homology and rational cuspidal curves,” *Forum of Mathematics, Sigma* **2** (2014), e28. DOI: <https://doi.org/10.1017/fms.2014.28>; arXiv: [1304.1062](https://arxiv.org/abs/1304.1062). Their semigroup infimal-convolution equalities supply the Heegaard-Floer constraint used in the one- and two-cusp results.
- András Némethi and Fernando Román, “The lattice cohomology of $S_{-d}^3(K)$,” in *Zeta Functions in Algebra and Geometry*, Contemporary Mathematics **566** (2012). DOI: <https://doi.org/10.1090/conm/566/11224>. Its surgery formulas underlie the lattice-cohomological reformulation and the proof for two cusps.
- Mariusz Koras and Karol Palka, “Complex planar curves homeomorphic to a line have at most four singular points,” *Journal de Mathématiques Pures et Appliquées* **158** (2022), 144–182. DOI: <https://doi.org/10.1016/j.matpur.2021.11.003>; arXiv: [1905.11376](https://arxiv.org/abs/1905.11376). This reduces the number of cusps to at most four and classifies the four-cusp case.
- Karol Palka and Tomasz Peřka, “Classification of planar rational cuspidal curves II. Log del Pezzo models,” *Proceedings of the London Mathematical Society* **120** (2020), 642–703. DOI: <https://doi.org/10.1112/plms.12300>; arXiv: [1810.08180](https://arxiv.org/abs/1810.08180). This gives a classification under the additional Negativity Conjecture, not an unconditional exhaustion of the curves in the problem.
- András Némethi, “Pairs of invariants of surface singularities,” in *Proceedings of the International Congress of Mathematicians 2018*, Vol. II (2018), 767–798, [direct PDF](#). Conjectures 6.4.5–6.4.6 restate the exact inequality and record the same $v \leq 2$ proof and the unresolved $v \geq 3$ gap.

11.5.3. Remaining open problems recorded by the database

11.5.3.1. Open problem 1

Let $C \subset \mathbb{CP}^2$ be a reduced, irreducible, rational plane curve whose singularities are all cusps, and assume that C has at least three cusps. For a cusp p , write

$$[m_0, m_1, \dots, m_r]$$

for the multiplicity sequence of the successive strict transforms of (C, p) in its minimal embedded resolution, with the terminal entries equal to 1 omitted; write q^s for s consecutive entries equal to q .

Prove or disprove that the degree and the cusp multiplicity sequences of C must occur in one of the following families:

1. $d \geq 4$, with integers $n \geq m \geq 1$ satisfying $n + m = d - 2$, and three cusps with multiplicity sequences

$$[d - 2], \quad [2^n], \quad [2^m].$$

2. $d = 2k + 3$ for some $k \geq 1$, and three cusps with multiplicity sequences

$$[d - 3, 2^k], \quad [3^k], \quad [2].$$

3. $d = 3k + 4$ for some $k \geq 1$, and three cusps with multiplicity sequences

$$[d - 4, 3^k], \quad [4^k, 2^2], \quad [2].$$

4. $d = 5$, and three cusps, each with multiplicity sequence $[2^2]$.

5. $d = 5$, and four cusps with multiplicity sequences

$$[2^3], \quad [2], \quad [2], \quad [2].$$

Reference supplied by the database. Jens Piontkowski, "On the Number of Cusps of Rational Cuspidal Plane Curves," *Experimental Mathematics* **16**(2) (2007), 251–255, Conjecture 4. DOI: <https://doi.org/10.1080/10586458.2007.10128996>.

11.5.4. Primary-source pointer

The originating ICM record is *Pairs of invariants of surface singularities* (2018). The source link retained by the dataset is <https://www.mathunion.org/fileadmin/ICM/Proceedings/uploads/ICM-2018-vol2-ver1-eb.pdf#page=840>.

Part II.

Audit and Reference Apparatus

A. Master coverage table

This table is the completeness certificate for the volume. Every one of the 278 Geometry records in the frozen API payload appears exactly once in the numbered body.

No.	ID	Year	Subfield	Open	Status
1	1958-geom-7	1958	Global/Riemannian geometry	1	Partially resolved
2	1970-geom-12	1970	Global/Riemannian geometry	1	Open / open-ended
3	1970-geom-13	1970	Global/Riemannian geometry	4	Partially resolved
4	1974-geom-20	1974	Global/Riemannian geometry	1	Partially resolved
5	1974-geom-23	1974	Global/Riemannian geometry	1	Partially resolved
6	1974-geom-300	1974	Global/Riemannian geometry	1	Open / open-ended
7	1978-geom-25	1978	Global/Riemannian geometry	2	Partially resolved
8	1978-geom-28	1978	Global/Riemannian geometry	1	Partially resolved
9	1978-geom-29	1978	Global/Riemannian geometry	1	Partially resolved
10	1978-geom-303	1978	Global/Riemannian geometry	0	Open / open-ended
11	1983-geom-32	1983	Global/Riemannian geometry	2	Partially resolved
12	1983-geom-33	1983	Global/Riemannian geometry	1	Partially resolved
13	1983-geom-35	1983	Global/Riemannian geometry	0	Partially resolved
14	1986-geom-290	1986	Global/Riemannian geometry	1	Open / open-ended
15	1986-geom-304	1986	Global/Riemannian geometry	1	Open / open-ended
16	1990-geom-305	1990	Global/Riemannian geometry	1	Open / open-ended
17	1990-geom-307	1990	Global/Riemannian geometry	2	Open / open-ended
18	1994-geom-51	1994	Global/Riemannian geometry	1	Partially resolved
19	1994-geom-59	1994	Global/Riemannian geometry	0	Partially resolved
20	1994-geom-308	1994	Global/Riemannian geometry	2	Open / open-ended
21	1994-geom-338	1994	Global/Riemannian geometry	1	Open / open-ended
22	1998-geom-63	1998	Global/Riemannian geometry	1	Open / open-ended
23	1998-geom-64	1998	Global/Riemannian geometry	1	Partially resolved
24	1998-geom-74	1998	Global/Riemannian geometry	1	Partially resolved
25	2002-geom-76	2002	Global/Riemannian geometry	0	Partially resolved
26	2002-geom-81	2002	Global/Riemannian geometry	2	Partially resolved
27	2002-ag-91	2002	Global/Riemannian geometry	1	Partially resolved
28	2002-ag-92	2002	Global/Riemannian geometry	0	Partially resolved
29	2002-ag-93	2002	Global/Riemannian geometry	3	Partially resolved
30	2002-geom-253	2002	Global/Riemannian geometry	2	Open / open-ended
31	2002-geom-254	2002	Global/Riemannian geometry	2	Open / open-ended
32	2002-geom-255	2002	Global/Riemannian geometry	0	Open / open-ended
33	2002-geom-276	2002	Global/Riemannian geometry	1	Open / open-ended
34	2002-geom-277	2002	Global/Riemannian geometry	1	Open / open-ended
35	2002-geom-280	2002	Global/Riemannian geometry	2	Open / open-ended
36	2002-geom-311	2002	Global/Riemannian geometry	2	Open / open-ended
37	2002-geom-312	2002	Global/Riemannian geometry	1	Open / open-ended
38	2002-geom-313	2002	Global/Riemannian geometry	1	Open / open-ended
39	2002-geom-314	2002	Global/Riemannian geometry	3	Open / open-ended
40	2002-geom-315	2002	Global/Riemannian geometry	2	Open / open-ended
41	2006-geom-91	2006	Global/Riemannian geometry	0	Open / open-ended
42	2006-geom-340	2006	Global/Riemannian geometry	1	Open / open-ended
43	2010-geom-114	2010	Global/Riemannian geometry	0	Partially resolved
44	2010-geom-115	2010	Global/Riemannian geometry	1	Partially resolved

No.	ID	Year	Subfield	Open	Status
45	2010-geom-116	2010	Global/Riemannian geometry	1	Partially resolved
46	2010-geom-118	2010	Global/Riemannian geometry	1	Partially resolved
47	2010-geom-119	2010	Global/Riemannian geometry	0	Partially resolved
48	2010-geom-286	2010	Global/Riemannian geometry	2	Open / open-ended
49	2010-geom-317	2010	Global/Riemannian geometry	4	Open / open-ended
50	2010-geom-318	2010	Global/Riemannian geometry	1	Open / open-ended
51	2010-geom-319	2010	Global/Riemannian geometry	2	Open / open-ended
52	2010-geom-320	2010	Global/Riemannian geometry	0	Open / open-ended
53	2010-geom-321	2010	Global/Riemannian geometry	1	Open / open-ended
54	2010-geom-322	2010	Global/Riemannian geometry	2	Open / open-ended
55	2010-geom-323	2010	Global/Riemannian geometry	3	Open / open-ended
56	2014-geom-122	2014	Global/Riemannian geometry	0	Partially resolved
57	2014-geom-128	2014	Global/Riemannian geometry	1	Partially resolved
58	2014-geom-129	2014	Global/Riemannian geometry	1	Partially resolved
59	2014-geom-130	2014	Global/Riemannian geometry	0	Partially resolved
60	2014-geom-135	2014	Global/Riemannian geometry	1	Partially resolved
61	2014-geom-136	2014	Global/Riemannian geometry	1	Partially resolved
62	2014-geom-137	2014	Global/Riemannian geometry	1	Partially resolved
63	2014-geom-294	2014	Global/Riemannian geometry	2	Open / open-ended
64	2014-geom-324	2014	Global/Riemannian geometry	2	Open / open-ended
65	2014-geom-325	2014	Global/Riemannian geometry	1	Open / open-ended
66	2014-geom-326	2014	Global/Riemannian geometry	3	Open / open-ended
67	2014-geom-327	2014	Global/Riemannian geometry	3	Open / open-ended
68	2014-geom-336	2014	Global/Riemannian geometry	2	Open / open-ended
69	2018-geom-138	2018	Global/Riemannian geometry	1	Partially resolved
70	2018-geom-298	2018	Global/Riemannian geometry	1	Open / open-ended
71	2018-geom-328	2018	Global/Riemannian geometry	2	Open / open-ended
72	2018-geom-329	2018	Global/Riemannian geometry	1	Open / open-ended
73	2022-geom-161	2022	Global/Riemannian geometry	2	Partially resolved
74	2022-geom-162	2022	Global/Riemannian geometry	1	Partially resolved
75	2022-geom-163	2022	Global/Riemannian geometry	3	Partially resolved
76	2022-geom-164	2022	Global/Riemannian geometry	1	Partially resolved
77	2022-geom-166	2022	Global/Riemannian geometry	2	Partially resolved
78	2022-geom-333	2022	Global/Riemannian geometry	1	Open / open-ended
79	1904-geom-218	1904	Discrete and polyhedral geometry	2	Open / open-ended
80	1950-geom-6	1950	Discrete and polyhedral geometry	2	Partially resolved
81	1970-geom-16	1970	Discrete and polyhedral geometry	1	Partially resolved
82	1978-geom-238	1978	Discrete and polyhedral geometry	2	Open / open-ended
83	1978-geom-302	1978	Discrete and polyhedral geometry	1	Open / open-ended
84	1986-geom-274	1986	Discrete and polyhedral geometry	2	Open / open-ended
85	1994-geom-52	1994	Discrete and polyhedral geometry	0	Partially resolved
86	1994-geom-53	1994	Discrete and polyhedral geometry	1	Partially resolved
87	1994-geom-54	1994	Discrete and polyhedral geometry	1	Partially resolved
88	1994-geom-55	1994	Discrete and polyhedral geometry	2	Partially resolved
89	1994-geom-241	1994	Discrete and polyhedral geometry	1	Open / open-ended
90	1994-geom-242	1994	Discrete and polyhedral geometry	3	Open / open-ended
91	1998-geom-244	1998	Discrete and polyhedral geometry	3	Open / open-ended
92	1998-geom-245	1998	Discrete and polyhedral geometry	3	Open / open-ended
93	1998-geom-246	1998	Discrete and polyhedral geometry	0	Open / open-ended
94	1998-geom-247	1998	Discrete and polyhedral geometry	1	Open / open-ended
95	1998-geom-248	1998	Discrete and polyhedral geometry	3	Open / open-ended
96	1998-geom-291	1998	Discrete and polyhedral geometry	0	Open / open-ended
97	2002-geom-77	2002	Discrete and polyhedral geometry	1	Partially resolved
98	2002-geom-78	2002	Discrete and polyhedral geometry	1	Partially resolved
99	2002-geom-87	2002	Discrete and polyhedral geometry	0	Partially resolved
100	2002-geom-88	2002	Discrete and polyhedral geometry	0	Partially resolved

No.	ID	Year	Subfield	Open	Status
101	2002-geom-250	2002	Discrete and polyhedral geometry	2	Open / open-ended
102	2002-geom-292	2002	Discrete and polyhedral geometry	3	Open / open-ended
103	2002-geom-293	2002	Discrete and polyhedral geometry	1	Open / open-ended
104	2010-geom-257	2010	Discrete and polyhedral geometry	2	Open / open-ended
105	2014-geom-260	2014	Discrete and polyhedral geometry	3	Open / open-ended
106	2018-geom-149	2018	Discrete and polyhedral geometry	1	Partially resolved
107	2018-geom-158	2018	Discrete and polyhedral geometry	1	Partially resolved
108	2018-geom-264	2018	Discrete and polyhedral geometry	3	Open / open-ended
109	2018-geom-295	2018	Discrete and polyhedral geometry	1	Open / open-ended
110	2022-geom-175	2022	Discrete and polyhedral geometry	1	Partially resolved
111	2022-geom-269	2022	Discrete and polyhedral geometry	3	Open / open-ended
112	2022-geom-270	2022	Discrete and polyhedral geometry	5	Open / open-ended
113	2022-geom-296	2022	Discrete and polyhedral geometry	3	Open / open-ended
114	1908-geom-219	1908	Classical and local differential geometry	1	Open / open-ended
115	1908-geom-220	1908	Classical and local differential geometry	1	Open / open-ended
116	1912-geom-4	1912	Classical and local differential geometry	0	Partially resolved
117	1920-geom-221	1920	Classical and local differential geometry	0	Open / open-ended
118	1924-geom-222	1924	Classical and local differential geometry	1	Open / open-ended
119	1924-geom-223	1924	Classical and local differential geometry	2	Open / open-ended
120	1928-geom-224	1928	Classical and local differential geometry	0	Open / open-ended
121	1928-geom-225	1928	Classical and local differential geometry	1	Open / open-ended
122	1932-geom-271	1932	Classical and local differential geometry	0	Open / open-ended
123	1936-geom-227	1936	Classical and local differential geometry	0	Open / open-ended
124	1958-geom-9	1958	Classical and local differential geometry	0	Partially resolved
125	1970-geom-14	1970	Classical and local differential geometry	5	Partially resolved
126	1978-geom-24	1978	Classical and local differential geometry	1	Partially resolved
127	1978-geom-26	1978	Classical and local differential geometry	1	Partially resolved
128	1983-geom-287	1983	Classical and local differential geometry	1	Open / open-ended
129	1986-geom-41	1986	Classical and local differential geometry	1	Partially resolved
130	1986-geom-42	1986	Classical and local differential geometry	1	Partially resolved
131	1986-geom-43	1986	Classical and local differential geometry	1	Partially resolved
132	1986-geom-44	1986	Classical and local differential geometry	0	Partially resolved
133	1990-geom-306	1990	Classical and local differential geometry	2	Open / open-ended
134	1998-geom-339	1998	Classical and local differential geometry	0	Open / open-ended
135	2002-geom-252	2002	Classical and local differential geometry	2	Open / open-ended
136	2006-ag-115	2006	Classical and local differential geometry	1	Partially resolved
137	2006-ag-116	2006	Classical and local differential geometry	1	Partially resolved
138	2010-geom-104	2010	Classical and local differential geometry	0	Partially resolved
139	2010-geom-105	2010	Classical and local differential geometry	1	Partially resolved
140	2010-geom-106	2010	Classical and local differential geometry	2	Partially resolved
141	2010-geom-107	2010	Classical and local differential geometry	3	Partially resolved
142	2010-geom-108	2010	Classical and local differential geometry	0	Partially resolved
143	2010-geom-109	2010	Classical and local differential geometry	1	Partially resolved
144	2010-geom-110	2010	Classical and local differential geometry	0	Partially resolved
145	2014-geom-127	2014	Classical and local differential geometry	1	Partially resolved
146	2022-ag-217	2022	Classical and local differential geometry	2	Partially resolved
147	1994-ag-53	1994	Symplectic and contact geometry	2	Partially resolved
148	1994-ag-54	1994	Symplectic and contact geometry	1	Partially resolved
149	1998-geom-62	1998	Symplectic and contact geometry	1	Partially resolved
150	1998-geom-69	1998	Symplectic and contact geometry	0	Partially resolved
151	1998-geom-73	1998	Symplectic and contact geometry	1	Partially resolved
152	1998-ag-74	1998	Symplectic and contact geometry	5	Partially resolved
153	1998-ag-80	1998	Symplectic and contact geometry	4	Partially resolved
154	1998-geom-275	1998	Symplectic and contact geometry	1	Open / open-ended
155	1998-geom-310	1998	Symplectic and contact geometry	1	Open / open-ended
156	2002-geom-82	2002	Symplectic and contact geometry	2	Partially resolved

No.	ID	Year	Subfield	Open	Status
157	2002-geom-83	2002	Symplectic and contact geometry	1	Partially resolved
158	2002-geom-84	2002	Symplectic and contact geometry	0	Partially resolved
159	2002-geom-85	2002	Symplectic and contact geometry	1	Partially resolved
160	2002-ag-94	2002	Symplectic and contact geometry	6	Partially resolved
161	2002-geom-278	2002	Symplectic and contact geometry	1	Open / open-ended
162	2002-geom-279	2002	Symplectic and contact geometry	2	Open / open-ended
163	2006-geom-102	2006	Symplectic and contact geometry	1	Partially resolved
164	2006-ag-108	2006	Symplectic and contact geometry	3	Partially resolved
165	2006-ag-109	2006	Symplectic and contact geometry	2	Partially resolved
166	2006-geom-282	2006	Symplectic and contact geometry	1	Open / open-ended
167	2006-geom-285	2006	Symplectic and contact geometry	0	Open / open-ended
168	2006-geom-297	2006	Symplectic and contact geometry	3	Open / open-ended
169	2006-geom-316	2006	Symplectic and contact geometry	1	Open / open-ended
170	2014-geom-124	2014	Symplectic and contact geometry	1	Partially resolved
171	2014-geom-132	2014	Symplectic and contact geometry	1	Partially resolved
172	2014-ag-152	2014	Symplectic and contact geometry	3	Partially resolved
173	2014-ag-173	2014	Symplectic and contact geometry	4	Partially resolved
174	2014-geom-283	2014	Symplectic and contact geometry	2	Open / open-ended
175	2018-geom-299	2018	Symplectic and contact geometry	2	Open / open-ended
176	2022-geom-288	2022	Symplectic and contact geometry	1	Open / open-ended
177	2022-geom-332	2022	Symplectic and contact geometry	2	Open / open-ended
178	1932-geom-226	1932	Metric and topological geometry	1	Open / open-ended
179	1954-geom-228	1954	Metric and topological geometry	3	Open / open-ended
180	1958-geom-8	1958	Metric and topological geometry	1	Partially resolved
181	1994-geom-61	1994	Metric and topological geometry	1	Partially resolved
182	1994-geom-240	1994	Metric and topological geometry	1	Open / open-ended
183	1998-geom-65	1998	Metric and topological geometry	0	Partially resolved
184	1998-geom-72	1998	Metric and topological geometry	2	Partially resolved
185	2002-geom-79	2002	Metric and topological geometry	1	Partially resolved
186	2002-geom-89	2002	Metric and topological geometry	1	Partially resolved
187	2002-geom-90	2002	Metric and topological geometry	1	Partially resolved
188	2006-geom-95	2006	Metric and topological geometry	2	Partially resolved
189	2006-geom-101	2006	Metric and topological geometry	0	Partially resolved
190	2006-geom-281	2006	Metric and topological geometry	1	Open / open-ended
191	2010-geom-121	2010	Metric and topological geometry	1	Partially resolved
192	2014-geom-126	2014	Metric and topological geometry	1	Partially resolved
193	2018-geom-144	2018	Metric and topological geometry	1	Partially resolved
194	2018-geom-145	2018	Metric and topological geometry	1	Open / open-ended
195	2018-geom-146	2018	Metric and topological geometry	0	Partially resolved
196	2018-geom-147	2018	Metric and topological geometry	2	Partially resolved
197	2018-geom-148	2018	Metric and topological geometry	1	Partially resolved
198	2018-geom-150	2018	Metric and topological geometry	1	Partially resolved
199	2018-geom-151	2018	Metric and topological geometry	0	Open / open-ended
200	2018-geom-152	2018	Metric and topological geometry	0	Open / open-ended
201	2018-geom-153	2018	Metric and topological geometry	1	Partially resolved
202	2018-geom-154	2018	Metric and topological geometry	1	Partially resolved
203	2018-geom-261	2018	Metric and topological geometry	2	Open / open-ended
204	2022-geom-267	2022	Metric and topological geometry	2	Open / open-ended
205	2022-geom-268	2022	Metric and topological geometry	1	Open / open-ended
206	1974-geom-301	1974	Manifolds and global analysis	2	Open / open-ended
207	1978-ag-18	1978	Manifolds and global analysis	1	Partially resolved
208	1978-geom-27	1978	Manifolds and global analysis	0	Partially resolved
209	1983-geom-34	1983	Manifolds and global analysis	0	Partially resolved
210	1990-geom-47	1990	Manifolds and global analysis	0	Partially resolved
211	1990-geom-239	1990	Manifolds and global analysis	1	Open / open-ended
212	1994-geom-60	1994	Manifolds and global analysis	1	Partially resolved

No.	ID	Year	Subfield	Open	Status
213	2002-ag-88	2002	Manifolds and global analysis	4	Partially resolved
214	2006-geom-97	2006	Manifolds and global analysis	2	Partially resolved
215	2006-geom-98	2006	Manifolds and global analysis	3	Partially resolved
216	2006-geom-99	2006	Manifolds and global analysis	2	Partially resolved
217	2010-geom-103	2010	Manifolds and global analysis	1	Partially resolved
218	2010-geom-111	2010	Manifolds and global analysis	1	Partially resolved
219	2014-ag-161	2014	Manifolds and global analysis	3	Partially resolved
220	2018-ag-177	2018	Manifolds and global analysis	0	Partially resolved
221	2022-geom-165	2022	Manifolds and global analysis	2	Partially resolved
222	2022-geom-168	2022	Manifolds and global analysis	0	Partially resolved
223	2022-geom-170	2022	Manifolds and global analysis	5	Partially resolved
224	2022-geom-172	2022	Manifolds and global analysis	5	Partially resolved
225	2022-geom-174	2022	Manifolds and global analysis	1	Partially resolved
226	2022-ag-215	2022	Manifolds and global analysis	3	Partially resolved
227	1994-geom-243	1994	Geometric flows and geometric PDE	2	Open / open-ended
228	1994-geom-309	1994	Geometric flows and geometric PDE	0	Open / open-ended
229	1998-geom-71	1998	Geometric flows and geometric PDE	0	Partially resolved
230	2002-geom-256	2002	Geometric flows and geometric PDE	1	Open / open-ended
231	2002-geom-335	2002	Geometric flows and geometric PDE	1	Open / open-ended
232	2010-ag-139	2010	Geometric flows and geometric PDE	1	Partially resolved
233	2014-geom-123	2014	Geometric flows and geometric PDE	1	Partially resolved
234	2014-geom-125	2014	Geometric flows and geometric PDE	1	Partially resolved
235	2014-geom-133	2014	Geometric flows and geometric PDE	2	Partially resolved
236	2014-geom-134	2014	Geometric flows and geometric PDE	1	Partially resolved
237	2014-ag-153	2014	Geometric flows and geometric PDE	3	Partially resolved
238	2014-geom-259	2014	Geometric flows and geometric PDE	1	Open / open-ended
239	2018-geom-262	2018	Geometric flows and geometric PDE	2	Open / open-ended
240	2018-geom-263	2018	Geometric flows and geometric PDE	1	Open / open-ended
241	2018-geom-284	2018	Geometric flows and geometric PDE	0	Open / open-ended
242	2022-geom-266	2022	Geometric flows and geometric PDE	3	Open / open-ended
243	2022-geom-331	2022	Geometric flows and geometric PDE	2	Open / open-ended
244	2022-geom-334	2022	Geometric flows and geometric PDE	4	Open / open-ended
245	2022-geom-337	2022	Geometric flows and geometric PDE	4	Open / open-ended
246	1974-geom-233	1974	Convex geometry	3	Open / open-ended
247	1974-geom-234	1974	Convex geometry	1	Open / open-ended
248	1974-geom-235	1974	Convex geometry	2	Open / open-ended
249	1974-geom-236	1974	Convex geometry	1	Open / open-ended
250	1974-geom-237	1974	Convex geometry	1	Open / open-ended
251	1974-geom-273	1974	Convex geometry	0	Open / open-ended
252	1990-geom-45	1990	Convex geometry	0	Partially resolved
253	2002-geom-251	2002	Convex geometry	1	Open / open-ended
254	2006-geom-96	2006	Convex geometry	1	Partially resolved
255	2006-geom-289	2006	Convex geometry	1	Open / open-ended
256	2022-geom-169	2022	Convex geometry	0	Partially resolved
257	2022-geom-265	2022	Convex geometry	2	Open / open-ended
258	1986-geom-38	1986	Applications of geometry / global analysis	0	Partially resolved
259	1994-geom-58	1994	Applications of geometry / global analysis	1	Partially resolved
260	1998-ag-82	1998	Applications of geometry / global analysis	3	Partially resolved
261	2002-ag-95	2002	Applications of geometry / global analysis	5	Partially resolved
262	2018-geom-140	2018	Applications of geometry / global analysis	1	Partially resolved
263	2018-geom-141	2018	Applications of geometry / global analysis	1	Partially resolved
264	2018-geom-142	2018	Applications of geometry / global analysis	1	Partially resolved
265	2018-geom-143	2018	Applications of geometry / global analysis	3	Partially resolved
266	1893-geom-1	1893	Incidence, axiomatic, and classical geometry	0	Partially resolved
267	1893-geom-2	1893	Incidence, axiomatic, and classical geometry	1	Partially resolved
268	1958-geom-272	1958	Incidence, axiomatic, and classical geometry	2	Open / open-ended

No.	ID	Year	Subfield	Open	Status
269	1970-geom-11	1970	Incidence, axiomatic, and classical geometry	5	Partially resolved
270	1986-geom-39	1986	Incidence, axiomatic, and classical geometry	1	Partially resolved
271	1998-geom-249	1998	Incidence, axiomatic, and classical geometry	3	Open / open-ended
272	2014-geom-258	2014	Incidence, axiomatic, and classical geometry	0	Open / open-ended
273	2022-geom-160	2022	Incidence, axiomatic, and classical geometry	1	Partially resolved
274	1970-geom-229	1970	Singularity theory	2	Open / open-ended
275	1974-geom-230	1974	Singularity theory	0	Open / open-ended
276	1974-geom-231	1974	Singularity theory	1	Open / open-ended
277	1974-geom-232	1974	Singularity theory	0	Open / open-ended
278	2018-geom-330	2018	Singularity theory	1	Open / open-ended

B. Original ICM source catalogue

The entries below catalogue the original ICM sources associated with the 278 records. Later research literature is retained in the problem-by-problem narratives.

1. **1893.** *CONSIDÉRATIONS GÉNÉRALES SUR LA MESURE DE LA SIMPLICITÉ DANS LES SCIENCES MATHÉMATIQUES ET APPLICATION L'ÉVALUATION THÉORIQUE DE LA SIMPLICITÉ DES TRACÉS GÉOMÉTRIQUES, OU GÉOMÉTROGRAPHIE.* <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1893/ICM1893.ocr.pdf#page=159> Associated records: 1893-geom-1.
2. **1893.** *V. Theory of Functions and Geometry.* <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1893.2/ICM1893.2.ocr.pdf#page=43> Associated records: 1893-geom-2.
3. **1904.** *Über die diskontinuierlichen und nicht-konvexen gleichseitig-gleichflächigen Polyeder.* <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1904/ICM1904.ocr.pdf#page=717> Associated records: 1904-geom-218.
4. **1908.** *Les origines, les méthodes et les problèmes de la Géométrie infinitésimale.* <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1908.1/ICM1908.1.ocr.pdf#page=109> Associated records: 1908-geom-219.
5. **1908.** *Sulle trasformazioni di Darboux delle superficie d'area minima.* <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1908.2/ICM1908.2.ocr.pdf#page=264> Associated records: 1908-geom-220.
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8. **1924.** *G., Sur les mouvements à deux paramètres doublement décomposables.* <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1924.1/ICM1924.1.ocr.pdf#page=913> Associated records: 1924-geom-222.
9. **1924.** *Riassunto di alcune ricerche di geometria proiettivo-differenziale.* <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1924.1/ICM1924.1.ocr.pdf#page=855> Associated records: 1924-geom-223.
10. **1928.** *Beispiele von Elementarkurven und Elementarflächen.* <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1928.4/ICM1928.4.ocr.pdf#page=197> Associated records: 1928-geom-224, 1928-geom-225.
11. **1932.** *Einige Bemerkungen über Winkelmetrik in Finslerschen Räumen.* <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM1932.2/ICM1932.2.ocr.pdf#page=199> Associated records: 1932-geom-271.
12. **1932.** *Un problème de M. Bricard.* <https://www.mathunion.org/fileadmin/ICM1932.2/ICM1932.2.ocr.pdf#page=199>

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160. **2022.** *Rigidity, lattices, and invariant measures beyond homogeneous dynamics.* <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-563-6-volume-5.pdf#page=129> Associated records: 2022-geom-168.
161. **2022.** *Slices and distances on two problems of Furstenberg and Falconer.* <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-562-9-volume-4.pdf#page=909> Associated records: 2022-geom-268.
162. **2022.** *Some questions related to the reverse Minkowski theorem.* <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-564-3-volume-6.pdf#page=747> Associated records: 2022-geom-169.
163. **2022.** *Some recent developments in Ricci flow.* <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-562-9-volume-4.pdf#page=75> Associated records: 2022-geom-334.
164. **2022.** *The magic of 8 and 24.* <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-559-9-volume-1.pdf#page=511> Associated records: 2022-geom-269, 2022-geom-270.
165. **2022.** *The regularity theory for the area functional (in geometric measure theory).* <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-560-5-volume-2.pdf#page=313> Associated records: 2022-geom-170, 2022-geom-172, 2022-geom-174.
166. **2022.** *The work of Maryna Viazovska.* <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-559-9-volume-1.pdf#page=101> Associated records: 2022-geom-296.
167. **2022.** *the upper bound side of the story.* <https://www.mathunion.org/fileadmin/ICM/Proceedings/ICM2022/978-3-98547-564-3-volume-6.pdf#page=471> Associated records: 2022-geom-175.

C. Index of separately recorded open problems

The website payload contains 393 separately enumerated descendant or residual questions in addition to the 278 main statements. Their full wording and supplied references appear in the relevant body sections; this appendix provides a compact locator.

No.	Record	Subfield	Count
1	1958-geom-7	Global/Riemannian geometry	1
2	1970-geom-12	Global/Riemannian geometry	1
3	1970-geom-13	Global/Riemannian geometry	4
4	1974-geom-20	Global/Riemannian geometry	1
5	1974-geom-23	Global/Riemannian geometry	1
6	1974-geom-300	Global/Riemannian geometry	1
7	1978-geom-25	Global/Riemannian geometry	2
8	1978-geom-28	Global/Riemannian geometry	1
9	1978-geom-29	Global/Riemannian geometry	1
11	1983-geom-32	Global/Riemannian geometry	2
12	1983-geom-33	Global/Riemannian geometry	1
14	1986-geom-290	Global/Riemannian geometry	1
15	1986-geom-304	Global/Riemannian geometry	1
16	1990-geom-305	Global/Riemannian geometry	1
17	1990-geom-307	Global/Riemannian geometry	2
18	1994-geom-51	Global/Riemannian geometry	1
20	1994-geom-308	Global/Riemannian geometry	2
21	1994-geom-338	Global/Riemannian geometry	1
22	1998-geom-63	Global/Riemannian geometry	1
23	1998-geom-64	Global/Riemannian geometry	1
24	1998-geom-74	Global/Riemannian geometry	1
26	2002-geom-81	Global/Riemannian geometry	2
27	2002-ag-91	Global/Riemannian geometry	1
29	2002-ag-93	Global/Riemannian geometry	3
30	2002-geom-253	Global/Riemannian geometry	2
31	2002-geom-254	Global/Riemannian geometry	2
33	2002-geom-276	Global/Riemannian geometry	1
34	2002-geom-277	Global/Riemannian geometry	1
35	2002-geom-280	Global/Riemannian geometry	2
36	2002-geom-311	Global/Riemannian geometry	2
37	2002-geom-312	Global/Riemannian geometry	1
38	2002-geom-313	Global/Riemannian geometry	1
39	2002-geom-314	Global/Riemannian geometry	3
40	2002-geom-315	Global/Riemannian geometry	2
42	2006-geom-340	Global/Riemannian geometry	1
44	2010-geom-115	Global/Riemannian geometry	1

No.	Record	Subfield	Count
45	2010-geom-116	Global/Riemannian geometry	1
46	2010-geom-118	Global/Riemannian geometry	1
48	2010-geom-286	Global/Riemannian geometry	2
49	2010-geom-317	Global/Riemannian geometry	4
50	2010-geom-318	Global/Riemannian geometry	1
51	2010-geom-319	Global/Riemannian geometry	2
53	2010-geom-321	Global/Riemannian geometry	1
54	2010-geom-322	Global/Riemannian geometry	2
55	2010-geom-323	Global/Riemannian geometry	3
57	2014-geom-128	Global/Riemannian geometry	1
58	2014-geom-129	Global/Riemannian geometry	1
60	2014-geom-135	Global/Riemannian geometry	1
61	2014-geom-136	Global/Riemannian geometry	1
62	2014-geom-137	Global/Riemannian geometry	1
63	2014-geom-294	Global/Riemannian geometry	2
64	2014-geom-324	Global/Riemannian geometry	2
65	2014-geom-325	Global/Riemannian geometry	1
66	2014-geom-326	Global/Riemannian geometry	3
67	2014-geom-327	Global/Riemannian geometry	3
68	2014-geom-336	Global/Riemannian geometry	2
69	2018-geom-138	Global/Riemannian geometry	1
70	2018-geom-298	Global/Riemannian geometry	1
71	2018-geom-328	Global/Riemannian geometry	2
72	2018-geom-329	Global/Riemannian geometry	1
73	2022-geom-161	Global/Riemannian geometry	2
74	2022-geom-162	Global/Riemannian geometry	1
75	2022-geom-163	Global/Riemannian geometry	3
76	2022-geom-164	Global/Riemannian geometry	1
77	2022-geom-166	Global/Riemannian geometry	2
78	2022-geom-333	Global/Riemannian geometry	1
79	1904-geom-218	Discrete and polyhedral geometry	2
80	1950-geom-6	Discrete and polyhedral geometry	2
81	1970-geom-16	Discrete and polyhedral geometry	1
82	1978-geom-238	Discrete and polyhedral geometry	2
83	1978-geom-302	Discrete and polyhedral geometry	1
84	1986-geom-274	Discrete and polyhedral geometry	2
86	1994-geom-53	Discrete and polyhedral geometry	1
87	1994-geom-54	Discrete and polyhedral geometry	1
88	1994-geom-55	Discrete and polyhedral geometry	2
89	1994-geom-241	Discrete and polyhedral geometry	1
90	1994-geom-242	Discrete and polyhedral geometry	3
91	1998-geom-244	Discrete and polyhedral geometry	3
92	1998-geom-245	Discrete and polyhedral geometry	3
94	1998-geom-247	Discrete and polyhedral geometry	1
95	1998-geom-248	Discrete and polyhedral geometry	3
97	2002-geom-77	Discrete and polyhedral geometry	1
98	2002-geom-78	Discrete and polyhedral geometry	1
101	2002-geom-250	Discrete and polyhedral geometry	2
102	2002-geom-292	Discrete and polyhedral geometry	3

No.	Record	Subfield	Count
103	2002-geom-293	Discrete and polyhedral geometry	1
104	2010-geom-257	Discrete and polyhedral geometry	2
105	2014-geom-260	Discrete and polyhedral geometry	3
106	2018-geom-149	Discrete and polyhedral geometry	1
107	2018-geom-158	Discrete and polyhedral geometry	1
108	2018-geom-264	Discrete and polyhedral geometry	3
109	2018-geom-295	Discrete and polyhedral geometry	1
110	2022-geom-175	Discrete and polyhedral geometry	1
111	2022-geom-269	Discrete and polyhedral geometry	3
112	2022-geom-270	Discrete and polyhedral geometry	5
113	2022-geom-296	Discrete and polyhedral geometry	3
114	1908-geom-219	Classical and local differential geometry	1
115	1908-geom-220	Classical and local differential geometry	1
118	1924-geom-222	Classical and local differential geometry	1
119	1924-geom-223	Classical and local differential geometry	2
121	1928-geom-225	Classical and local differential geometry	1
125	1970-geom-14	Classical and local differential geometry	5
126	1978-geom-24	Classical and local differential geometry	1
127	1978-geom-26	Classical and local differential geometry	1
128	1983-geom-287	Classical and local differential geometry	1
129	1986-geom-41	Classical and local differential geometry	1
130	1986-geom-42	Classical and local differential geometry	1
131	1986-geom-43	Classical and local differential geometry	1
133	1990-geom-306	Classical and local differential geometry	2
135	2002-geom-252	Classical and local differential geometry	2
136	2006-ag-115	Classical and local differential geometry	1
137	2006-ag-116	Classical and local differential geometry	1
139	2010-geom-105	Classical and local differential geometry	1
140	2010-geom-106	Classical and local differential geometry	2
141	2010-geom-107	Classical and local differential geometry	3
143	2010-geom-109	Classical and local differential geometry	1
145	2014-geom-127	Classical and local differential geometry	1
146	2022-ag-217	Classical and local differential geometry	2
147	1994-ag-53	Symplectic and contact geometry	2
148	1994-ag-54	Symplectic and contact geometry	1
149	1998-geom-62	Symplectic and contact geometry	1
151	1998-geom-73	Symplectic and contact geometry	1
152	1998-ag-74	Symplectic and contact geometry	5
153	1998-ag-80	Symplectic and contact geometry	4
154	1998-geom-275	Symplectic and contact geometry	1
155	1998-geom-310	Symplectic and contact geometry	1
156	2002-geom-82	Symplectic and contact geometry	2
157	2002-geom-83	Symplectic and contact geometry	1
159	2002-geom-85	Symplectic and contact geometry	1
160	2002-ag-94	Symplectic and contact geometry	6
161	2002-geom-278	Symplectic and contact geometry	1
162	2002-geom-279	Symplectic and contact geometry	2
163	2006-geom-102	Symplectic and contact geometry	1
164	2006-ag-108	Symplectic and contact geometry	3

No.	Record	Subfield	Count
165	2006-ag-109	Symplectic and contact geometry	2
166	2006-geom-282	Symplectic and contact geometry	1
168	2006-geom-297	Symplectic and contact geometry	3
169	2006-geom-316	Symplectic and contact geometry	1
170	2014-geom-124	Symplectic and contact geometry	1
171	2014-geom-132	Symplectic and contact geometry	1
172	2014-ag-152	Symplectic and contact geometry	3
173	2014-ag-173	Symplectic and contact geometry	4
174	2014-geom-283	Symplectic and contact geometry	2
175	2018-geom-299	Symplectic and contact geometry	2
176	2022-geom-288	Symplectic and contact geometry	1
177	2022-geom-332	Symplectic and contact geometry	2
178	1932-geom-226	Metric and topological geometry	1
179	1954-geom-228	Metric and topological geometry	3
180	1958-geom-8	Metric and topological geometry	1
181	1994-geom-61	Metric and topological geometry	1
182	1994-geom-240	Metric and topological geometry	1
184	1998-geom-72	Metric and topological geometry	2
185	2002-geom-79	Metric and topological geometry	1
186	2002-geom-89	Metric and topological geometry	1
187	2002-geom-90	Metric and topological geometry	1
188	2006-geom-95	Metric and topological geometry	2
190	2006-geom-281	Metric and topological geometry	1
191	2010-geom-121	Metric and topological geometry	1
192	2014-geom-126	Metric and topological geometry	1
193	2018-geom-144	Metric and topological geometry	1
194	2018-geom-145	Metric and topological geometry	1
196	2018-geom-147	Metric and topological geometry	2
197	2018-geom-148	Metric and topological geometry	1
198	2018-geom-150	Metric and topological geometry	1
201	2018-geom-153	Metric and topological geometry	1
202	2018-geom-154	Metric and topological geometry	1
203	2018-geom-261	Metric and topological geometry	2
204	2022-geom-267	Metric and topological geometry	2
205	2022-geom-268	Metric and topological geometry	1
206	1974-geom-301	Manifolds and global analysis	2
207	1978-ag-18	Manifolds and global analysis	1
211	1990-geom-239	Manifolds and global analysis	1
212	1994-geom-60	Manifolds and global analysis	1
213	2002-ag-88	Manifolds and global analysis	4
214	2006-geom-97	Manifolds and global analysis	2
215	2006-geom-98	Manifolds and global analysis	3
216	2006-geom-99	Manifolds and global analysis	2
217	2010-geom-103	Manifolds and global analysis	1
218	2010-geom-111	Manifolds and global analysis	1
219	2014-ag-161	Manifolds and global analysis	3
221	2022-geom-165	Manifolds and global analysis	2
223	2022-geom-170	Manifolds and global analysis	5
224	2022-geom-172	Manifolds and global analysis	5

No.	Record	Subfield	Count
225	2022-geom-174	Manifolds and global analysis	1
226	2022-ag-215	Manifolds and global analysis	3
227	1994-geom-243	Geometric flows and geometric PDE	2
230	2002-geom-256	Geometric flows and geometric PDE	1
231	2002-geom-335	Geometric flows and geometric PDE	1
232	2010-ag-139	Geometric flows and geometric PDE	1
233	2014-geom-123	Geometric flows and geometric PDE	1
234	2014-geom-125	Geometric flows and geometric PDE	1
235	2014-geom-133	Geometric flows and geometric PDE	2
236	2014-geom-134	Geometric flows and geometric PDE	1
237	2014-ag-153	Geometric flows and geometric PDE	3
238	2014-geom-259	Geometric flows and geometric PDE	1
239	2018-geom-262	Geometric flows and geometric PDE	2
240	2018-geom-263	Geometric flows and geometric PDE	1
242	2022-geom-266	Geometric flows and geometric PDE	3
243	2022-geom-331	Geometric flows and geometric PDE	2
244	2022-geom-334	Geometric flows and geometric PDE	4
245	2022-geom-337	Geometric flows and geometric PDE	4
246	1974-geom-233	Convex geometry	3
247	1974-geom-234	Convex geometry	1
248	1974-geom-235	Convex geometry	2
249	1974-geom-236	Convex geometry	1
250	1974-geom-237	Convex geometry	1
253	2002-geom-251	Convex geometry	1
254	2006-geom-96	Convex geometry	1
255	2006-geom-289	Convex geometry	1
257	2022-geom-265	Convex geometry	2
259	1994-geom-58	Applications of geometry / global analysis	1
260	1998-ag-82	Applications of geometry / global analysis	3
261	2002-ag-95	Applications of geometry / global analysis	5
262	2018-geom-140	Applications of geometry / global analysis	1
263	2018-geom-141	Applications of geometry / global analysis	1
264	2018-geom-142	Applications of geometry / global analysis	1
265	2018-geom-143	Applications of geometry / global analysis	3
267	1893-geom-2	Incidence, axiomatic, and classical geometry	1
268	1958-geom-272	Incidence, axiomatic, and classical geometry	2
269	1970-geom-11	Incidence, axiomatic, and classical geometry	5
270	1986-geom-39	Incidence, axiomatic, and classical geometry	1
271	1998-geom-249	Incidence, axiomatic, and classical geometry	3
273	2022-geom-160	Incidence, axiomatic, and classical geometry	1
274	1970-geom-229	Singularity theory	2
276	1974-geom-231	Singularity theory	1
278	2018-geom-330	Singularity theory	1

D. Data provenance, reproducibility, and limitations

D.1. Included project files

data/geometry_current_snapshot.json

The exact Geometry-only records used to generate the body.

data/icm_records_full_2026-08-14.json

The complete API payload frozen at collection time.

data/coverage_current.csv

One audit row per body problem, including status, review flag, open-problem count, source metadata, and text lengths.

data/source_catalog.csv

Unique original sources and the record identifiers attached to them.

data/link_inventory.tsv

Every distinct URL extracted from statements, progress narratives, and remaining-problem references, with referring record identifiers.

scripts/generate_survey.py

The deterministic generator for chapters and audit appendices.

D.2. Rebuilding

From the project directory, run:

```
python3 scripts/generate_survey.py
latexmk -xelatex -interaction=nonstopmode -halt-on-error main.tex
```

This requires Python 3, Pandoc 3, and a TeX distribution containing XeLaTeX and `latexmk`. Regeneration reads the included frozen JSON snapshot and does not contact the live website.

The generator asserts that all 278 Geometry records and all 11 subfield counts agree with the website’s category metadata. A failed assertion stops generation rather than silently producing an incomplete volume.

D.3. Interpretive limitations

This edition records a rapidly changing research landscape at a specific timestamp. Four limitations should remain visible.

1. A website status is not a substitute for a referee’s verification of the cited theorem.

2. “No later resolution found” is evidence about a literature search, not a theorem of nonexistence.
3. Several ICM questions were originally programs rather than sharply delimited conjectures; later authors may formulate their surviving cores differently.
4. New preprints or accepted papers after the snapshot date can change a verdict without changing the historical value of the earlier record.

D.4. Recommended scholarly use

Use this survey as a map into the primary literature. Before quoting a status claim in new research, follow the supplied DOI or arXiv link, check the exact hypotheses and publication version, and search for work appearing after 14 August 2026. The explicit record identifier and frozen snapshot make corrections and future editions traceable.

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